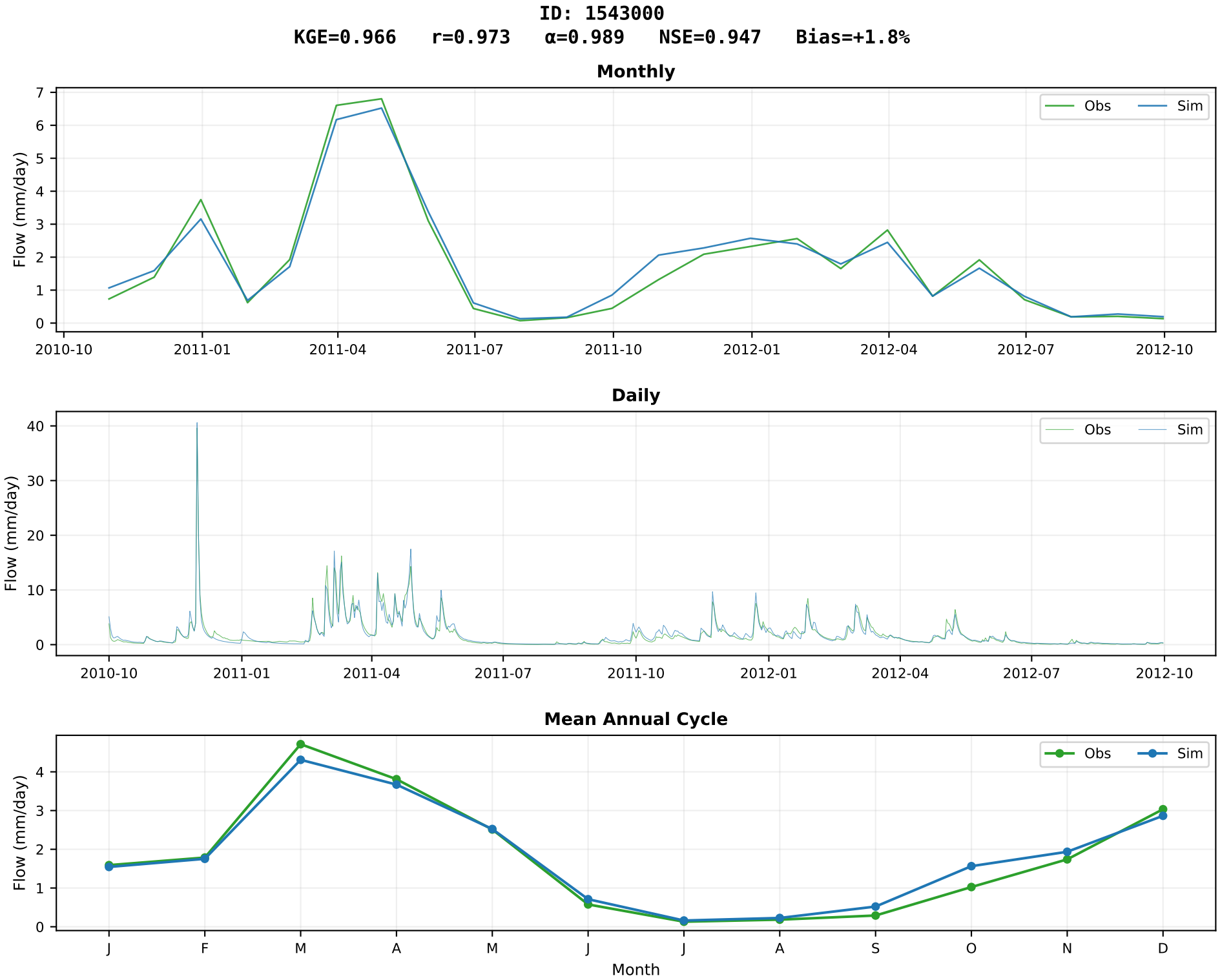
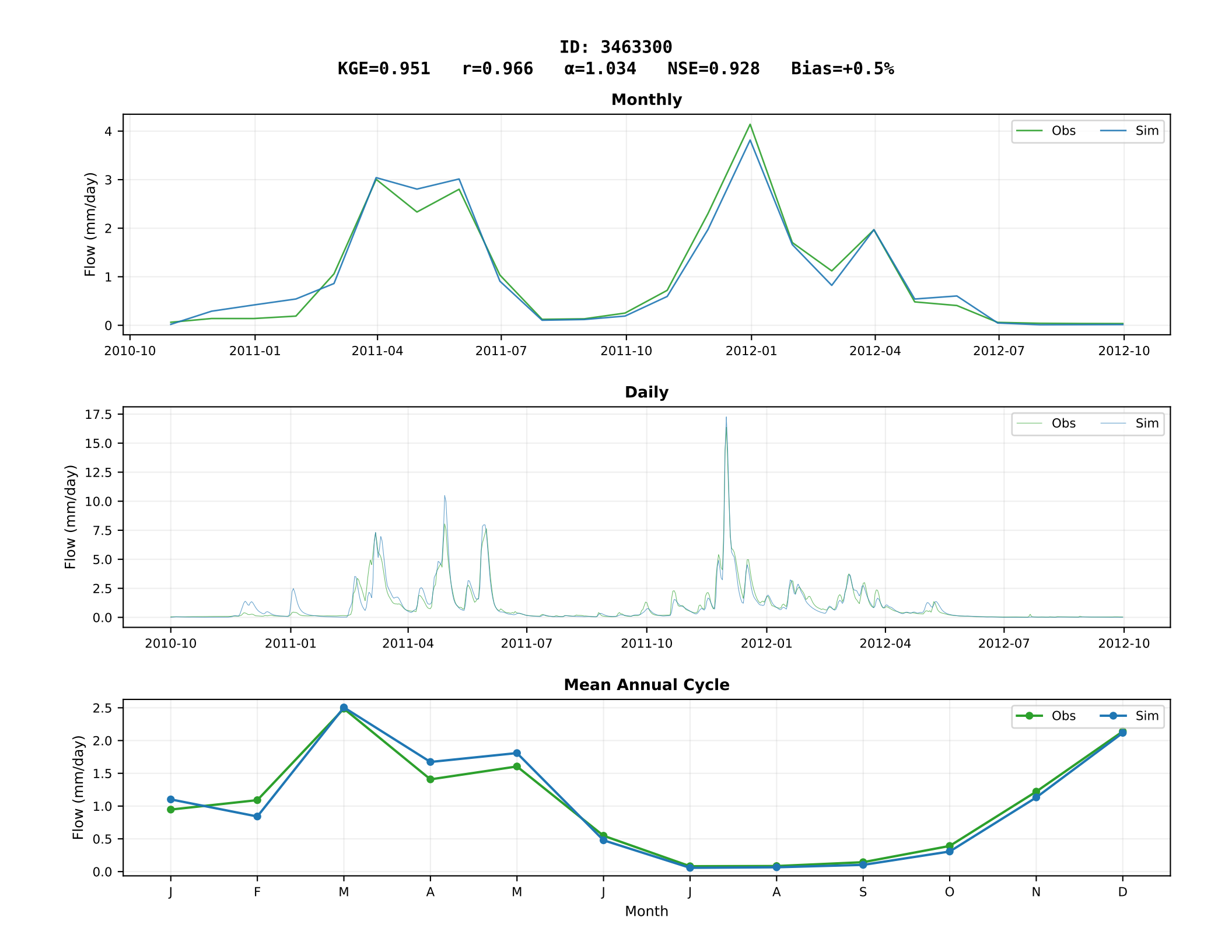
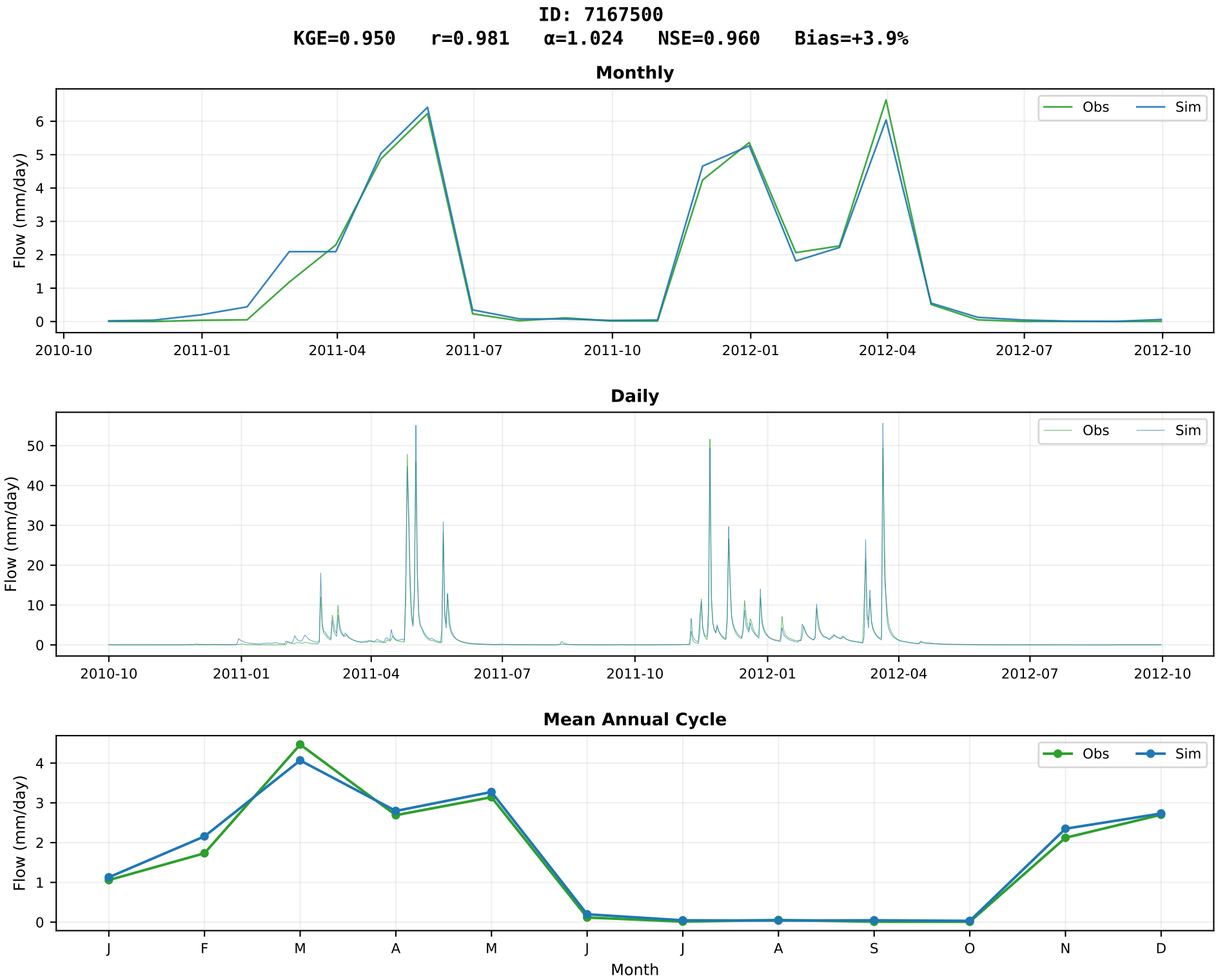
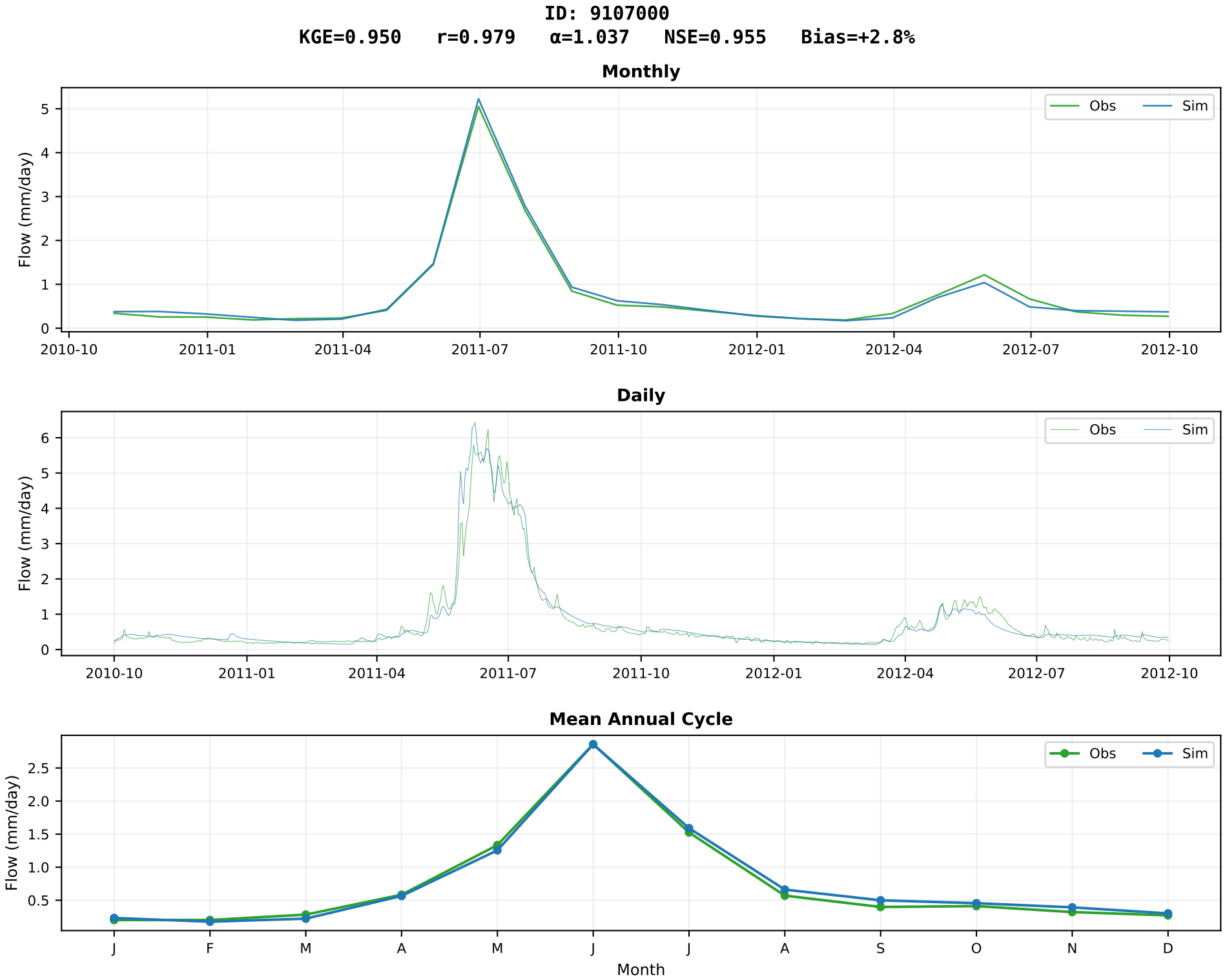
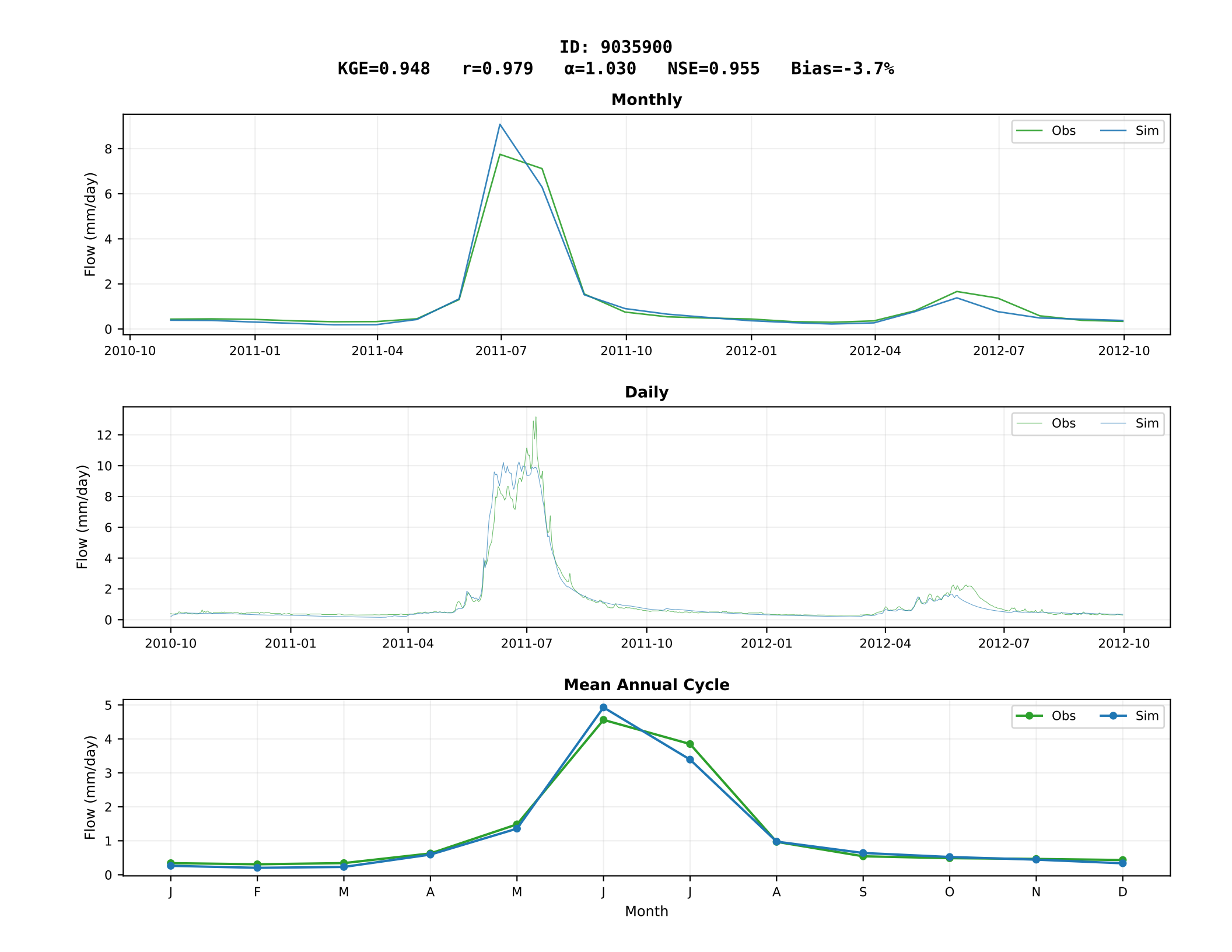
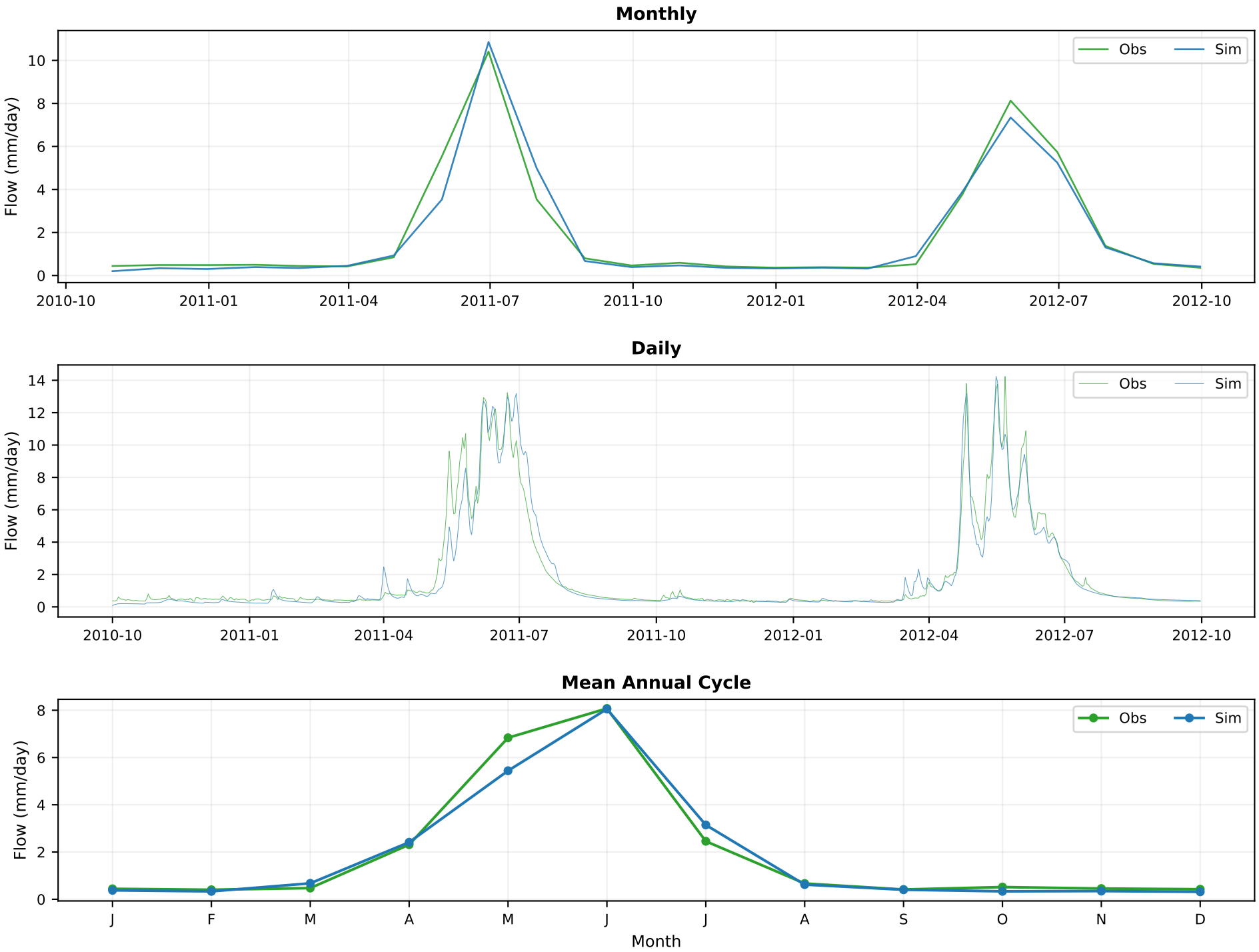

ID: 1543000
KGE=0.966 r=0.973 α=0.989 NSE=0.947 Bias=+1.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

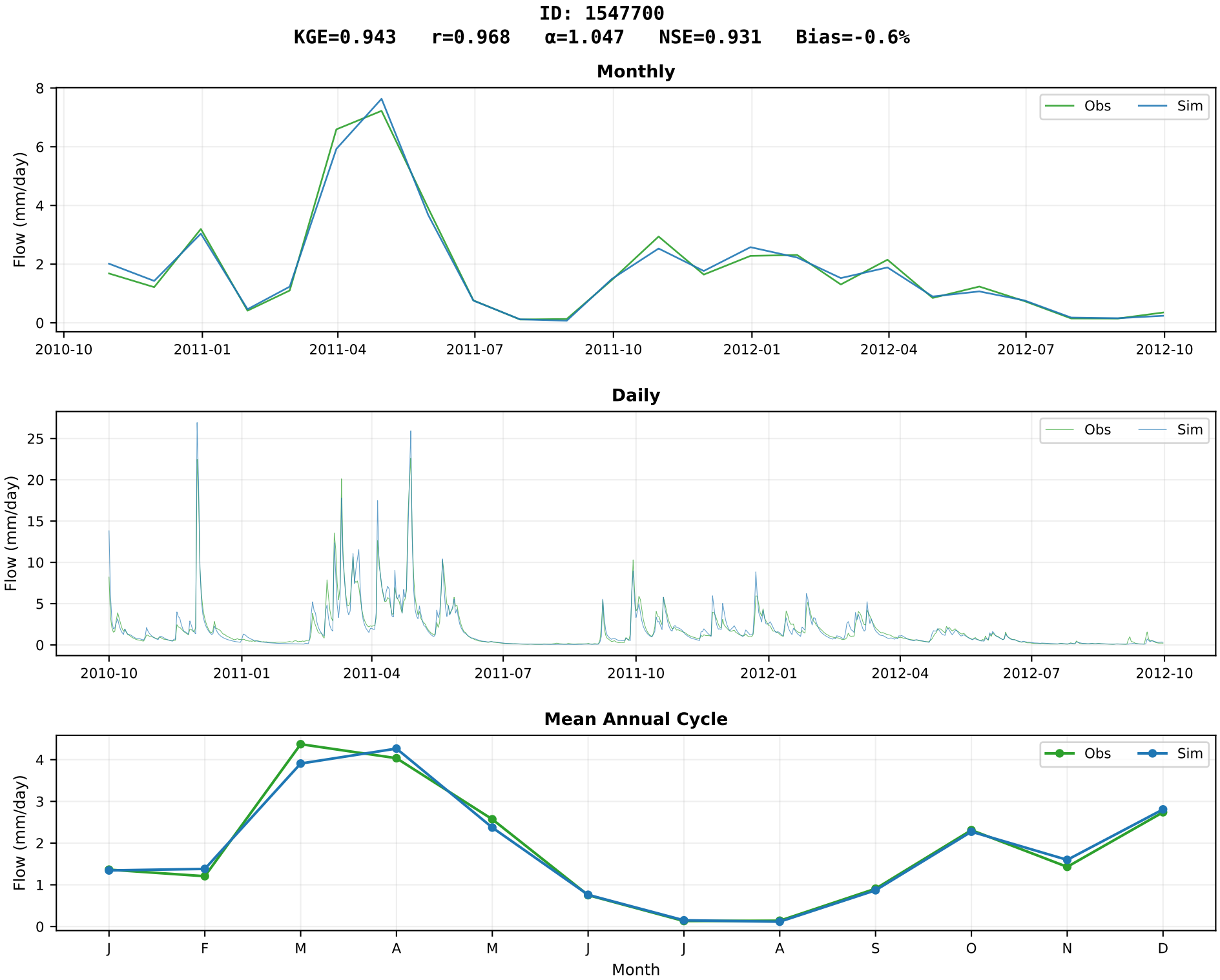

ID: 3463300
KGE=0.951 r=0.966 α=1.034 NSE=0.928 Bias=+0.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Mean Annual Cycle
Flow (mm/day)
J
F
M
A
M
J
J
A
S
O
N
D
Month

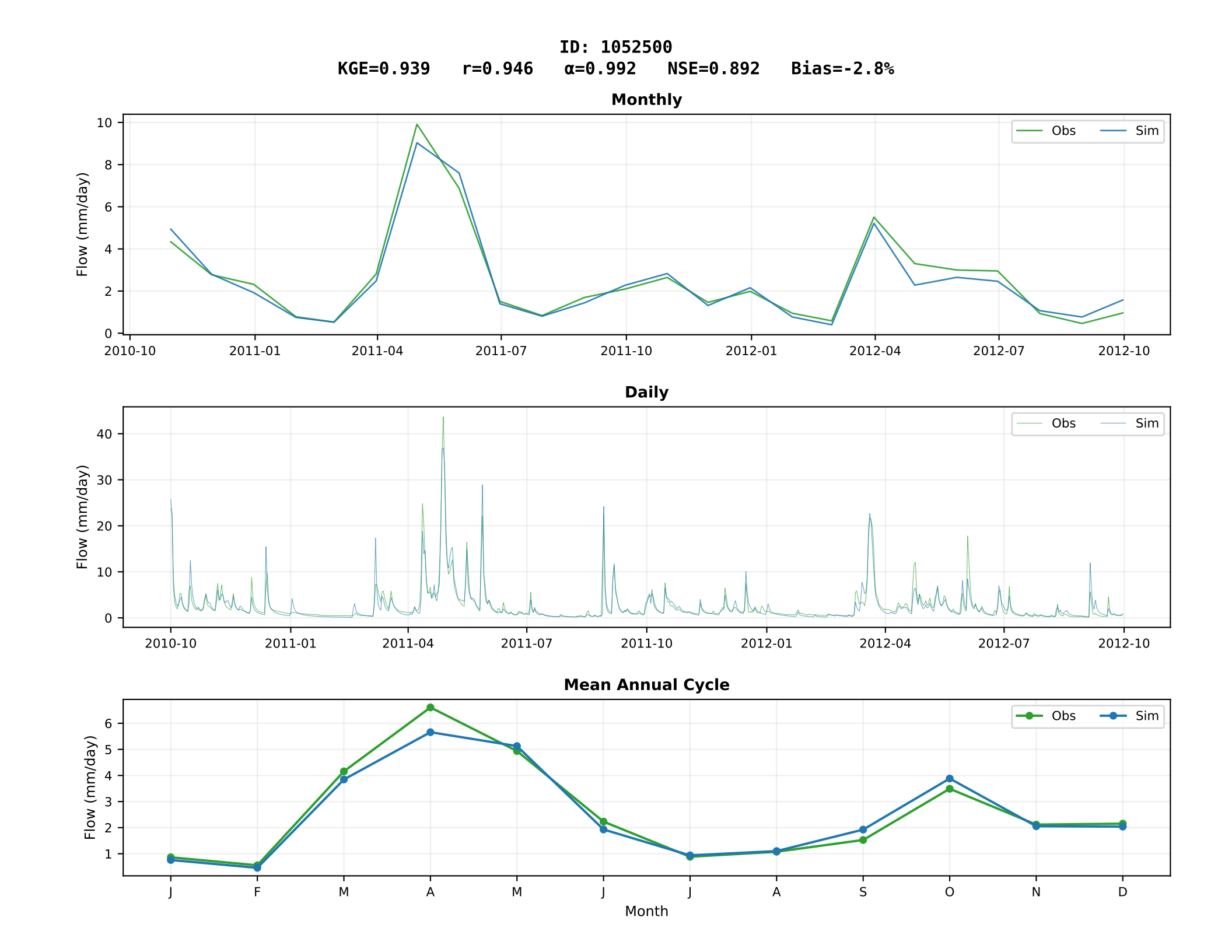

ID: 7167500
KGE=0.950 r=0.981 α=1.024 NSE=0.960 Bias=+3.9%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

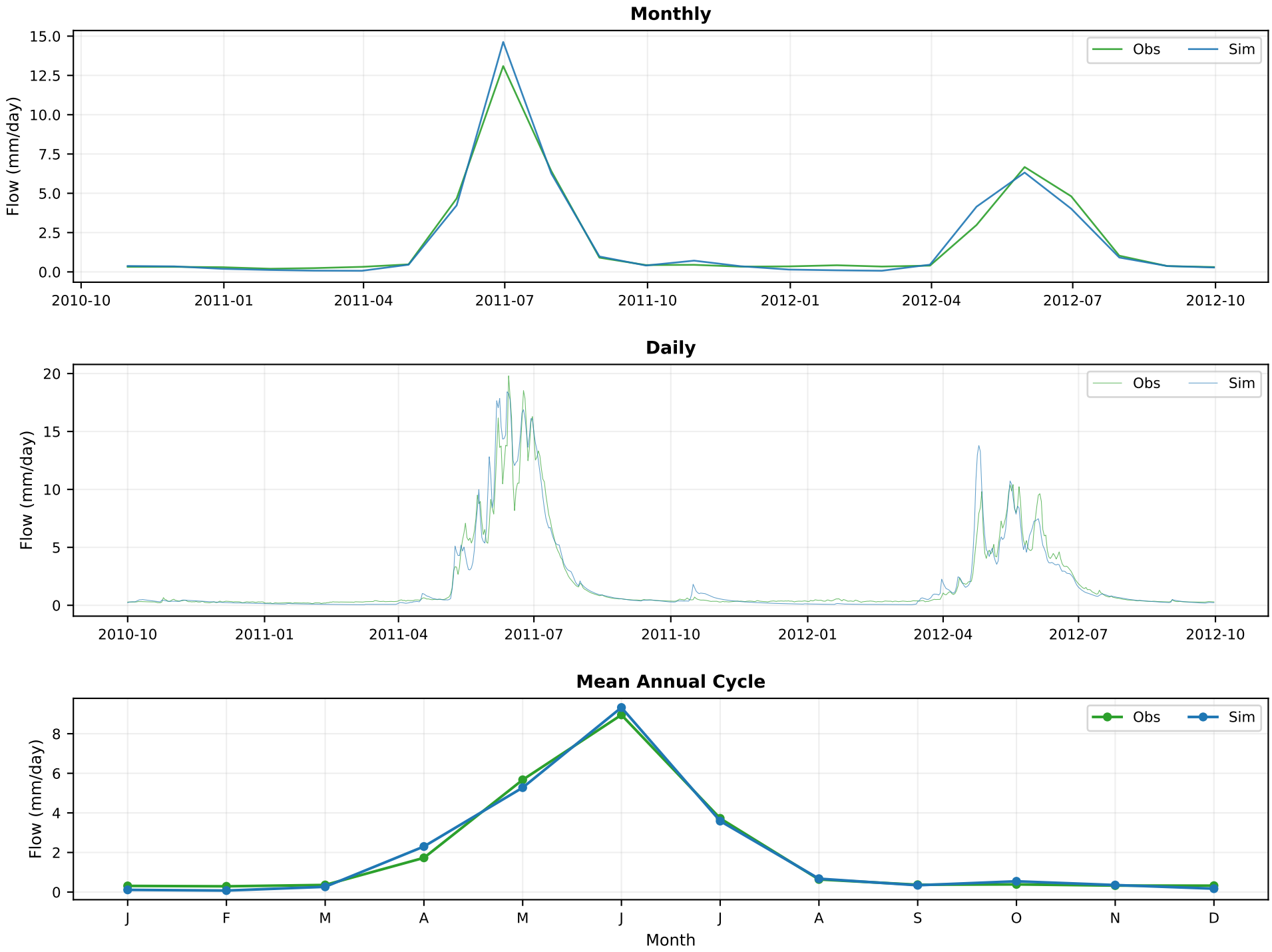

ID: 9107000
KGE=0.950 r=0.979 α=1.037 NSE=0.955 Bias=+2.8%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.5
1.0
1.5
2.0
2.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

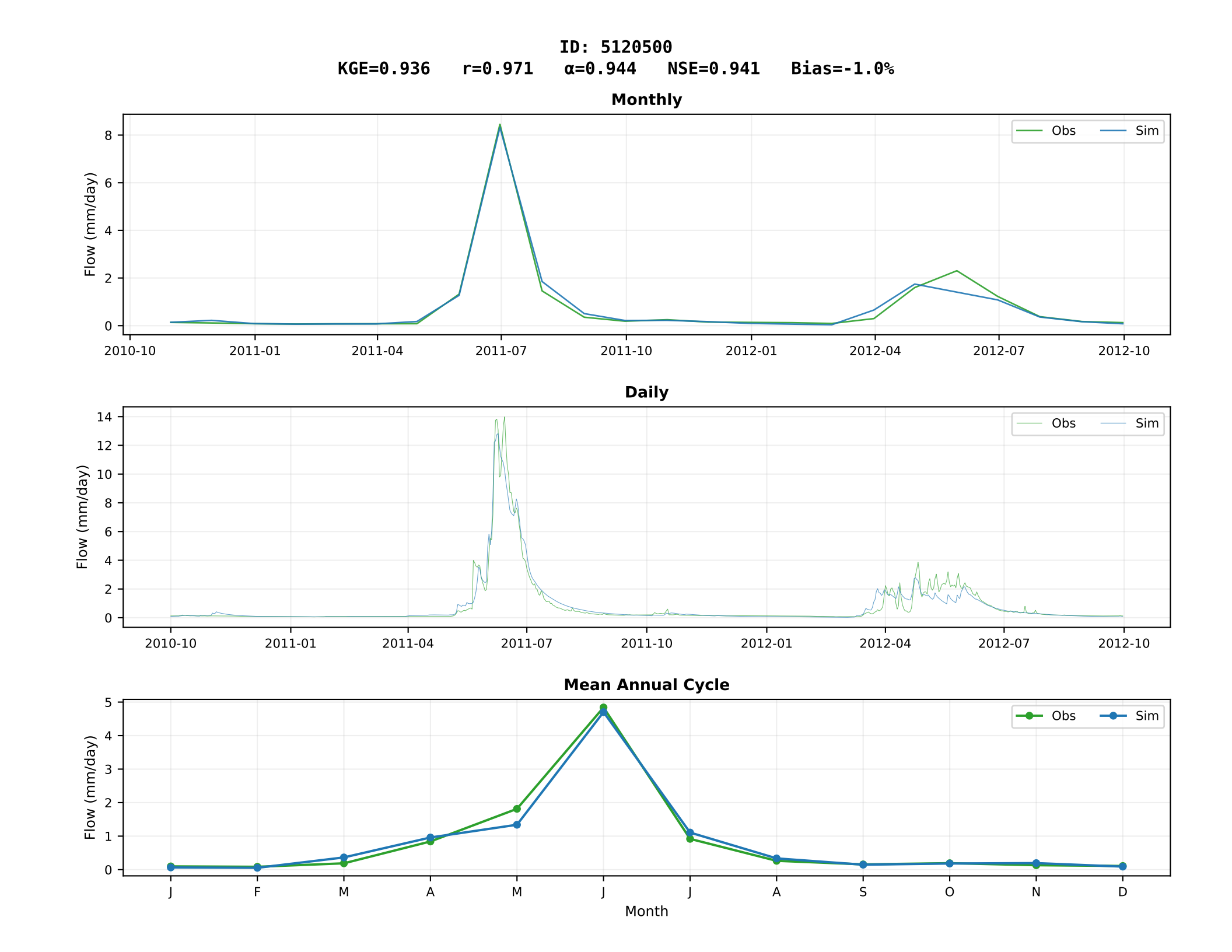

ID: 9035900
KGE=0.948 r=0.979 α=1.030 NSE=0.955 Bias=-3.7%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
12
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

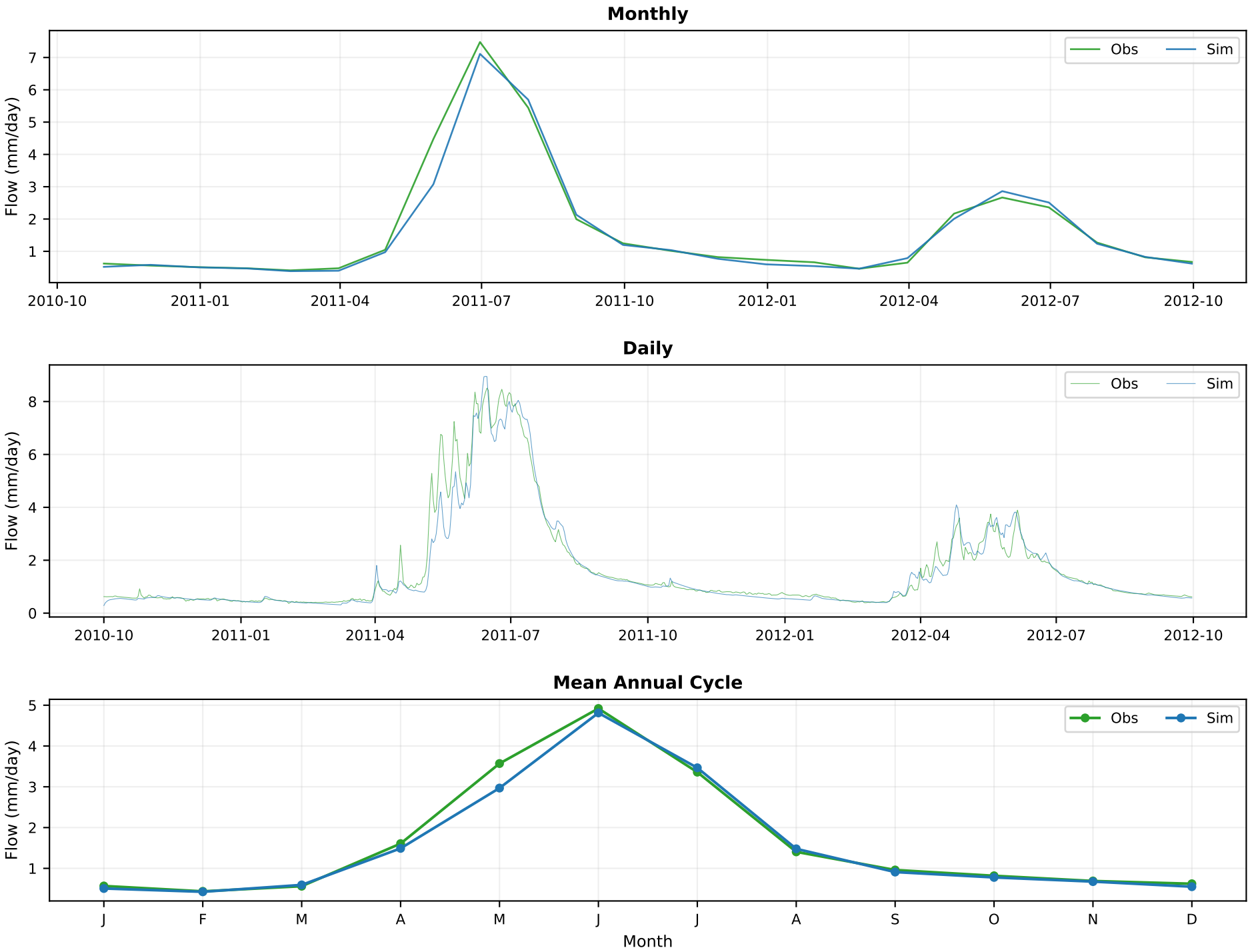

WEST FORK COW CREEK NEAR GLENDALE, OR (ID=14309500)
KGE=0.945 r=0.967 α=0.996 NSE=0.933 Bias=-4.4%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

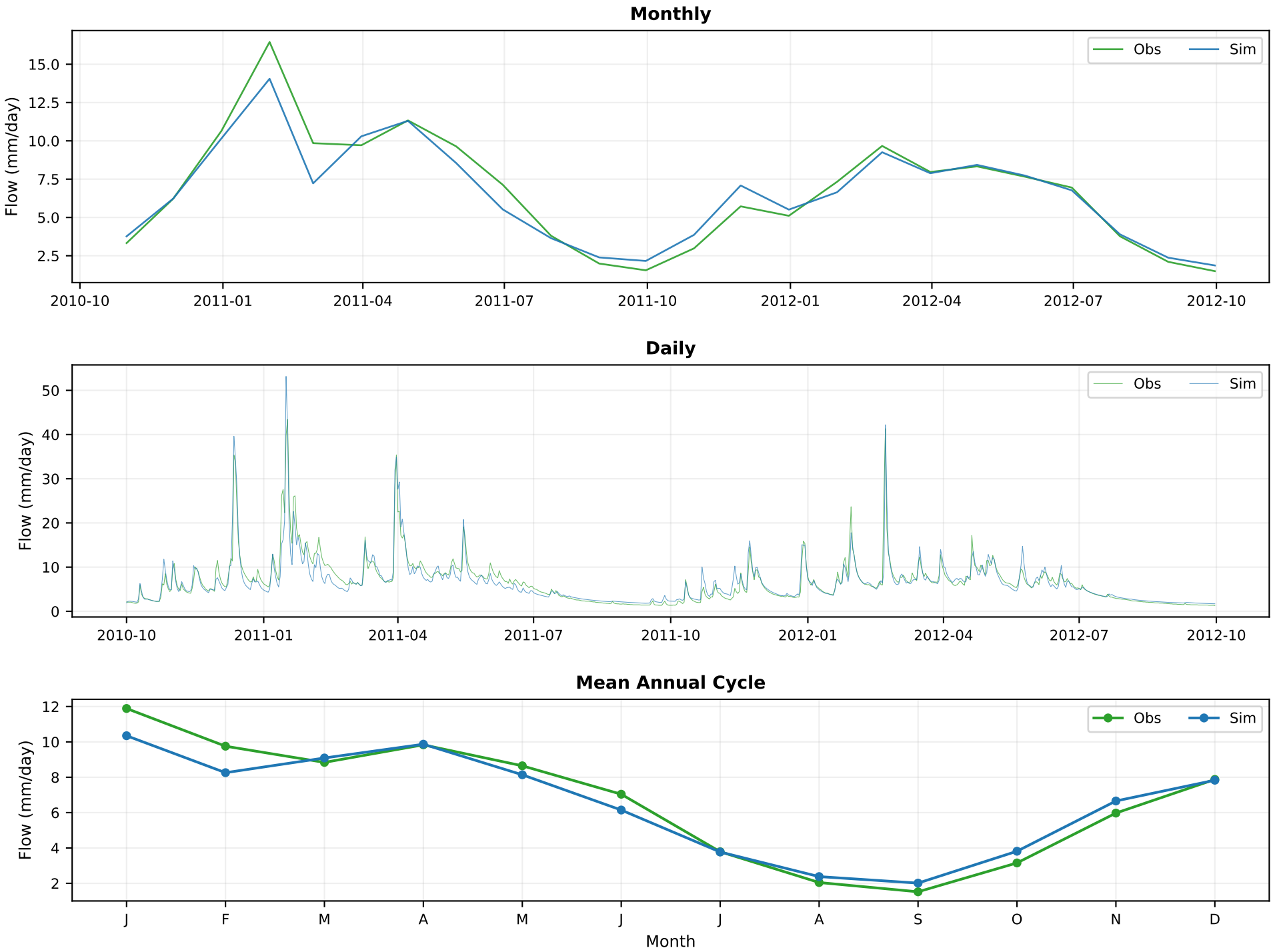

ID: 1547700
KGE=0.943 r=0.968 α=1.047 NSE=0.931 Bias=-0.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

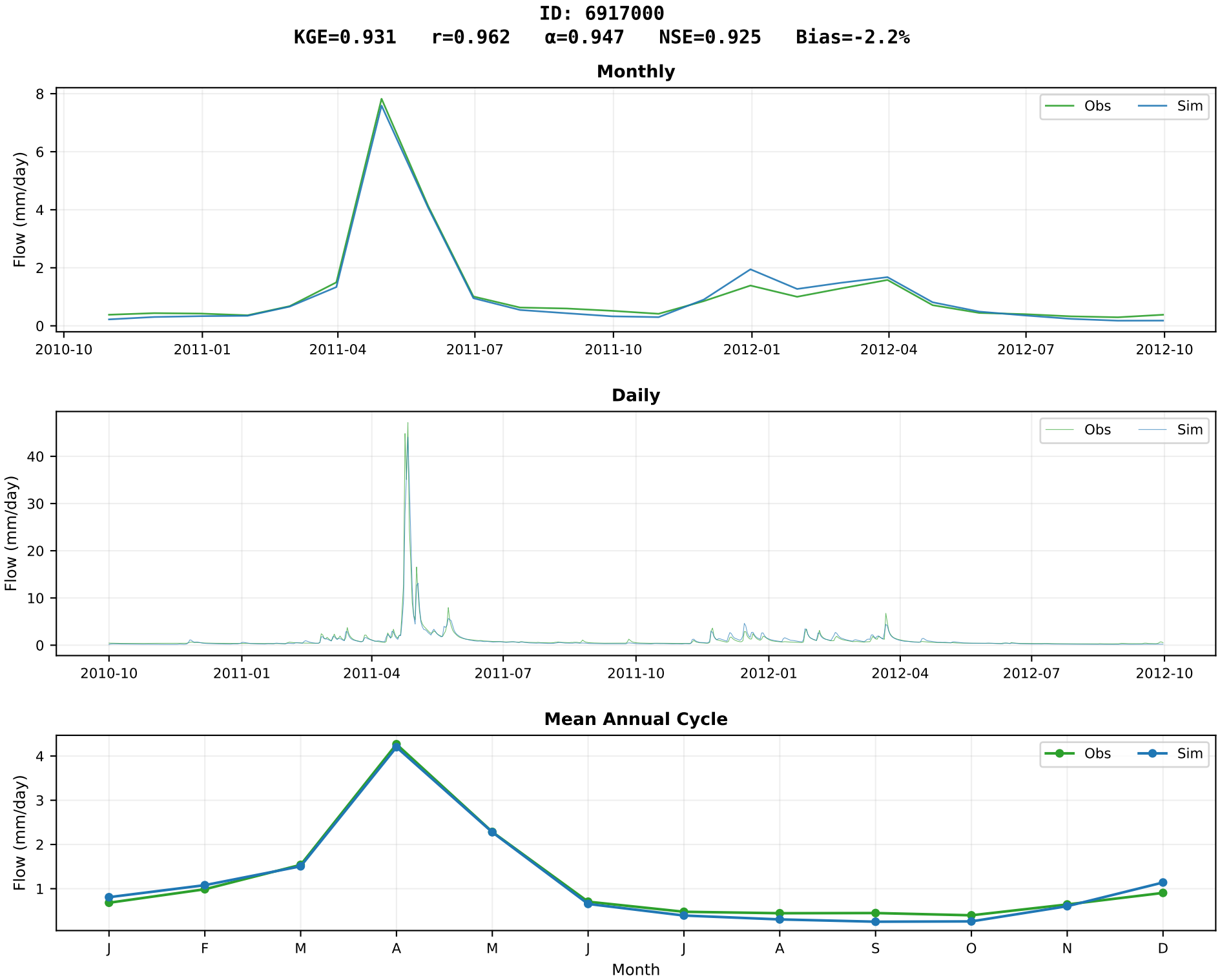

ID: 1052500
KGE=0.939 r=0.946 α=0.992 NSE=0.892 Bias=-2.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

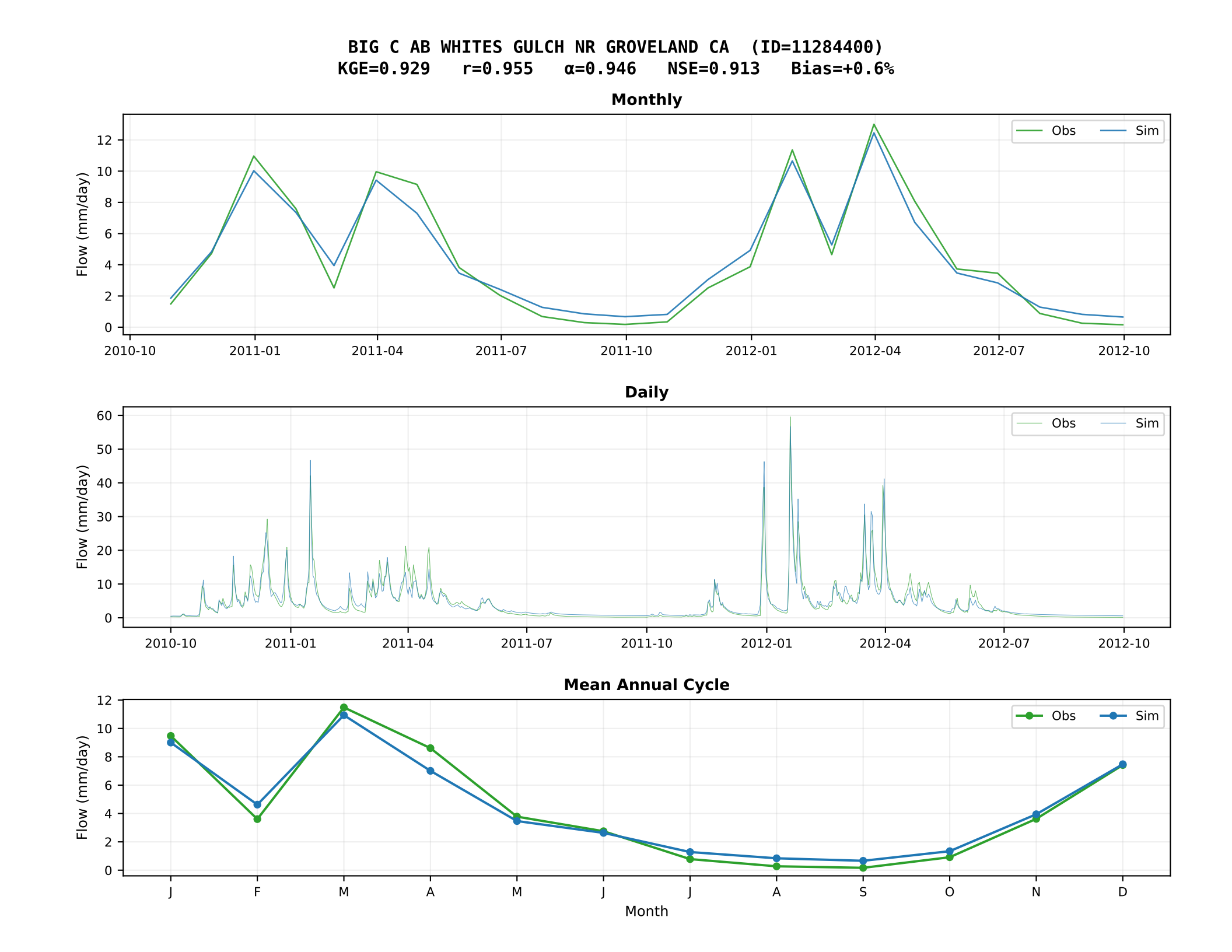

TILTON RIVER AB BEAR CANYON CREEK NEAR CINEBAR, WA (ID=14236200)
KGE=0.936 r=0.972 α=1.057 NSE=0.937 Bias=-0.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

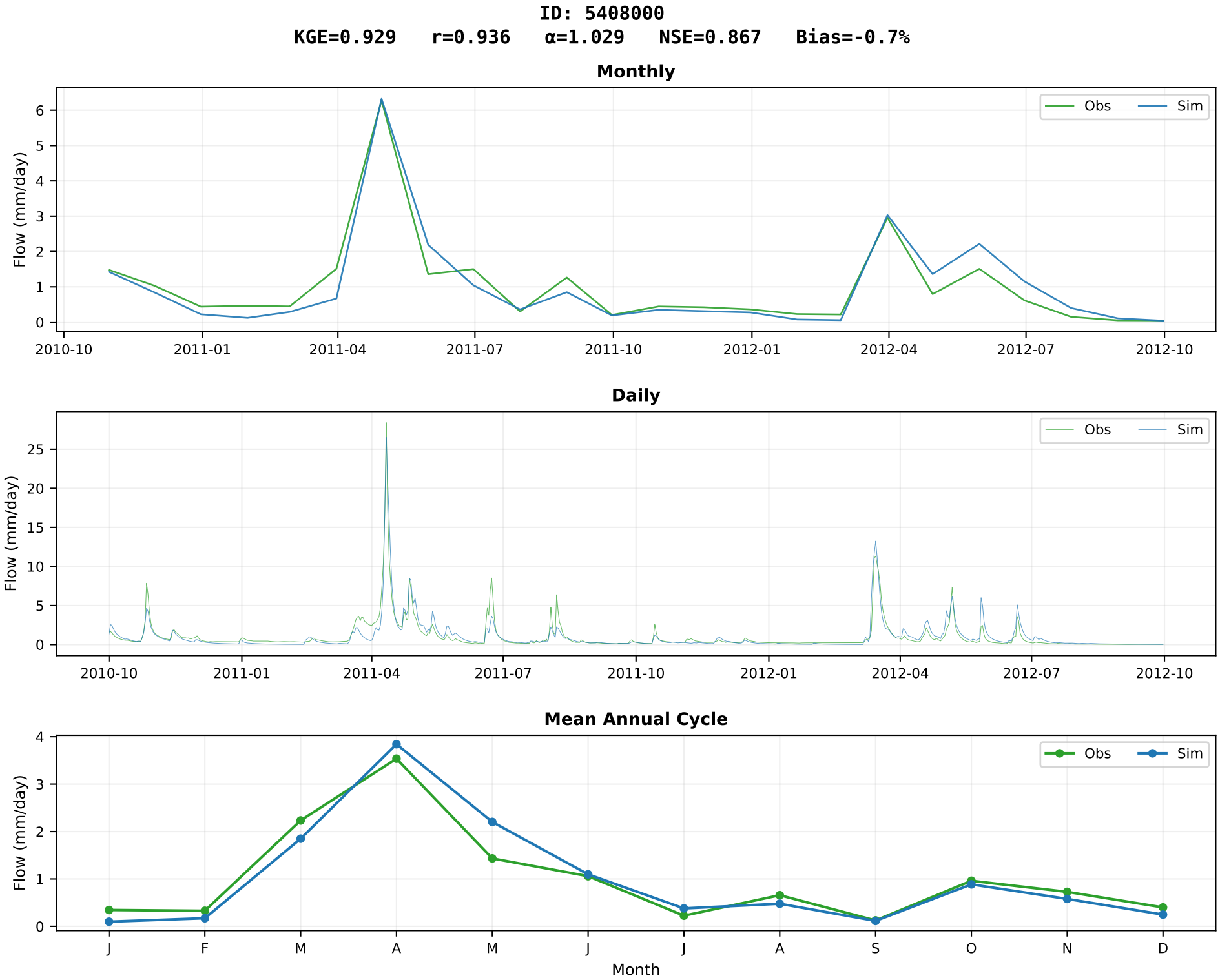

ID: 5120500
KGE=0.936 r=0.971 α=0.944 NSE=0.941 Bias=-1.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


SILETZ RIVER AT SILETZ, OR (ID=14305500)
KGE=0.935 r=0.975 α=0.960 NSE=0.950 Bias=-4.4%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

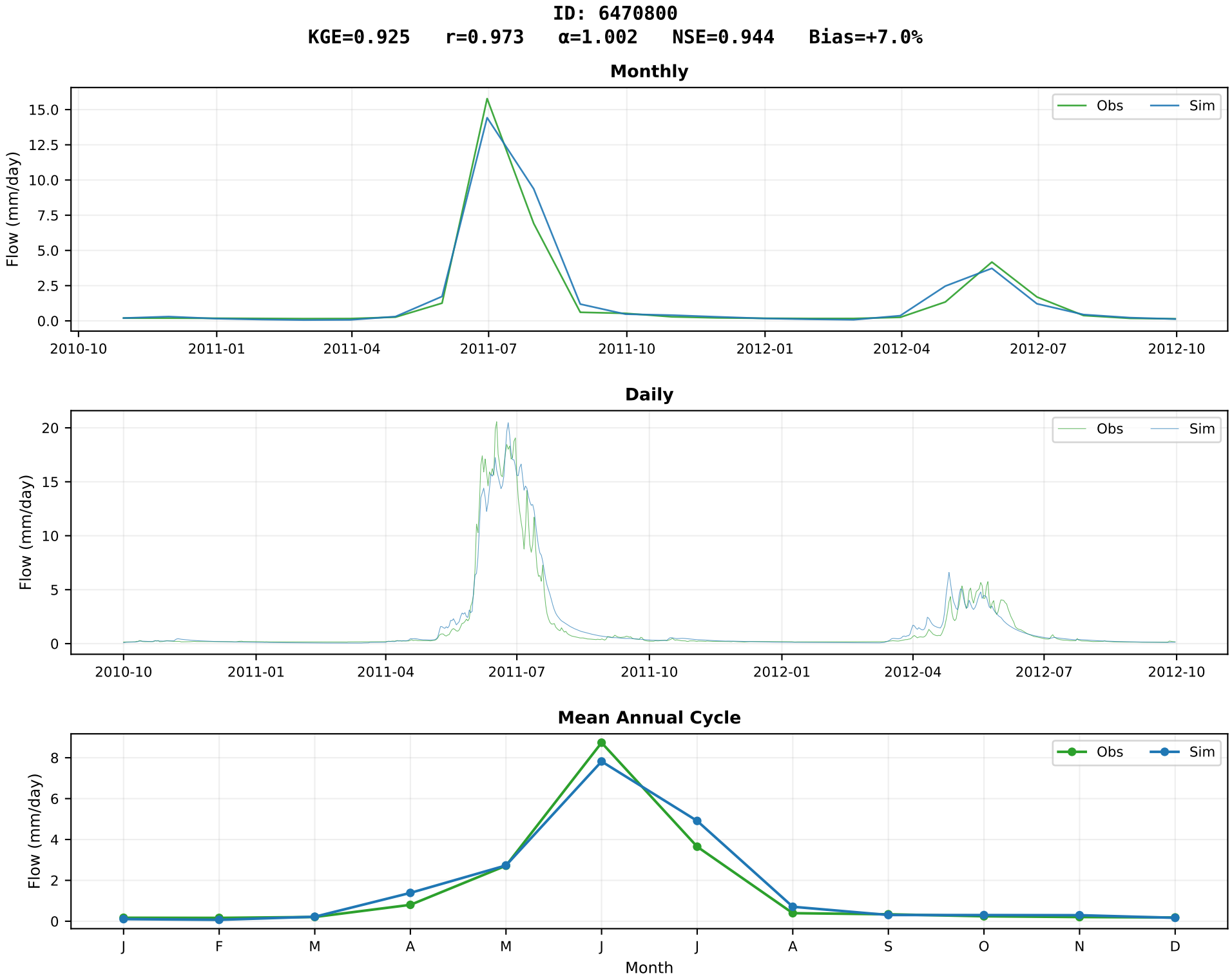

JOHNSON CREEK AT YELLOW PINE ID (ID=13313000)
KGE=0.933 r=0.941 α=0.980 NSE=0.883 Bias=-2.5%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

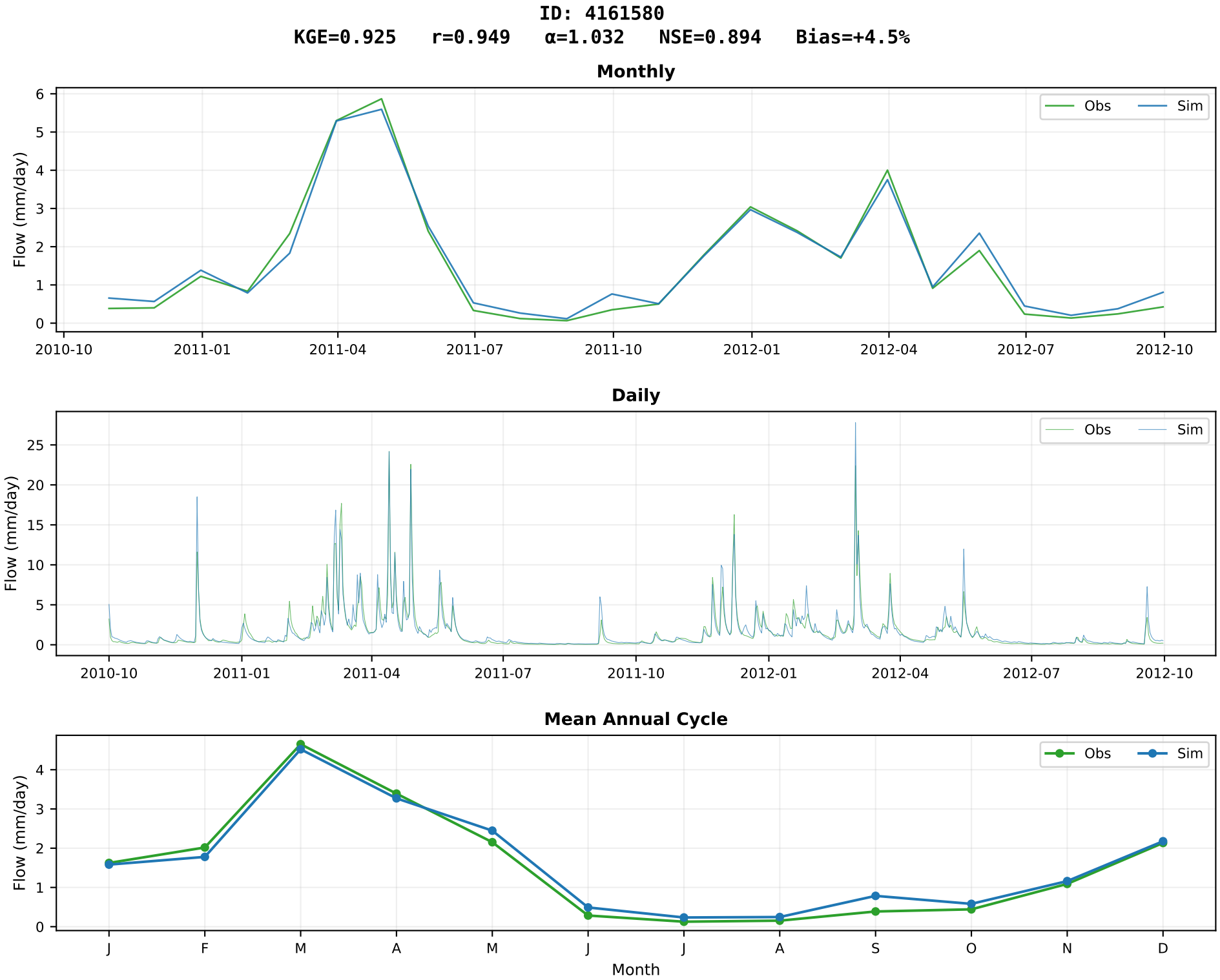

ID: 6917000
KGE=0.931 r=0.962 α=0.947 NSE=0.925 Bias=-2.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

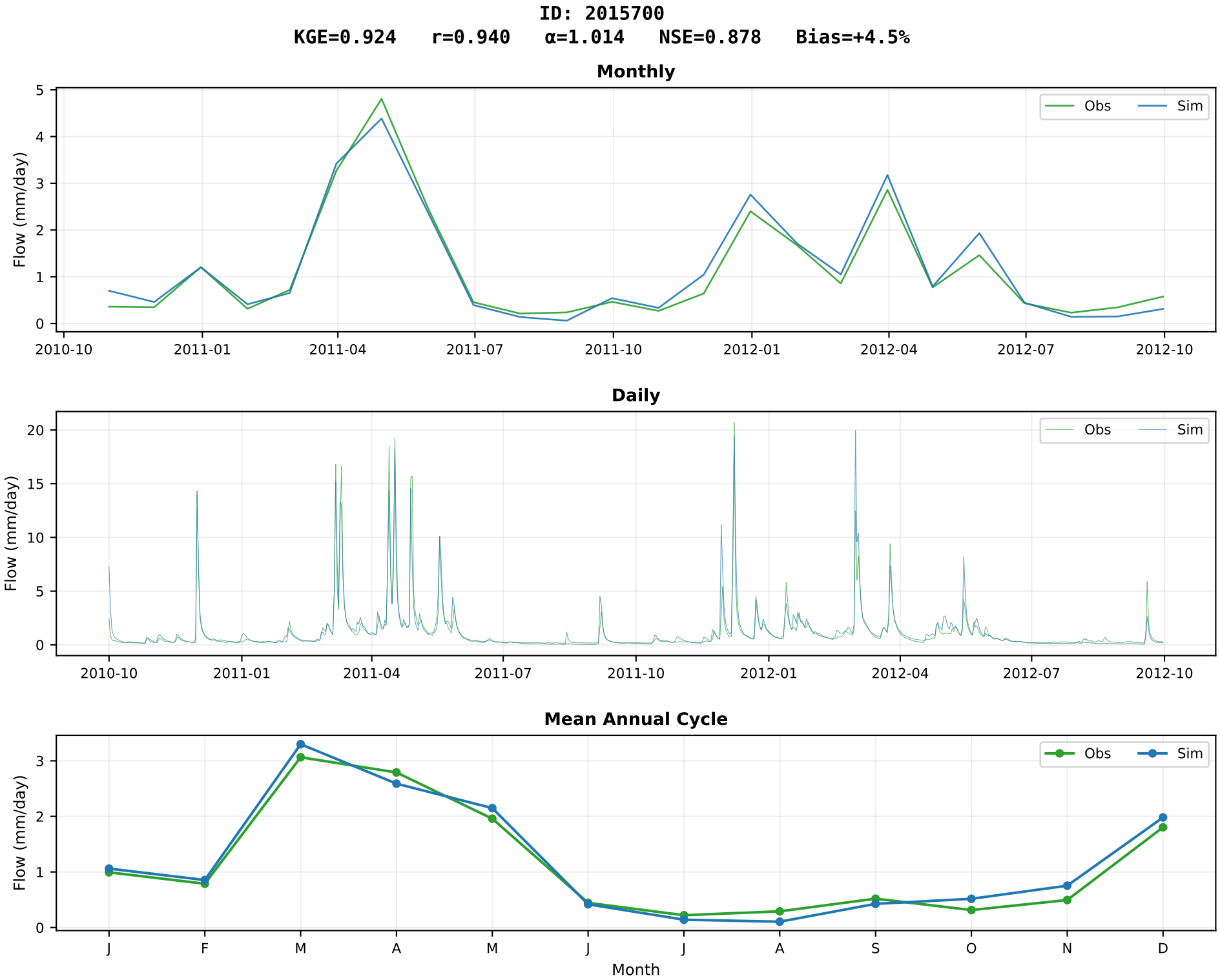

BIG C AB WHITES GULCH NR GROVELAND CA (ID=11284400)
KGE=0.929 r=0.955 α=0.946 NSE=0.913 Bias=+0.6%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

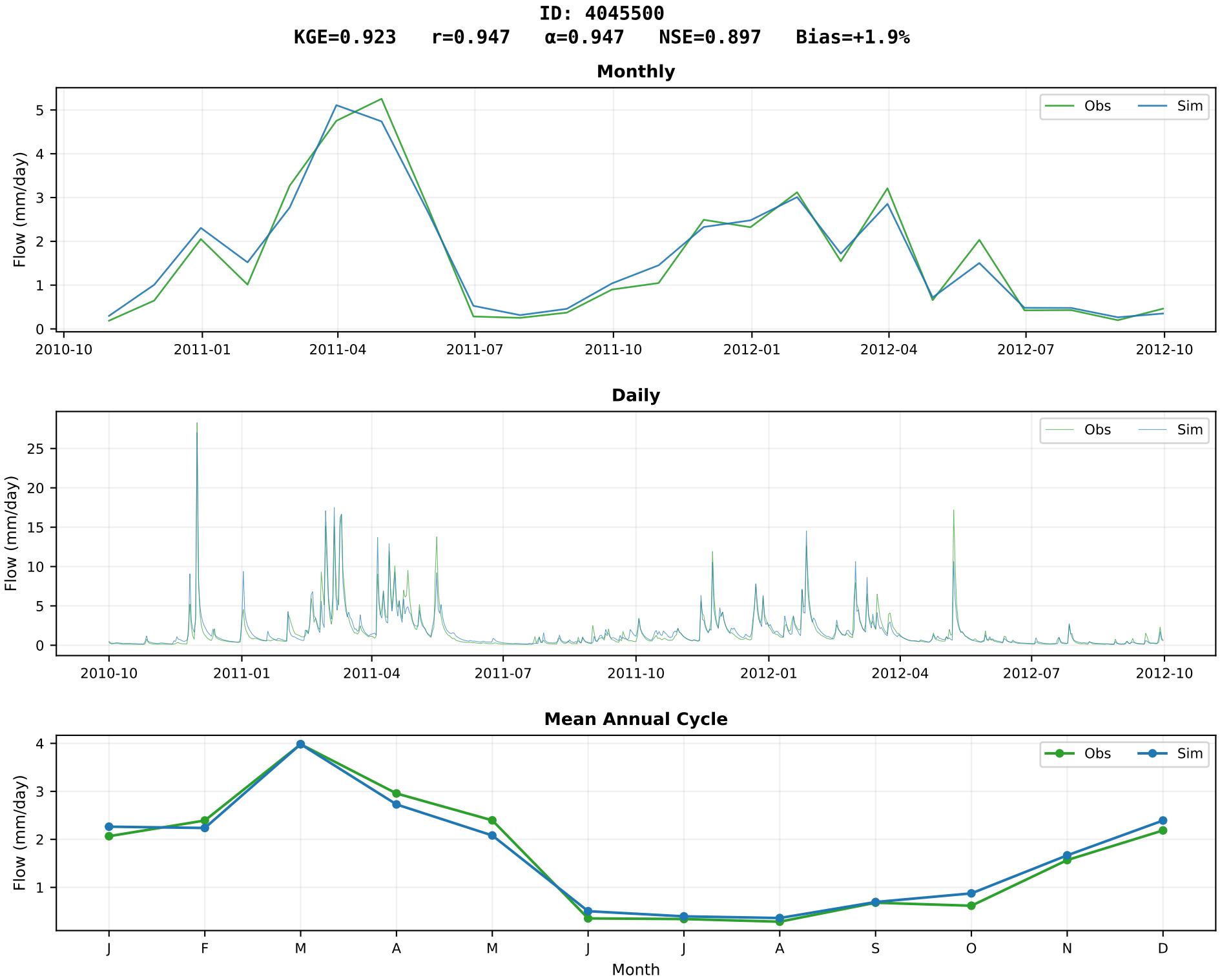

ID: 5408000
KGE=0.929 r=0.936 α=1.029 NSE=0.867 Bias=-0.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

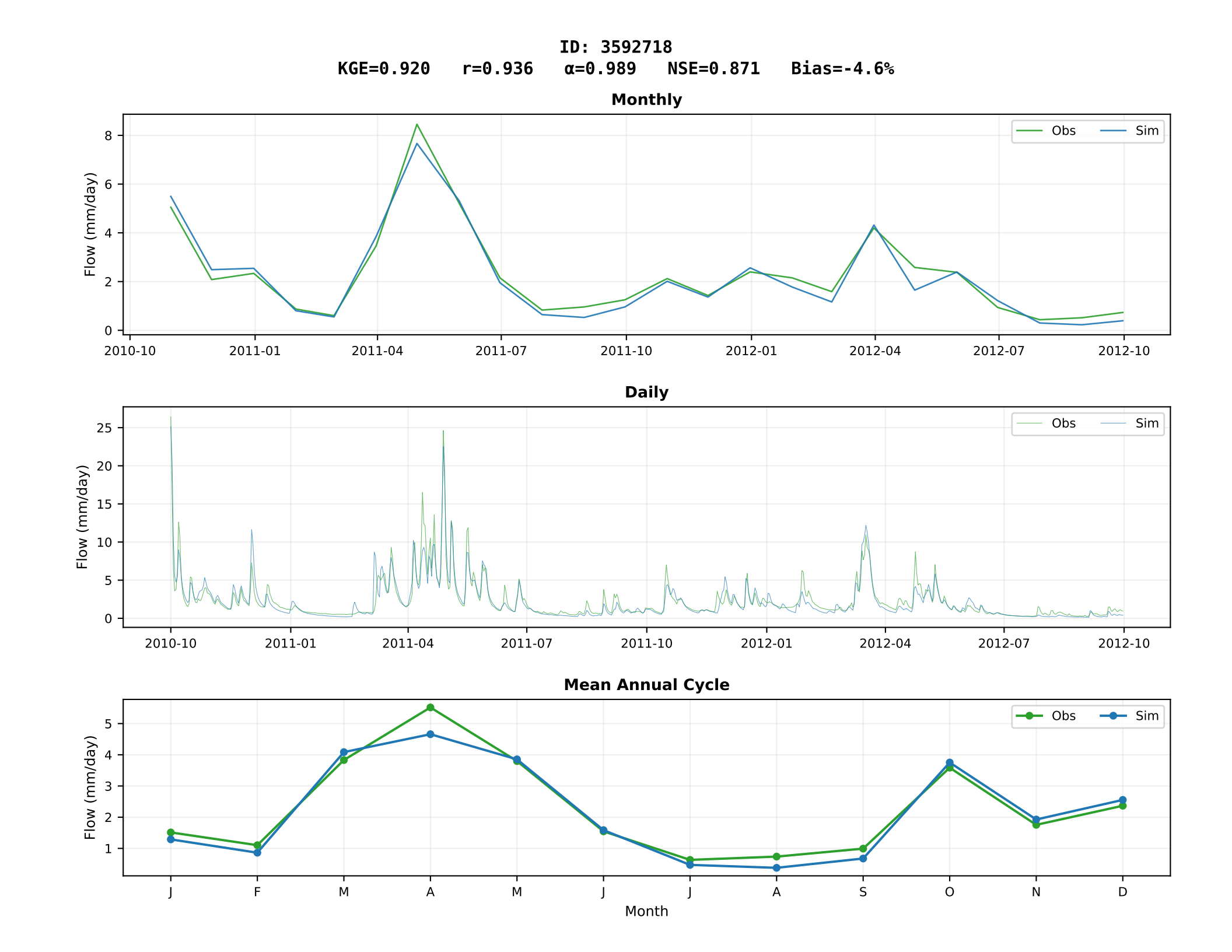

UMATILLA RIVER ABOVE MEACHAM CREEK, NR GIBBON, OR (ID=14020000)
KGE=0.926 r=0.935 α=0.968 NSE=0.873 Bias=+1.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

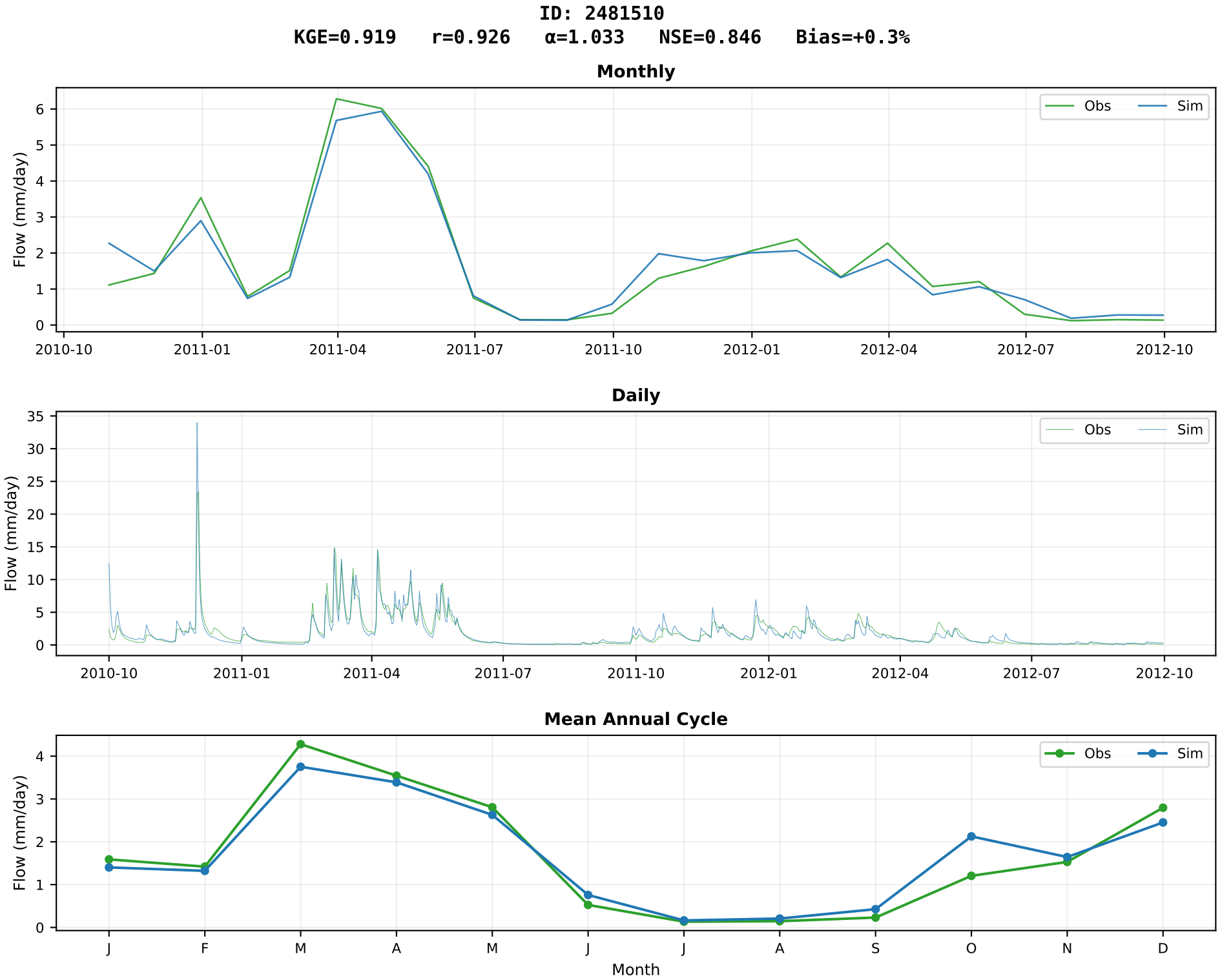

ID: 6470800
KGE=0.925 r=0.973 α=1.002 NSE=0.944 Bias=+7.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

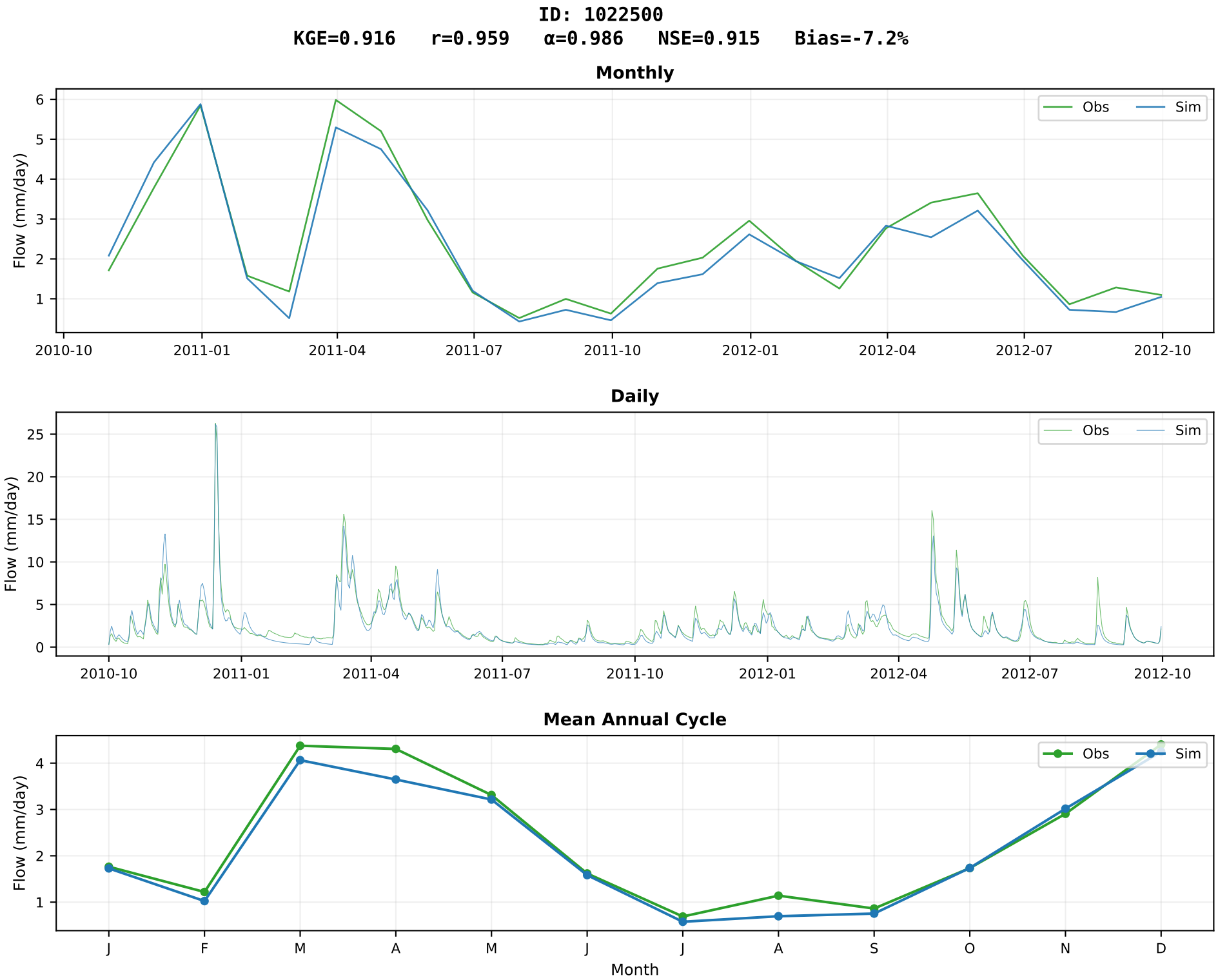

ID: 4161580
KGE=0.925 r=0.949 α=1.032 NSE=0.894 Bias=+4.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

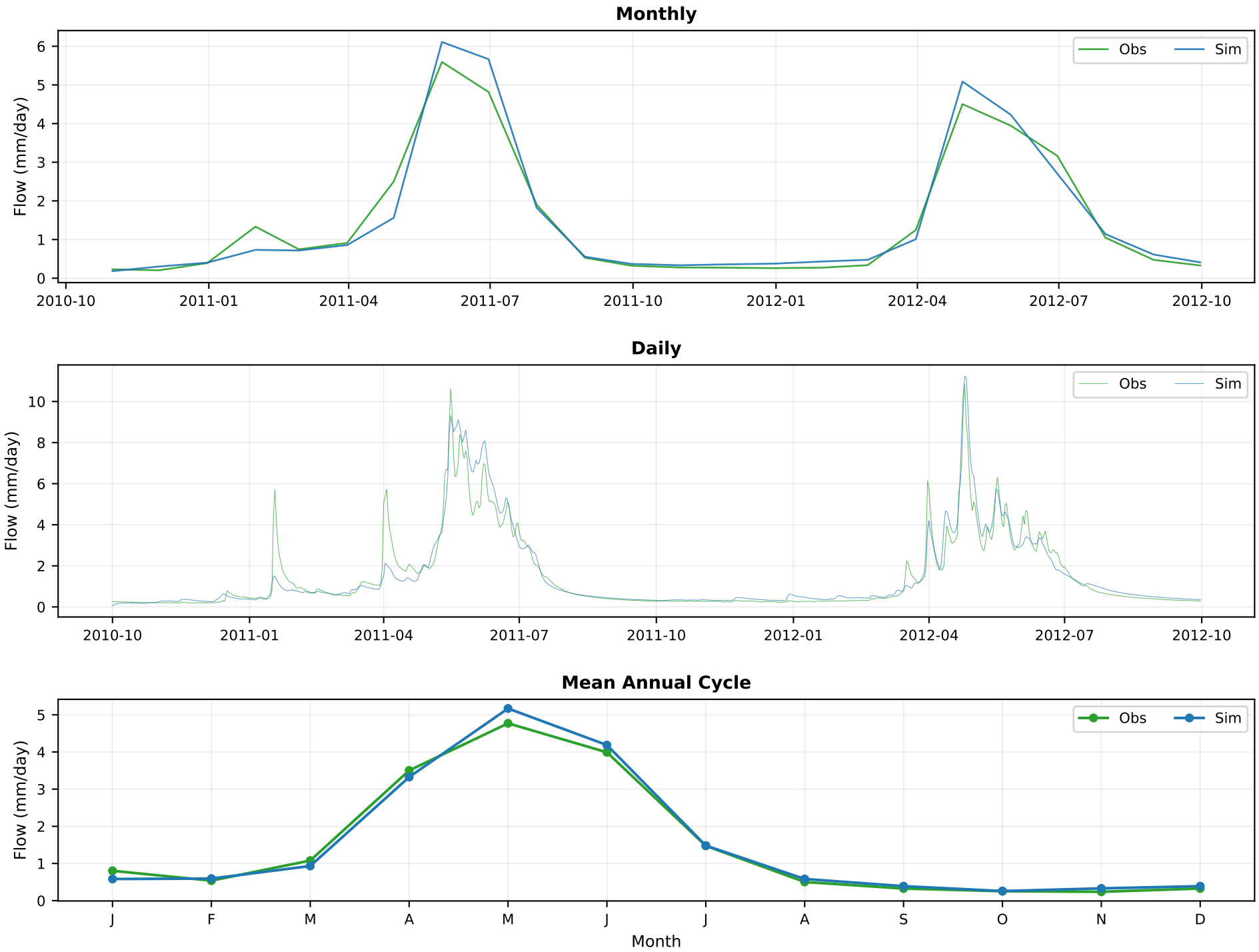

ID: 2015700
KGE=0.924 r=0.940 α=1.014 NSE=0.878 Bias=+4.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

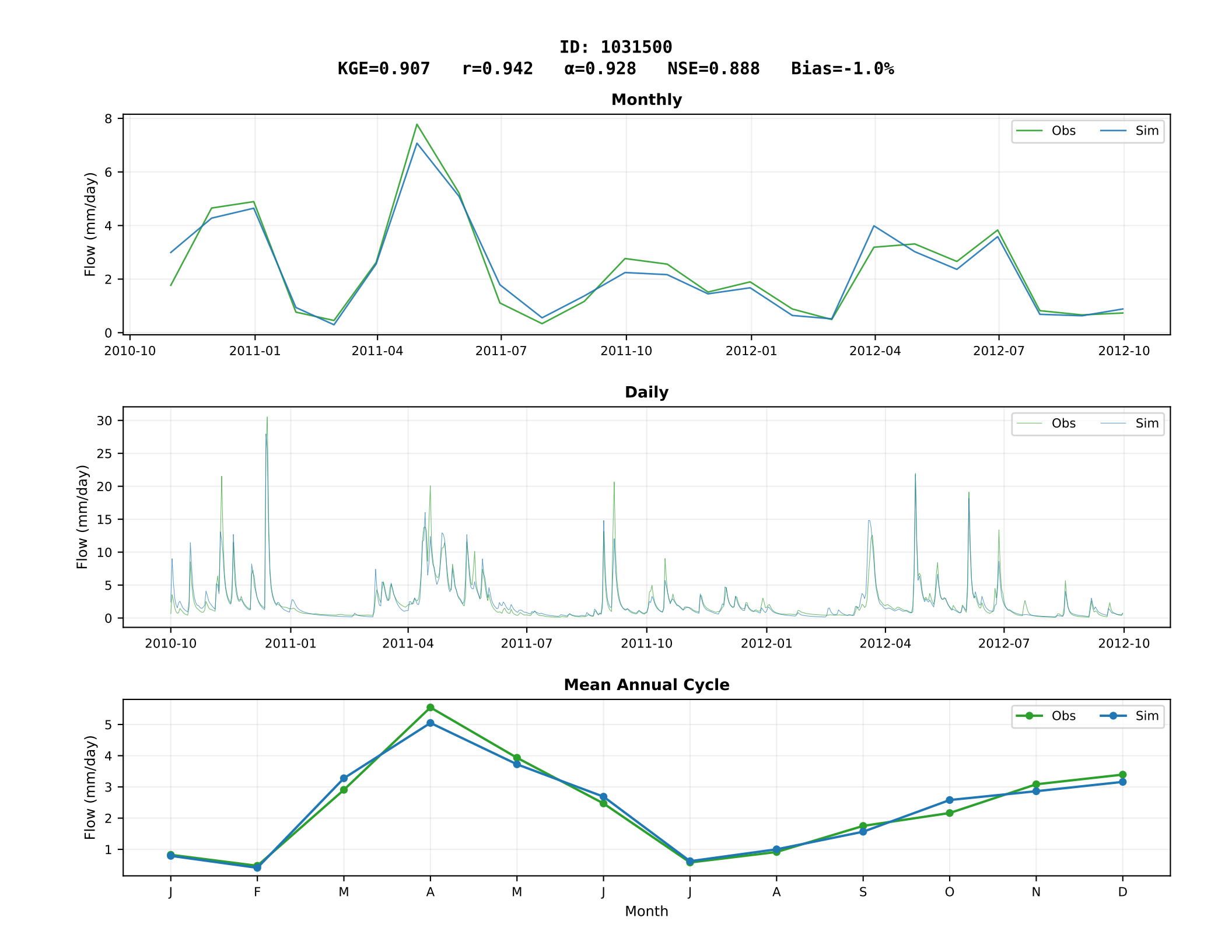

ID: 4045500
KGE=0.923 r=0.947 α=0.947 NSE=0.897 Bias=+1.9%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

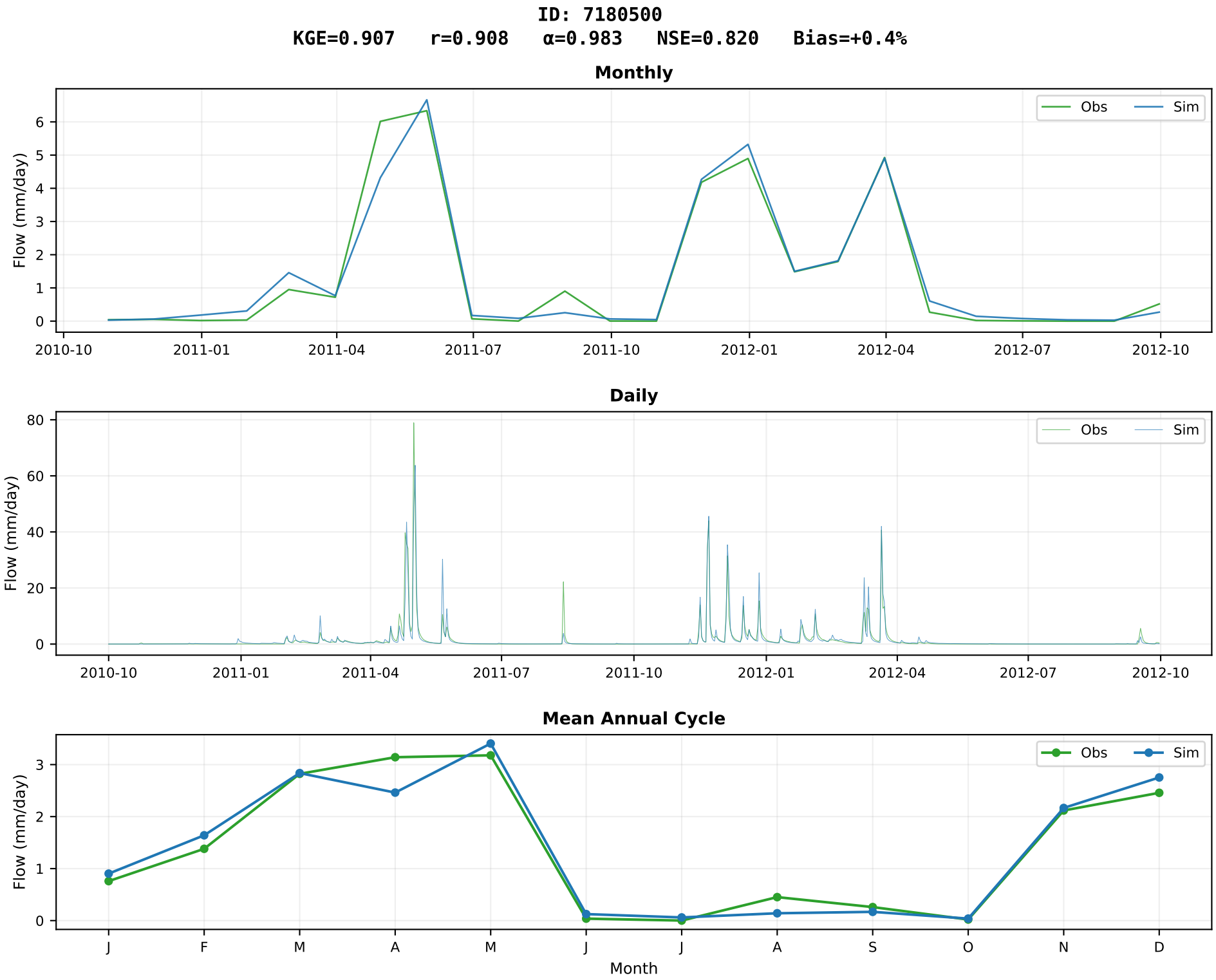

ID: 3592718
KGE=0.920 r=0.936 α=0.989 NSE=0.871 Bias=-4.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

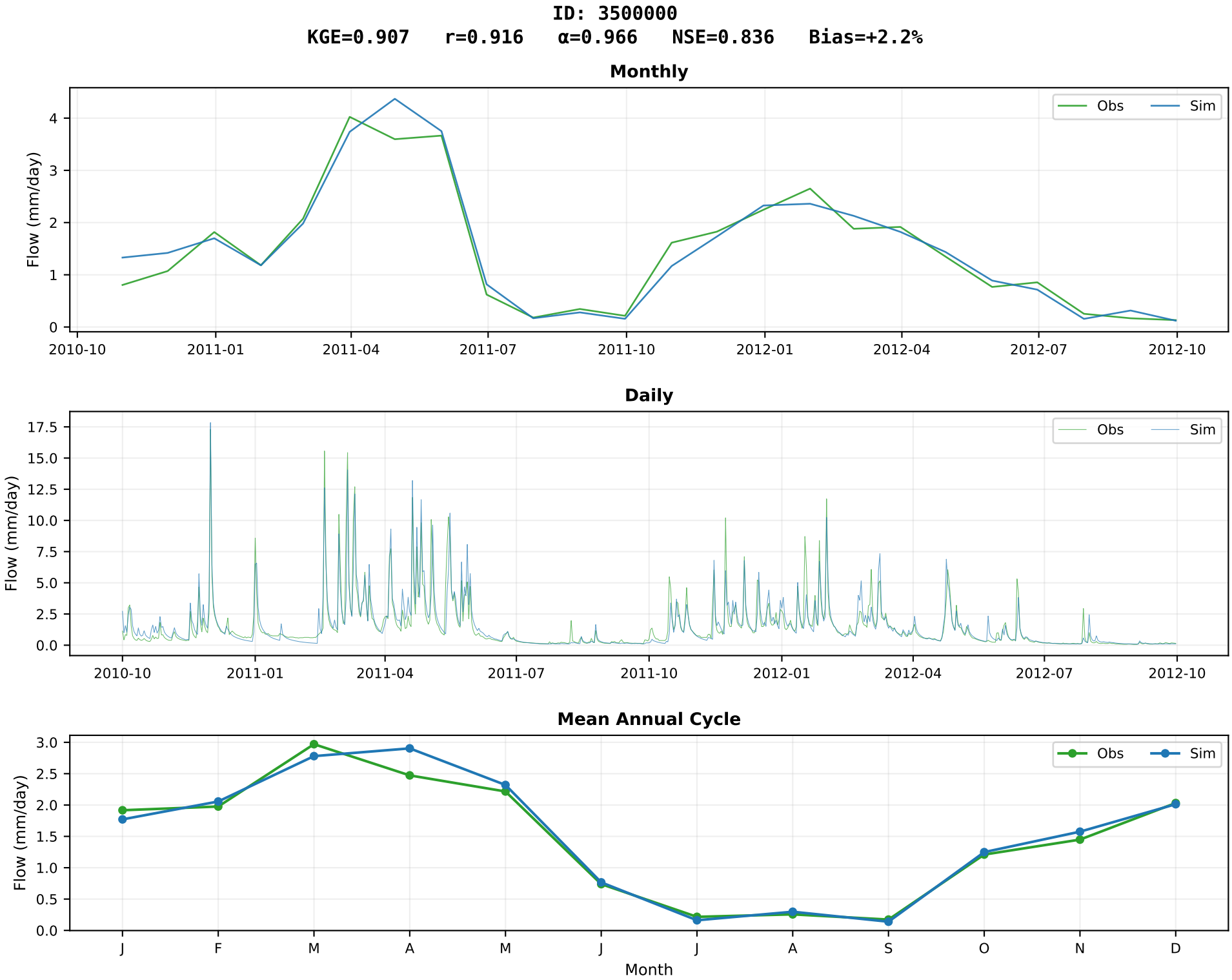

ID: 2481510
KGE=0.919 r=0.926 α=1.033 NSE=0.846 Bias=+0.3%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
35
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

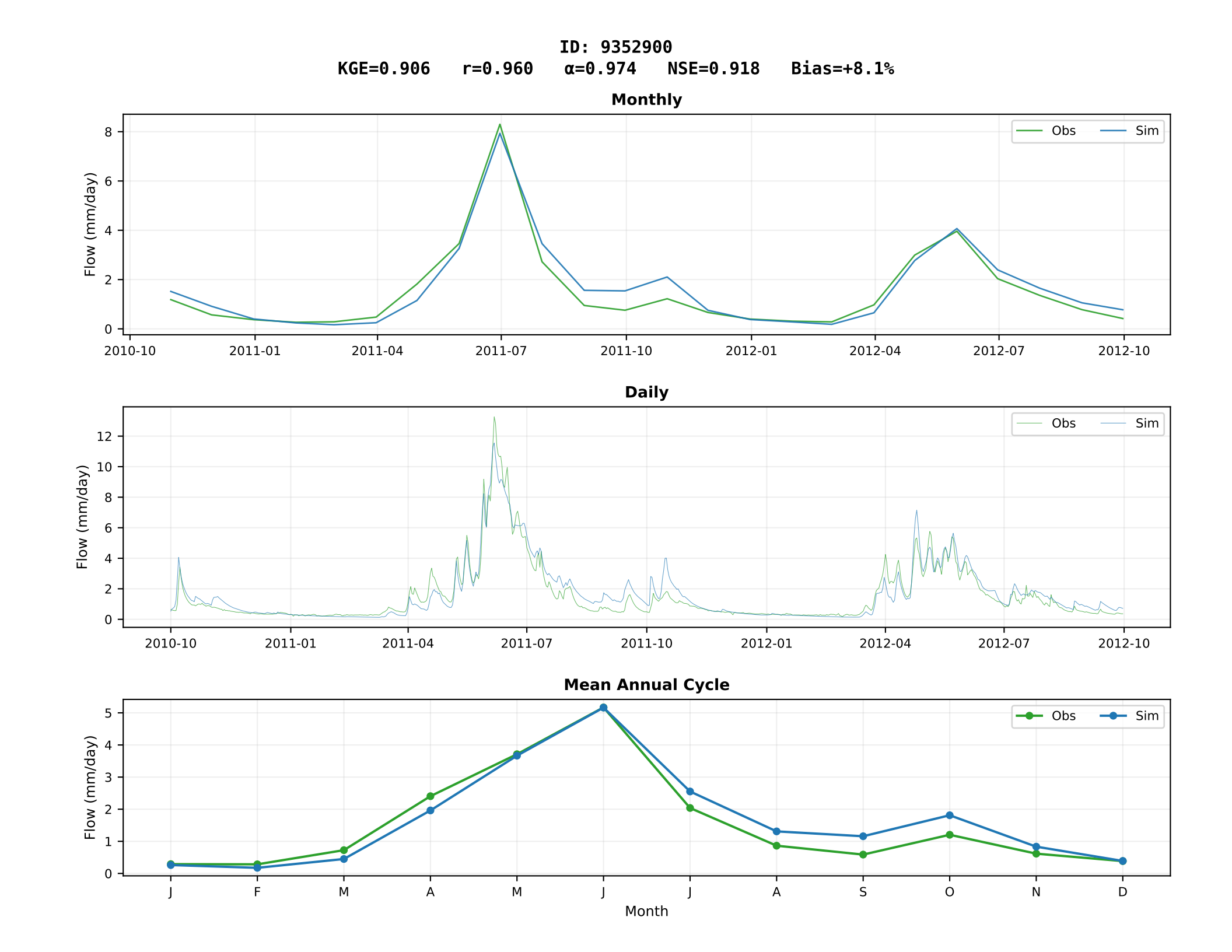

ID: 1022500
KGE=0.916 r=0.959 α=0.986 NSE=0.915 Bias=-7.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

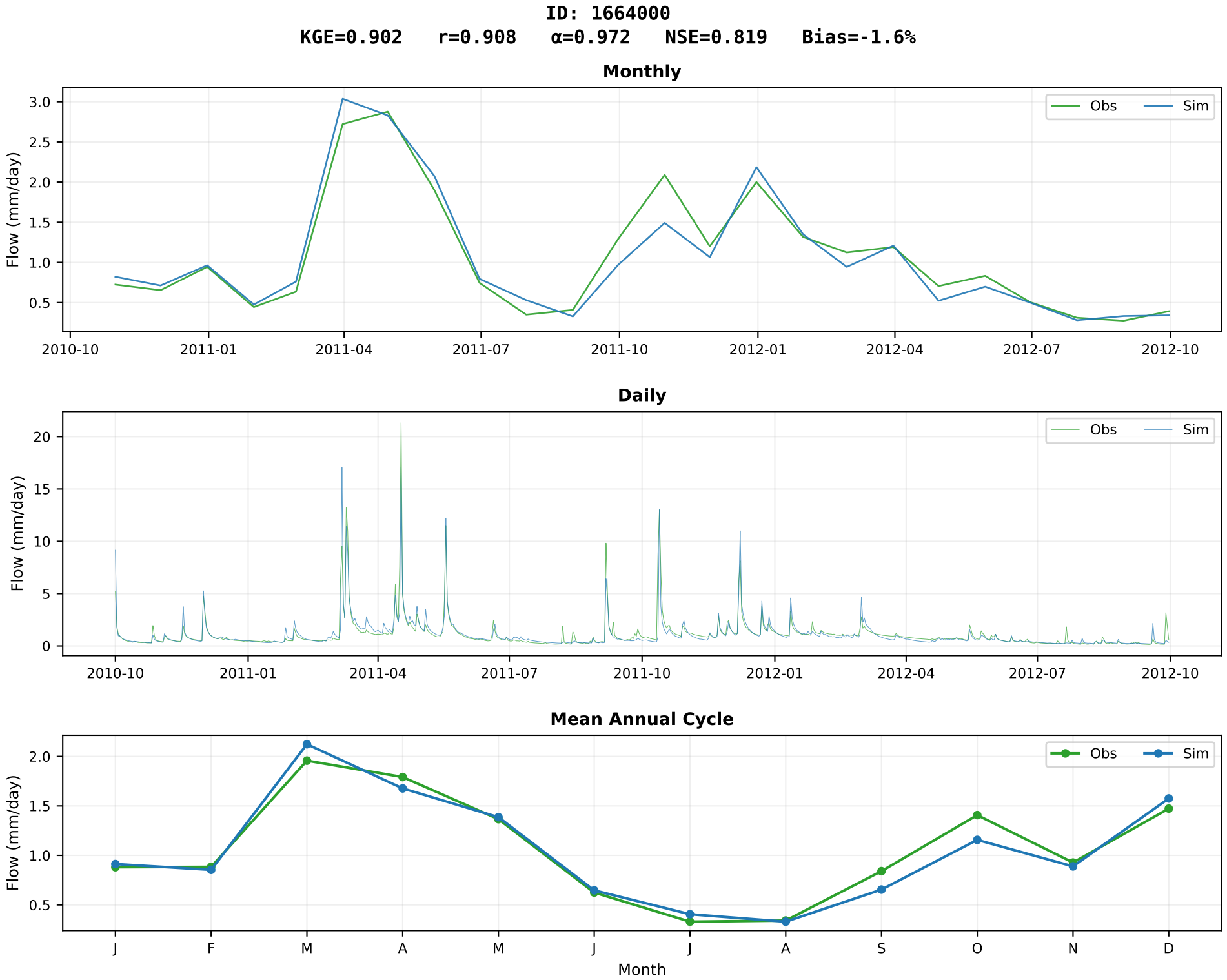

SOUTH SANTIAM RIVER BELOW CASCADIA, OR (ID=14185000)
KGE=0.915 r=0.955 α=1.068 NSE=0.898 Bias=+2.3%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

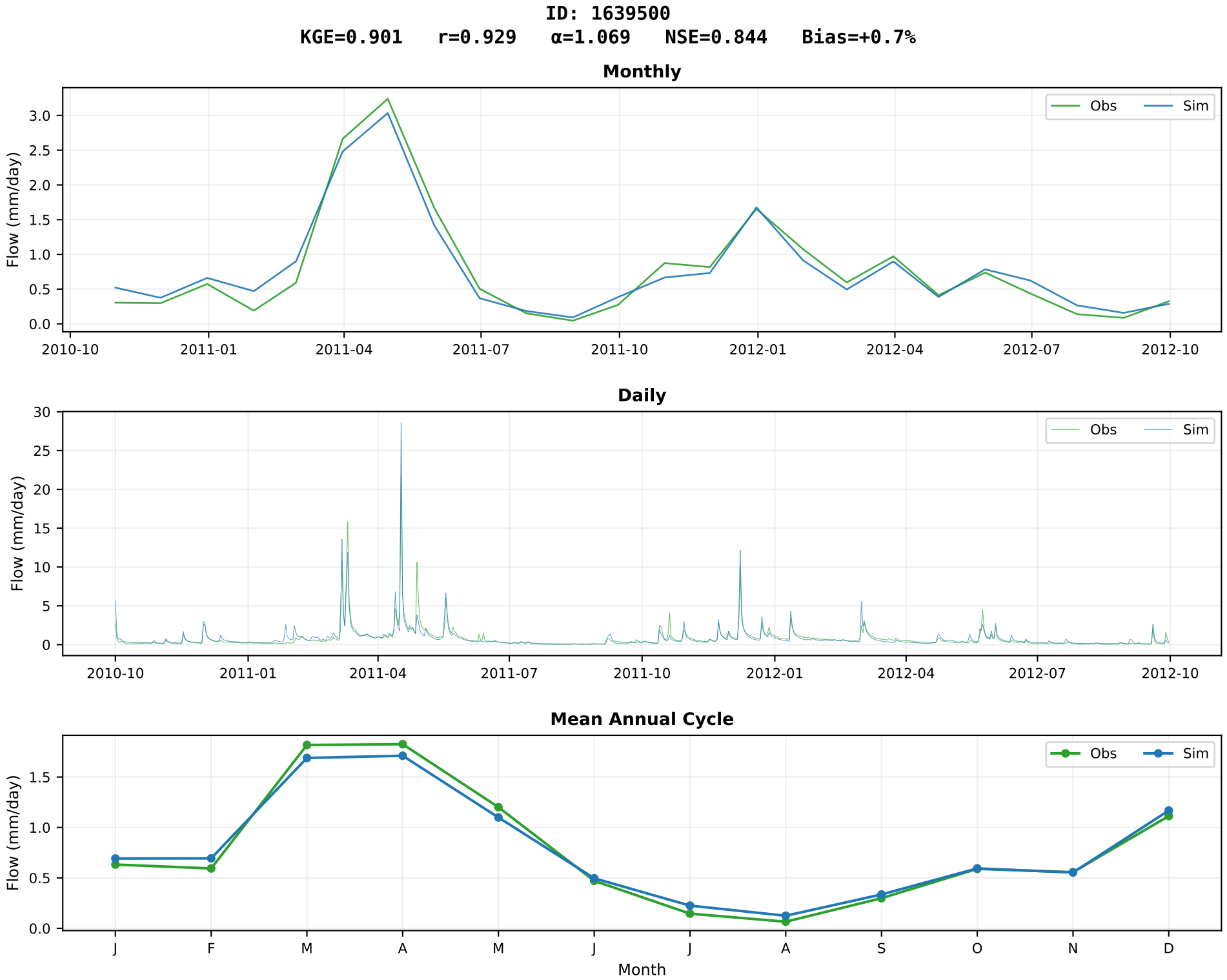

ID: 1031500
KGE=0.907 r=0.942 α=0.928 NSE=0.888 Bias=-1.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

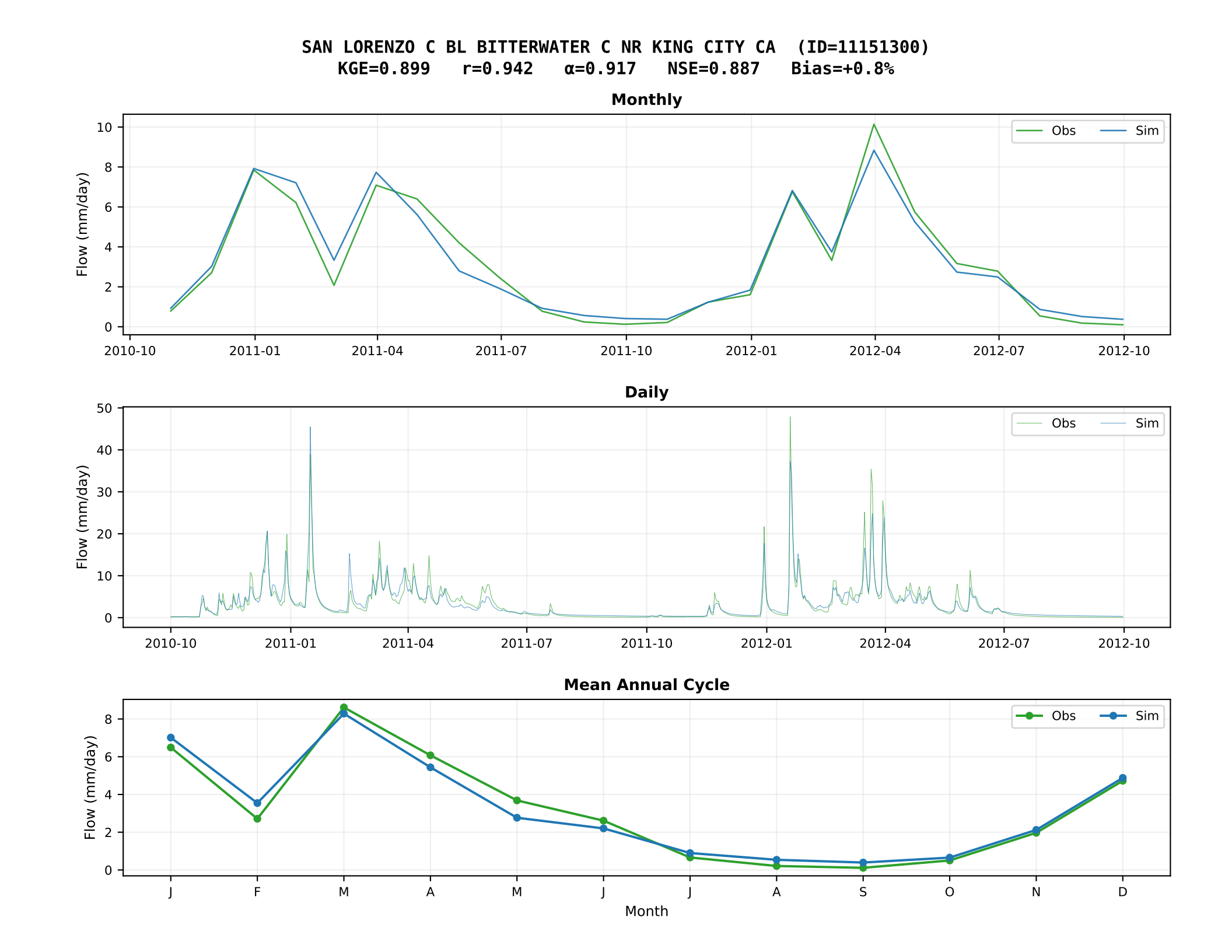

ID: 7180500
KGE=0.907 r=0.908 α=0.983 NSE=0.820 Bias=+0.4%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

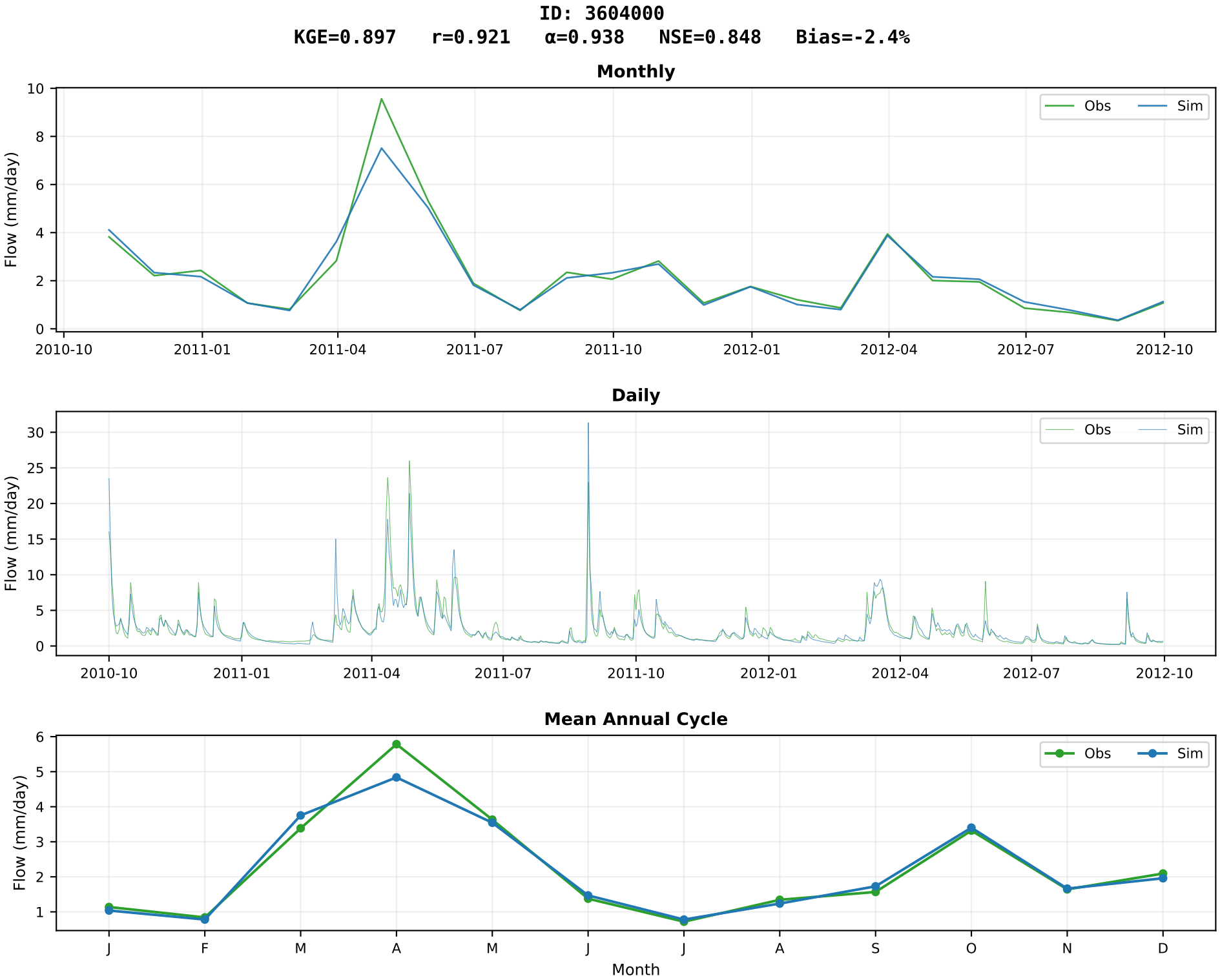

ID: 3500000
KGE=0.907 r=0.916 α=0.966 NSE=0.836 Bias=+2.2%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

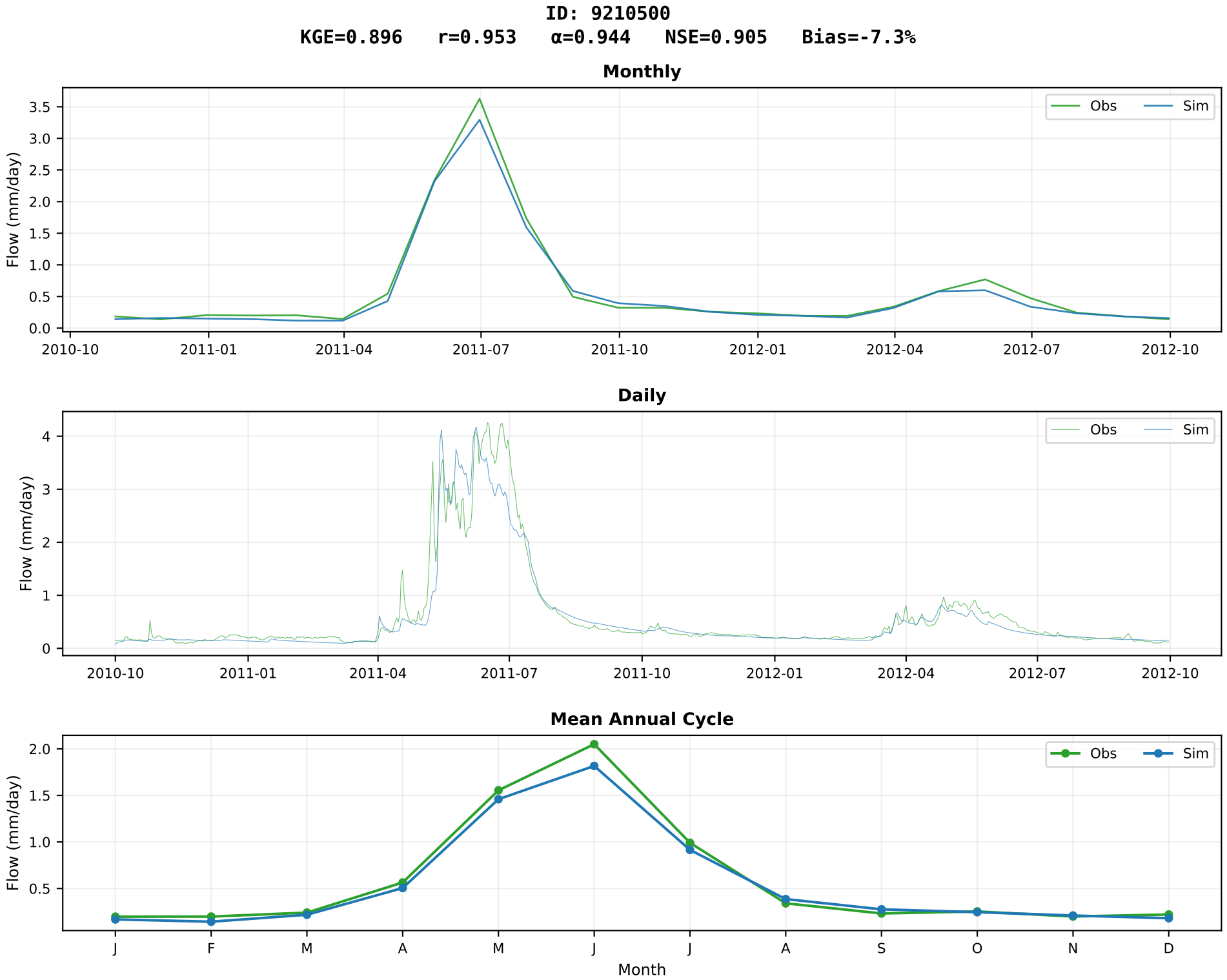

ID: 9352900
KGE=0.906 r=0.960 α=0.974 NSE=0.918 Bias=+8.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

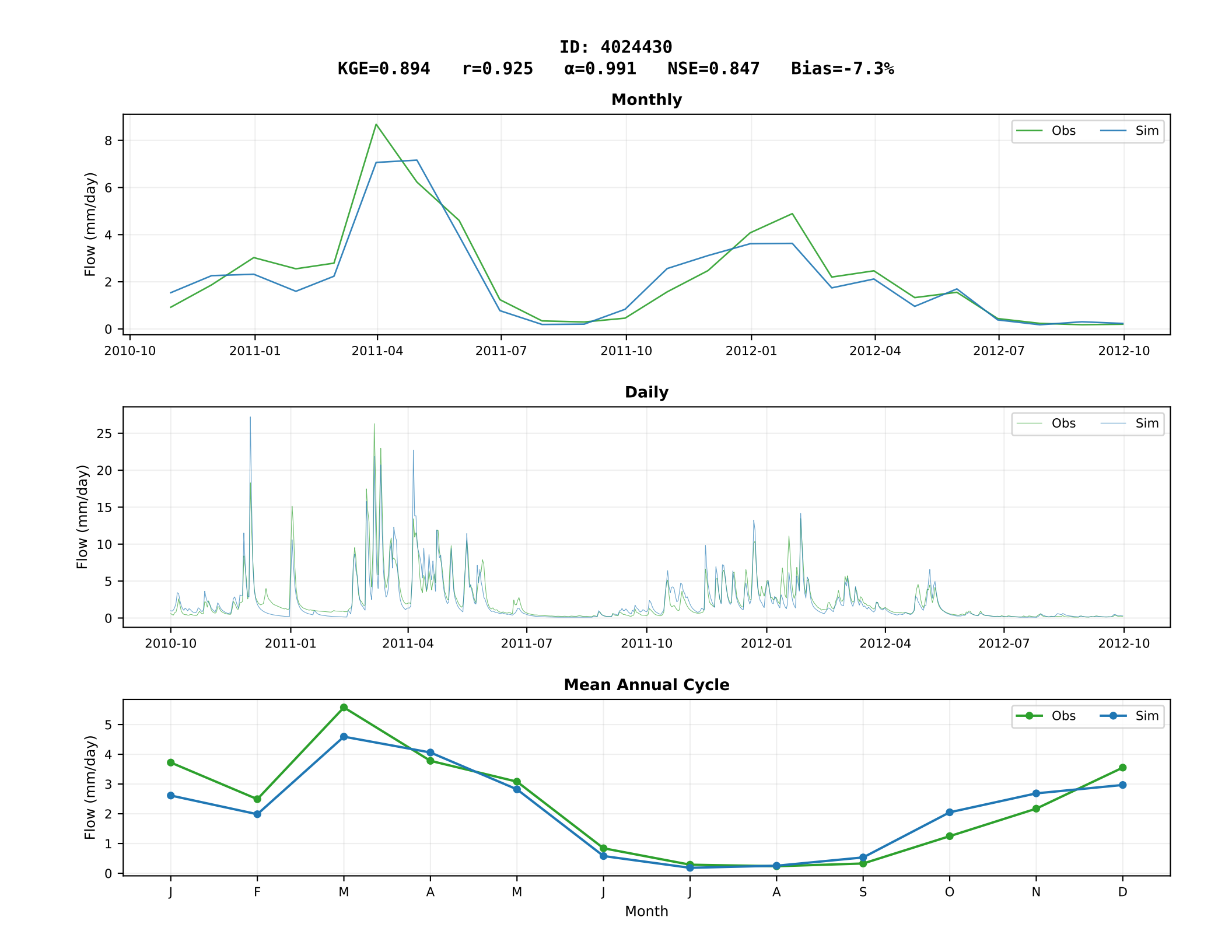

ID: 1664000
KGE=0.902 r=0.908 α=0.972 NSE=0.819 Bias=-1.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

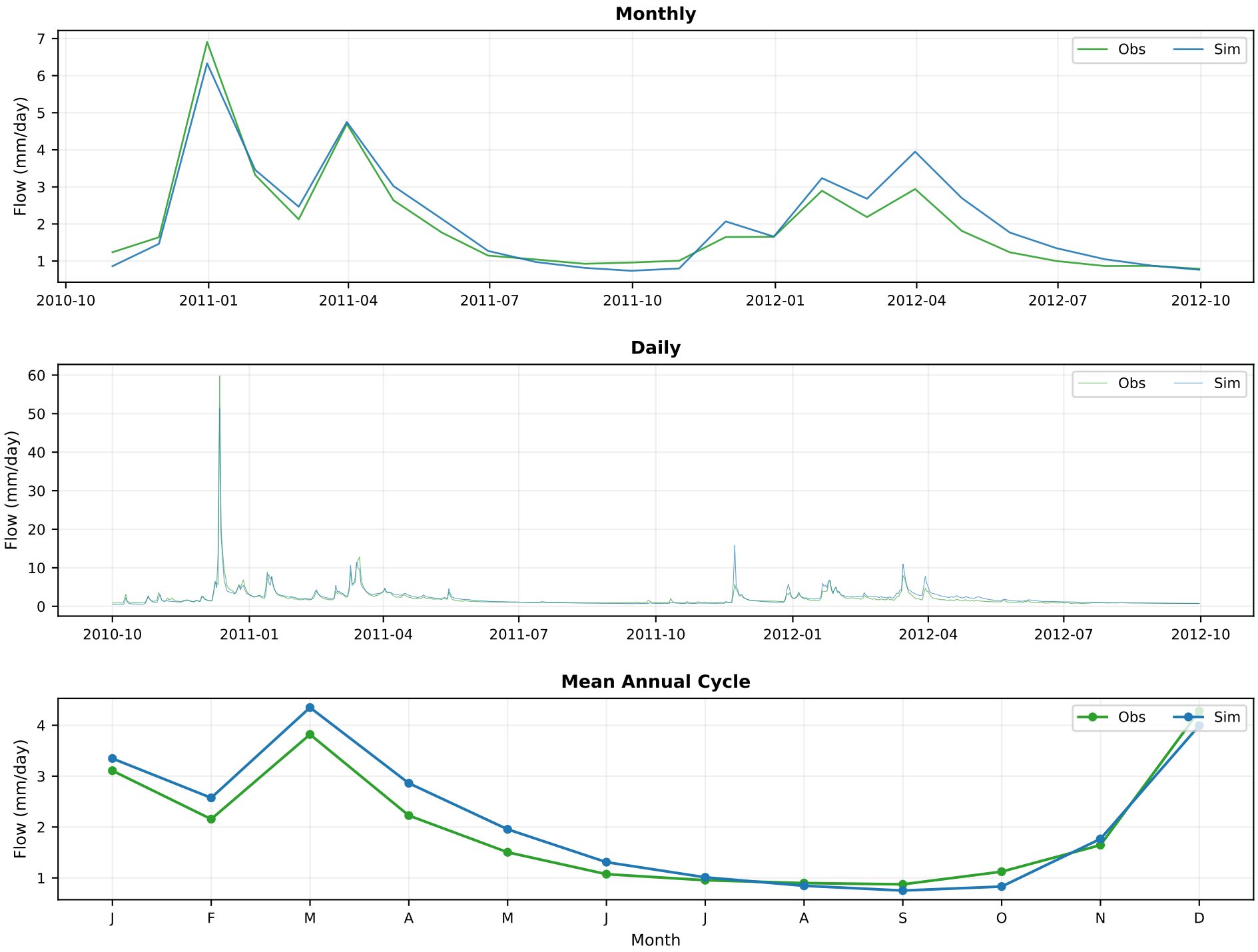

ID: 1639500
KGE=0.901 r=0.929 α=1.069 NSE=0.844 Bias=+0.7%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

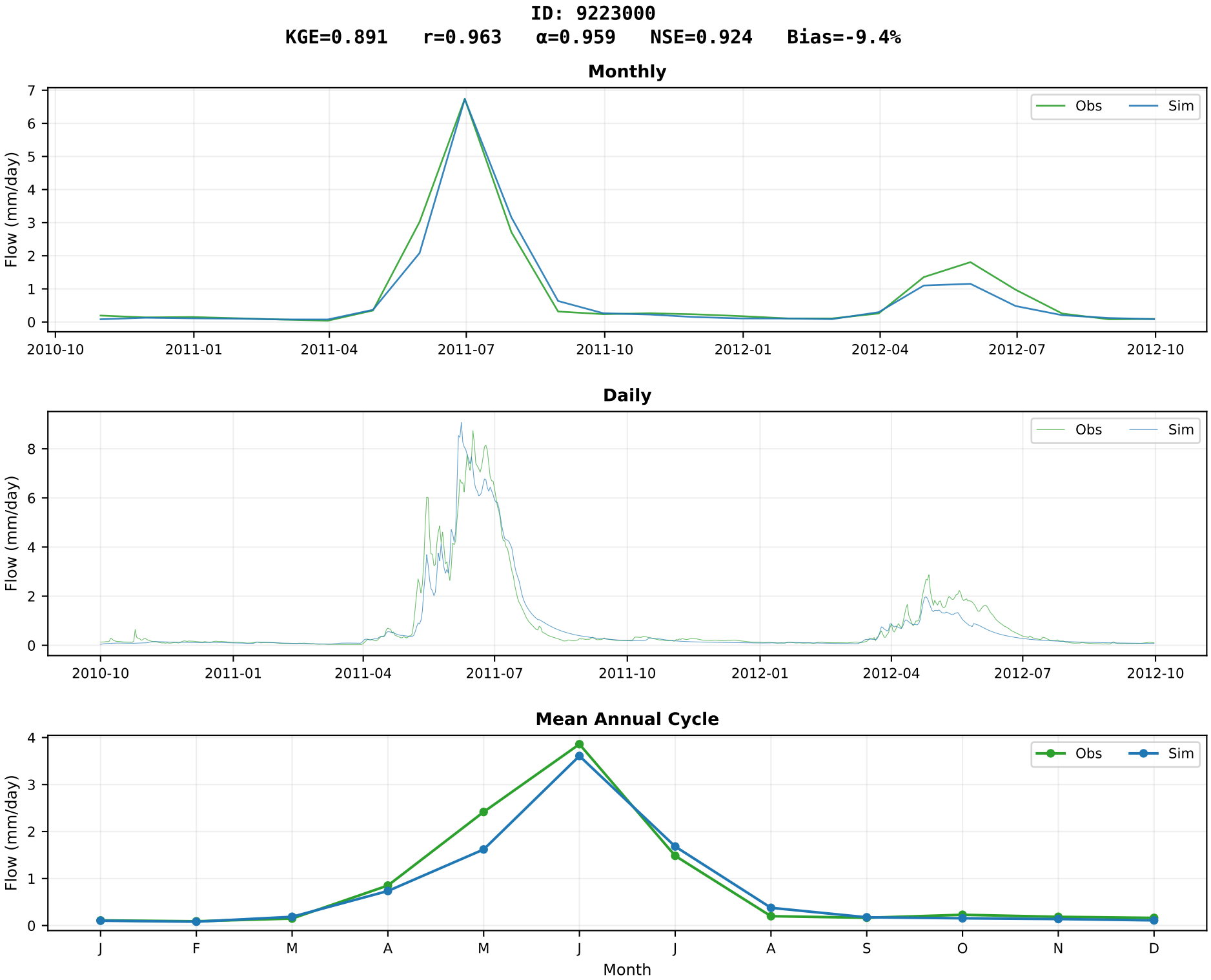

SAN LORENZO C BL BITTERWATER C NR KING CITY CA (ID=11151300)
KGE=0.899 r=0.942 α=0.917 NSE=0.887 Bias=+0.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

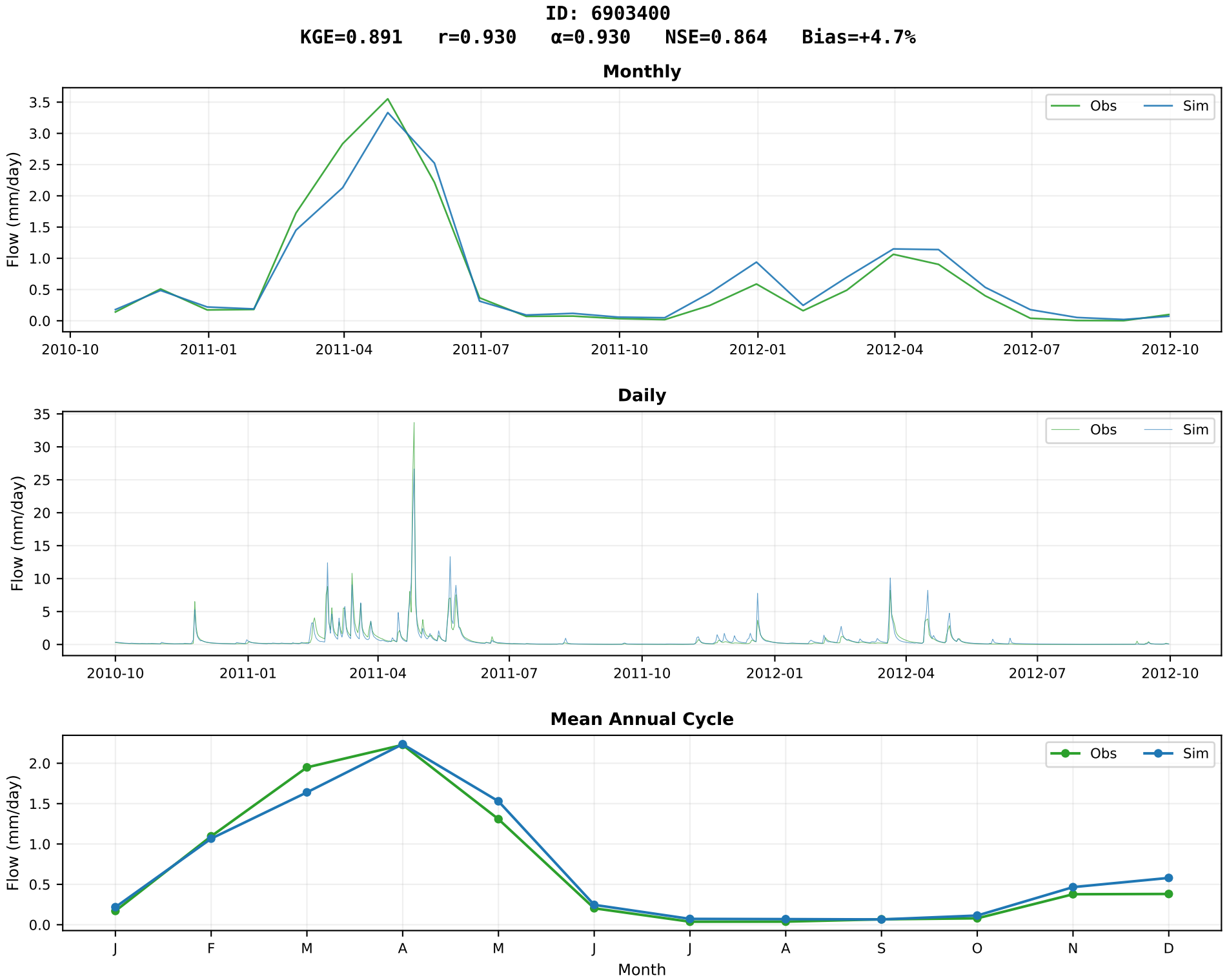

ID: 3604000
KGE=0.897 r=0.921 α=0.938 NSE=0.848 Bias=-2.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

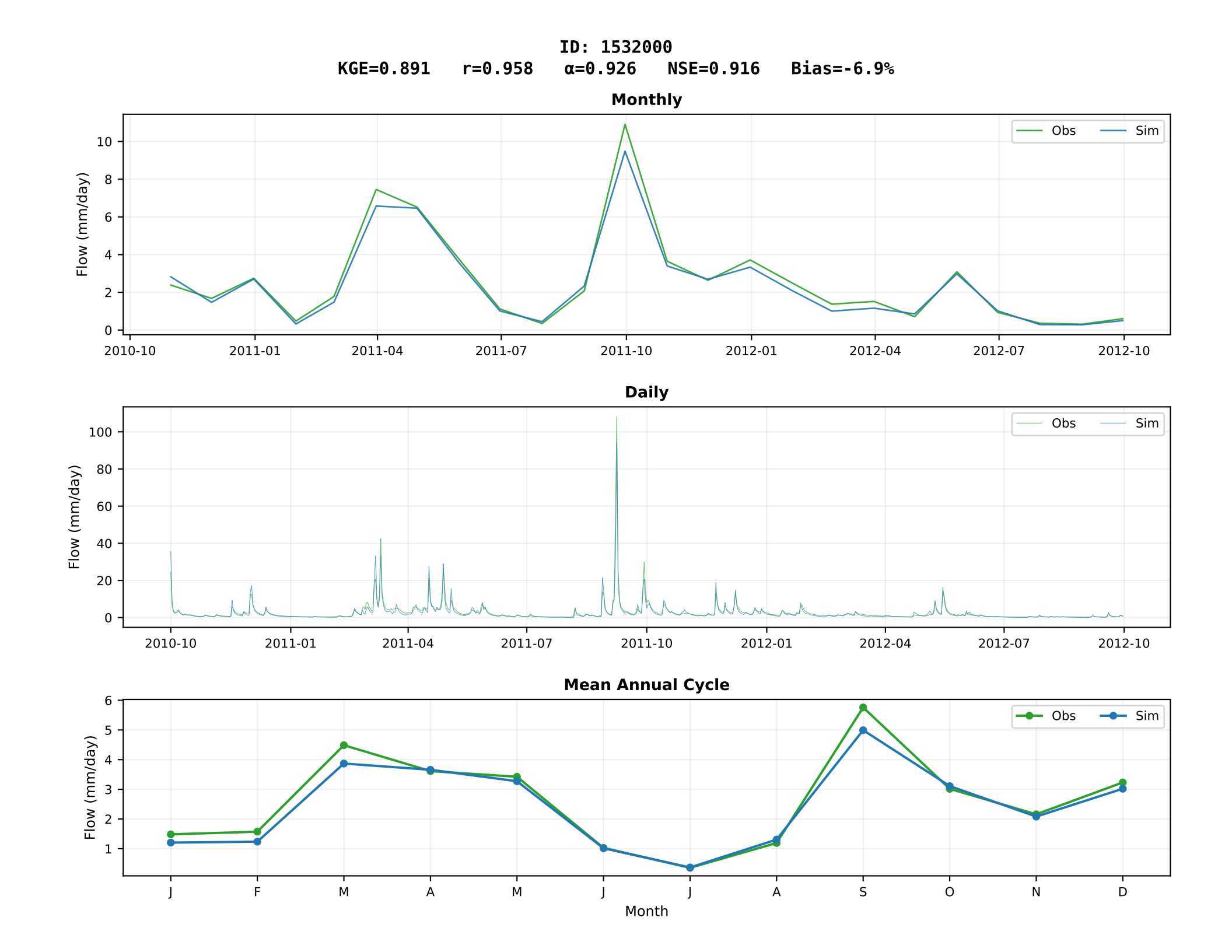

ID: 9210500
KGE=0.896 r=0.953 α=0.944 NSE=0.905 Bias=-7.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

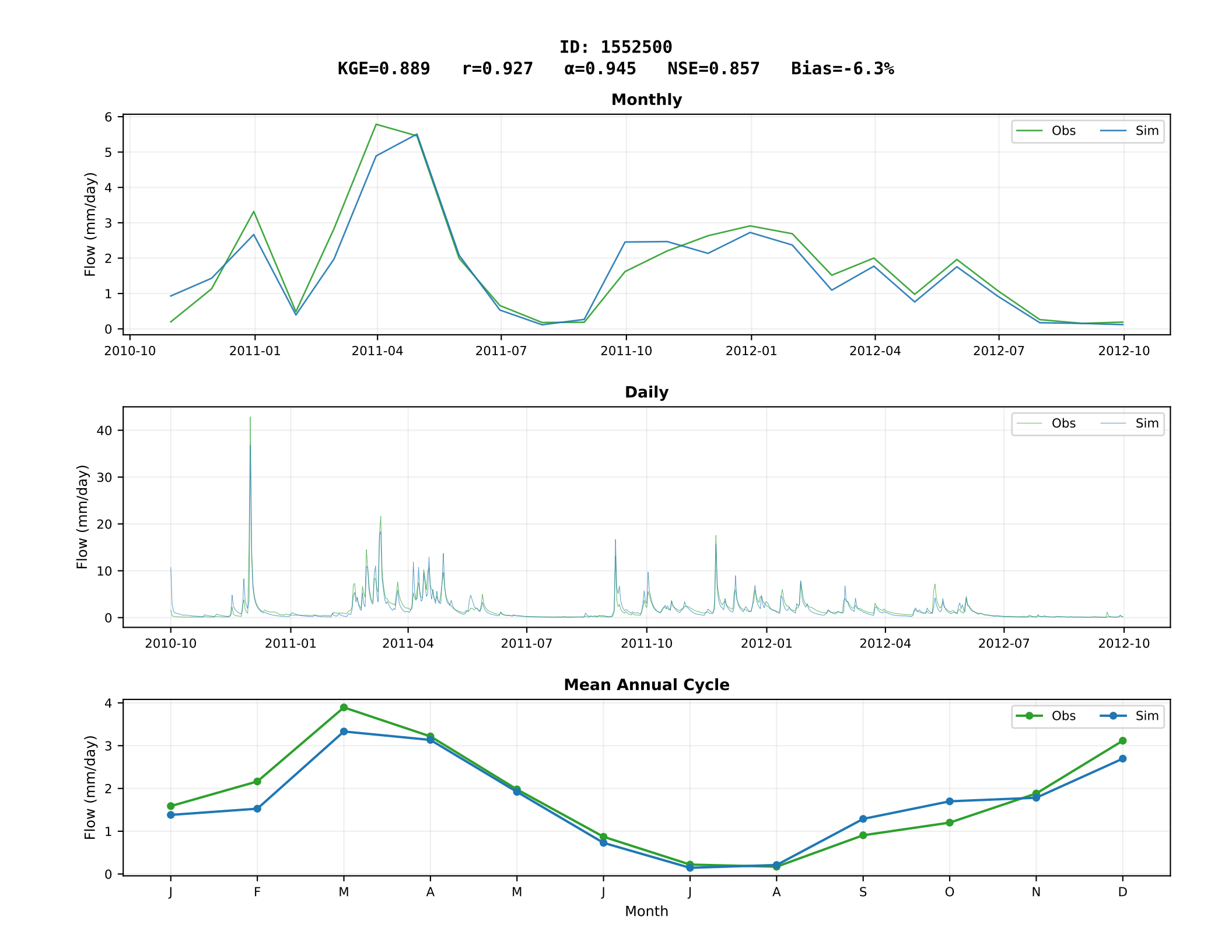

ID: 4024430
KGE=0.894 r=0.925 α=0.991 NSE=0.847 Bias=-7.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

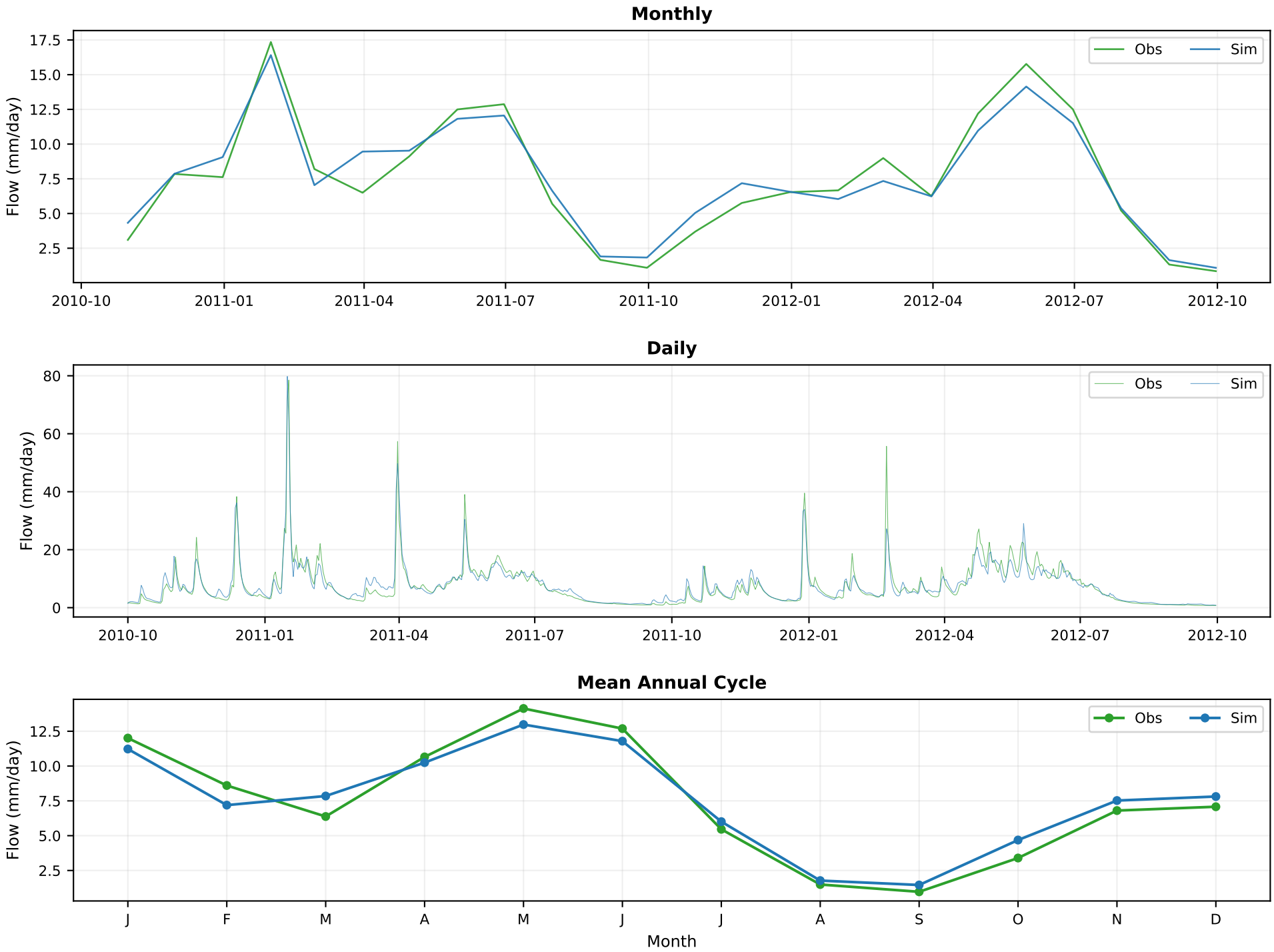

CACHE CREEK NEAR JACKSON, WY (ID=13018300)
KGE=0.893 r=0.957 α=0.944 NSE=0.912 Bias=+8.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

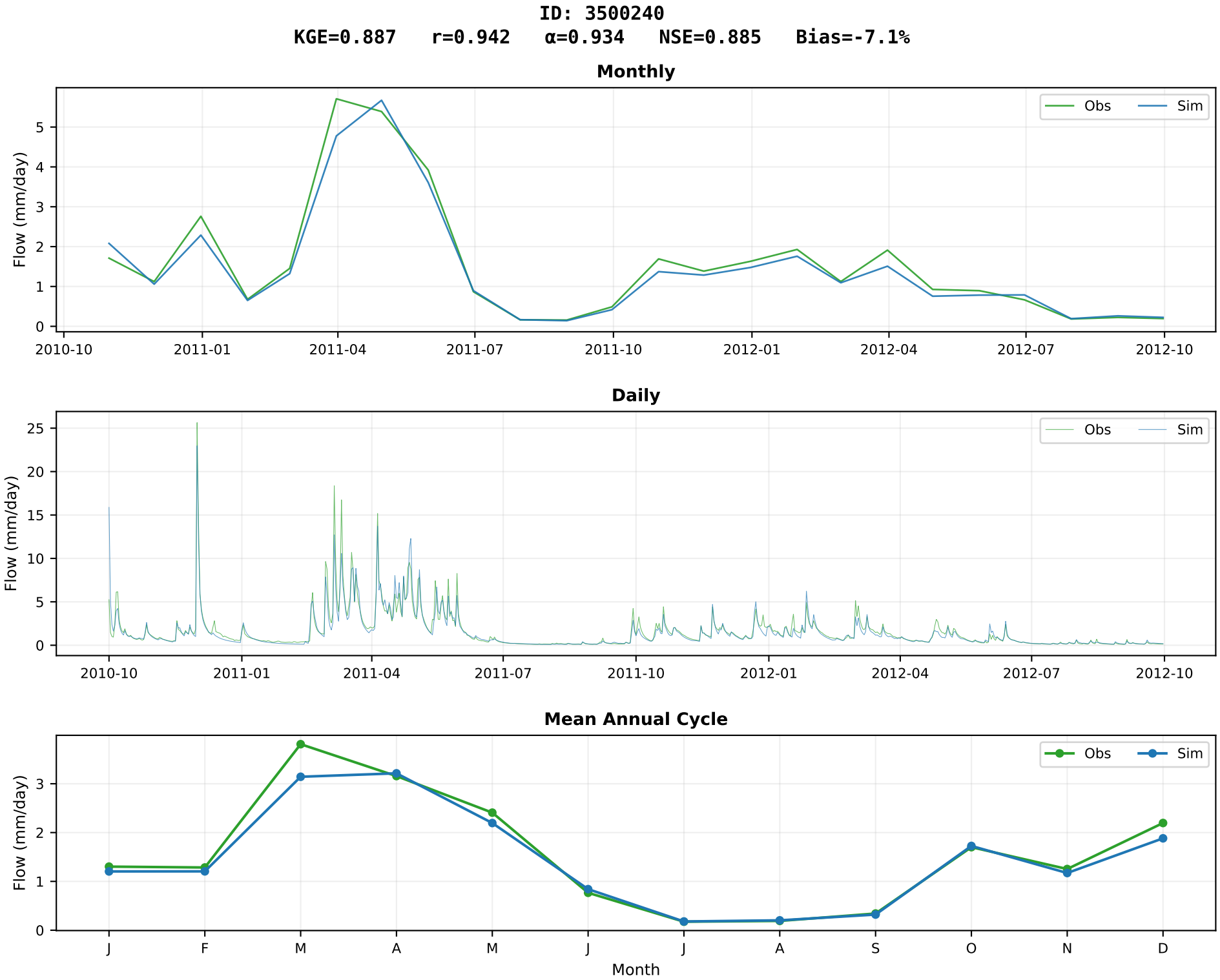

ID: 9223000
KGE=0.891 r=0.963 α=0.959 NSE=0.924 Bias=-9.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

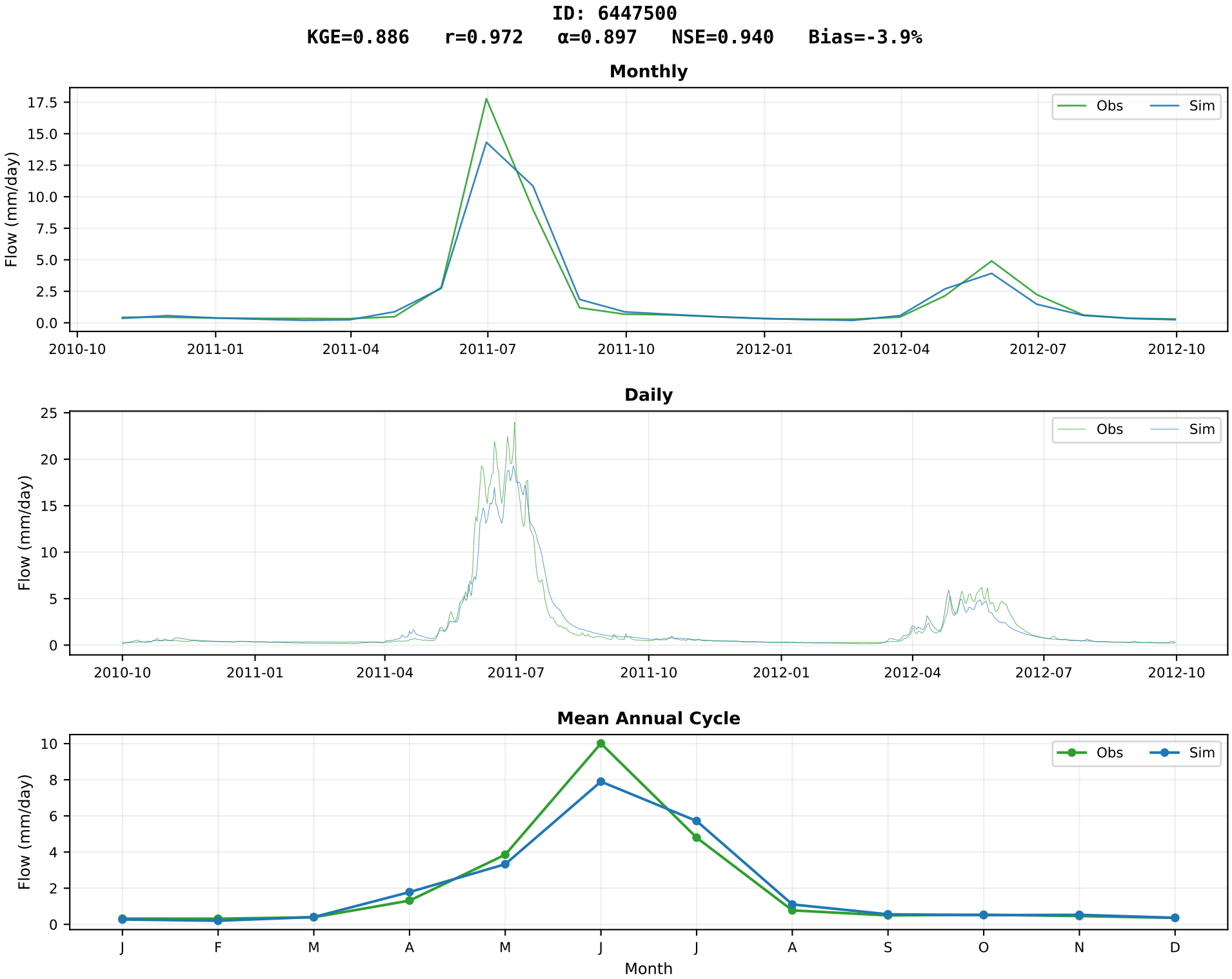

ID: 6903400
KGE=0.891 r=0.930 α=0.930 NSE=0.864 Bias=+4.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

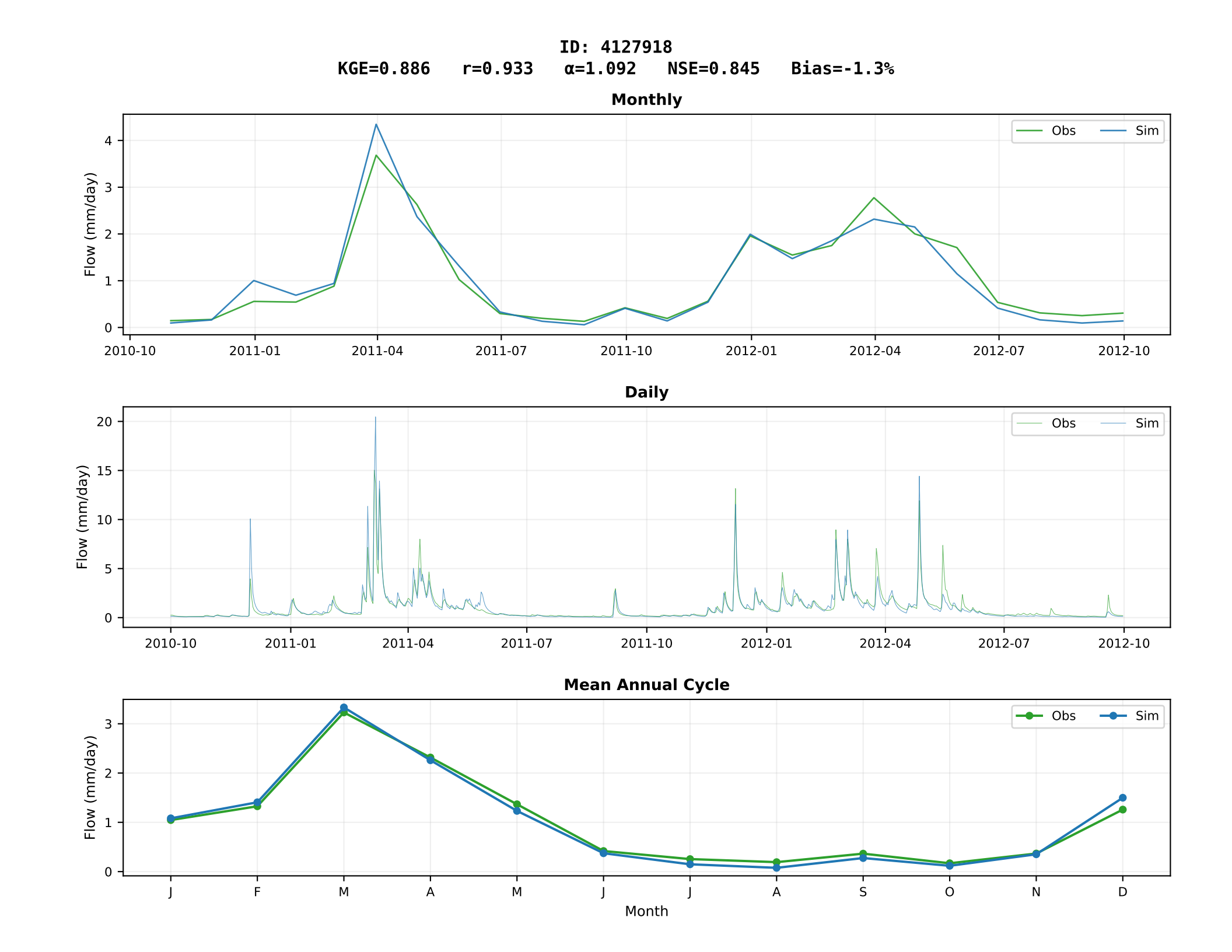

ID: 1532000
KGE=0.891 r=0.958 α=0.926 NSE=0.916 Bias=-6.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

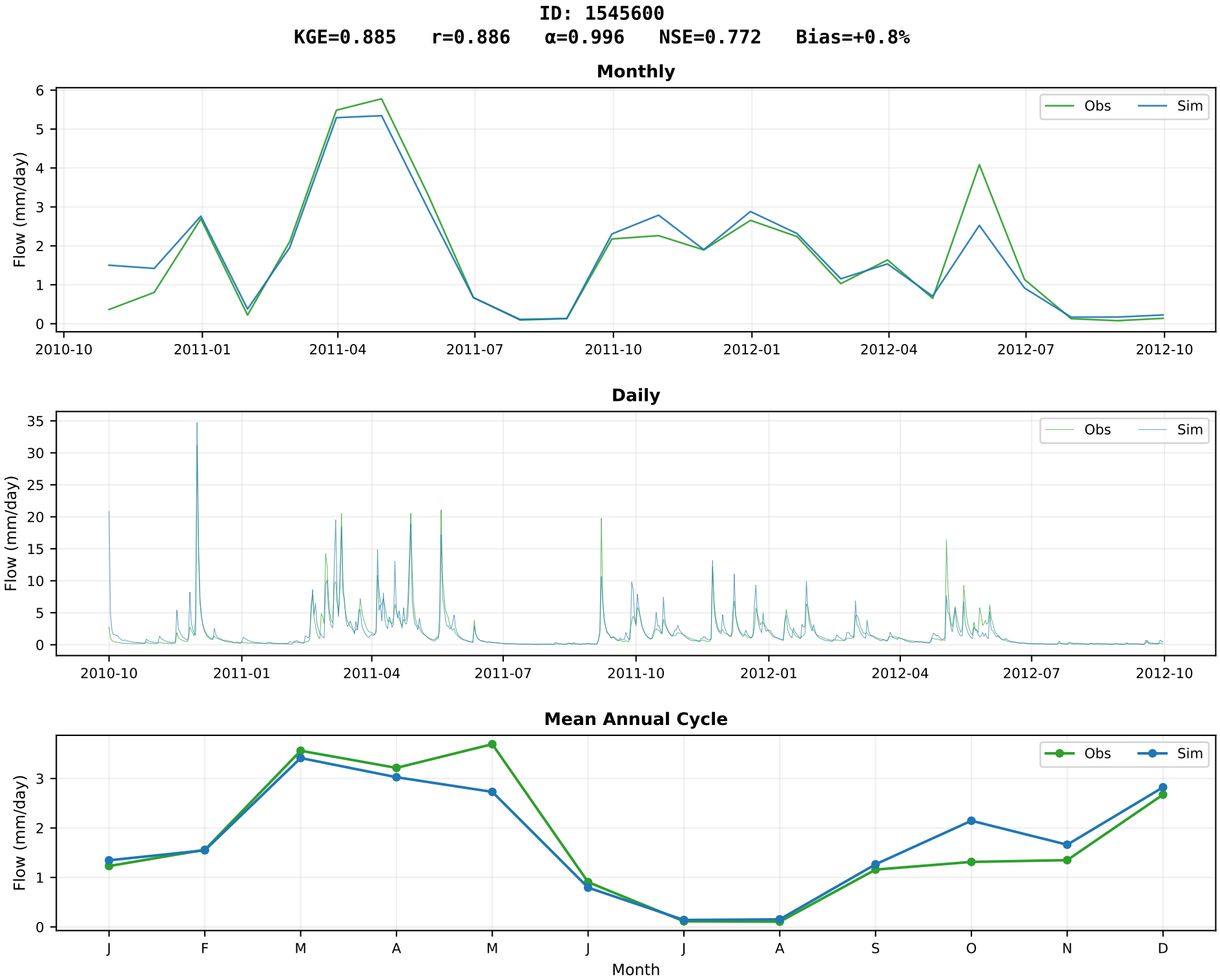

ID: 1552500
KGE=0.889 r=0.927 α=0.945 NSE=0.857 Bias=-6.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

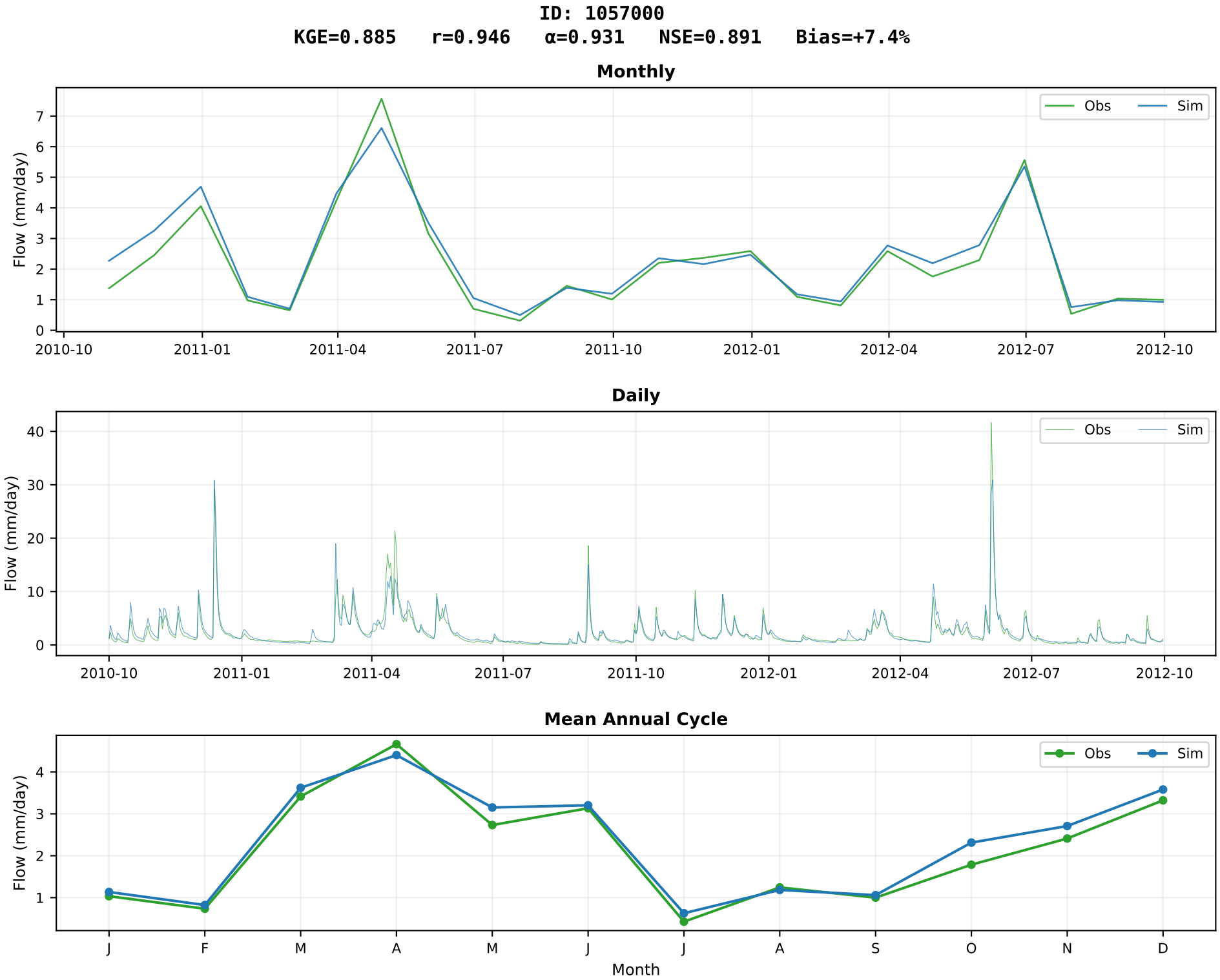

LAKE FORK PAYETTE RIVER AB JUMBO CR NR MCCALL ID (ID=13240000)
KGE=0.888 r=0.944 α=0.904 NSE=0.889 Bias=+1.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

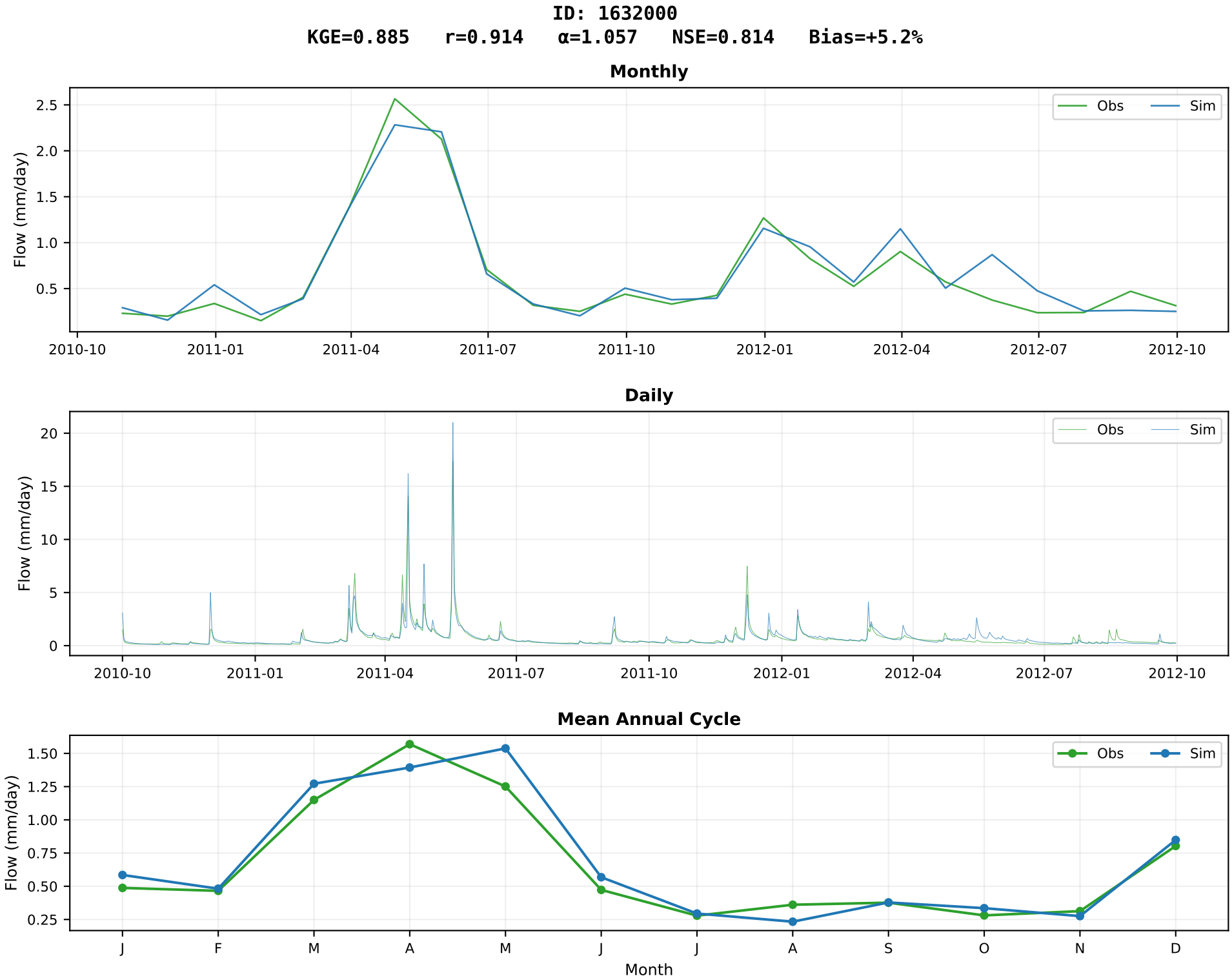

ID: 3500240
KGE=0.887 r=0.942 α=0.934 NSE=0.885 Bias=-7.1%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

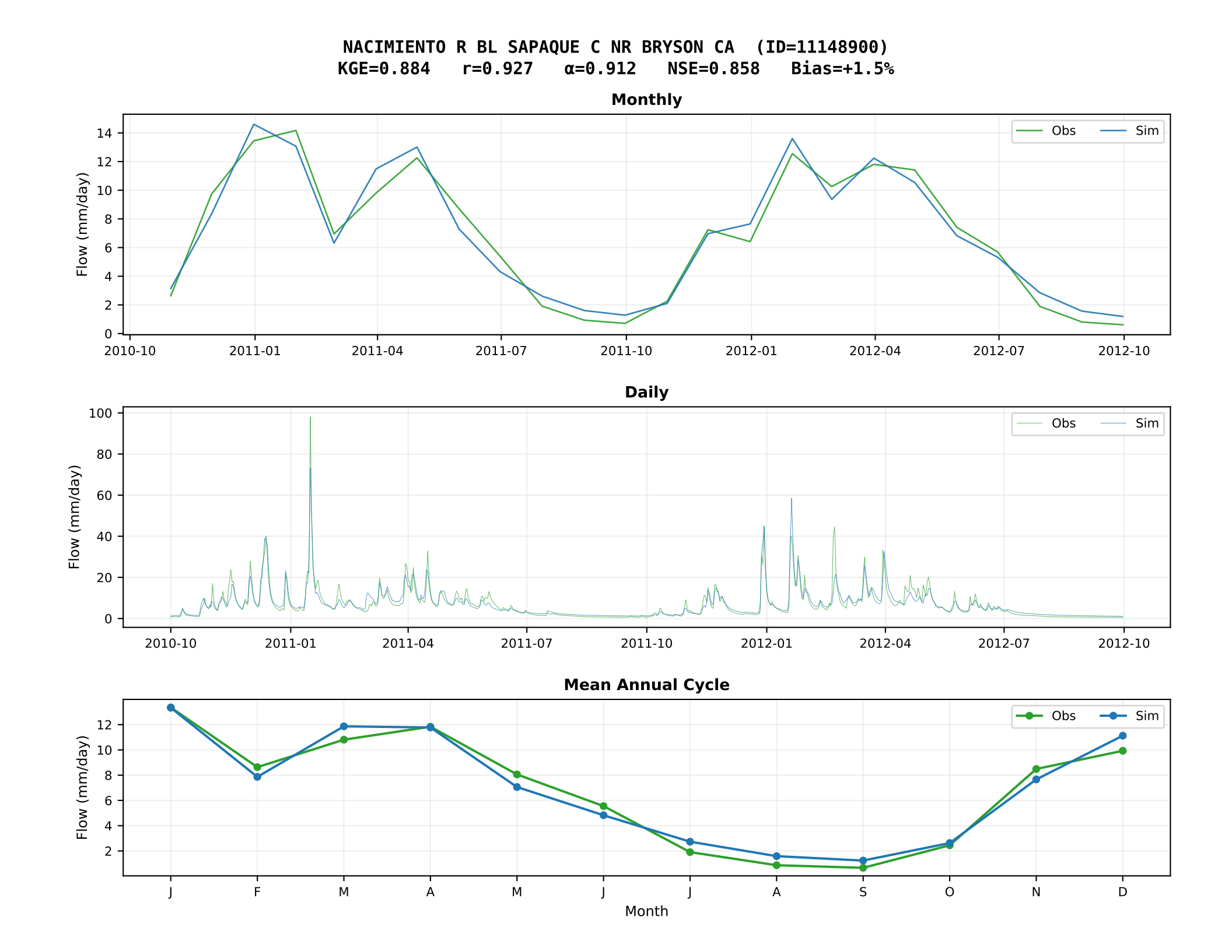

ID: 6447500
KGE=0.886 r=0.972 α=0.897 NSE=0.940 Bias=-3.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

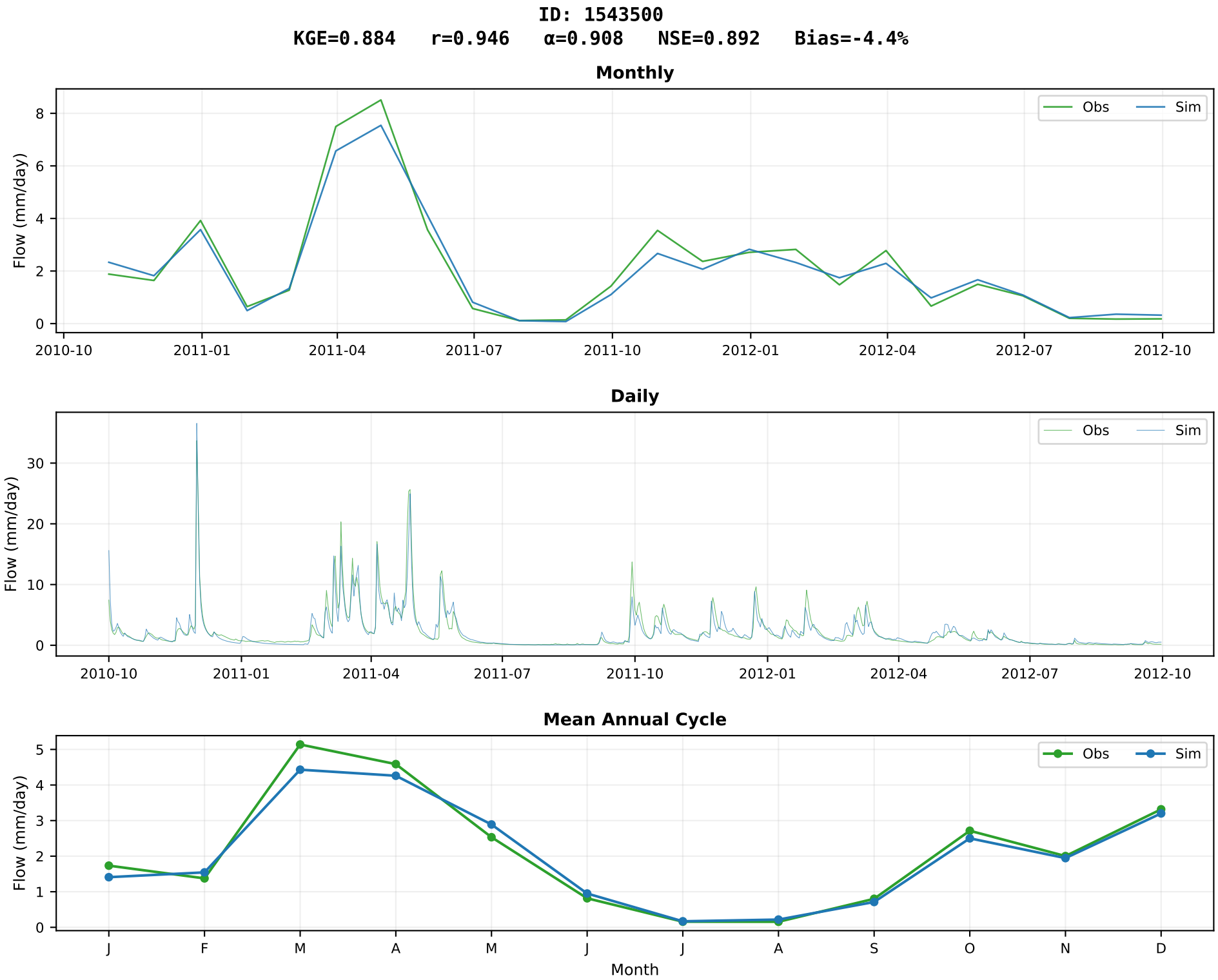

ID: 4127918
KGE=0.886 r=0.933 α=1.092 NSE=0.845 Bias=-1.3%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

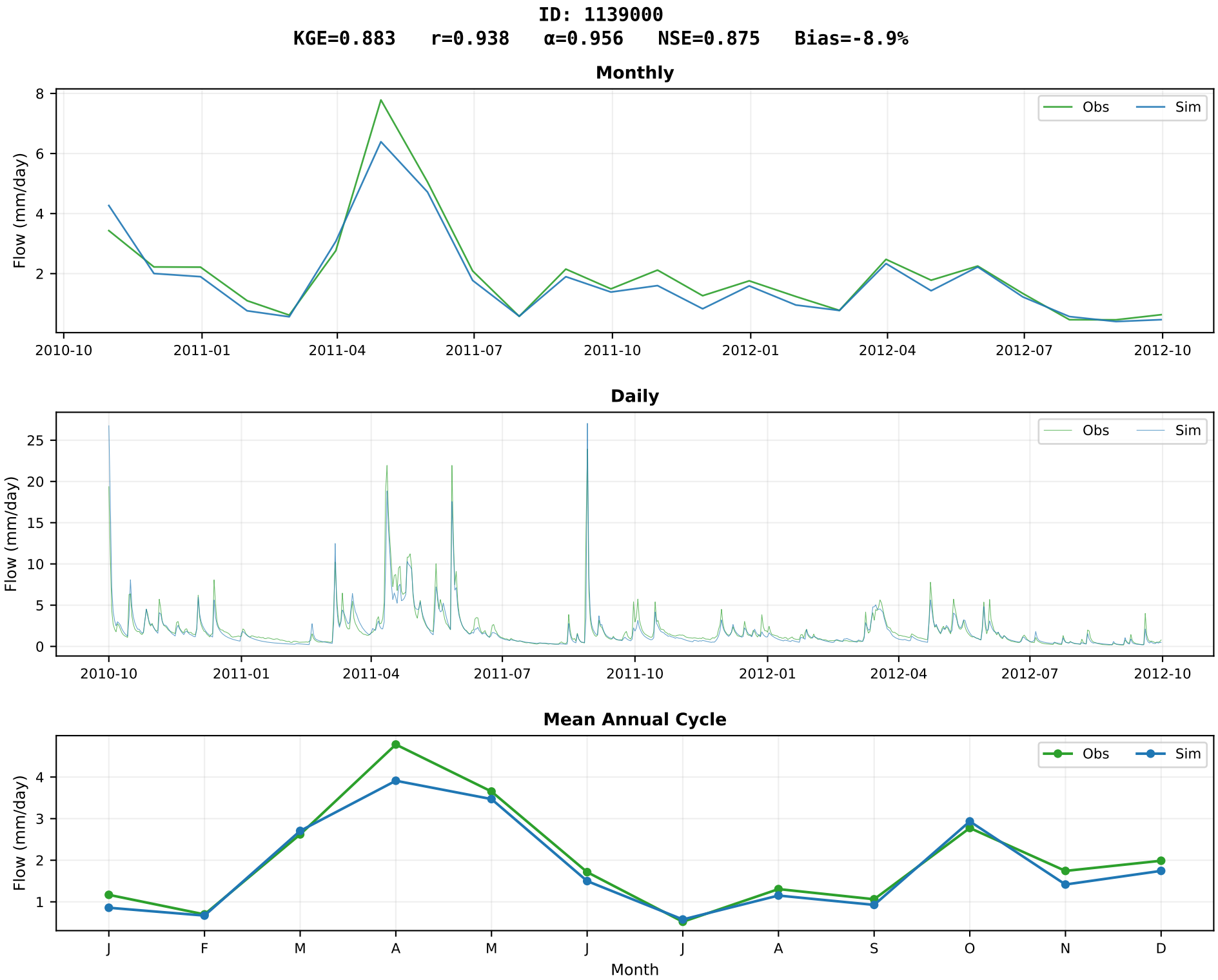

ID: 1545600
KGE=0.885 r=0.886 α=0.996 NSE=0.772 Bias=+0.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

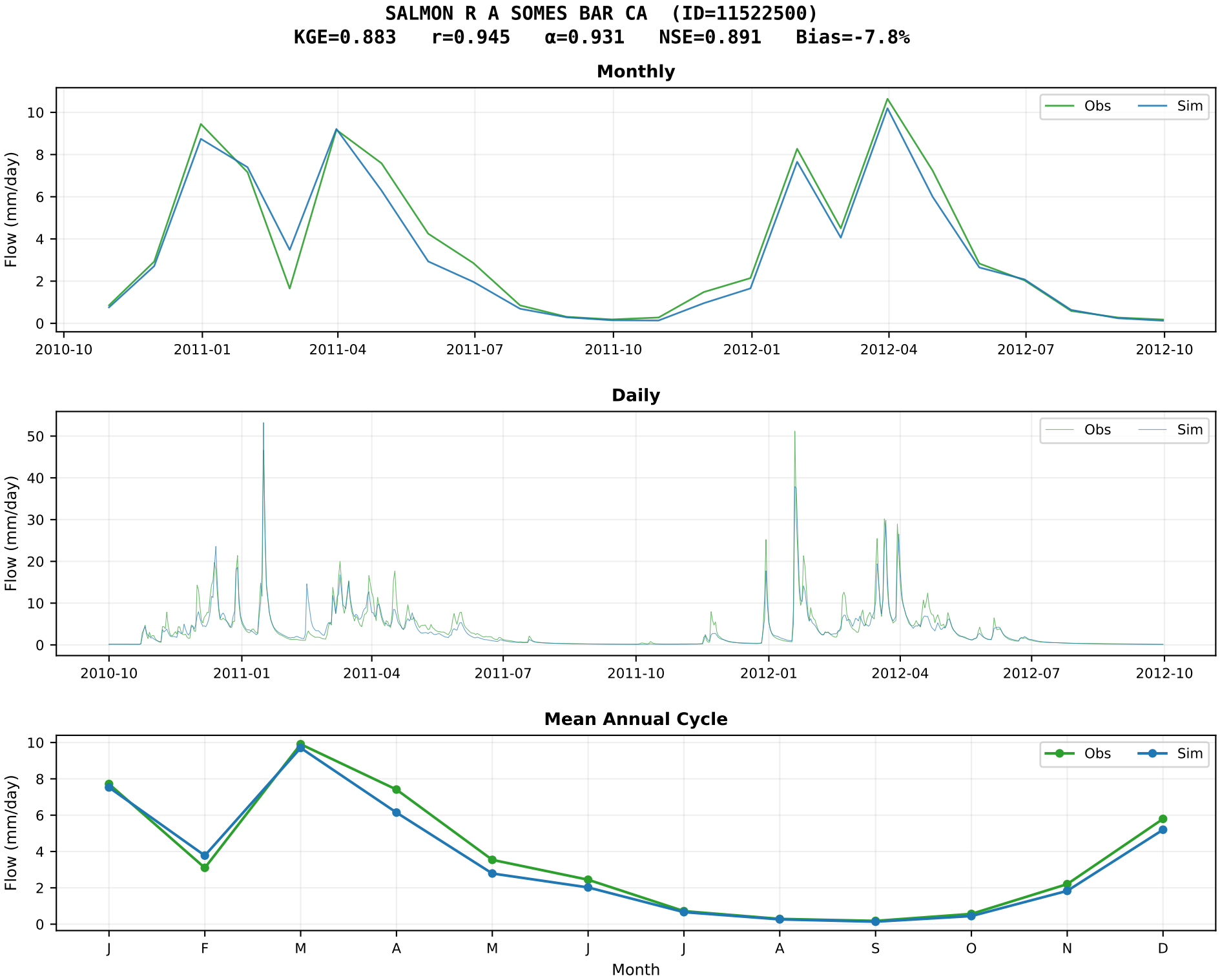

ID: 1057000
KGE=0.885 r=0.946 α=0.931 NSE=0.891 Bias=+7.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

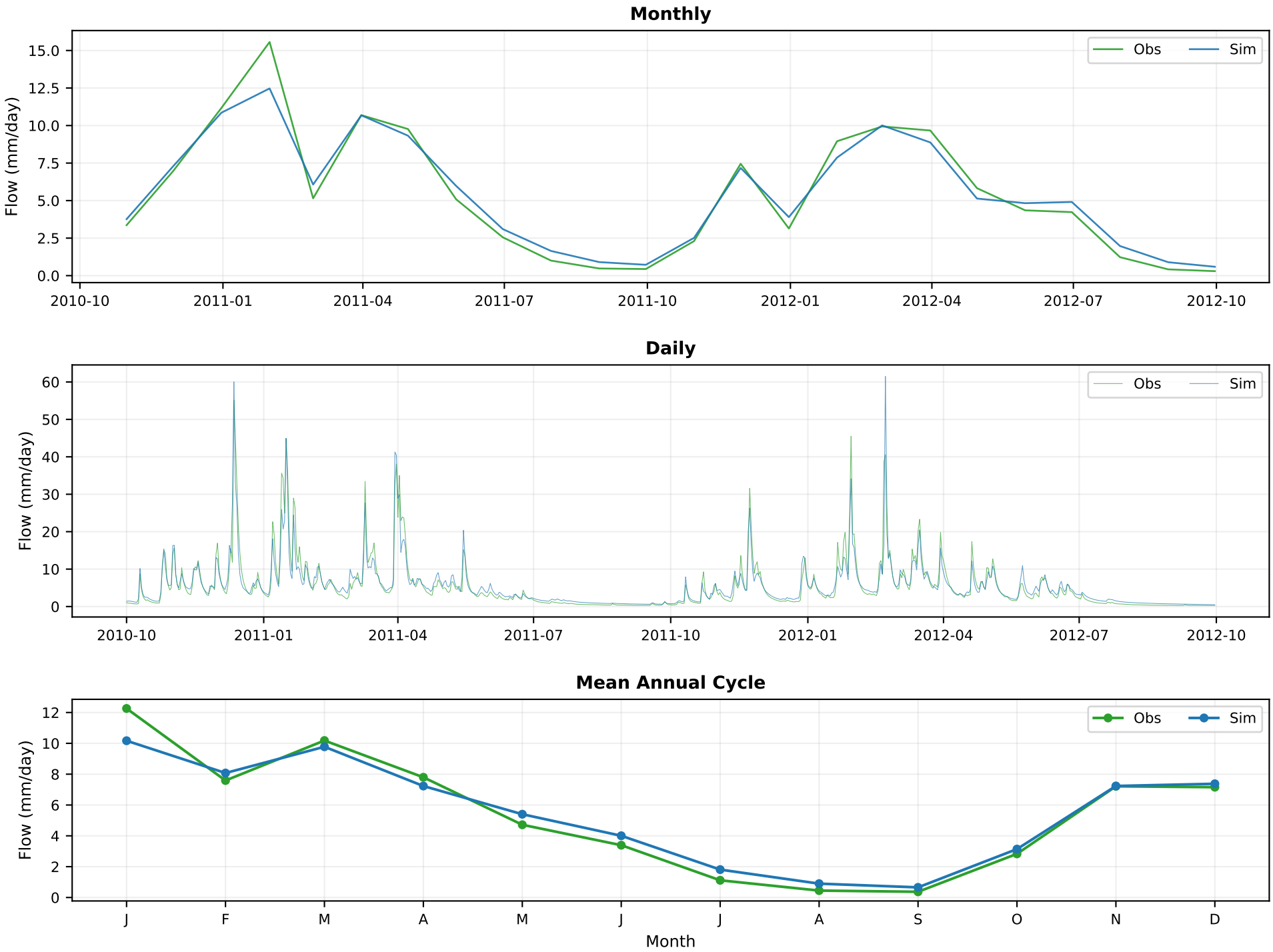

ID: 1632000
KGE=0.885 r=0.914 α=1.057 NSE=0.814 Bias=+5.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

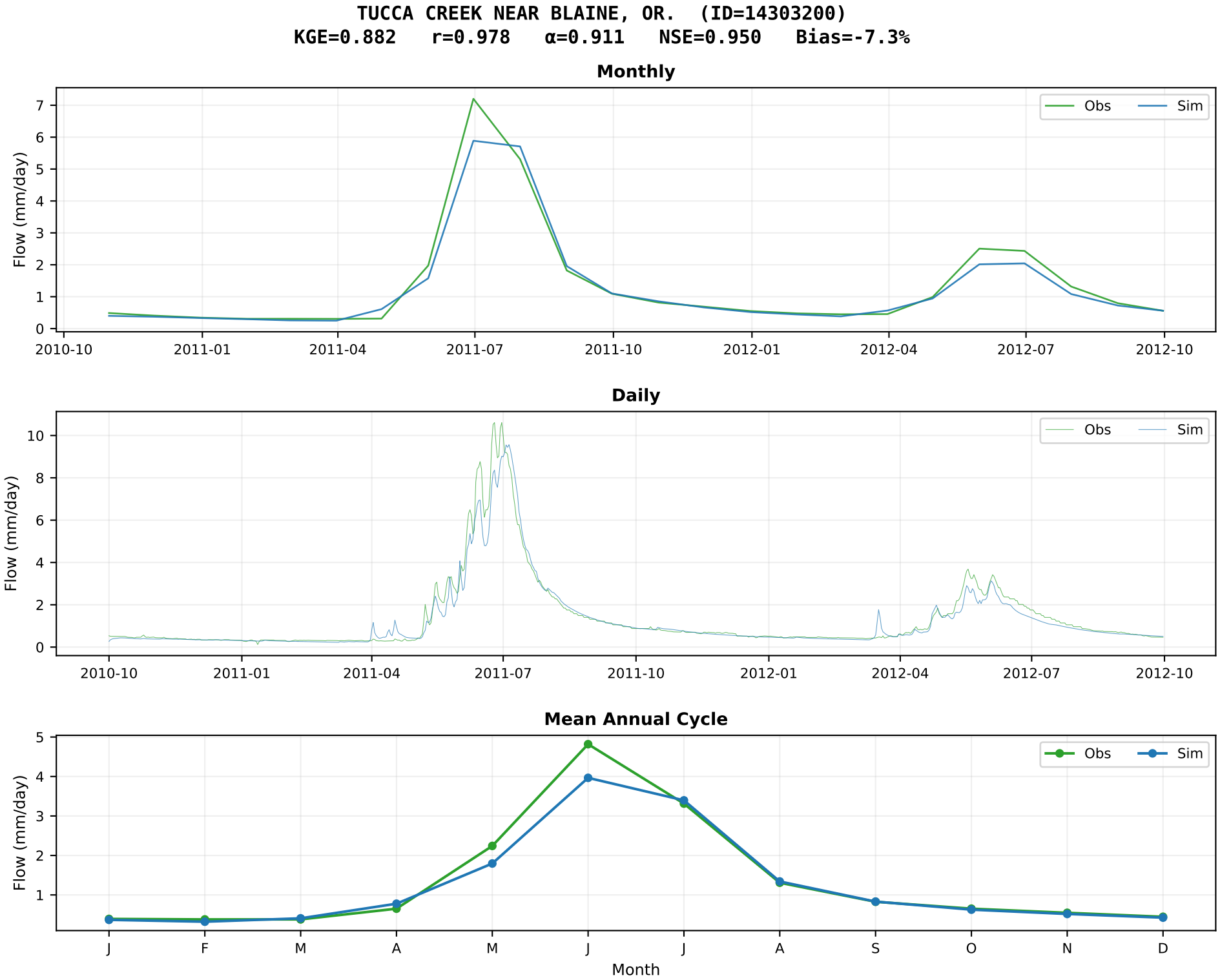

NACIMIENTO R BL SAPAQUE C NR BRYSON CA (ID=11148900)
KGE=0.884 r=0.927 α=0.912 NSE=0.858 Bias=+1.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

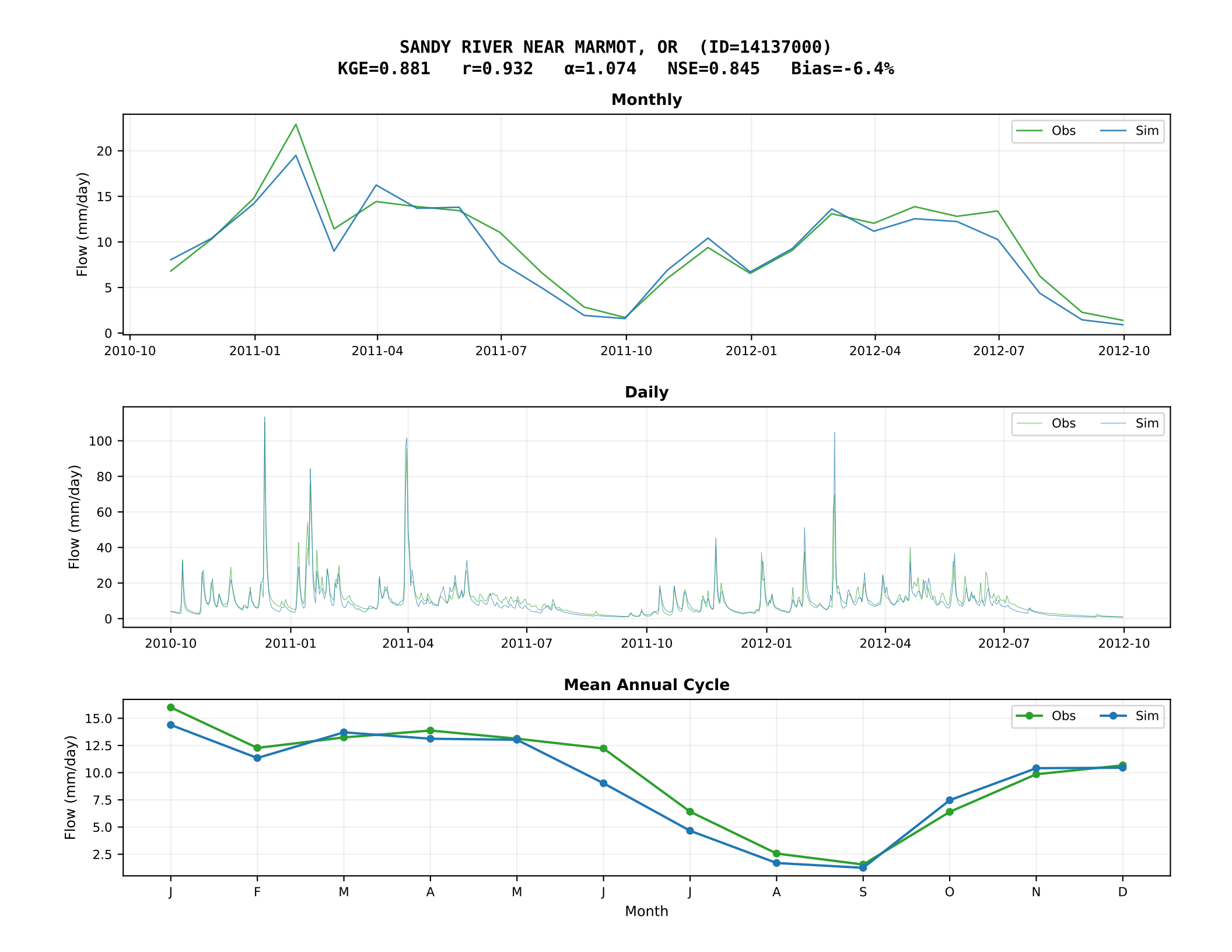

ID: 1543500
KGE=0.884 r=0.946 α=0.908 NSE=0.892 Bias=-4.4%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

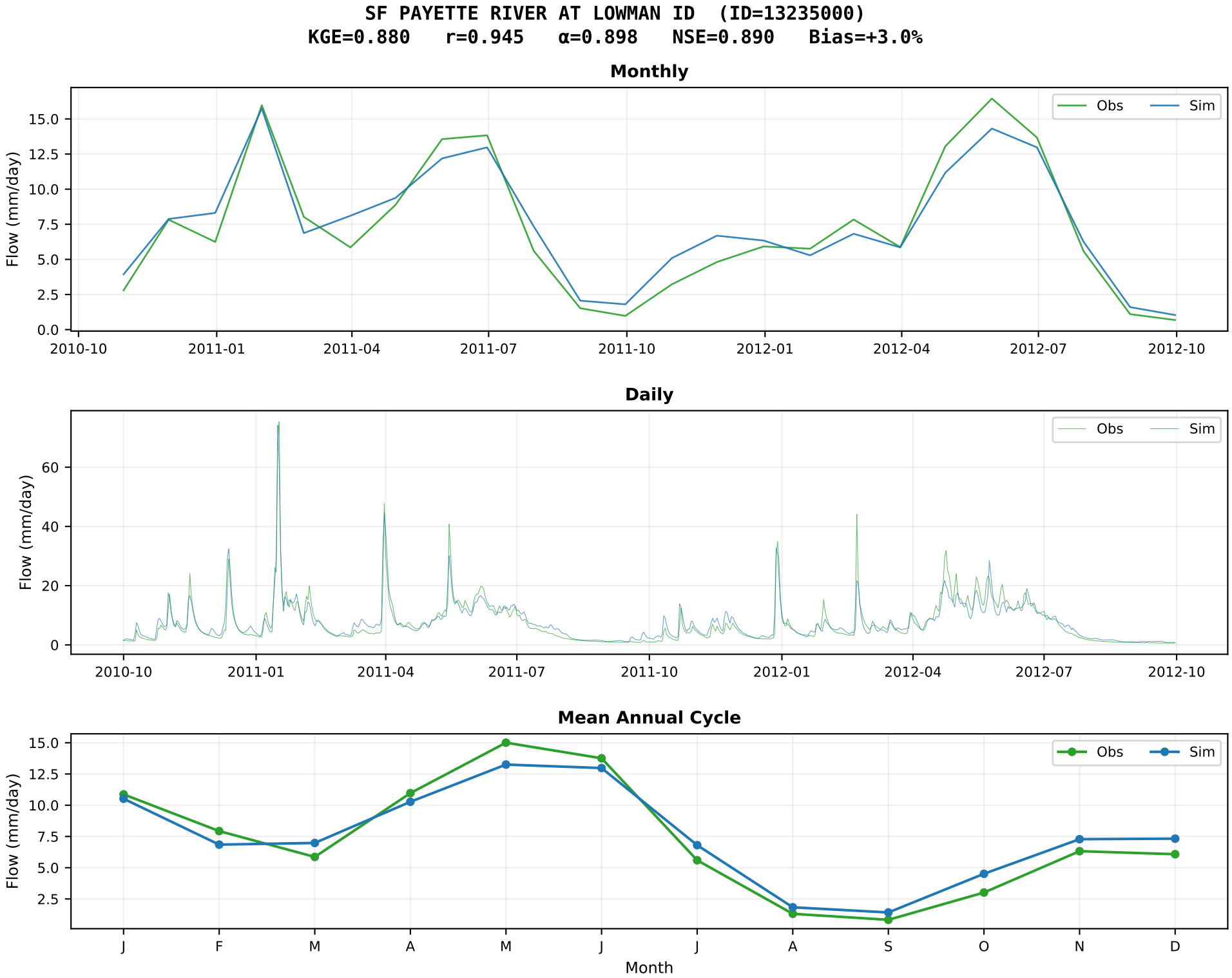

ID: 1139000
KGE=0.883 r=0.938 α=0.956 NSE=0.875 Bias=-8.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

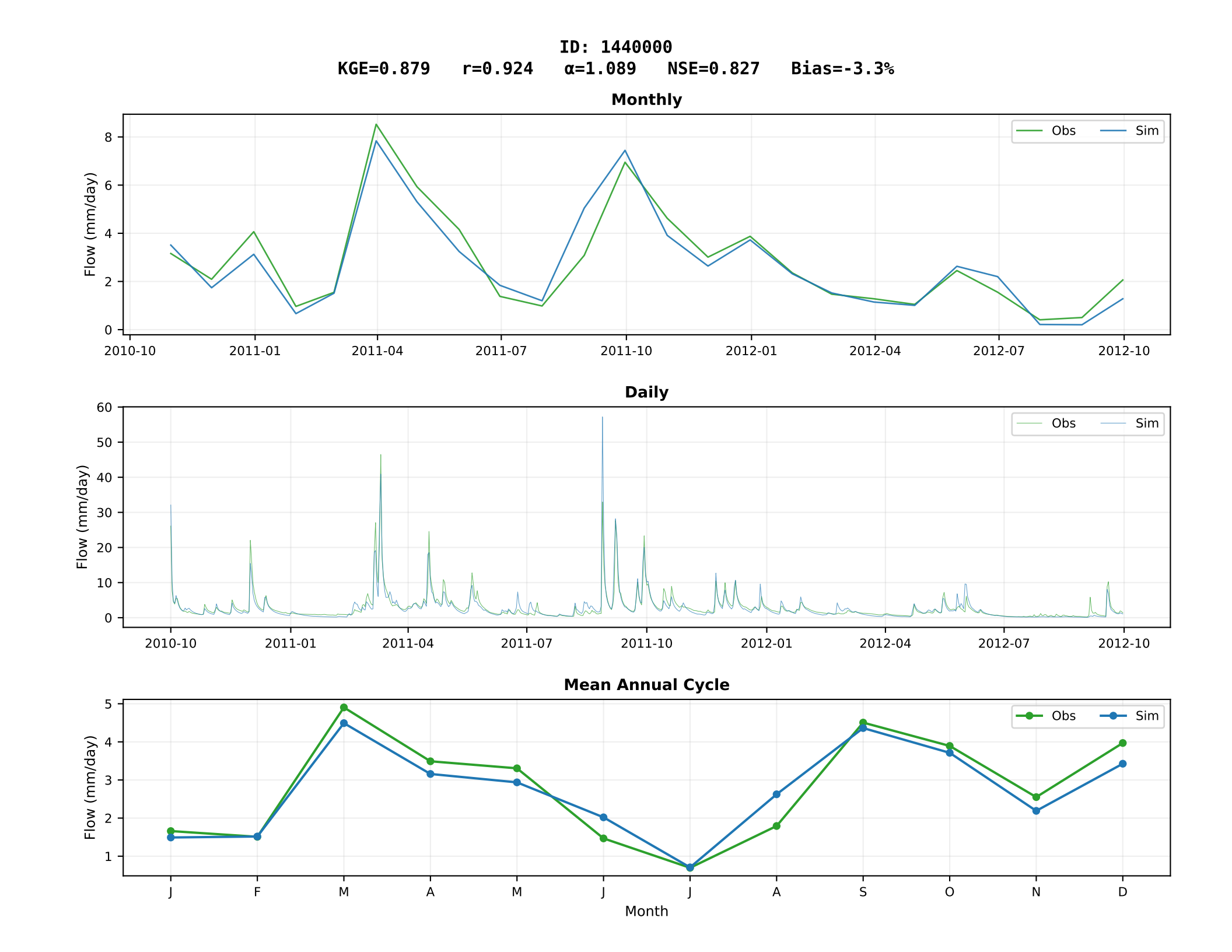

SALMON R A SOMES BAR CA (ID=11522500)
KGE=0.883 r=0.945 α=0.931 NSE=0.891 Bias=-7.8%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
J
F
M
A
M
J
J
A
S
O
N
D
Month

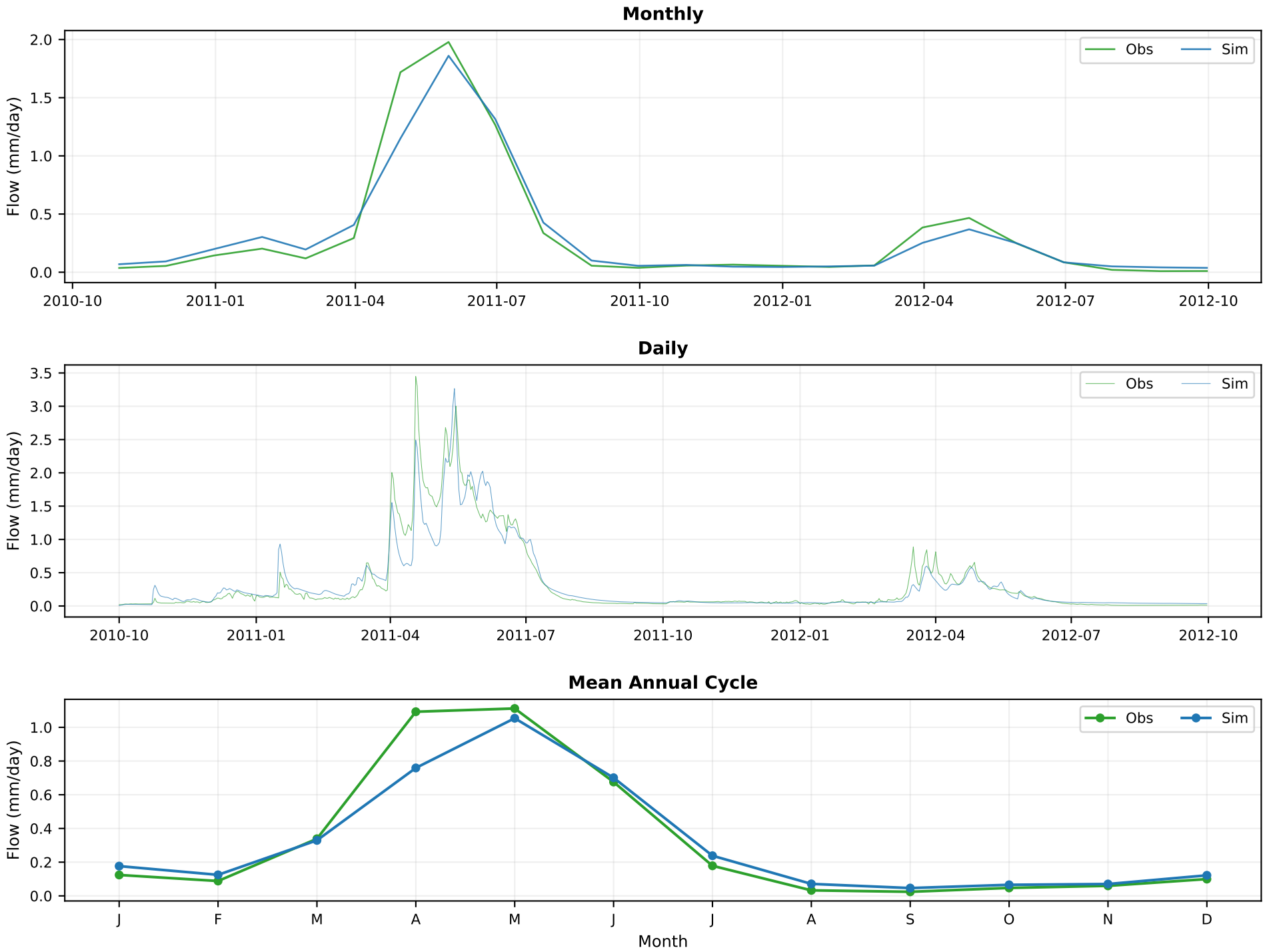

BEAVER CREEK, BLW QUARTZ CR, NR SIMNASHO, OR. (ID=14096850)
KGE=0.883 r=0.943 α=0.898 NSE=0.887 Bias=+1.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

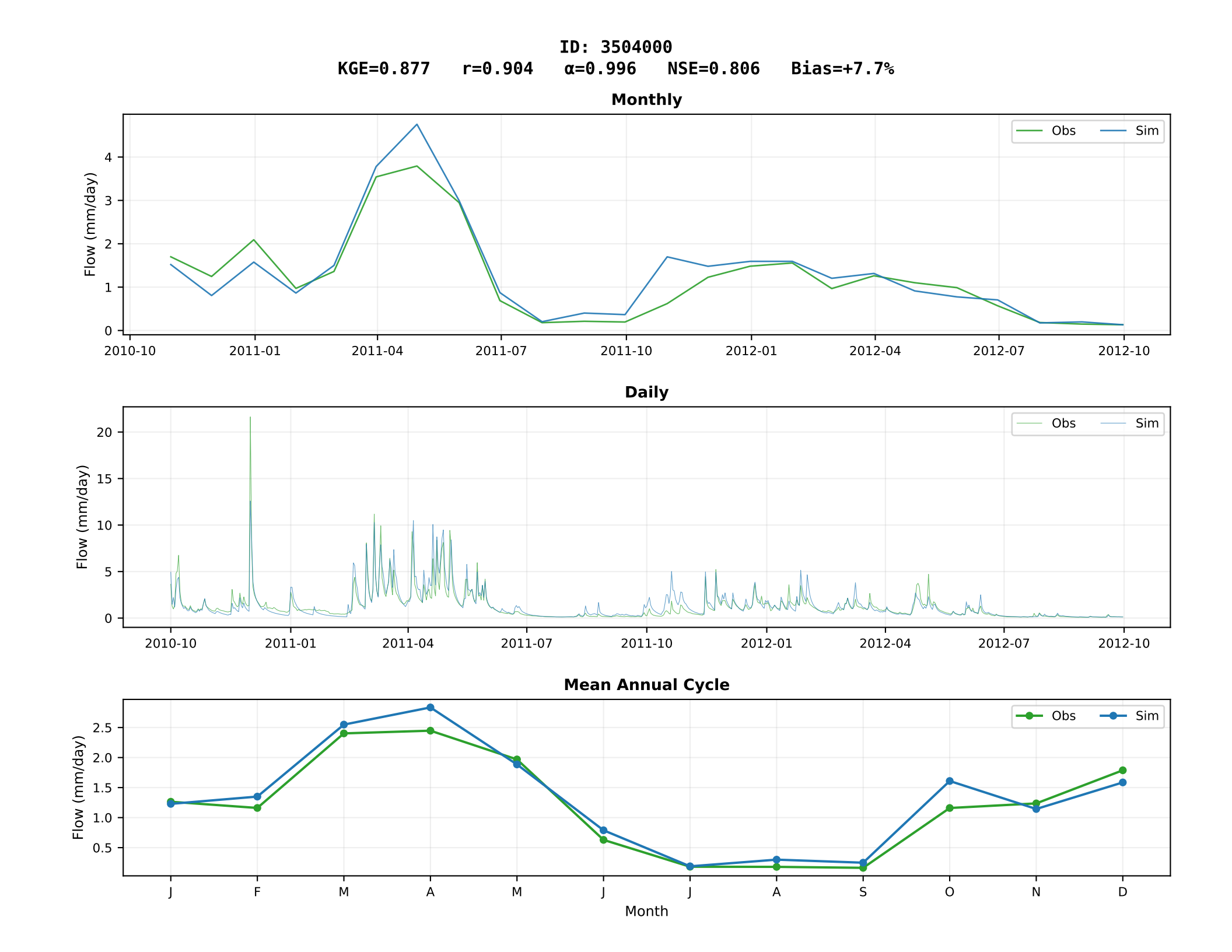

TUCCA CREEK NEAR BLAINE, OR. (ID=14303200)
KGE=0.882 r=0.978 α=0.911 NSE=0.950 Bias=-7.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

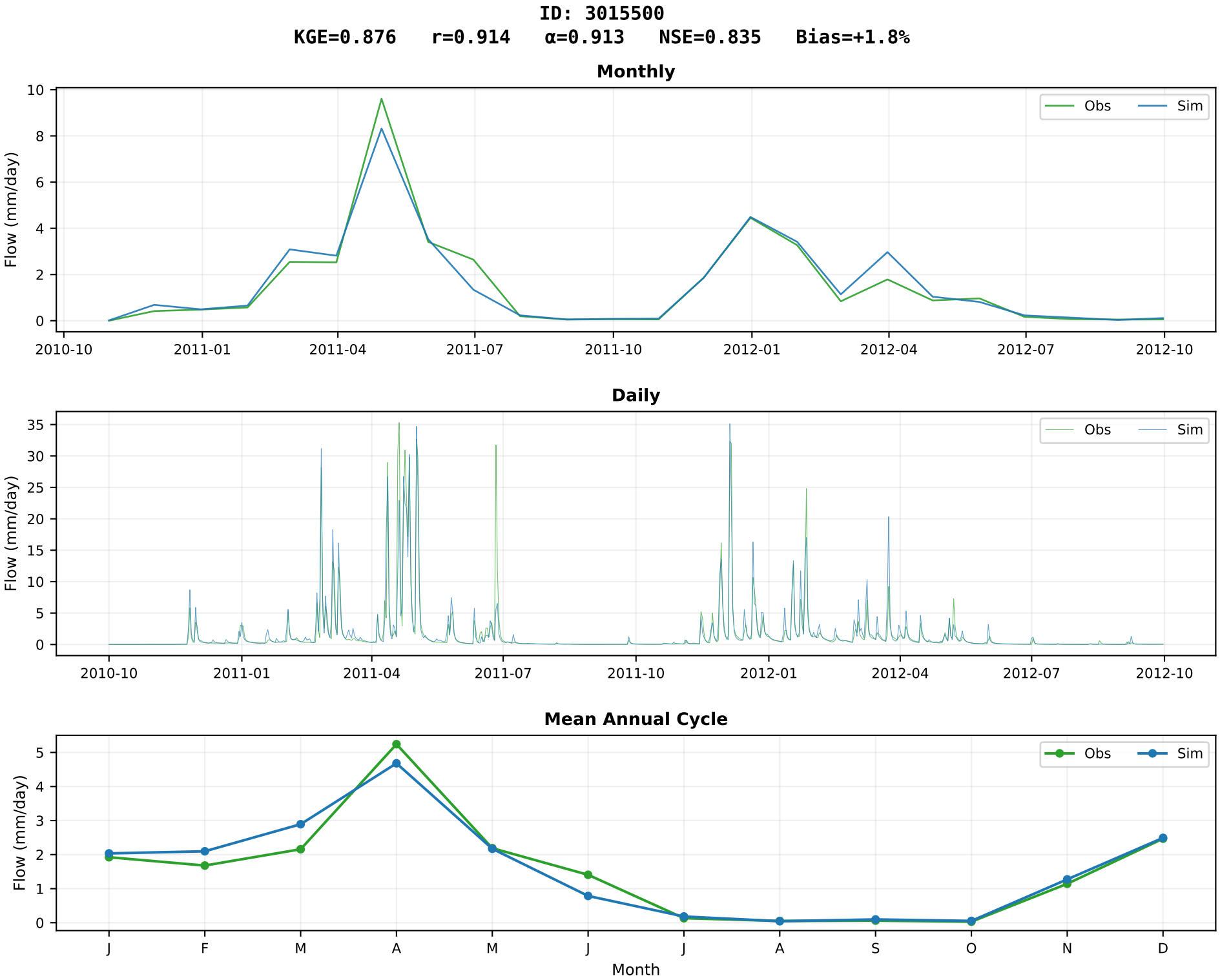

SANDY RIVER NEAR MARMOT, OR (ID=14137000)
KGE=0.881 r=0.932 α=1.074 NSE=0.845 Bias=-6.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

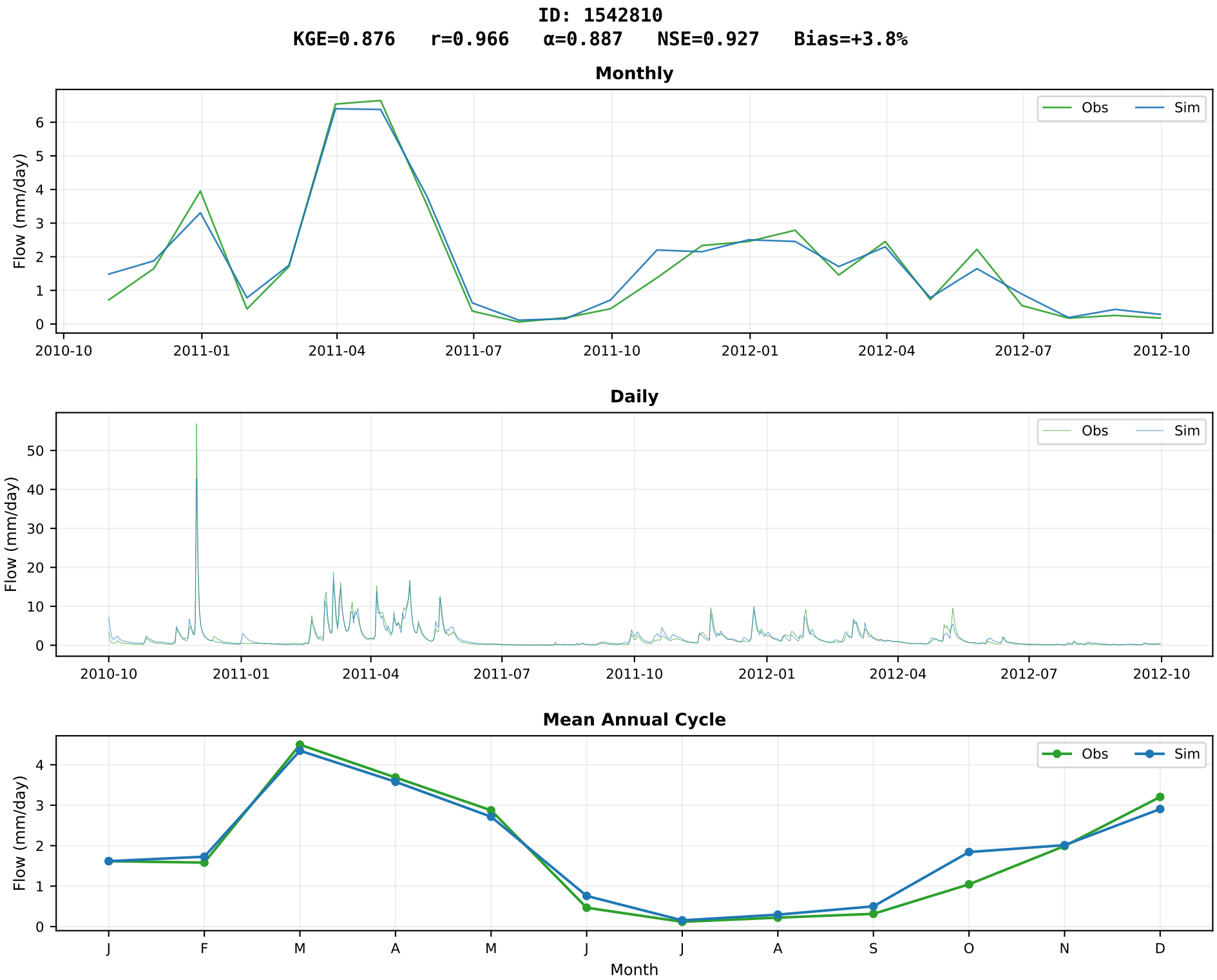

SF PAYETTE RIVER AT LOWMAN ID (ID=13235000)
KGE=0.880 r=0.945 α=0.898 NSE=0.890 Bias=+3.0%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 1440000
KGE=0.879 r=0.924 α=1.089 NSE=0.827 Bias=-3.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

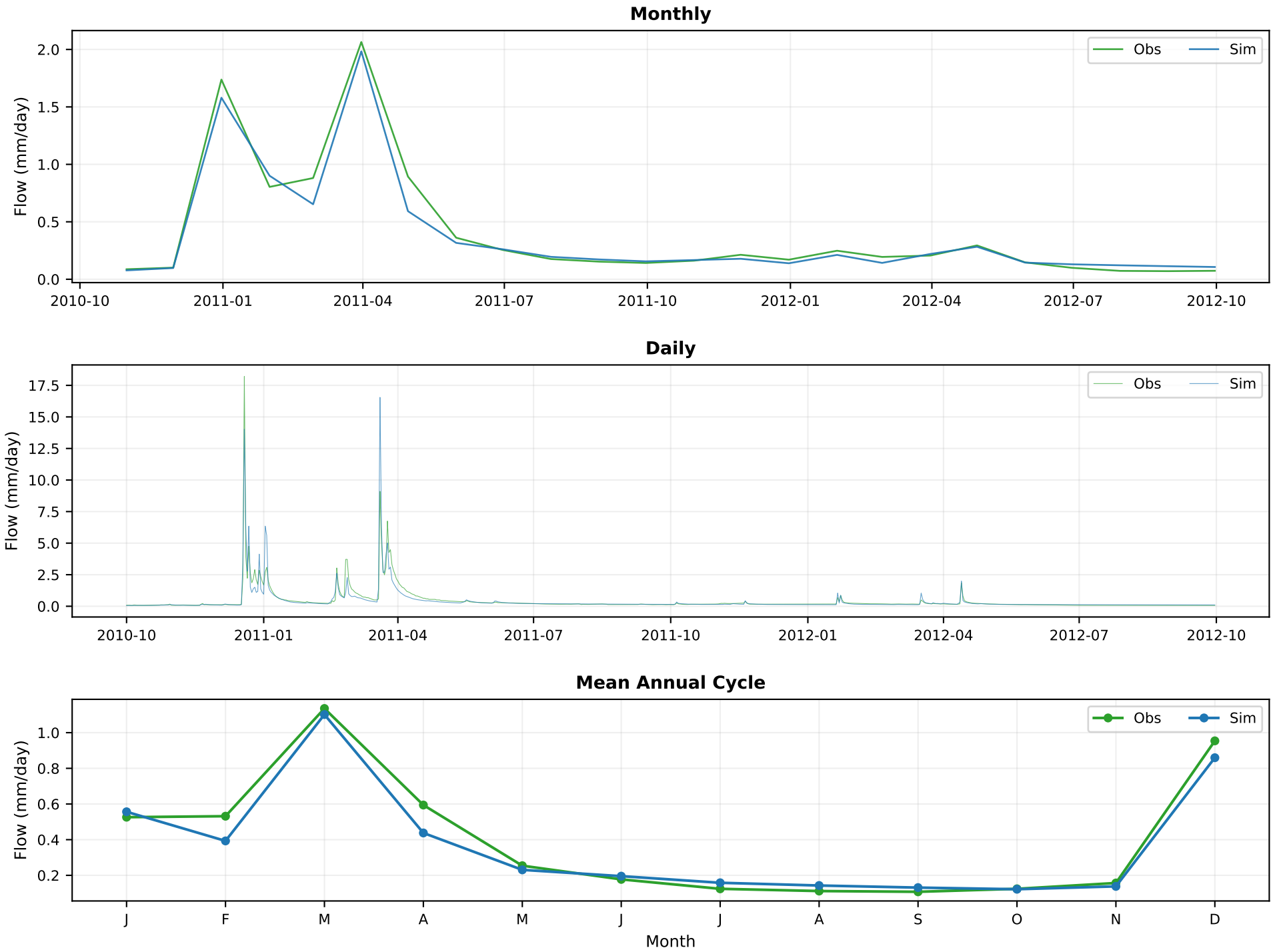

EAST FORK LOBSTER CREEK NEAR ALSEA, OR. (ID=14306340)
KGE=0.878 r=0.951 α=0.892 NSE=0.901 Bias=-2.9%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

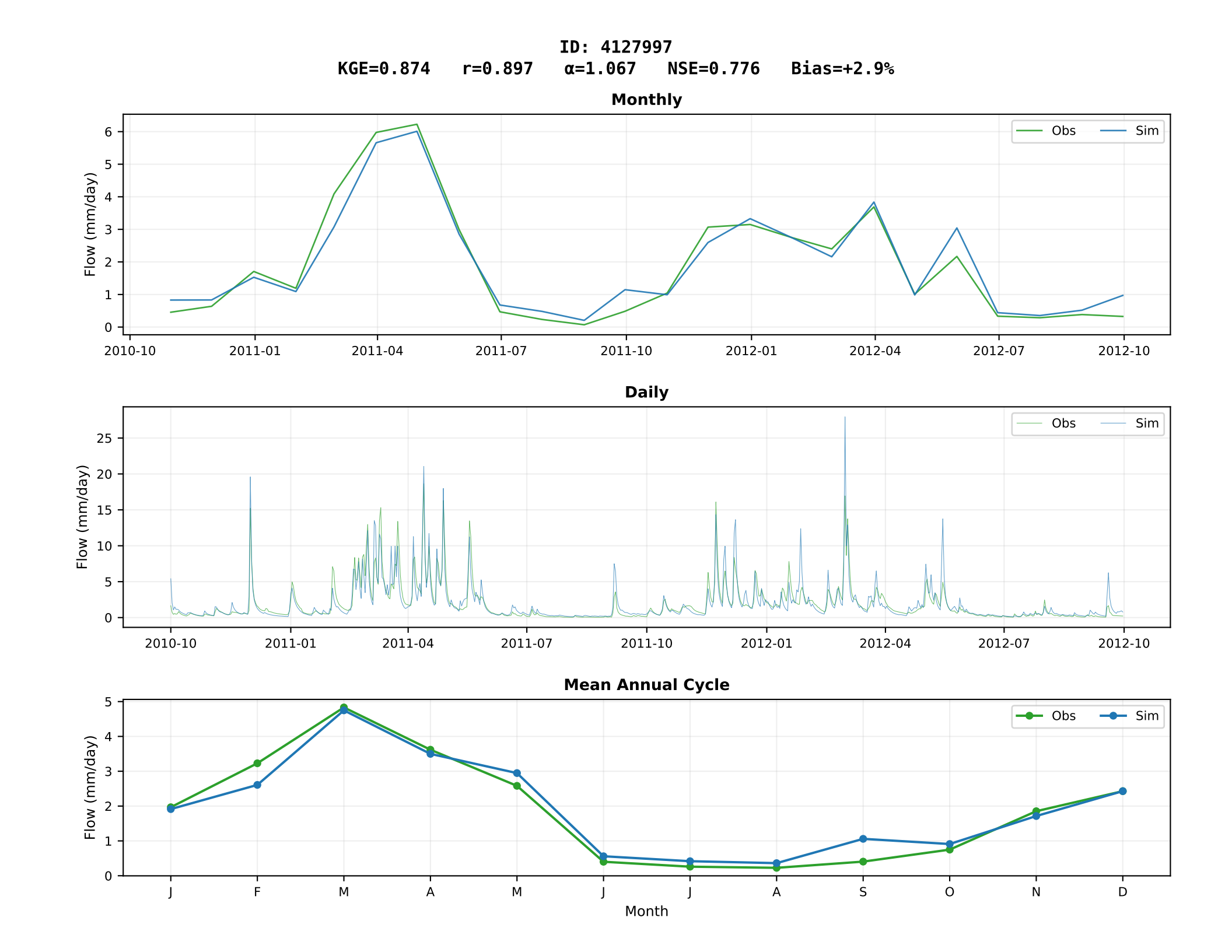

ID: 3504000
KGE=0.877 r=0.904 α=0.996 NSE=0.806 Bias=+7.7%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.5
1.0
1.5
2.0
2.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

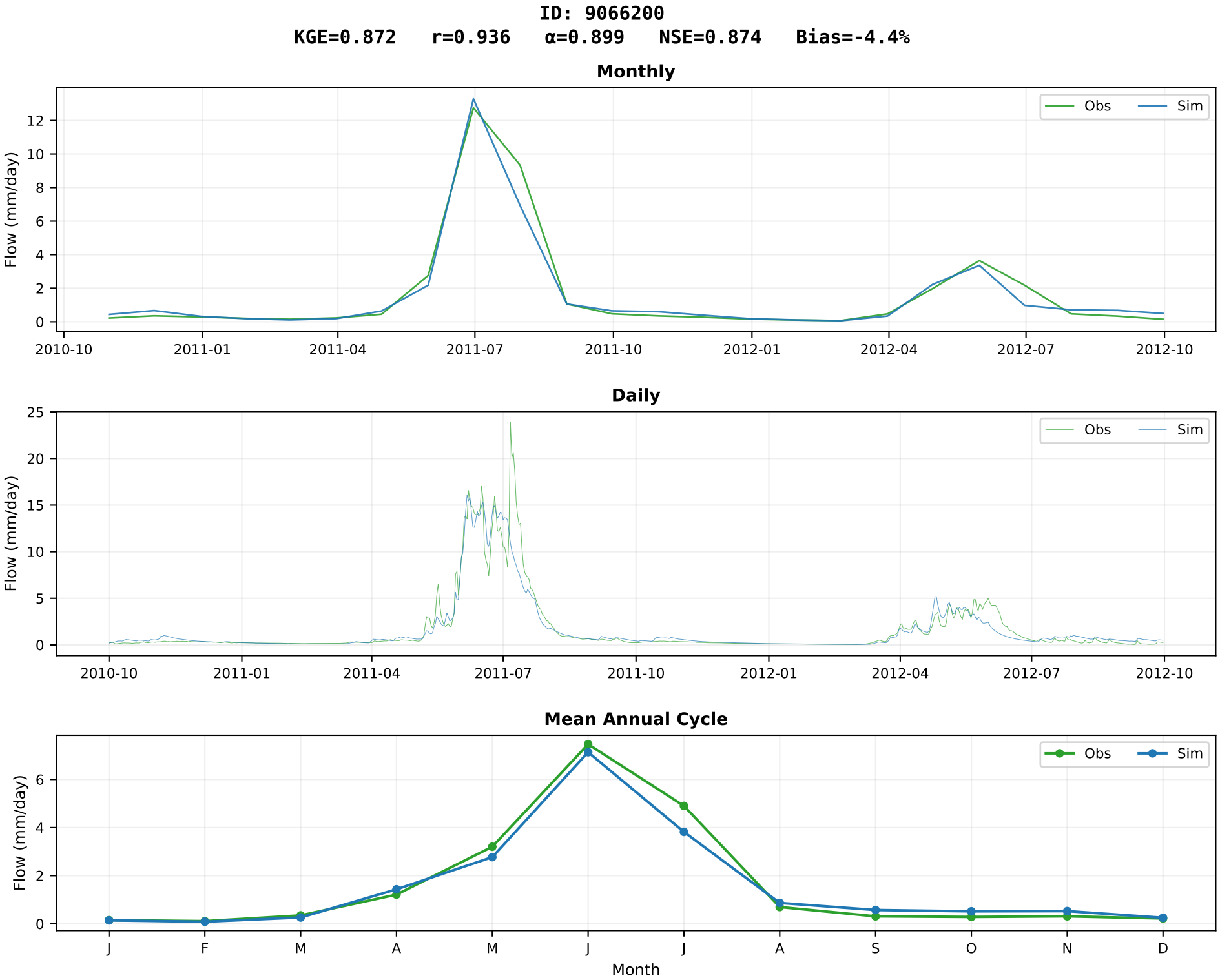

ID: 3015500
KGE=0.876 r=0.914 α=0.913 NSE=0.835 Bias=+1.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

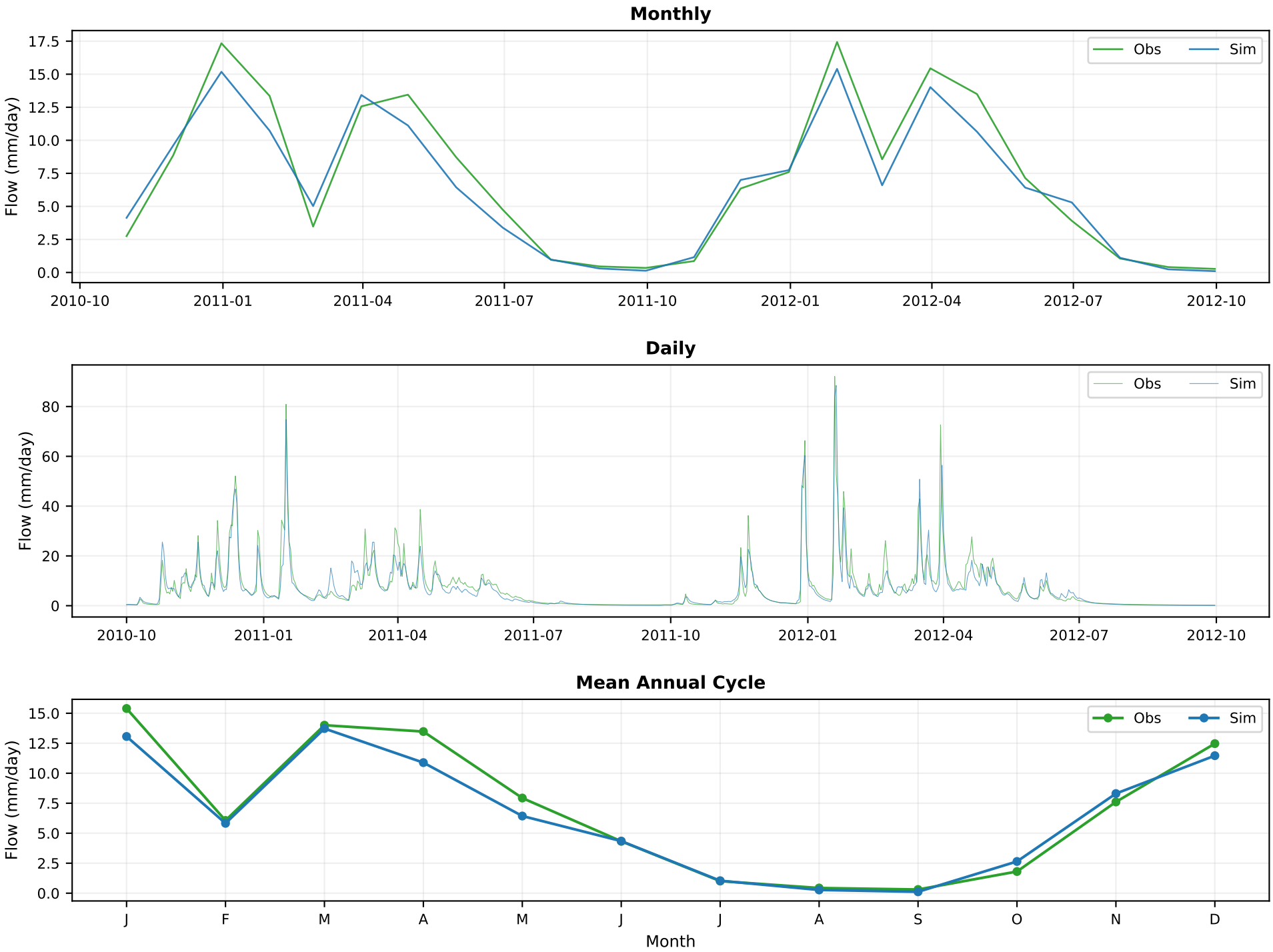

ID: 1542810
KGE=0.876 r=0.966 α=0.887 NSE=0.927 Bias=+3.8%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

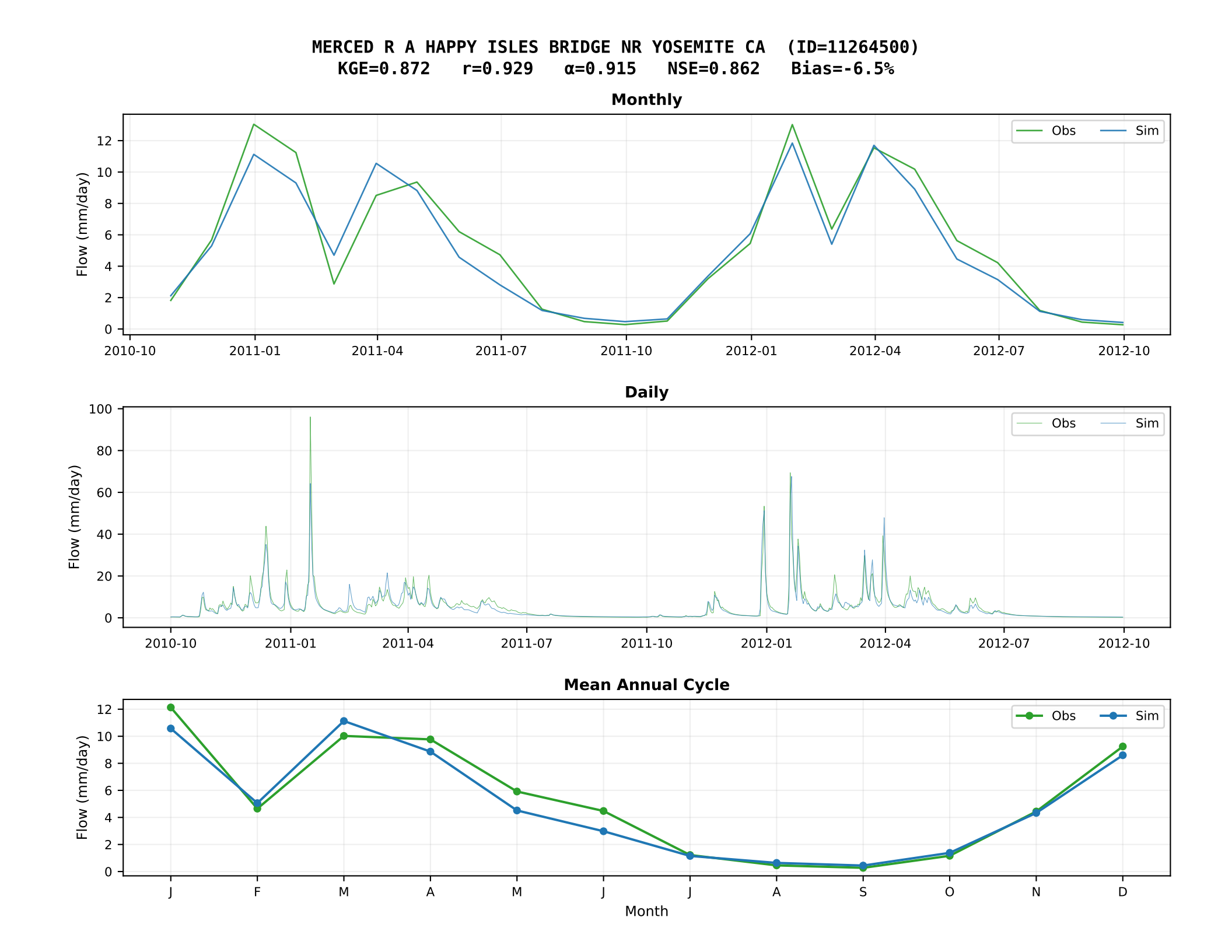

ID: 3439000
KGE=0.876 r=0.912 α=0.978 NSE=0.788 Bias=+8.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

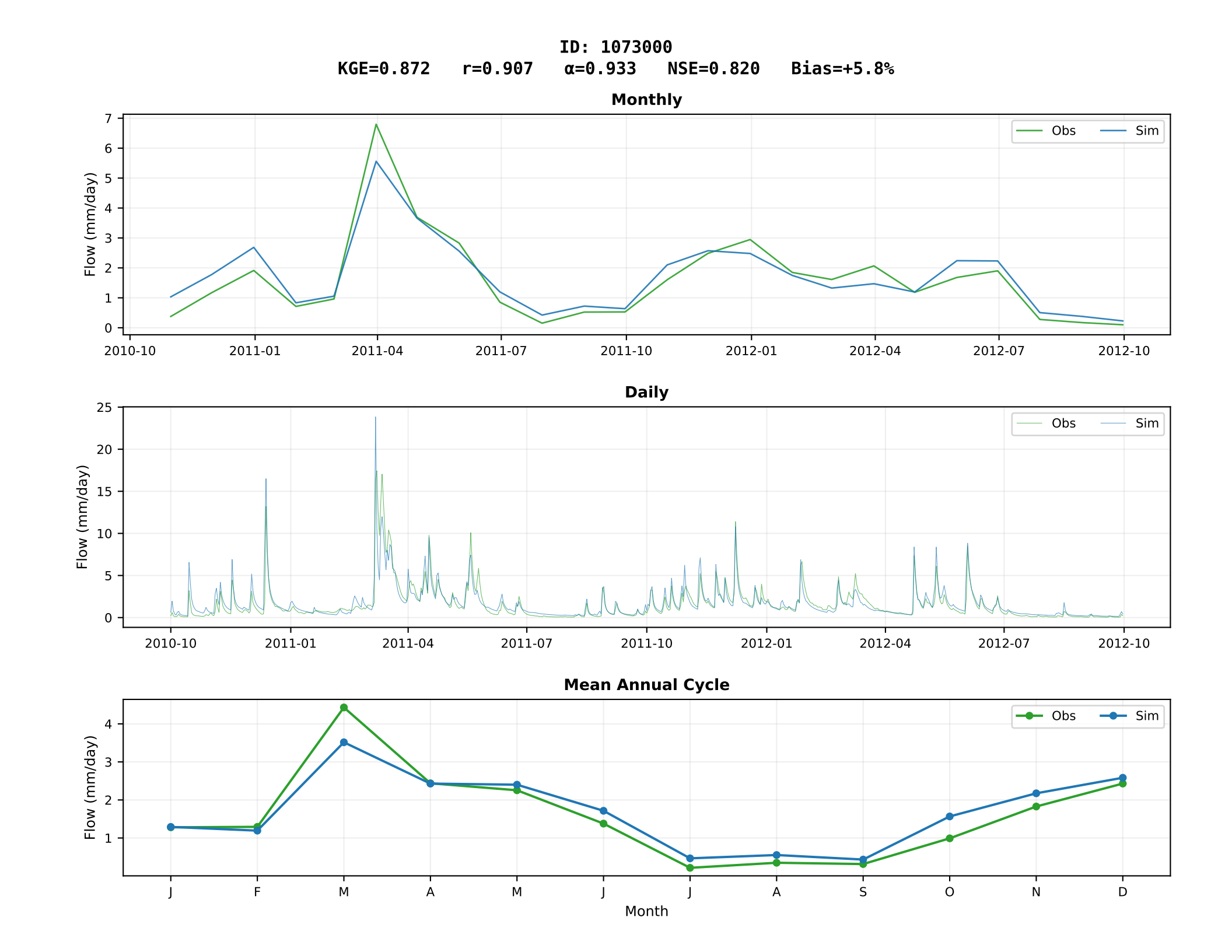

CHEHALIS RIVER NEAR DOTY, WA (ID=12020000)
KGE=0.876 r=0.904 α=1.042 NSE=0.798 Bias=-6.7%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.0
0.8
0.6
0.4
0.2
J
F
M
A
M
J
J
A
S
O
N
D
Month

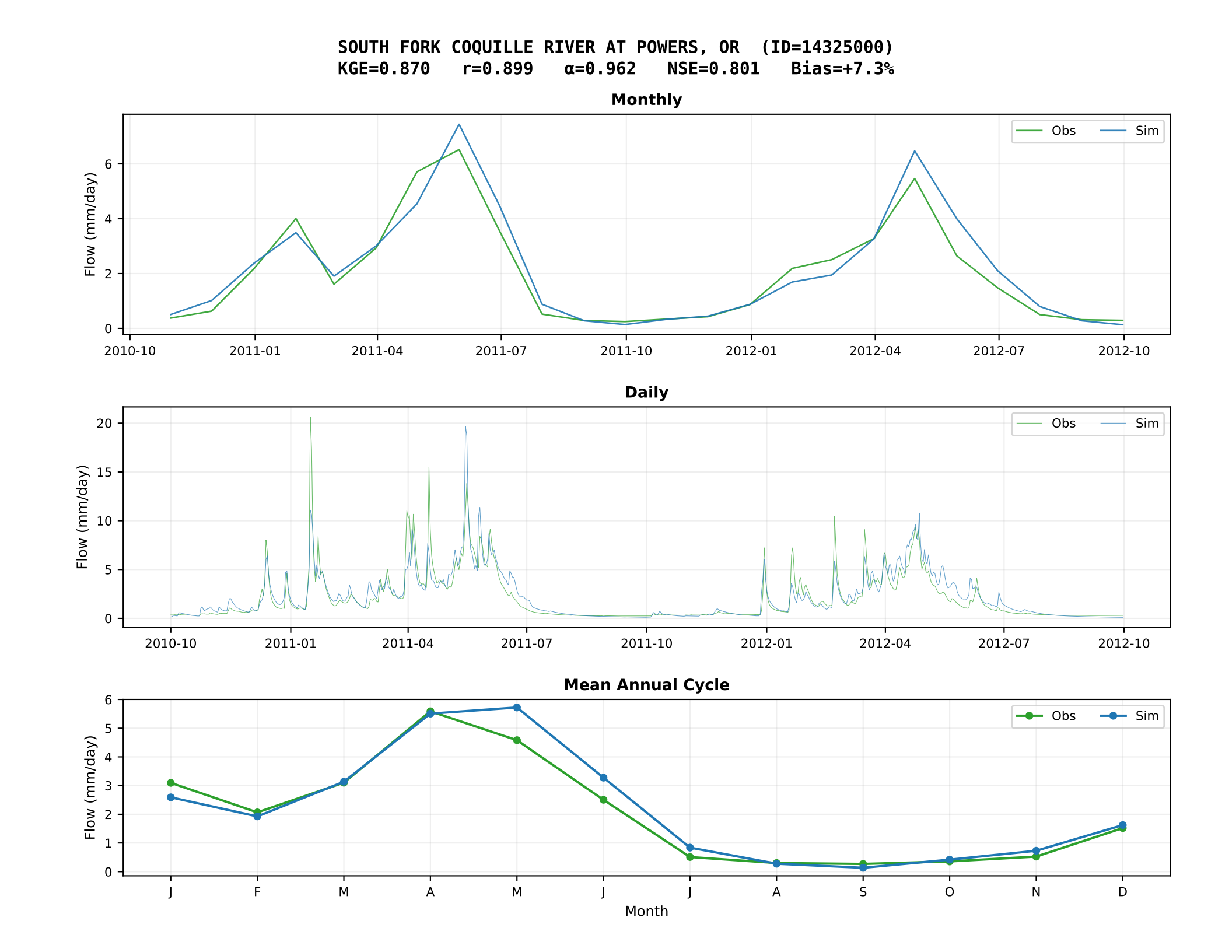

ID: 4127997
KGE=0.874 r=0.897 α=1.067 NSE=0.776 Bias=+2.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

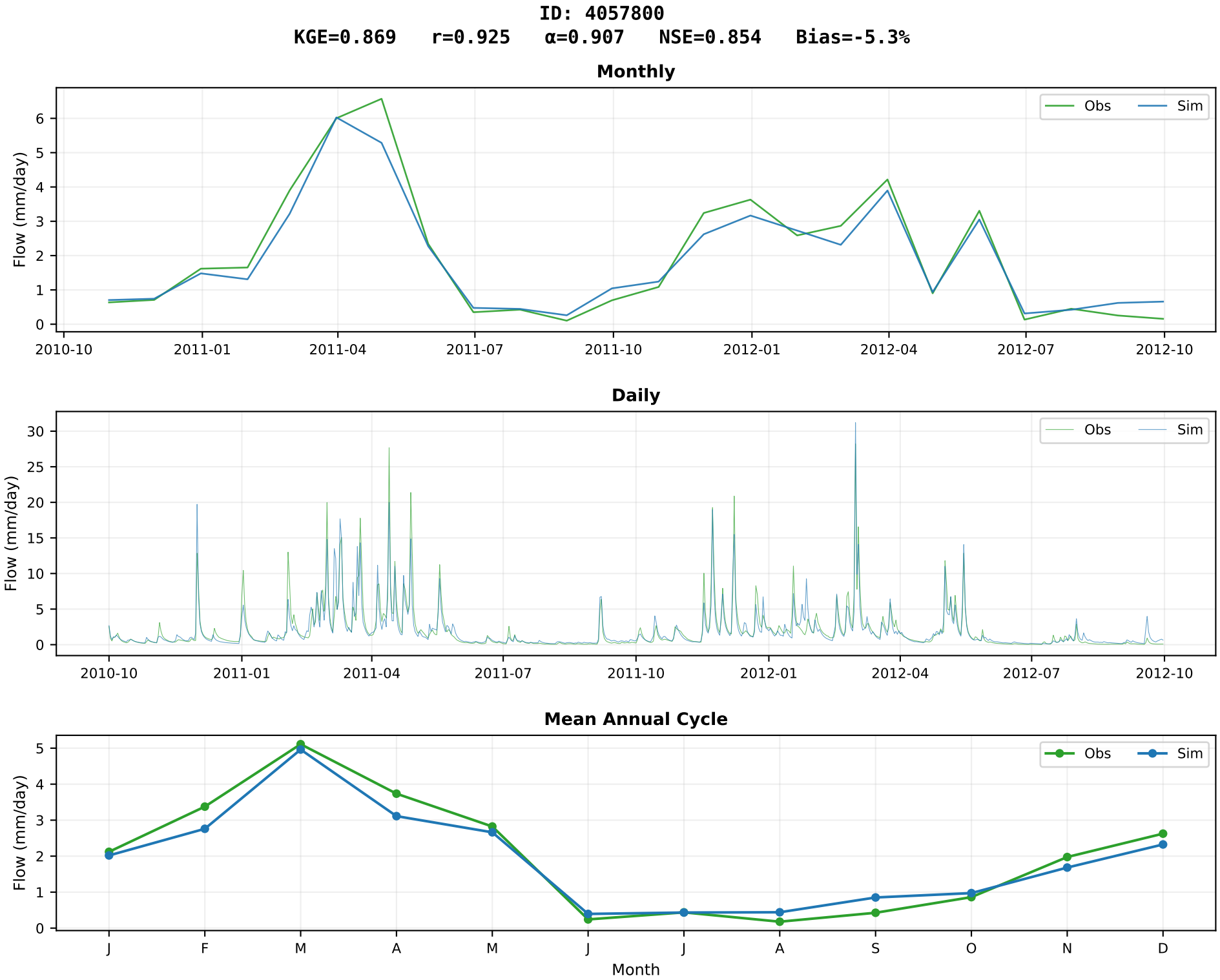

ID: 9066200
KGE=0.872 r=0.936 α=0.899 NSE=0.874 Bias=-4.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

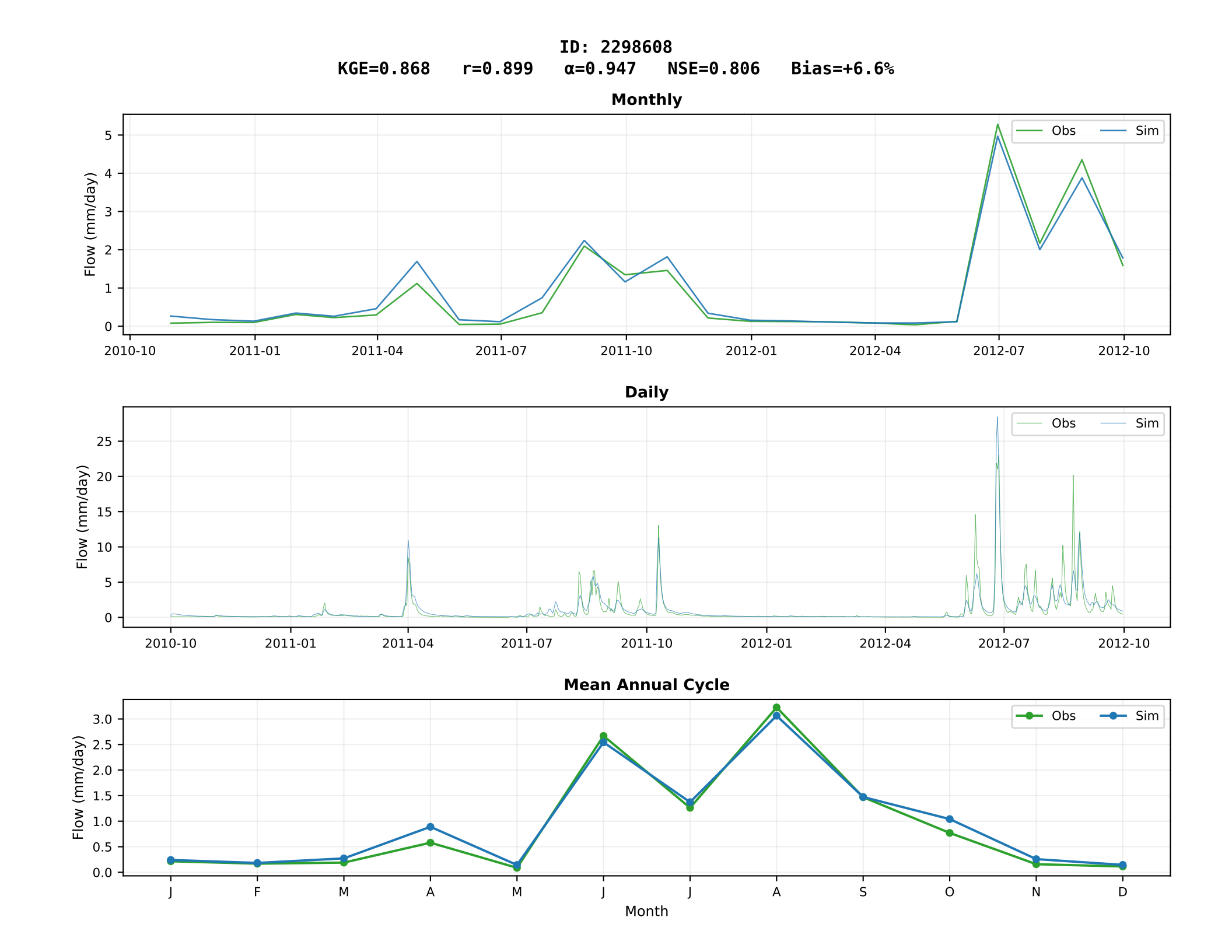

MERCED R A POHONO BRIDGE NR YOSEMITE CA (ID=11266500)
KGE=0.872 r=0.929 α=0.930 NSE=0.859 Bias=-7.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

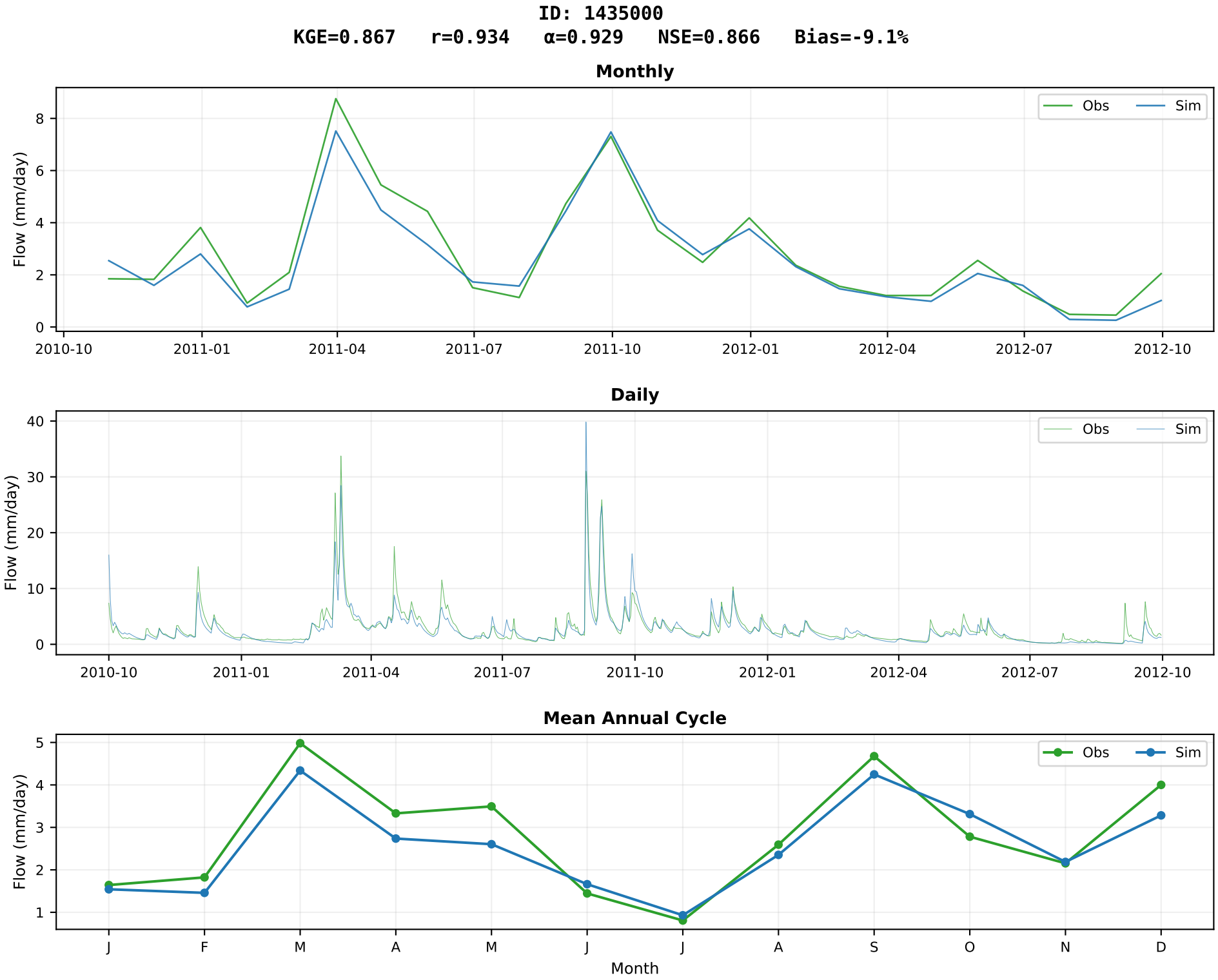

MERCED R A HAPPY ISLES BRIDGE NR YOSEMITE CA (ID=11264500)
KGE=0.872 r=0.929 α=0.915 NSE=0.862 Bias=-6.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

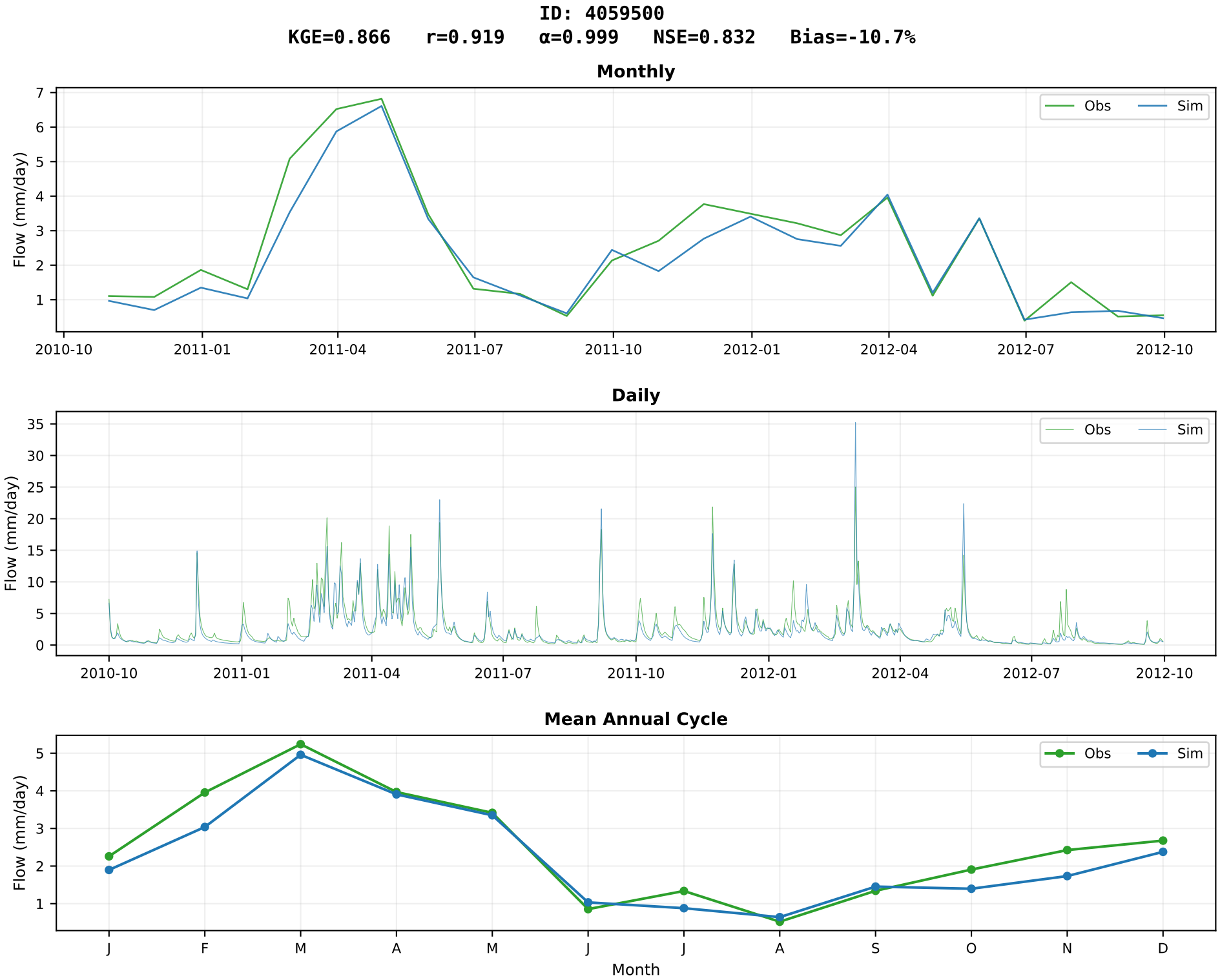

ID: 1073000
KGE=0.872 r=0.907 α=0.933 NSE=0.820 Bias=+5.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

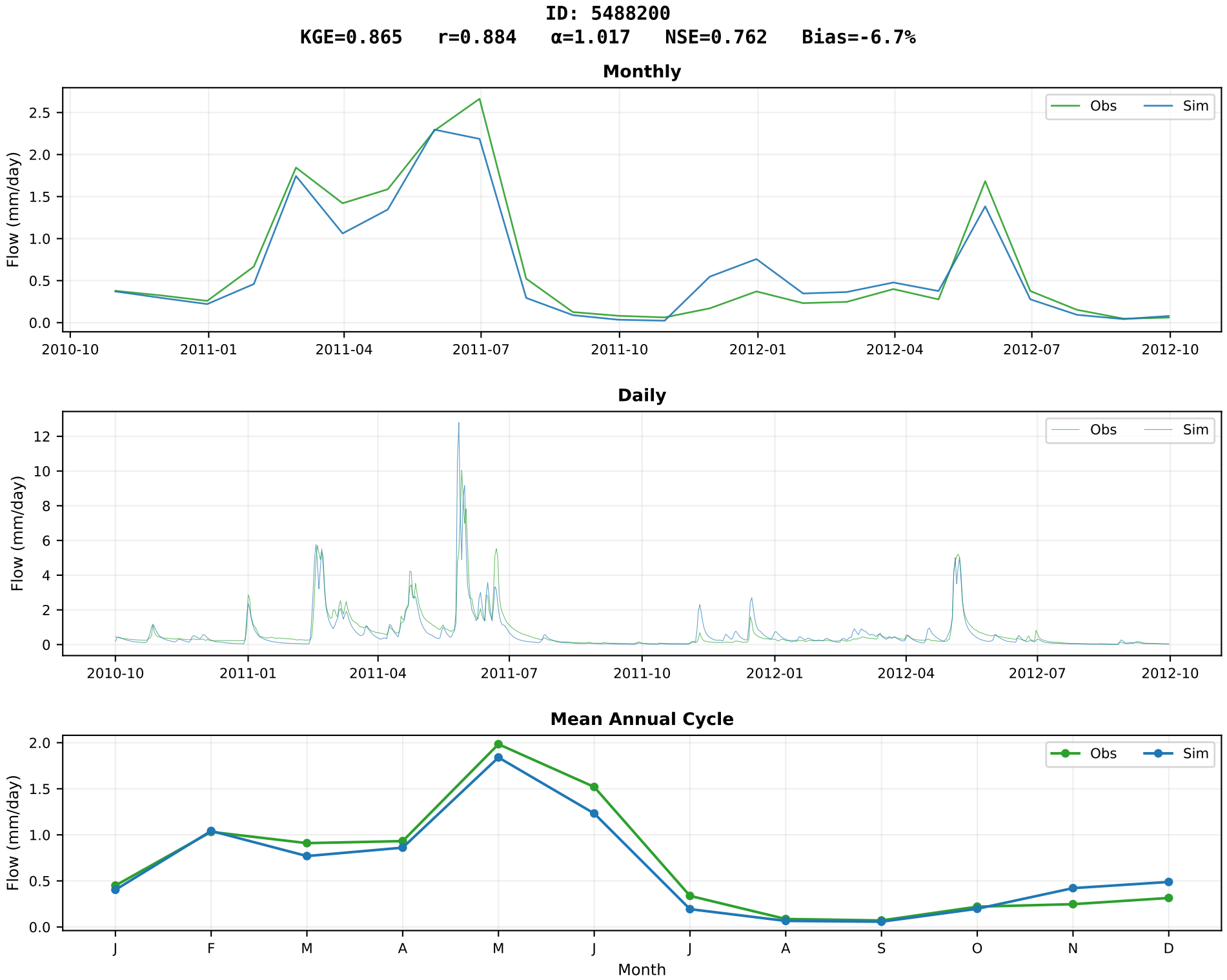

SOUTH FORK COQUILLE RIVER AT POWERS, OR (ID=14325000)
KGE=0.870 r=0.899 α=0.962 NSE=0.801 Bias=+7.3%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

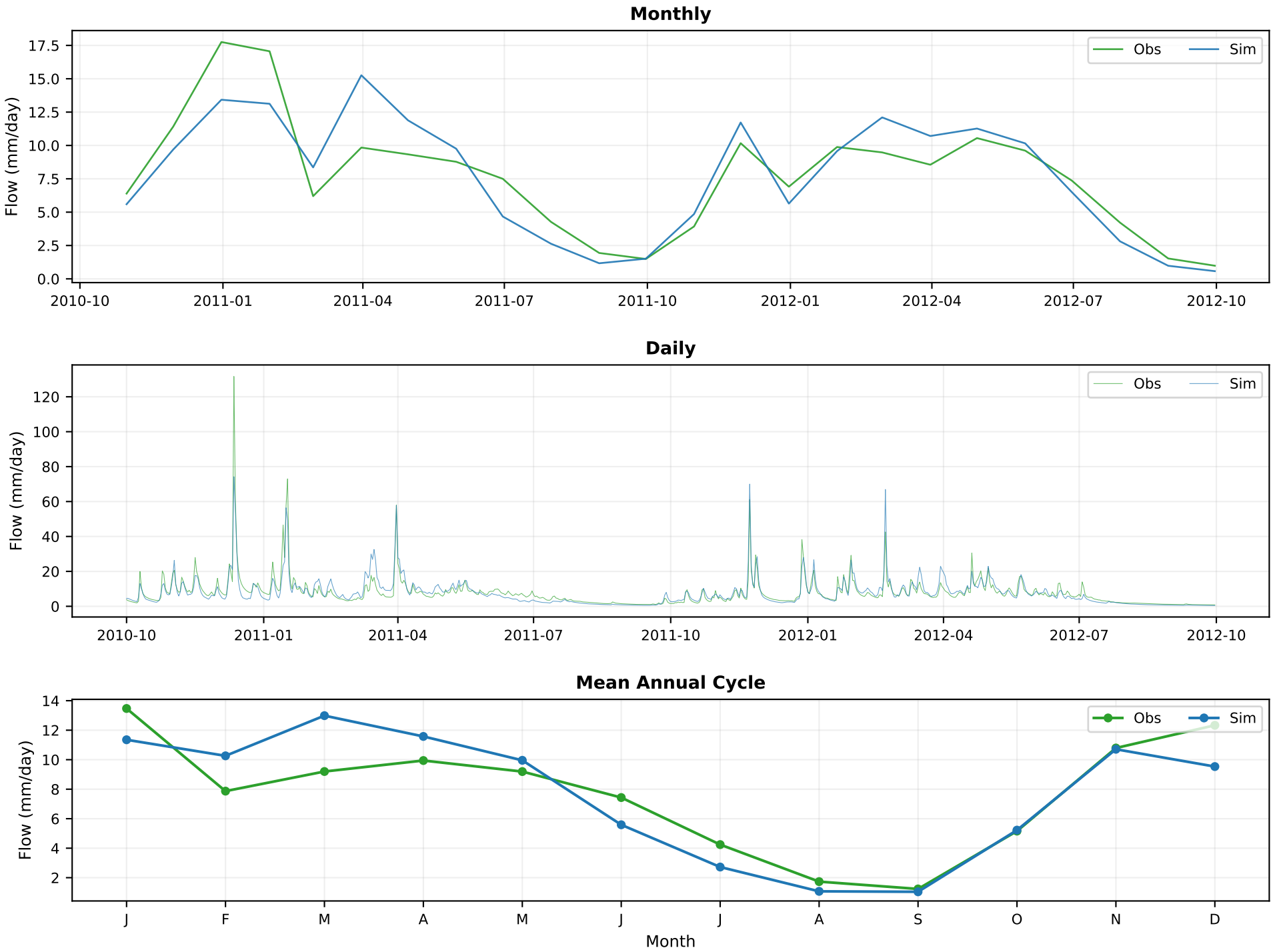

ID: 4057800
KGE=0.869 r=0.925 α=0.907 NSE=0.854 Bias=-5.3%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

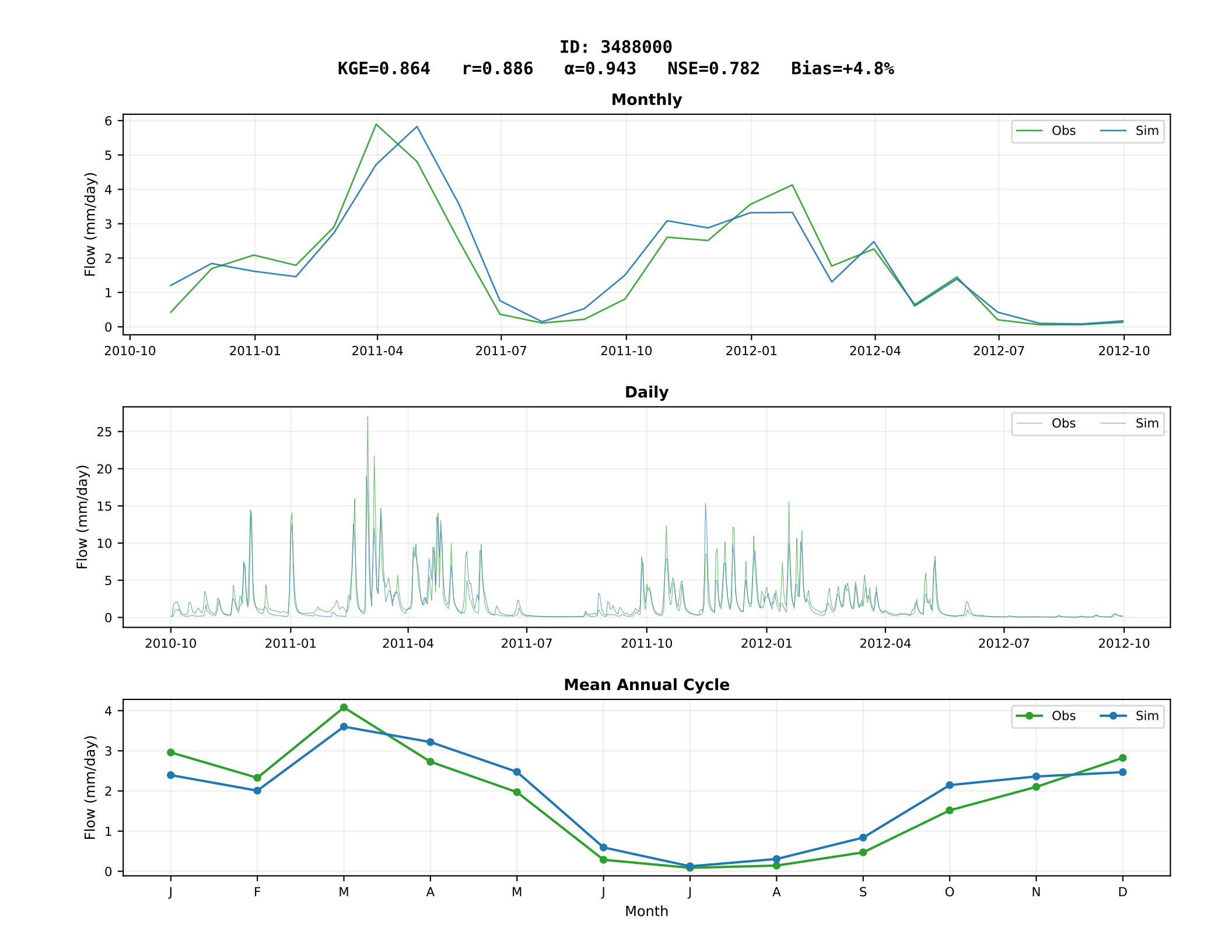

ID: 2298608
KGE=0.868 r=0.899 α=0.947 NSE=0.806 Bias=+6.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

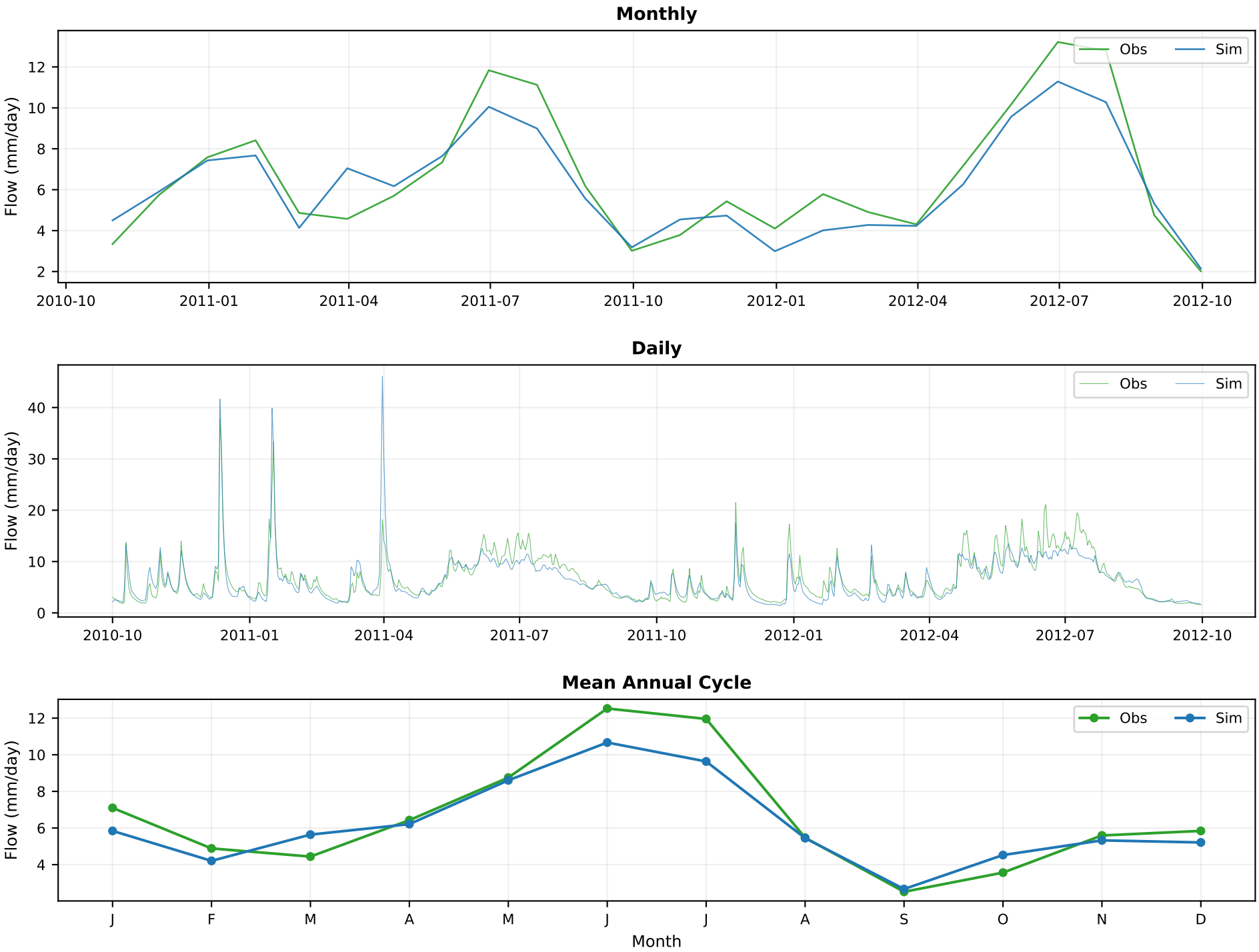

ID: 1435000
KGE=0.867 r=0.934 α=0.929 NSE=0.866 Bias=-9.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

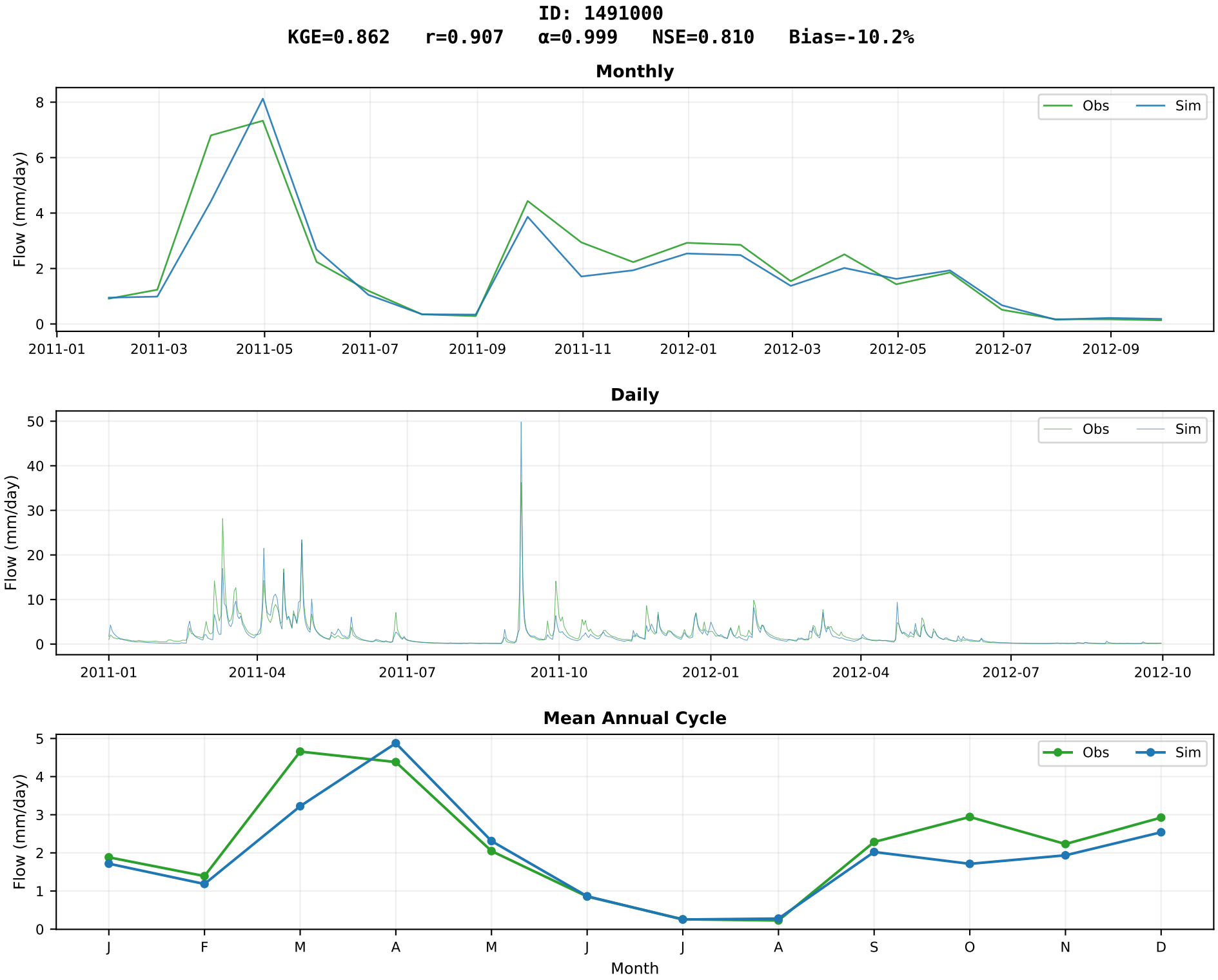

ID: 4059500
KGE=0.866 r=0.919 α=0.999 NSE=0.832 Bias=-10.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

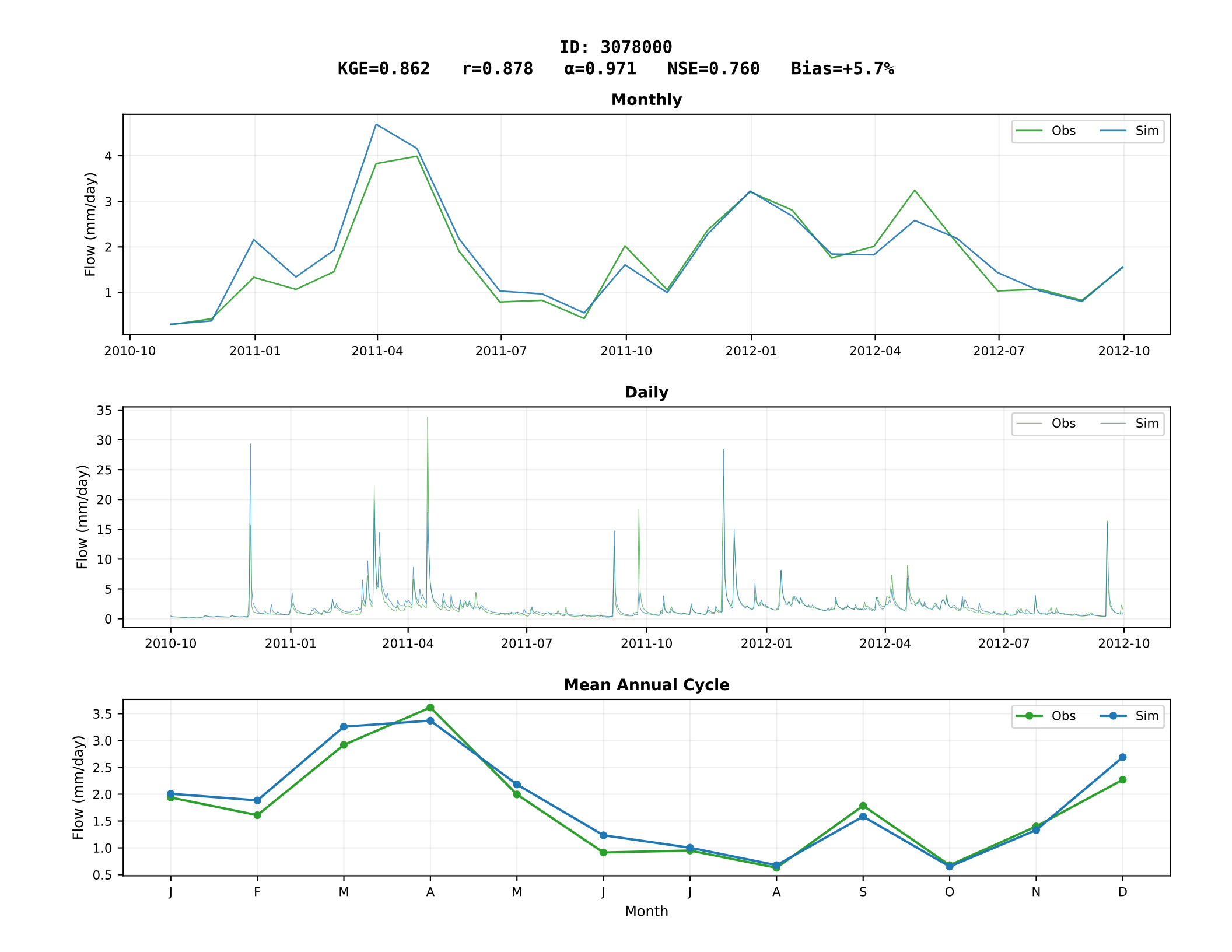

ID: 5488200
KGE=0.865 r=0.884 α=1.017 NSE=0.762 Bias=-6.7%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

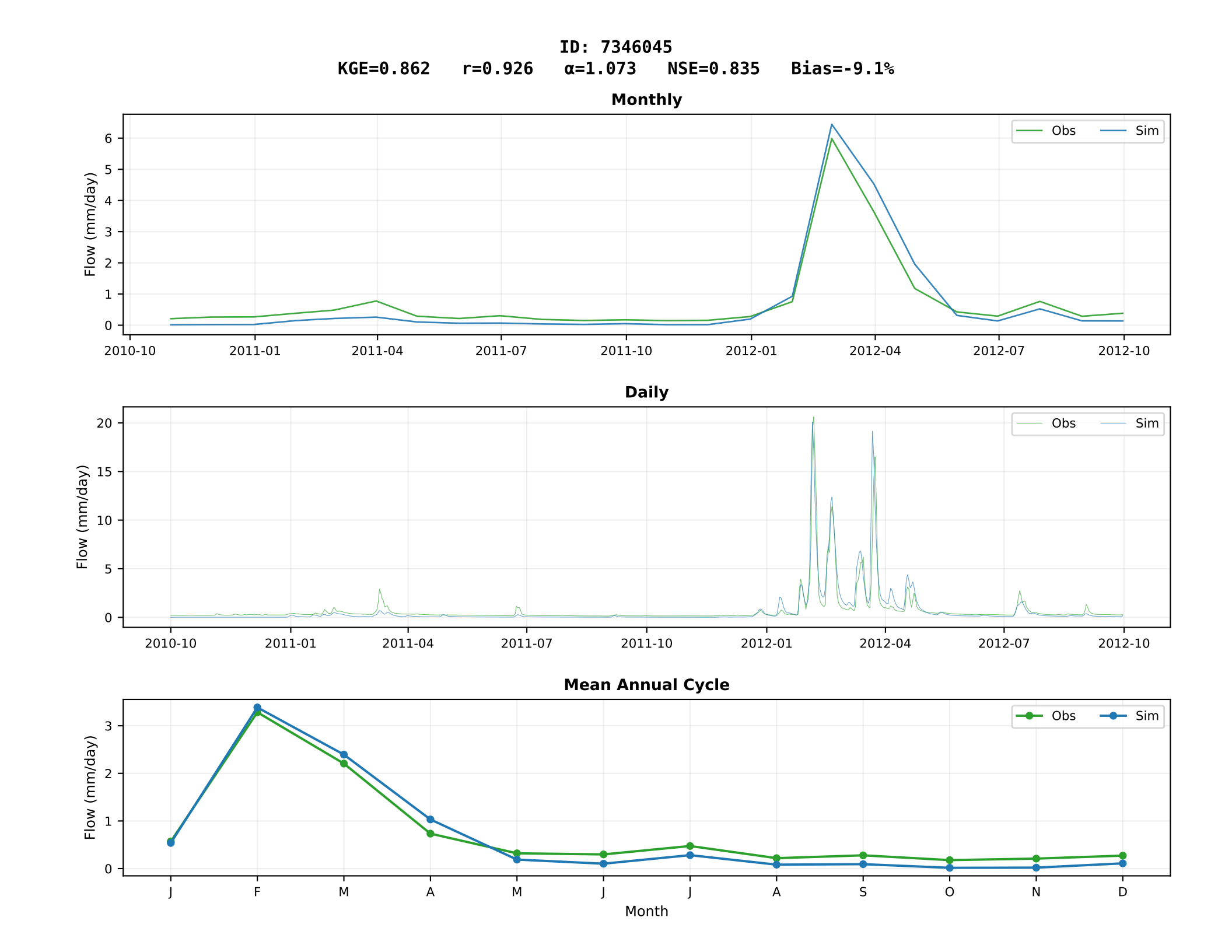

BLAZED ALDER CREEK NEAR RHODODENDRON, OREG. (ID=14138800)
KGE=0.864 r=0.891 α=0.919 NSE=0.793 Bias=-0.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

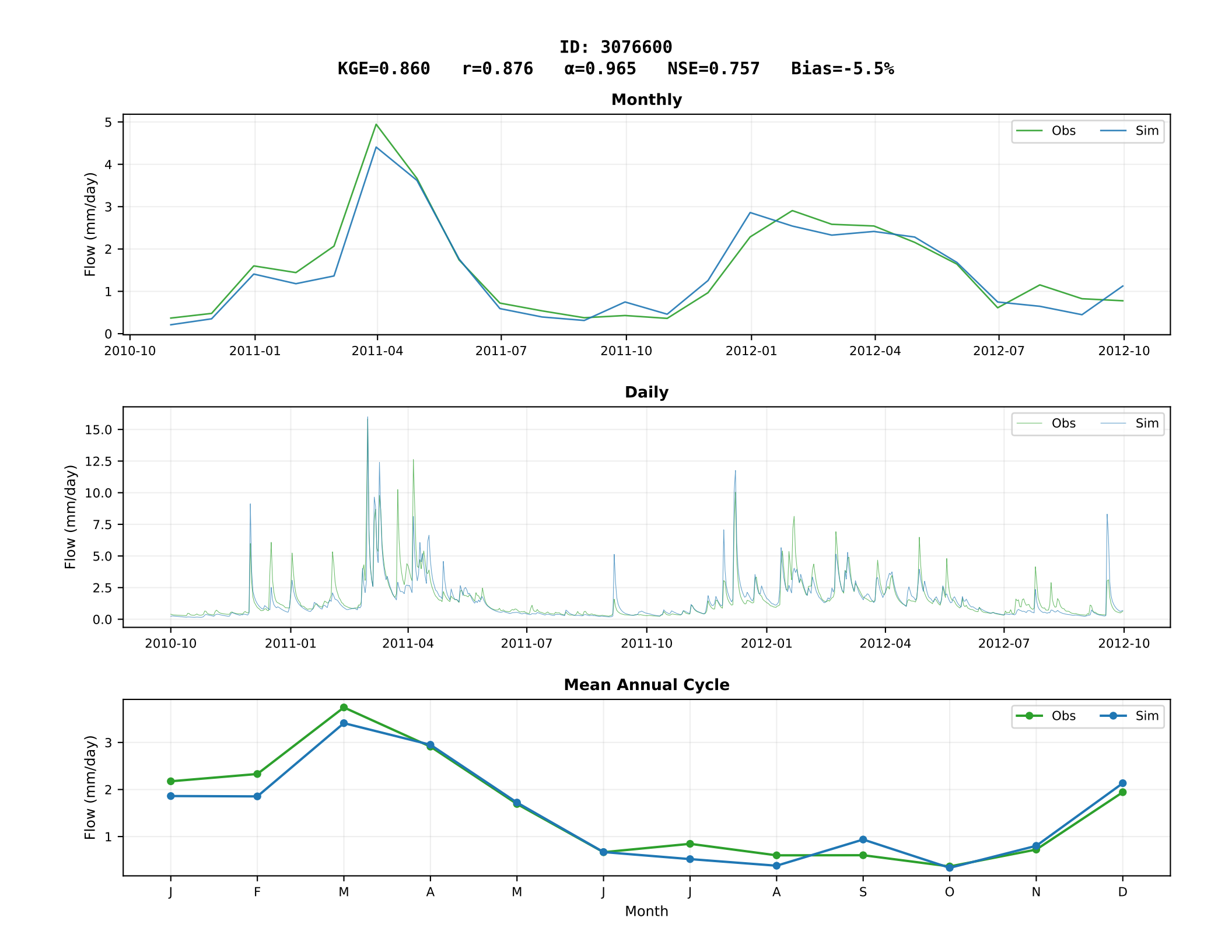

ID: 3488000
KGE=0.864 r=0.886 α=0.943 NSE=0.782 Bias=+4.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

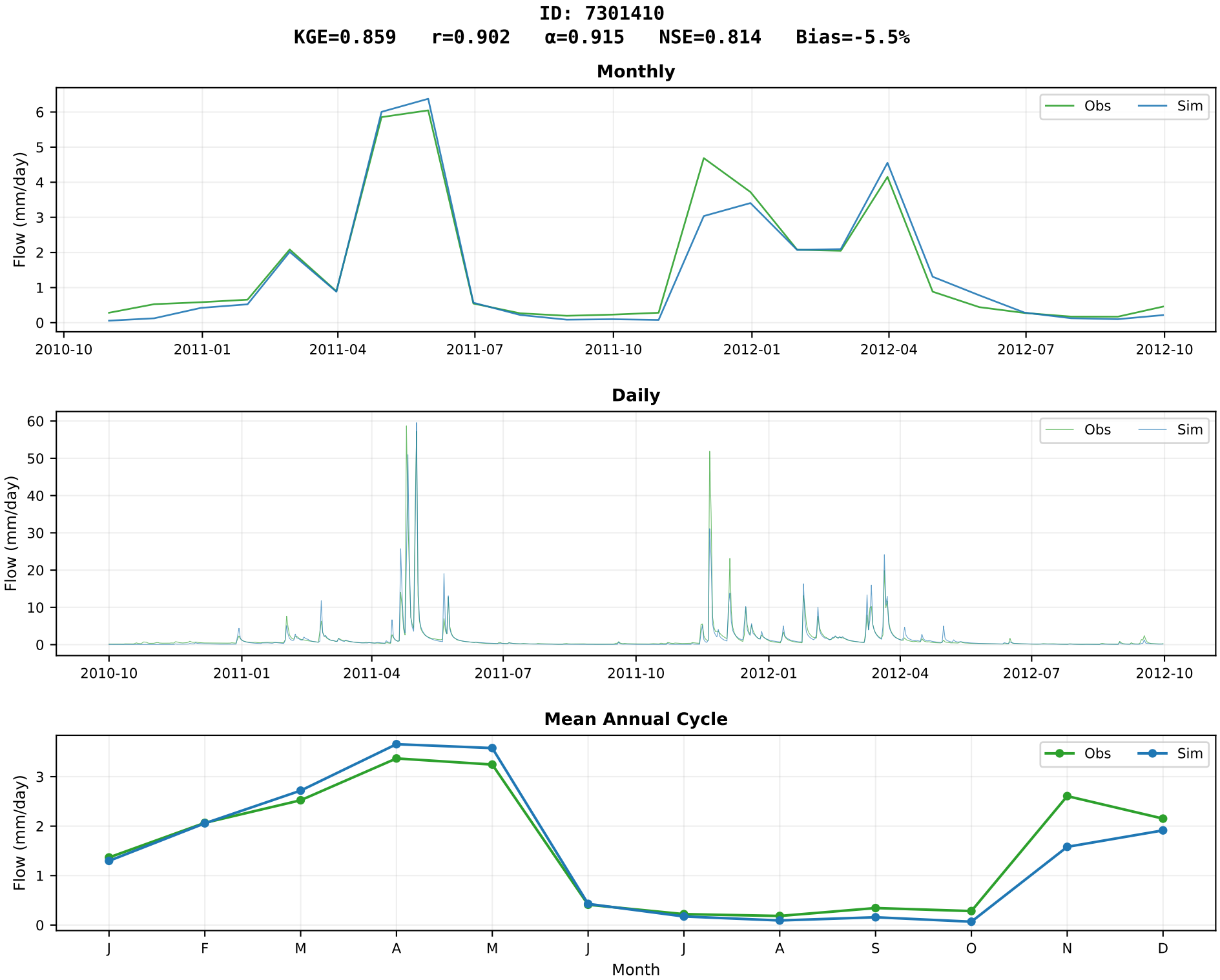

LITTLE SANDY RIVER NEAR BULL RUN, OREG. (ID=14141500)
KGE=0.862 r=0.878 α=0.994 NSE=0.749 Bias=-6.4%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

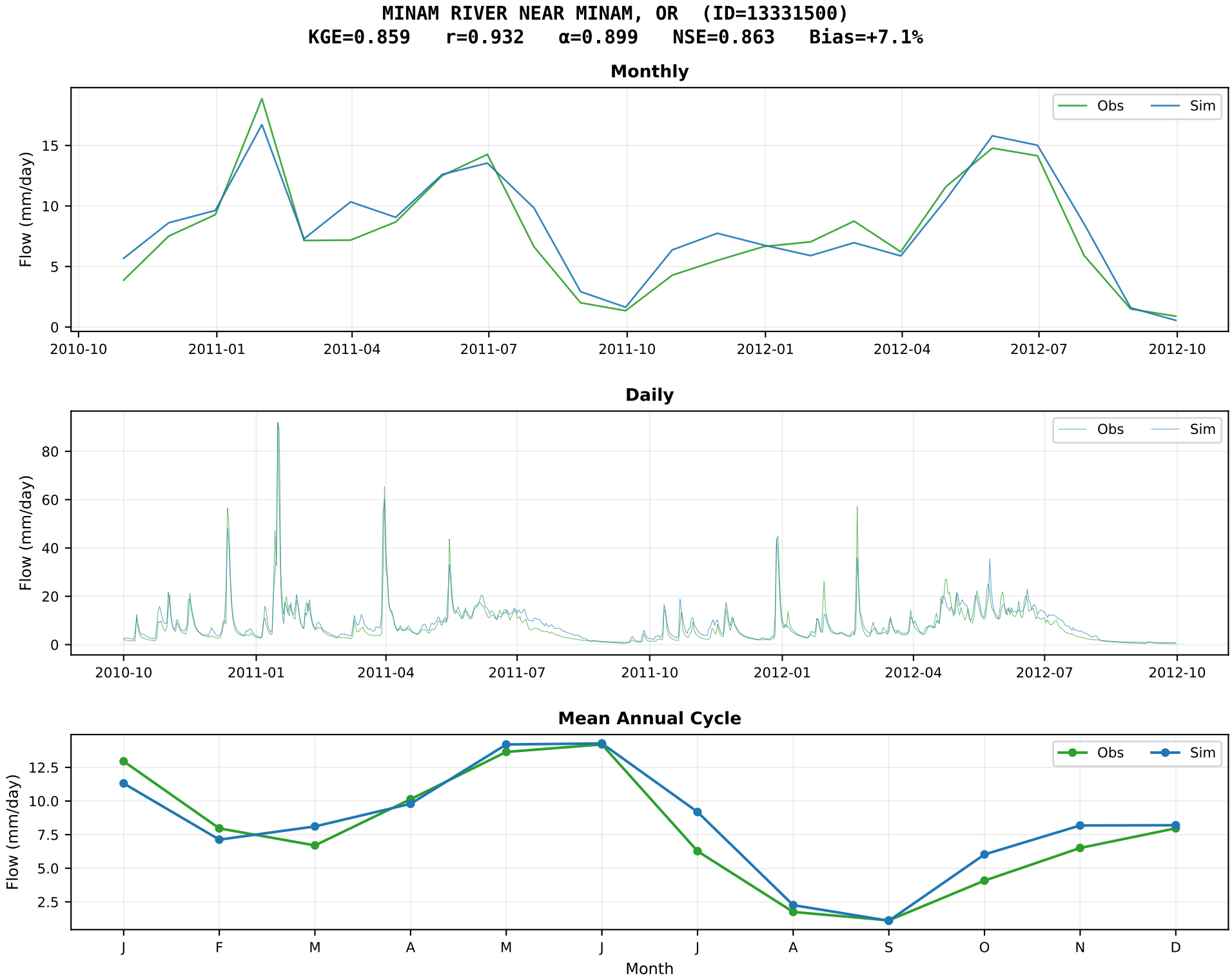

ID: 1491000
KGE=0.862 r=0.907 α=0.999 NSE=0.810 Bias=-10.2%
Monthly
Flow (mm/day)
Obs
Sim
2011-01
2011-03
2011-05
2011-07
2011-09
2011-11
2012-01
2012-03
2012-05
2012-07
2012-09
Daily
Flow (mm/day)
Obs
Sim
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

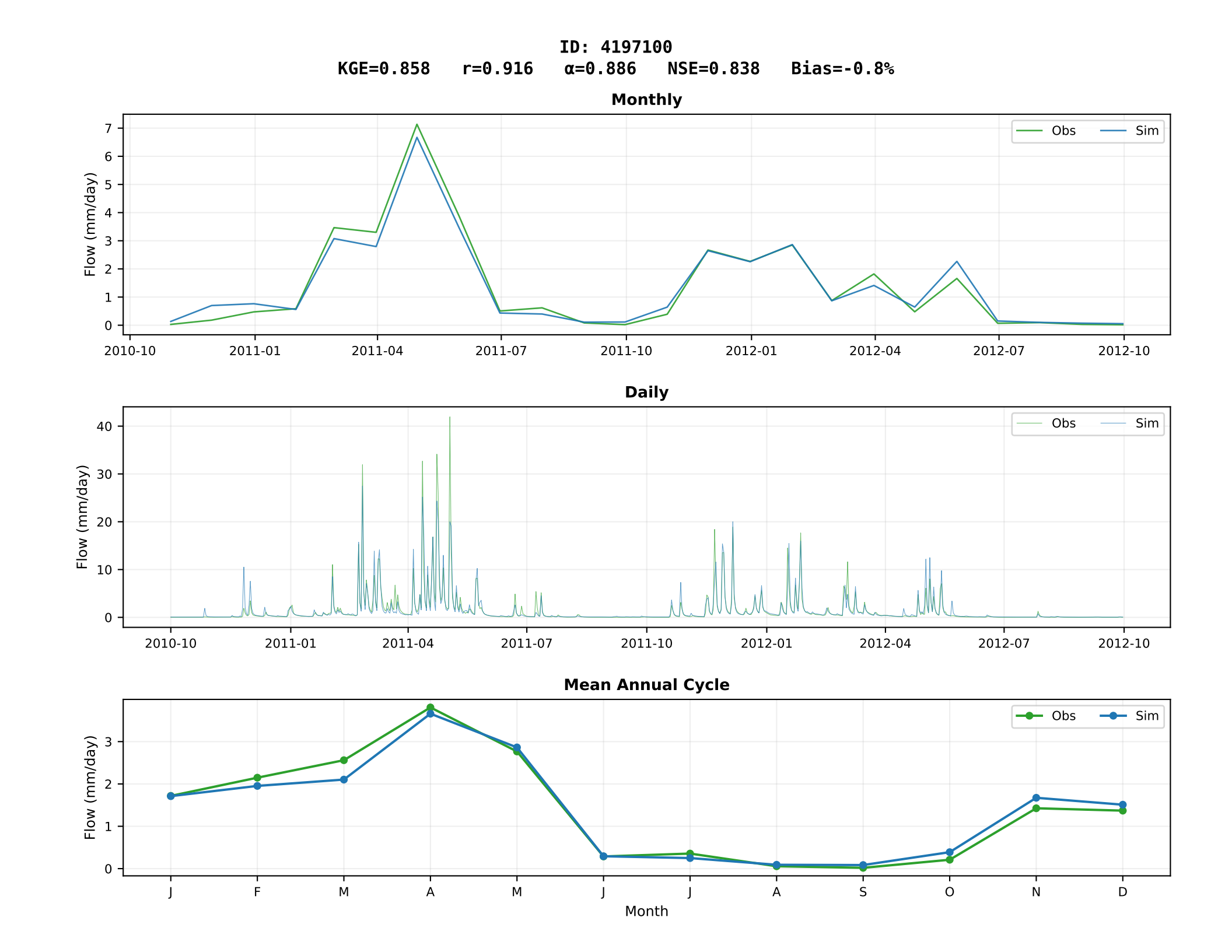

ID: 3078000
KGE=0.862 r=0.878 α=0.971 NSE=0.760 Bias=+5.7%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

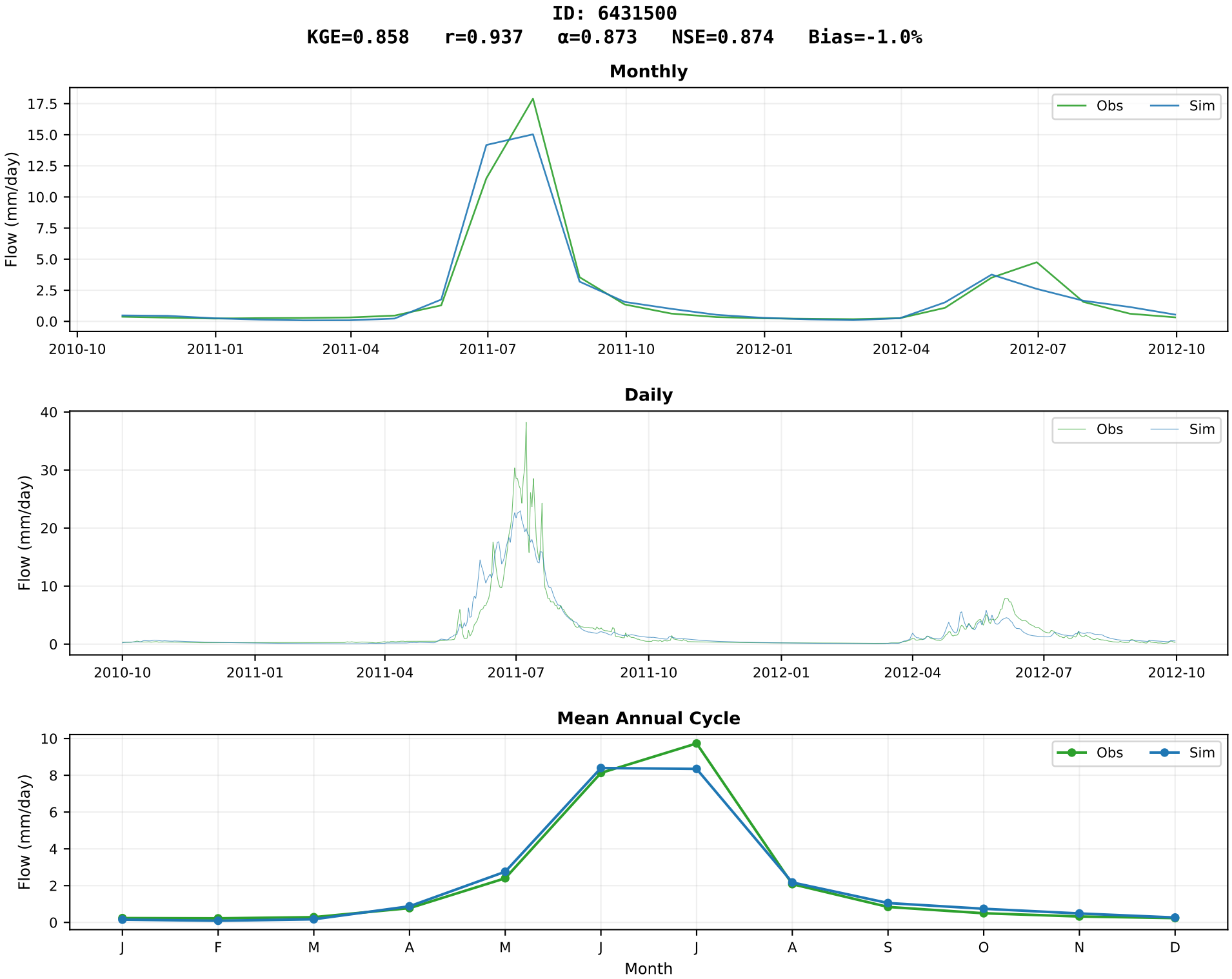

ID: 7346045
KGE=0.862 r=0.926 α=1.073 NSE=0.835 Bias=-9.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

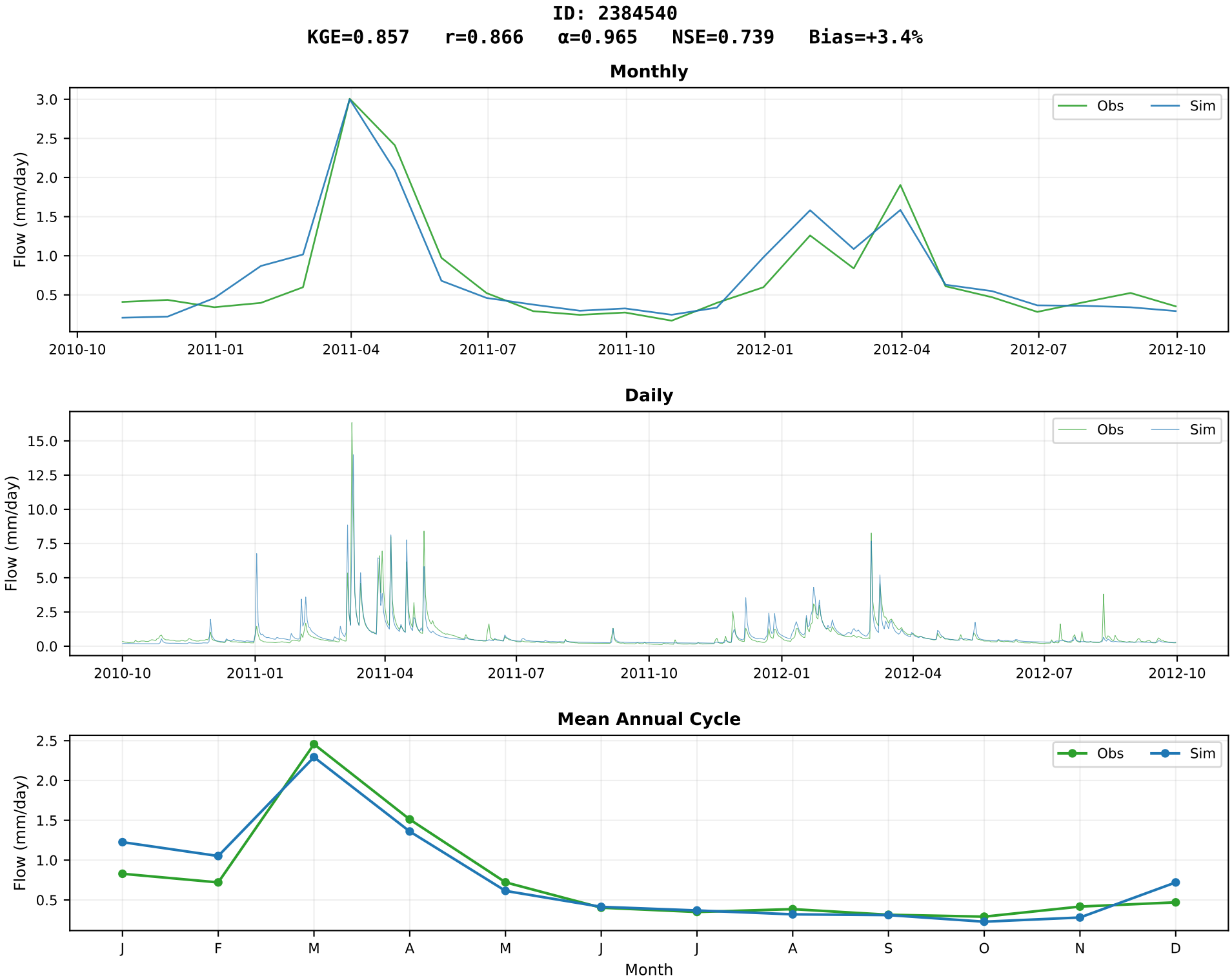

ID: 3076600
KGE=0.860 r=0.876 α=0.965 NSE=0.757 Bias=-5.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

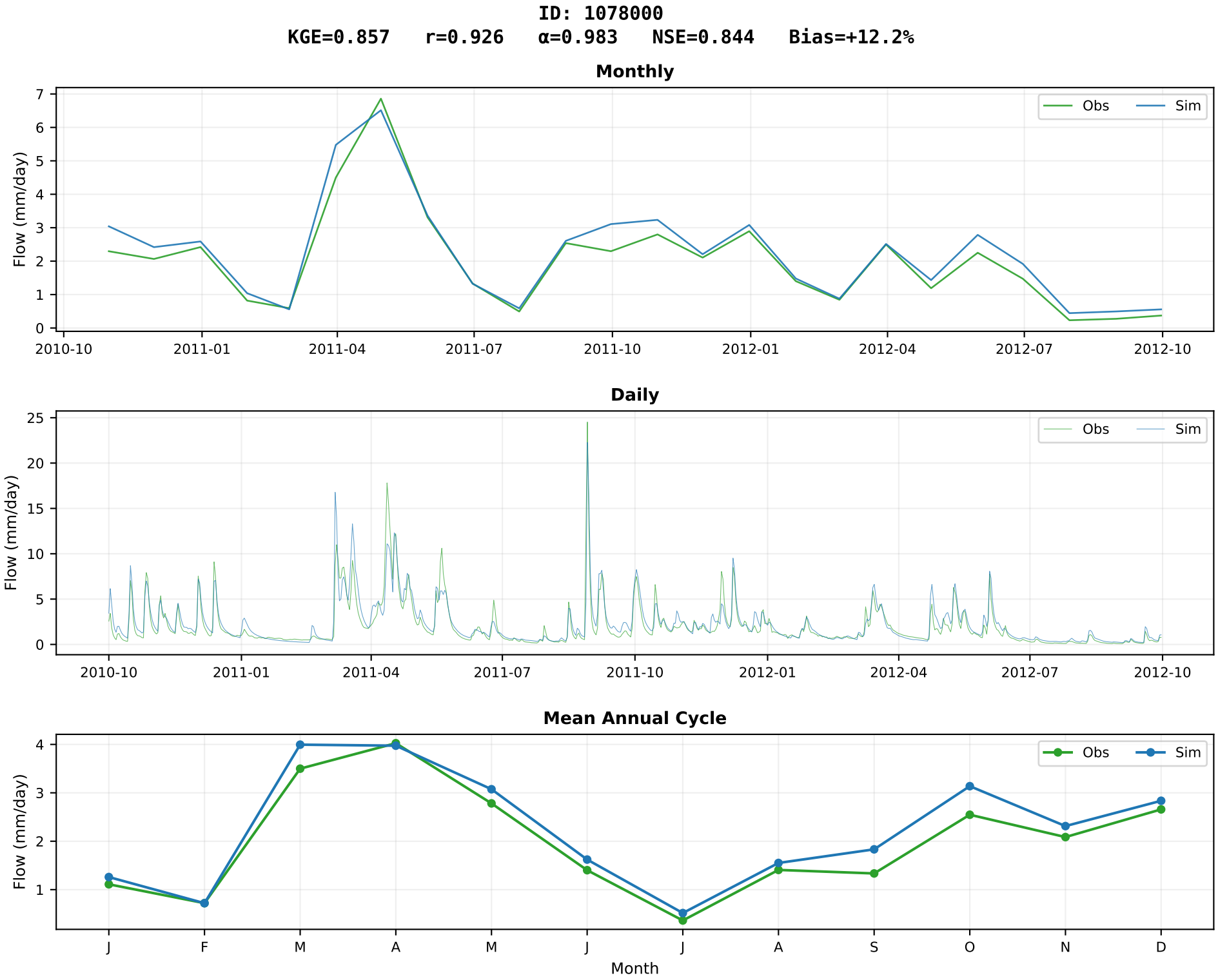

ID: 7301410
KGE=0.859 r=0.902 α=0.915 NSE=0.814 Bias=-5.5%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

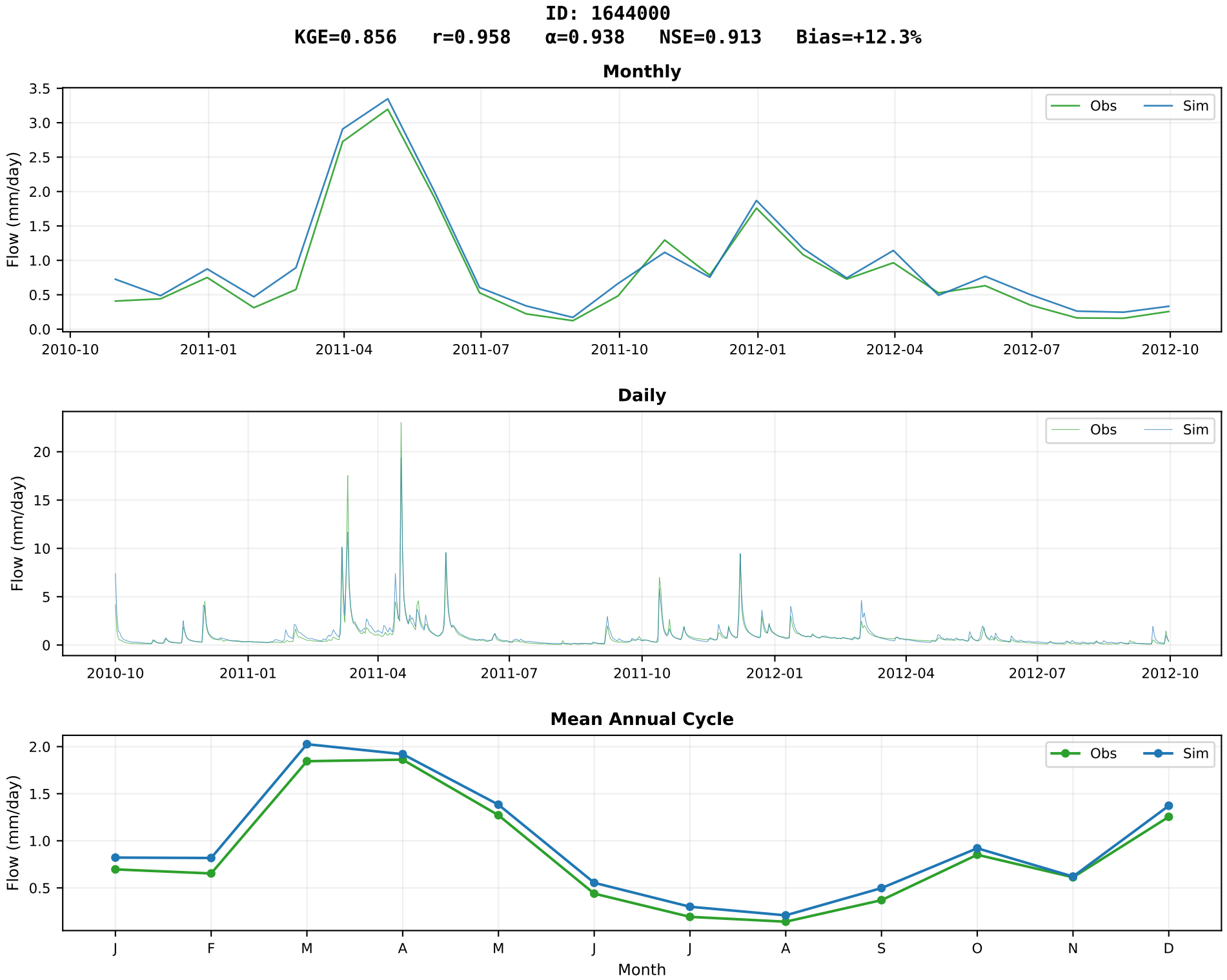

MINAM RIVER NEAR MINAM, OR (ID=13331500)
KGE=0.859 r=0.932 α=0.899 NSE=0.863 Bias=+7.1%
Monthly
Obs
Sim
Flow (mm/day)
0
5
10
15
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.5
5.0
7.5
10.0
12.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

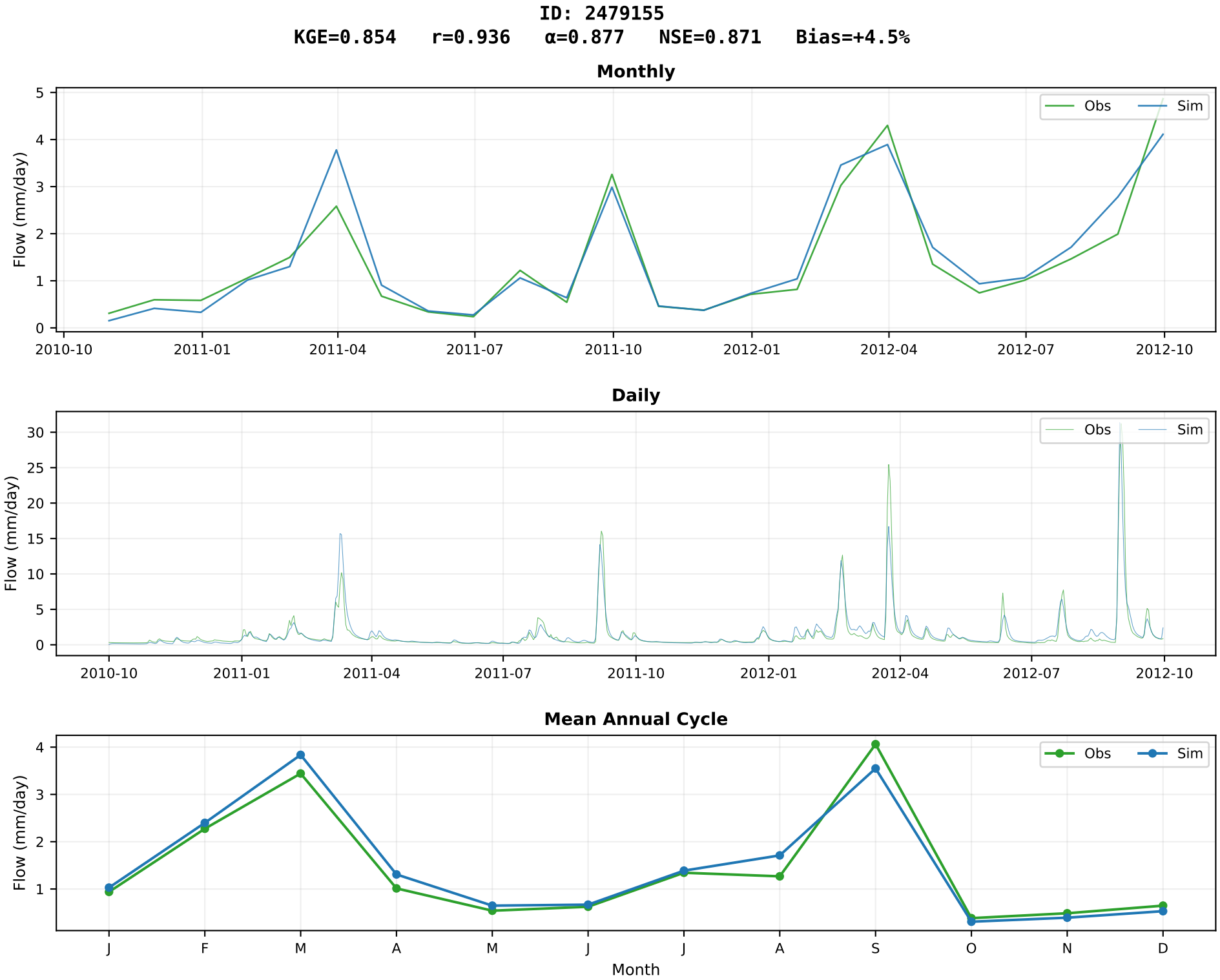

ID: 4197100
KGE=0.858 r=0.916 α=0.886 NSE=0.838 Bias=-0.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

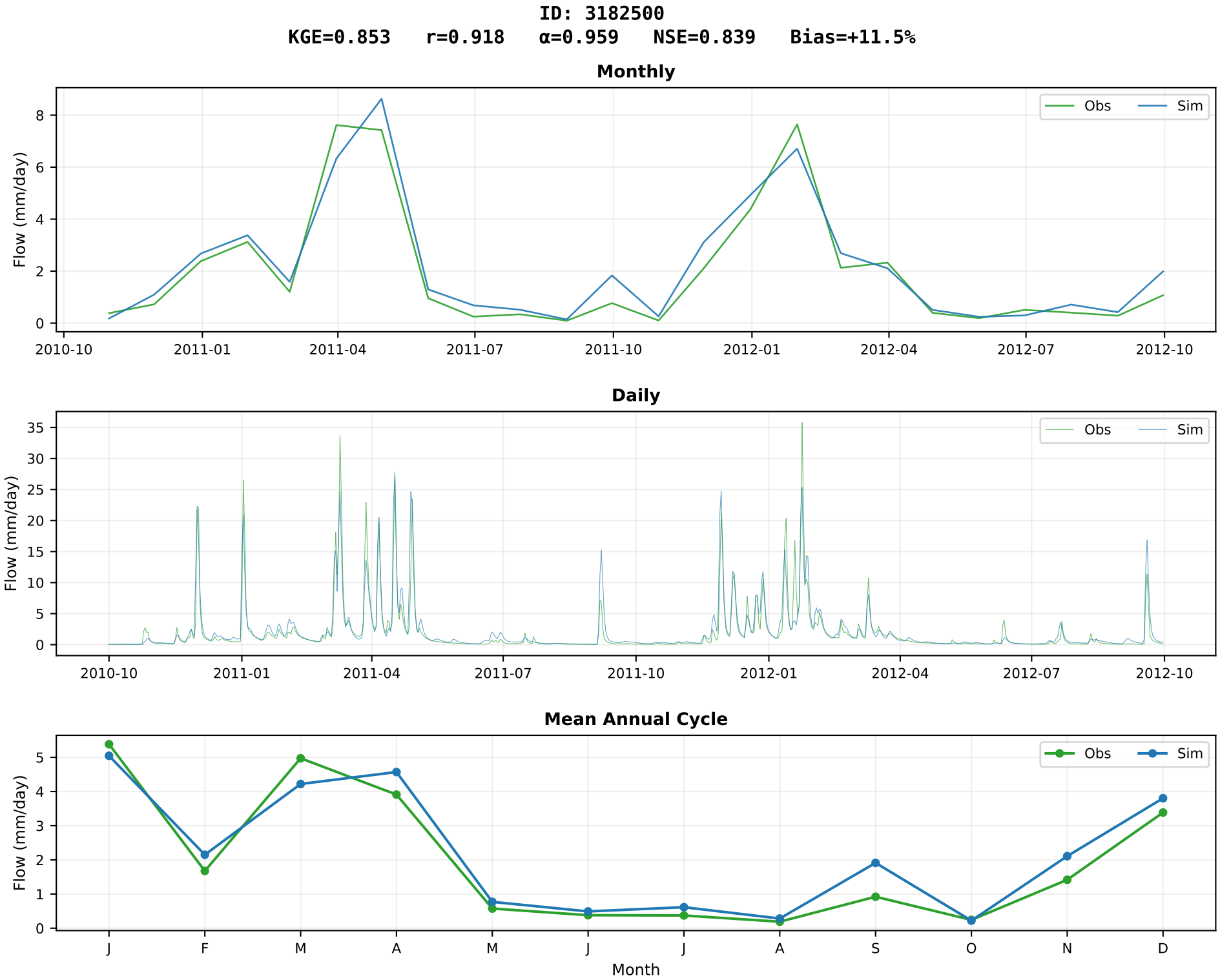

ID: 6431500
KGE=0.858 r=0.937 α=0.873 NSE=0.874 Bias=-1.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

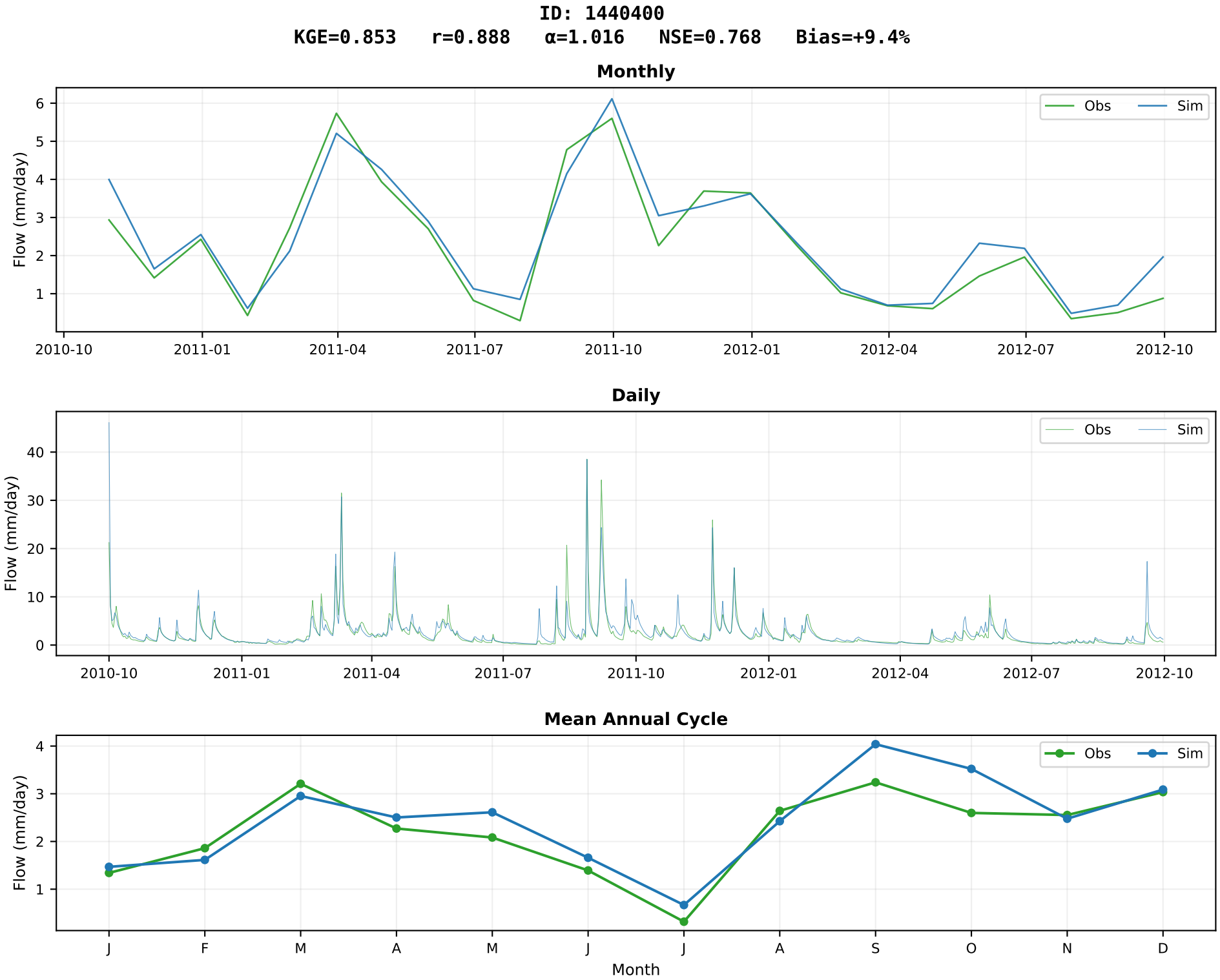

ID: 2384540
KGE=0.857 r=0.866 α=0.965 NSE=0.739 Bias=+3.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

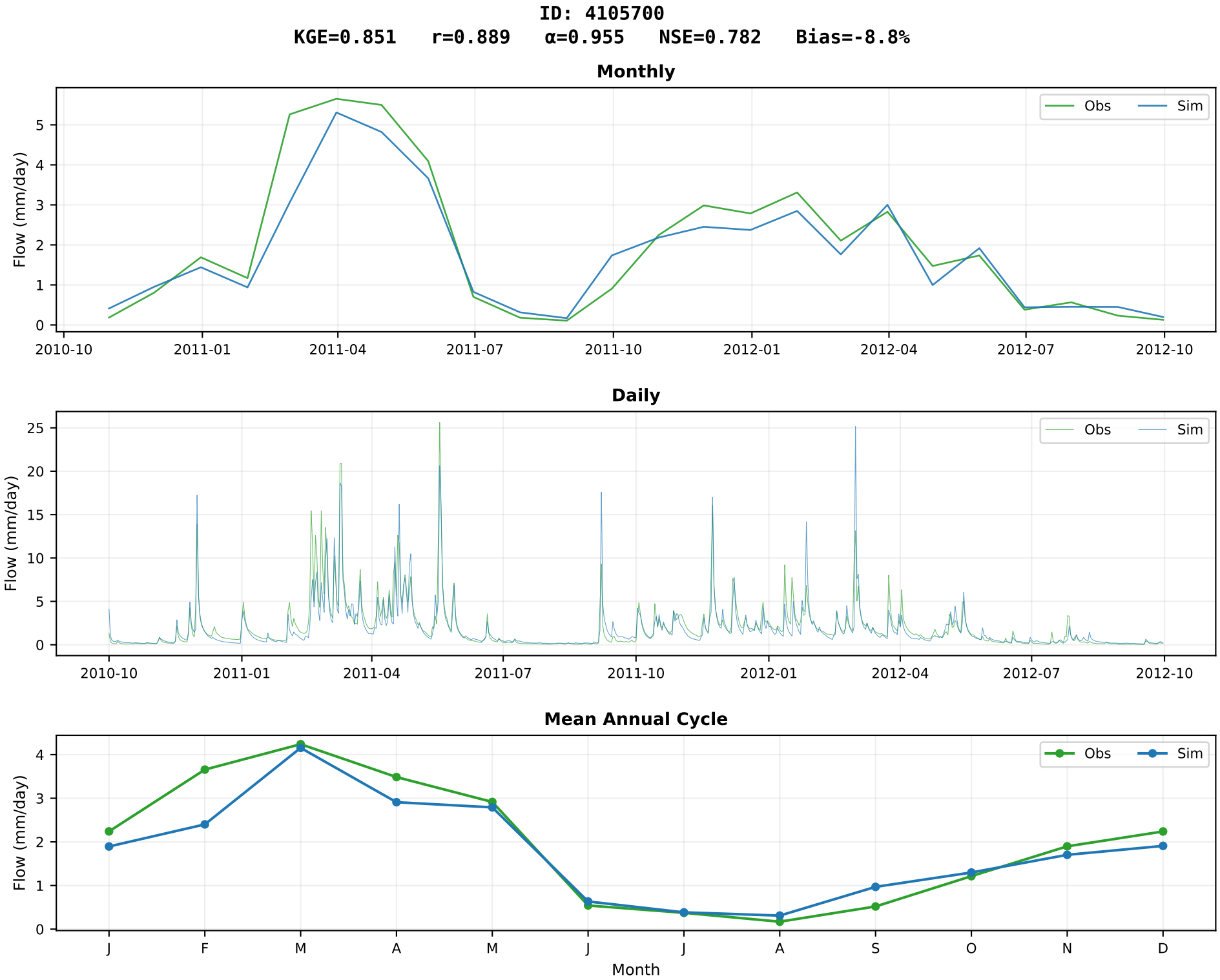

ID: 1078000
KGE=0.857 r=0.926 α=0.983 NSE=0.844 Bias=+12.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

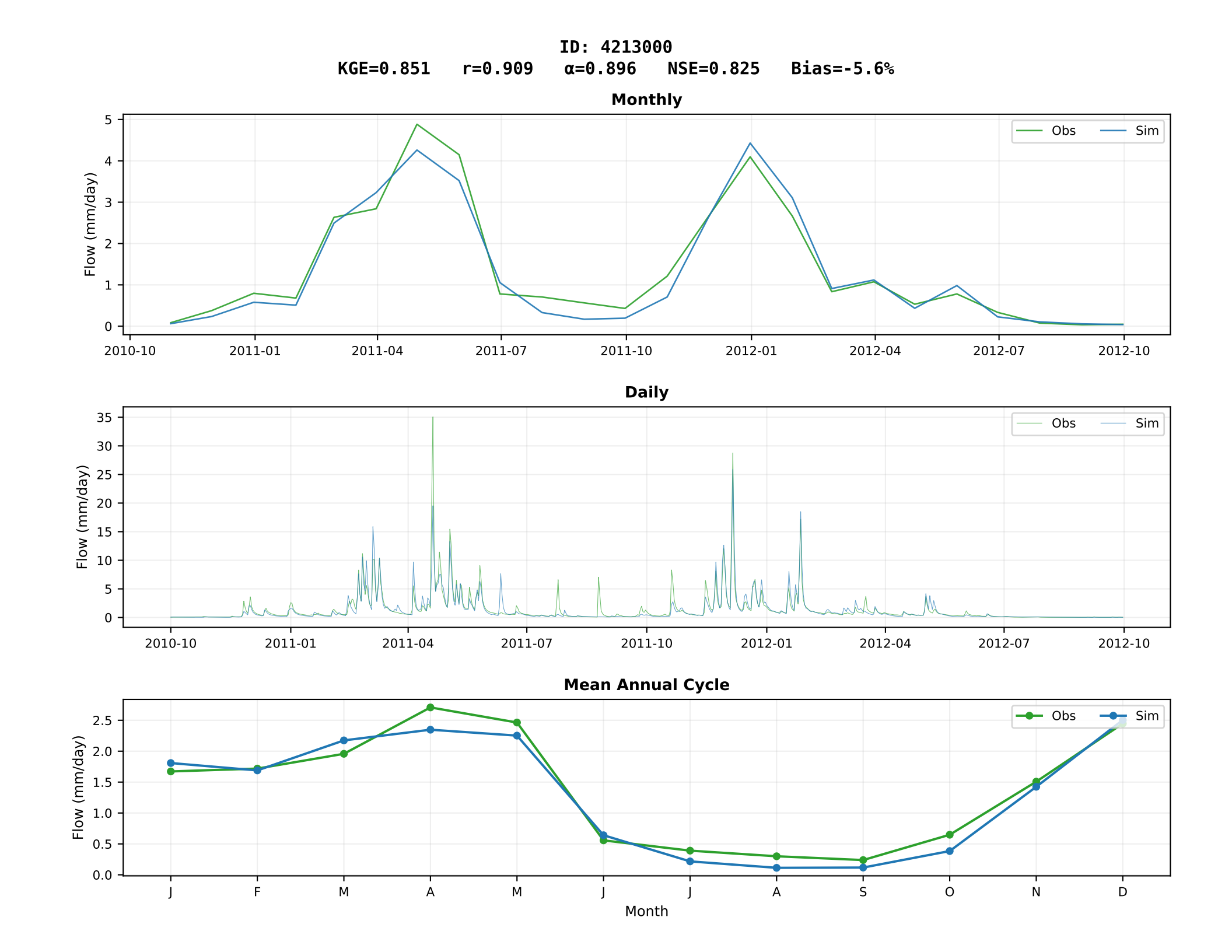

ID: 1644000
KGE=0.856 r=0.958 α=0.938 NSE=0.913 Bias=+12.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

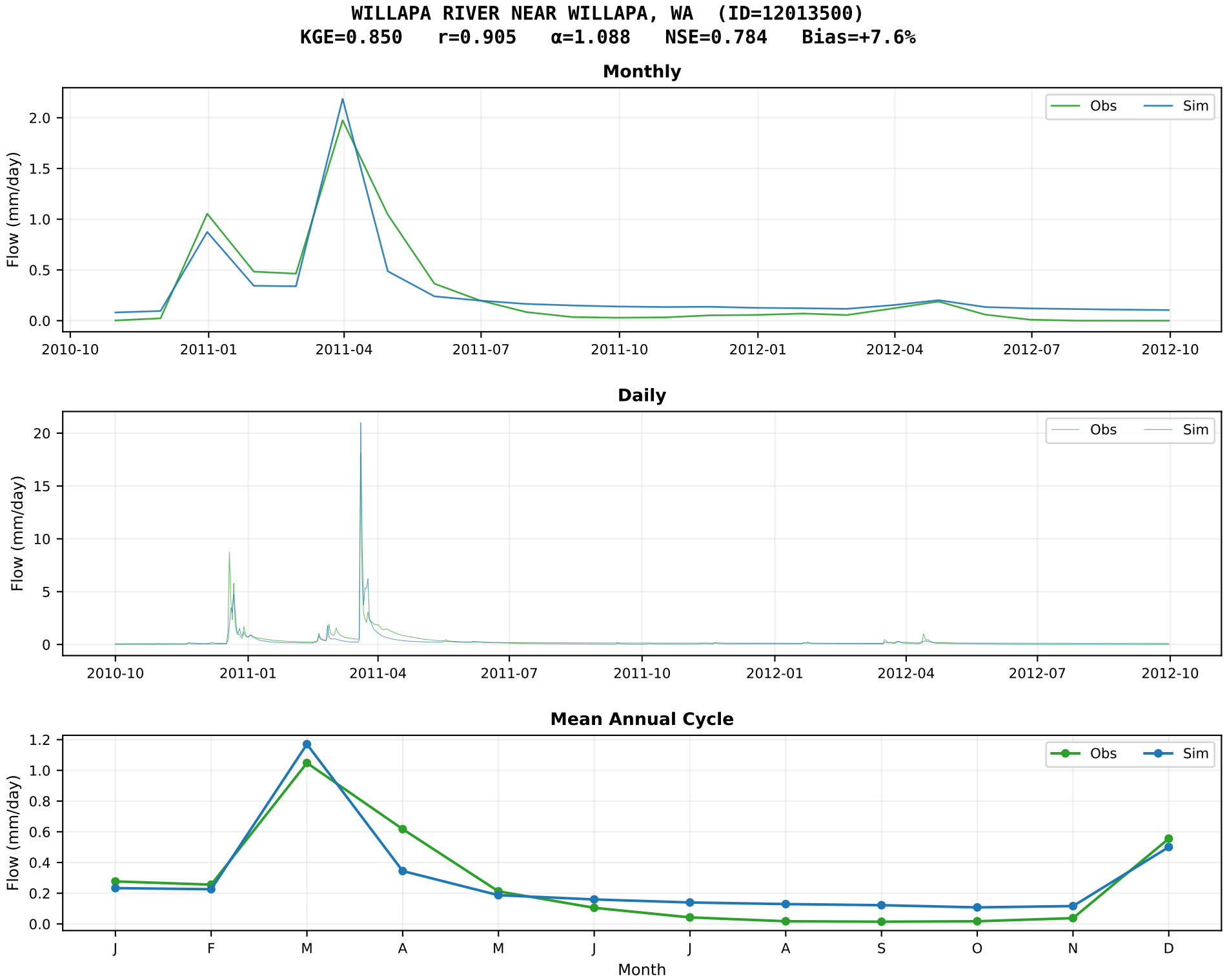

ID: 2479155
KGE=0.854 r=0.936 α=0.877 NSE=0.871 Bias=+4.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

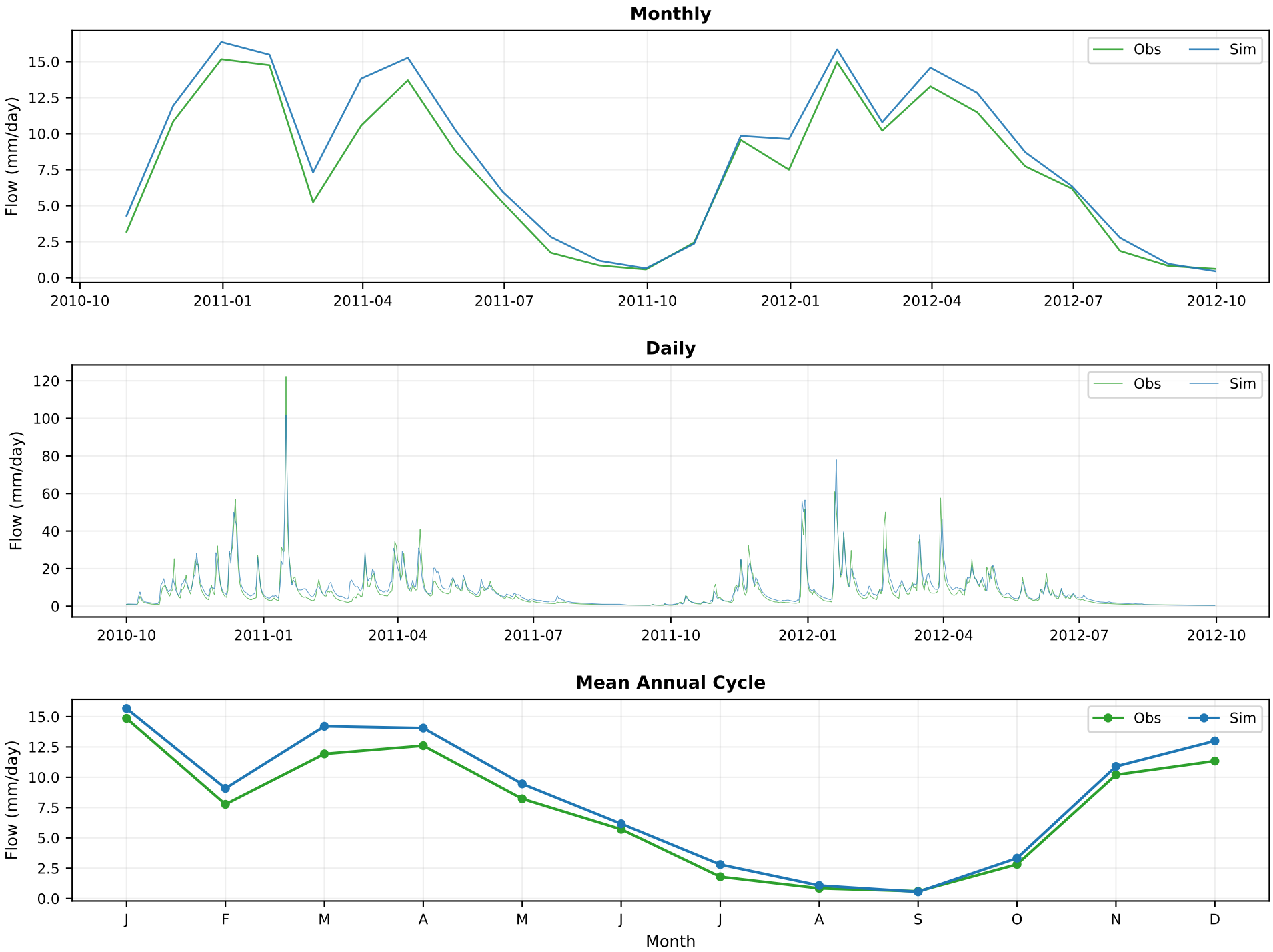

ID: 3182500
KGE=0.853 r=0.918 α=0.959 NSE=0.839 Bias=+11.5%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
35
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

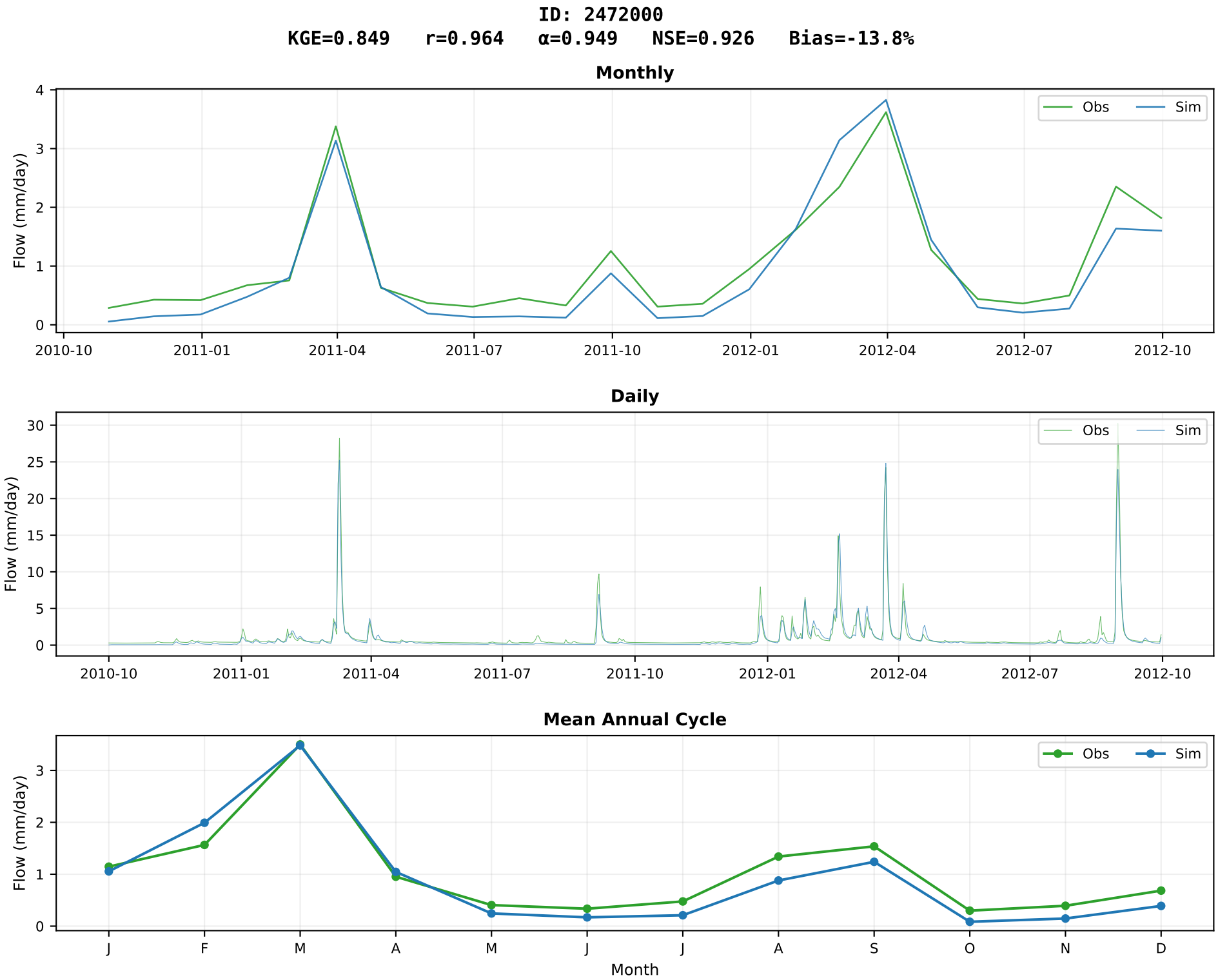

ID: 1440400
KGE=0.853 r=0.888 α=1.016 NSE=0.768 Bias=+9.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

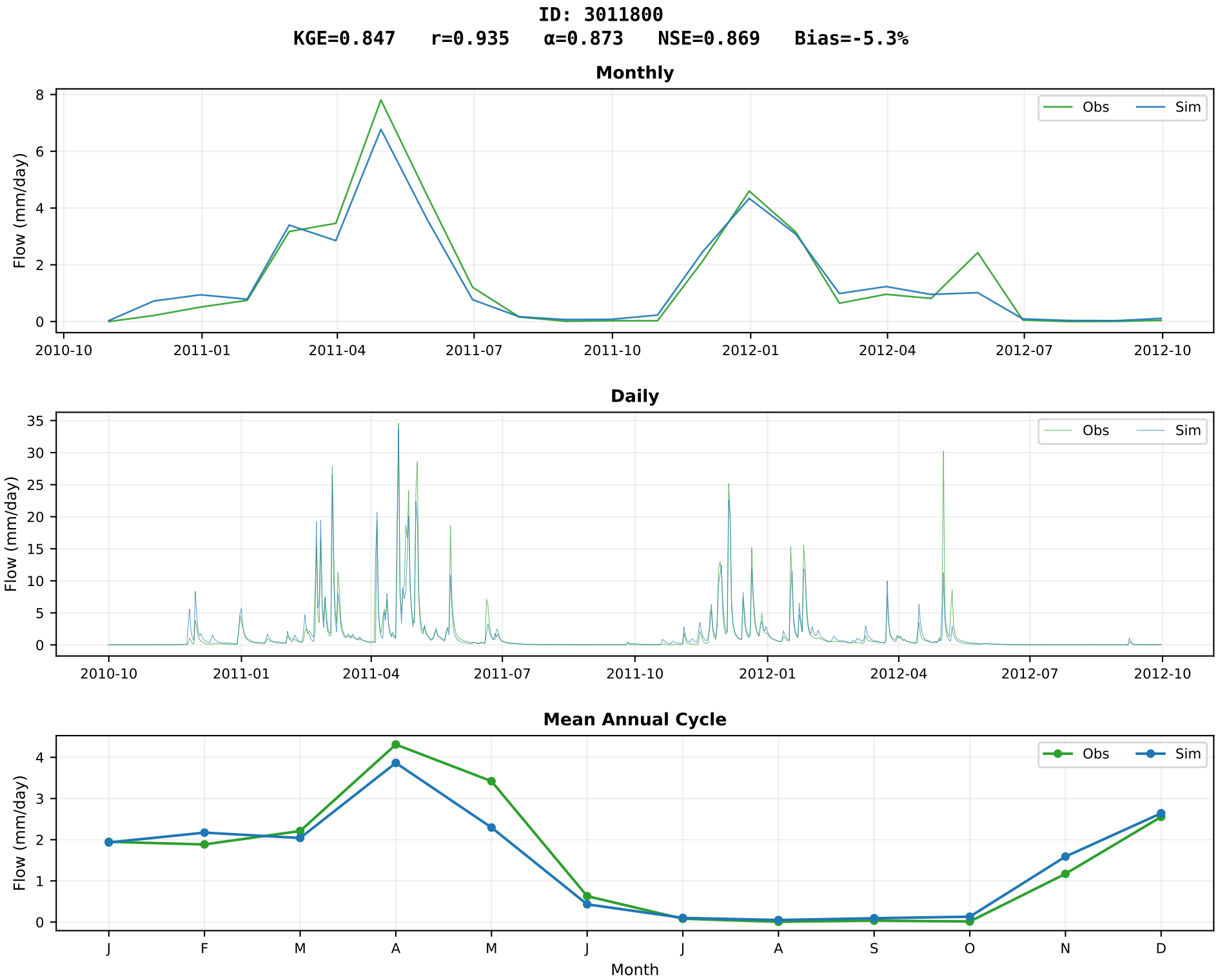

ID: 4105700
KGE=0.851 r=0.889 α=0.955 NSE=0.782 Bias=-8.8%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

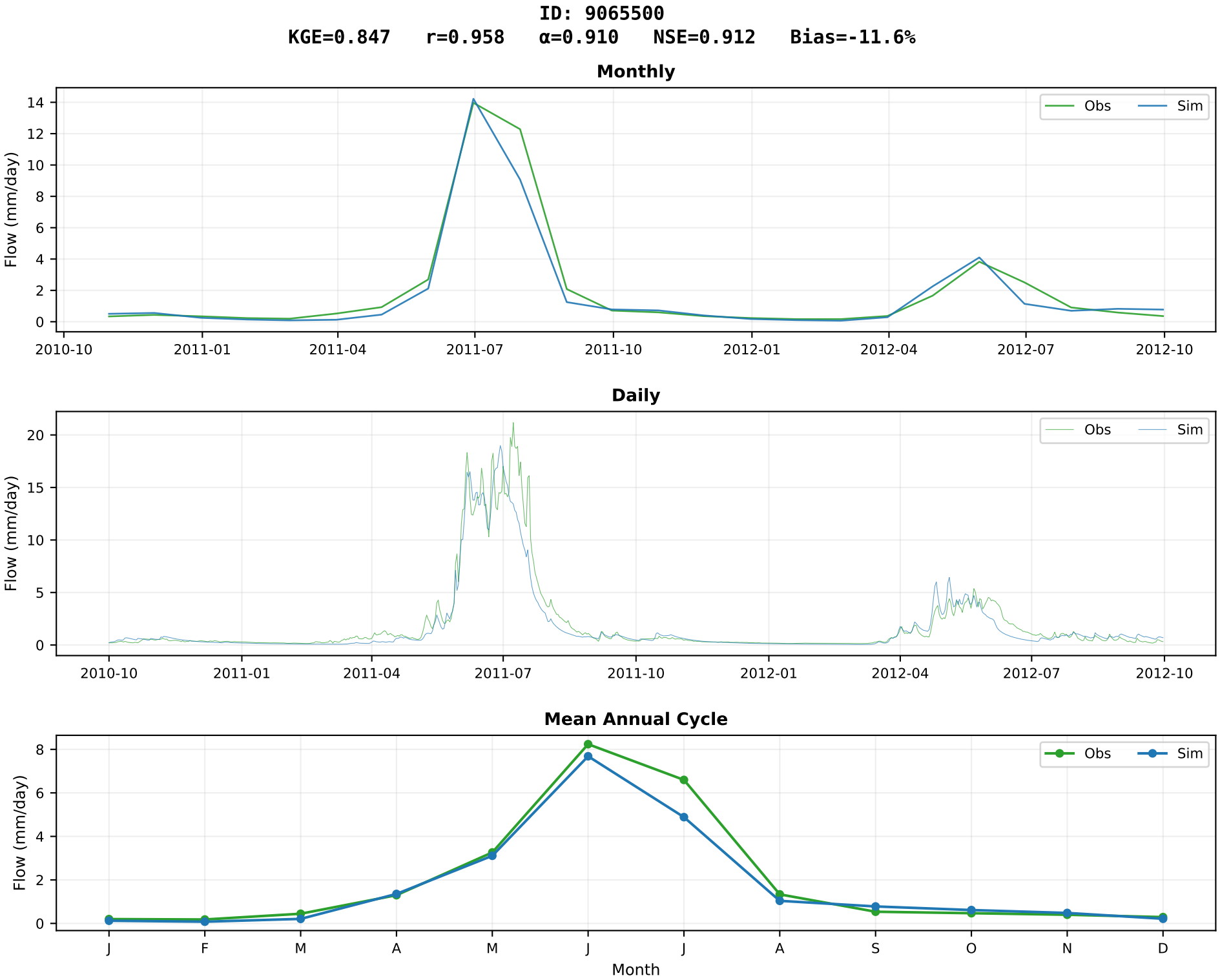

ID: 4213000
KGE=0.851 r=0.909 α=0.896 NSE=0.825 Bias=-5.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

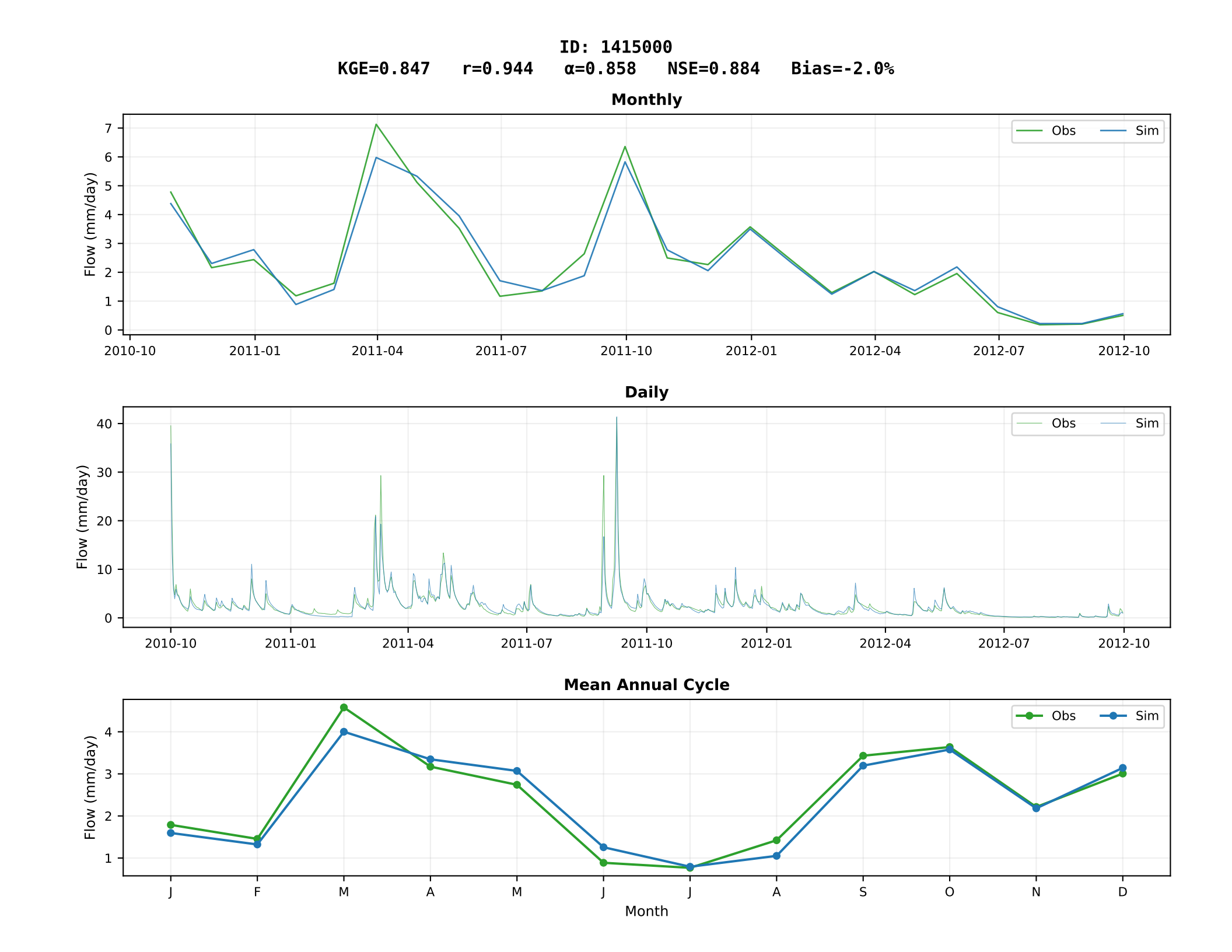

WILLAPA RIVER NEAR WILLAPA, WA (ID=12013500)
KGE=0.850 r=0.905 α=1.088 NSE=0.784 Bias=+7.6%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

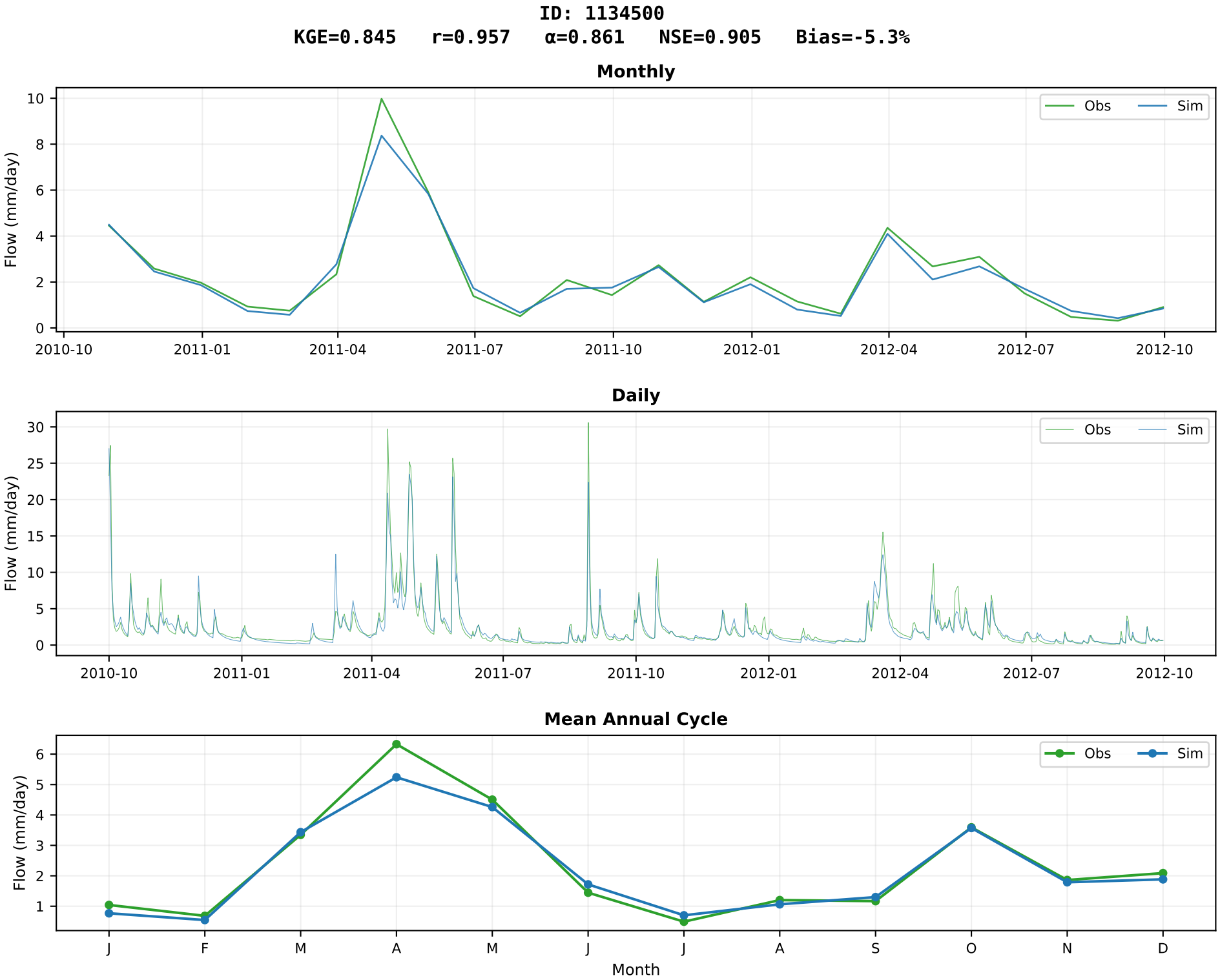

SANTA CRUZ C NR SANTA YNEZ CA (ID=11124500)
KGE=0.849 r=0.933 α=0.967 NSE=0.860 Bias=+13.1%
Monthly
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
100
120
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

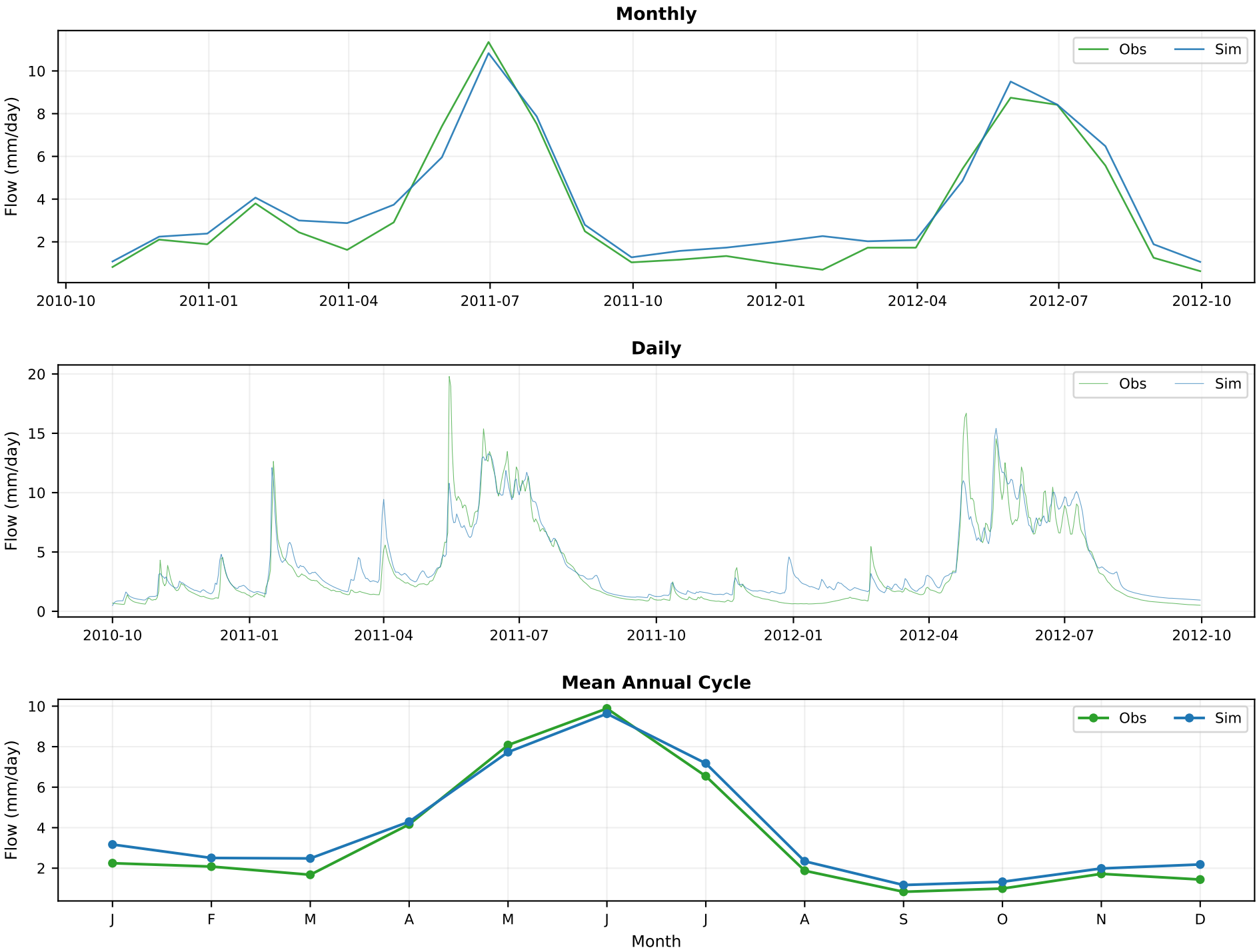

ID: 2472000
KGE=0.849 r=0.964 α=0.949 NSE=0.926 Bias=-13.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

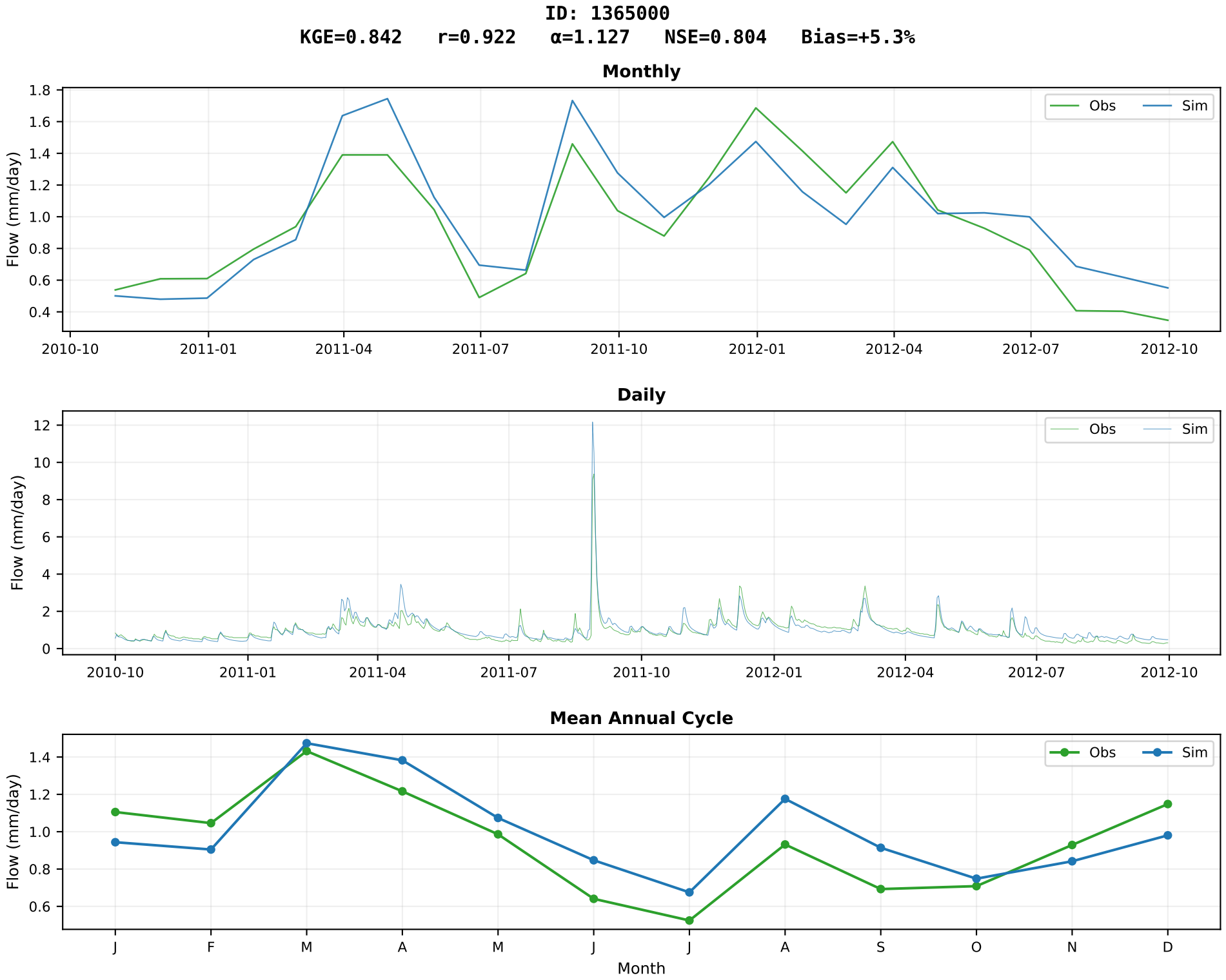

ID: 3011800
KGE=0.847 r=0.935 α=0.873 NSE=0.869 Bias=-5.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

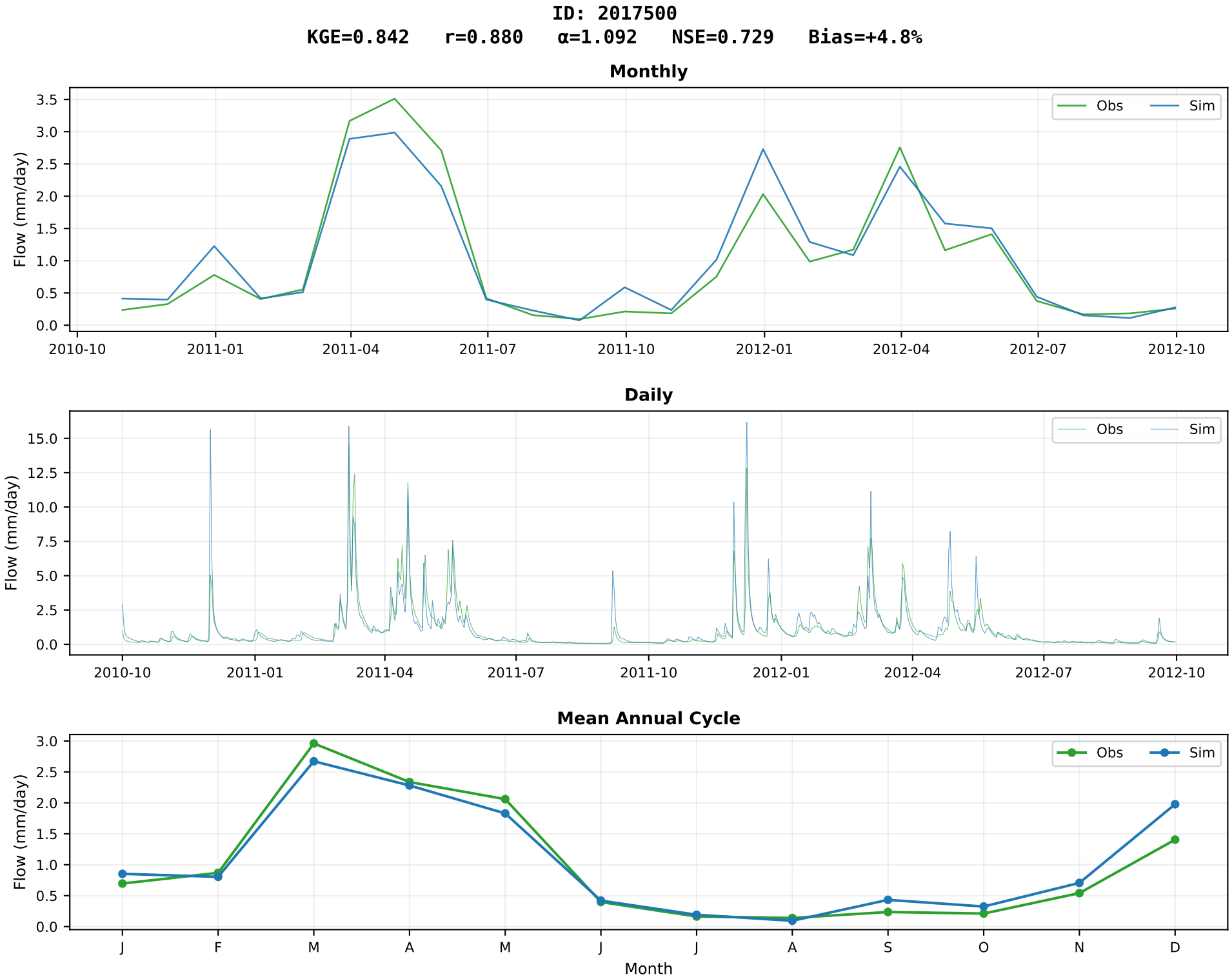

ID: 9065500
KGE=0.847 r=0.958 α=0.910 NSE=0.912 Bias=-11.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

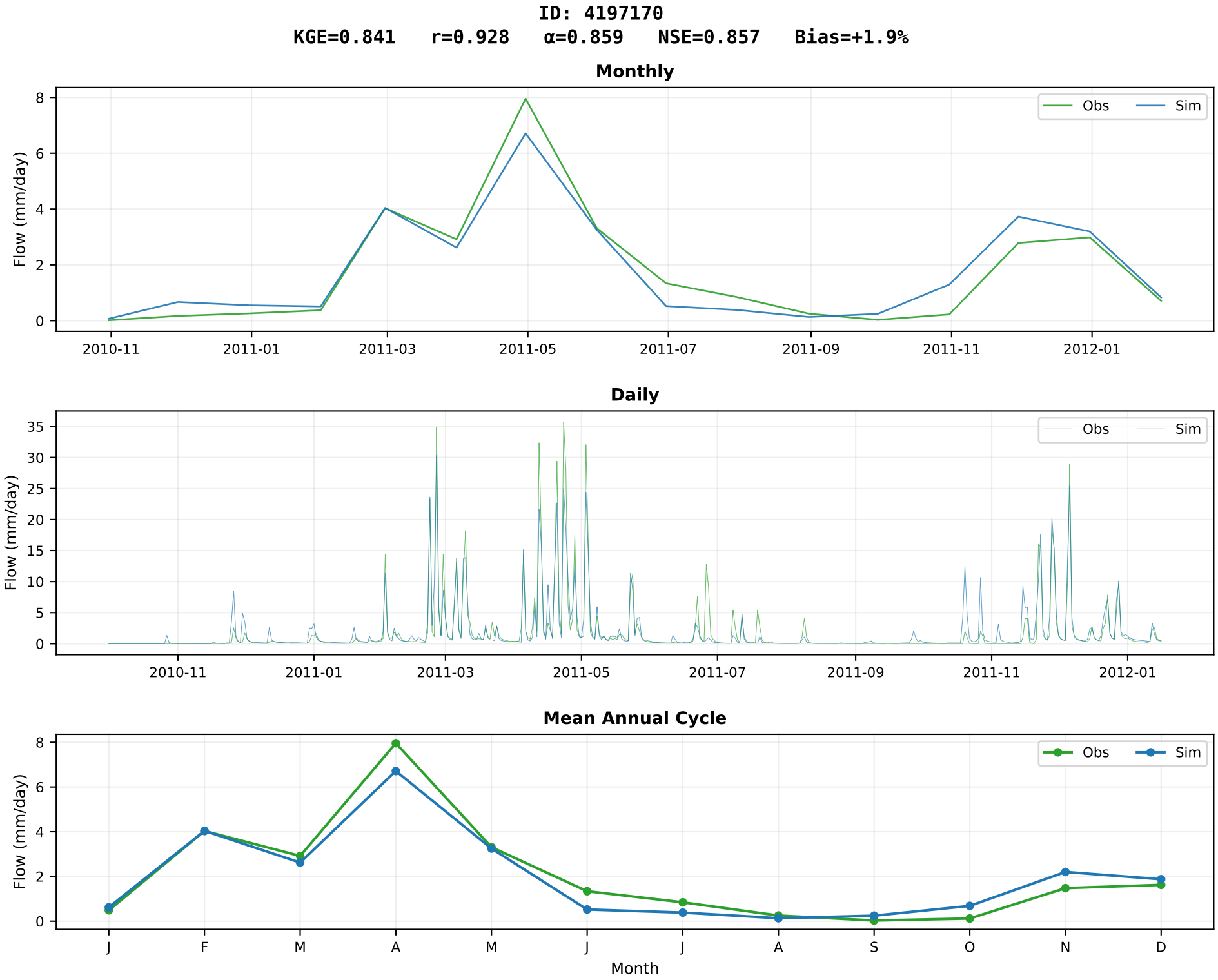

ID: 1415000
KGE=0.847 r=0.944 α=0.858 NSE=0.884 Bias=-2.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

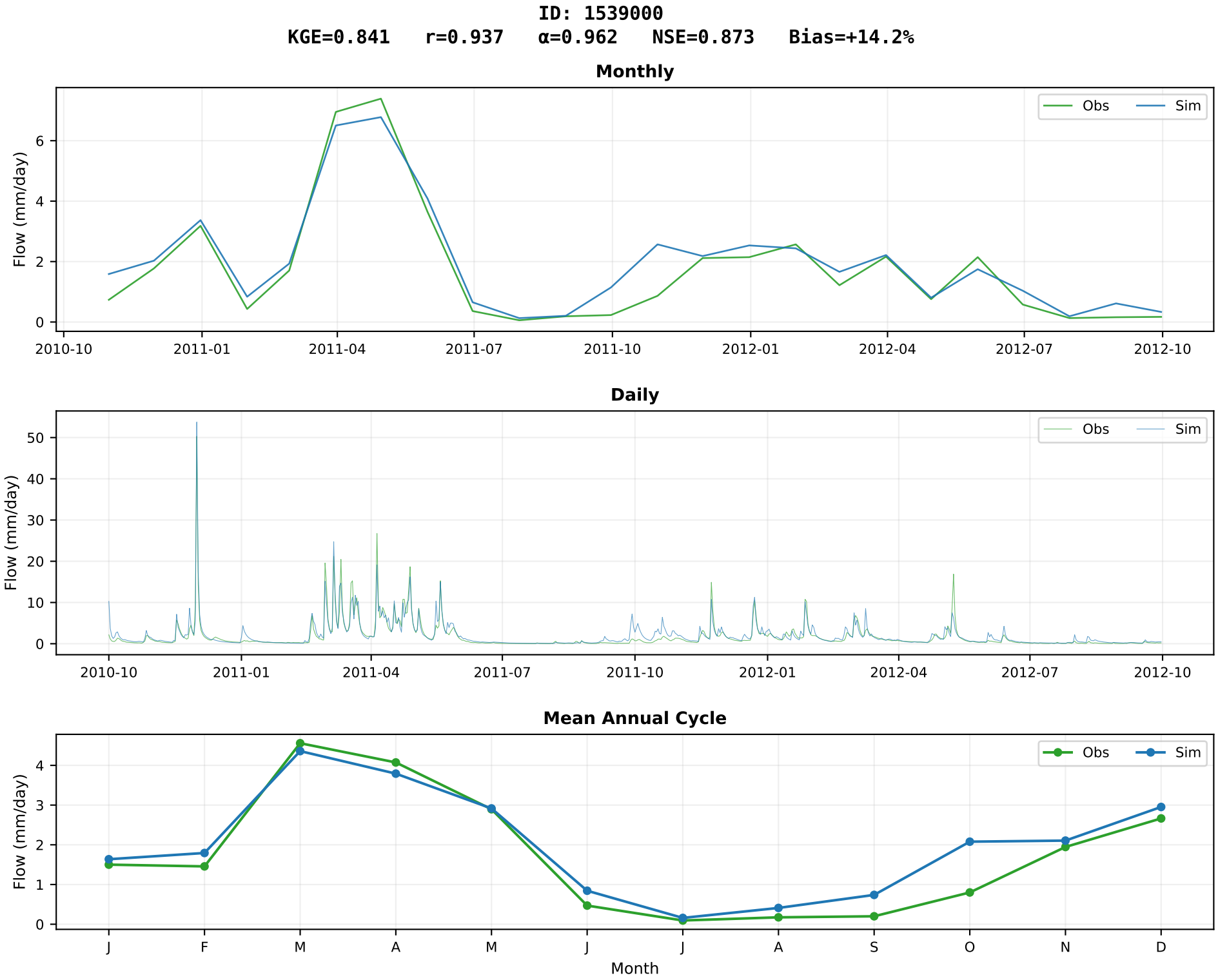

ID: 1134500
KGE=0.845 r=0.957 α=0.861 NSE=0.905 Bias=-5.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

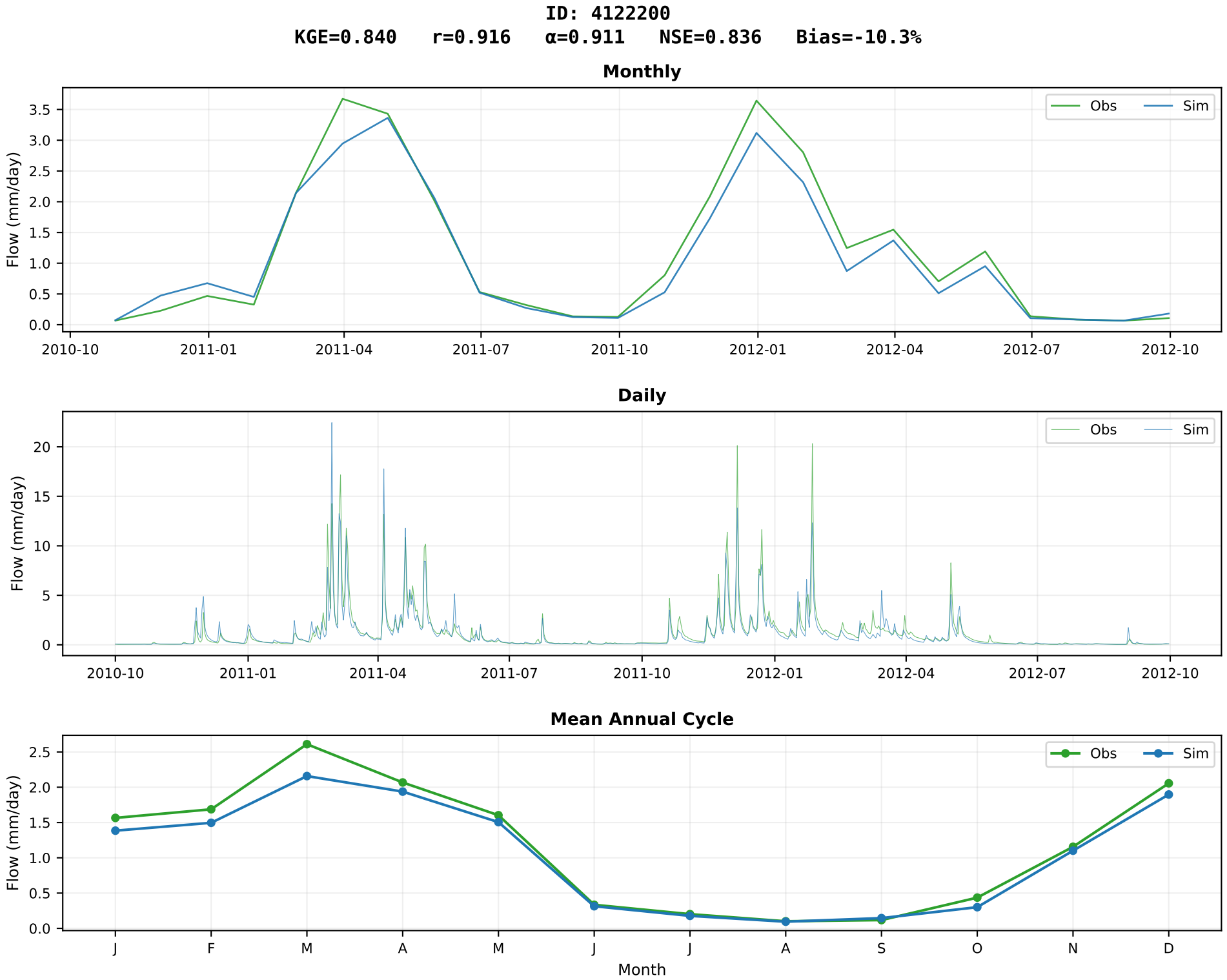

EAST FORK LEWIS RIVER NEAR HEISSON, WA (ID=14222500)
KGE=0.842 r=0.946 α=0.899 NSE=0.882 Bias=+10.8%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

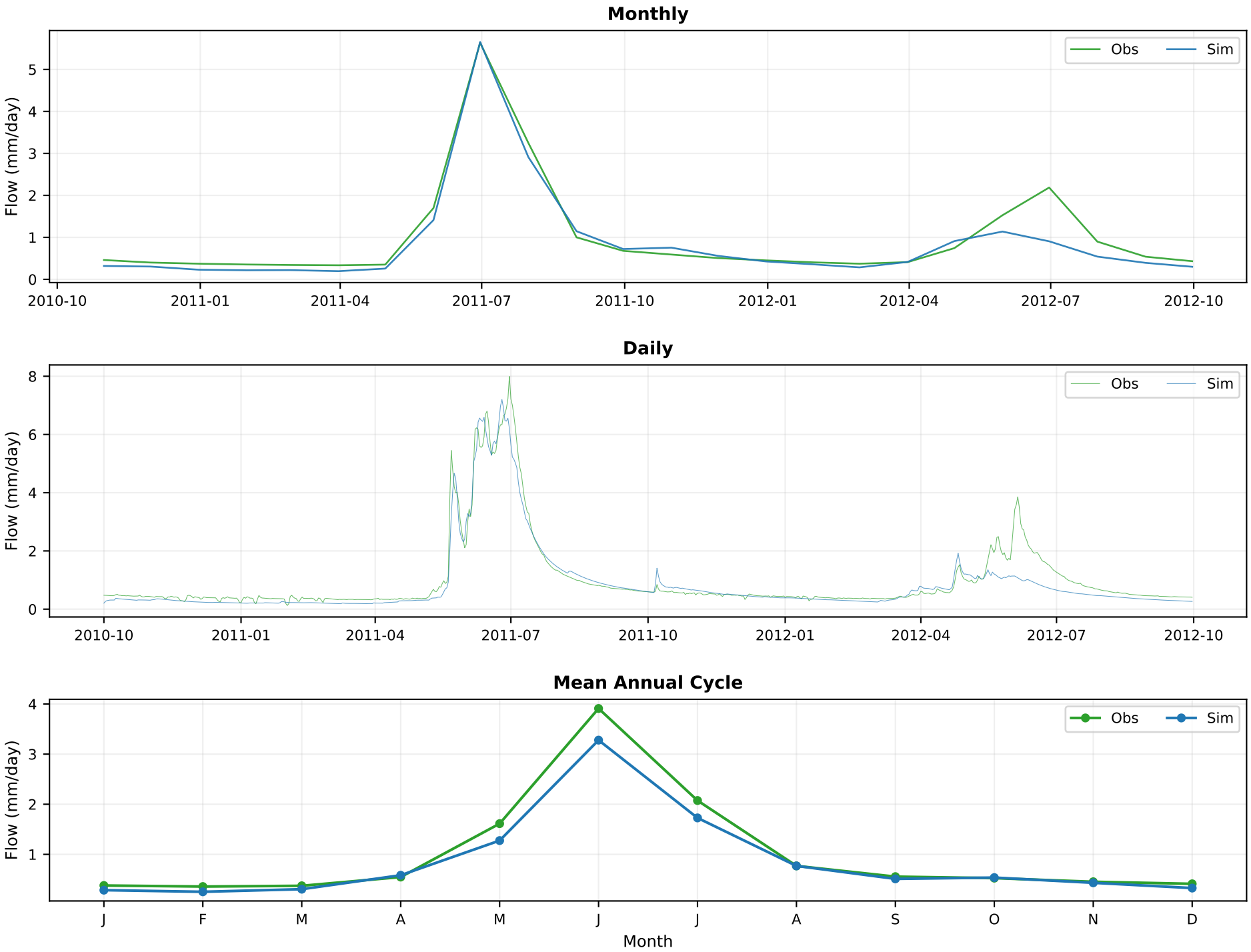

ID: 1365000
KGE=0.842 r=0.922 α=1.127 NSE=0.804 Bias=+5.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

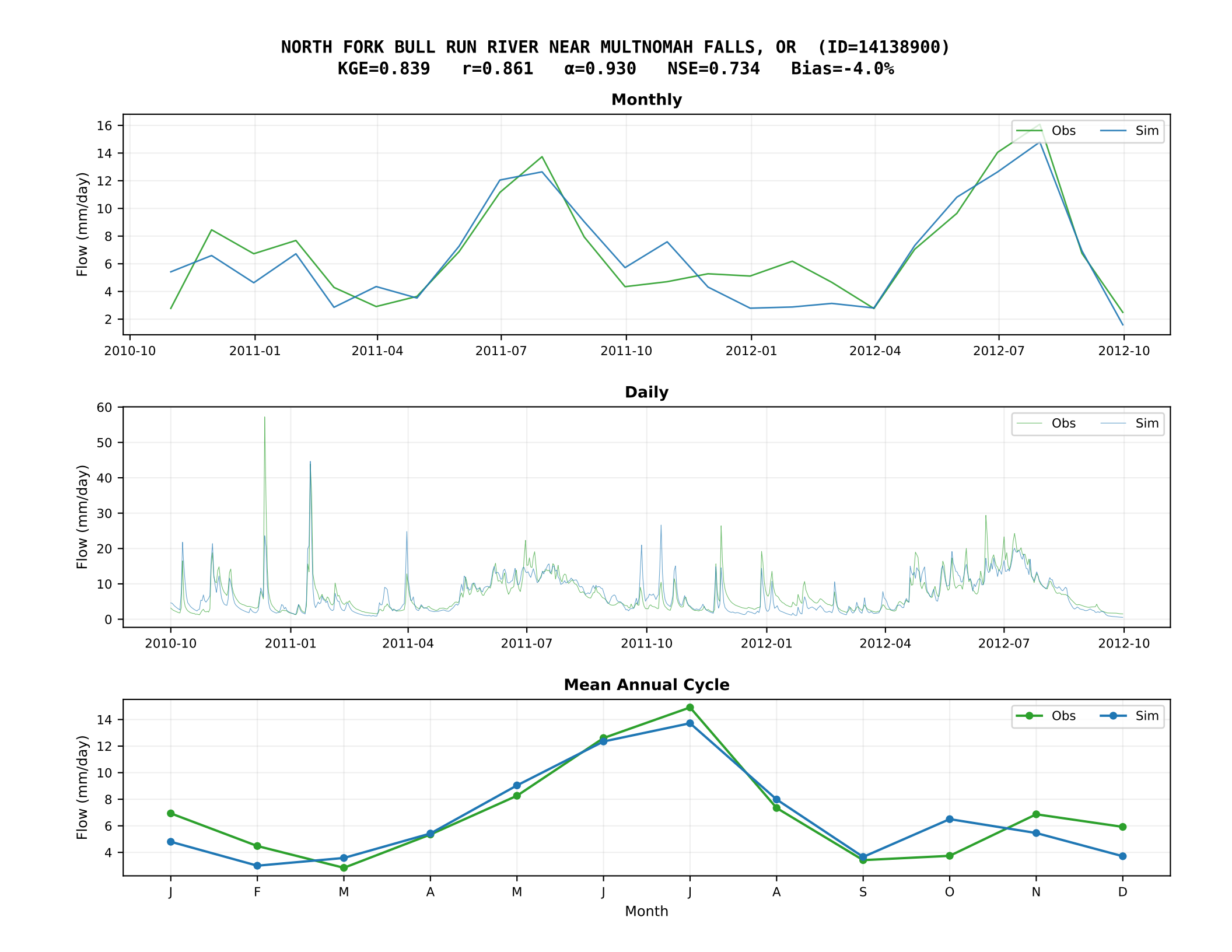

ID: 2017500
KGE=0.842 r=0.880 α=1.092 NSE=0.729 Bias=+4.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Mean Annual Cycle
Flow (mm/day)
J
F
M
A
M
J
J
A
S
O
N
D
Month

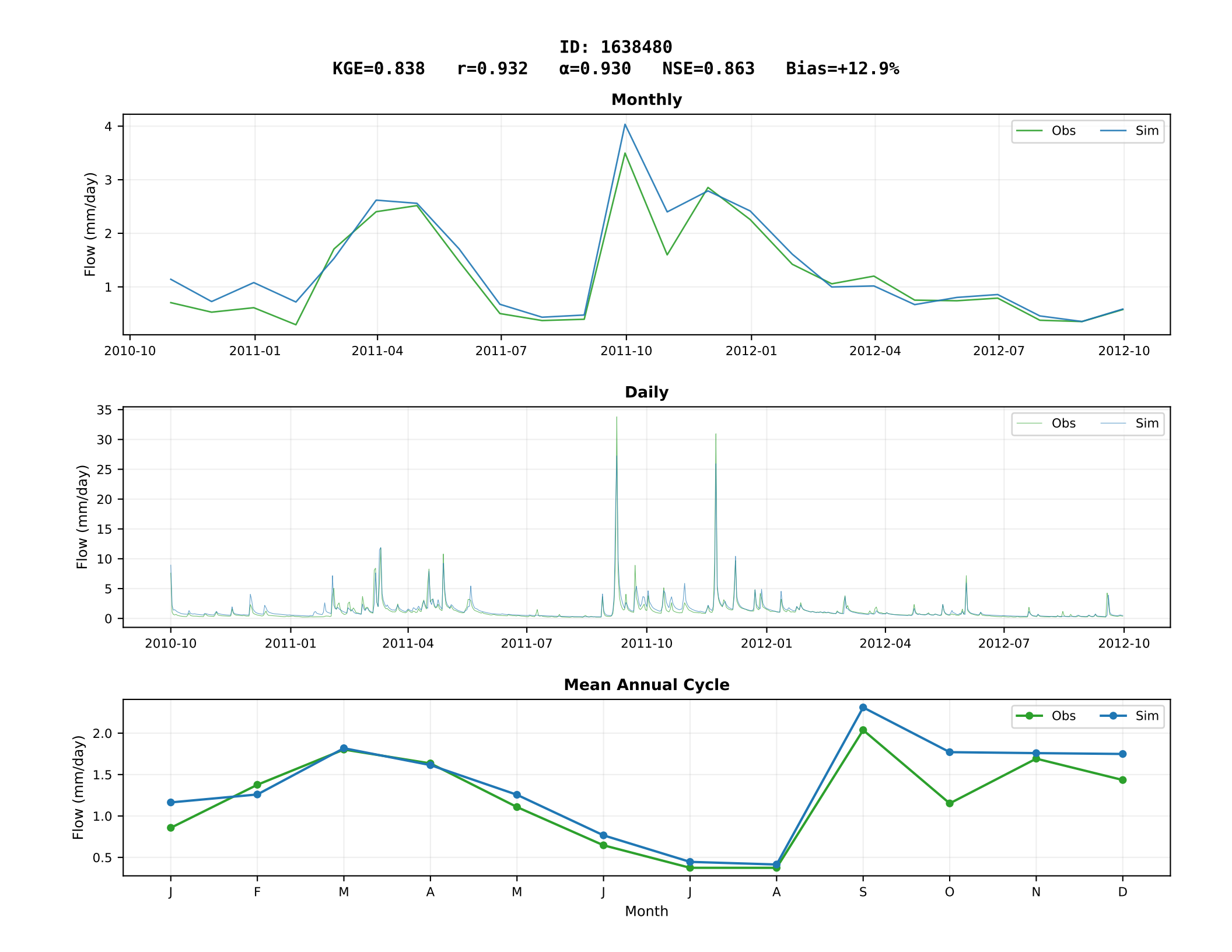

ID: 4197170
KGE=0.841 r=0.928 α=0.859 NSE=0.857 Bias=+1.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-11
2011-01
2011-03
2011-05
2011-07
2011-09
2011-11
2012-01
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

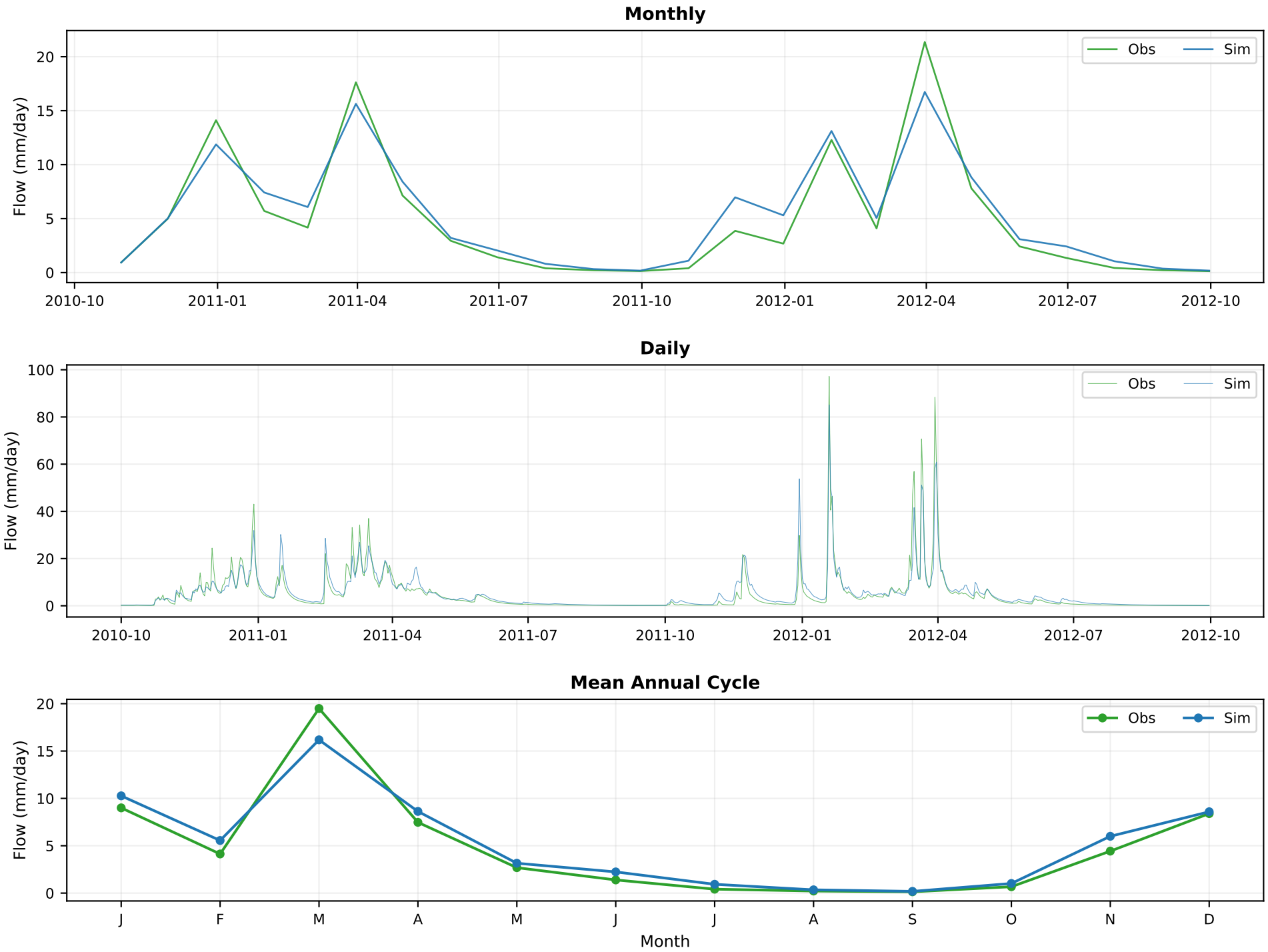

ID: 1539000
KGE=0.841 r=0.937 α=0.962 NSE=0.873 Bias=+14.2%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

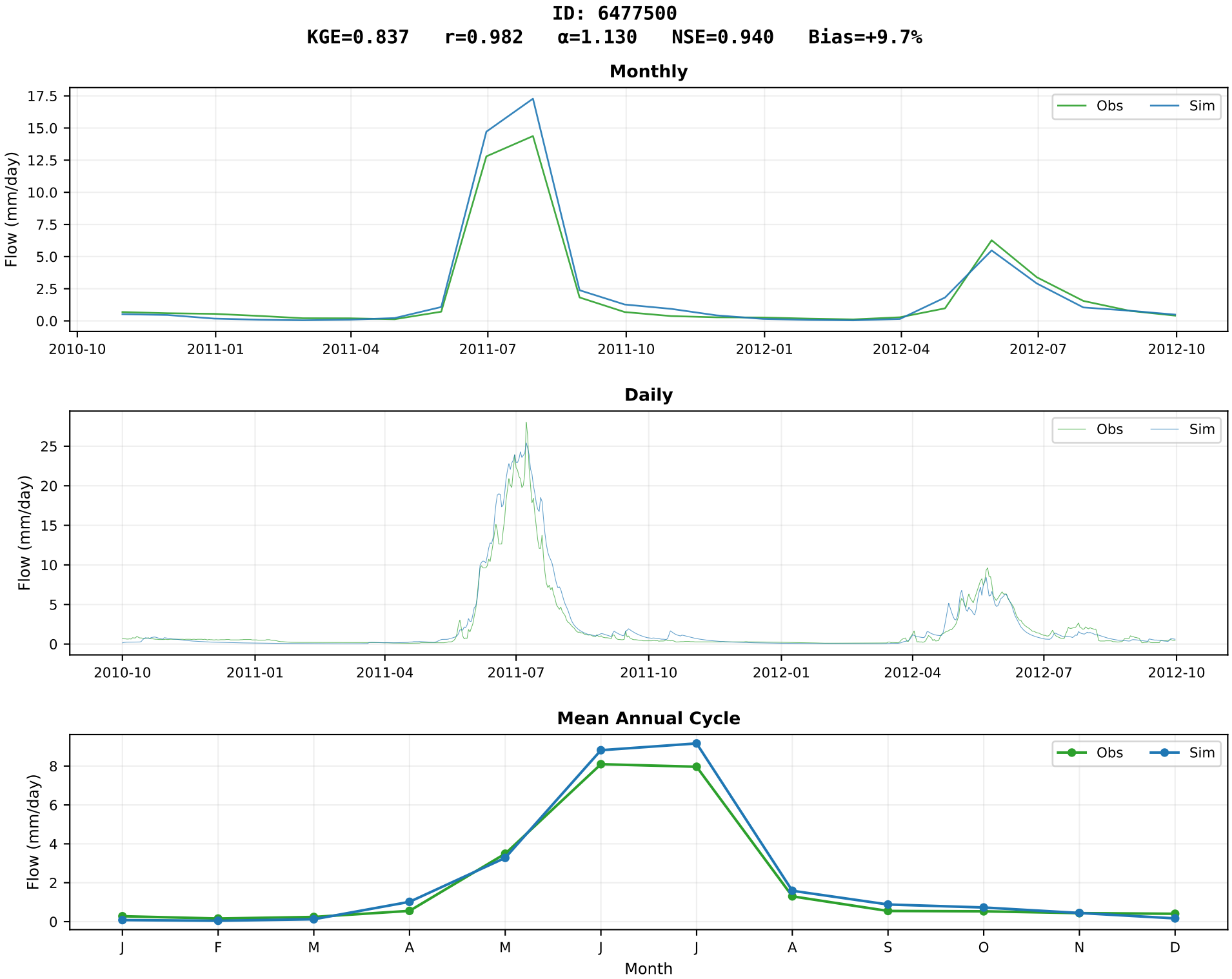

ID: 4122200
KGE=0.840 r=0.916 α=0.911 NSE=0.836 Bias=-10.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

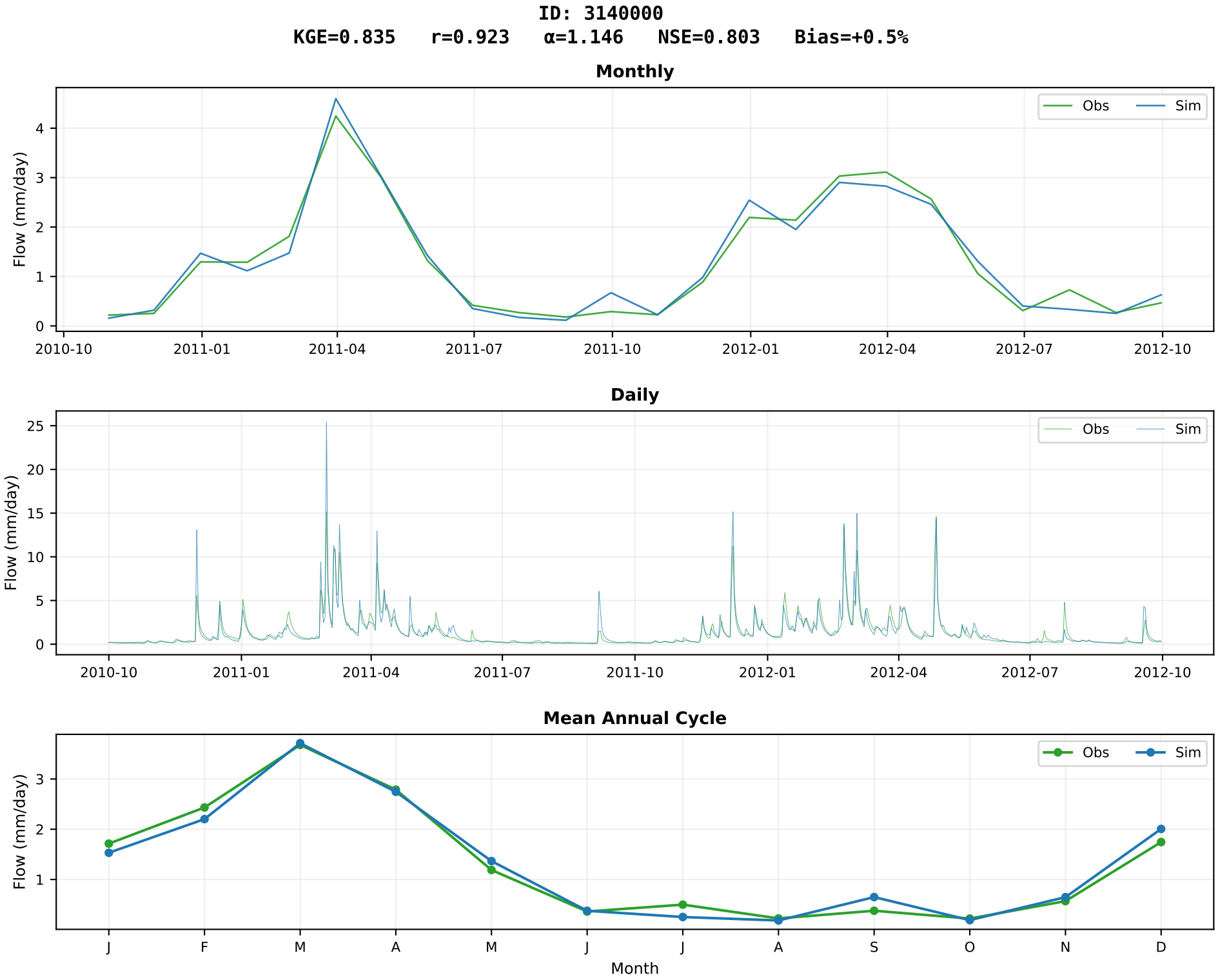

ID: 8014500
KGE=0.840 r=0.957 α=0.936 NSE=0.904 Bias=-14.1%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

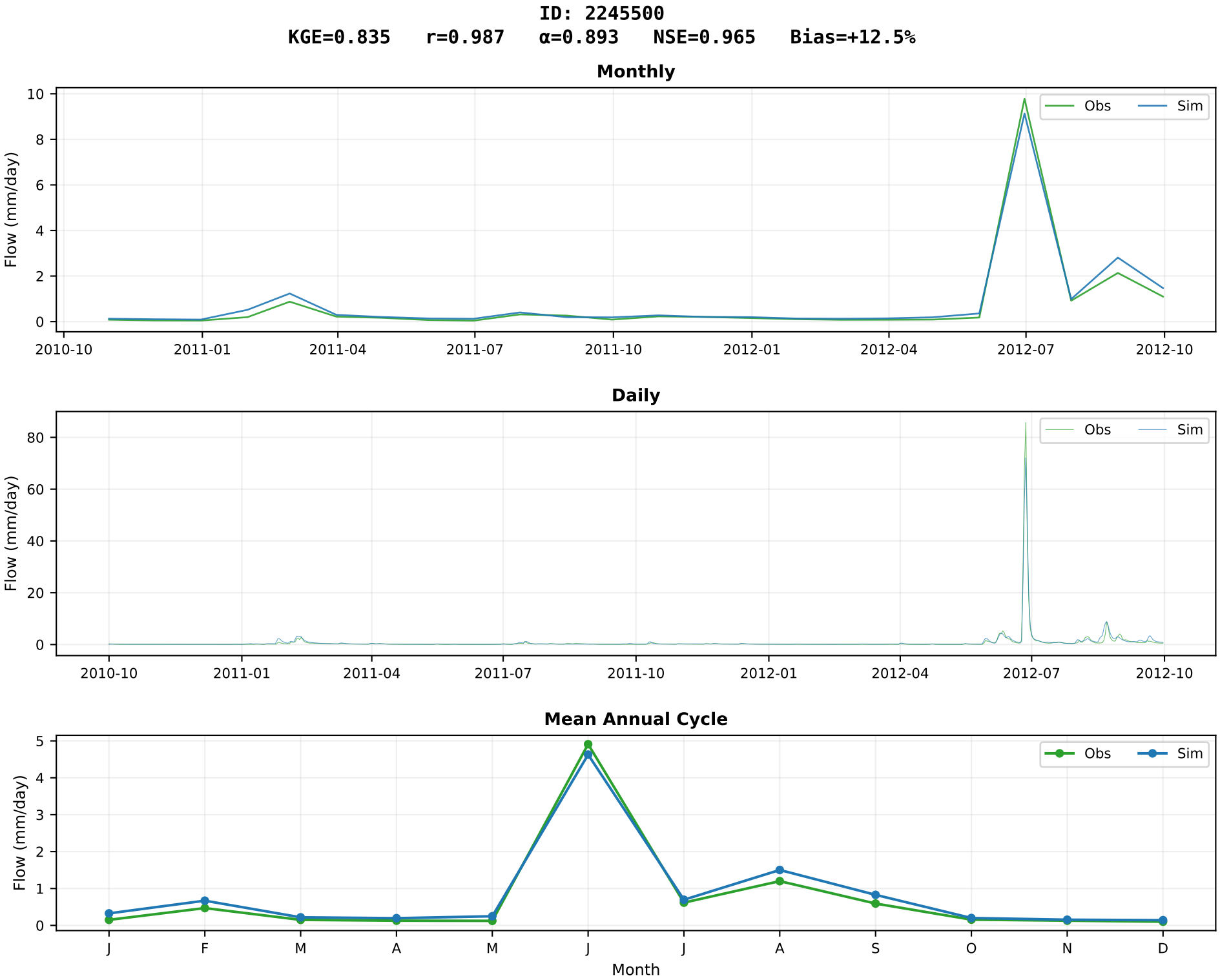

NORTH FORK BULL RUN RIVER NEAR MULTNOMAH FALLS, OR (ID=14138900)
KGE=0.839 r=0.861 α=0.930 NSE=0.734 Bias=-4.0%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

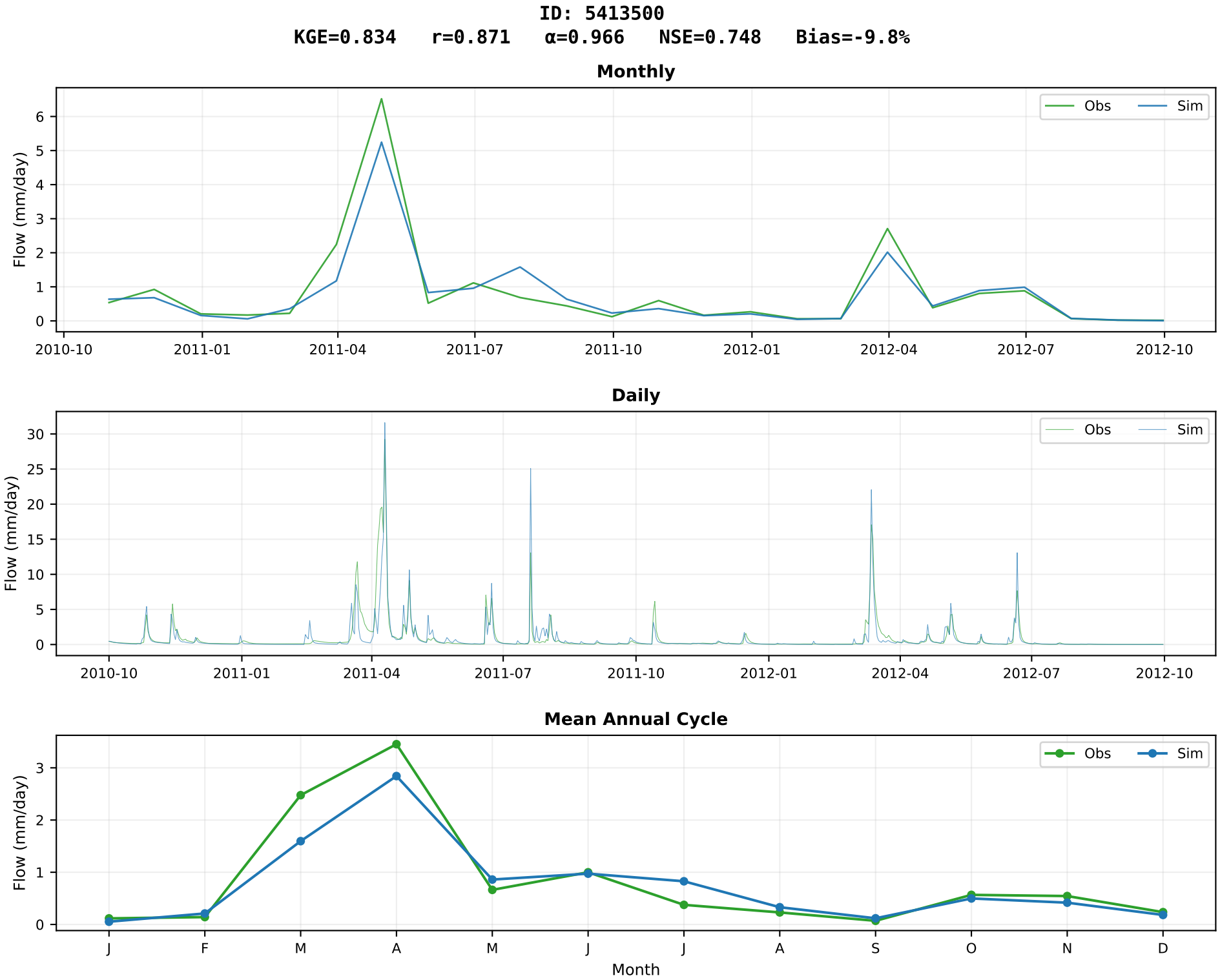

ID: 1638480
KGE=0.838 r=0.932 α=0.930 NSE=0.863 Bias=+12.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

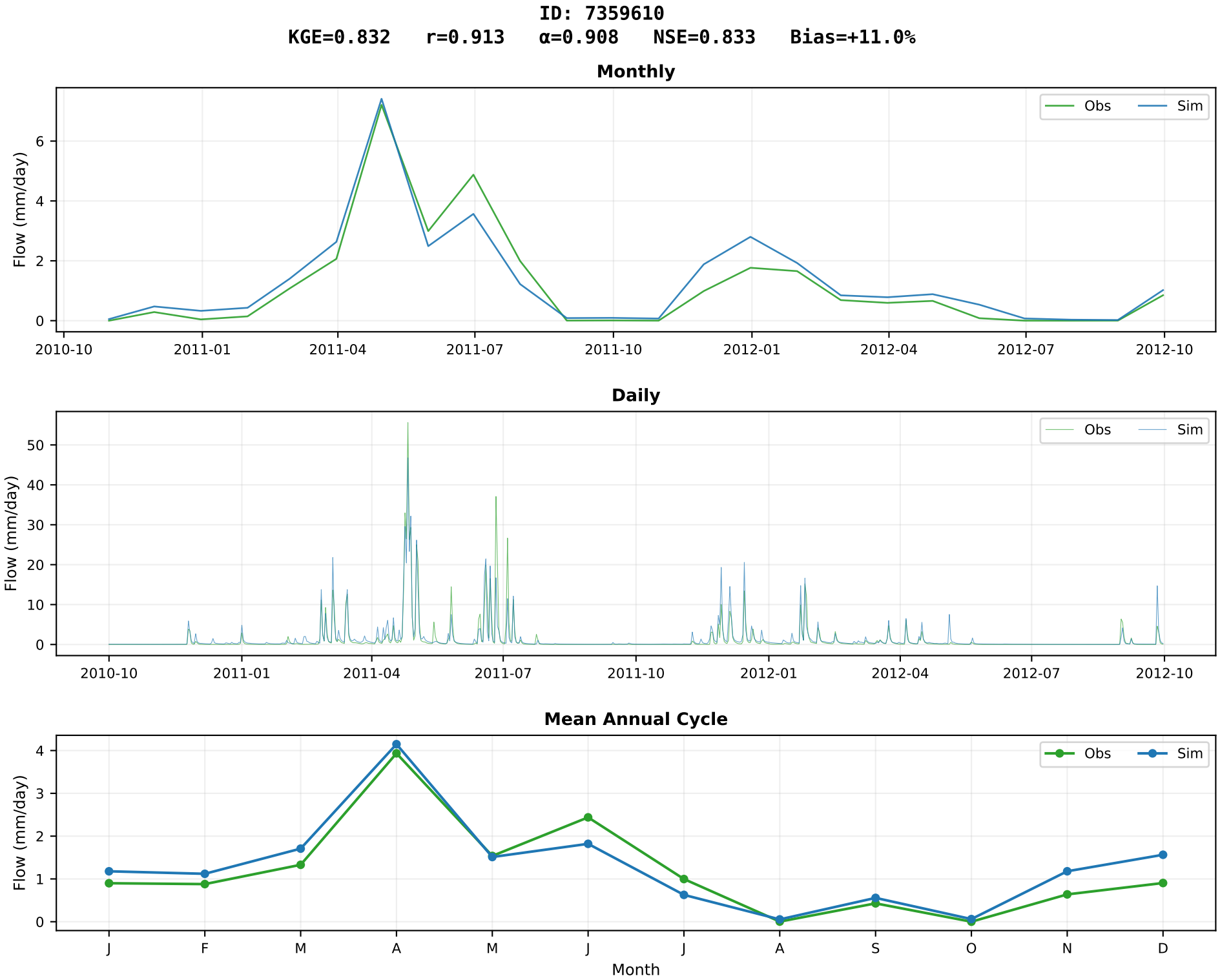

TRINITY R AB COFFEE C NR TRINITY CTR CA (ID=11523200)
KGE=0.838 r=0.942 α=0.869 NSE=0.880 Bias=+7.7%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

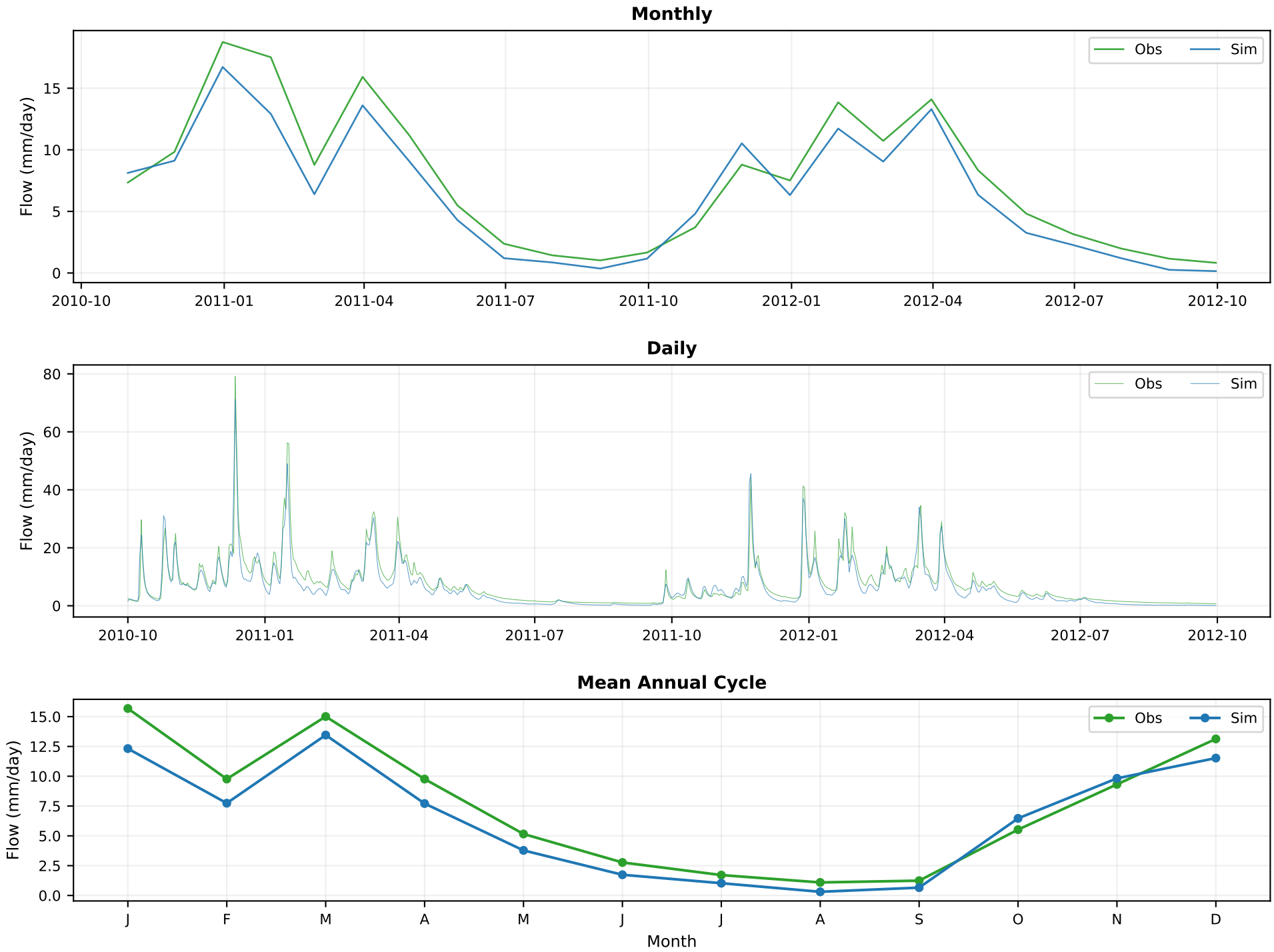

ID: 6477500
KGE=0.837 r=0.982 α=1.130 NSE=0.940 Bias=+9.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

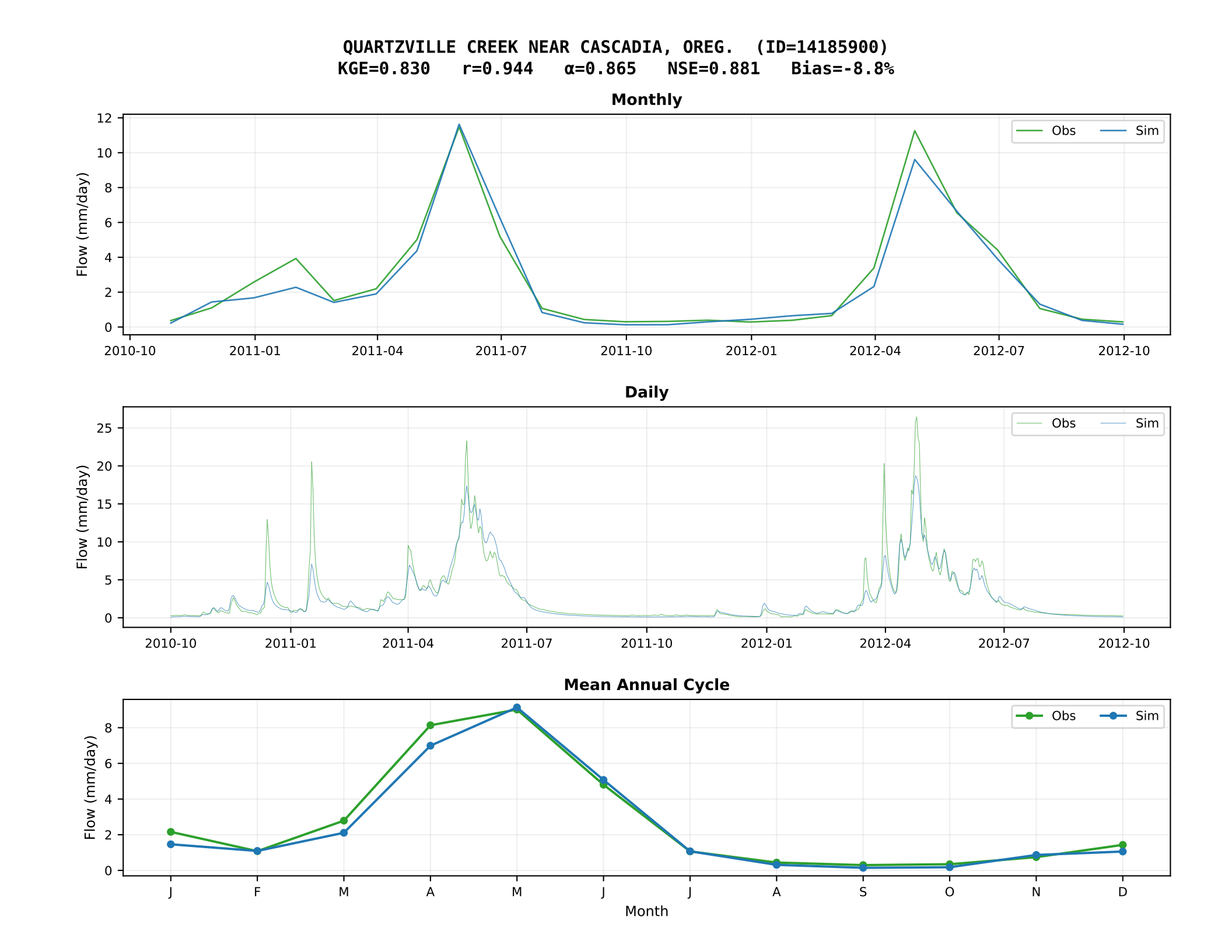

ID: 3140000
KGE=0.835 r=0.923 α=1.146 NSE=0.803 Bias=+0.5%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

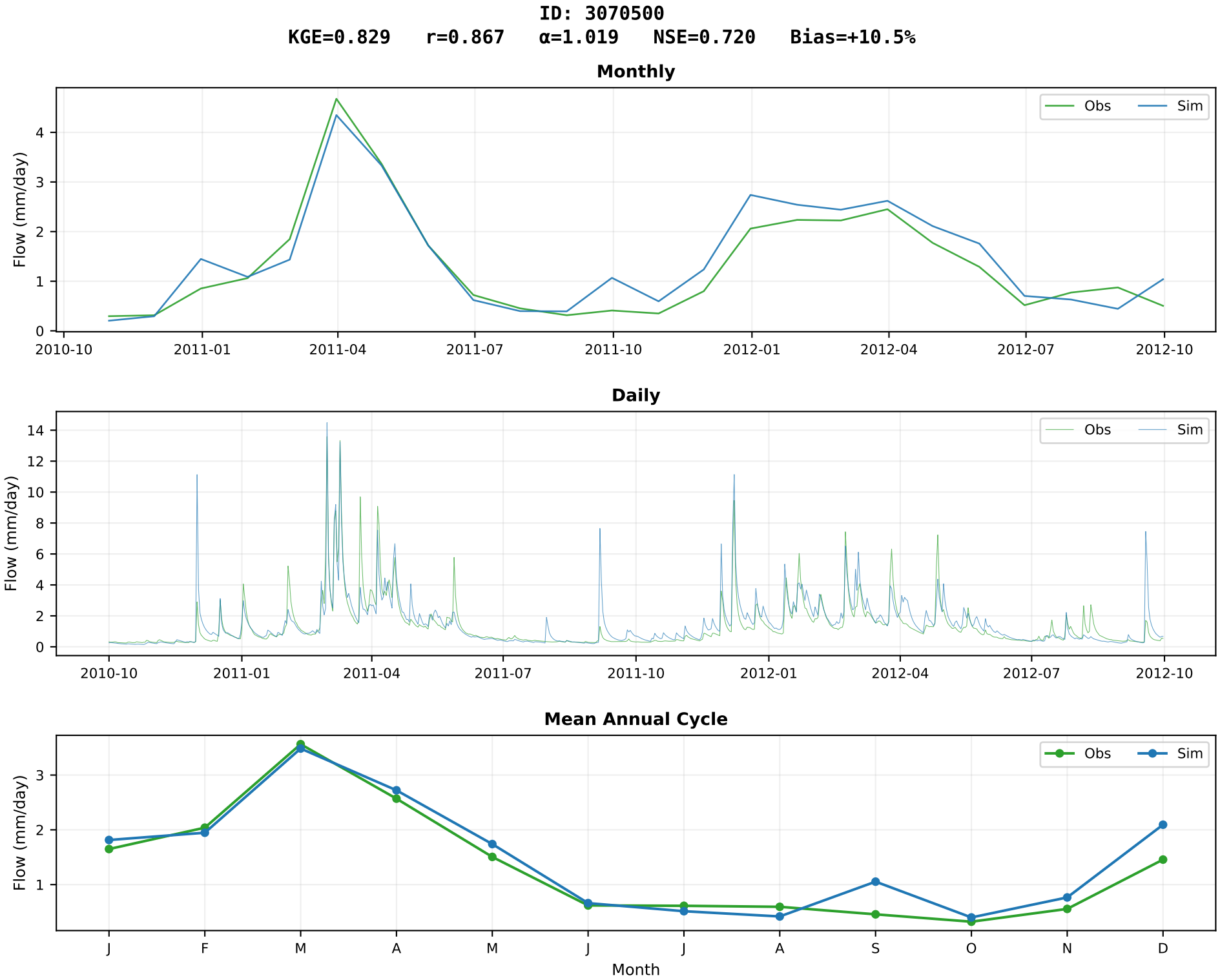

ID: 2245500
KGE=0.835 r=0.987 α=0.893 NSE=0.965 Bias=+12.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

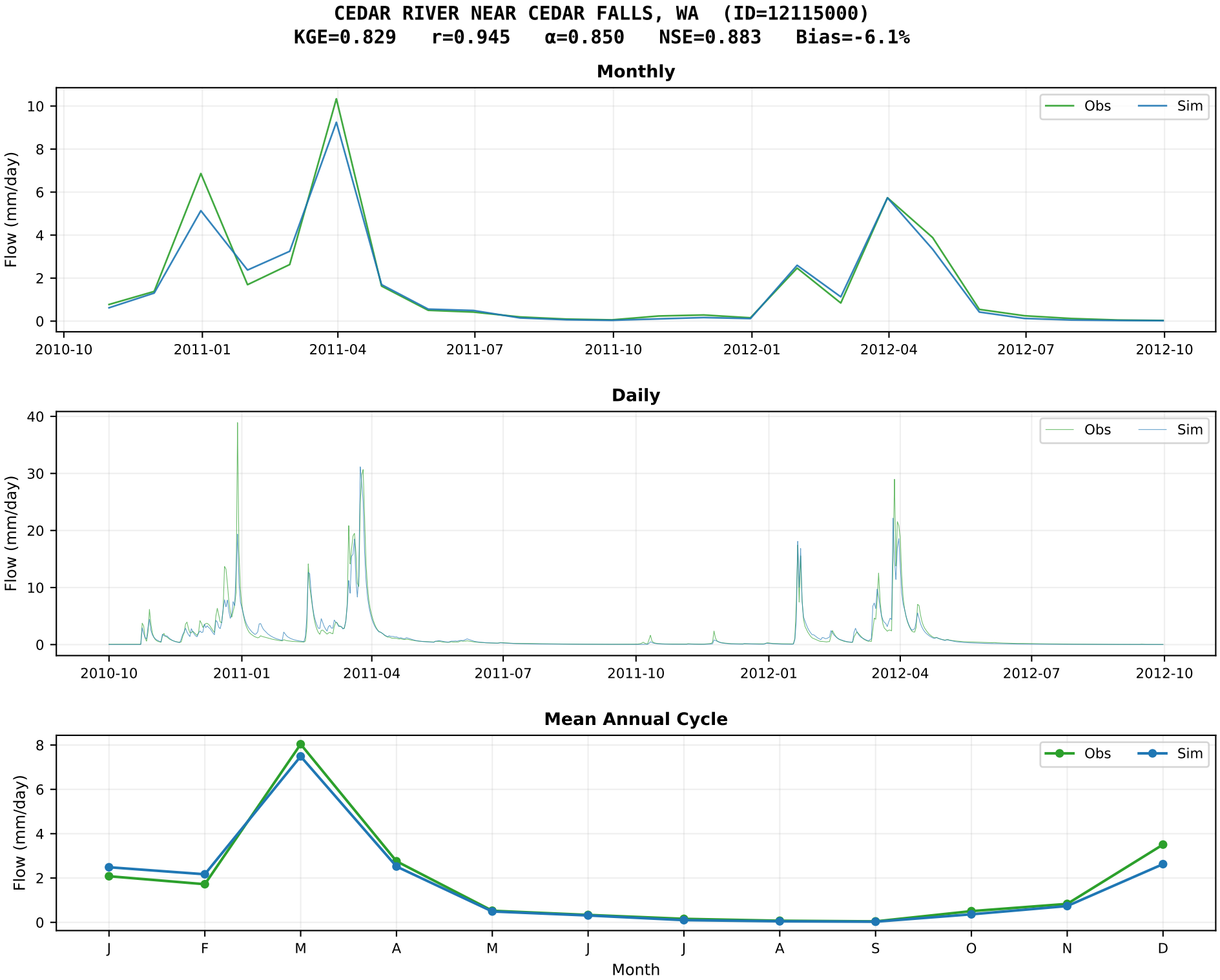

ID: 5413500
KGE=0.834 r=0.871 α=0.966 NSE=0.748 Bias=-9.8%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

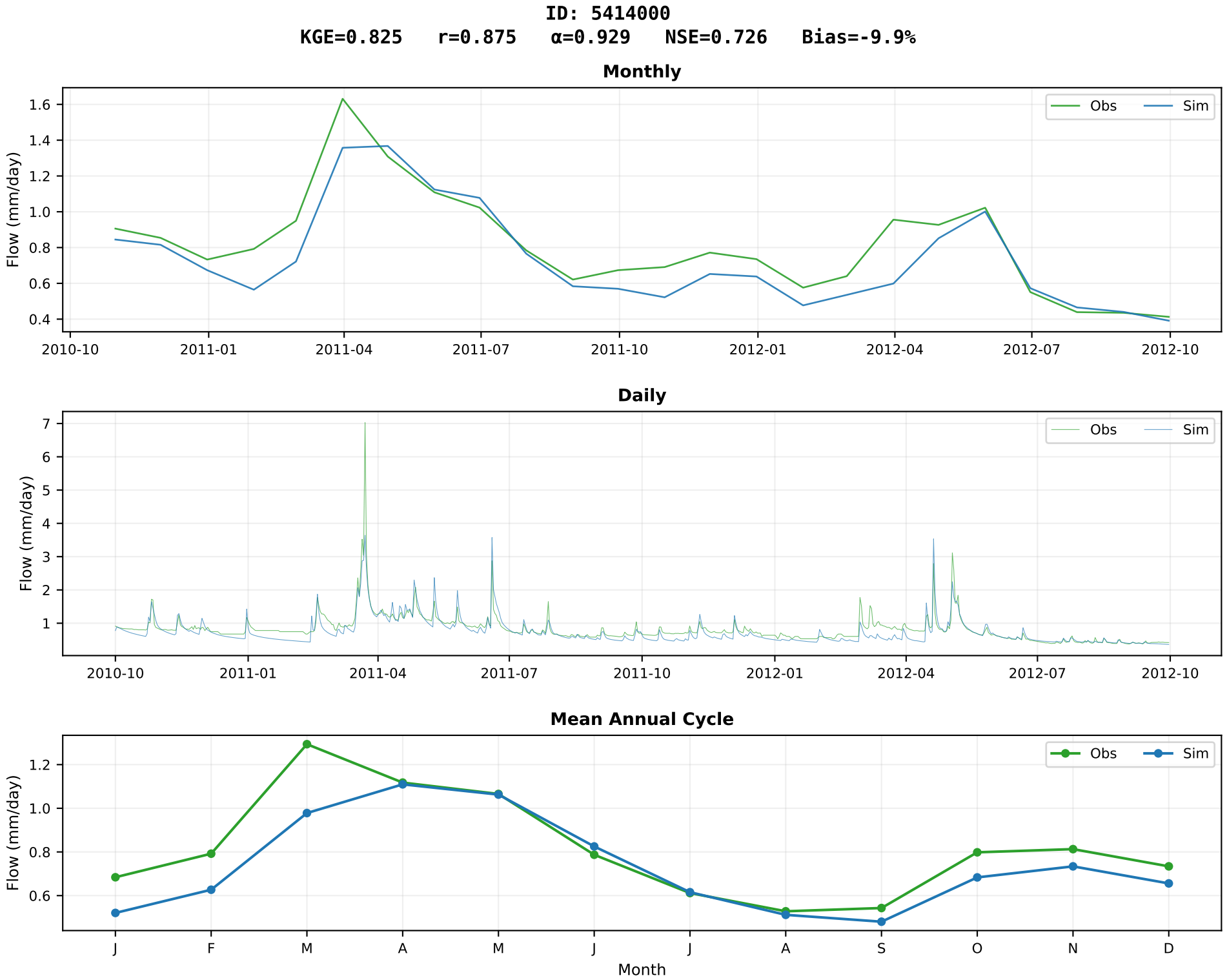

ID: 7359610
KGE=0.832 r=0.913 α=0.908 NSE=0.833 Bias=+11.0%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

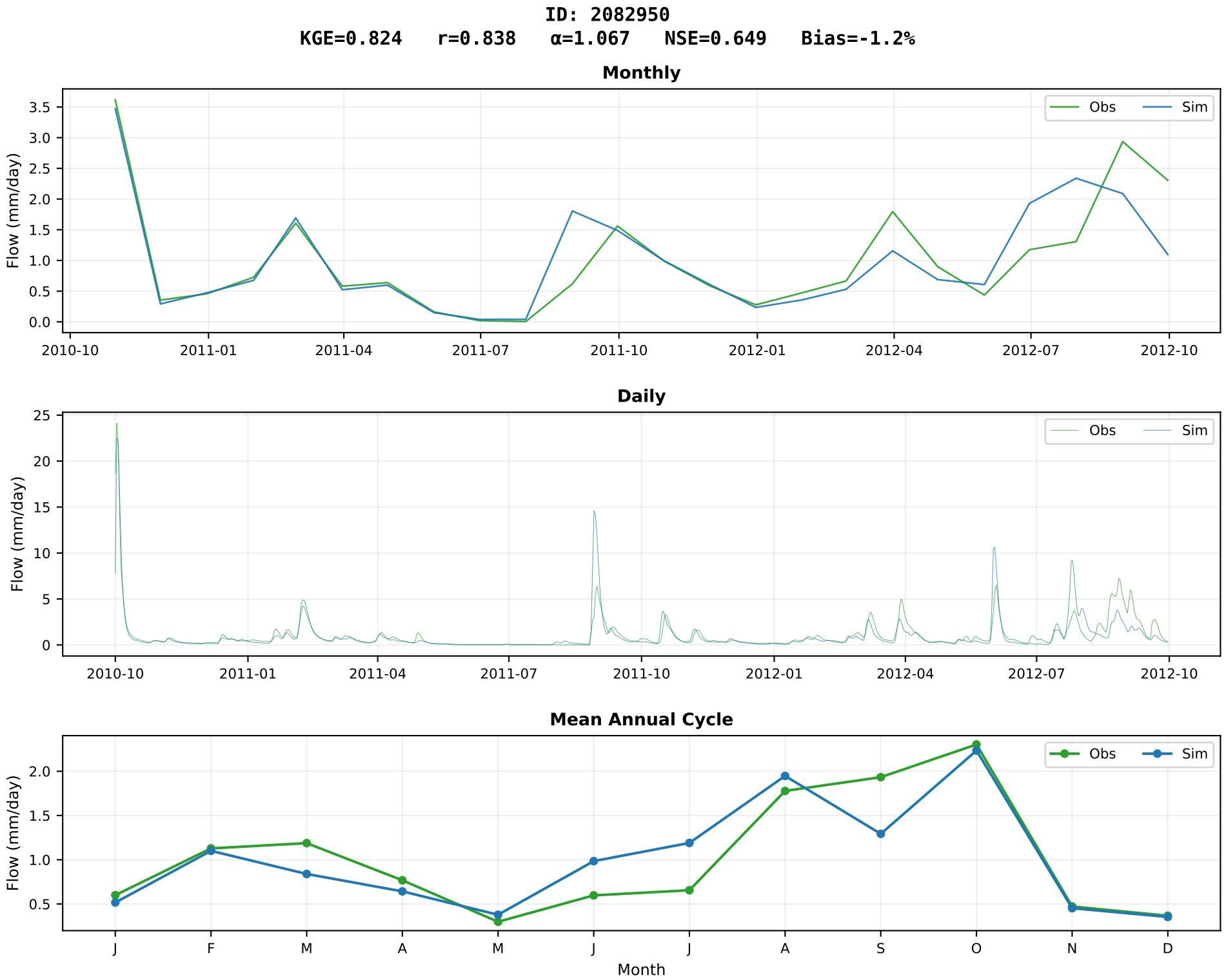

NF COEUR D ALENE R AB SHOSHONE CK NR PRICHARD ID (ID=12411000)
KGE=0.830 r=0.952 α=0.939 NSE=0.886 Bias=-15.1%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

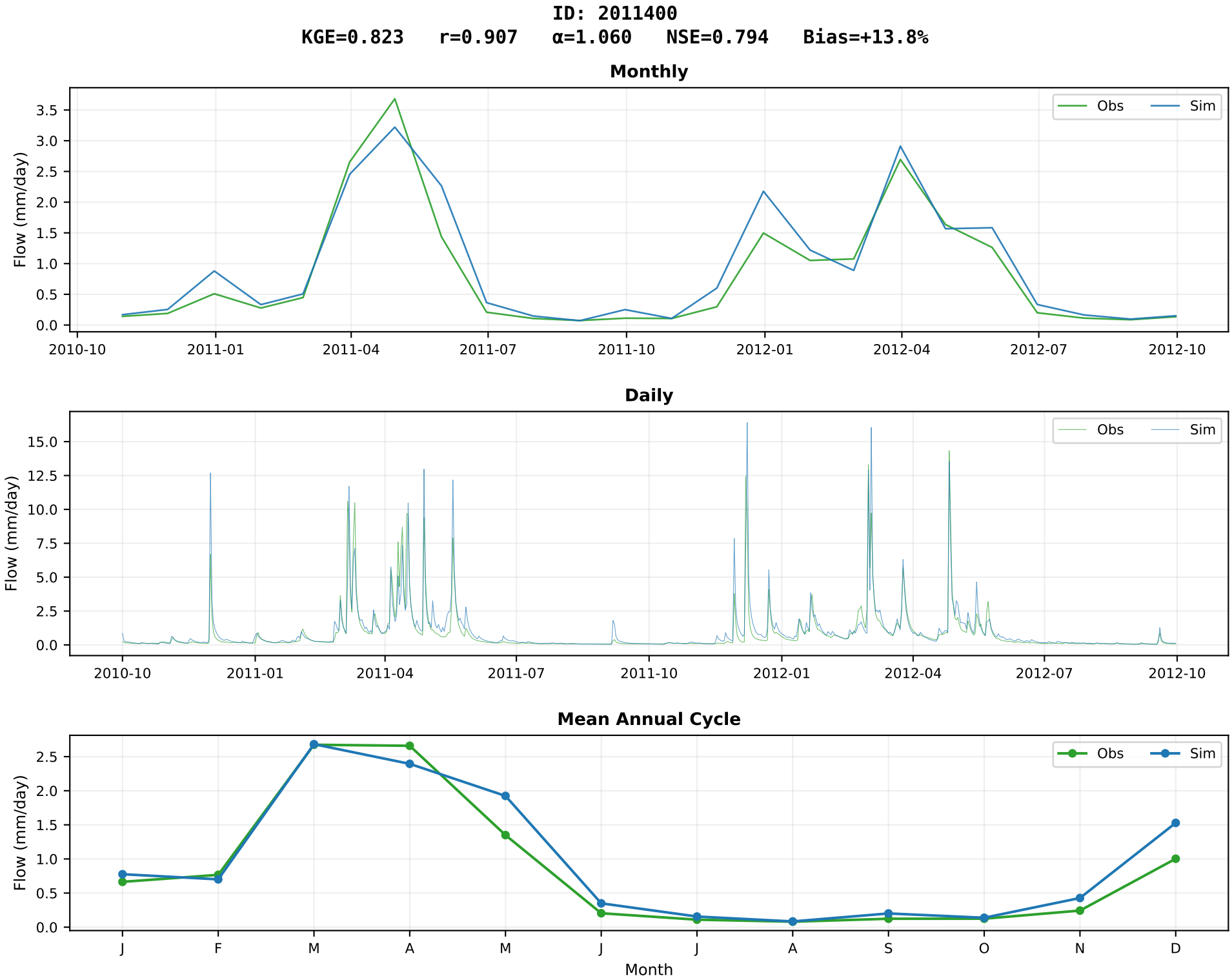

QUARTZVILLE CREEK NEAR CASCADIA, OREG. (ID=14185900)
KGE=0.830 r=0.944 α=0.865 NSE=0.881 Bias=-8.8%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

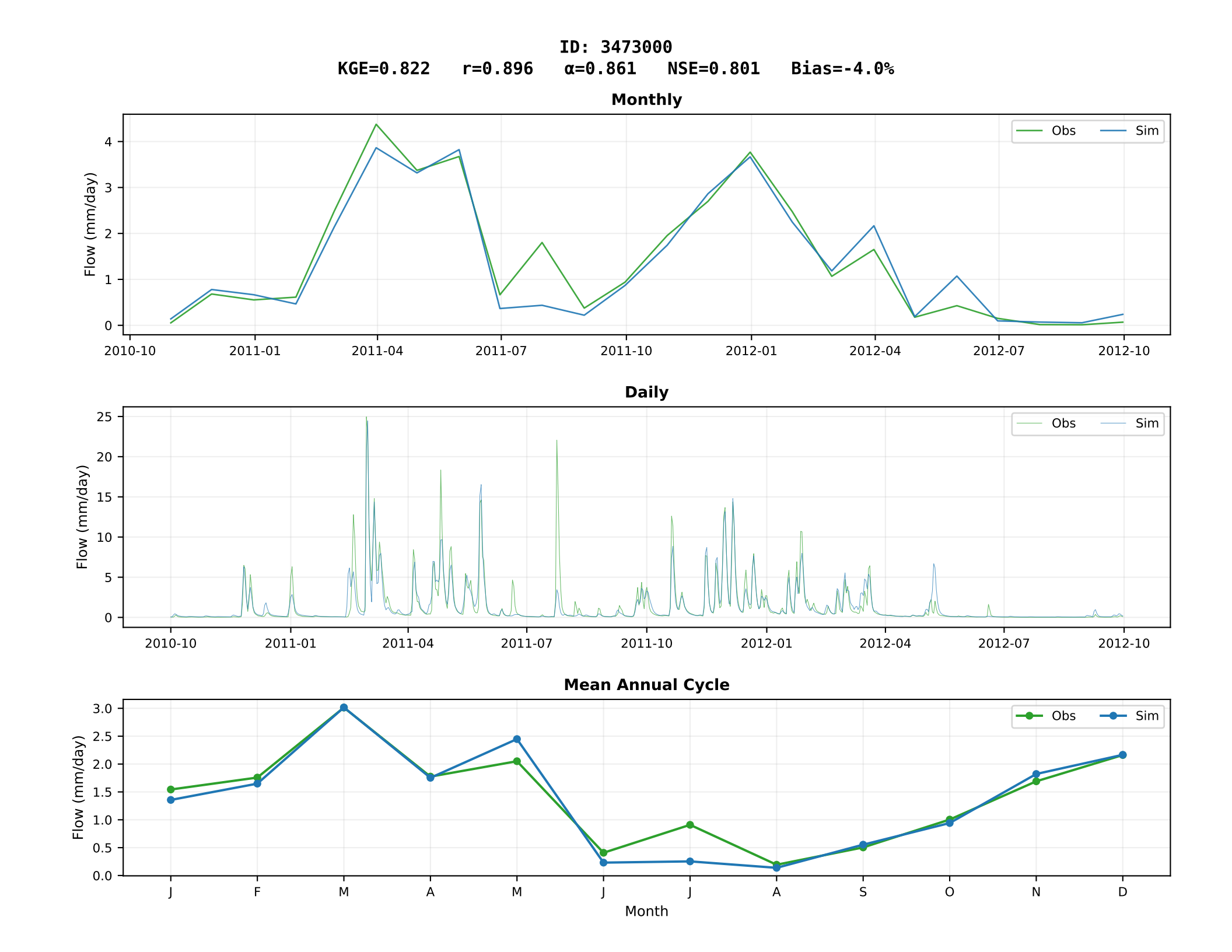

ID: 3070500
KGE=0.829 r=0.867 α=1.019 NSE=0.720 Bias=+10.5%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
12
14
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

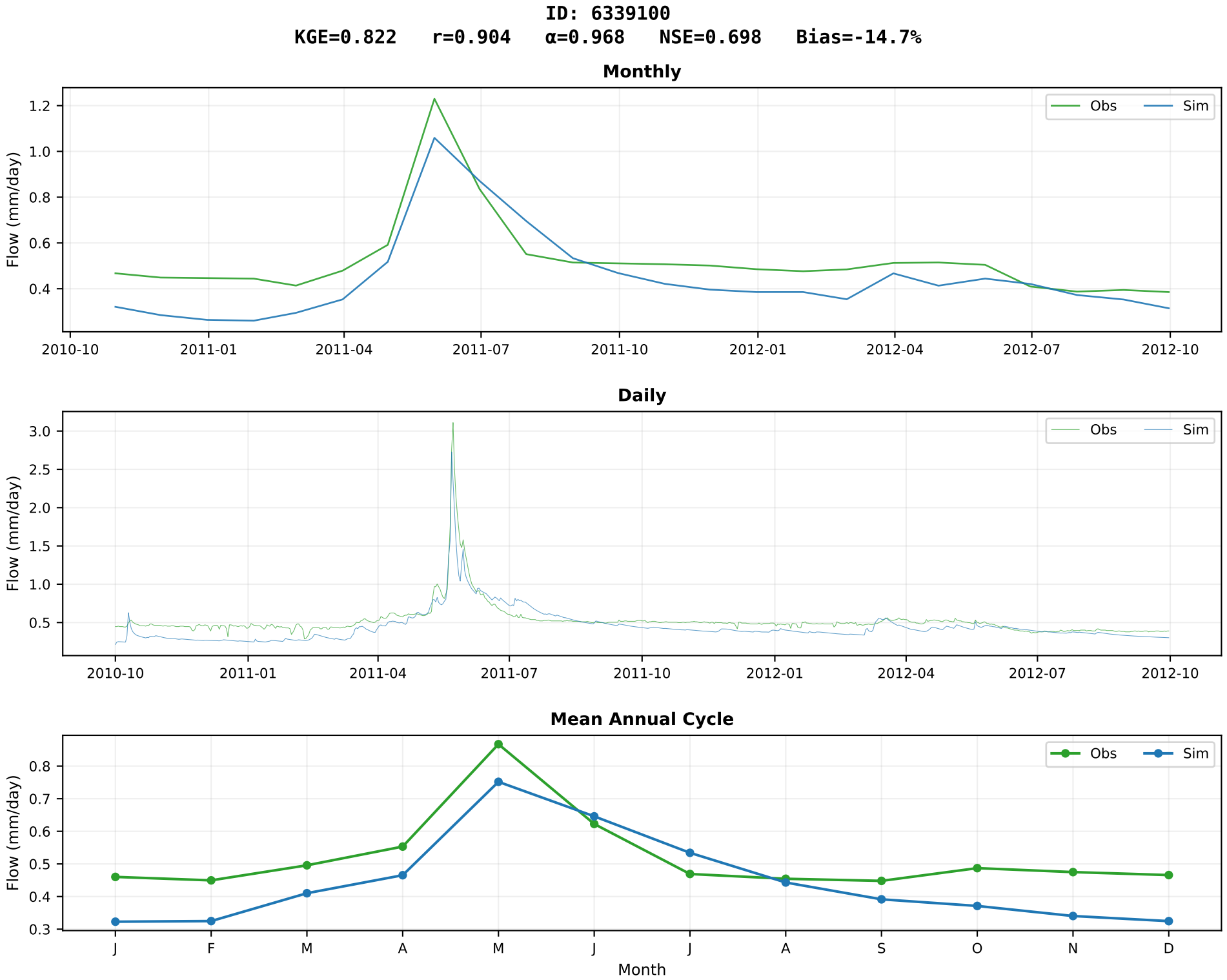

CEDAR RIVER NEAR CEDAR FALLS, WA (ID=12115000)
KGE=0.829 r=0.945 α=0.850 NSE=0.883 Bias=-6.1%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

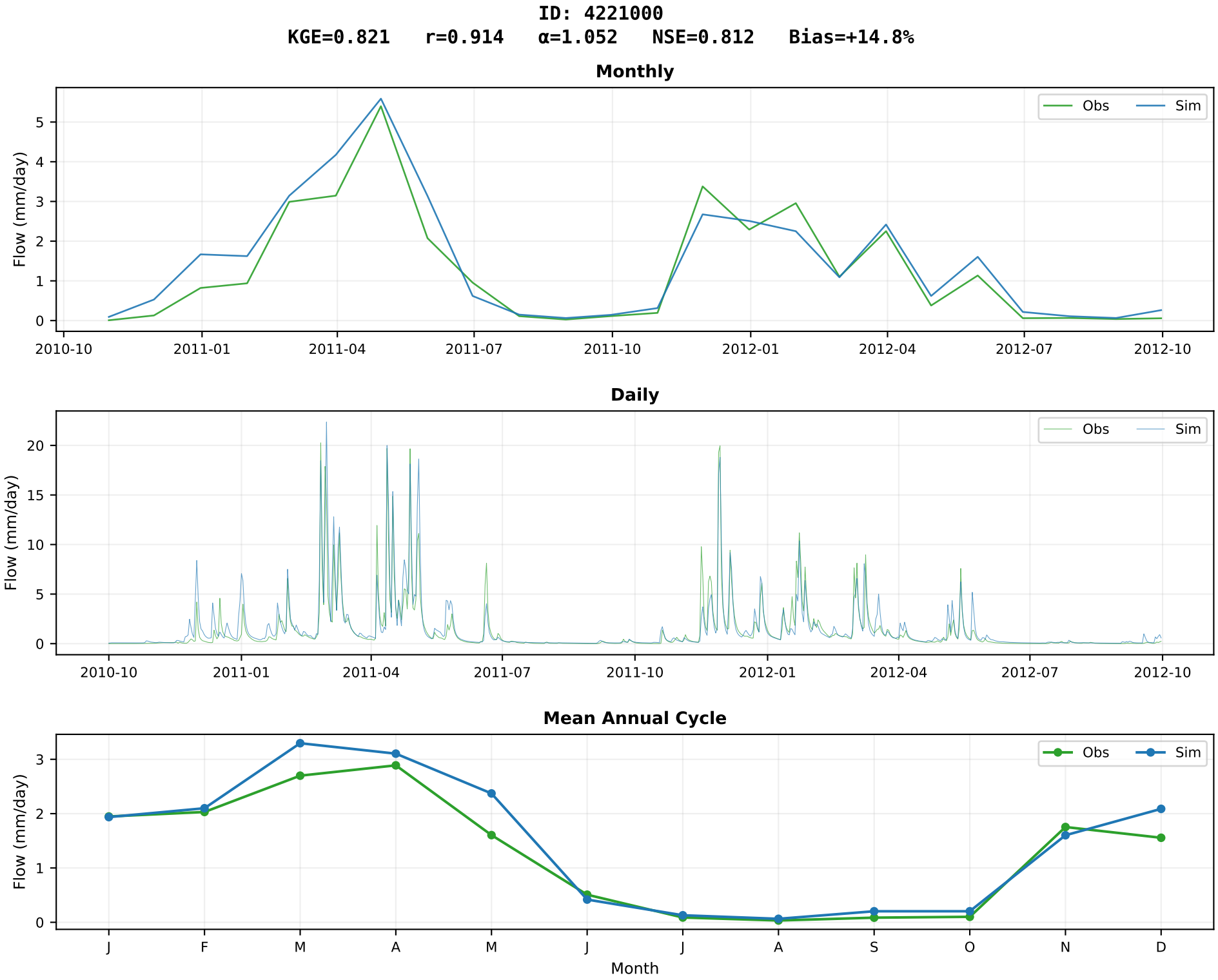

ID: 5414000
KGE=0.825 r=0.875 α=0.929 NSE=0.726 Bias=-9.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

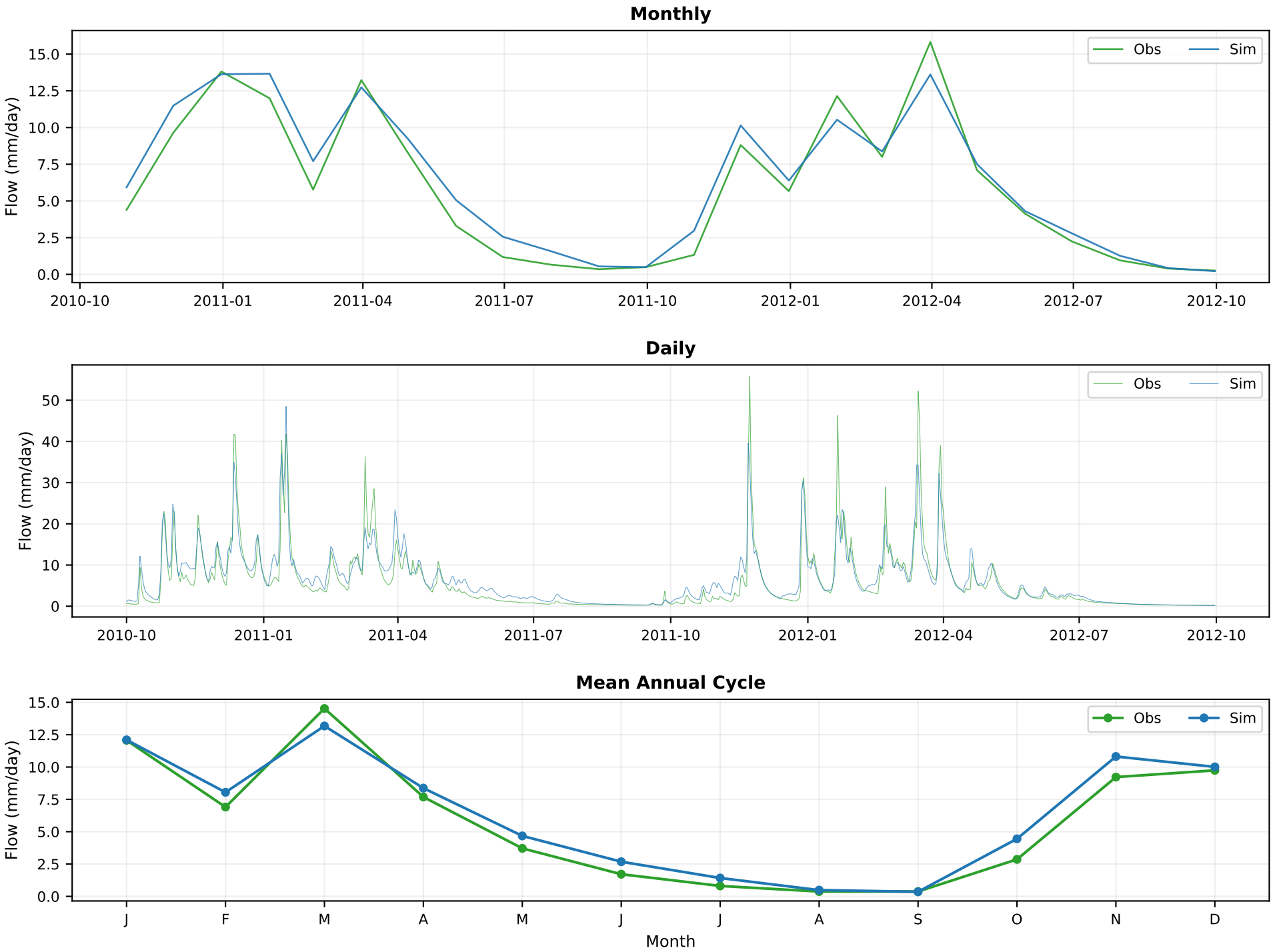

ID: 2082950
KGE=0.824 r=0.838 α=1.067 NSE=0.649 Bias=-1.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

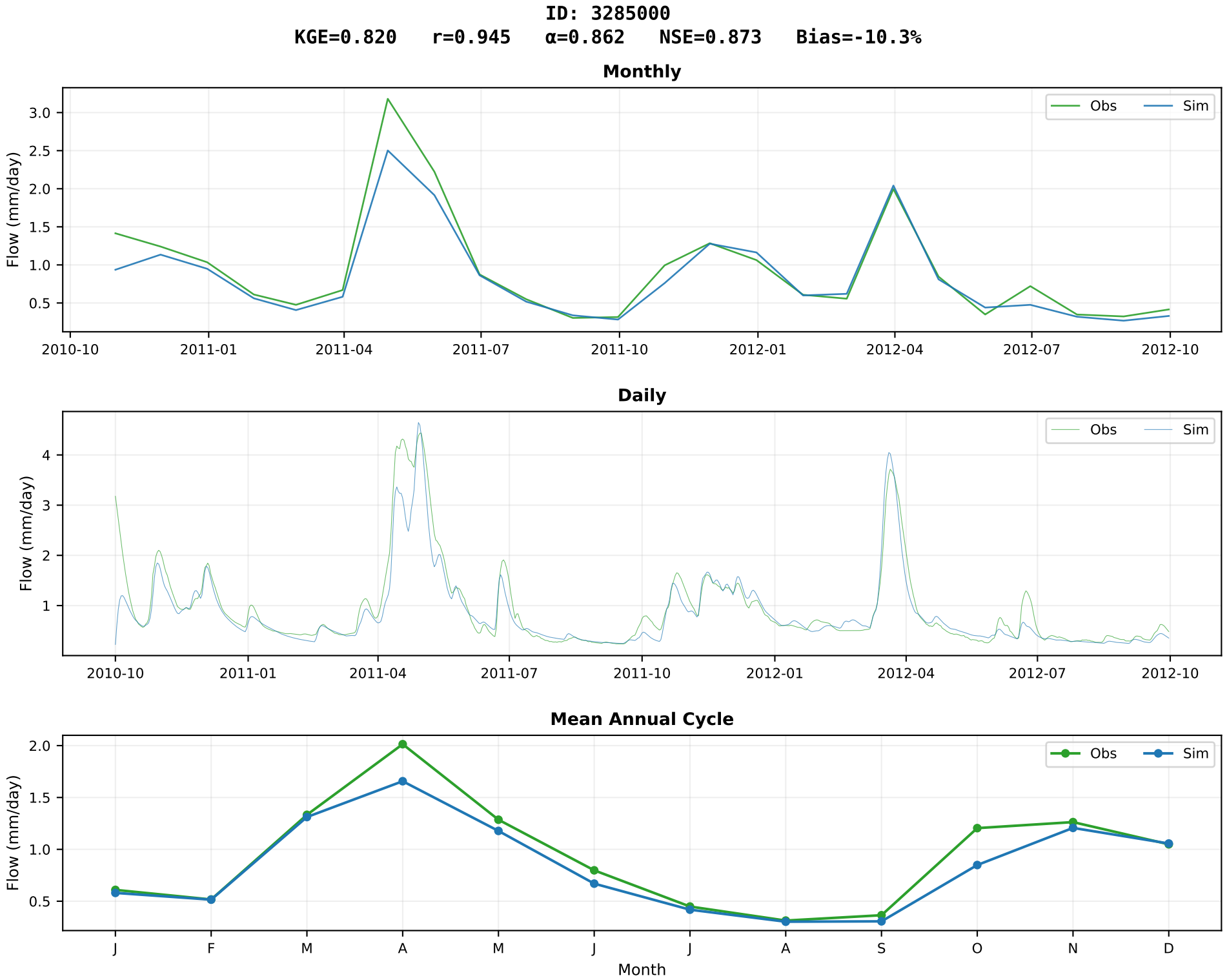

ID: 2011400
KGE=0.823 r=0.907 α=1.060 NSE=0.794 Bias=+13.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

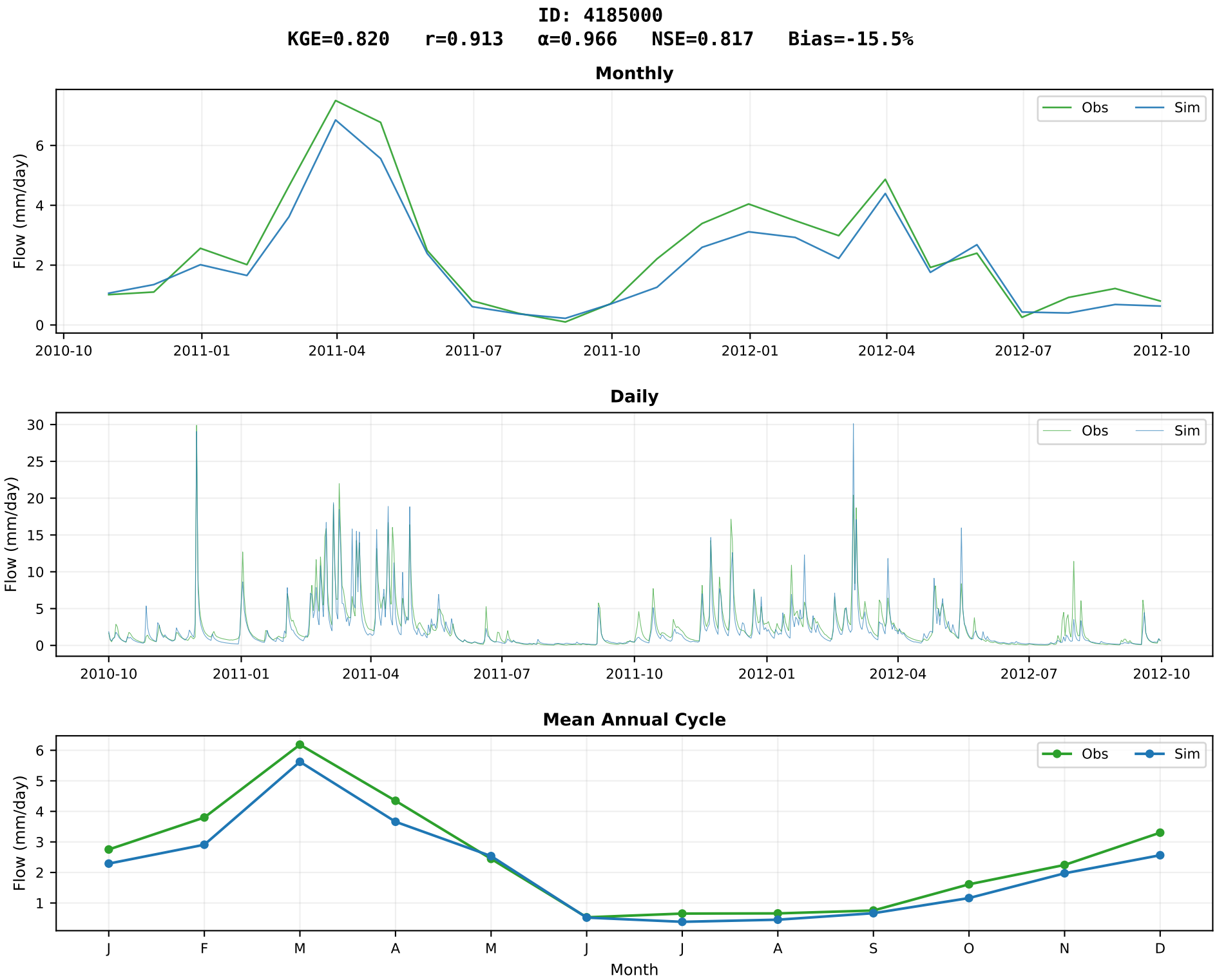

ID: 3473000
KGE=0.822 r=0.896 α=0.861 NSE=0.801 Bias=-4.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

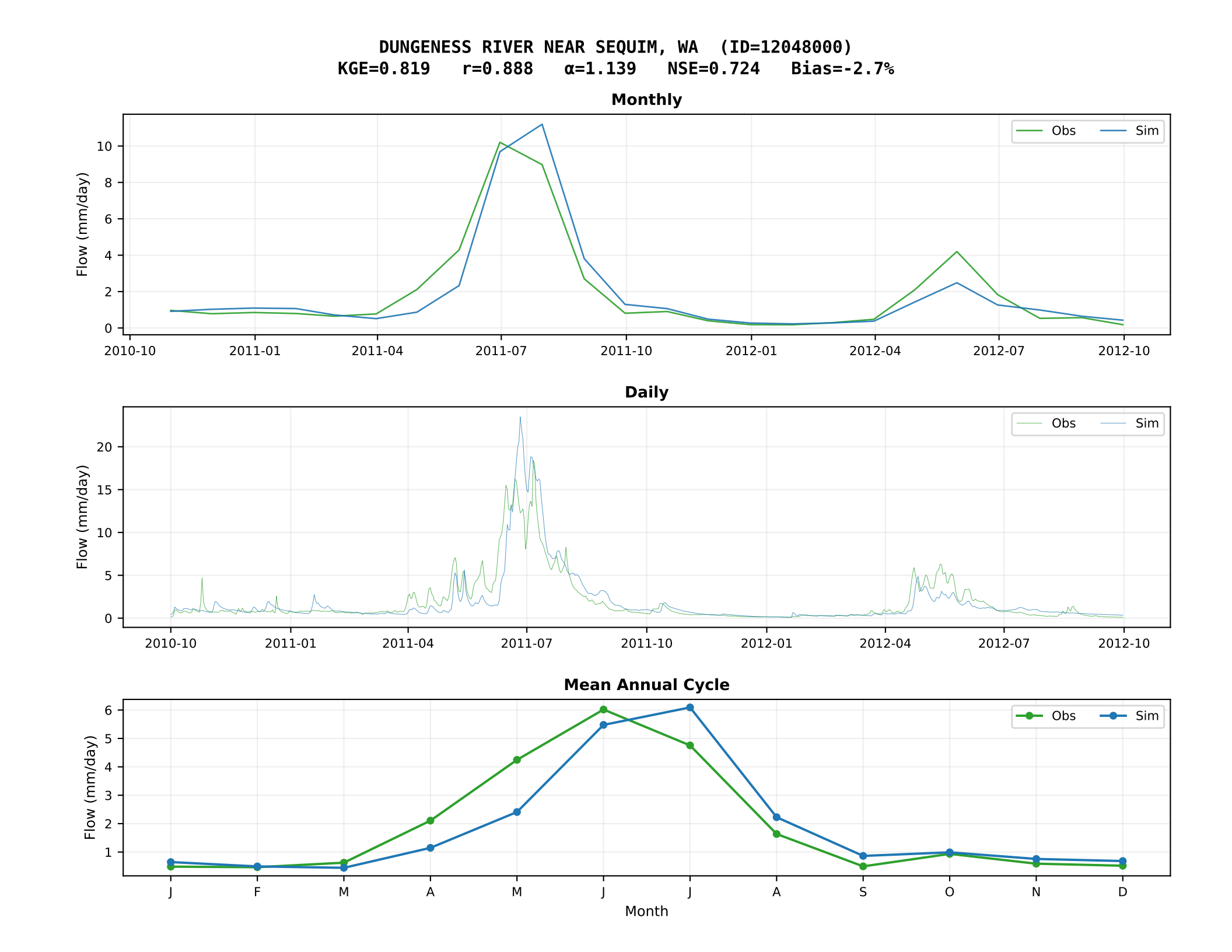

ID: 6339100
KGE=0.822 r=0.904 α=0.968 NSE=0.698 Bias=-14.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

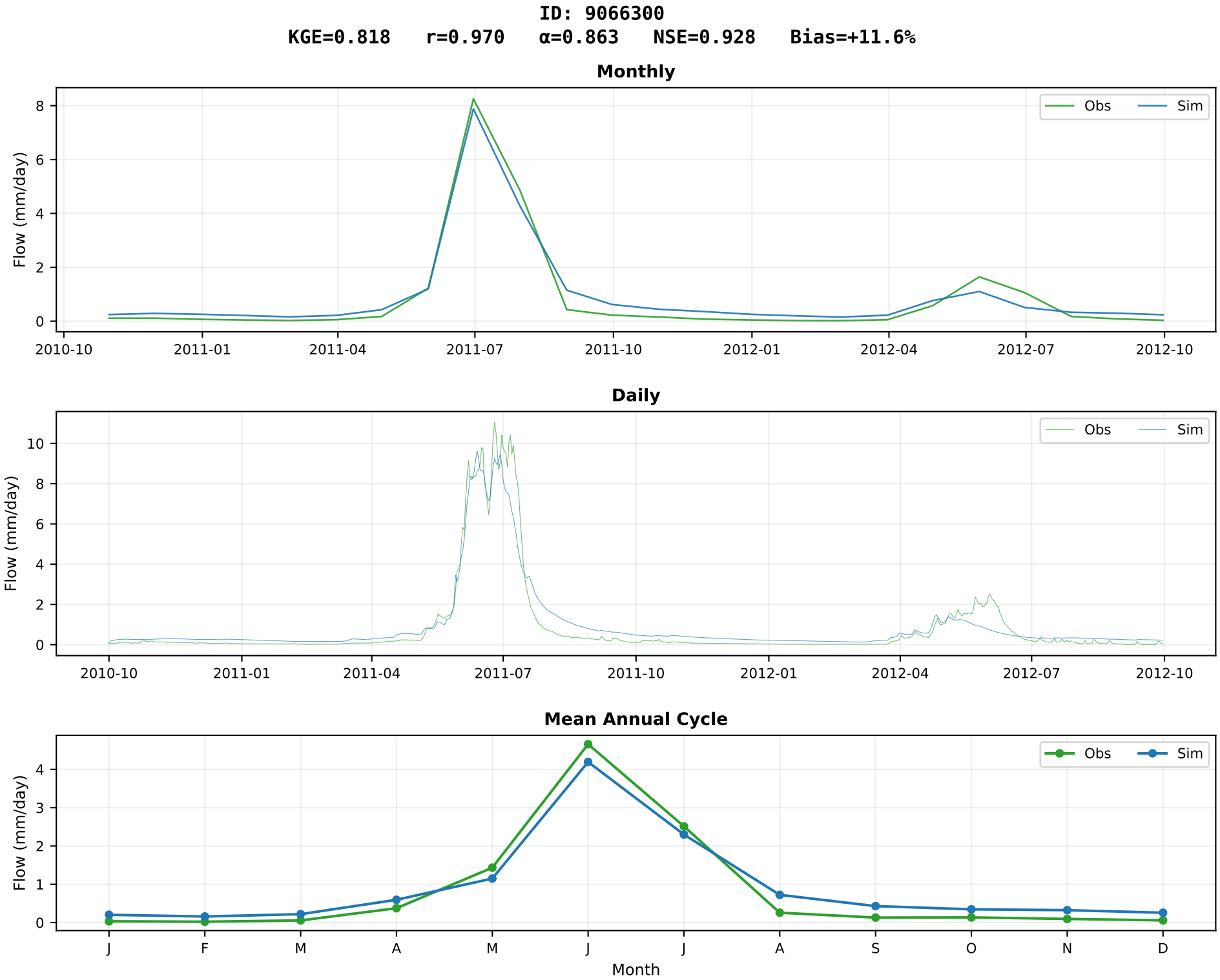

ID: 4221000
KGE=0.821 r=0.914 α=1.052 NSE=0.812 Bias=+14.8%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

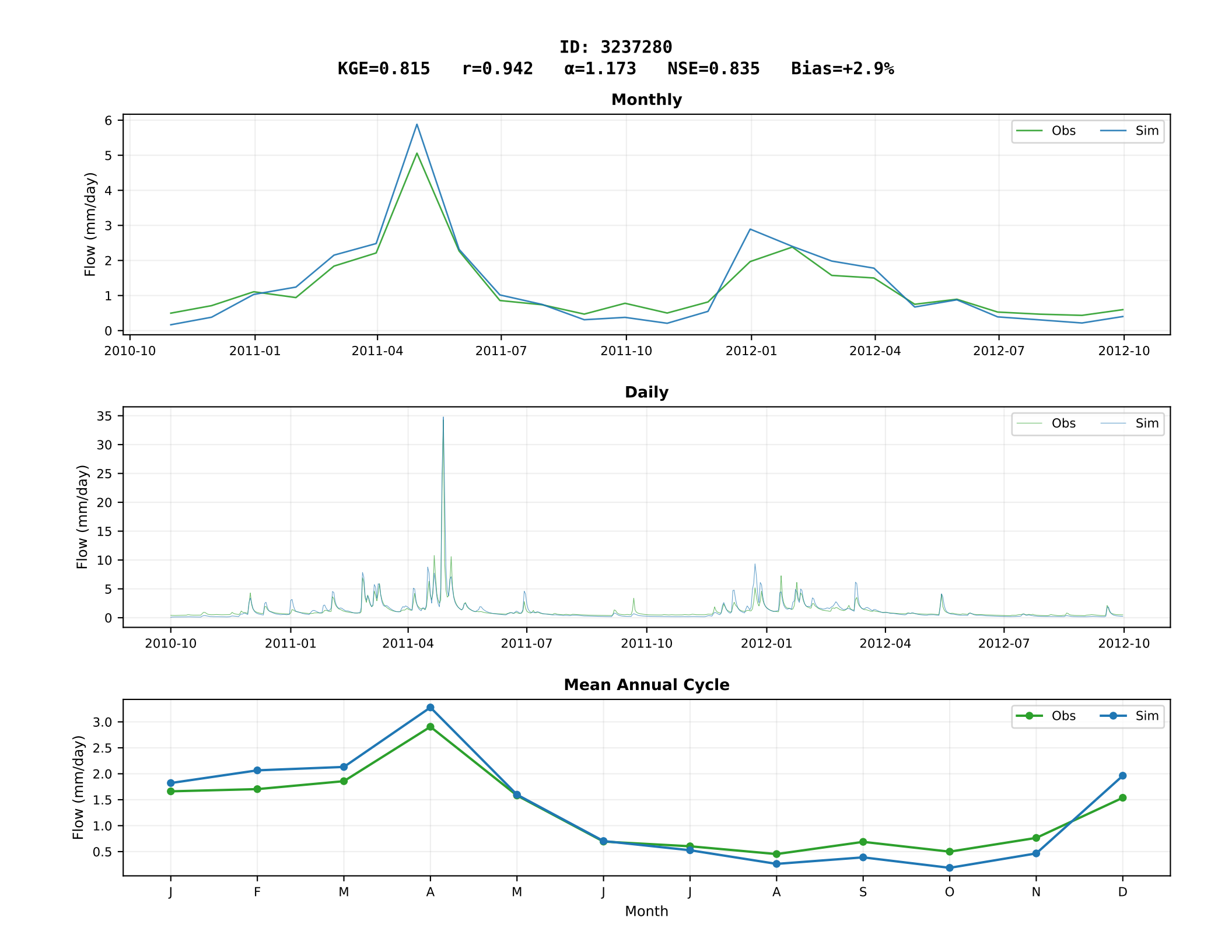

South Fork Jocko River near Arlee MT (ID=12381400)
KGE=0.821 r=0.924 α=0.868 NSE=0.845 Bias=+9.3%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

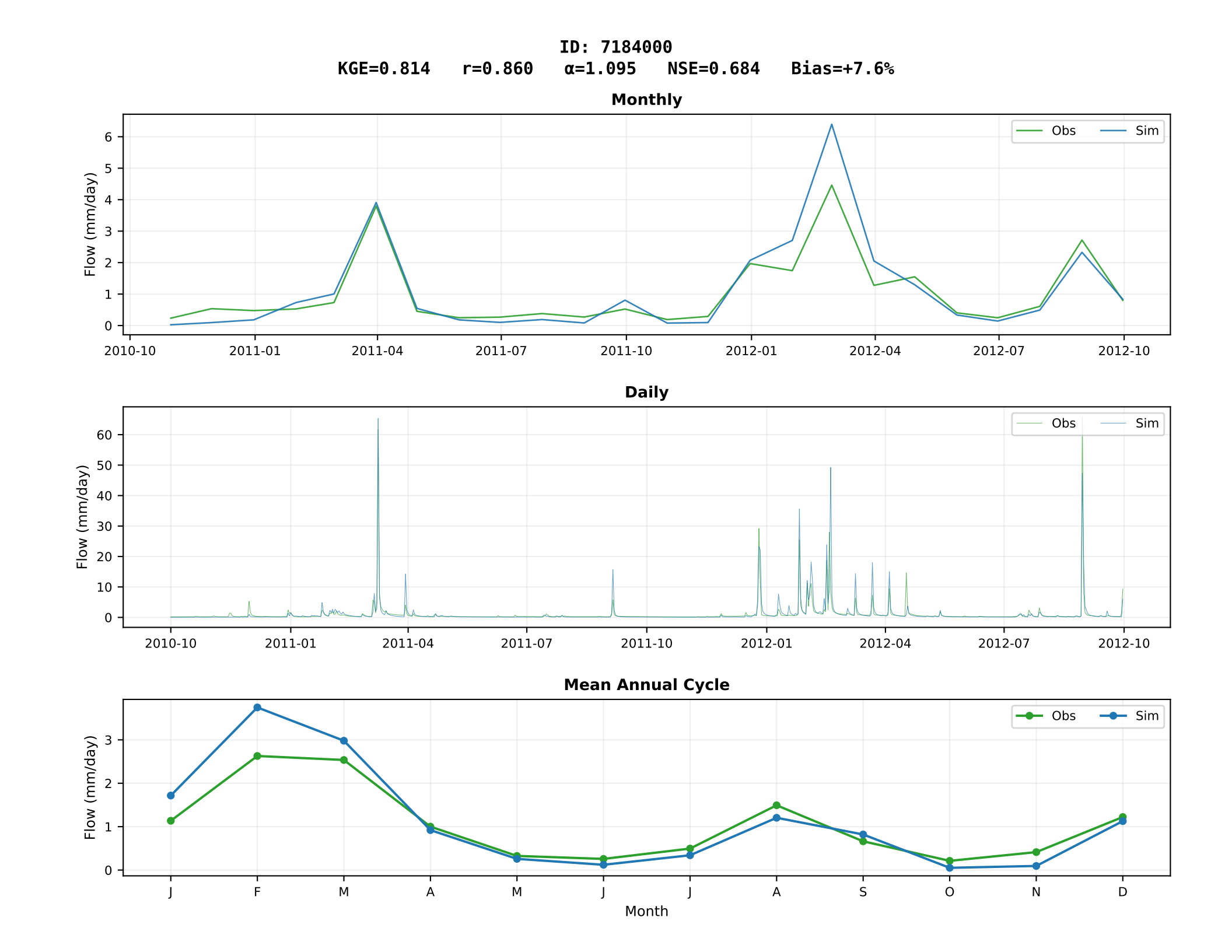

ID: 3285000
KGE=0.820 r=0.945 α=0.862 NSE=0.873 Bias=-10.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

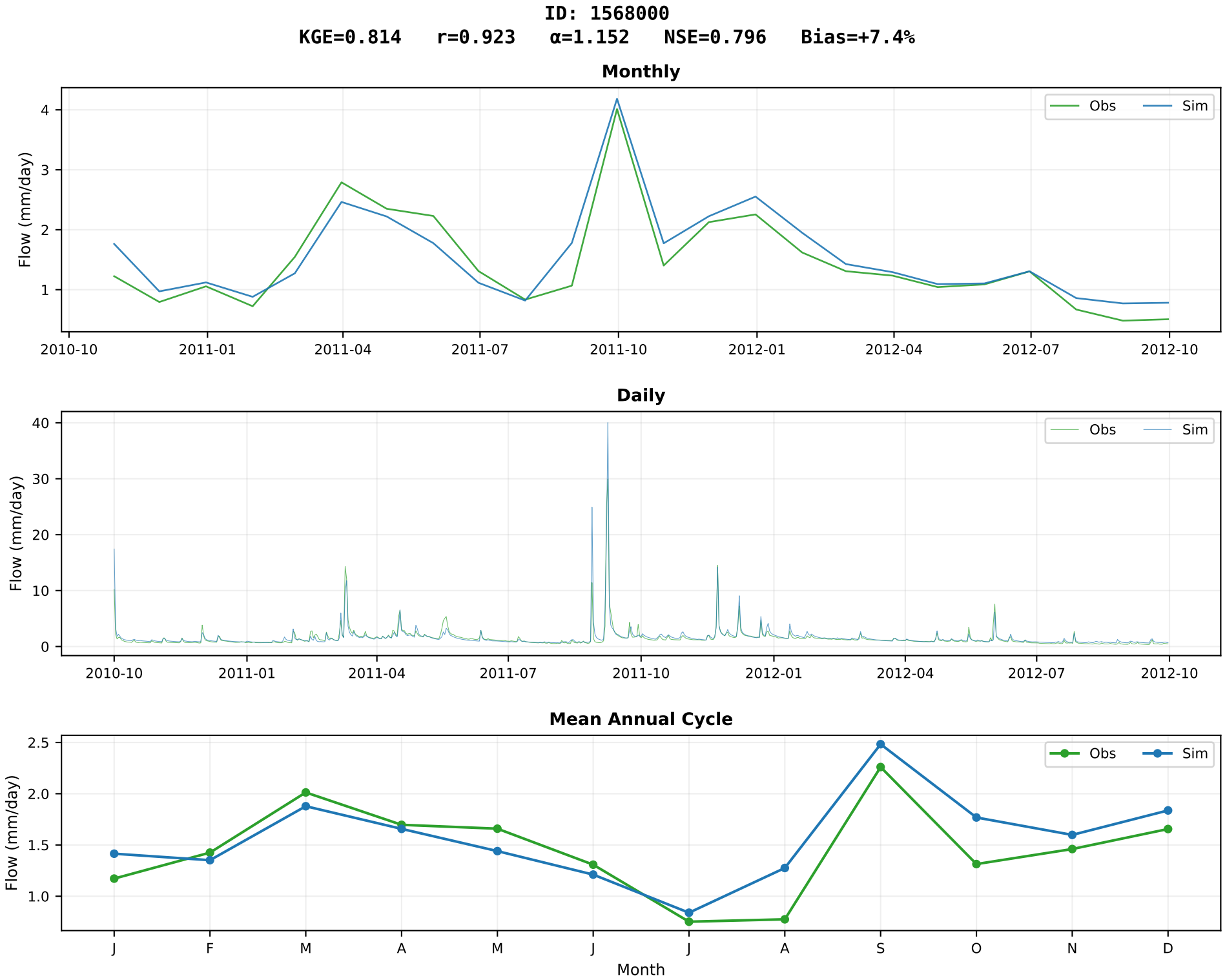

ID: 4185000
KGE=0.820 r=0.913 α=0.966 NSE=0.817 Bias=-15.5%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
5
6
J
F
M
A
M
J
J
A
S
O
N
D
Month

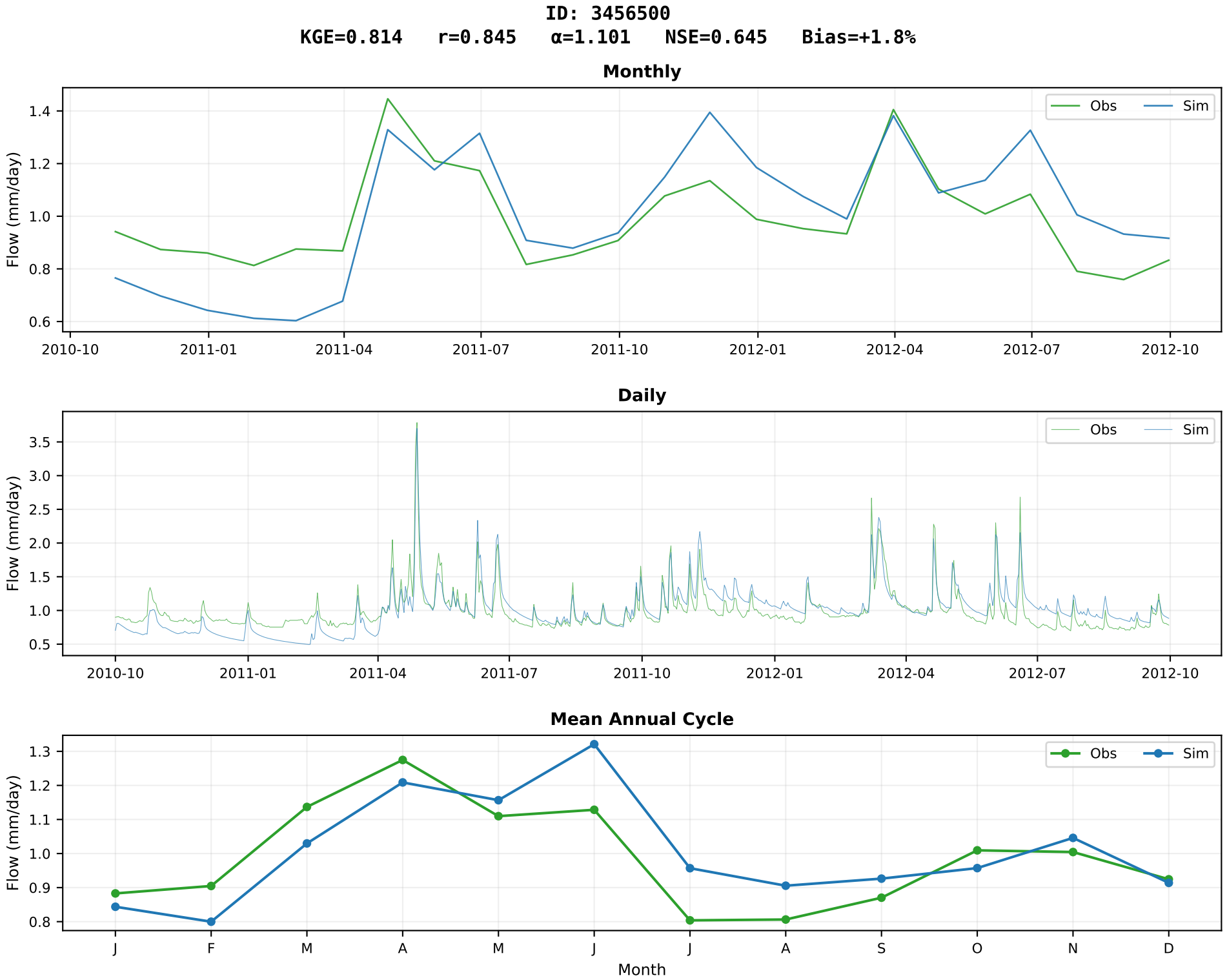

DUNGENESS RIVER NEAR SEQUIM, WA (ID=12048000)
KGE=0.819 r=0.888 α=1.139 NSE=0.724 Bias=-2.7%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

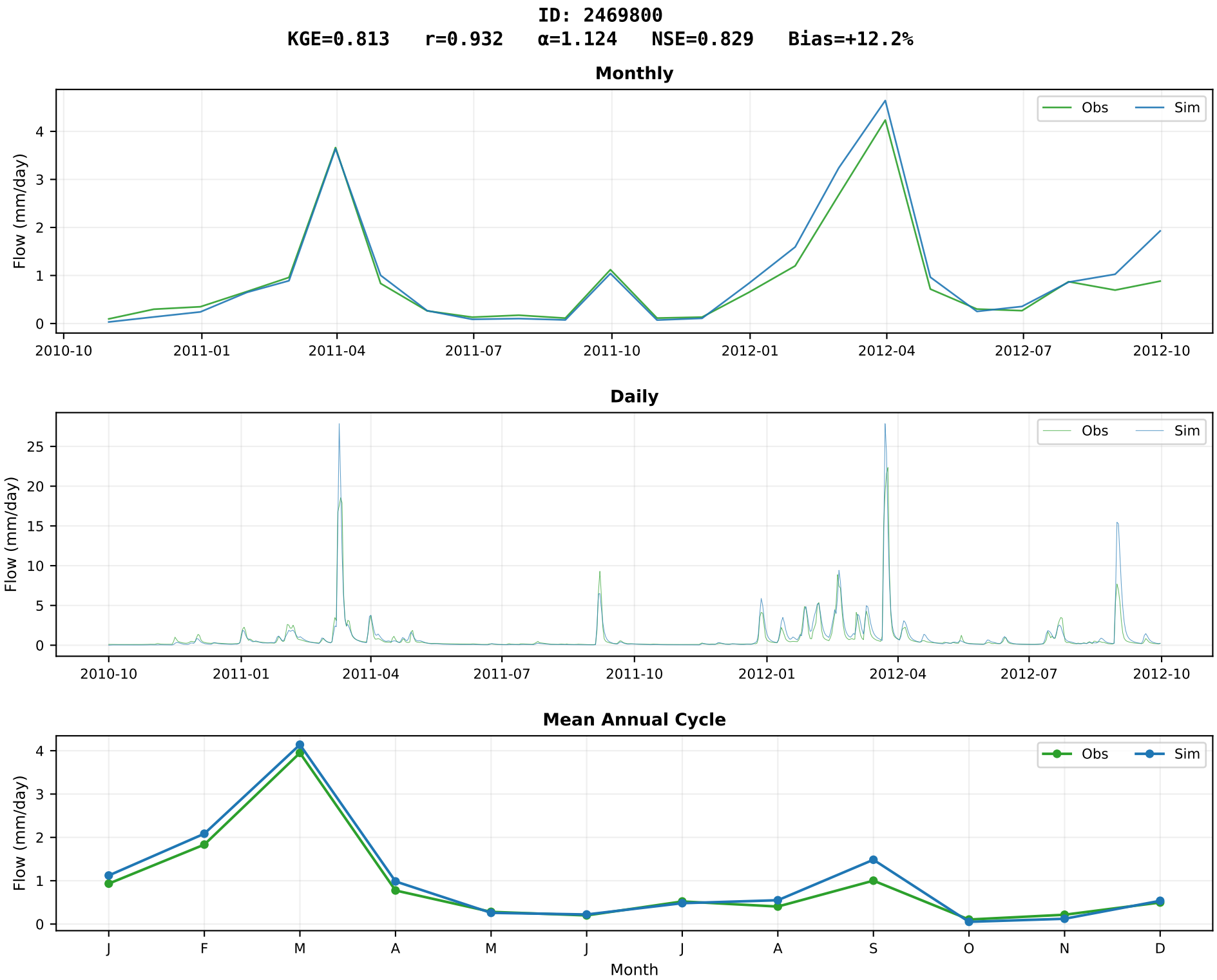

ID: 9066300
KGE=0.818 r=0.970 α=0.863 NSE=0.928 Bias=+11.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

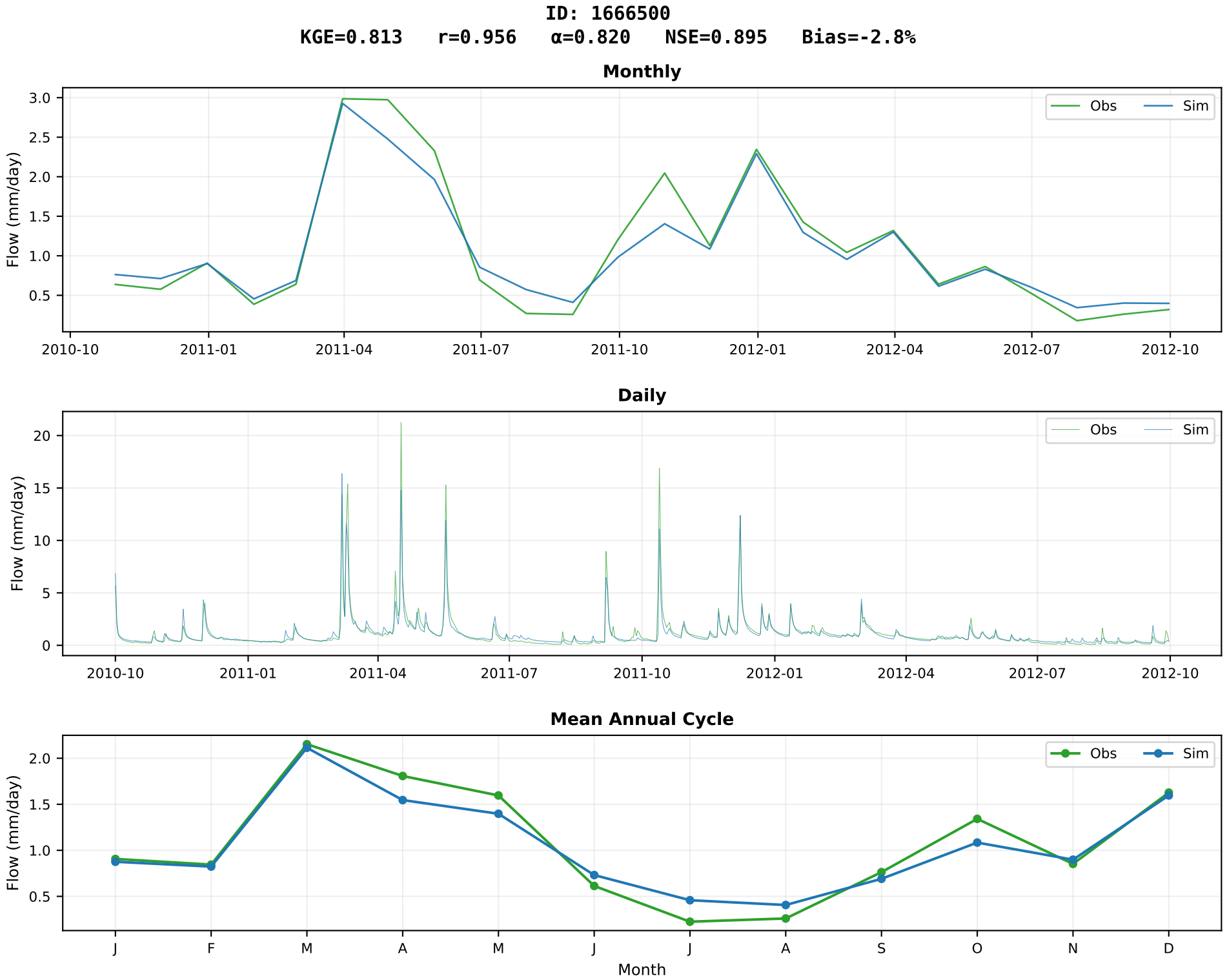

ID: 3237280
KGE=0.815 r=0.942 α=1.173 NSE=0.835 Bias=+2.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

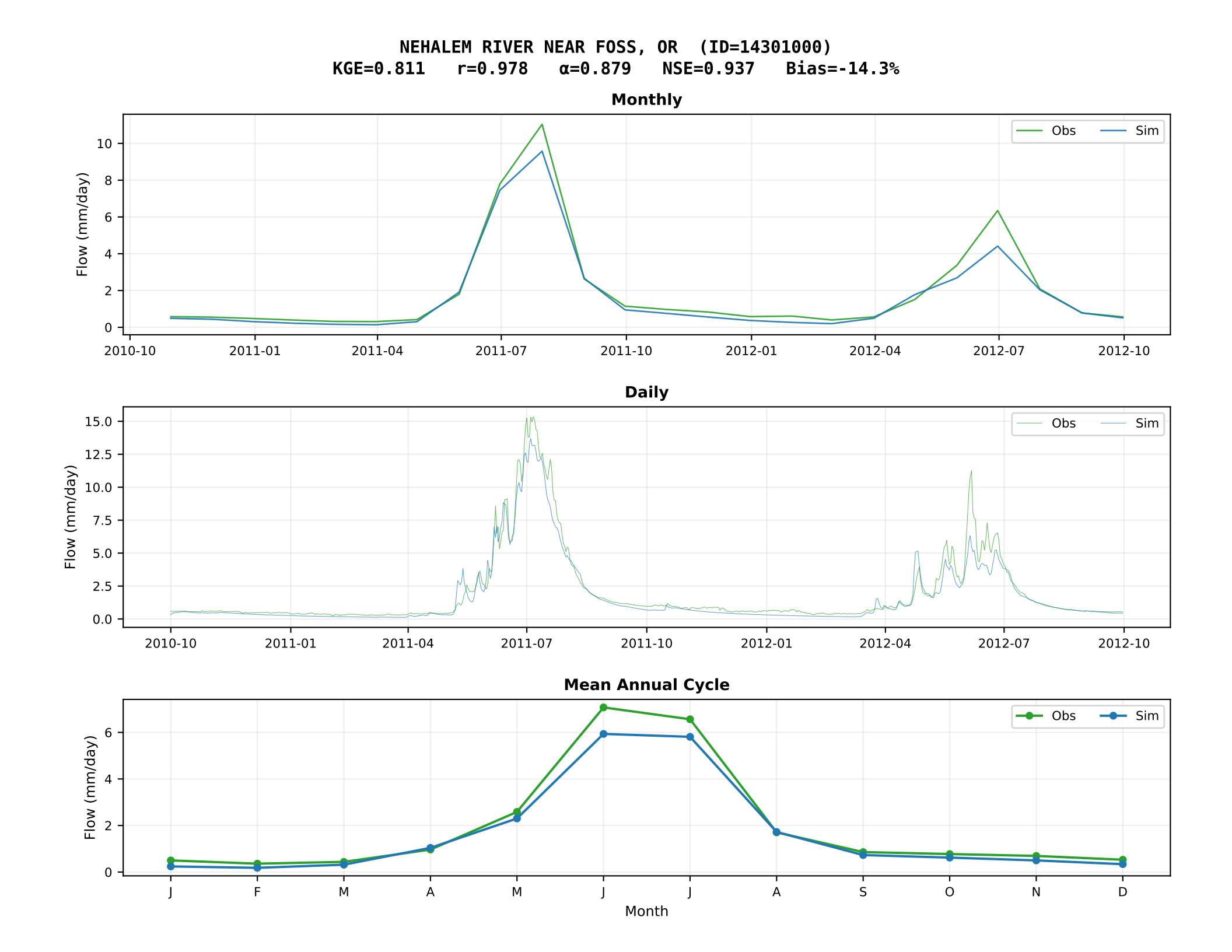

ID: 7184000
KGE=0.814 r=0.860 α=1.095 NSE=0.684 Bias=+7.6%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

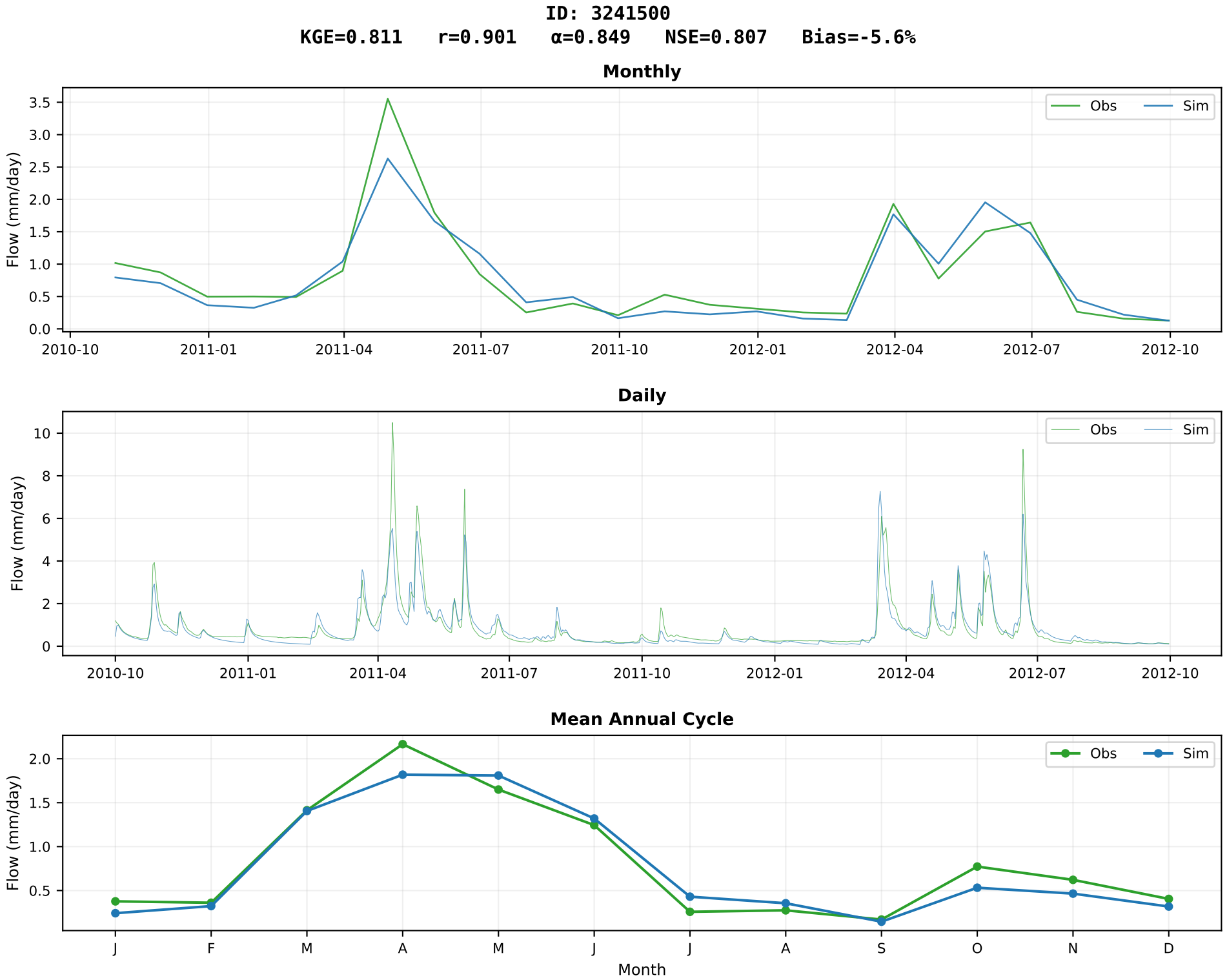

ID: 1568000
KGE=0.814 r=0.923 α=1.152 NSE=0.796 Bias=+7.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

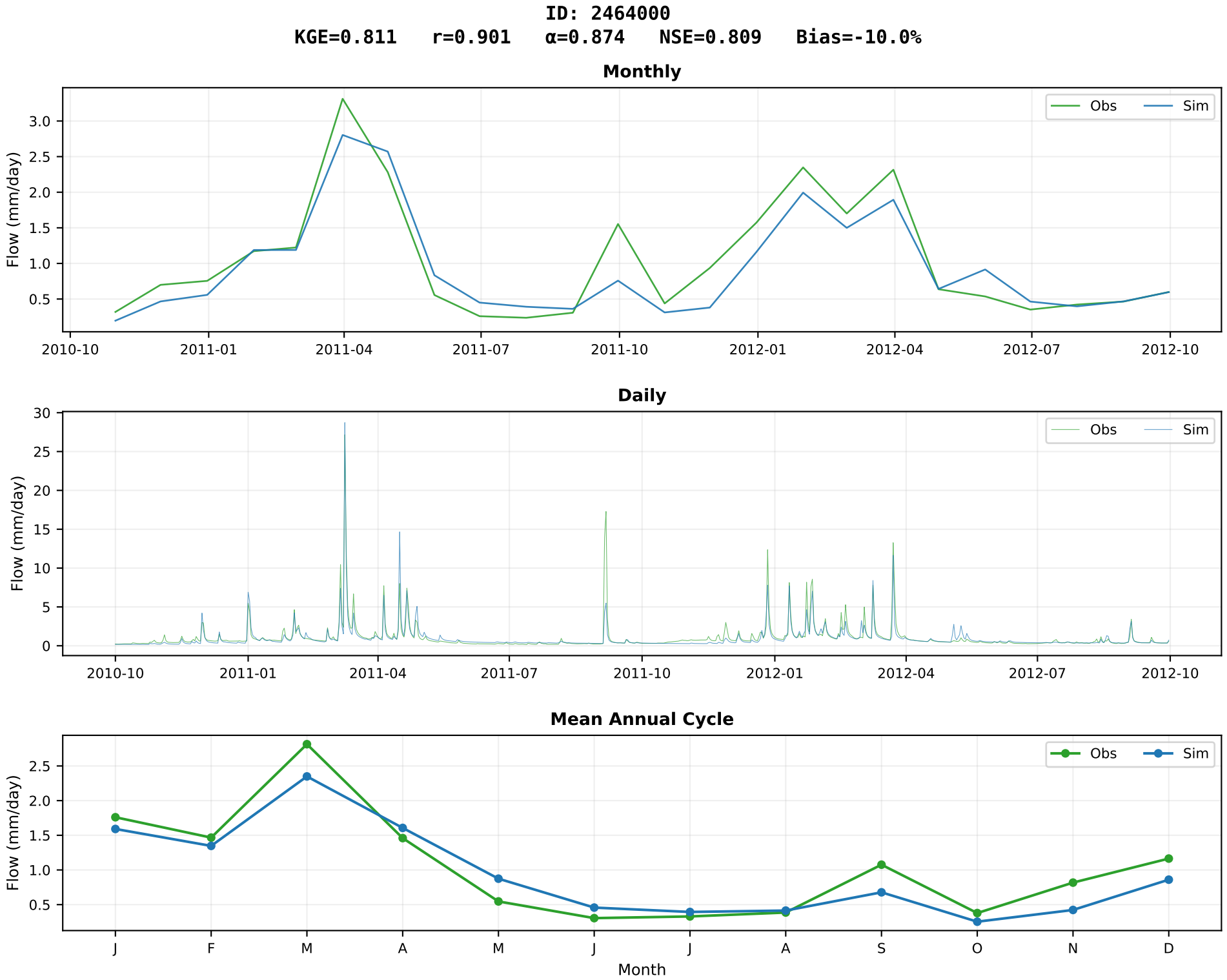

ID: 3456500
KGE=0.814 r=0.845 α=1.101 NSE=0.645 Bias=+1.8%
Monthly
Obs
Sim
Flow (mm/day)
1.4
1.2
1.0
0.8
0.6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.3
1.2
1.1
1.0
0.9
0.8
J
F
M
A
M
J
J
A
S
O
N
D
Month

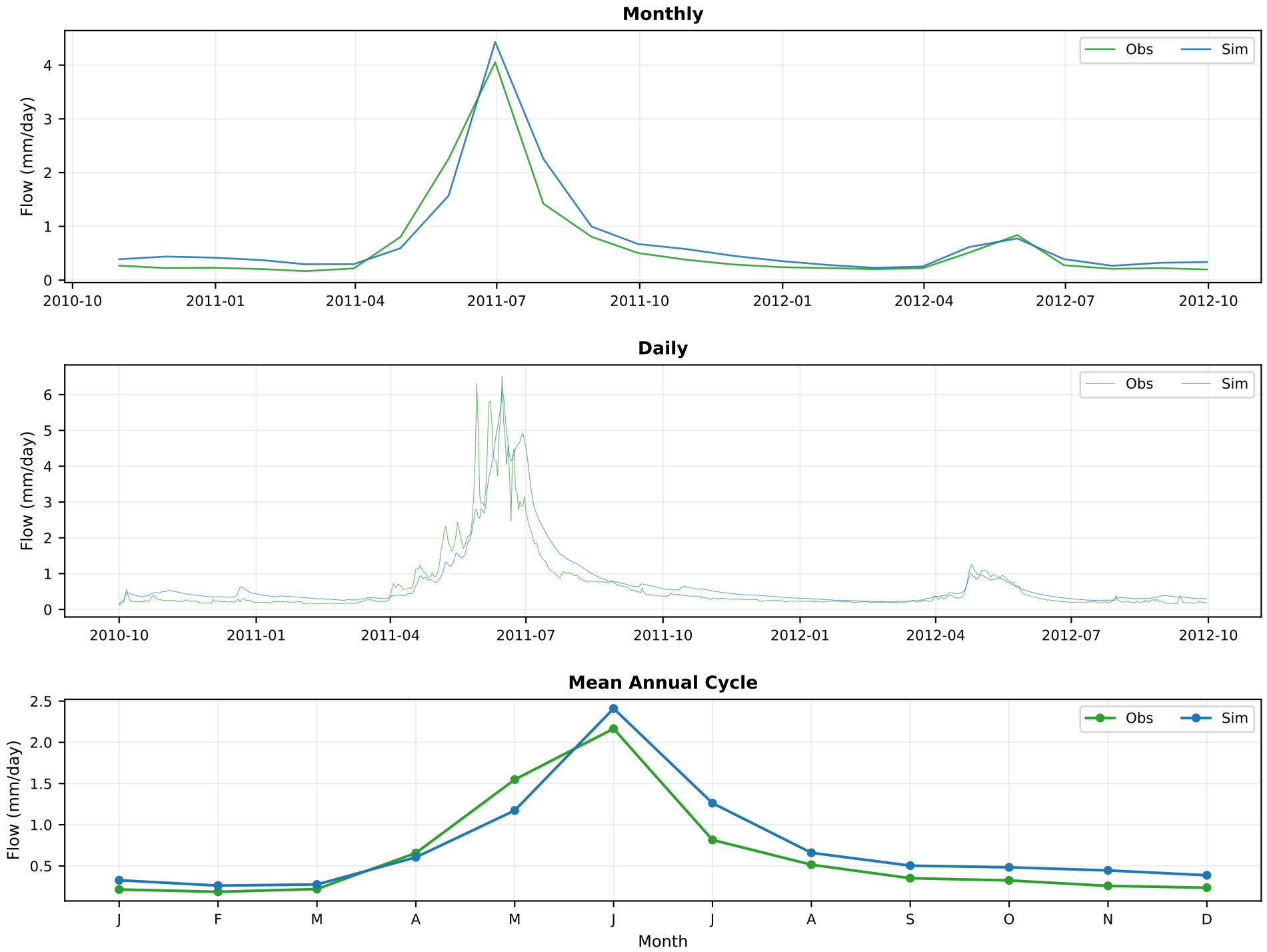

ID: 2469800
KGE=0.813 r=0.932 α=1.124 NSE=0.829 Bias=+12.2%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

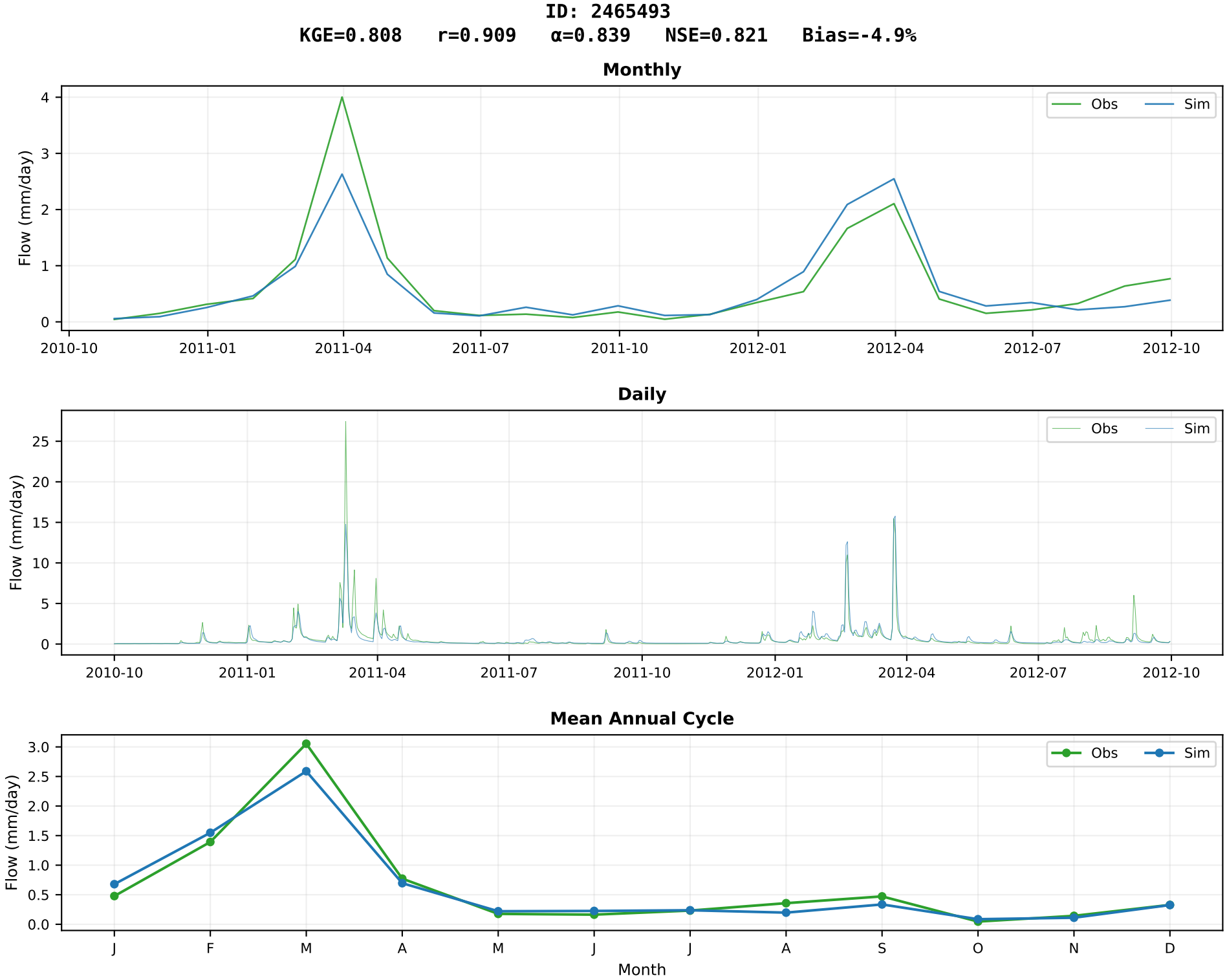

ID: 1666500
KGE=0.813 r=0.956 α=0.820 NSE=0.895 Bias=-2.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

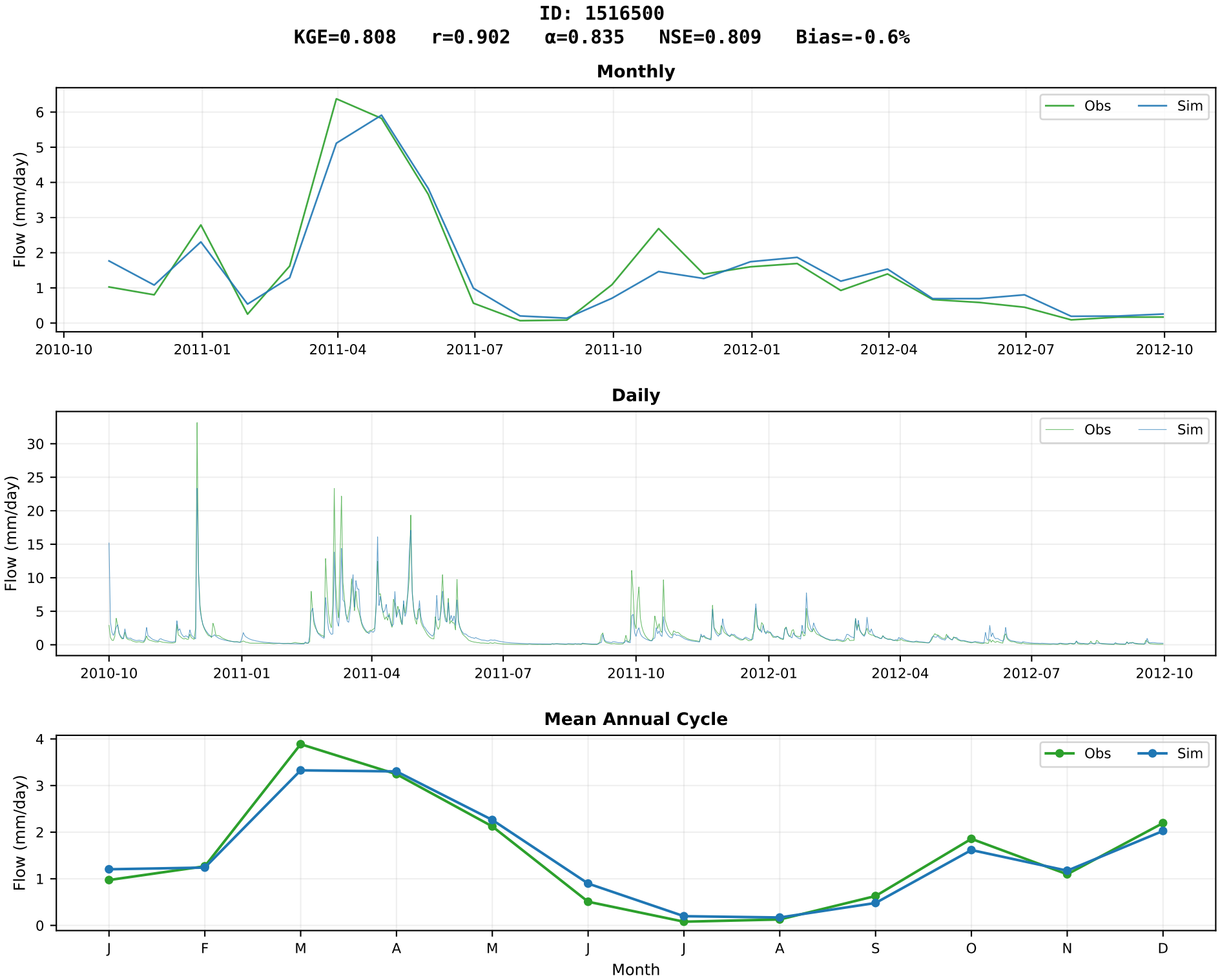

NEHALEM RIVER NEAR FOSS, OR (ID=14301000)
KGE=0.811 r=0.978 α=0.879 NSE=0.937 Bias=-14.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

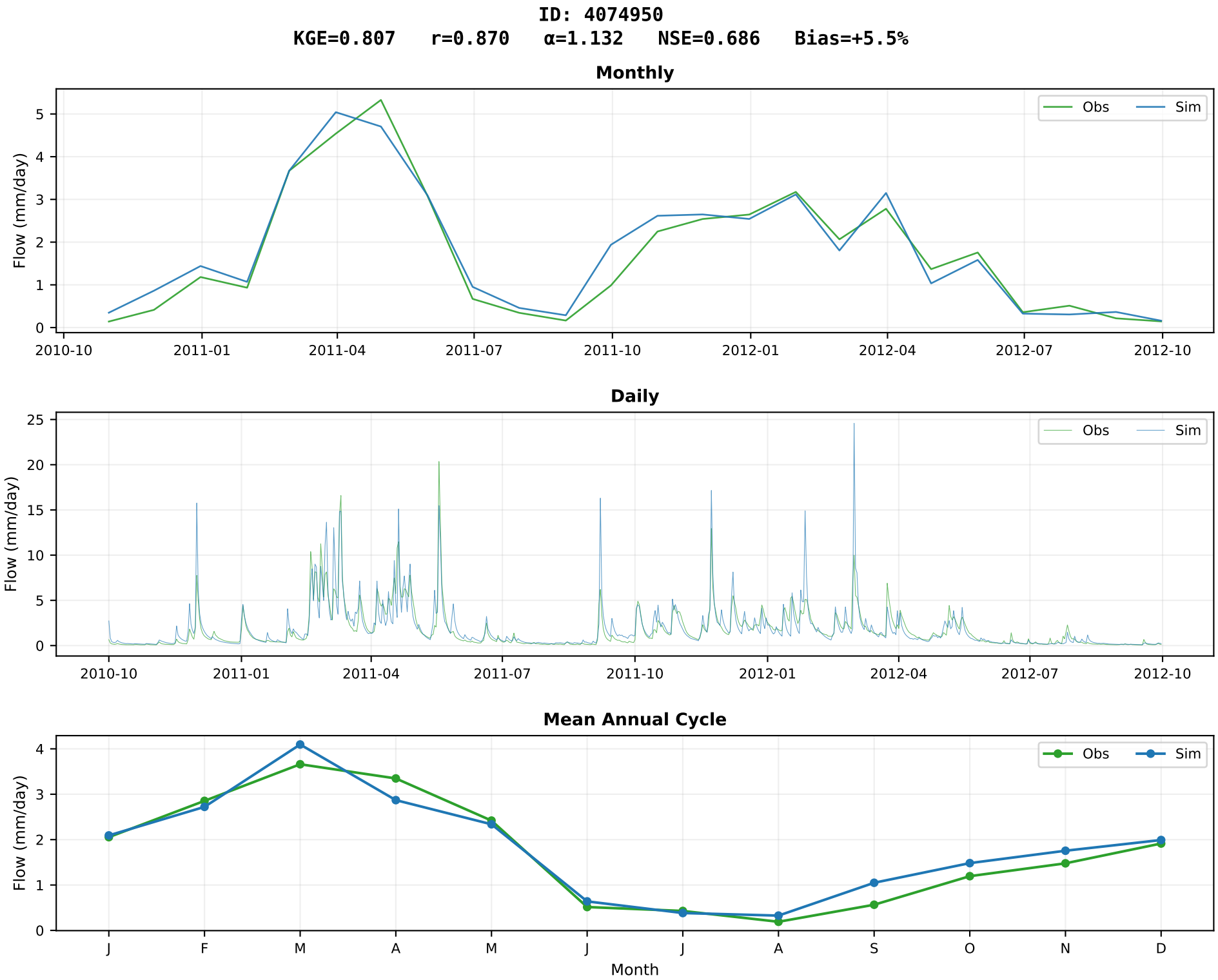

ID: 3241500
KGE=0.811 r=0.901 α=0.849 NSE=0.807 Bias=-5.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

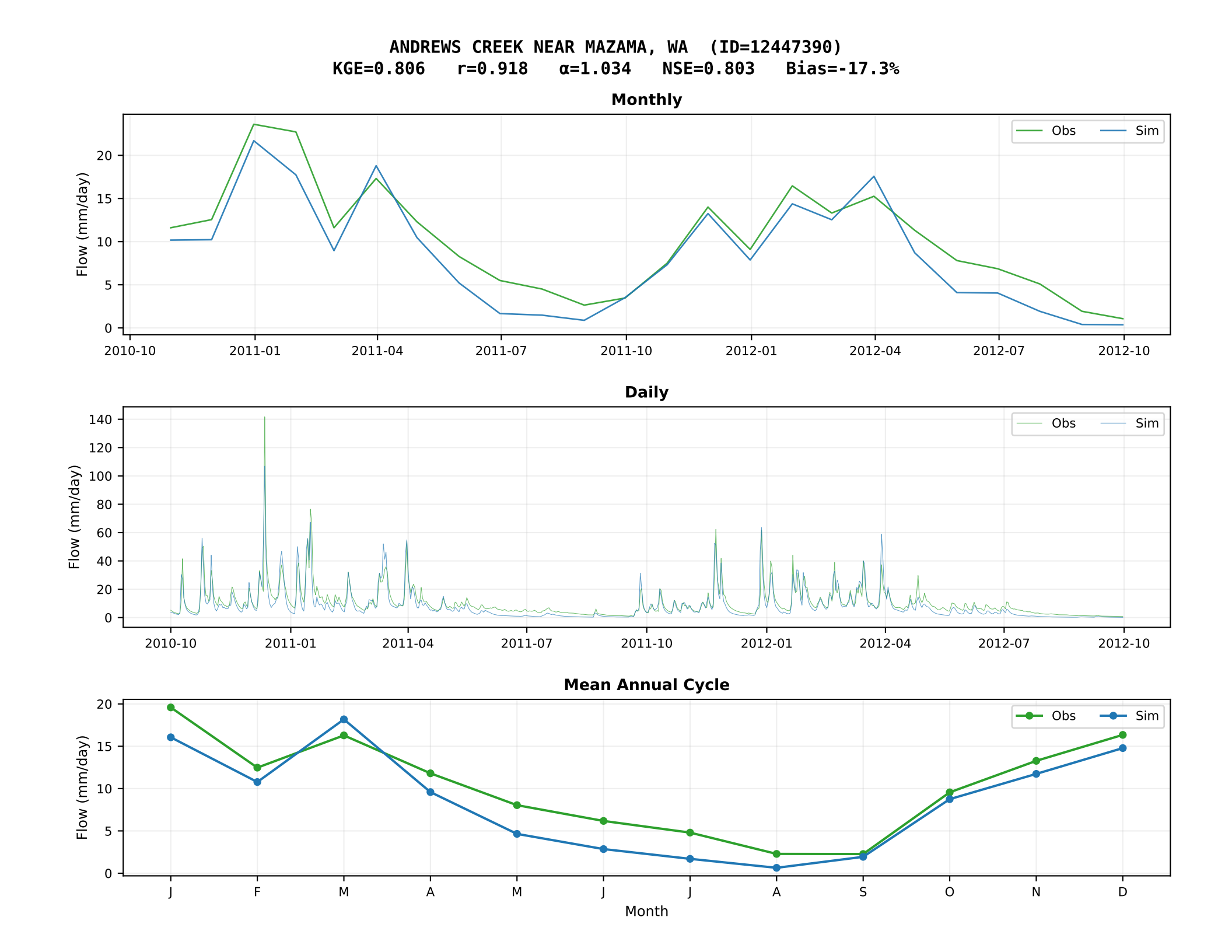

ID: 2464000
KGE=0.811 r=0.901 α=0.874 NSE=0.809 Bias=-10.0%
Monthly
Obs
Sim
Flow (mm/day)
3.0
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
30
25
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

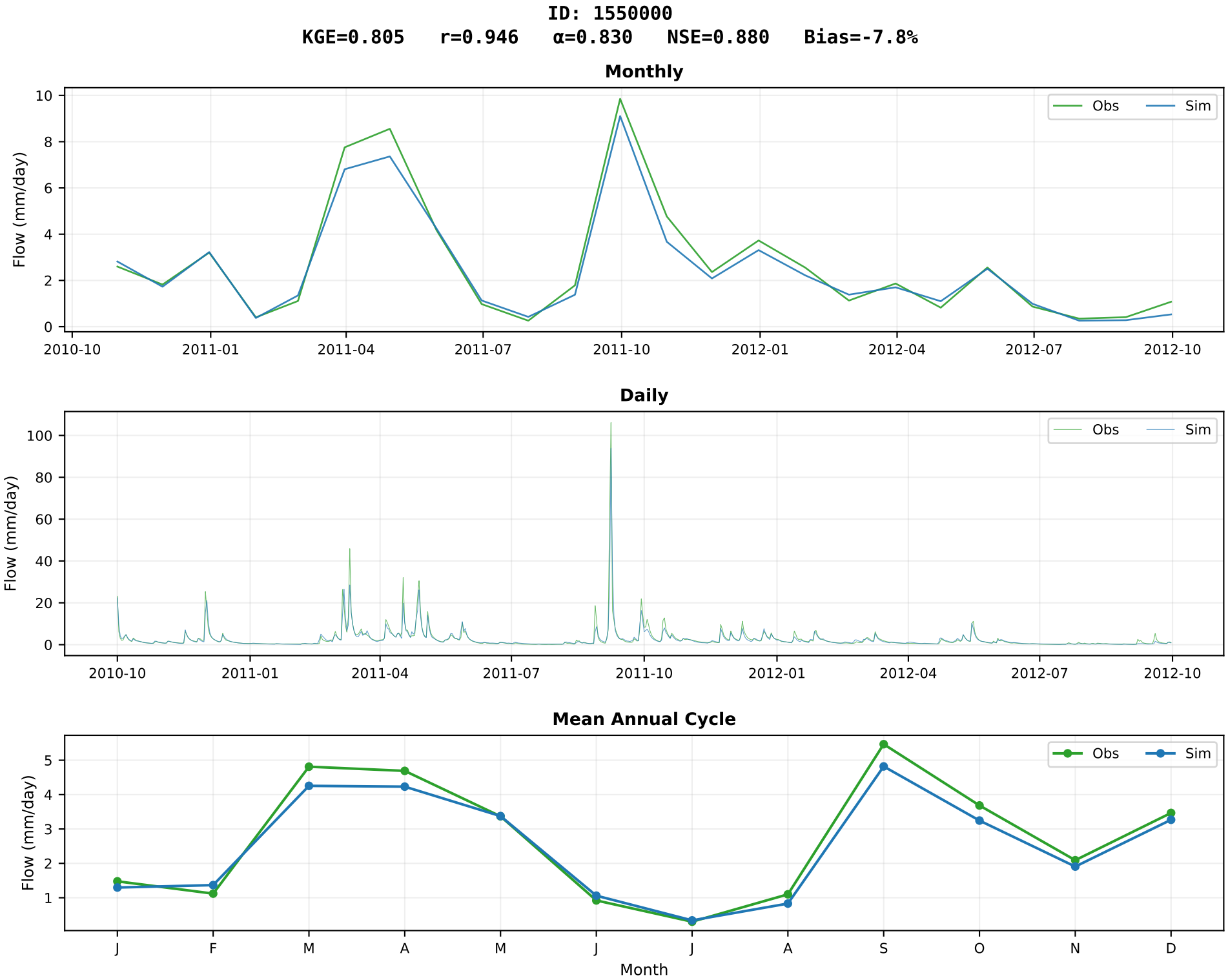

BEAVER RIVER NEAR BEAVER, UT (ID=10234500)
KGE=0.809 r=0.922 α=1.019 NSE=0.826 Bias=+17.4%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.5
1.0
1.5
2.0
2.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

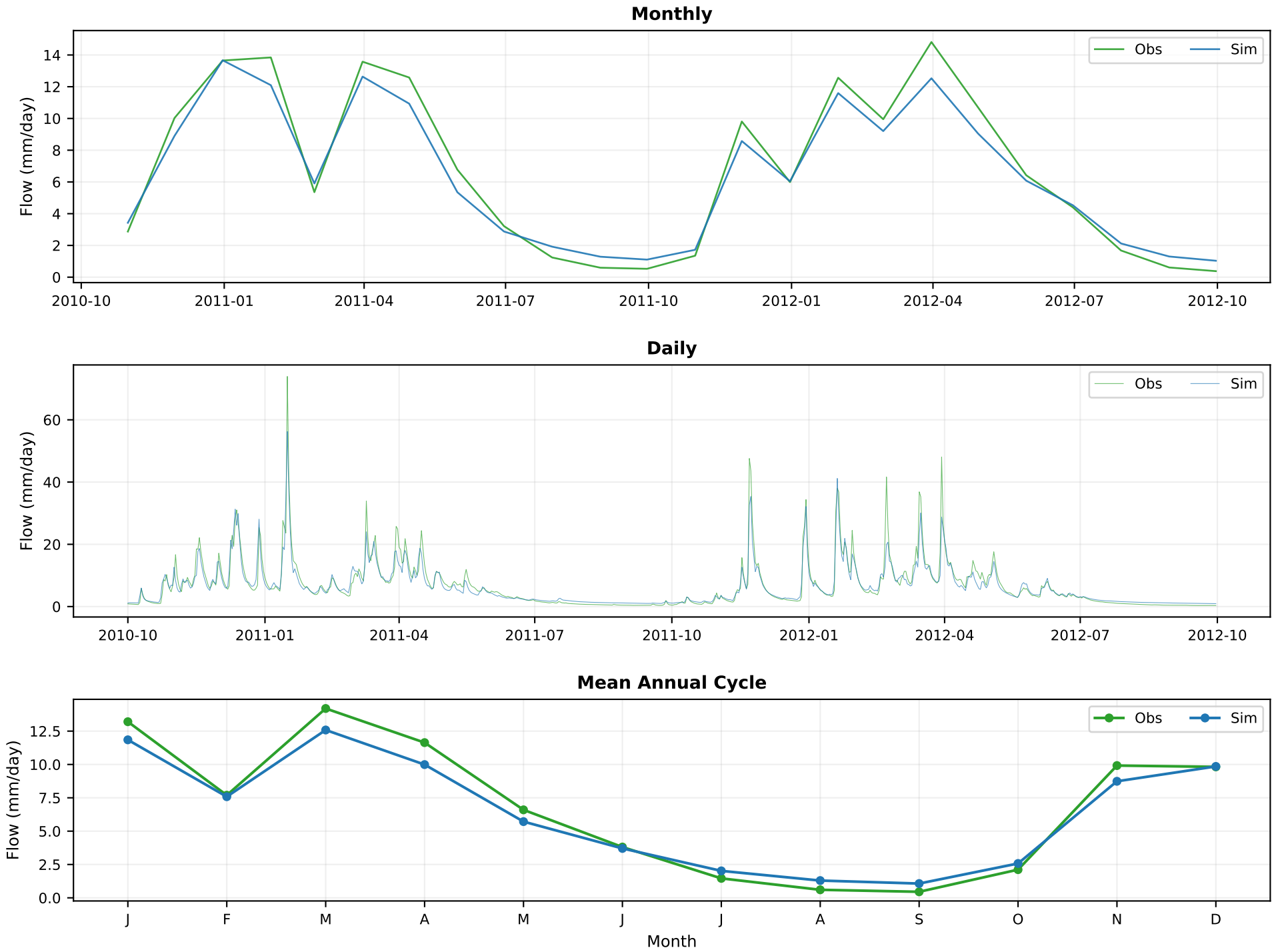

ID: 2465493
KGE=0.808 r=0.909 α=0.839 NSE=0.821 Bias=-4.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

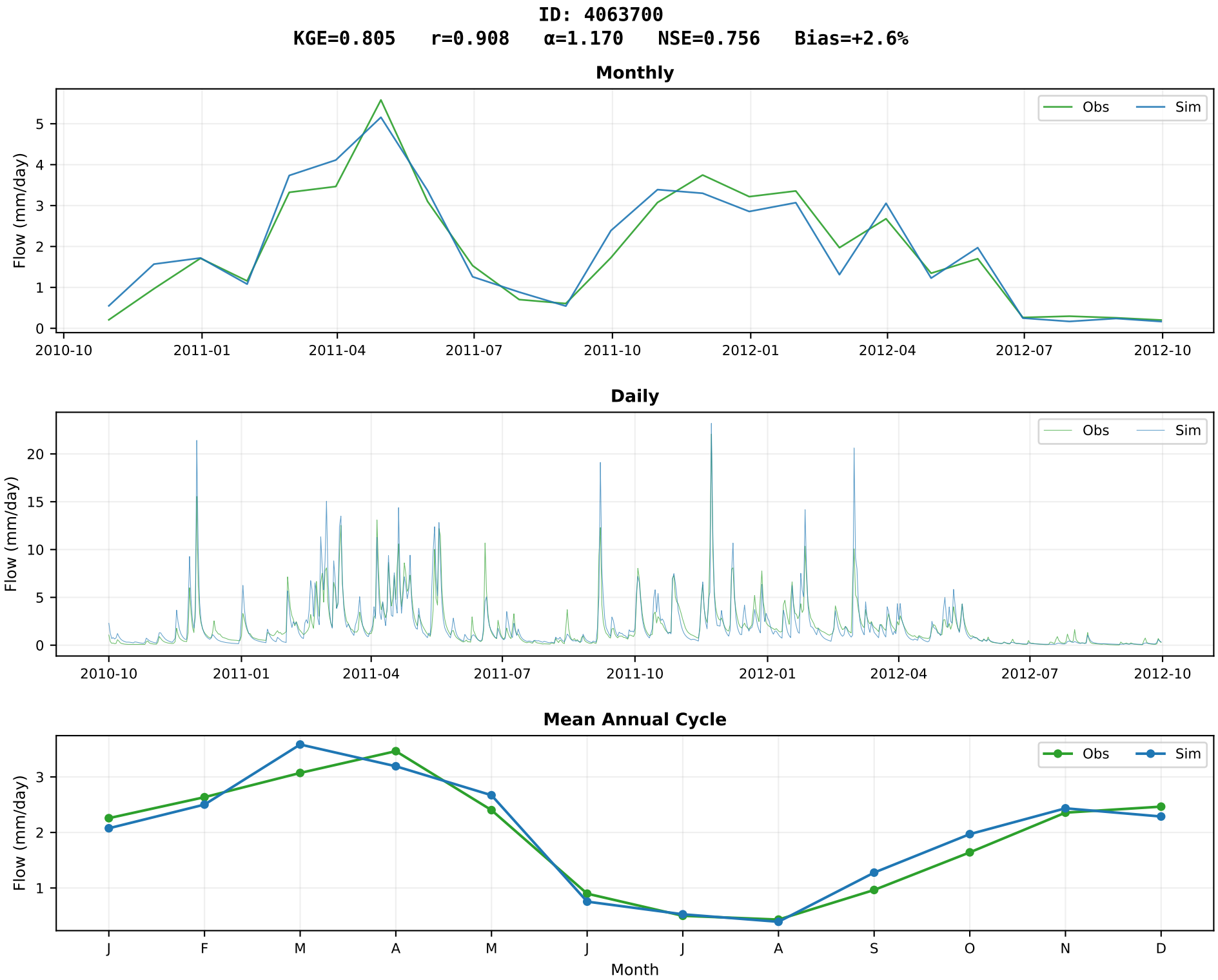

ID: 1516500
KGE=0.808 r=0.902 α=0.835 NSE=0.809 Bias=-0.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

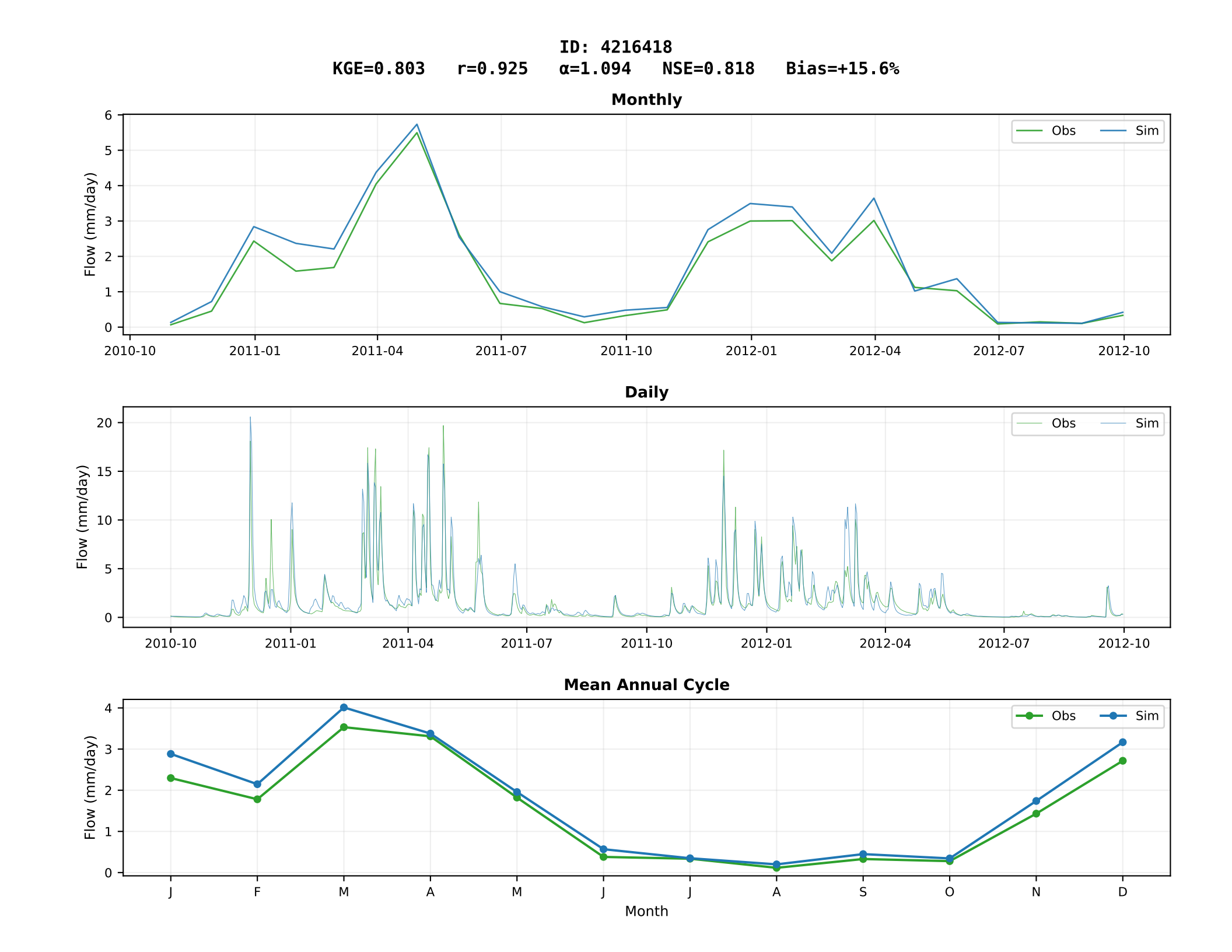

ID: 4074950
KGE=0.807 r=0.870 α=1.132 NSE=0.686 Bias=+5.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

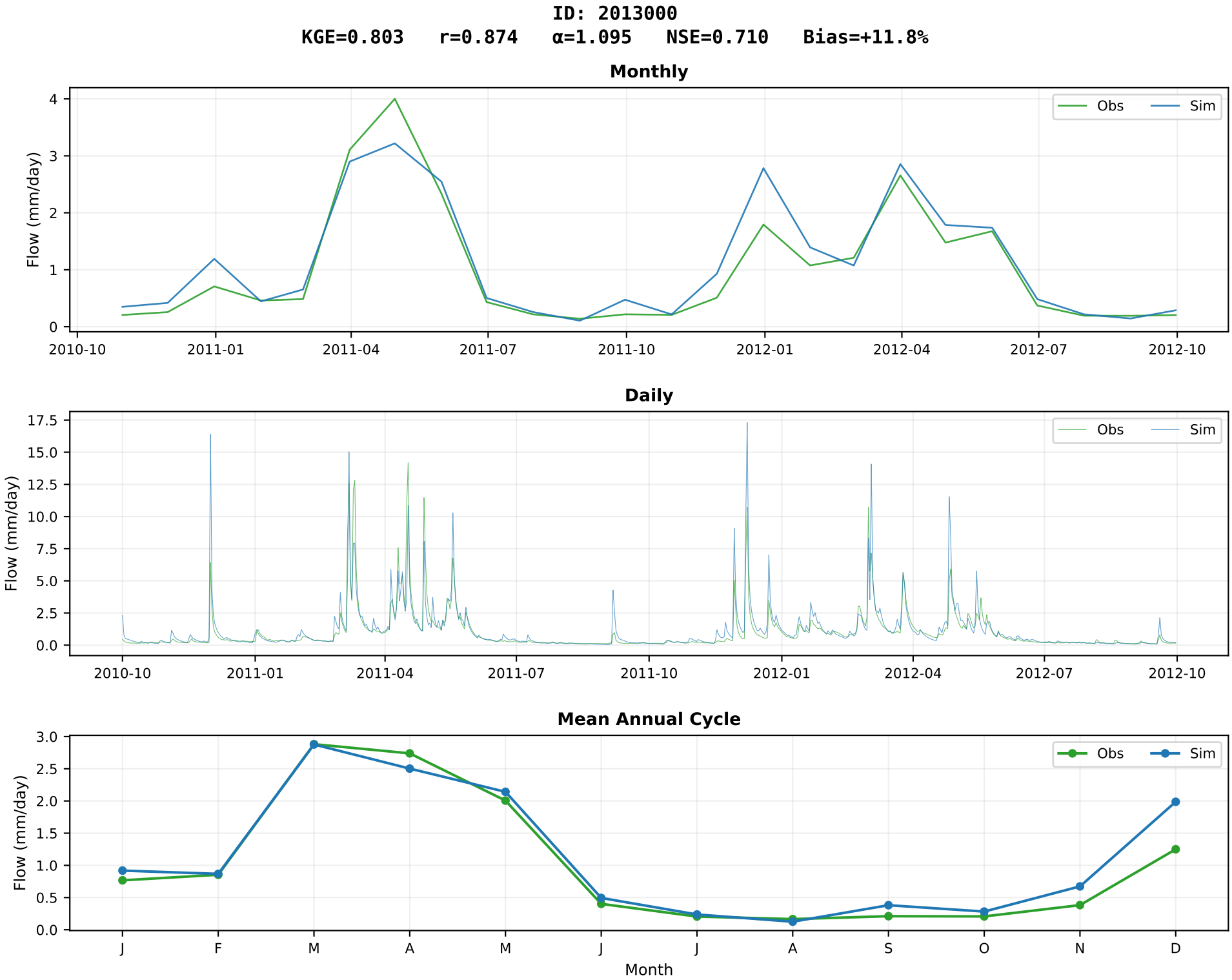

ANDREWS CREEK NEAR MAZAMA, WA (ID=12447390)
KGE=0.806 r=0.918 α=1.034 NSE=0.803 Bias=-17.3%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

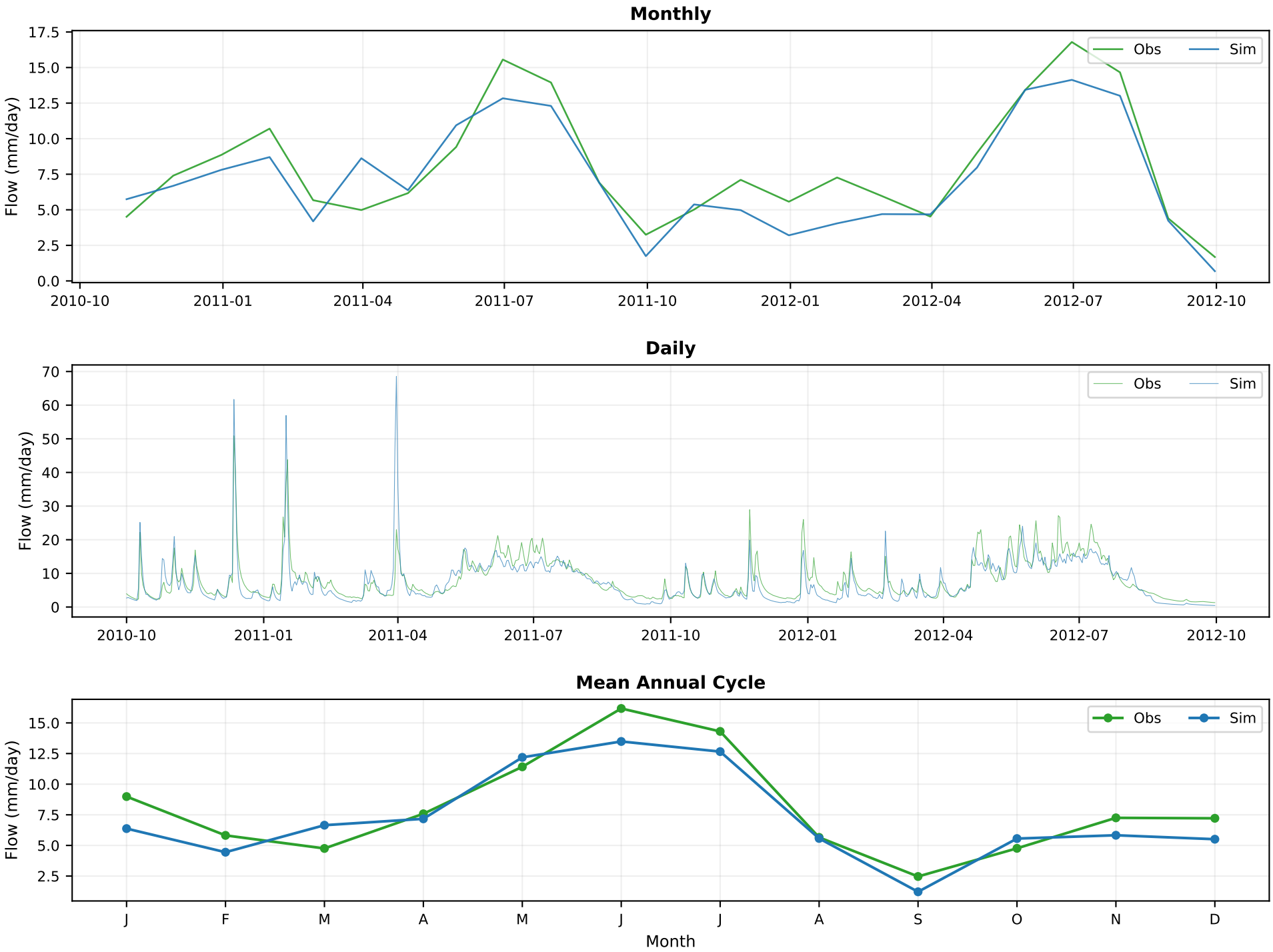

ID: 1550000
KGE=0.805 r=0.946 α=0.830 NSE=0.880 Bias=-7.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

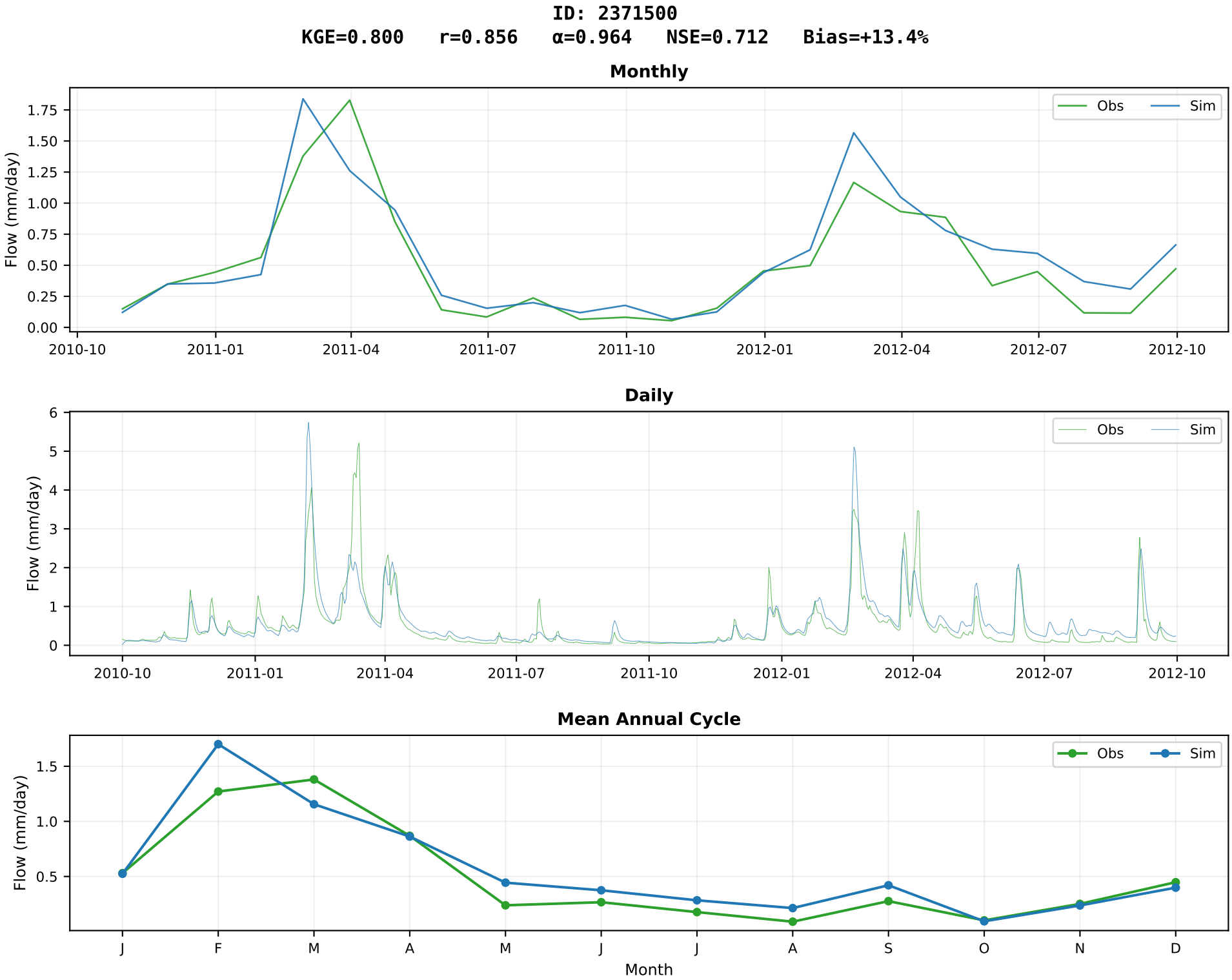

NF CACHE C A HOUGH SPRING NR CLEARLAKE OAKS CA (ID=11451100)
KGE=0.805 r=0.960 α=0.817 NSE=0.899 Bias=-5.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

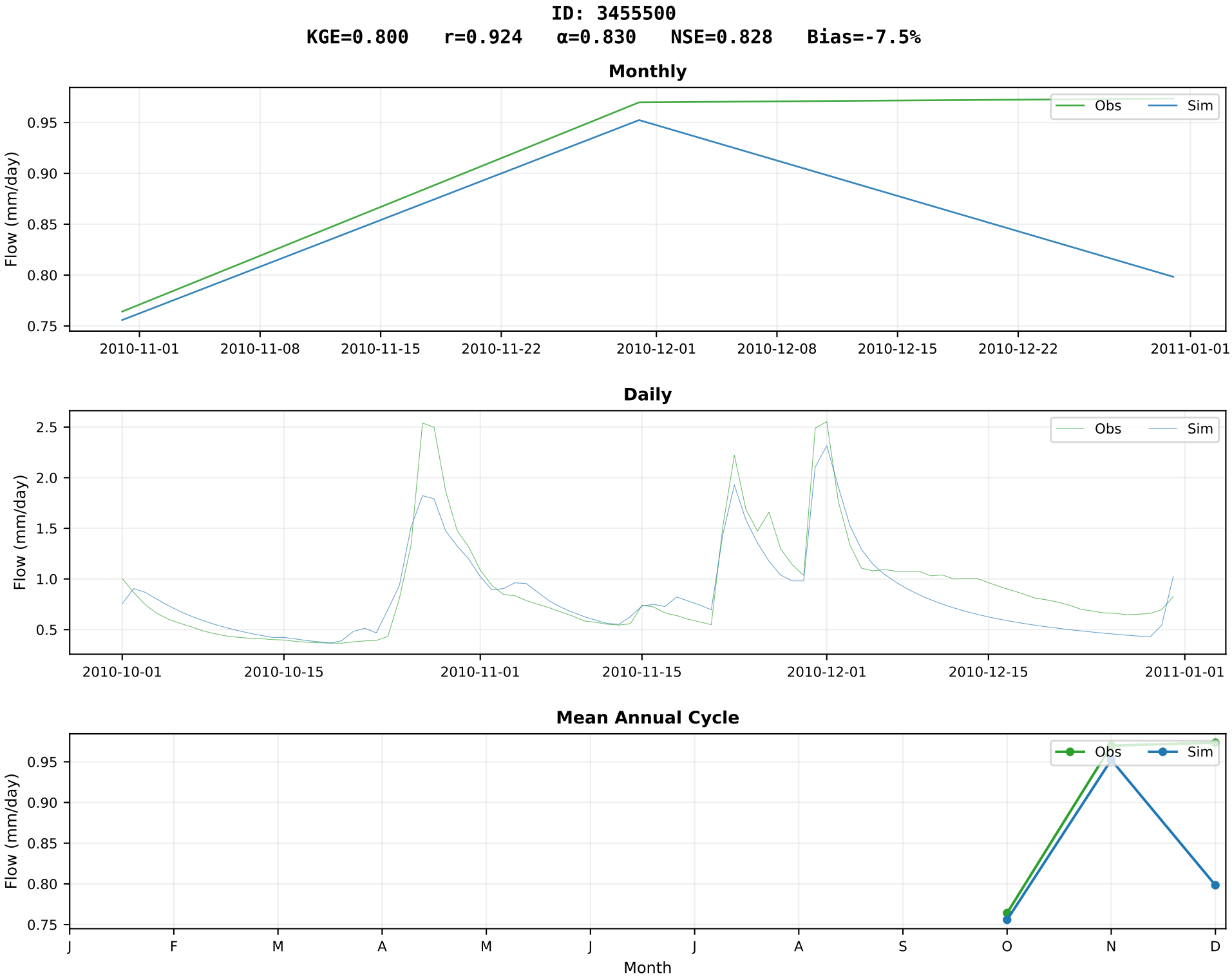

ID: 4063700
KGE=0.805 r=0.908 α=1.170 NSE=0.756 Bias=+2.6%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

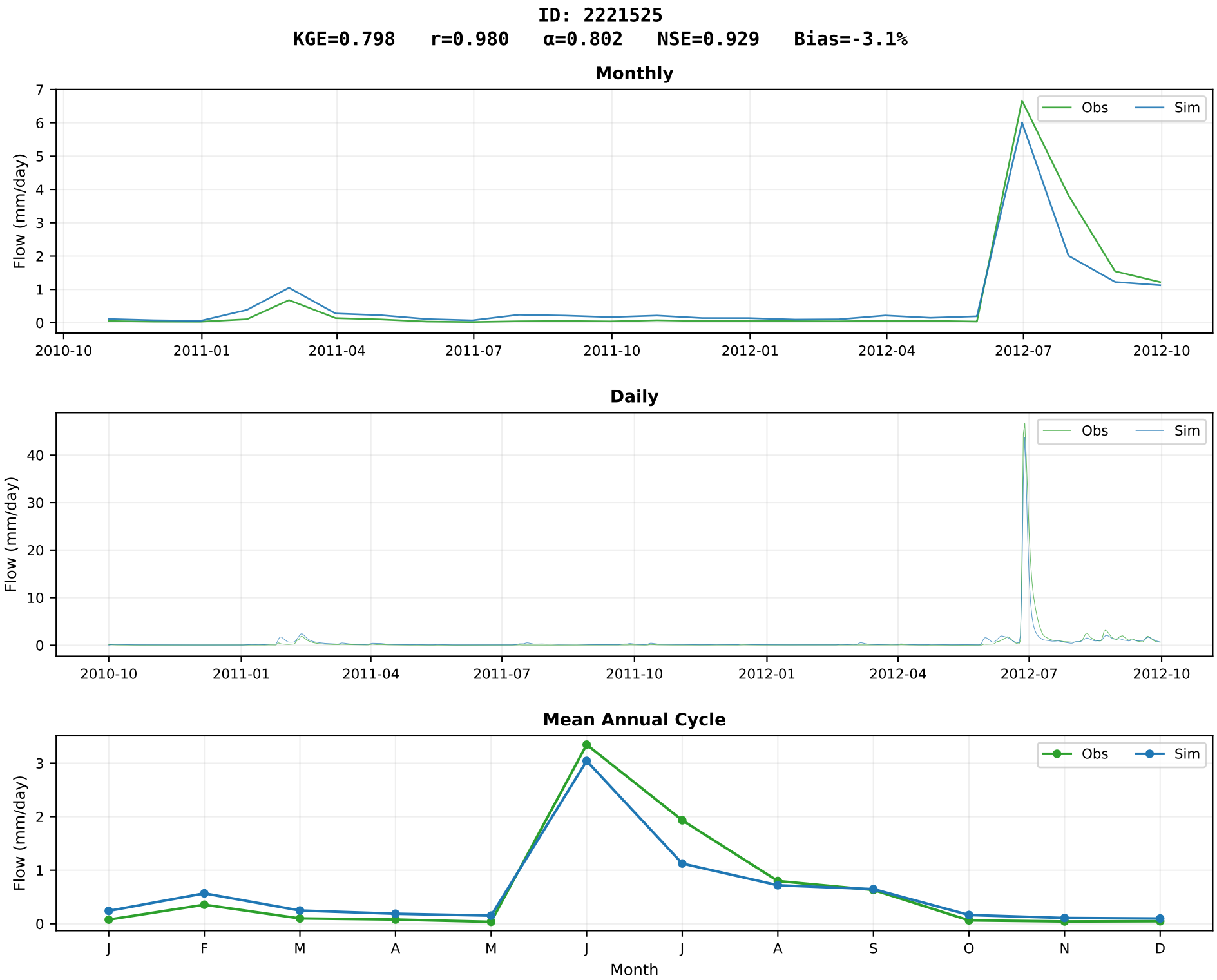

ID: 4216418
KGE=0.803 r=0.925 α=1.094 NSE=0.818 Bias=+15.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

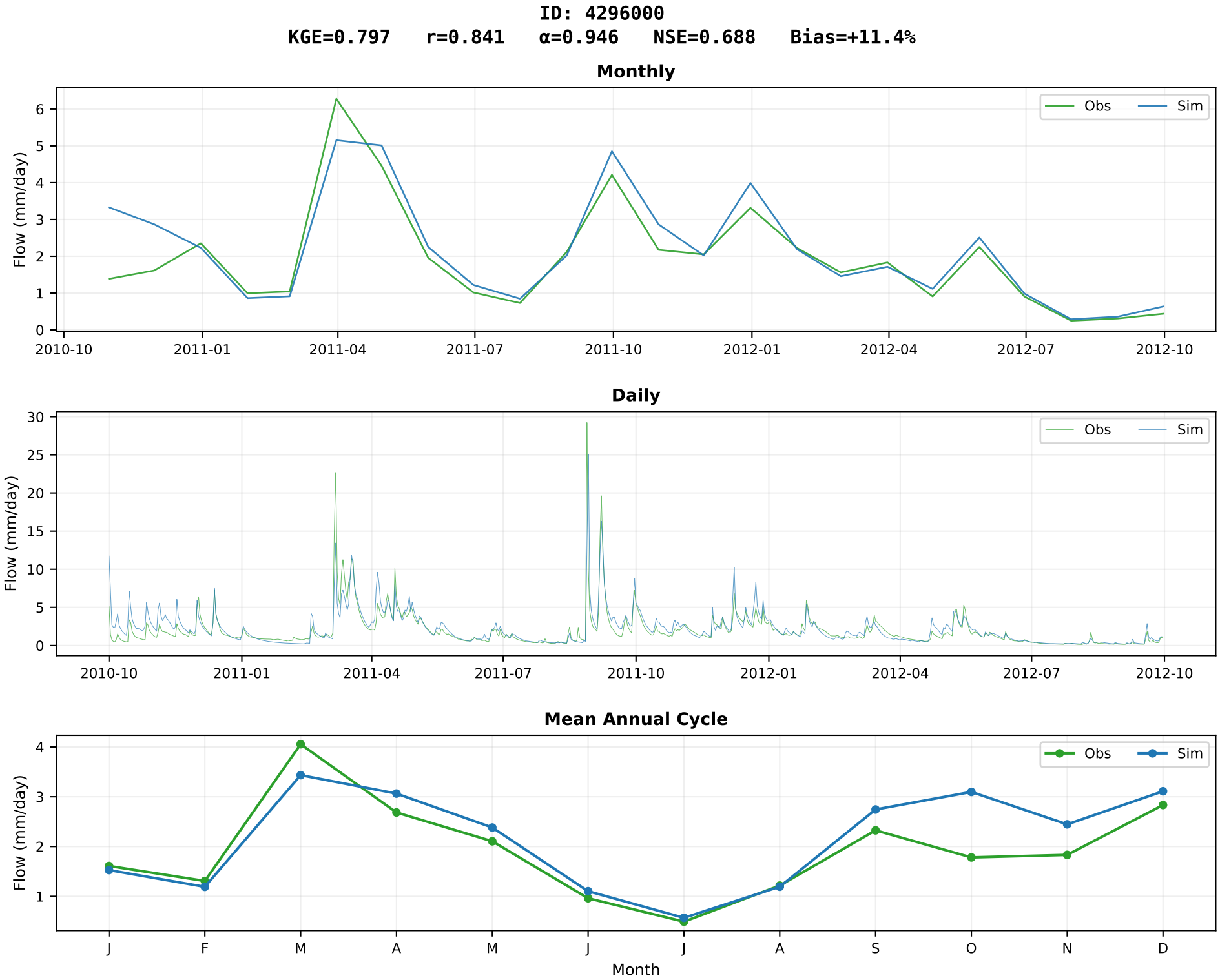

ID: 2013000
KGE=0.803 r=0.874 α=1.095 NSE=0.710 Bias=+11.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

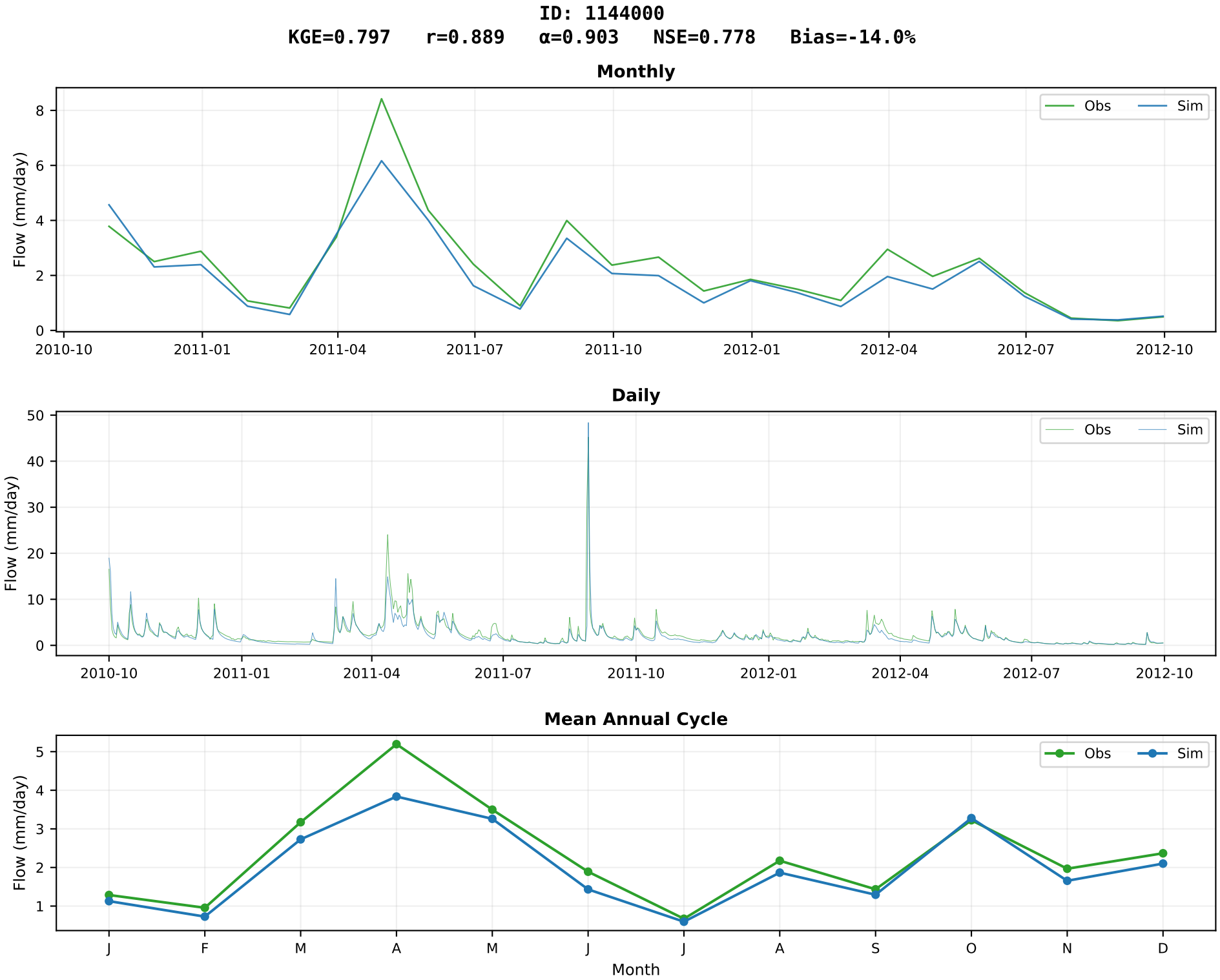

SOUTH FORK BULL RUN RIVER NEAR BULL RUN, OR (ID=14139800)
KGE=0.802 r=0.845 α=1.074 NSE=0.645 Bias=-10.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2371500
KGE=0.800 r=0.856 α=0.964 NSE=0.712 Bias=+13.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

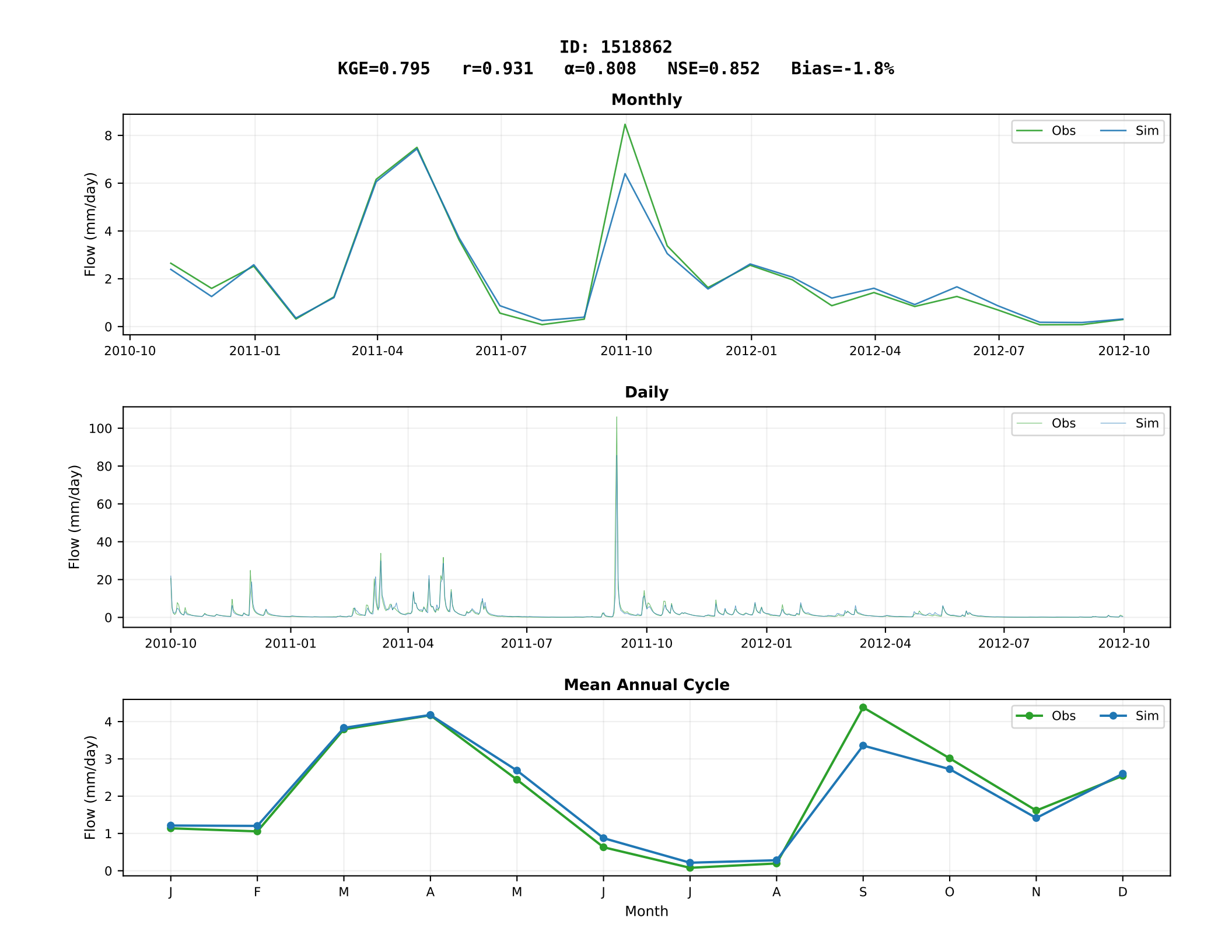

ID: 3455500
KGE=0.800 r=0.924 α=0.830 NSE=0.828 Bias=-7.5%
Monthly
Obs
Sim
Flow (mm/day)
0.95
0.90
0.85
0.80
0.75
2010-11-01
2010-11-08
2010-11-15
2010-11-22
2010-12-01
2010-12-08
2010-12-15
2010-12-22
2011-01-01
Daily
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
2010-10-01
2010-10-15
2010-11-01
2010-11-15
2010-12-01
2010-12-15
2011-01-01
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.95
0.90
0.85
0.80
0.75
J
F
M
A
M
J
J
A
S
O
N
D
Month

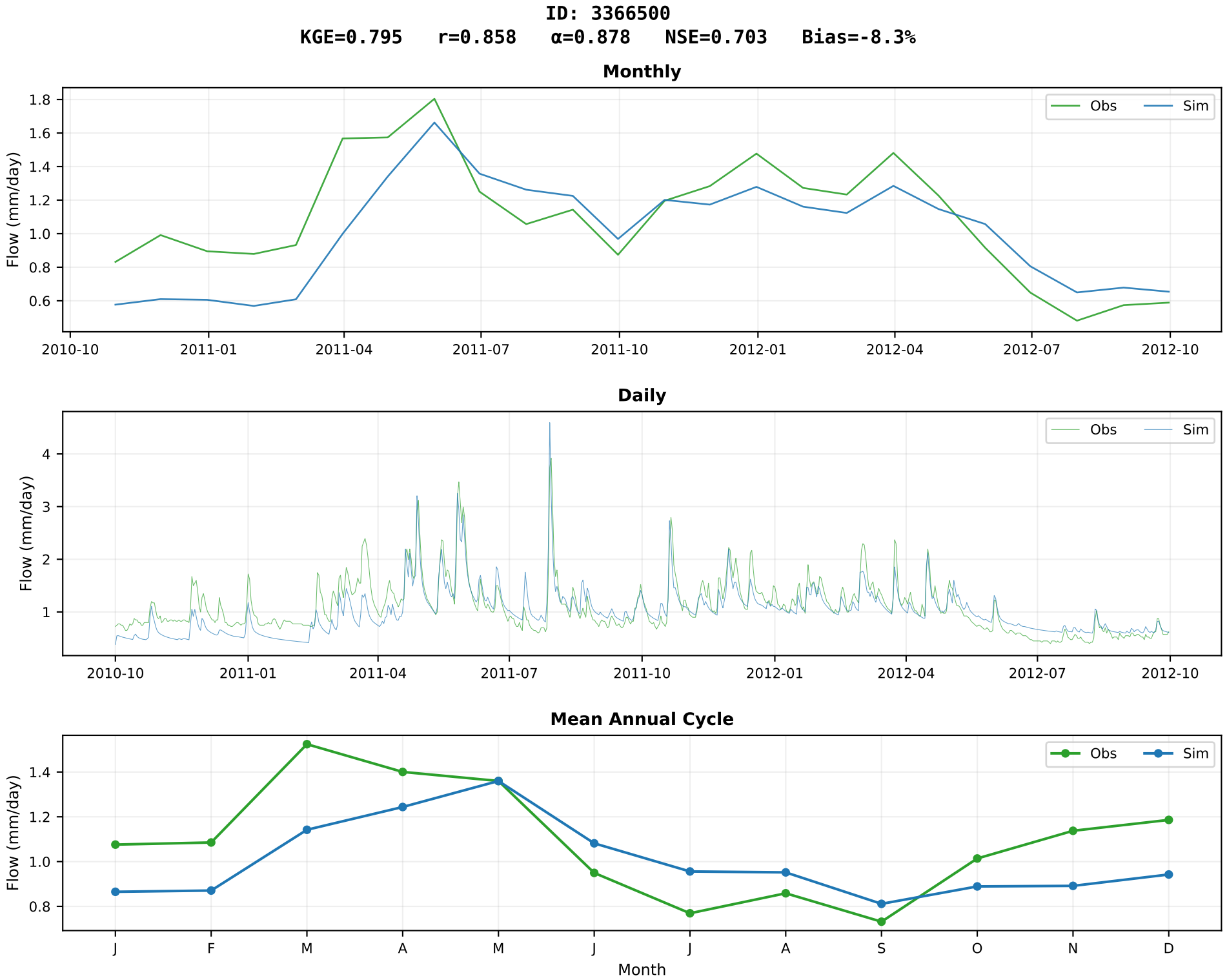

ID: 2221525
KGE=0.798 r=0.980 α=0.802 NSE=0.929 Bias=-3.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

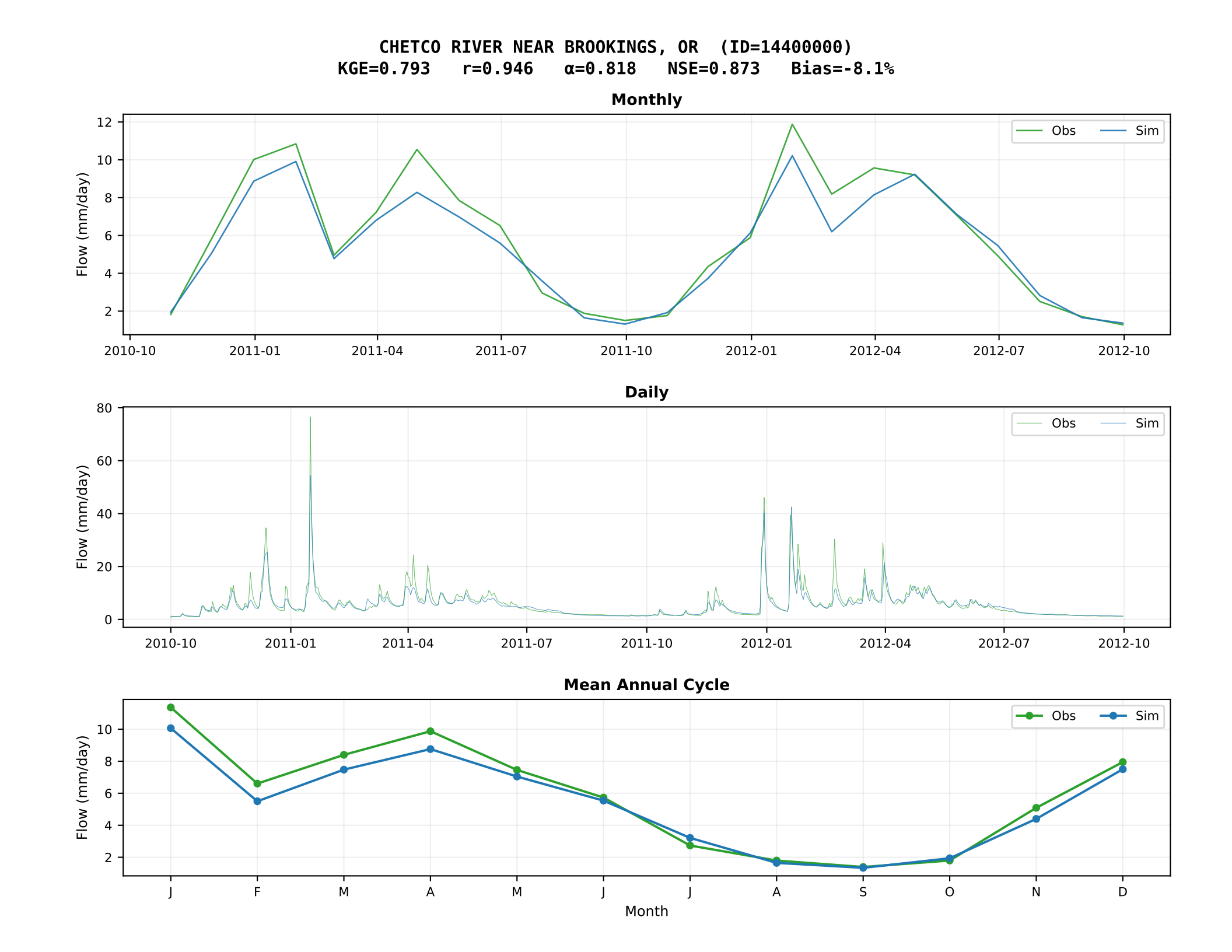

ID: 4296000
KGE=0.797 r=0.841 α=0.946 NSE=0.688 Bias=+11.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

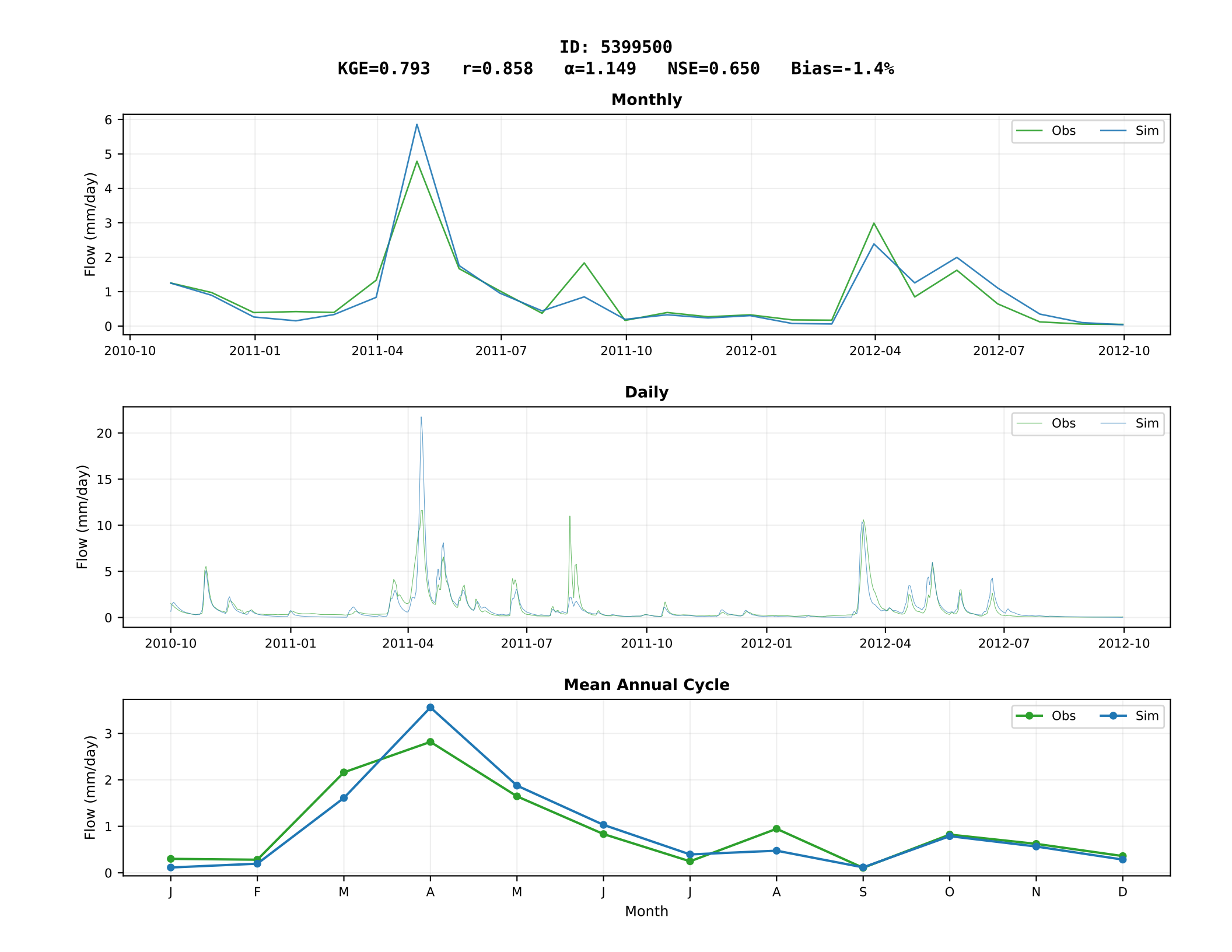

ID: 1144000
KGE=0.797 r=0.889 α=0.903 NSE=0.778 Bias=-14.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

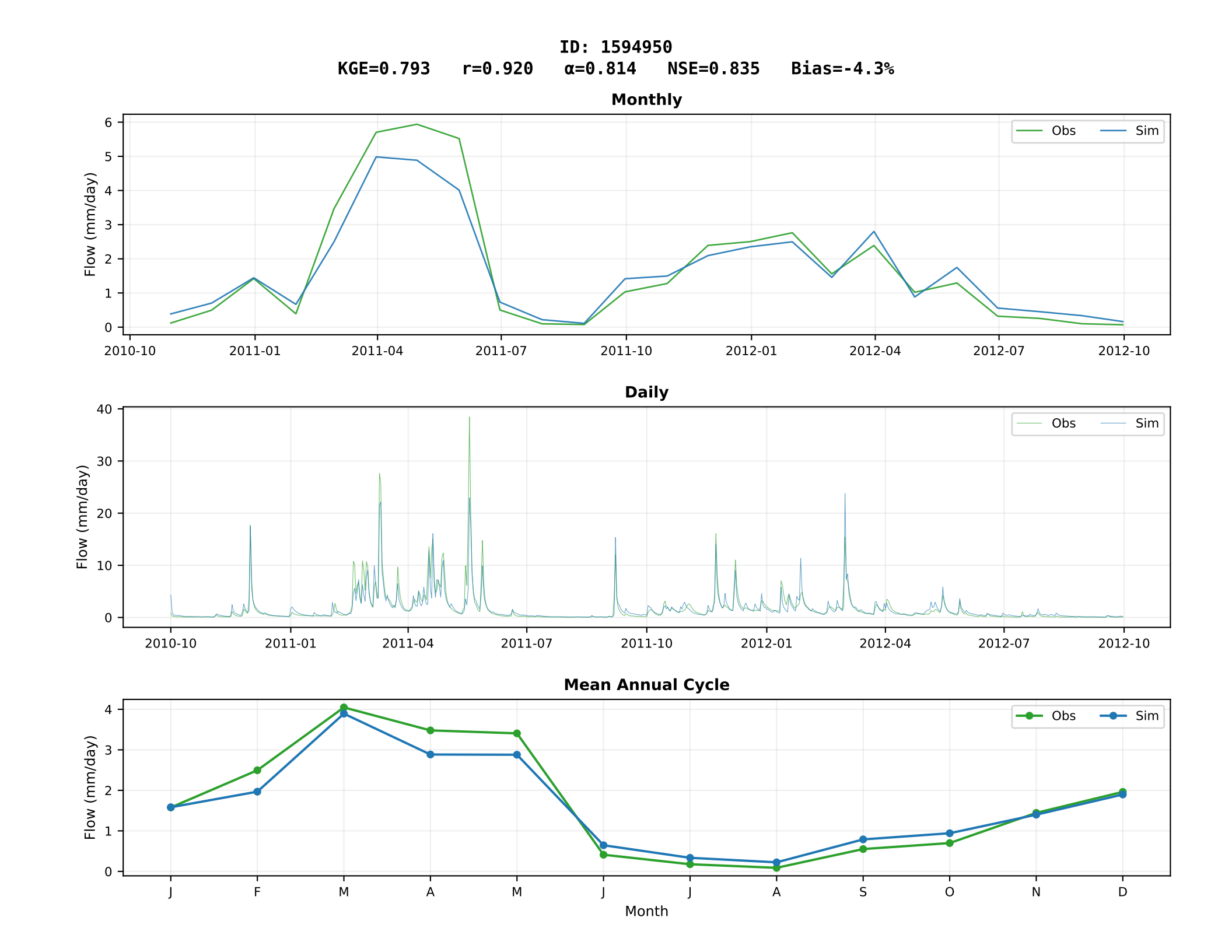

ID: 3238500
KGE=0.796 r=0.853 α=0.867 NSE=0.727 Bias=-5.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

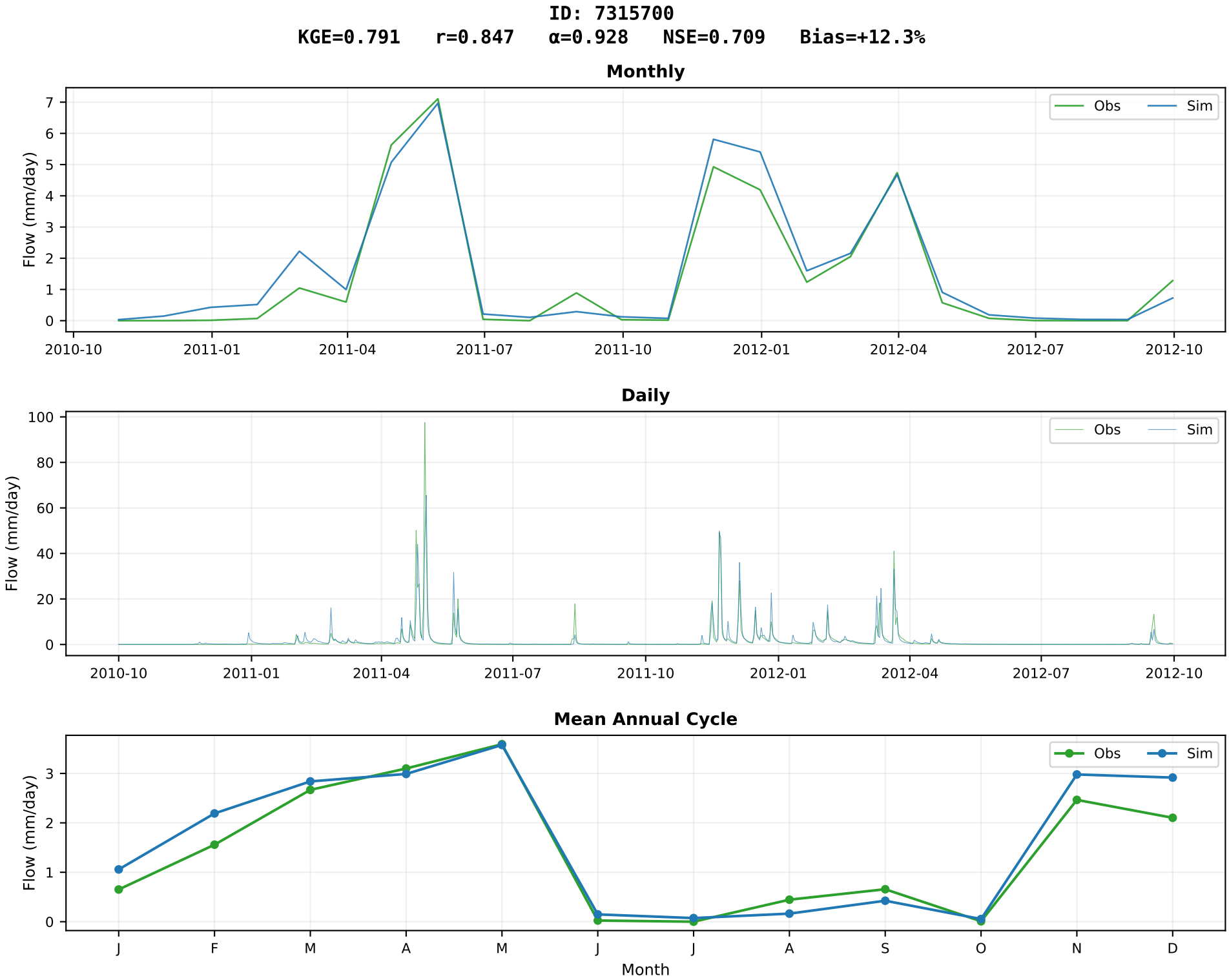

ID: 1518862
KGE=0.795 r=0.931 α=0.808 NSE=0.852 Bias=-1.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

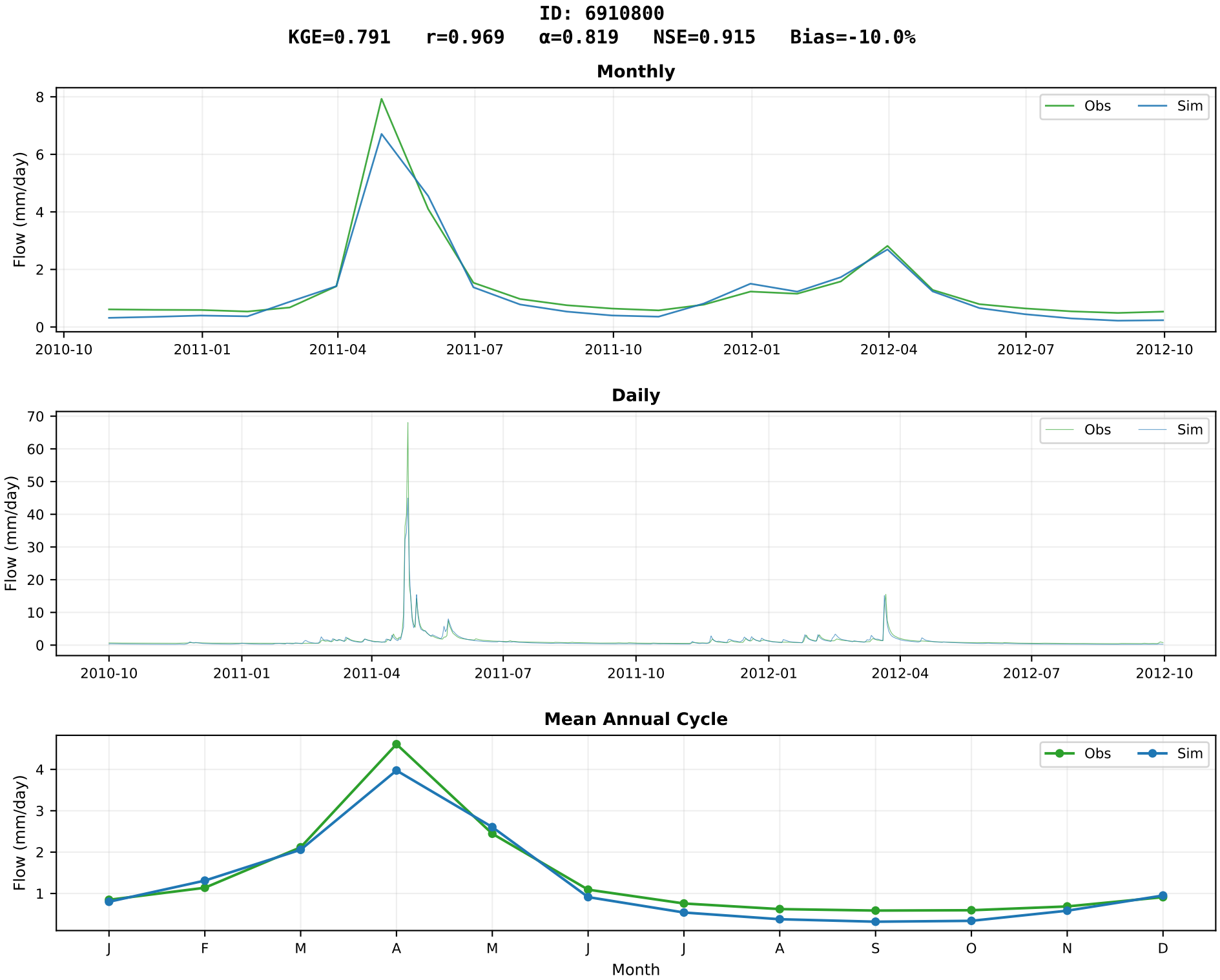

ID: 3366500
KGE=0.795 r=0.858 α=0.878 NSE=0.703 Bias=-8.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

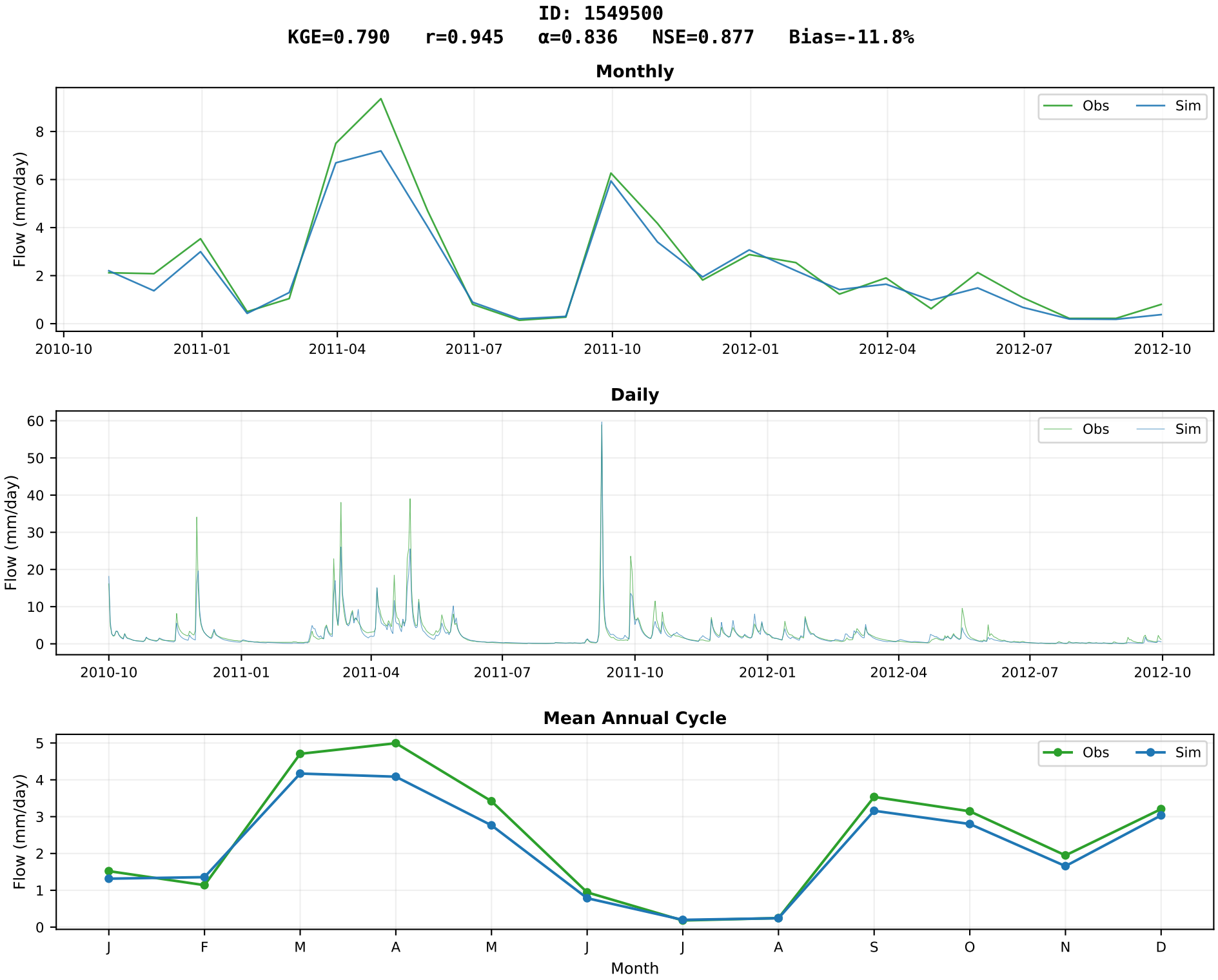

CHETCO RIVER NEAR BROOKINGS, OR (ID=14400000)
KGE=0.793 r=0.946 α=0.818 NSE=0.873 Bias=-8.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

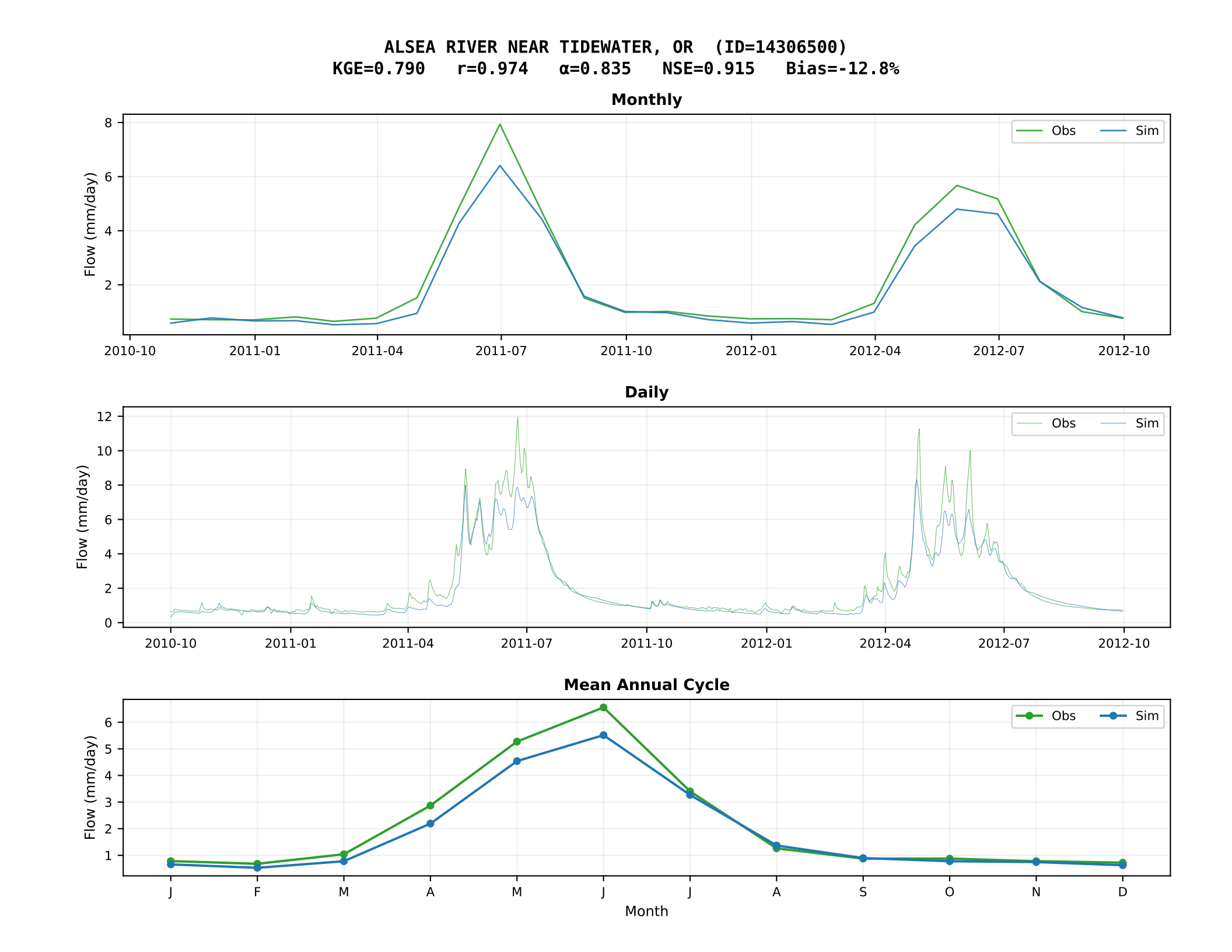

ID: 5399500
KGE=0.793 r=0.858 α=1.149 NSE=0.650 Bias=-1.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

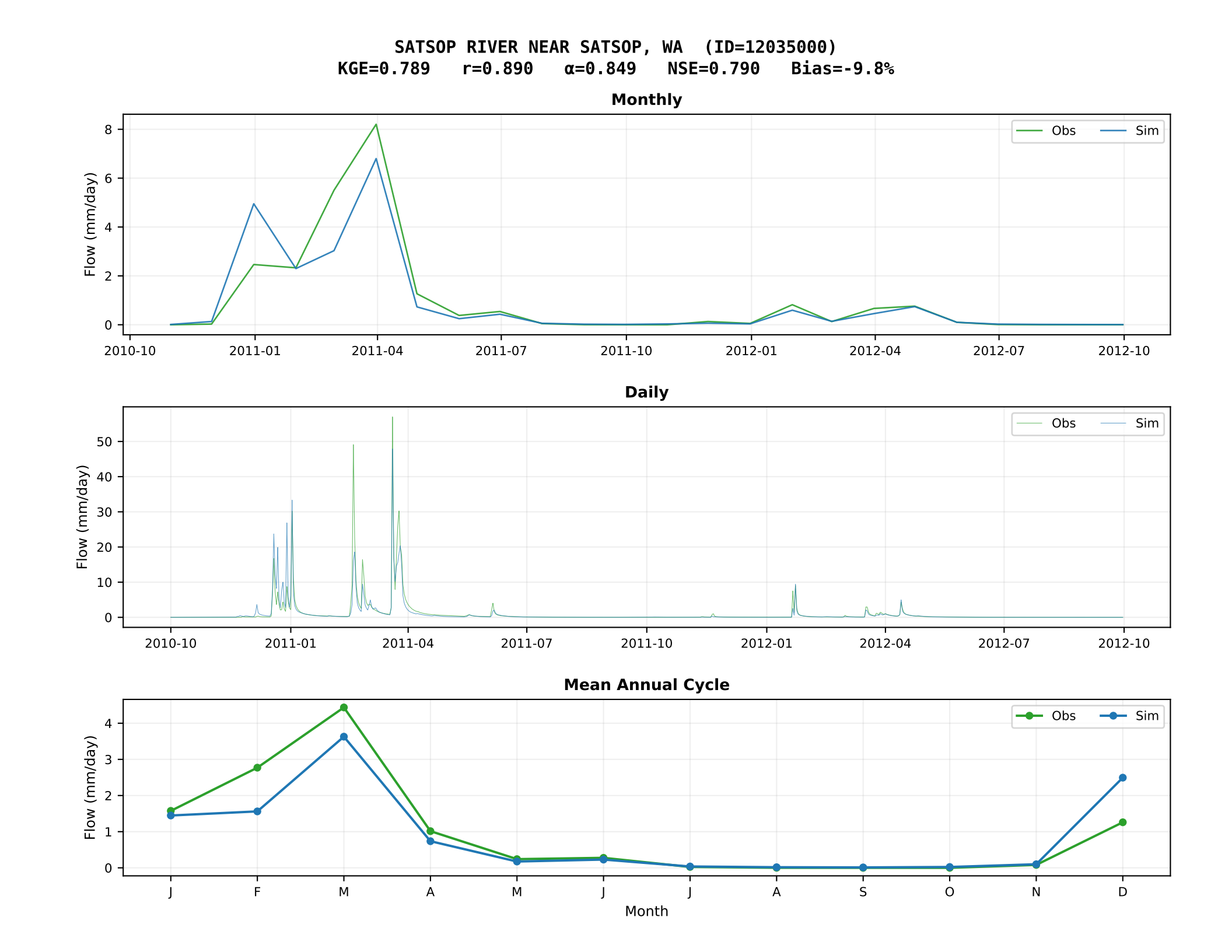

ID: 1594950
KGE=0.793 r=0.920 α=0.814 NSE=0.835 Bias=-4.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

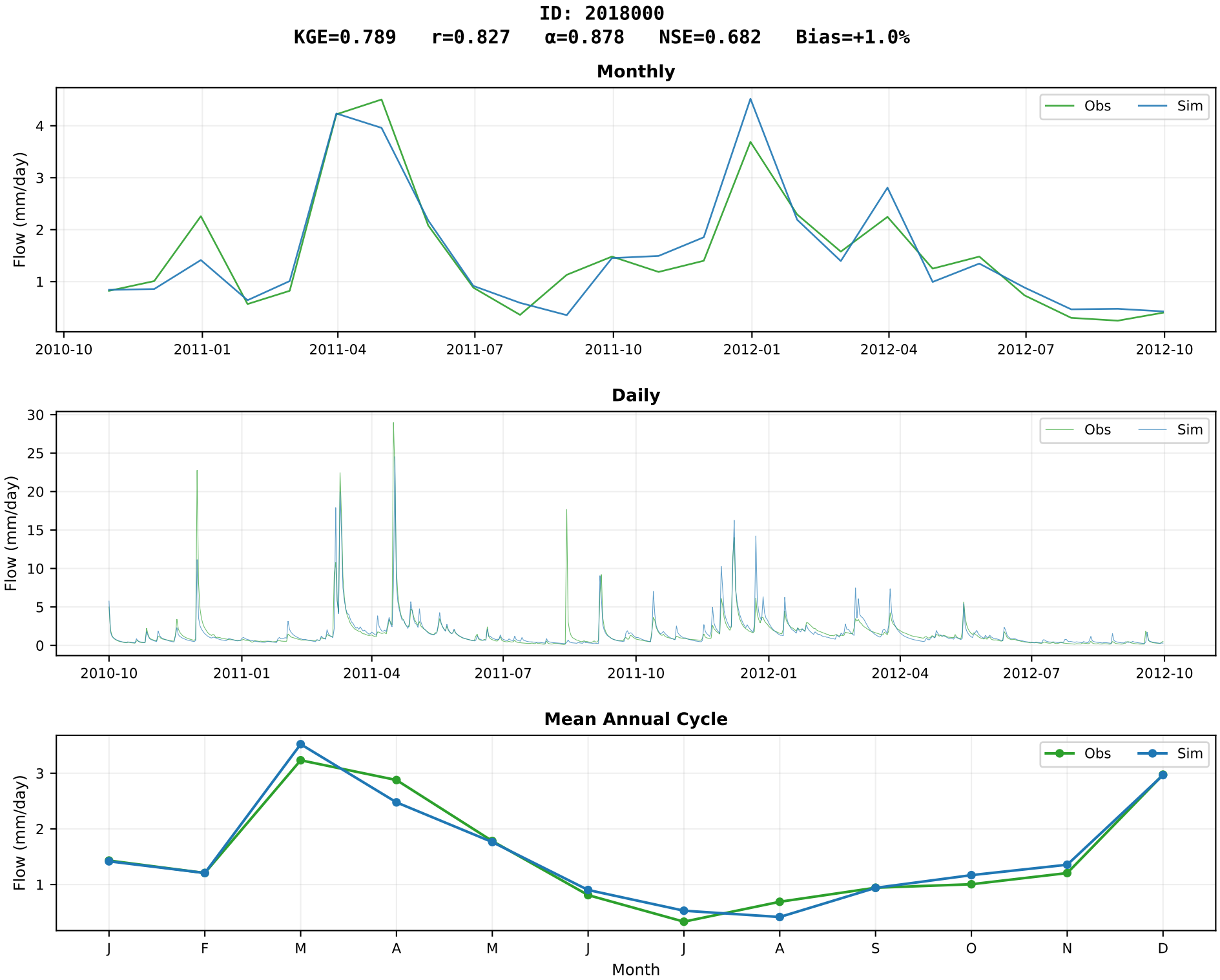

ID: 7315700
KGE=0.791 r=0.847 α=0.928 NSE=0.709 Bias=+12.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

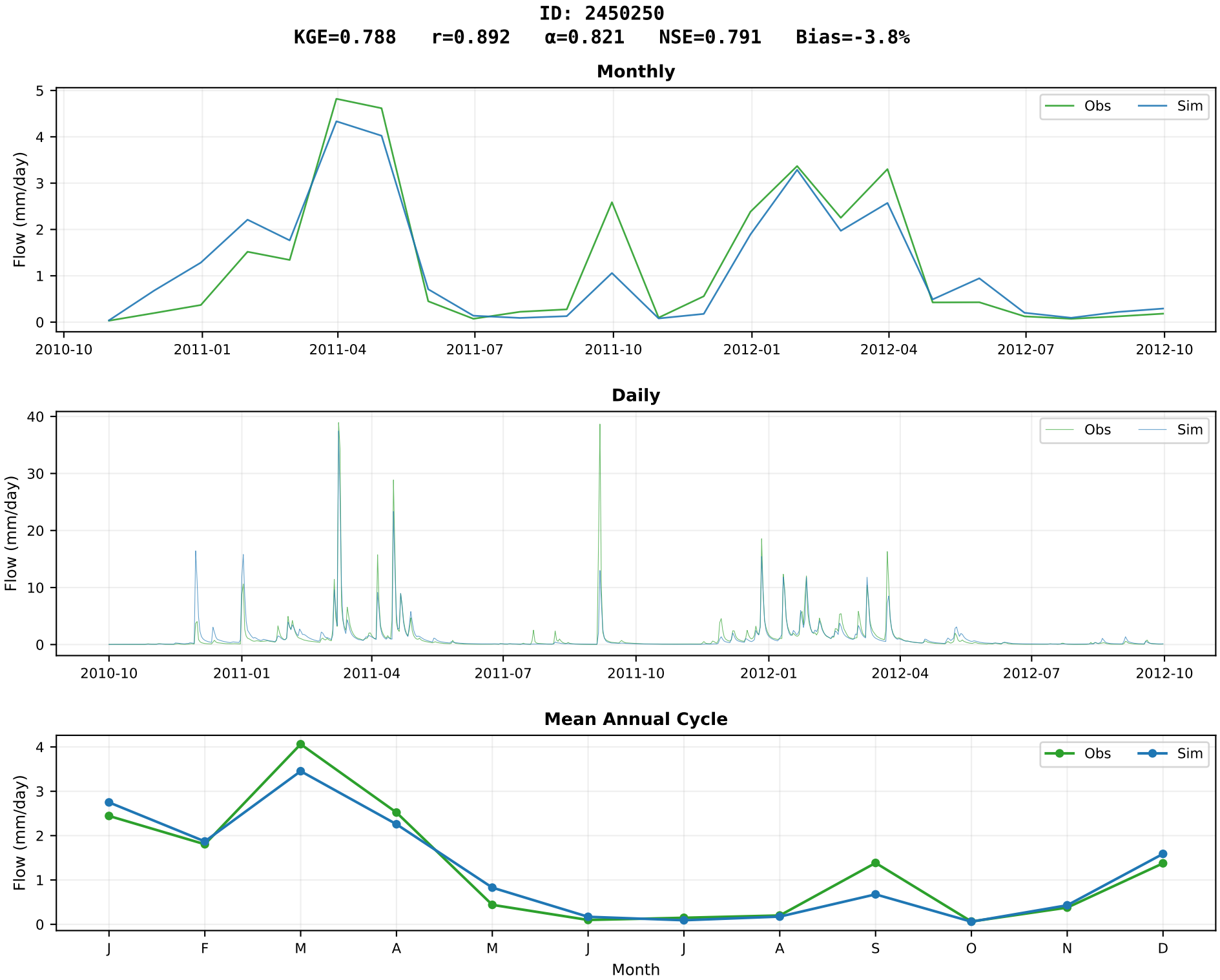

ID: 6910800
KGE=0.791 r=0.969 α=0.819 NSE=0.915 Bias=-10.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

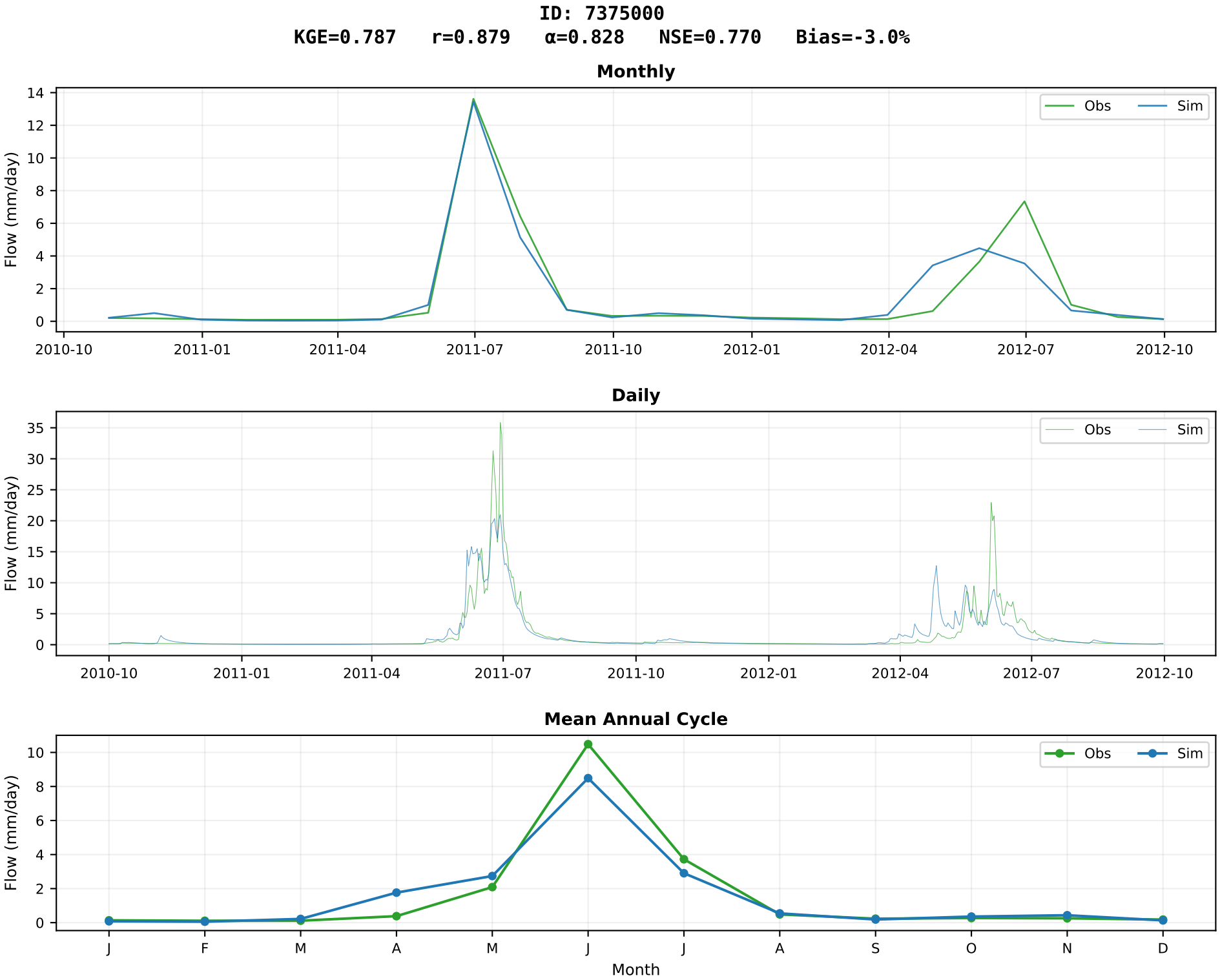

ID: 1549500
KGE=0.790 r=0.945 α=0.836 NSE=0.877 Bias=-11.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

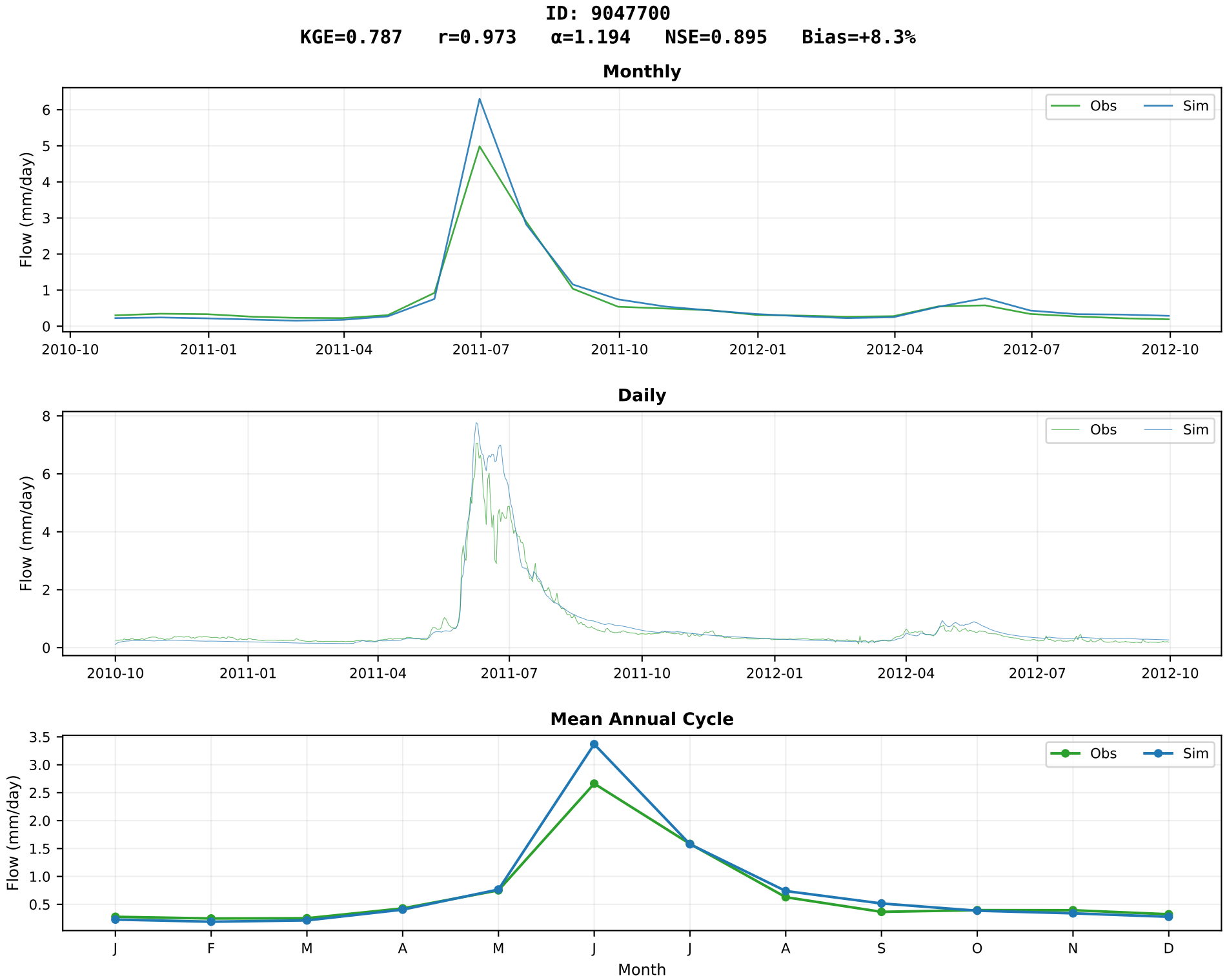

ALSEA RIVER NEAR TIDEWATER, OR (ID=14306500)
KGE=0.790 r=0.974 α=0.835 NSE=0.915 Bias=-12.8%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

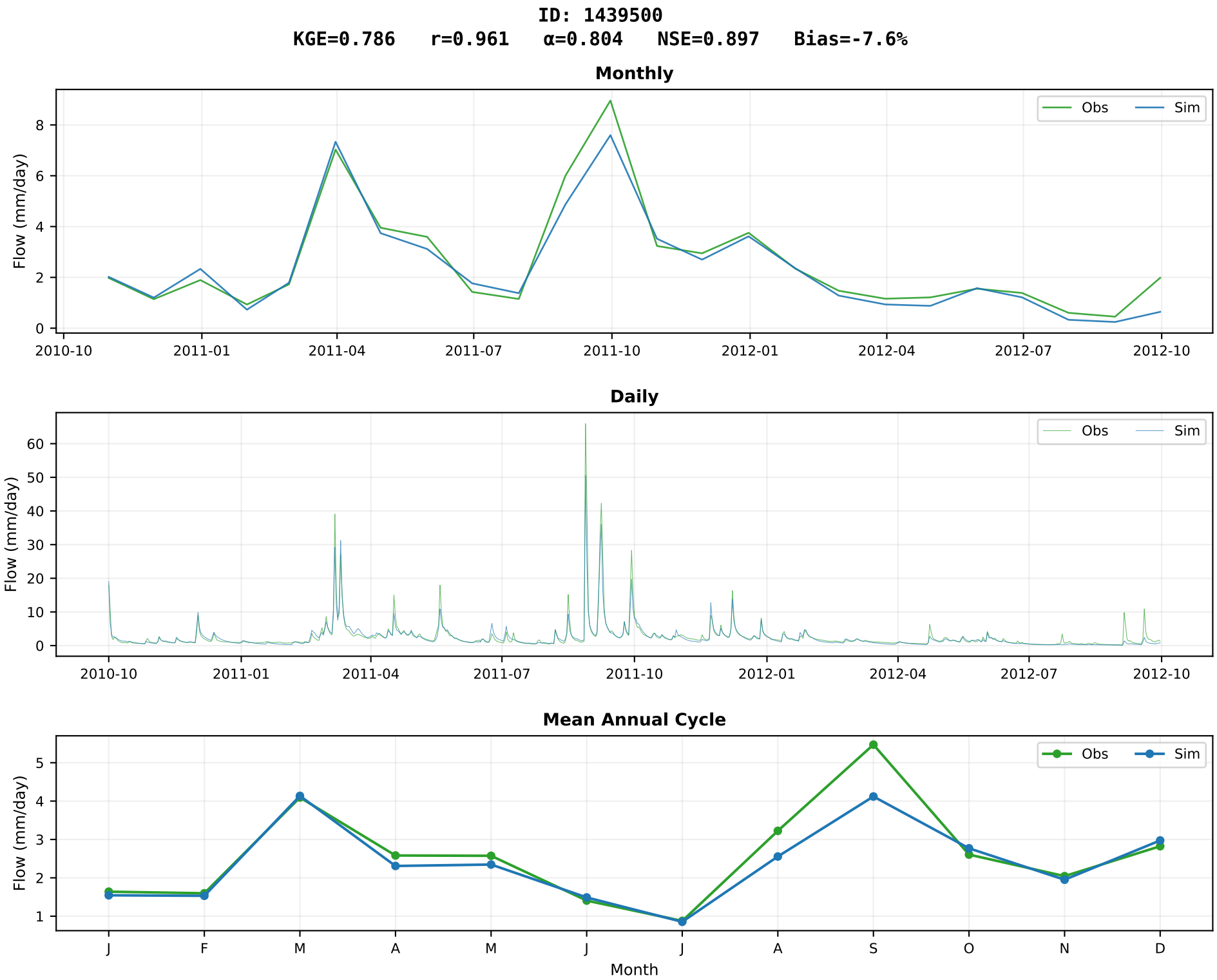

SATSOP RIVER NEAR SATSOP, WA (ID=12035000)
KGE=0.789 r=0.890 α=0.849 NSE=0.790 Bias=-9.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

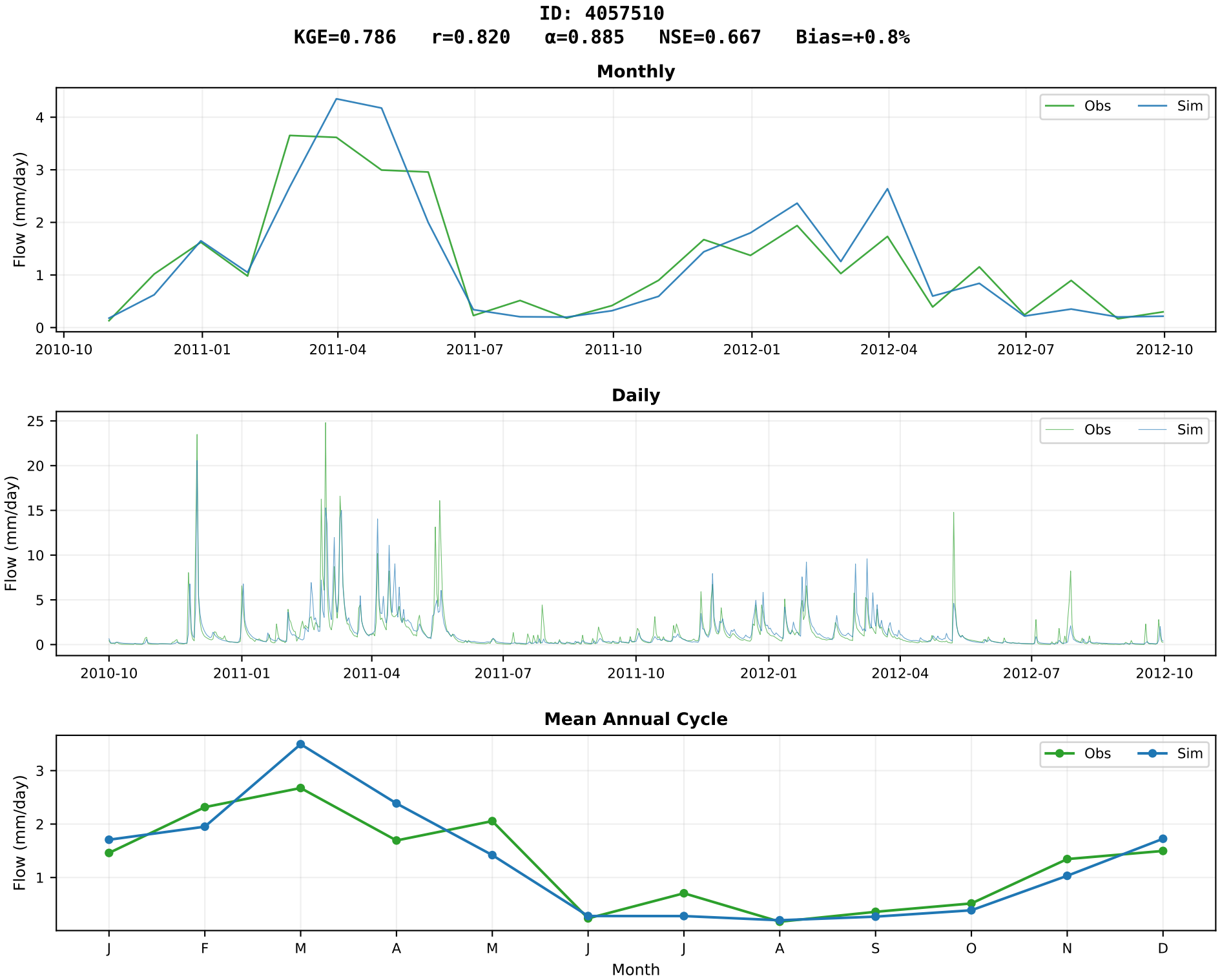

ID: 2018000
KGE=0.789 r=0.827 α=0.878 NSE=0.682 Bias=+1.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

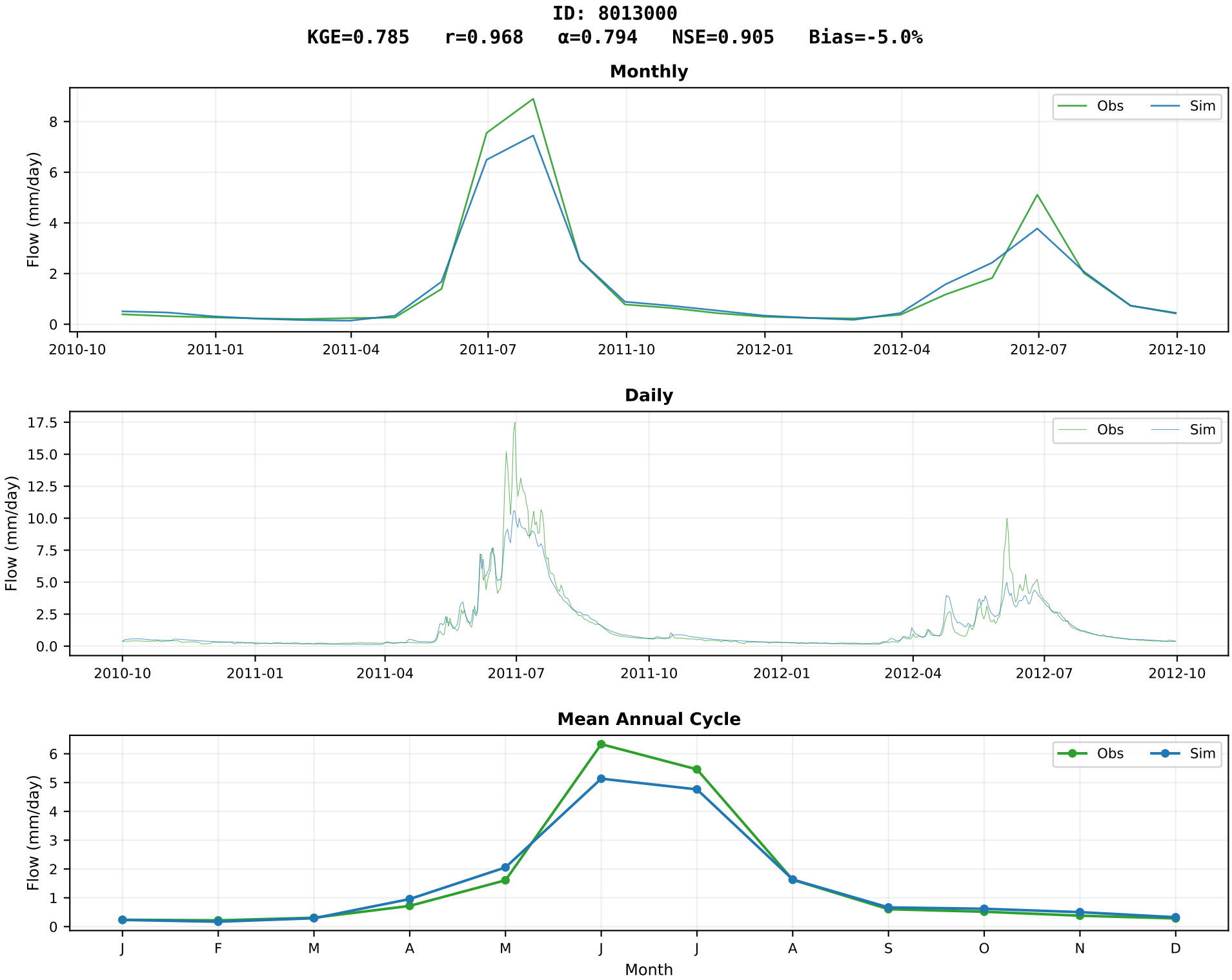

ID: 2450250
KGE=0.788 r=0.892 α=0.821 NSE=0.791 Bias=-3.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

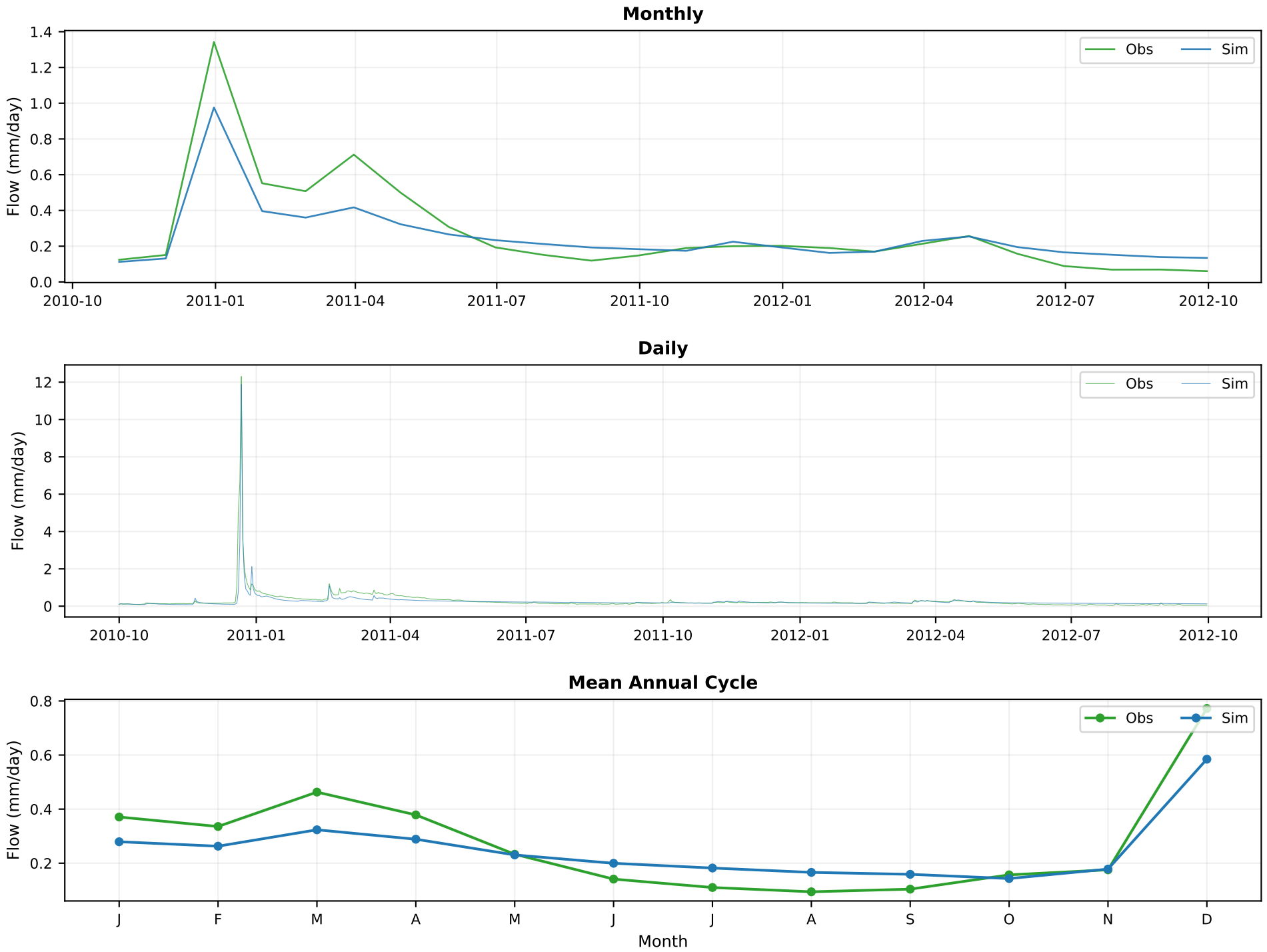

ID: 7375000
KGE=0.787 r=0.879 α=0.828 NSE=0.770 Bias=-3.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

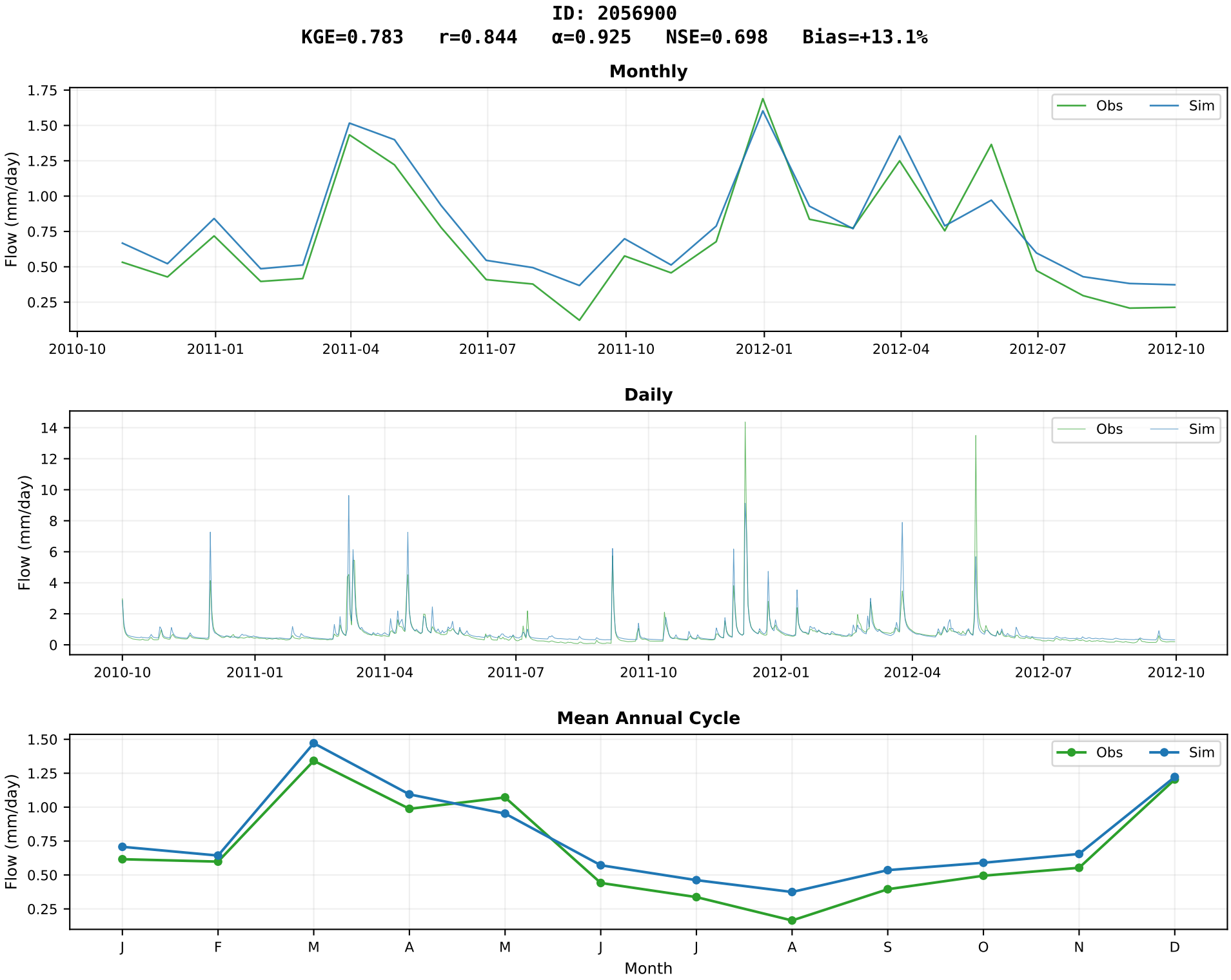

ID: 9047700
KGE=0.787 r=0.973 α=1.194 NSE=0.895 Bias=+8.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

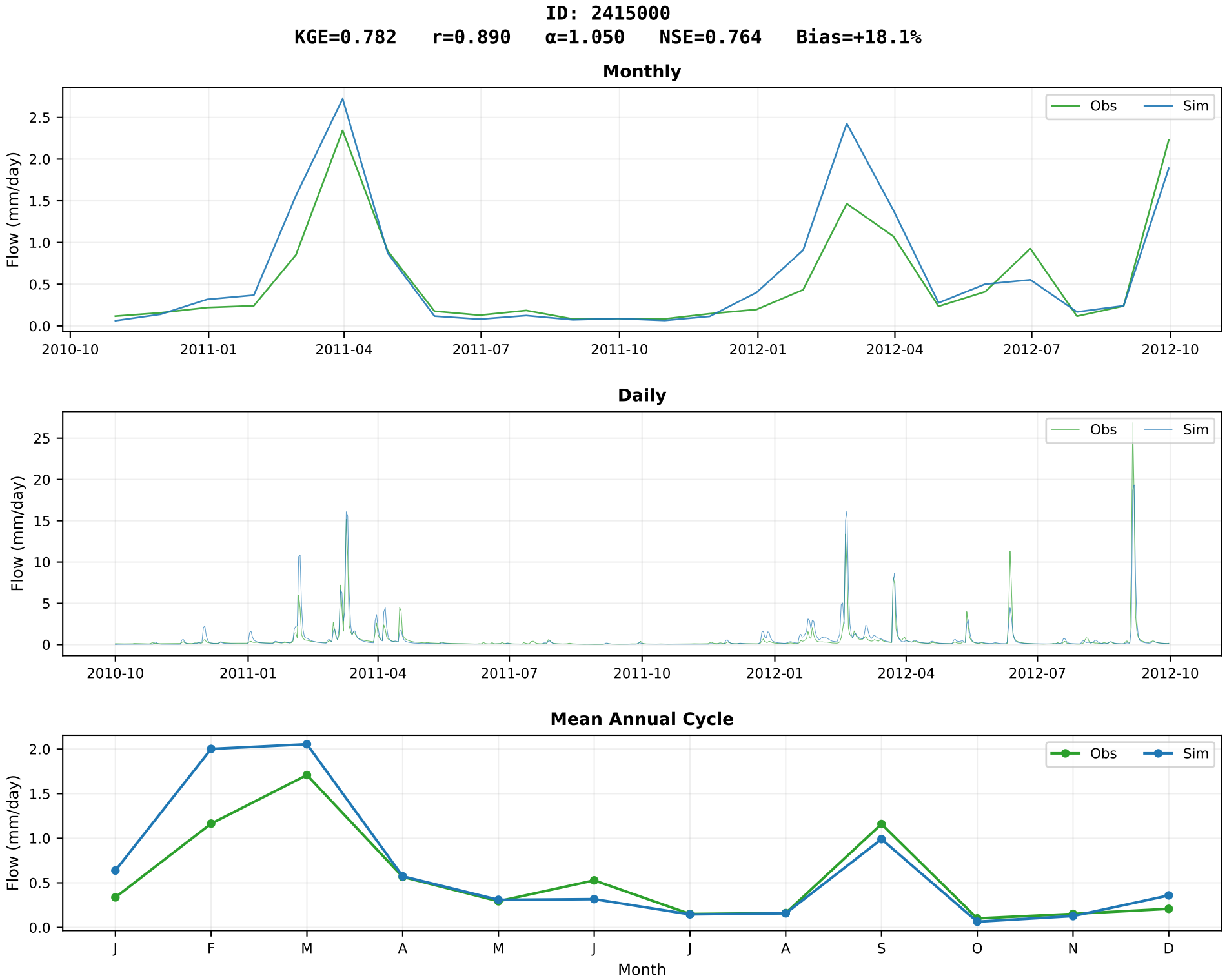

ID: 1439500
KGE=0.786 r=0.961 α=0.804 NSE=0.897 Bias=-7.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

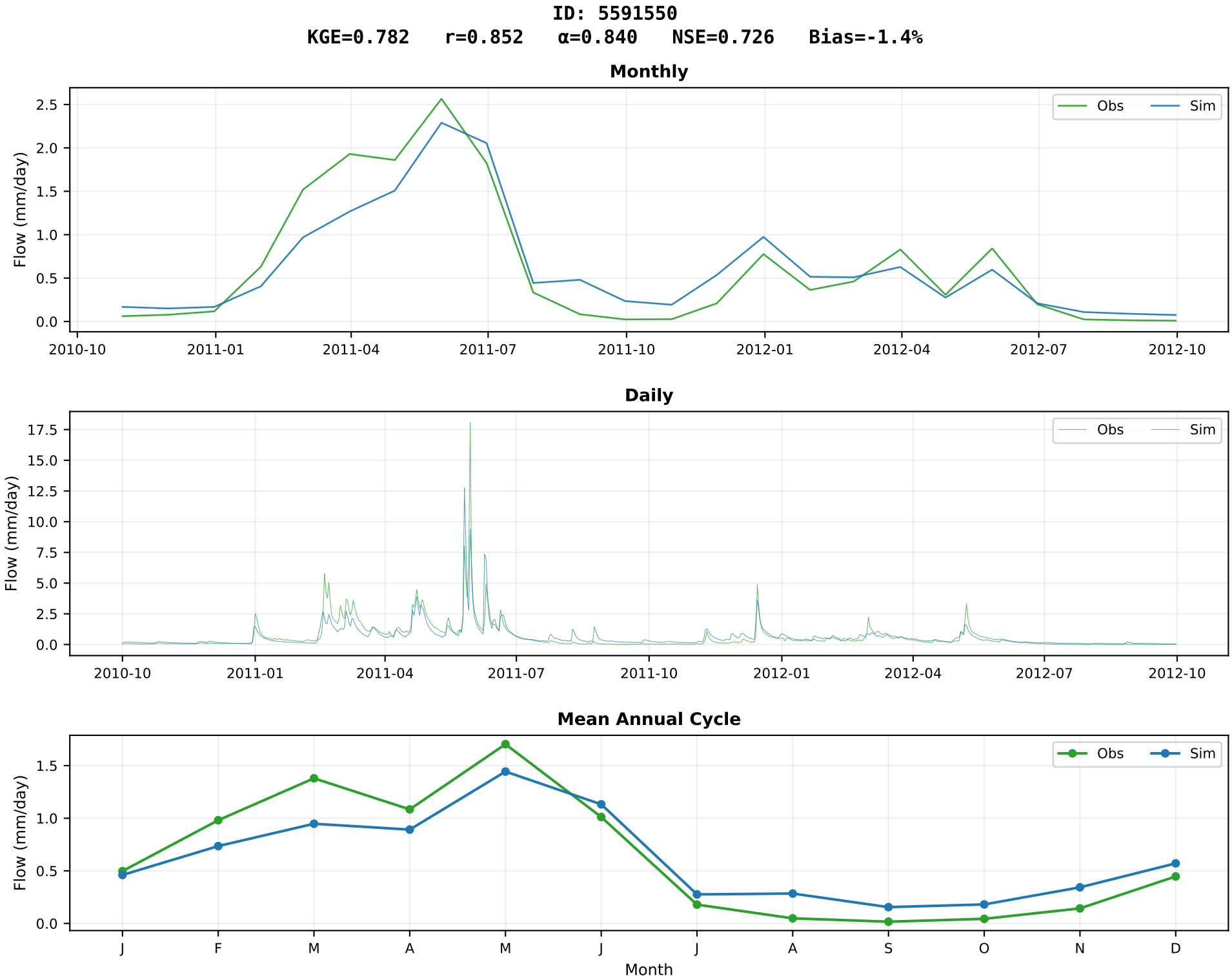

ID: 4057510
KGE=0.786 r=0.820 α=0.885 NSE=0.667 Bias=+0.8%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

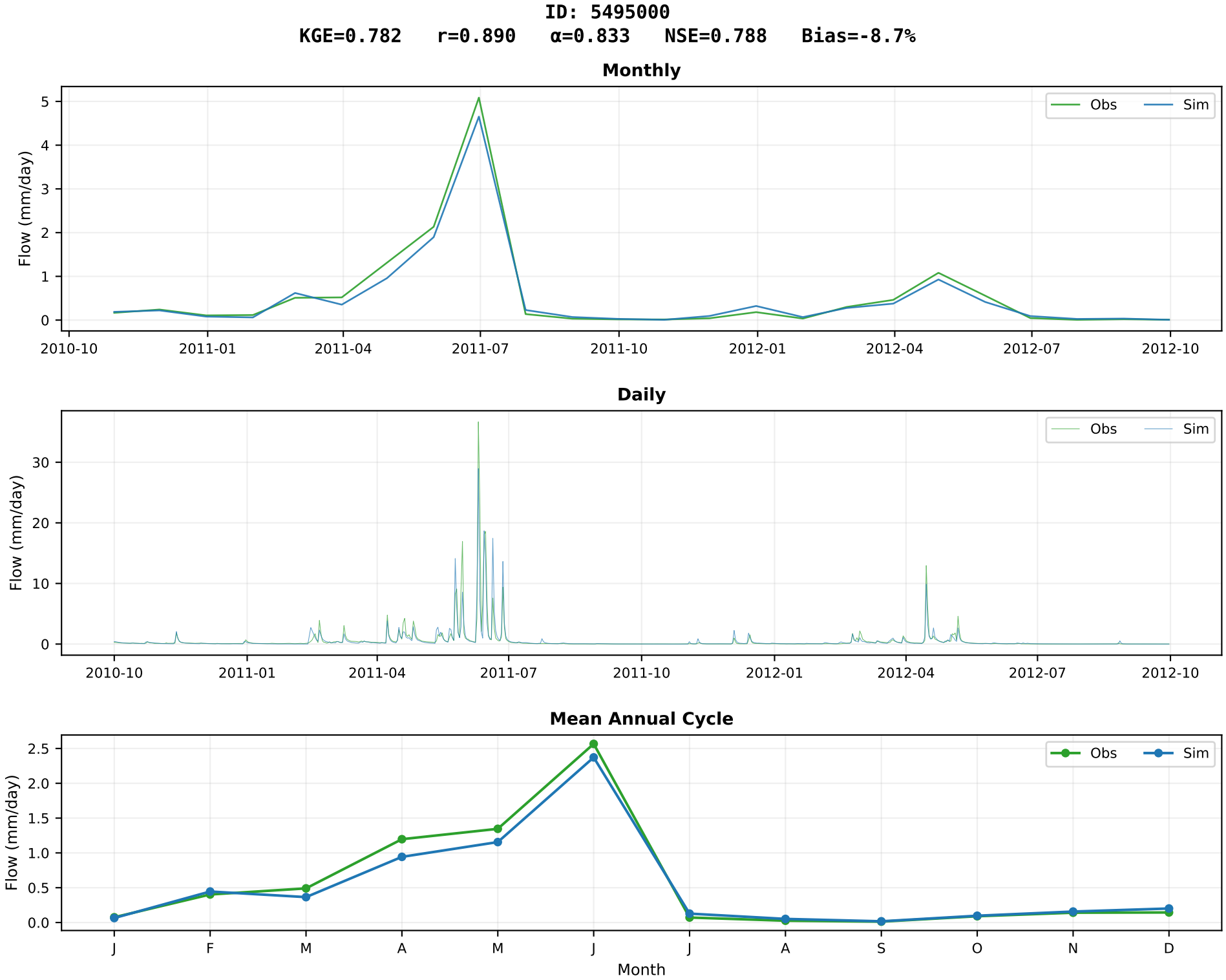

ID: 8013000
KGE=0.785 r=0.968 α=0.794 NSE=0.905 Bias=-5.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

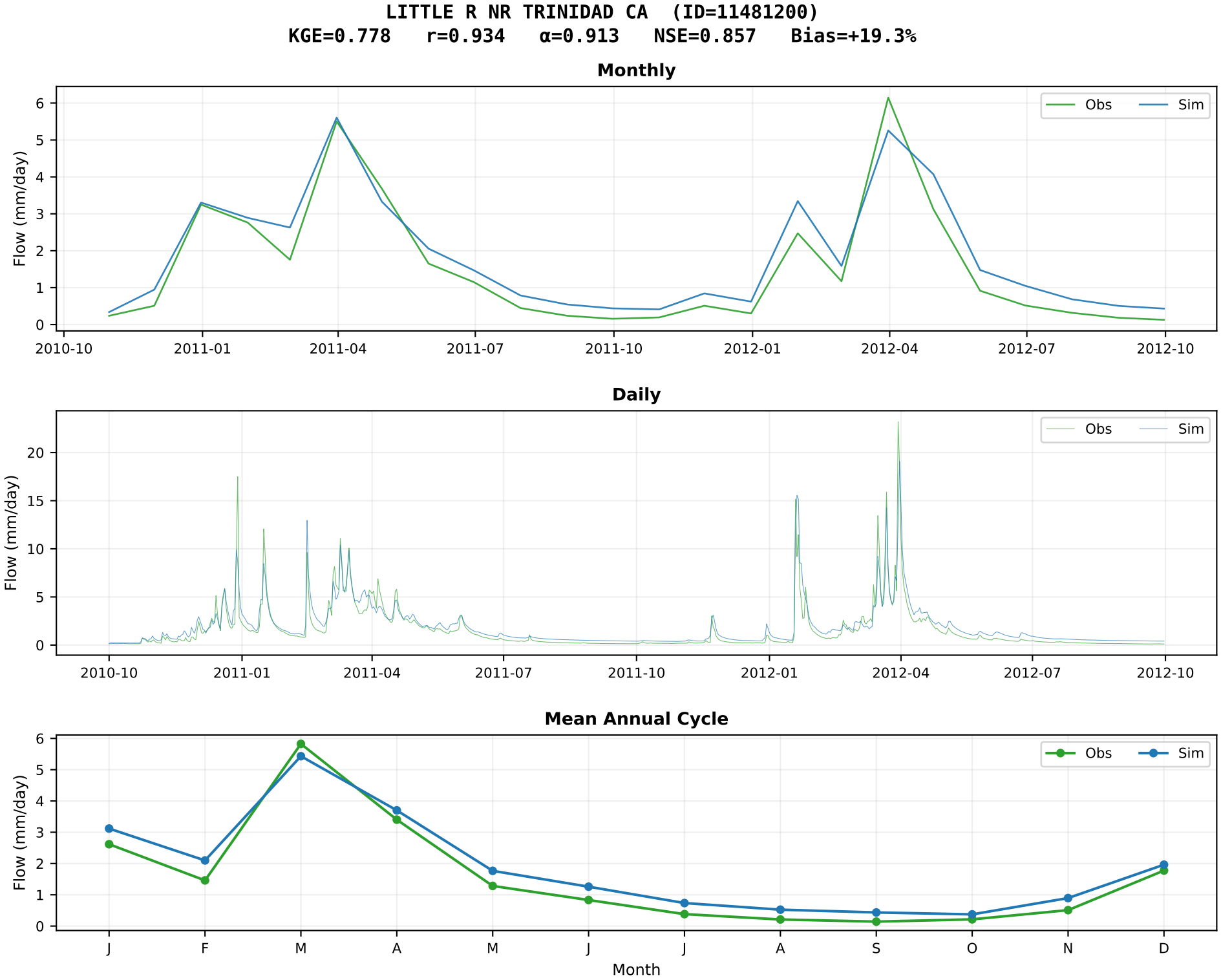

GENERAL C NR MEEKS BAY CA (ID=10336645)
KGE=0.784 r=0.926 α=0.824 NSE=0.845 Bias=-10.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

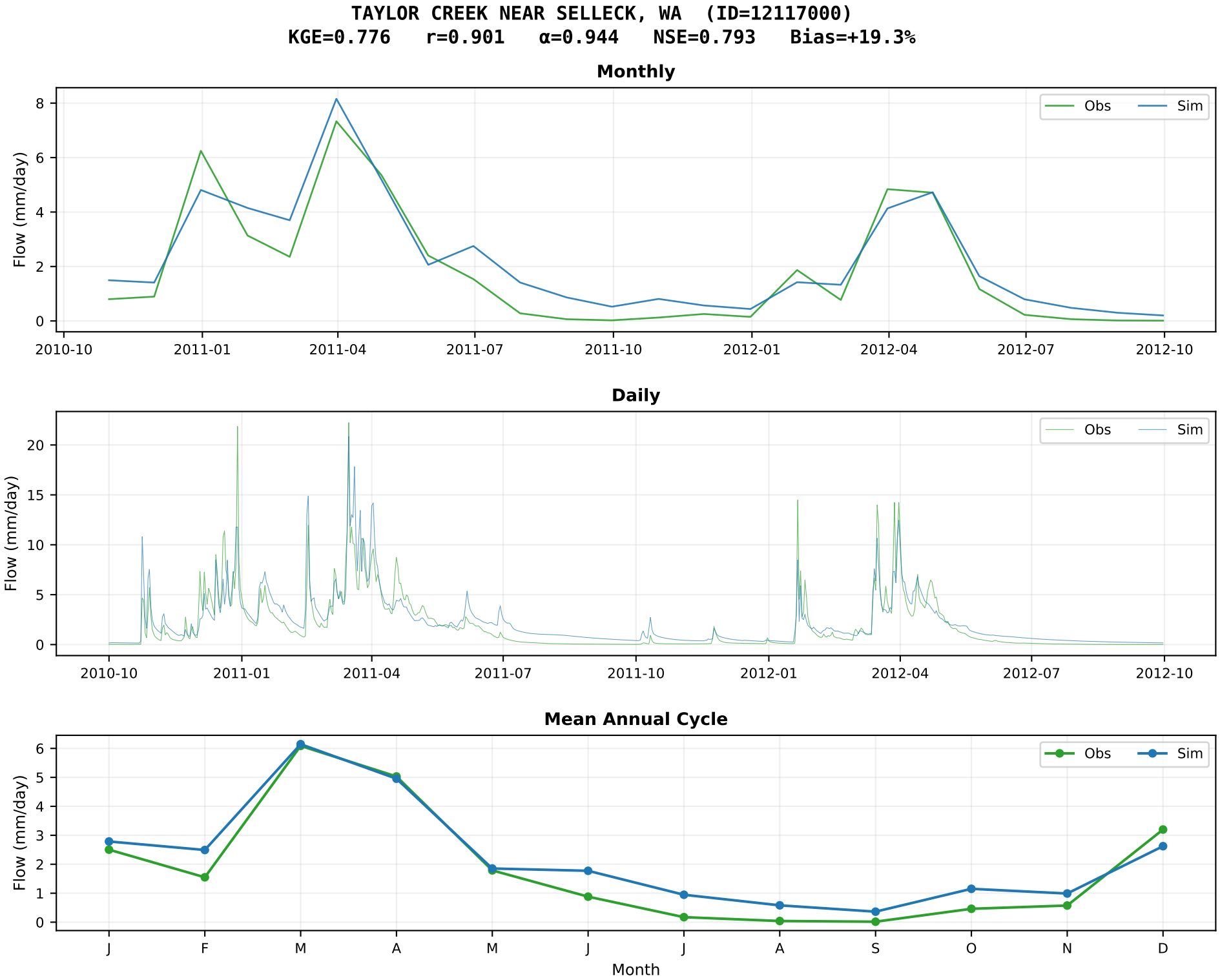

ID: 2056900
KGE=0.783 r=0.844 α=0.925 NSE=0.698 Bias=+13.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

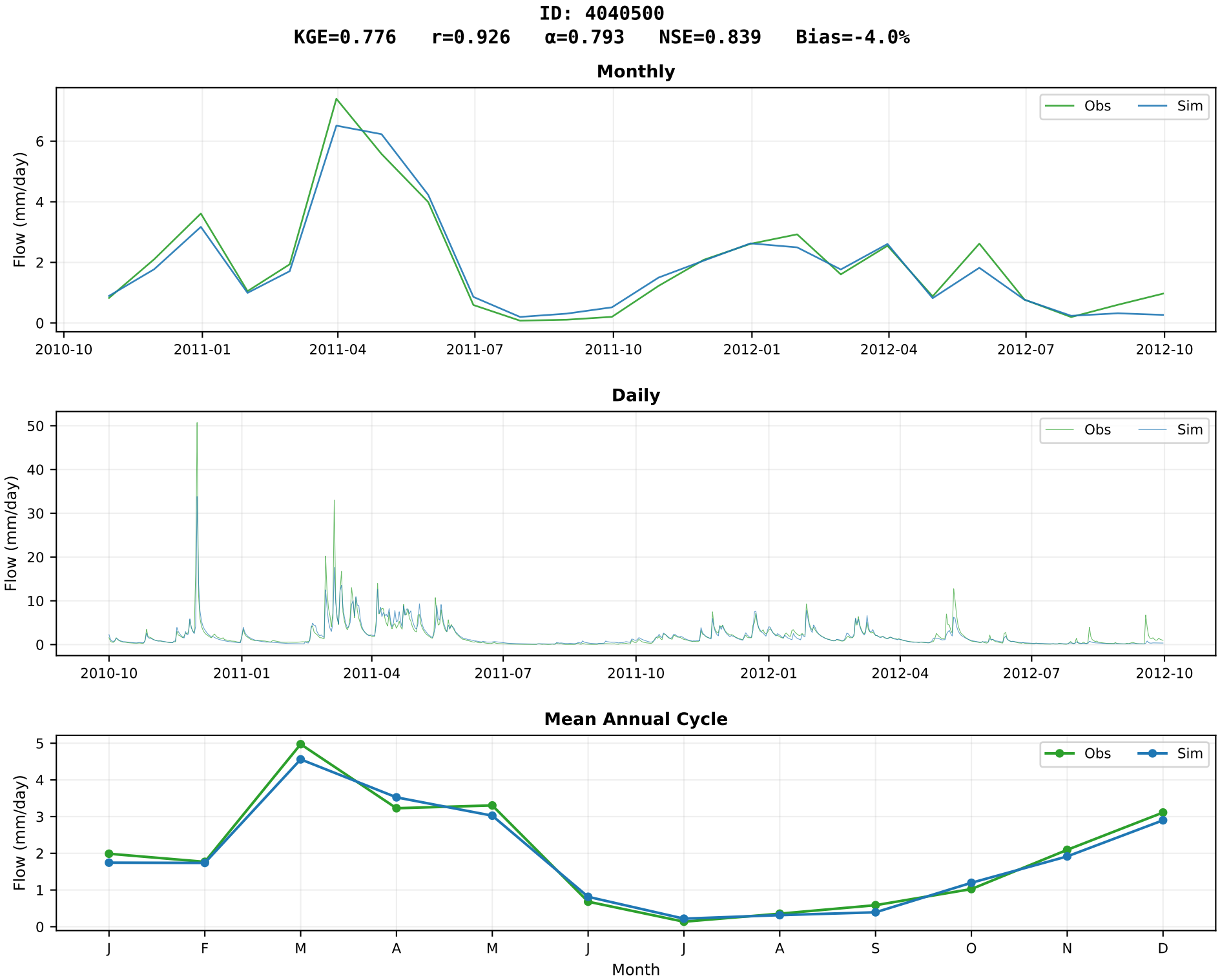

ID: 2415000
KGE=0.782 r=0.890 α=1.050 NSE=0.764 Bias=+18.1%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

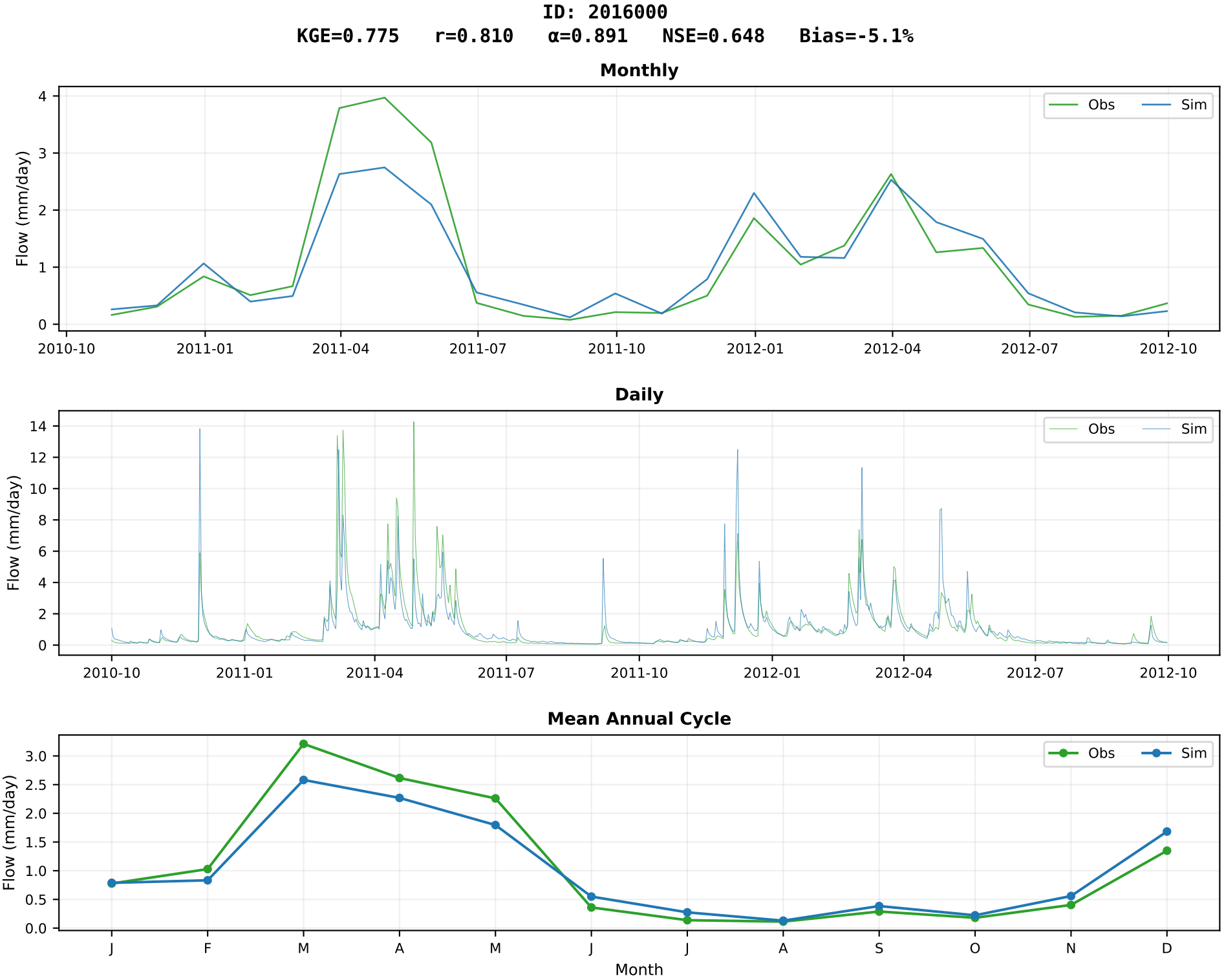

ID: 5591550
KGE=0.782 r=0.852 α=0.840 NSE=0.726 Bias=-1.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

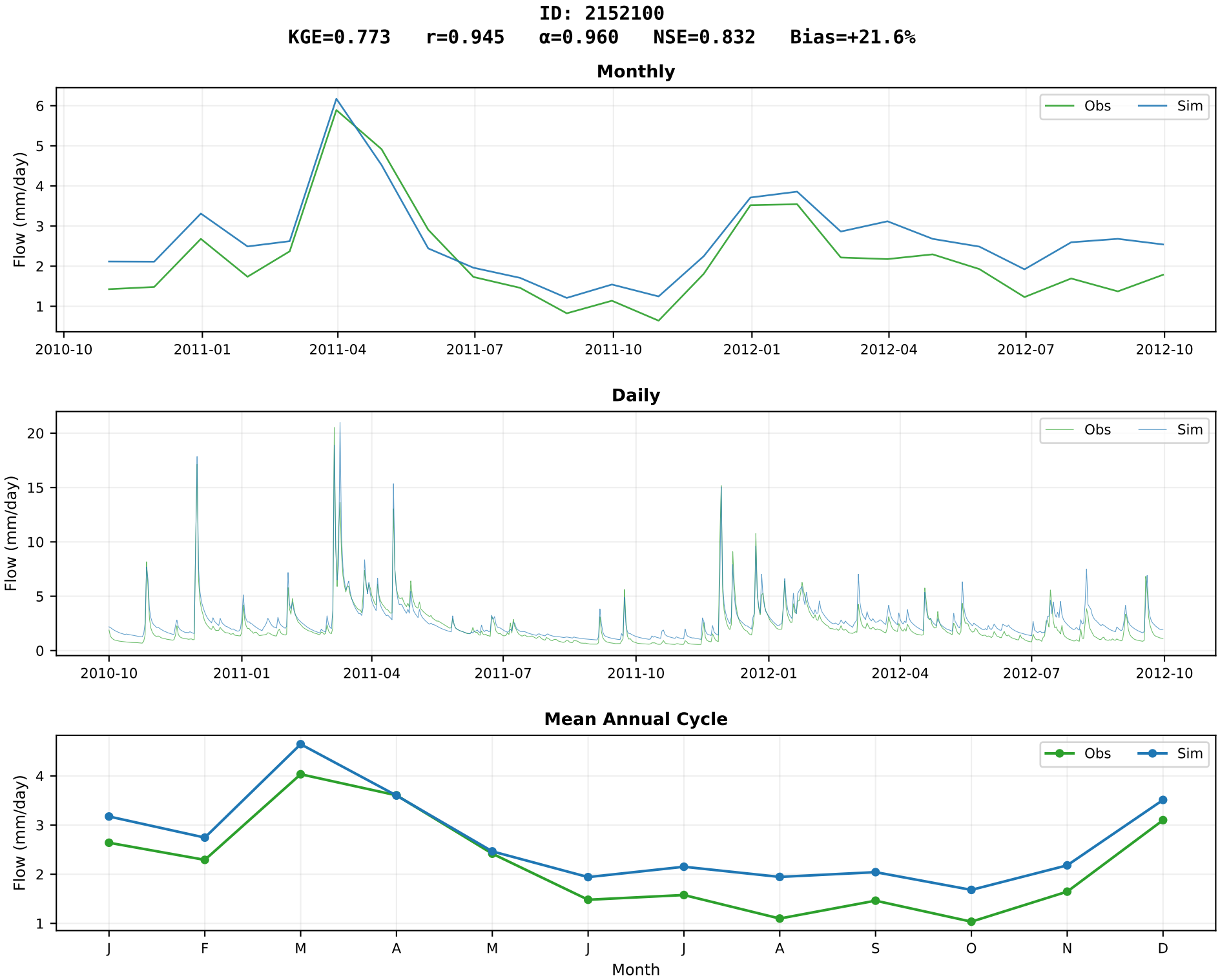

ID: 5495000
KGE=0.782 r=0.890 α=0.833 NSE=0.788 Bias=-8.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

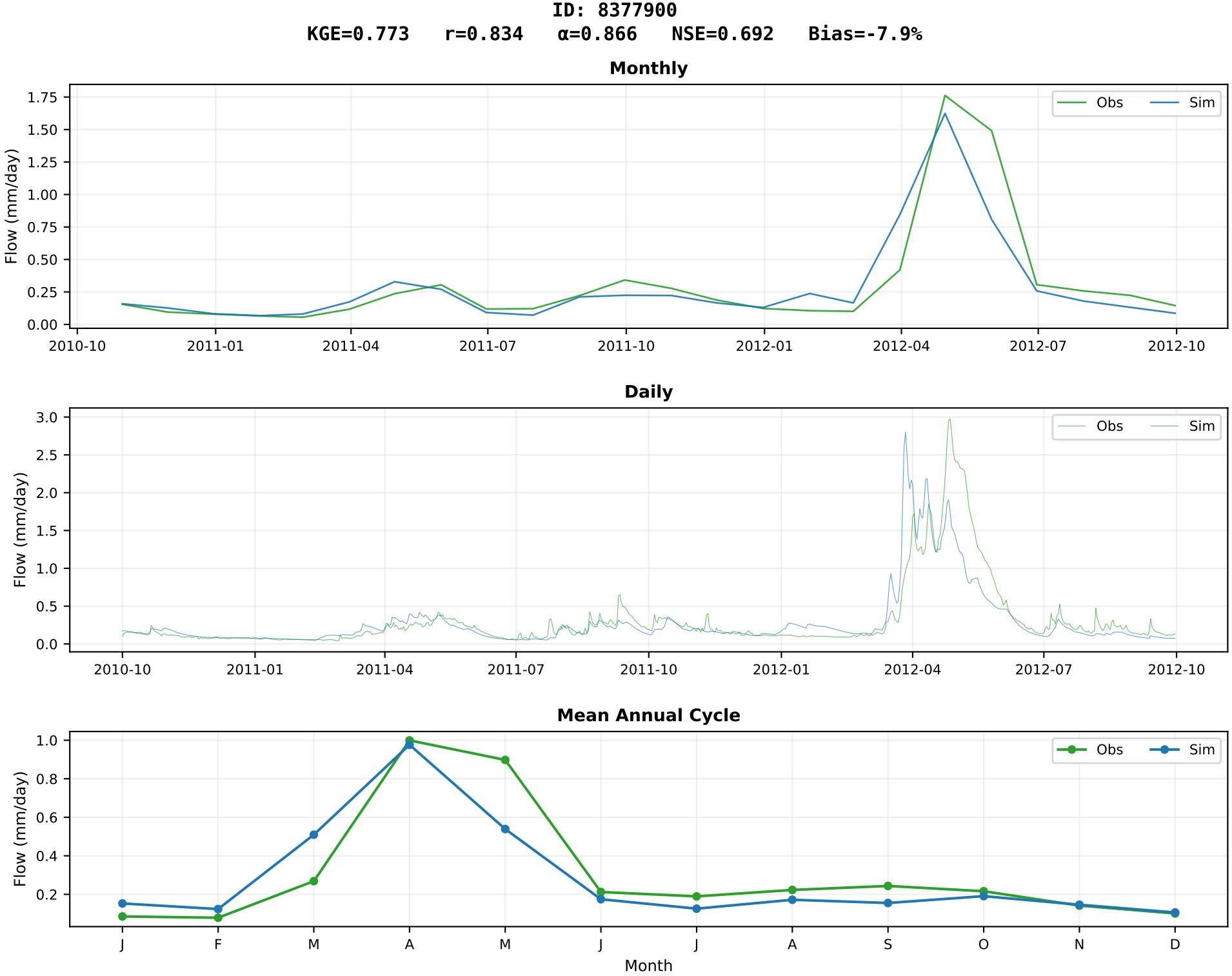

LITTLE R NR TRINIDAD CA (ID=11481200)
KGE=0.778 r=0.934 α=0.913 NSE=0.857 Bias=+19.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

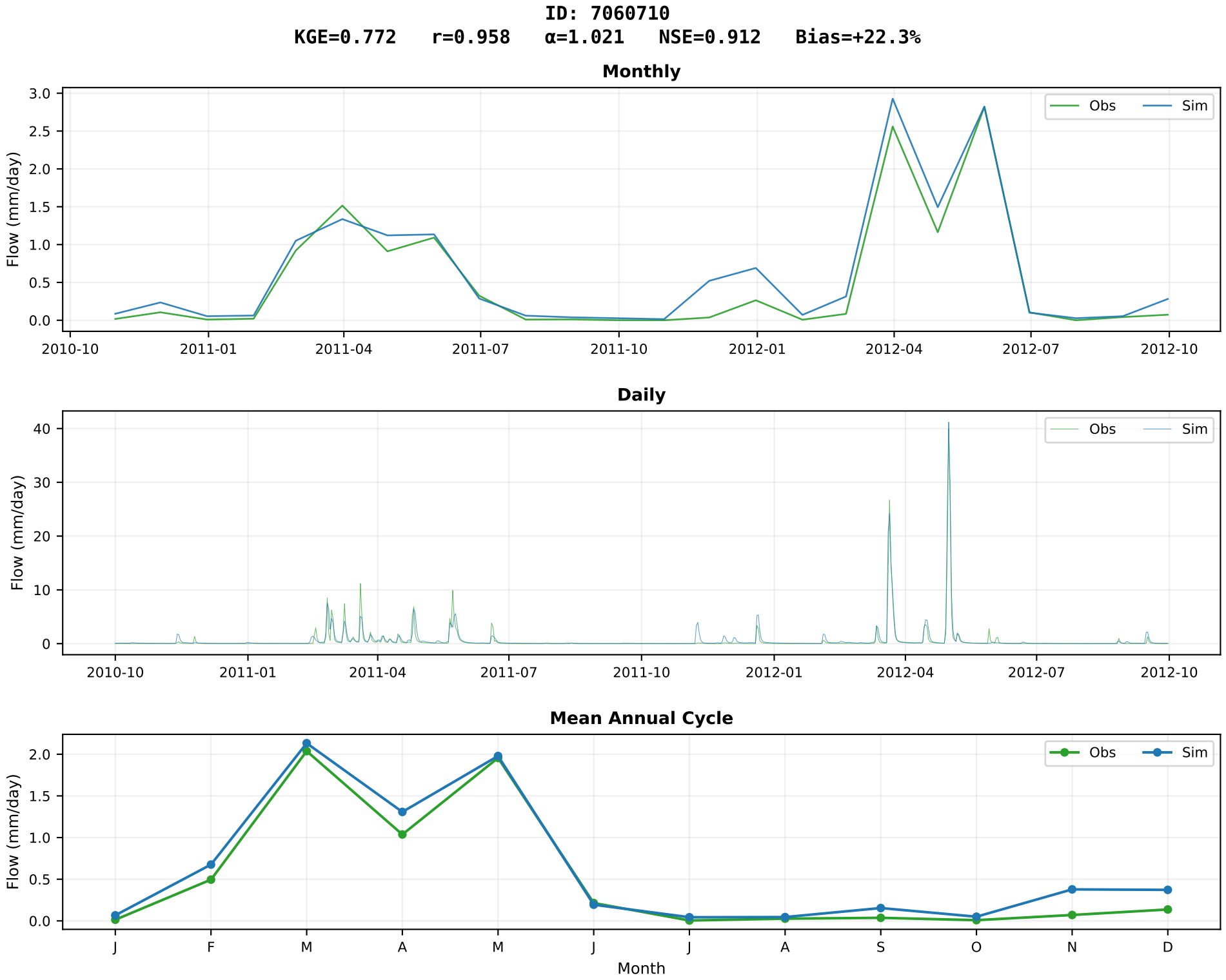

TAYLOR CREEK NEAR SELLECK, WA (ID=12117000)
KGE=0.776 r=0.901 α=0.944 NSE=0.793 Bias=+19.3%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

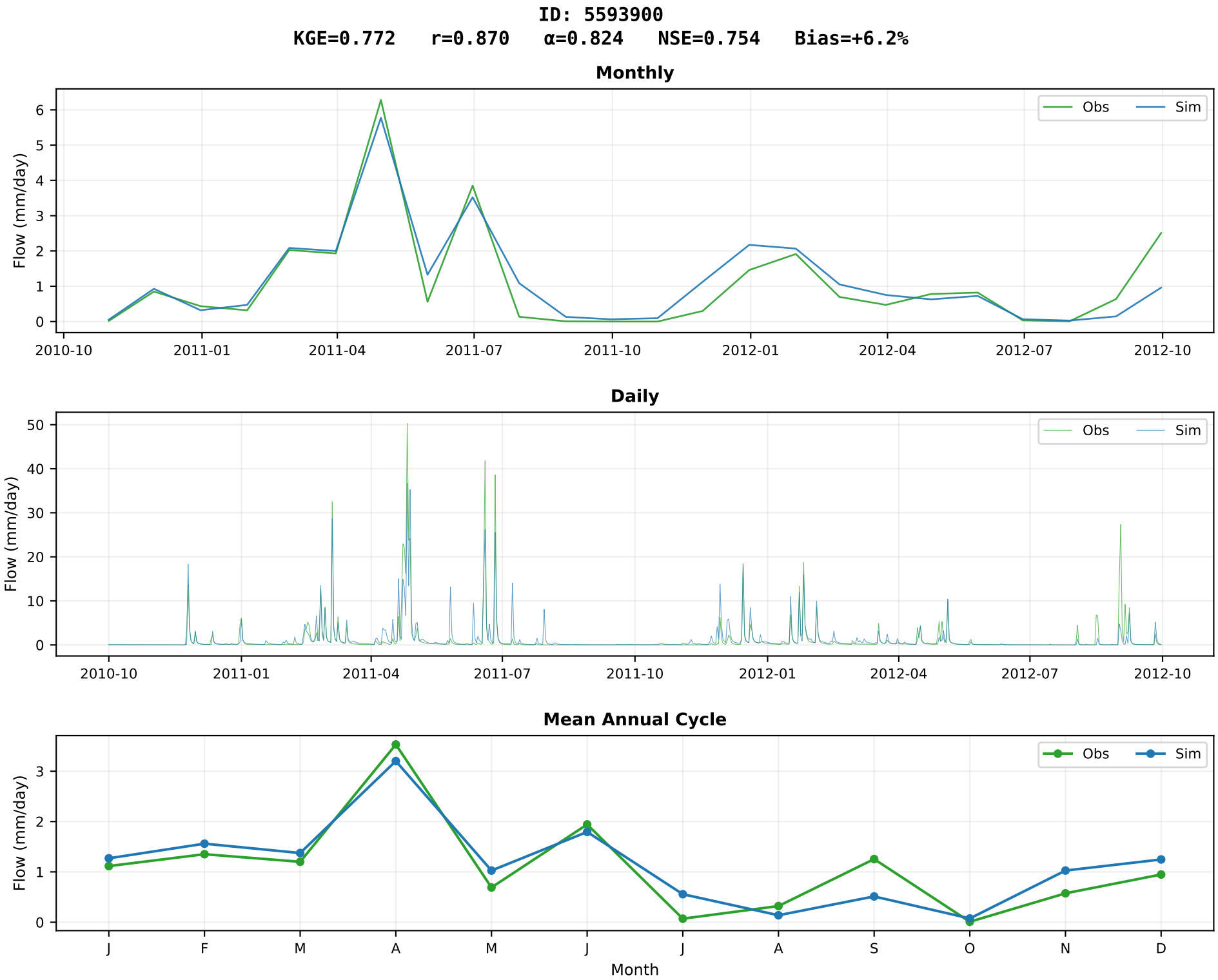

ID: 4040500
KGE=0.776 r=0.926 α=0.793 NSE=0.839 Bias=-4.0%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

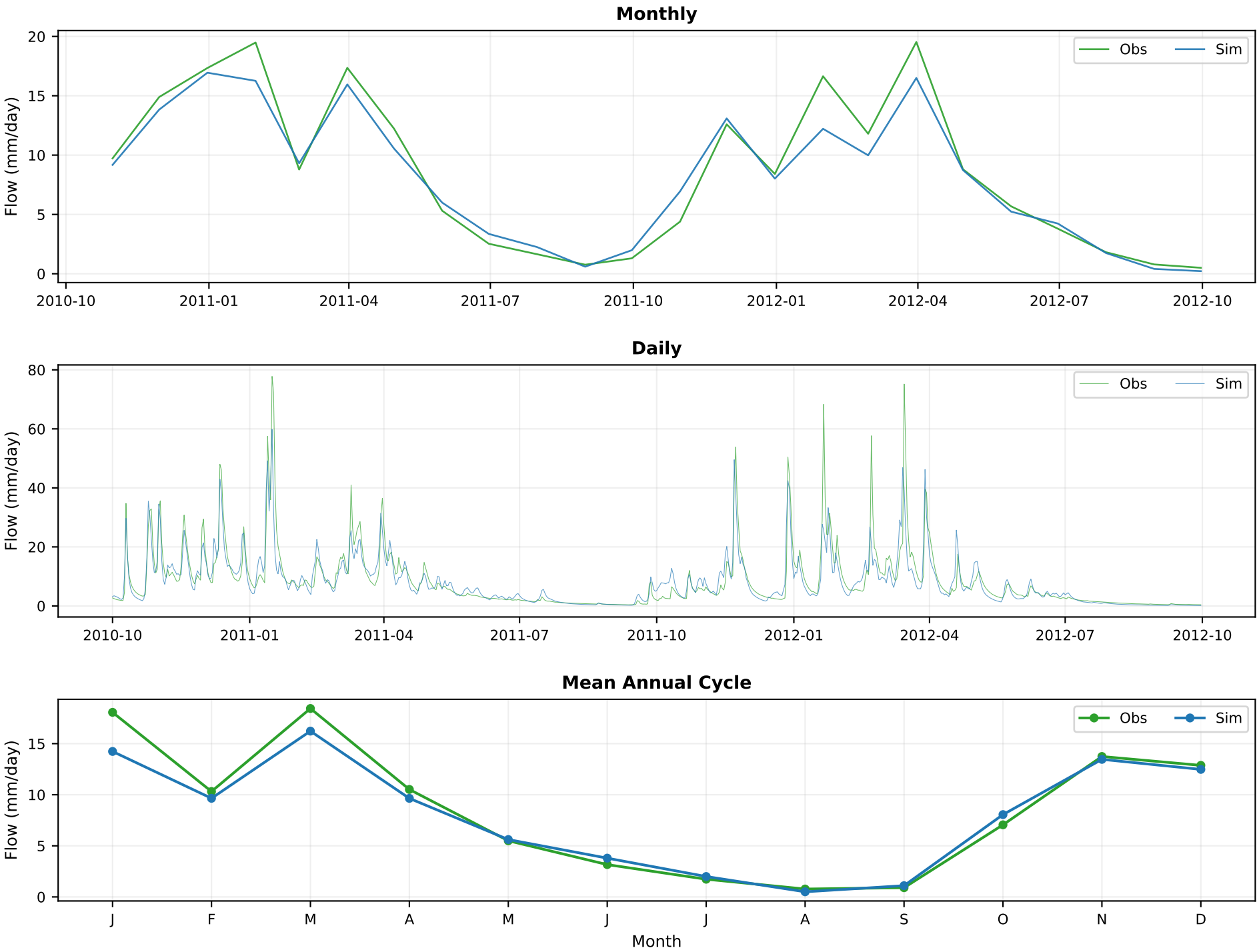

ID: 2016000
KGE=0.775 r=0.810 α=0.891 NSE=0.648 Bias=-5.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2152100
KGE=0.773 r=0.945 α=0.960 NSE=0.832 Bias=+21.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 8377900
KGE=0.773 r=0.834 α=0.866 NSE=0.692 Bias=-7.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 7060710
KGE=0.772 r=0.958 α=1.021 NSE=0.912 Bias=+22.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

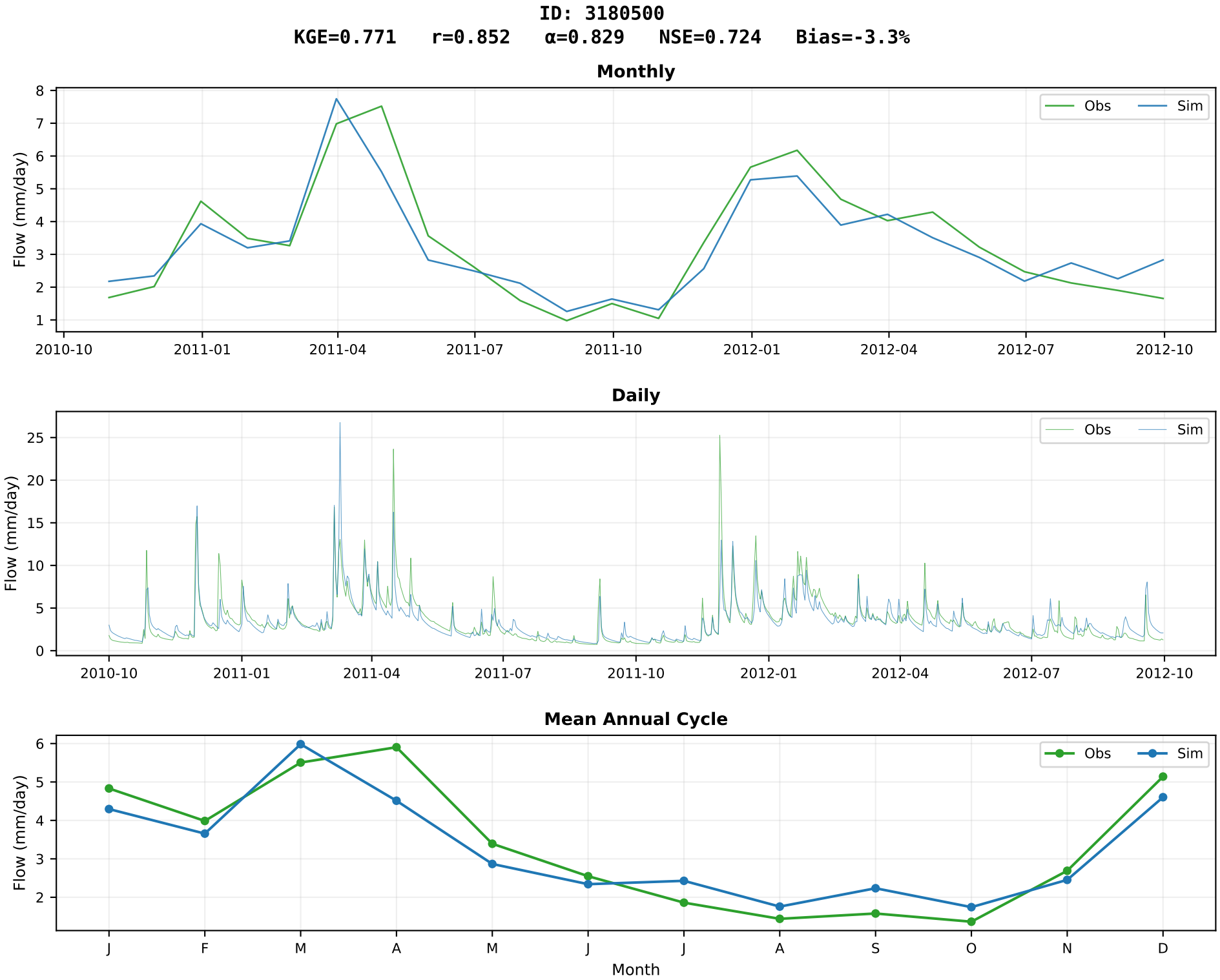

ID: 5593900
KGE=0.772 r=0.870 α=0.824 NSE=0.754 Bias=+6.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

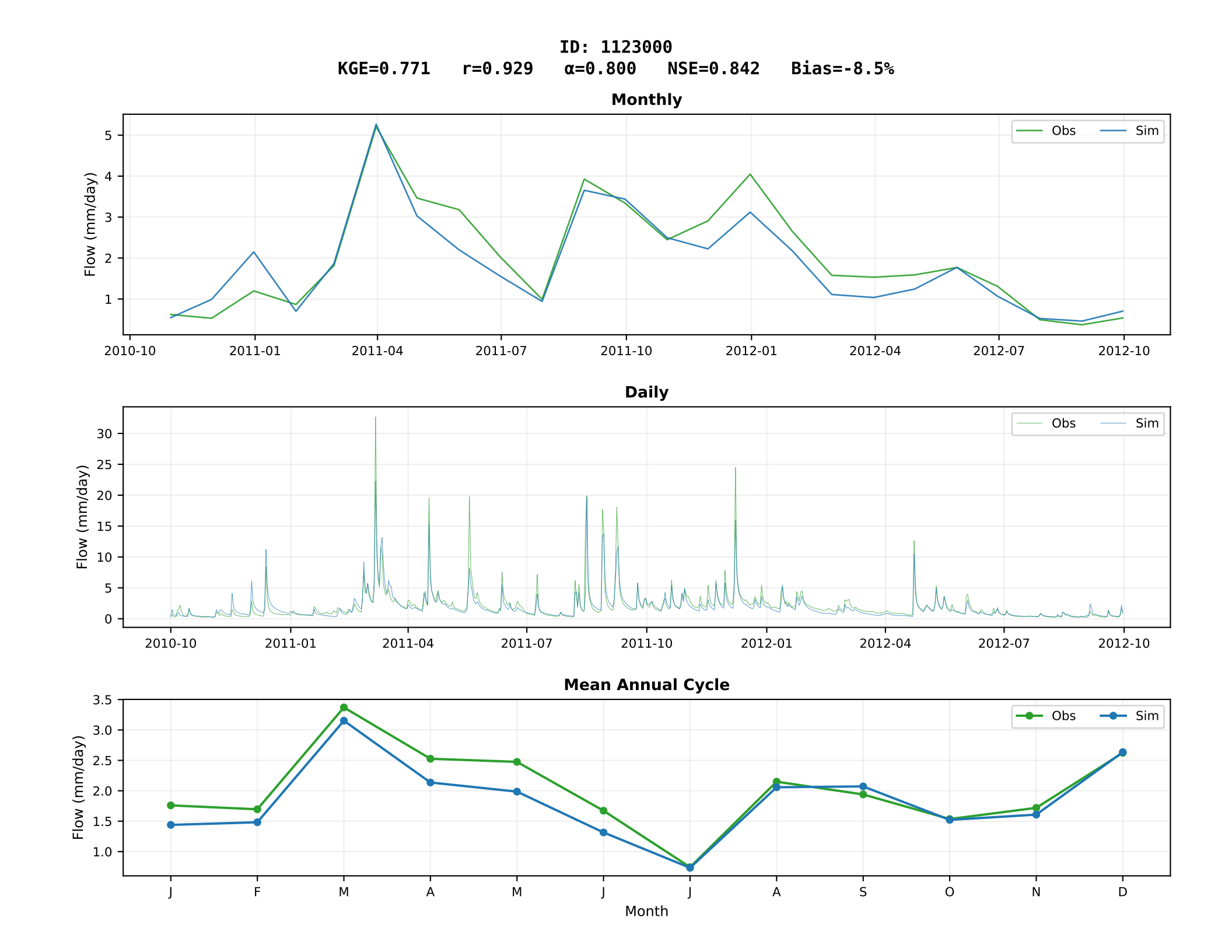

Mission Cr ab Reservoir nr ST Ignatius MT (ID=12377150)
KGE=0.771 r=0.888 α=0.810 NSE=0.780 Bias=-6.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

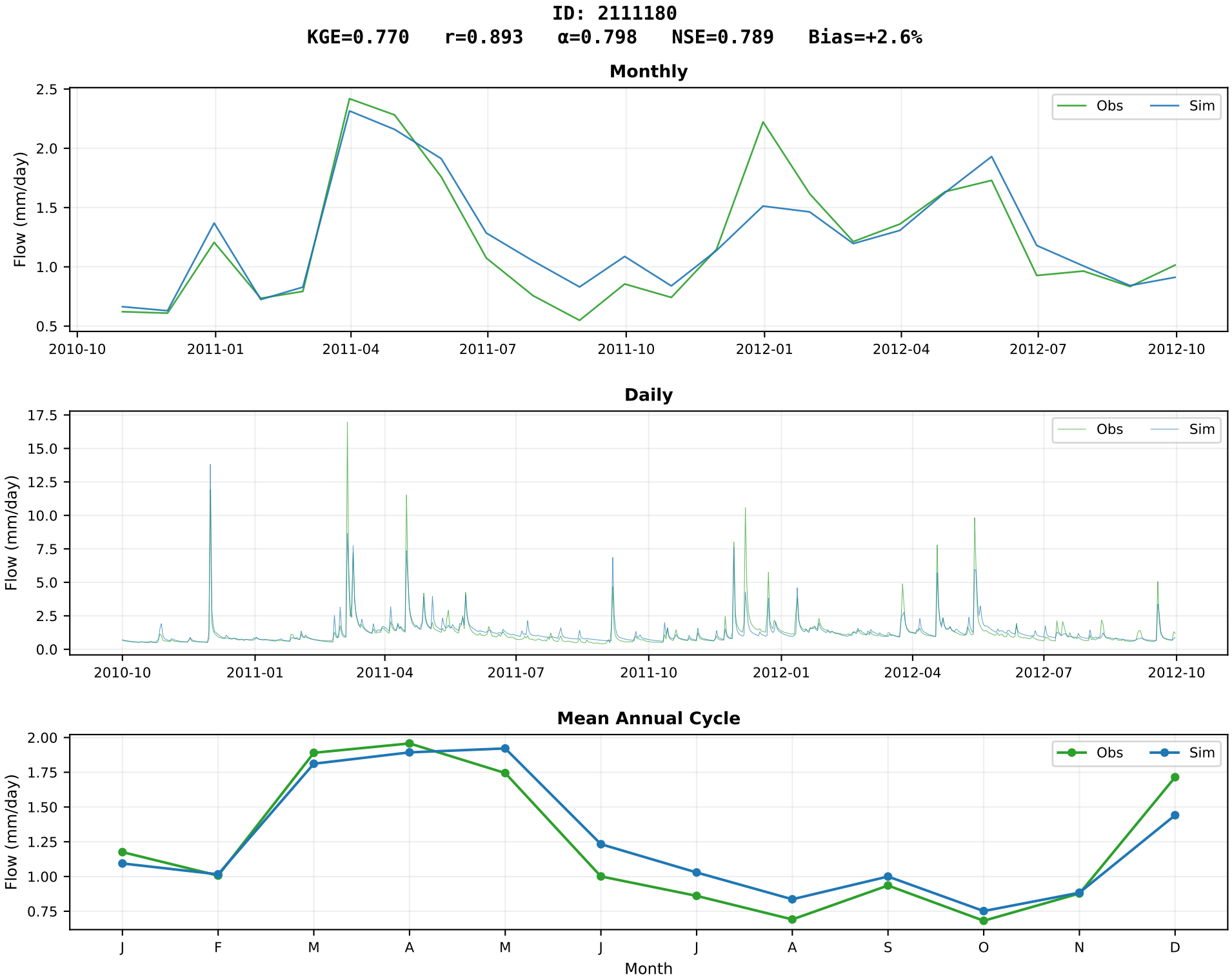
ID: 3368000

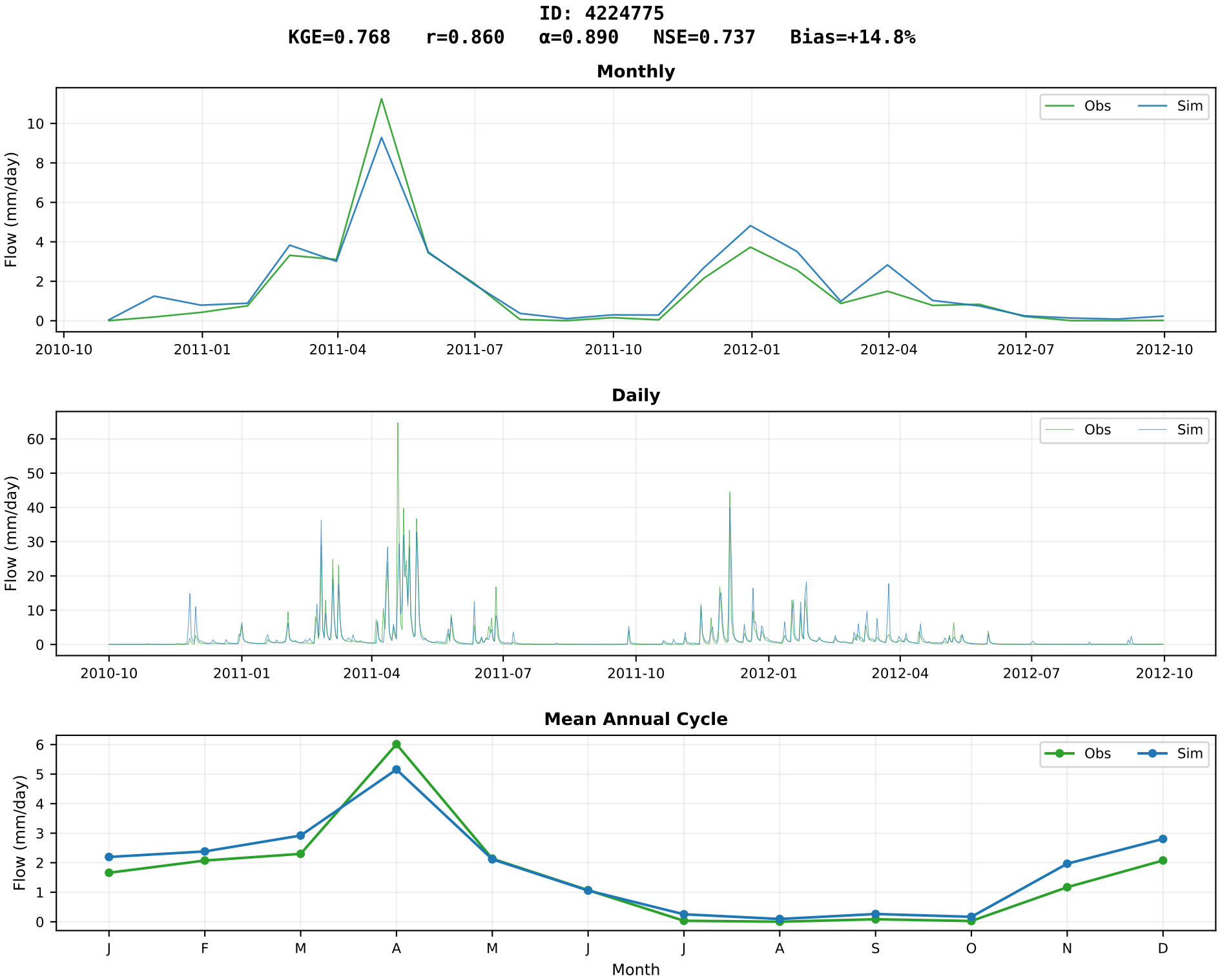
KGE=0.771 r=0.899 α=1.008 NSE=0.678 Bias=+20.5%

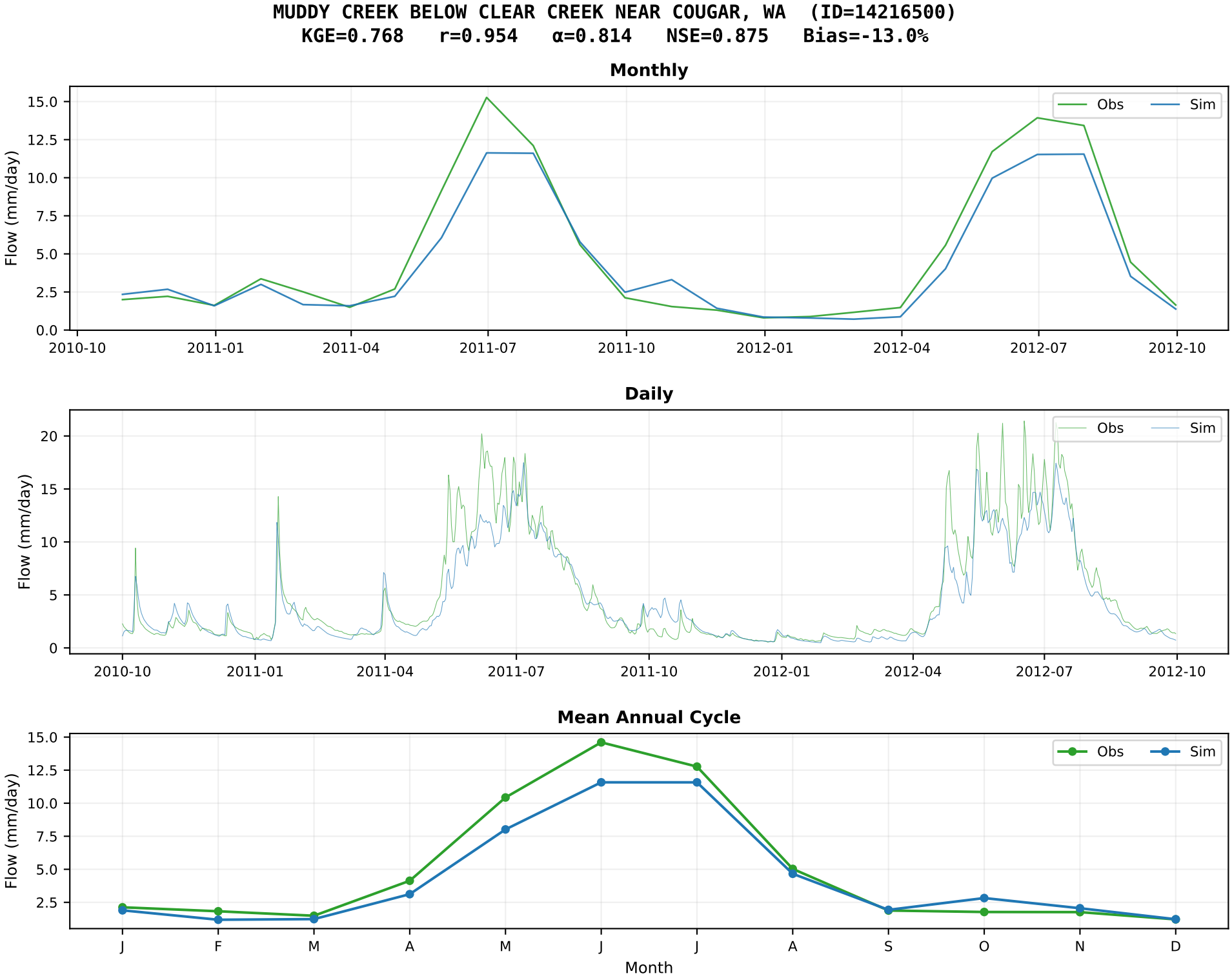


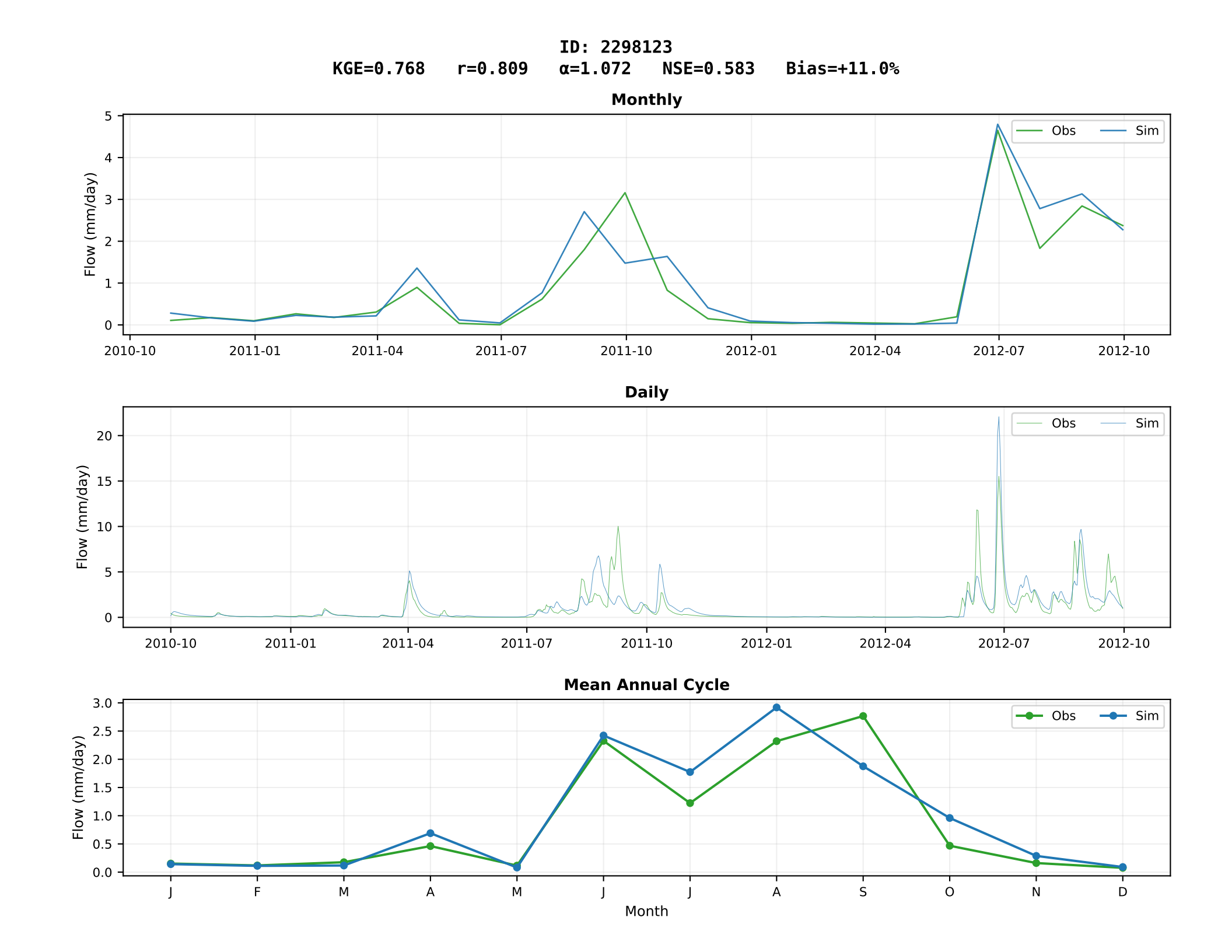

ID: 3180500
KGE=0.771 r=0.852 α=0.829 NSE=0.724 Bias=-3.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

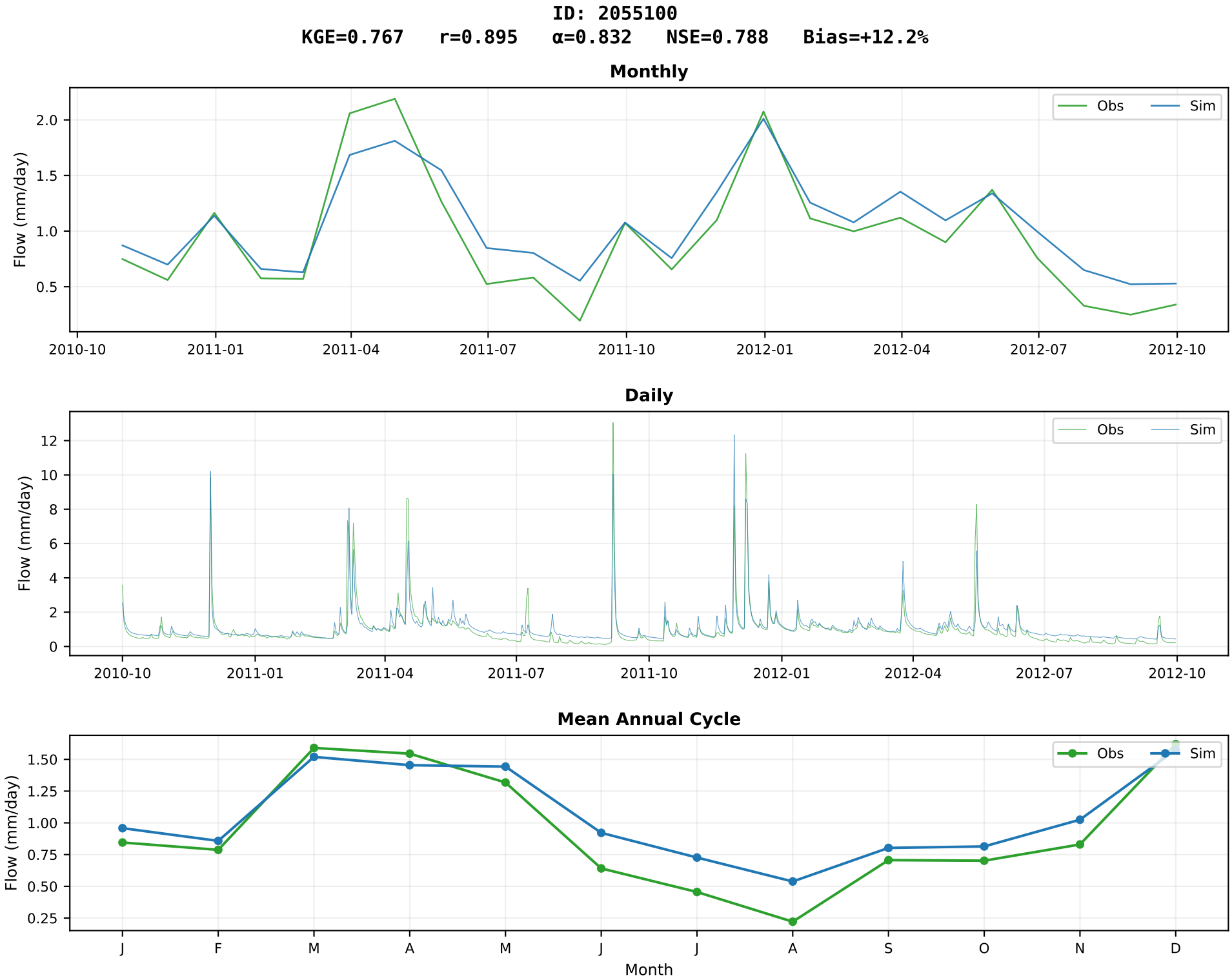

ID: 1123000
KGE=0.771 r=0.929 α=0.800 NSE=0.842 Bias=-8.5%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

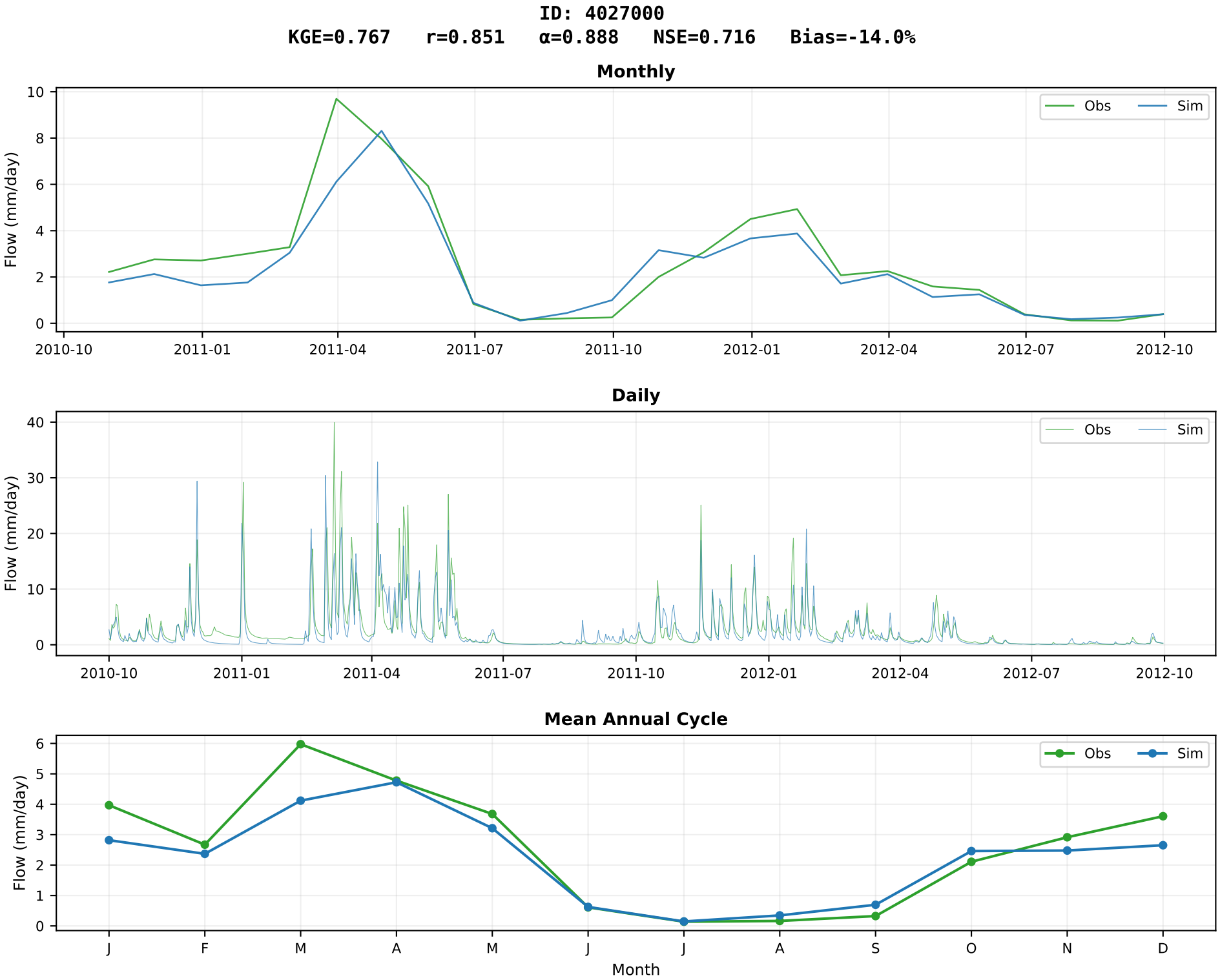

ID: 2111180
KGE=0.770 r=0.893 α=0.798 NSE=0.789 Bias=+2.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

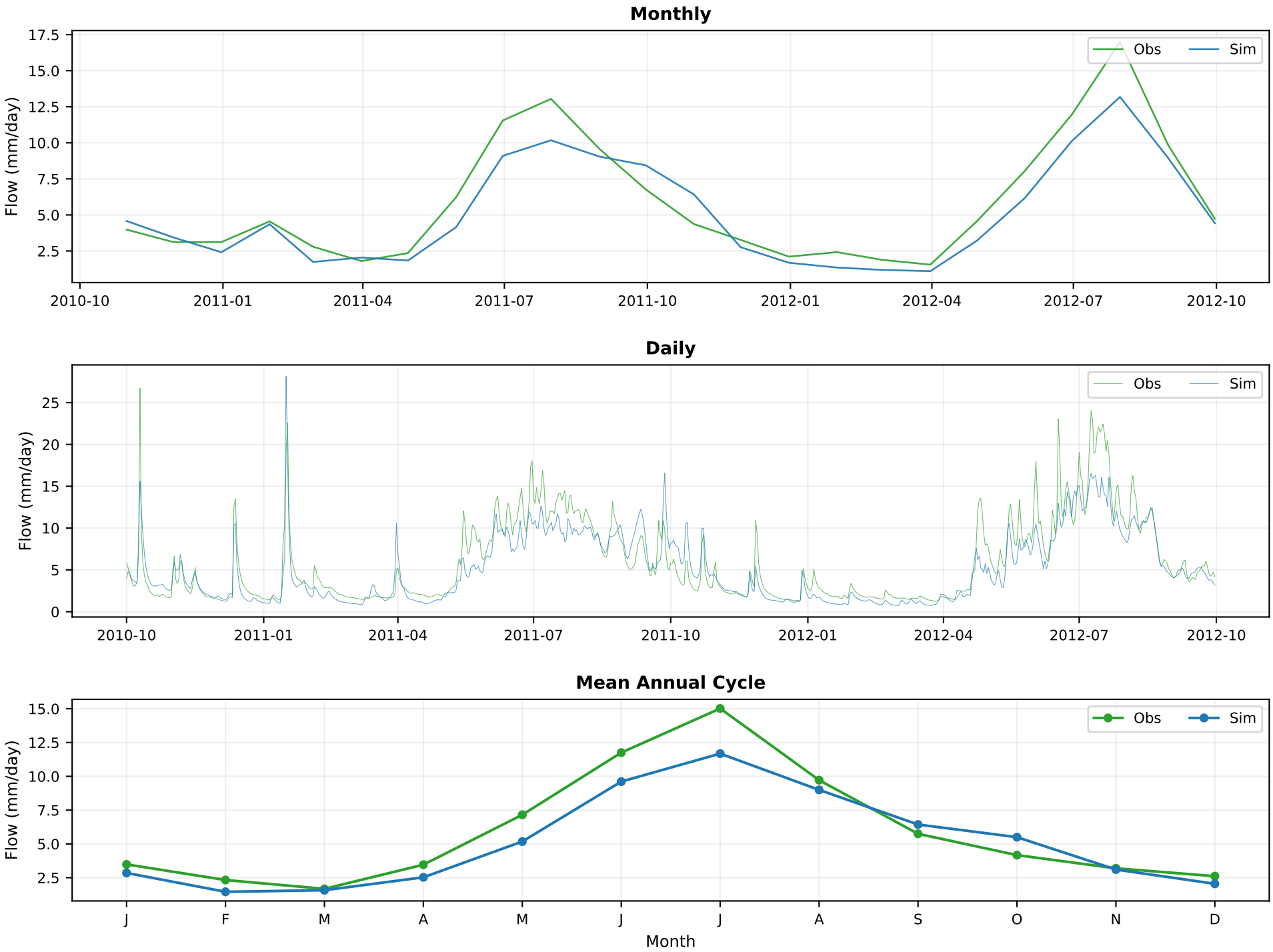

ID: 4224775
KGE=0.768 r=0.860 α=0.890 NSE=0.737 Bias=+14.8%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
J
F
M
A
M
J
J
A
S
O
N
D
Month

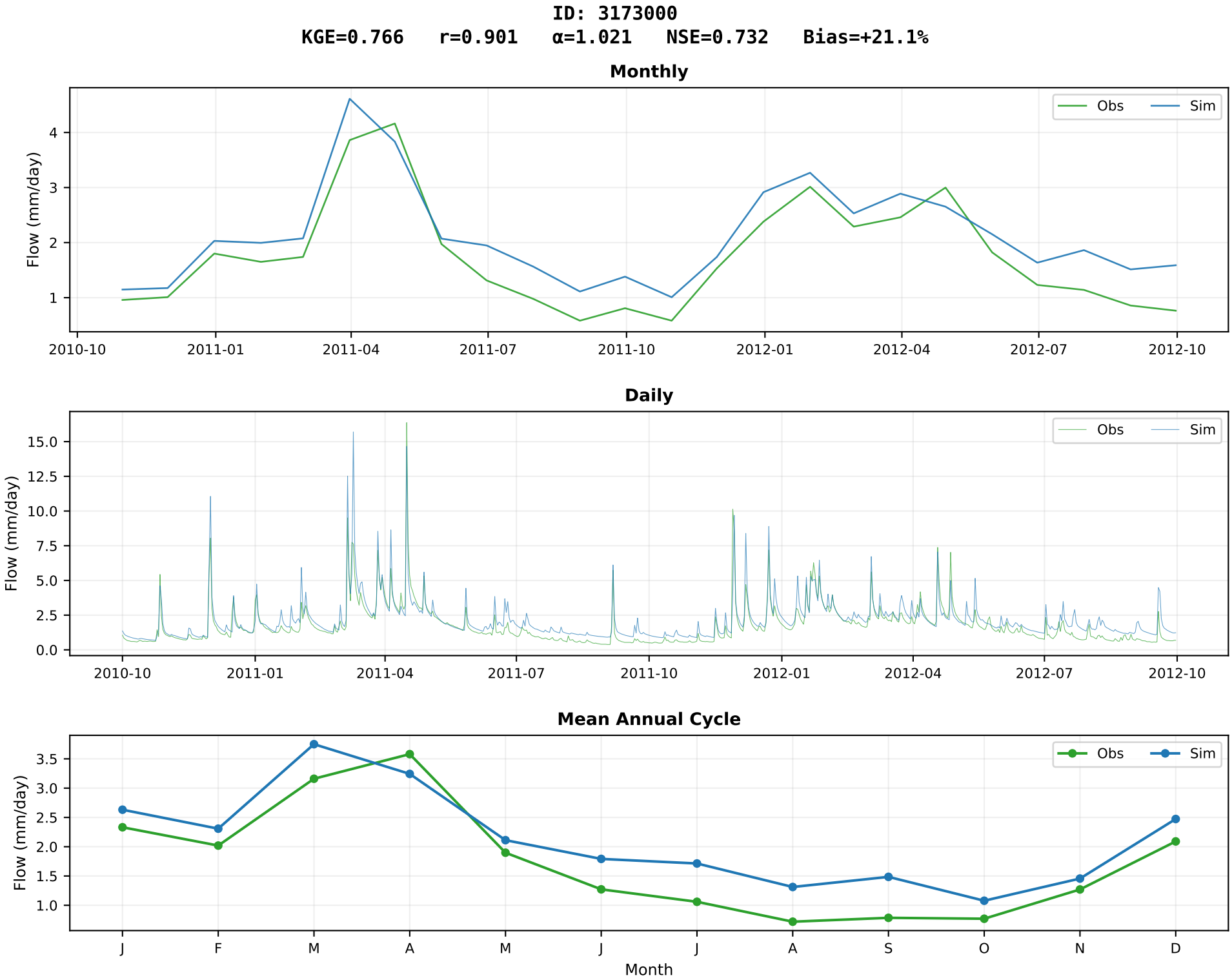

MUDDY CREEK BELOW CLEAR CREEK NEAR COUGAR, WA (ID=14216500)
KGE=0.768 r=0.954 α=0.814 NSE=0.875 Bias=-13.0%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

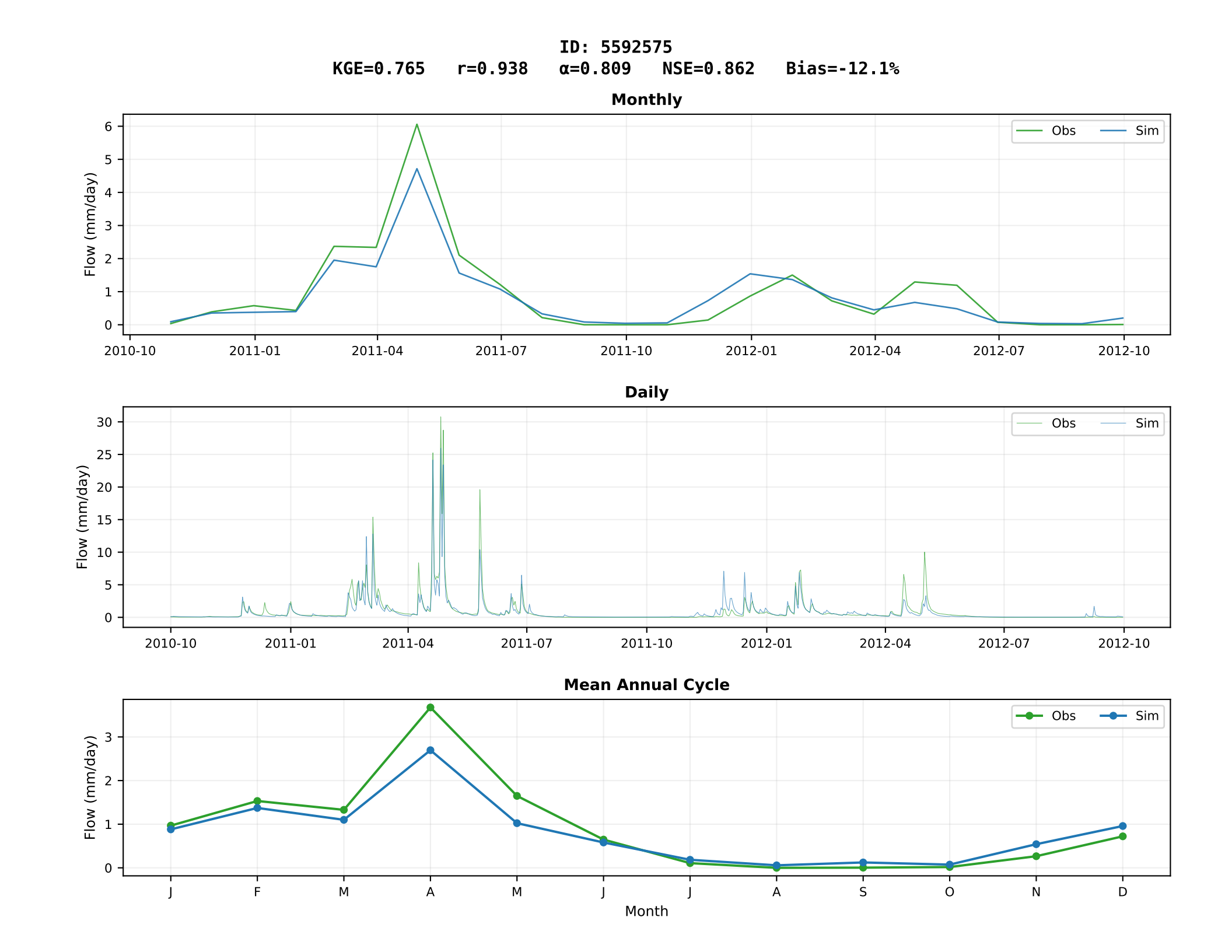

ID: 2298123
KGE=0.768 r=0.809 α=1.072 NSE=0.583 Bias=+11.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

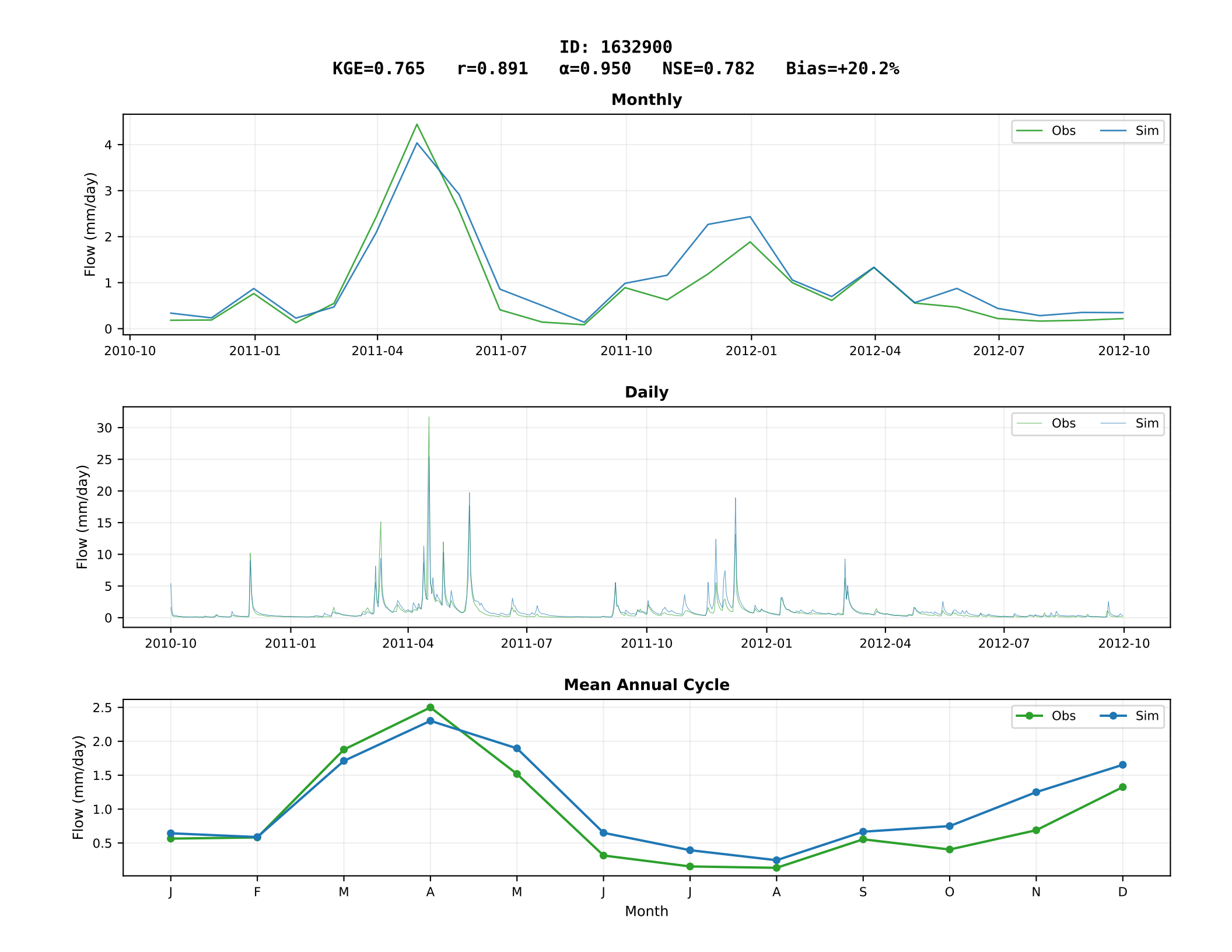

ID: 2055100
KGE=0.767 r=0.895 α=0.832 NSE=0.788 Bias=+12.2%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
12
10
8
6
4
2
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.50
1.25
1.00
0.75
0.50
0.25
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 4027000
KGE=0.767 r=0.851 α=0.888 NSE=0.716 Bias=-14.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

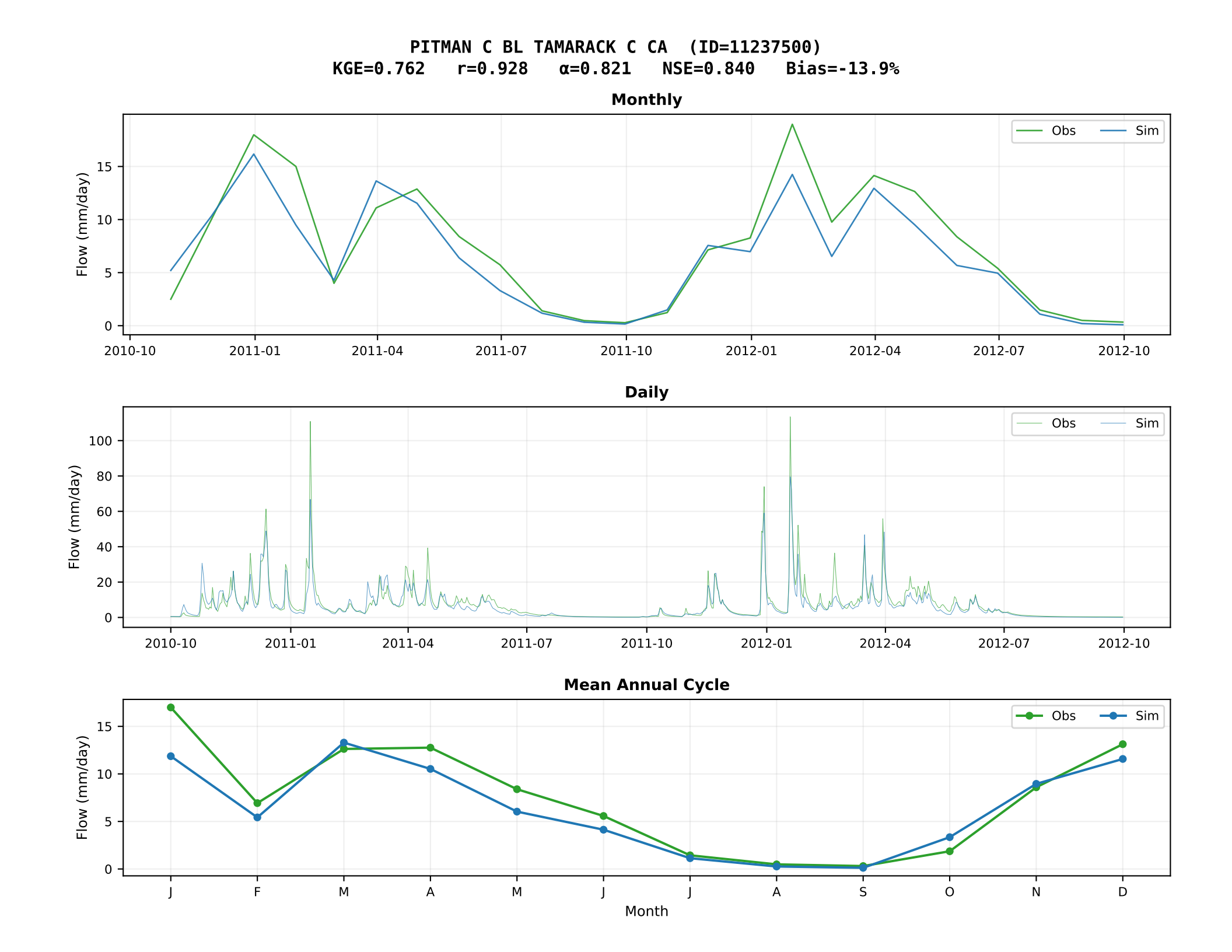

FIR CREEK NEAR BRIGHTWOOD, OR (ID=14138870)
KGE=0.767 r=0.911 α=0.830 NSE=0.797 Bias=-13.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

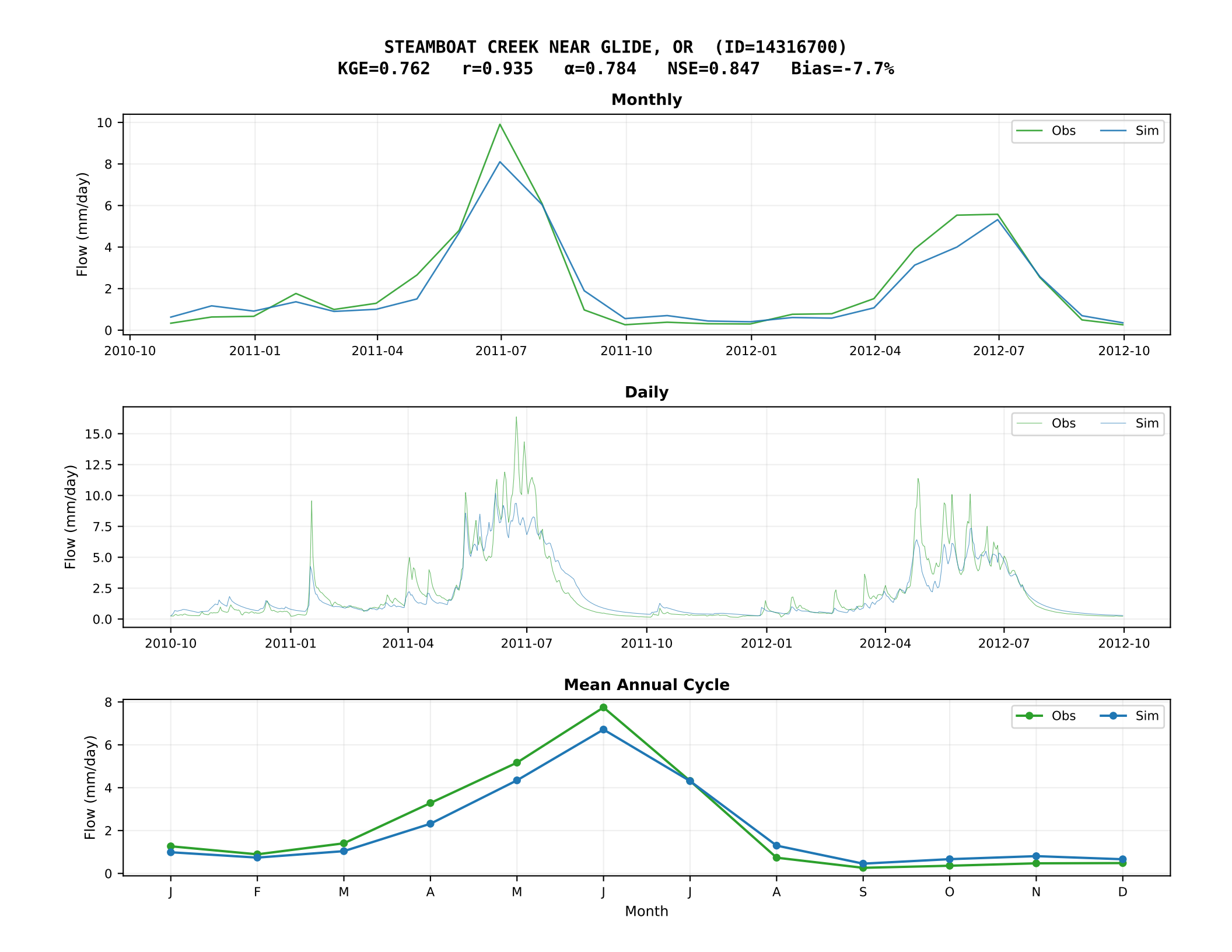

ID: 3173000
KGE=0.766 r=0.901 α=1.021 NSE=0.732 Bias=+21.1%
Monthly
Obs
Sim
Flow (mm/day)
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.0
1.5
2.0
2.5
3.0
3.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

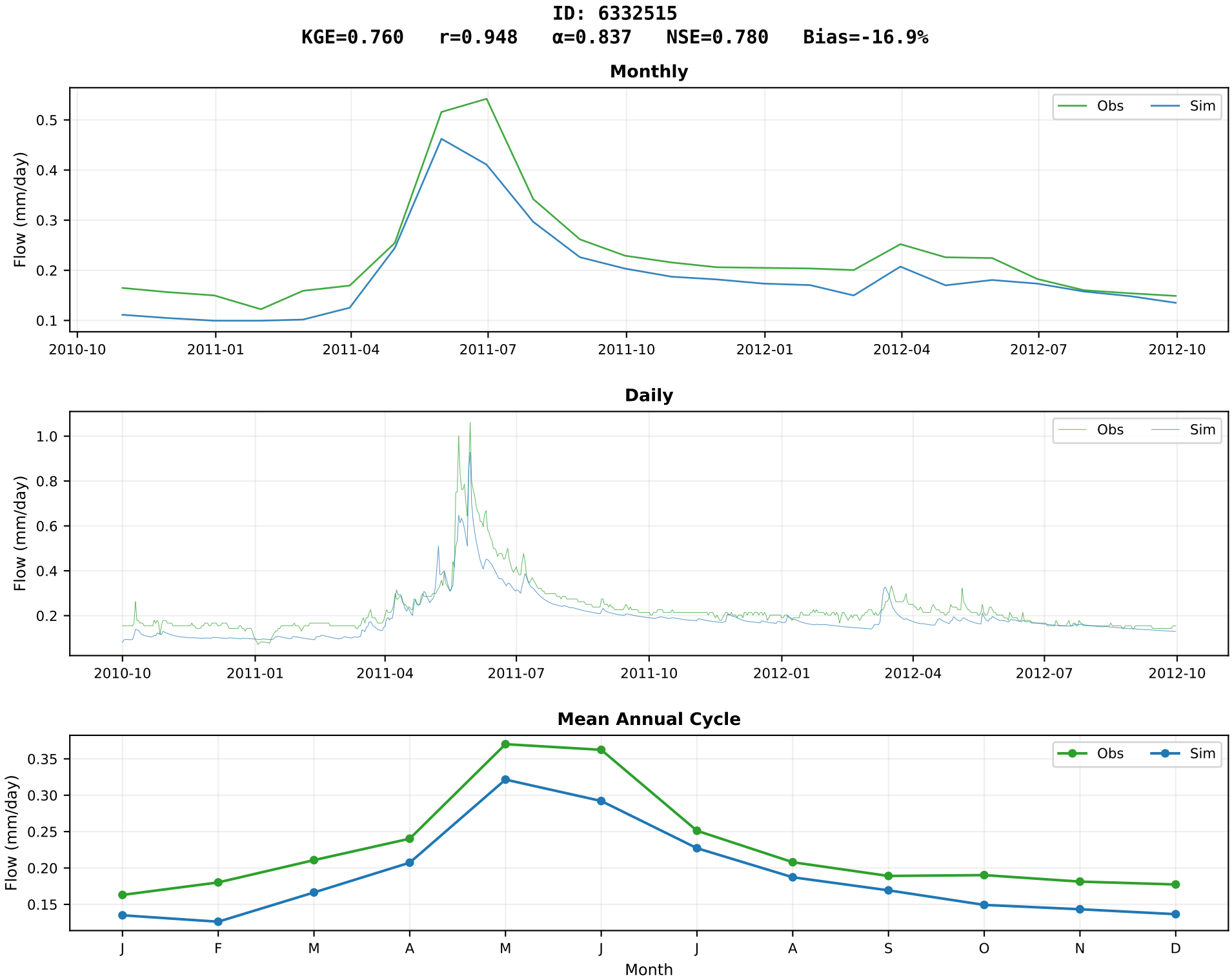

ID: 5592575
KGE=0.765 r=0.938 α=0.809 NSE=0.862 Bias=-12.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

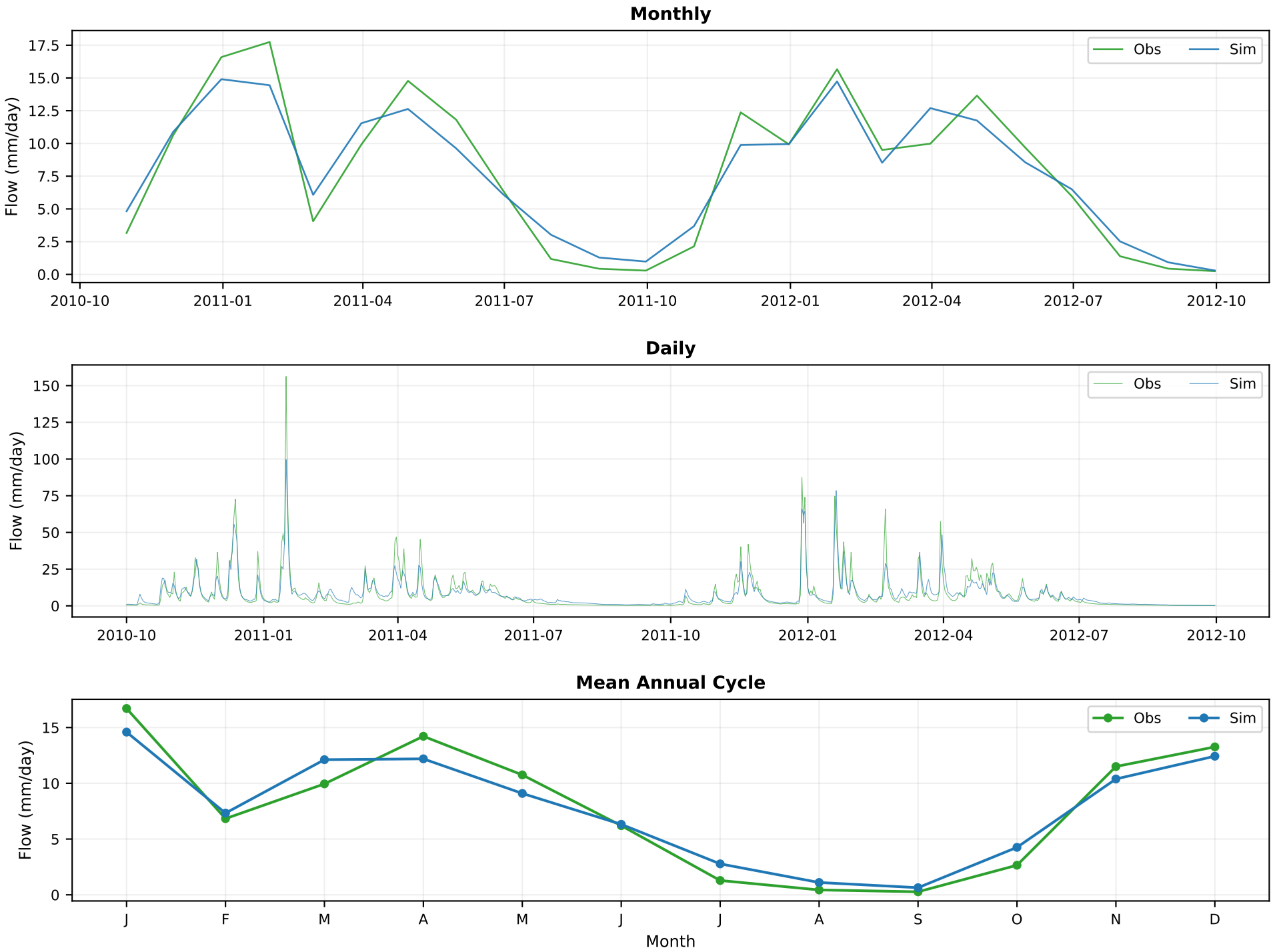

ID: 1632900
KGE=0.765 r=0.891 α=0.950 NSE=0.782 Bias=+20.2%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.5
1.0
1.5
2.0
2.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

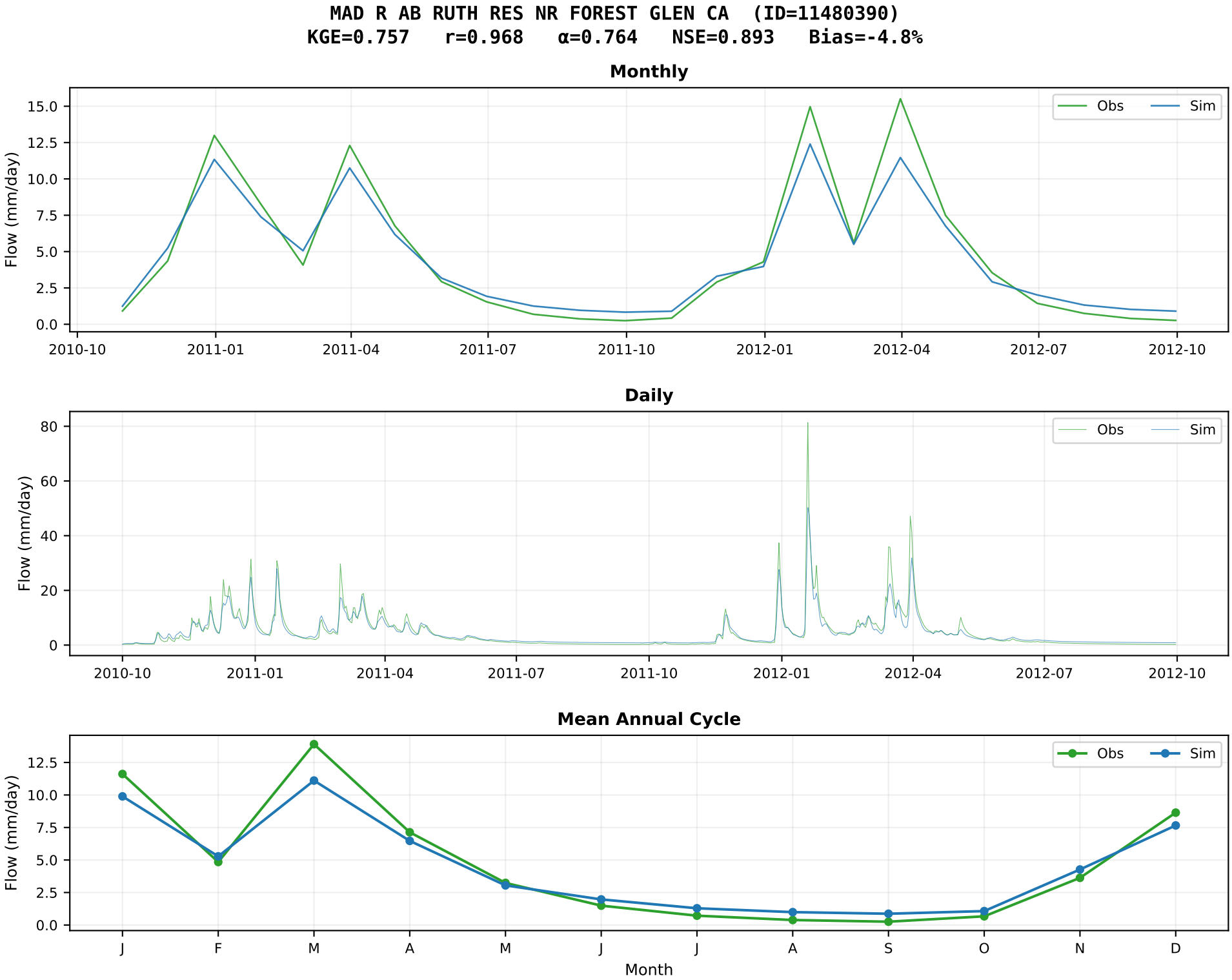

ID: 3471500
KGE=0.764 r=0.895 α=0.802 NSE=0.792 Bias=-7.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

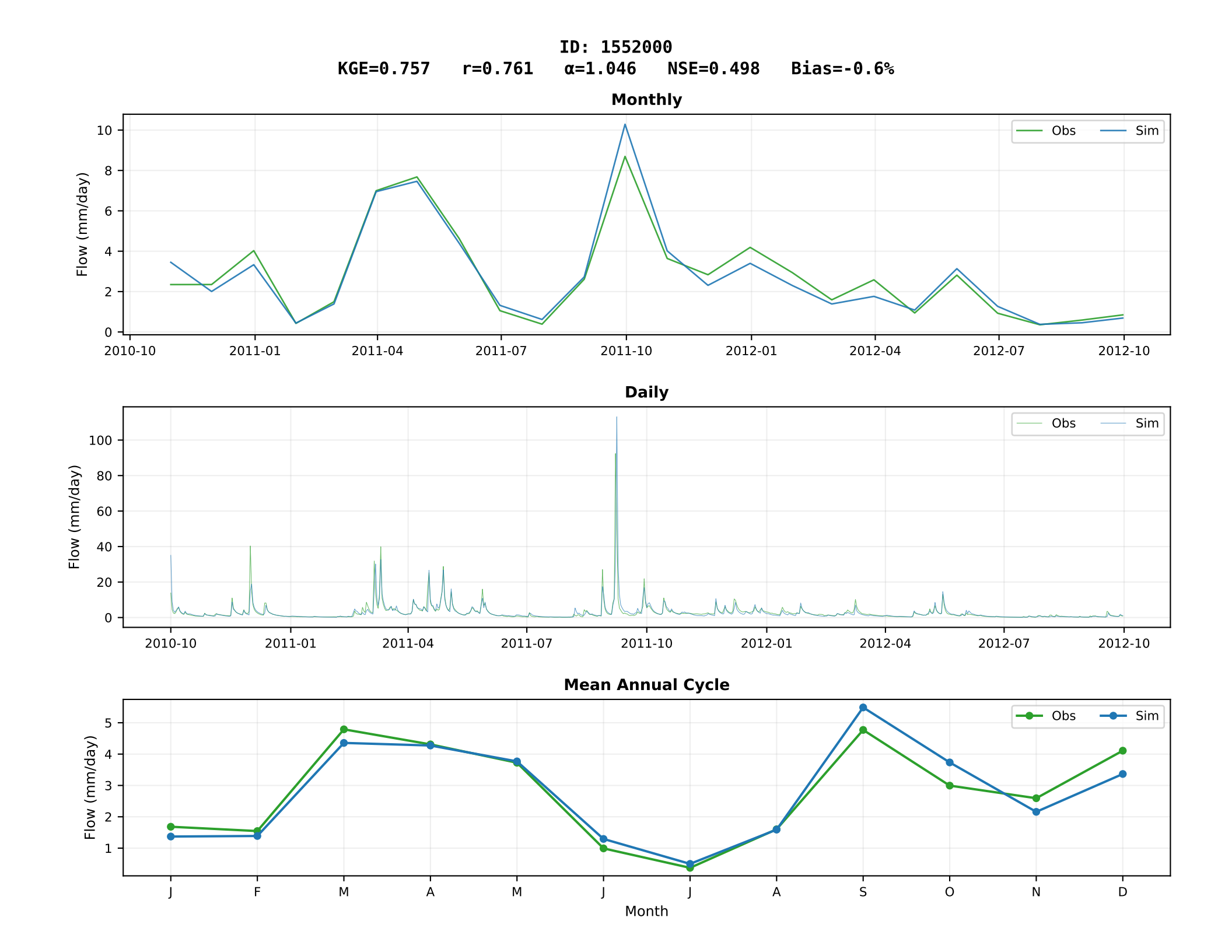

PITMAN C BL TAMARACK C CA (ID=11237500)
KGE=0.762 r=0.928 α=0.821 NSE=0.840 Bias=-13.9%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D


STEAMBOAT CREEK NEAR GLIDE, OR (ID=14316700)
KGE=0.762 r=0.935 α=0.784 NSE=0.847 Bias=-7.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

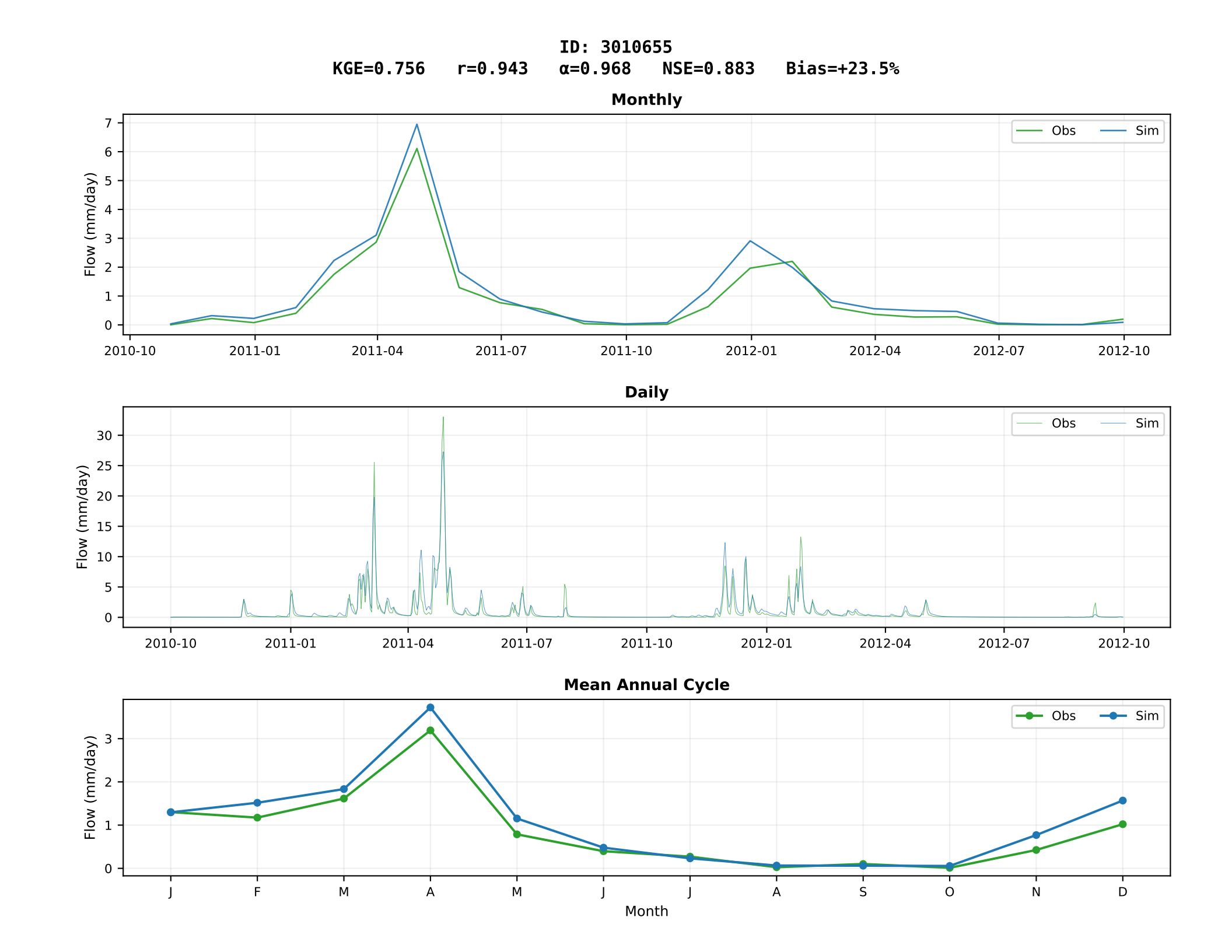

ID: 6332515
KGE=0.760 r=0.948 α=0.837 NSE=0.780 Bias=-16.9%
Monthly
Obs
Sim
Flow (mm/day)
0.1
0.2
0.3
0.4
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.2
0.4
0.6
0.8
1.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.15
0.20
0.25
0.30
0.35
J
F
M
A
M
J
J
A
S
O
N
D
Month

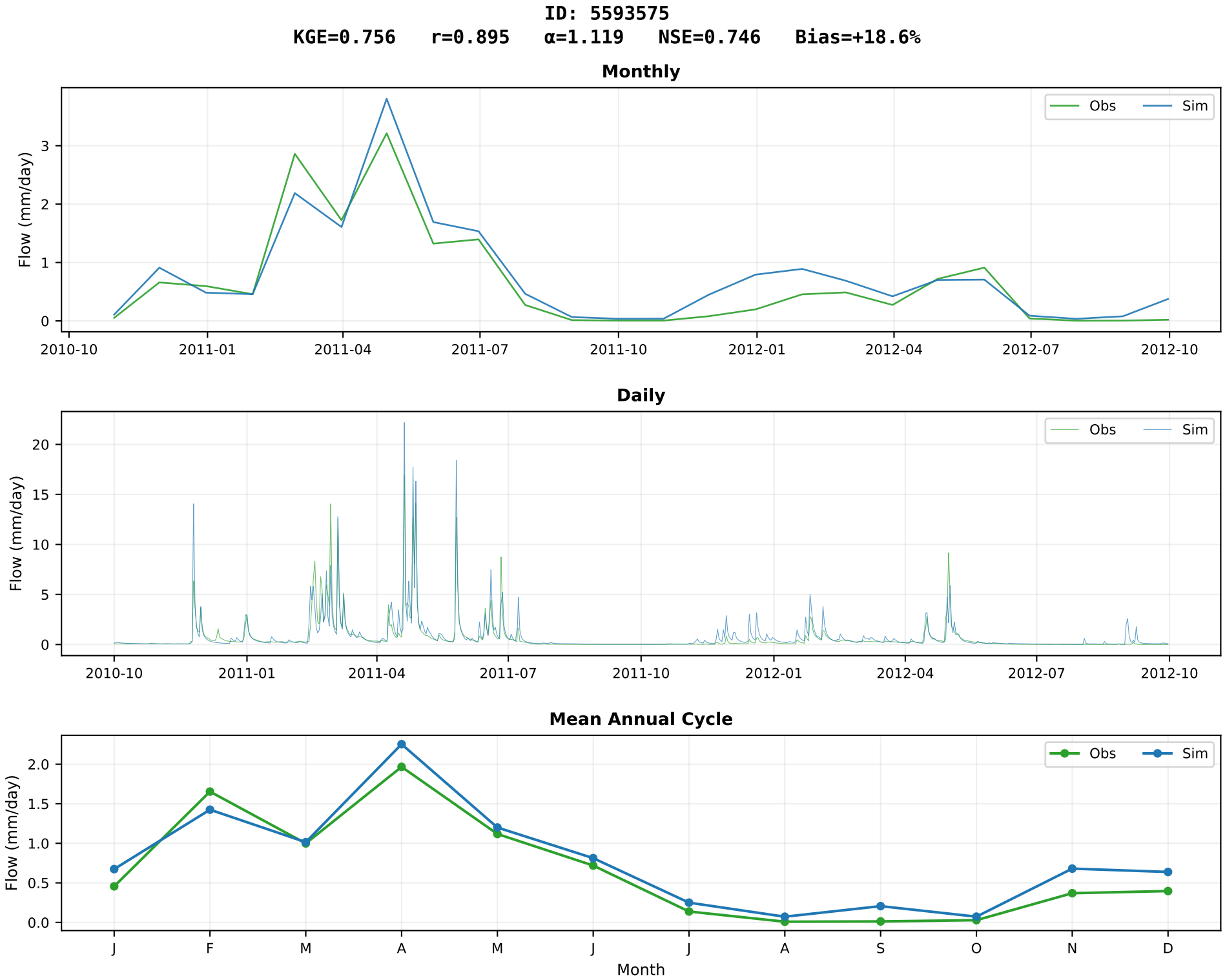

ANDREAS C NR PALM SPRINGS CA (ID=10259000)
KGE=0.757 r=0.921 α=0.771 NSE=0.826 Bias=-0.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

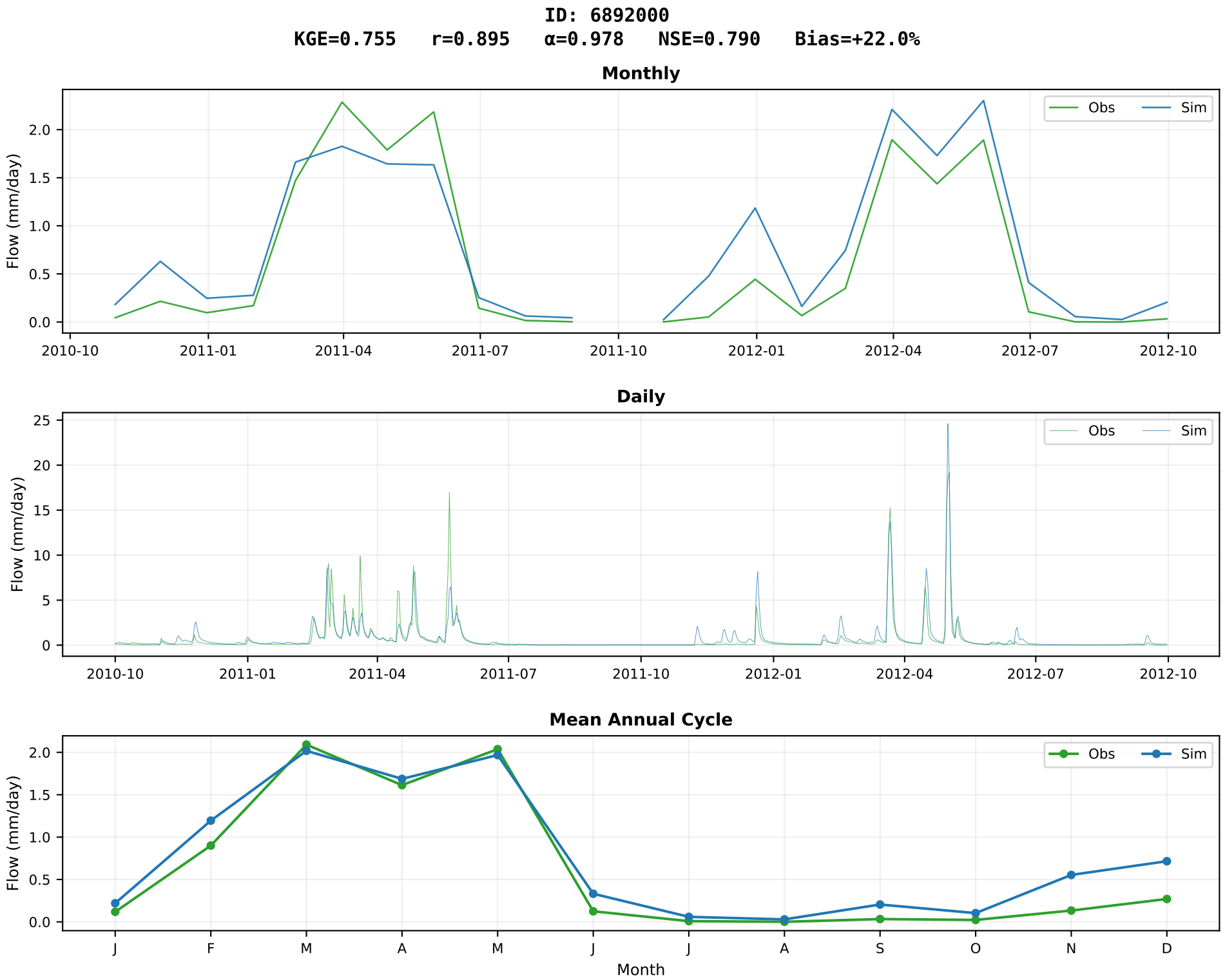

MAD R AB RUTH RES NR FOREST GLEN CA (ID=11480390)
KGE=0.757 r=0.968 α=0.764 NSE=0.893 Bias=-4.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

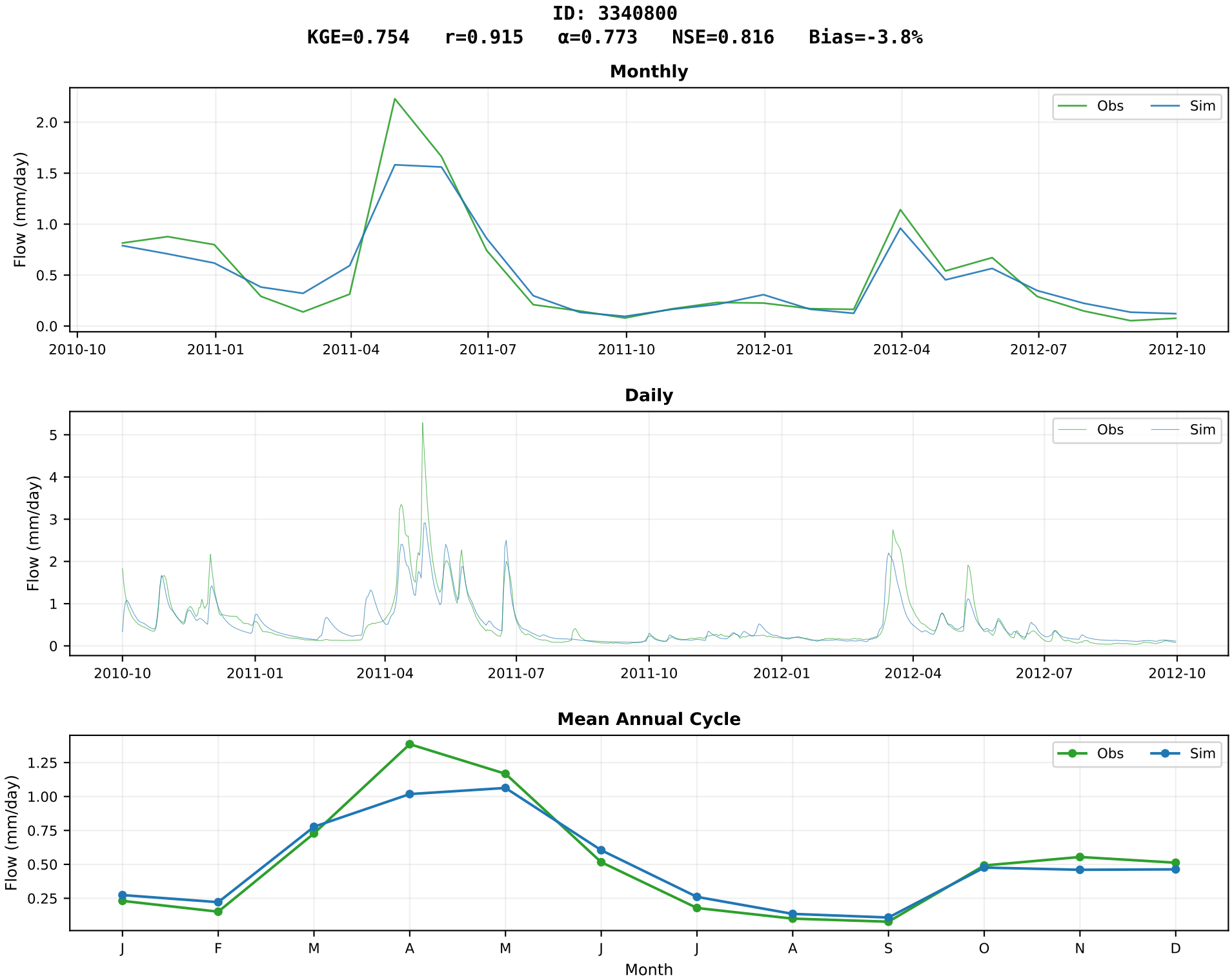

ID: 1552000
KGE=0.757 r=0.761 α=1.046 NSE=0.498 Bias=-0.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

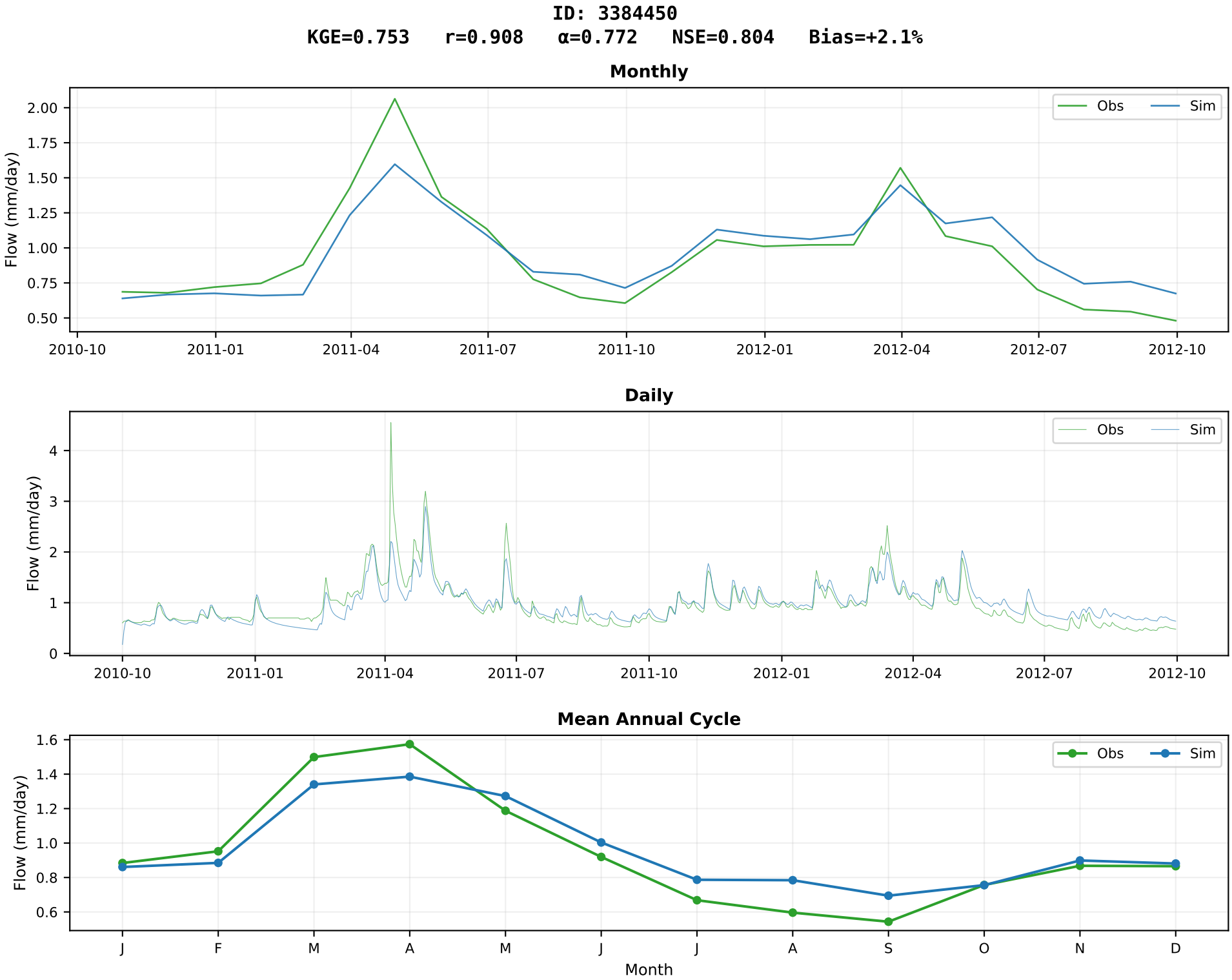

GREYS RIVER AB RESERVOIR NR ALPINE WY (ID=13023000)
KGE=0.756 r=0.917 α=0.808 NSE=0.787 Bias=-12.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

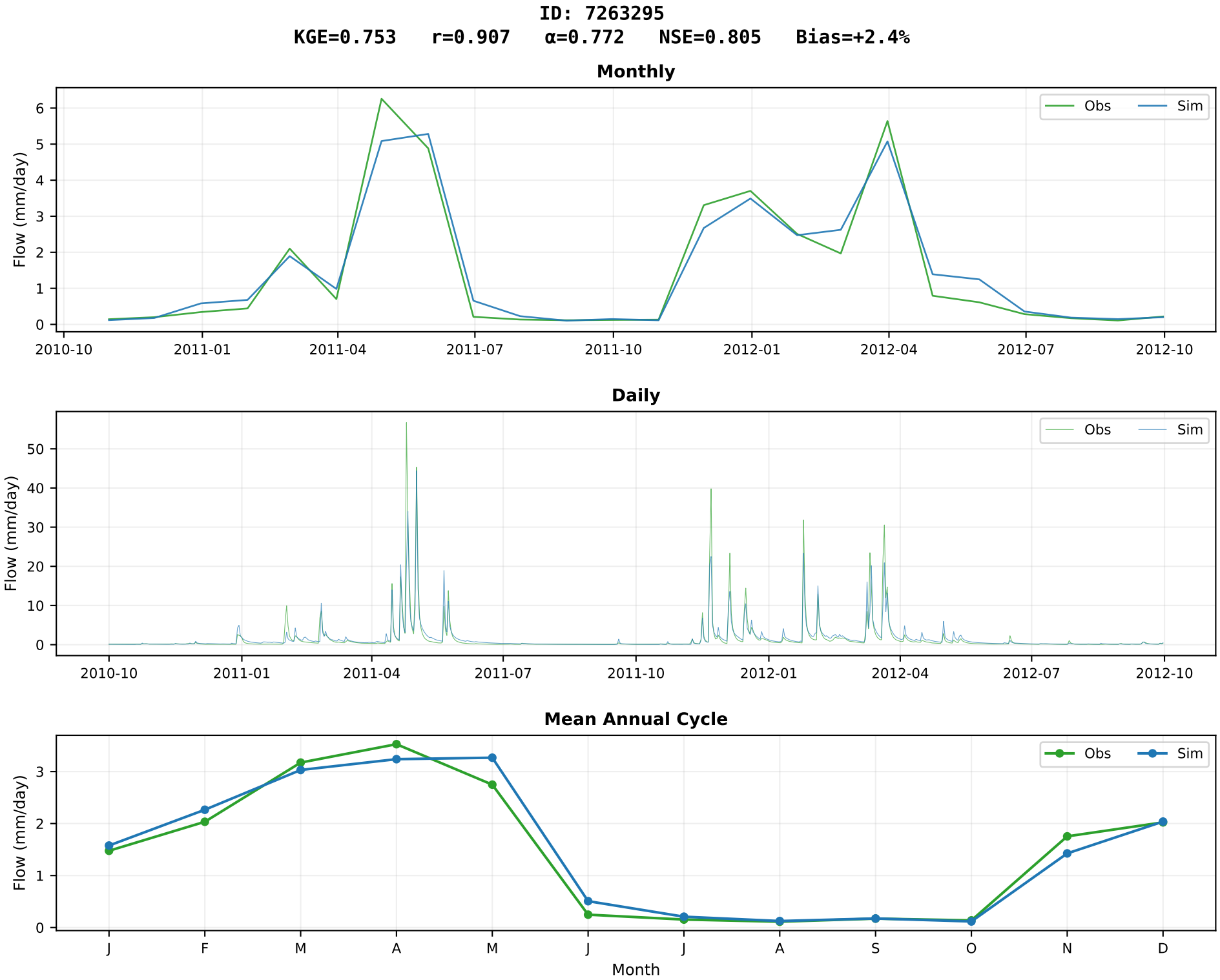

ID: 3010655
KGE=0.756 r=0.943 α=0.968 NSE=0.883 Bias=+23.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

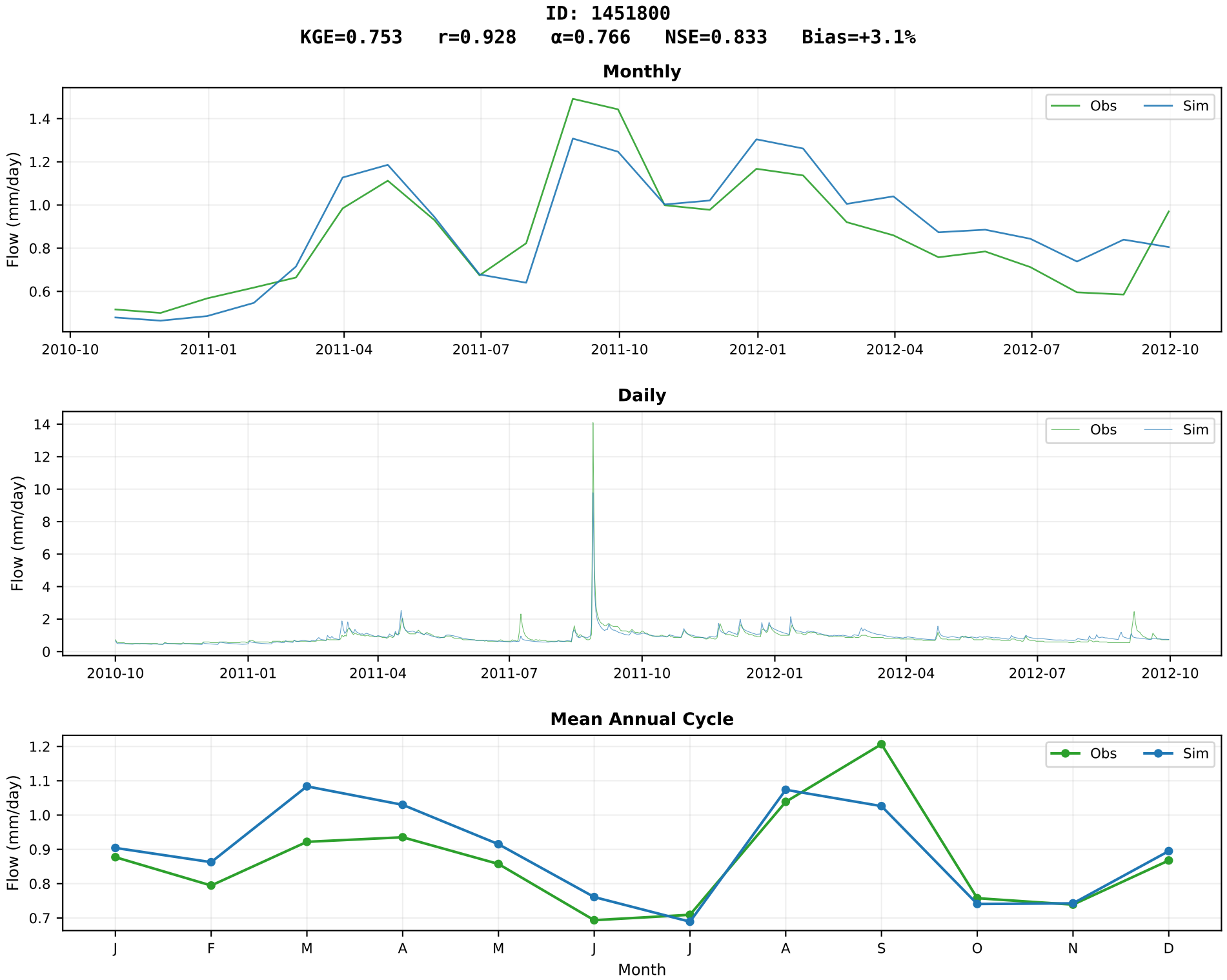

ID: 5593575
KGE=0.756 r=0.895 α=1.119 NSE=0.746 Bias=+18.6%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

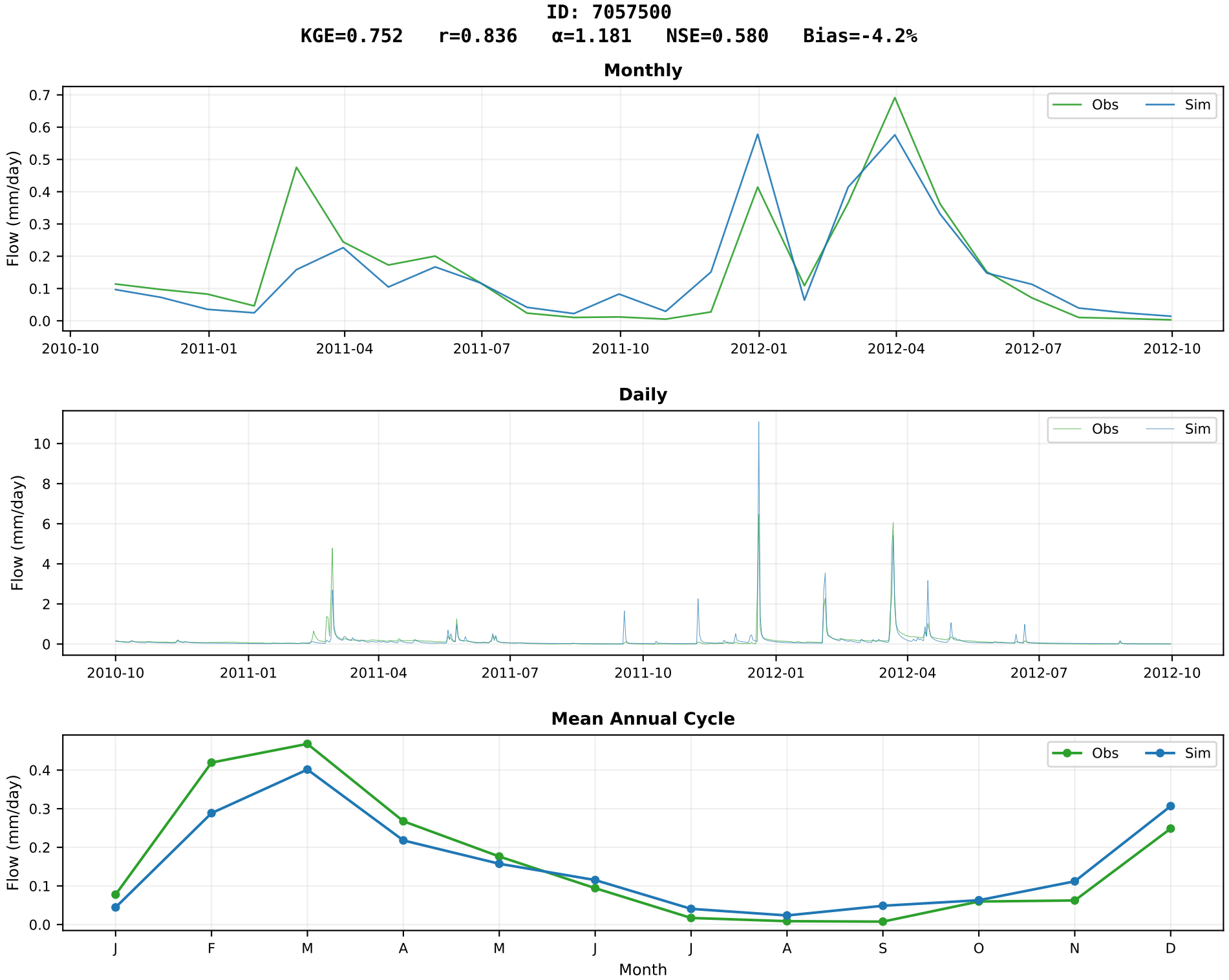

ID: 6892000
KGE=0.755 r=0.895 α=0.978 NSE=0.790 Bias=+22.0%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

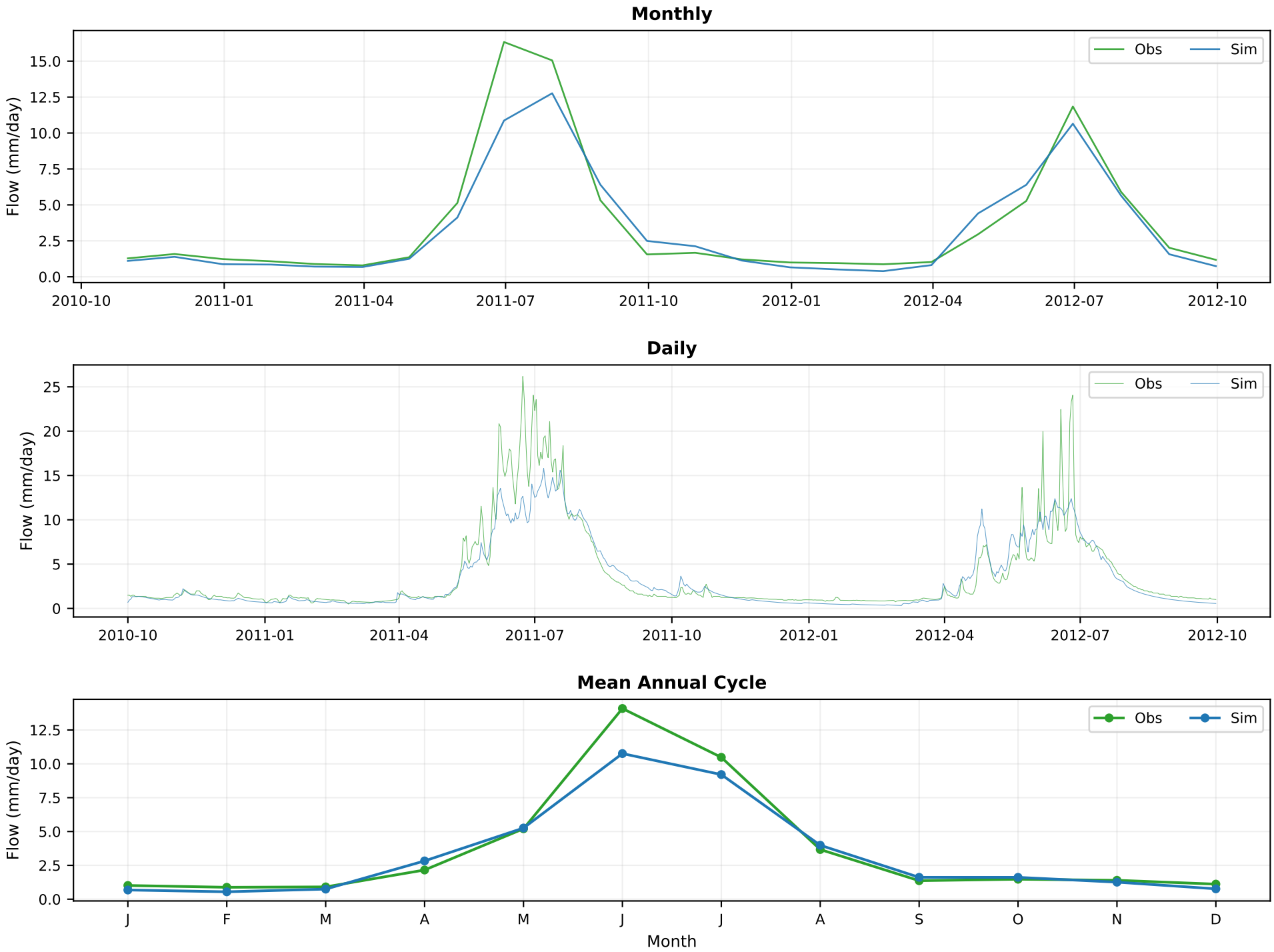

ID: 3340800
KGE=0.754 r=0.915 α=0.773 NSE=0.816 Bias=-3.8%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
5
4
3
2
1
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.25
1.00
0.75
0.50
0.25
J
F
M
A
M
J
J
A
S
O
N
D
Month

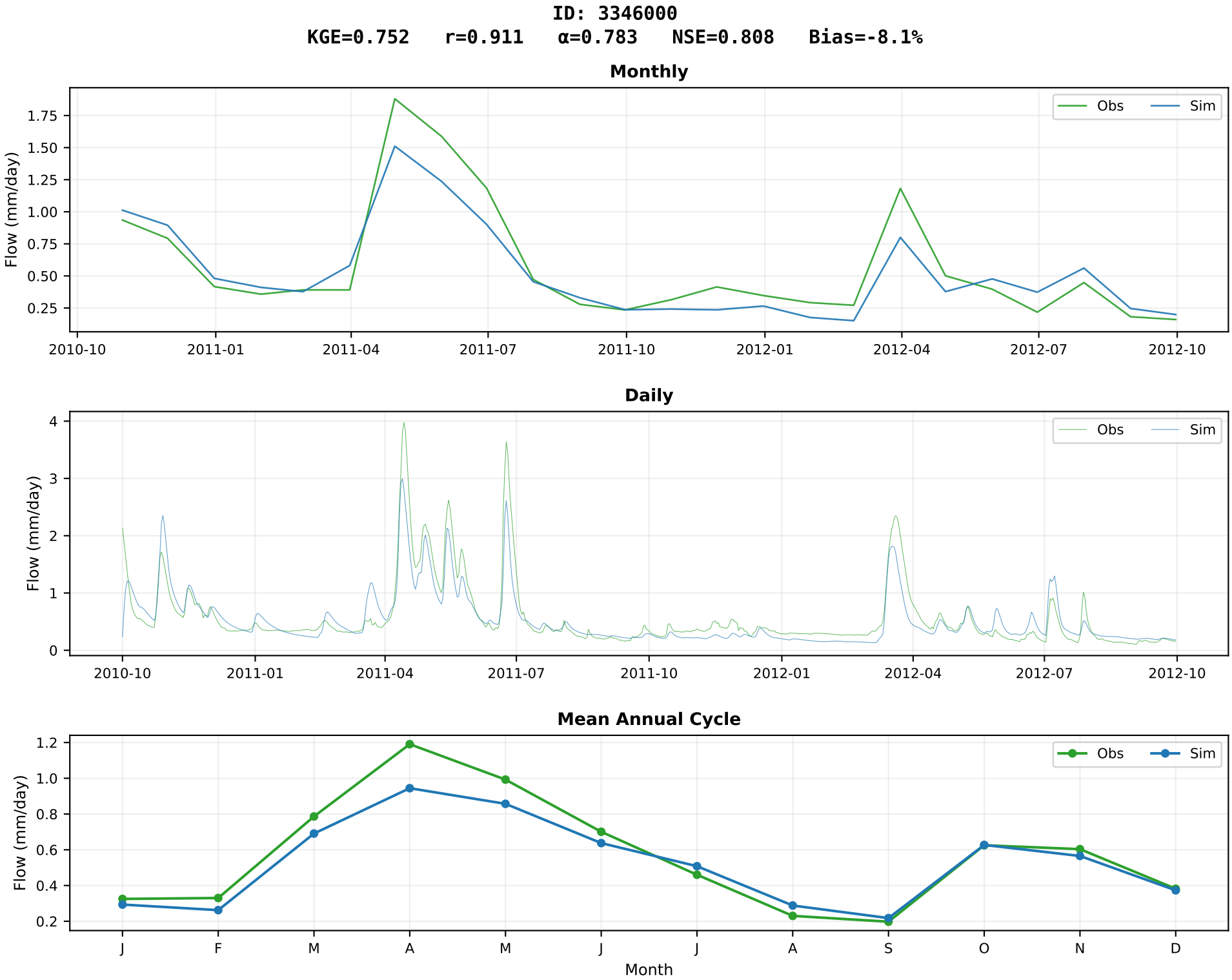

ID: 3384450
KGE=0.753 r=0.908 α=0.772 NSE=0.804 Bias=+2.1%
Monthly
Flow (mm/day)
Obs
Sim
2.00
1.75
1.50
1.25
1.00
0.75
0.50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
4
3
2
1
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
1.6
1.4
1.2
1.0
0.8
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month

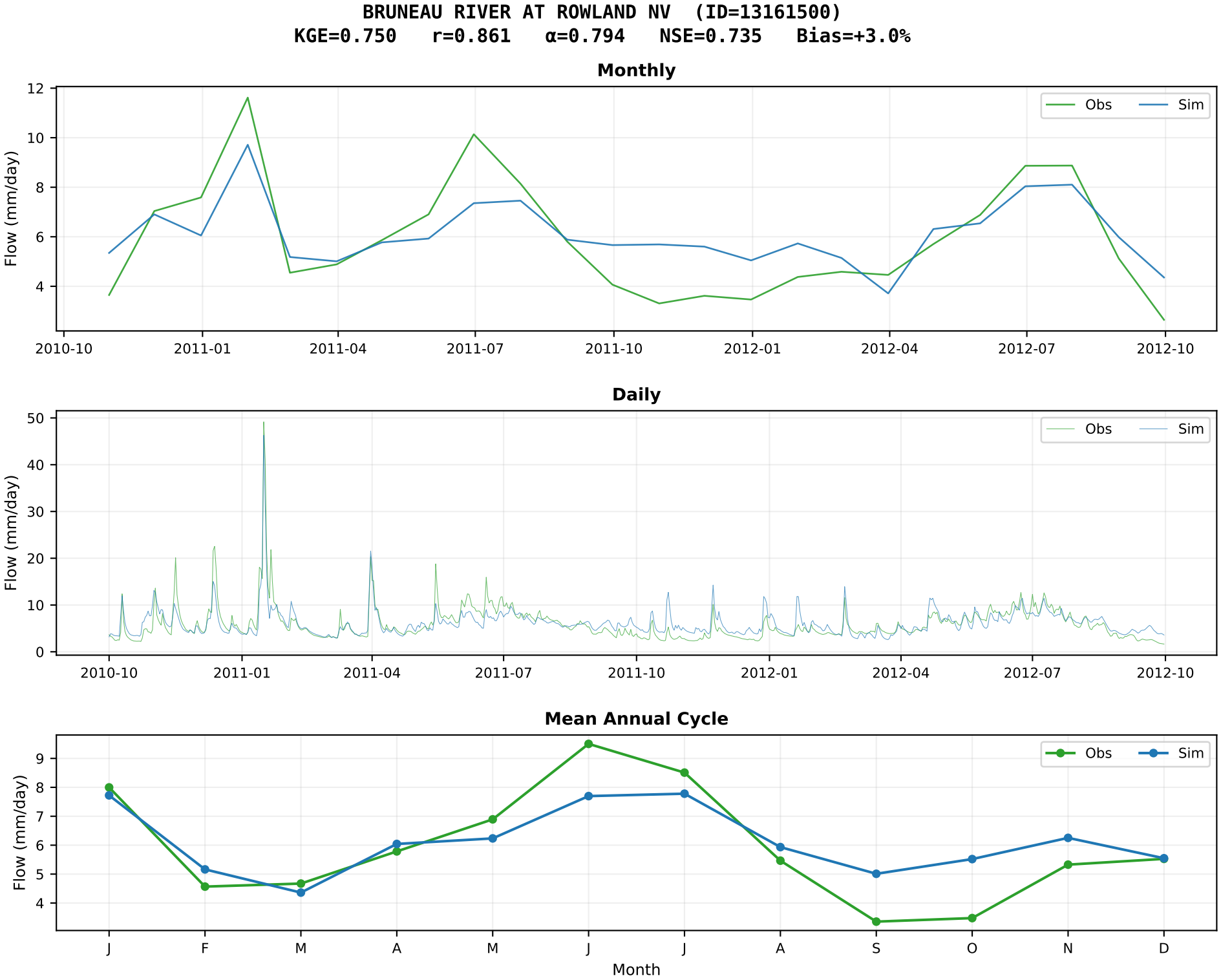

ID: 7263295
KGE=0.753 r=0.907 α=0.772 NSE=0.805 Bias=+2.4%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

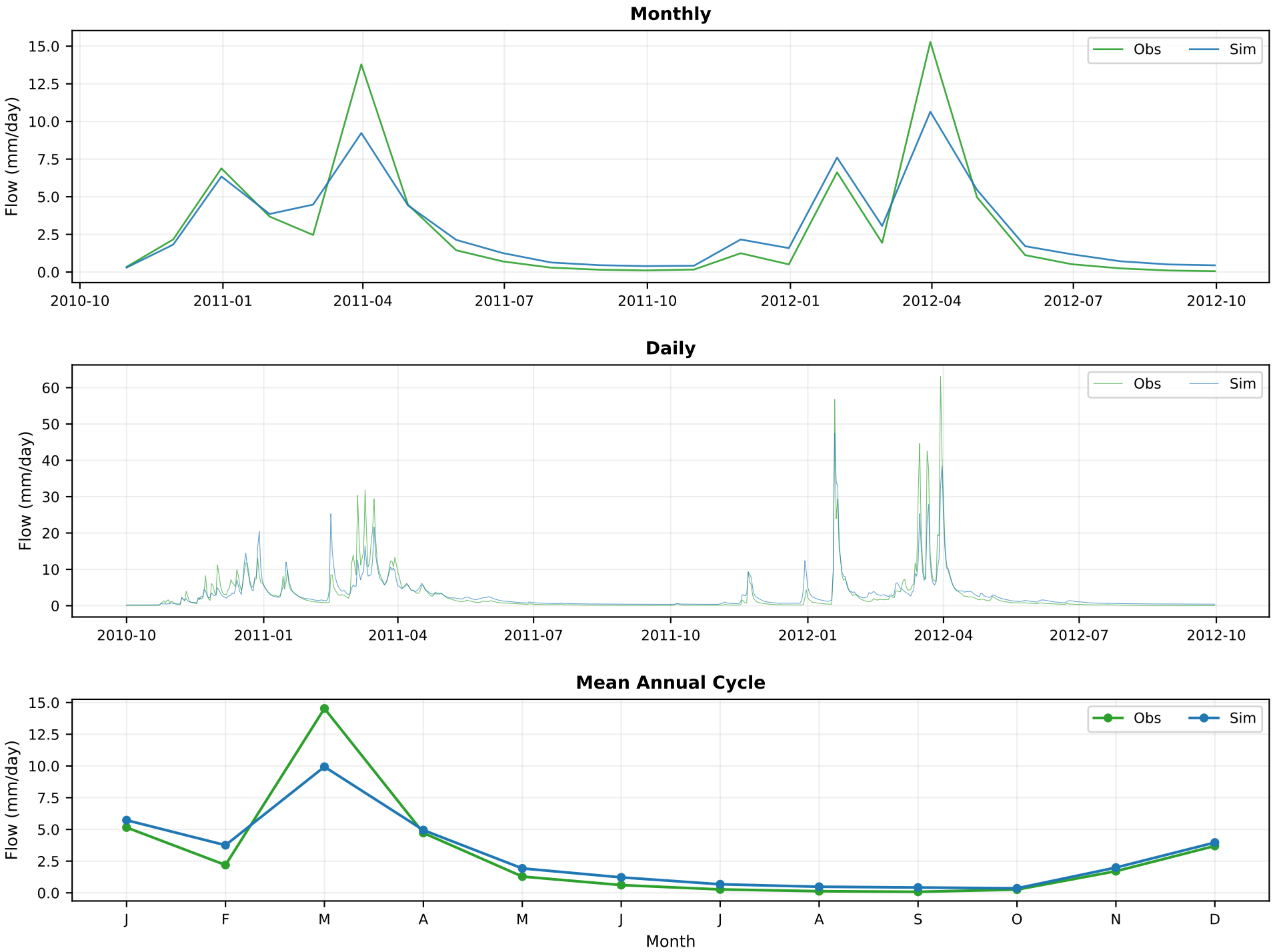

ID: 1451800
KGE=0.753 r=0.928 α=0.766 NSE=0.833 Bias=+3.1%
Monthly
Obs
Sim
Flow (mm/day)
1.4
1.2
1.0
0.8
0.6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
14
12
10
8
6
4
2
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.2
1.1
1.0
0.9
0.8
0.7
J
F
M
A
M
J
J
A
S
O
N
D
Month

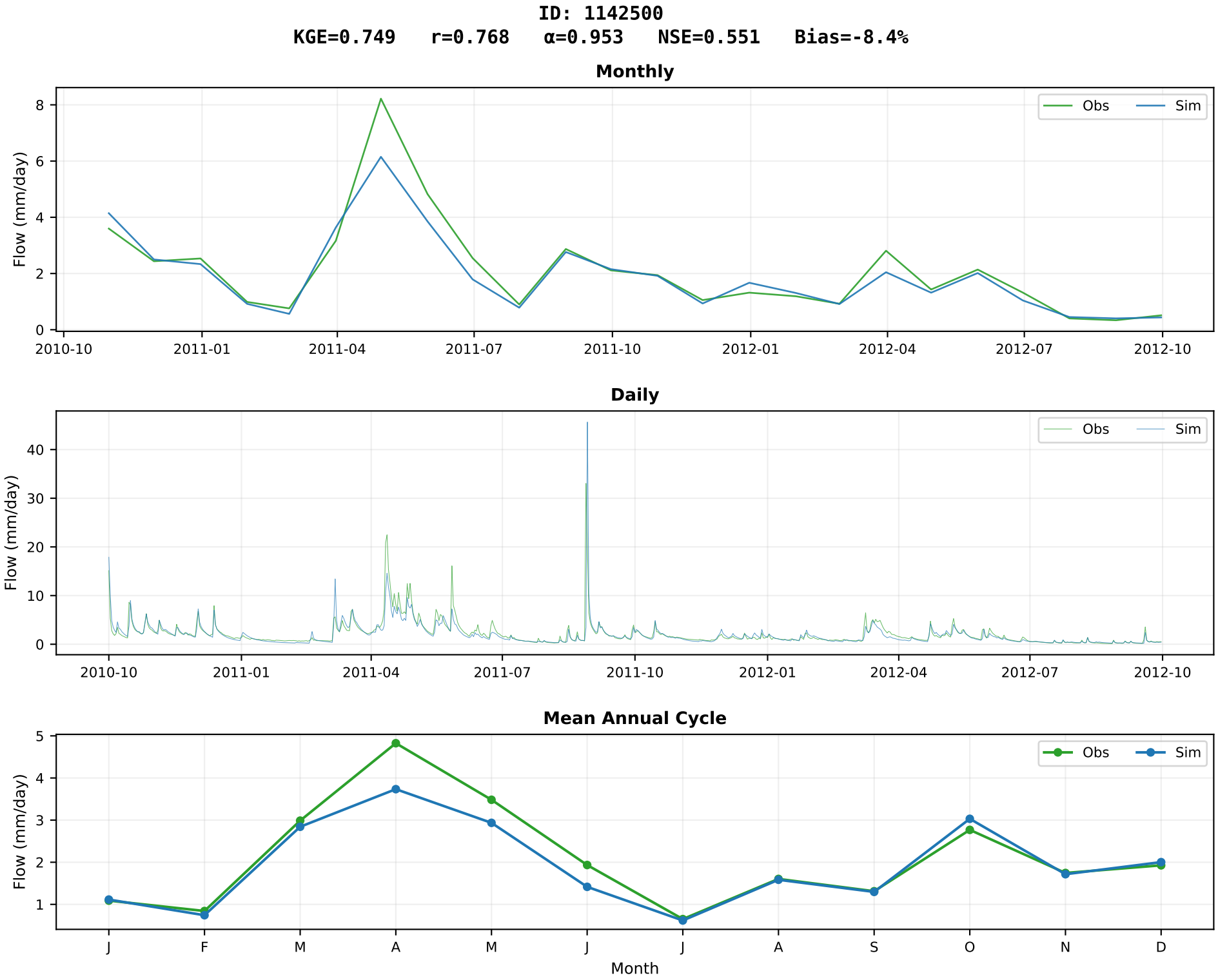

ID: 7057500
KGE=0.752 r=0.836 α=1.181 NSE=0.580 Bias=-4.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

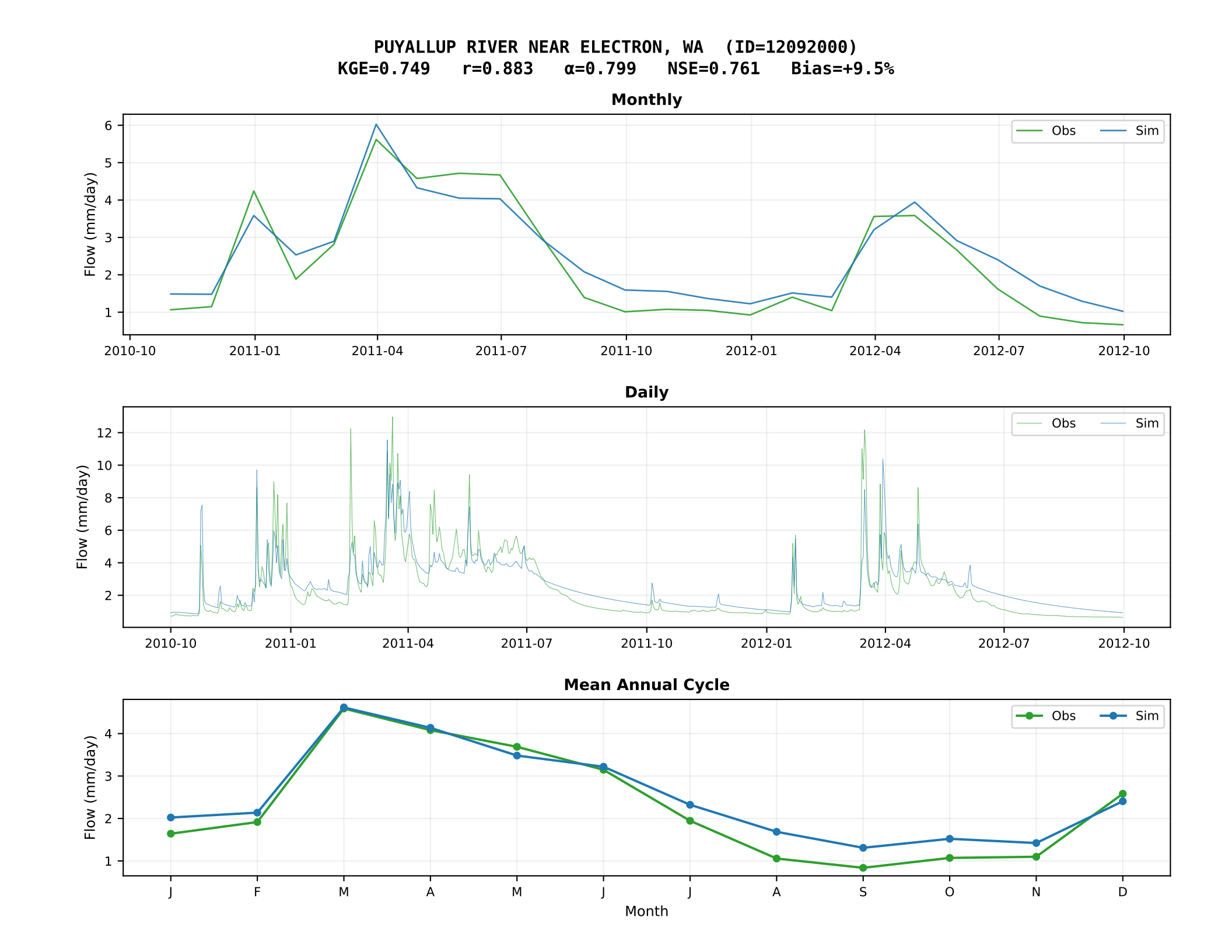

ROW RIVER ABOVE PITCHER CREEK NEAR, DORENA, OREG (ID=14154500)
KGE=0.752 r=0.923 α=0.787 NSE=0.828 Bias=-10.2%
Monthly
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

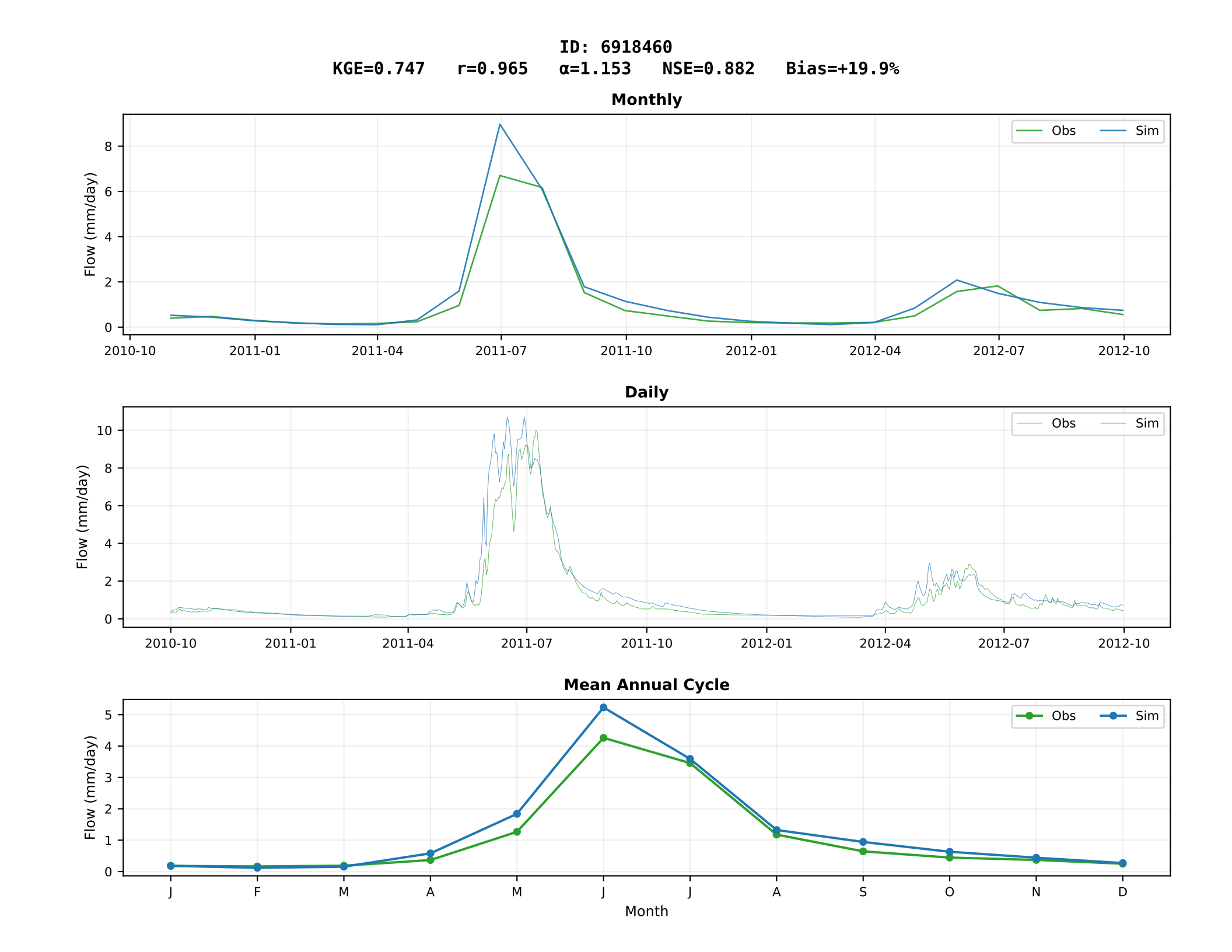

ID: 3346000
KGE=0.752 r=0.911 α=0.783 NSE=0.808 Bias=-8.1%
Monthly
Flow (mm/day)
Obs
Sim
1.75
1.50
1.25
1.00
0.75
0.50
0.25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
4
3
2
1
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
1.2
1.0
0.8
0.6
0.4
0.2
J
F
M
A
M
J
J
A
S
O
N
D
Month

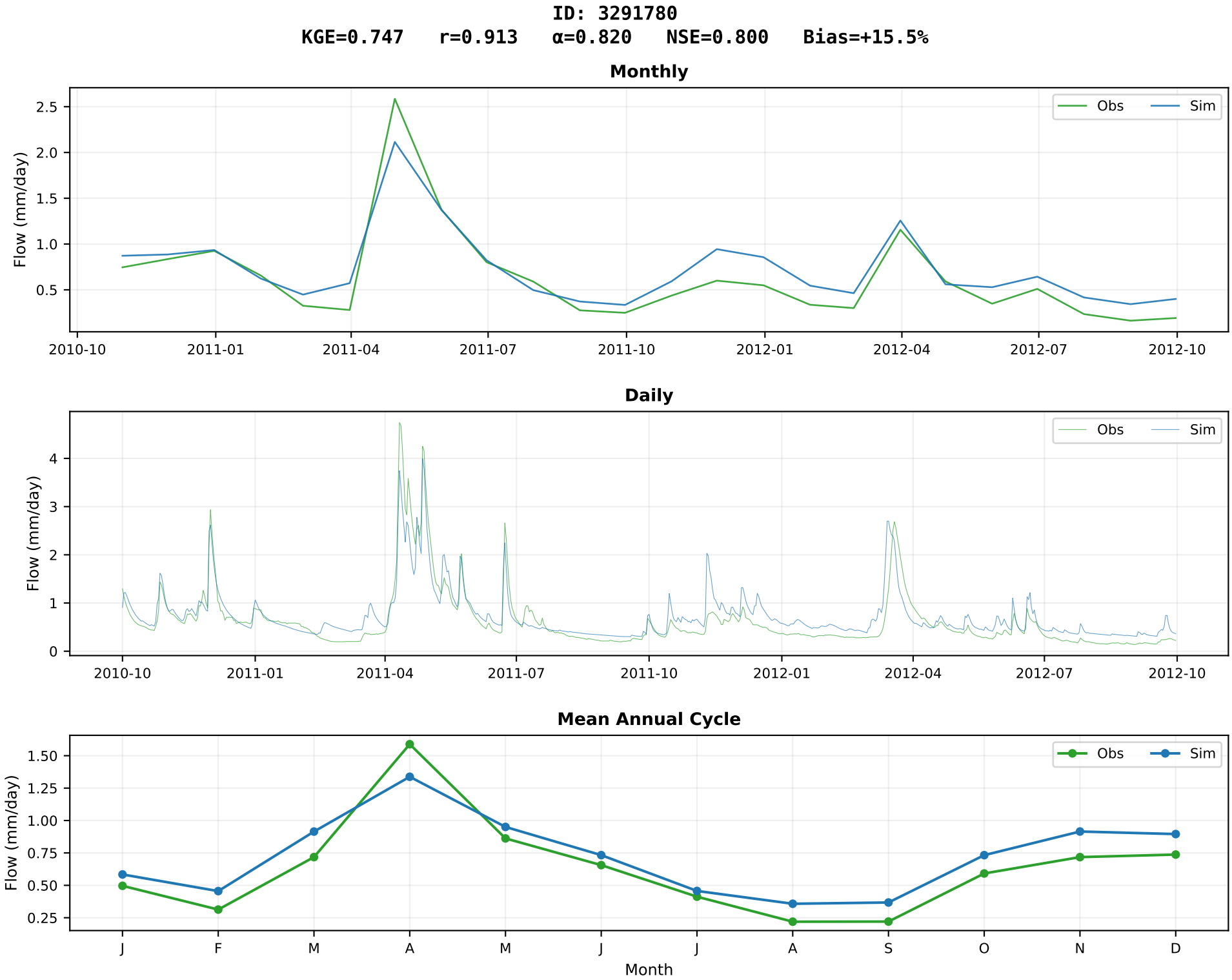

BRUNEAU RIVER AT ROWLAND NV (ID=13161500)
KGE=0.750 r=0.861 α=0.794 NSE=0.735 Bias=+3.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

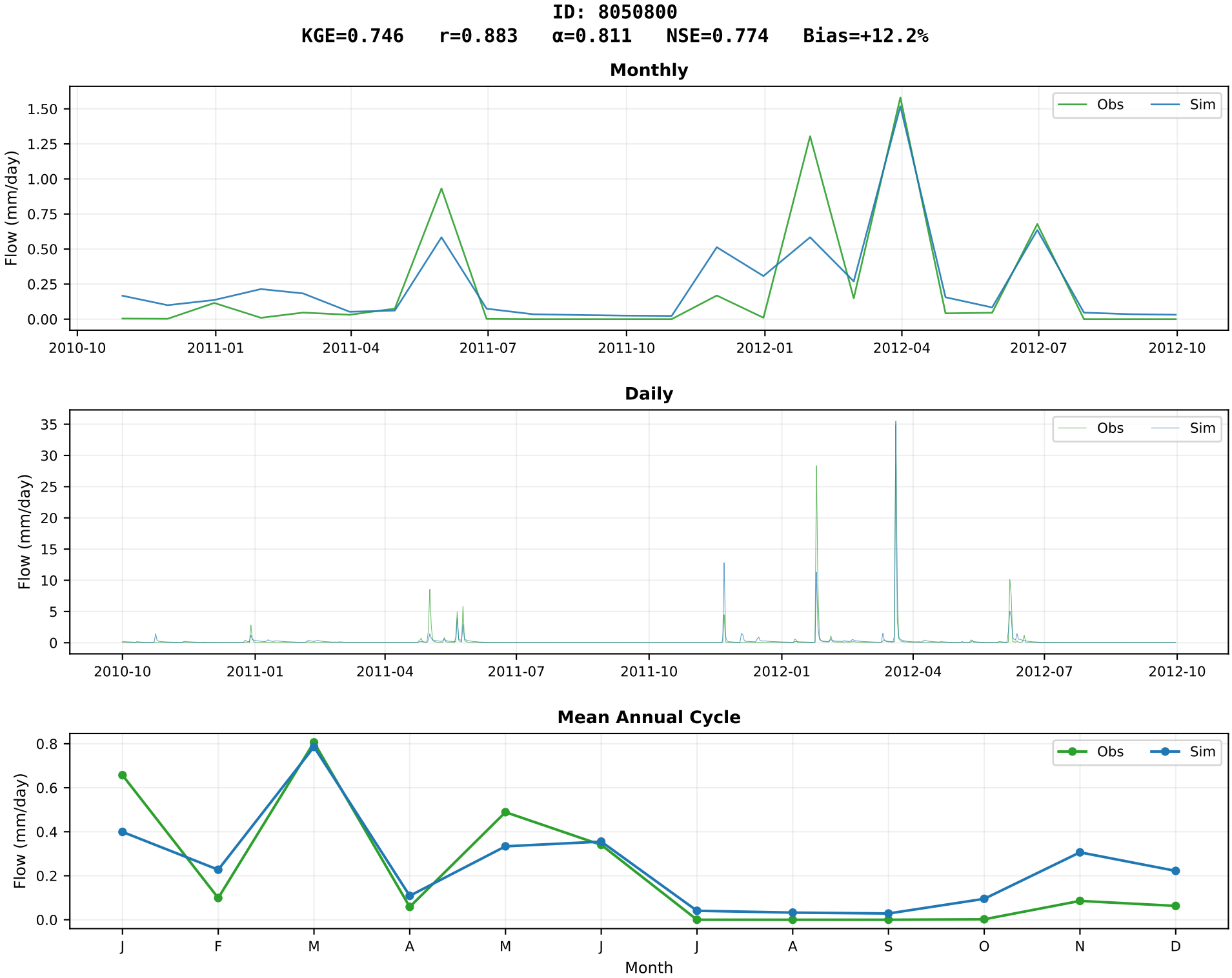

REDWOOD C A ORICK CA (ID=11482500)
KGE=0.749 r=0.913 α=0.765 NSE=0.811 Bias=+1.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

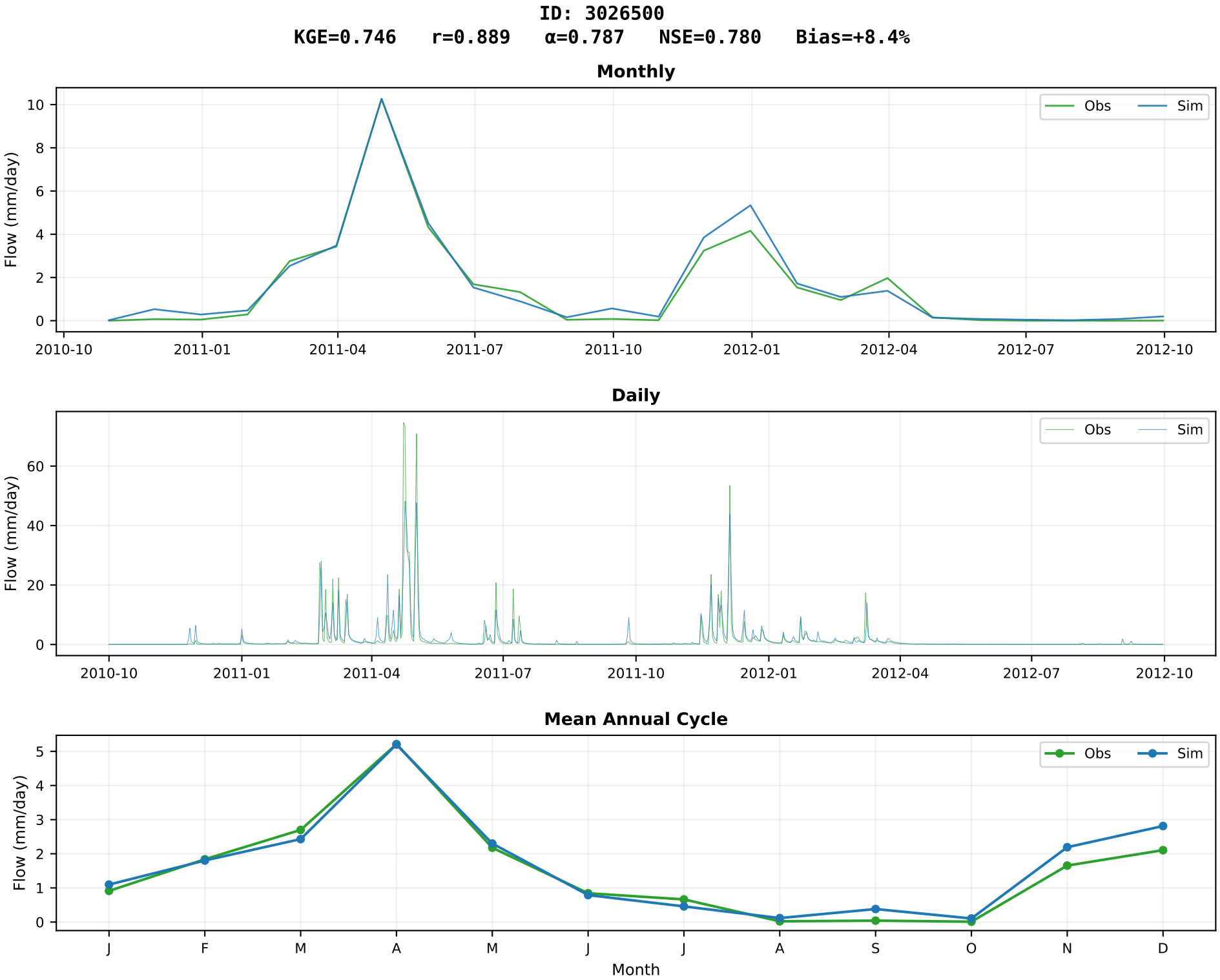

ID: 1142500
KGE=0.749 r=0.768 α=0.953 NSE=0.551 Bias=-8.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

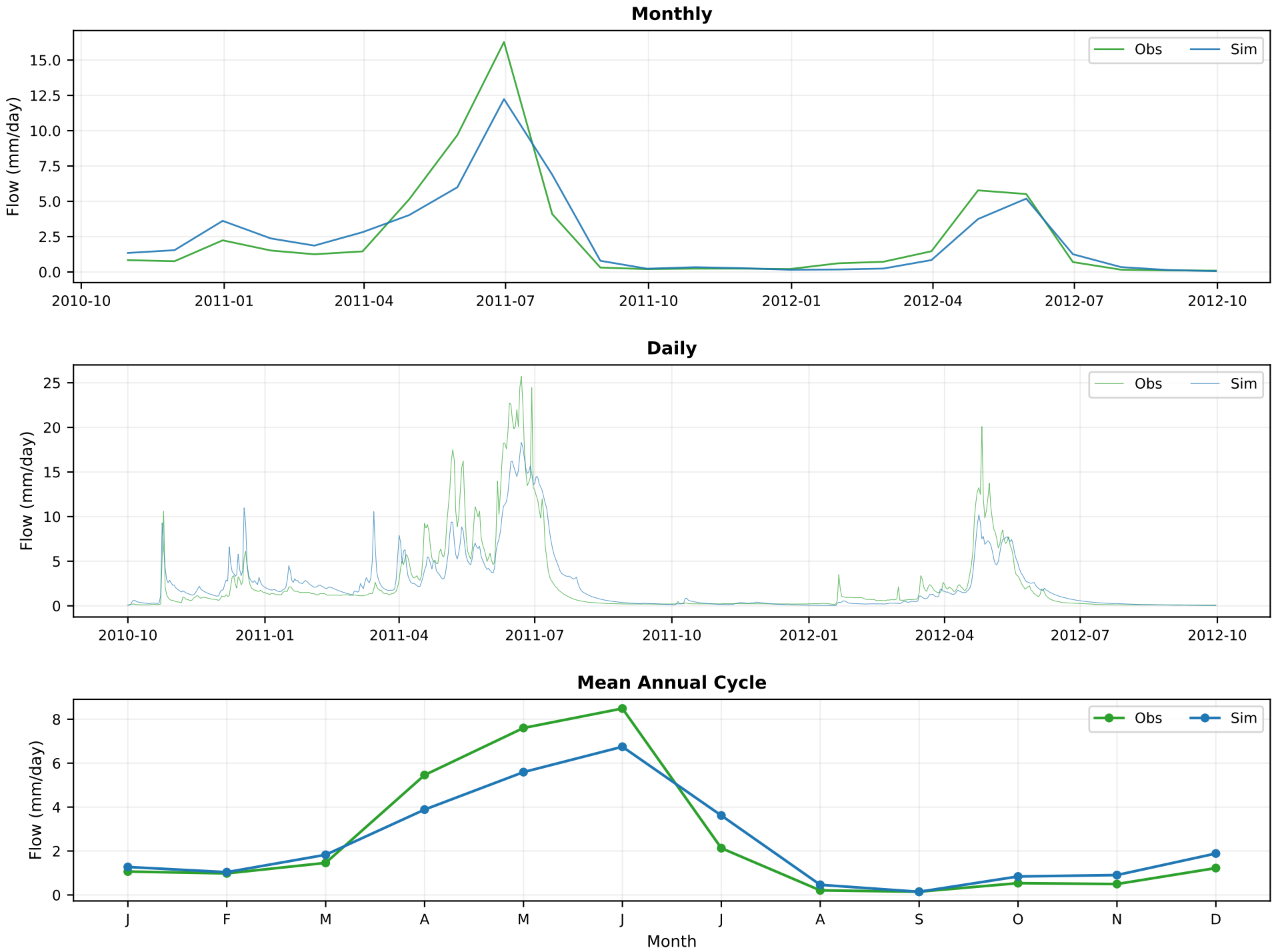

PUYALLUP RIVER NEAR ELECTRON, WA (ID=12092000)
KGE=0.749 r=0.883 α=0.799 NSE=0.761 Bias=+9.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 6918460
KGE=0.747 r=0.965 α=1.153 NSE=0.882 Bias=+19.9%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 3291780
KGE=0.747 r=0.913 α=0.820 NSE=0.800 Bias=+15.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 8050800
KGE=0.746 r=0.883 α=0.811 NSE=0.774 Bias=+12.2%
Monthly
Flow (mm/day)
Obs
Sim
1.50
1.25
1.00
0.75
0.50
0.25
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
35
30
25
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.8
0.6
0.4
0.2
0.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

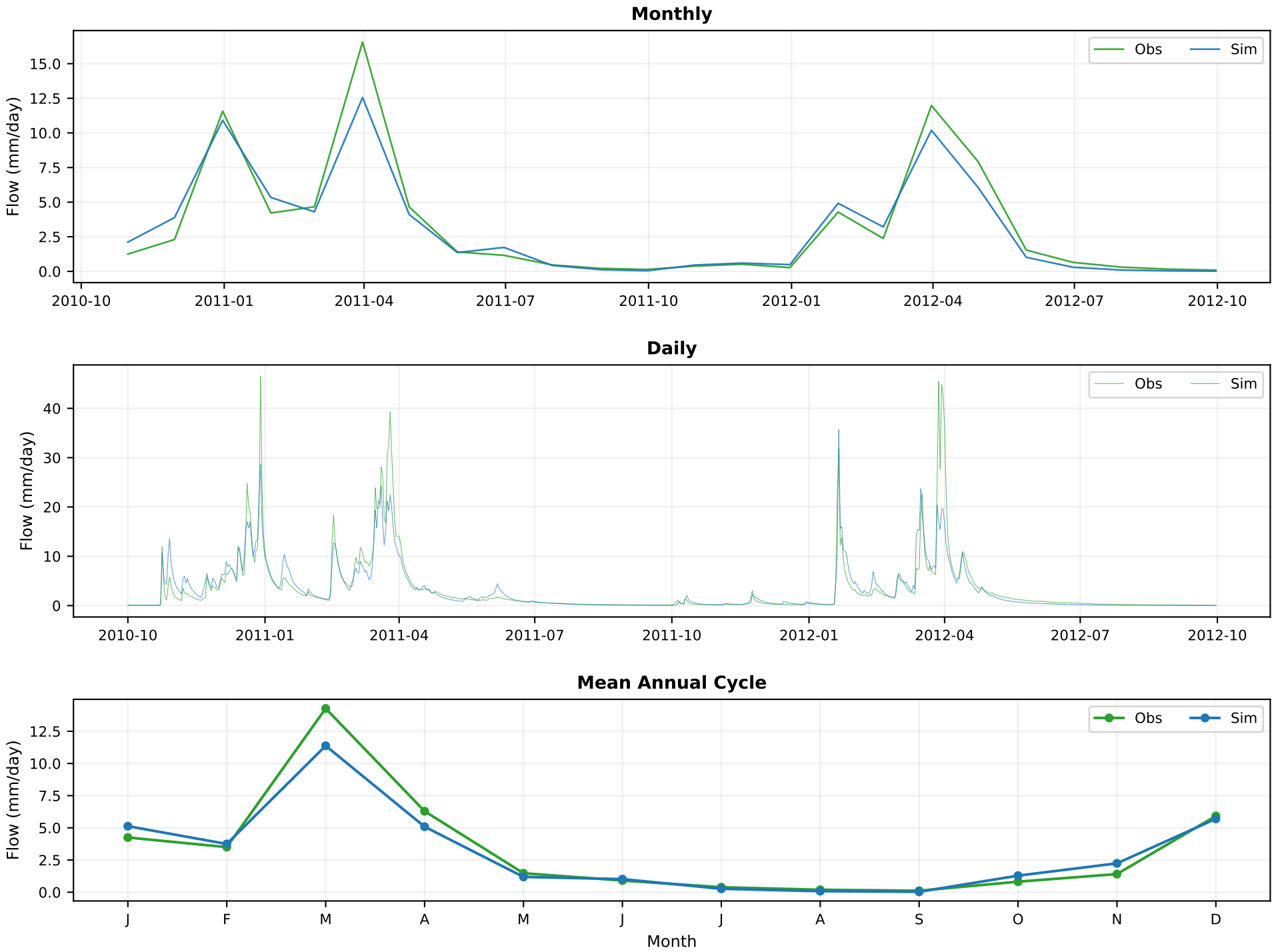

ID: 3026500
KGE=0.746 r=0.889 α=0.787 NSE=0.780 Bias=+8.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

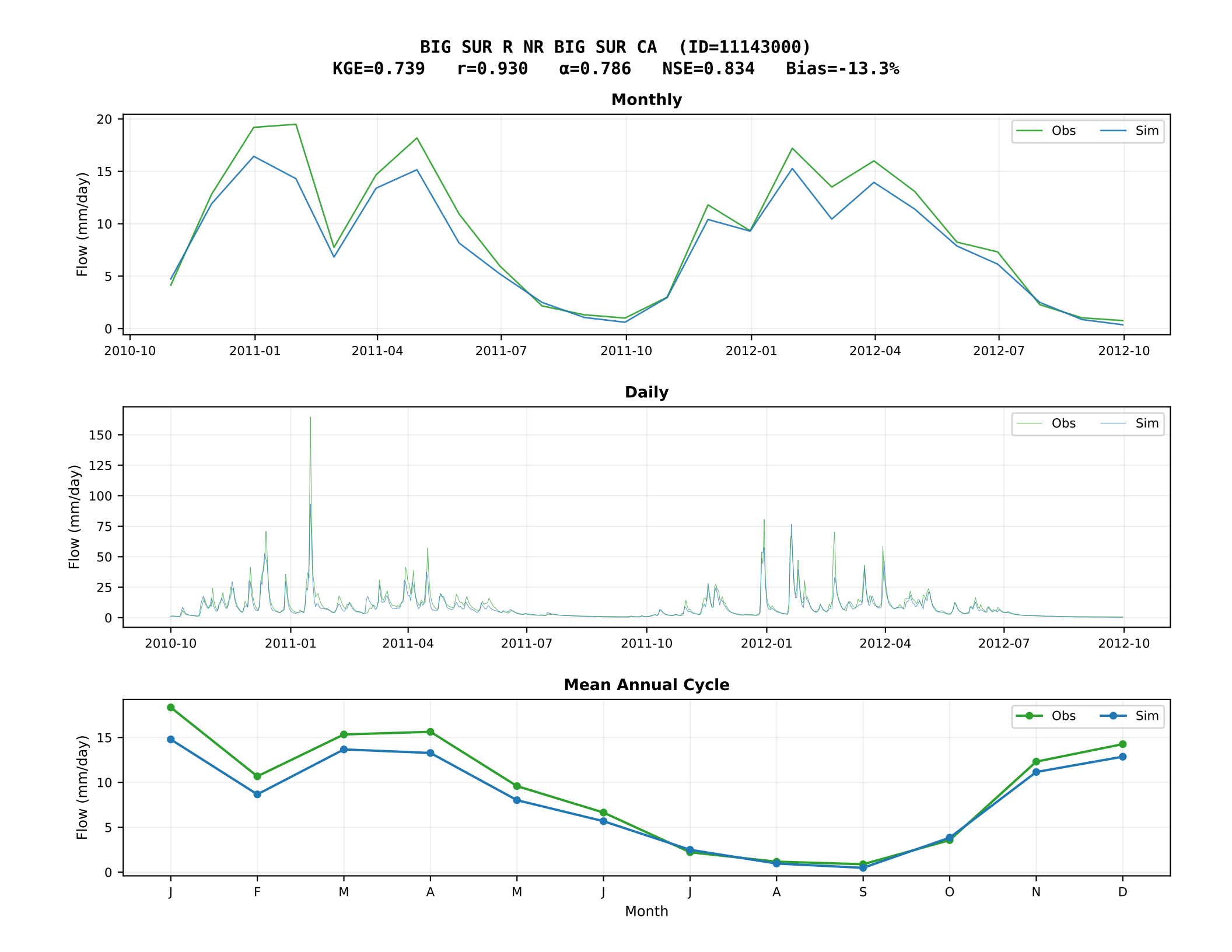

BLACKWOOD C NR TAHOE CITY CA (ID=10336660)
KGE=0.745 r=0.906 α=0.768 NSE=0.801 Bias=-5.1%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

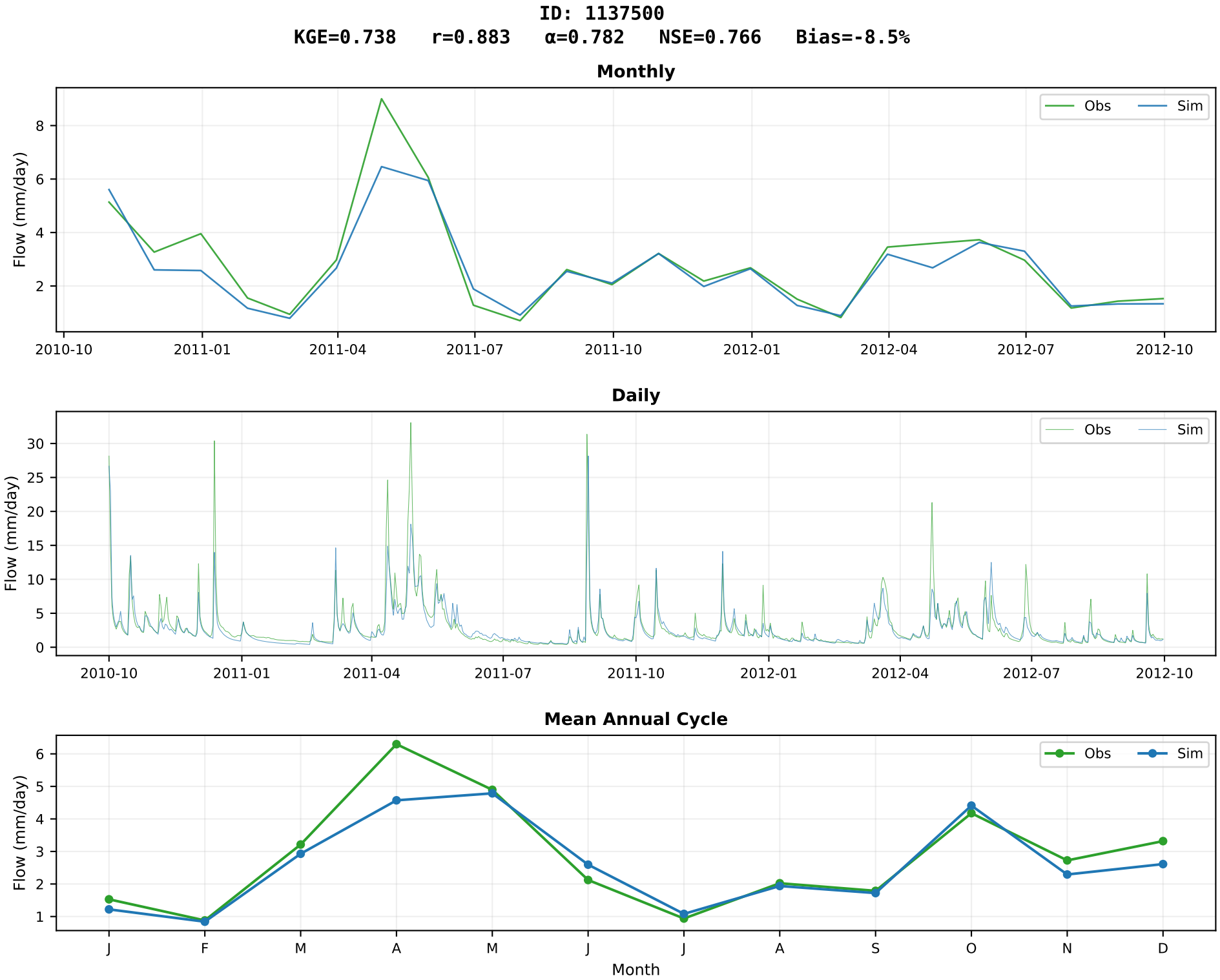
ID: 2074500

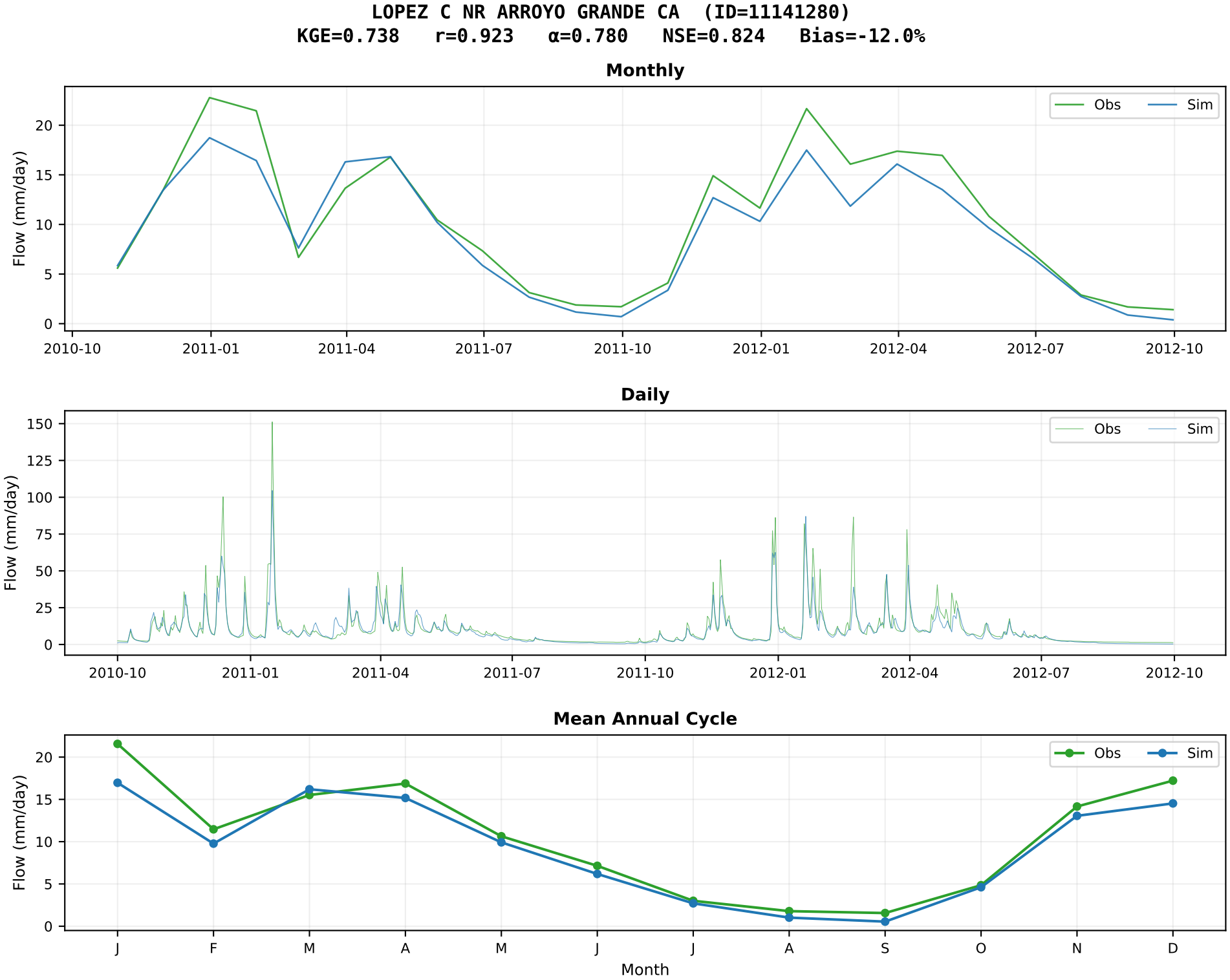
KGE=0.745 r=0.798 α=1.117 NSE=0.532 Bias=+10.4%

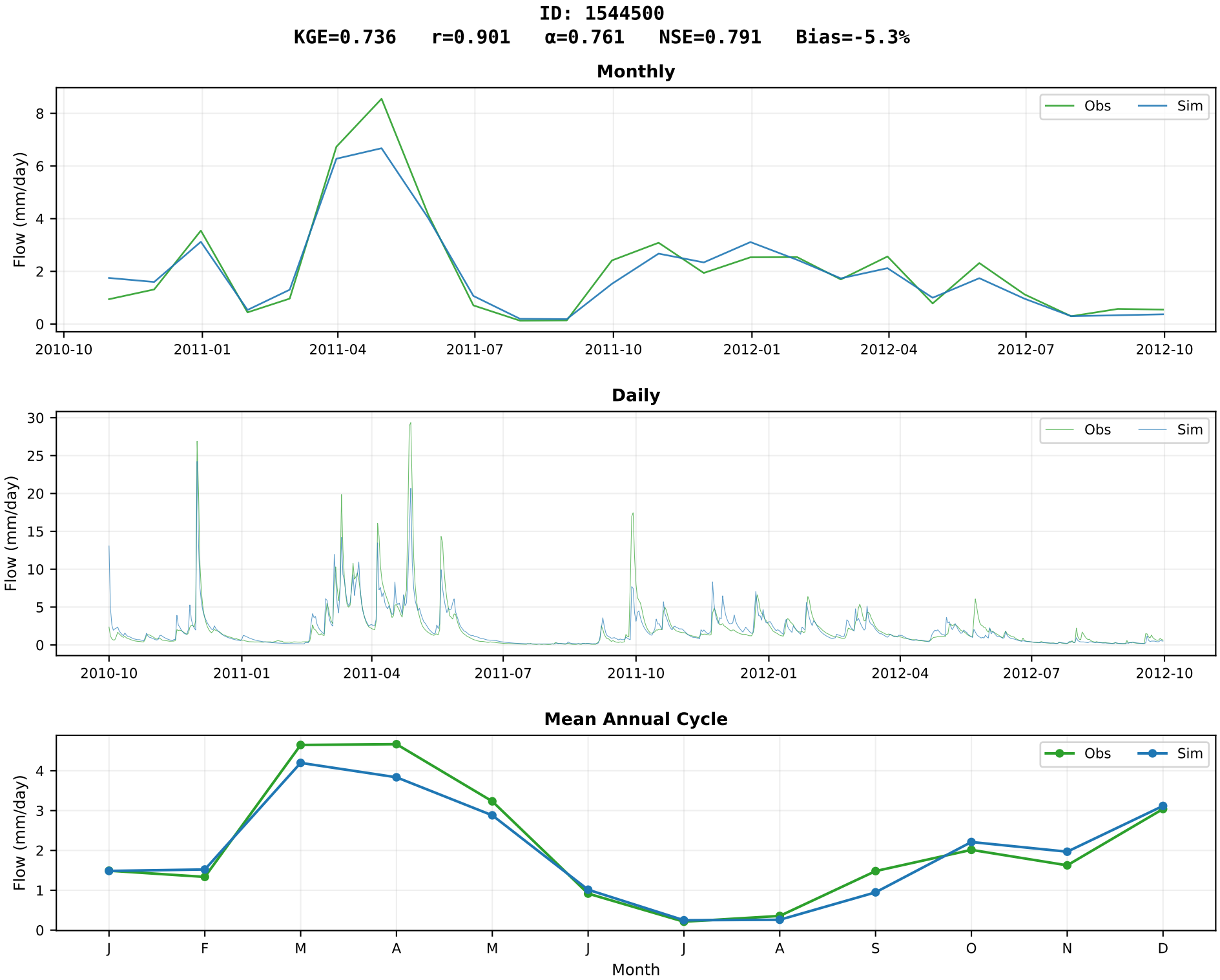


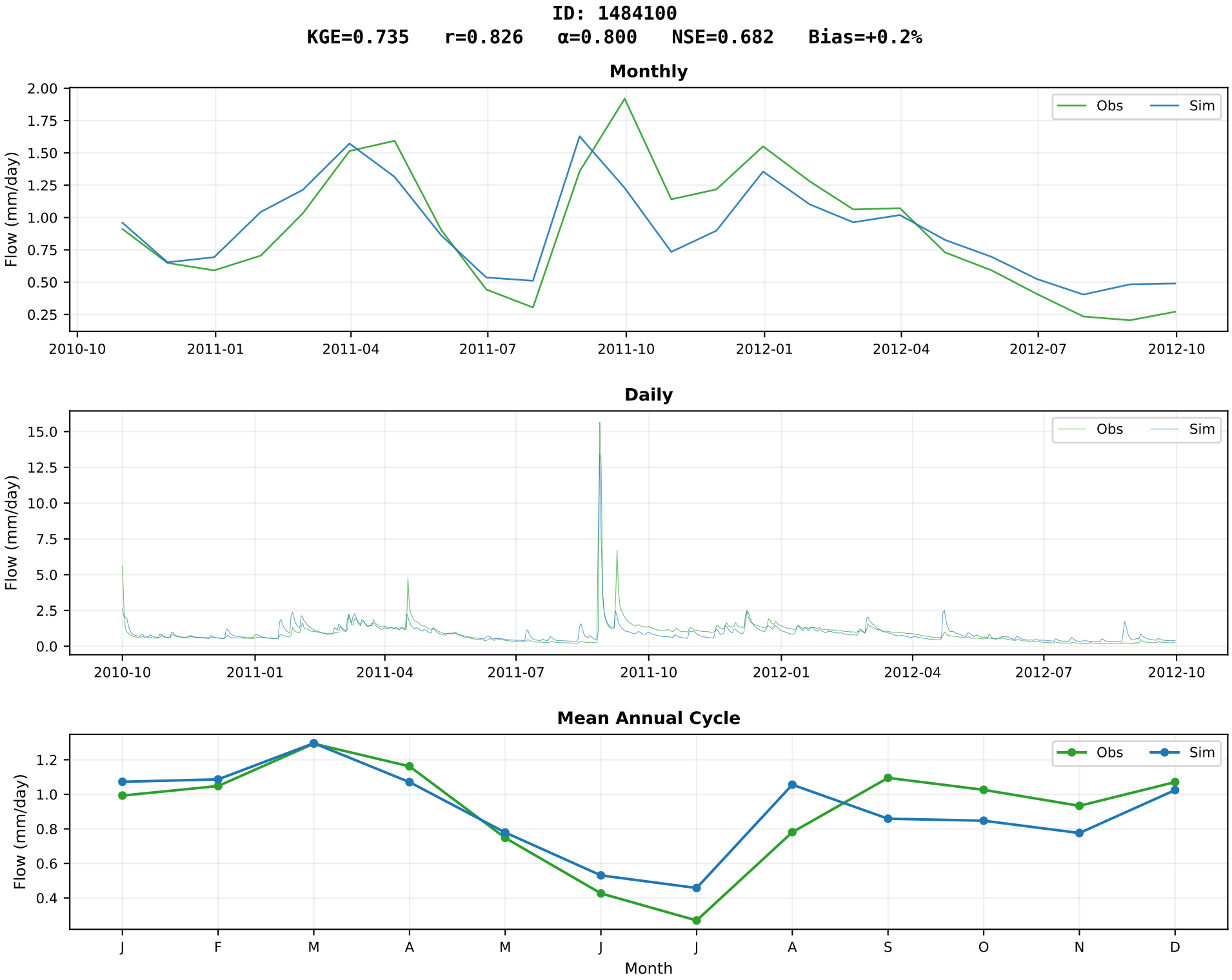

SF SNOQUALMIE RIVER AT EDGEWICK, WA (ID=12143600)
KGE=0.740 r=0.911 α=0.764 NSE=0.808 Bias=-6.1%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

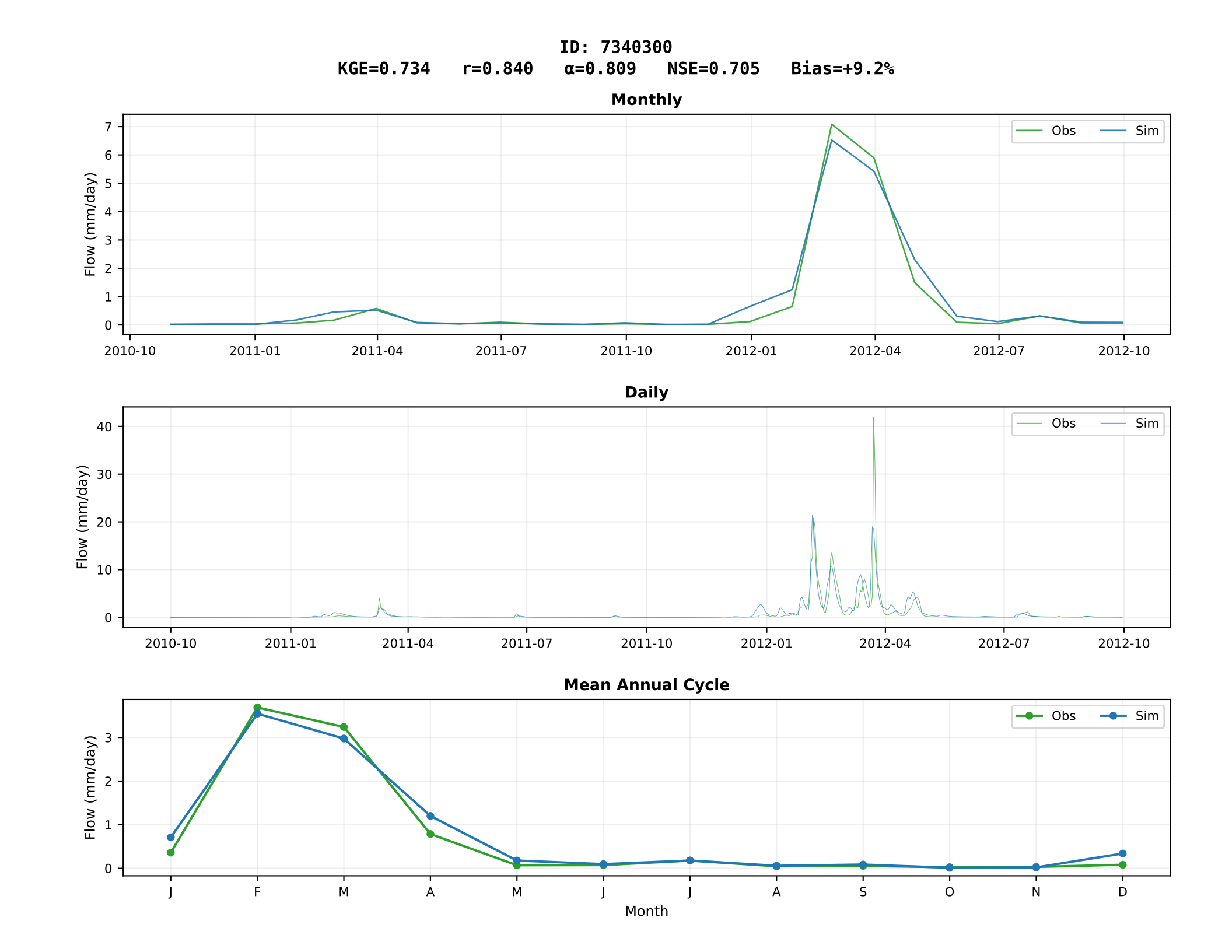

BIG SUR R NR BIG SUR CA (ID=11143000)
KGE=0.739 r=0.930 α=0.786 NSE=0.834 Bias=-13.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

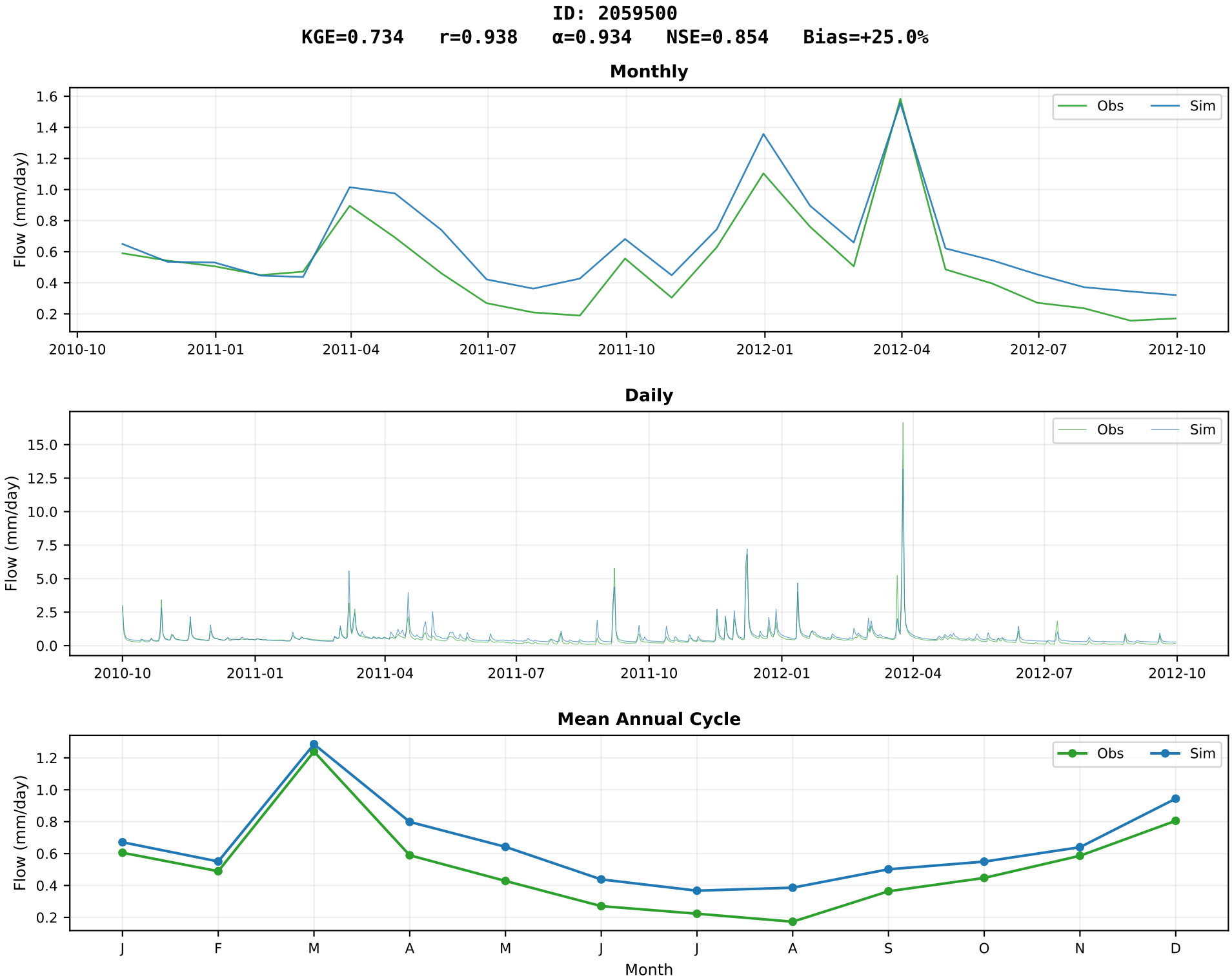

ID: 1137500
KGE=0.738 r=0.883 α=0.782 NSE=0.766 Bias=-8.5%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

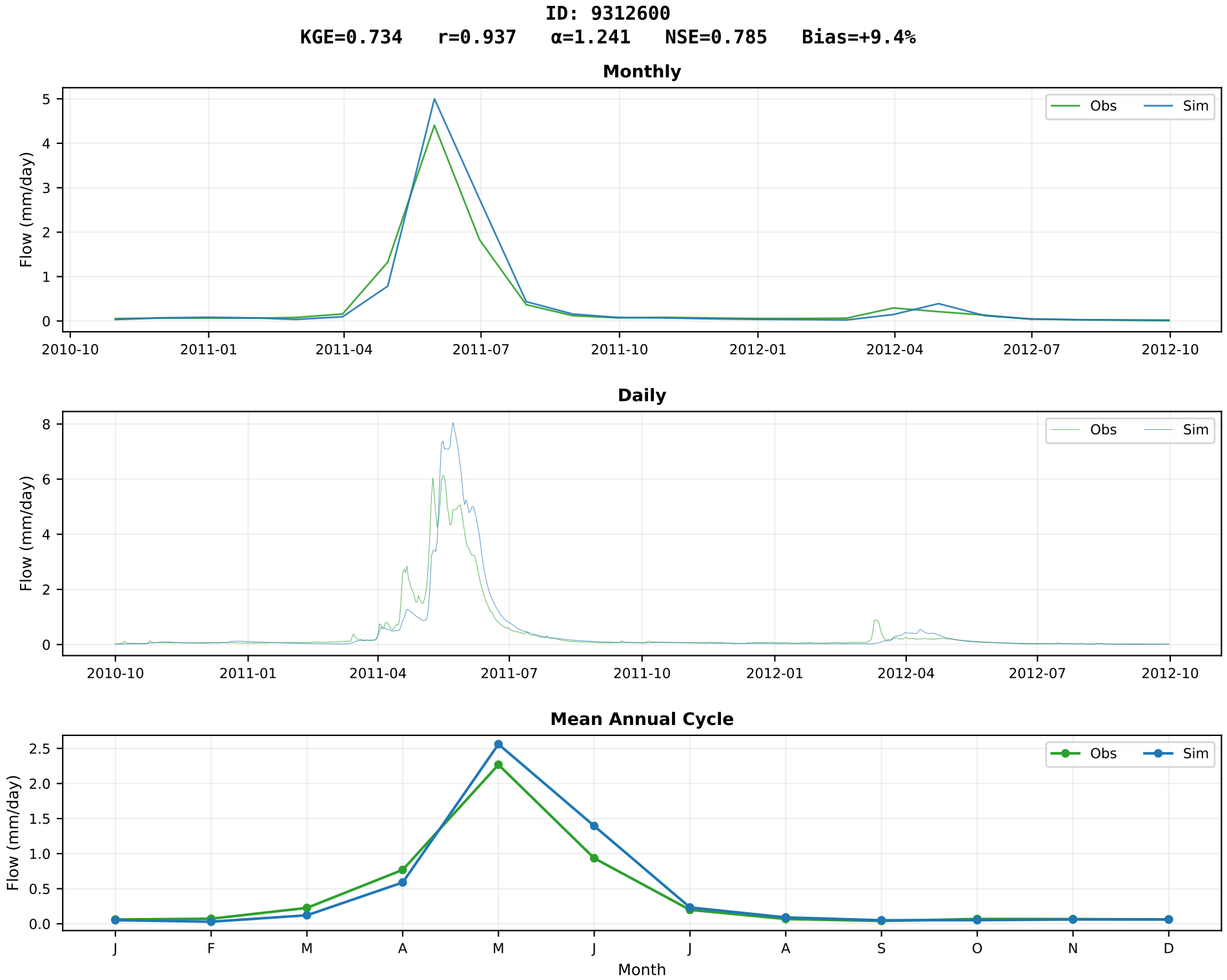

LOPEZ C NR ARROYO GRANDE CA (ID=11141280)
KGE=0.738 r=0.923 α=0.780 NSE=0.824 Bias=-12.0%
Monthly
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
25
50
75
100
125
150
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
5
10
15
20
J
F
M
A
M
J
J
A
S
O
N
D
Month

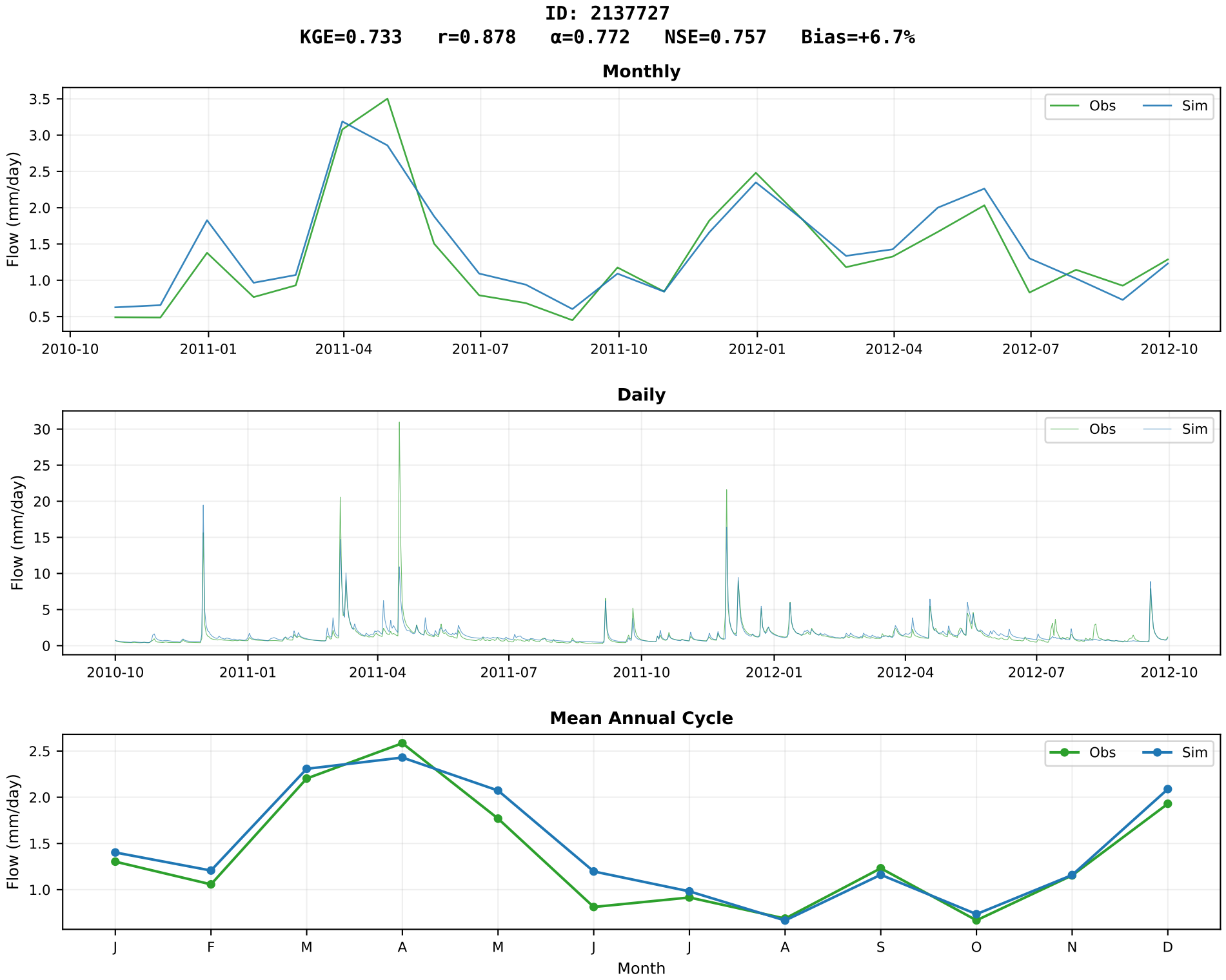

ID: 1544500
KGE=0.736 r=0.901 α=0.761 NSE=0.791 Bias=-5.3%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

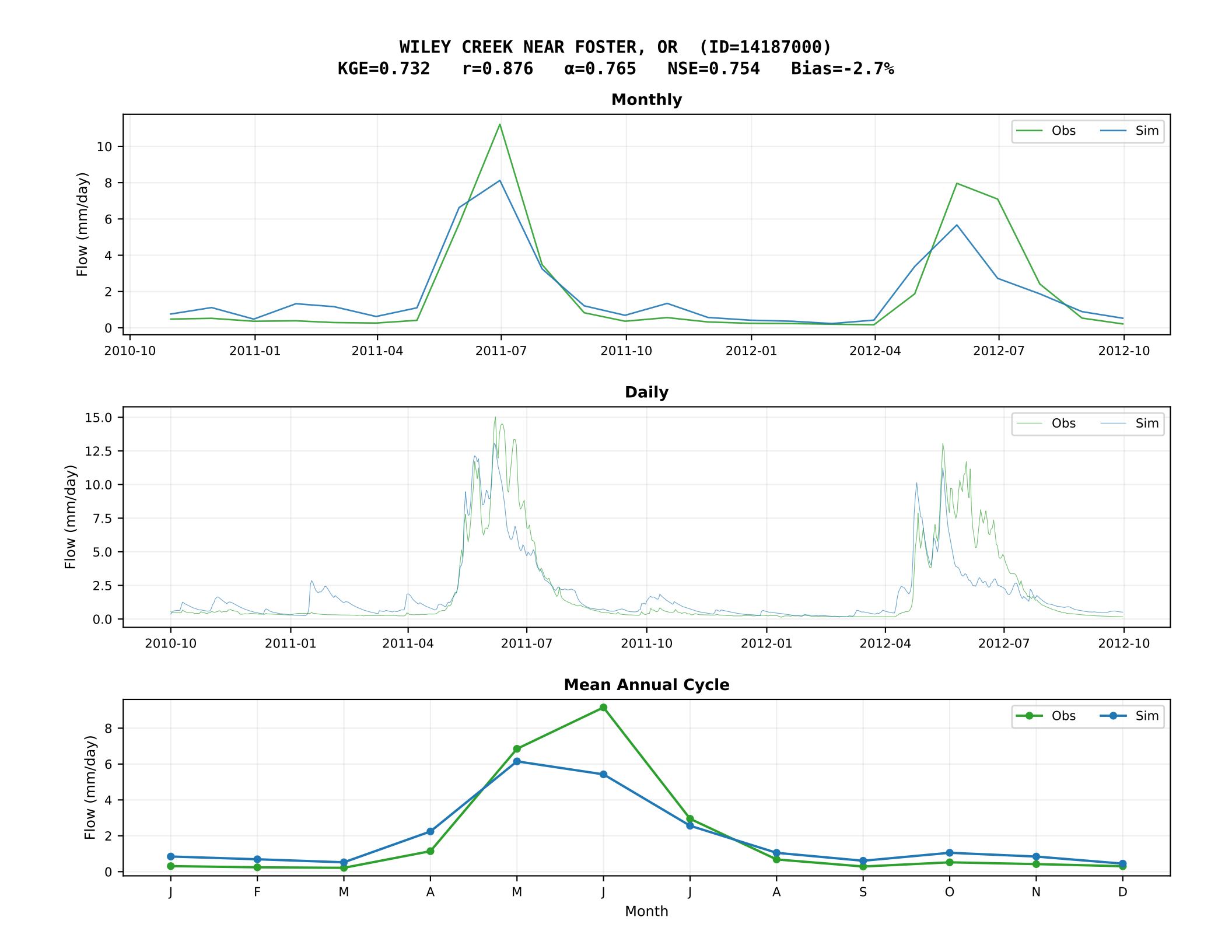

ID: 1484100
KGE=0.735 r=0.826 α=0.800 NSE=0.682 Bias=+0.2%
Monthly
Flow (mm/day)
Obs
Sim
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
1.2
1.0
0.8
0.6
0.4
J
F
M
A
M
J
J
A
S
O
N
D
Month

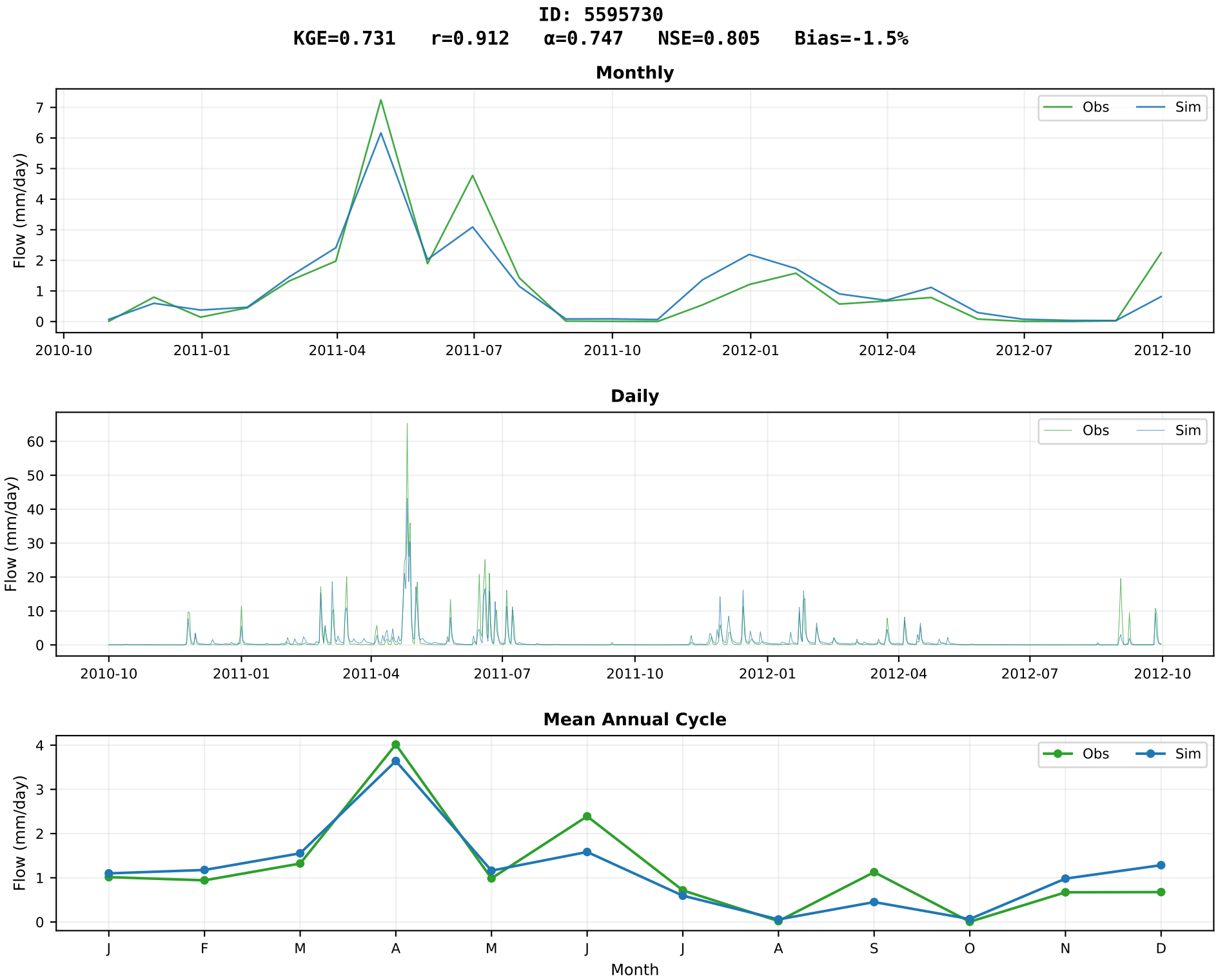

ID: 7340300
KGE=0.734 r=0.840 α=0.809 NSE=0.705 Bias=+9.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

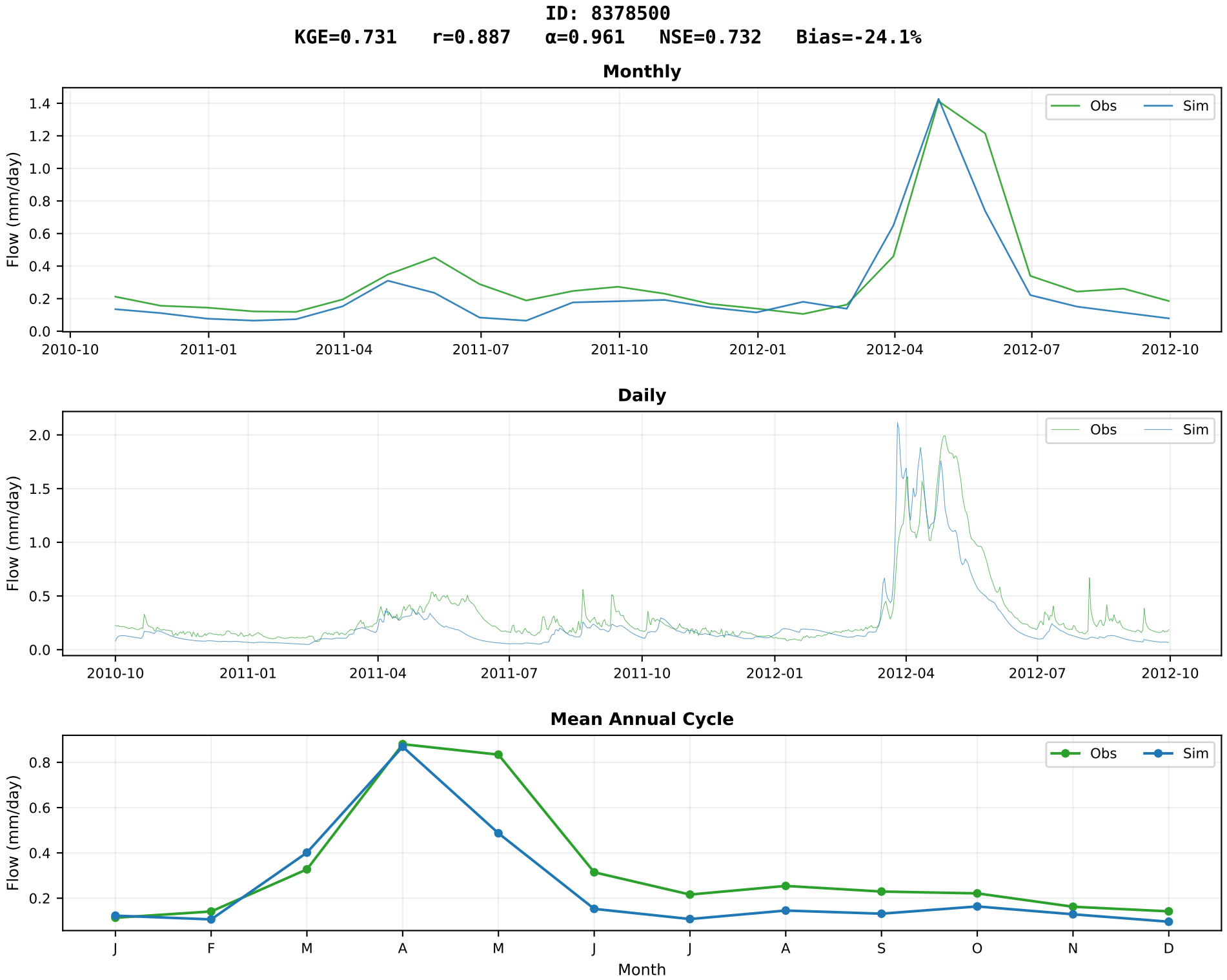

ID: 2059500
KGE=0.734 r=0.938 α=0.934 NSE=0.854 Bias=+25.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Mean Annual Cycle
Flow (mm/day)
J
F
M
A
M
J
J
A
S
O
N
D
Month

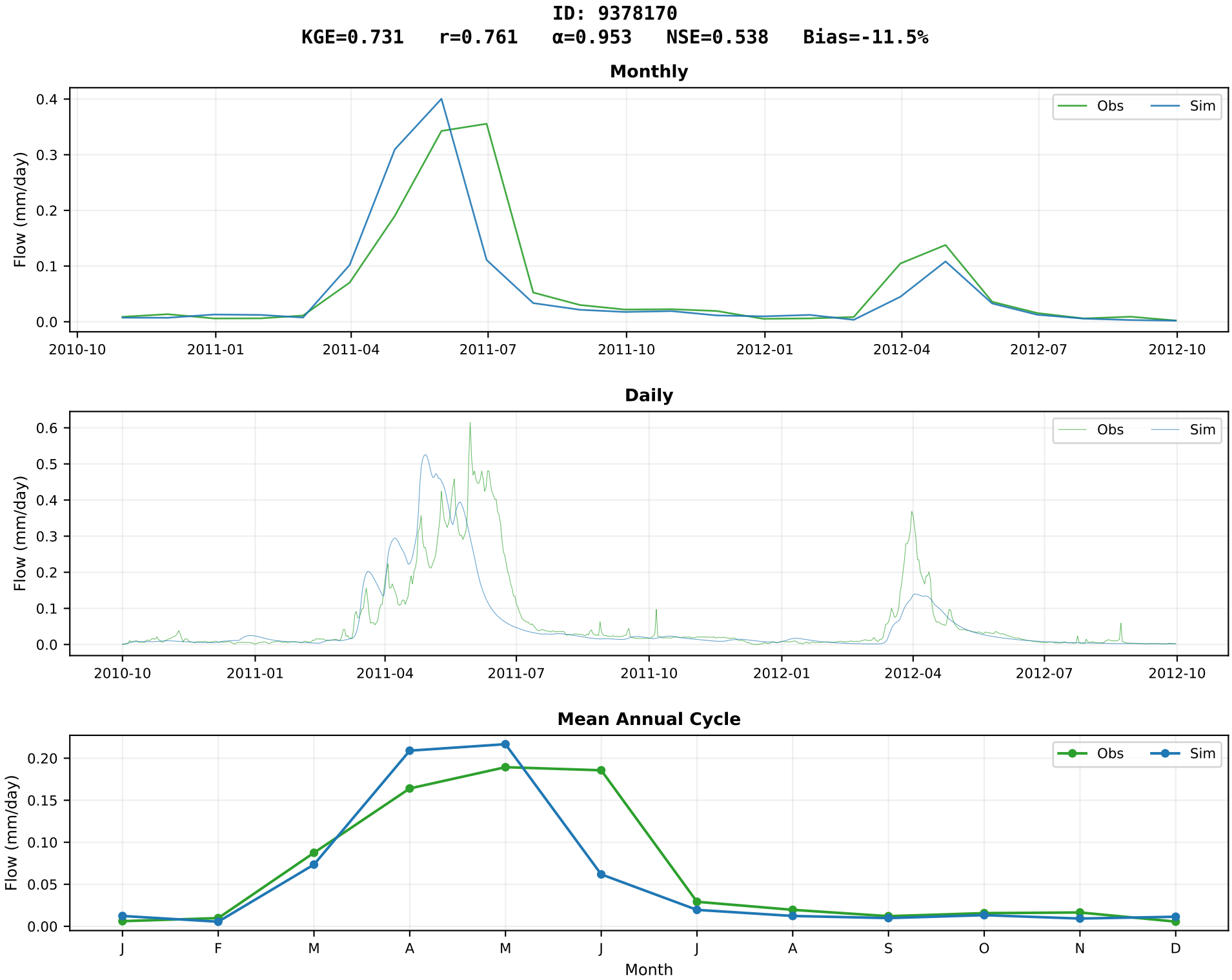

ID: 9312600
KGE=0.734 r=0.937 α=1.241 NSE=0.785 Bias=+9.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

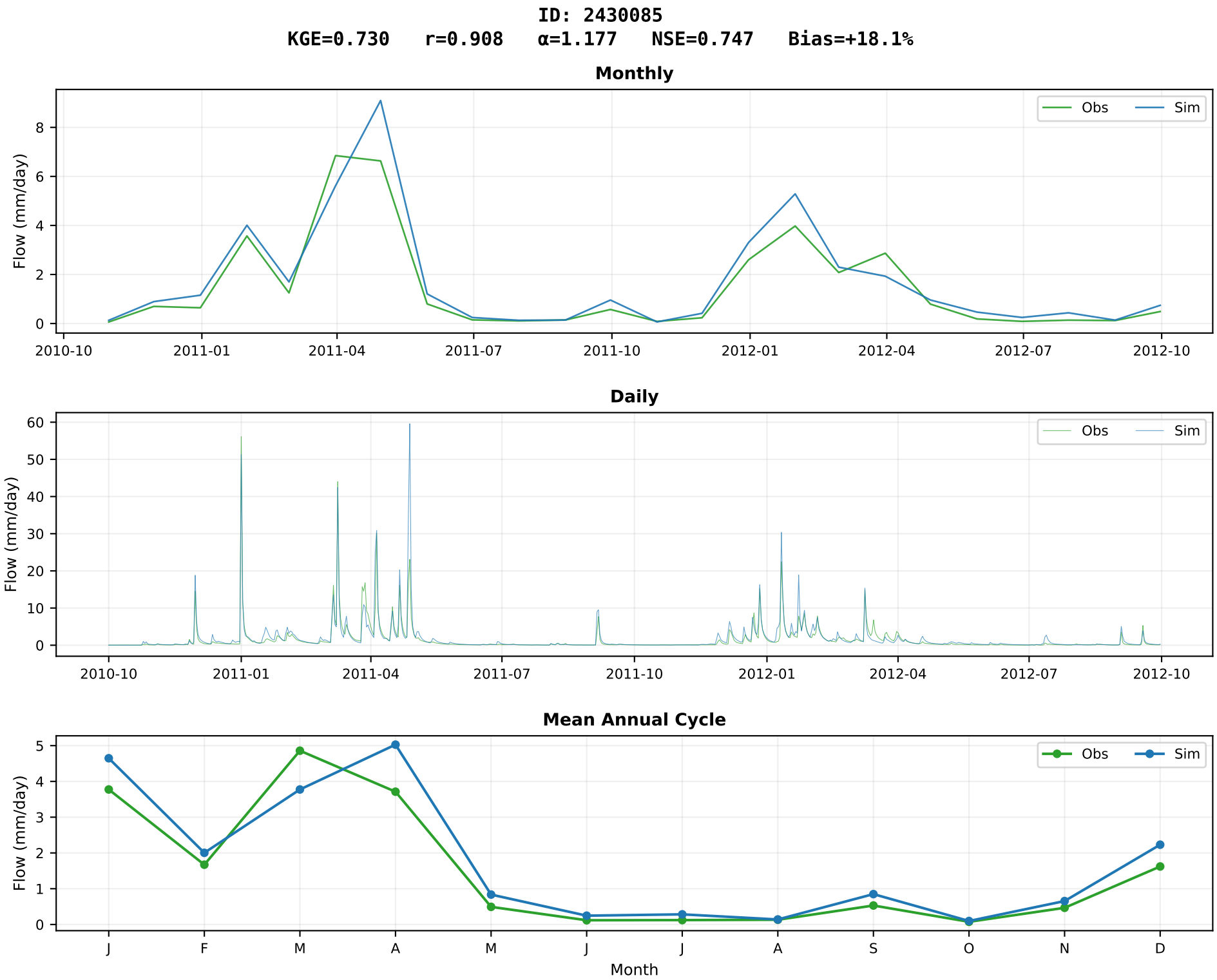

ID: 2137727
KGE=0.733 r=0.878 α=0.772 NSE=0.757 Bias=+6.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

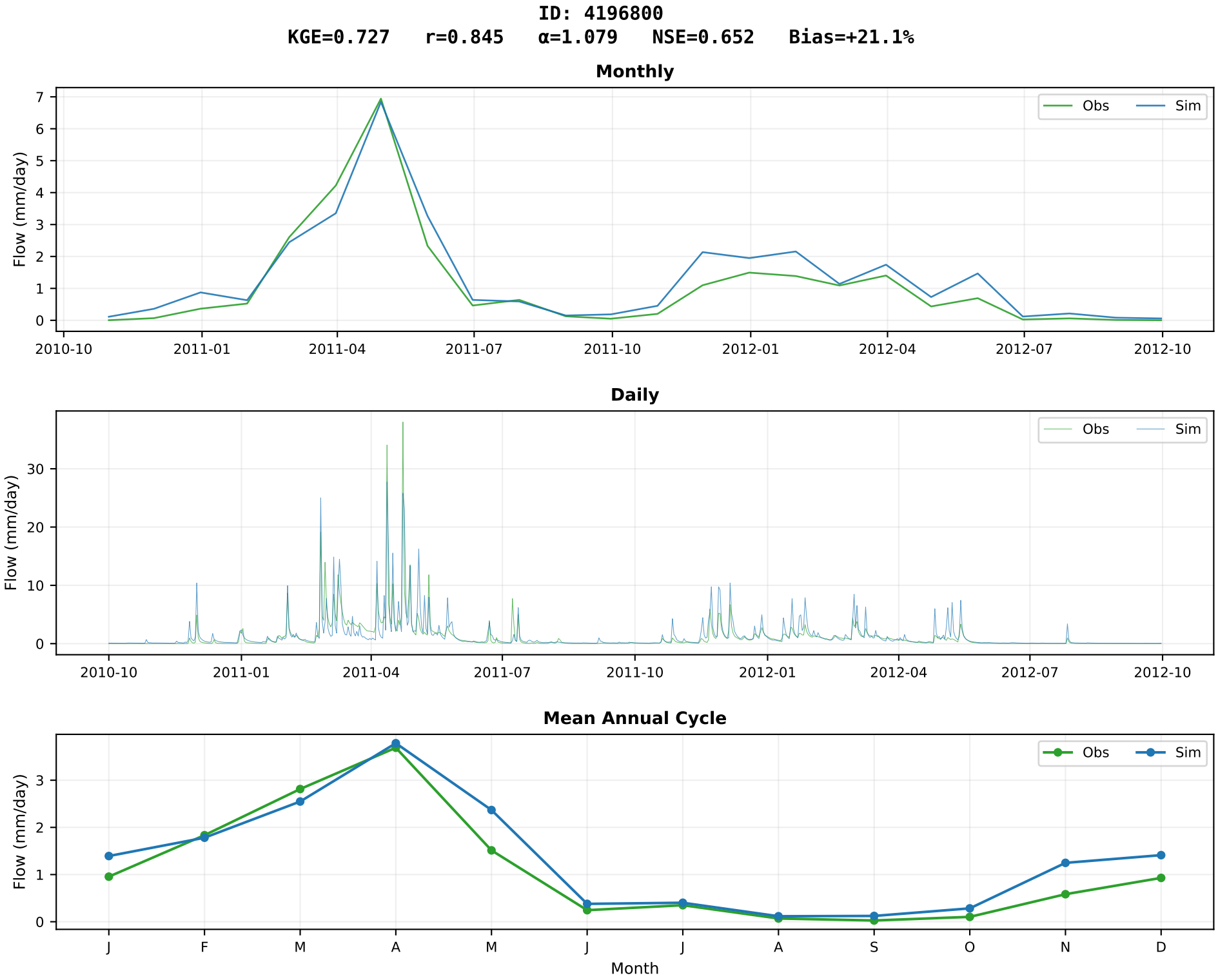

WILEY CREEK NEAR FOSTER, OR (ID=14187000)
KGE=0.732 r=0.876 α=0.765 NSE=0.754 Bias=-2.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Mean Annual Cycle
Flow (mm/day)
J
F
M
A
M
J
J
A
S
O
N
D
Month

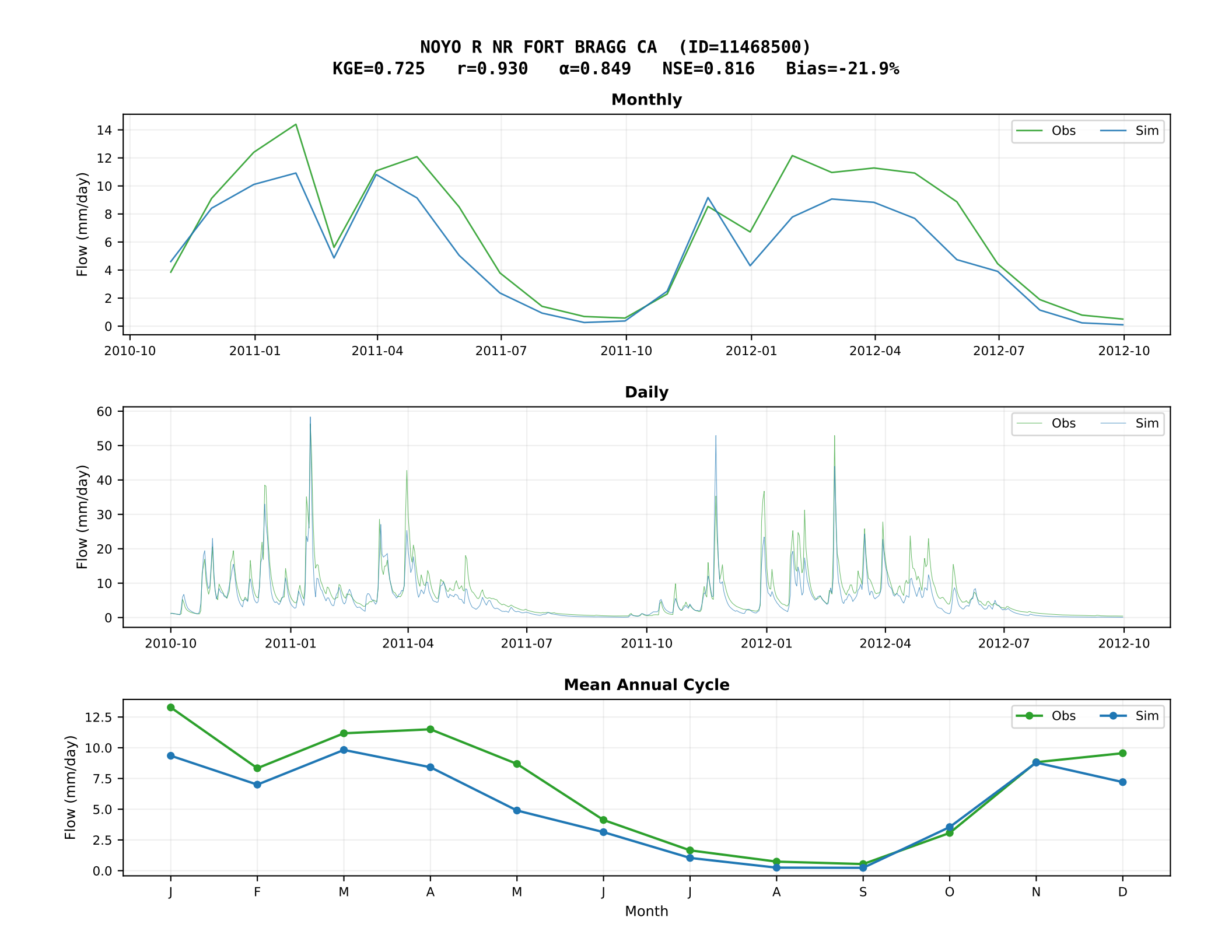

ID: 5595730
KGE=0.731 r=0.912 α=0.747 NSE=0.805 Bias=-1.5%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
7
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

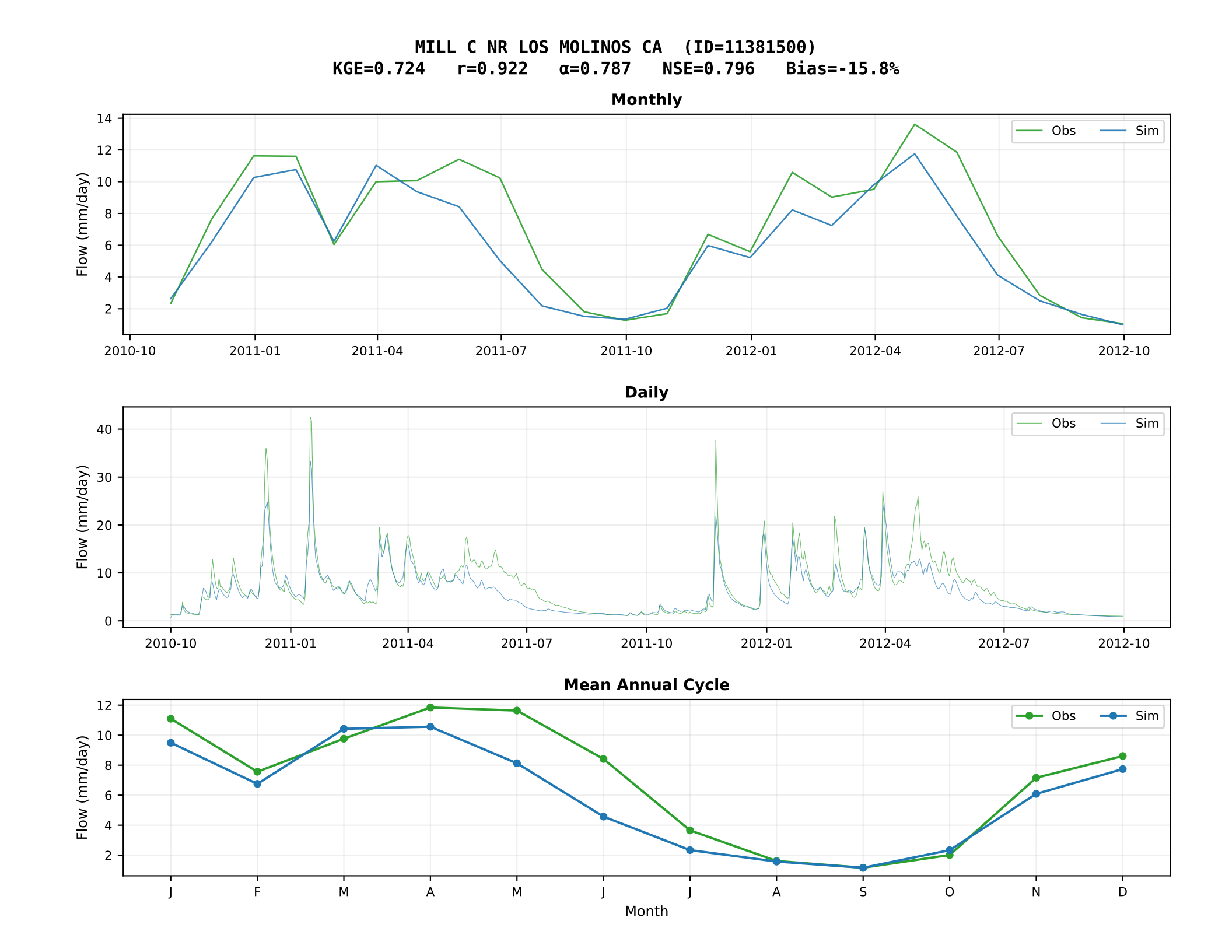

ID: 8378500
KGE=0.731 r=0.887 α=0.961 NSE=0.732 Bias=-24.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

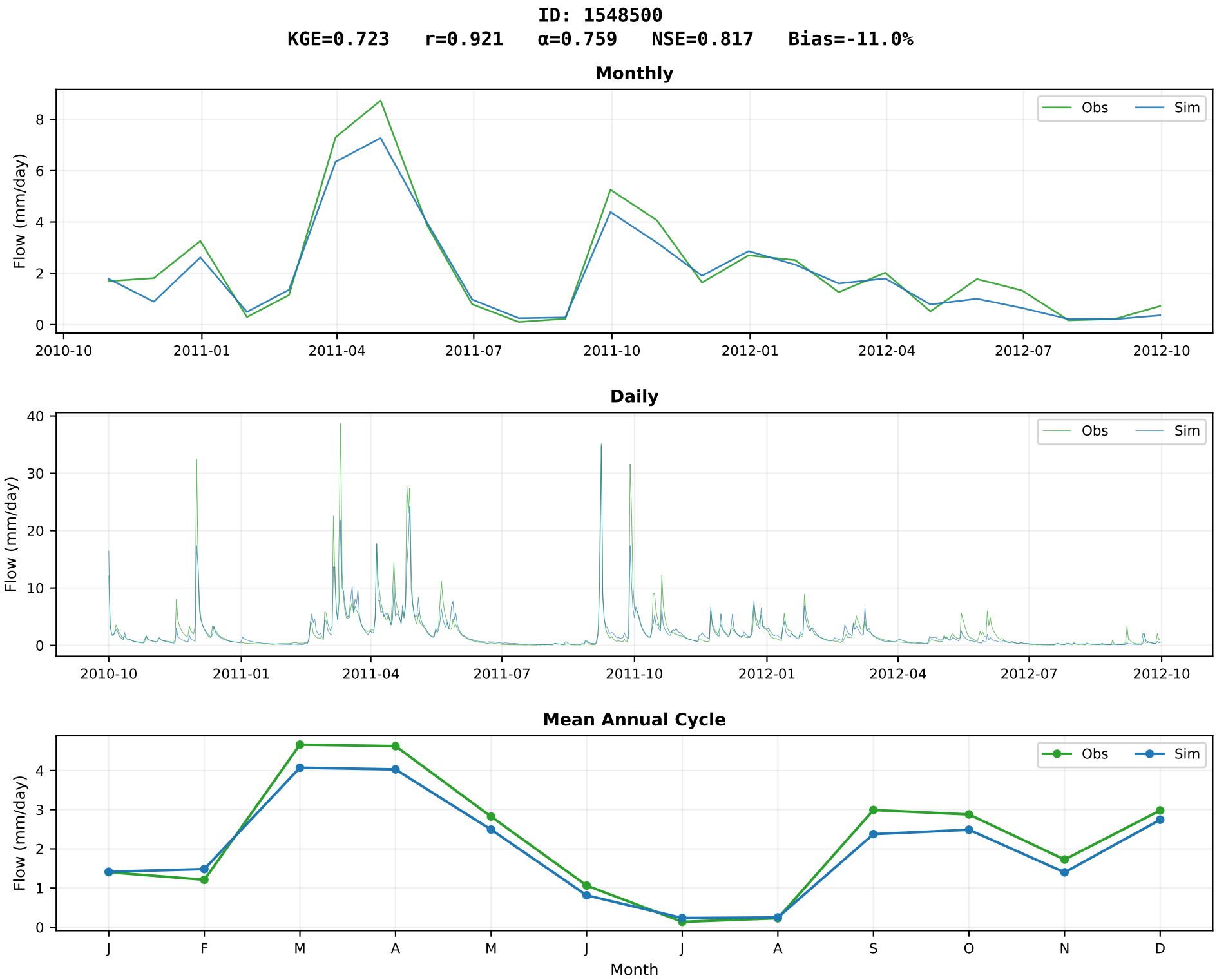

ID: 9378170
KGE=0.731 r=0.761 α=0.953 NSE=0.538 Bias=-11.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

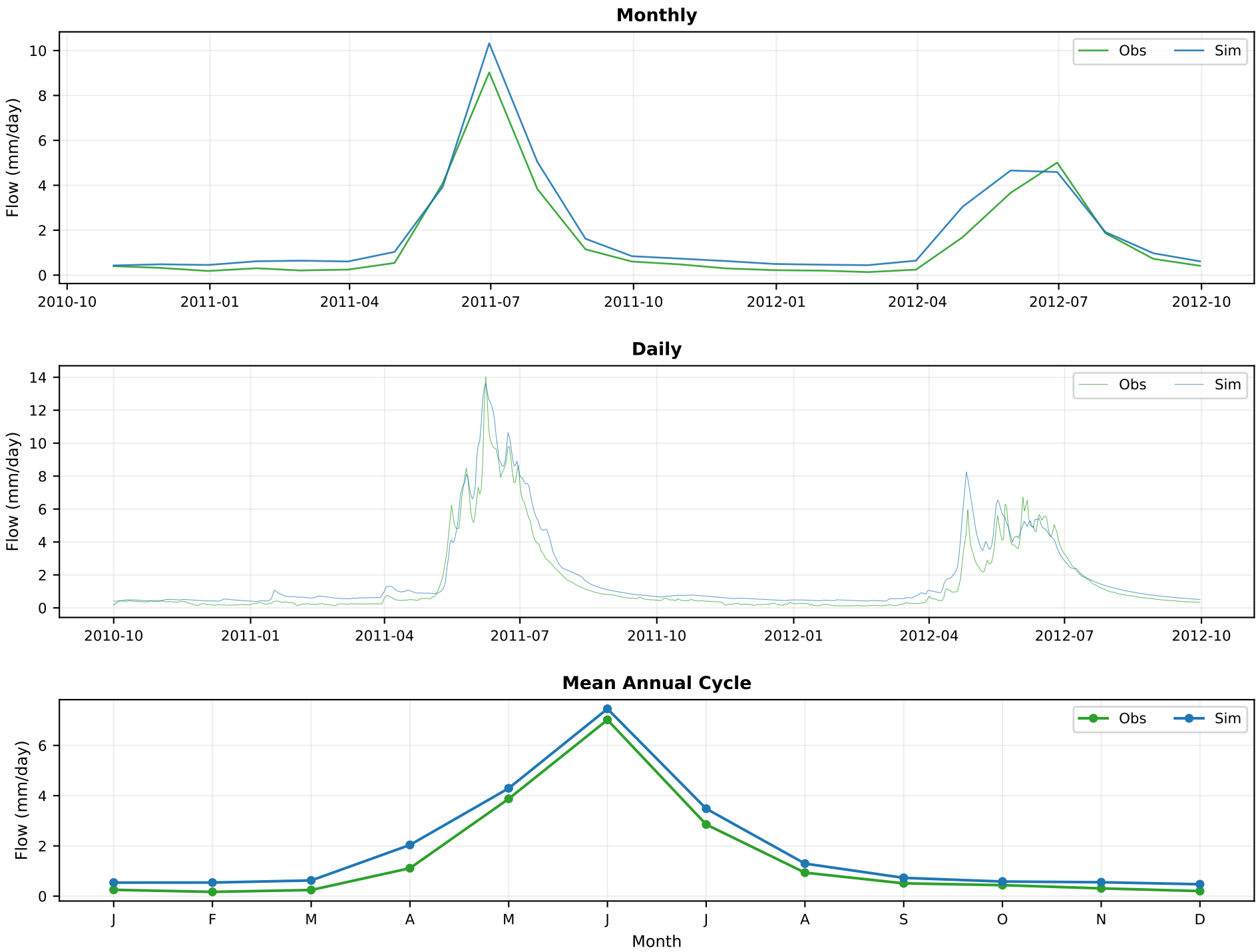

ID: 2430085
KGE=0.730 r=0.908 α=1.177 NSE=0.747 Bias=+18.1%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

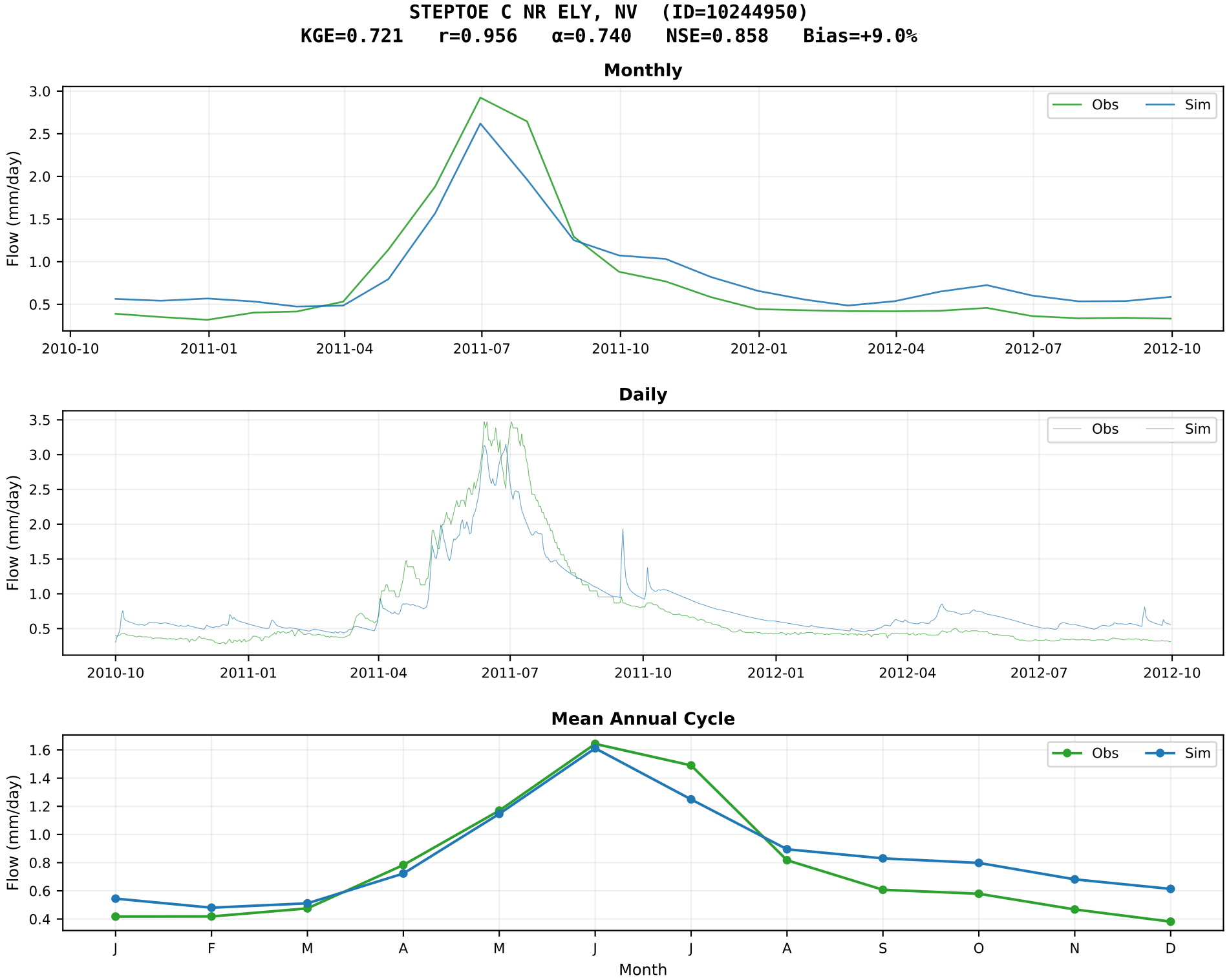

ID: 4196800
KGE=0.727 r=0.845 α=1.079 NSE=0.652 Bias=+21.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

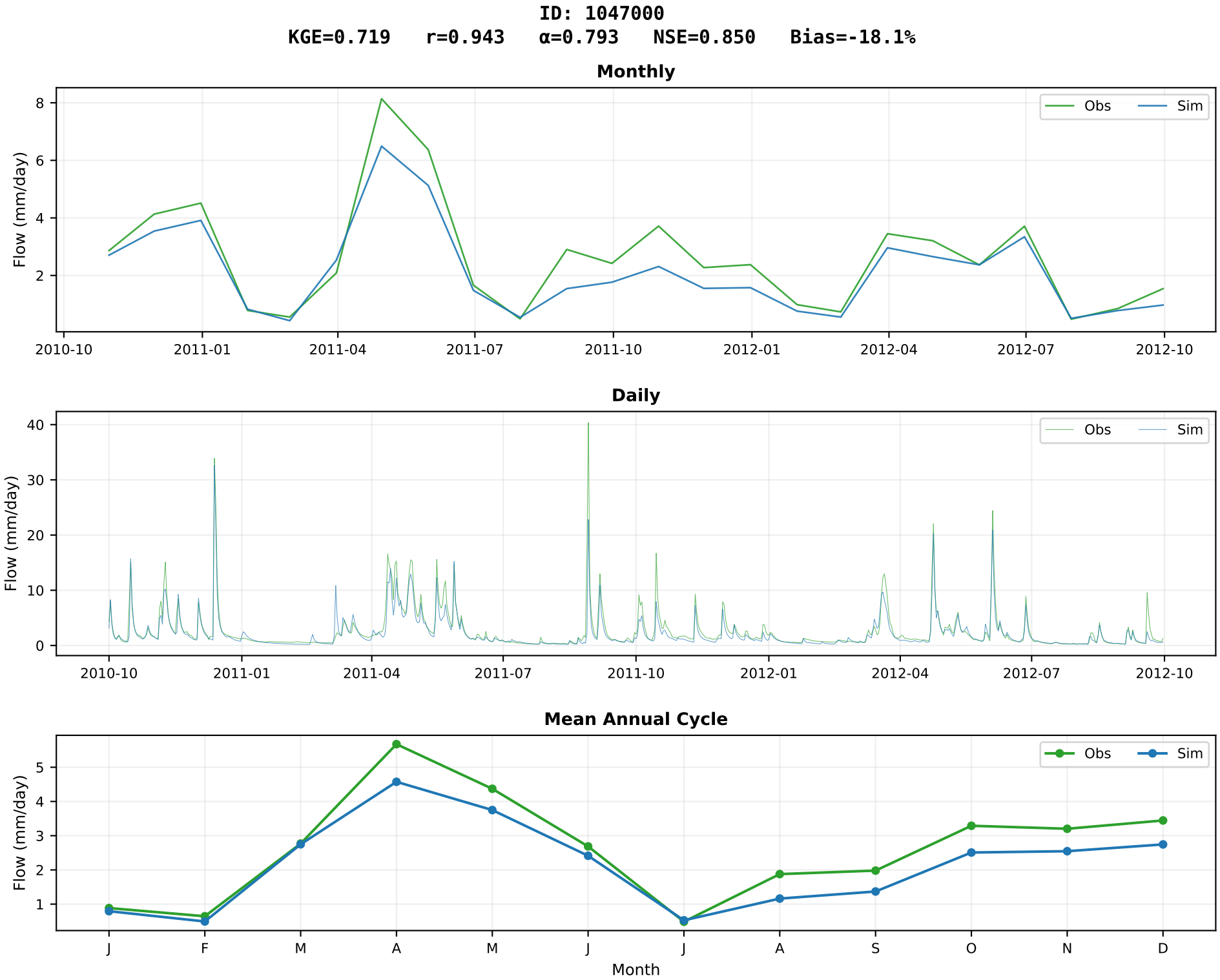

NOYO R NR FORT BRAGG CA (ID=11468500)
KGE=0.725 r=0.930 α=0.849 NSE=0.816 Bias=-21.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

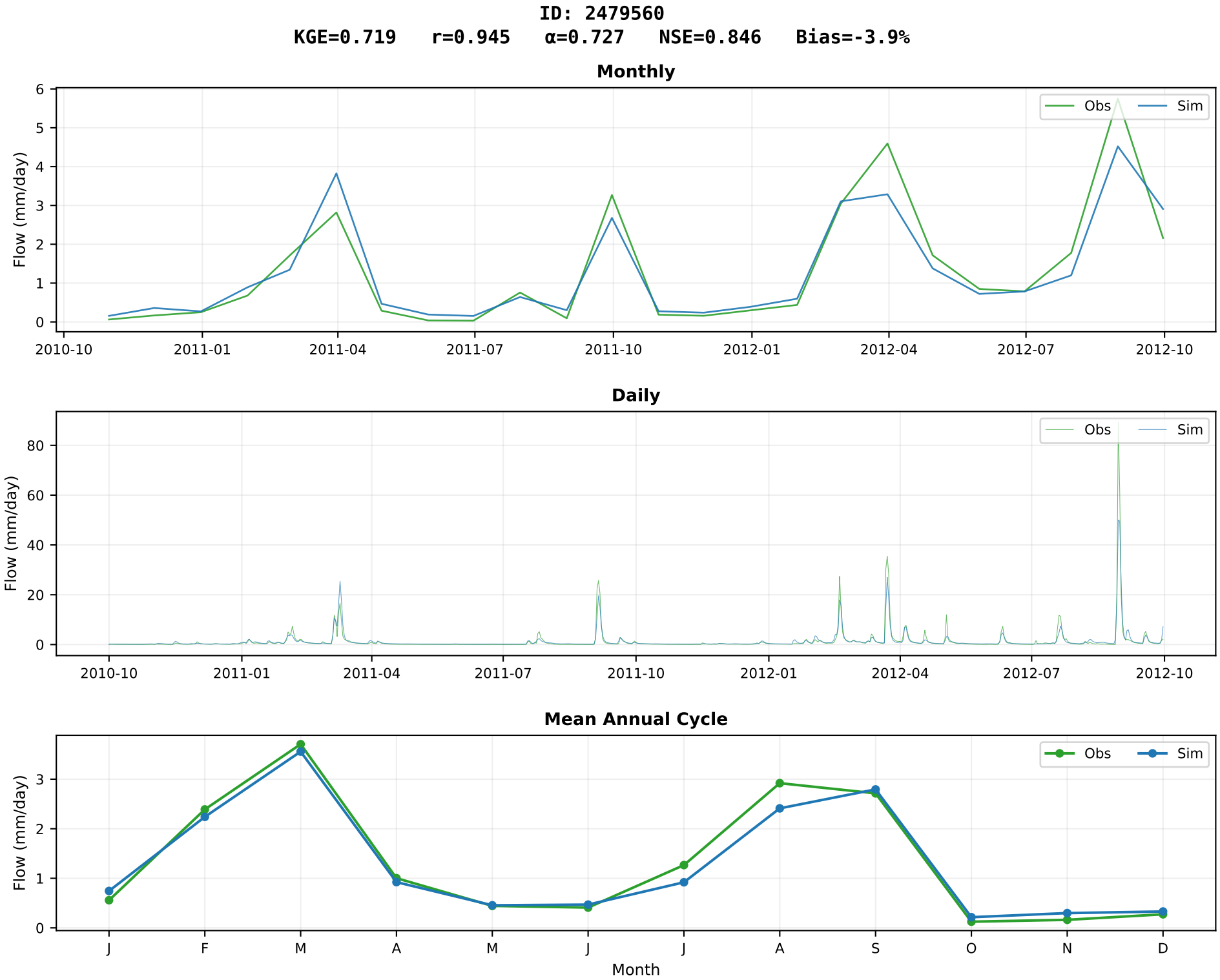

MILL C NR LOS MOLINOS CA (ID=11381500)
KGE=0.724 r=0.922 α=0.787 NSE=0.796 Bias=-15.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

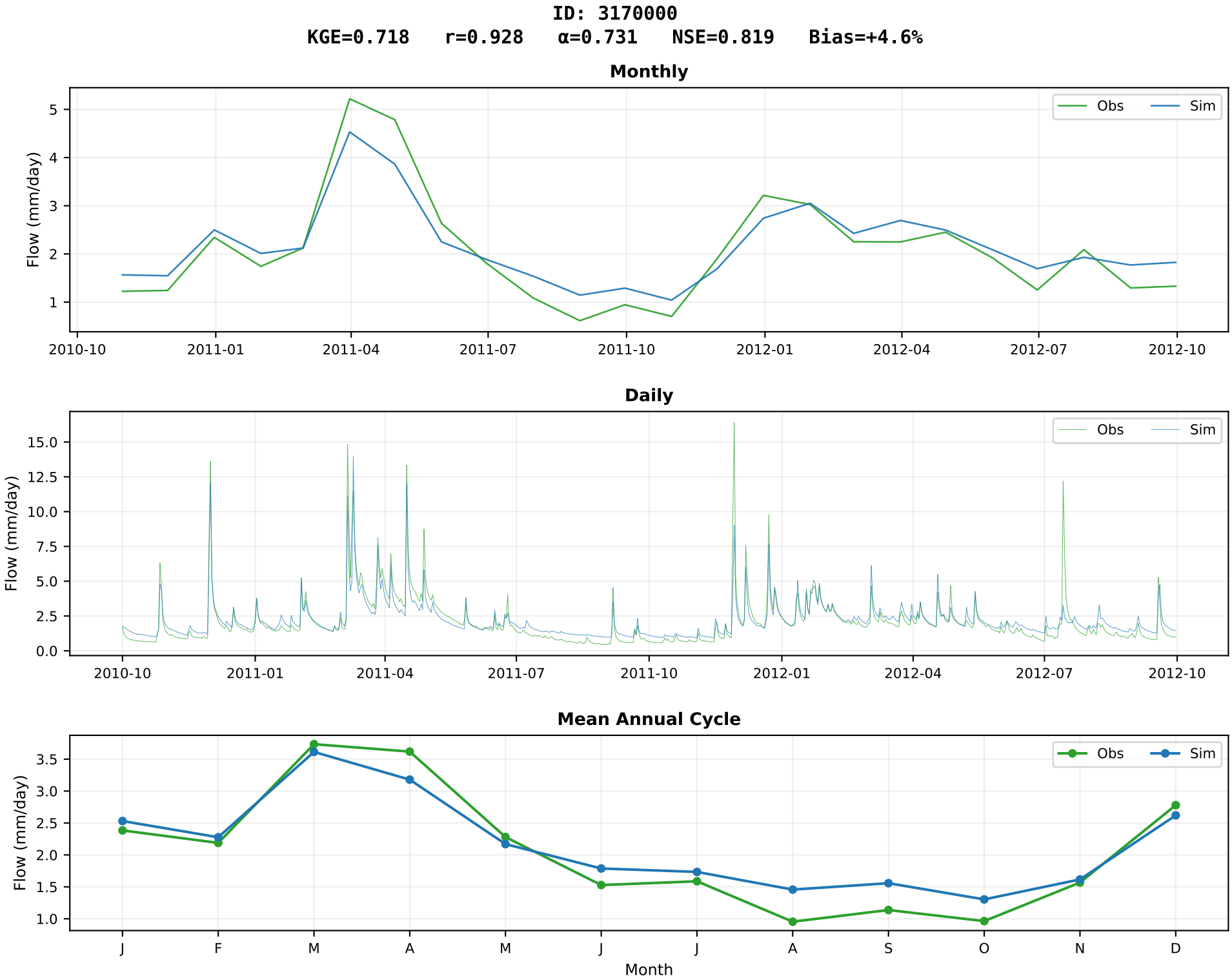

ID: 1548500
KGE=0.723 r=0.921 α=0.759 NSE=0.817 Bias=-11.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


LONG TOM RIVER NEAR NOTI, OREG. (ID=14166500)
KGE=0.721 r=0.972 α=1.093 NSE=0.902 Bias=+26.1%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D


STEPTOE C NR ELY, NV (ID=10244950)
KGE=0.721 r=0.956 α=0.740 NSE=0.858 Bias=+9.0%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

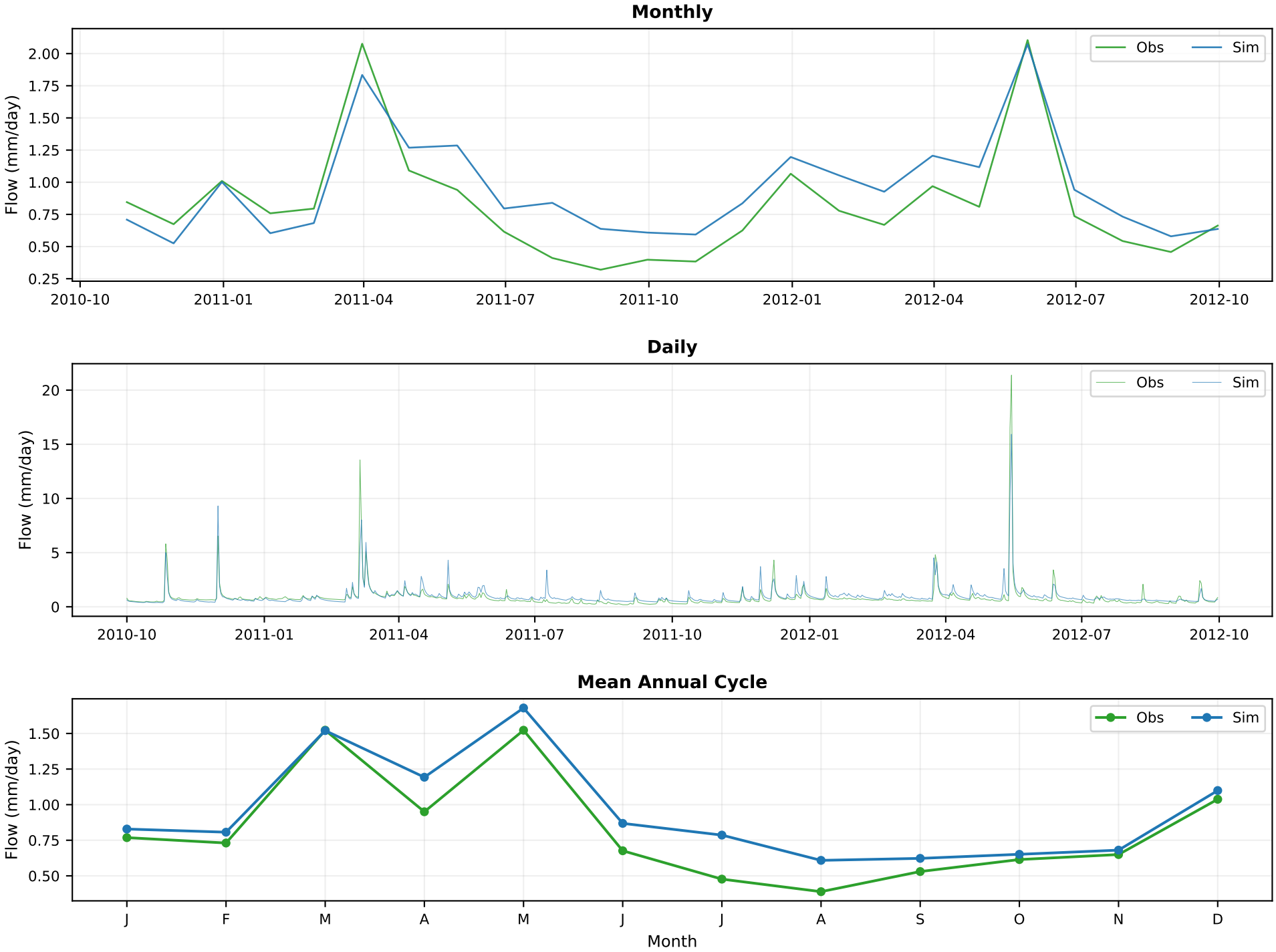

ID: 1047000
KGE=0.719 r=0.943 α=0.793 NSE=0.850 Bias=-18.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

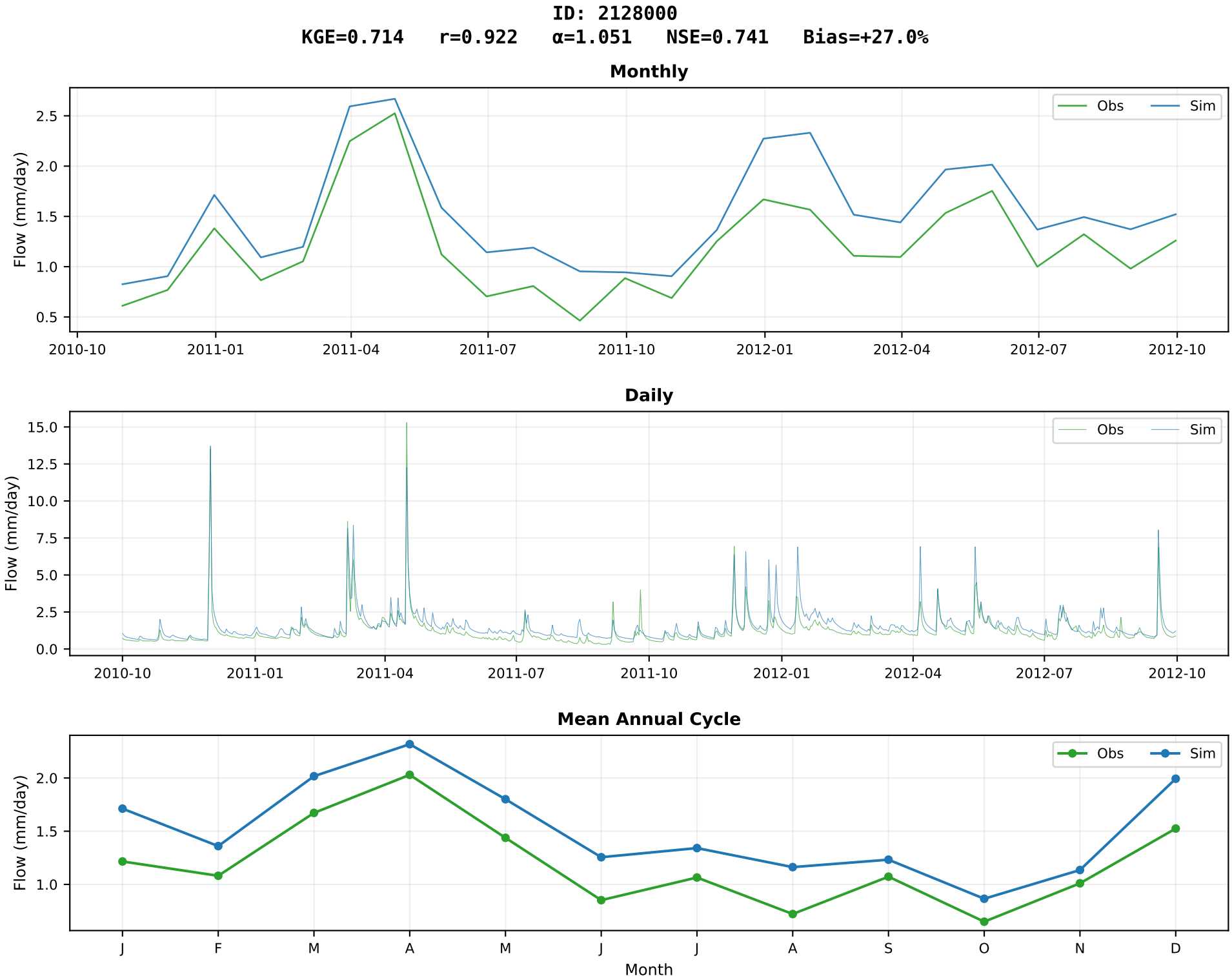

ID: 2479560
KGE=0.719 r=0.945 α=0.727 NSE=0.846 Bias=-3.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

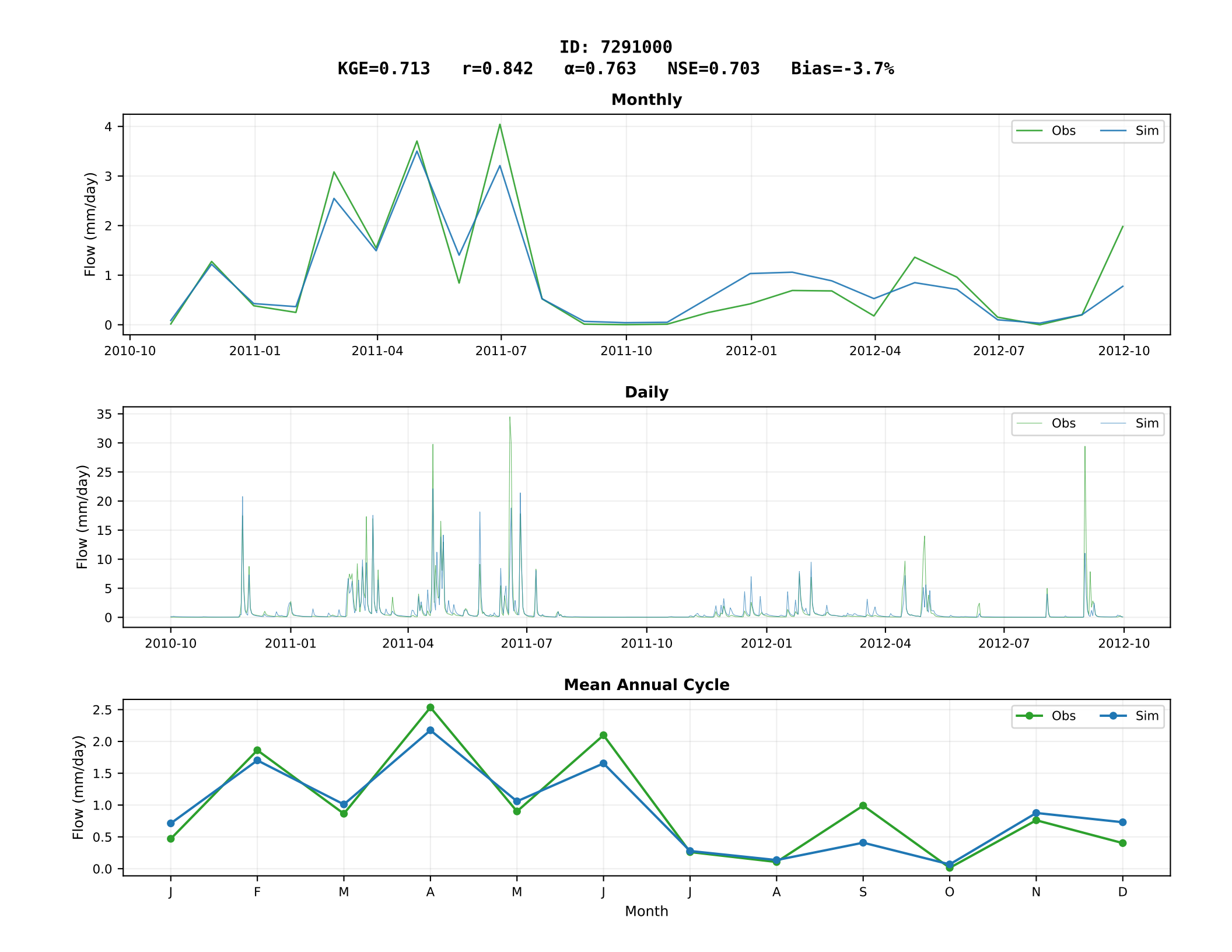

ID: 3170000
KGE=0.718 r=0.928 α=0.731 NSE=0.819 Bias=+4.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

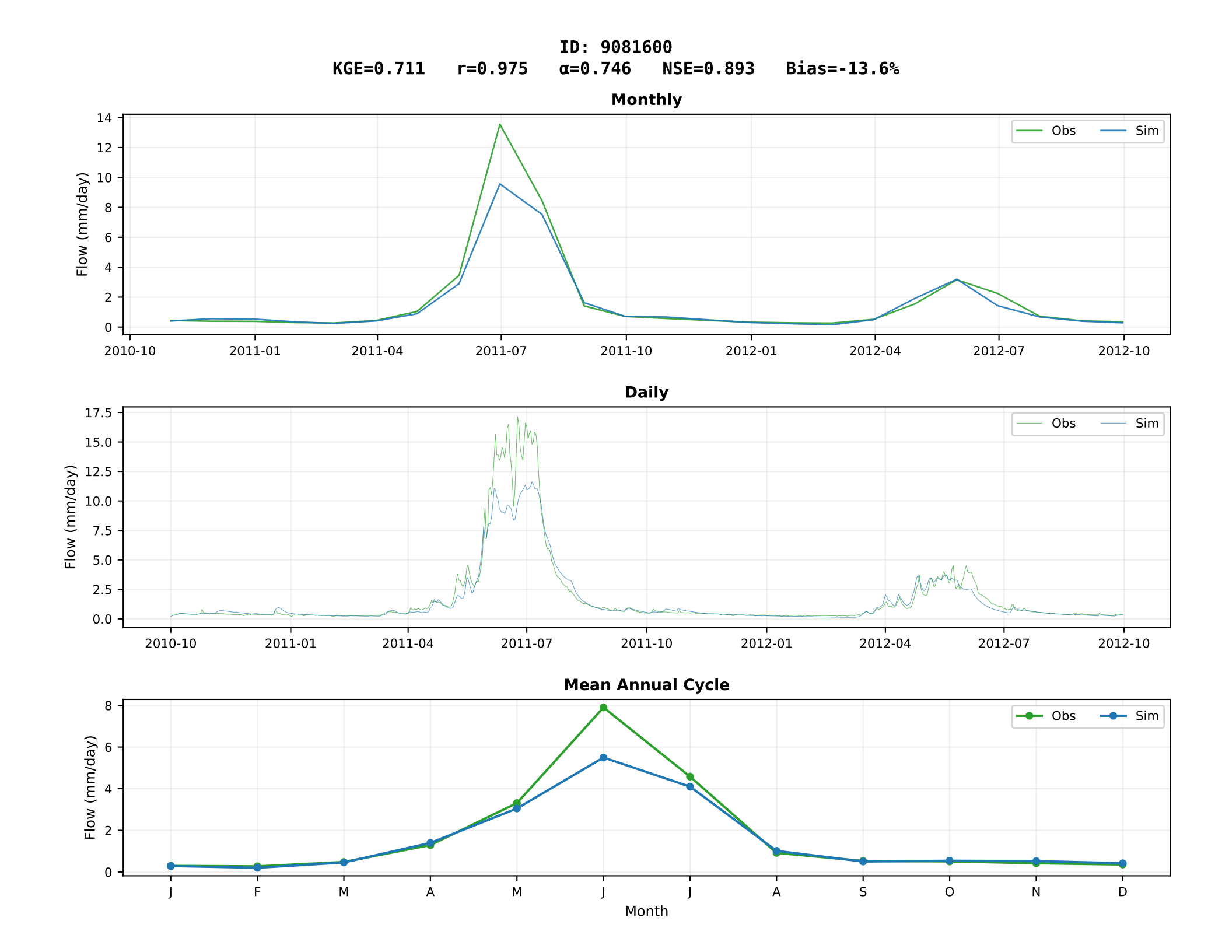
ID: 2112360

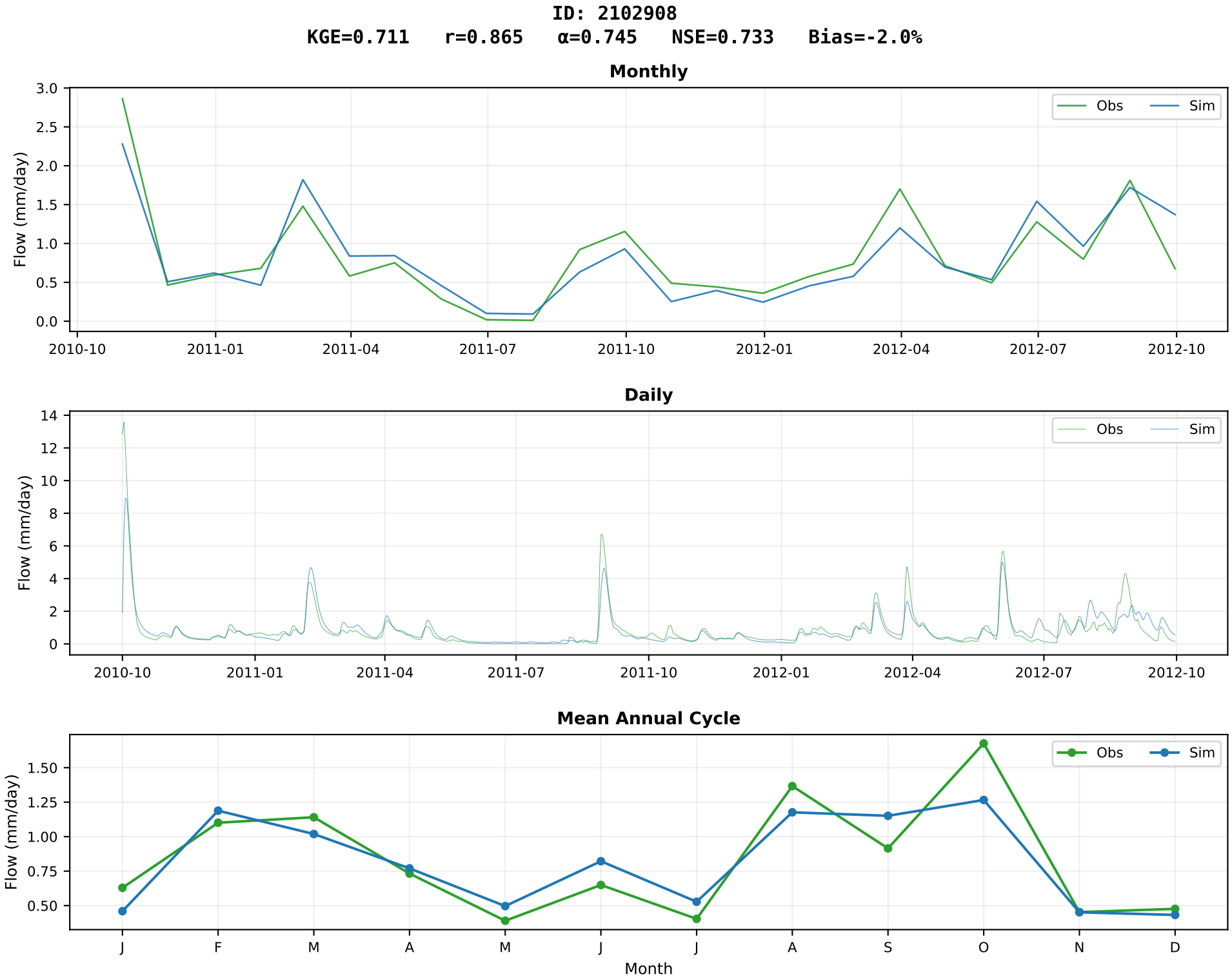
KGE=0.714 r=0.900 α=0.778 NSE=0.786 Bias=+14.9%

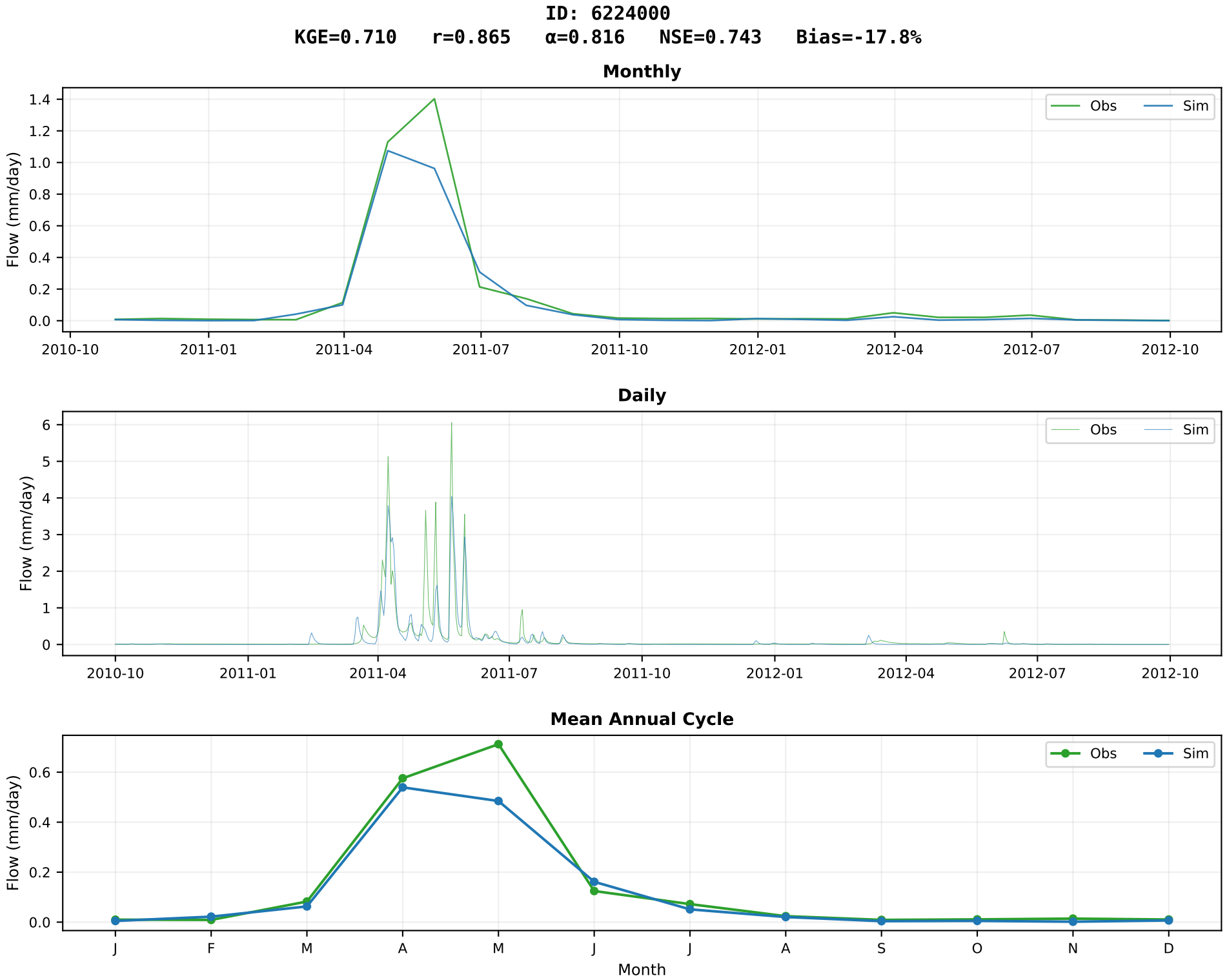


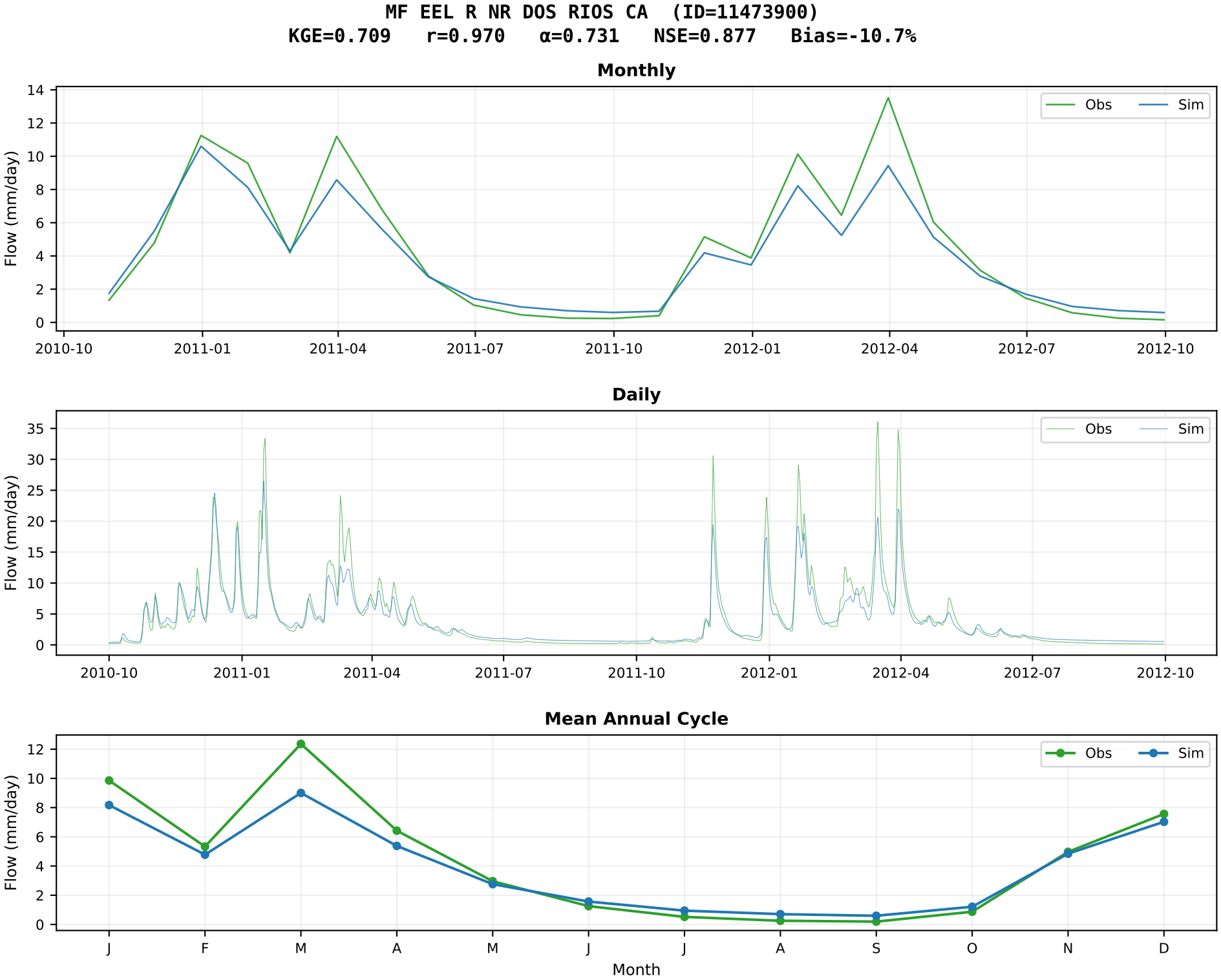

ID: 2128000
KGE=0.714 r=0.922 α=1.051 NSE=0.741 Bias=+27.0%
Monthly
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

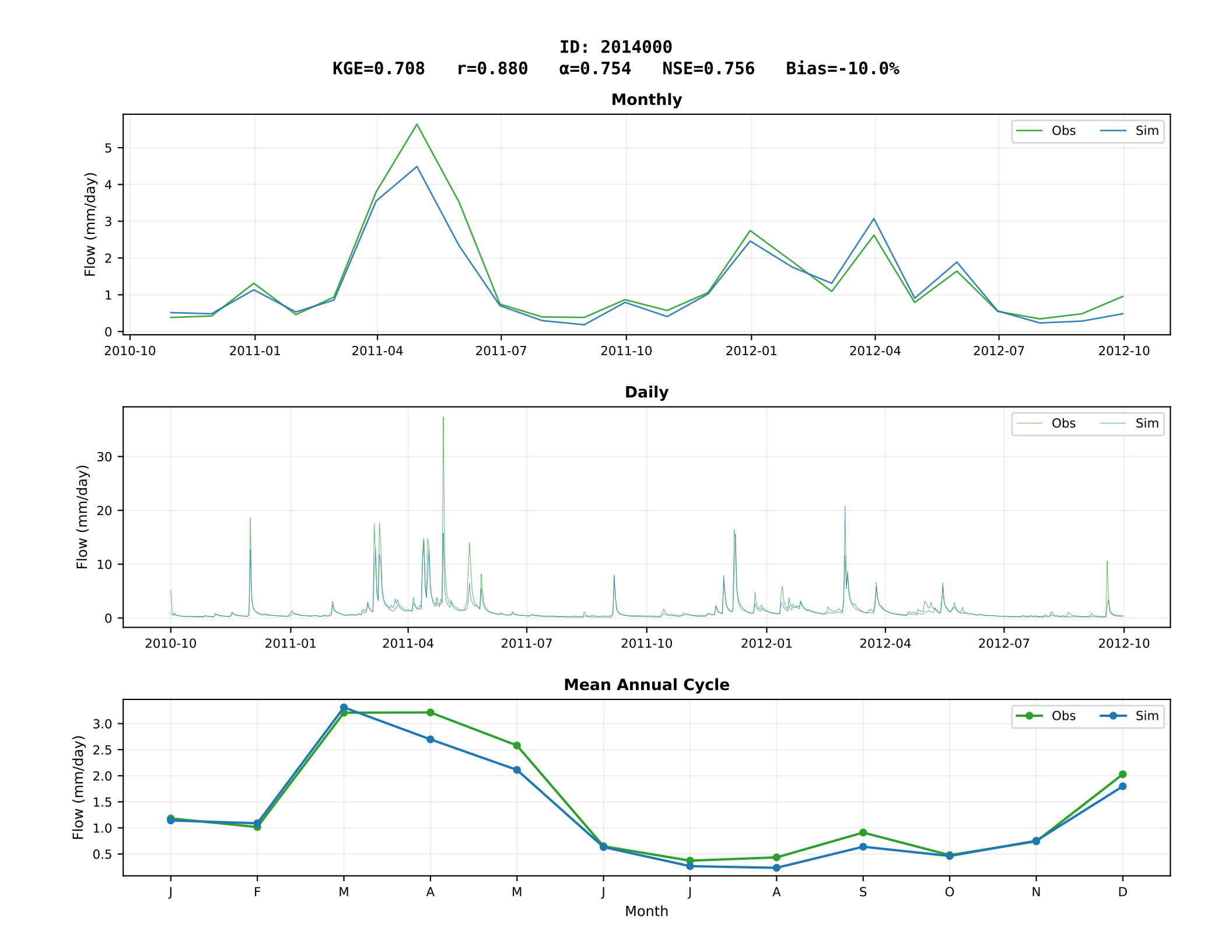

ID: 7291000
KGE=0.713 r=0.842 α=0.763 NSE=0.703 Bias=-3.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

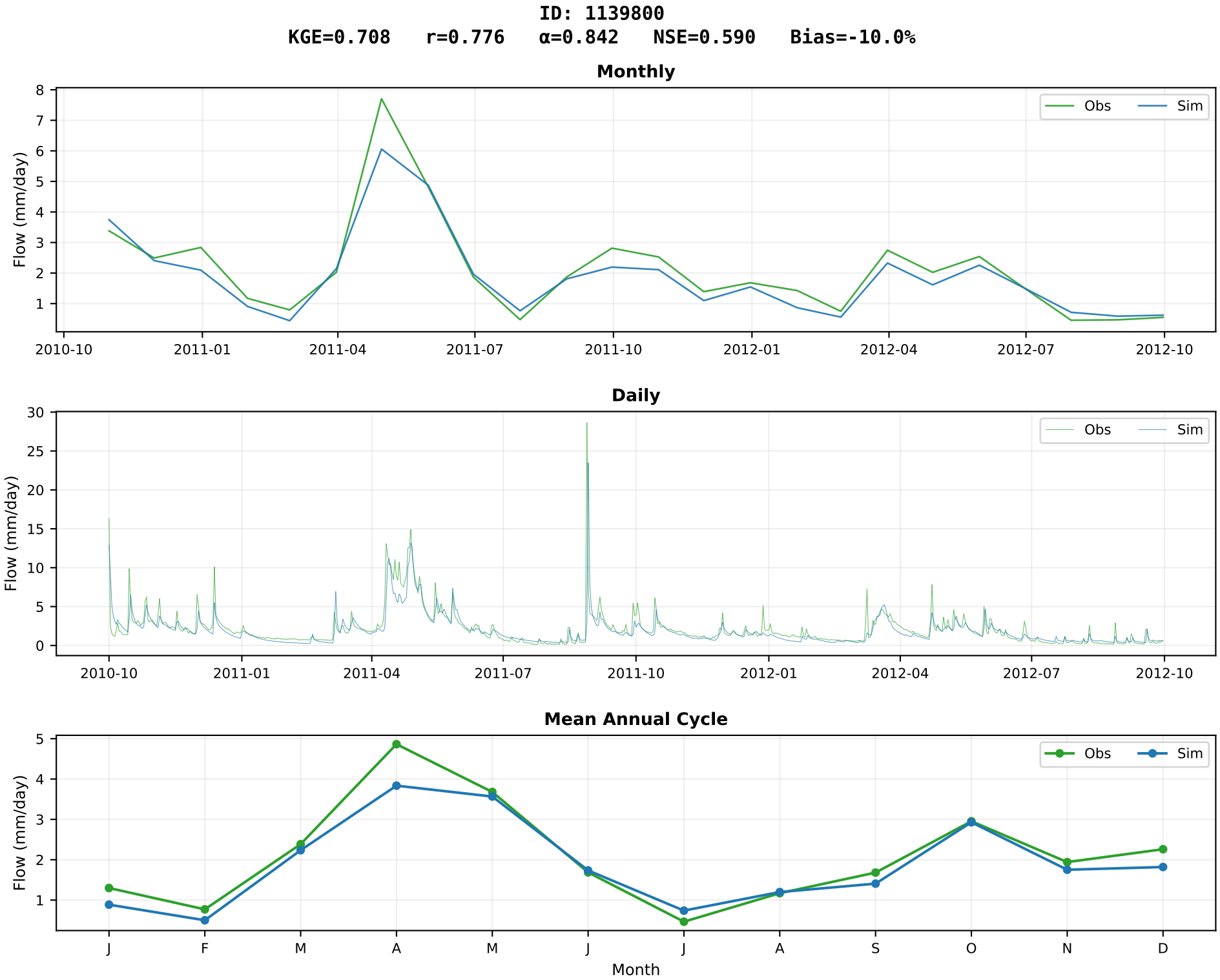

ID: 9081600
KGE=0.711 r=0.975 α=0.746 NSE=0.893 Bias=-13.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

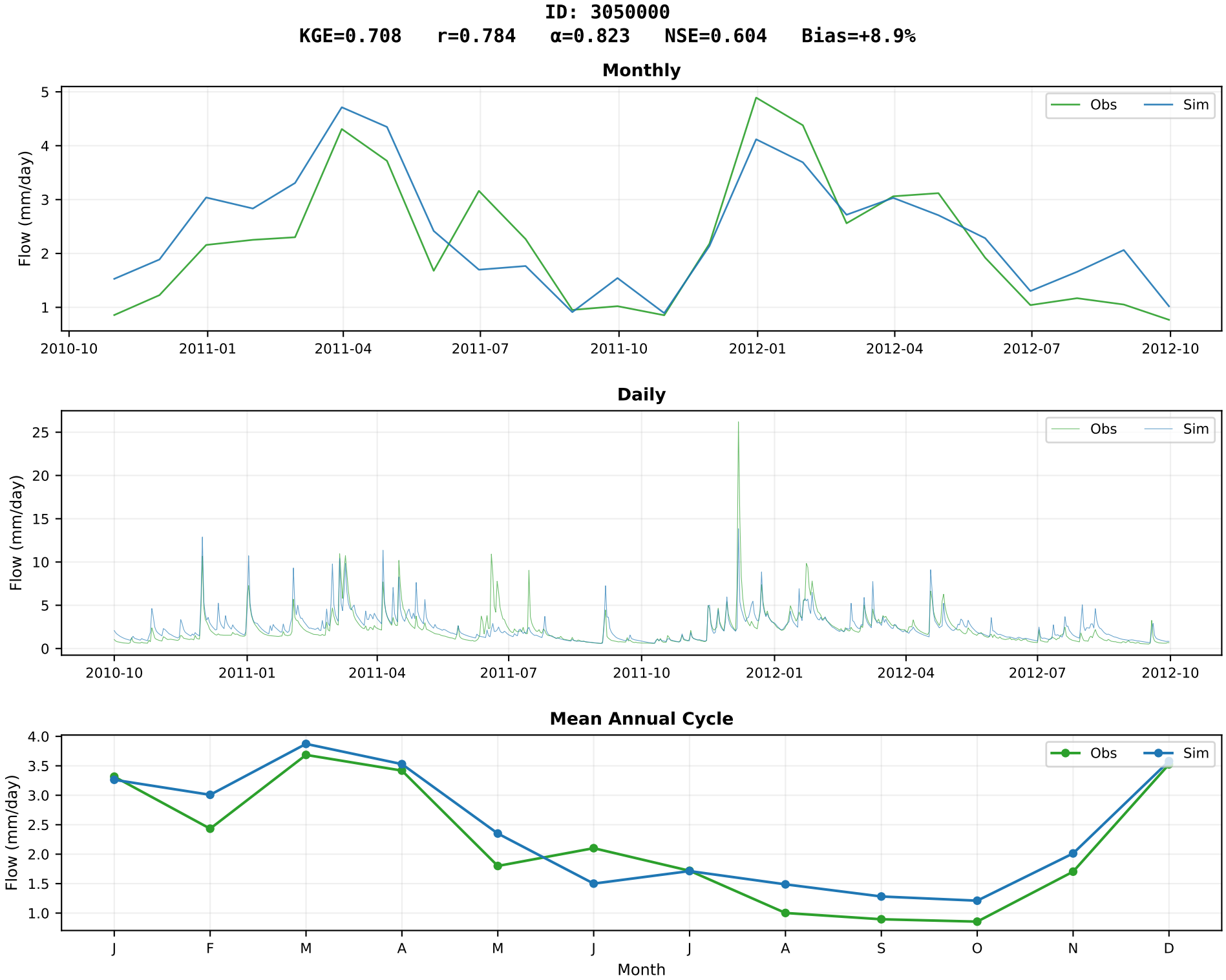

ID: 2102908
KGE=0.711 r=0.865 α=0.745 NSE=0.733 Bias=-2.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

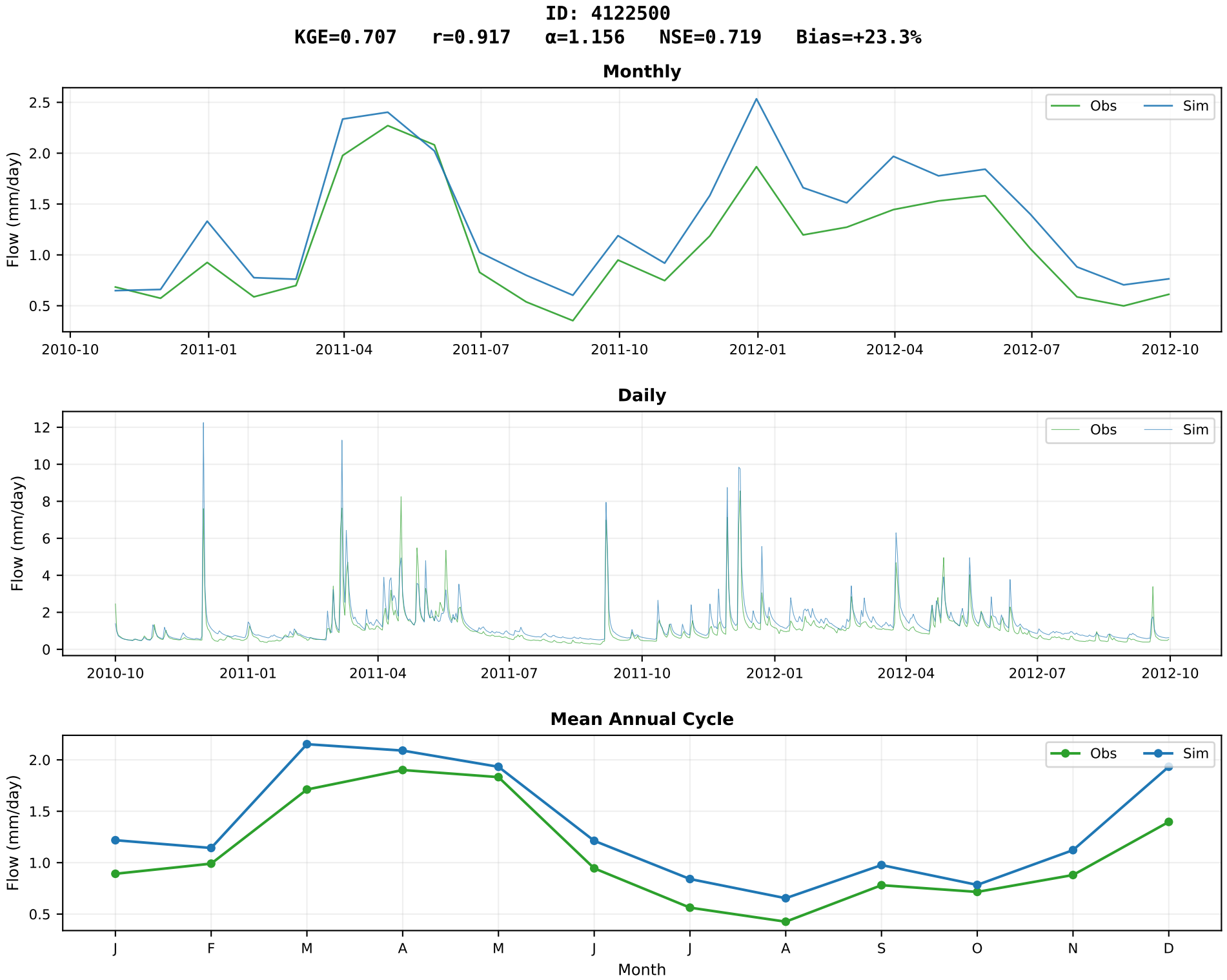

ID: 6224000
KGE=0.710 r=0.865 α=0.816 NSE=0.743 Bias=-17.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

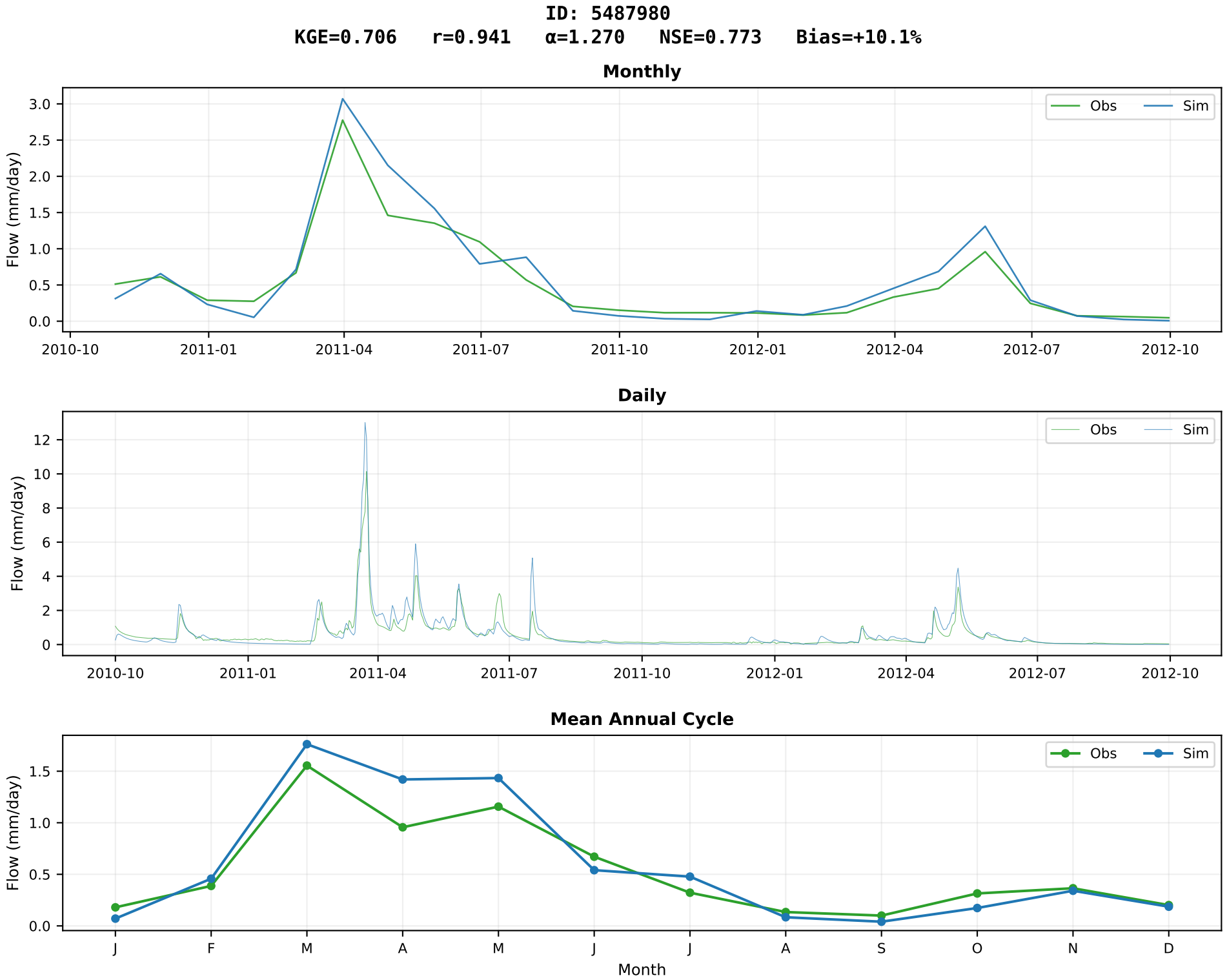

MF EEL R NR DOS RIOS CA (ID=11473900)
KGE=0.709 r=0.970 α=0.731 NSE=0.877 Bias=-10.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

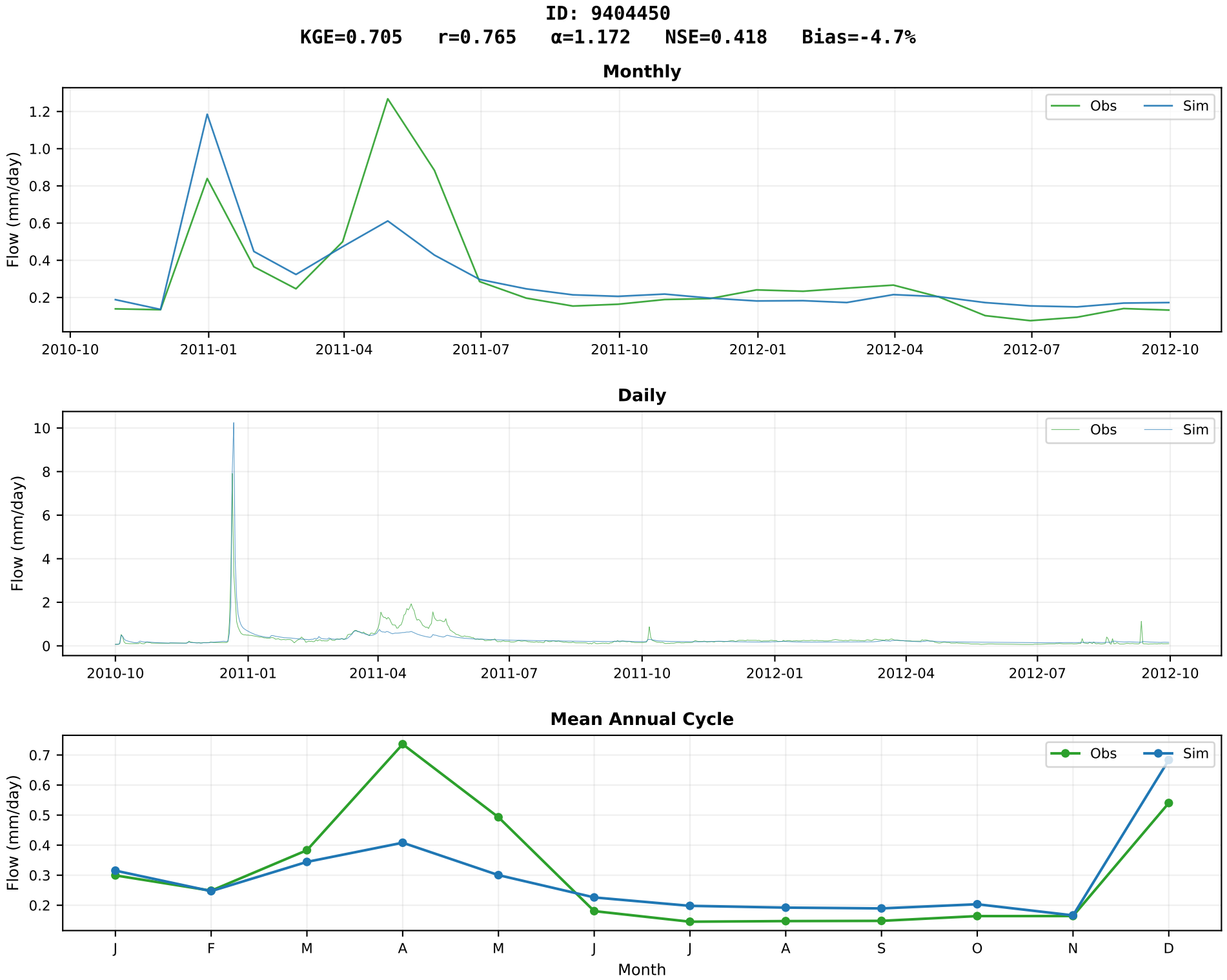

ID: 2014000
KGE=0.708 r=0.880 α=0.754 NSE=0.756 Bias=-10.0%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.5
1.0
1.5
2.0
2.5
3.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

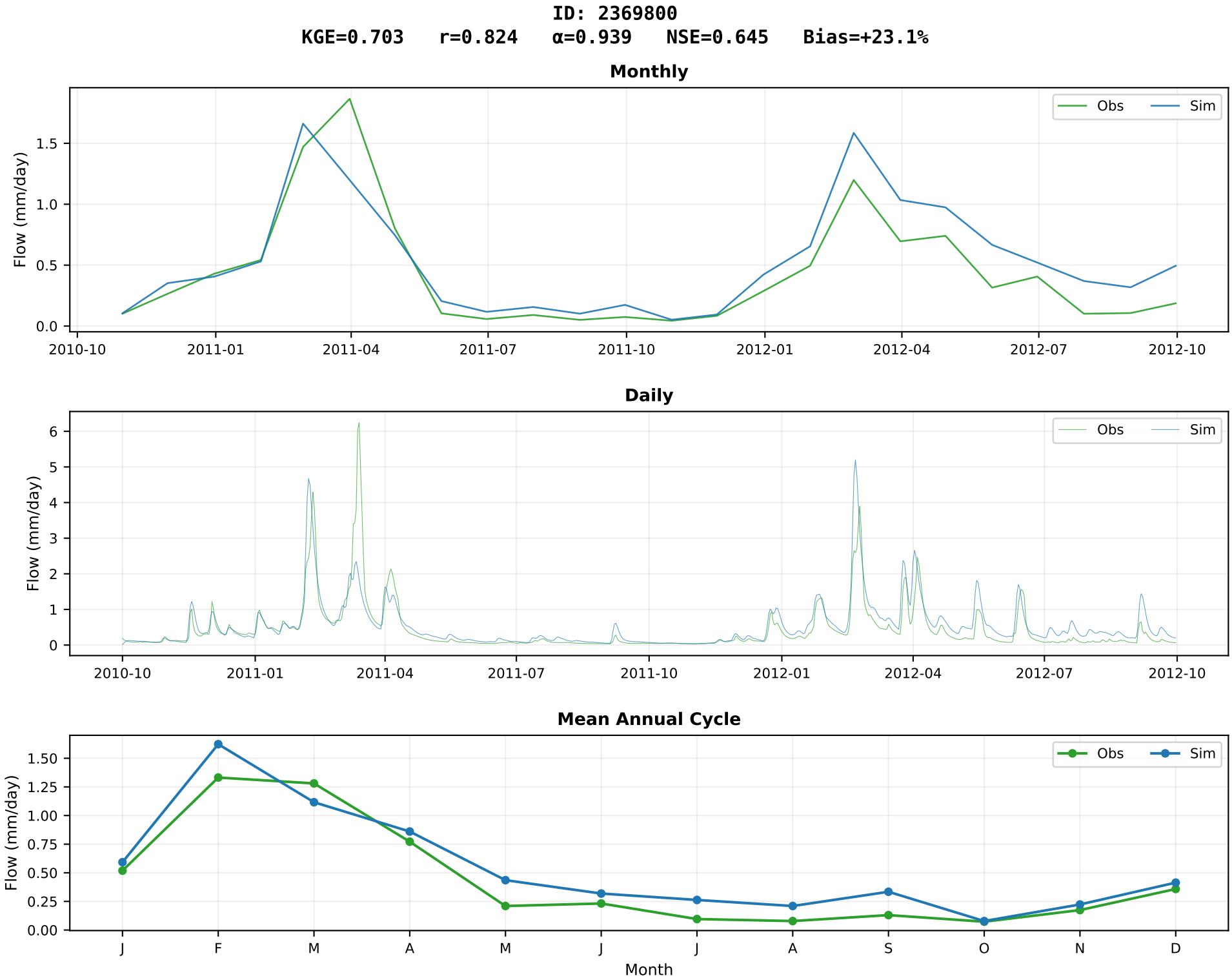

ID: 1139800
KGE=0.708 r=0.776 α=0.842 NSE=0.590 Bias=-10.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

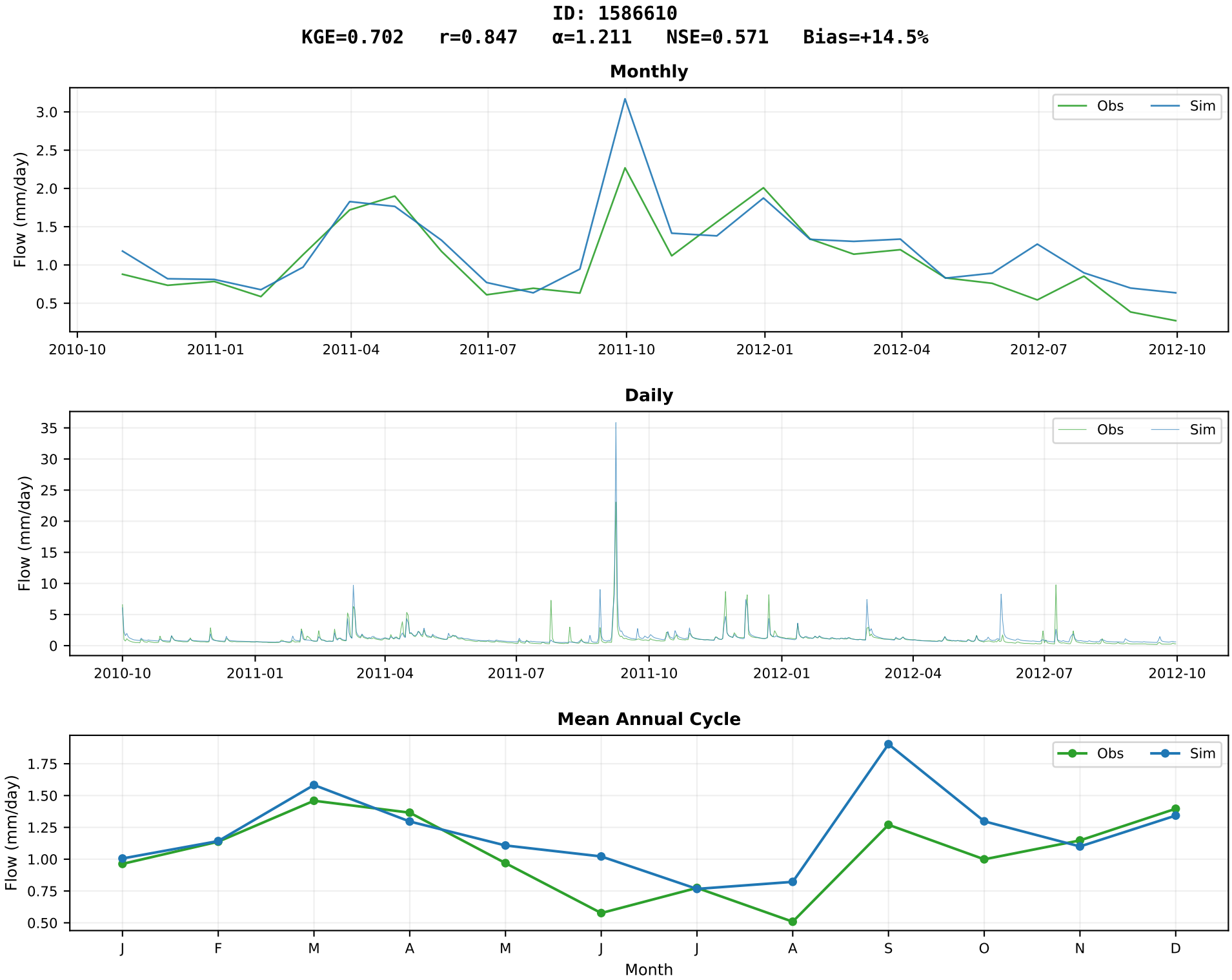

ID: 3050000
KGE=0.708 r=0.784 α=0.823 NSE=0.604 Bias=+8.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

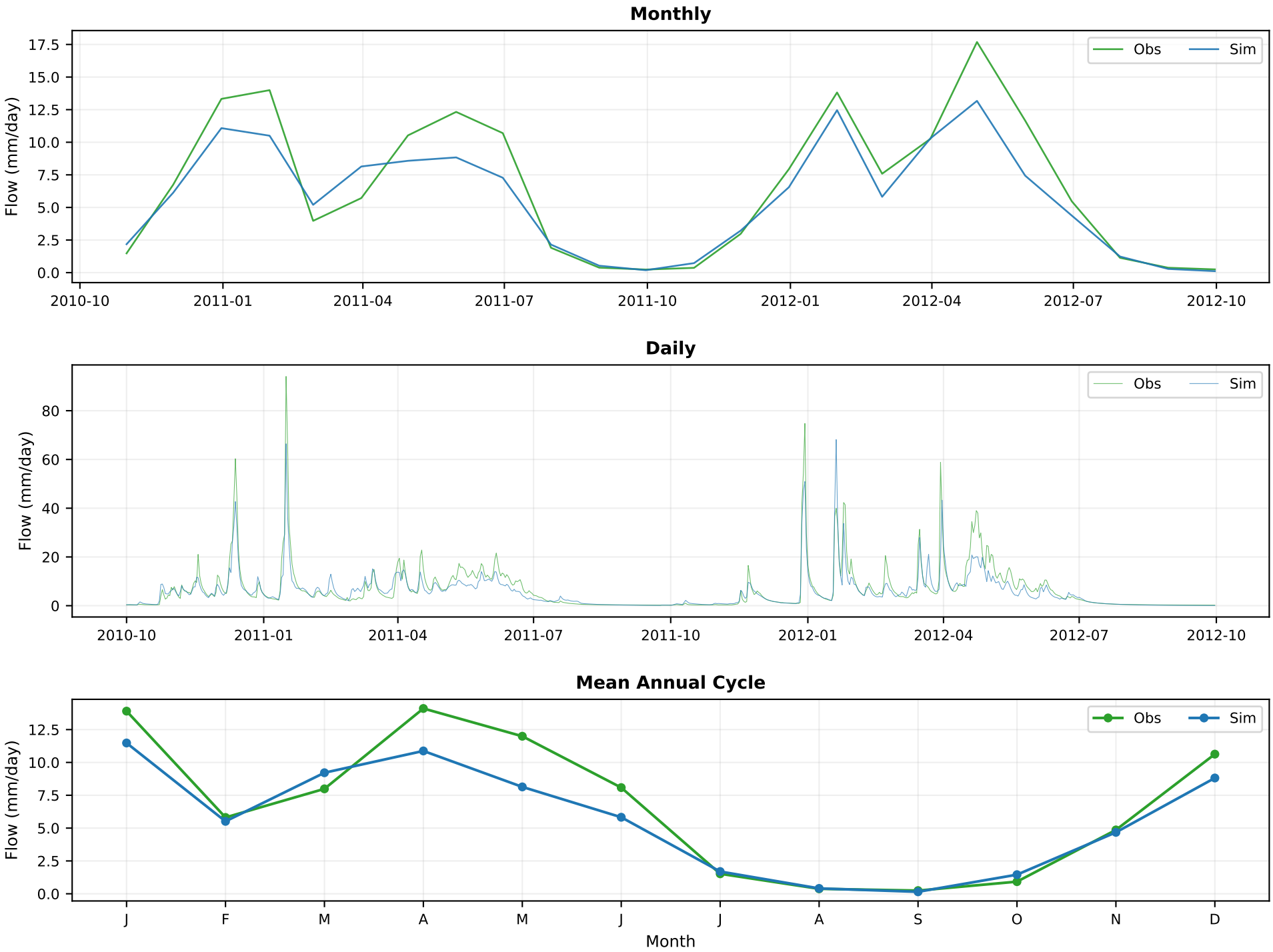

ID: 4122500
KGE=0.707 r=0.917 α=1.156 NSE=0.719 Bias=+23.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

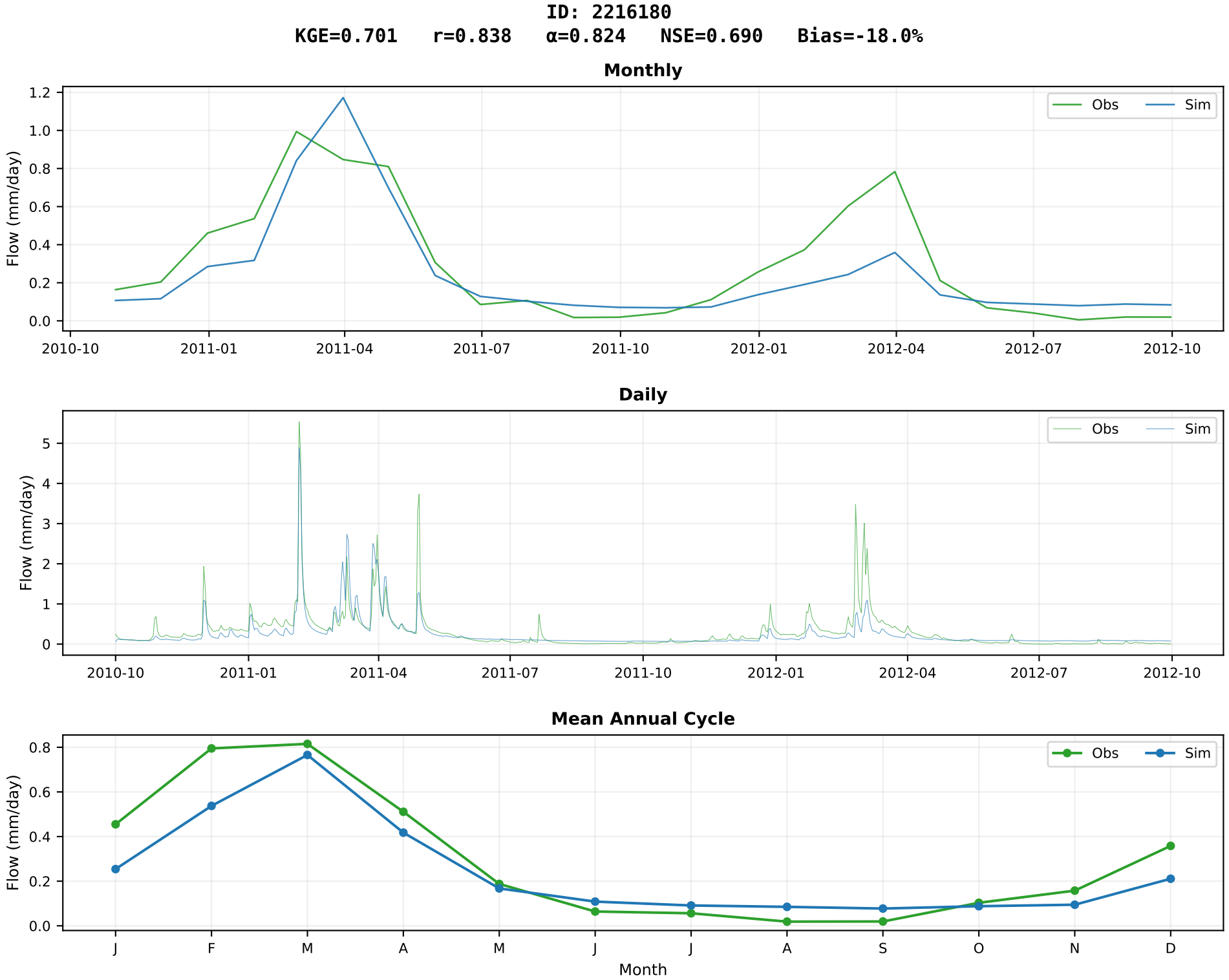

ID: 5487980
KGE=0.706 r=0.941 α=1.270 NSE=0.773 Bias=+10.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

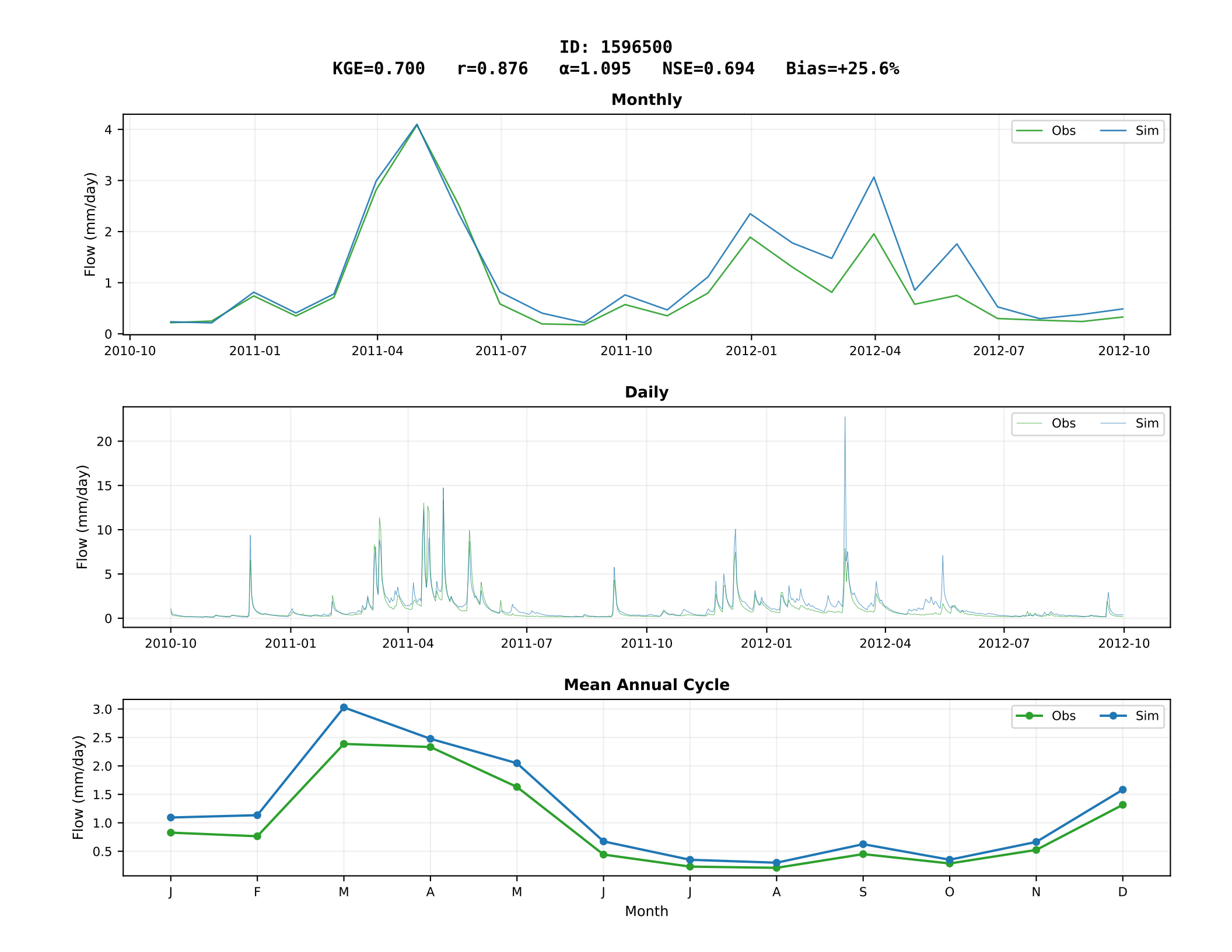

ID: 9404450
KGE=0.705 r=0.765 α=1.172 NSE=0.418 Bias=-4.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

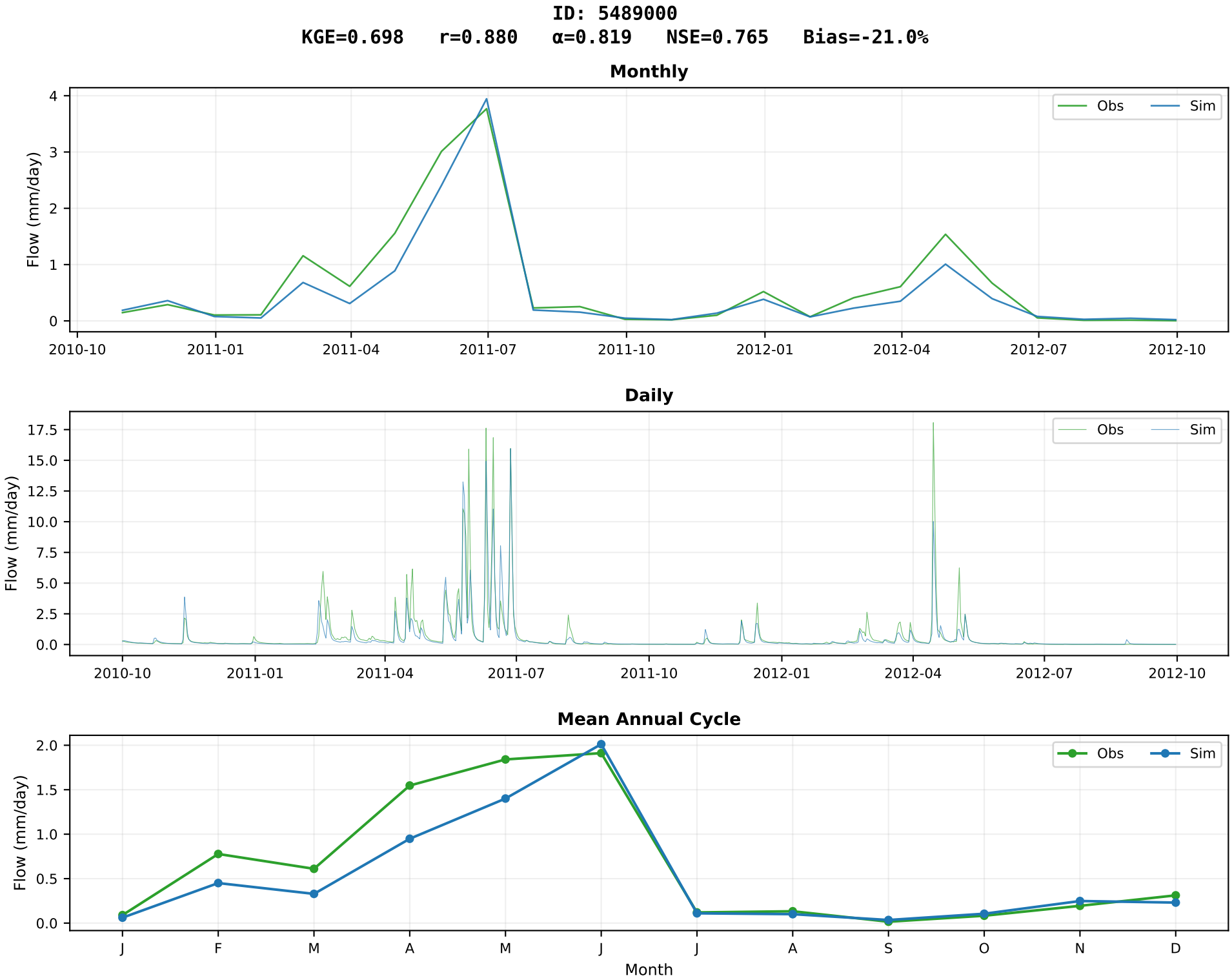

ID: 2369800
KGE=0.703 r=0.824 α=0.939 NSE=0.645 Bias=+23.1%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
J
F
M
A
M
J
J
A
S
O
N
D
Month

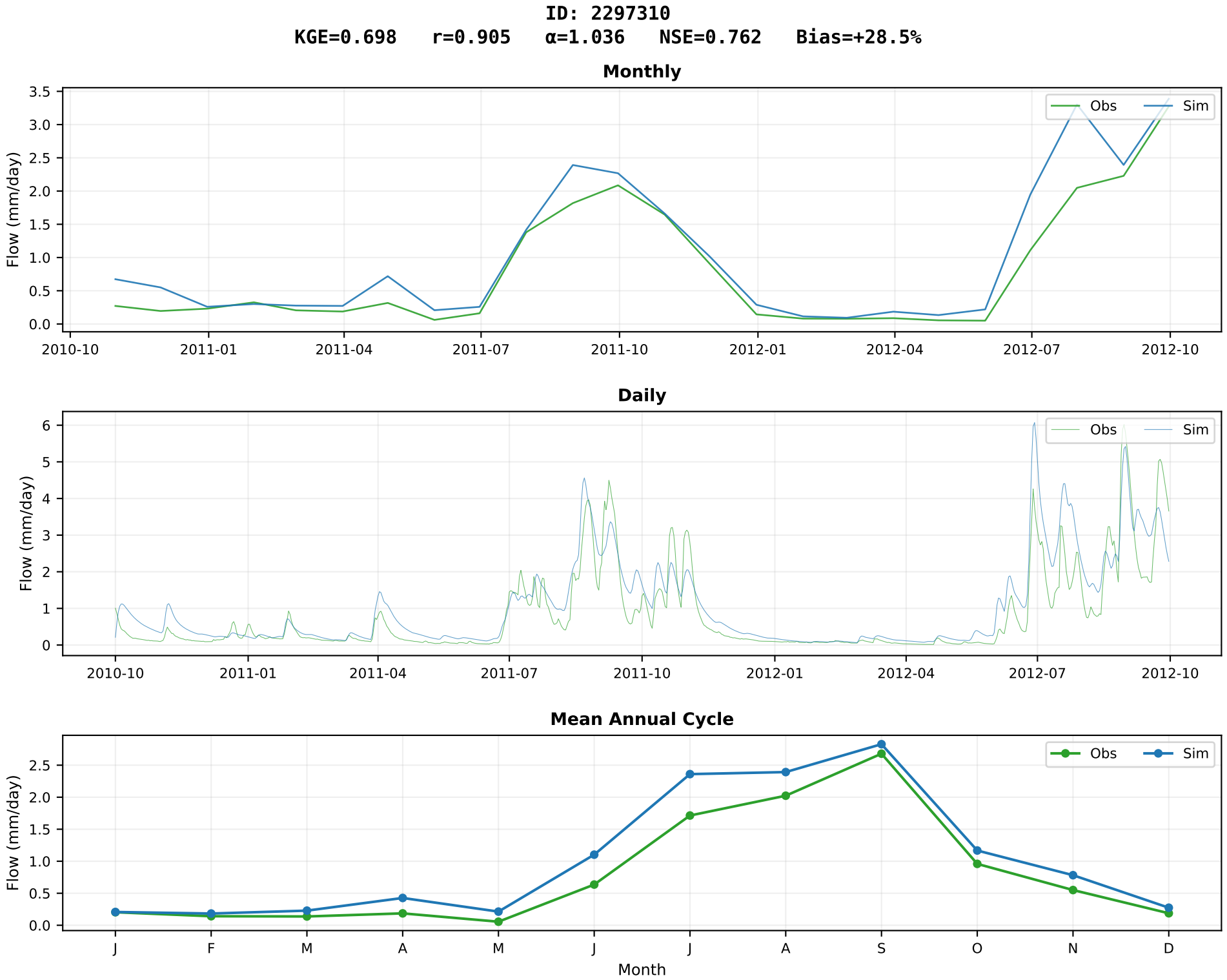

ID: 1586610
KGE=0.702 r=0.847 α=1.211 NSE=0.571 Bias=+14.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D


ARROYO VALLE BL LANG CN NR LIVERMORE CA (ID=11176400)
KGE=0.701 r=0.918 α=0.756 NSE=0.804 Bias=-15.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2216180
KGE=0.701 r=0.838 α=0.824 NSE=0.690 Bias=-18.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

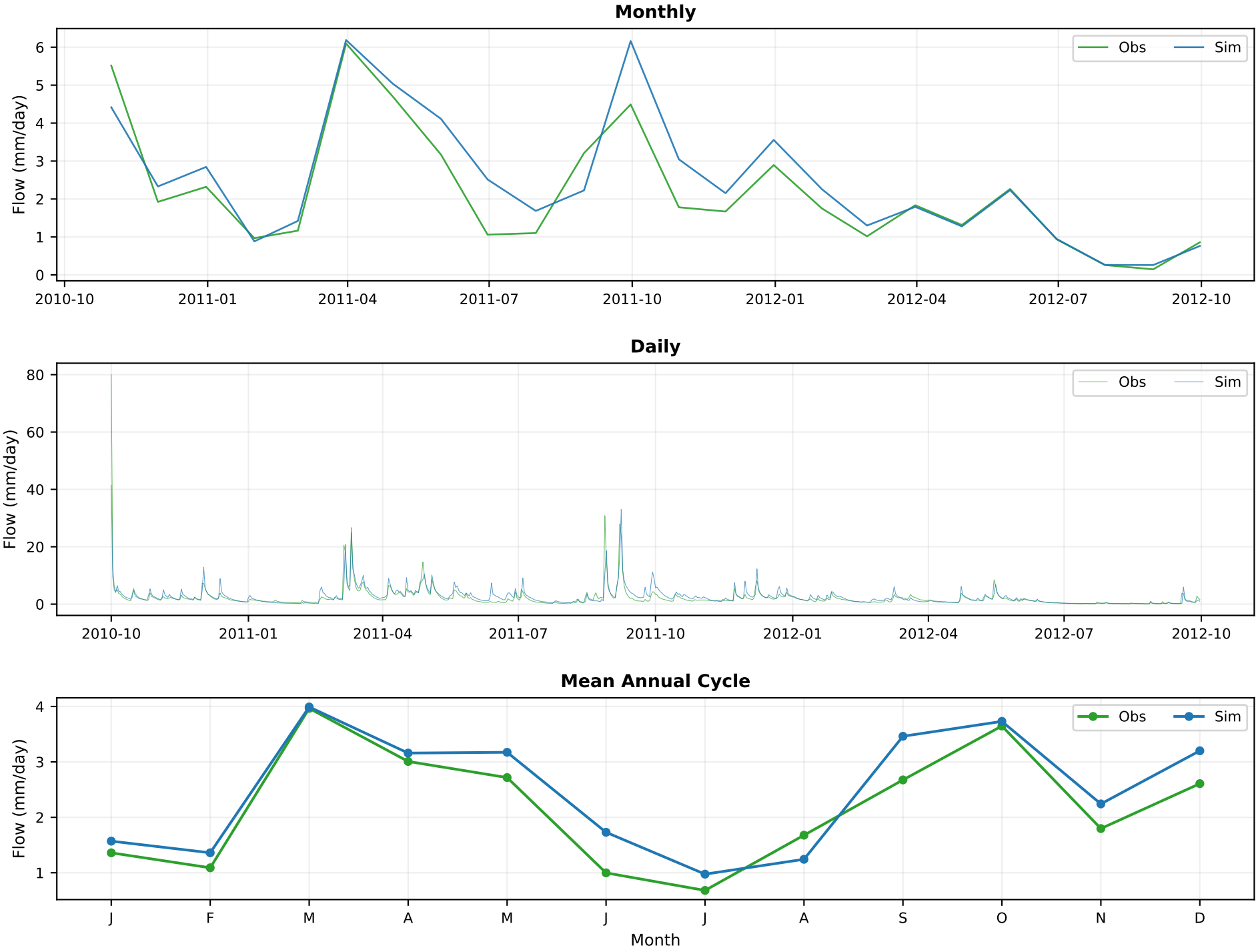

ID: 1596500
KGE=0.700 r=0.876 α=1.095 NSE=0.694 Bias=+25.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

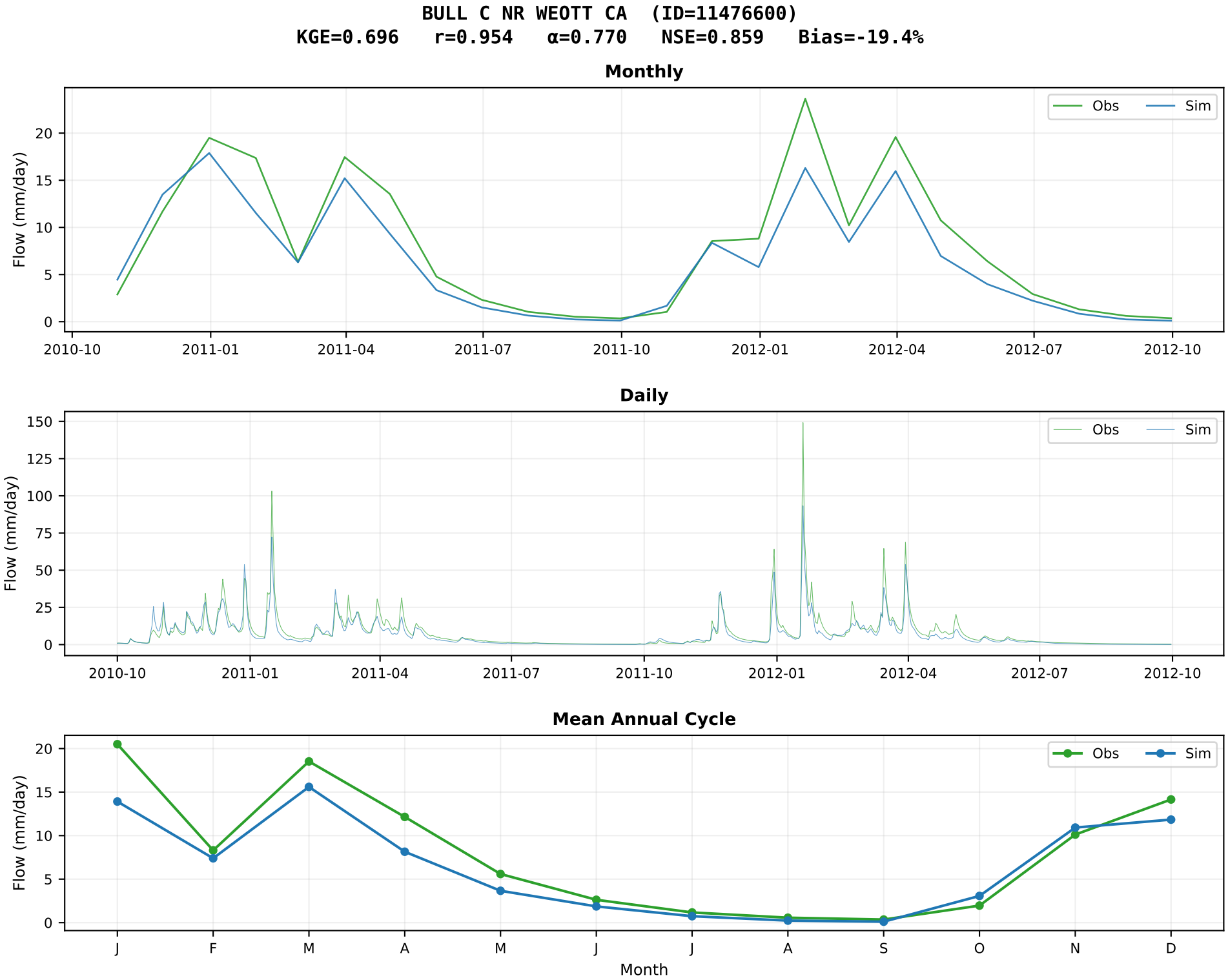

ID: 5489000
KGE=0.698 r=0.880 α=0.819 NSE=0.765 Bias=-21.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

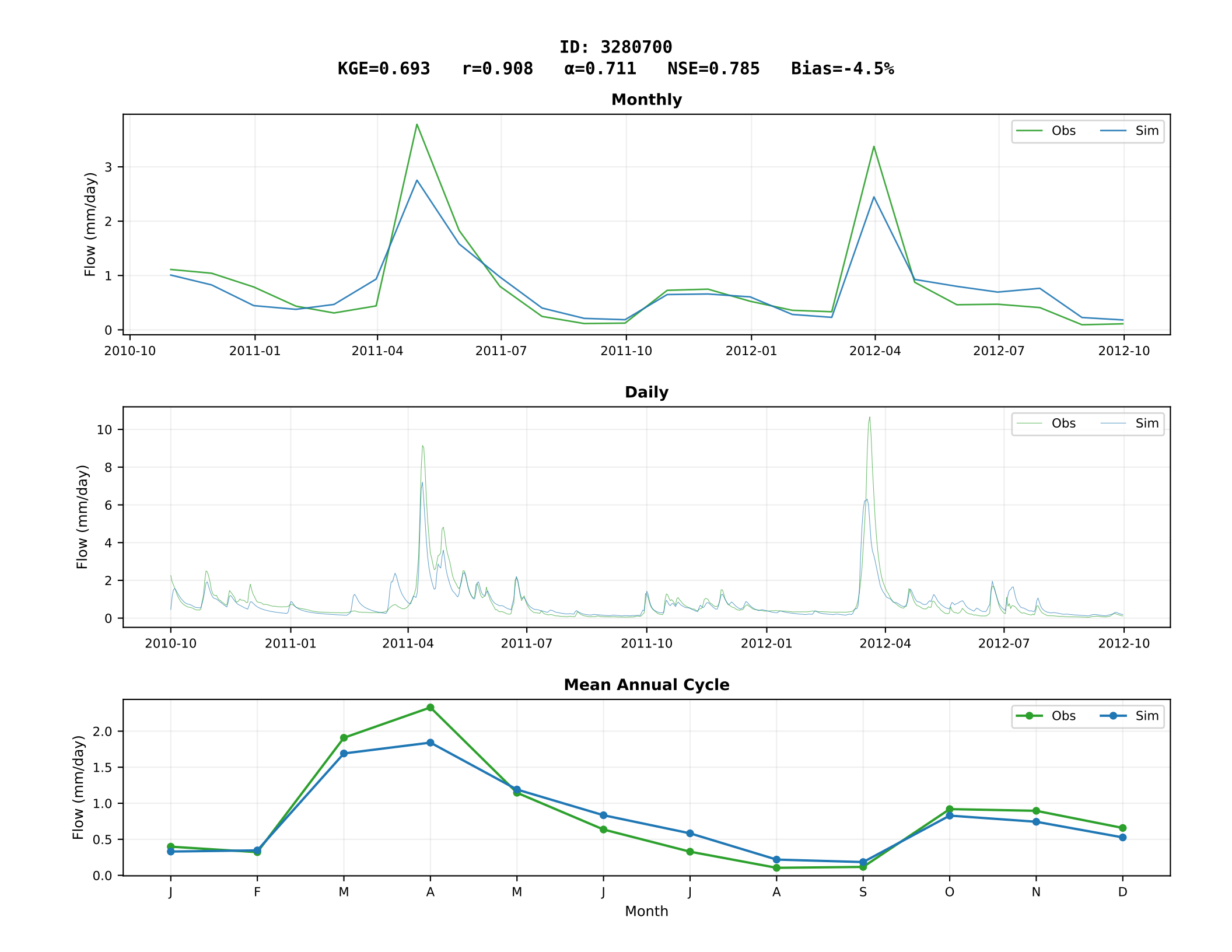

ID: 2297310
KGE=0.698 r=0.905 α=1.036 NSE=0.762 Bias=+28.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

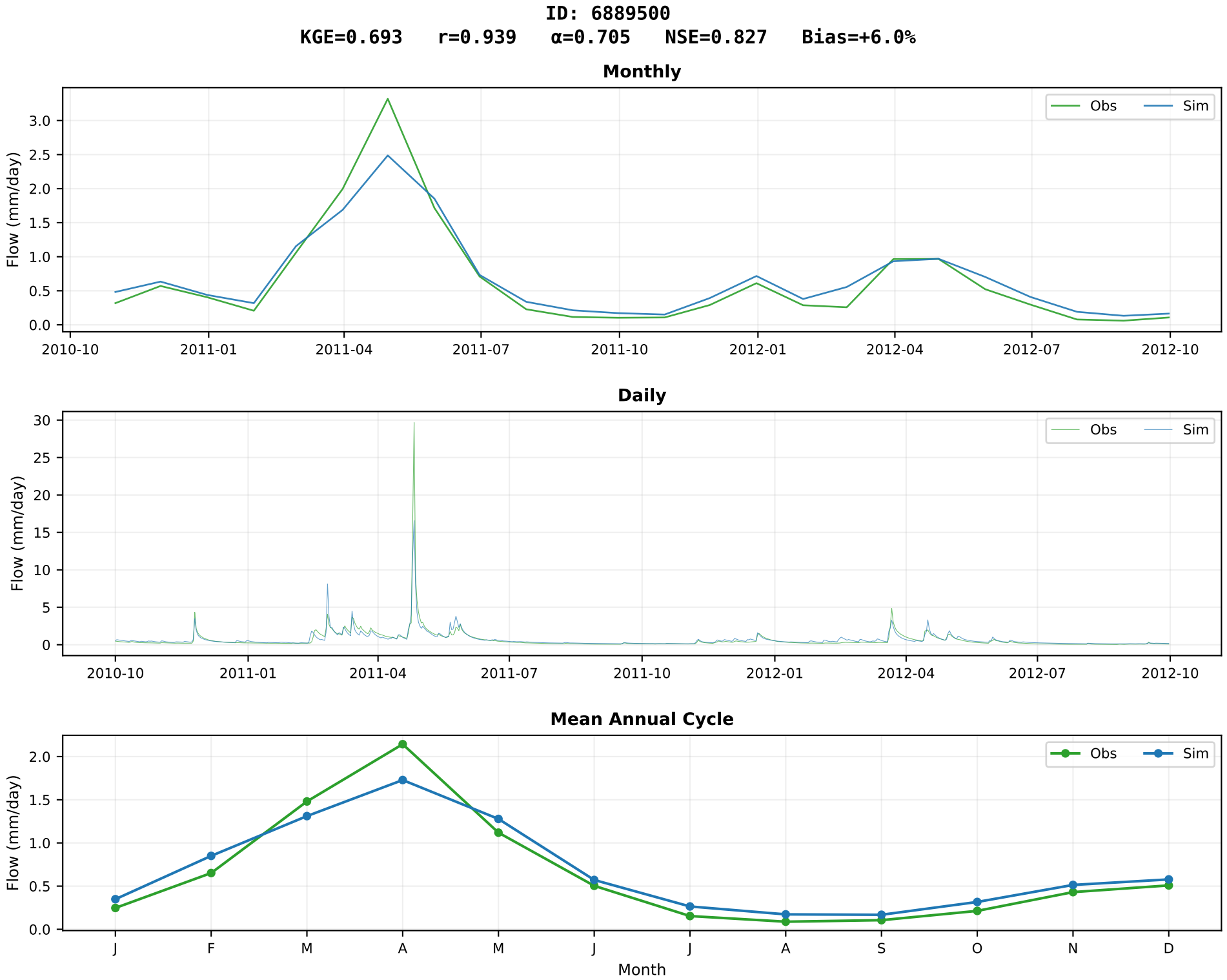
ID: 1414500

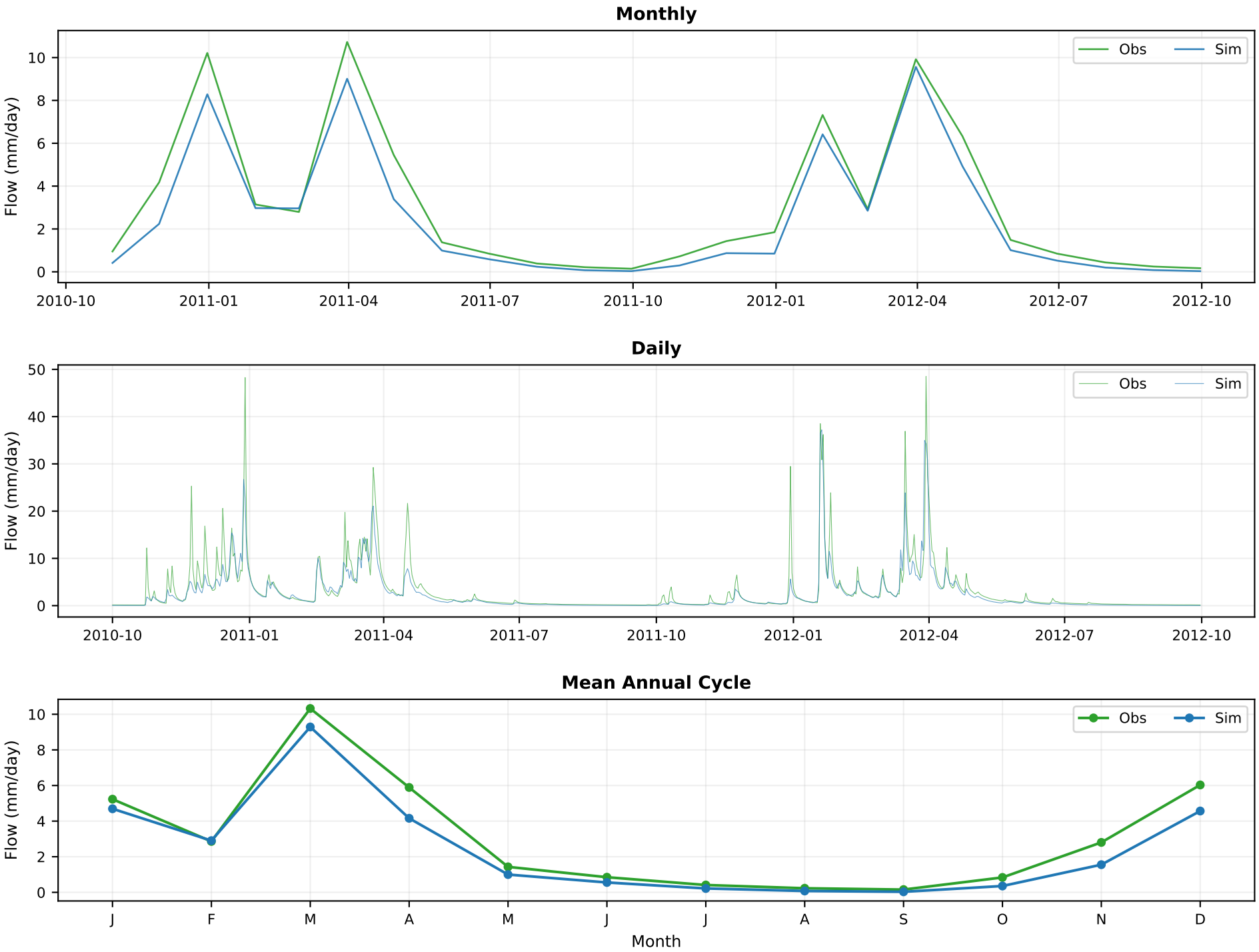
KGE=0.697 r=0.862 α=0.768 NSE=0.728 Bias=+13.6%

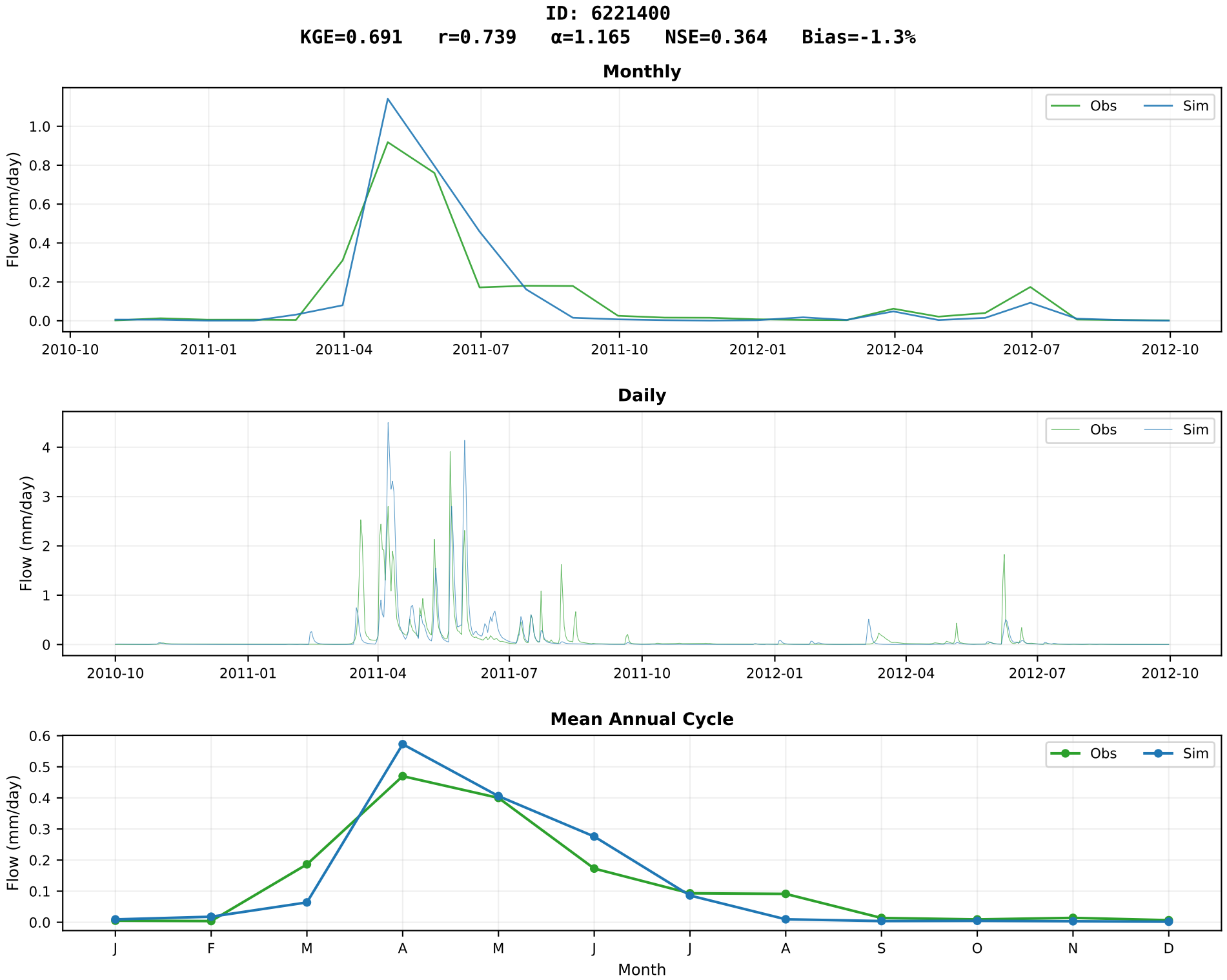


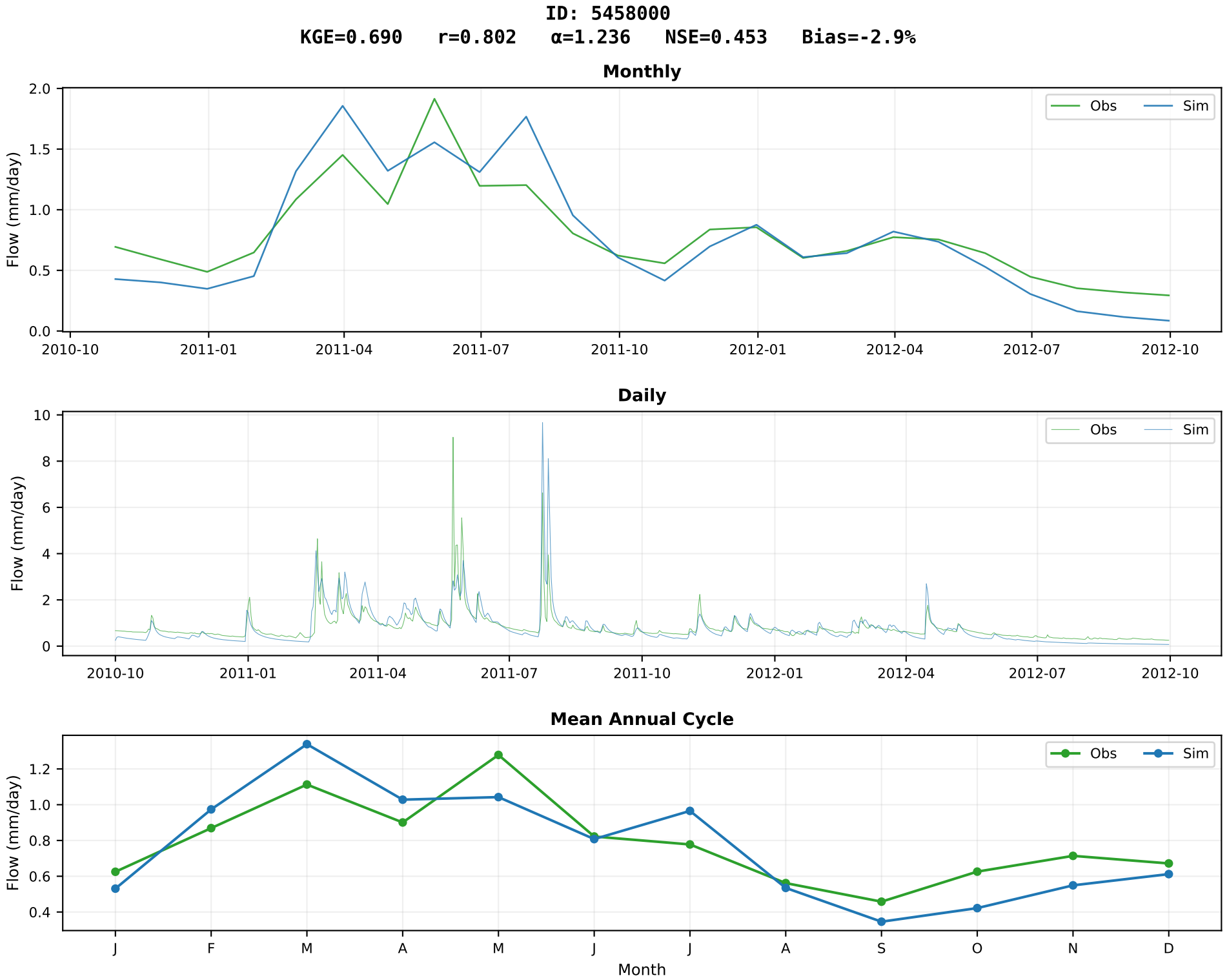

BULL C NR WEOTT CA (ID=11476600)
KGE=0.696 r=0.954 α=0.770 NSE=0.859 Bias=-19.4%
Monthly
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
25
50
75
100
125
150
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
5
10
15
20
J
F
M
A
M
J
J
A
S
O
N
D
Month

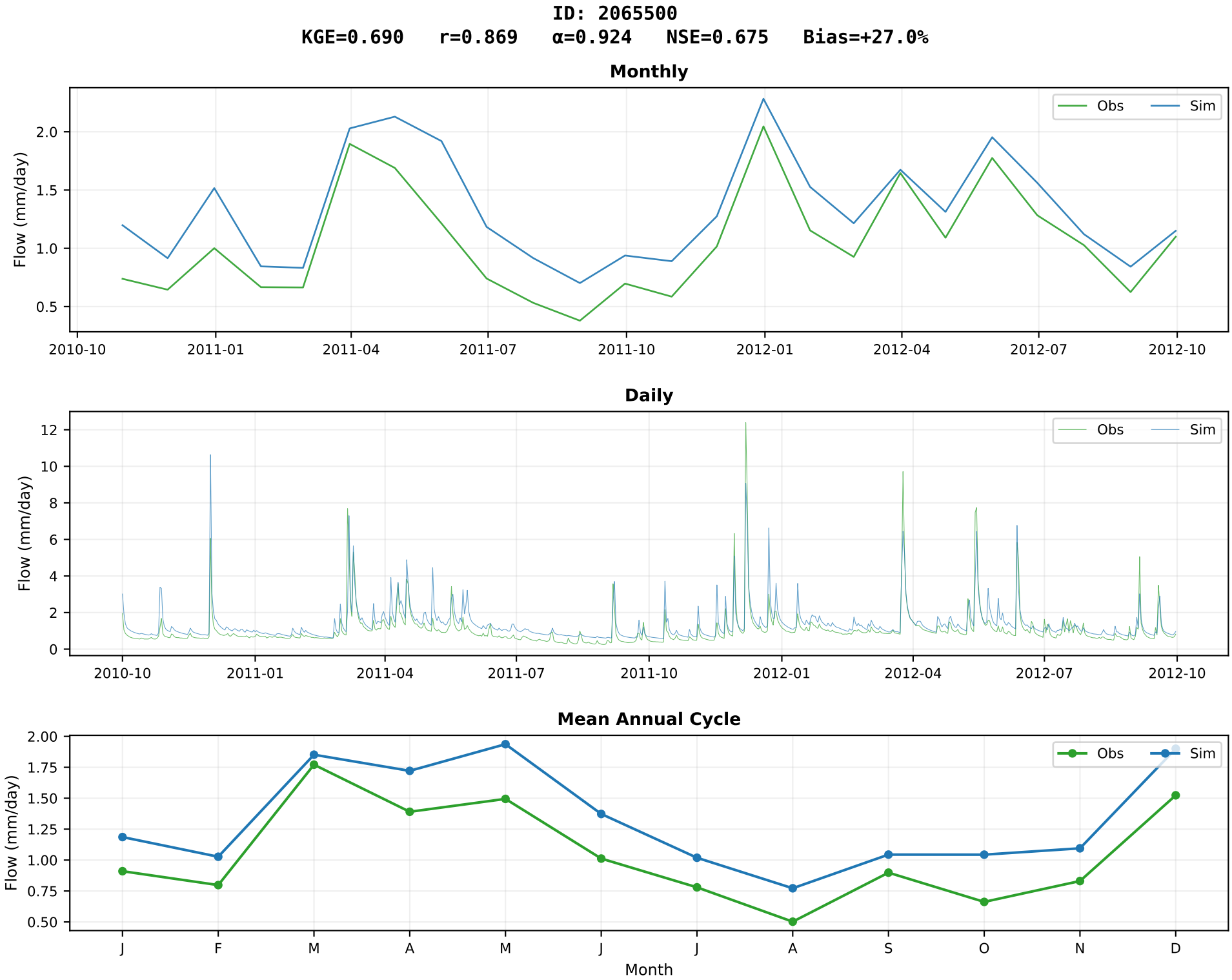

ID: 3280700
KGE=0.693 r=0.908 α=0.711 NSE=0.785 Bias=-4.5%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

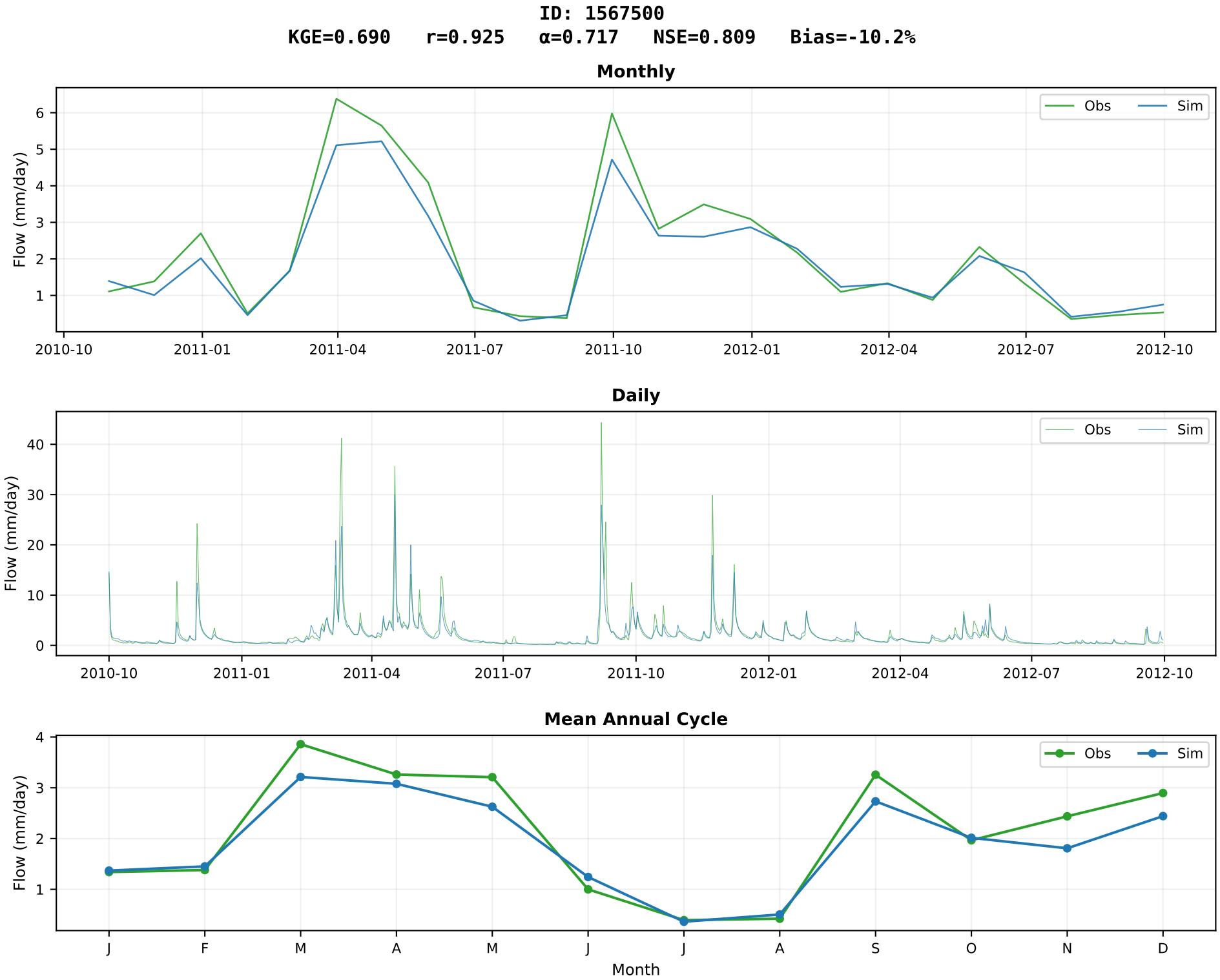

ID: 6889500
KGE=0.693 r=0.939 α=0.705 NSE=0.827 Bias=+6.0%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

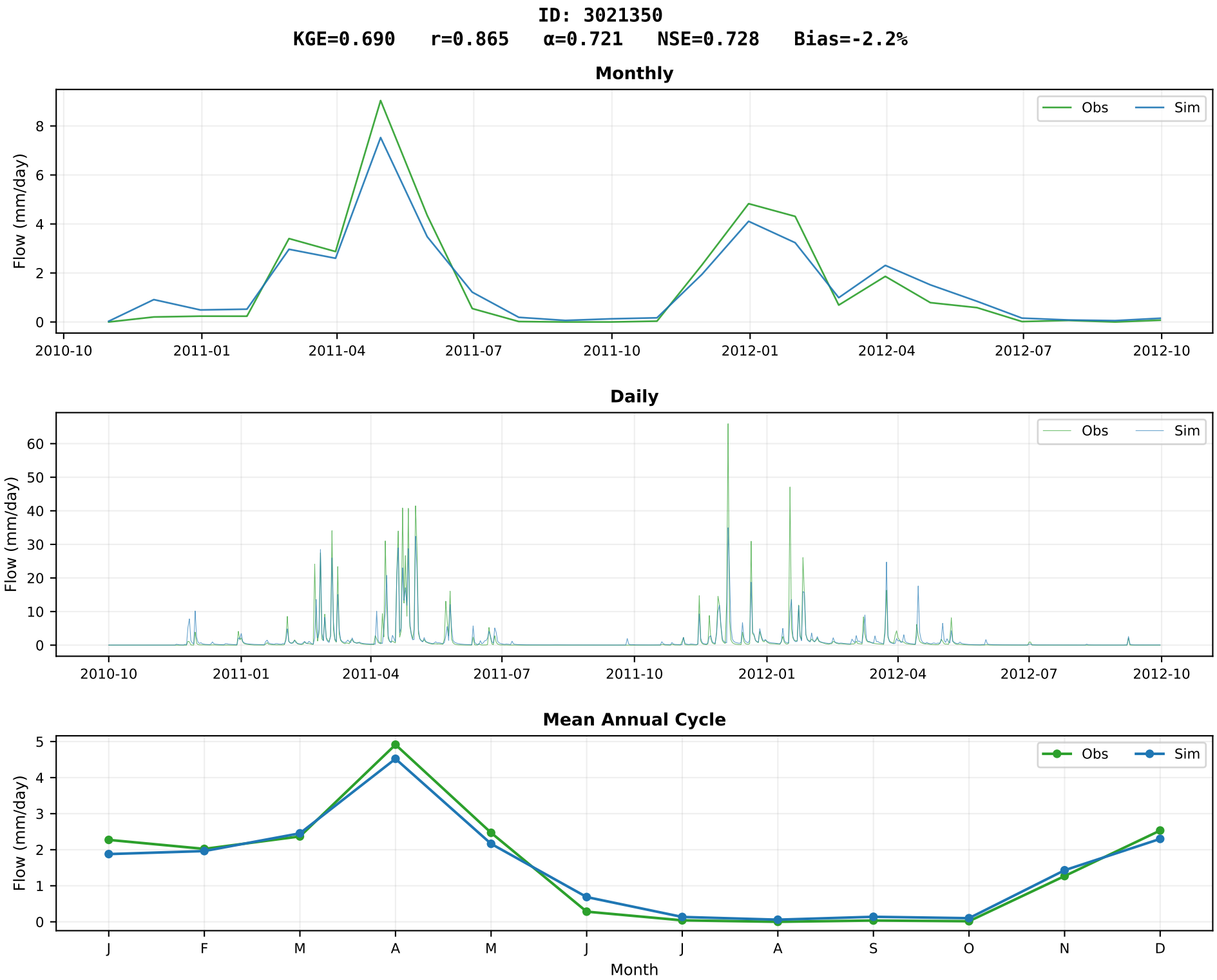

NF STILLAGUAMISH RIVER NEAR ARLINGTON, WA (ID=12167000)
KGE=0.693 r=0.884 α=0.804 NSE=0.762 Bias=-20.7%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

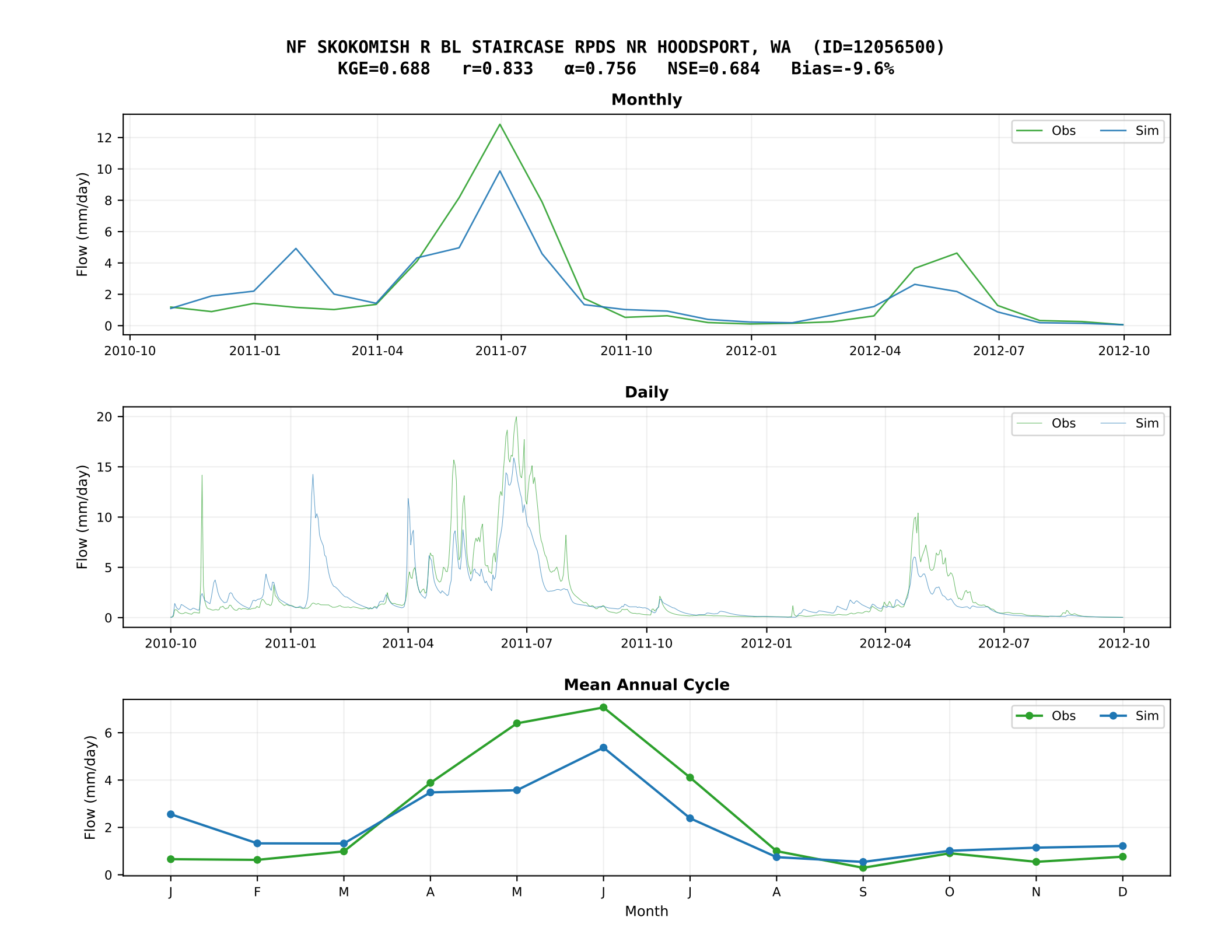

ID: 6221400
KGE=0.691 r=0.739 α=1.165 NSE=0.364 Bias=-1.3%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month

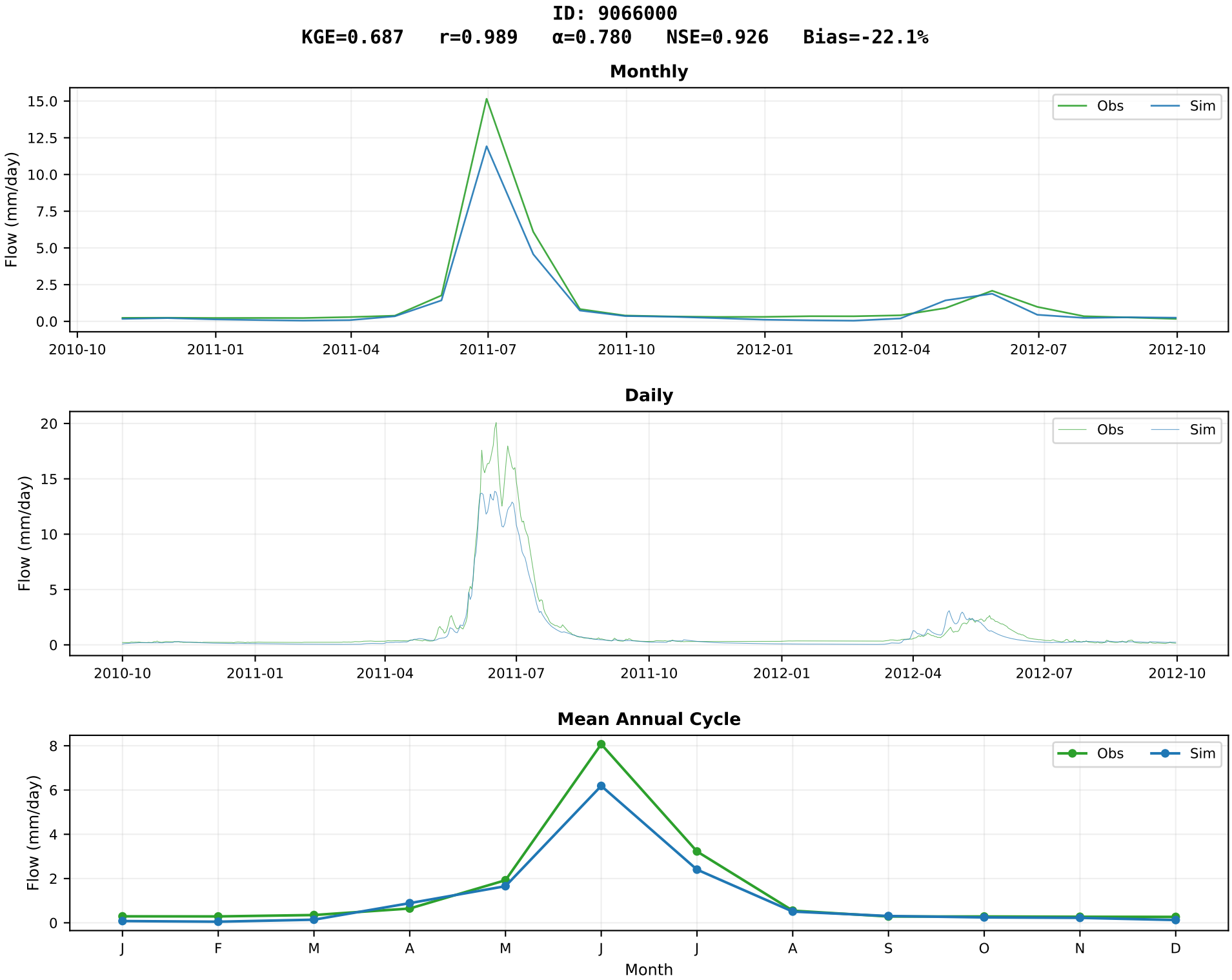

ID: 5458000
KGE=0.690 r=0.802 α=1.236 NSE=0.453 Bias=-2.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

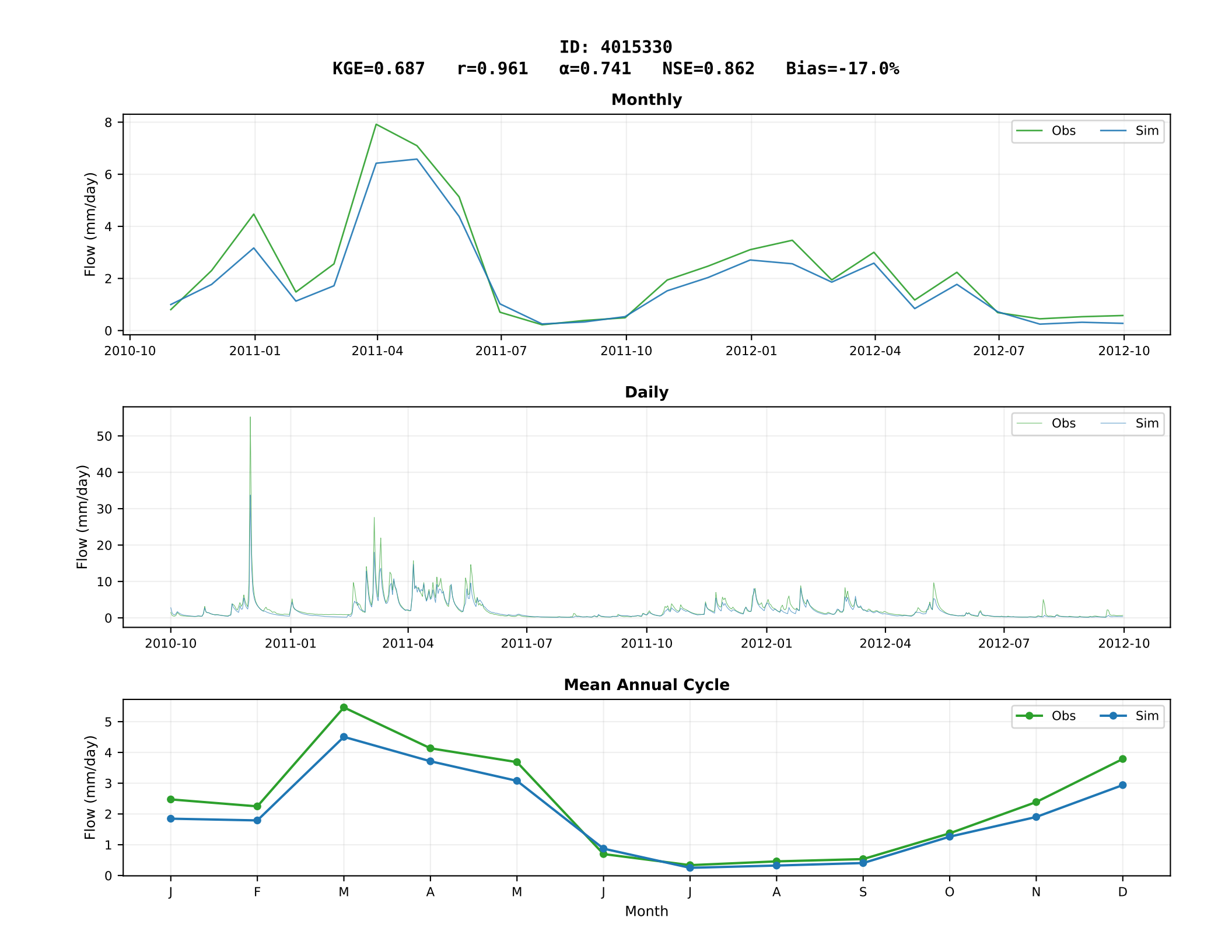

ID: 2065500
KGE=0.690 r=0.869 α=0.924 NSE=0.675 Bias=+27.0%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
12
10
8
6
4
2
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
J
F
M
A
M
J
J
A
S
O
N
D
Month

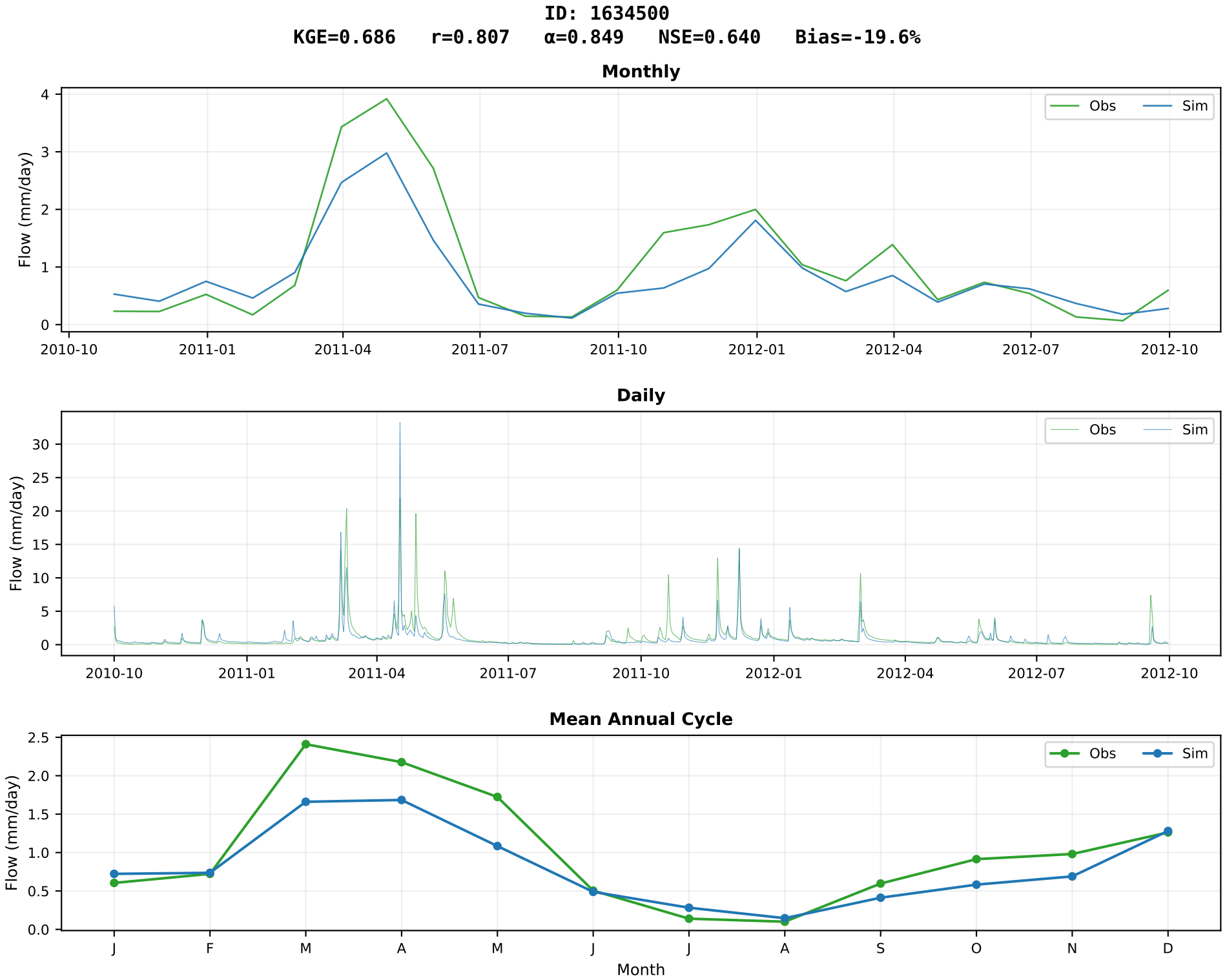

ID: 1567500
KGE=0.690 r=0.925 α=0.717 NSE=0.809 Bias=-10.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

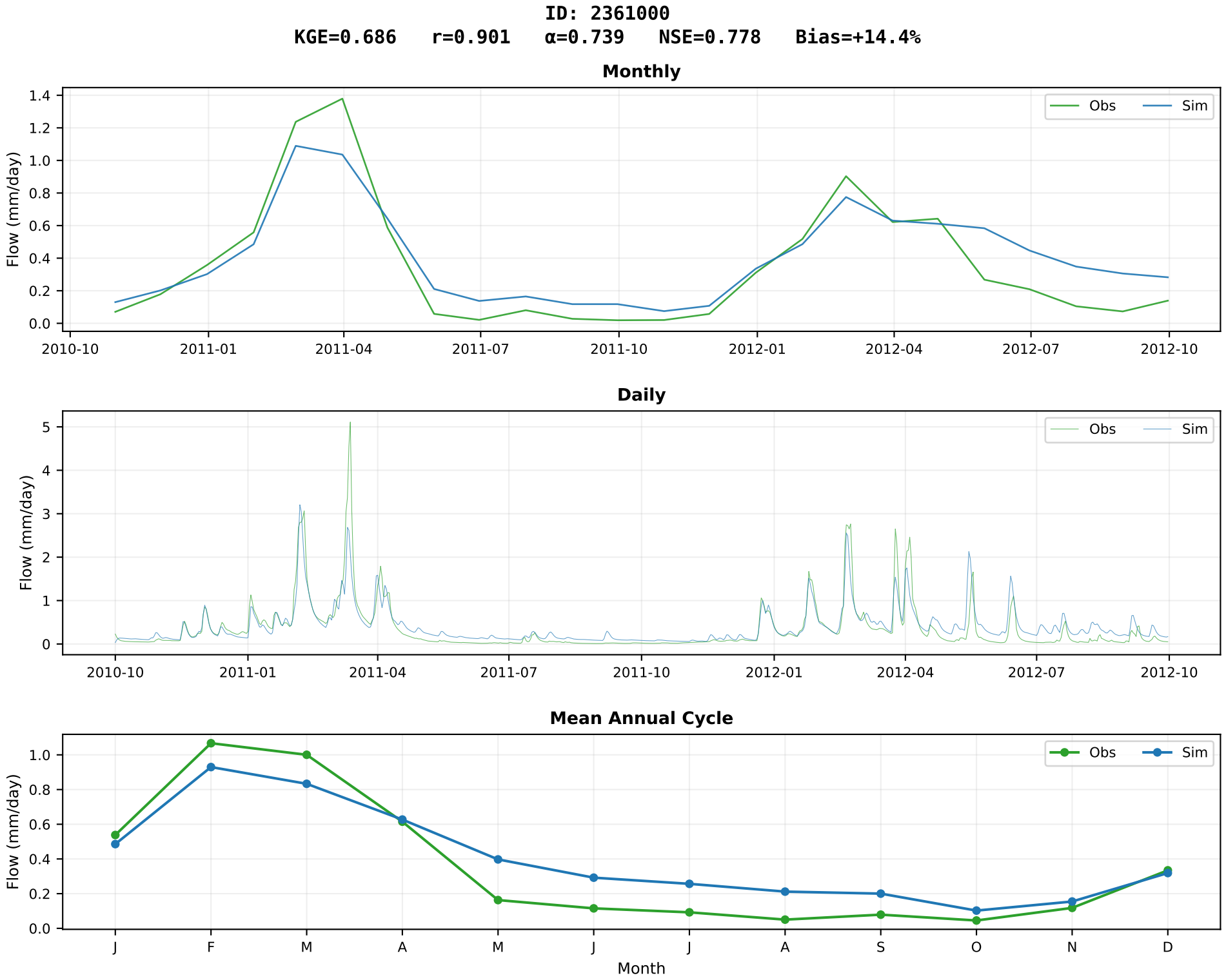

ID: 3021350
KGE=0.690 r=0.865 α=0.721 NSE=0.728 Bias=-2.2%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

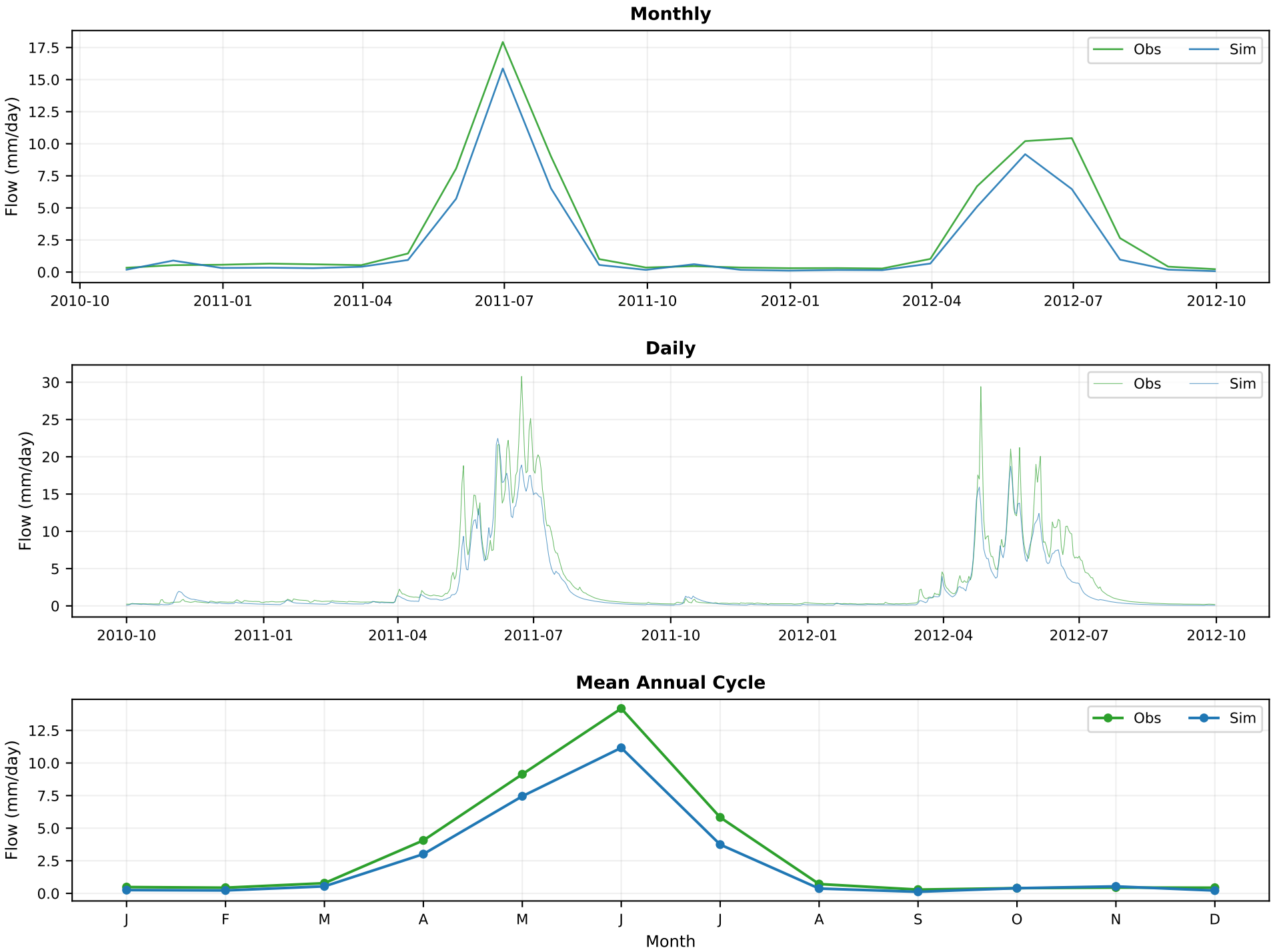

NF SKOKOMISH R BL STAIRCASE RPDS NR HOODSPORT, WA (ID=12056500)
KGE=0.688 r=0.833 α=0.756 NSE=0.684 Bias=-9.6%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

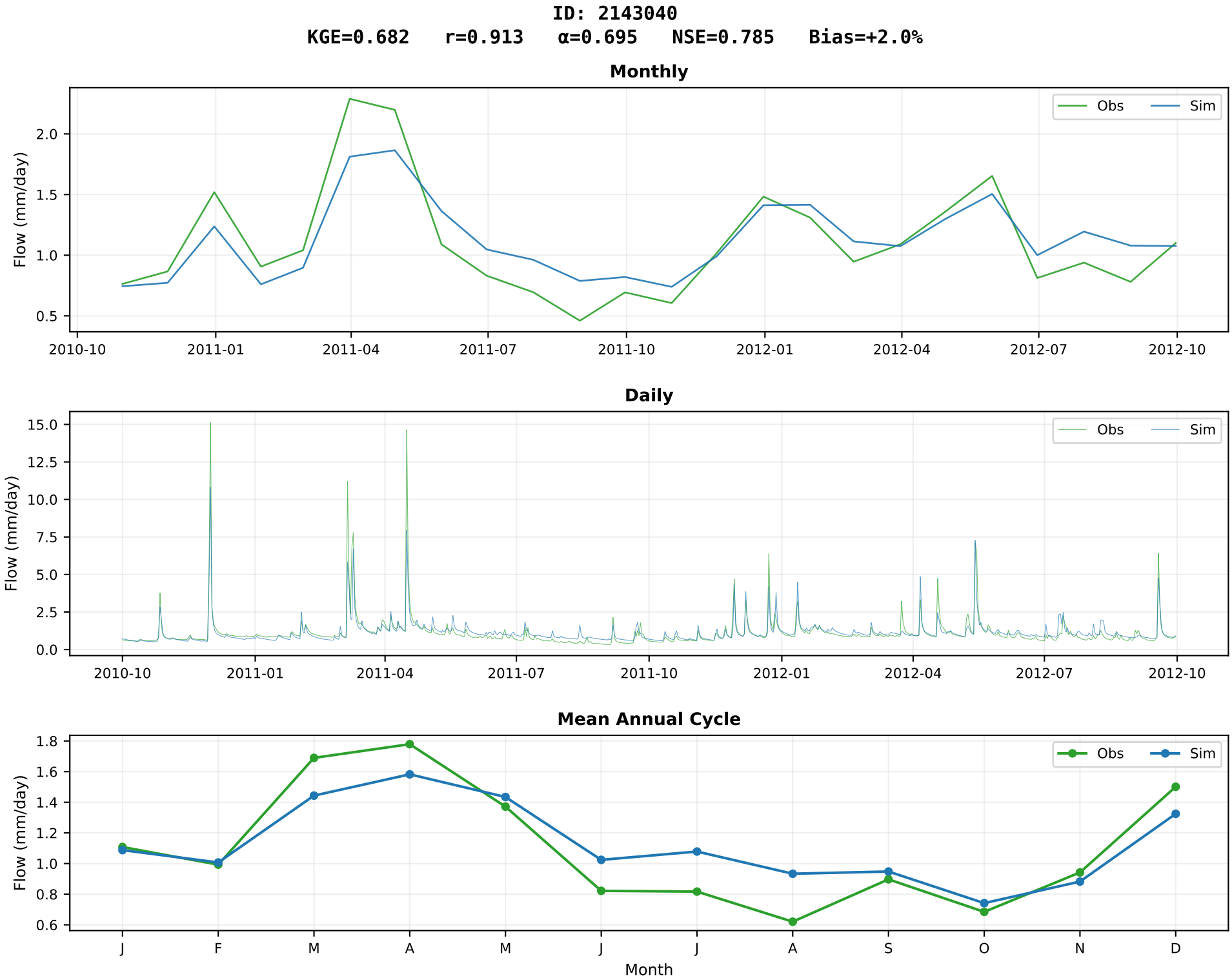

ID: 9066000
KGE=0.687 r=0.989 α=0.780 NSE=0.926 Bias=-22.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

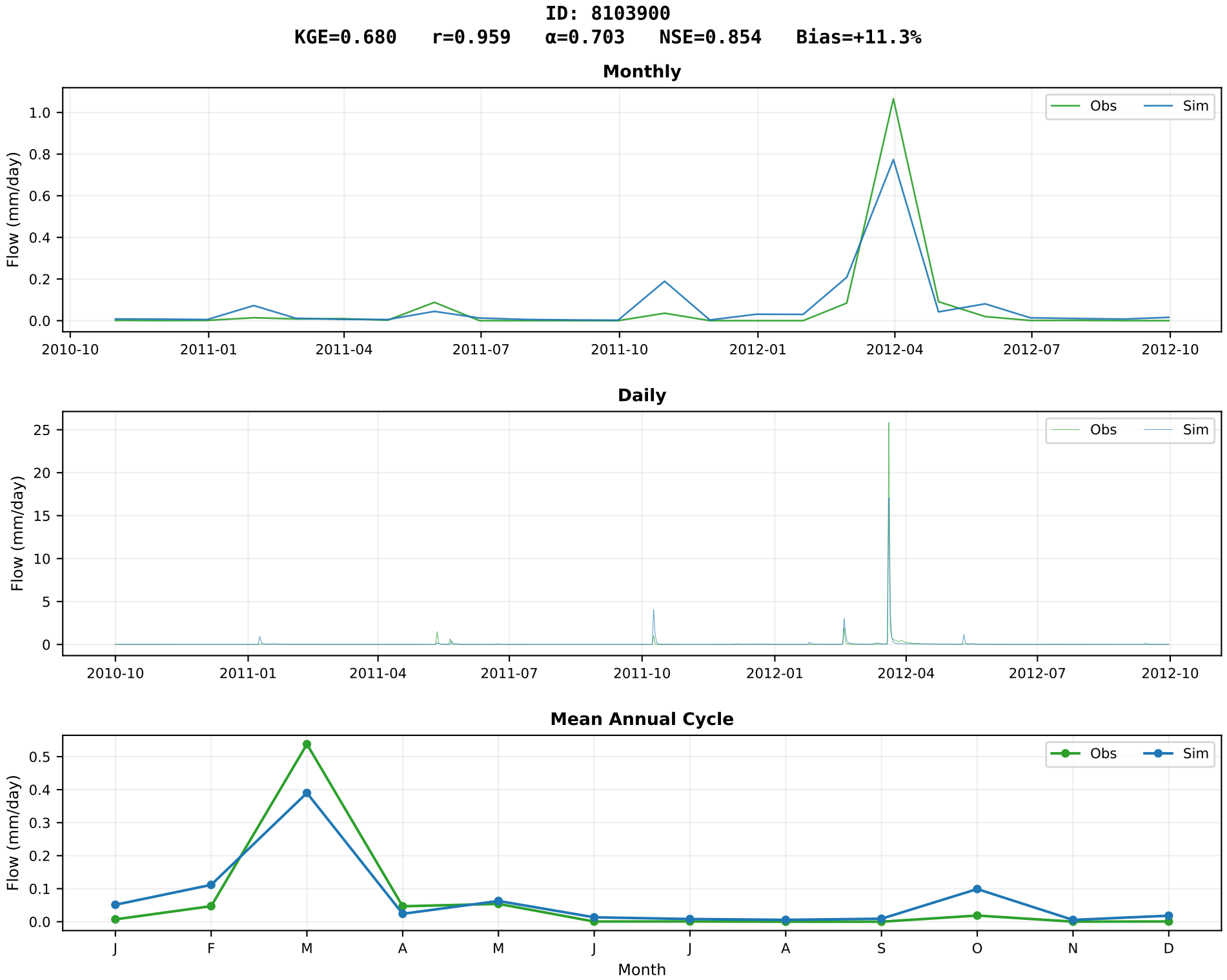

ID: 4015330
KGE=0.687 r=0.961 α=0.741 NSE=0.862 Bias=-17.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

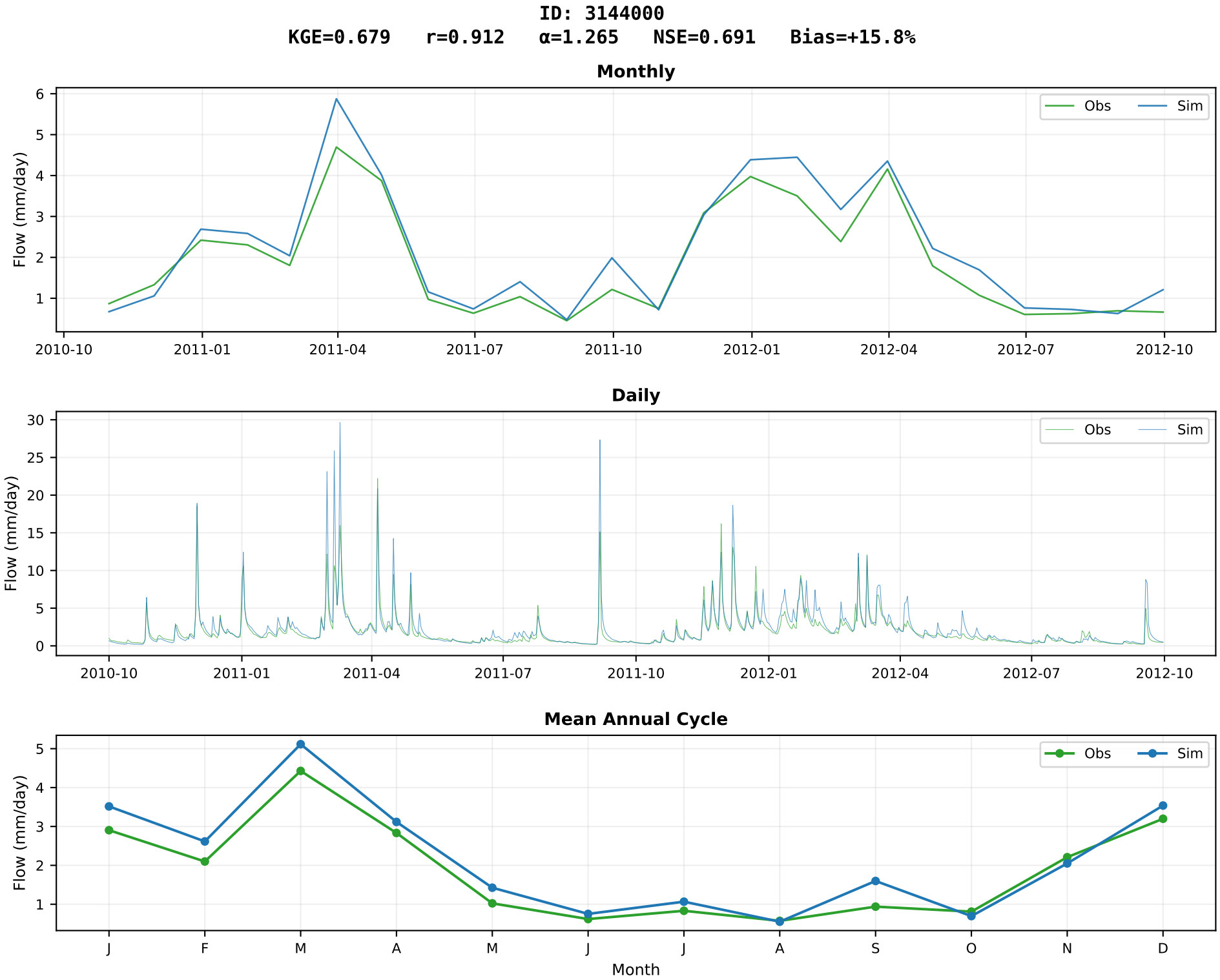

ID: 1634500
KGE=0.686 r=0.807 α=0.849 NSE=0.640 Bias=-19.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

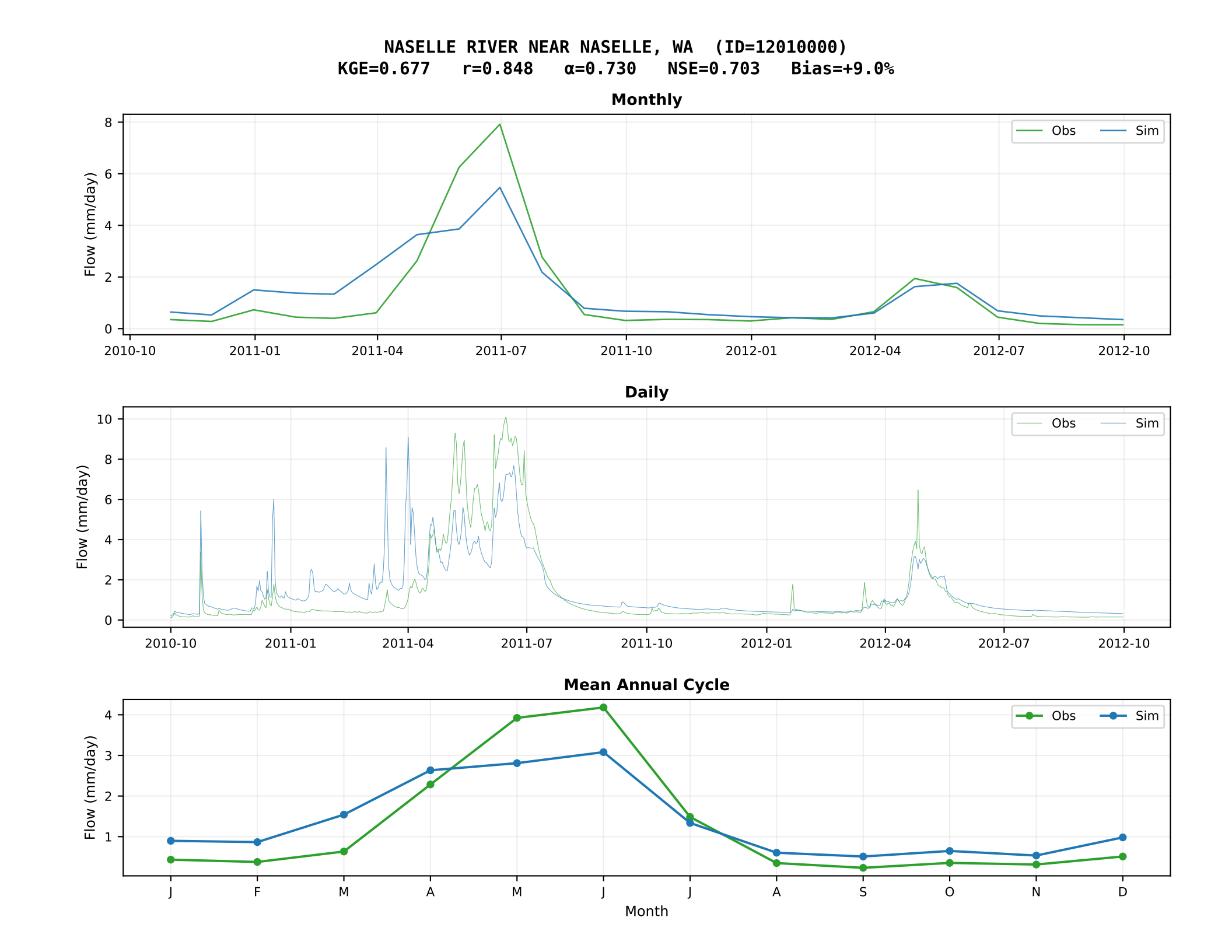

ID: 2361000
KGE=0.686 r=0.901 α=0.739 NSE=0.778 Bias=+14.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

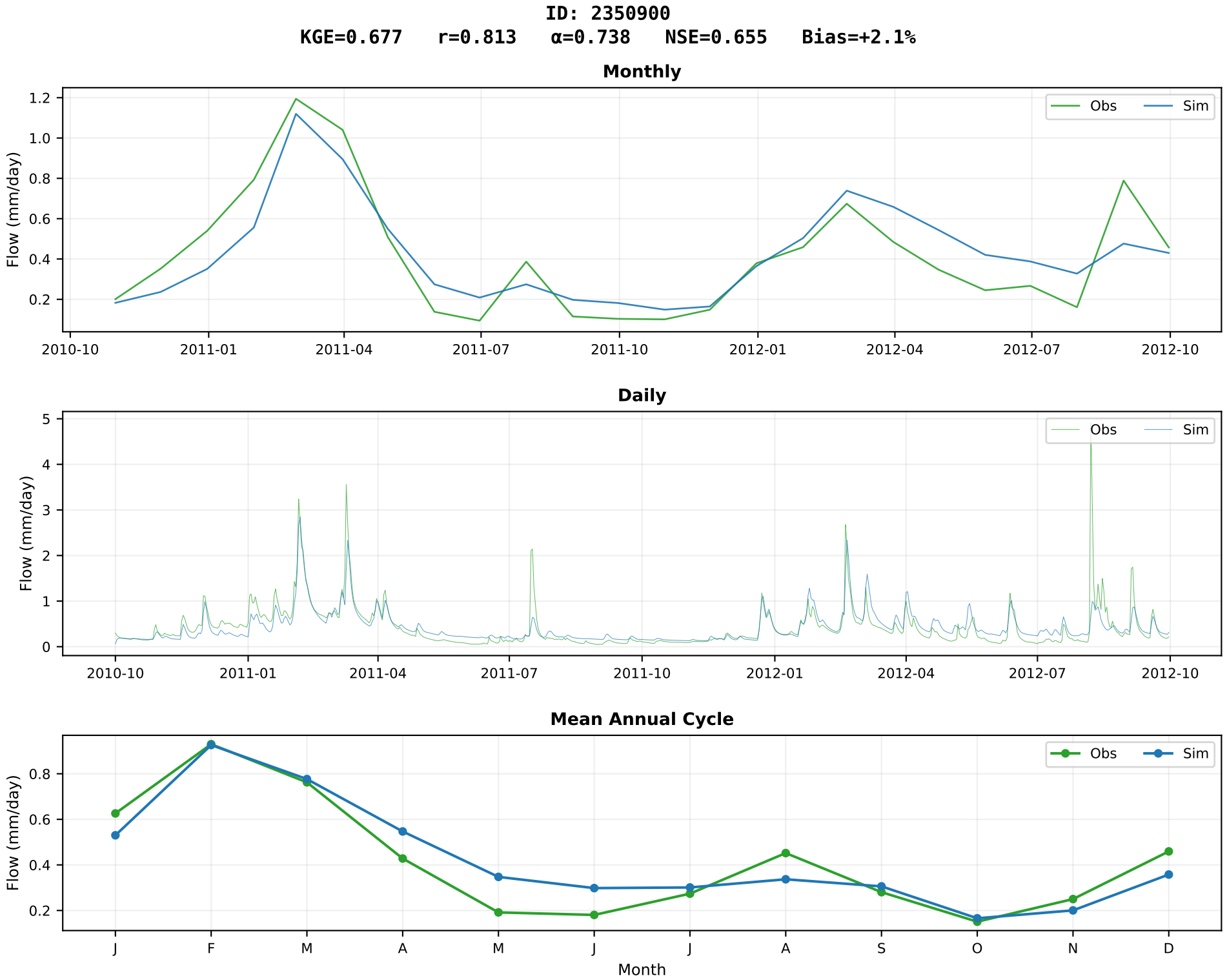

COW CREEK ABV GALESVILLE RES, NR AZALEA, OR. (ID=14308990)
KGE=0.686 r=0.956 α=0.811 NSE=0.873 Bias=-24.7%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

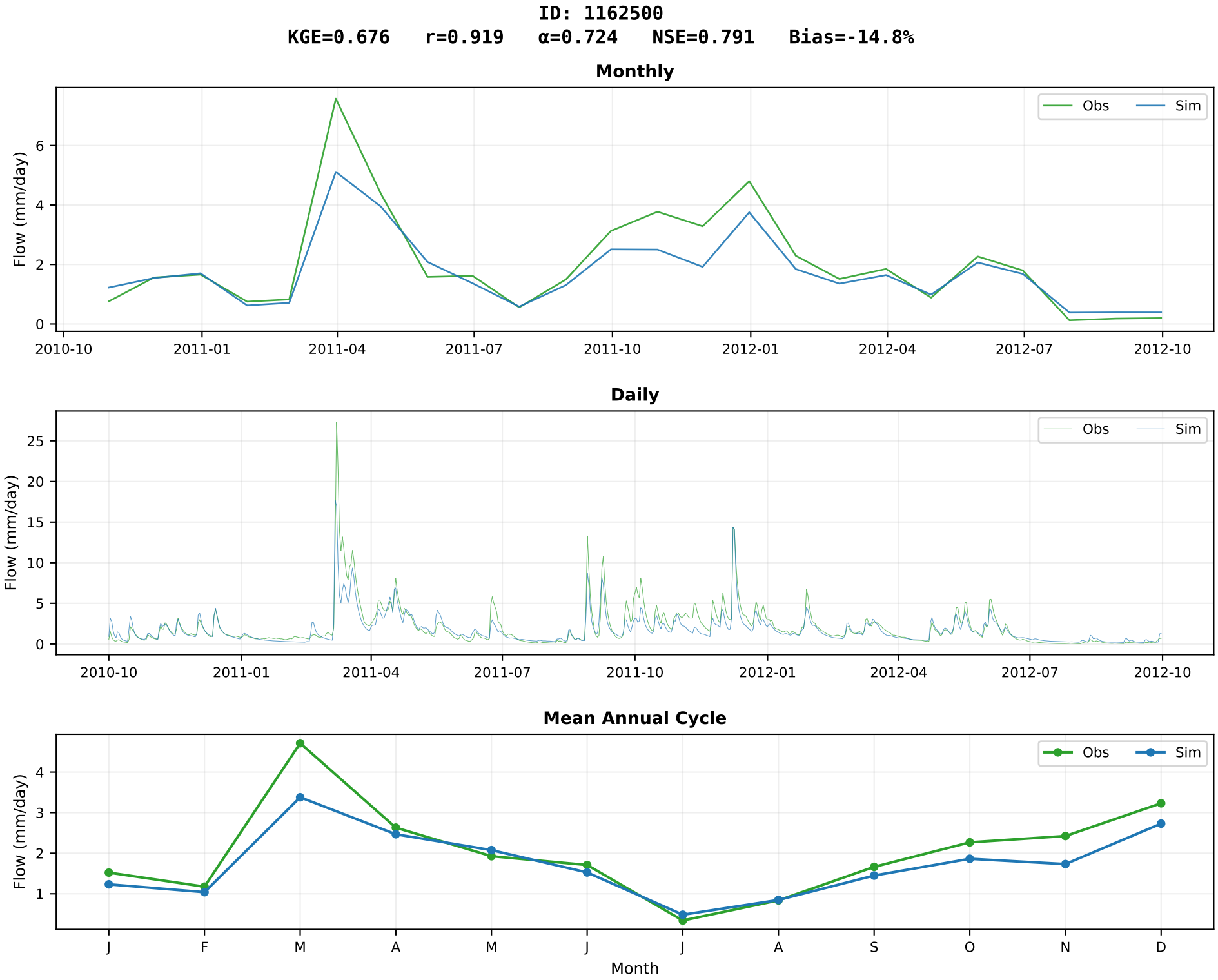

ID: 2143040
KGE=0.682 r=0.913 α=0.695 NSE=0.785 Bias=+2.0%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.8
1.6
1.4
1.2
1.0
0.8
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month

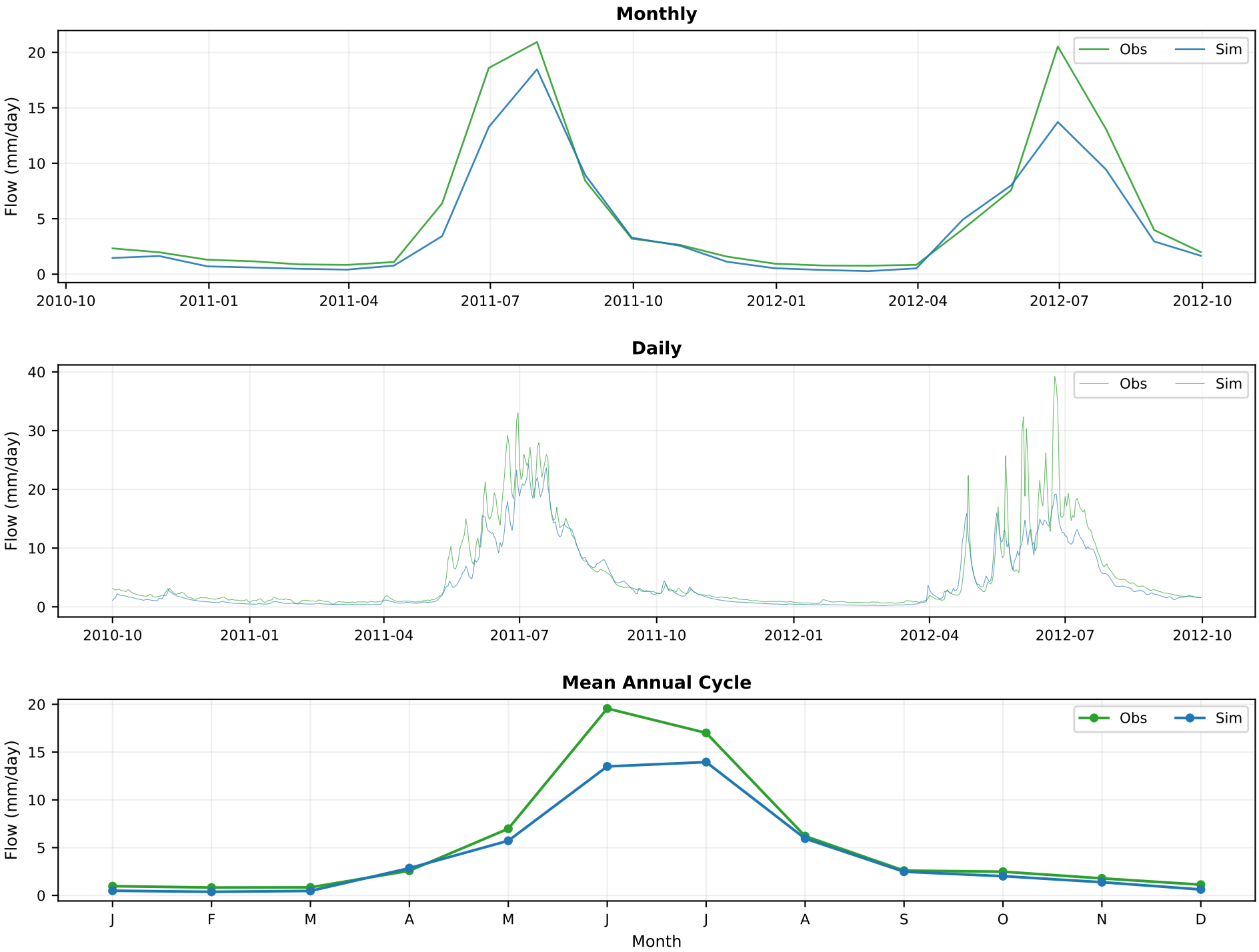

ID: 8103900
KGE=0.680 r=0.959 α=0.703 NSE=0.854 Bias=+11.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

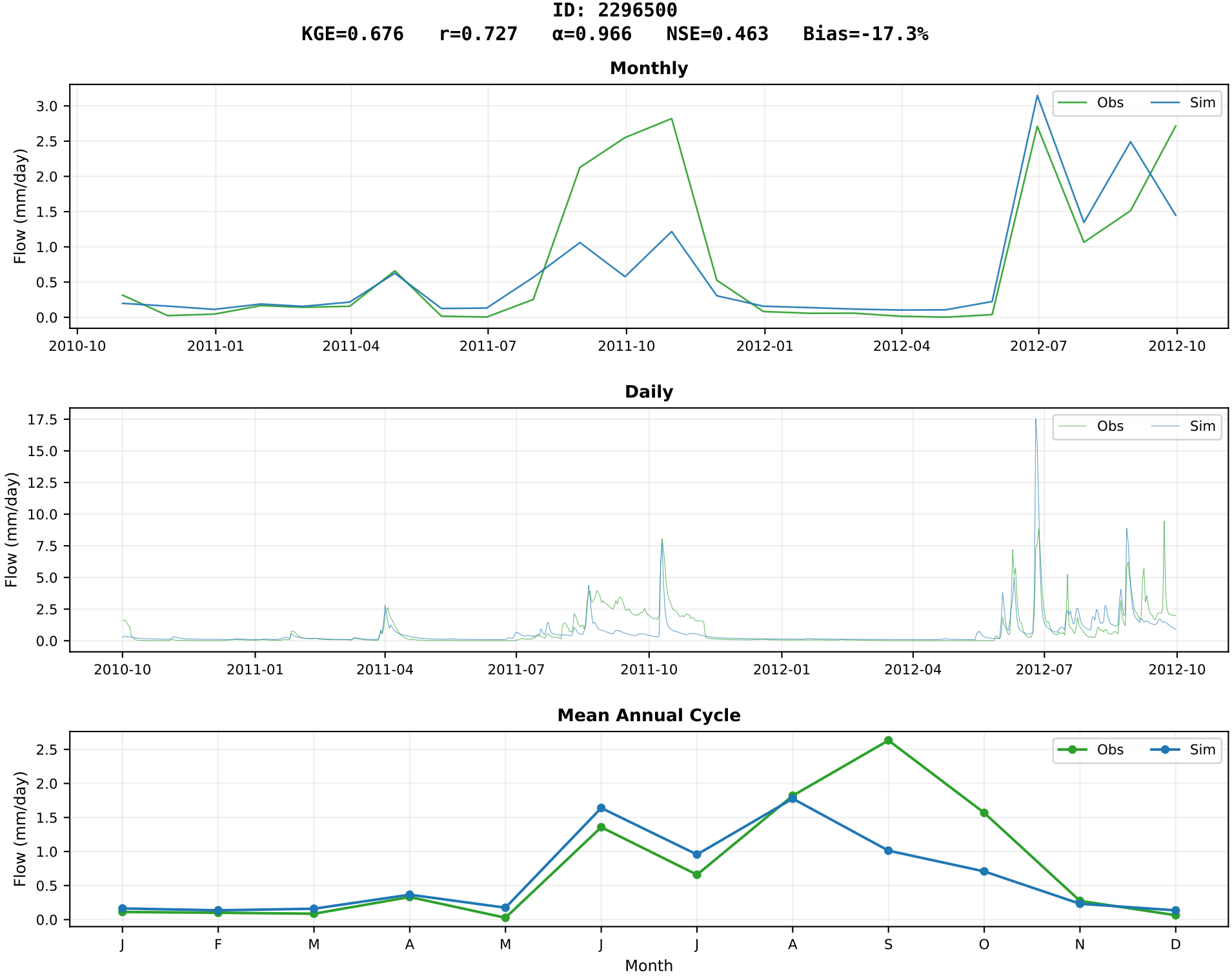

ID: 3144000
KGE=0.679 r=0.912 α=1.265 NSE=0.691 Bias=+15.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

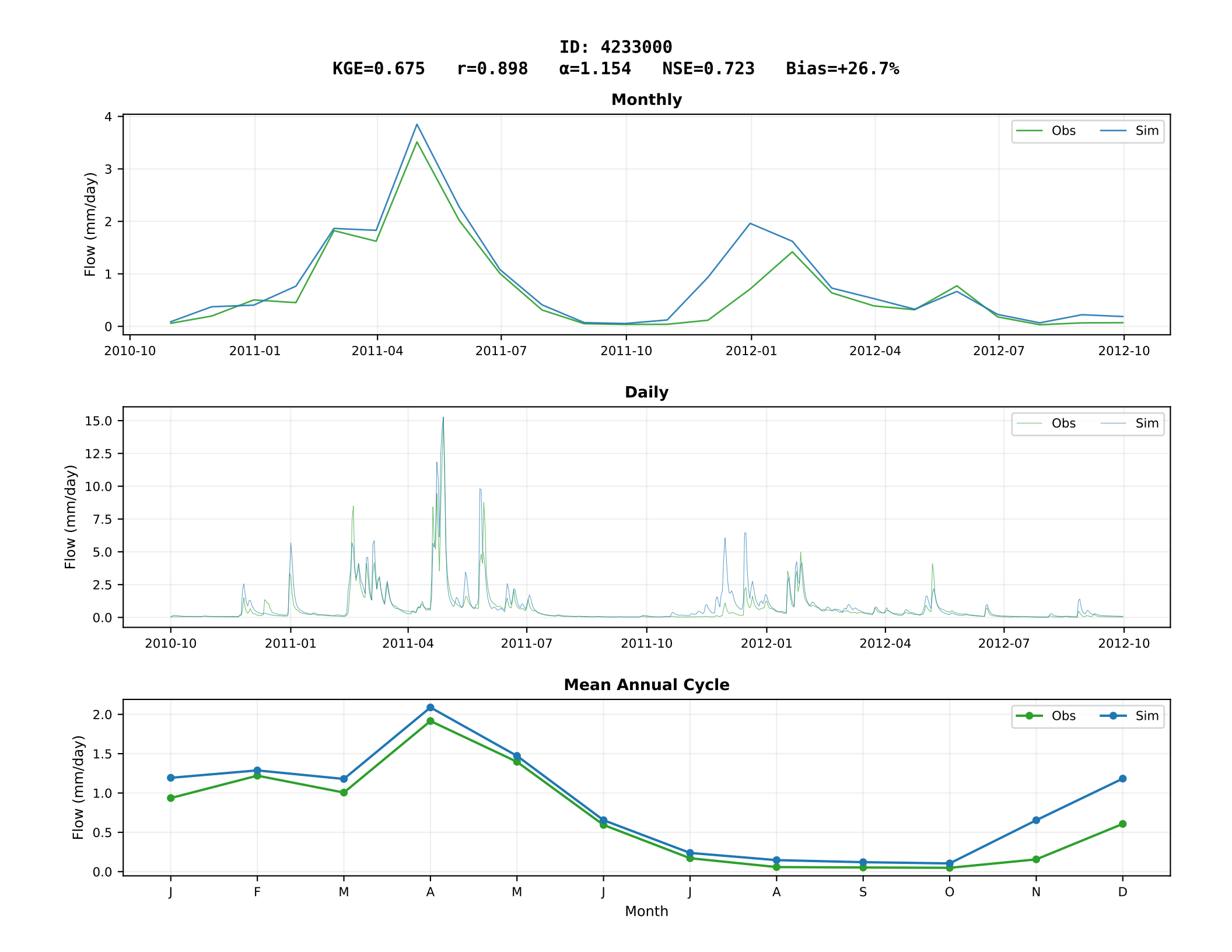

NASELLE RIVER NEAR NASELLE, WA (ID=12010000)
KGE=0.677 r=0.848 α=0.730 NSE=0.703 Bias=+9.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

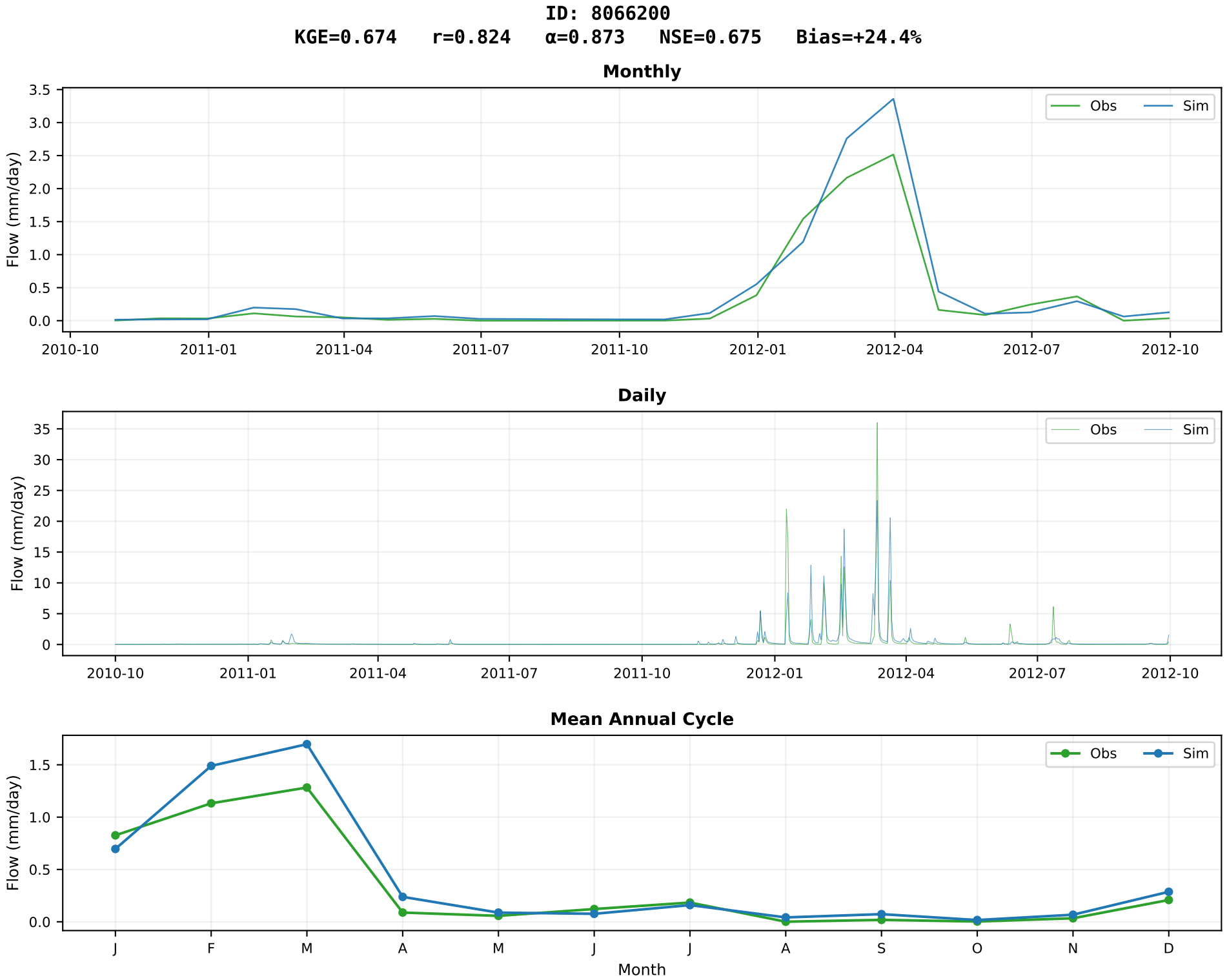

ID: 2350900
KGE=0.677 r=0.813 α=0.738 NSE=0.655 Bias=+2.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

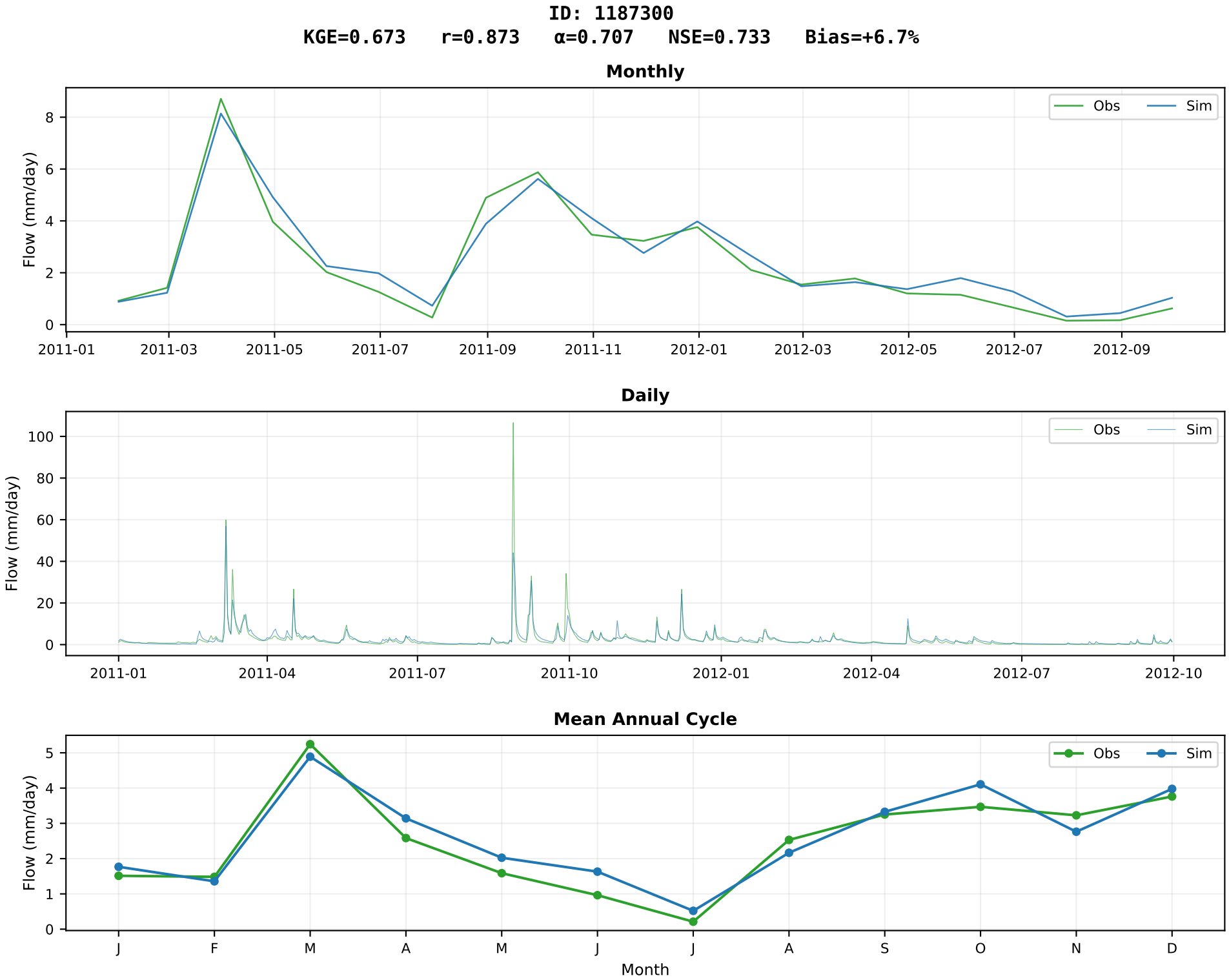

ID: 1162500
KGE=0.676 r=0.919 α=0.724 NSE=0.791 Bias=-14.8%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

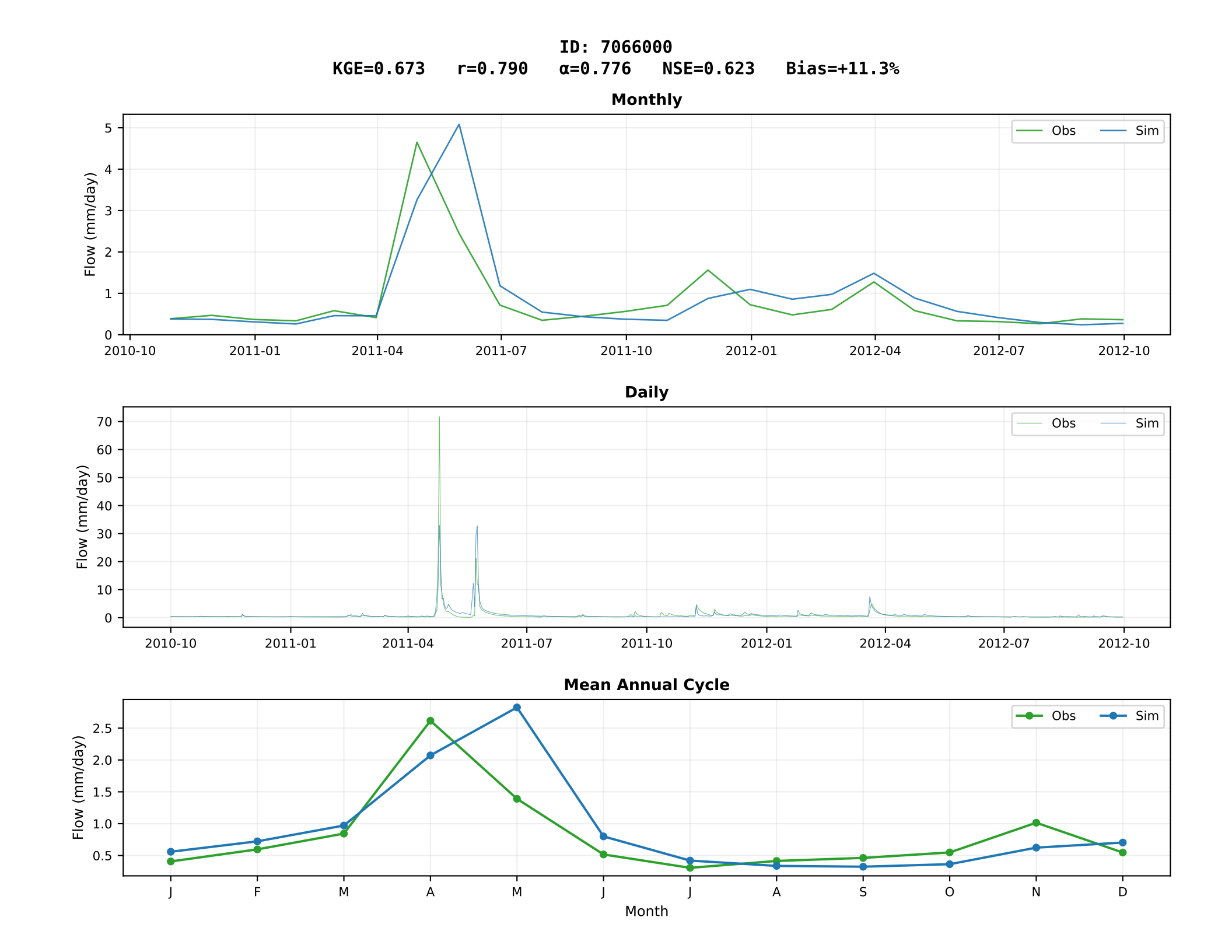

SMITH R AB SMITH R RES NR BELKNAP SPRGS,OREG. (ID=14158790)
KGE=0.676 r=0.937 α=0.759 NSE=0.822 Bias=-20.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

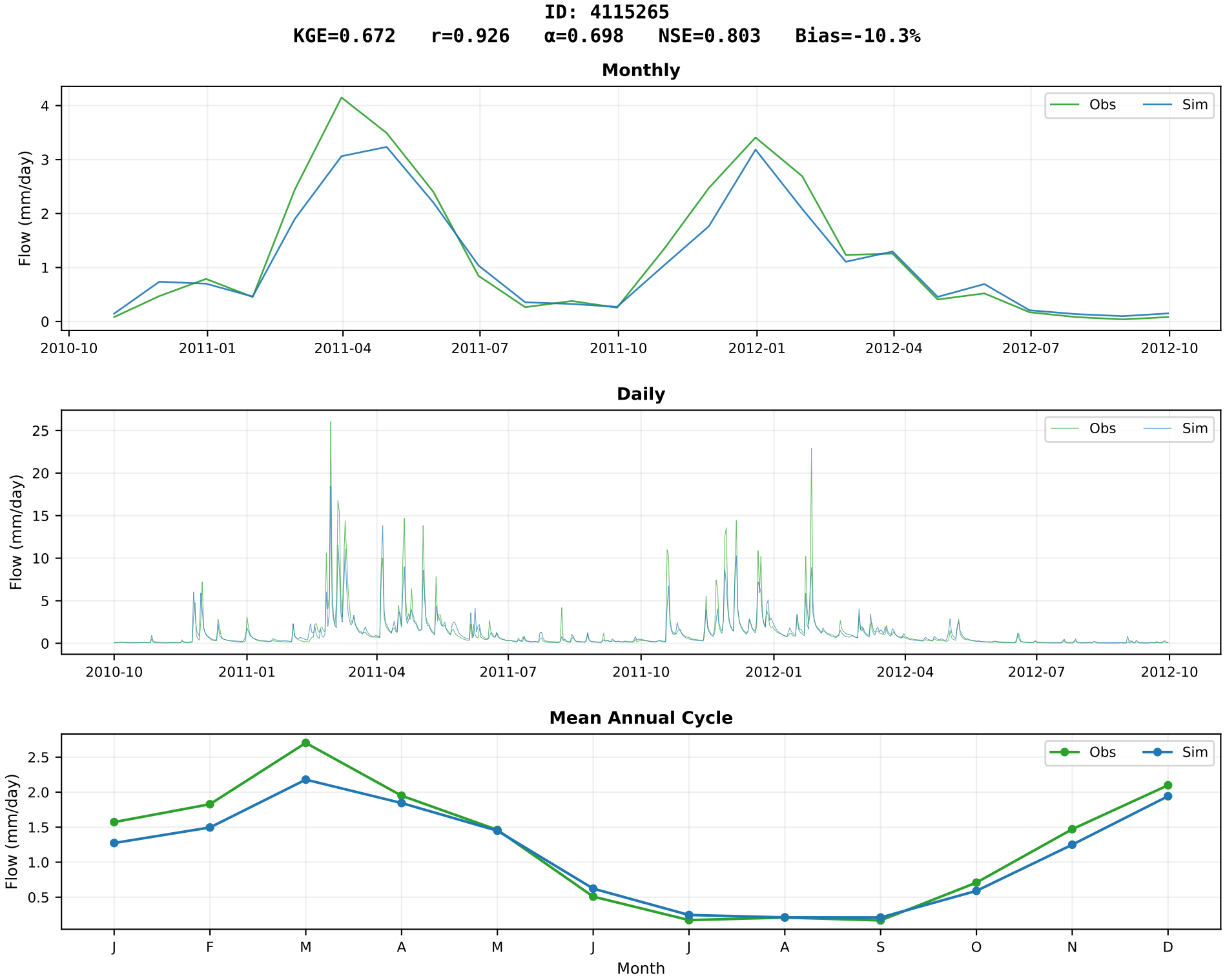

ID: 2296500
KGE=0.676 r=0.727 α=0.966 NSE=0.463 Bias=-17.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

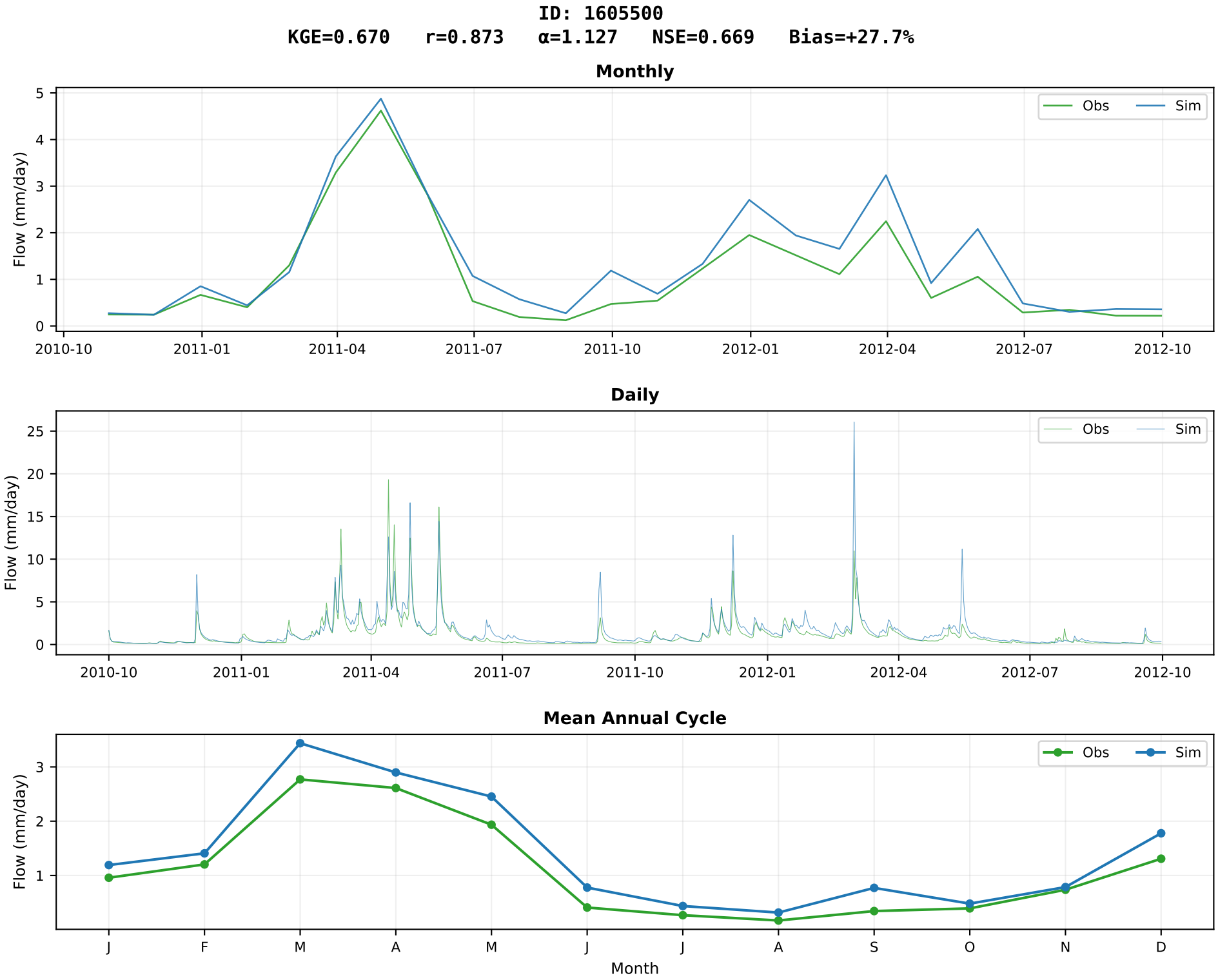

ID: 4233000
KGE=0.675 r=0.898 α=1.154 NSE=0.723 Bias=+26.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

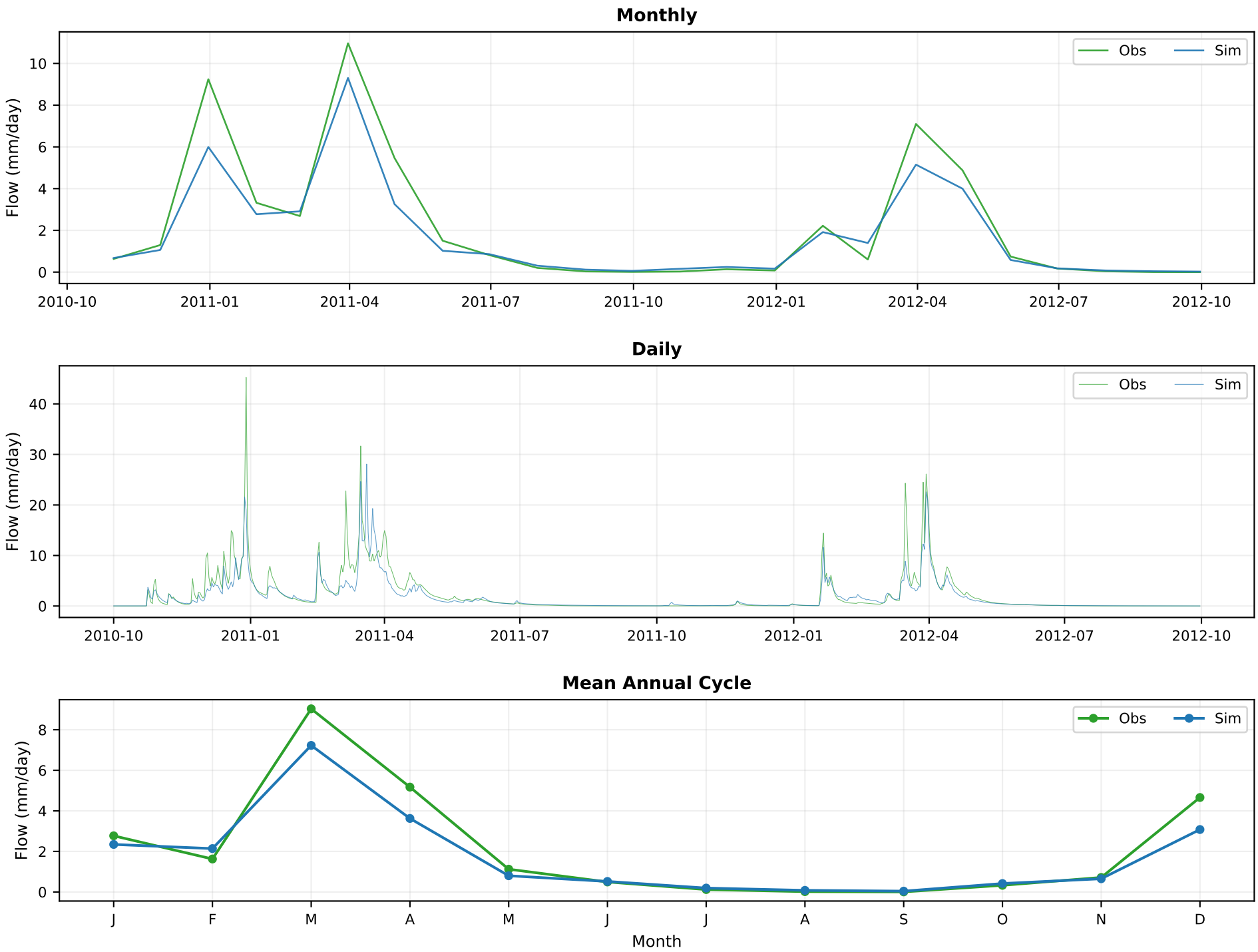

ID: 8066200
KGE=0.674 r=0.824 α=0.873 NSE=0.675 Bias=+24.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

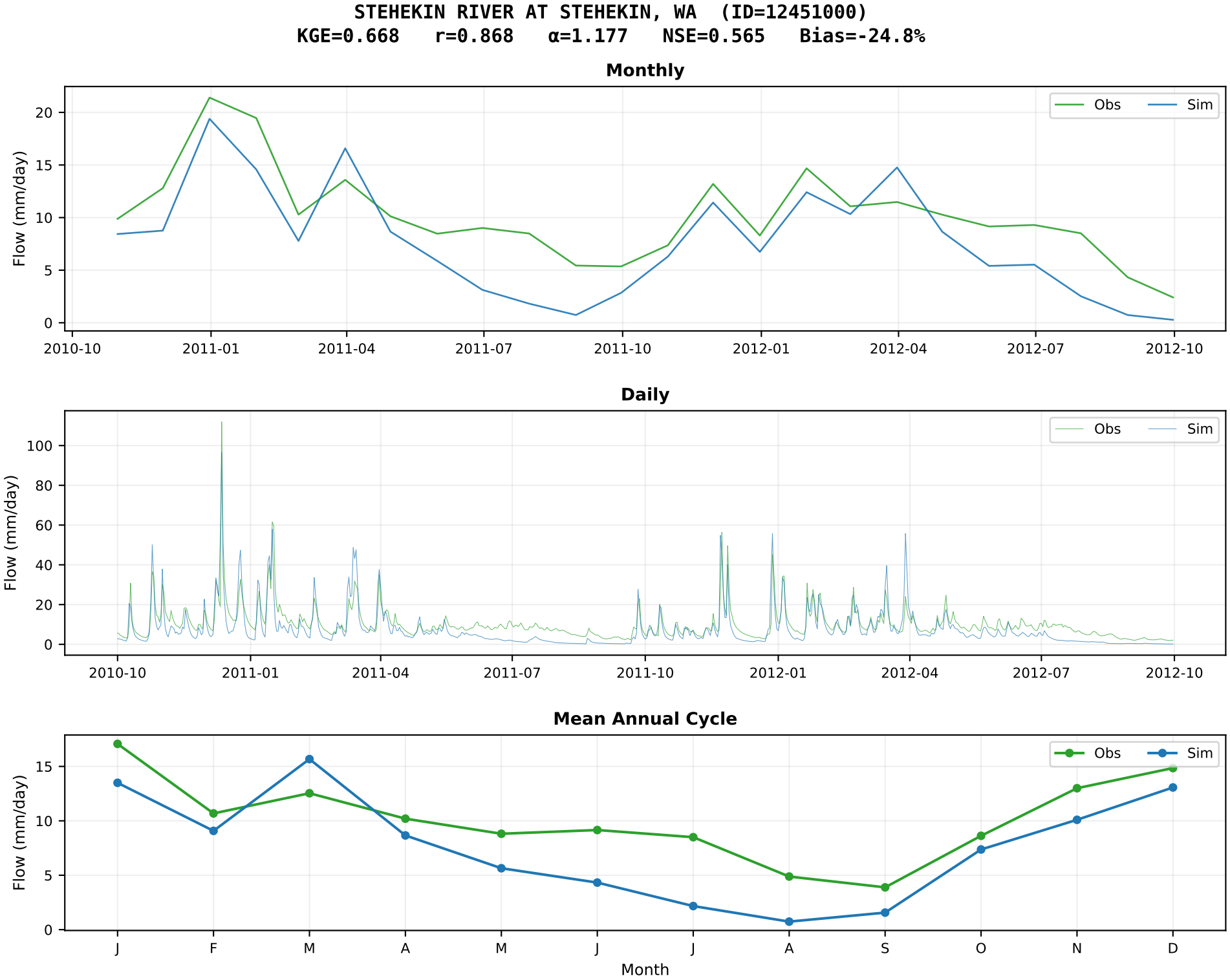

ID: 1187300
KGE=0.673 r=0.873 α=0.707 NSE=0.733 Bias=+6.7%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2011-01
2011-03
2011-05
2011-07
2011-09
2011-11
2012-01
2012-03
2012-05
2012-07
2012-09
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
100
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

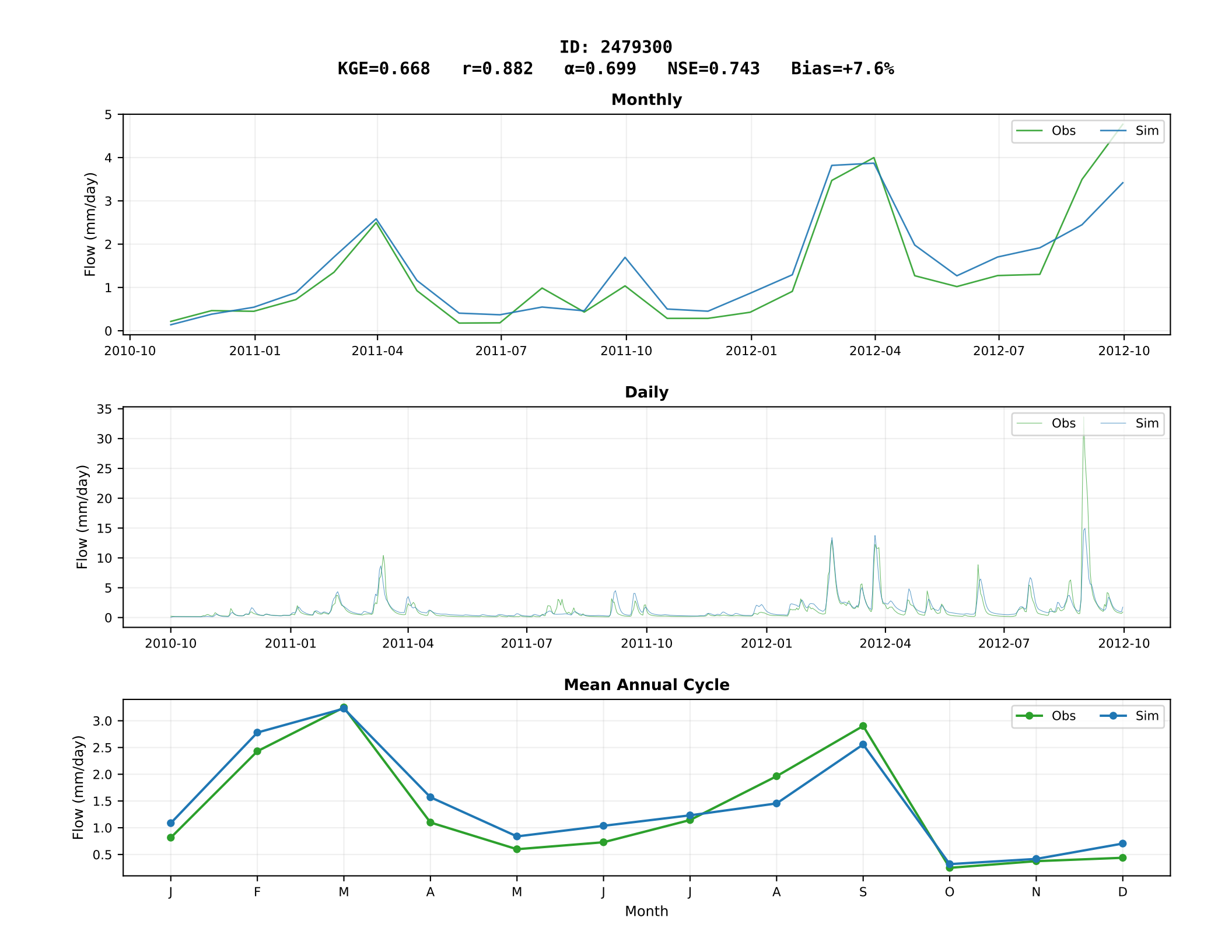

ID: 7066000
KGE=0.673 r=0.790 α=0.776 NSE=0.623 Bias=+11.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

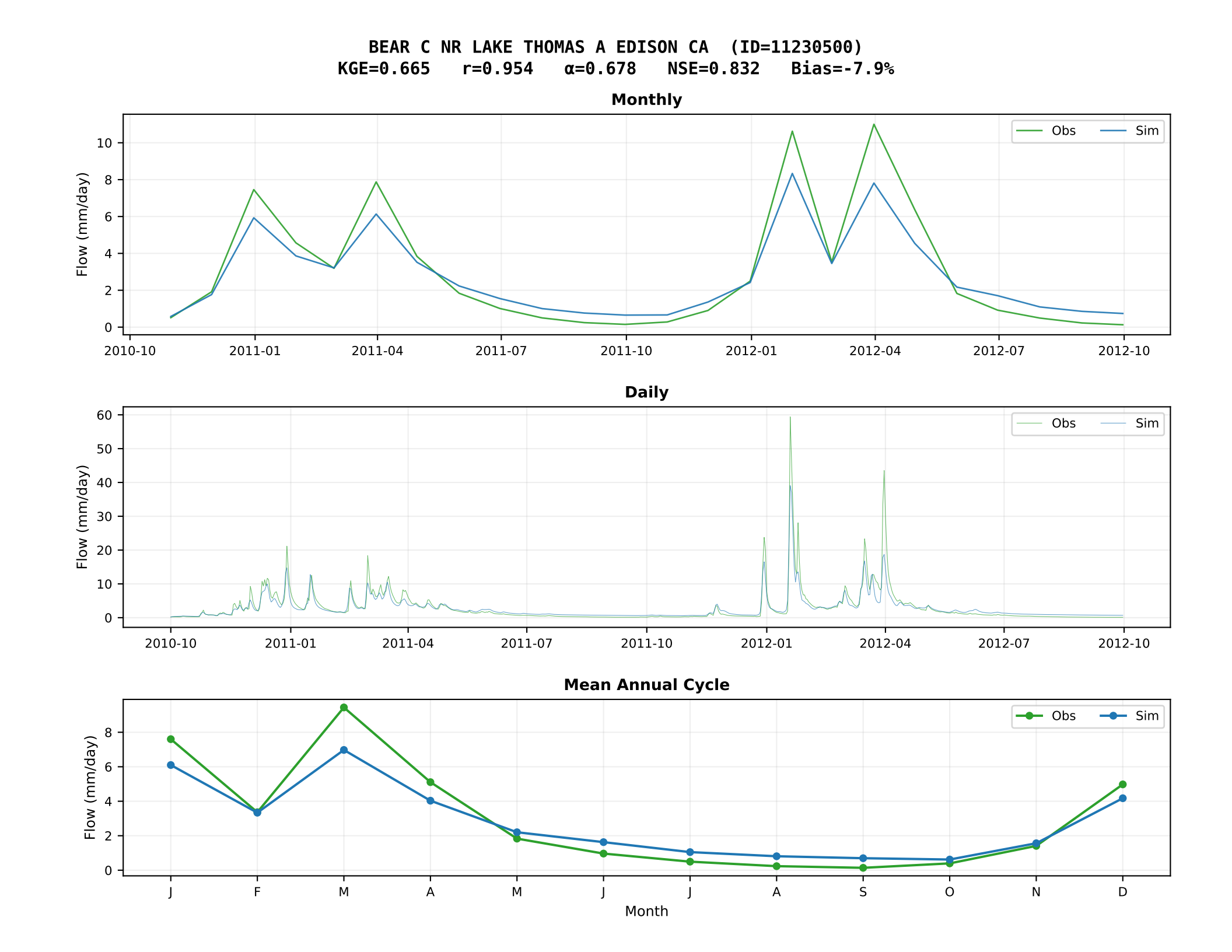

ID: 4115265
KGE=0.672 r=0.926 α=0.698 NSE=0.803 Bias=-10.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

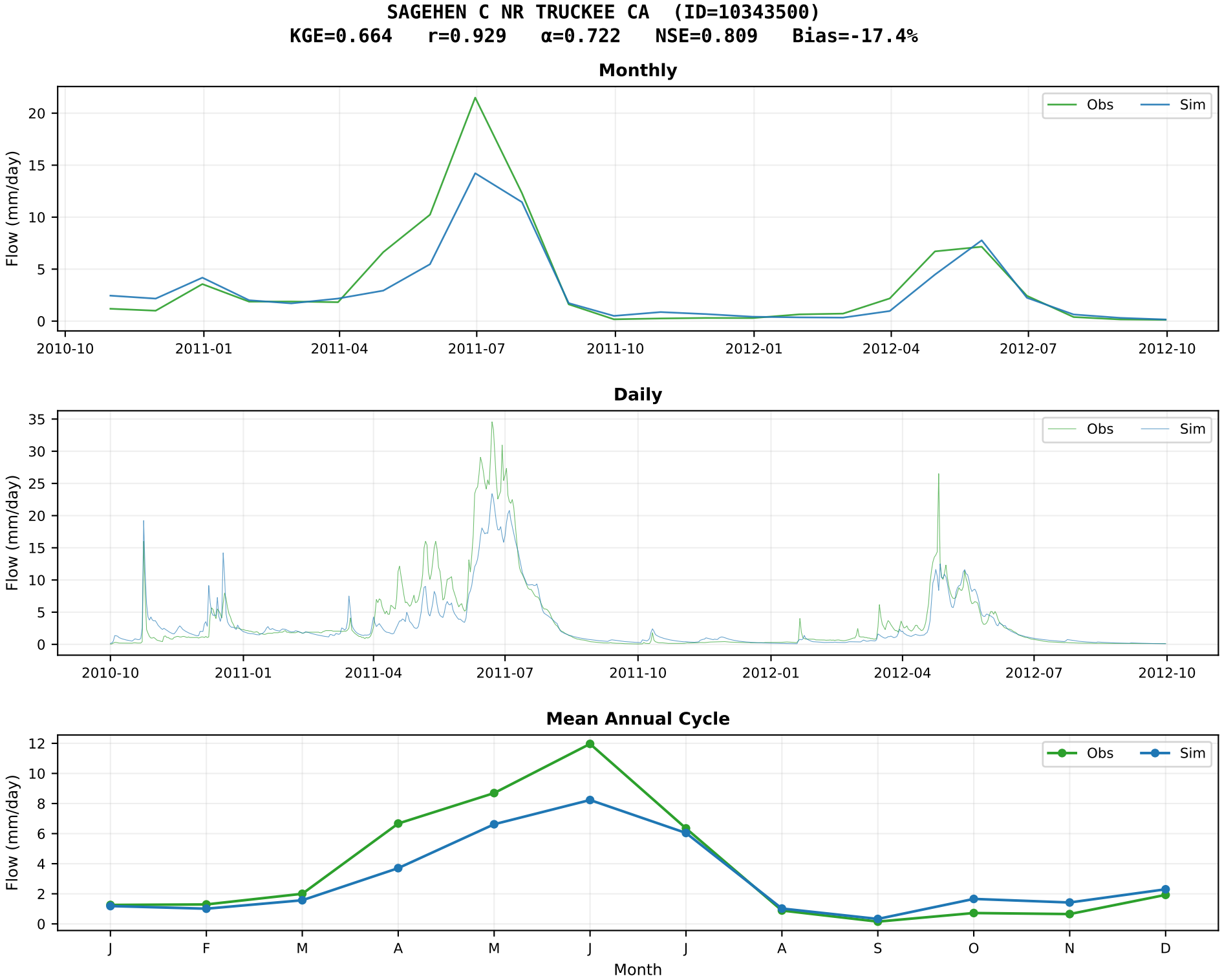

ID: 1605500
KGE=0.670 r=0.873 α=1.127 NSE=0.669 Bias=+27.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

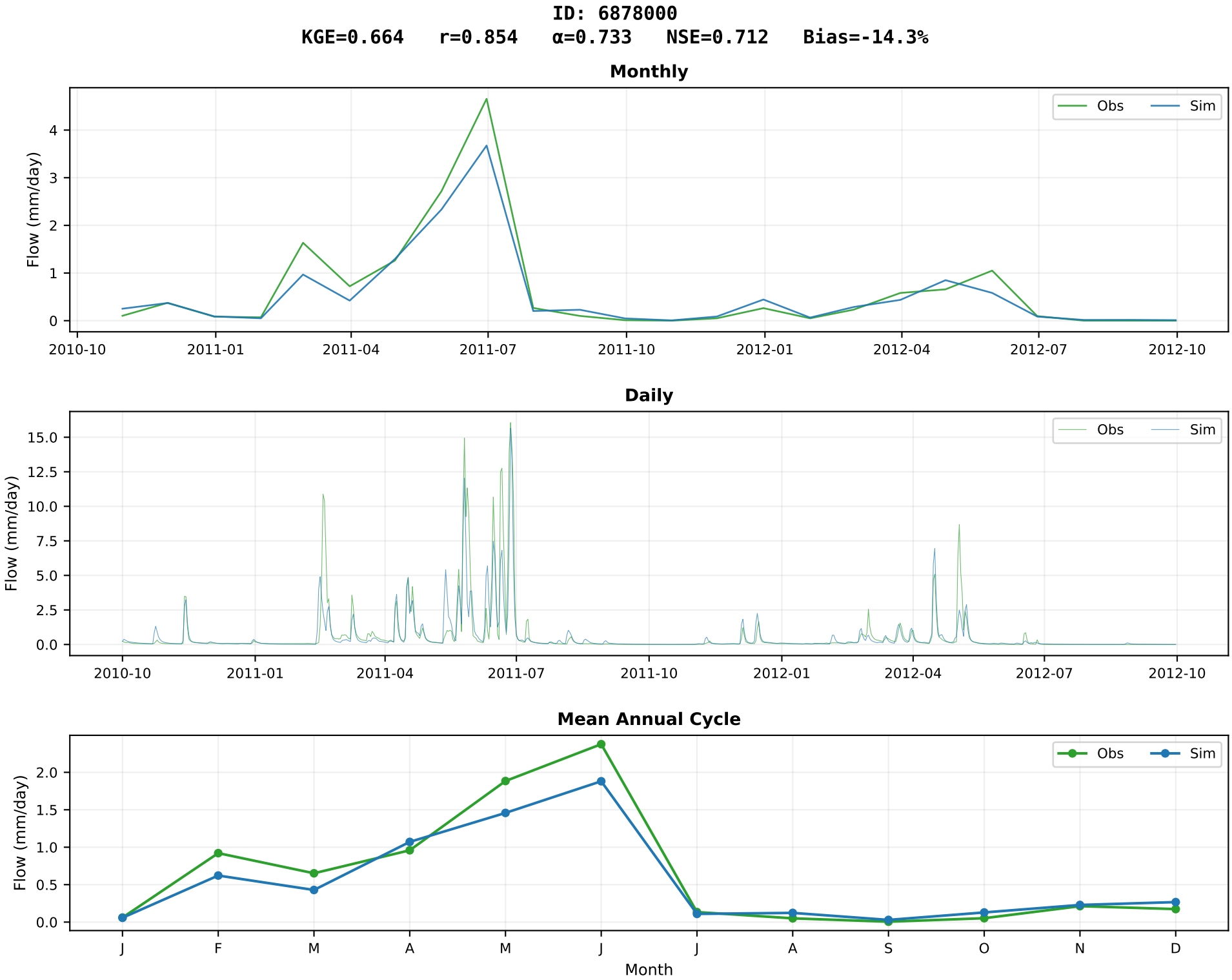

NORTH FORK TOLT RIVER NEAR CARNATION, WA (ID=12147500)
KGE=0.669 r=0.874 α=0.762 NSE=0.741 Bias=-19.2%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
2
4
6
8
J
F
M
A
M
J
J
A
S
O
N
D
Month

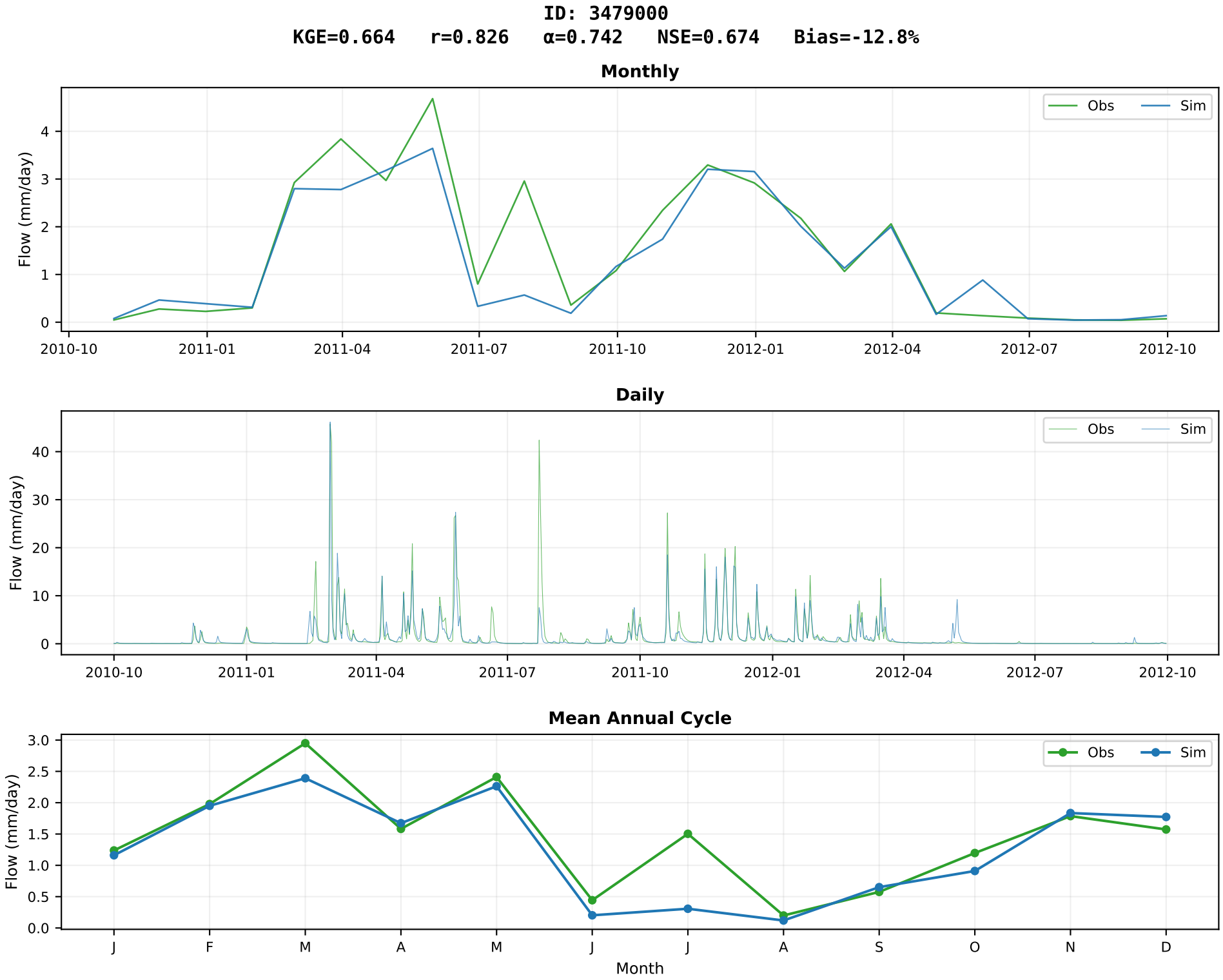

STEHEKIN RIVER AT STEHEKIN, WA (ID=12451000)
KGE=0.668 r=0.868 α=1.177 NSE=0.565 Bias=-24.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

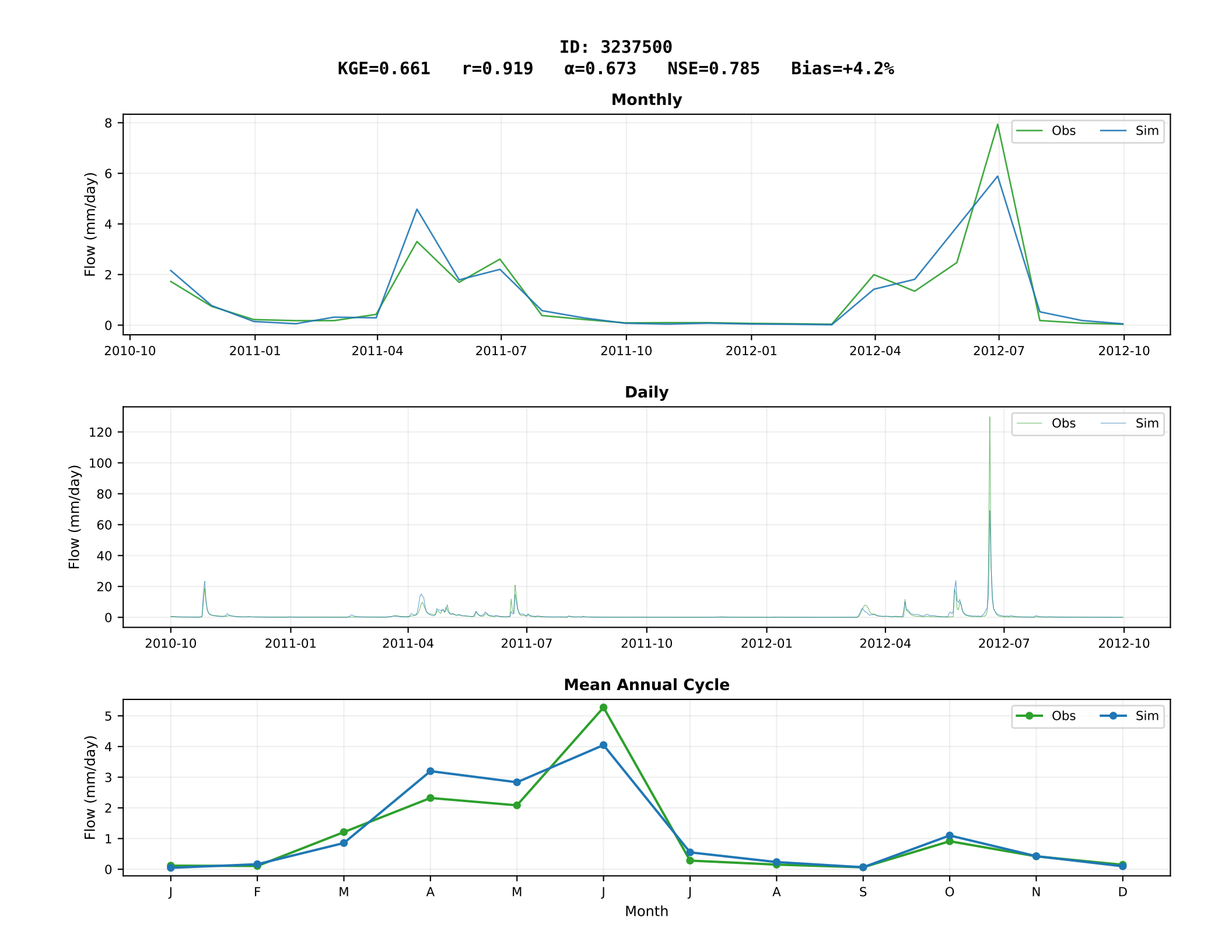

ID: 2479300
KGE=0.668 r=0.882 α=0.699 NSE=0.743 Bias=+7.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

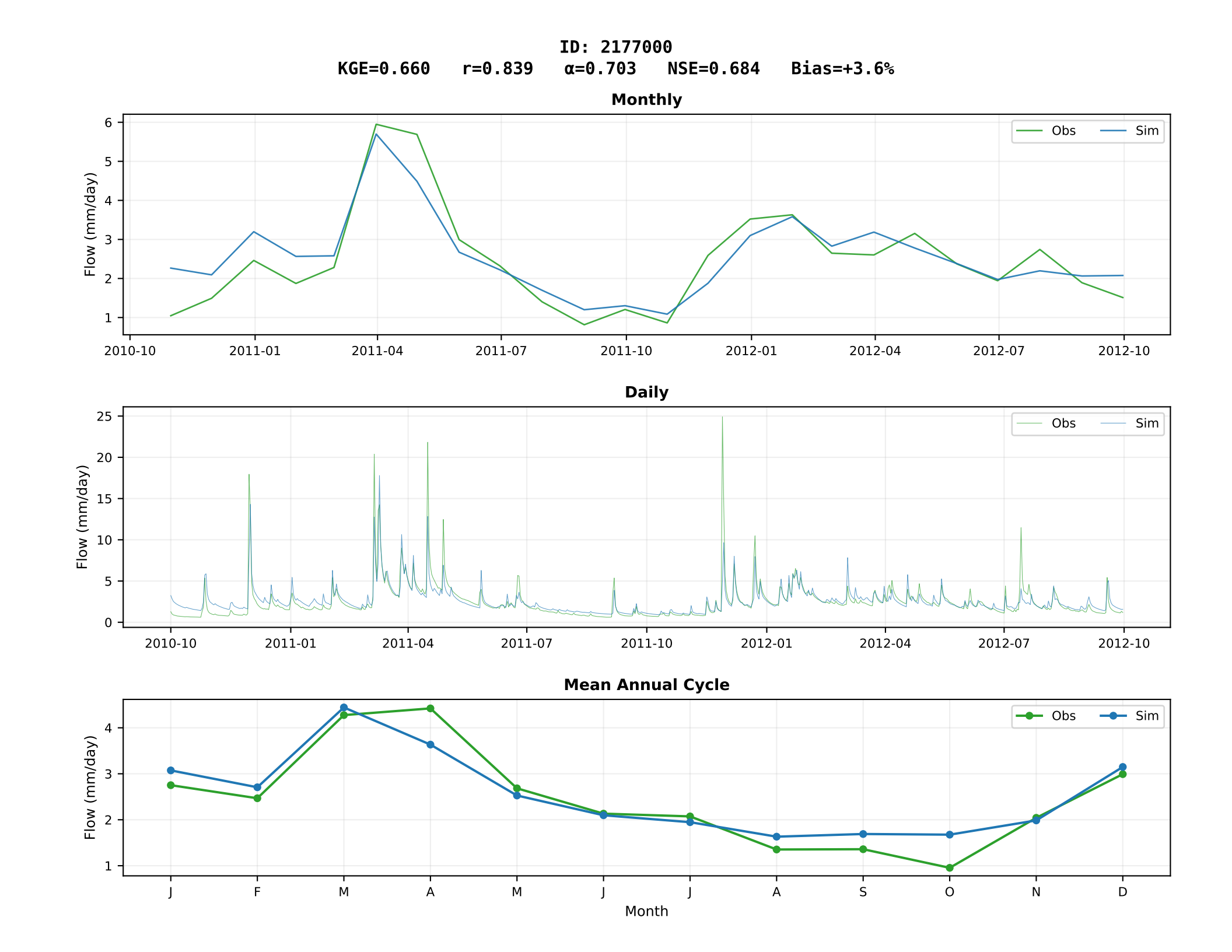

BEAR C NR LAKE THOMAS A EDISON CA (ID=11230500)
KGE=0.665 r=0.954 α=0.678 NSE=0.832 Bias=-7.9%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

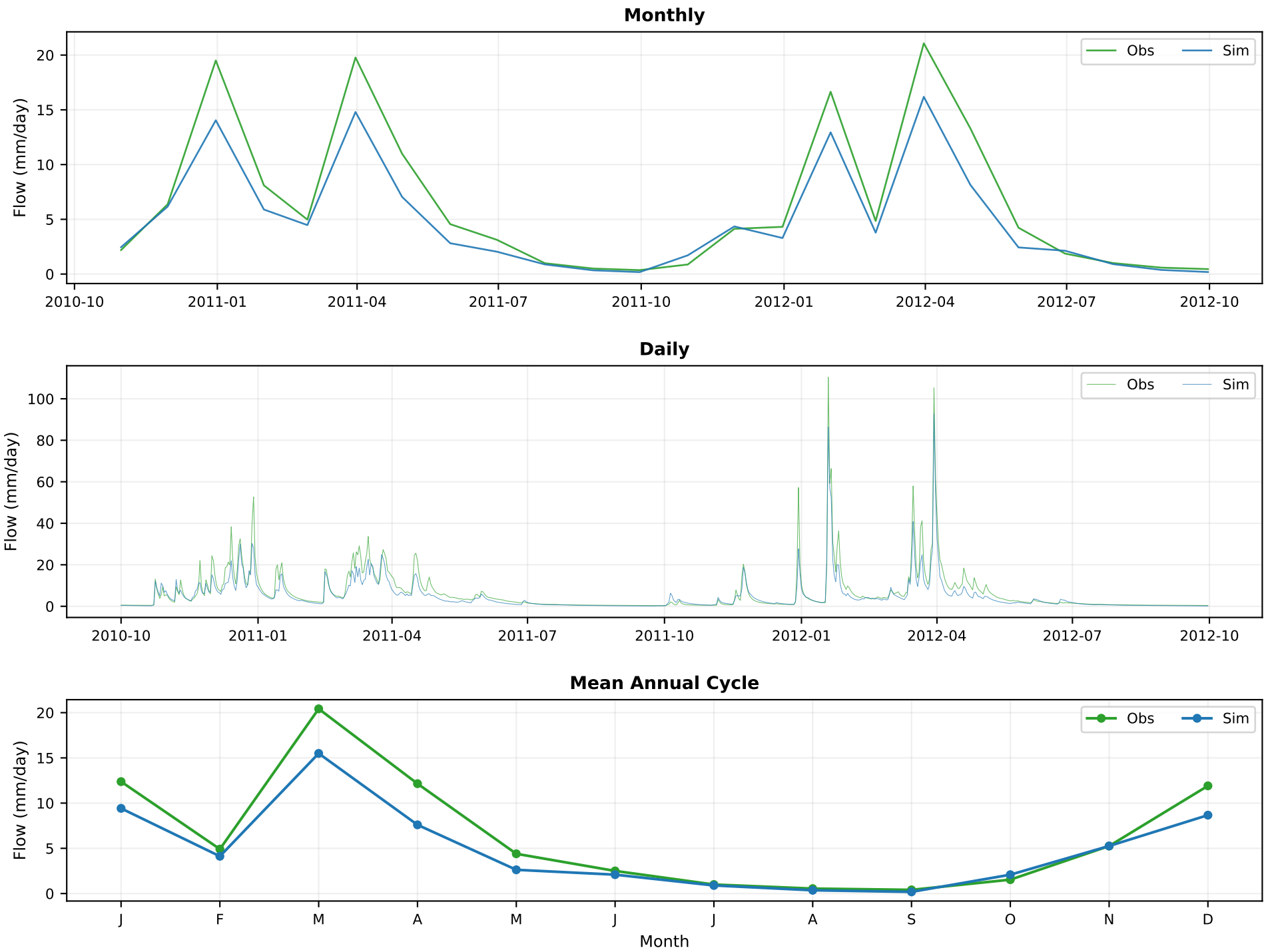

SAGEHEN C NR TRUCKEE CA (ID=10343500)
KGE=0.664 r=0.929 α=0.722 NSE=0.809 Bias=-17.4%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

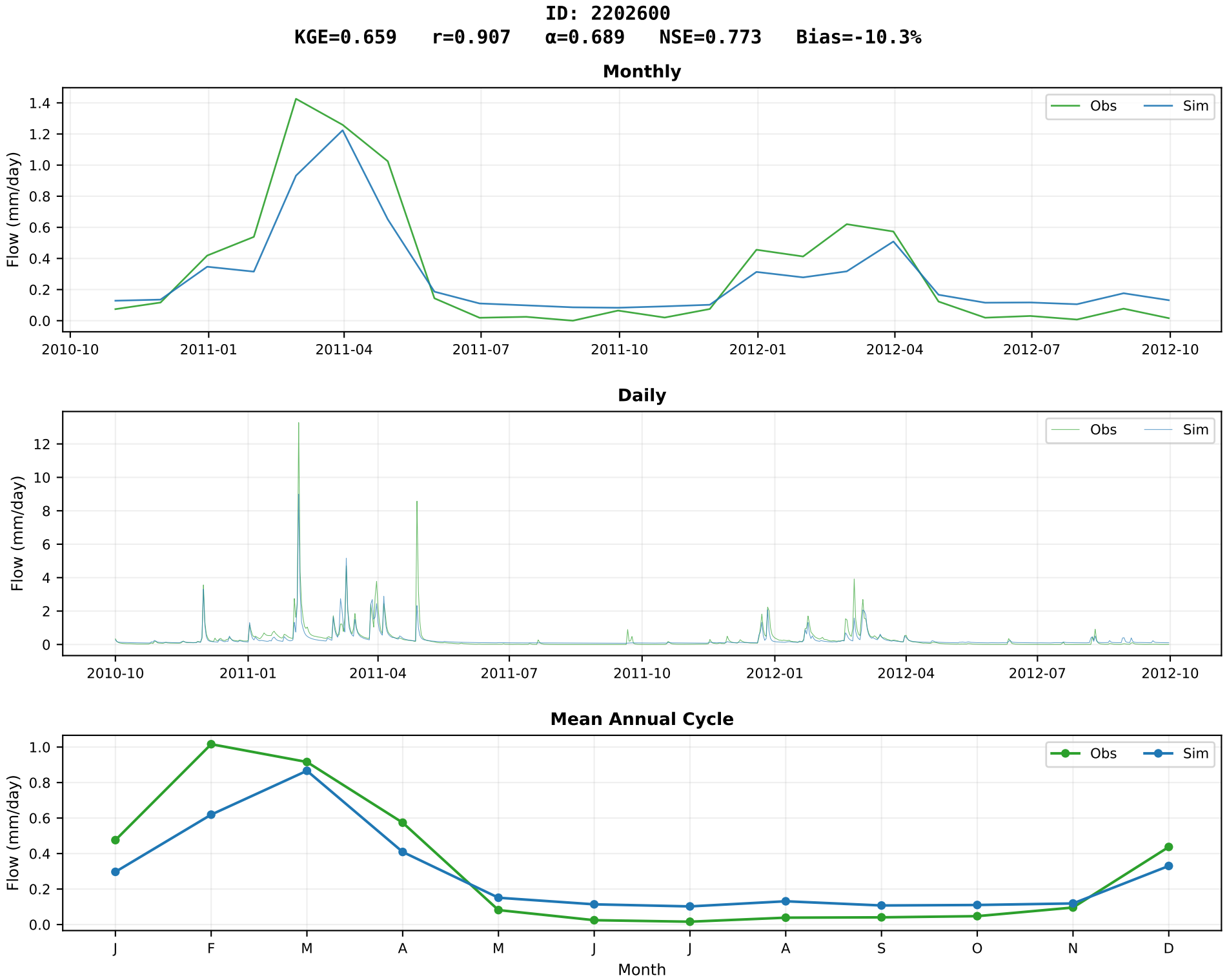

ID: 6878000
KGE=0.664 r=0.854 α=0.733 NSE=0.712 Bias=-14.3%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

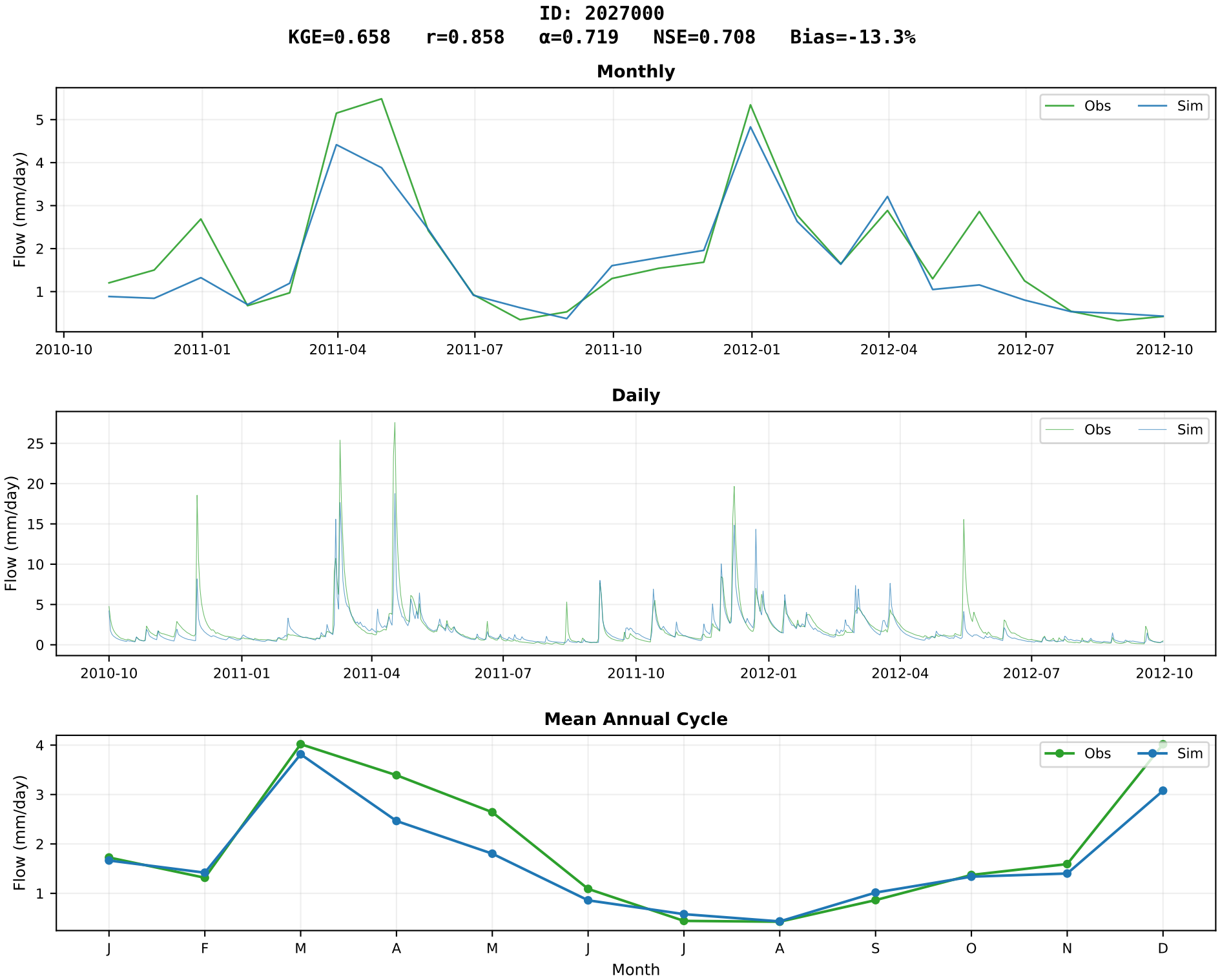

ID: 3479000
KGE=0.664 r=0.826 α=0.742 NSE=0.674 Bias=-12.8%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

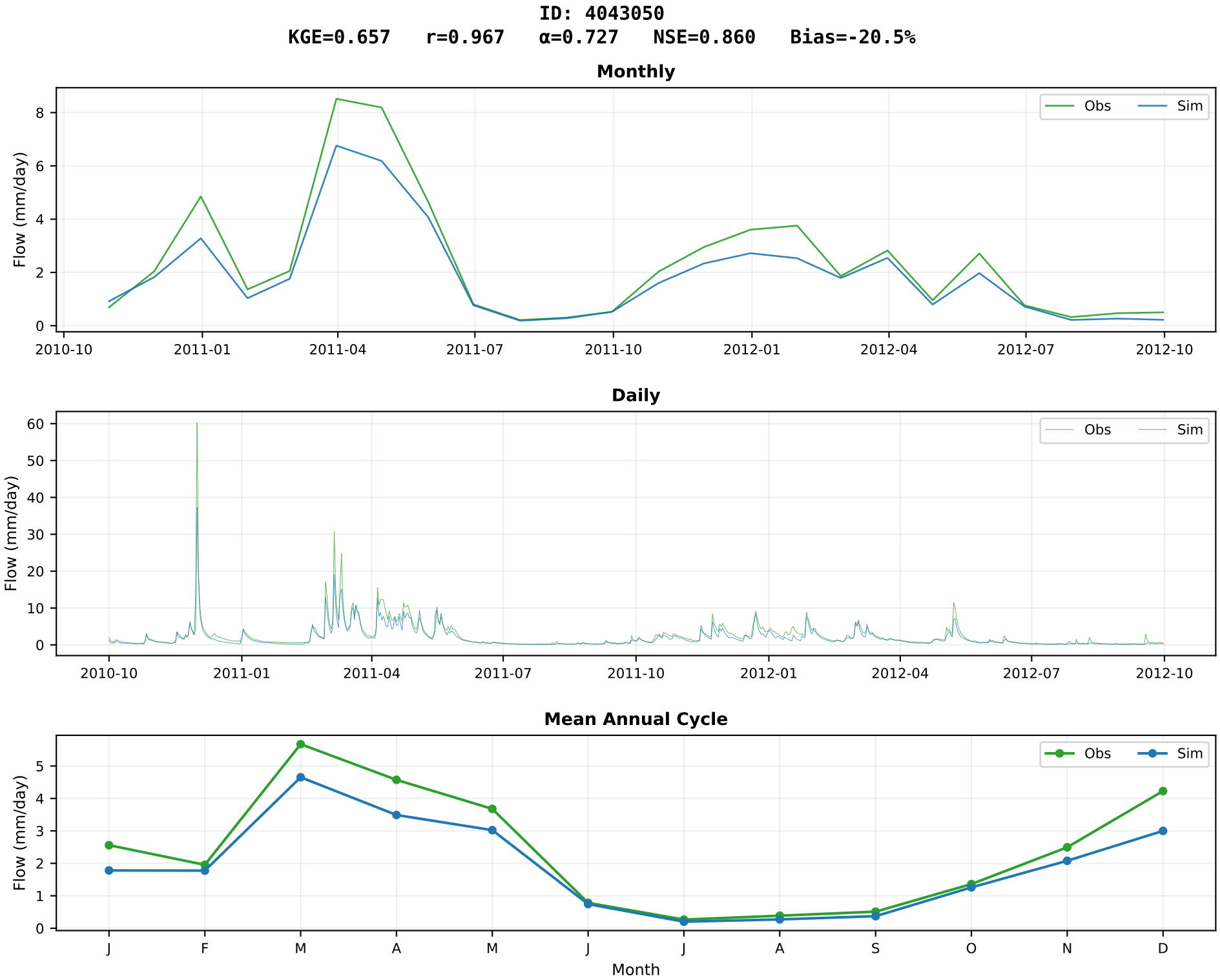

ID: 3237500
KGE=0.661 r=0.919 α=0.673 NSE=0.785 Bias=+4.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

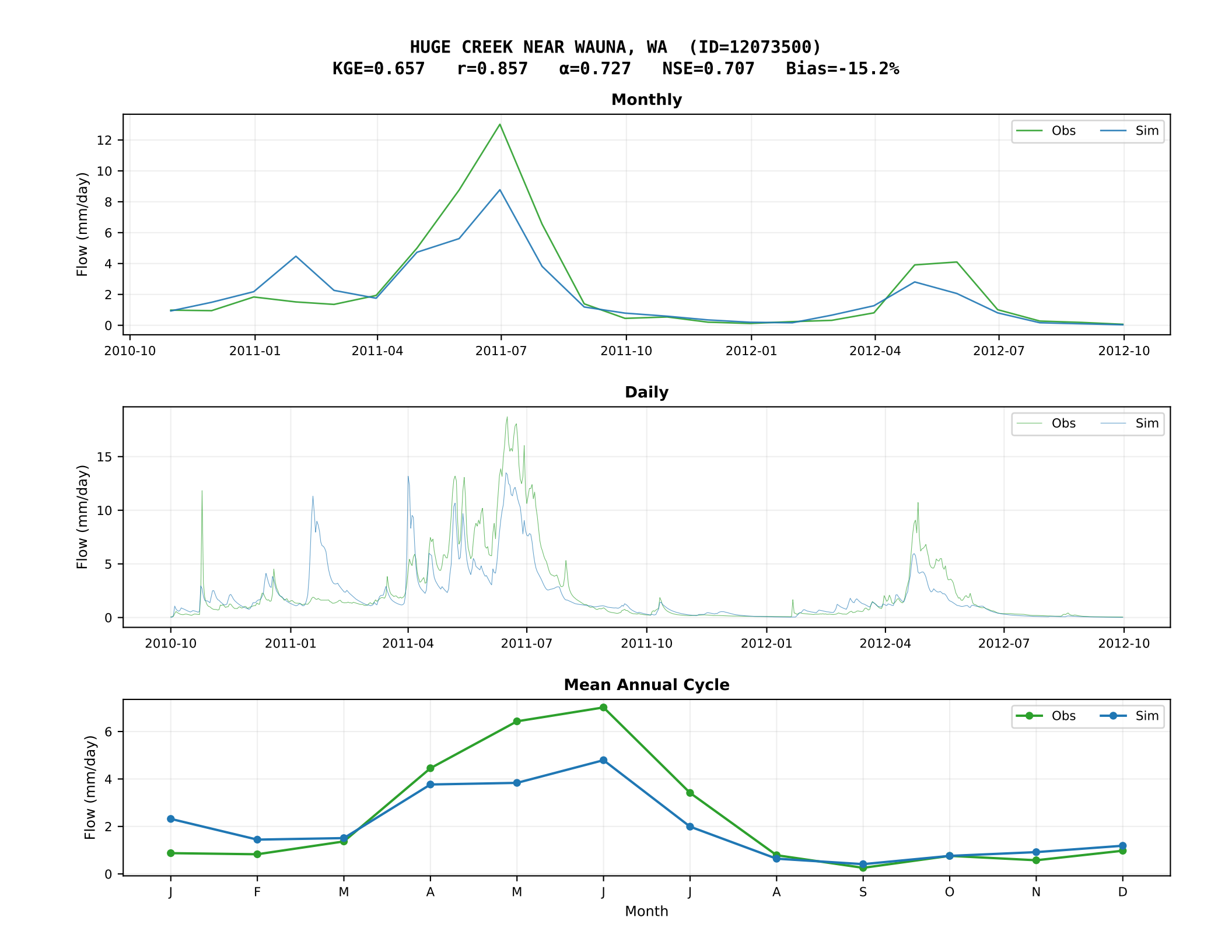

ID: 2177000
KGE=0.660 r=0.839 α=0.703 NSE=0.684 Bias=+3.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

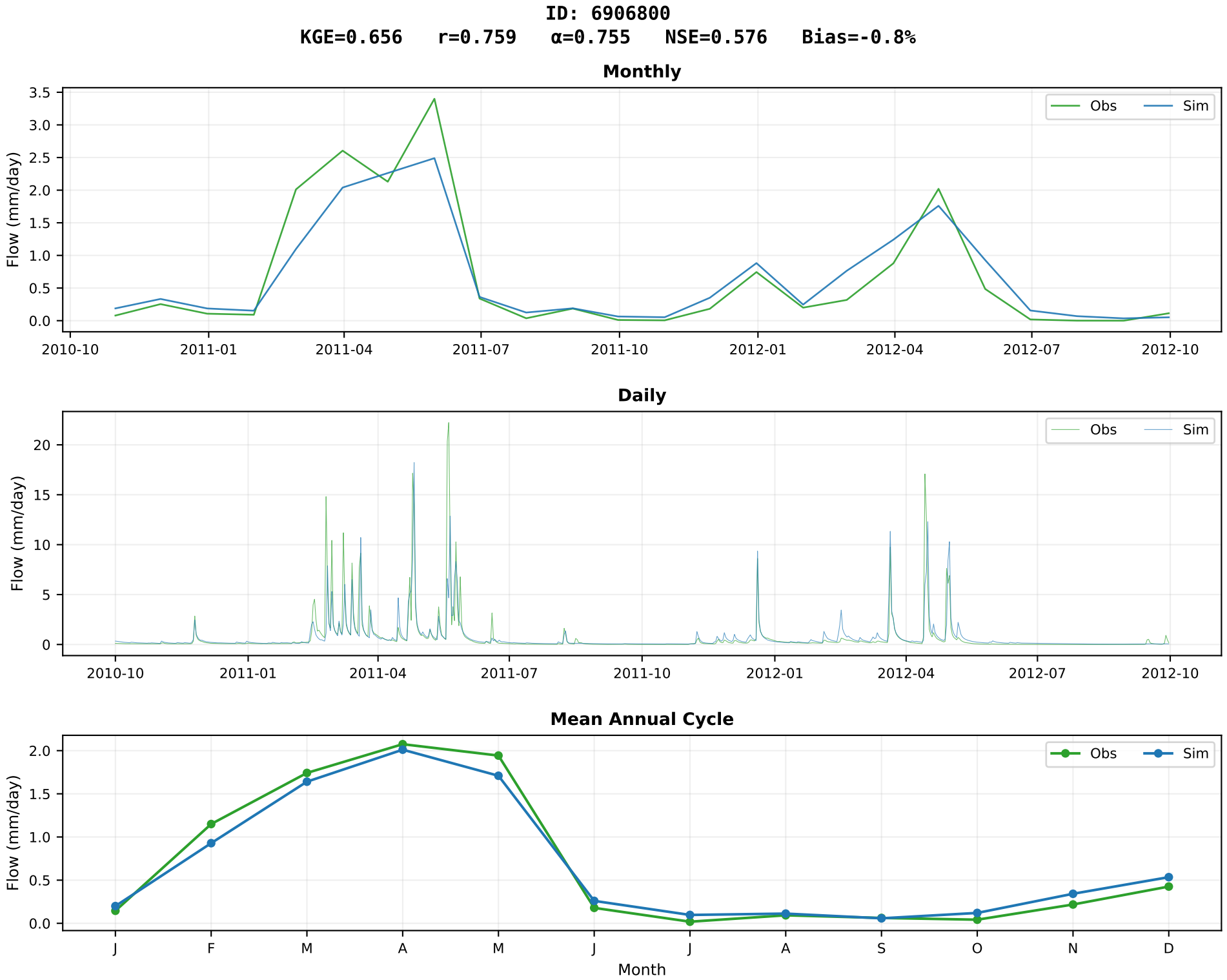

South Crow Creek near Ronan MT (ID=12375900)
KGE=0.660 r=0.969 α=0.762 NSE=0.873 Bias=-24.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

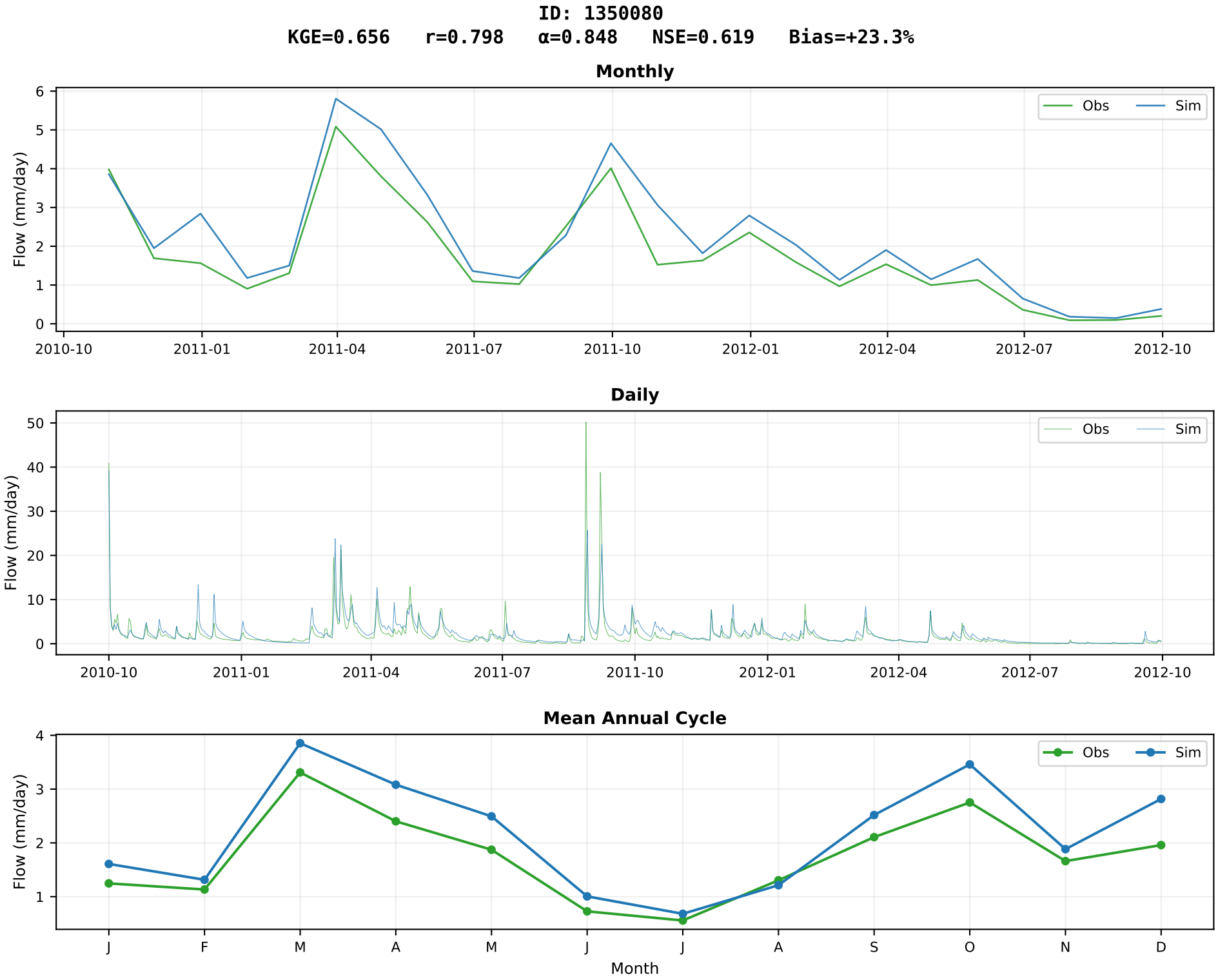

ID: 2202600
KGE=0.659 r=0.907 α=0.689 NSE=0.773 Bias=-10.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

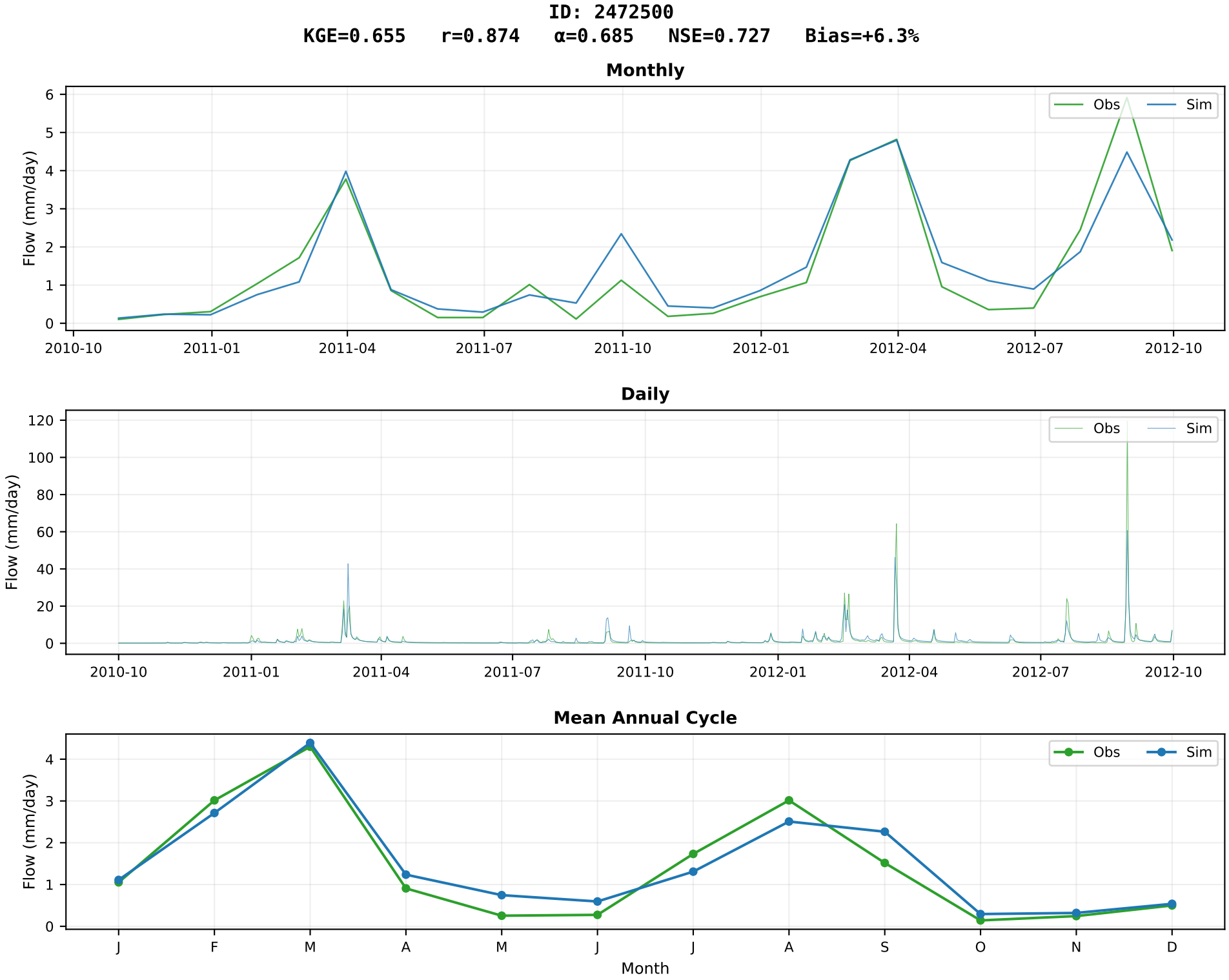

ID: 2027000
KGE=0.658 r=0.858 α=0.719 NSE=0.708 Bias=-13.3%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

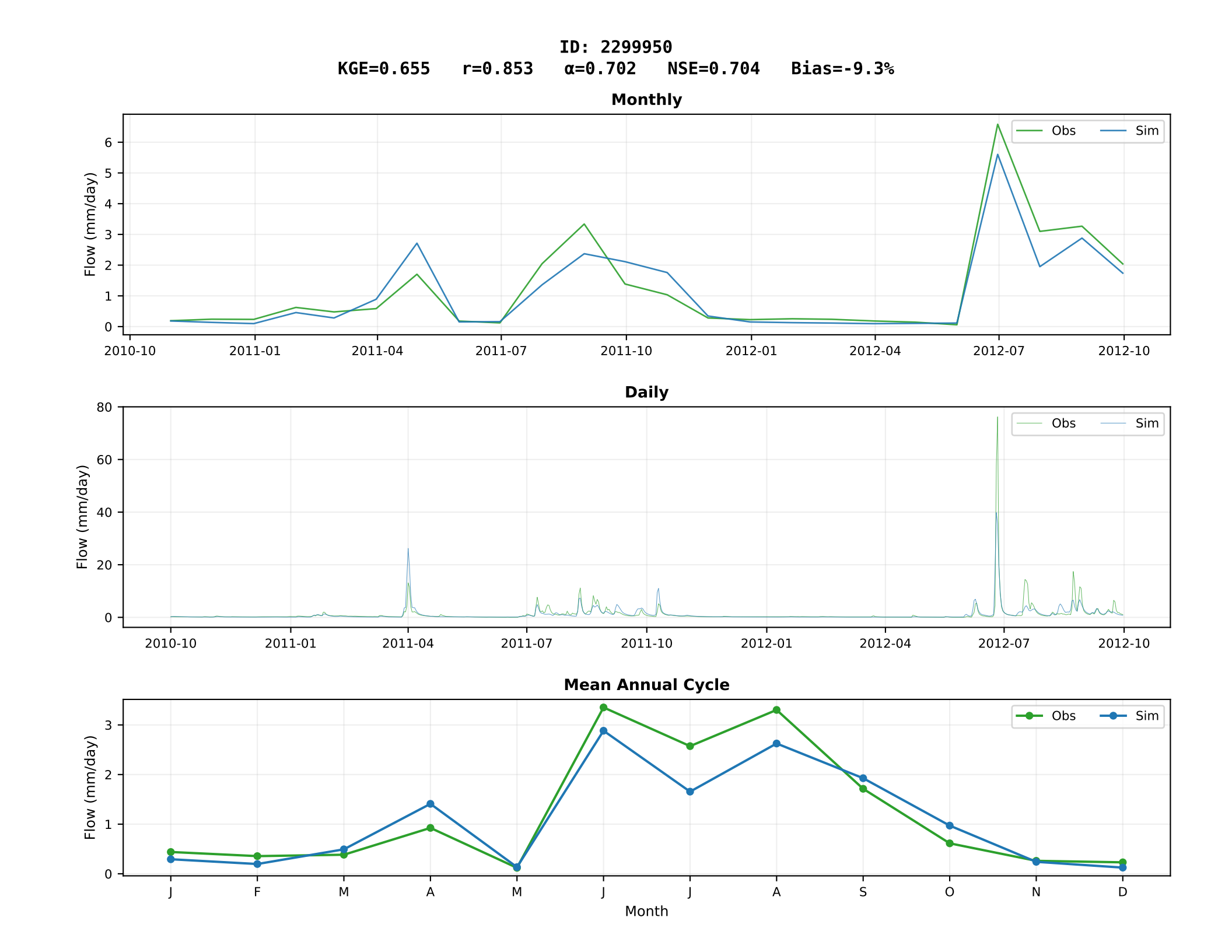

ID: 4043050
KGE=0.657 r=0.967 α=0.727 NSE=0.860 Bias=-20.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


HUGE CREEK NEAR WAUNA, WA (ID=12073500)
KGE=0.657 r=0.857 α=0.727 NSE=0.707 Bias=-15.2%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D


ID: 6906800
KGE=0.656 r=0.759 α=0.755 NSE=0.576 Bias=-0.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

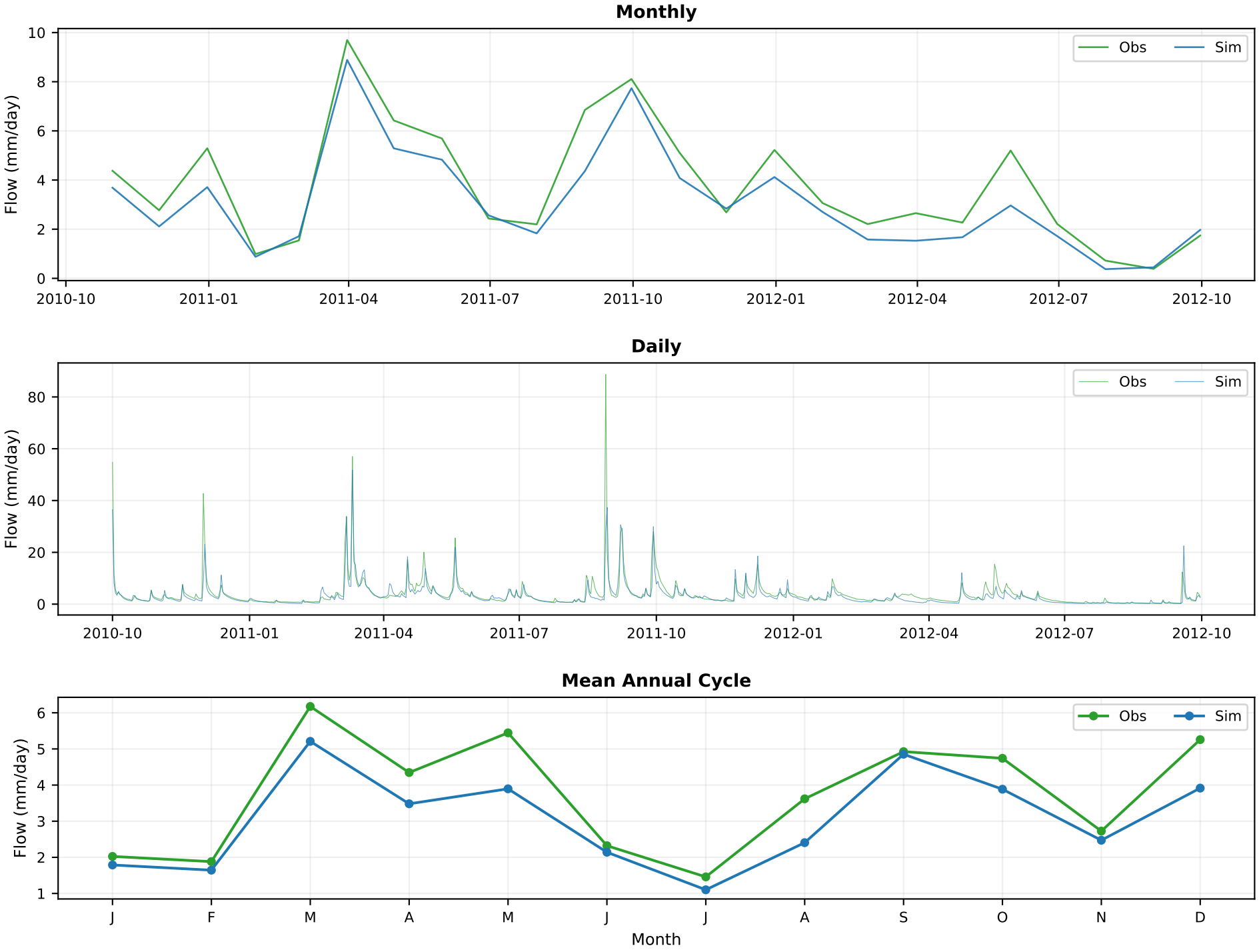

ID: 1350080
KGE=0.656 r=0.798 α=0.848 NSE=0.619 Bias=+23.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

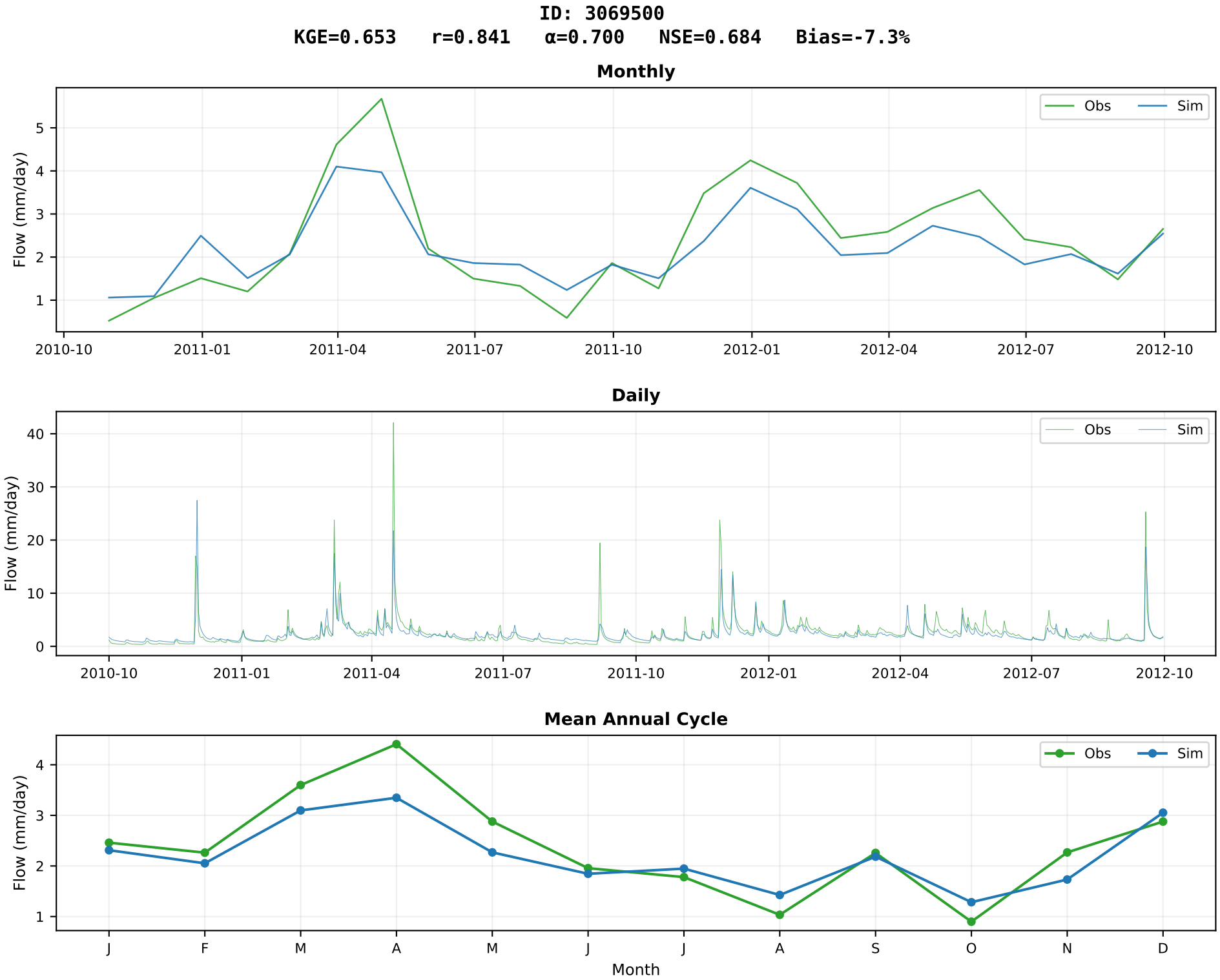

ID: 2472500
KGE=0.655 r=0.874 α=0.685 NSE=0.727 Bias=+6.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

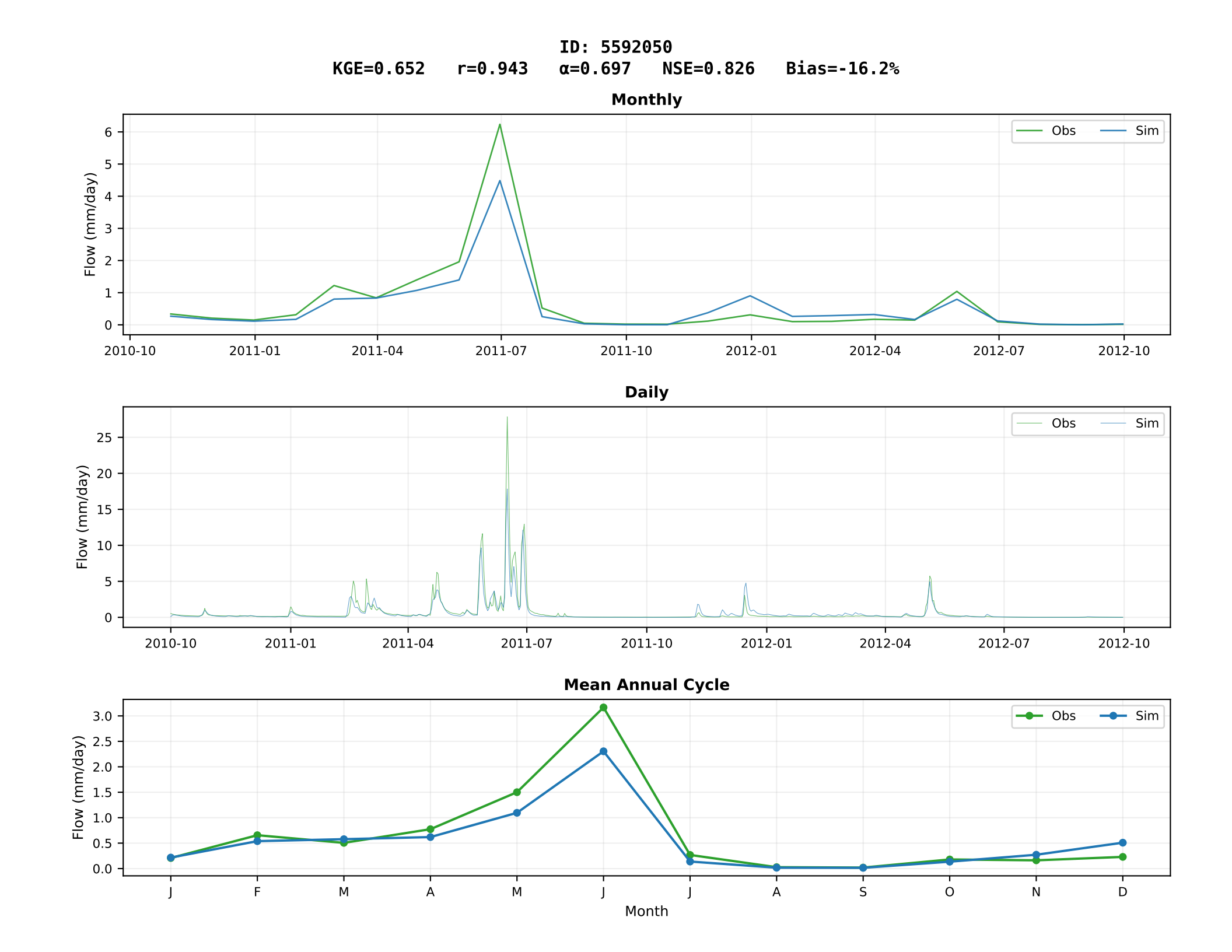

ID: 2299950
KGE=0.655 r=0.853 α=0.702 NSE=0.704 Bias=-9.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

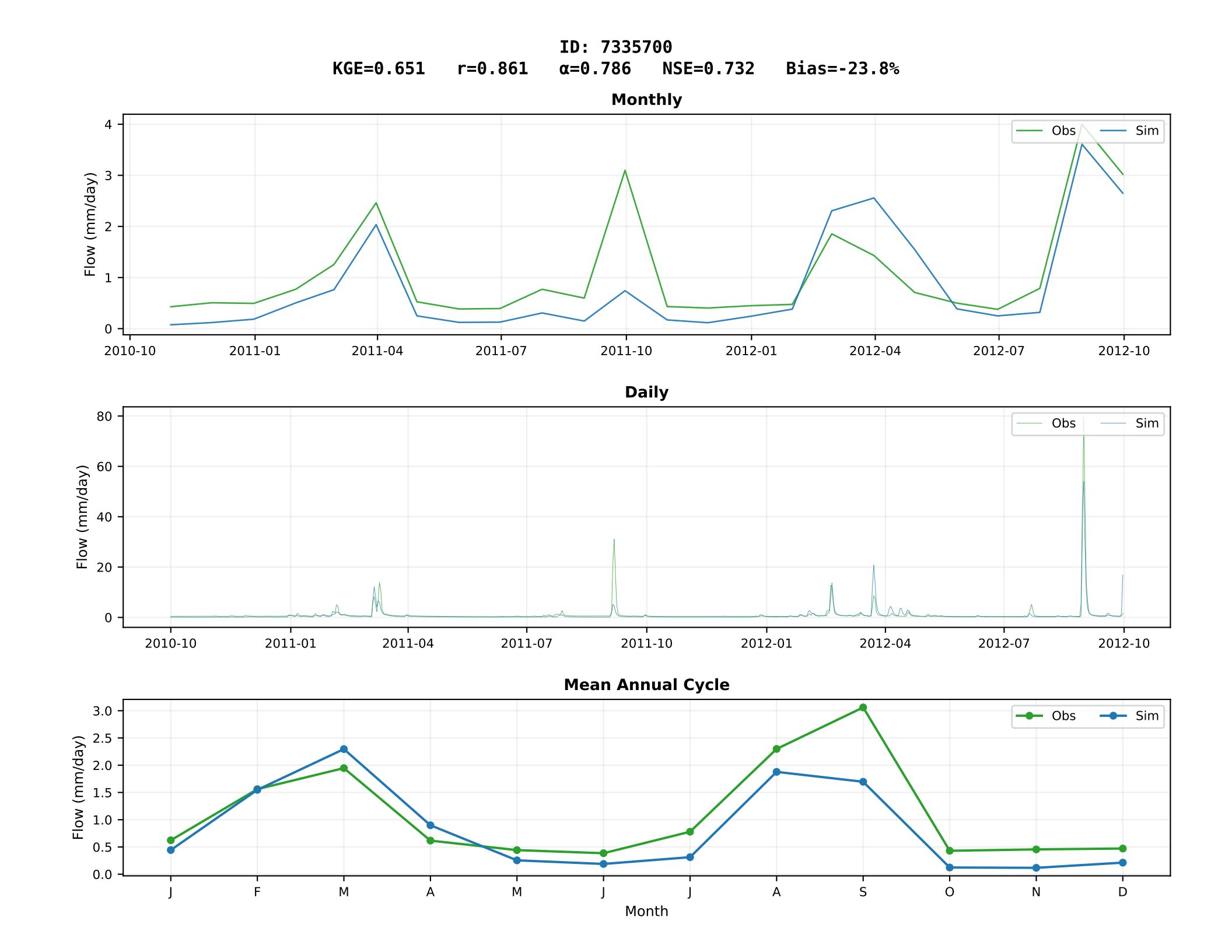
ID: 1350140

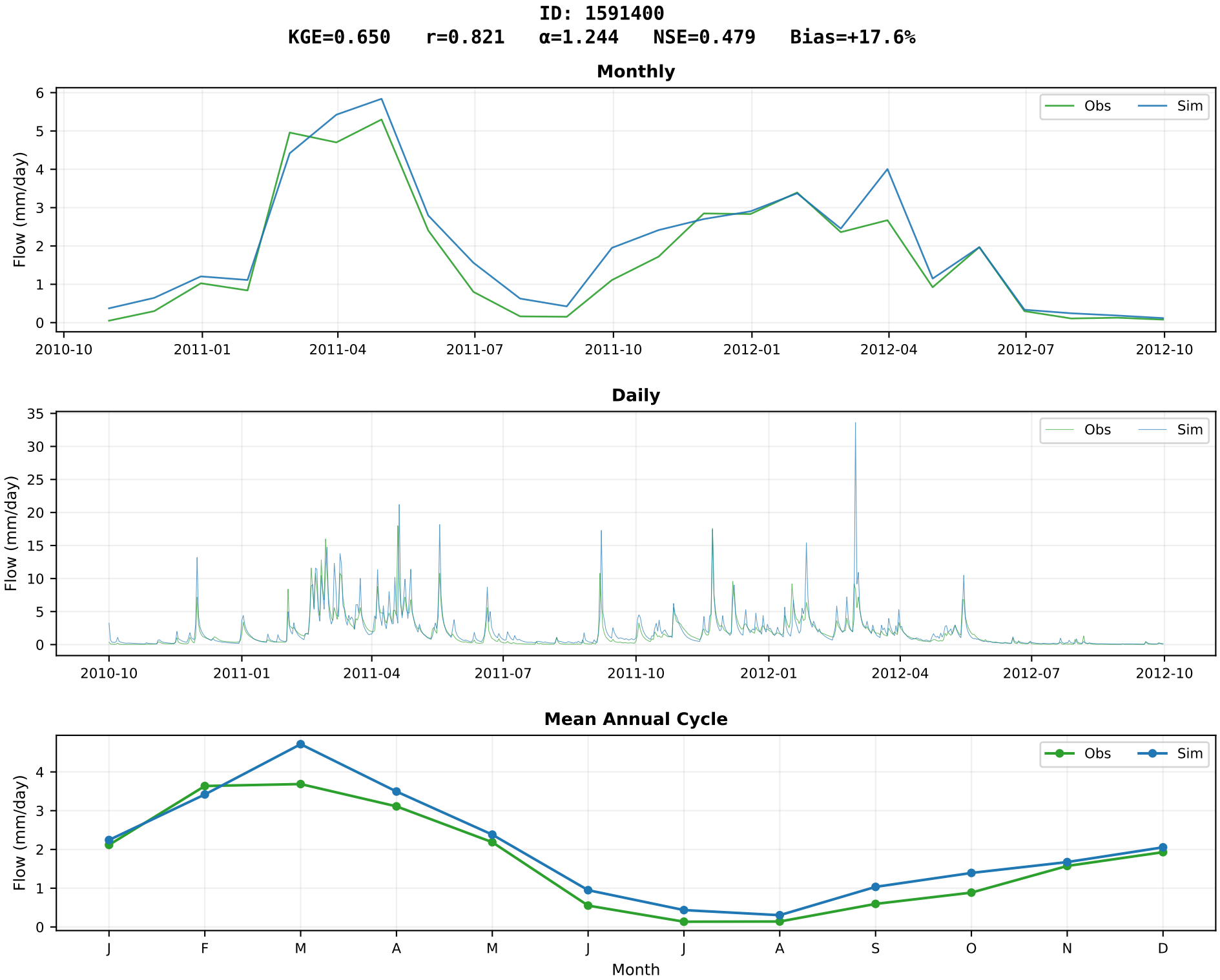
KGE=0.654 r=0.839 α=0.754 NSE=0.682 Bias=-18.2%

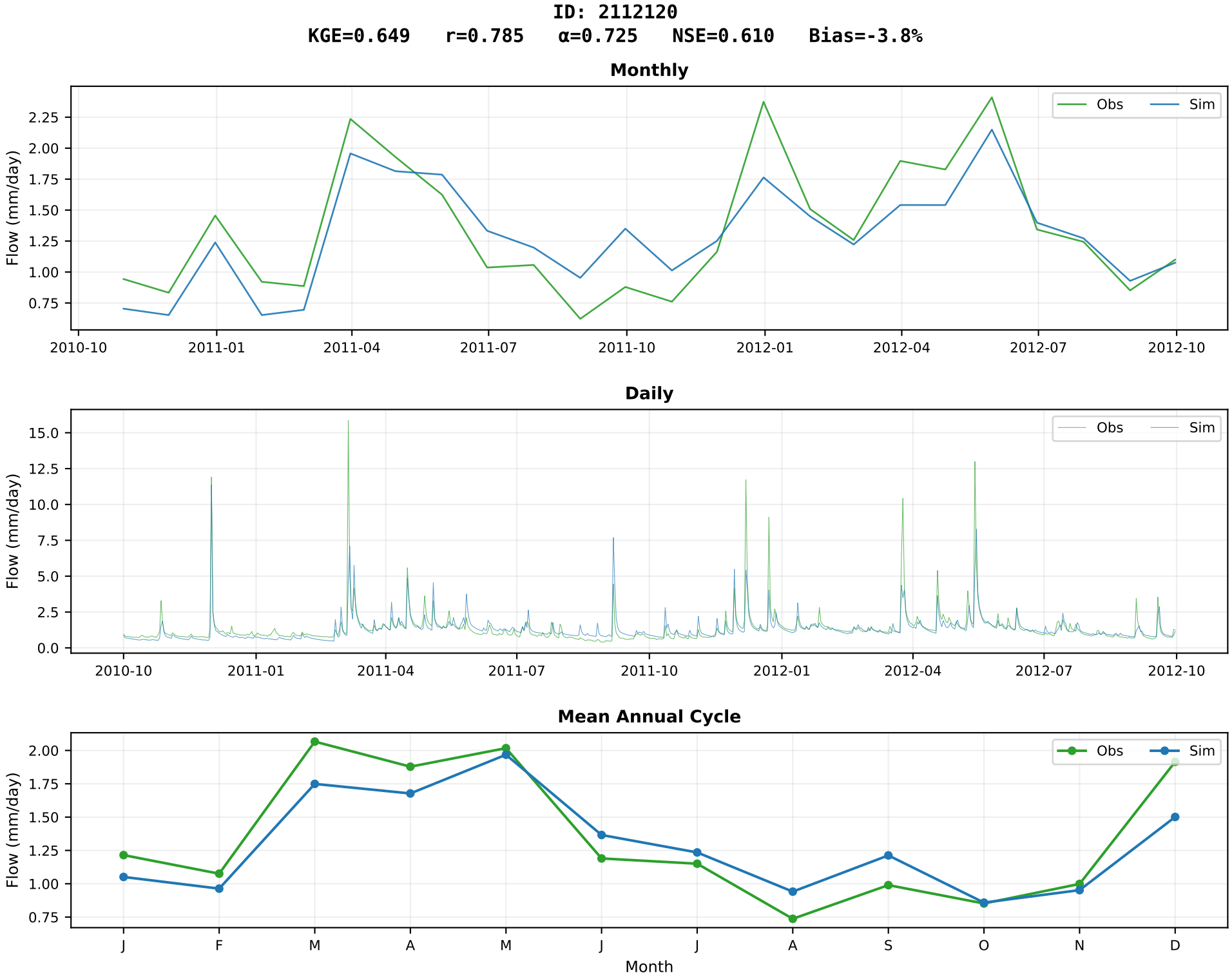


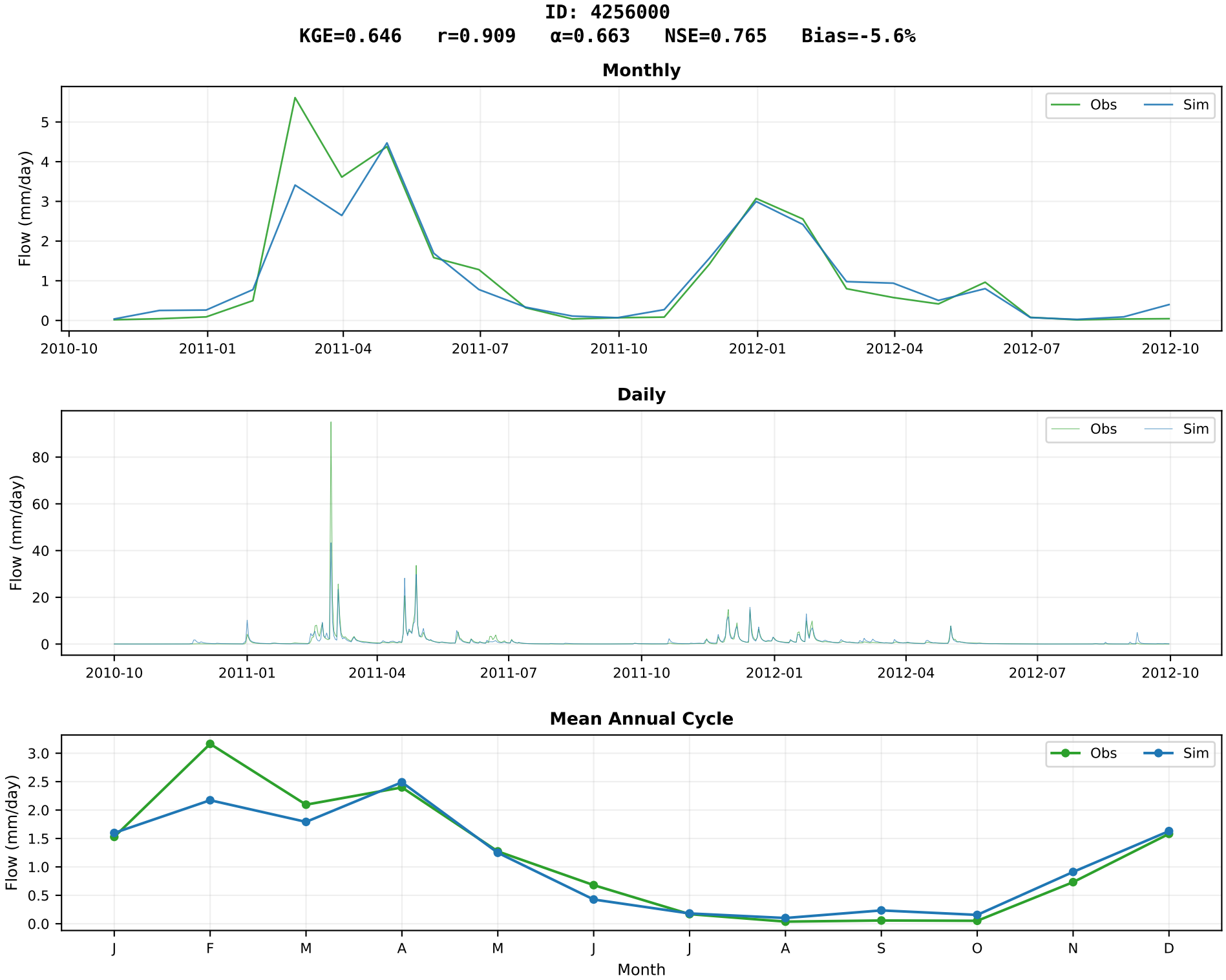

ID: 3069500
KGE=0.653 r=0.841 α=0.700 NSE=0.684 Bias=-7.3%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

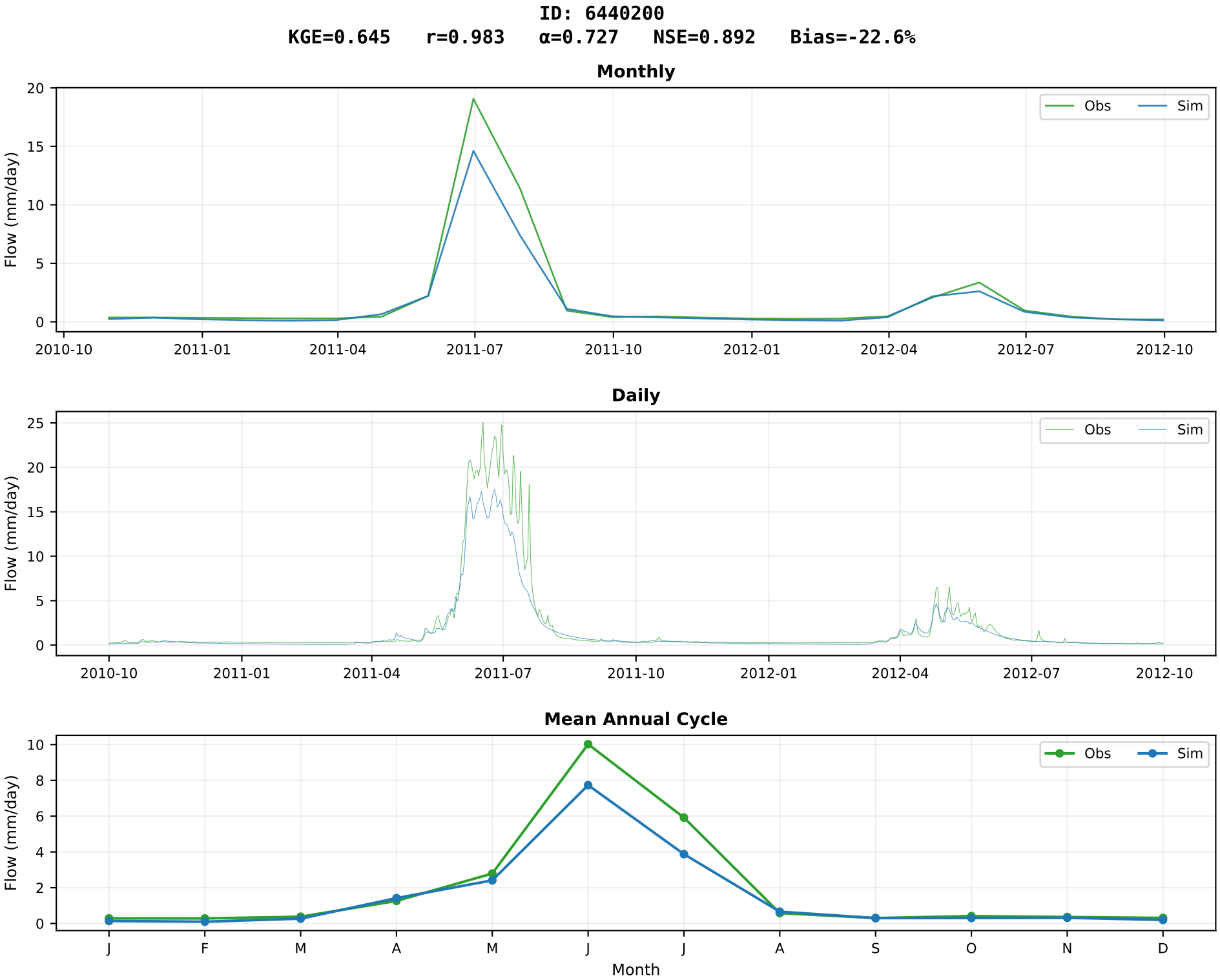

ID: 5592050
KGE=0.652 r=0.943 α=0.697 NSE=0.826 Bias=-16.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

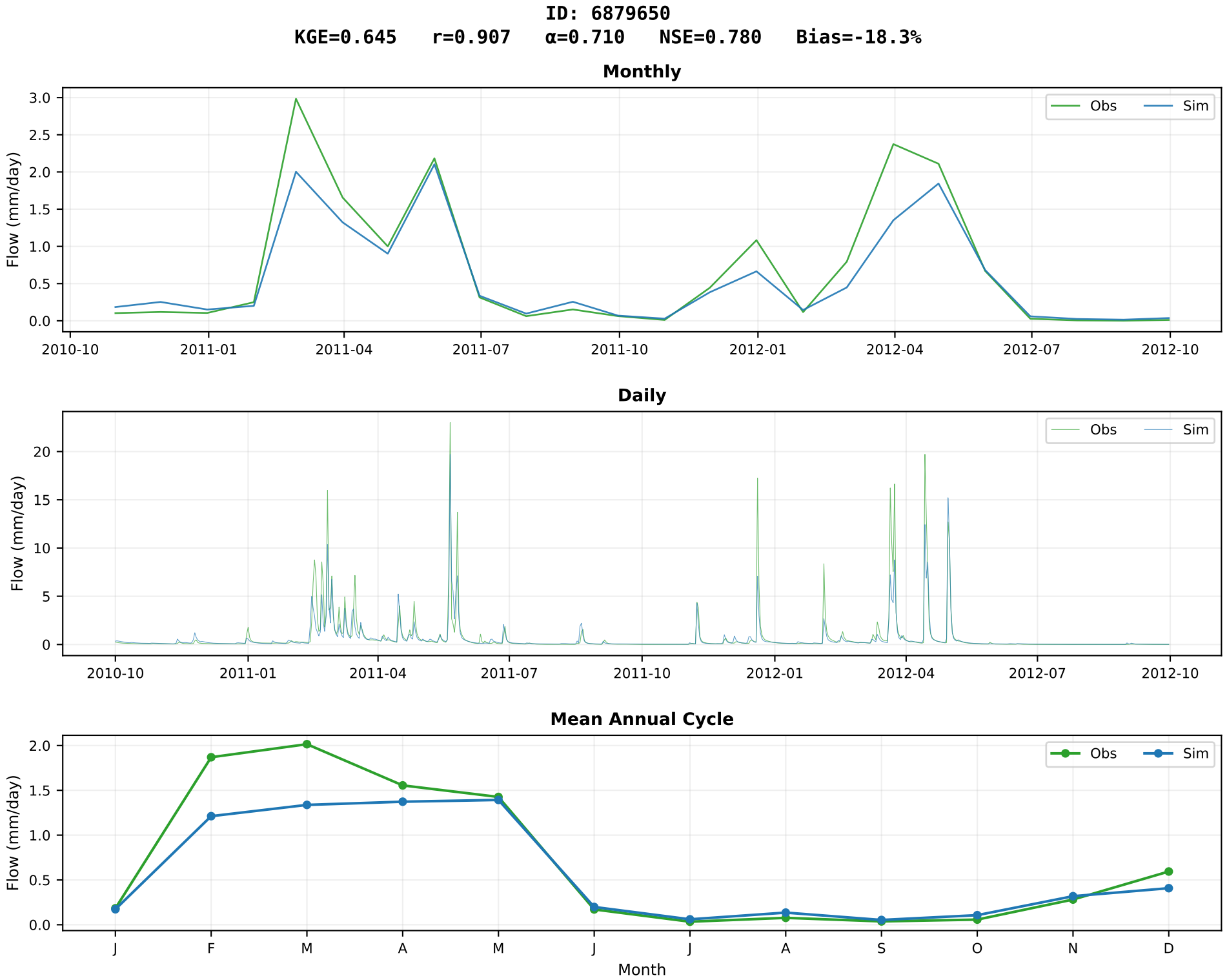

ID: 7335700
KGE=0.651 r=0.861 α=0.786 NSE=0.732 Bias=-23.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

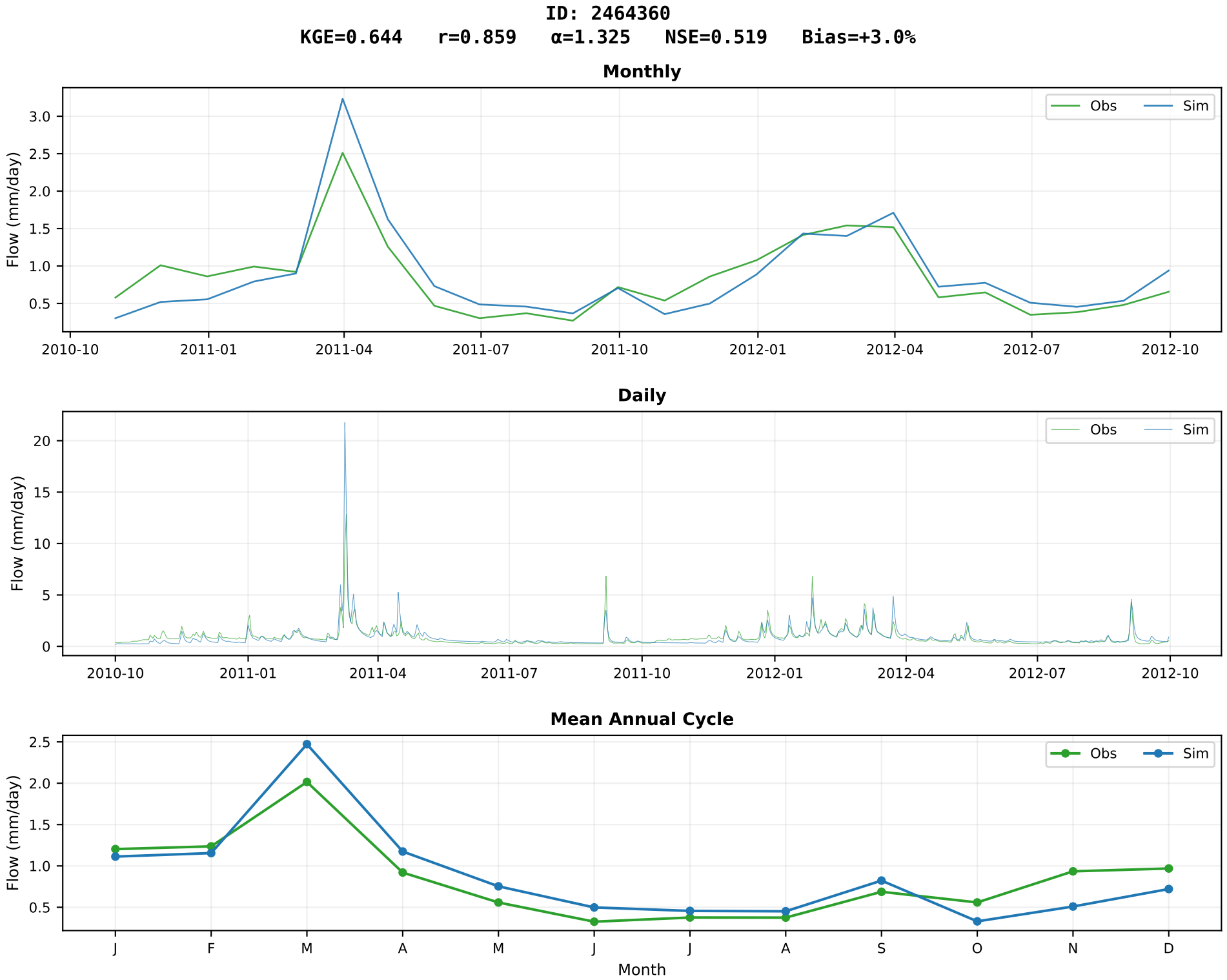

ID: 1591400
KGE=0.650 r=0.821 α=1.244 NSE=0.479 Bias=+17.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

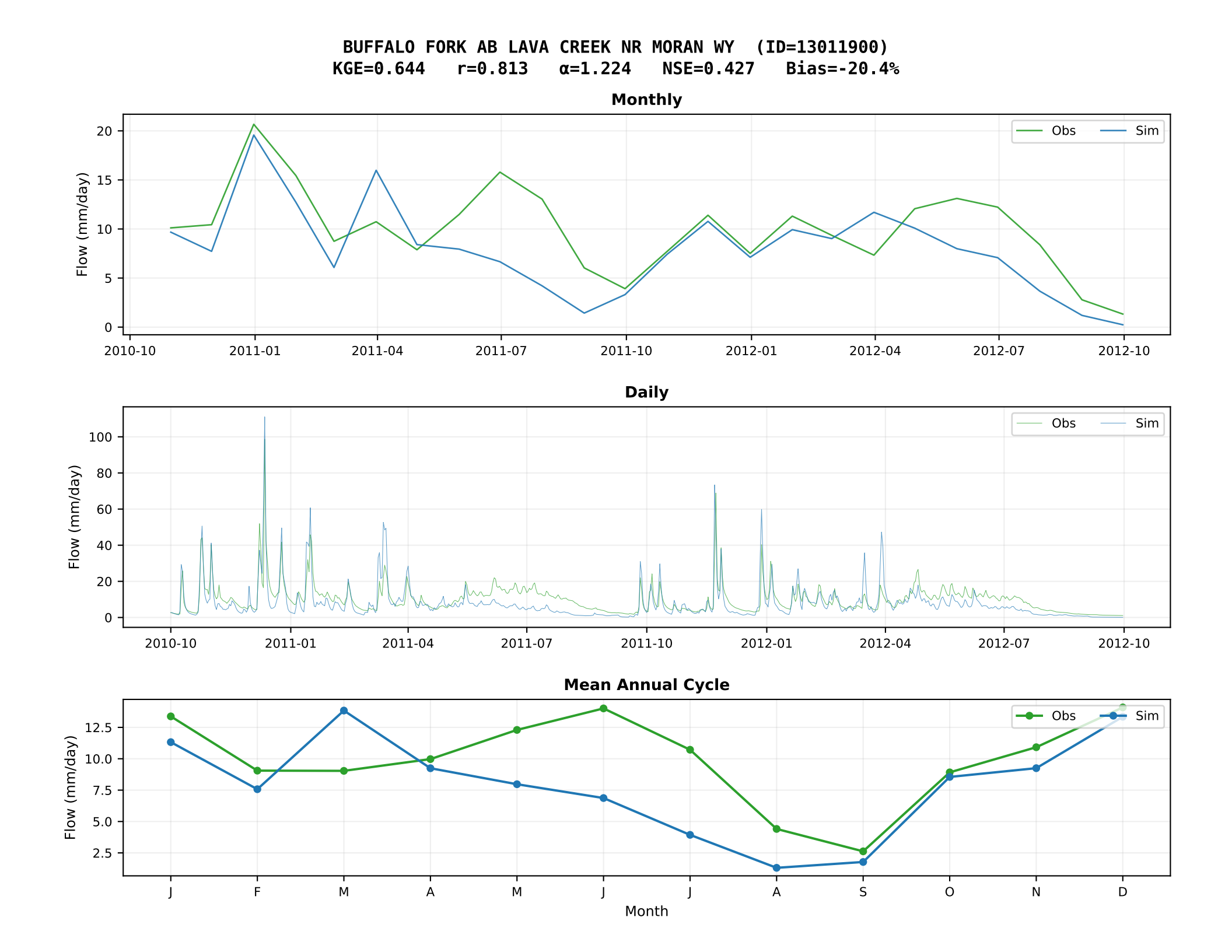

ID: 2112120
KGE=0.649 r=0.785 α=0.725 NSE=0.610 Bias=-3.8%
Monthly
Flow (mm/day)
Obs
Sim
2.25
2.00
1.75
1.50
1.25
1.00
0.75
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
2.00
1.75
1.50
1.25
1.00
0.75
J
F
M
A
M
J
J
A
S
O
N
D
Month

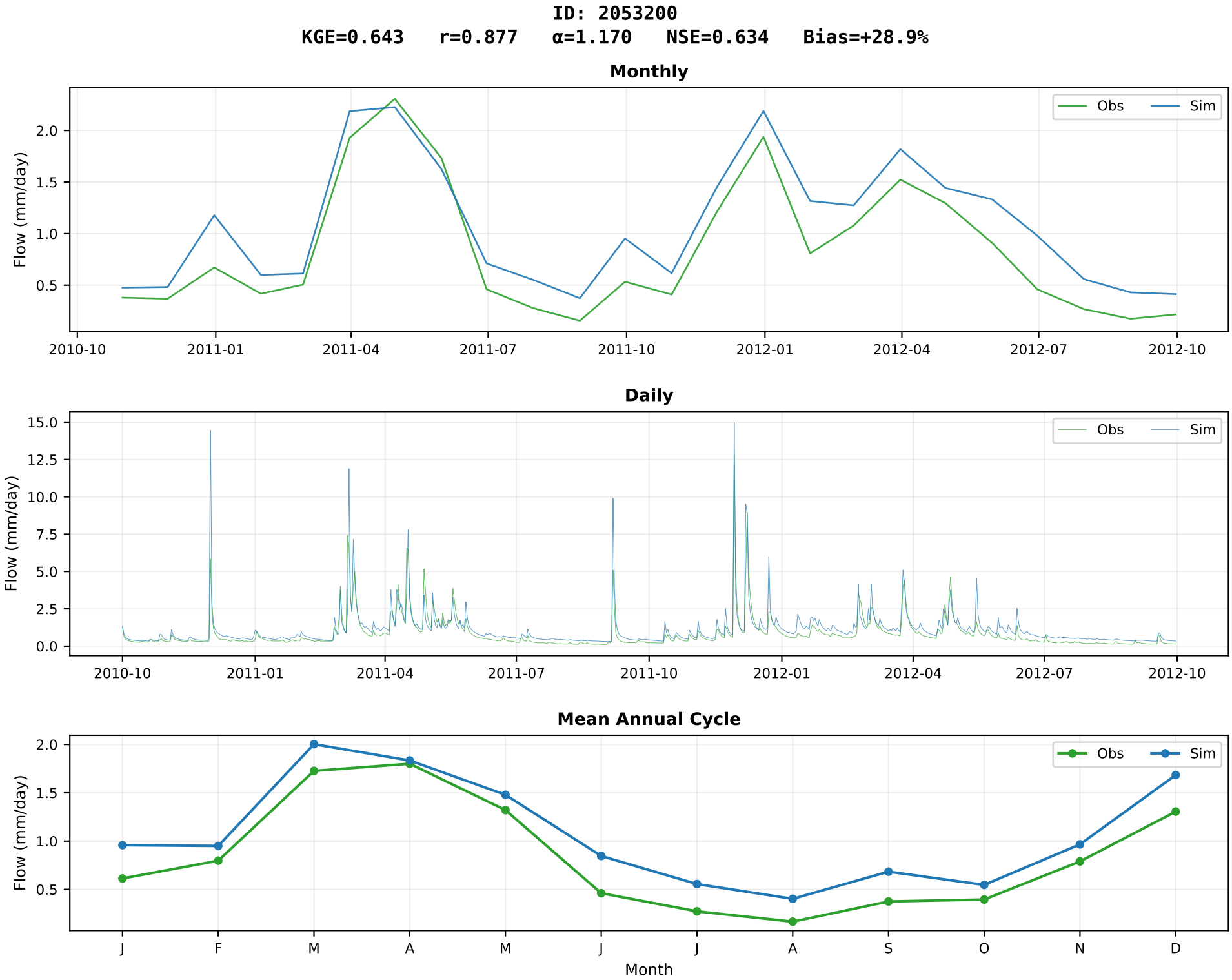

ID: 4256000
KGE=0.646 r=0.909 α=0.663 NSE=0.765 Bias=-5.6%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

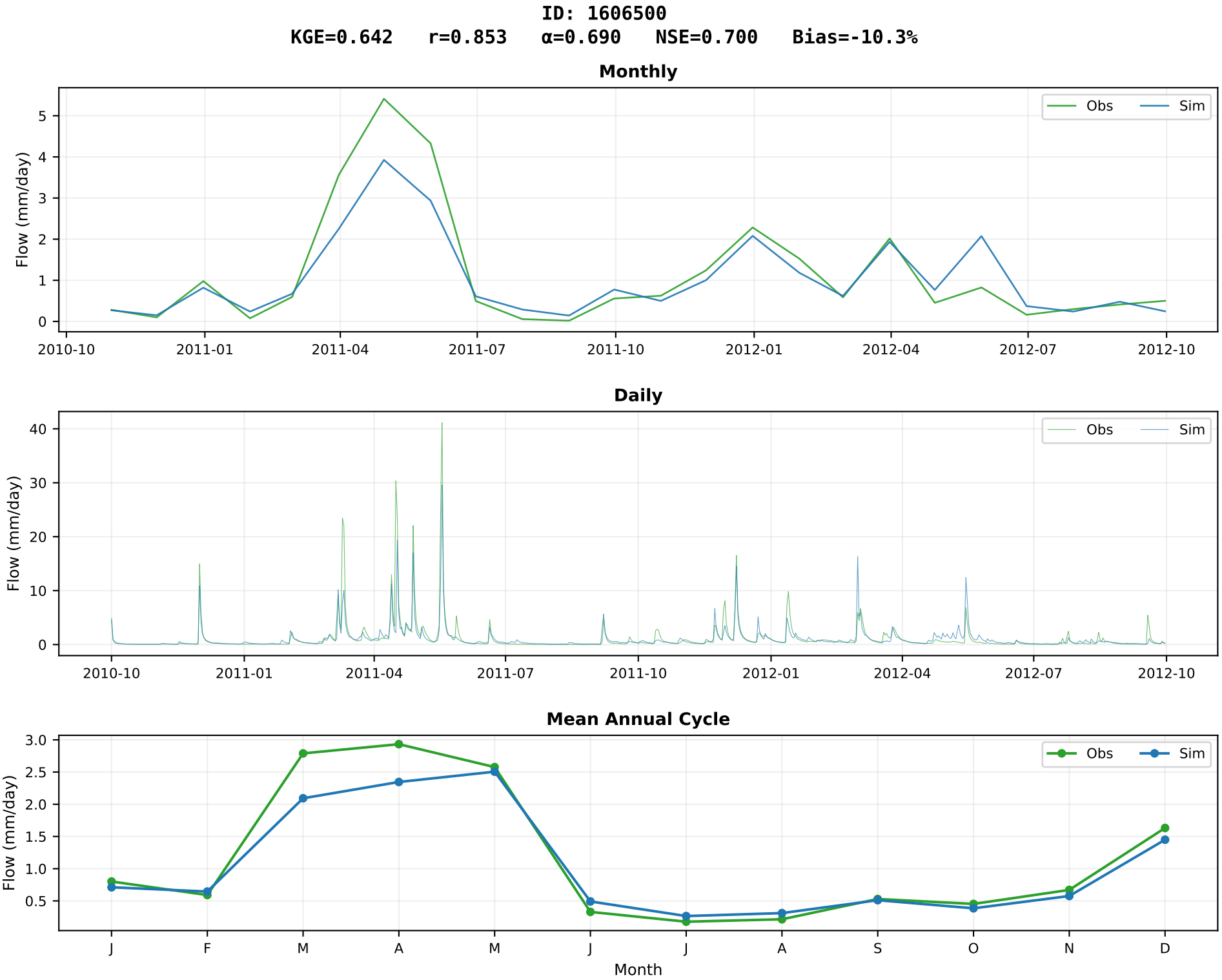

ID: 6440200
KGE=0.645 r=0.983 α=0.727 NSE=0.892 Bias=-22.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

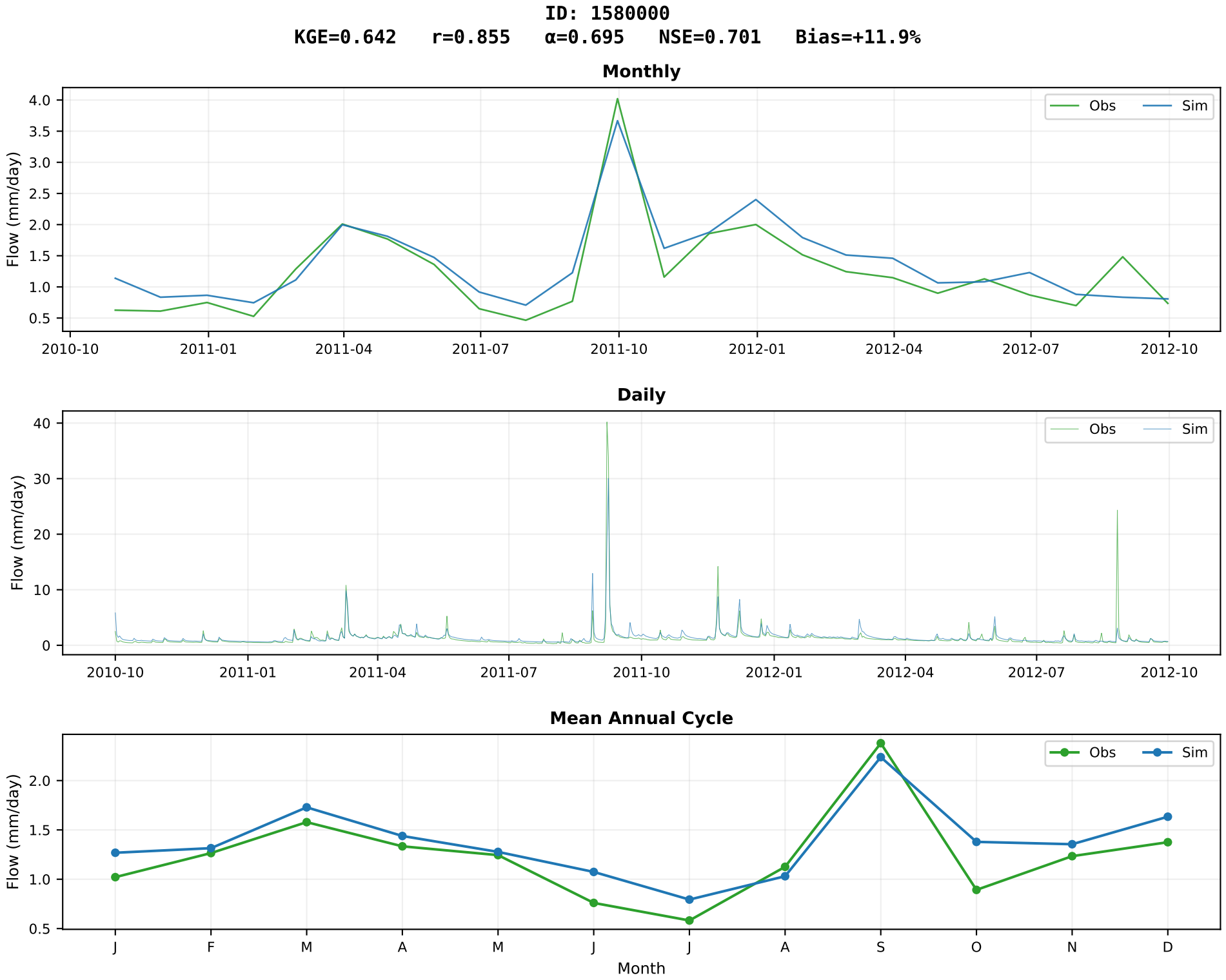

ID: 6879650
KGE=0.645 r=0.907 α=0.710 NSE=0.780 Bias=-18.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

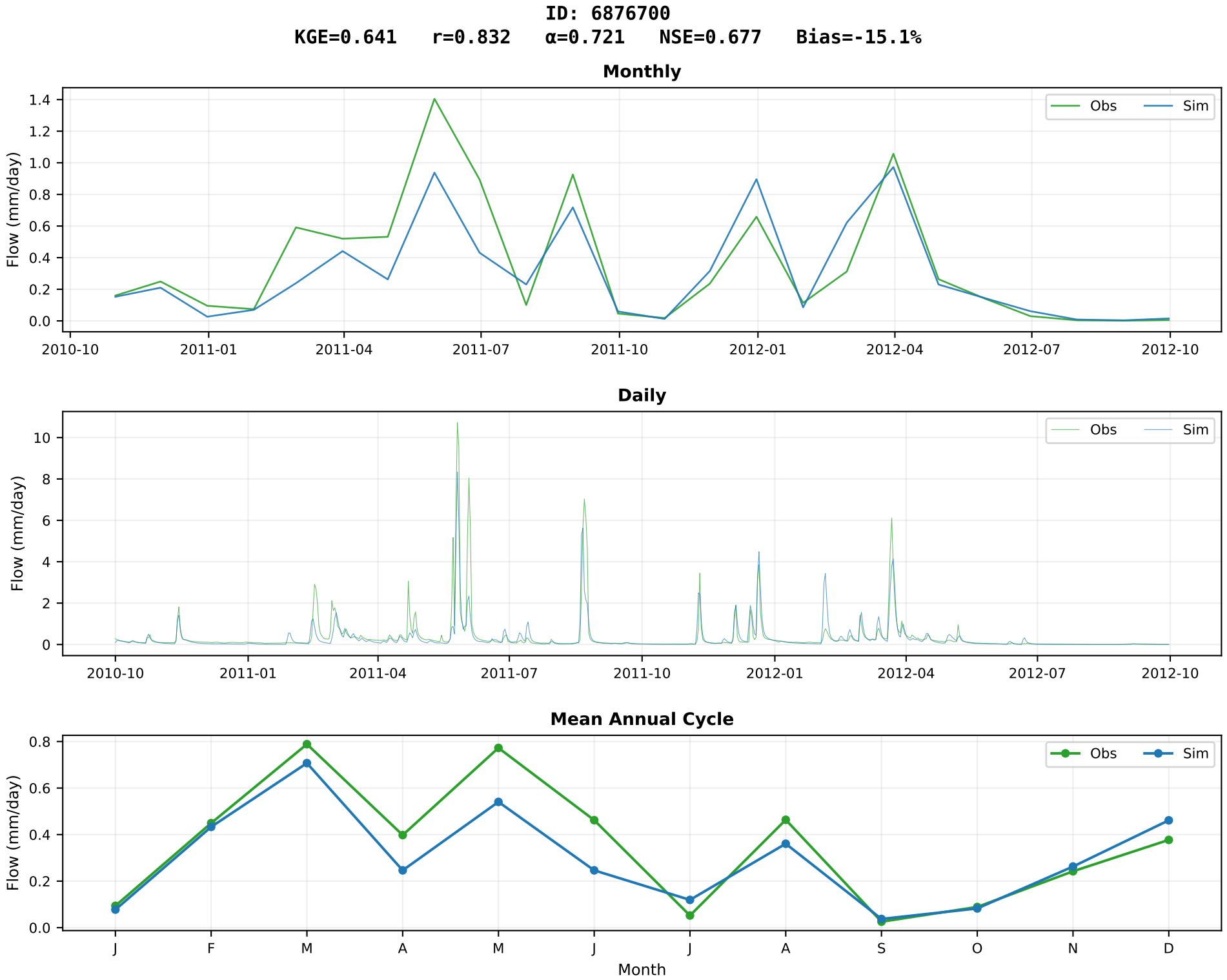

ID: 2464360
KGE=0.644 r=0.859 α=1.325 NSE=0.519 Bias=+3.0%
Monthly
Obs
Sim
Flow (mm/day)
3.0
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

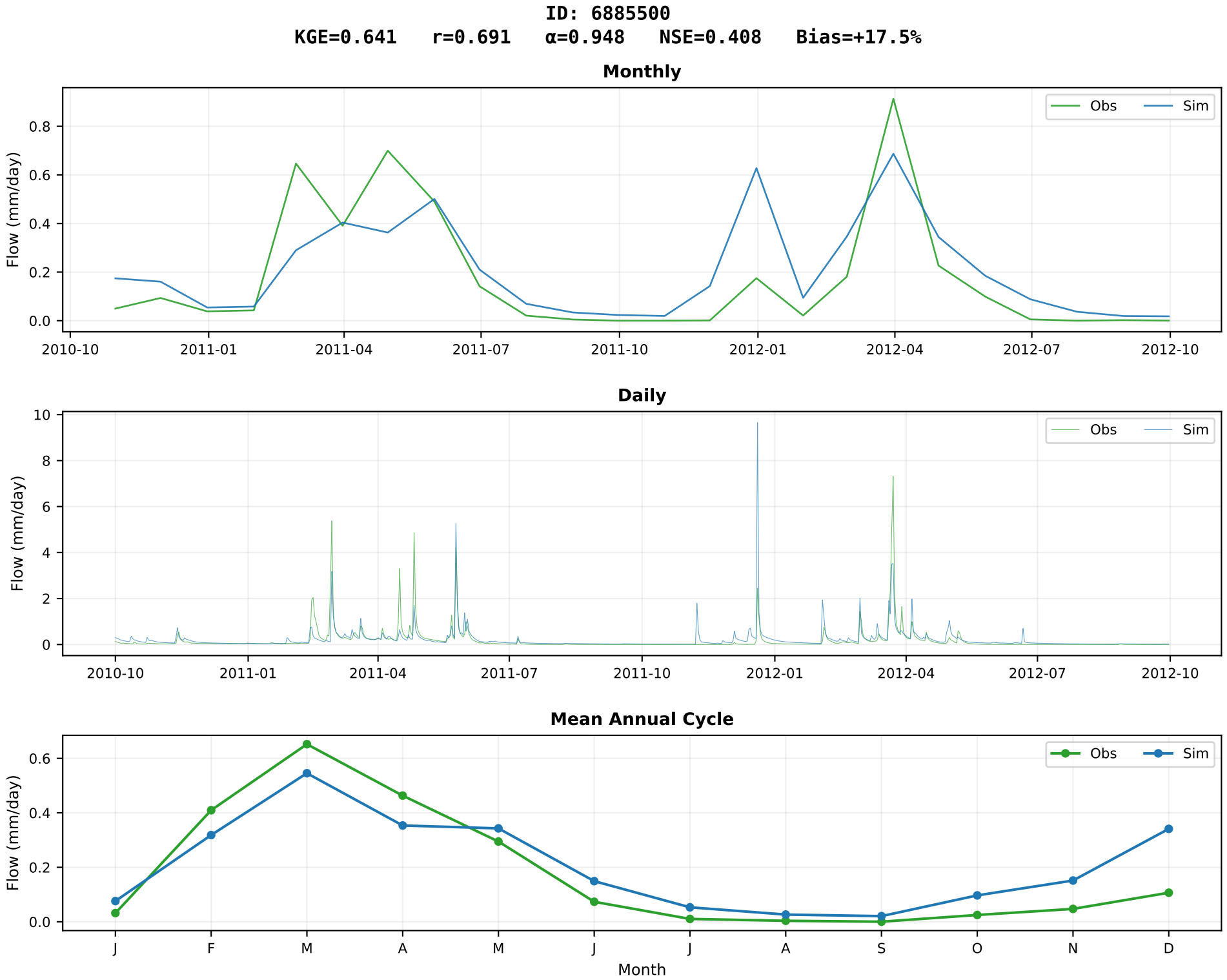

BUFFALO FORK AB LAVA CREEK NR MORAN WY (ID=13011900)
KGE=0.644 r=0.813 α=1.224 NSE=0.427 Bias=-20.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

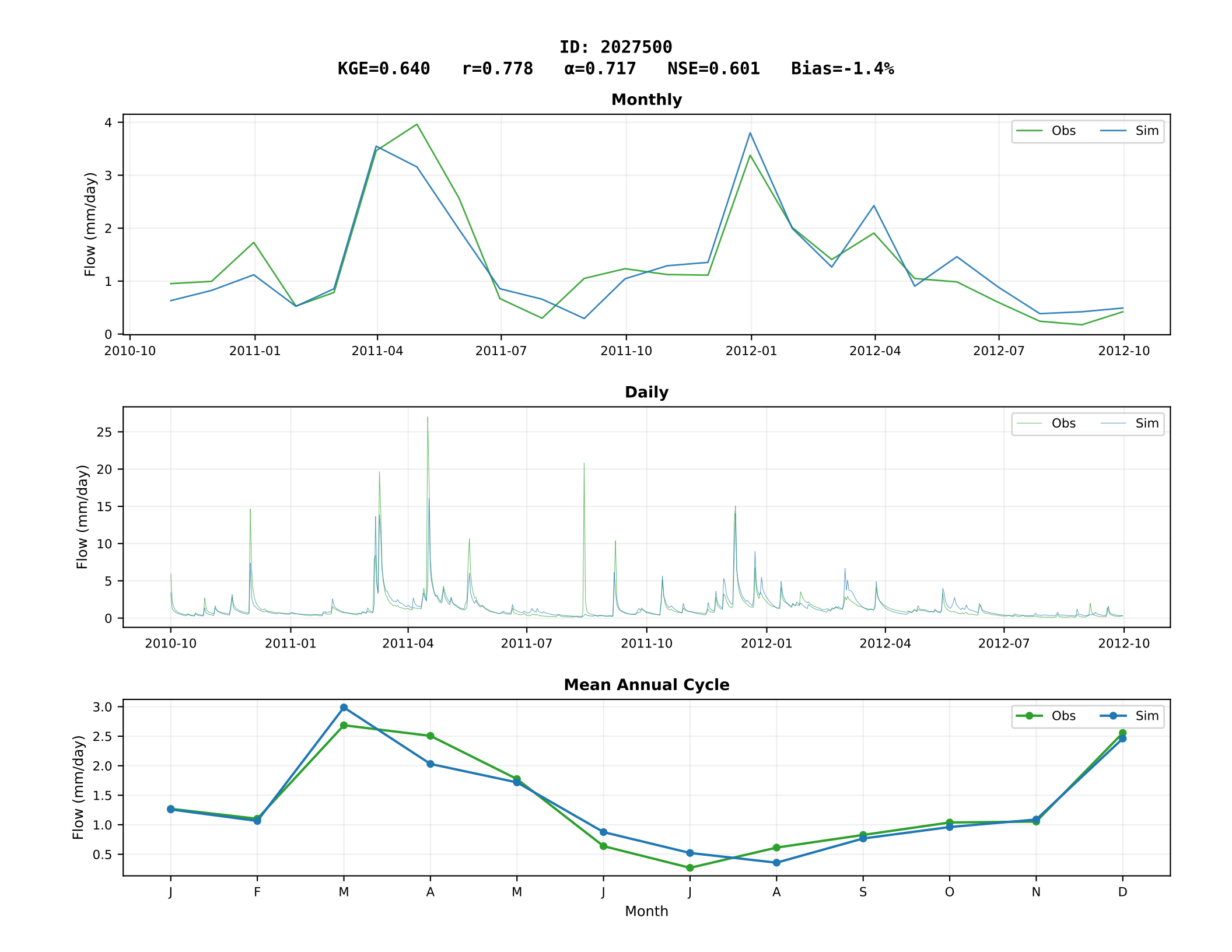

ID: 2053200
KGE=0.643 r=0.877 α=1.170 NSE=0.634 Bias=+28.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

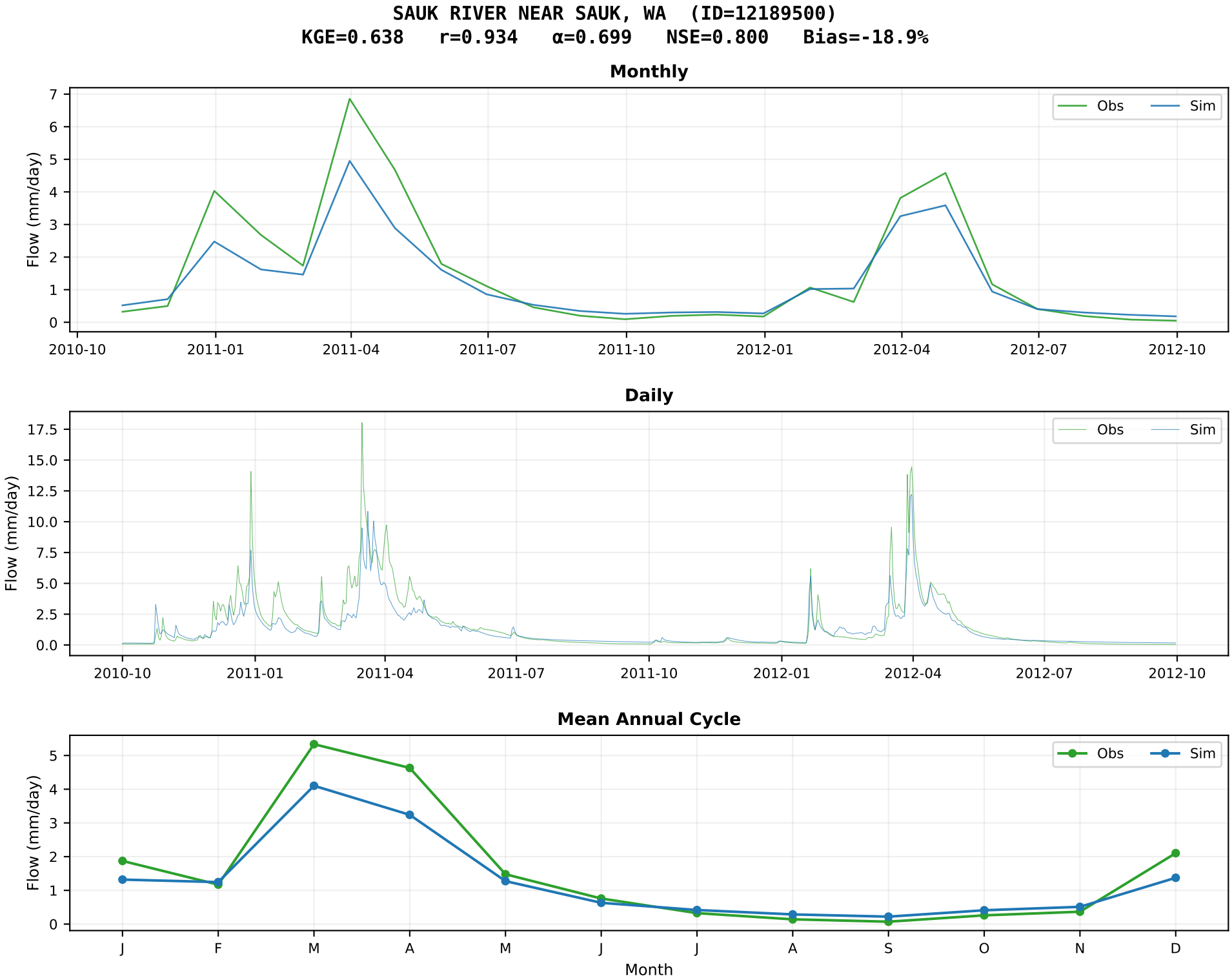

ID: 1606500
KGE=0.642 r=0.853 α=0.690 NSE=0.700 Bias=-10.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

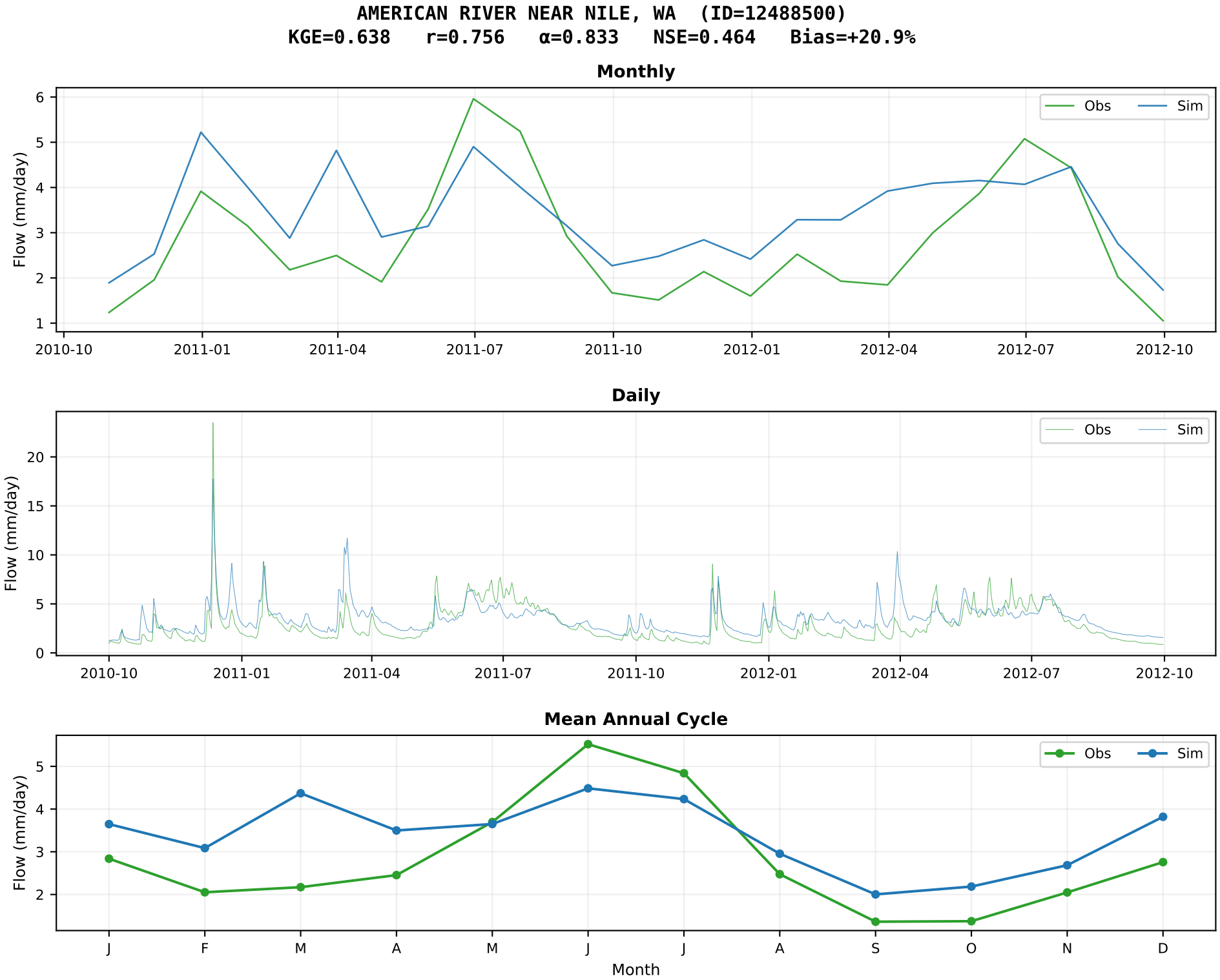

ID: 1580000
KGE=0.642 r=0.855 α=0.695 NSE=0.701 Bias=+11.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

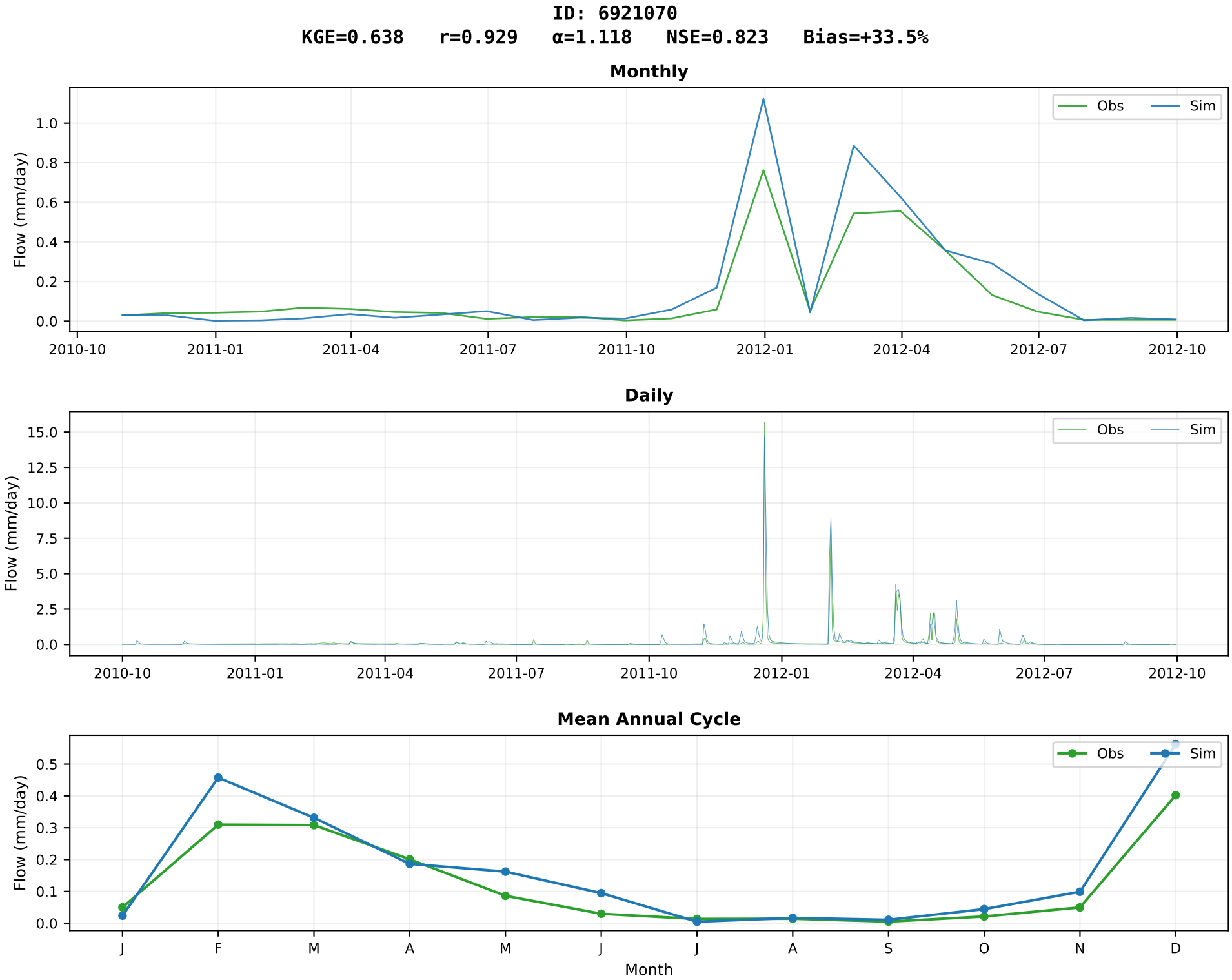

ID: 6876700
KGE=0.641 r=0.832 α=0.721 NSE=0.677 Bias=-15.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D


ID: 6885500
KGE=0.641 r=0.691 α=0.948 NSE=0.408 Bias=+17.5%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2027500
KGE=0.640 r=0.778 α=0.717 NSE=0.601 Bias=-1.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

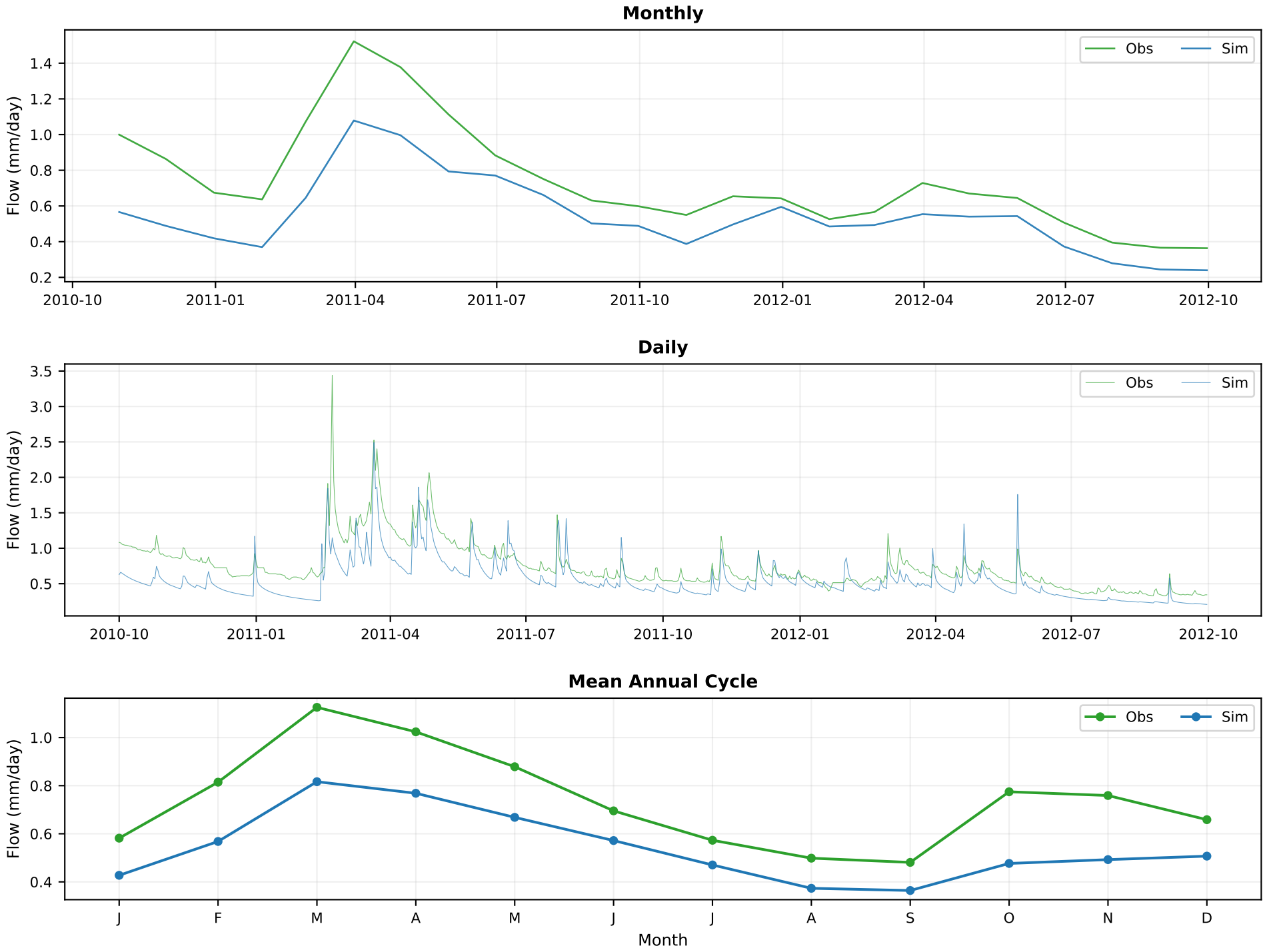

SAUK RIVER NEAR SAUK, WA (ID=12189500)
KGE=0.638 r=0.934 α=0.699 NSE=0.800 Bias=-18.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

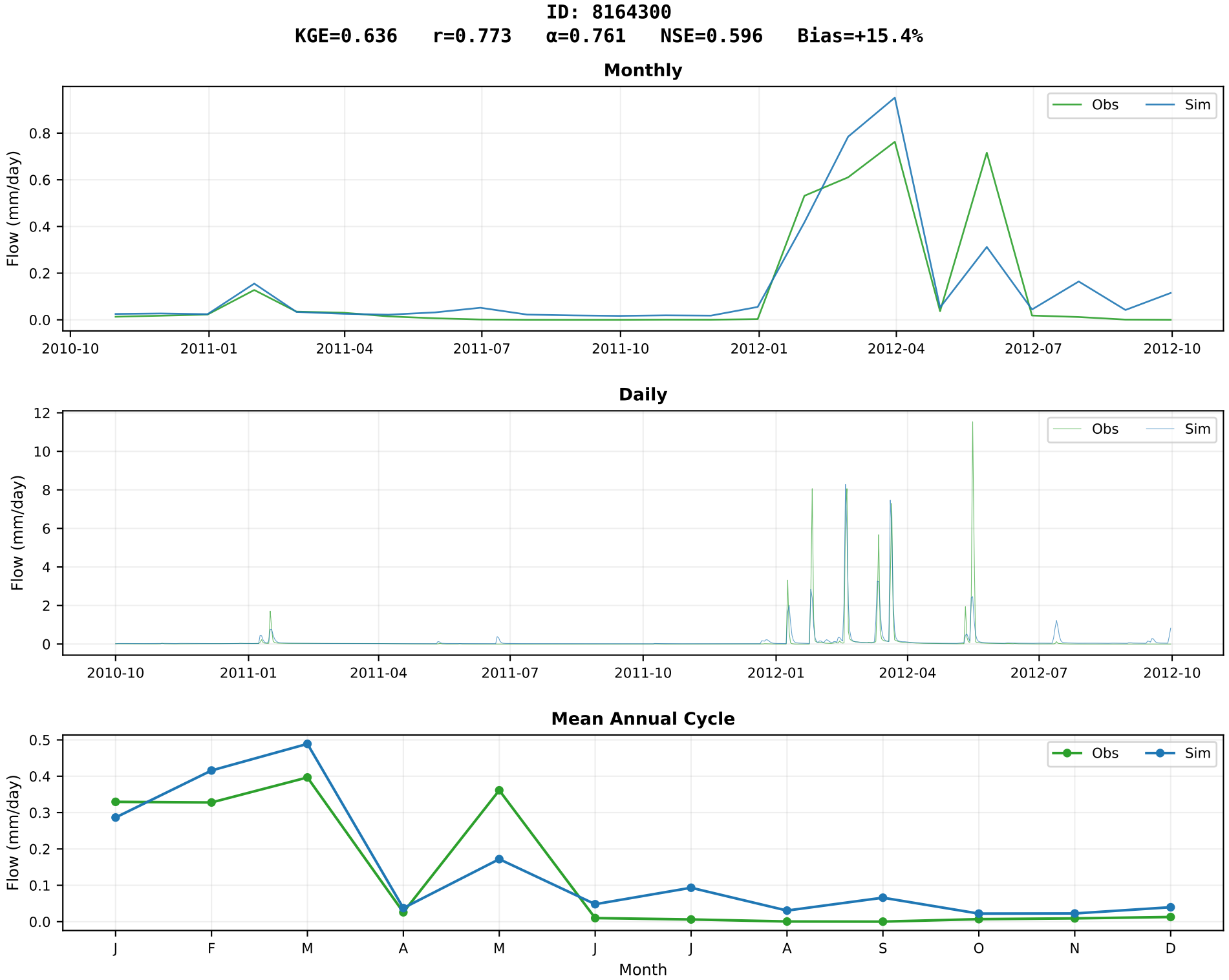

AMERICAN RIVER NEAR NILE, WA (ID=12488500)
KGE=0.638 r=0.756 α=0.833 NSE=0.464 Bias=+20.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

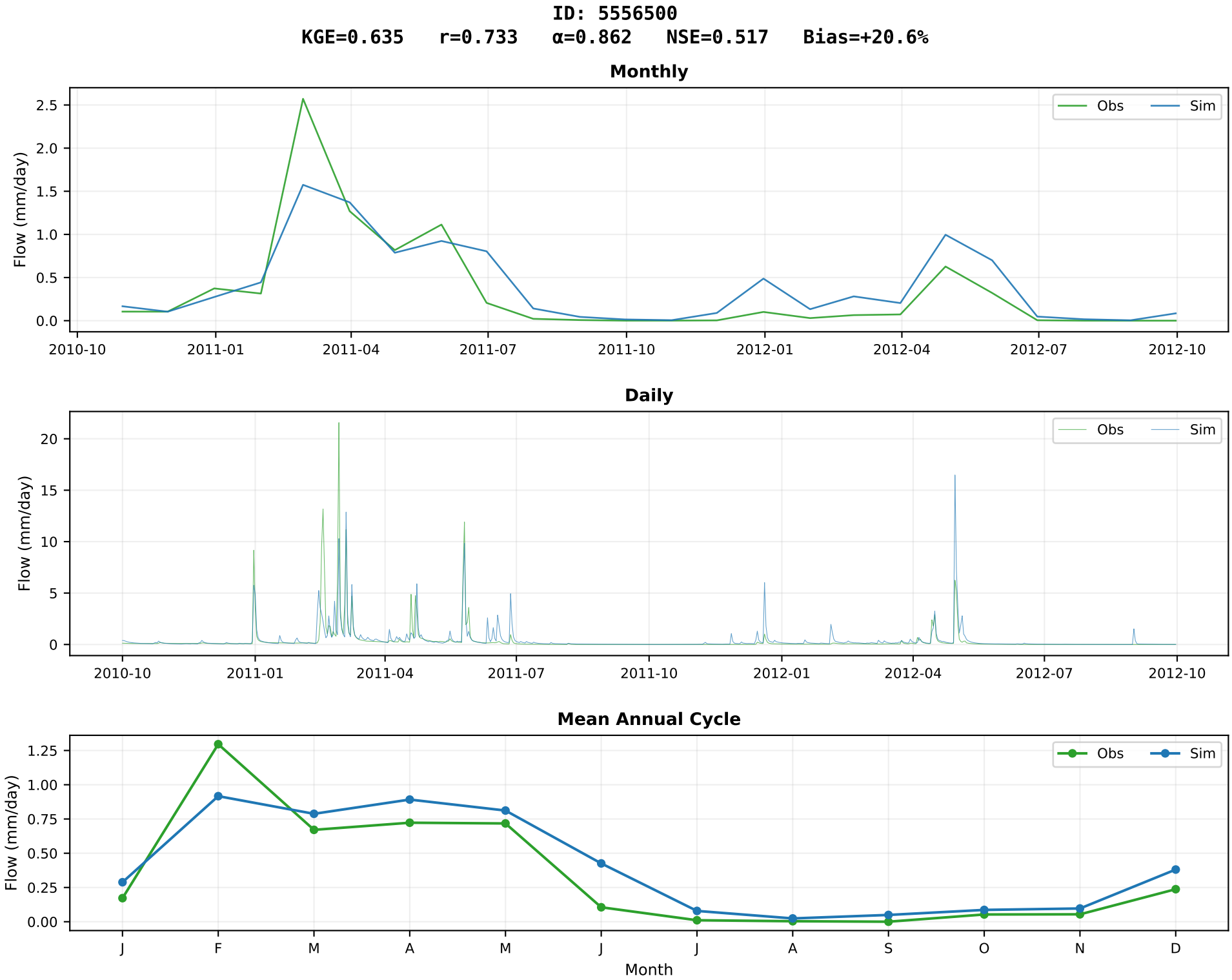

ID: 6921070
KGE=0.638 r=0.929 α=1.118 NSE=0.823 Bias=+33.5%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
0.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

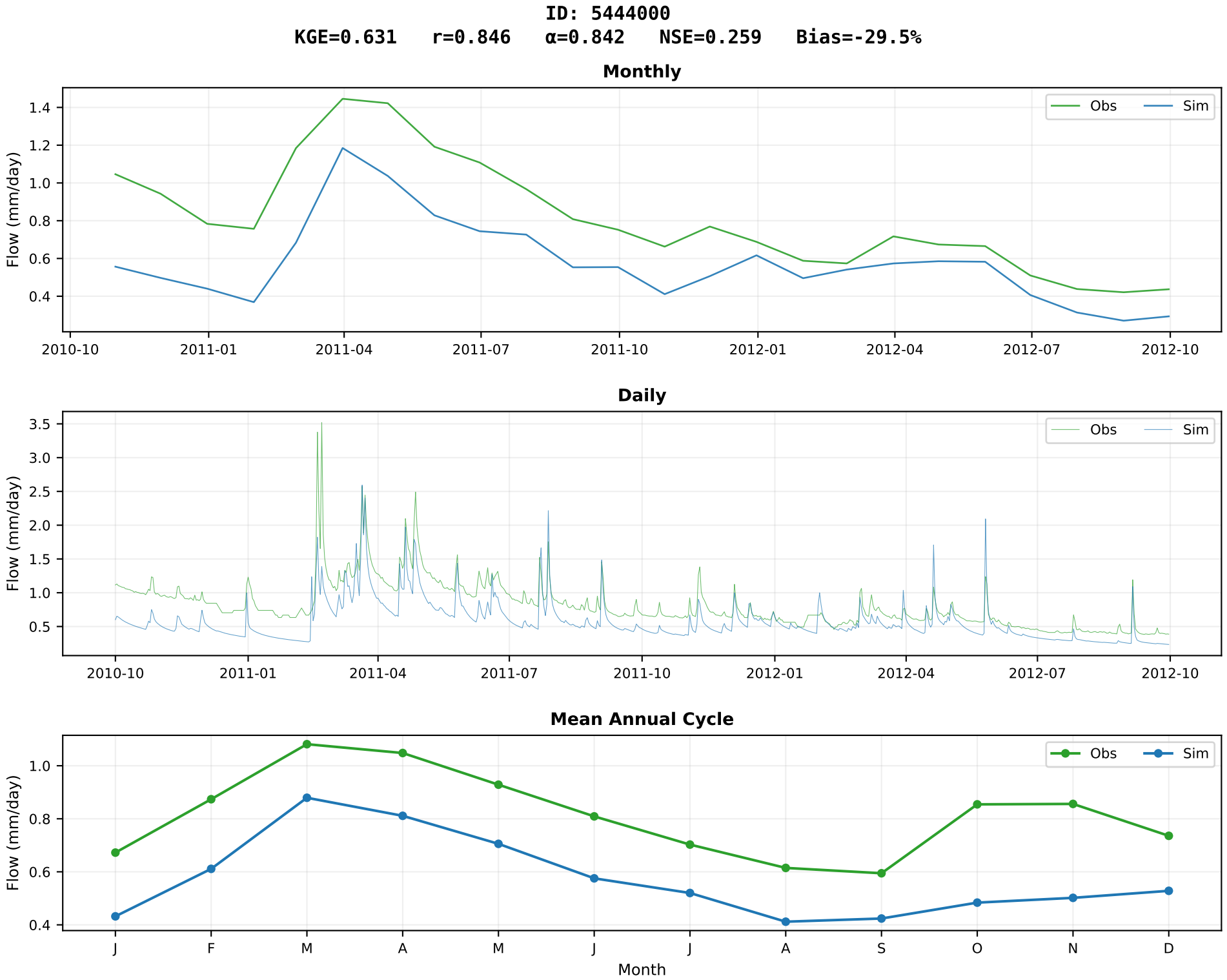
ID: 5454000

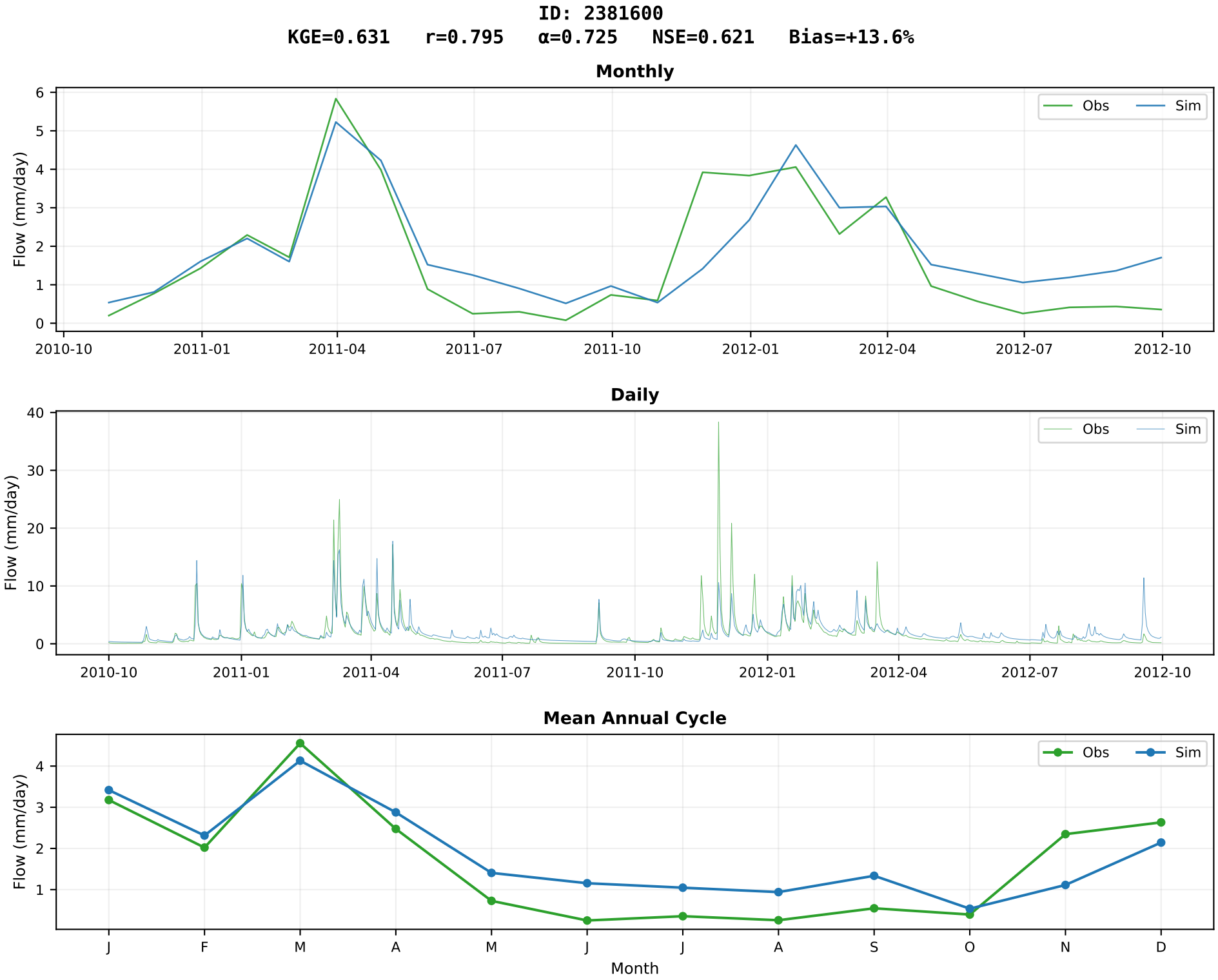
KGE=0.637 r=0.845 α=0.808 NSE=0.377 Bias=-26.6%




ID: 8164300
KGE=0.636 r=0.773 α=0.761 NSE=0.596 Bias=+15.4%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
12
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
0.5
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 5556500
KGE=0.635 r=0.733 α=0.862 NSE=0.517 Bias=+20.6%
Monthly
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.25
1.00
0.75
0.50
0.25
0.00
J
F
M
A
M
J
J
A
S
O
N
D
Month

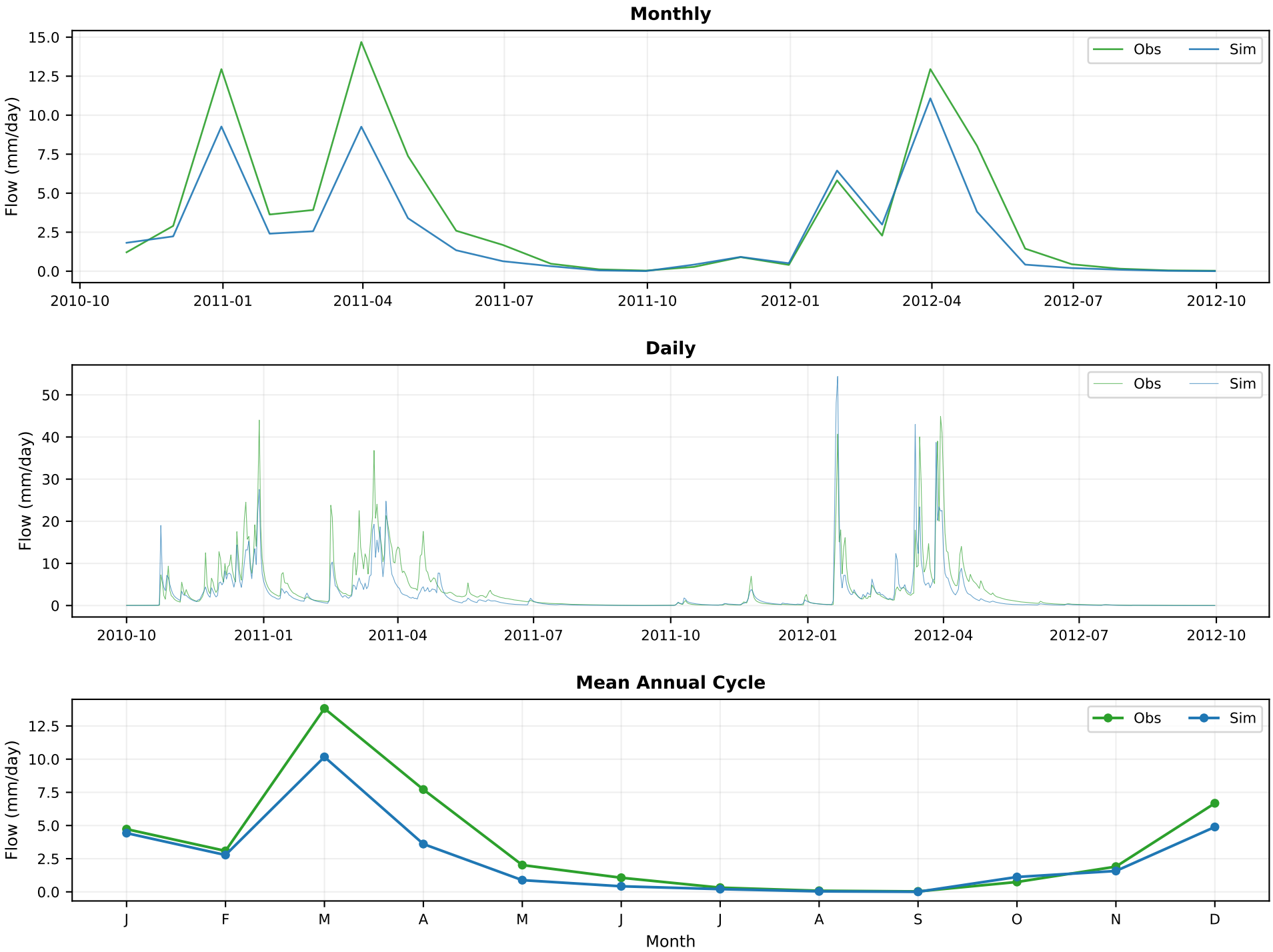

ID: 5444000
KGE=0.631 r=0.846 α=0.842 NSE=0.259 Bias=-29.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

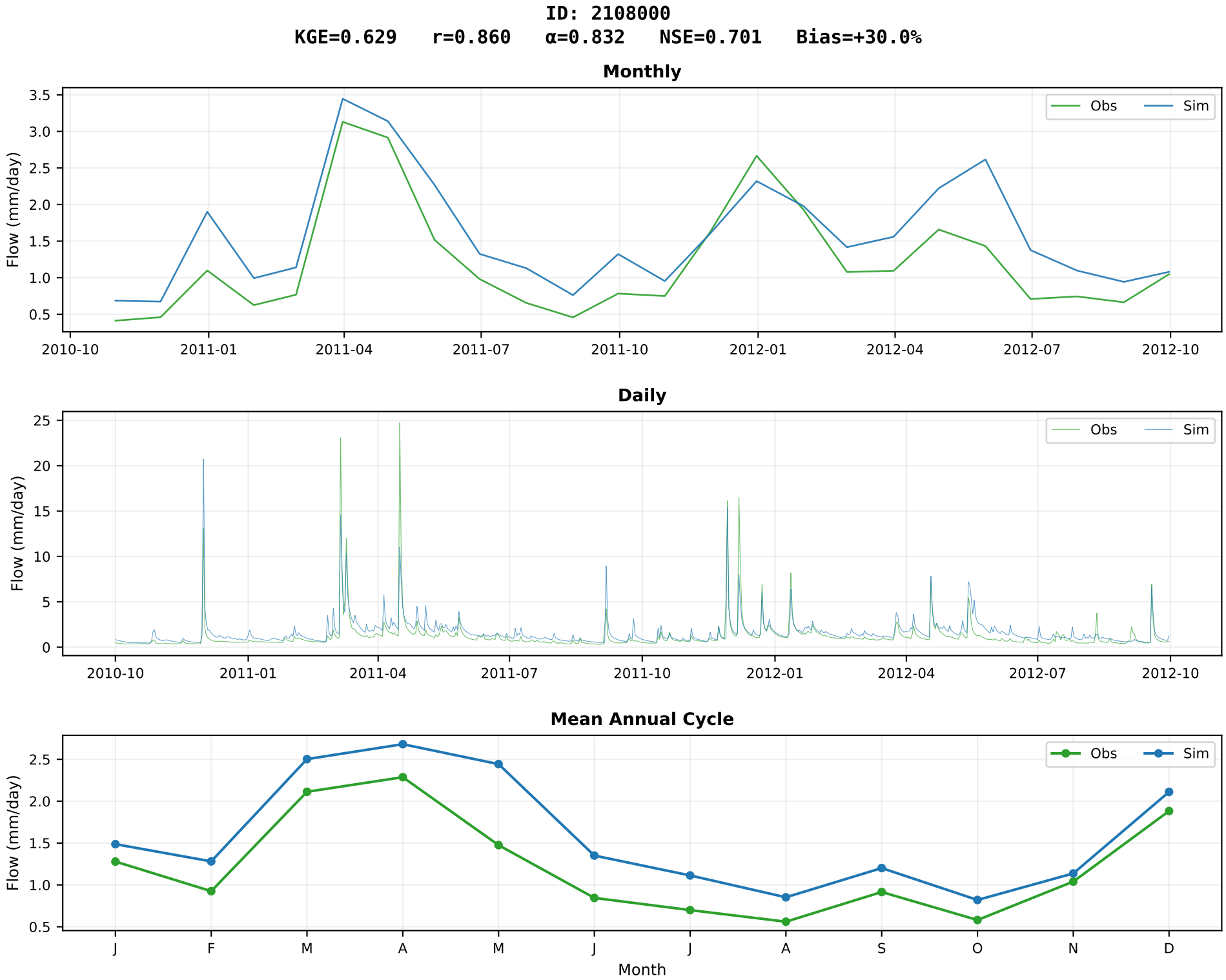

ID: 2381600
KGE=0.631 r=0.795 α=0.725 NSE=0.621 Bias=+13.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

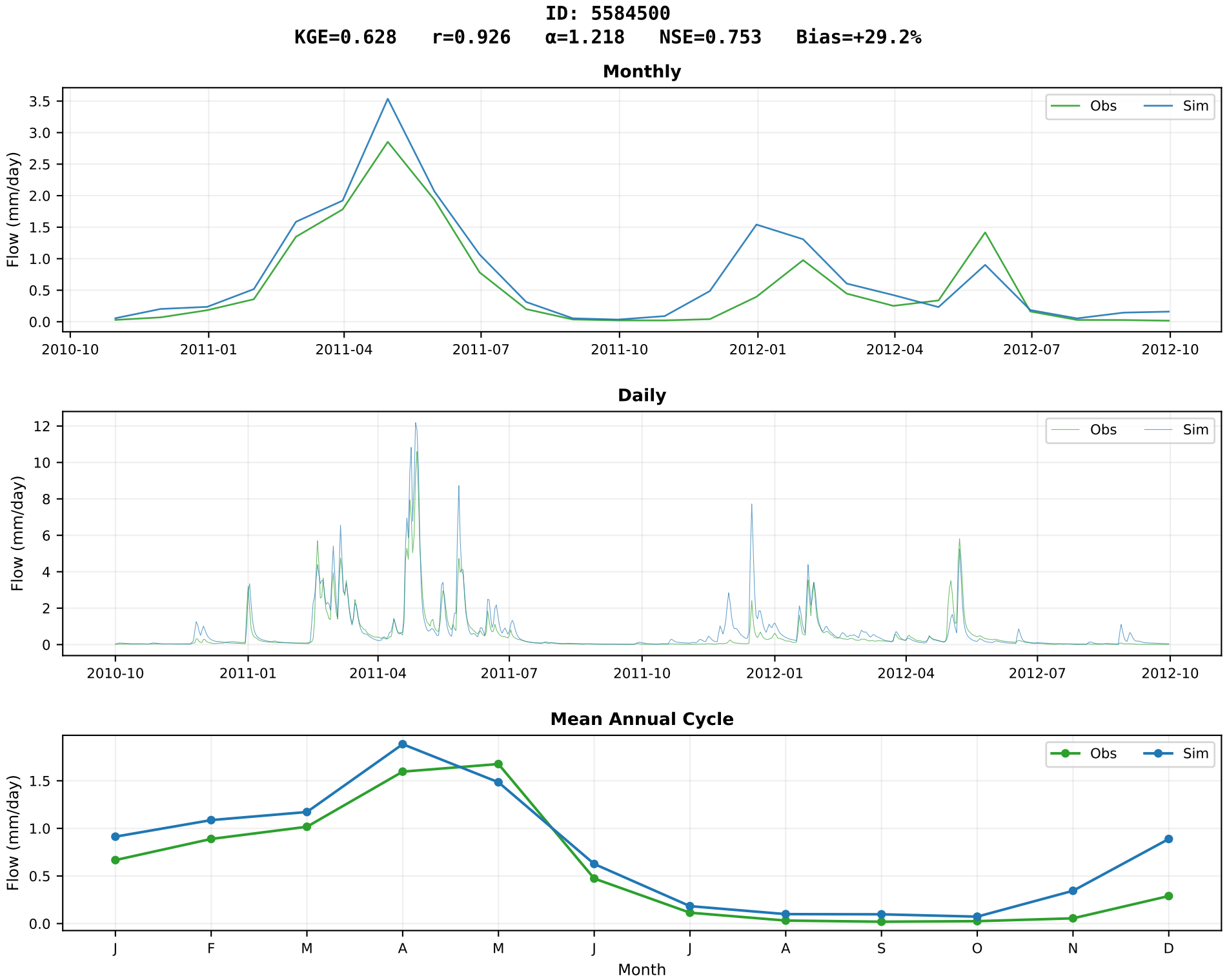
ID: 2053800

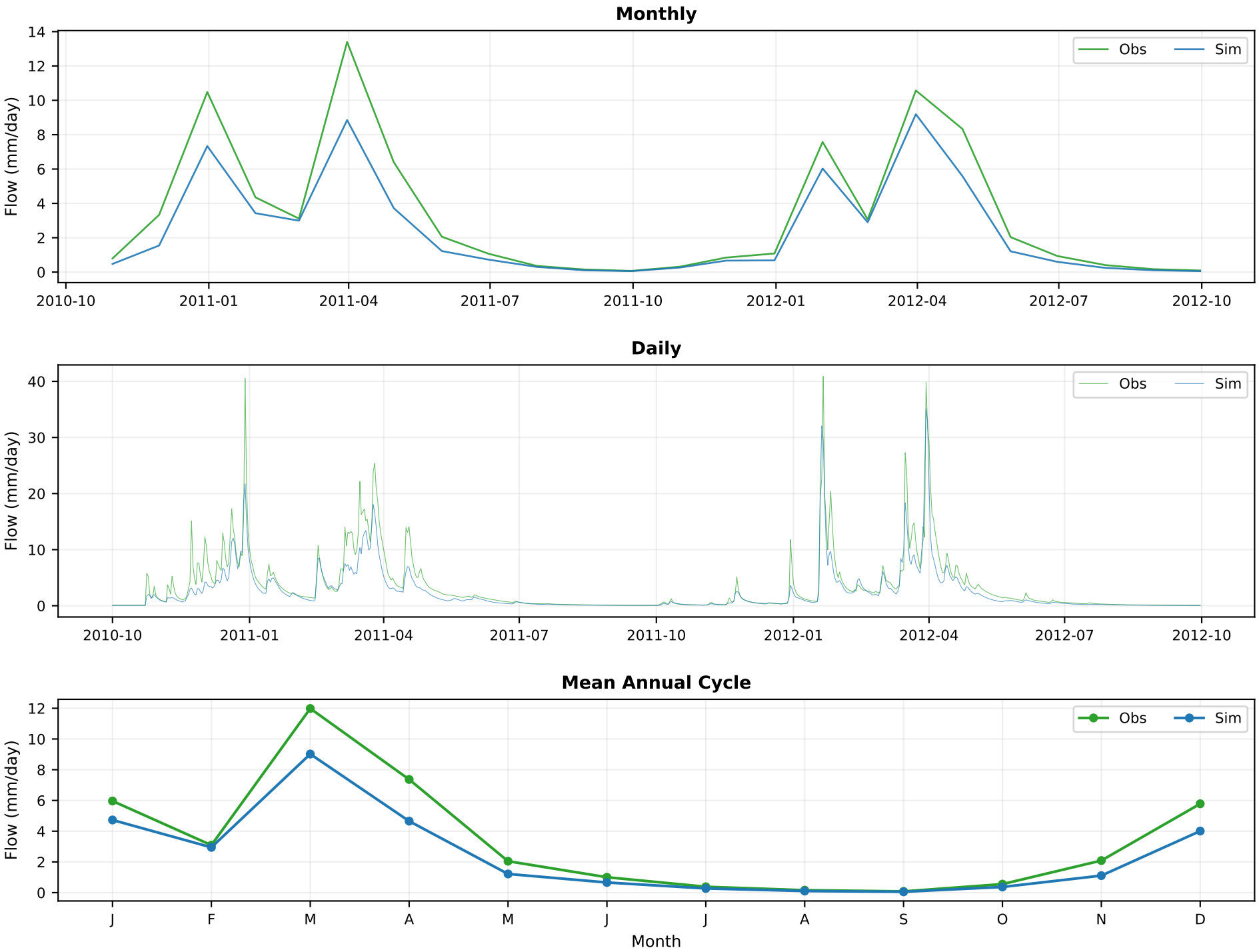
KGE=0.631 r=0.813 α=1.273 NSE=0.431 Bias=+16.3%

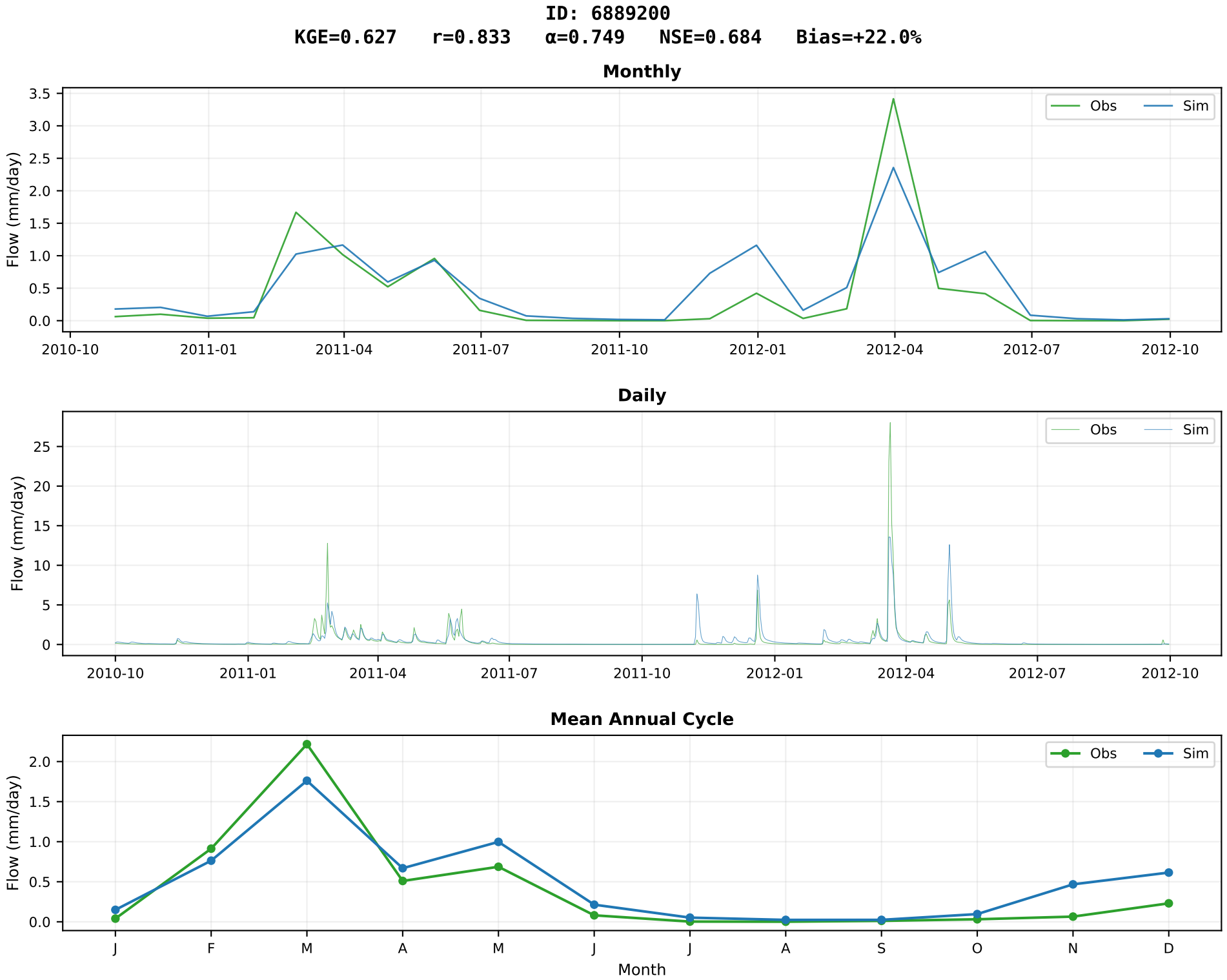


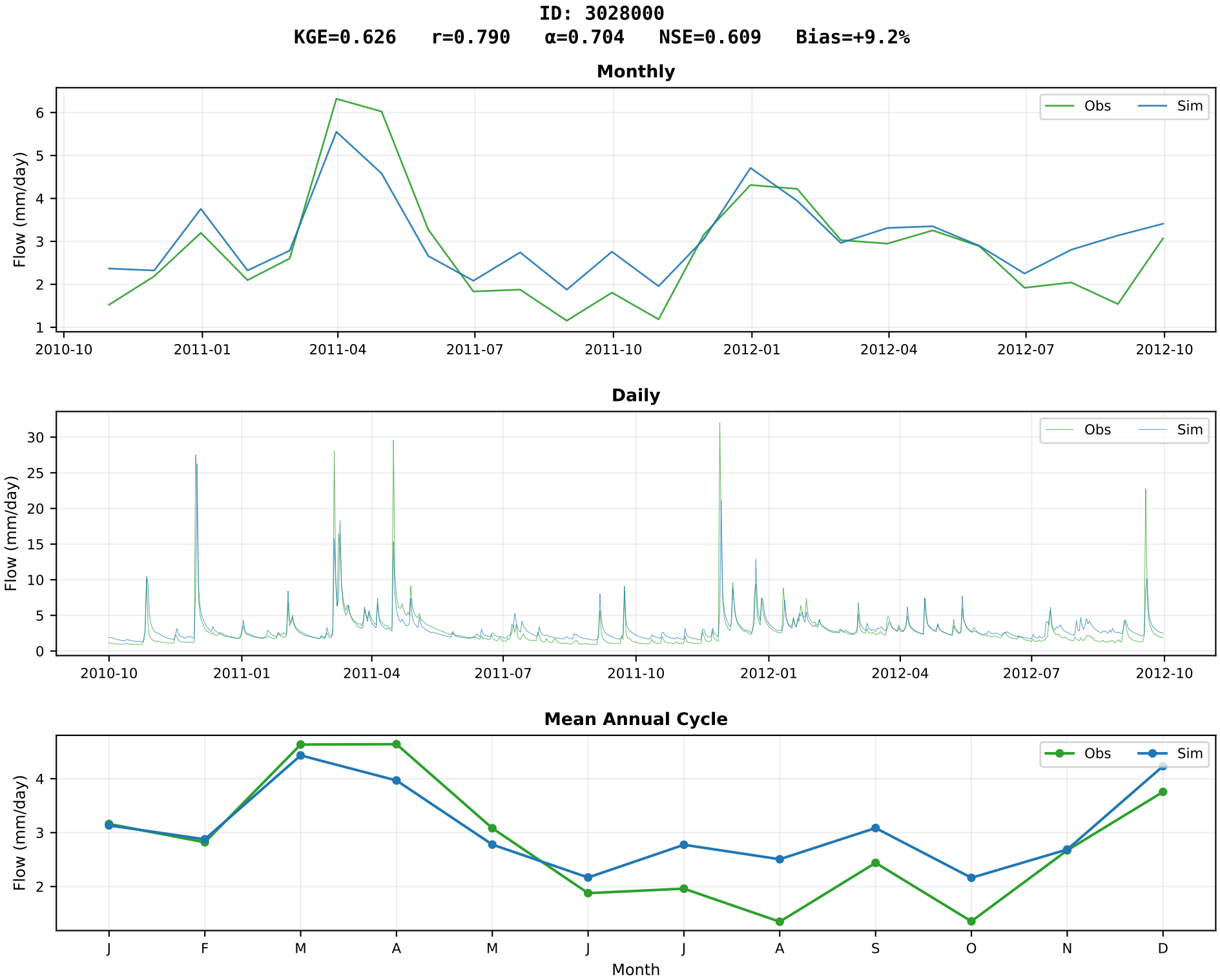

RAGING RIVER NEAR FALL CITY, WA (ID=12145500)
KGE=0.631 r=0.837 α=0.831 NSE=0.674 Bias=-28.6%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

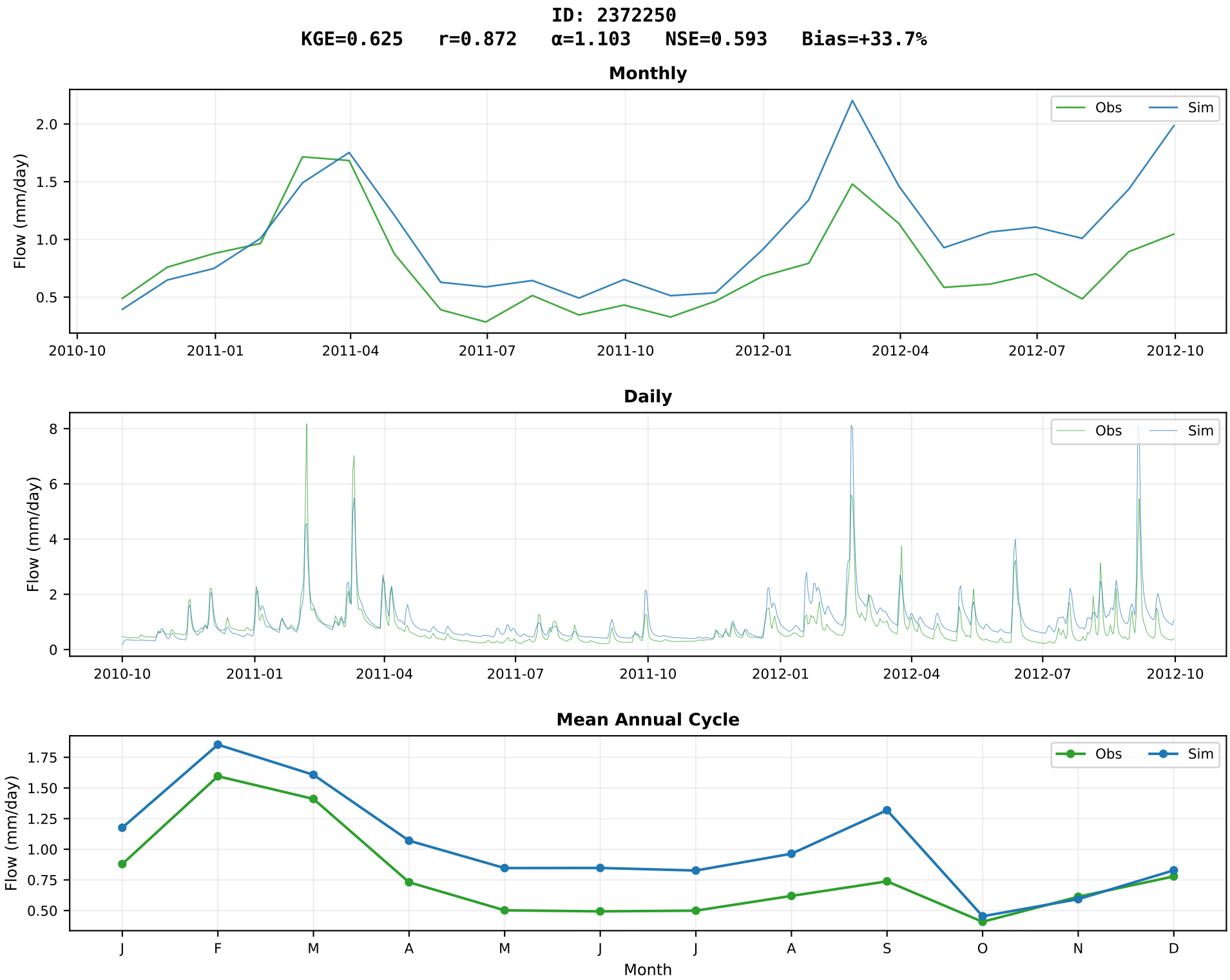

ID: 2108000
KGE=0.629 r=0.860 α=0.832 NSE=0.701 Bias=+30.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

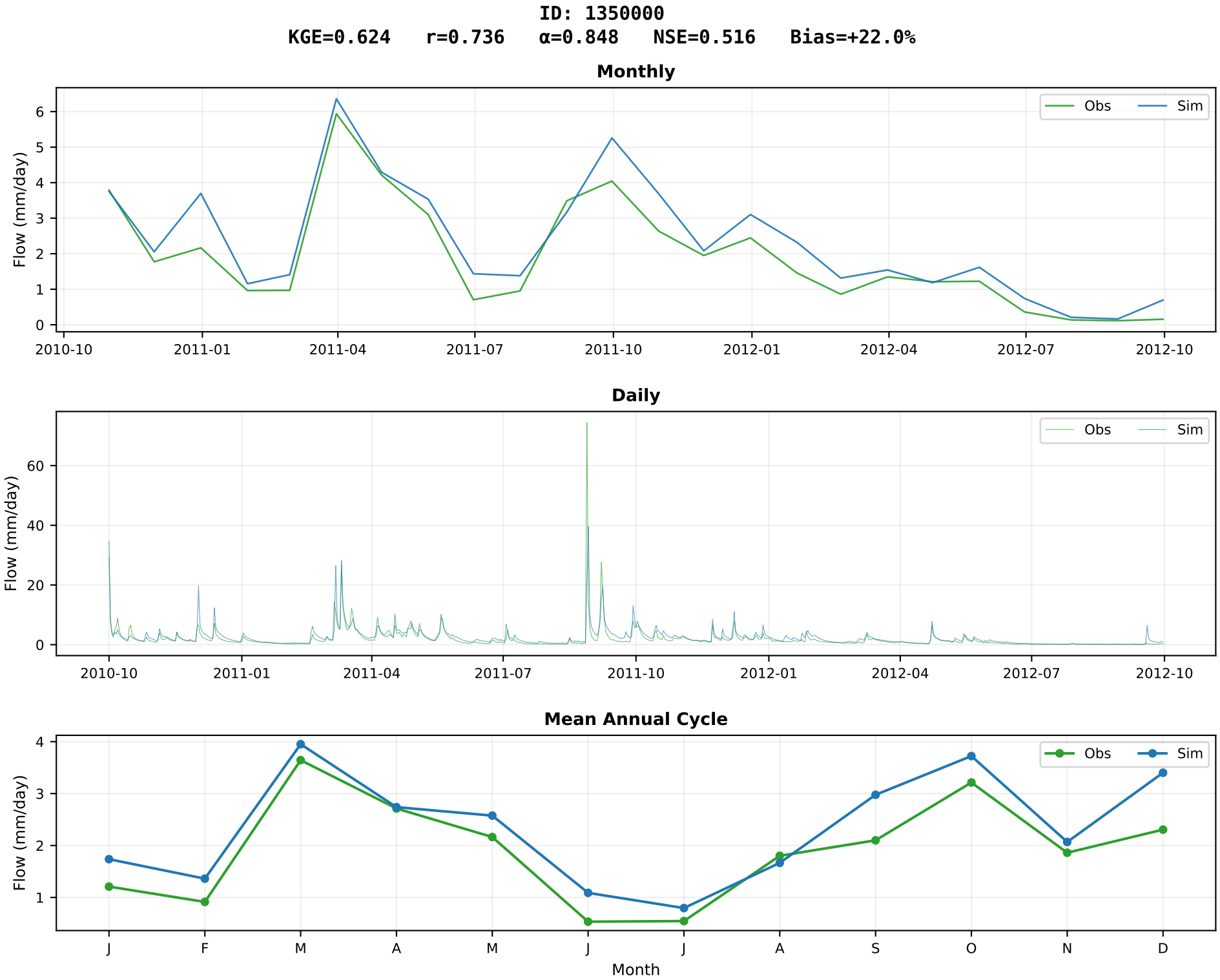

ID: 5584500
KGE=0.628 r=0.926 α=1.218 NSE=0.753 Bias=+29.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

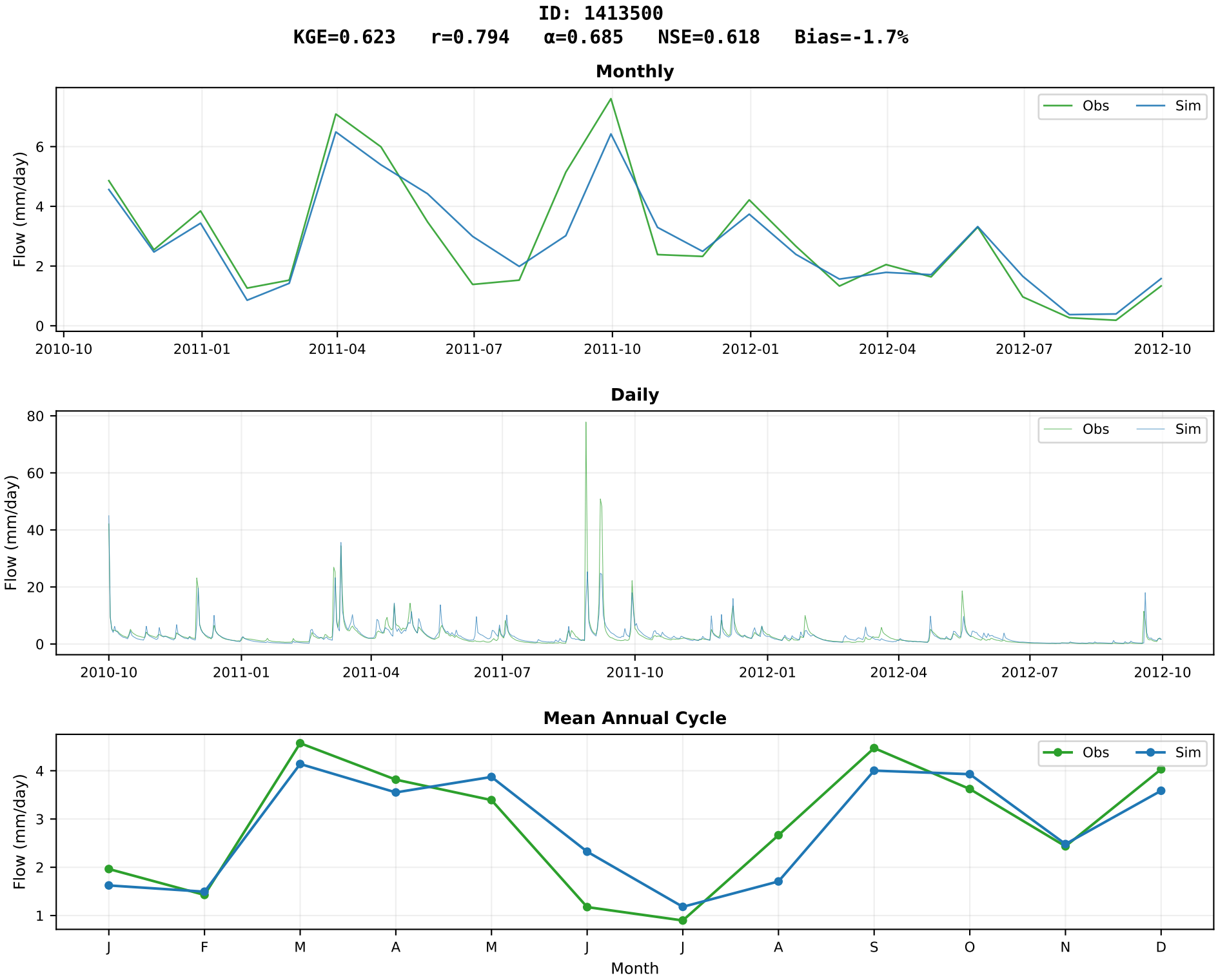

THUNDER CREEK NEAR NEWHALEM, WA (ID=12175500)
KGE=0.628 r=0.936 α=0.765 NSE=0.814 Bias=-28.1%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

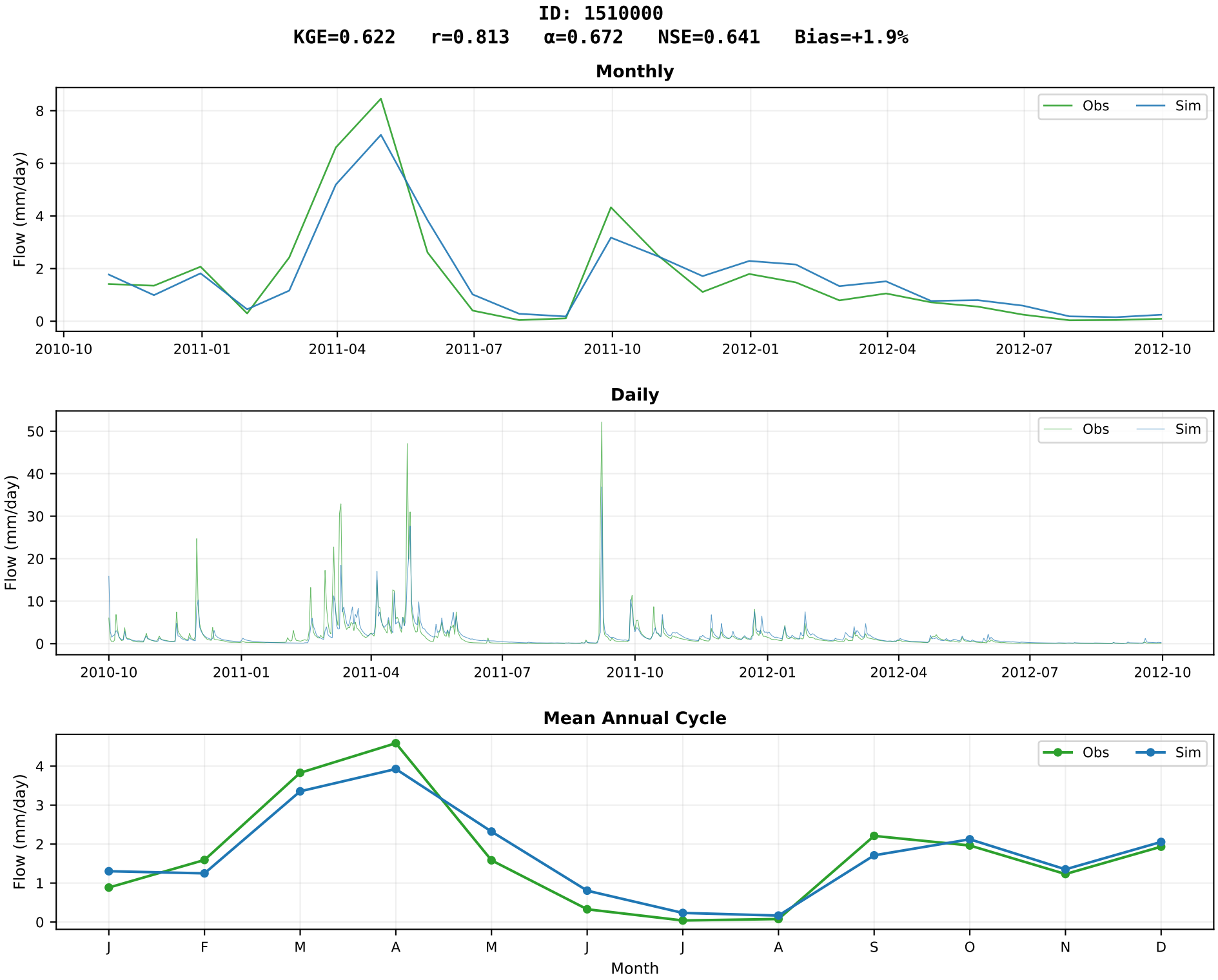

ID: 6889200
KGE=0.627 r=0.833 α=0.749 NSE=0.684 Bias=+22.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

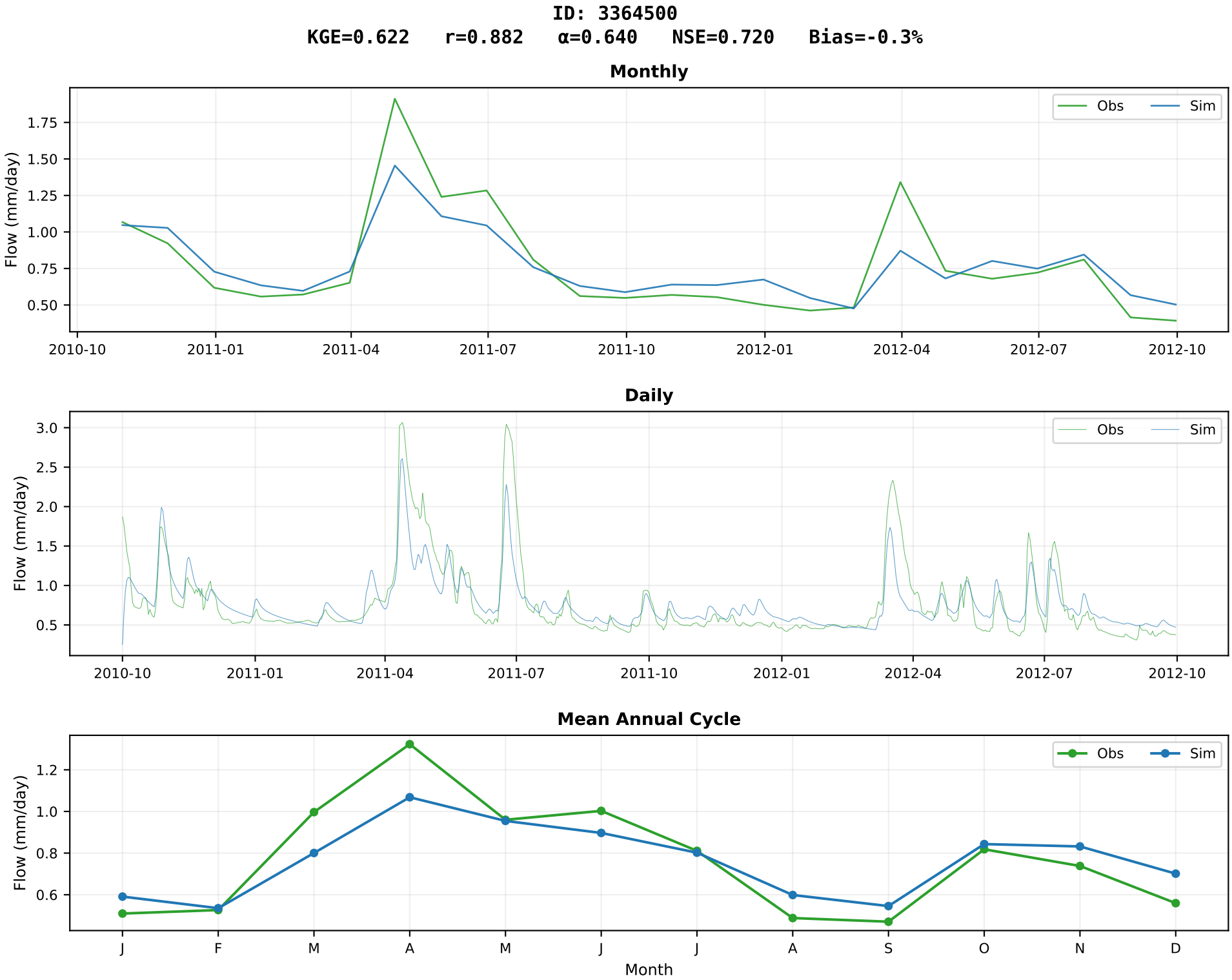

ID: 3028000
KGE=0.626 r=0.790 α=0.704 NSE=0.609 Bias=+9.2%
Monthly
Obs
Sim
Flow (mm/day)
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

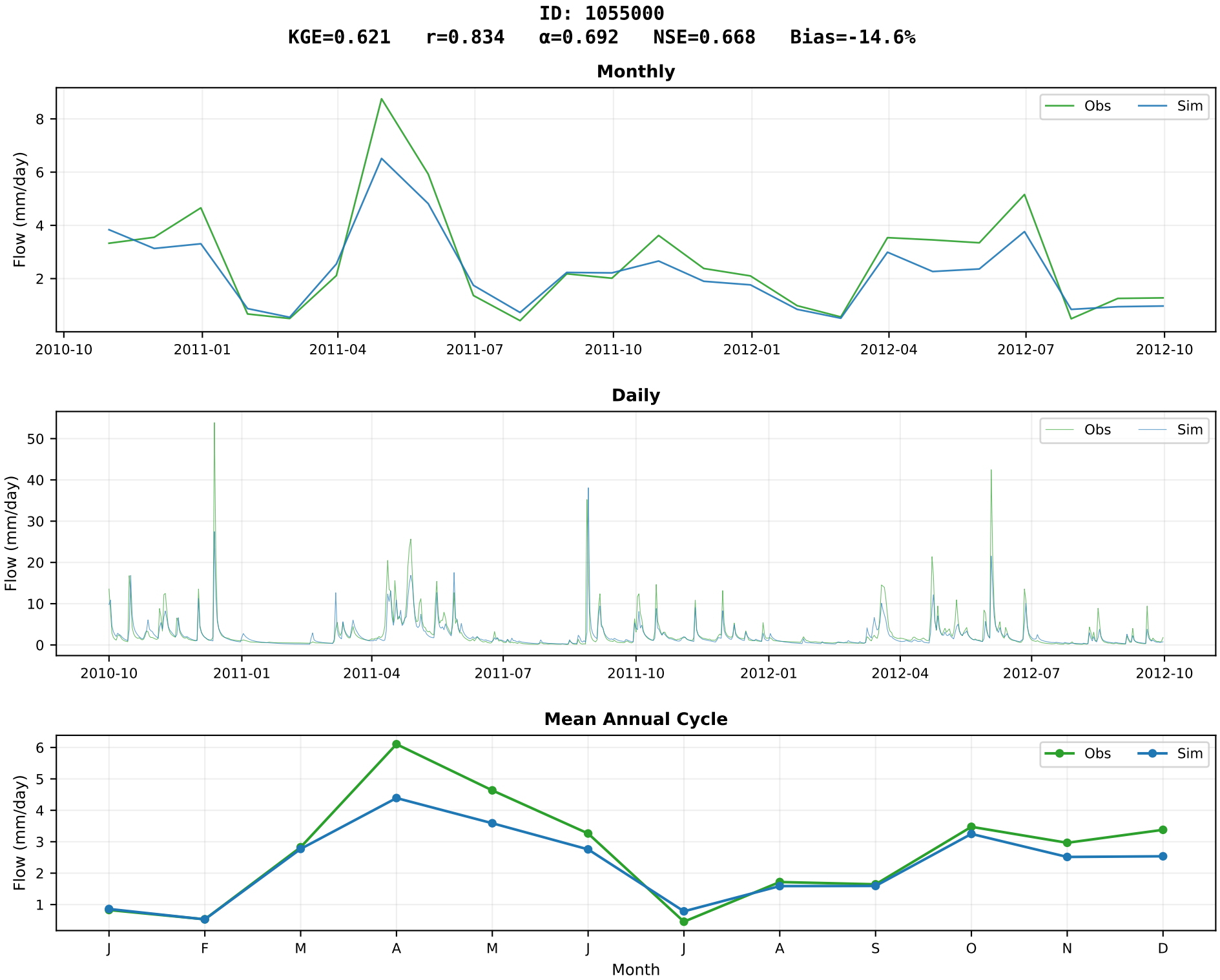

ID: 2372250
KGE=0.625 r=0.872 α=1.103 NSE=0.593 Bias=+33.7%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
8
6
4
2
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.75
1.50
1.25
1.00
0.75
0.50
J
F
M
A
M
J
J
A
S
O
N
D
Month

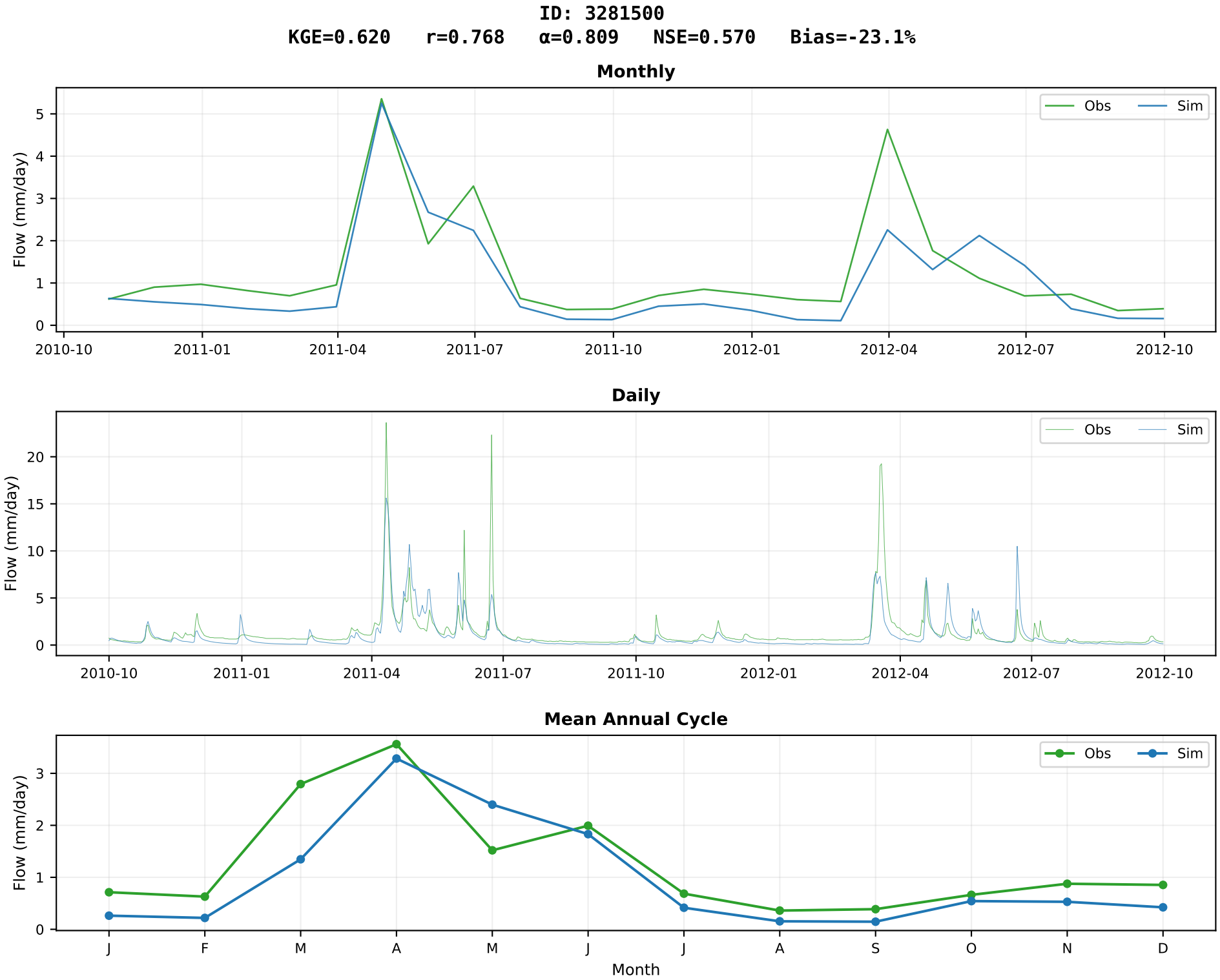

ID: 1350000
KGE=0.624 r=0.736 α=0.848 NSE=0.516 Bias=+22.0%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

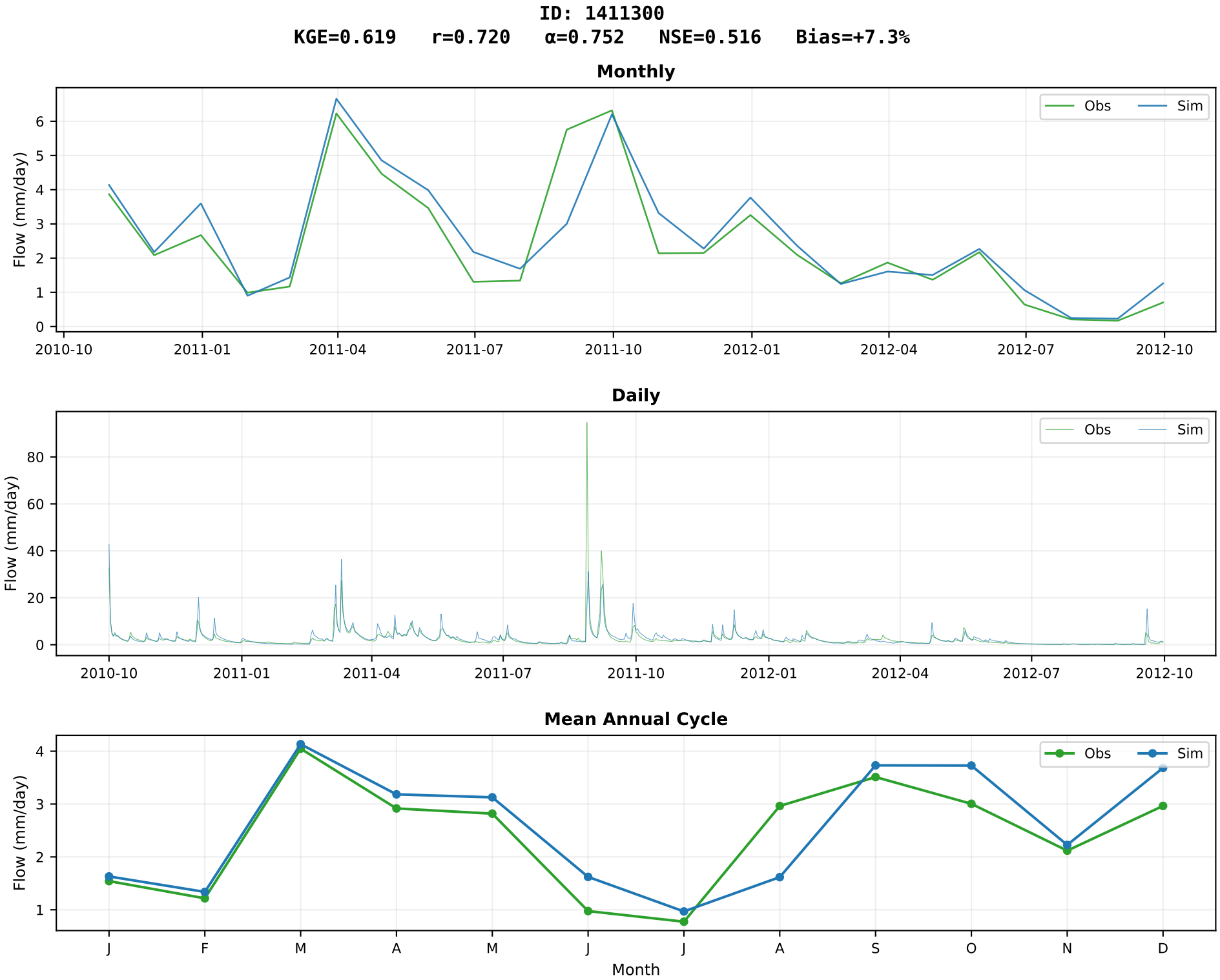

ID: 1413500
KGE=0.623 r=0.794 α=0.685 NSE=0.618 Bias=-1.7%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

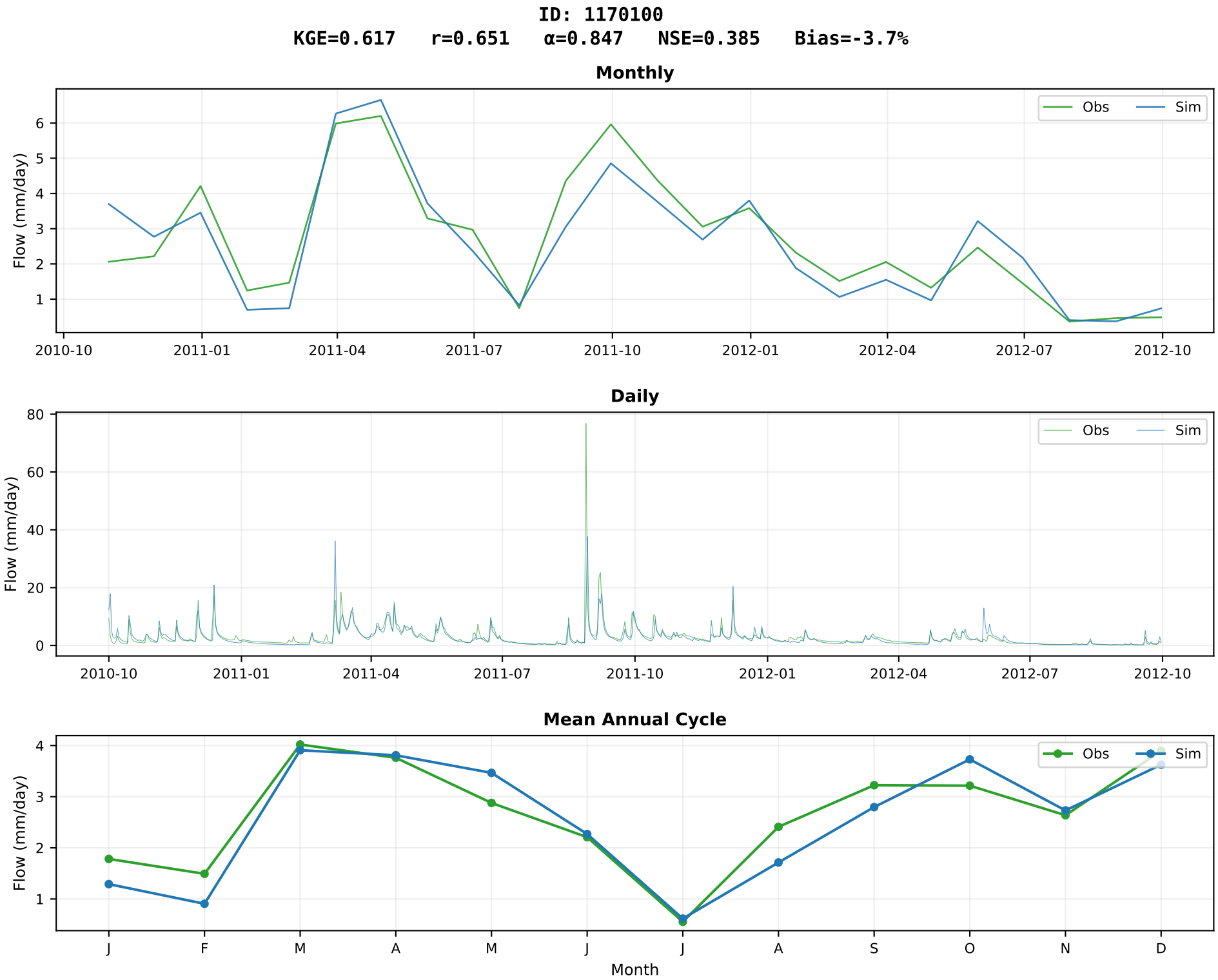

ID: 1510000
KGE=0.622 r=0.813 α=0.672 NSE=0.641 Bias=+1.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

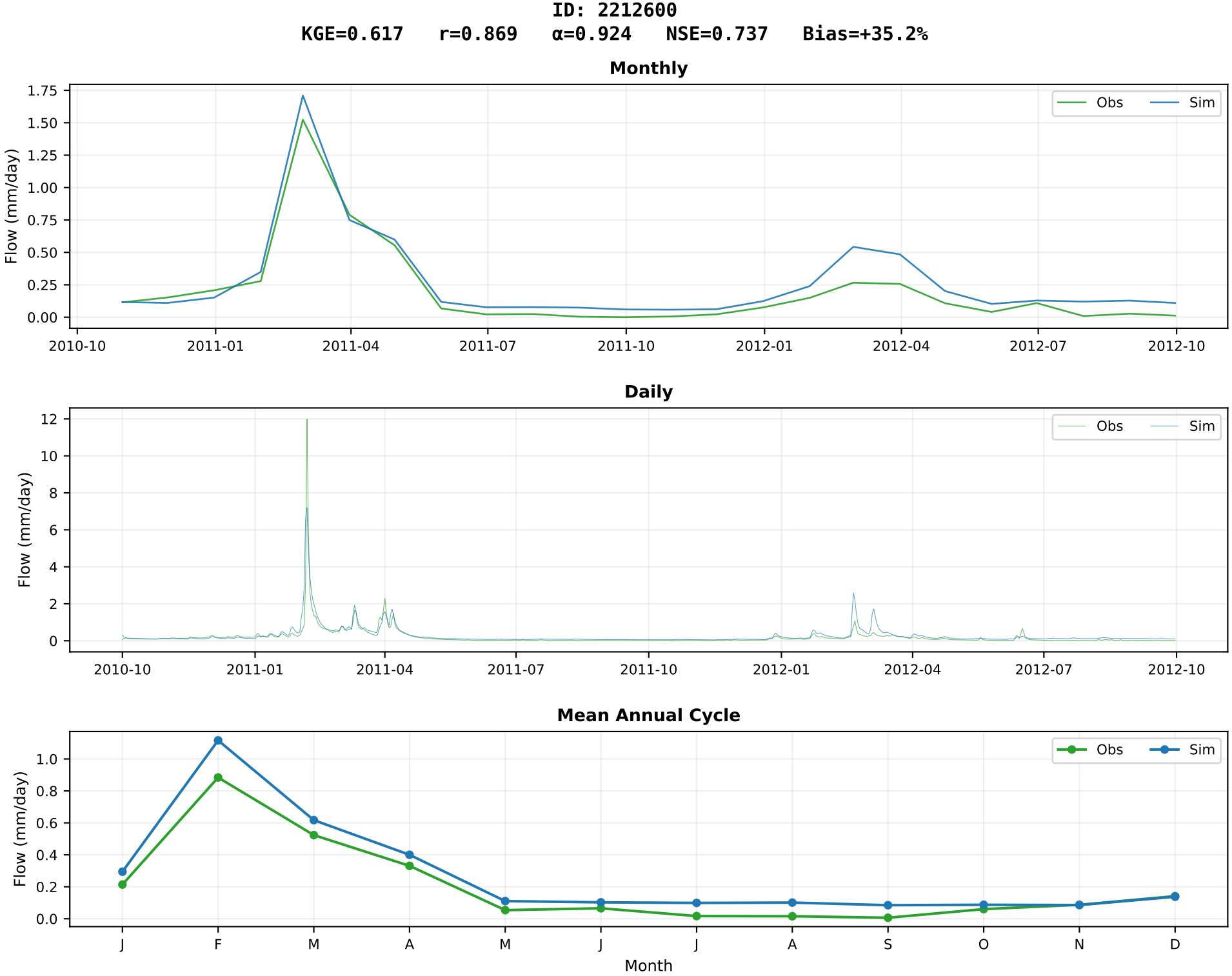

ID: 3364500
KGE=0.622 r=0.882 α=0.640 NSE=0.720 Bias=-0.3%
Monthly
Obs
Sim
Flow (mm/day)
1.75
1.50
1.25
1.00
0.75
0.50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
3.0
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.2
1.0
0.8
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month

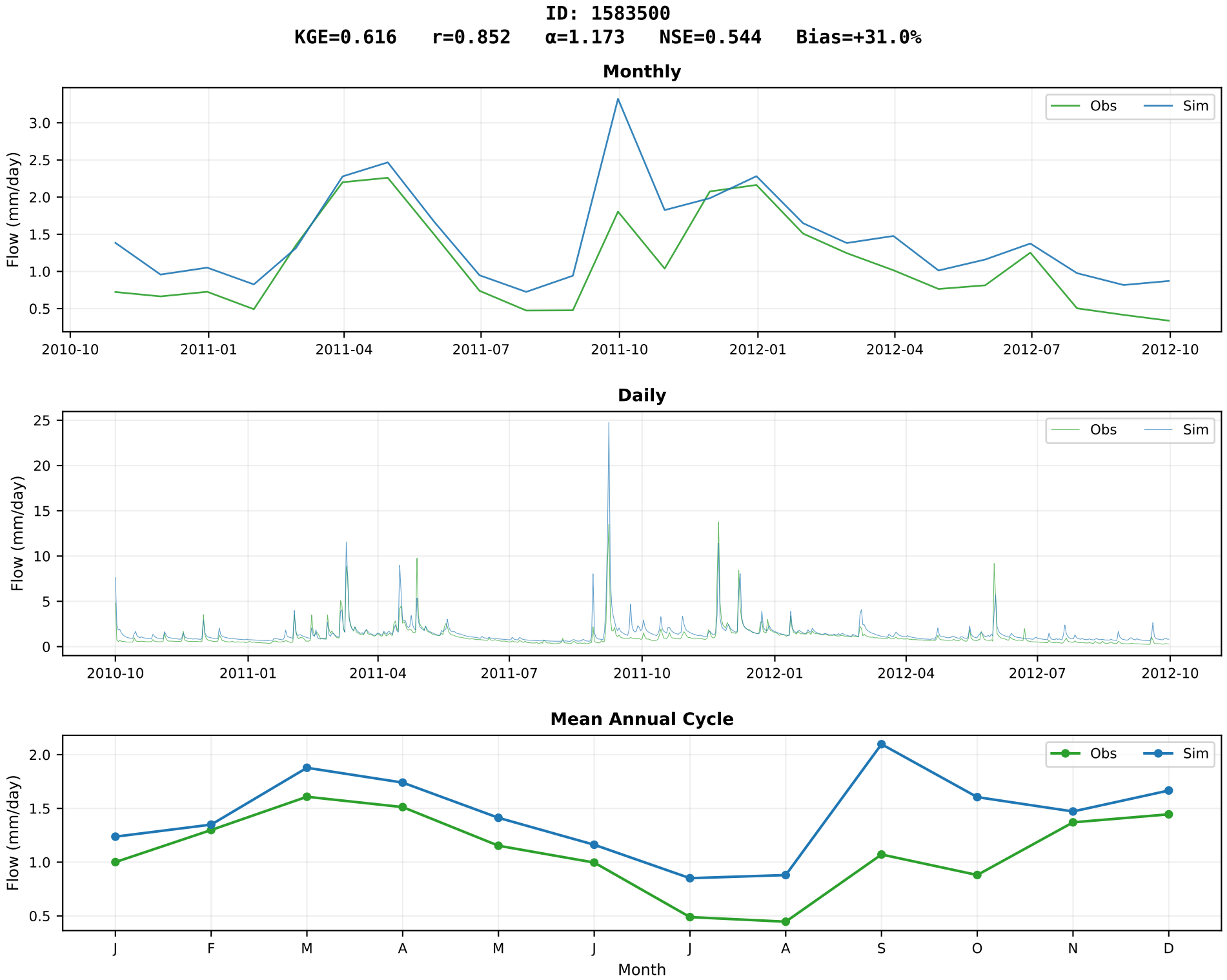

ID: 1055000
KGE=0.621 r=0.834 α=0.692 NSE=0.668 Bias=-14.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

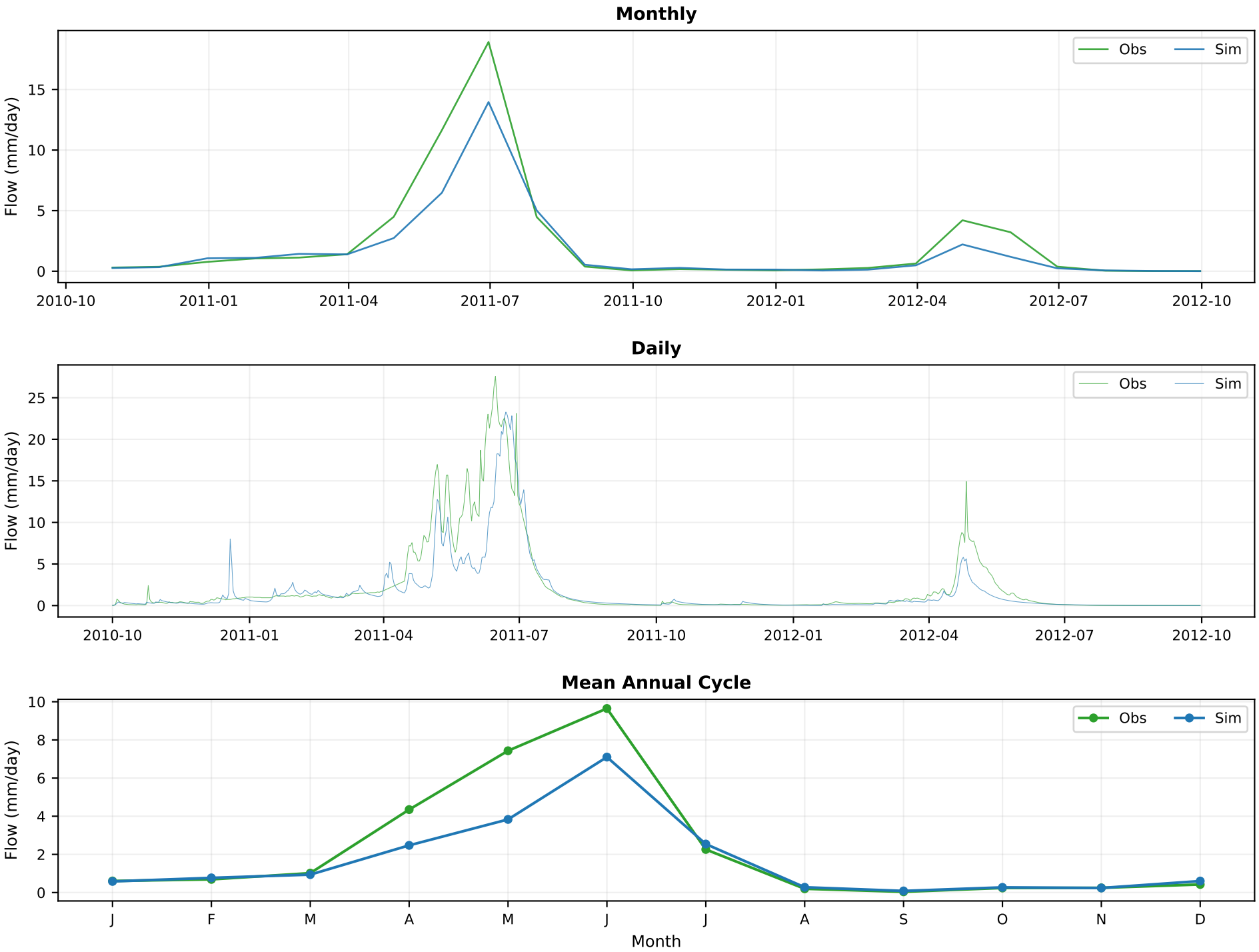

ID: 3281500
KGE=0.620 r=0.768 α=0.809 NSE=0.570 Bias=-23.1%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

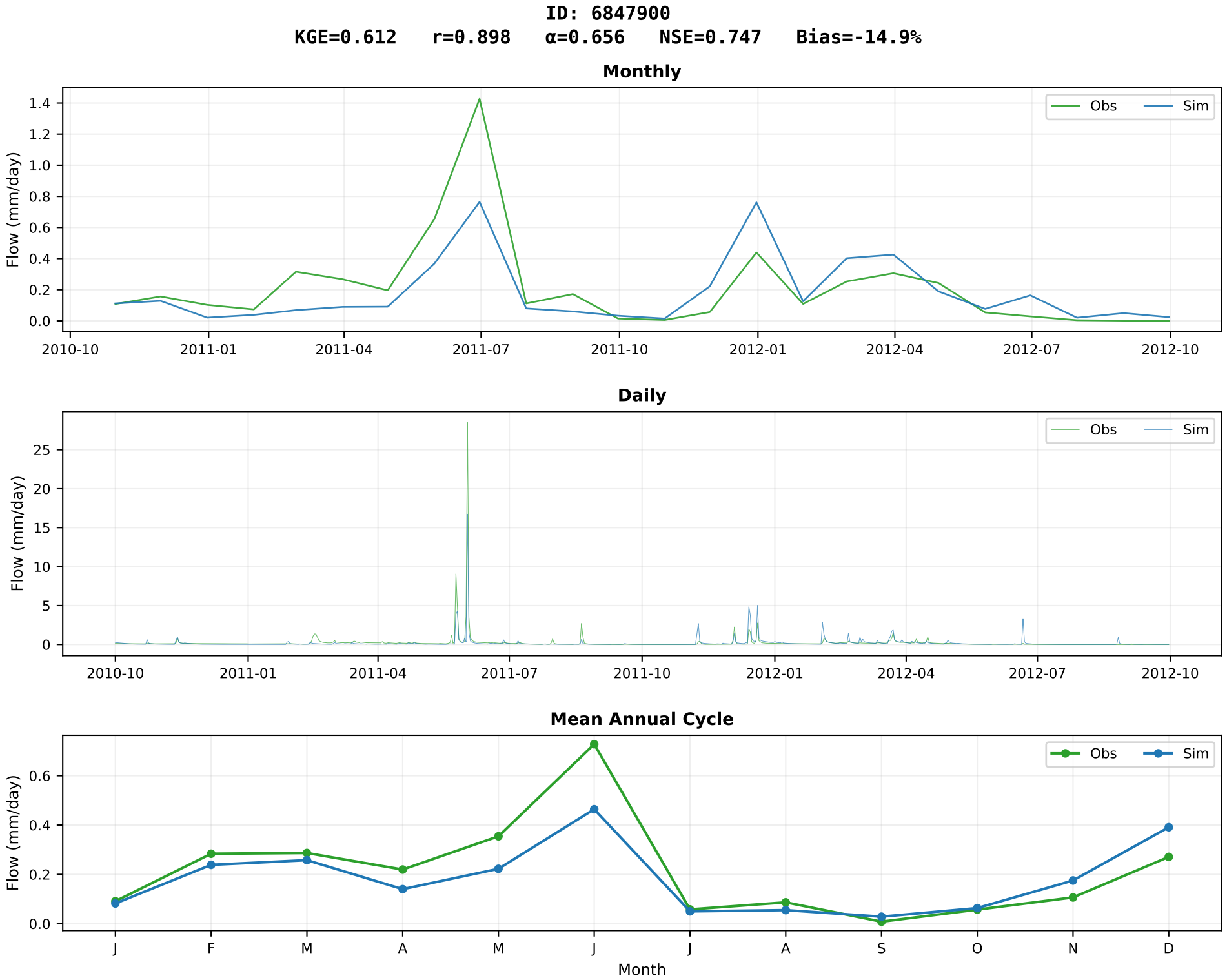

ID: 1411300
KGE=0.619 r=0.720 α=0.752 NSE=0.516 Bias=+7.3%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

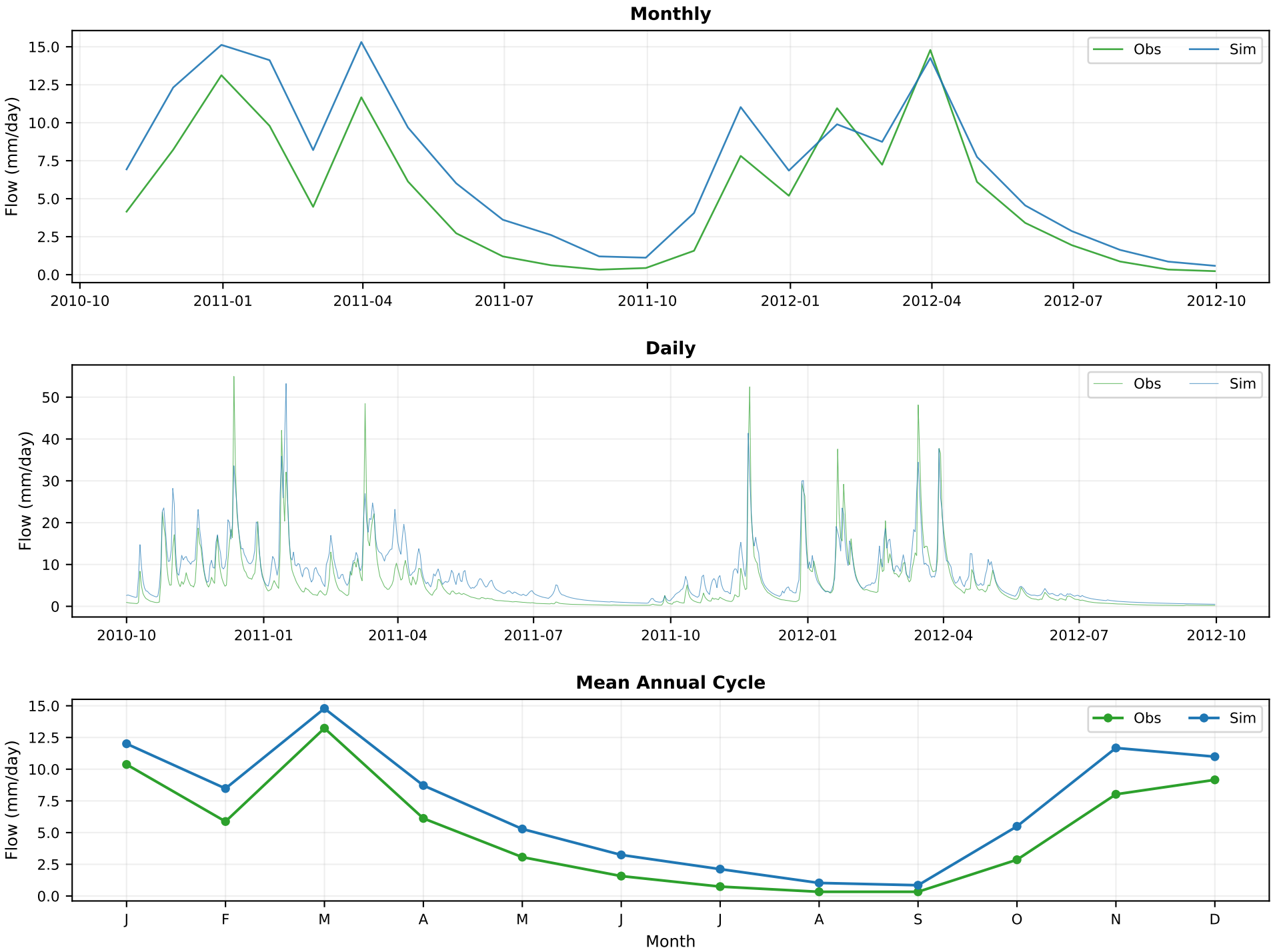

ID: 1170100
KGE=0.617 r=0.651 α=0.847 NSE=0.385 Bias=-3.7%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

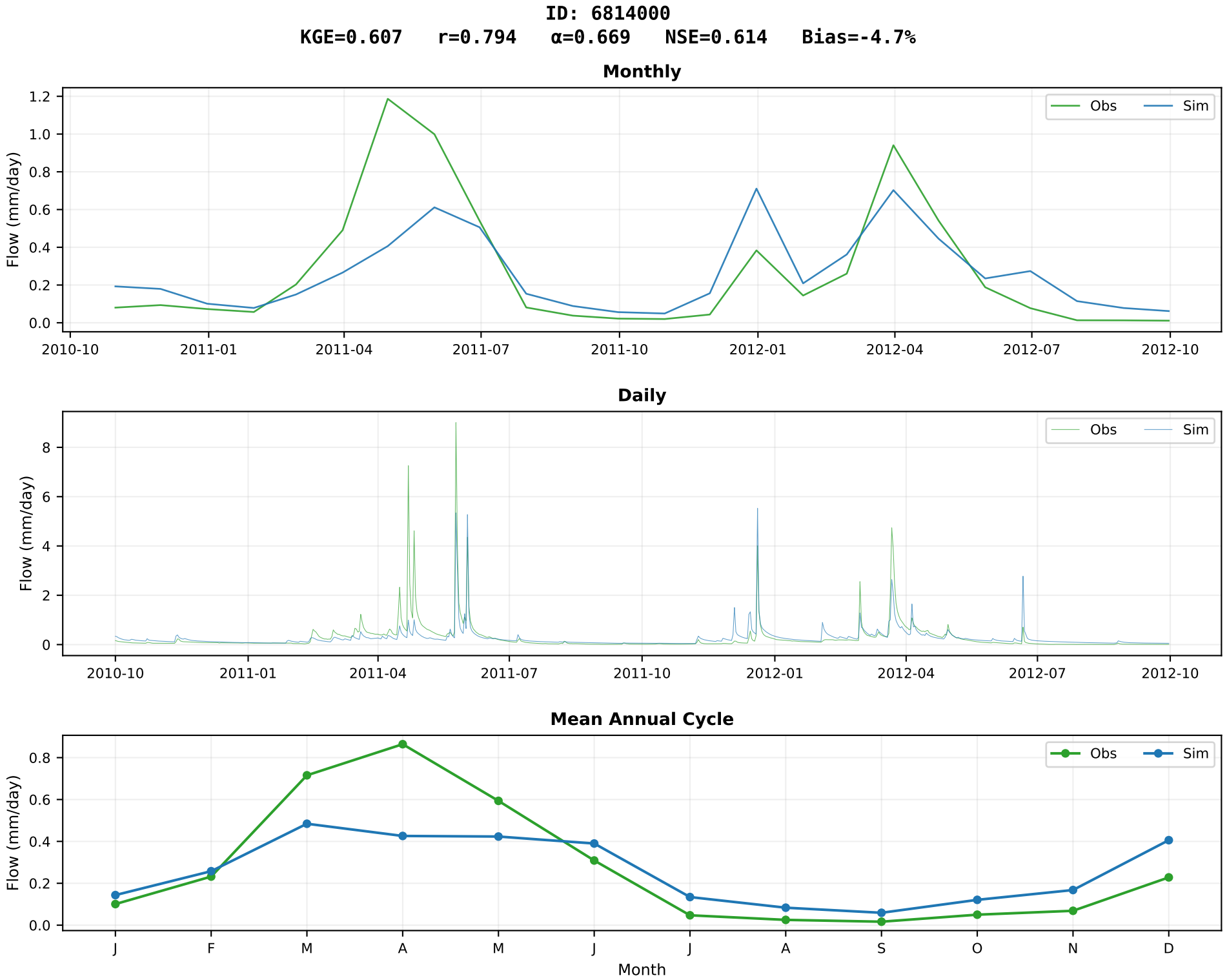

ID: 2212600
KGE=0.617 r=0.869 α=0.924 NSE=0.737 Bias=+35.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

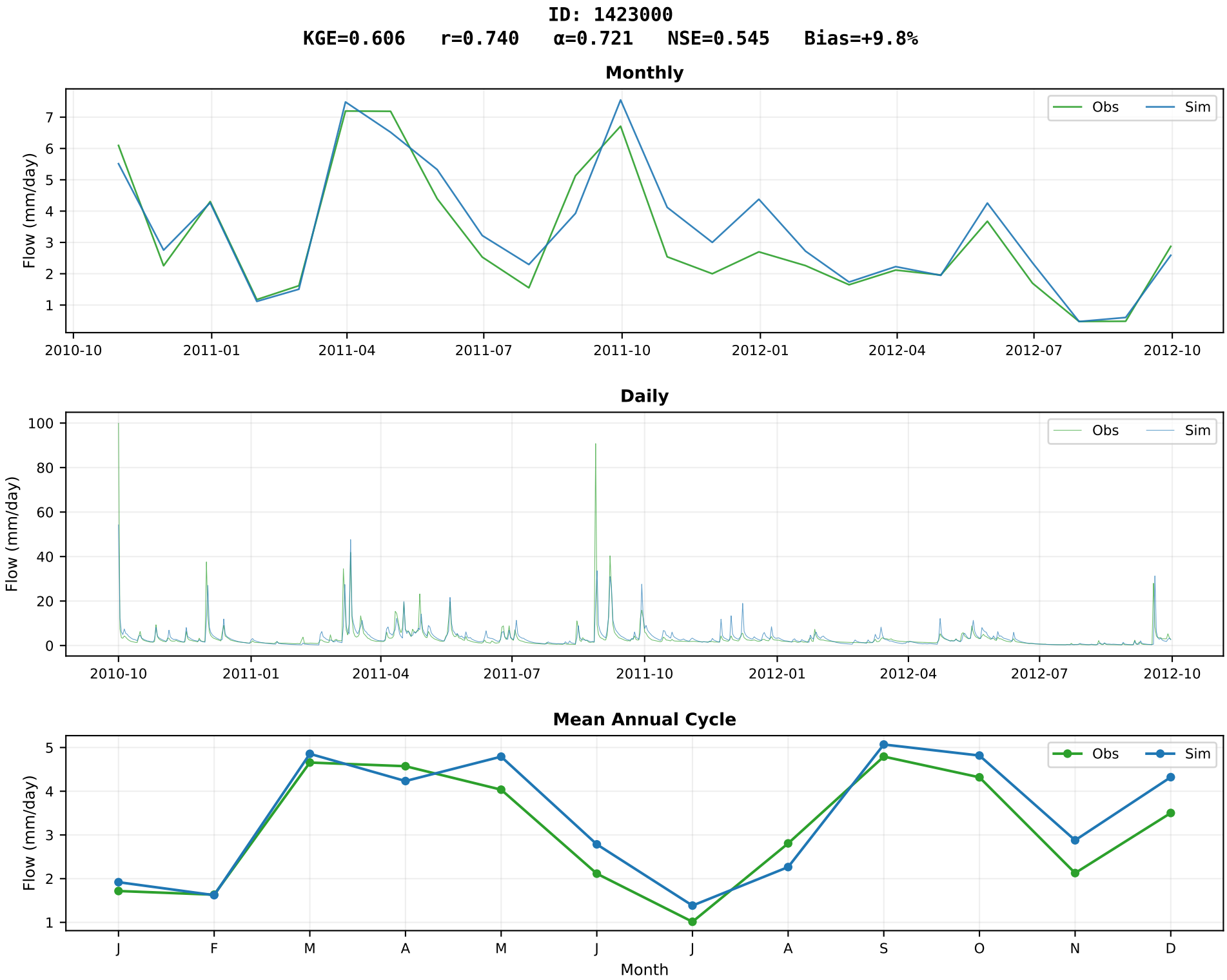

ID: 1583500
KGE=0.616 r=0.852 α=1.173 NSE=0.544 Bias=+31.0%
Monthly
Obs
Sim
Flow (mm/day)
3.0
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
25
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

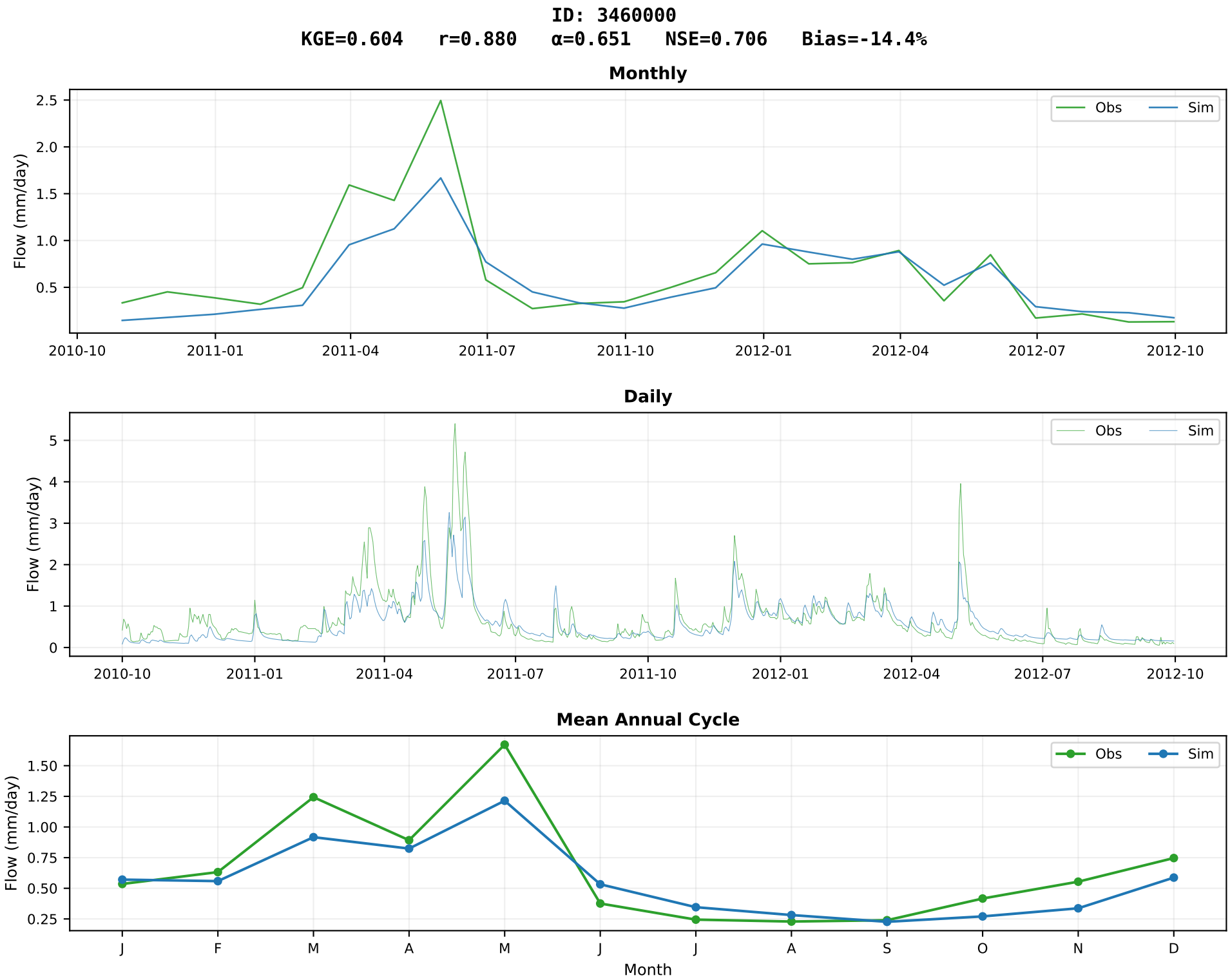

DUCKABUSH RIVER NEAR BRINNON, WA (ID=12054000)
KGE=0.616 r=0.877 α=0.760 NSE=0.738 Bias=-27.4%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

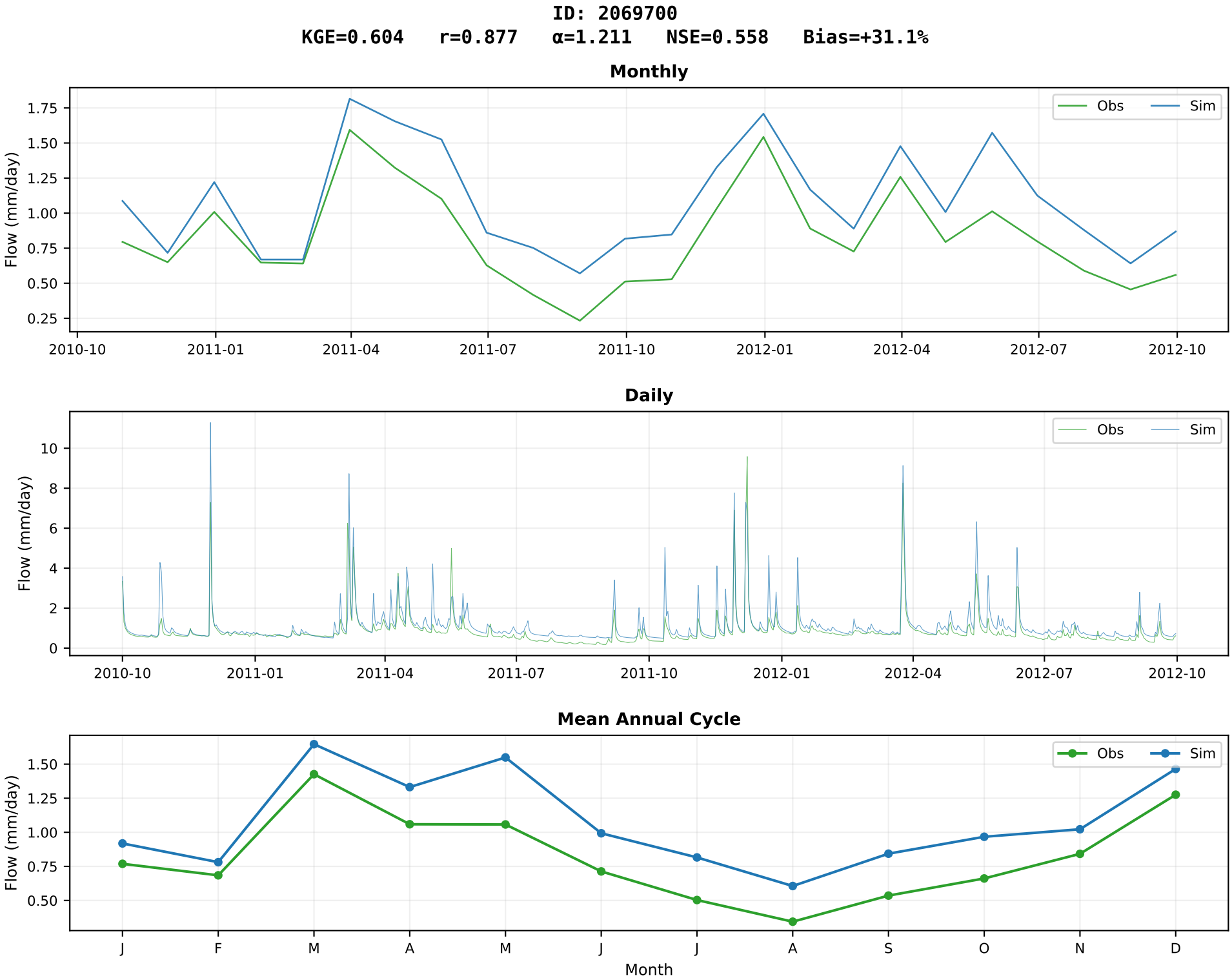

ID: 6847900
KGE=0.612 r=0.898 α=0.656 NSE=0.747 Bias=-14.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

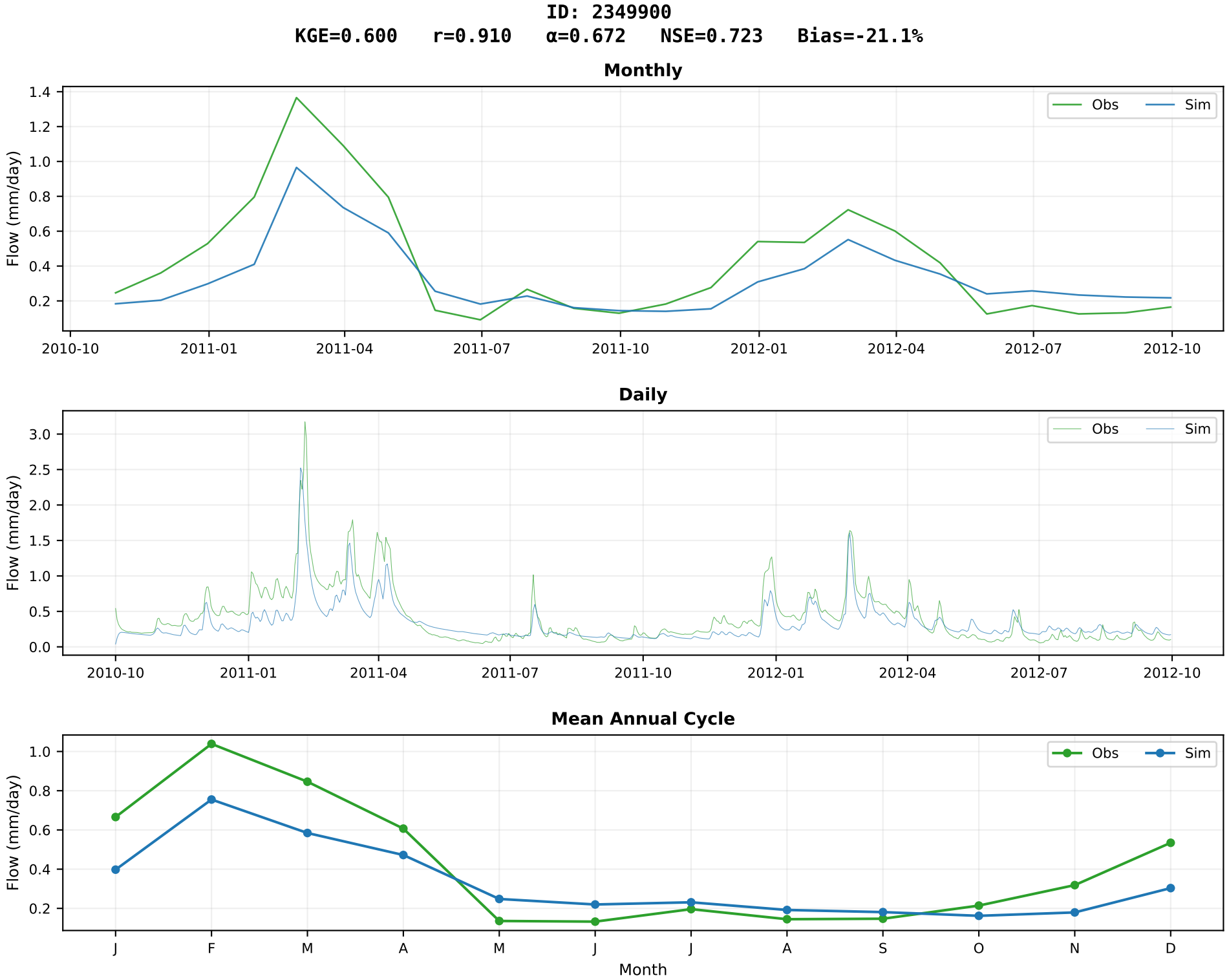

Big Knife Creek near Arlee MT (ID=12383500)
KGE=0.610 r=0.887 α=0.958 NSE=0.708 Bias=+37.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

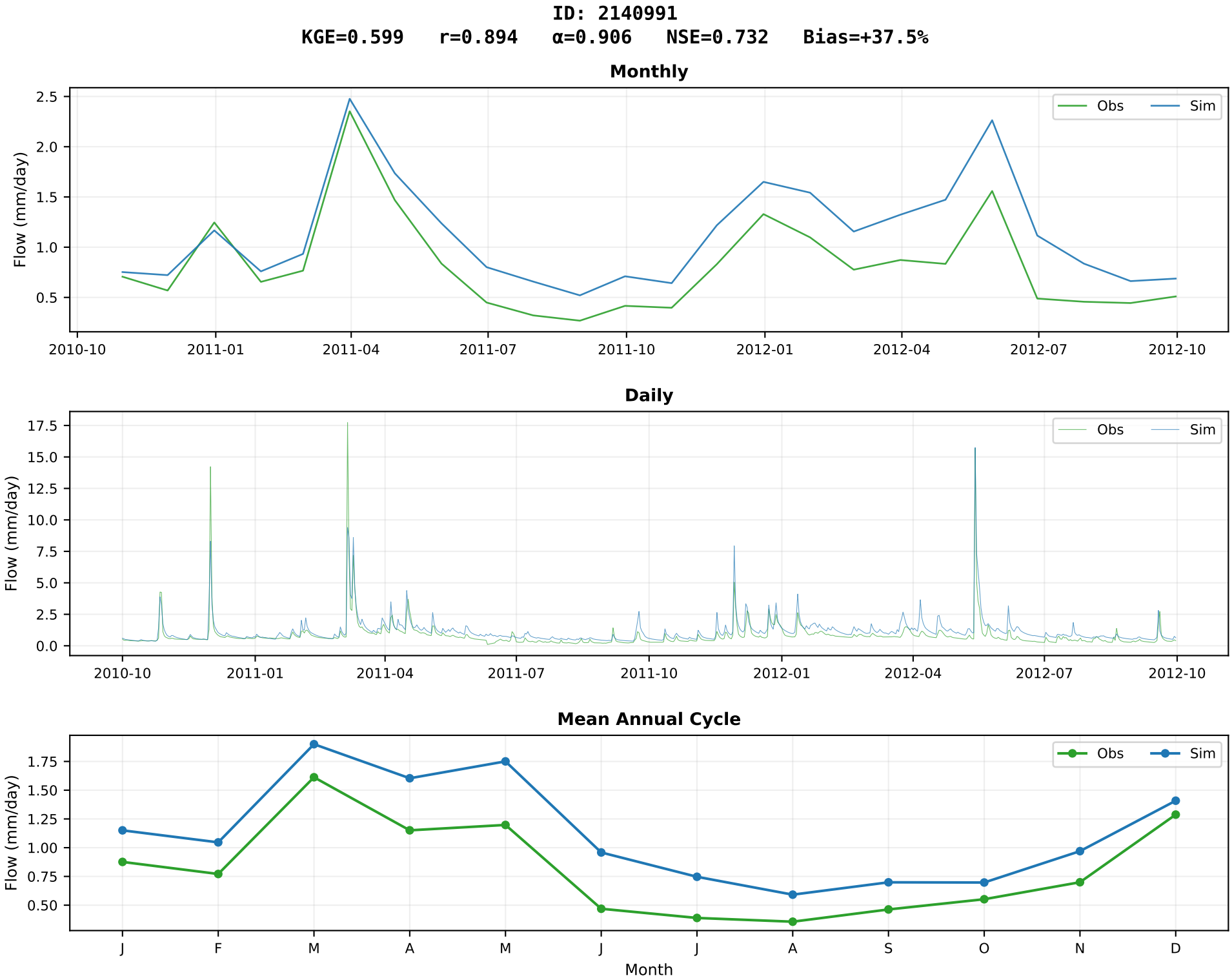

ID: 6814000
KGE=0.607 r=0.794 α=0.669 NSE=0.614 Bias=-4.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

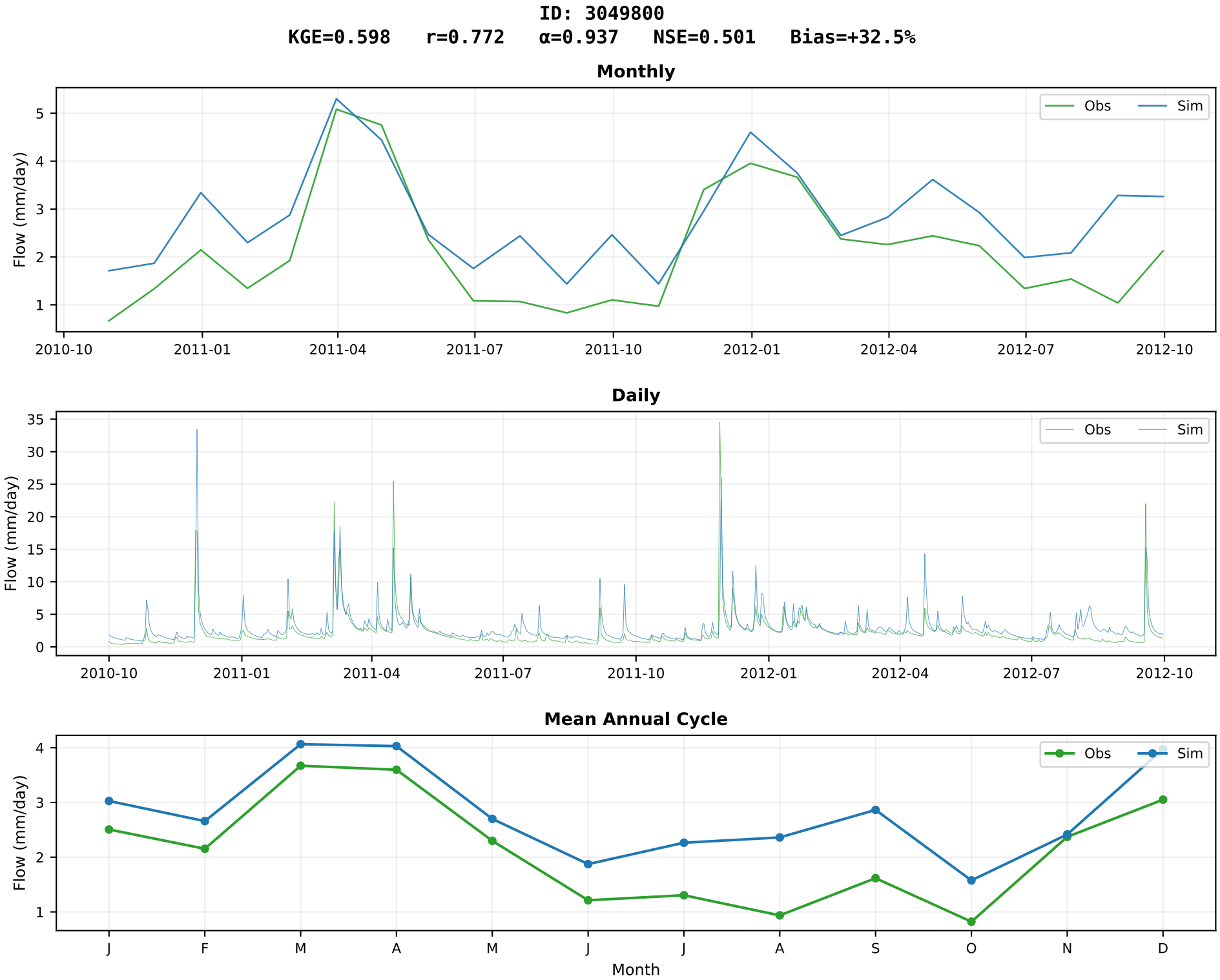

ID: 1423000
KGE=0.606 r=0.740 α=0.721 NSE=0.545 Bias=+9.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

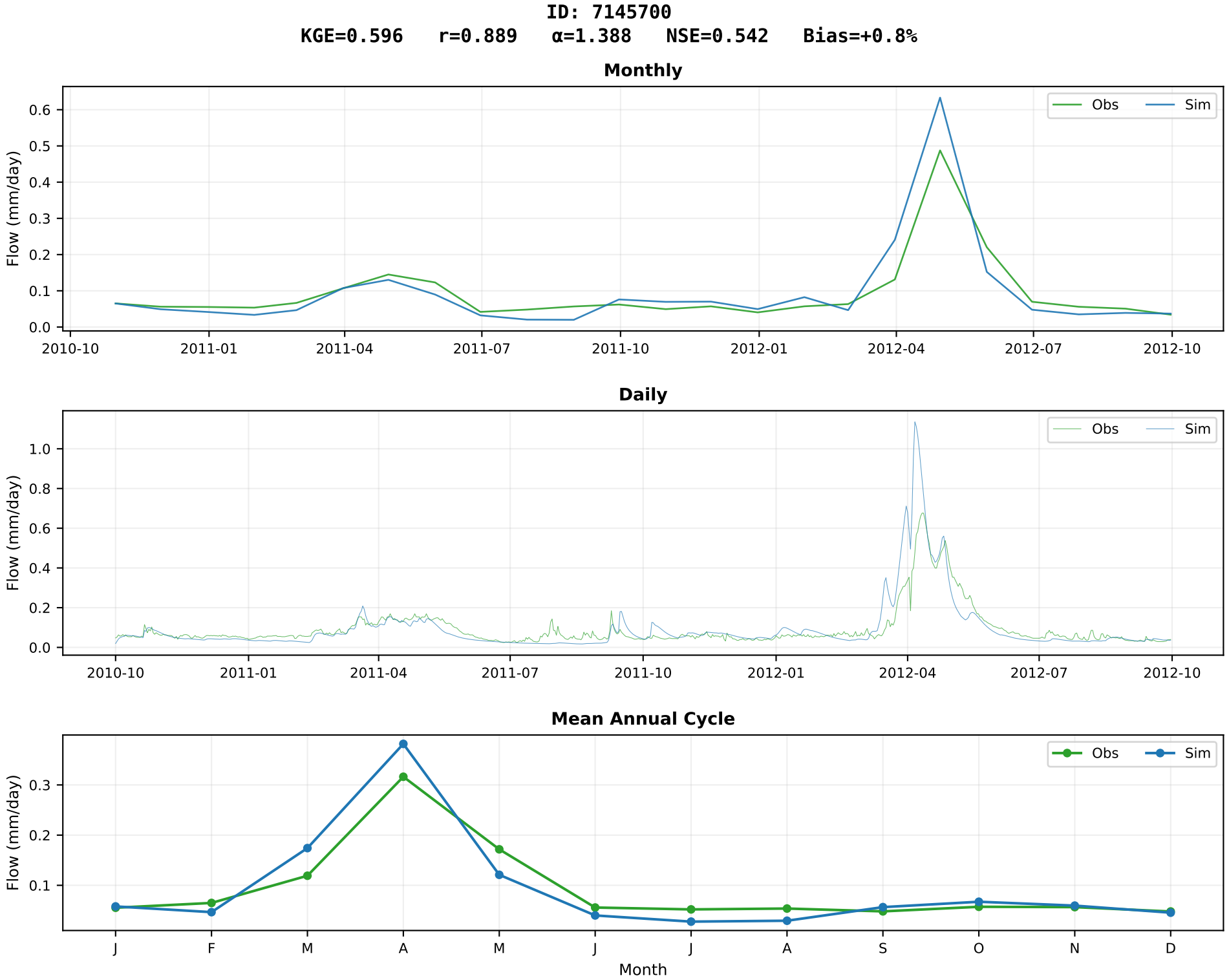

ID: 3460000
KGE=0.604 r=0.880 α=0.651 NSE=0.706 Bias=-14.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

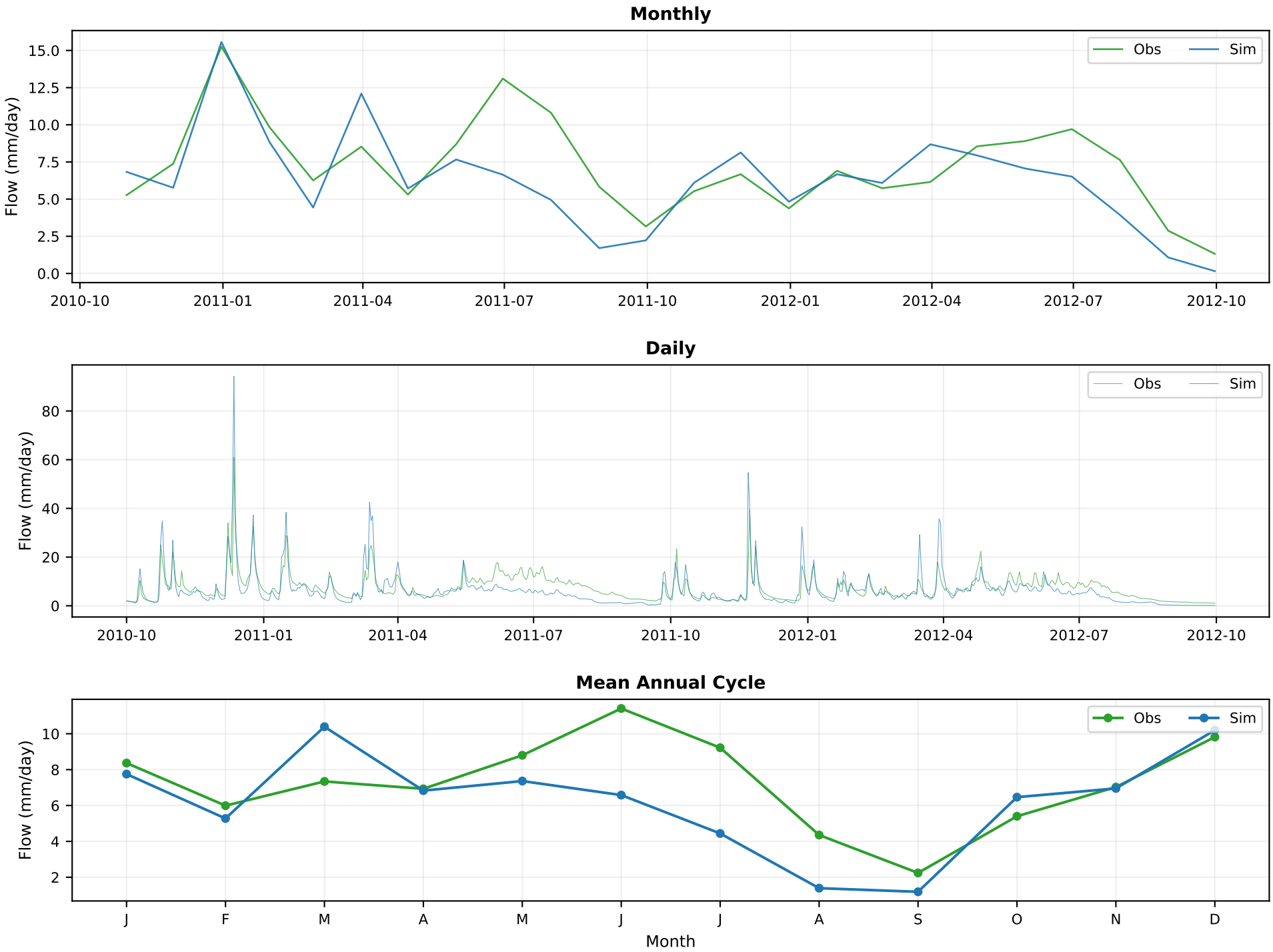

ID: 2069700
KGE=0.604 r=0.877 α=1.211 NSE=0.558 Bias=+31.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

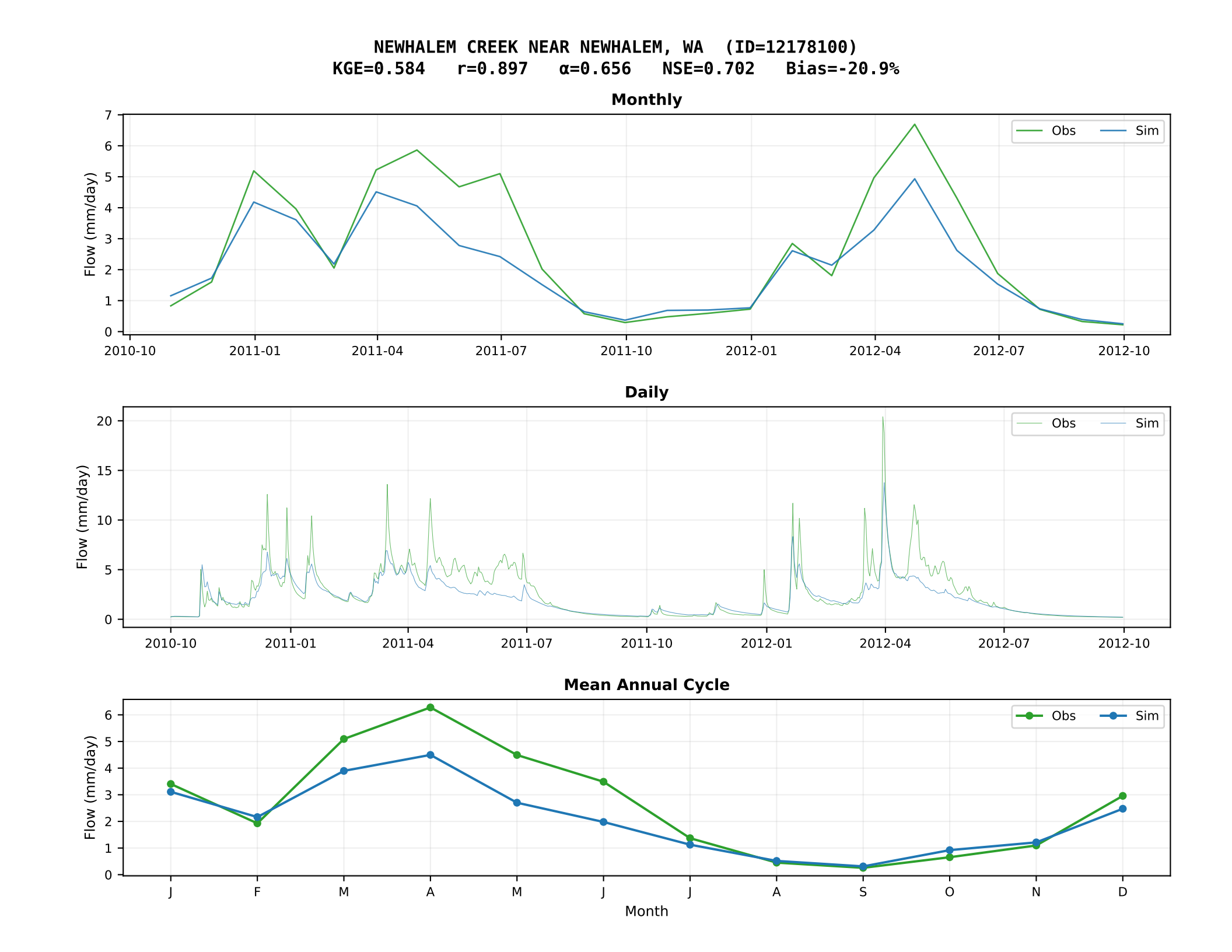

ID: 2349900
KGE=0.600 r=0.910 α=0.672 NSE=0.723 Bias=-21.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

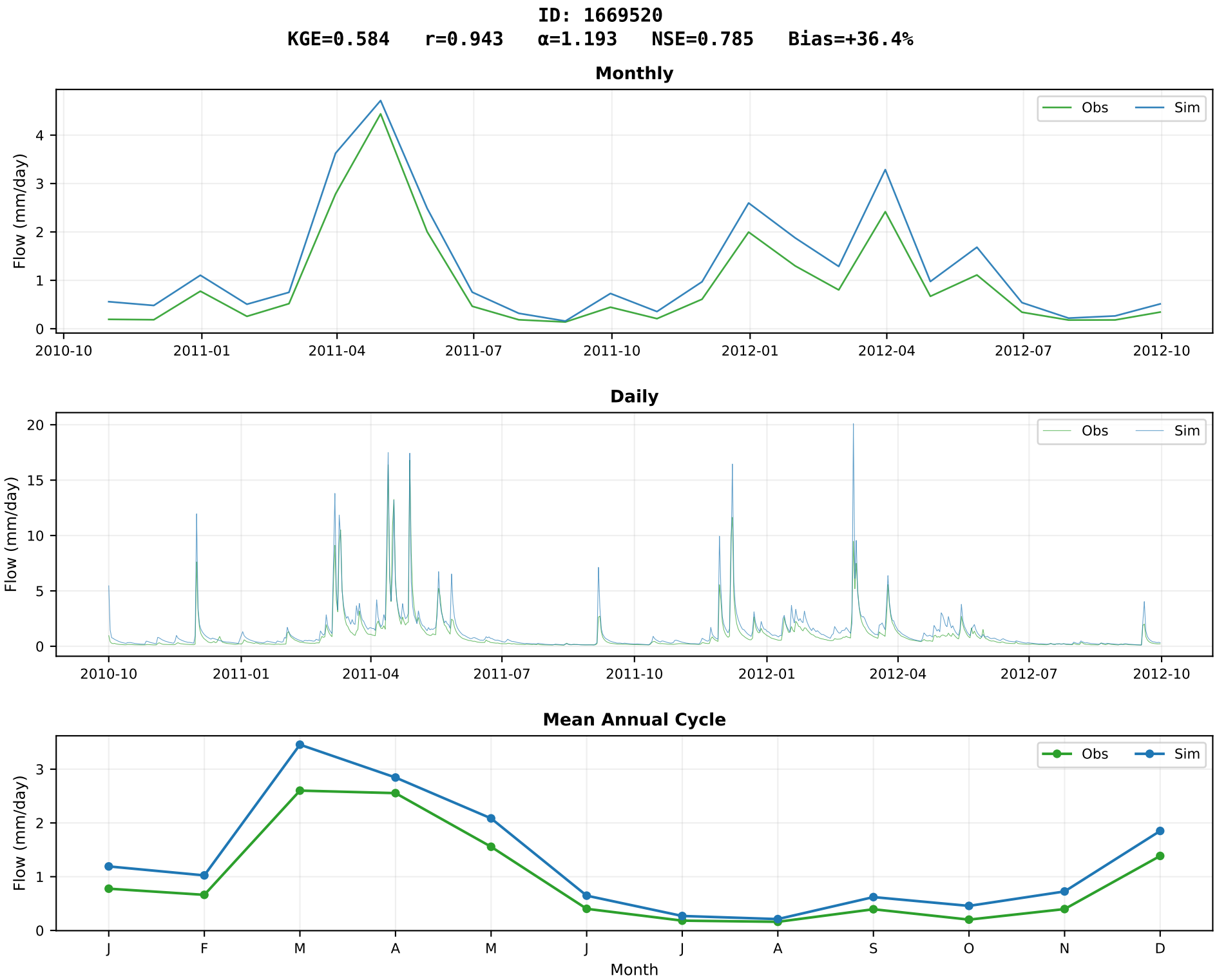

ID: 2140991
KGE=0.599 r=0.894 α=0.906 NSE=0.732 Bias=+37.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

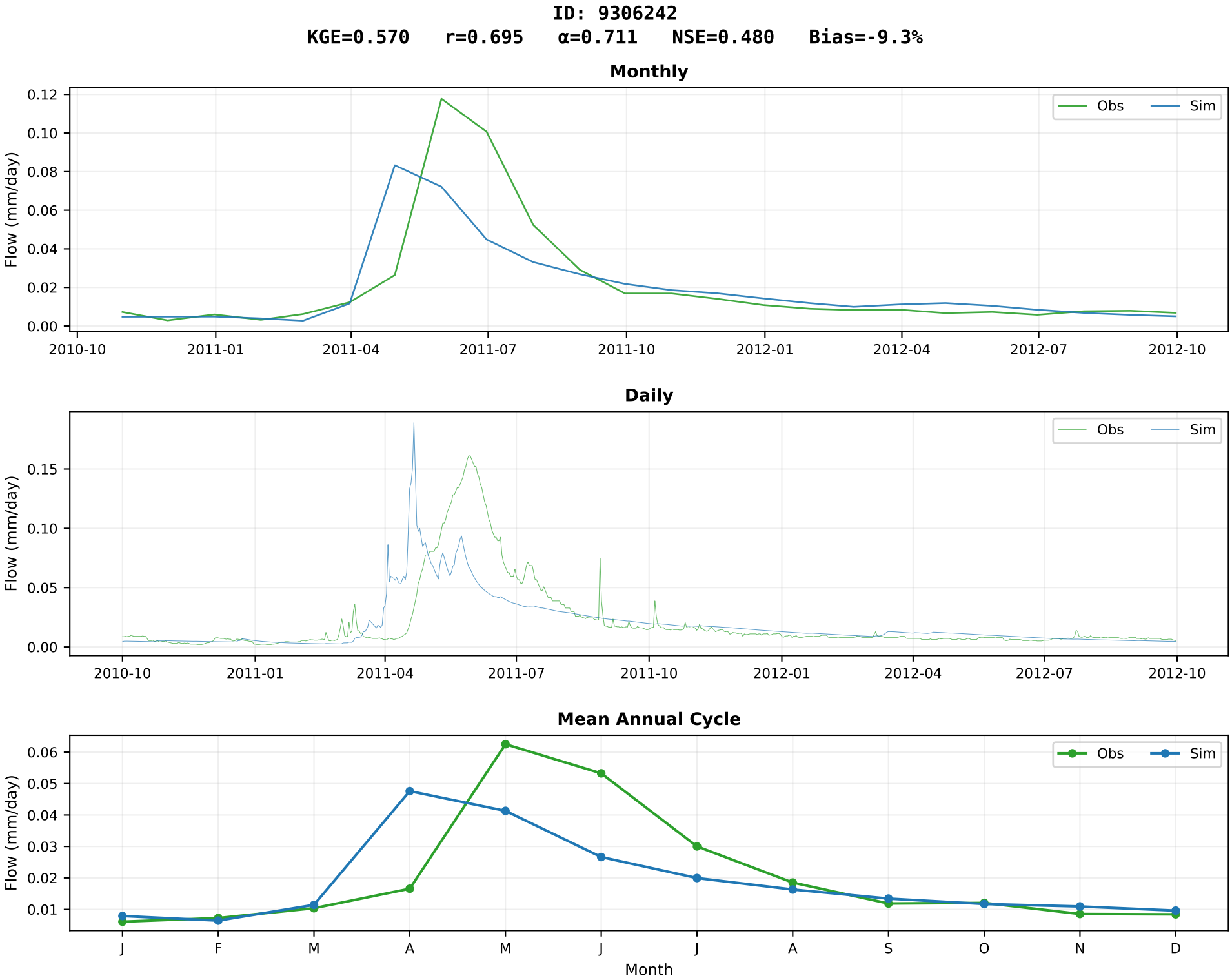

ID: 3049800
KGE=0.598 r=0.772 α=0.937 NSE=0.501 Bias=+32.5%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

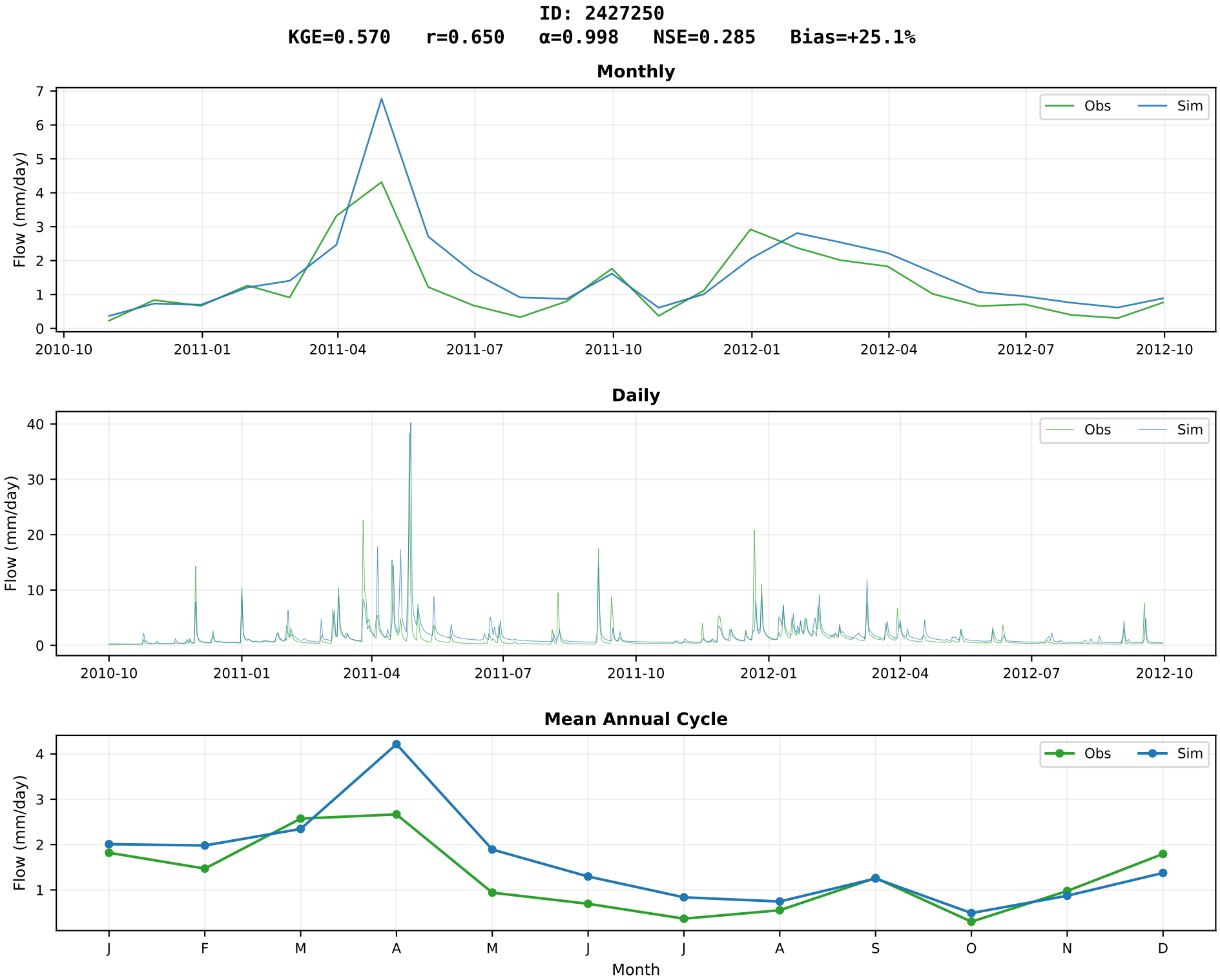

ID: 7145700
KGE=0.596 r=0.889 α=1.388 NSE=0.542 Bias=+0.8%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.1
0.2
0.3
J
F
M
A
M
J
J
A
S
O
N
D
Month

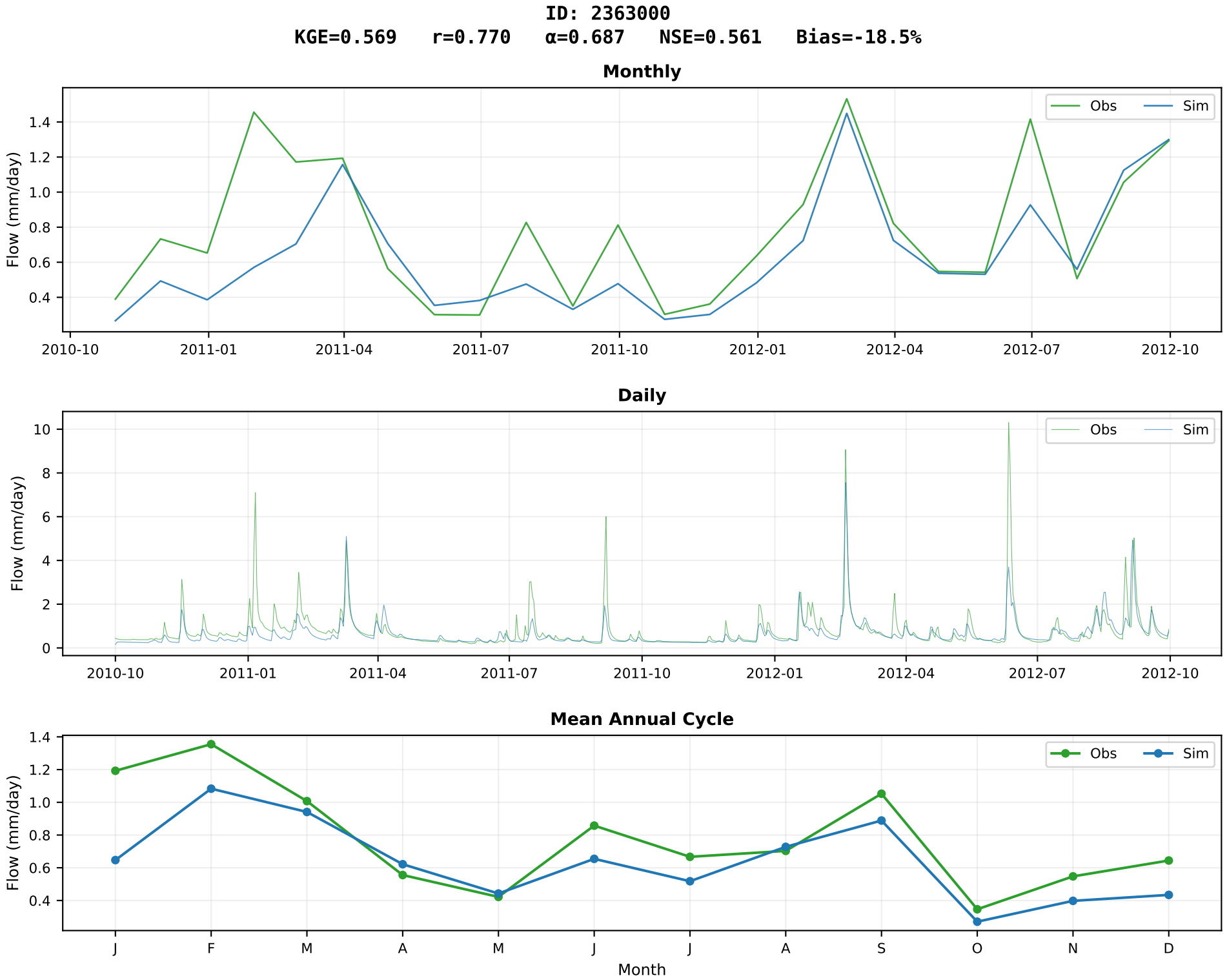

PACIFIC CREEK AT MORAN WY (ID=13011500)
KGE=0.585 r=0.812 α=1.343 NSE=0.341 Bias=-13.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

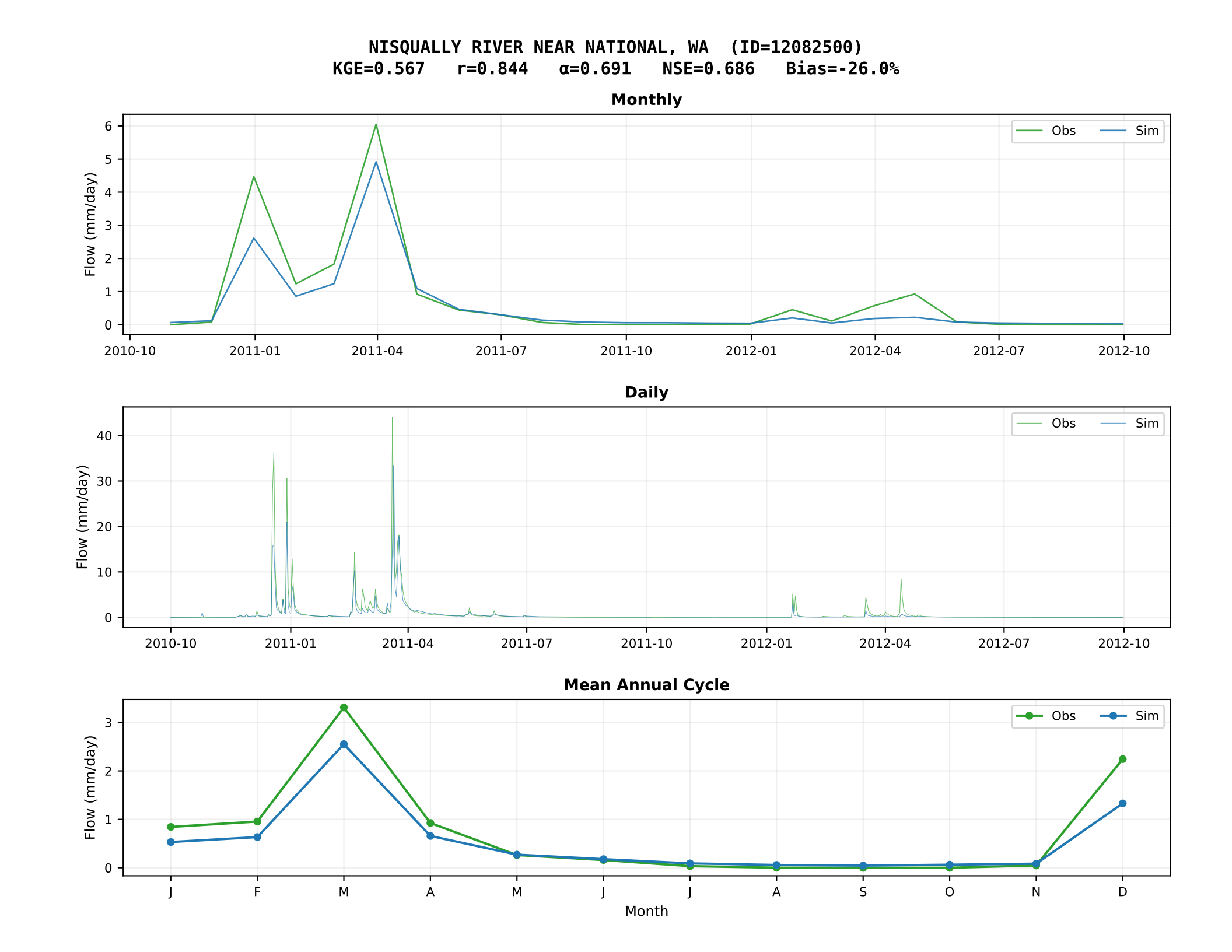

NEWHALEM CREEK NEAR NEWHALEM, WA (ID=12178100)
KGE=0.584 r=0.897 α=0.656 NSE=0.702 Bias=-20.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

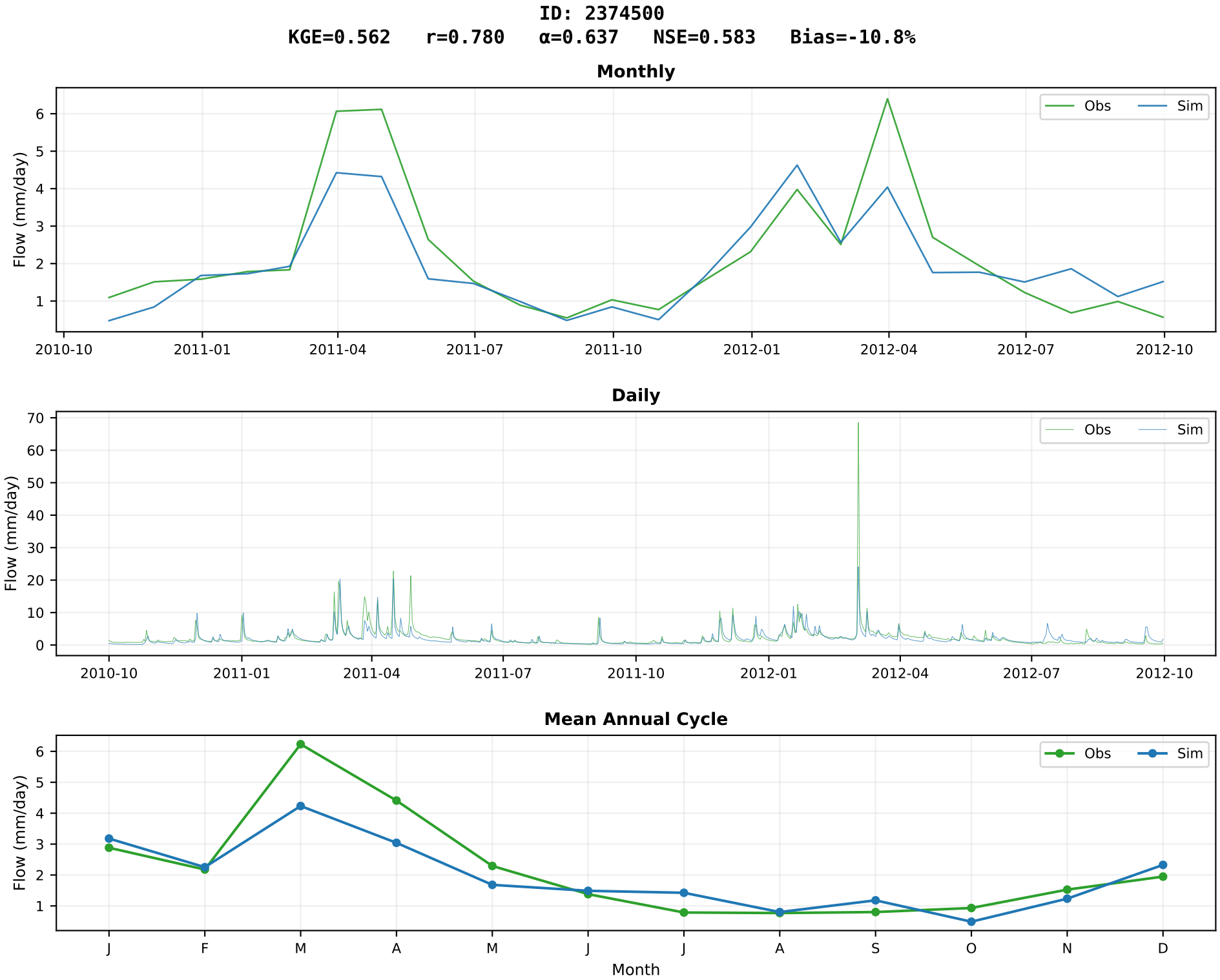

ID: 1669520
KGE=0.584 r=0.943 α=1.193 NSE=0.785 Bias=+36.4%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
J
F
M
A
M
J
J
A
S
O
N
D
Month

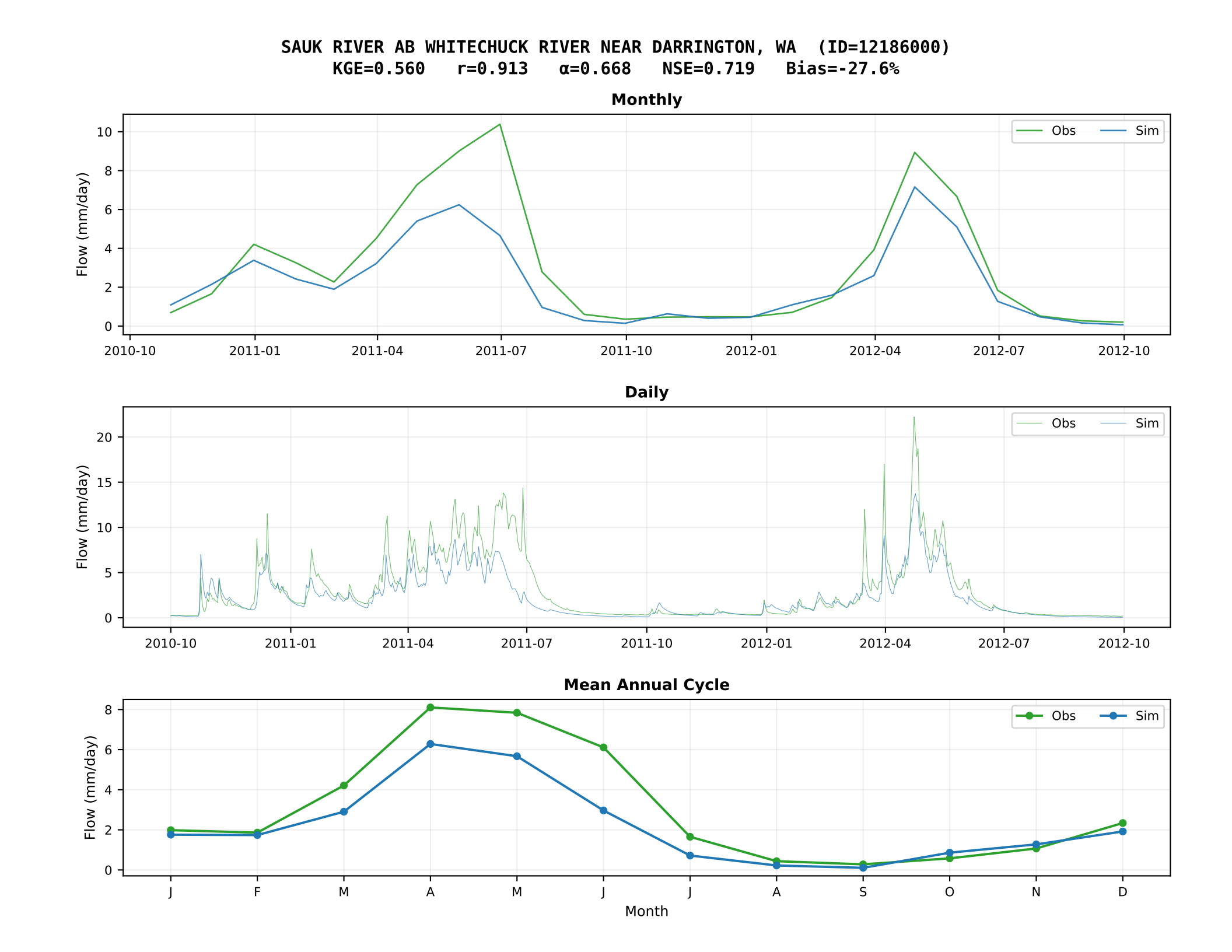

ID: 9306242
KGE=0.570 r=0.695 α=0.711 NSE=0.480 Bias=-9.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

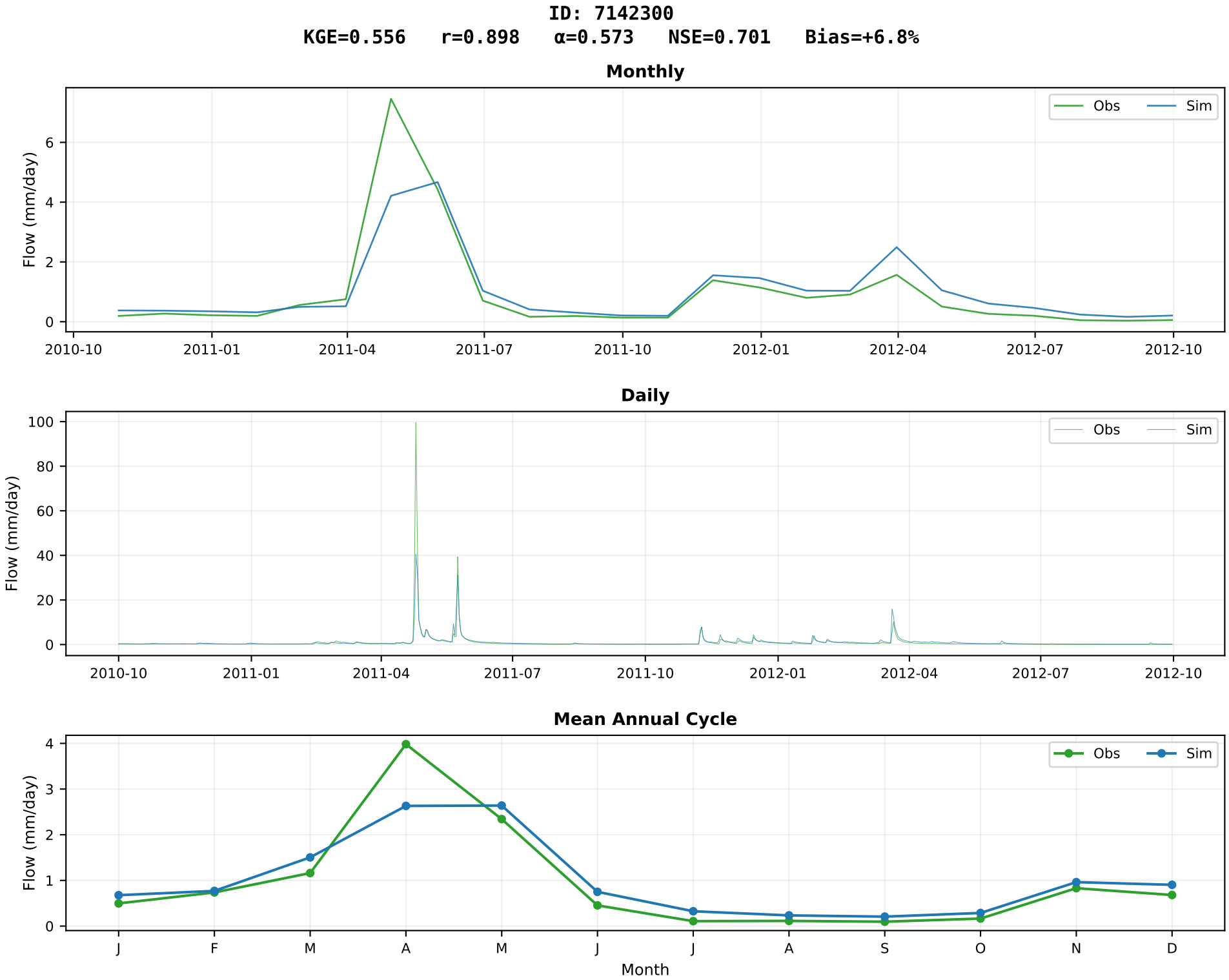

ID: 2427250
KGE=0.570 r=0.650 α=0.998 NSE=0.285 Bias=+25.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

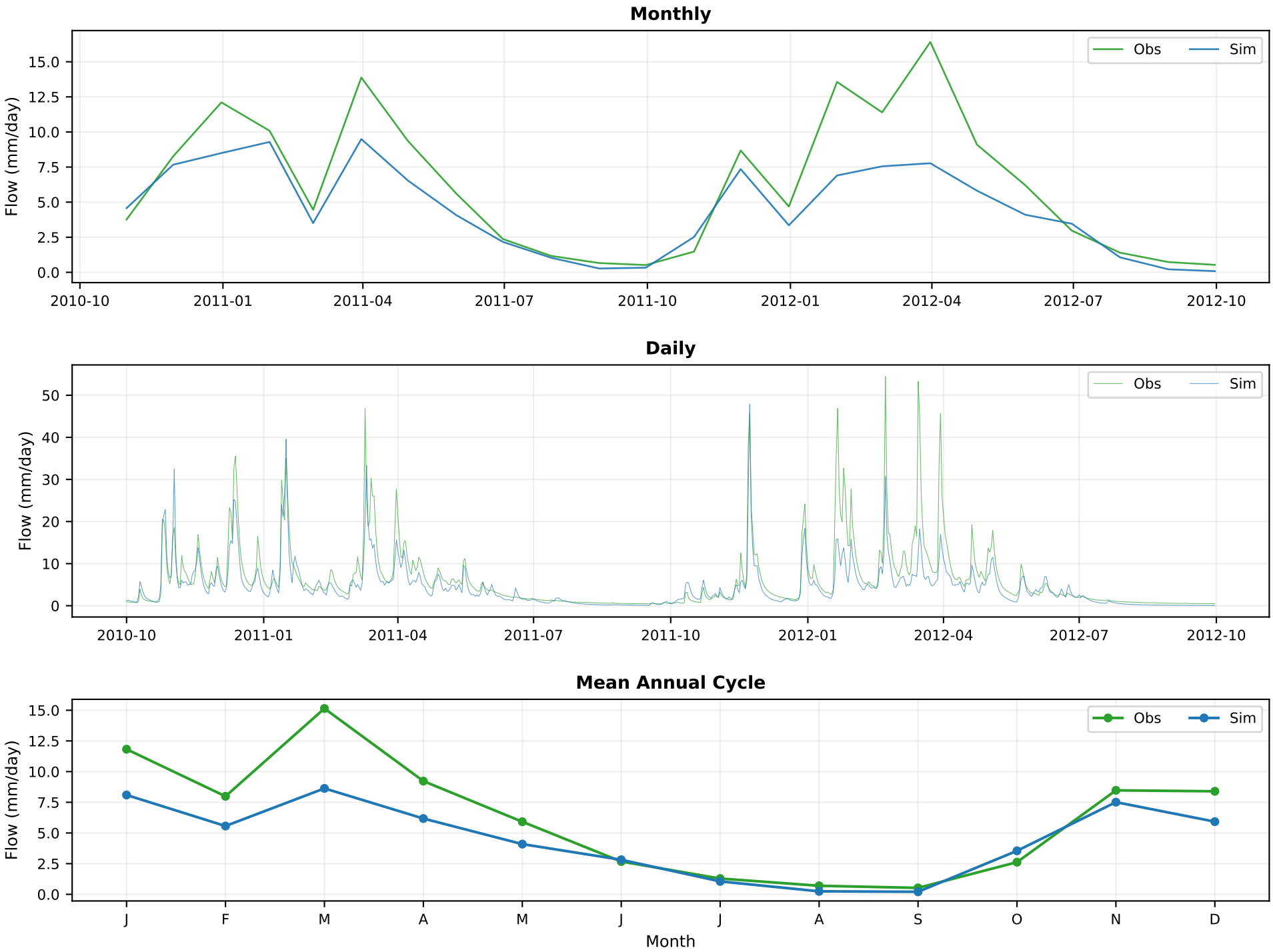

ID: 2363000
KGE=0.569 r=0.770 α=0.687 NSE=0.561 Bias=-18.5%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

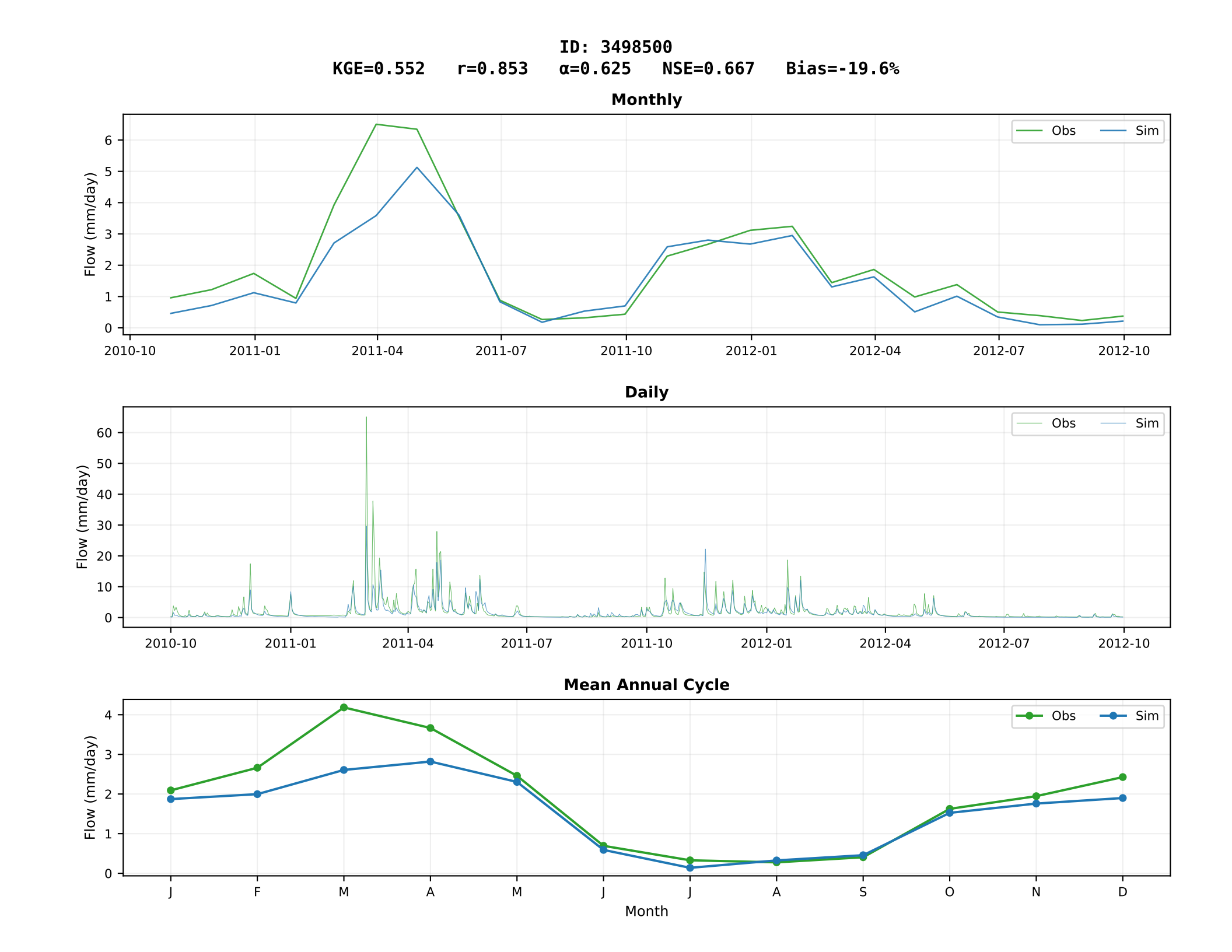

NISQUALLY RIVER NEAR NATIONAL, WA (ID=12082500)
KGE=0.567 r=0.844 α=0.691 NSE=0.686 Bias=-26.0%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D
Month

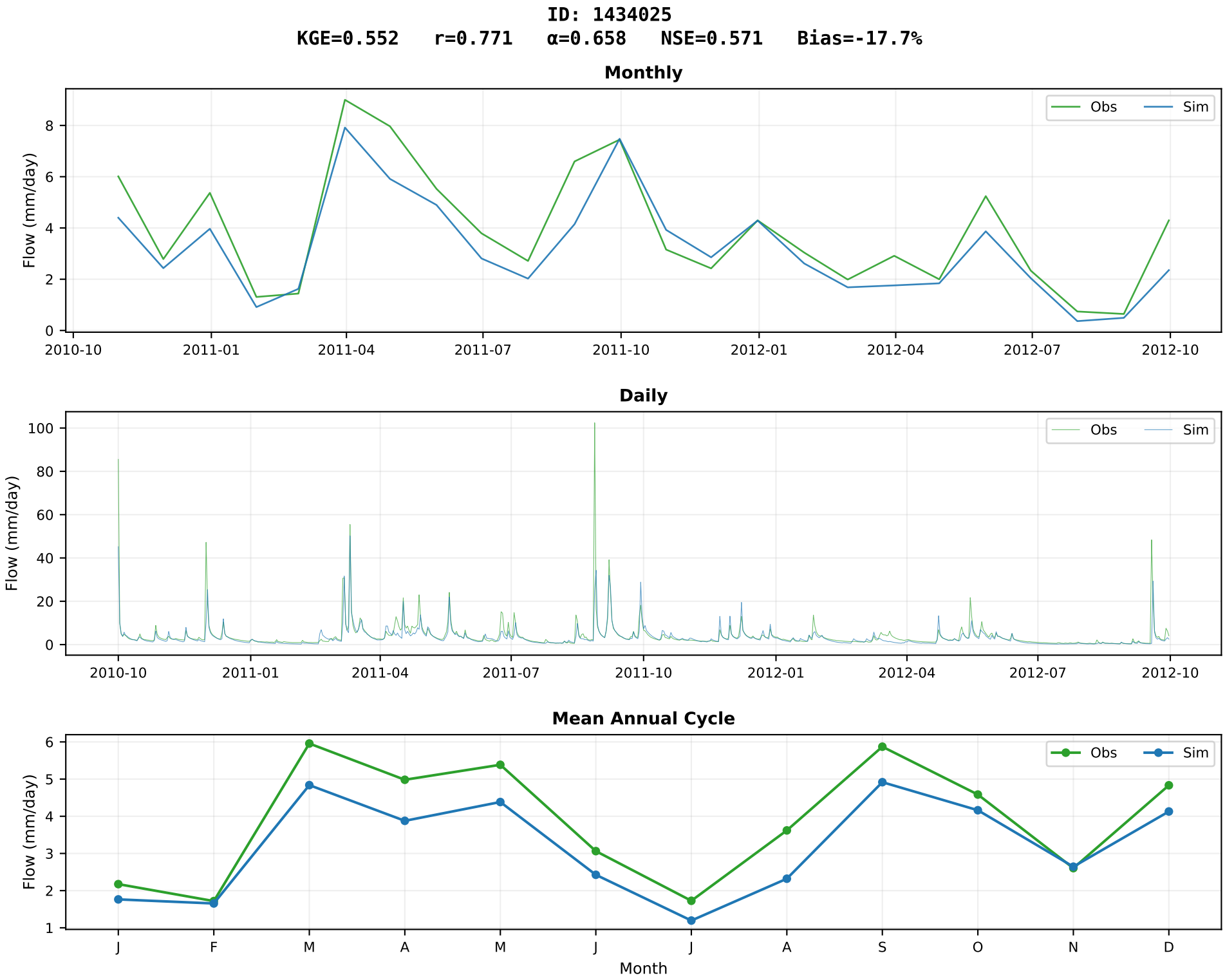

ID: 2374500
KGE=0.562 r=0.780 α=0.637 NSE=0.583 Bias=-10.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

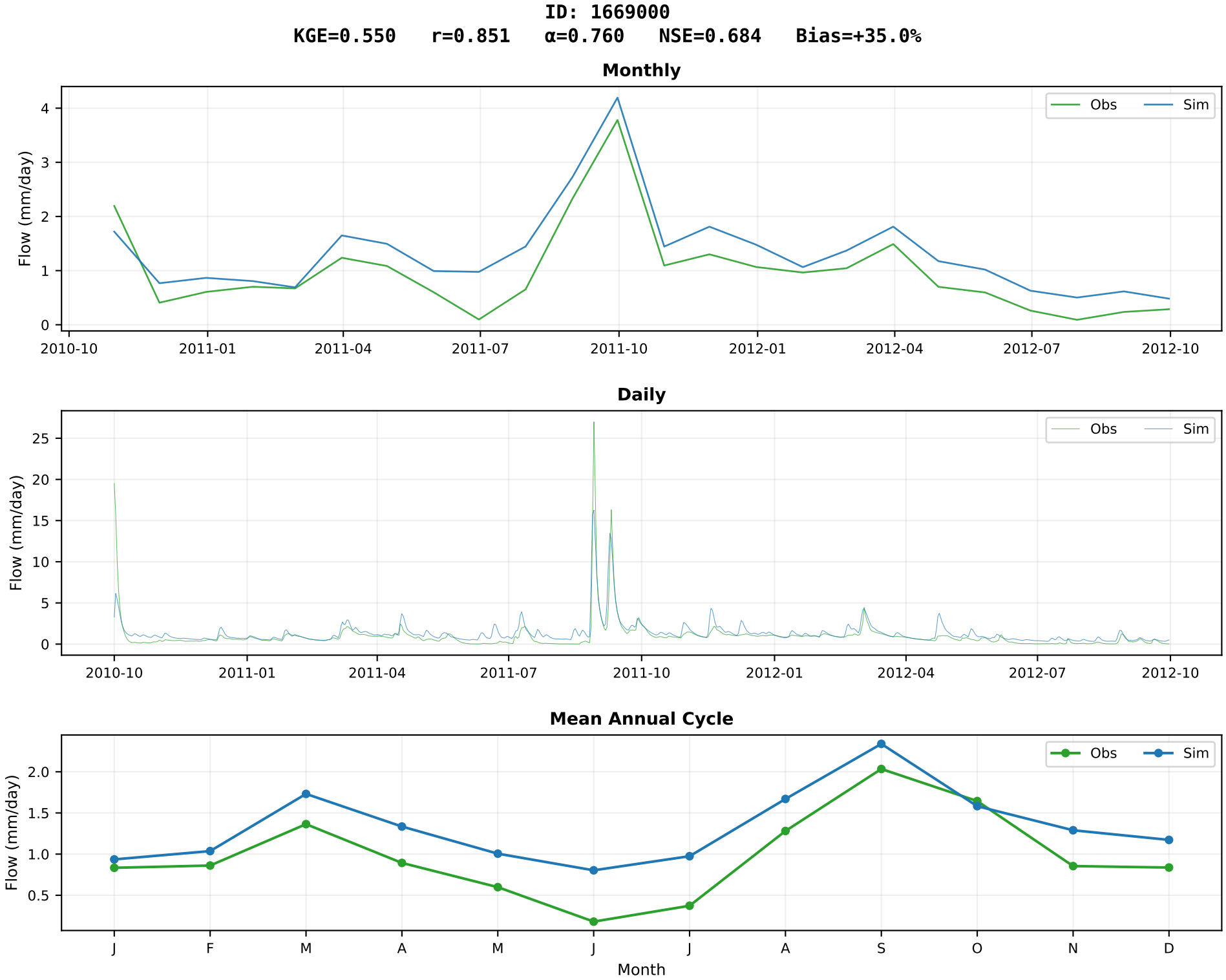

SAUK RIVER AB WHITECHUCK RIVER NEAR DARRINGTON, WA (ID=12186000)
KGE=0.560 r=0.913 α=0.668 NSE=0.719 Bias=-27.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

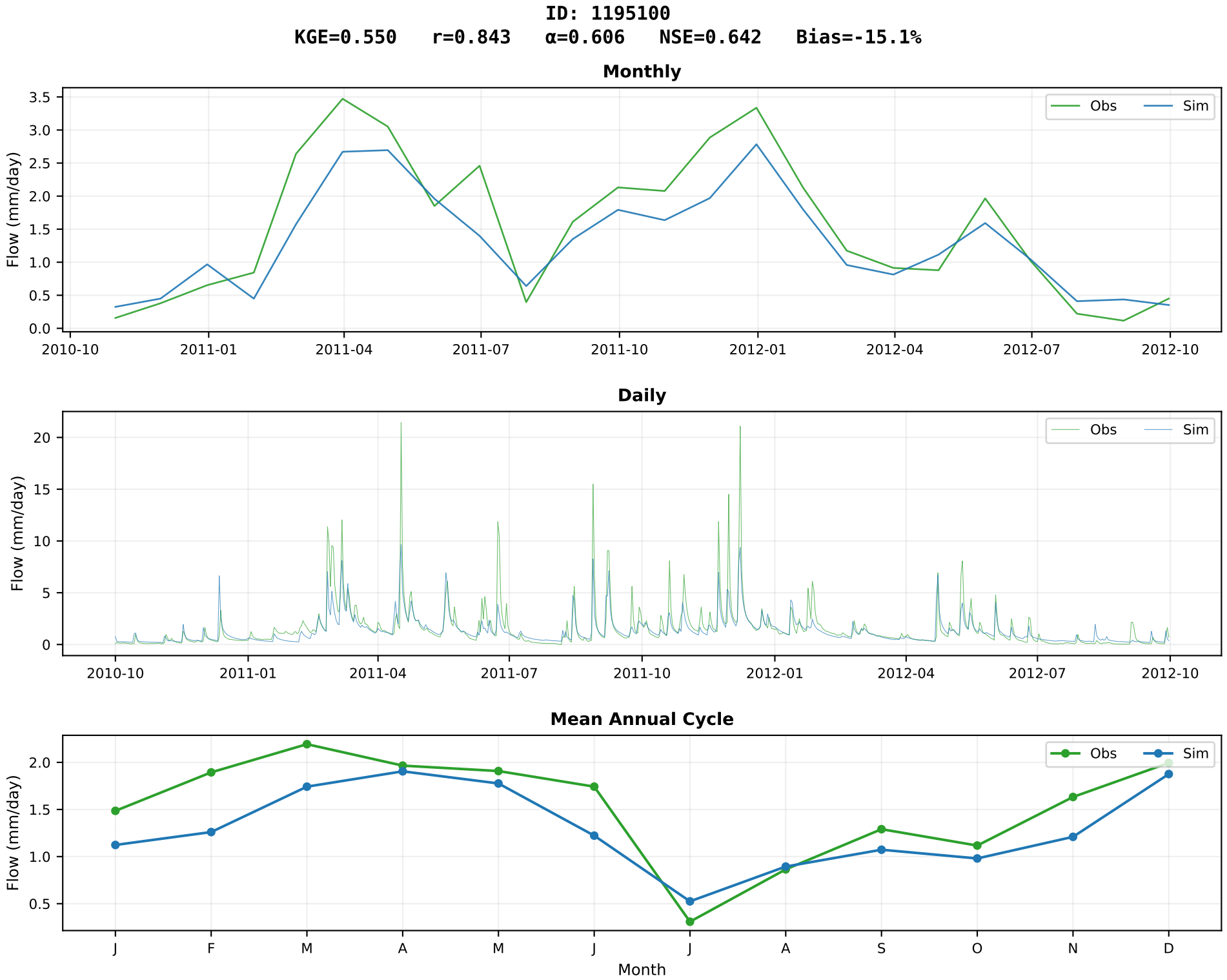

ID: 7142300
KGE=0.556 r=0.898 α=0.573 NSE=0.701 Bias=+6.8%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
100
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month

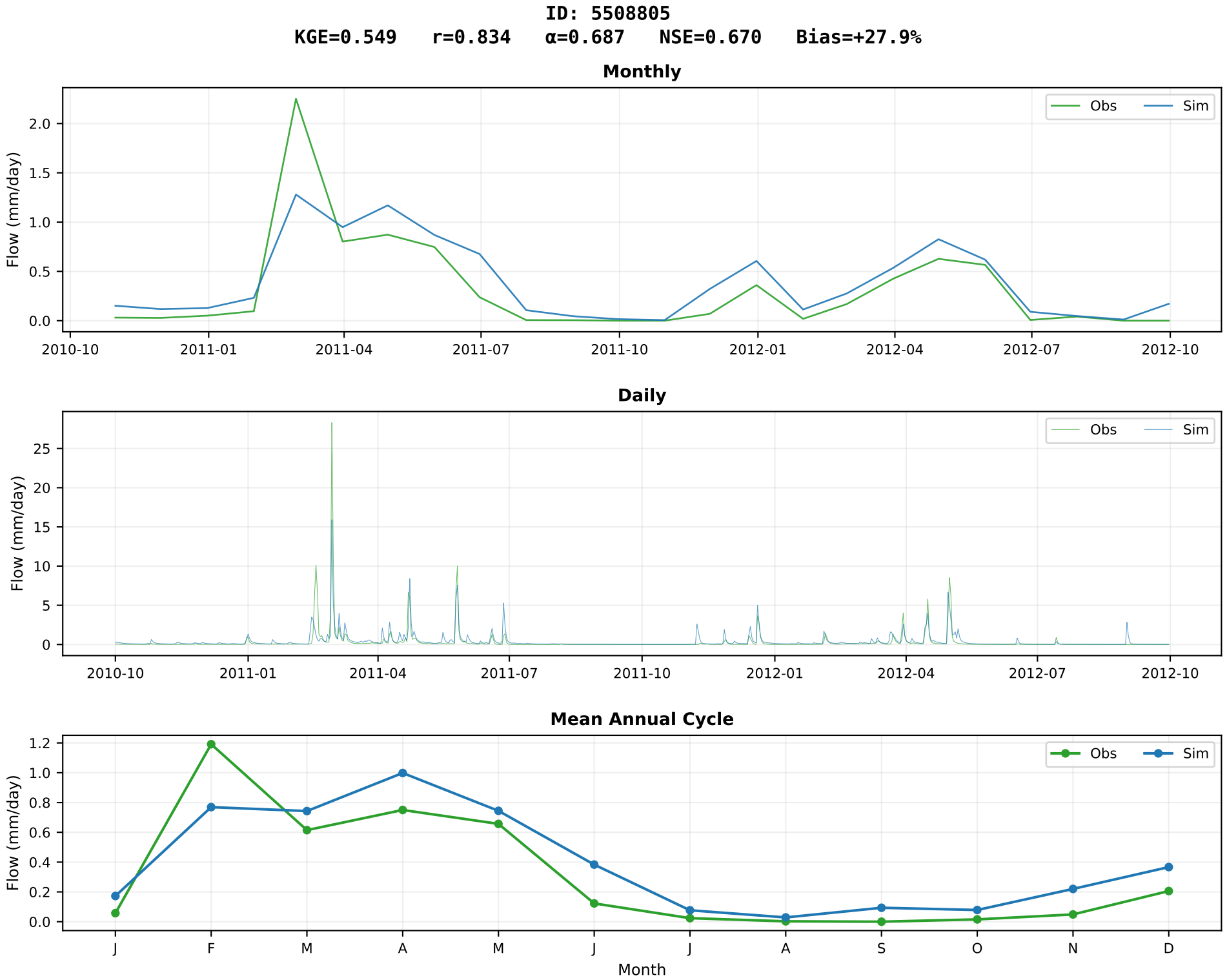

Prospect Creek at Thompson Falls MT (ID=12390700)
KGE=0.555 r=0.881 α=0.675 NSE=0.683 Bias=-28.0%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

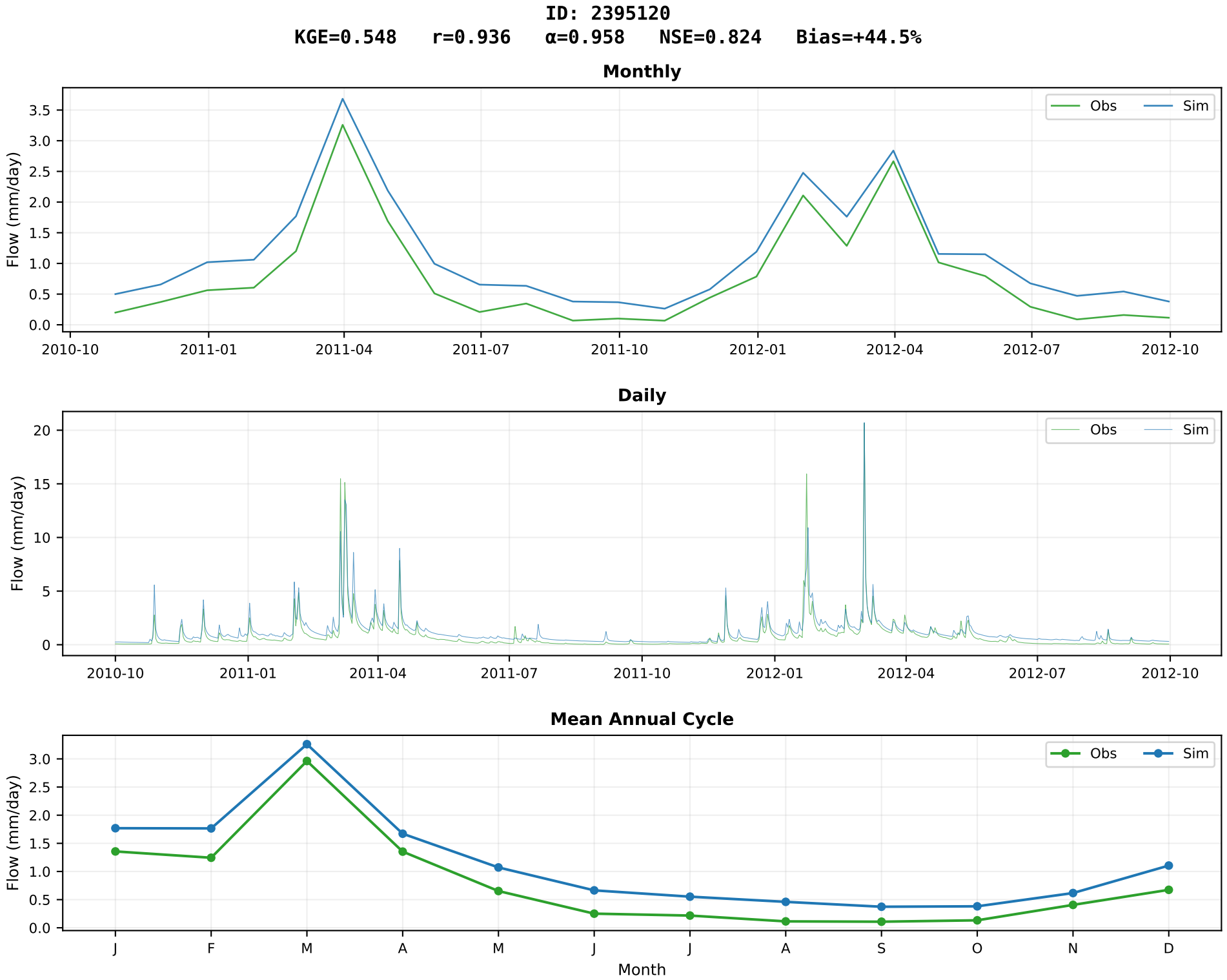

ID: 3498500
KGE=0.552 r=0.853 α=0.625 NSE=0.667 Bias=-19.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

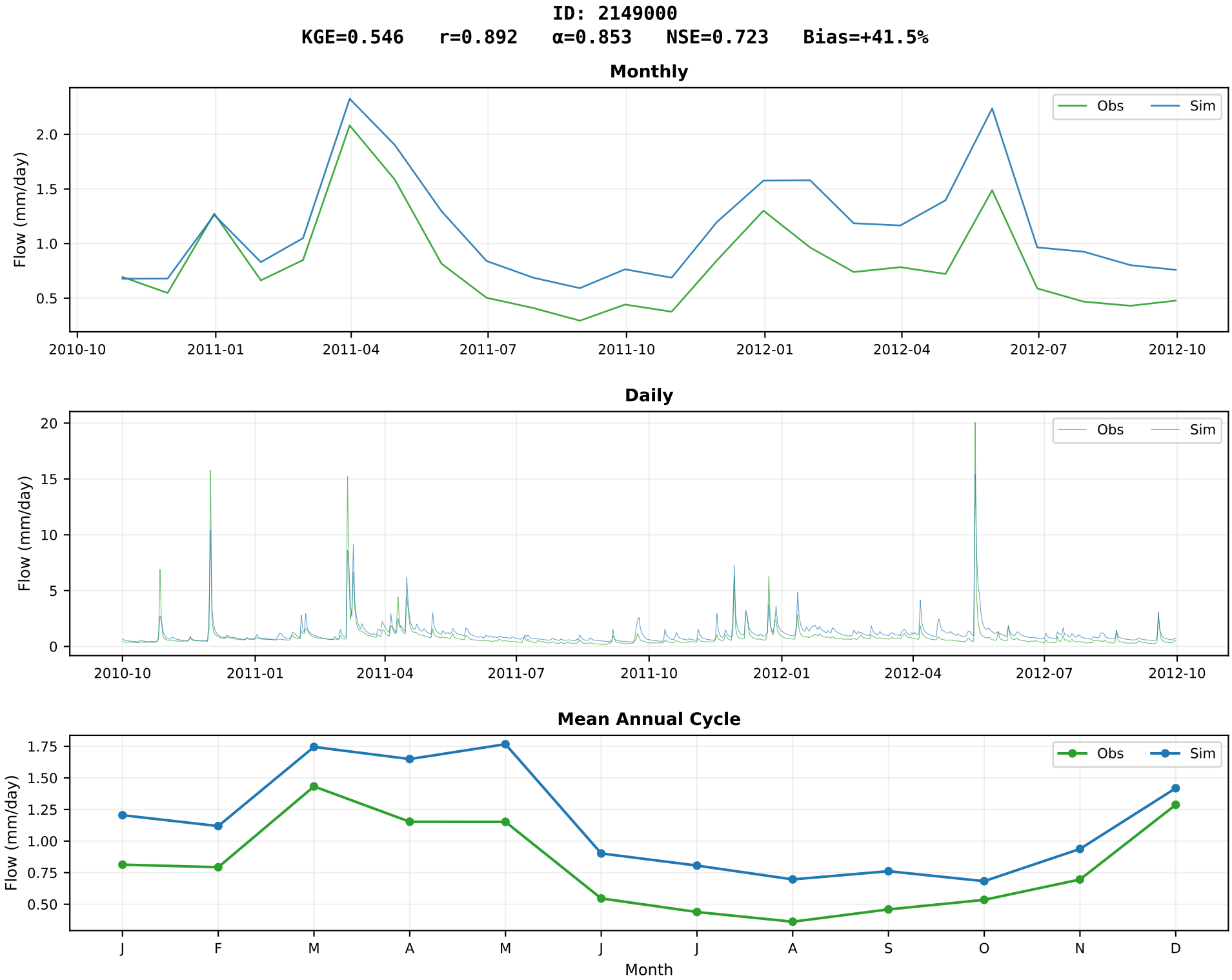

ID: 1434025
KGE=0.552 r=0.771 α=0.658 NSE=0.571 Bias=-17.7%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
20
40
60
80
100
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1
2
3
4
5
6
J
F
M
A
M
J
J
A
S
O
N
D
Month

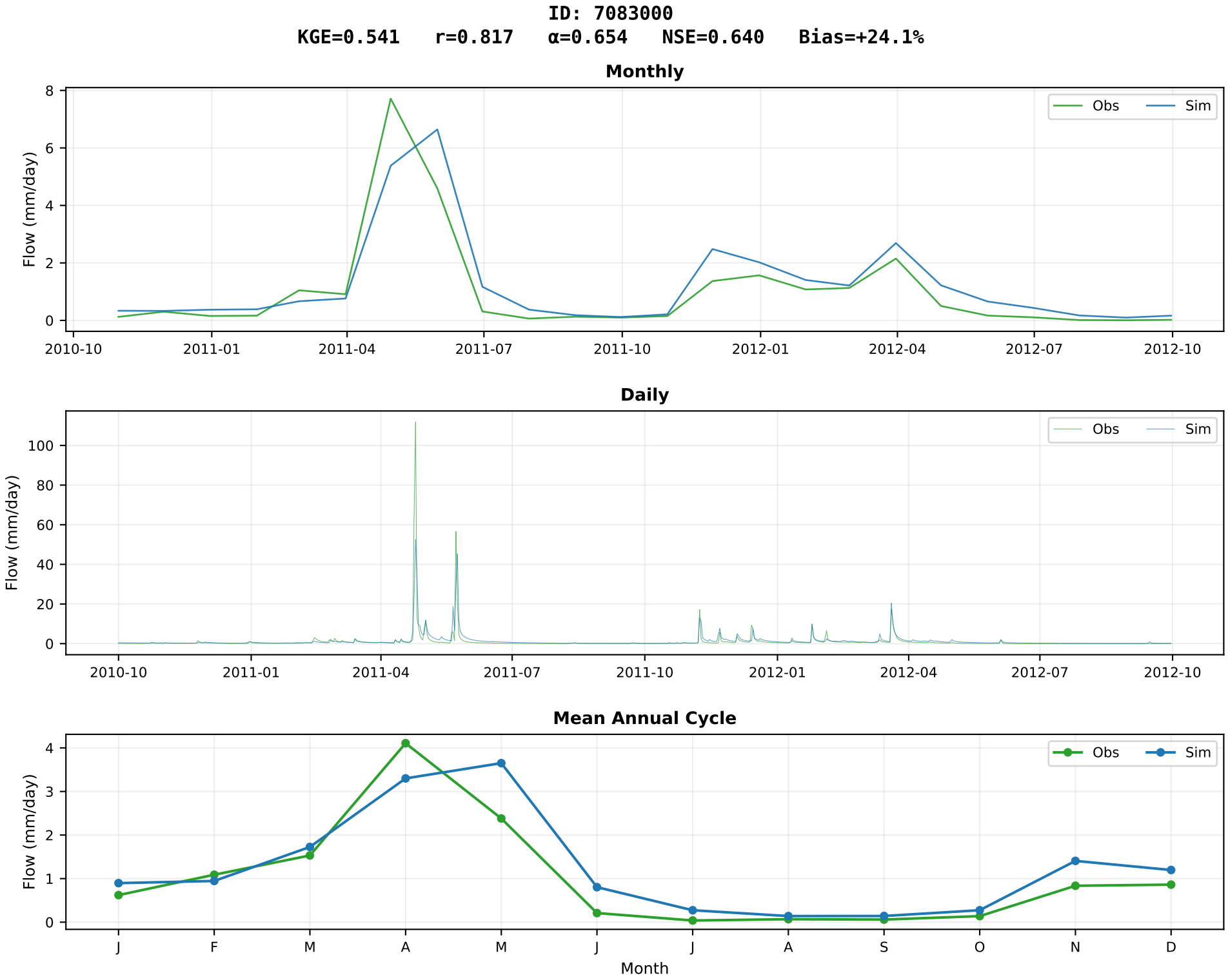

ID: 1669000
KGE=0.550 r=0.851 α=0.760 NSE=0.684 Bias=+35.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

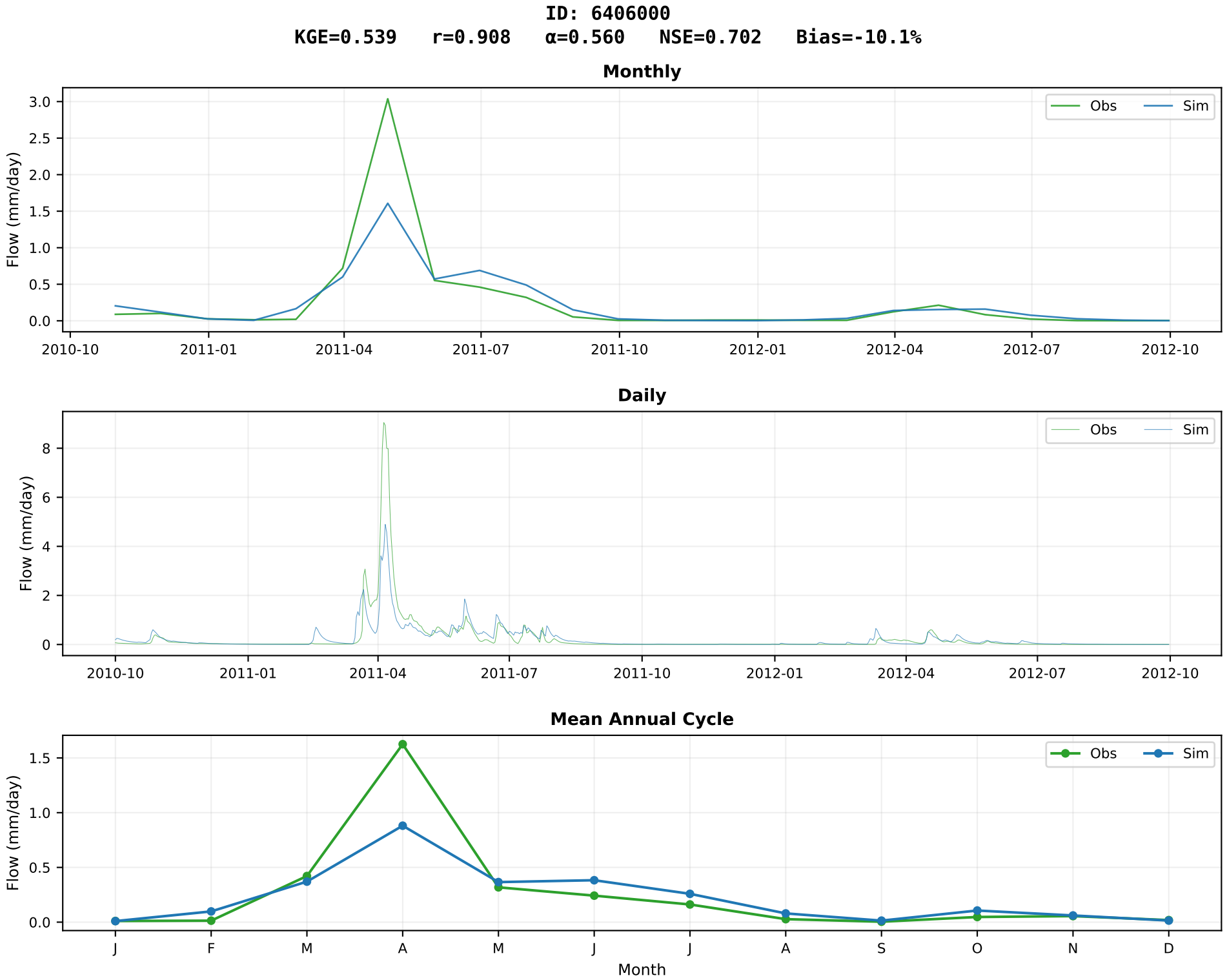

ID: 1195100
KGE=0.550 r=0.843 α=0.606 NSE=0.642 Bias=-15.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

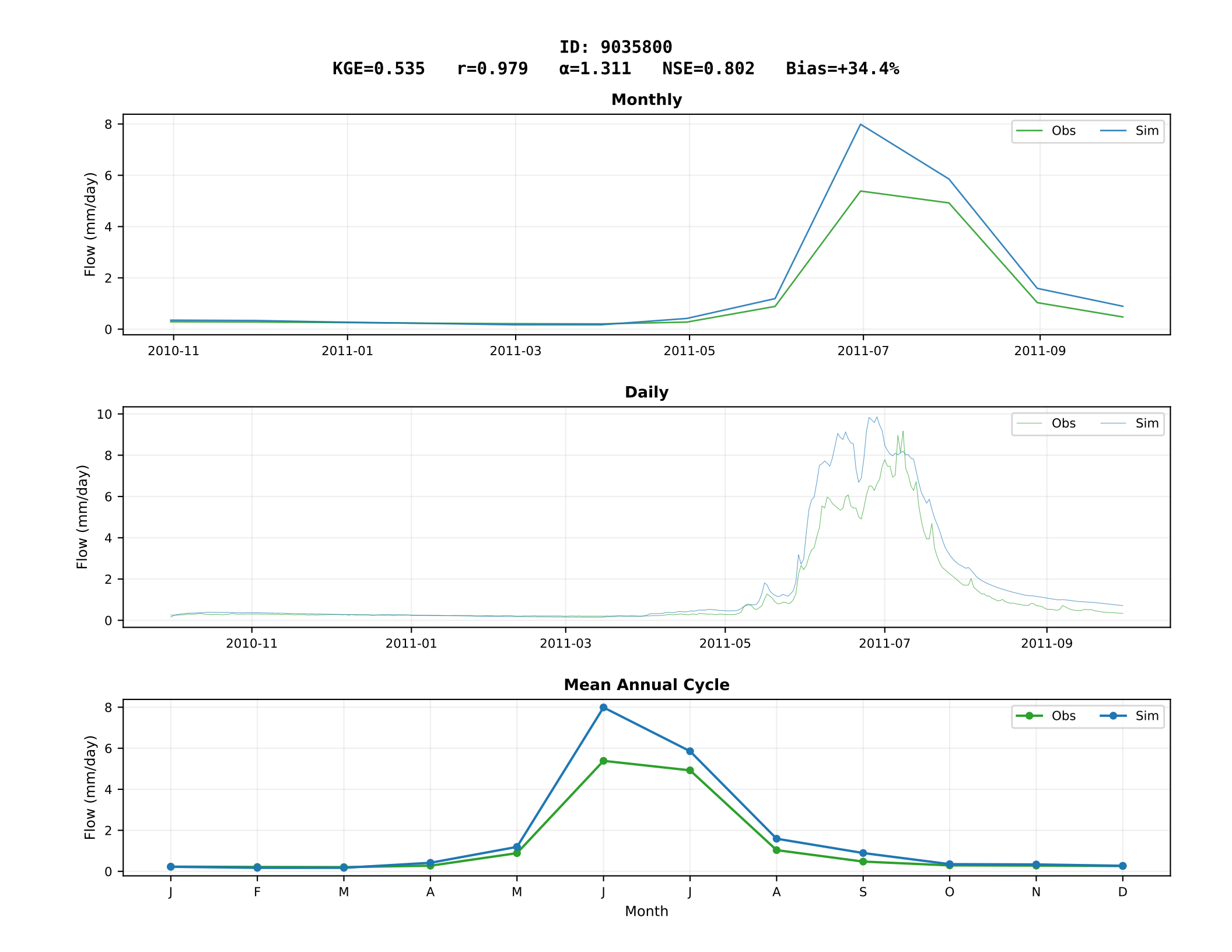

ID: 5508805
KGE=0.549 r=0.834 α=0.687 NSE=0.670 Bias=+27.9%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
25
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

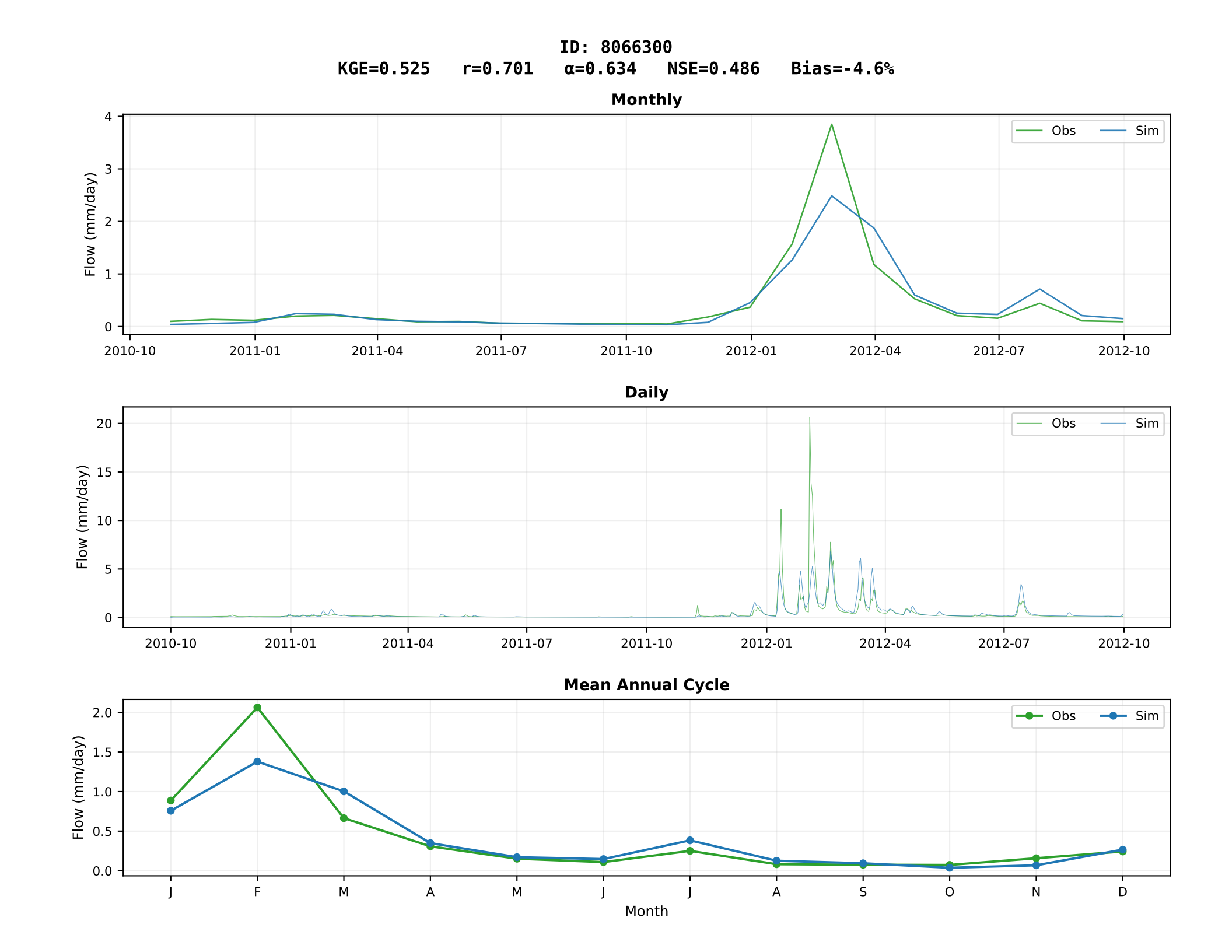

ID: 2395120
KGE=0.548 r=0.936 α=0.958 NSE=0.824 Bias=+44.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

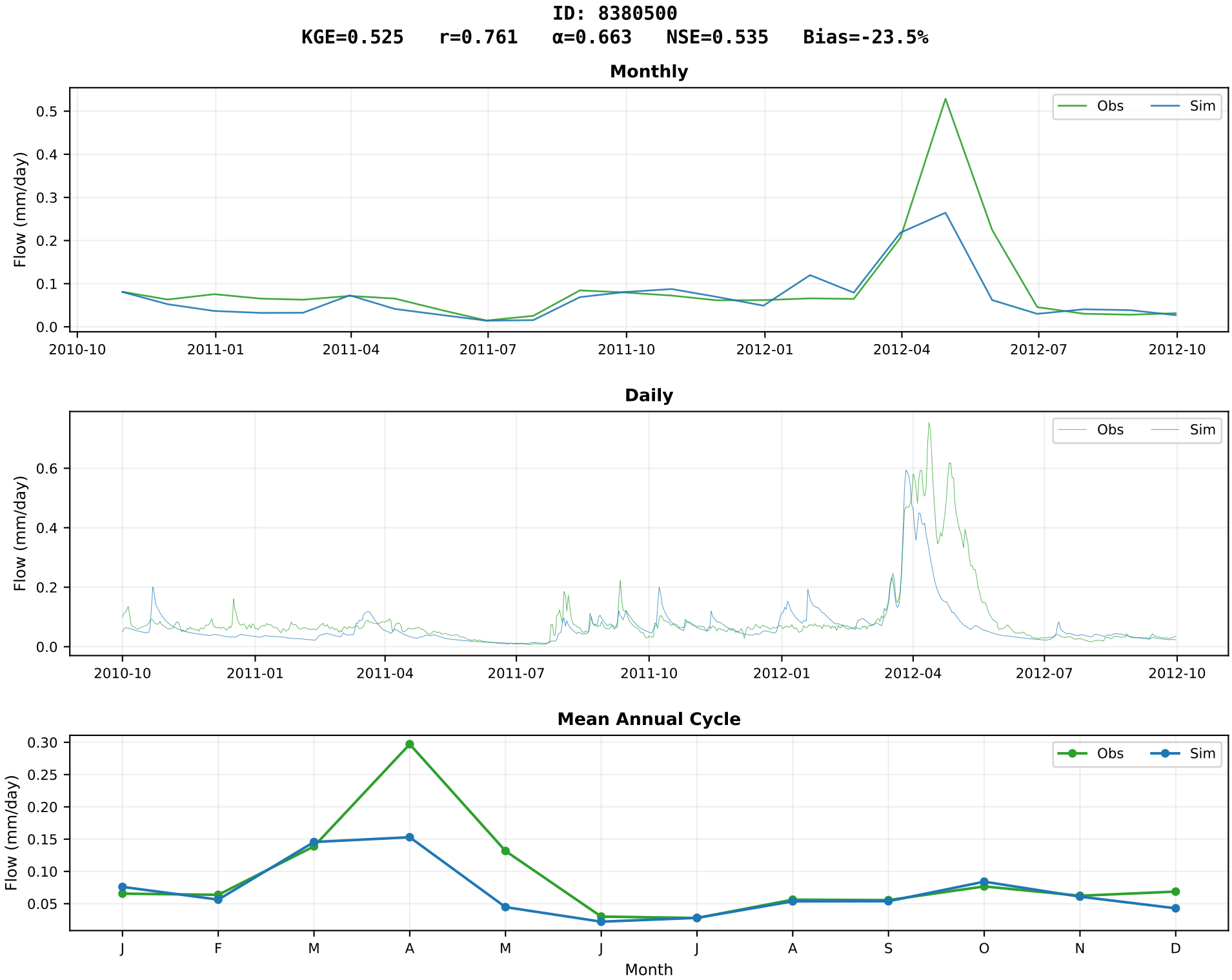

ID: 2149000
KGE=0.546 r=0.892 α=0.853 NSE=0.723 Bias=+41.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

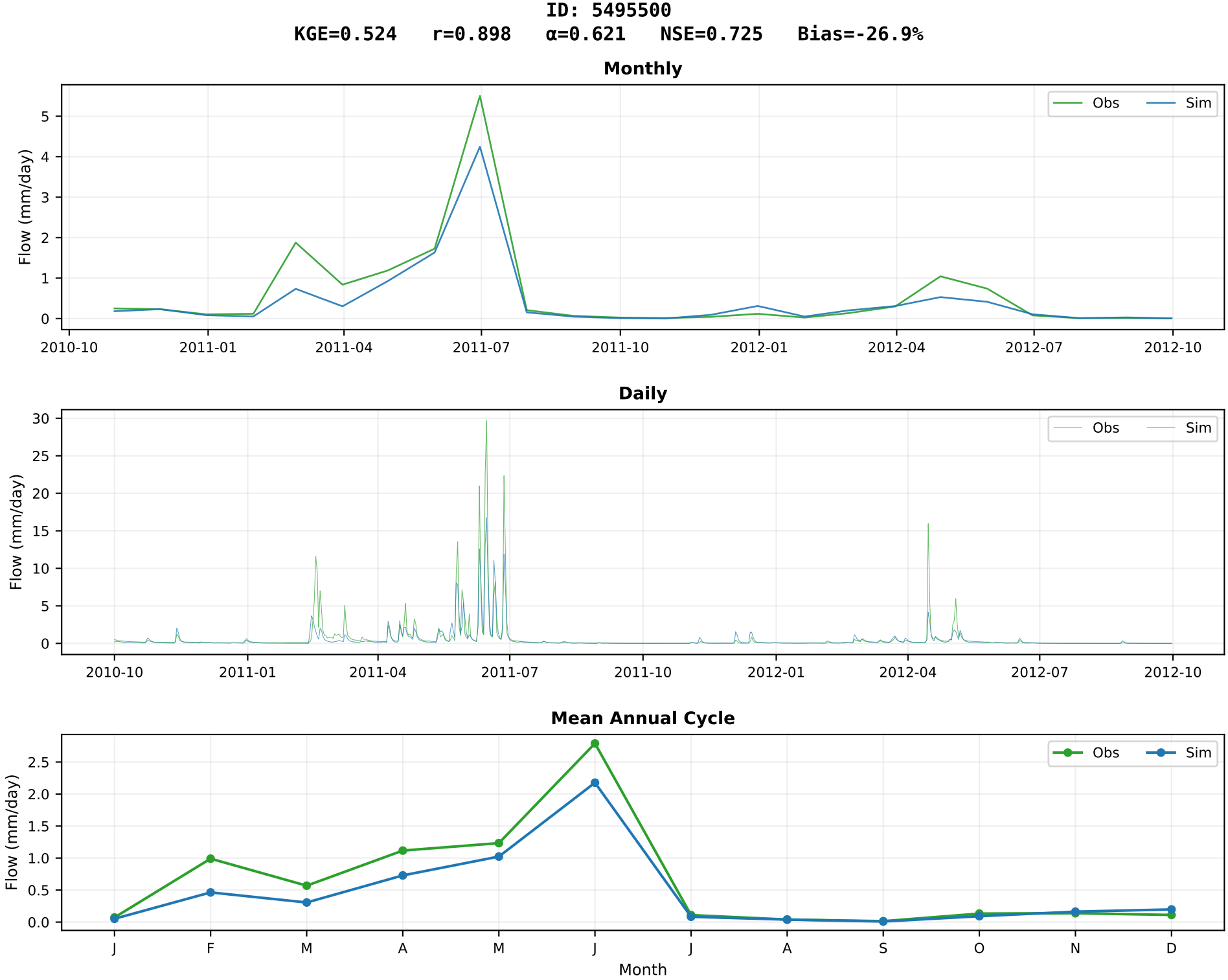

ID: 7083000
KGE=0.541 r=0.817 α=0.654 NSE=0.640 Bias=+24.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 6406000
KGE=0.539 r=0.908 α=0.560 NSE=0.702 Bias=-10.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 9035800
KGE=0.535 r=0.979 α=1.311 NSE=0.802 Bias=+34.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-11
2011-01
2011-03
2011-05
2011-07
2011-09
Daily
Flow (mm/day)
Obs
Sim
2010-11
2011-01
2011-03
2011-05
2011-07
2011-09
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 8066300
KGE=0.525 r=0.701 α=0.634 NSE=0.486 Bias=-4.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

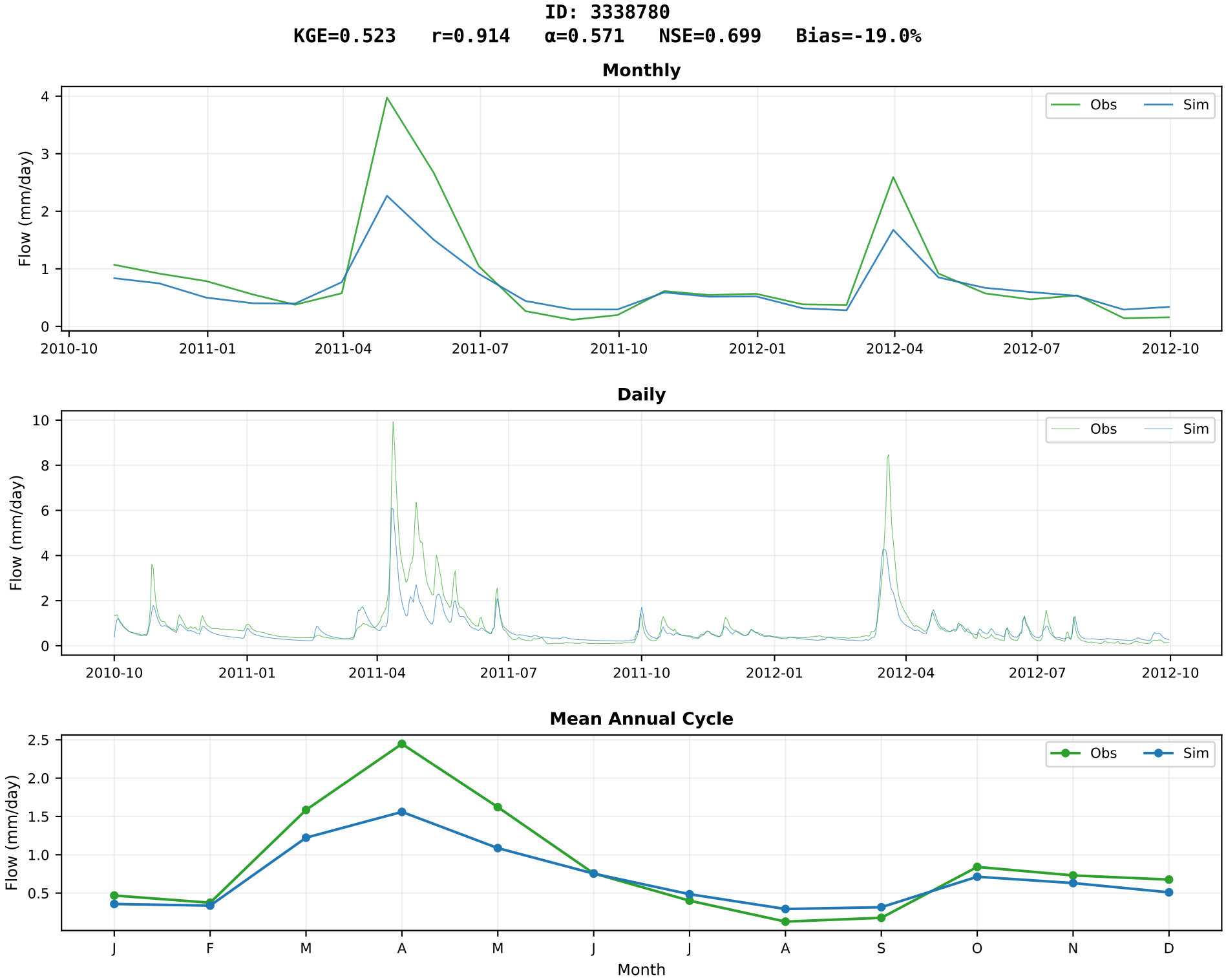

ID: 8380500
KGE=0.525 r=0.761 α=0.663 NSE=0.535 Bias=-23.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

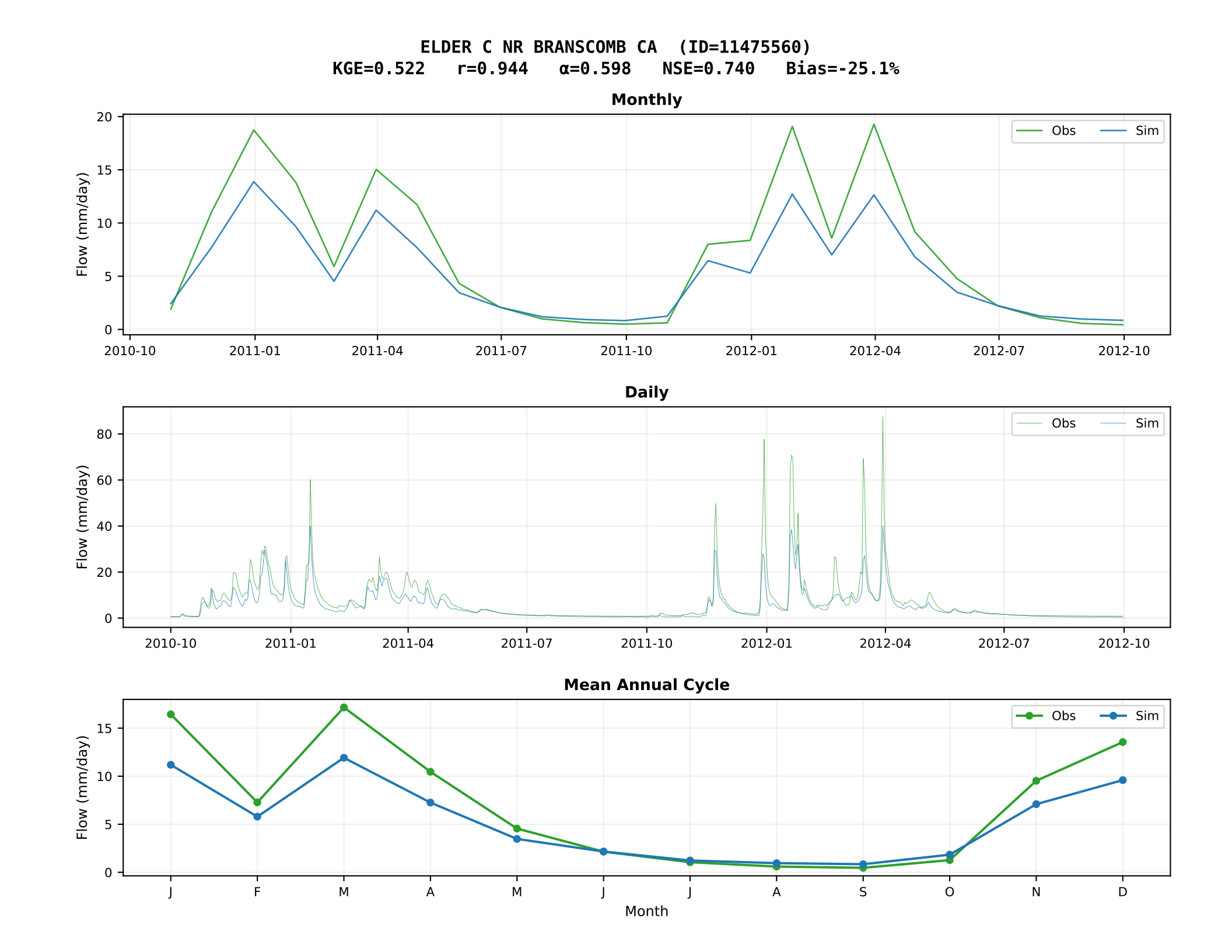

ID: 5495500
KGE=0.524 r=0.898 α=0.621 NSE=0.725 Bias=-26.9%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
25
30
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

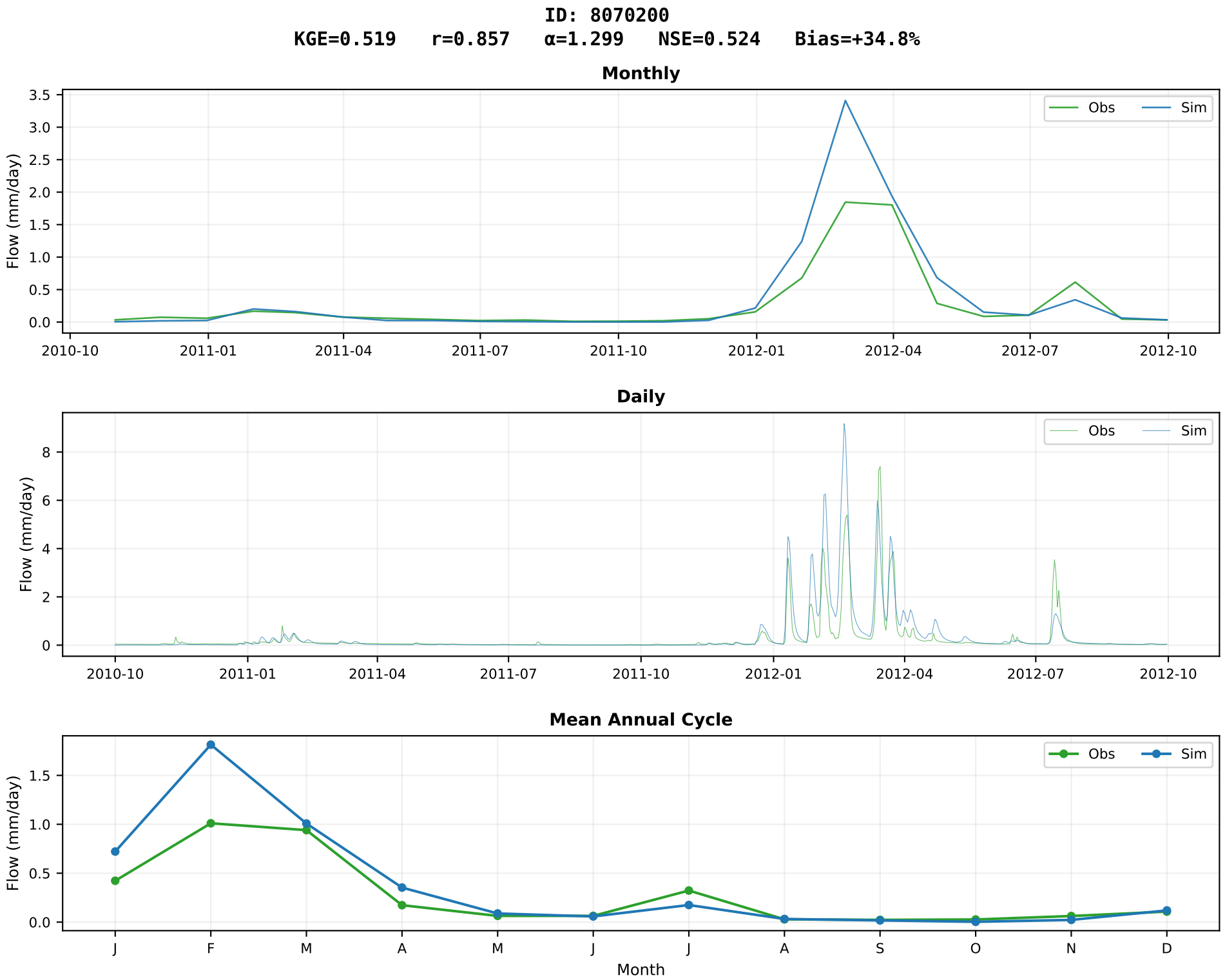
ID: 1169000

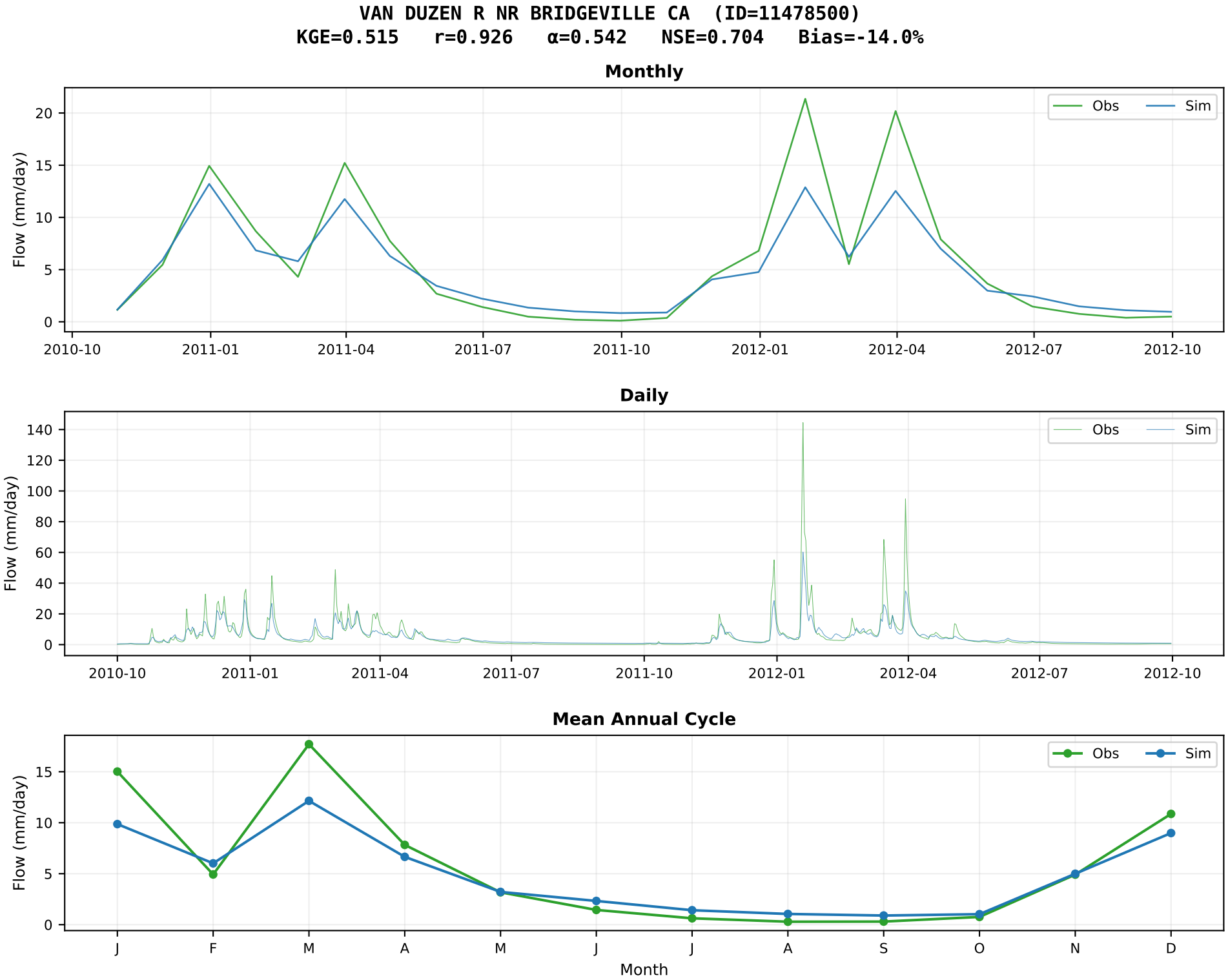
KGE=0.524 r=0.689 α=0.672 NSE=0.467 Bias=-14.9%

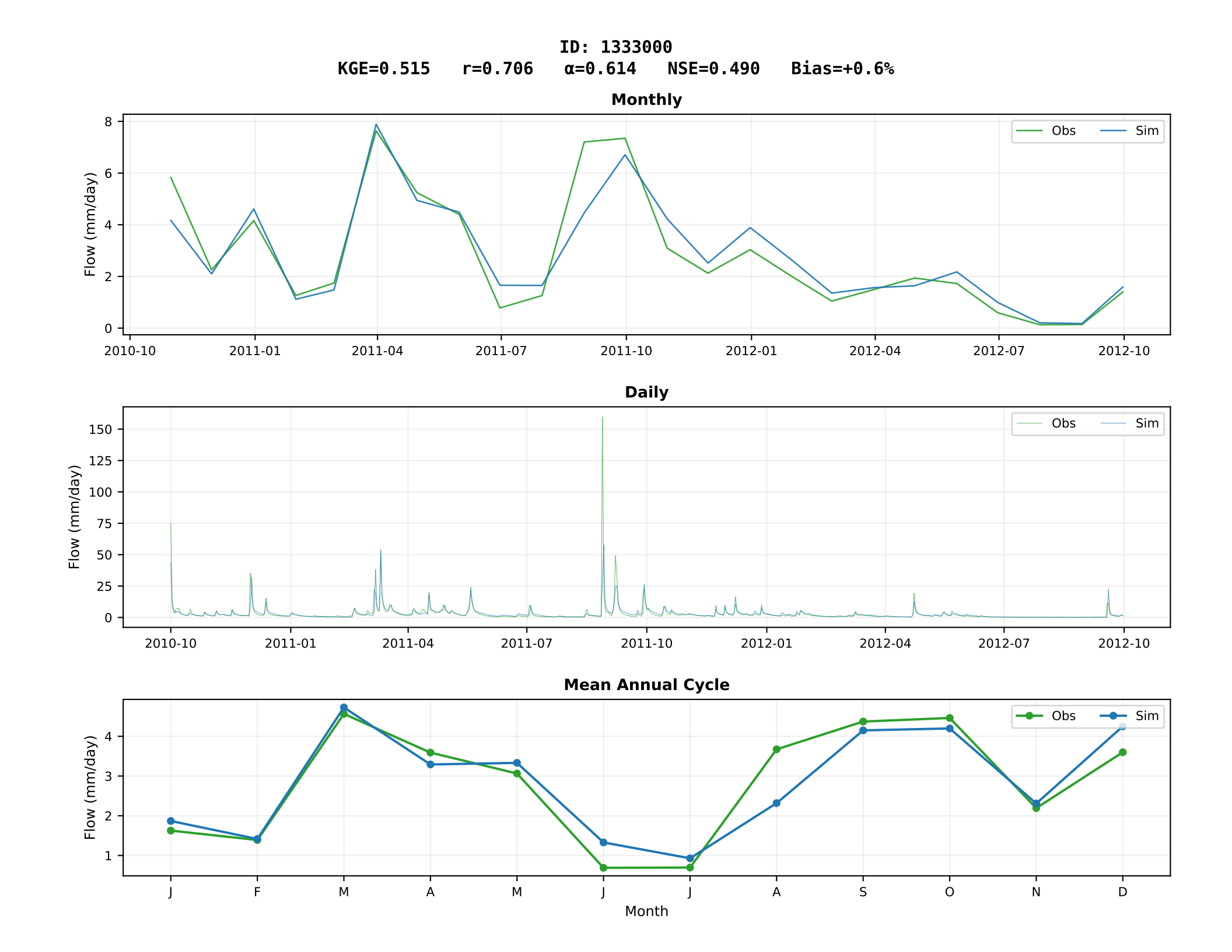


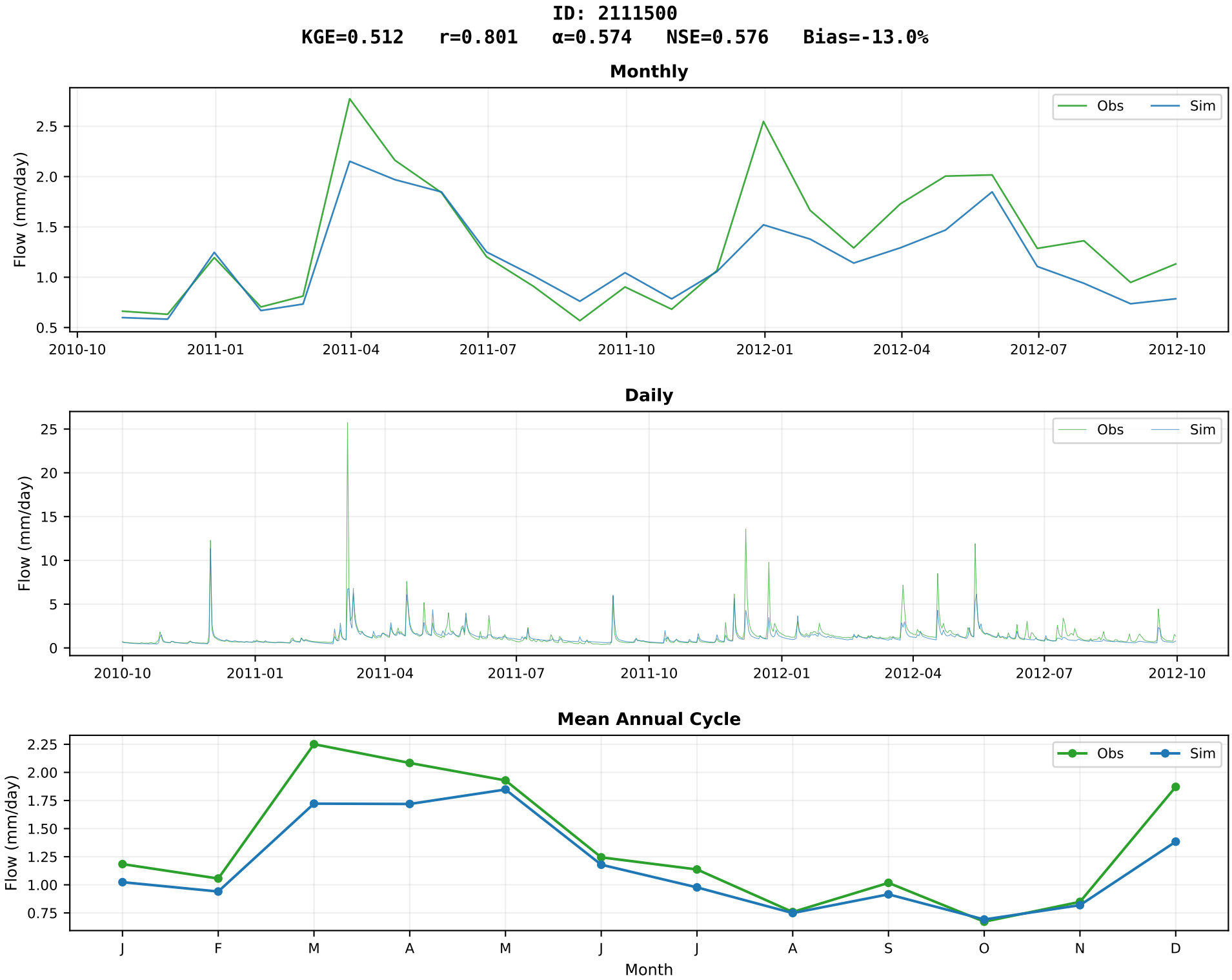

ID: 3338780
KGE=0.523 r=0.914 α=0.571 NSE=0.699 Bias=-19.0%
Monthly
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.5
1.0
1.5
2.0
2.5
J
F
M
A
M
J
J
A
S
O
N
D
Month


ELDER C NR BRANSCOMB CA (ID=11475560)
KGE=0.522 r=0.944 α=0.598 NSE=0.740 Bias=-25.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D


ID: 8070200
KGE=0.519 r=0.857 α=1.299 NSE=0.524 Bias=+34.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


VAN DUZEN R NR BRIDGEVILLE CA (ID=11478500)
KGE=0.515 r=0.926 α=0.542 NSE=0.704 Bias=-14.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

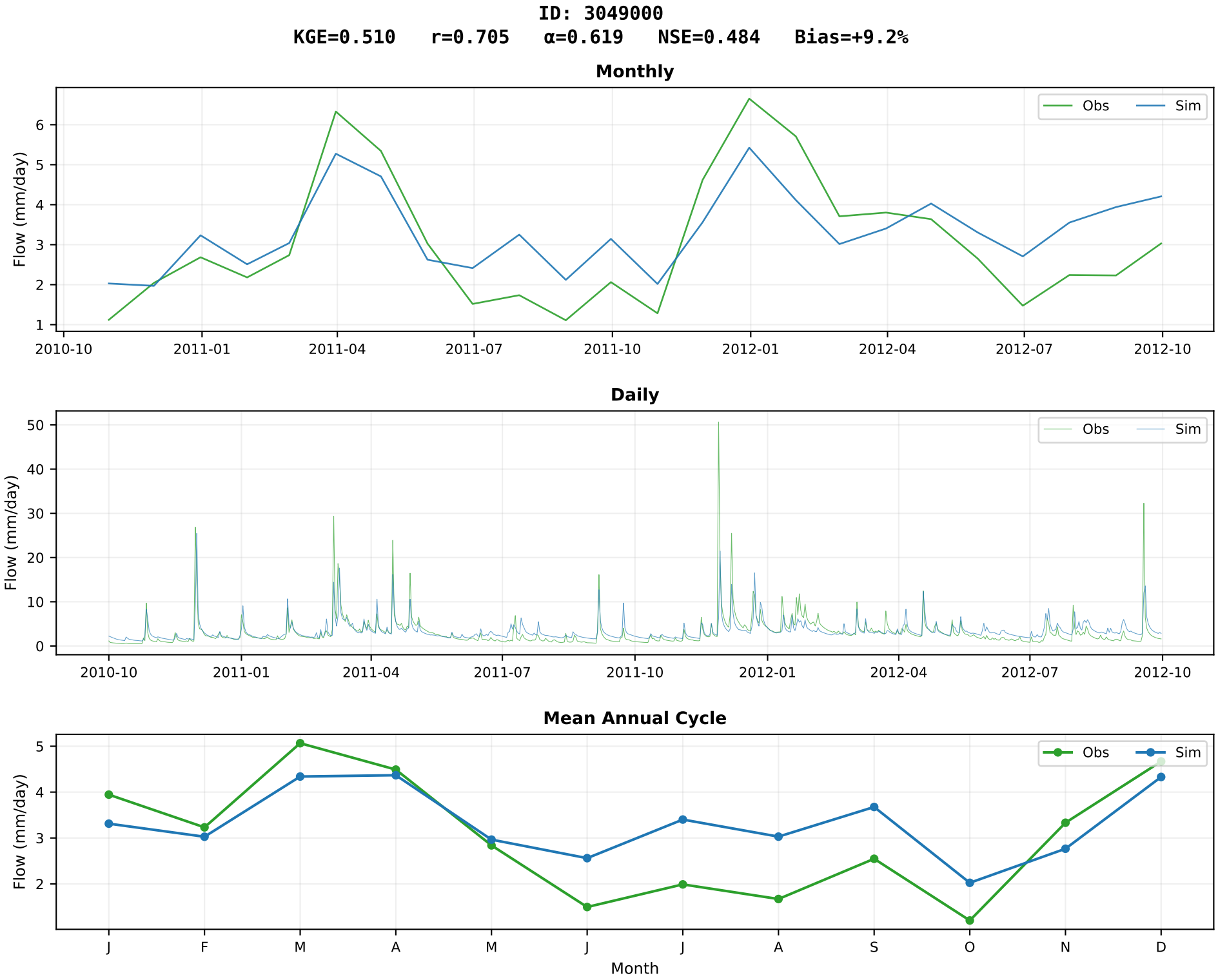

ID: 1333000
KGE=0.515 r=0.706 α=0.614 NSE=0.490 Bias=+0.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

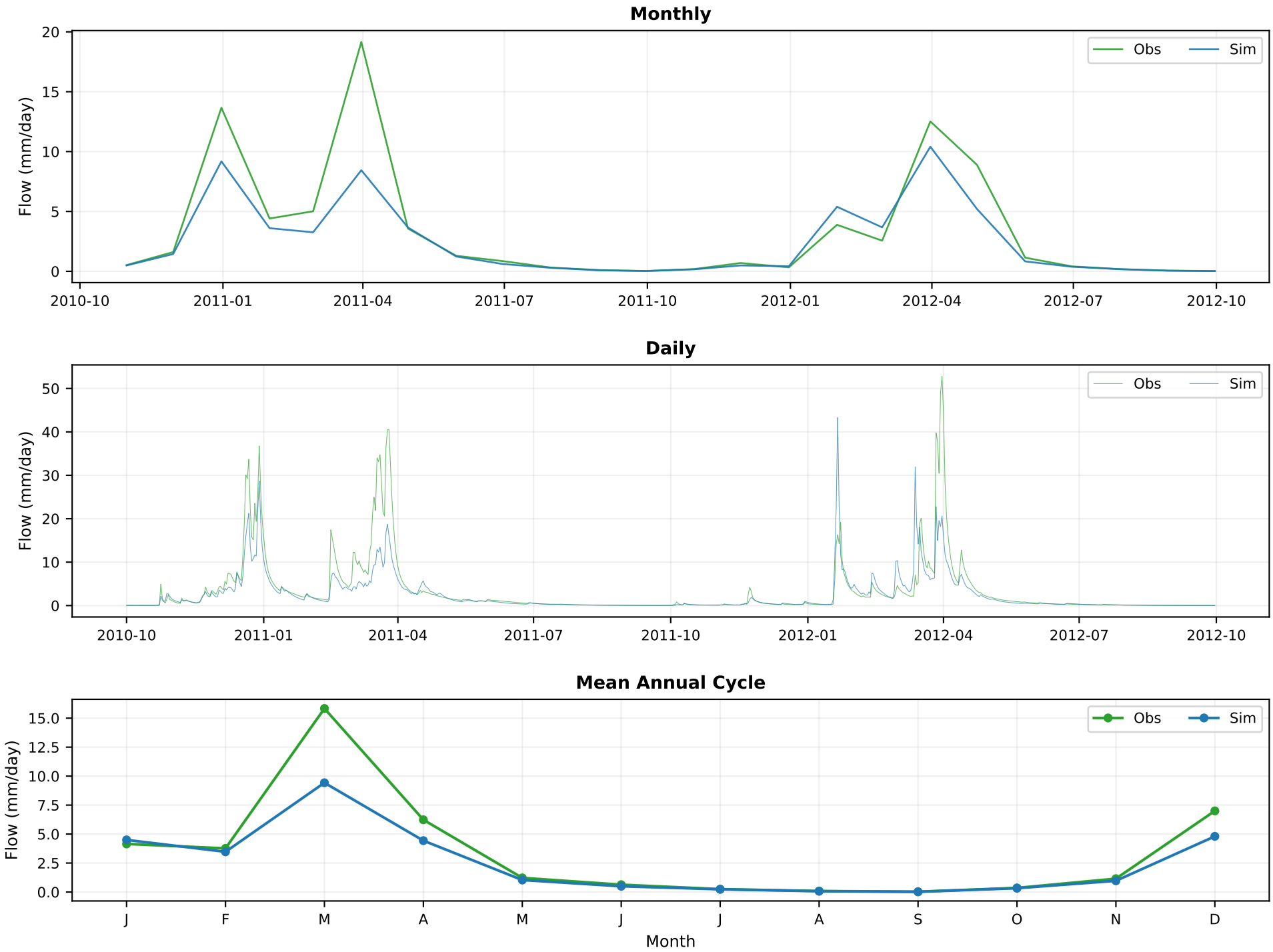

ID: 2111500
KGE=0.512 r=0.801 α=0.574 NSE=0.576 Bias=-13.0%
Monthly
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
25
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.25
2.00
1.75
1.50
1.25
1.00
0.75
J
F
M
A
M
J
J
A
S
O
N
D
Month

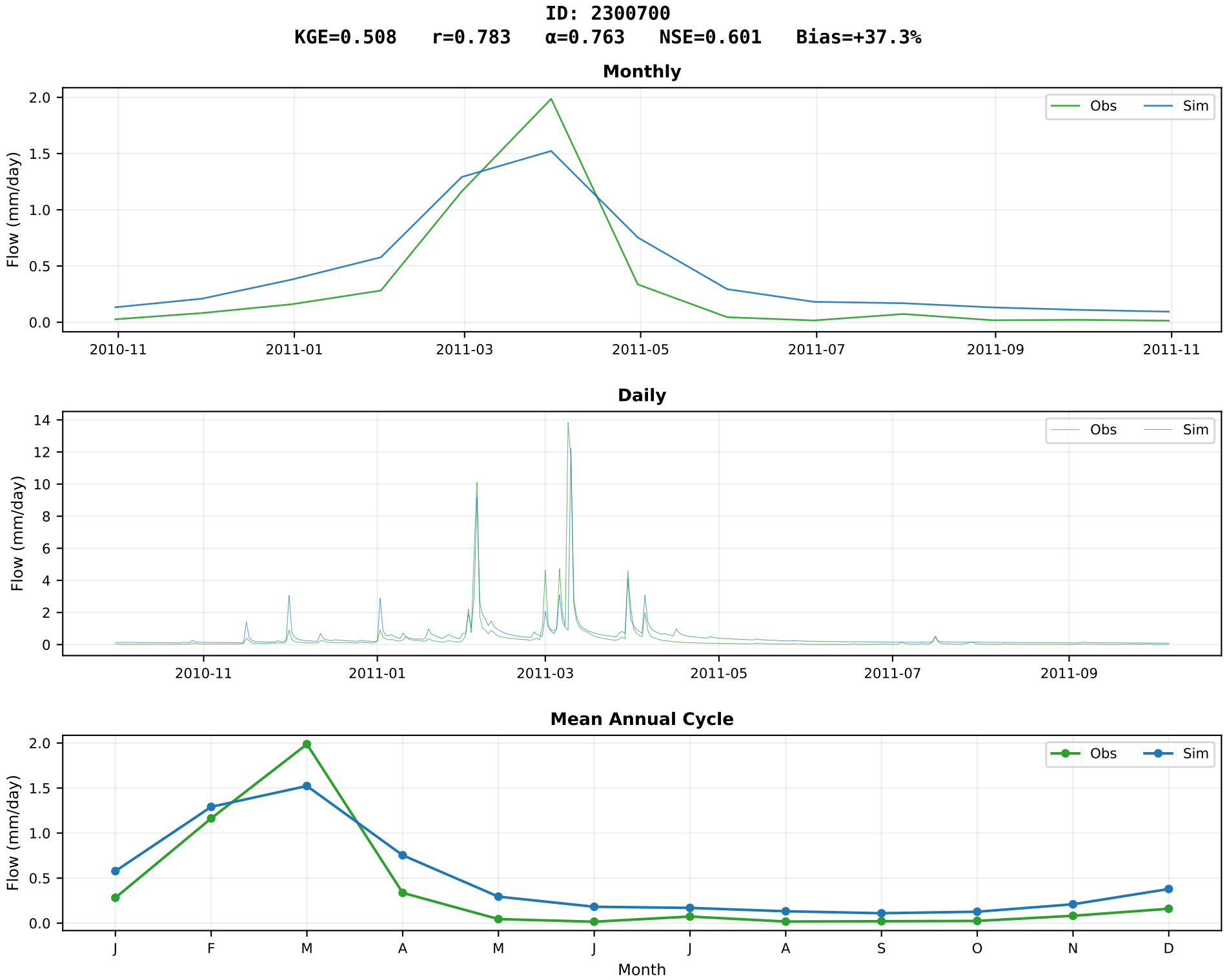
ID: 2096846

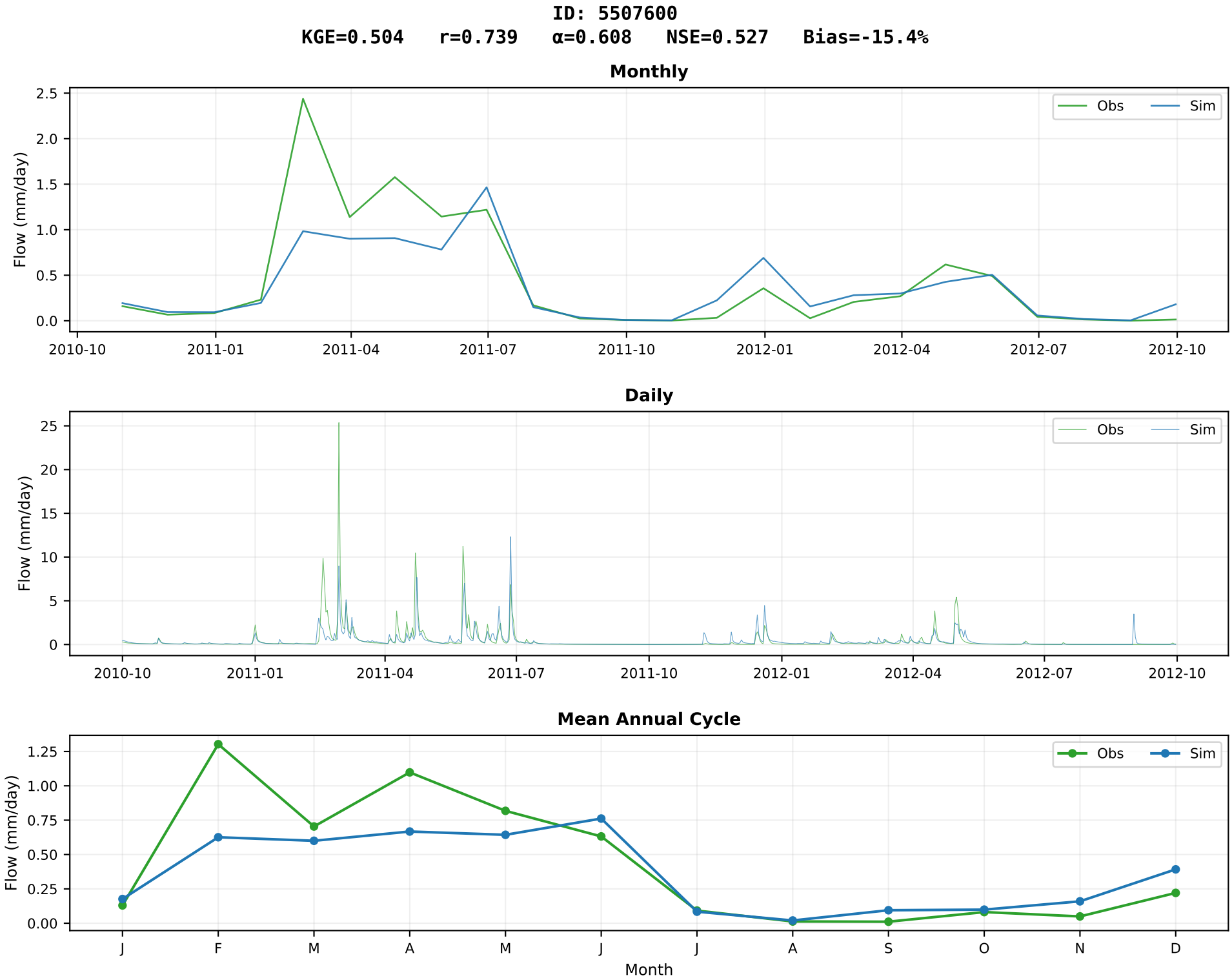
KGE=0.510 r=0.607 α=0.730 NSE=0.315 Bias=+11.4%

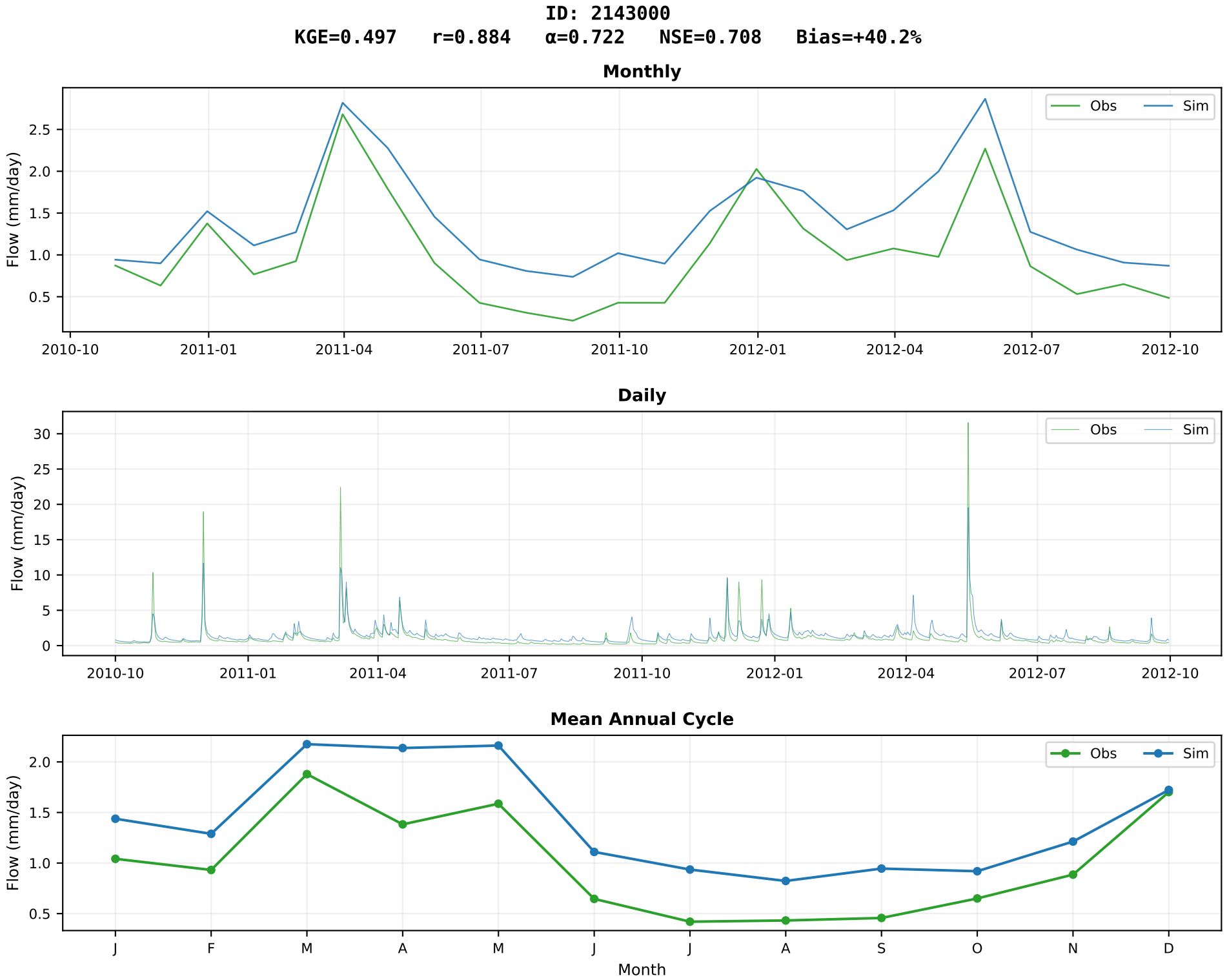



ID: 3049000
KGE=0.510 r=0.705 α=0.619 NSE=0.484 Bias=+9.2%
Monthly
Obs
Sim
Flow (mm/day)
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month


SF SNOQUALMIE RIVER AT NORTH BEND, WA (ID=12144000)
KGE=0.508 r=0.836 α=0.623 NSE=0.636 Bias=-27.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

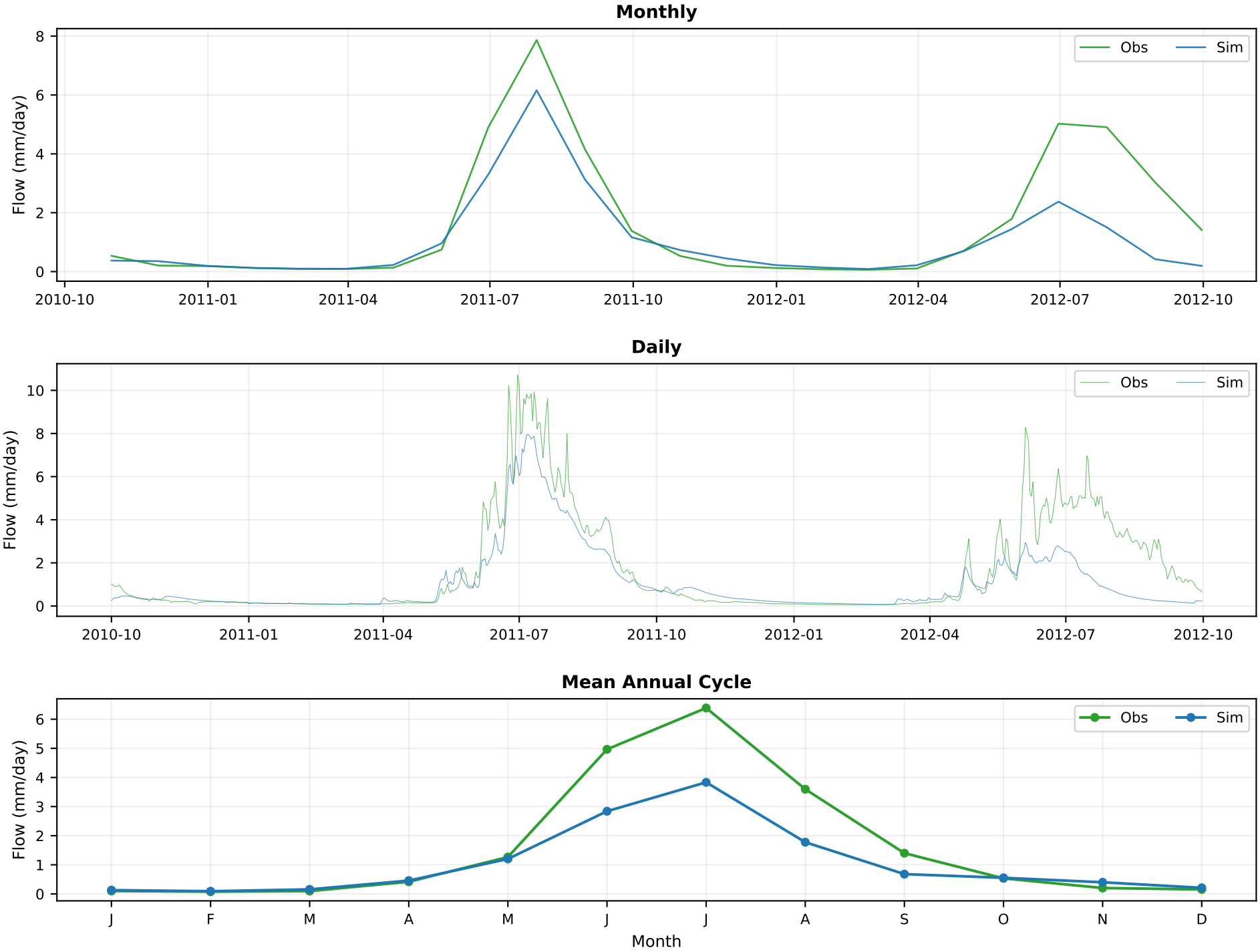

ID: 2300700
KGE=0.508 r=0.783 α=0.763 NSE=0.601 Bias=+37.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-11
2011-01
2011-03
2011-05
2011-07
2011-09
2011-11
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

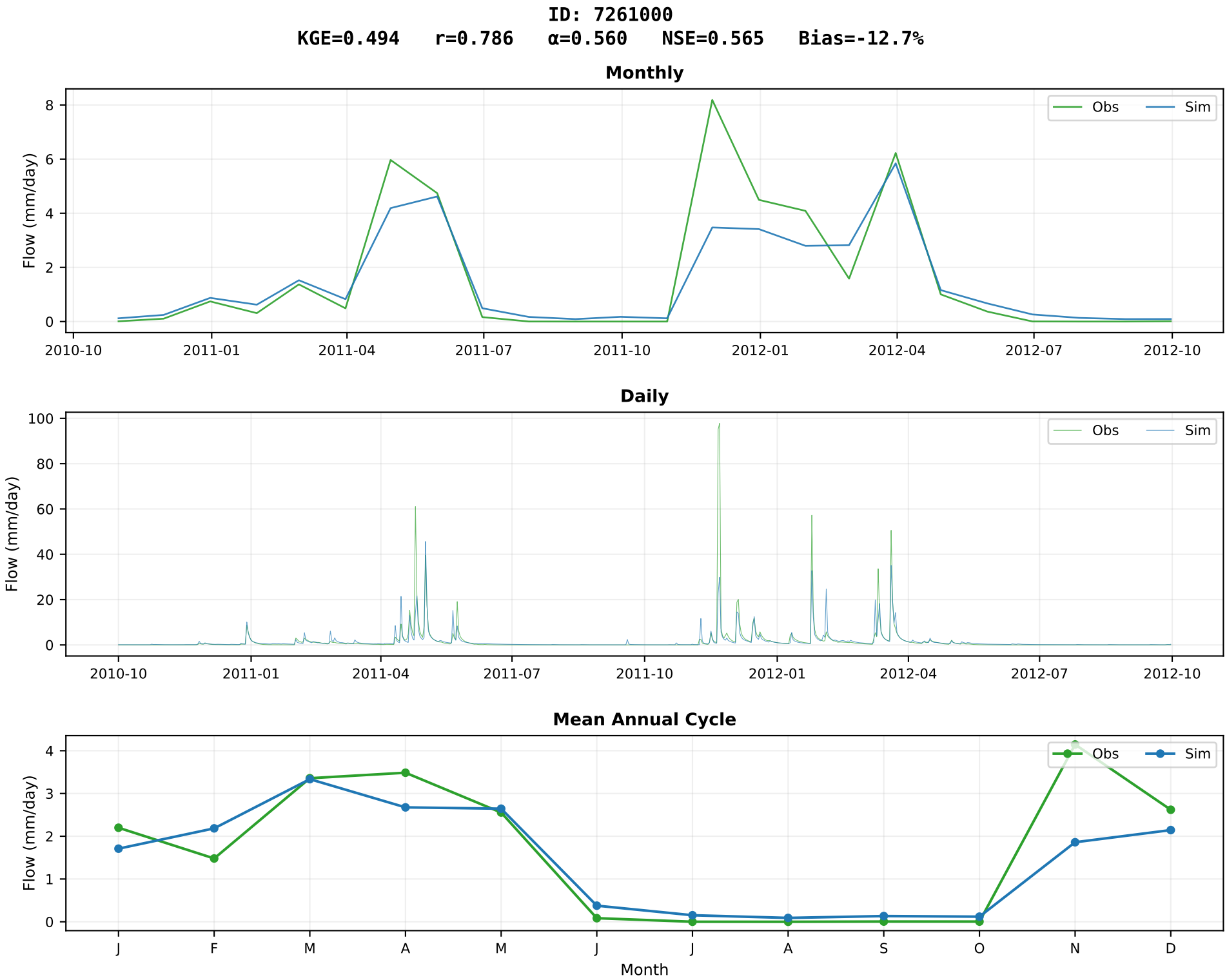

ID: 5507600
KGE=0.504 r=0.739 α=0.608 NSE=0.527 Bias=-15.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

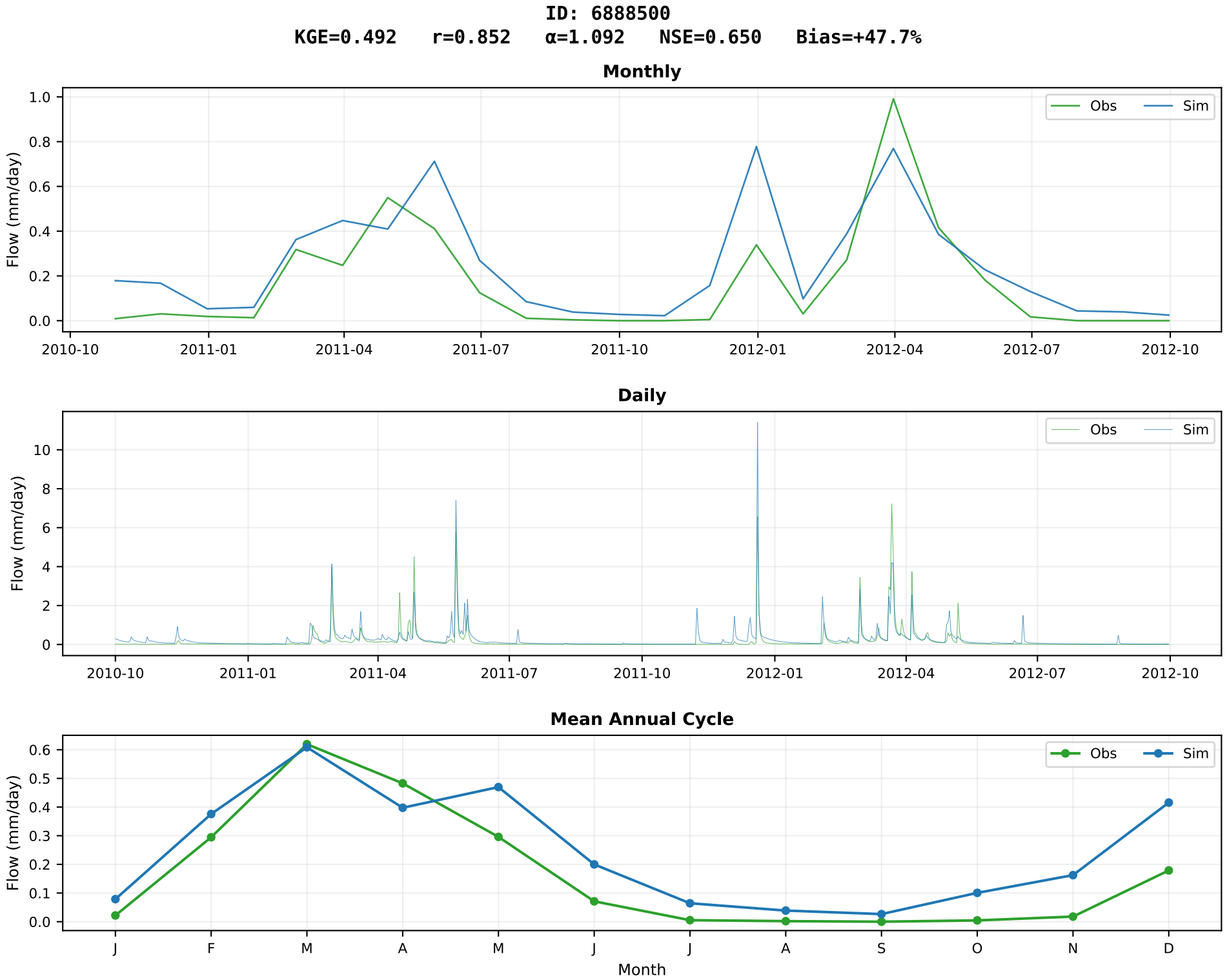

ID: 2143000
KGE=0.497 r=0.884 α=0.722 NSE=0.708 Bias=+40.2%
Monthly
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
30
25
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
J
F
M
A
M
J
J
A
S
O
N
D
Month

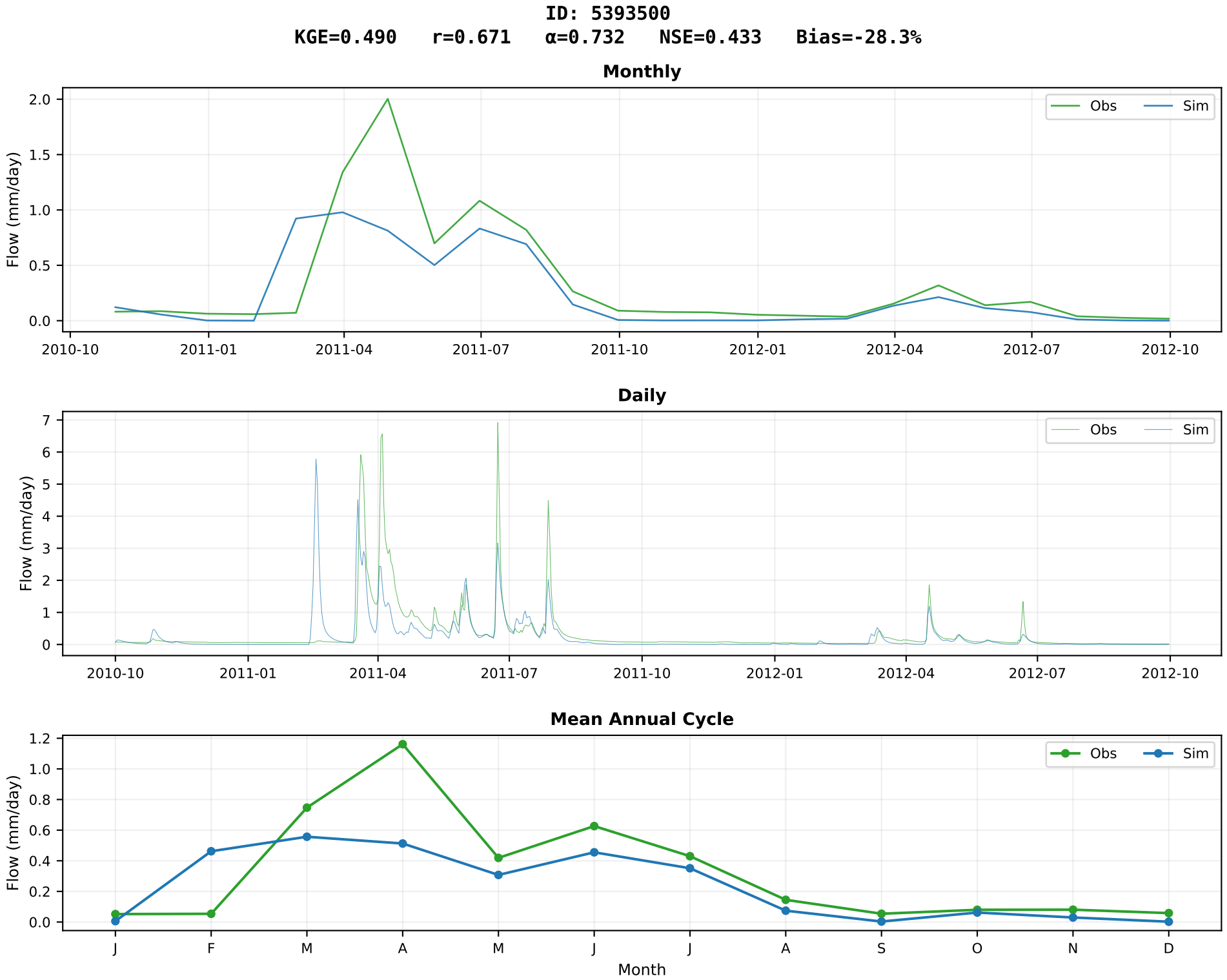
ID: 7362100

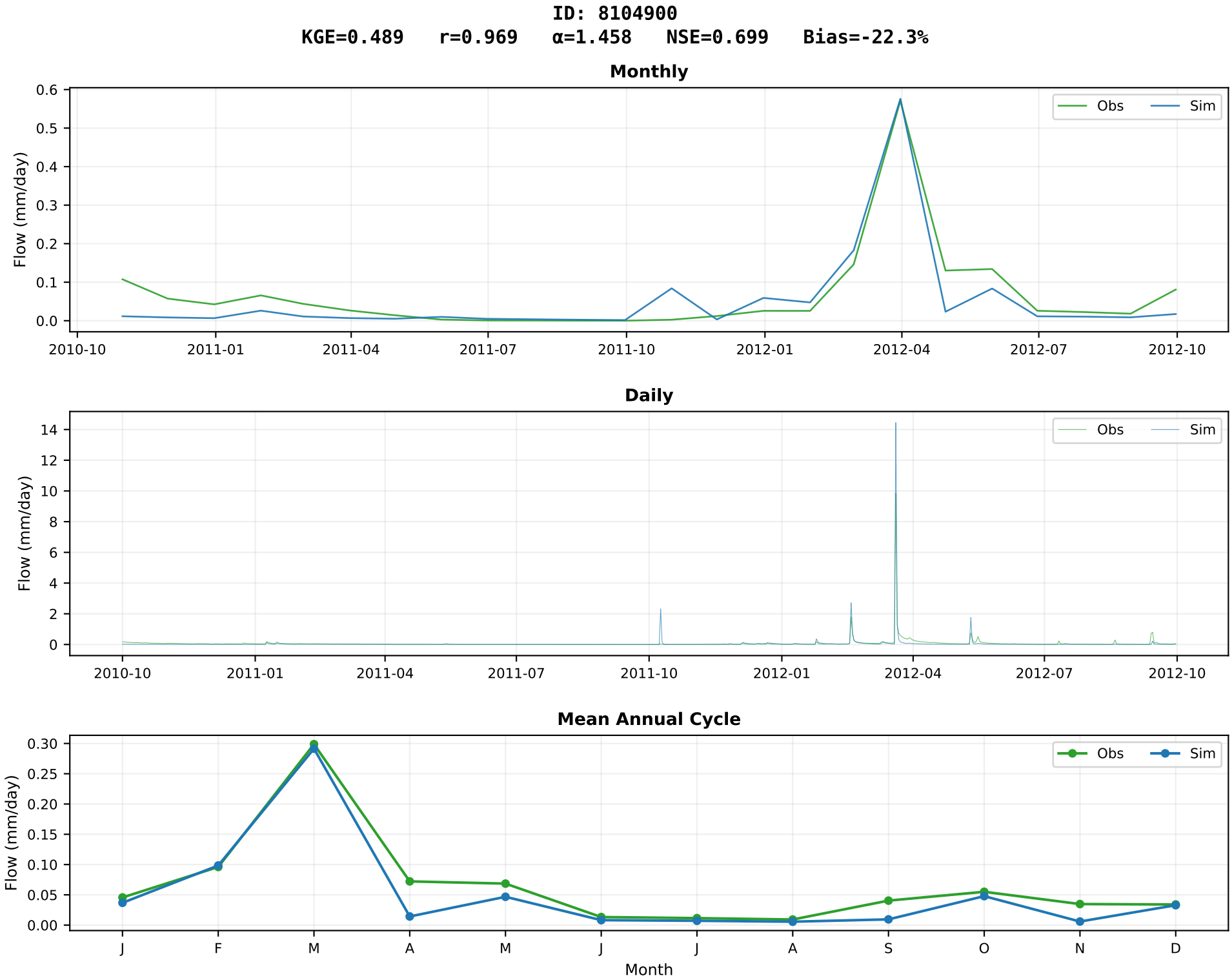
KGE=0.497 r=0.894 α=0.663 NSE=0.682 Bias=-35.8%

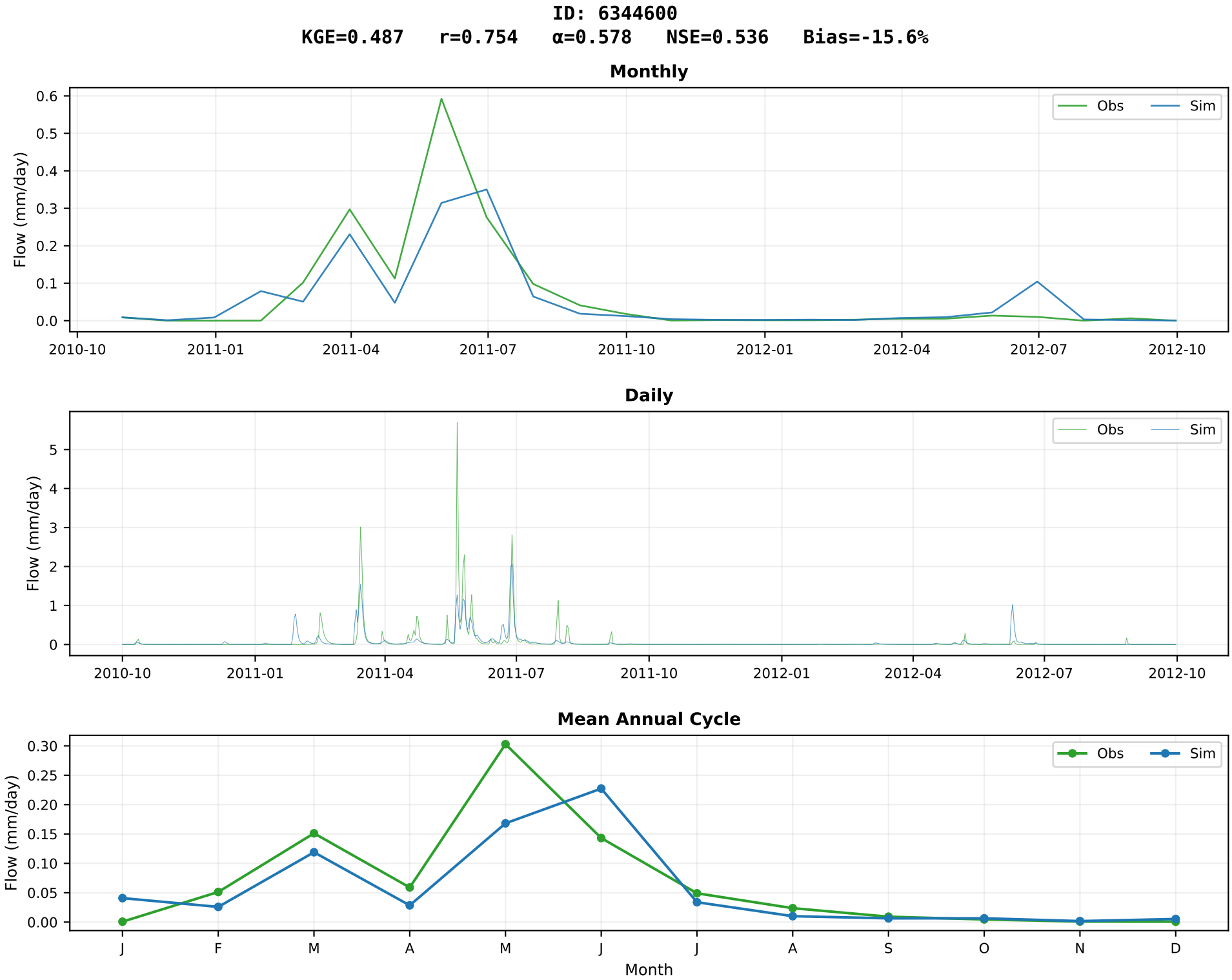


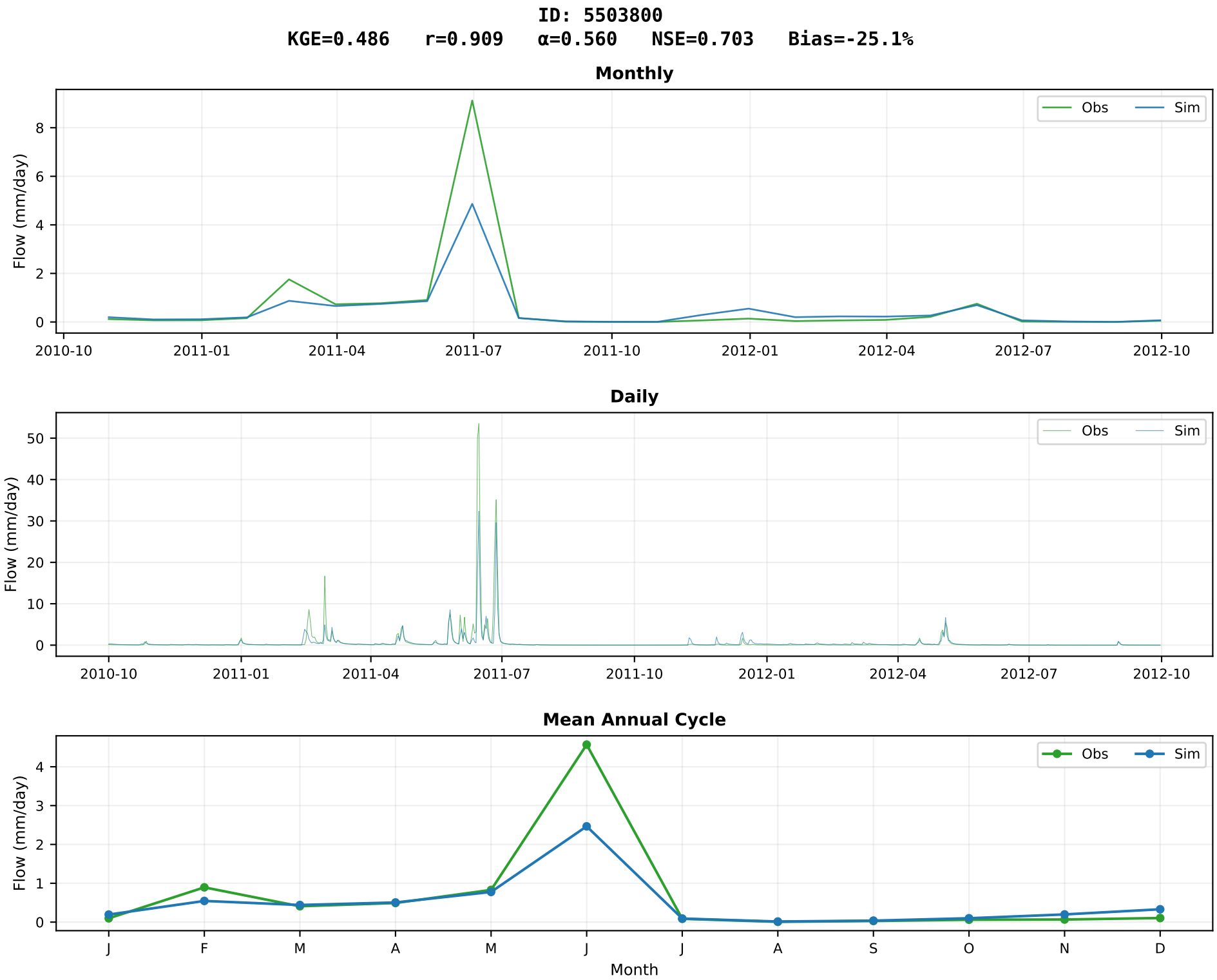

ID: 7261000
KGE=0.494 r=0.786 α=0.560 NSE=0.565 Bias=-12.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

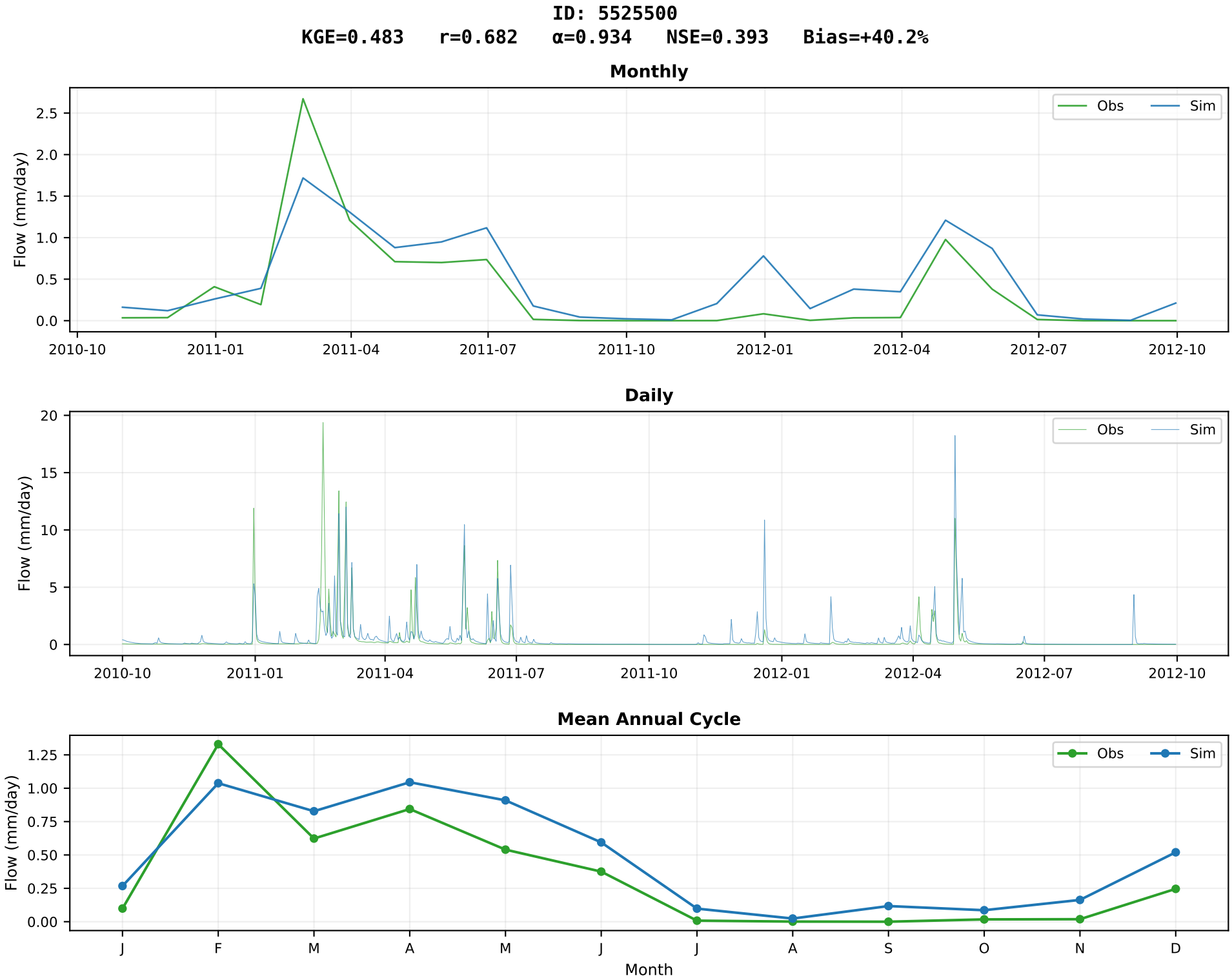

ID: 6888500
KGE=0.492 r=0.852 α=1.092 NSE=0.650 Bias=+47.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

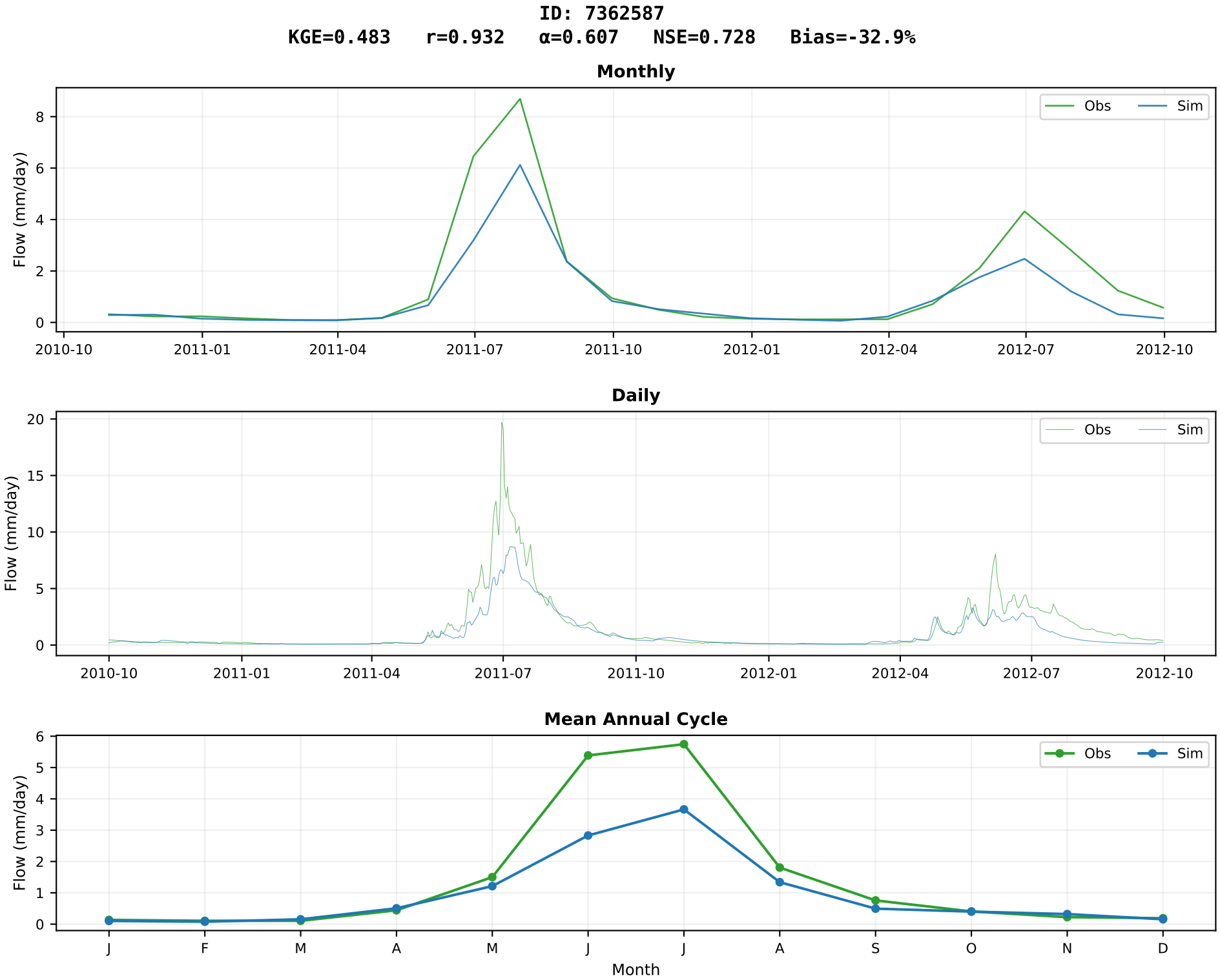

ID: 5393500
KGE=0.490 r=0.671 α=0.732 NSE=0.433 Bias=-28.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

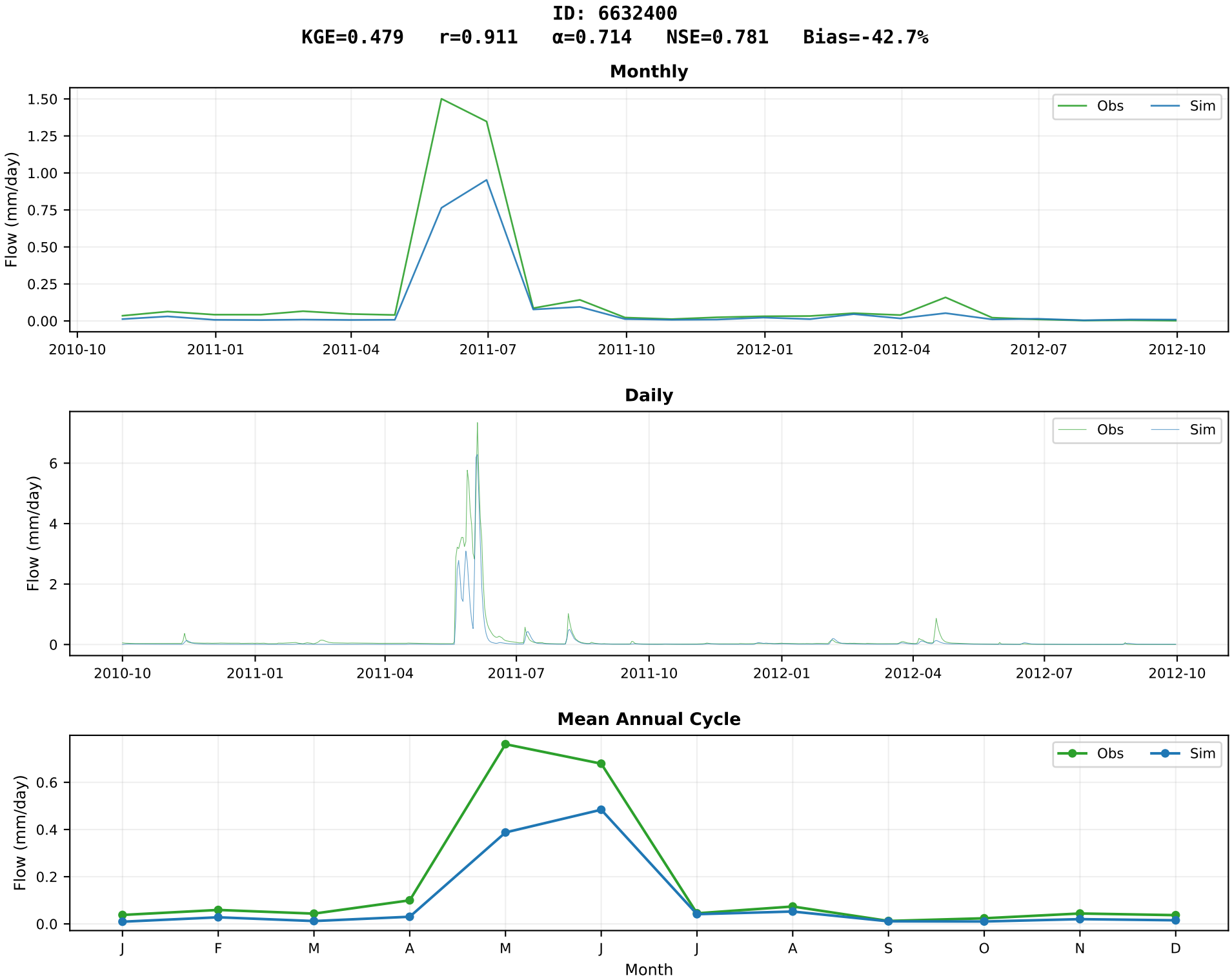

ID: 8104900
KGE=0.489 r=0.969 α=1.458 NSE=0.699 Bias=-22.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D


ID: 6344600
KGE=0.487 r=0.754 α=0.578 NSE=0.536 Bias=-15.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 5503800
KGE=0.486 r=0.909 α=0.560 NSE=0.703 Bias=-25.1%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 5525500
KGE=0.483 r=0.682 α=0.934 NSE=0.393 Bias=+40.2%
Monthly
Obs
Sim
Flow (mm/day)
2.5
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.25
1.00
0.75
0.50
0.25
0.00
J
F
M
A
M
J
J
A
S
O
N
D
Month

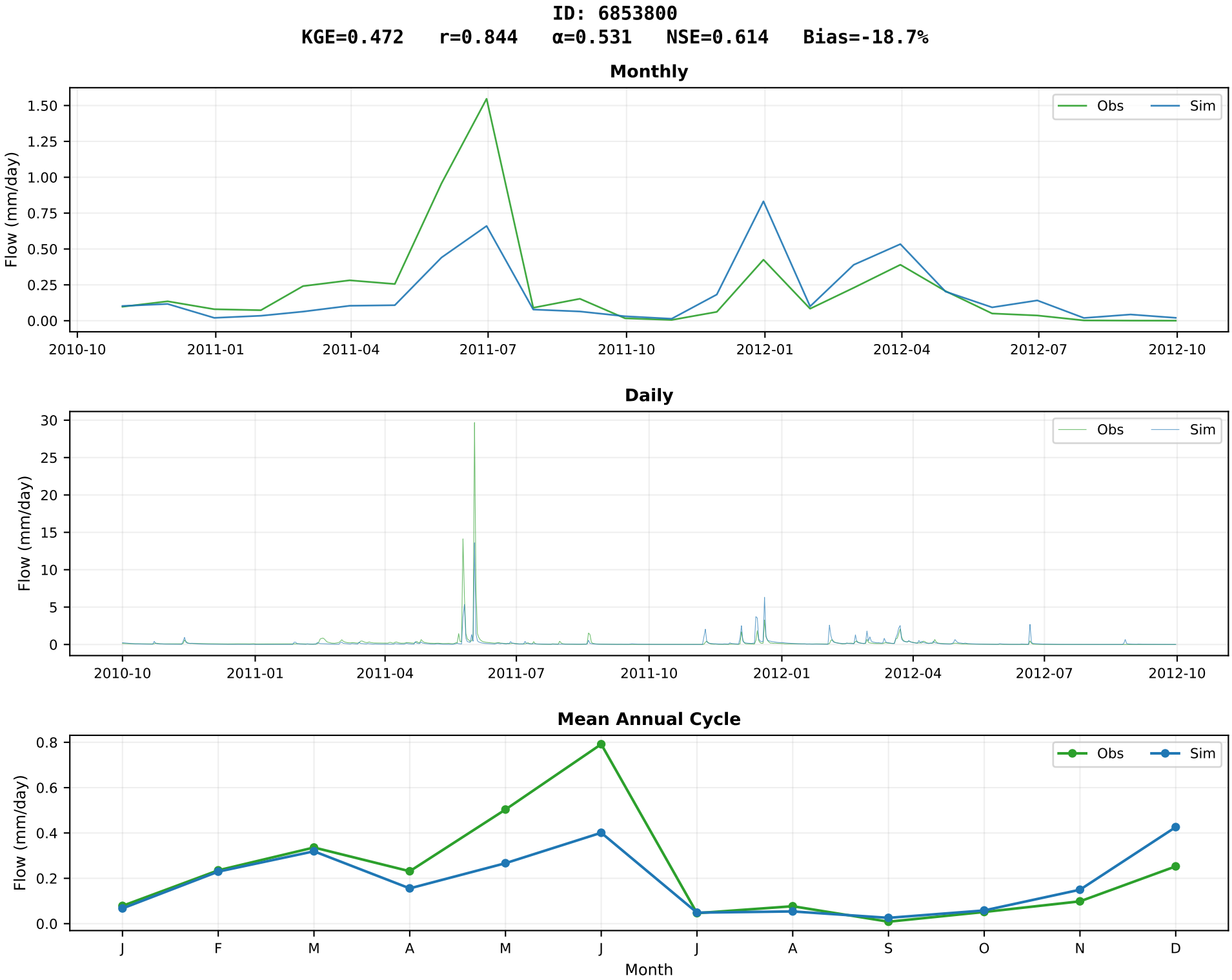

ID: 7362587
KGE=0.483 r=0.932 α=0.607 NSE=0.728 Bias=-32.9%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
5
10
15
20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
J
F
M
A
M
J
J
A
S
O
N
D
Month

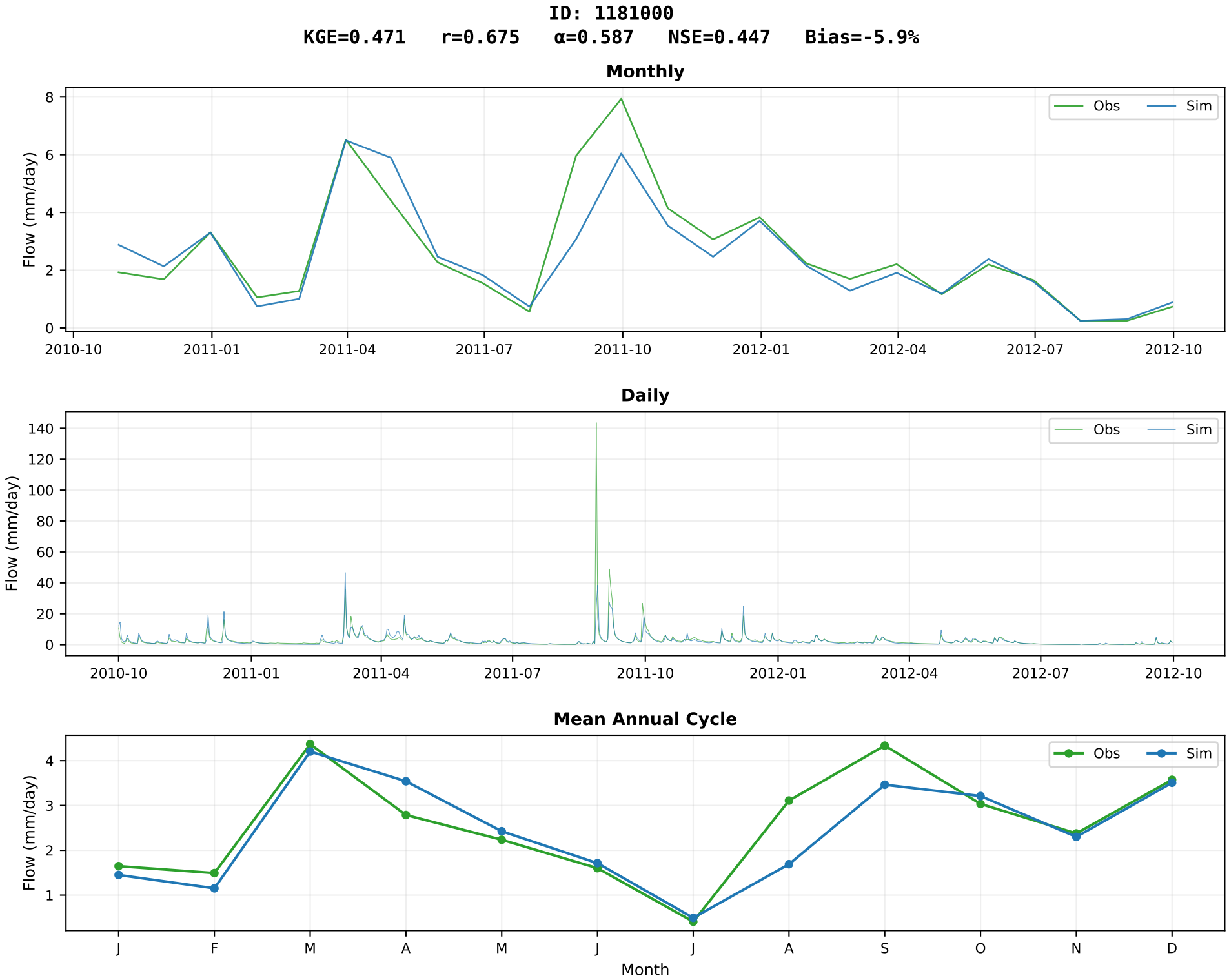

ID: 6632400
KGE=0.479 r=0.911 α=0.714 NSE=0.781 Bias=-42.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

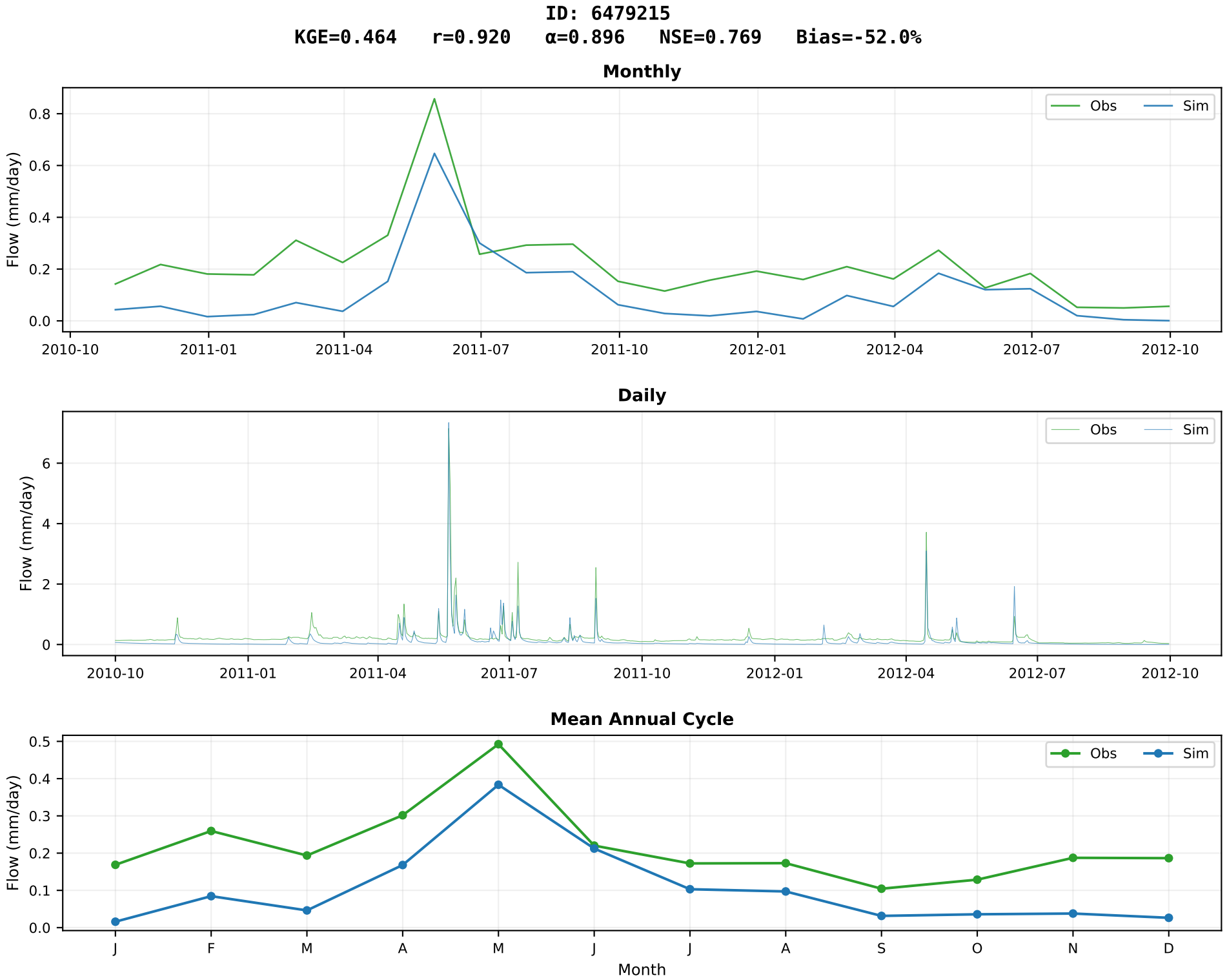
ID: 2051500

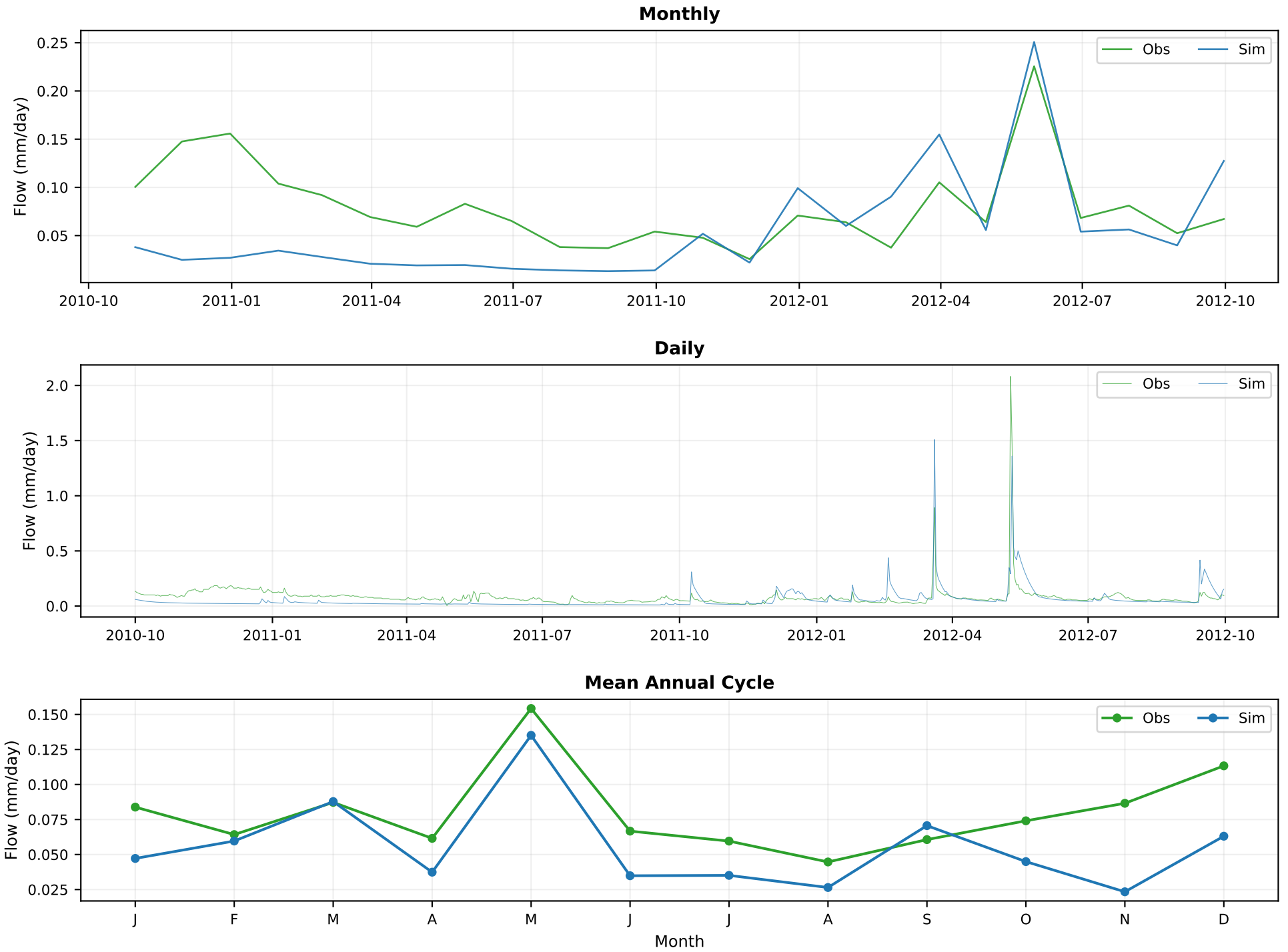
KGE=0.477 r=0.767 α=0.532 NSE=0.533 Bias=-1.8%

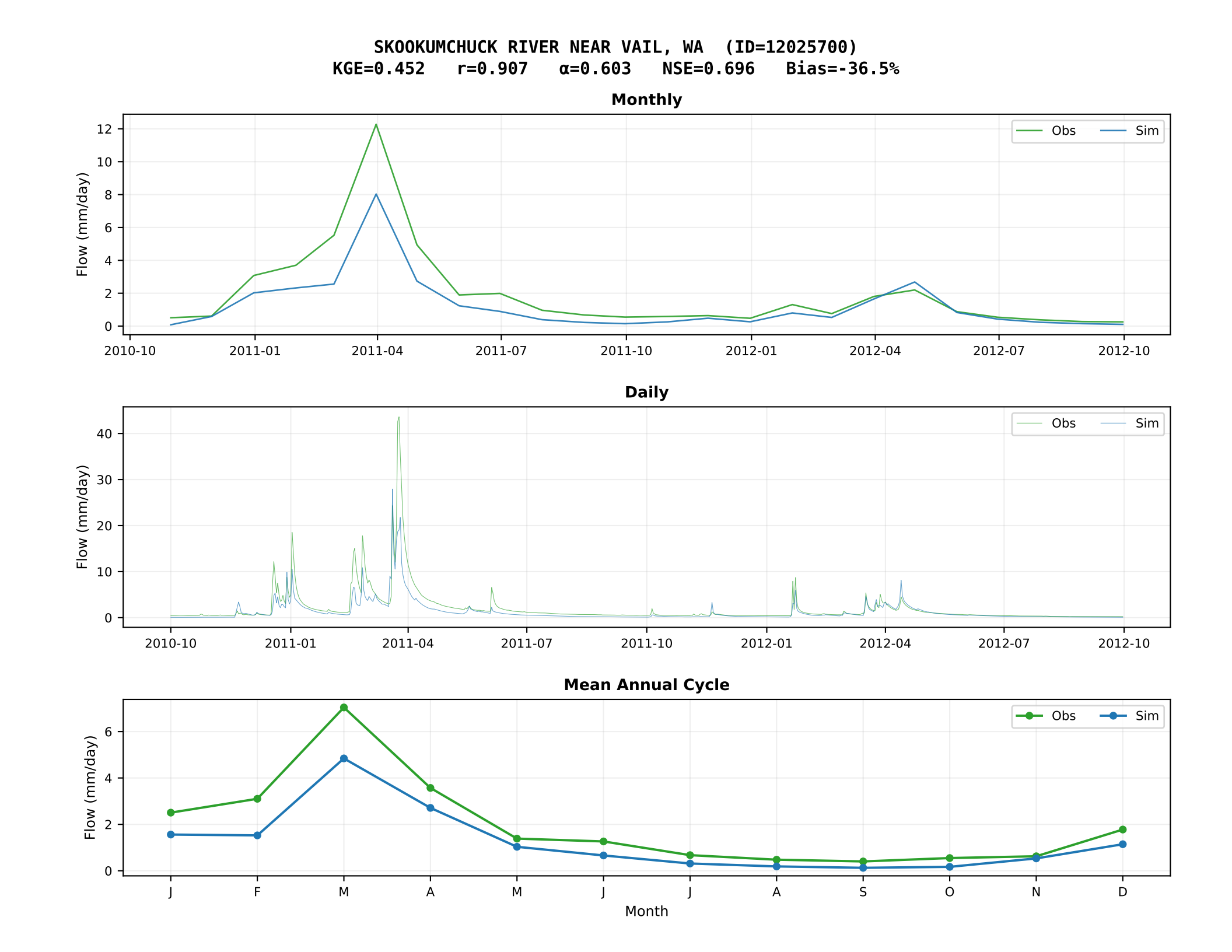


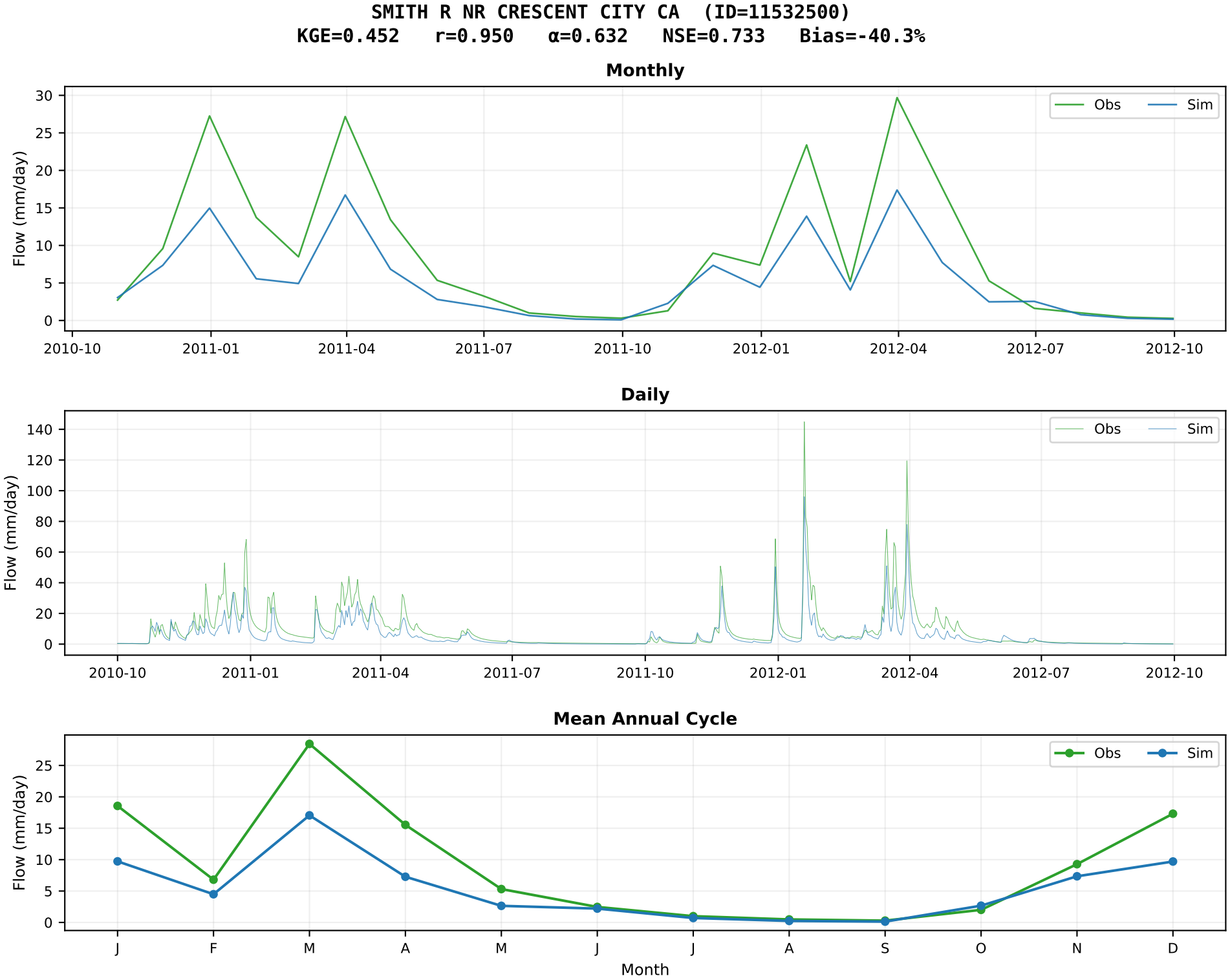

ID: 6853800
KGE=0.472 r=0.844 α=0.531 NSE=0.614 Bias=-18.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

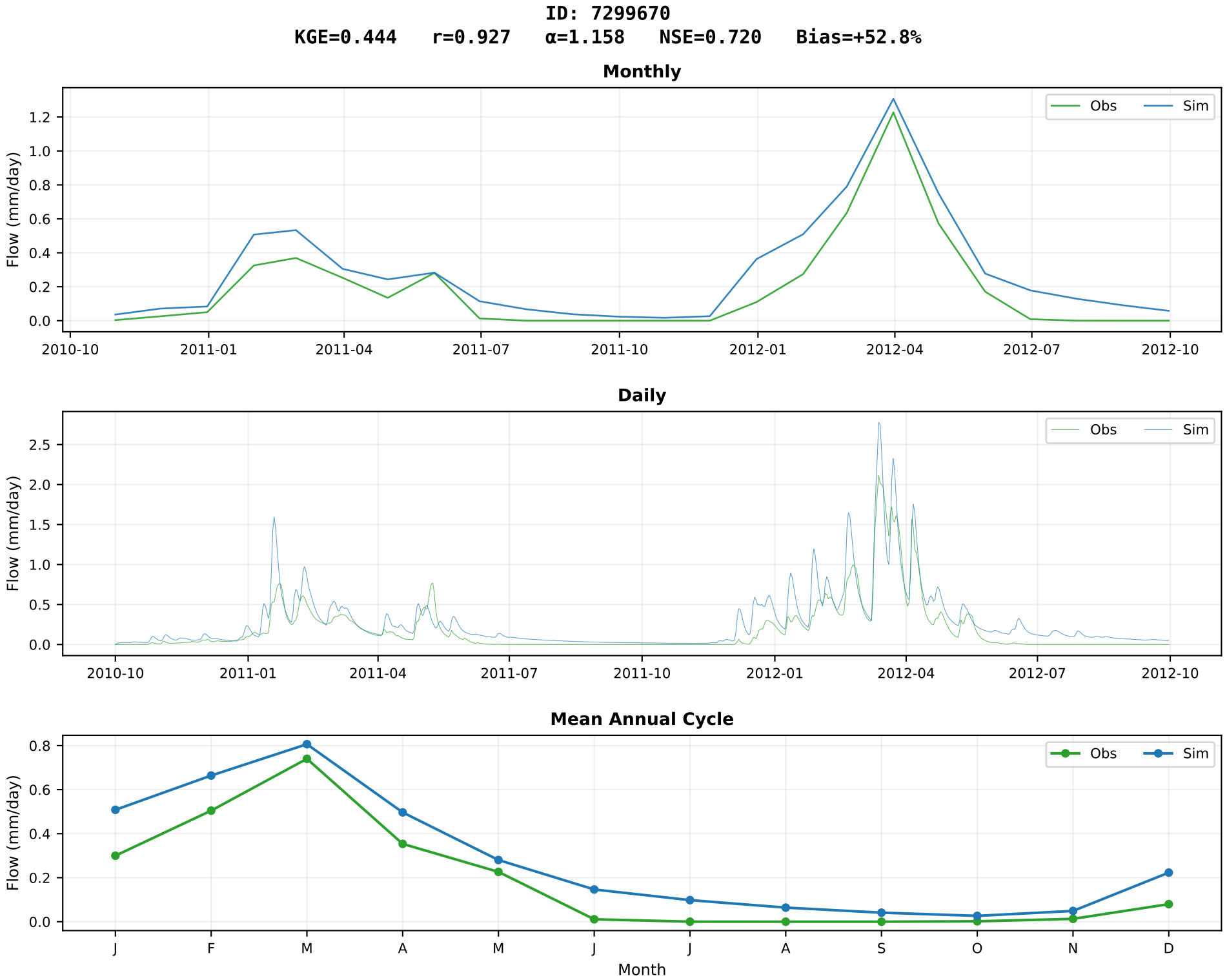

ID: 1181000
KGE=0.471 r=0.675 α=0.587 NSE=0.447 Bias=-5.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

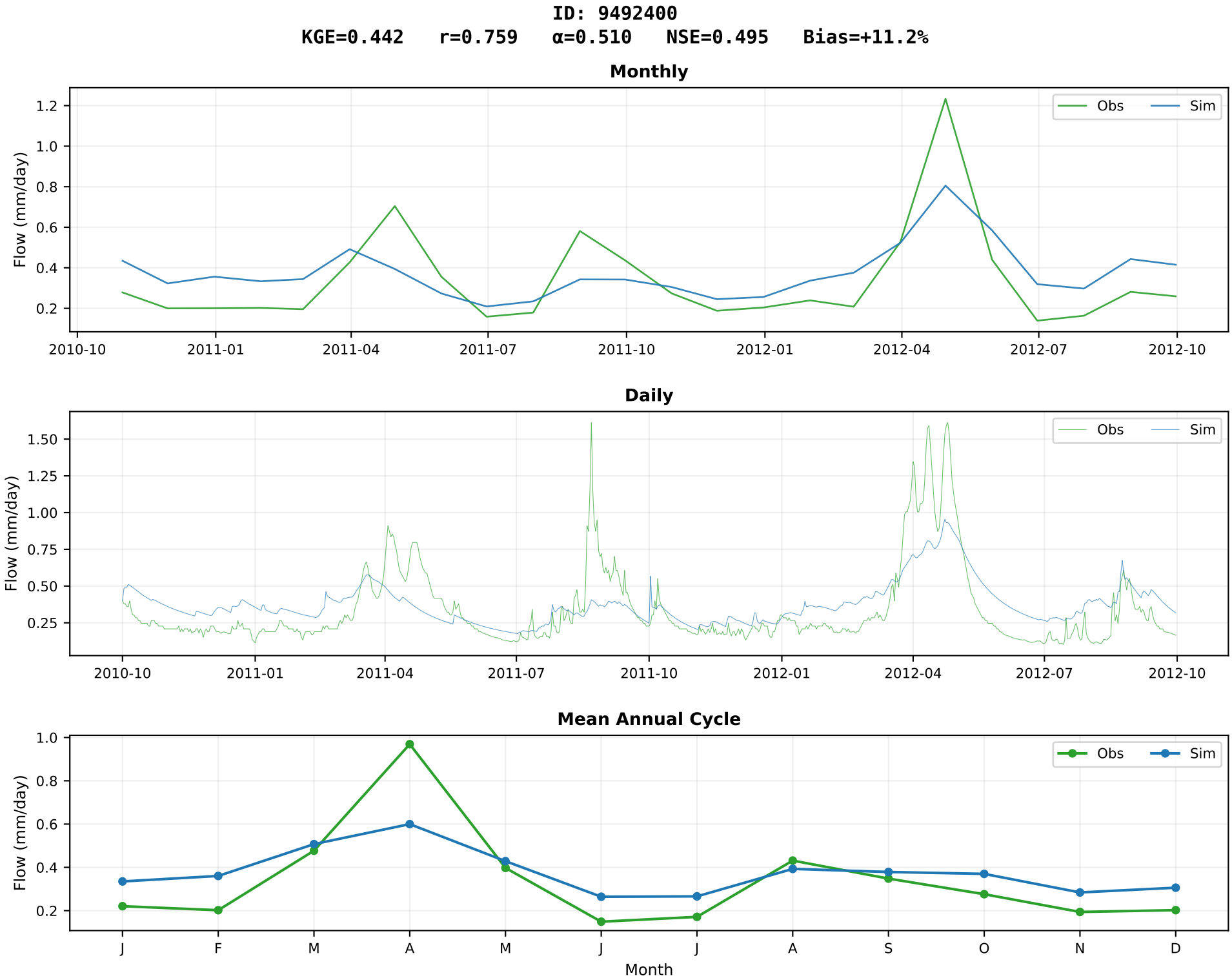

ID: 6479215
KGE=0.464 r=0.920 α=0.896 NSE=0.769 Bias=-52.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

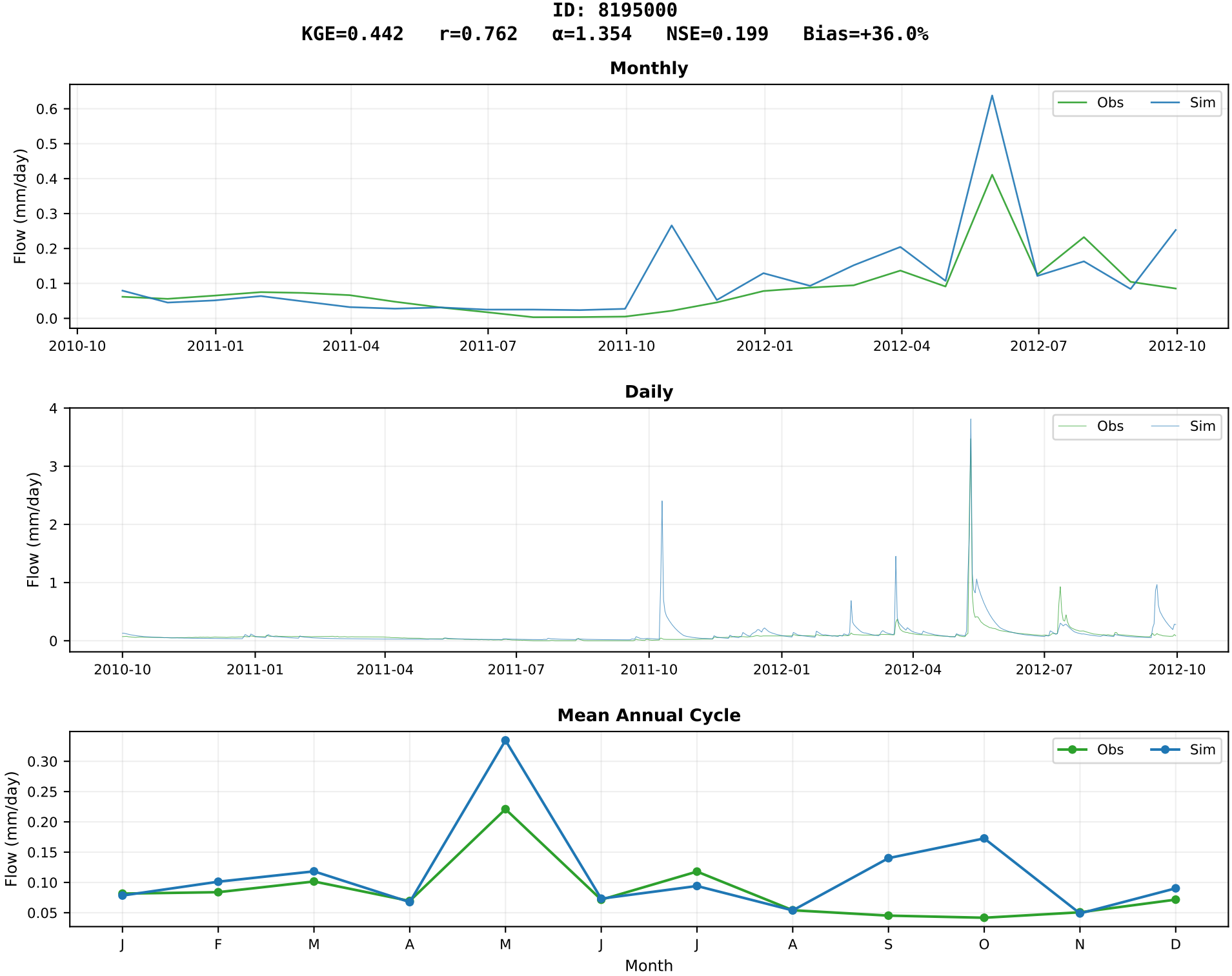

ID: 8165300
KGE=0.459 r=0.556 α=0.950 NSE=0.098 Bias=-30.5%
Monthly
Flow (mm/day)
Obs
Sim
0.25
0.20
0.15
0.10
0.05
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.150
0.125
0.100
0.075
0.050
0.025
J
F
M
A
M
J
J
A
S
O
N
D
Month

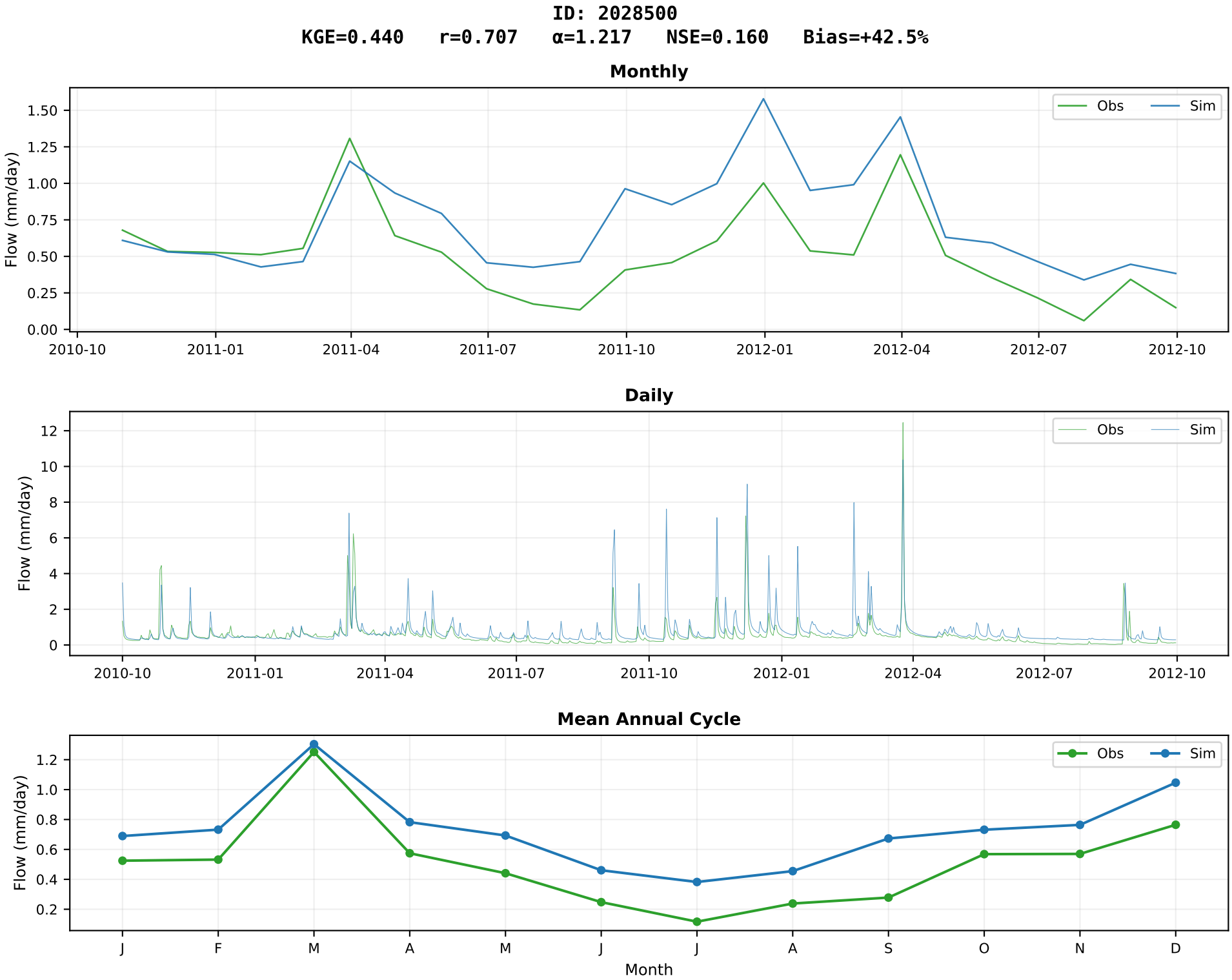

SKOOKUMCHUCK RIVER NEAR VAIL, WA (ID=12025700)
KGE=0.452 r=0.907 α=0.603 NSE=0.696 Bias=-36.5%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

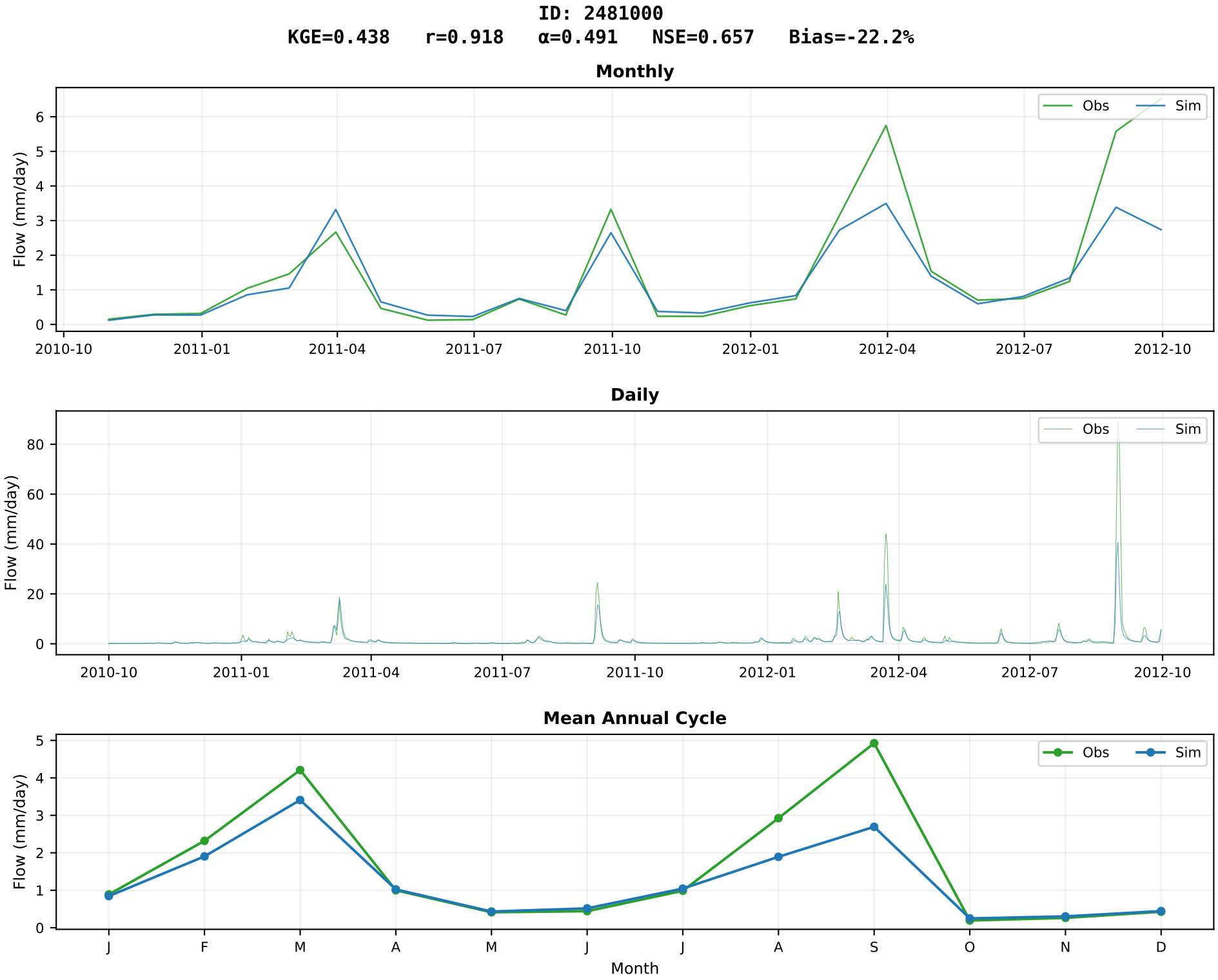

SMITH R NR CRESCENT CITY CA (ID=11532500)
KGE=0.452 r=0.950 α=0.632 NSE=0.733 Bias=-40.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

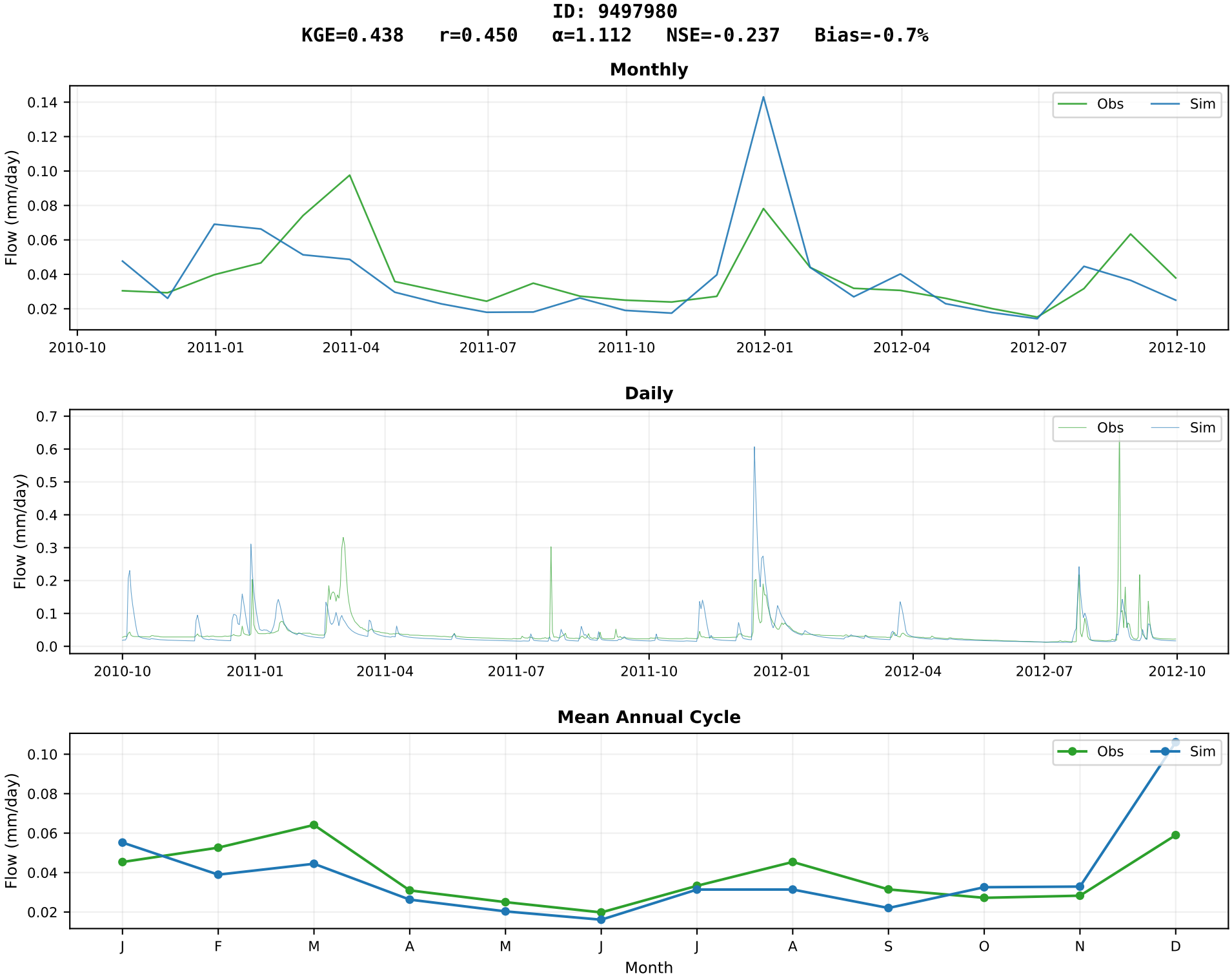

ID: 7299670
KGE=0.444 r=0.927 α=1.158 NSE=0.720 Bias=+52.8%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
J
F
M
A
M
J
J
A
S
O
N
D
Month

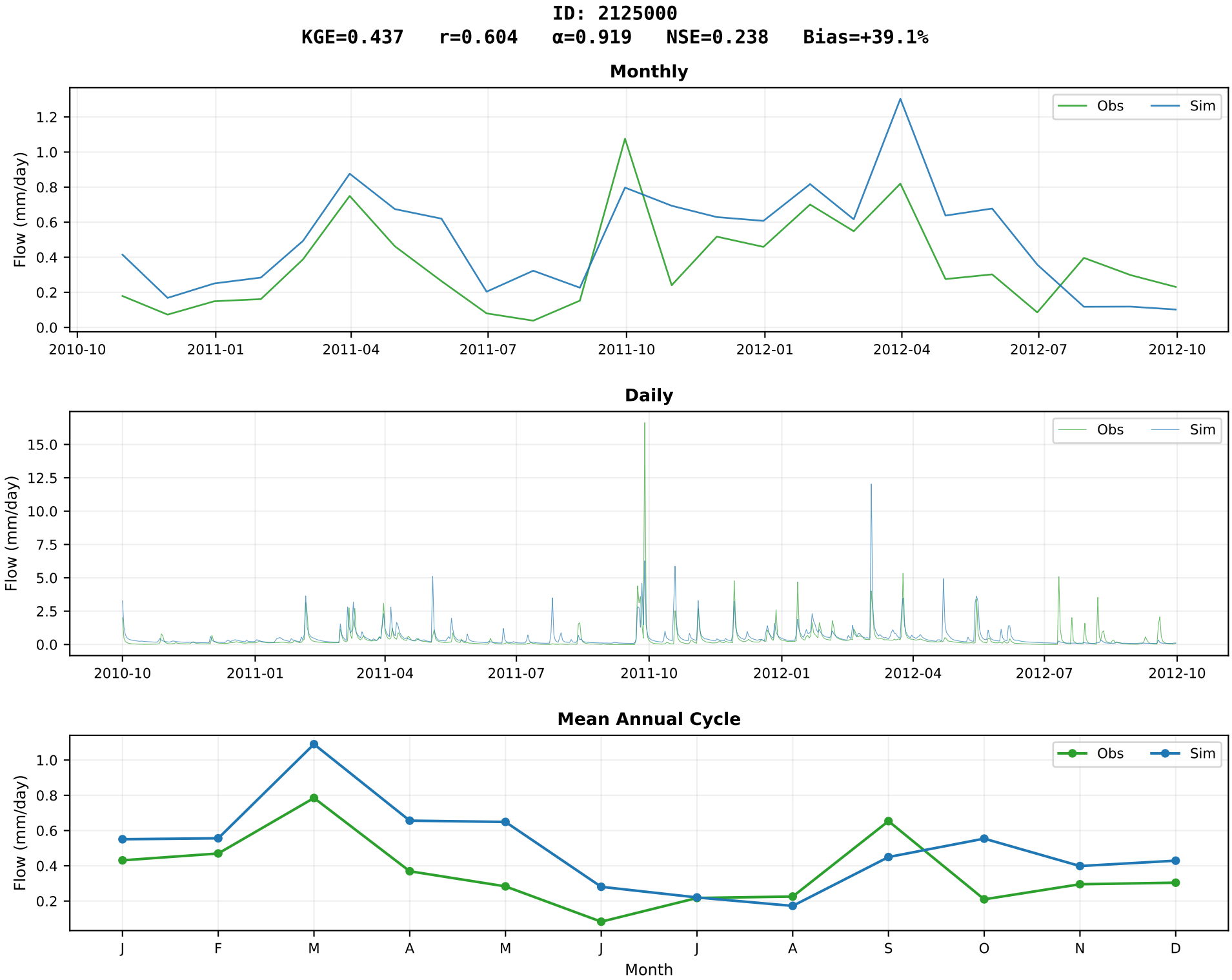

ID: 9492400
KGE=0.442 r=0.759 α=0.510 NSE=0.495 Bias=+11.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

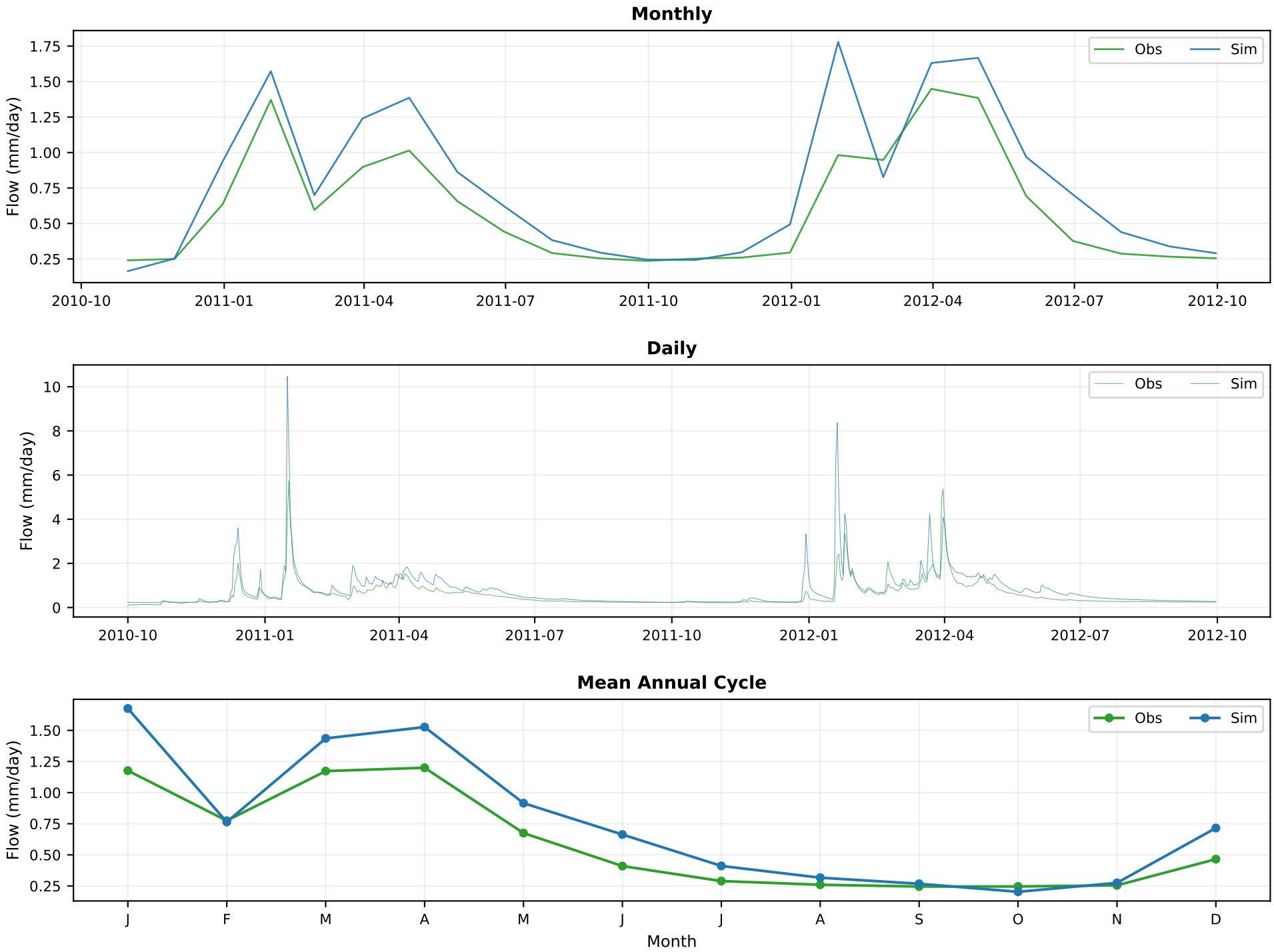

ID: 8195000
KGE=0.442 r=0.762 α=1.354 NSE=0.199 Bias=+36.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

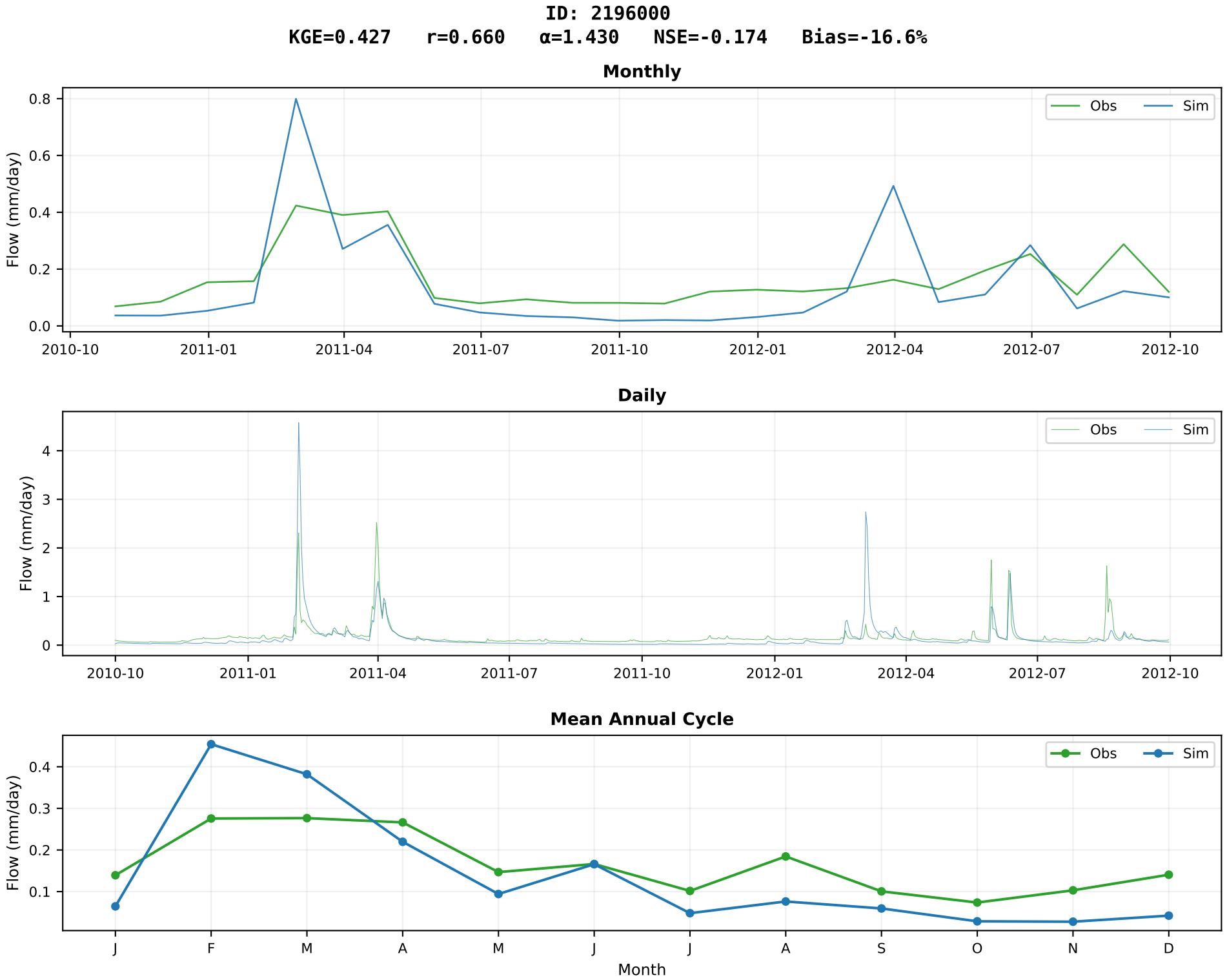

ID: 2028500
KGE=0.440 r=0.707 α=1.217 NSE=0.160 Bias=+42.5%
Monthly
Obs
Sim
Flow (mm/day)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
12
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.2
0.4
0.6
0.8
1.0
1.2
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2481000
KGE=0.438 r=0.918 α=0.491 NSE=0.657 Bias=-22.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 9497980
KGE=0.438 r=0.450 α=1.112 NSE=-0.237 Bias=-0.7%
Monthly
Flow (mm/day)
Obs
Sim
0.14
0.12
0.10
0.08
0.06
0.04
0.02
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.10
0.08
0.06
0.04
0.02
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2125000
KGE=0.437 r=0.604 α=0.919 NSE=0.238 Bias=+39.1%
Monthly
Flow (mm/day)
Obs
Sim
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0.0
2.5
5.0
7.5
10.0
12.5
15.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.2
0.4
0.6
0.8
1.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

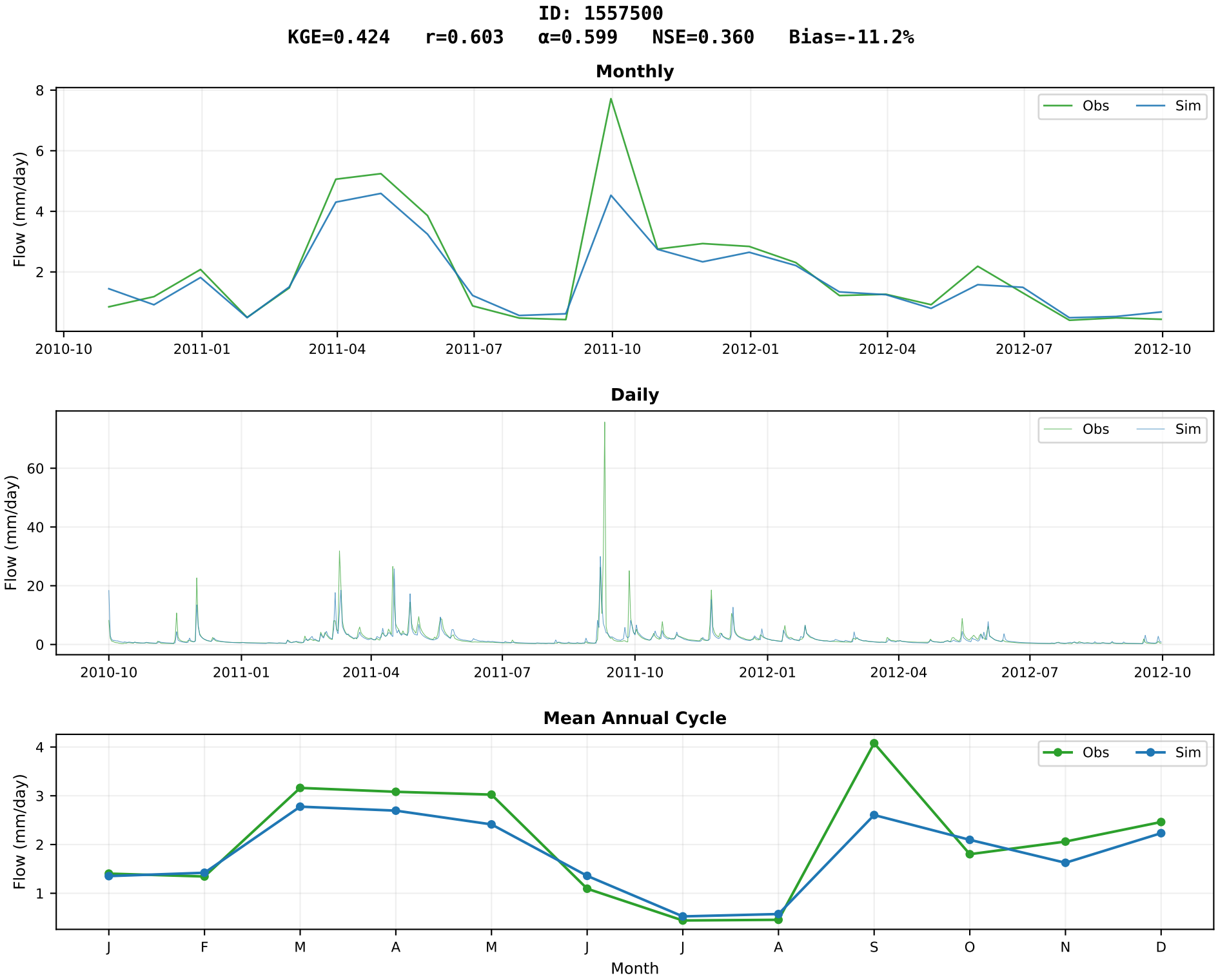

STAR GULCH NEAR RUCH, OR. (ID=14362250)
KGE=0.436 r=0.823 α=1.456 NSE=0.195 Bias=+28.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

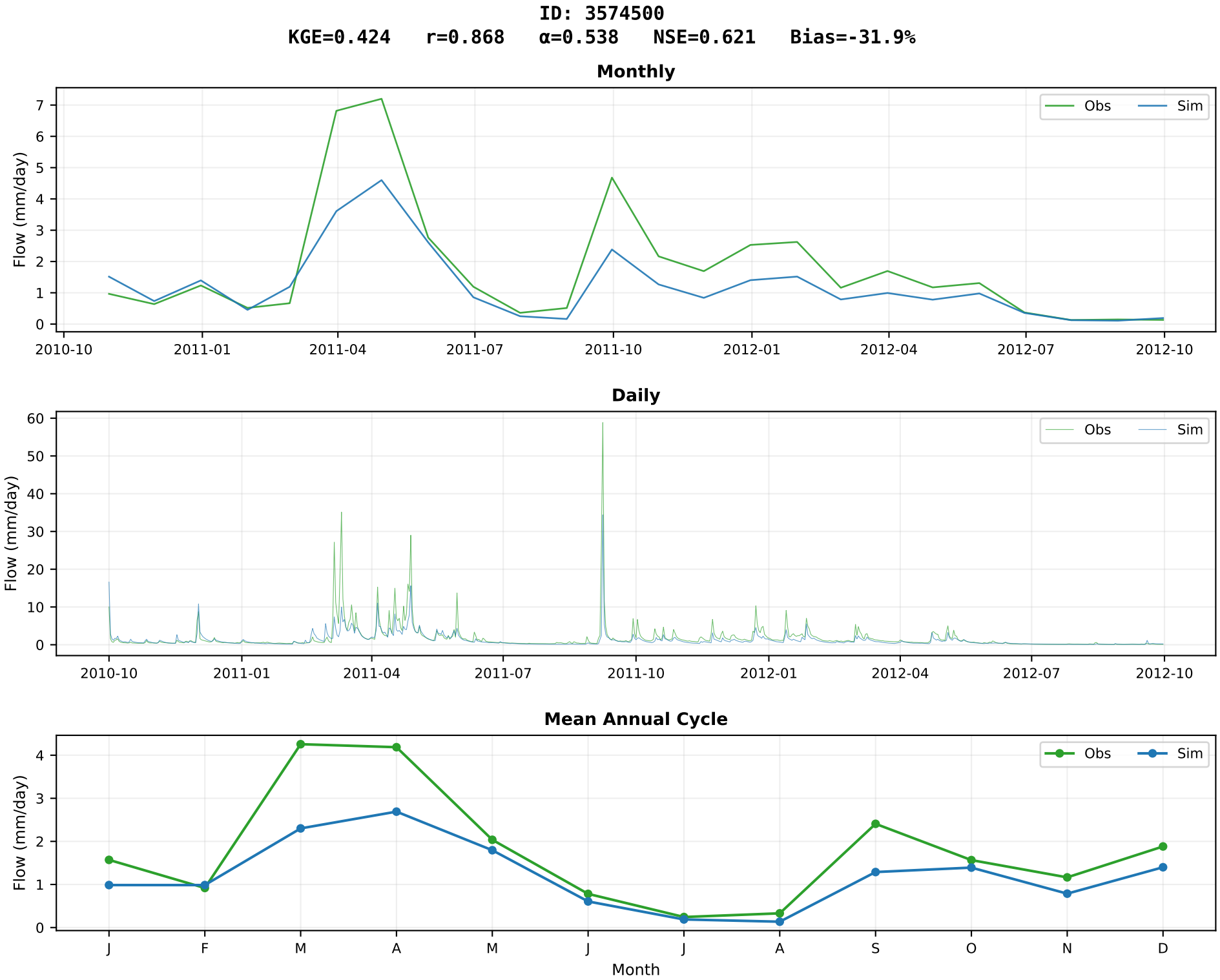

ID: 2196000
KGE=0.427 r=0.660 α=1.430 NSE=-0.174 Bias=-16.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

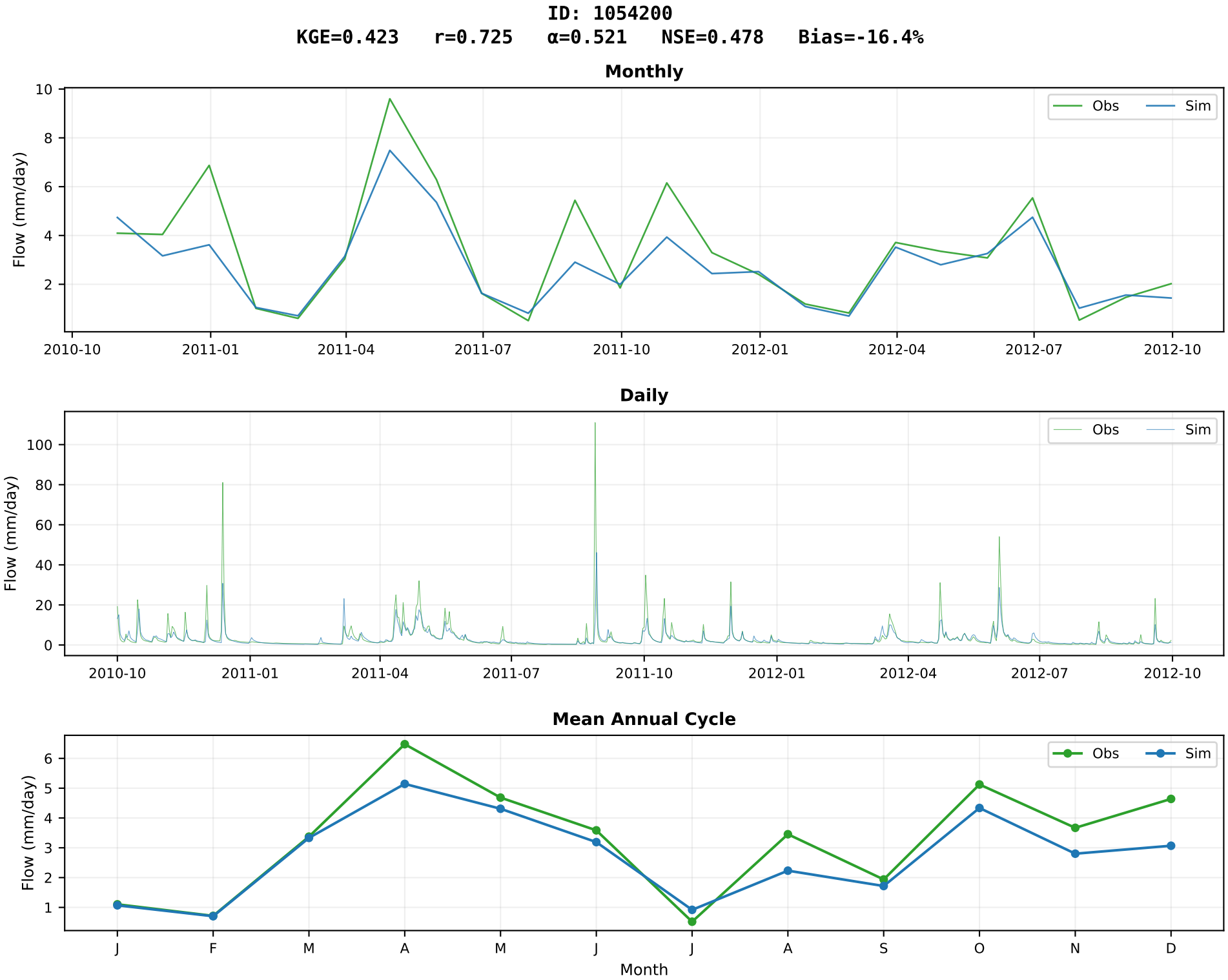
ID: 2077200

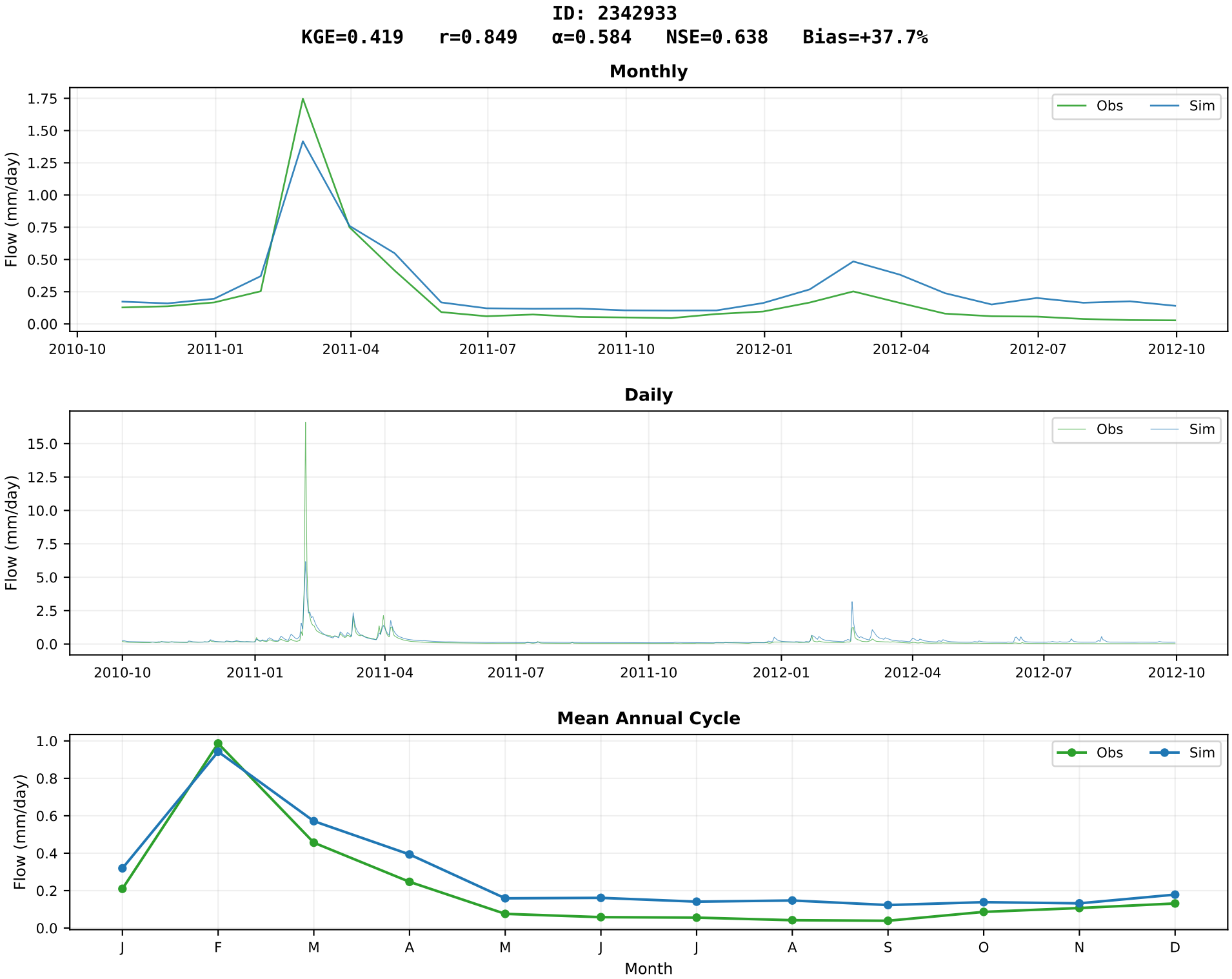
KGE=0.425 r=0.874 α=1.057 NSE=0.682 Bias=+55.8%

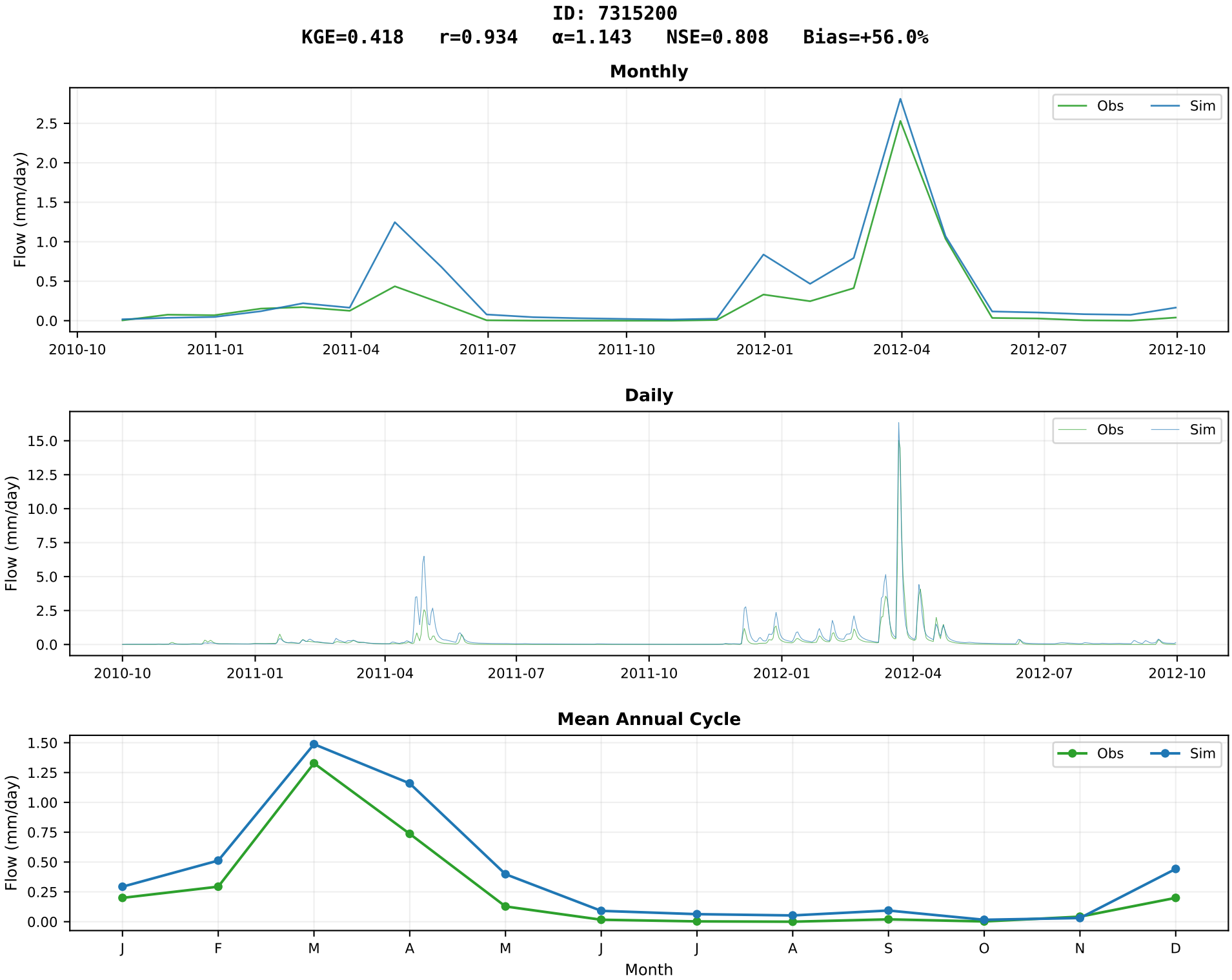


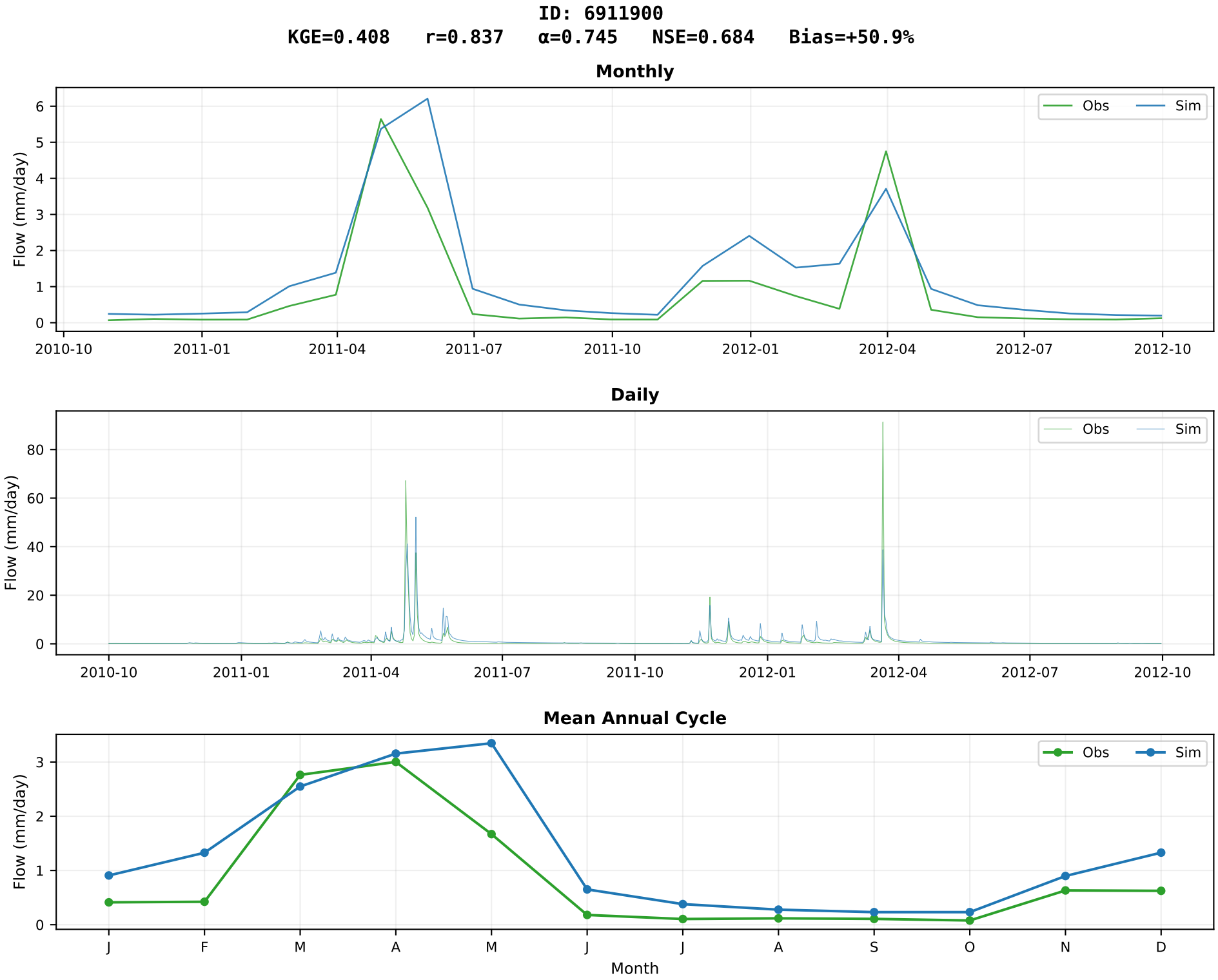

ID: 1557500
KGE=0.424 r=0.603 α=0.599 NSE=0.360 Bias=-11.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

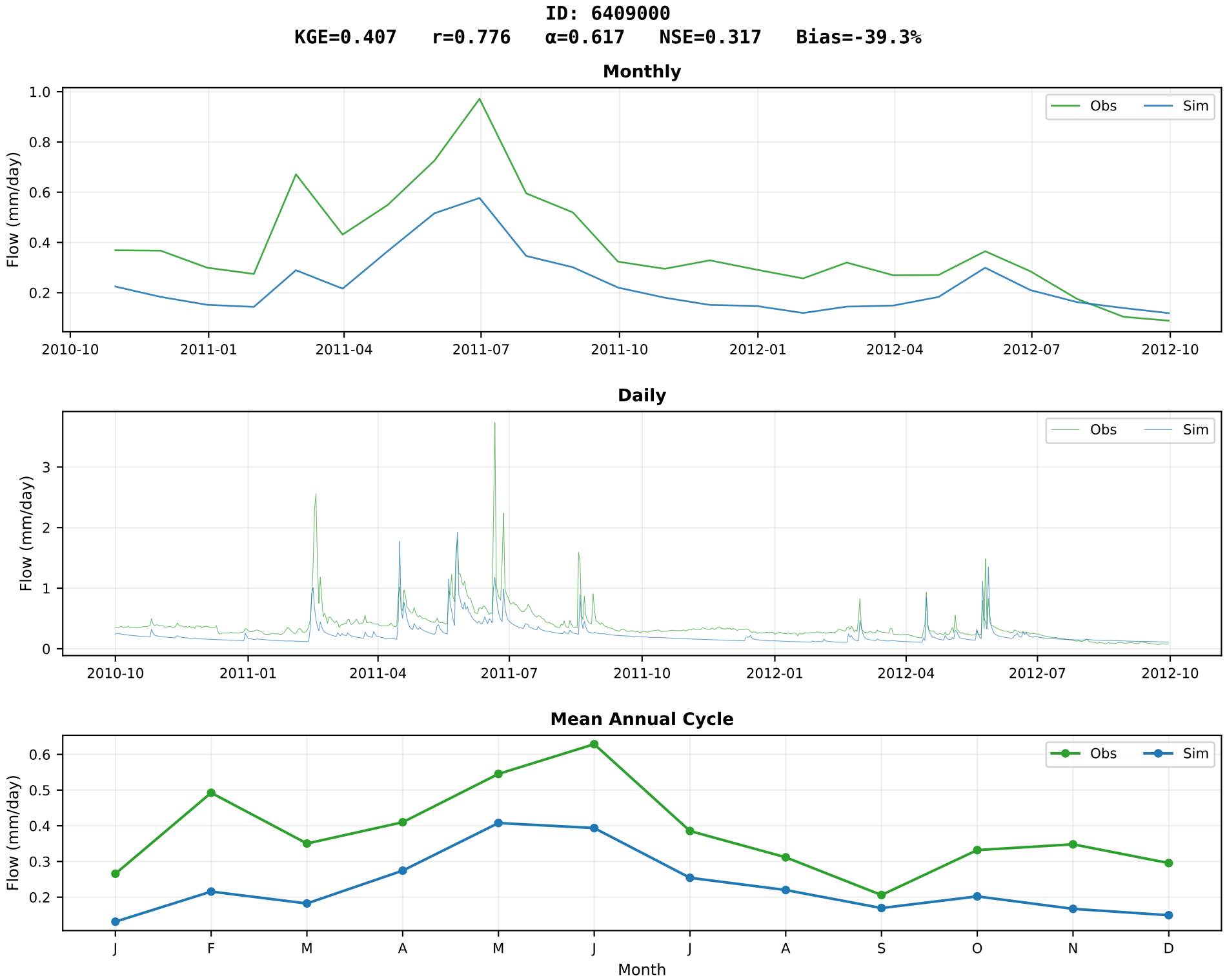

ID: 3574500
KGE=0.424 r=0.868 α=0.538 NSE=0.621 Bias=-31.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

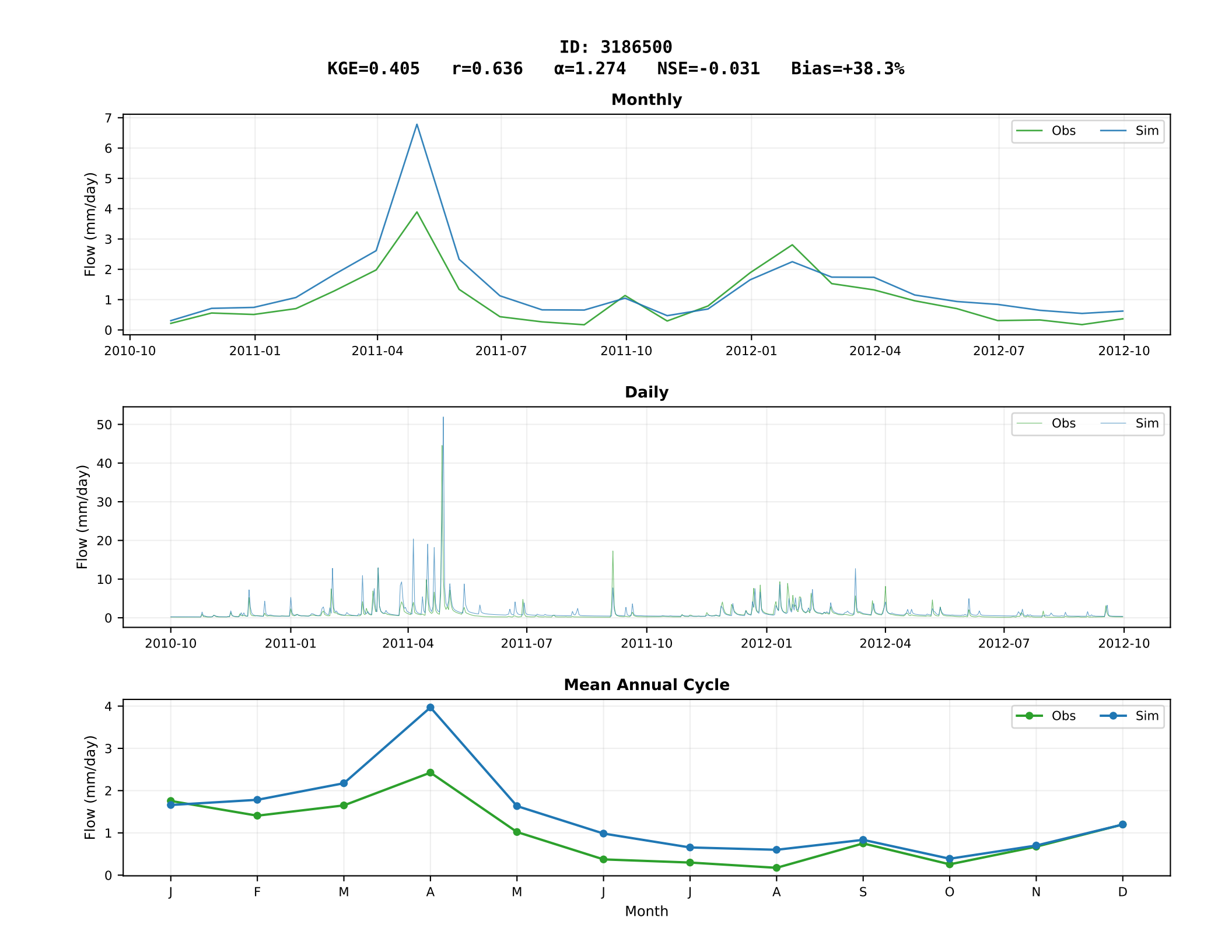

ID: 1054200
KGE=0.423 r=0.725 α=0.521 NSE=0.478 Bias=-16.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

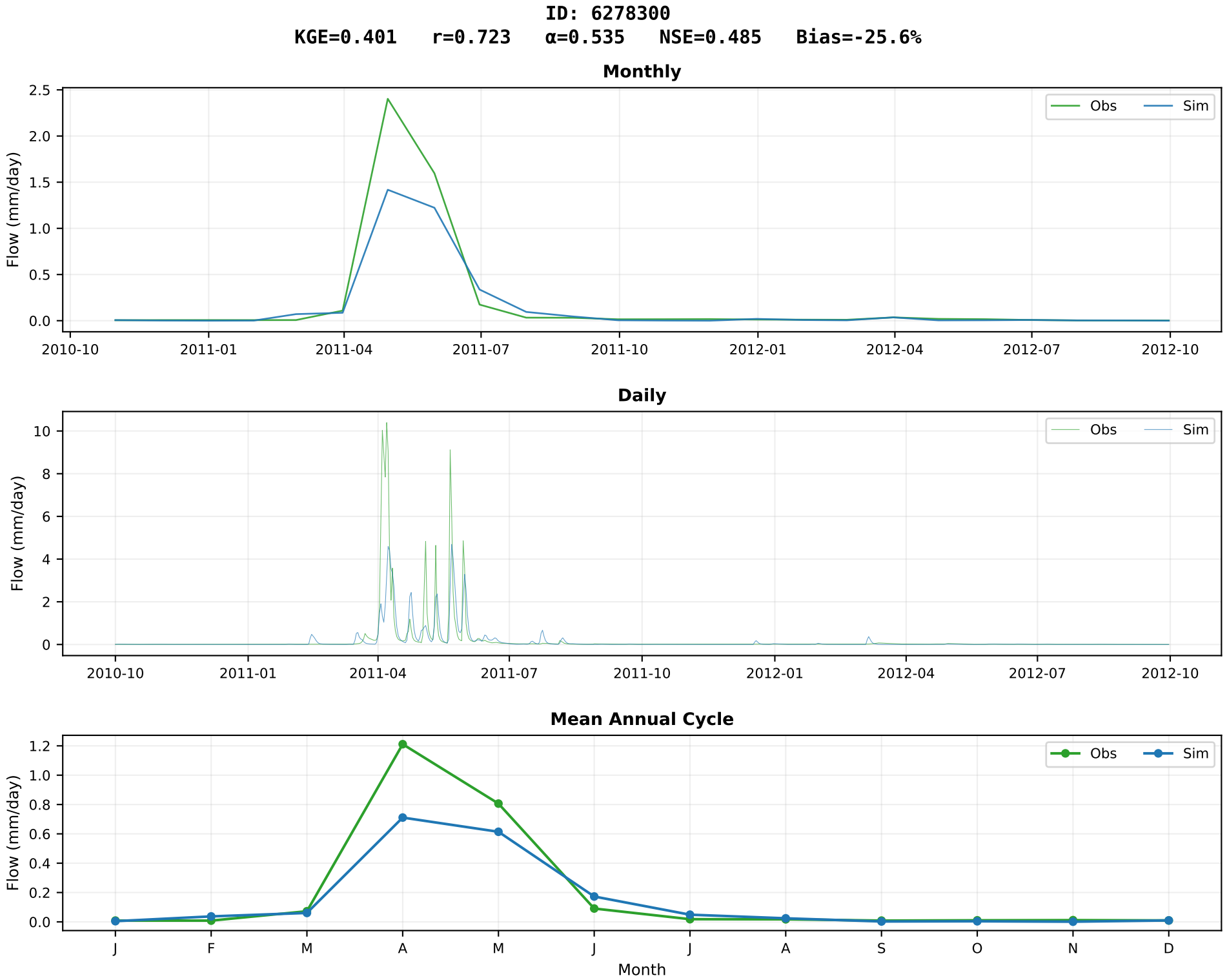

ID: 2342933
KGE=0.419 r=0.849 α=0.584 NSE=0.638 Bias=+37.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

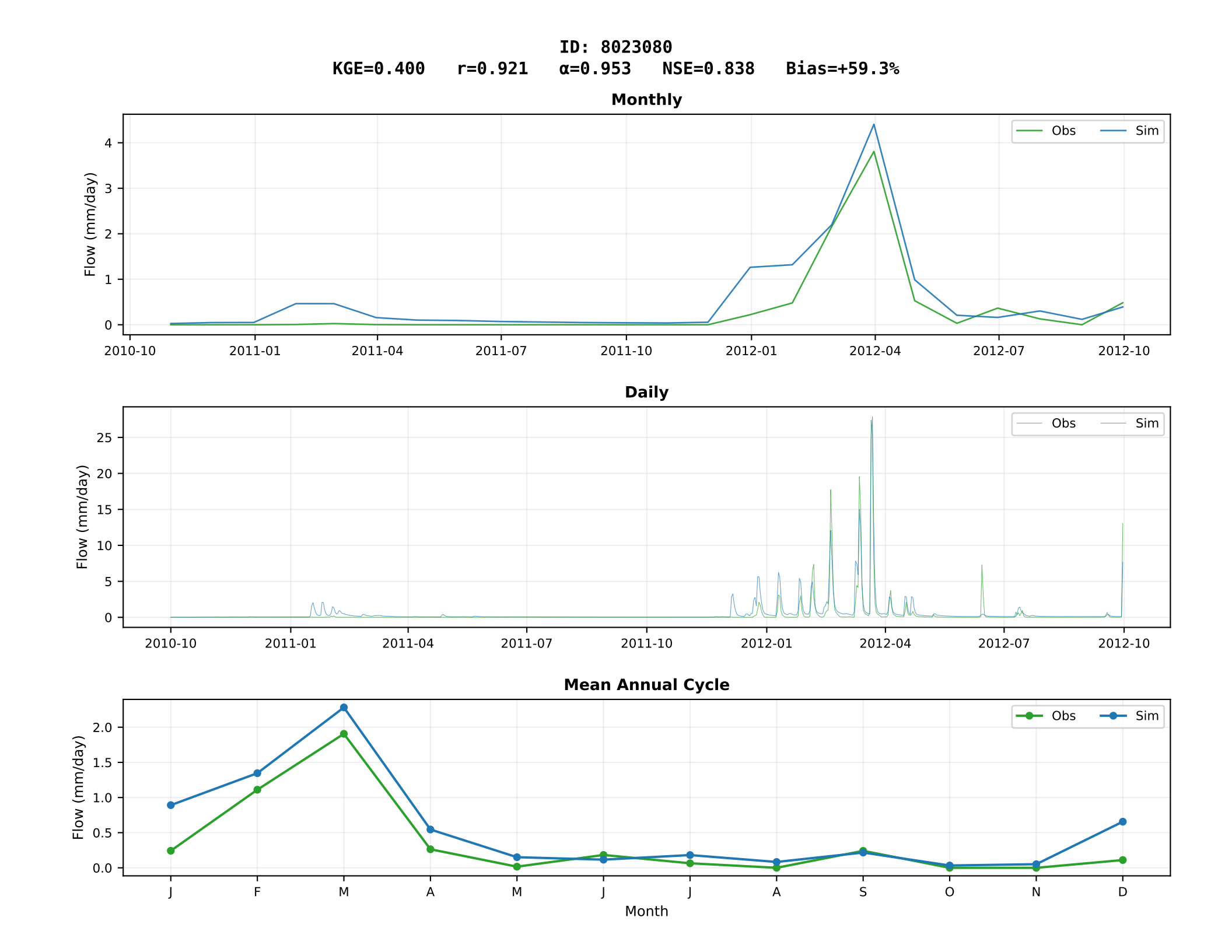

ID: 7315200
KGE=0.418 r=0.934 α=1.143 NSE=0.808 Bias=+56.0%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

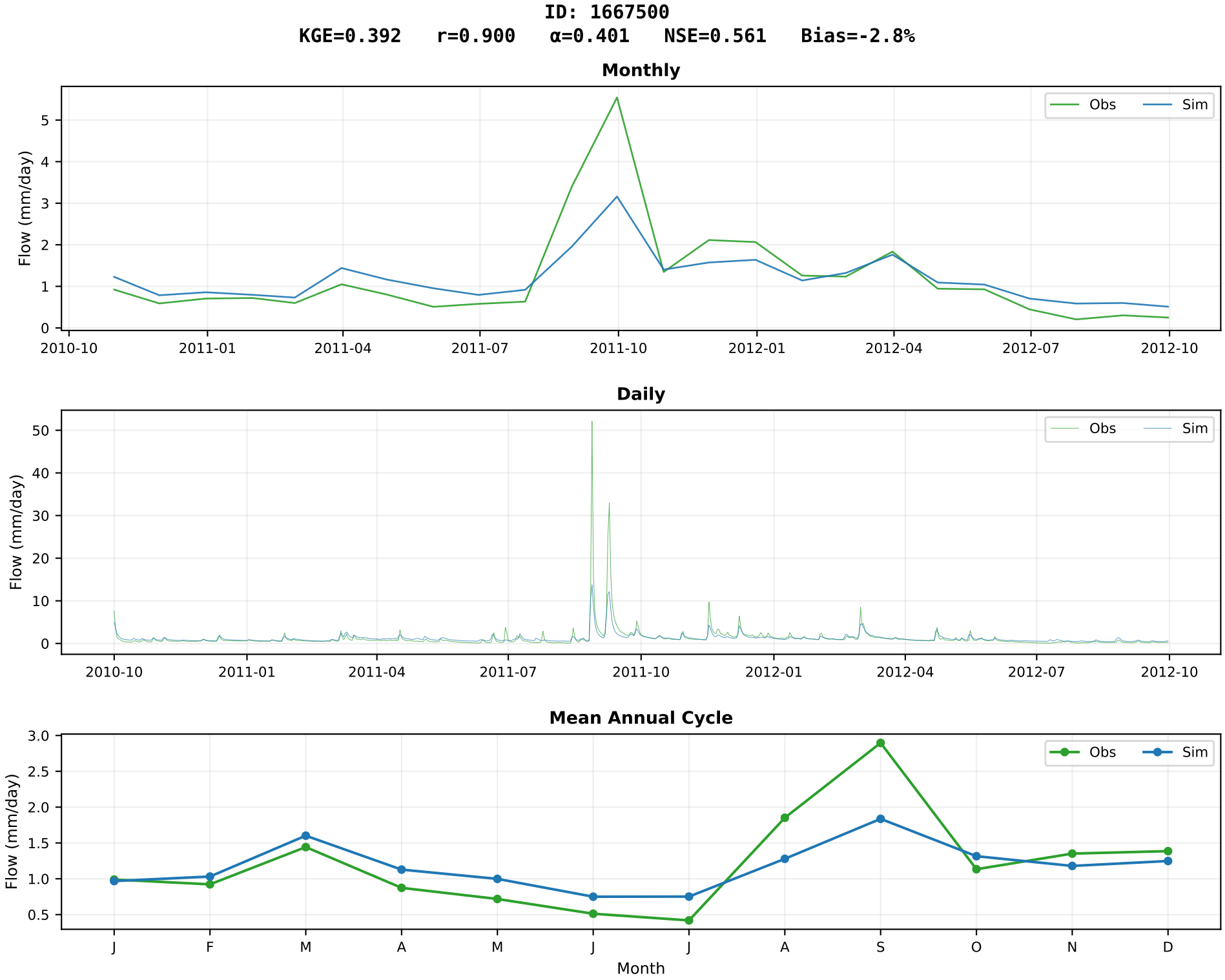

ID: 6911900
KGE=0.408 r=0.837 α=0.745 NSE=0.684 Bias=+50.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

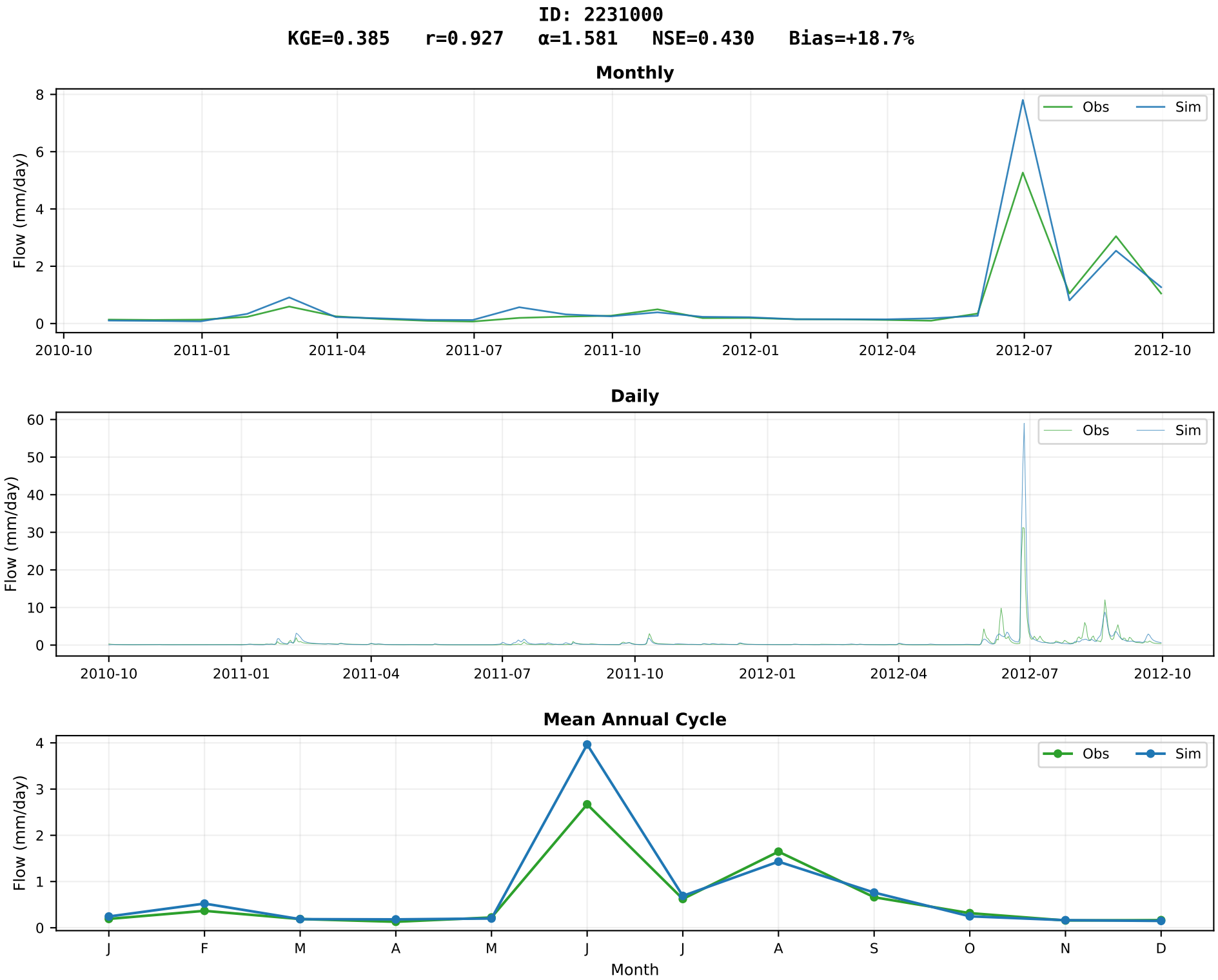

ID: 6409000
KGE=0.407 r=0.776 α=0.617 NSE=0.317 Bias=-39.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

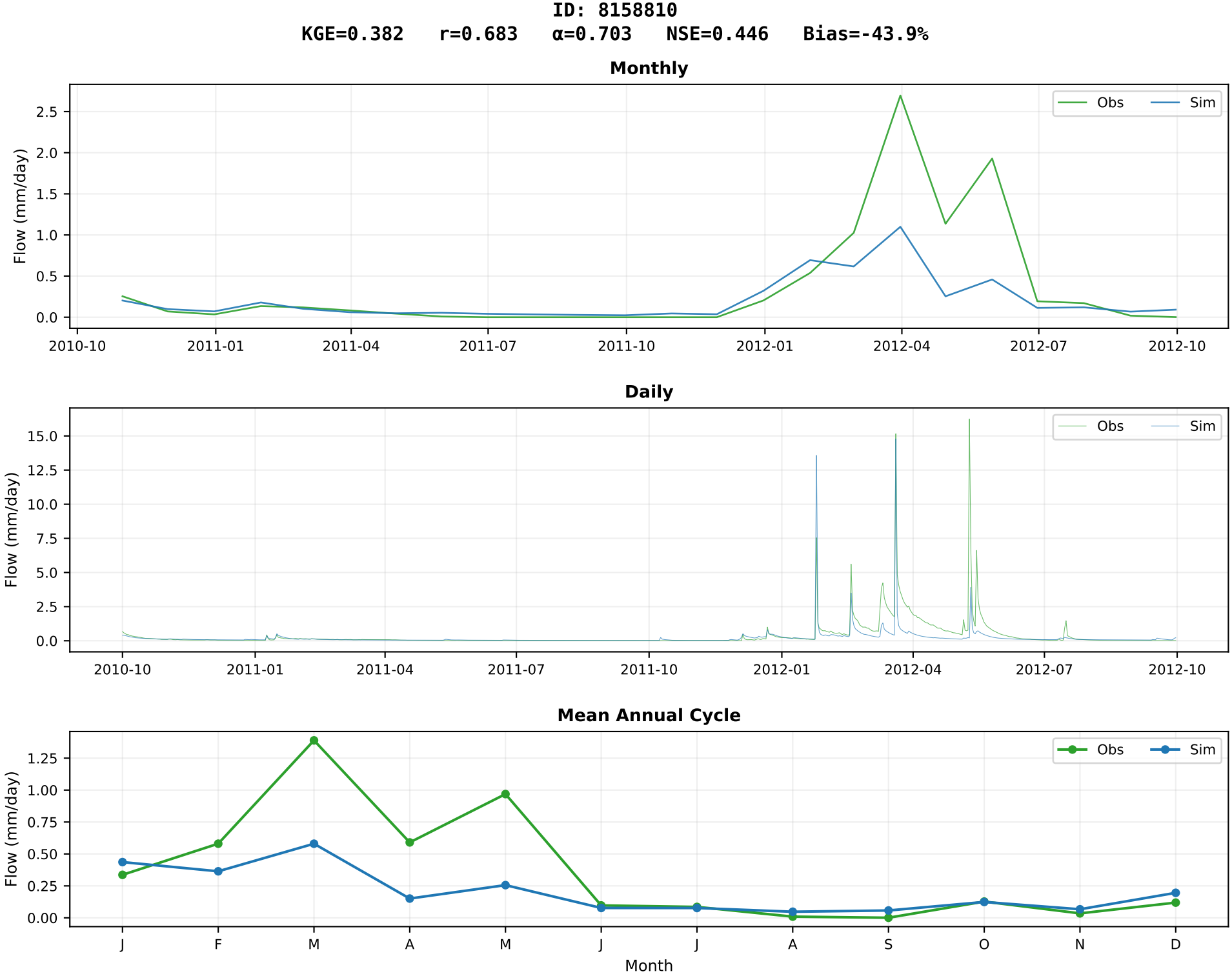

ID: 3186500
KGE=0.405 r=0.636 α=1.274 NSE=-0.031 Bias=+38.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

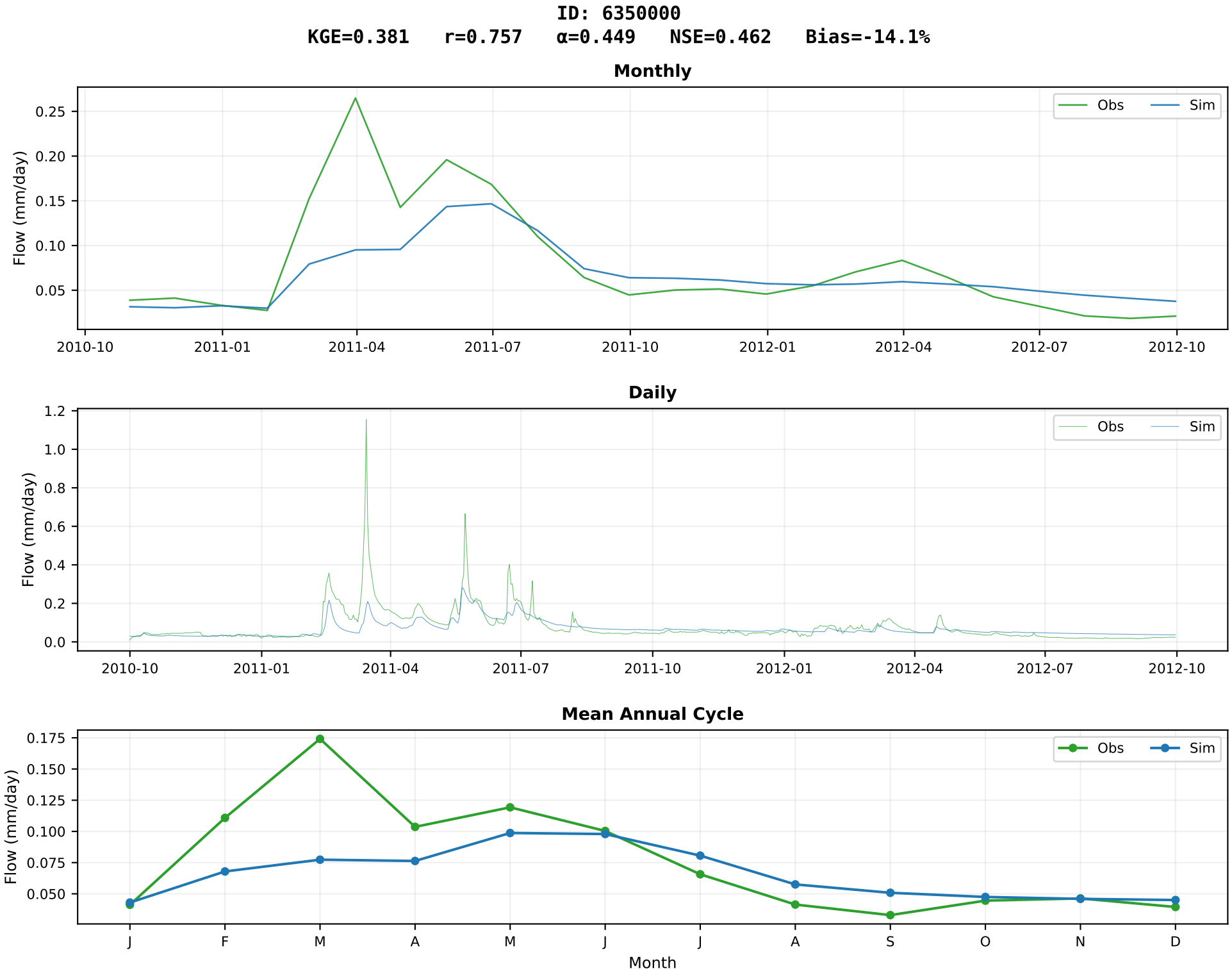

ID: 6278300
KGE=0.401 r=0.723 α=0.535 NSE=0.485 Bias=-25.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

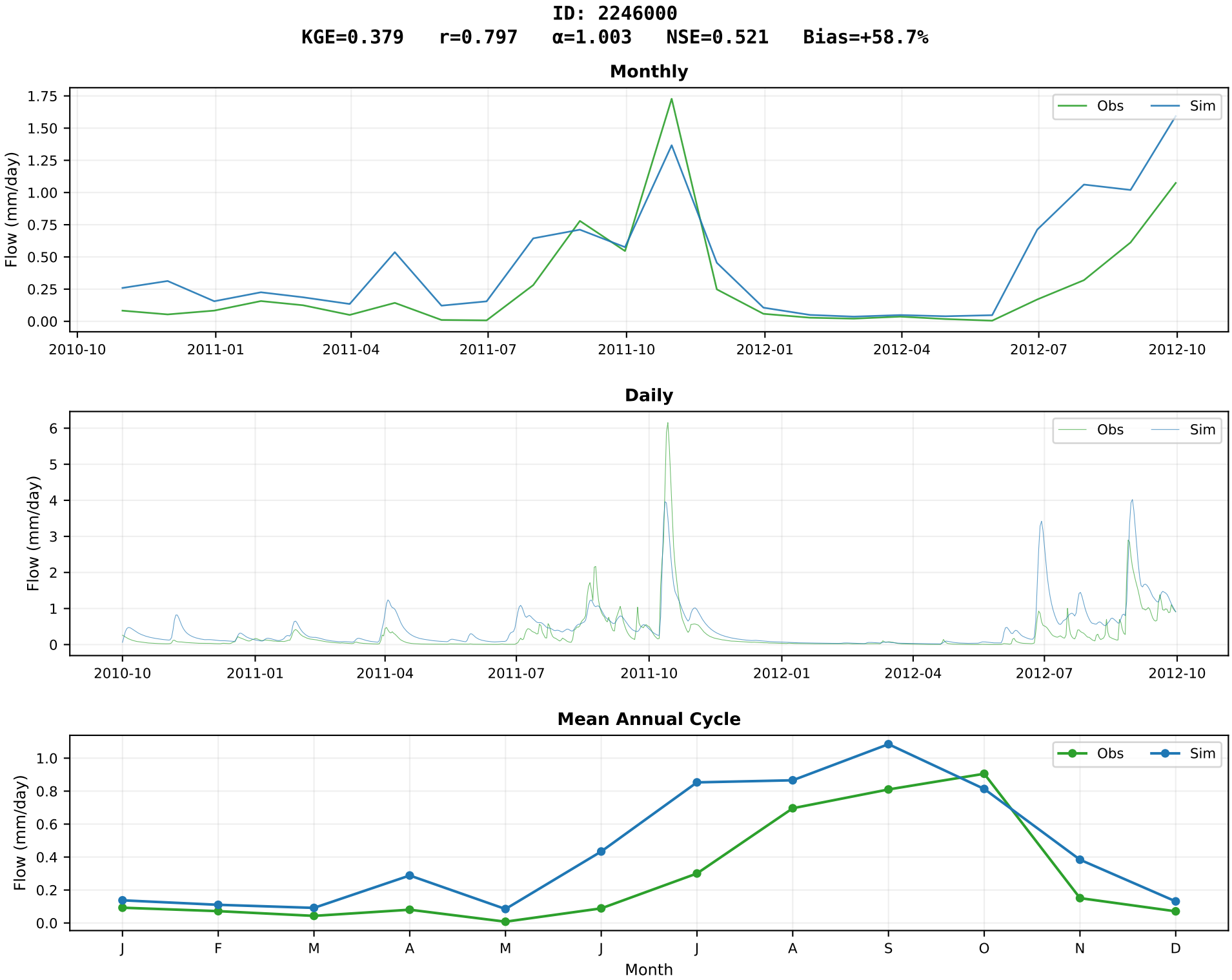

ID: 8023080
KGE=0.400 r=0.921 α=0.953 NSE=0.838 Bias=+59.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

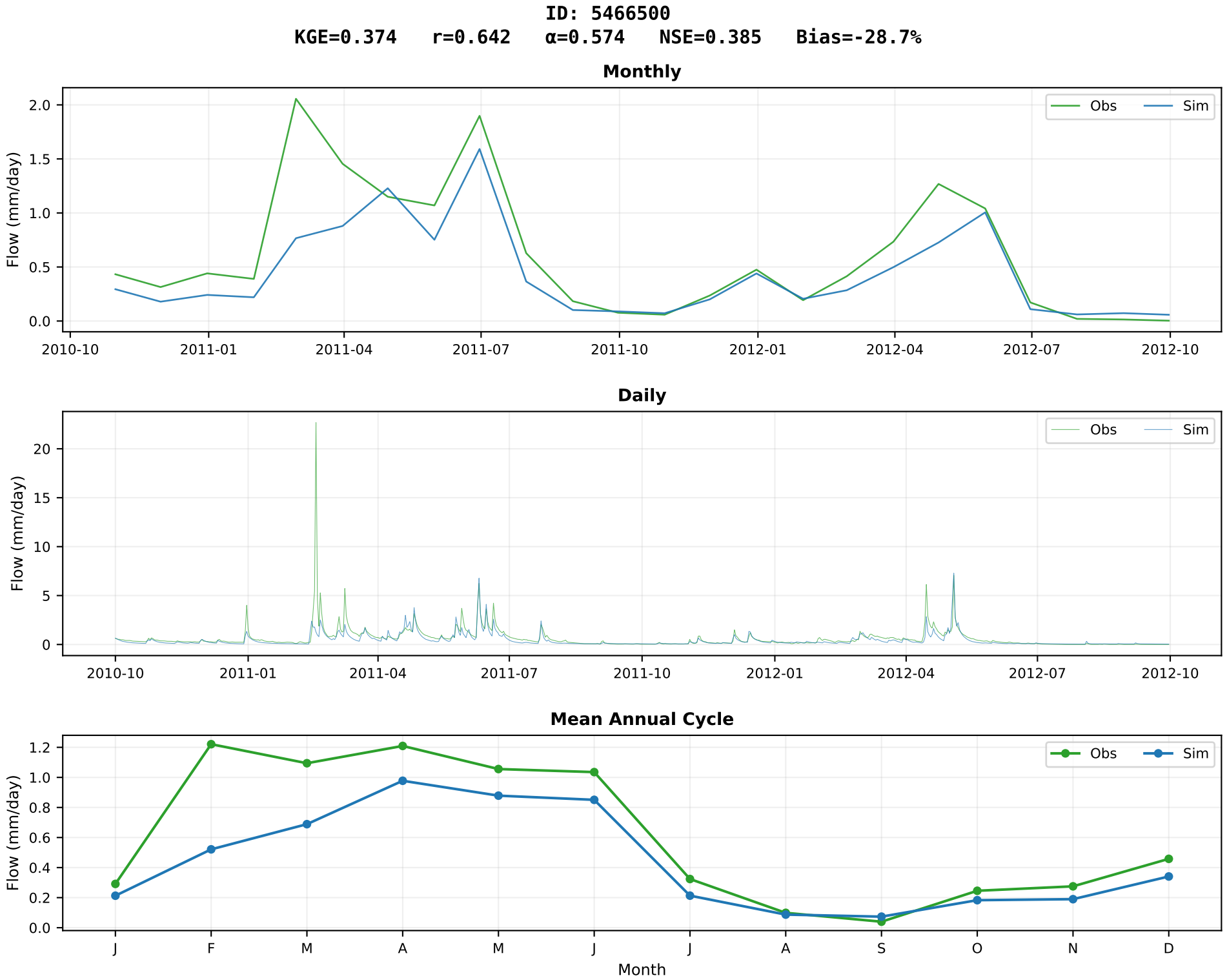

ID: 1667500
KGE=0.392 r=0.900 α=0.401 NSE=0.561 Bias=-2.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

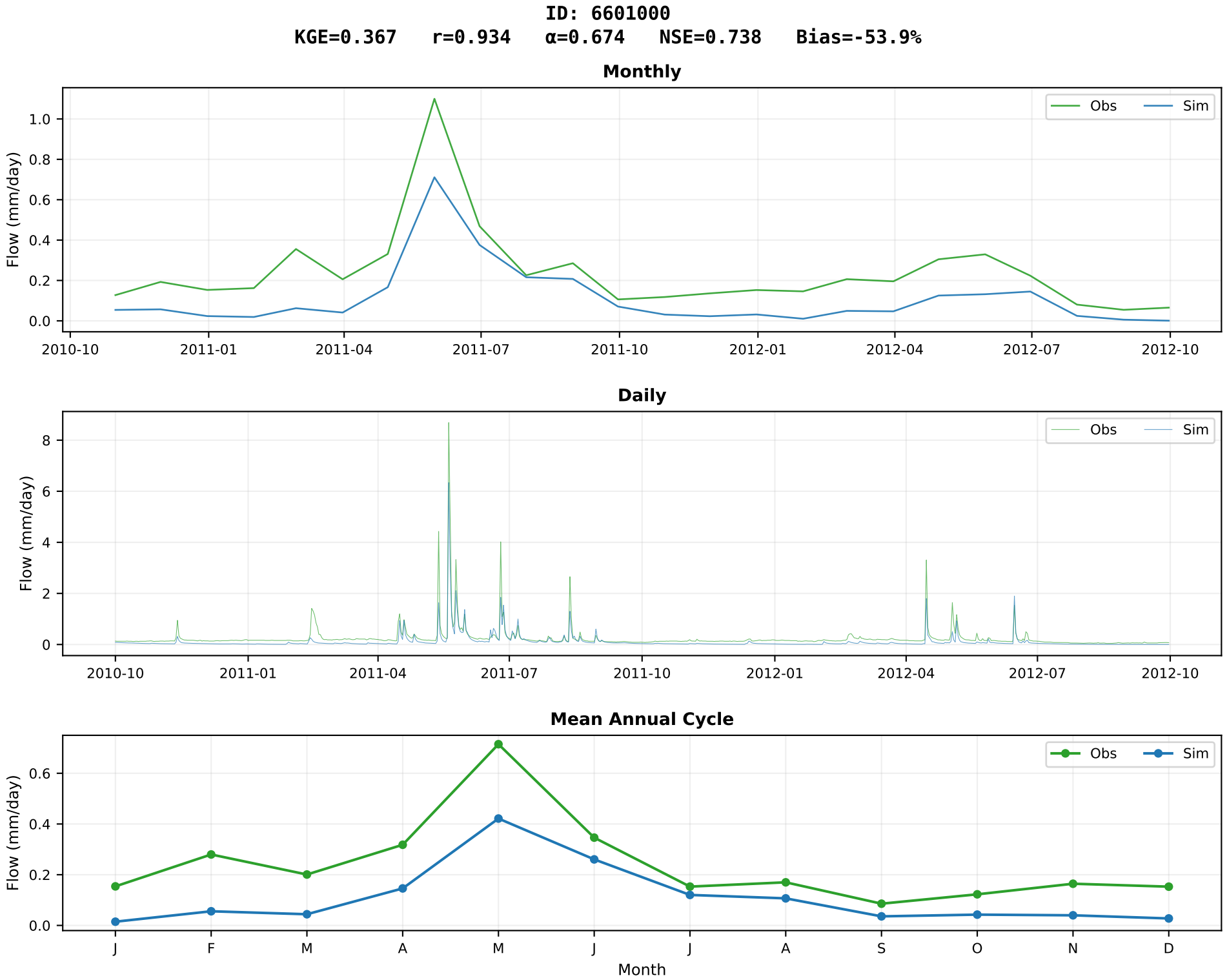

ID: 2231000
KGE=0.385 r=0.927 α=1.581 NSE=0.430 Bias=+18.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

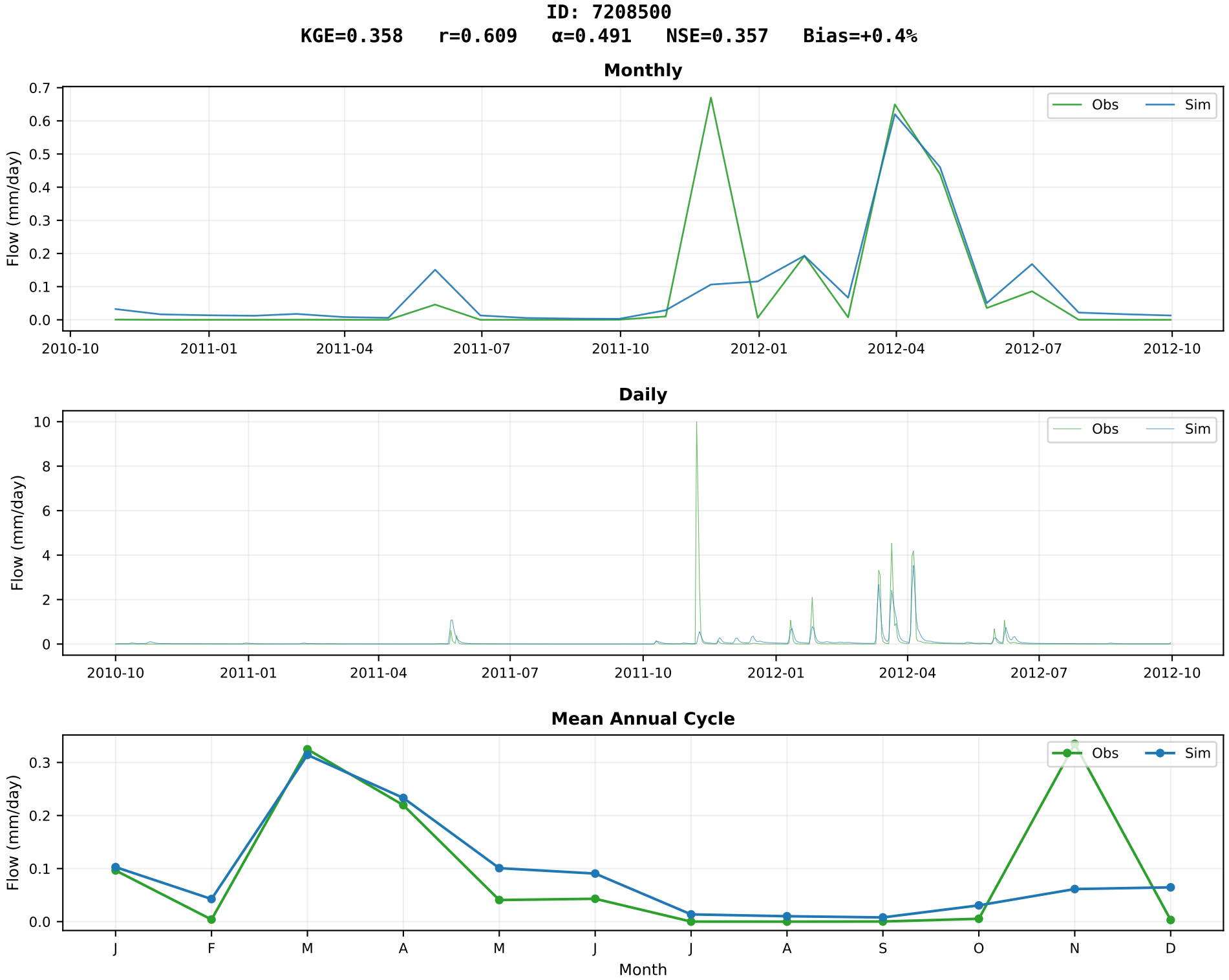

ID: 8158810
KGE=0.382 r=0.683 α=0.703 NSE=0.446 Bias=-43.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

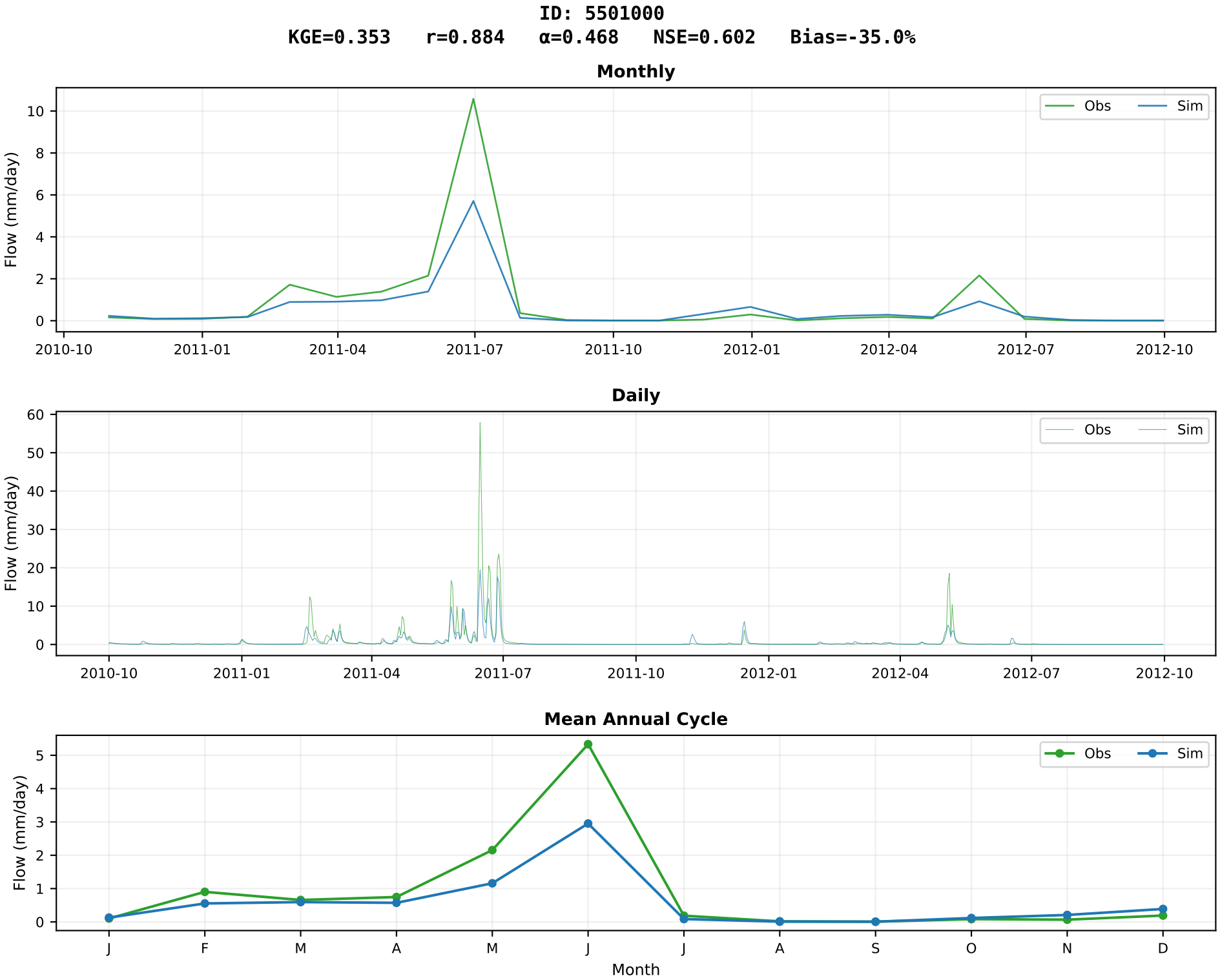

ID: 6350000
KGE=0.381 r=0.757 α=0.449 NSE=0.462 Bias=-14.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

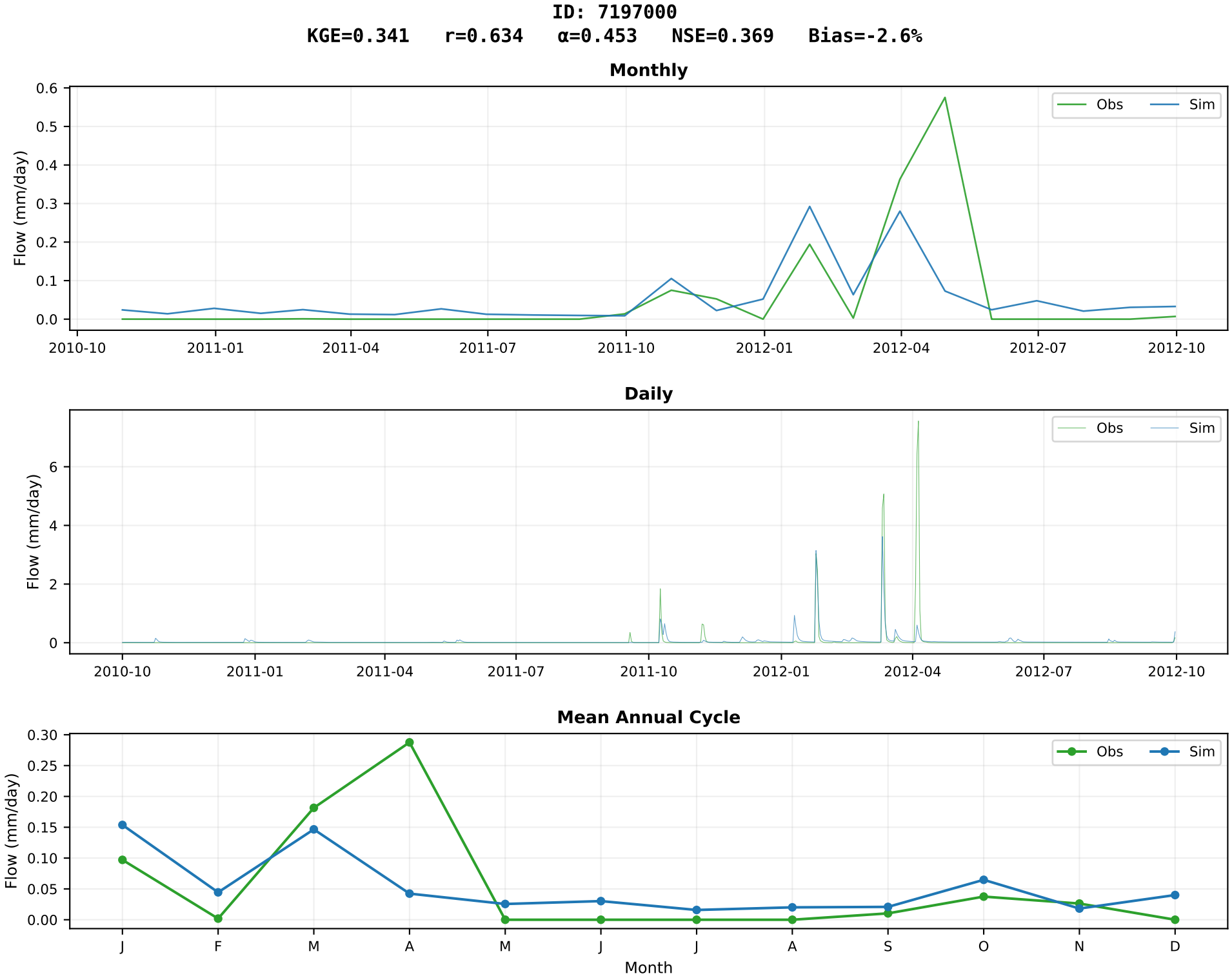

ID: 2246000
KGE=0.379 r=0.797 α=1.003 NSE=0.521 Bias=+58.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

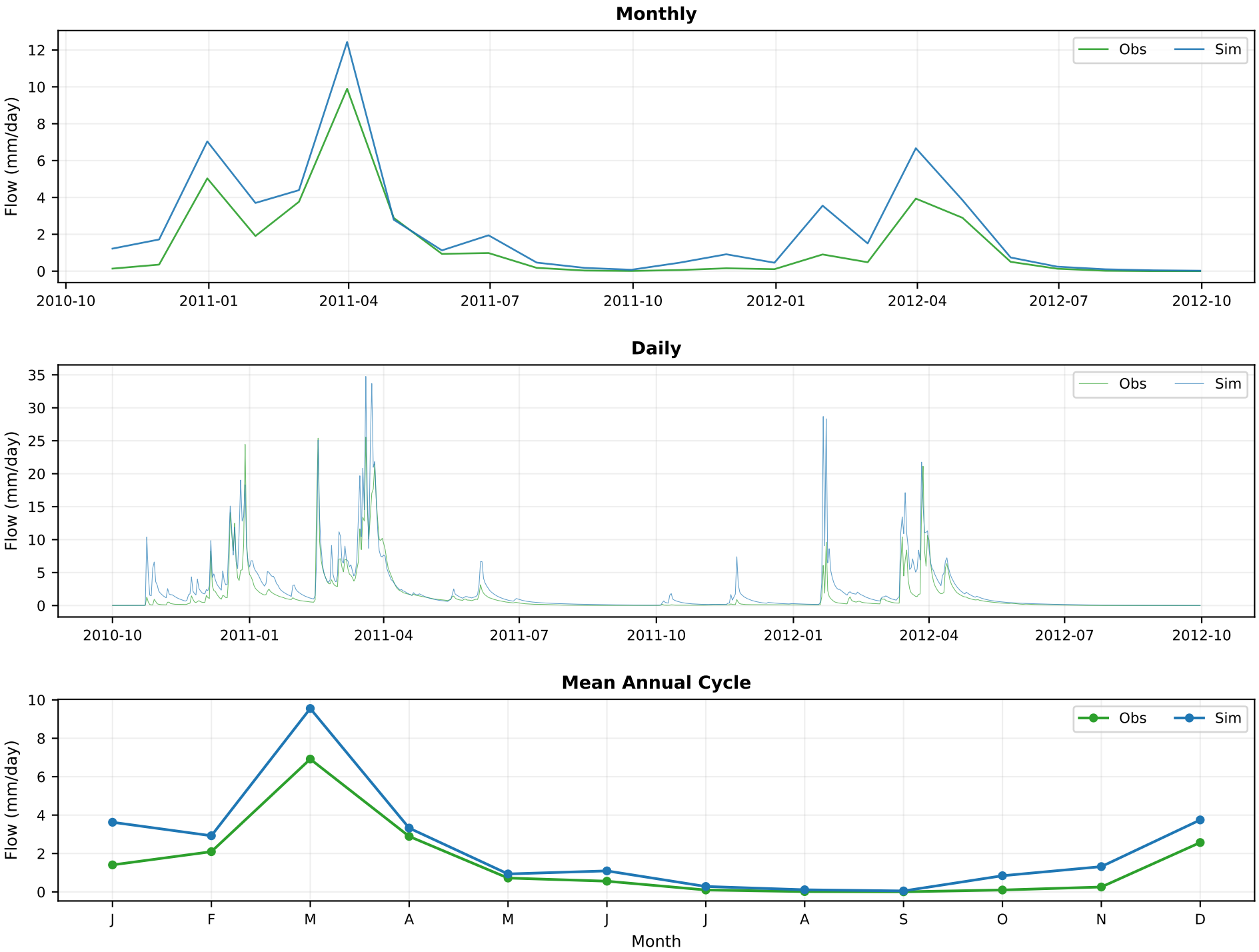

ID: 5466500
KGE=0.374 r=0.642 α=0.574 NSE=0.385 Bias=-28.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

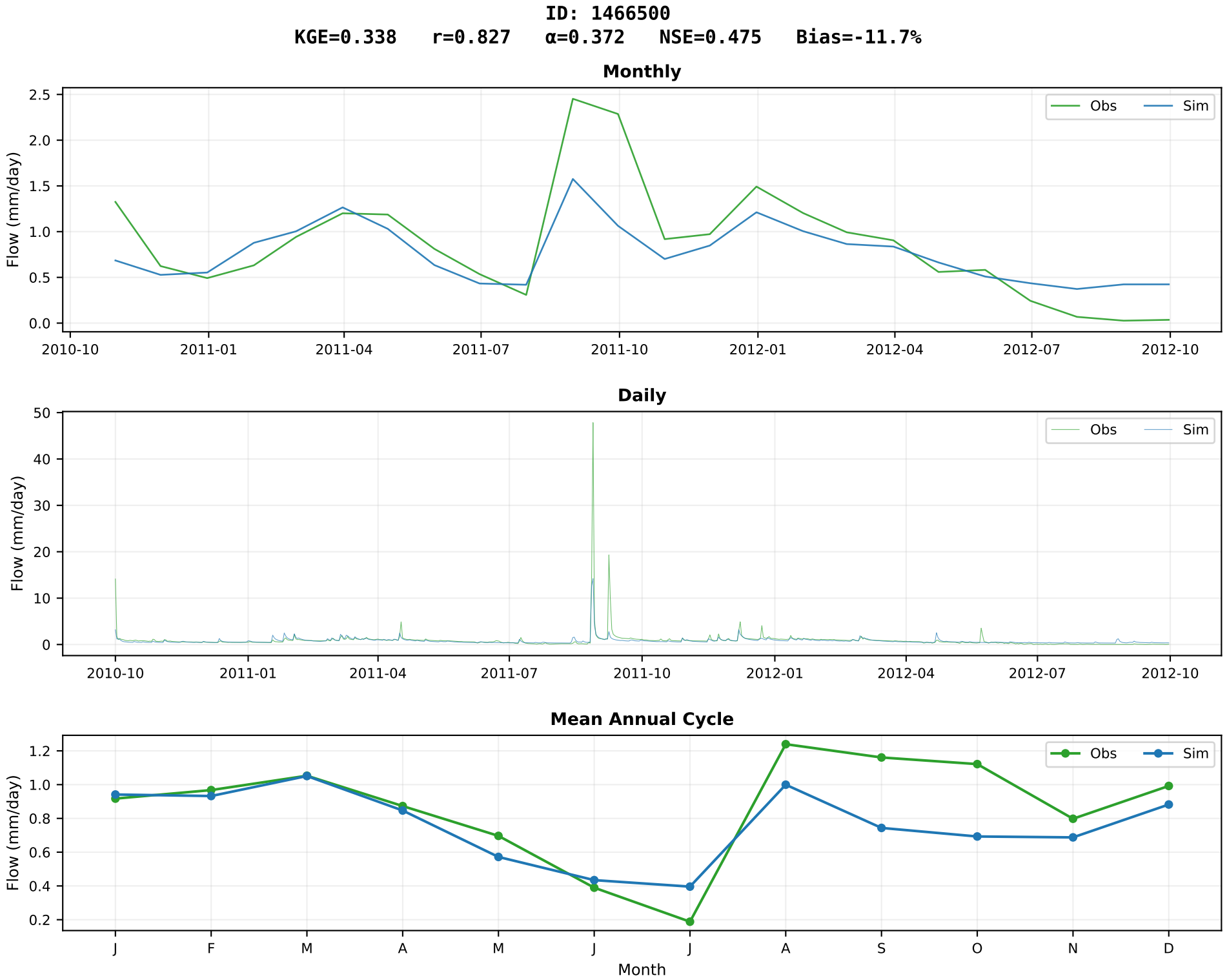

ID: 6601000
KGE=0.367 r=0.934 α=0.674 NSE=0.738 Bias=-53.9%
Monthly
Obs
Sim
Flow (mm/day)
1.0
0.8
0.6
0.4
0.2
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
8
6
4
2
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.6
0.4
0.2
0.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

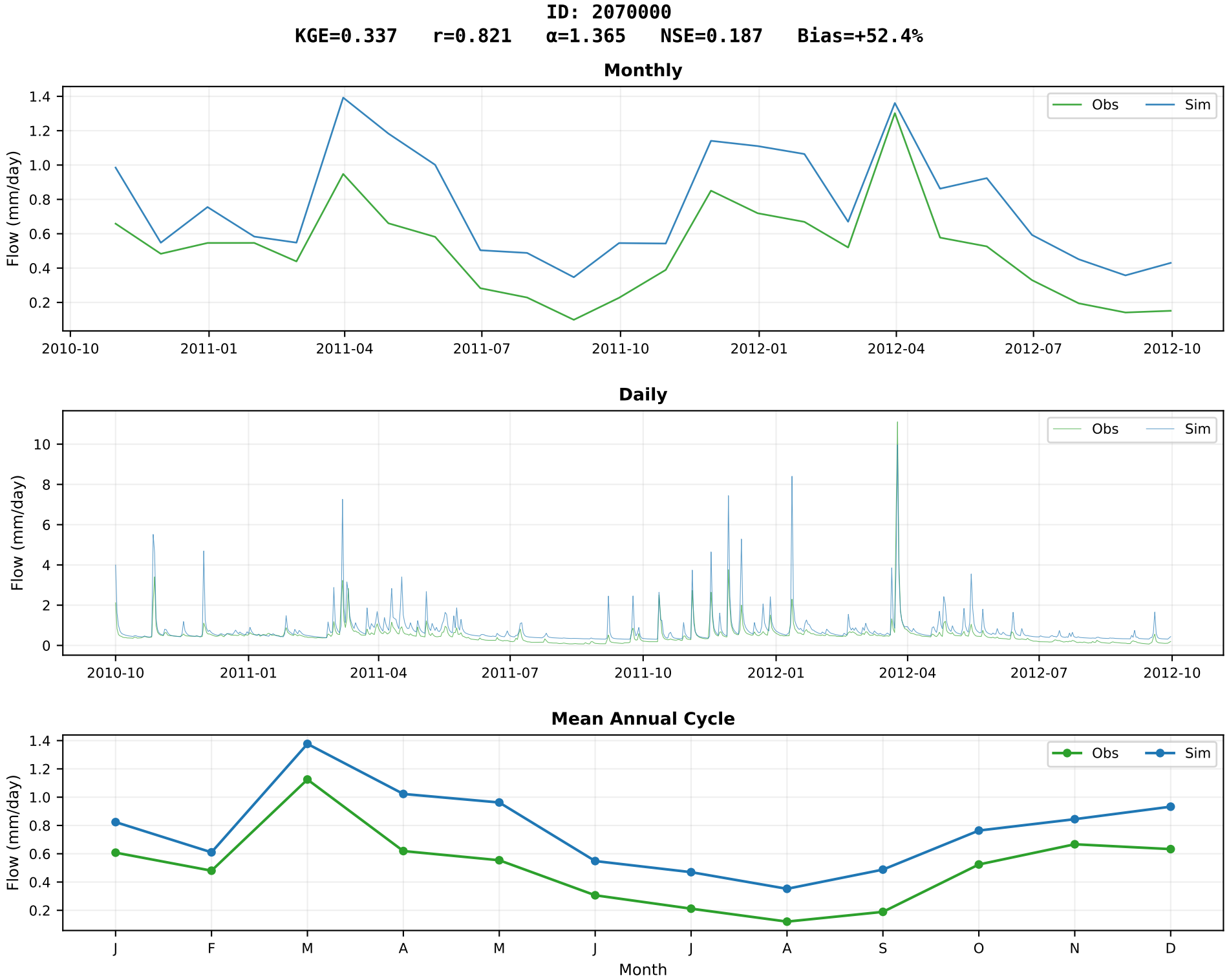

ID: 7208500
KGE=0.358 r=0.609 α=0.491 NSE=0.357 Bias=+0.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 5501000
KGE=0.353 r=0.884 α=0.468 NSE=0.602 Bias=-35.0%
Monthly
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
10
20
30
40
50
60
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
J
F
M
A
M
J
J
A
S
O
N
D
Month

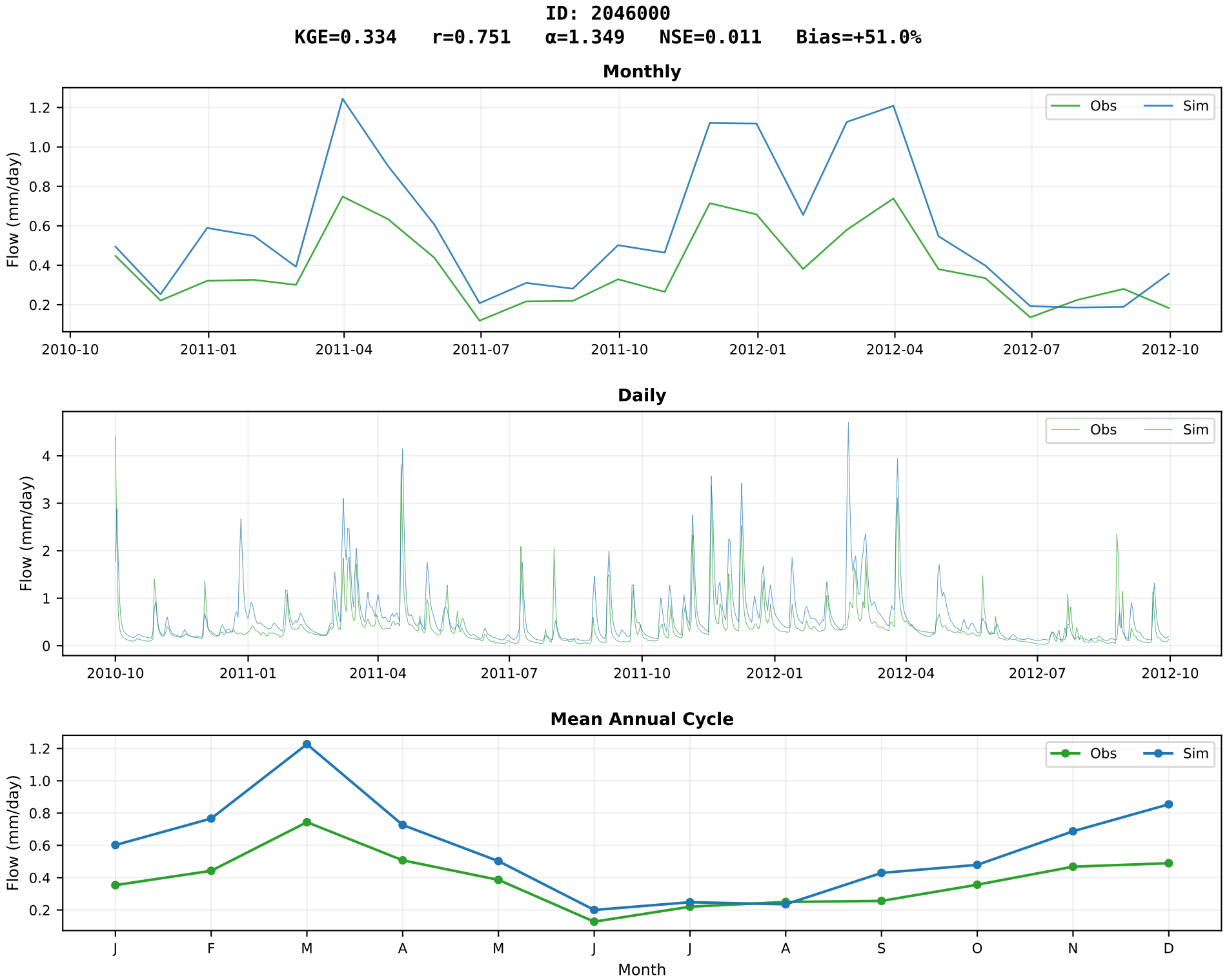

ID: 7197000
KGE=0.341 r=0.634 α=0.453 NSE=0.369 Bias=-2.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

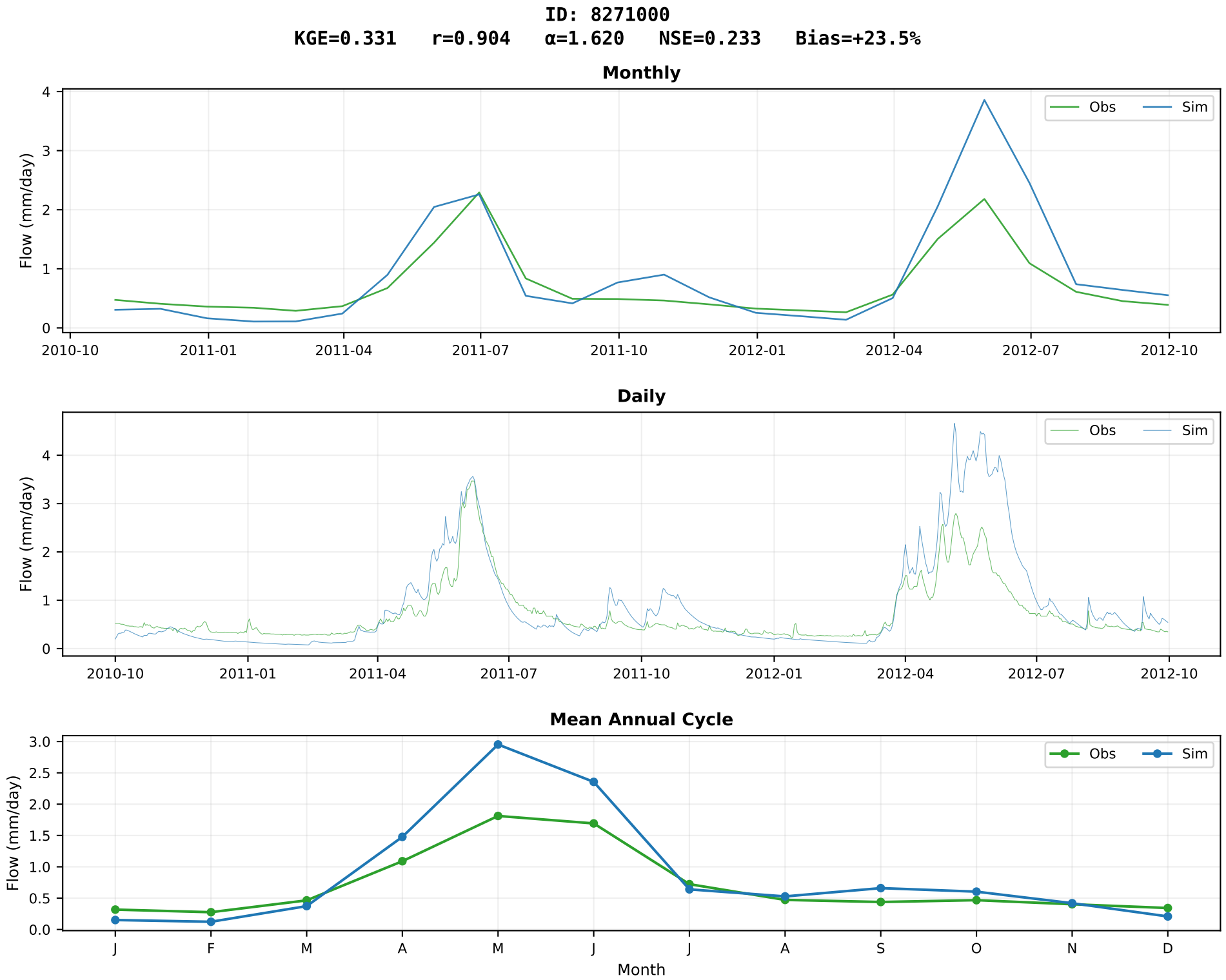

CEDAR RIVER BELOW BEAR CREEK NEAR CEDAR FALLS, WA (ID=12114500)
KGE=0.339 r=0.886 α=1.298 NSE=0.545 Bias=+57.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

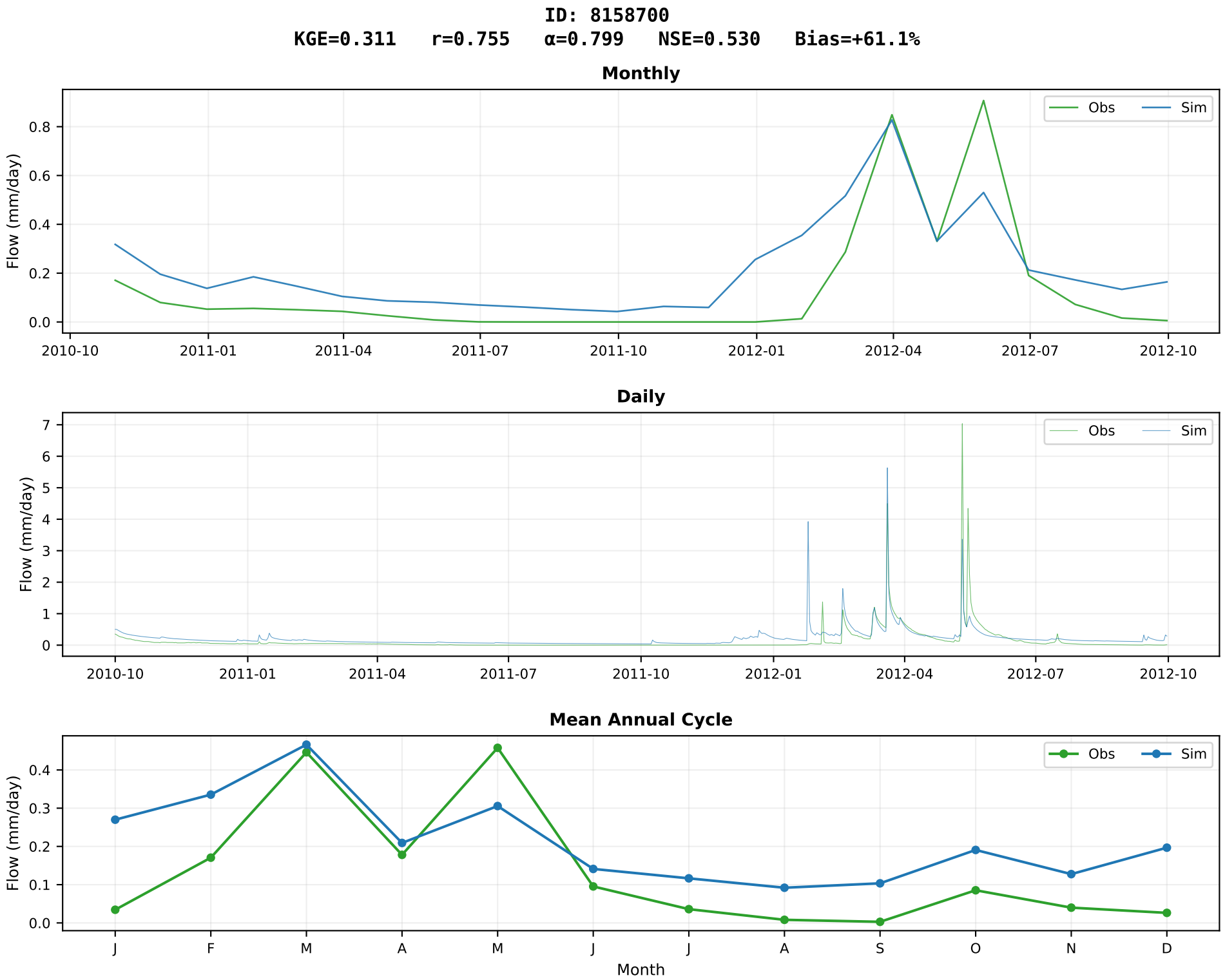

ID: 1466500
KGE=0.338 r=0.827 α=0.372 NSE=0.475 Bias=-11.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

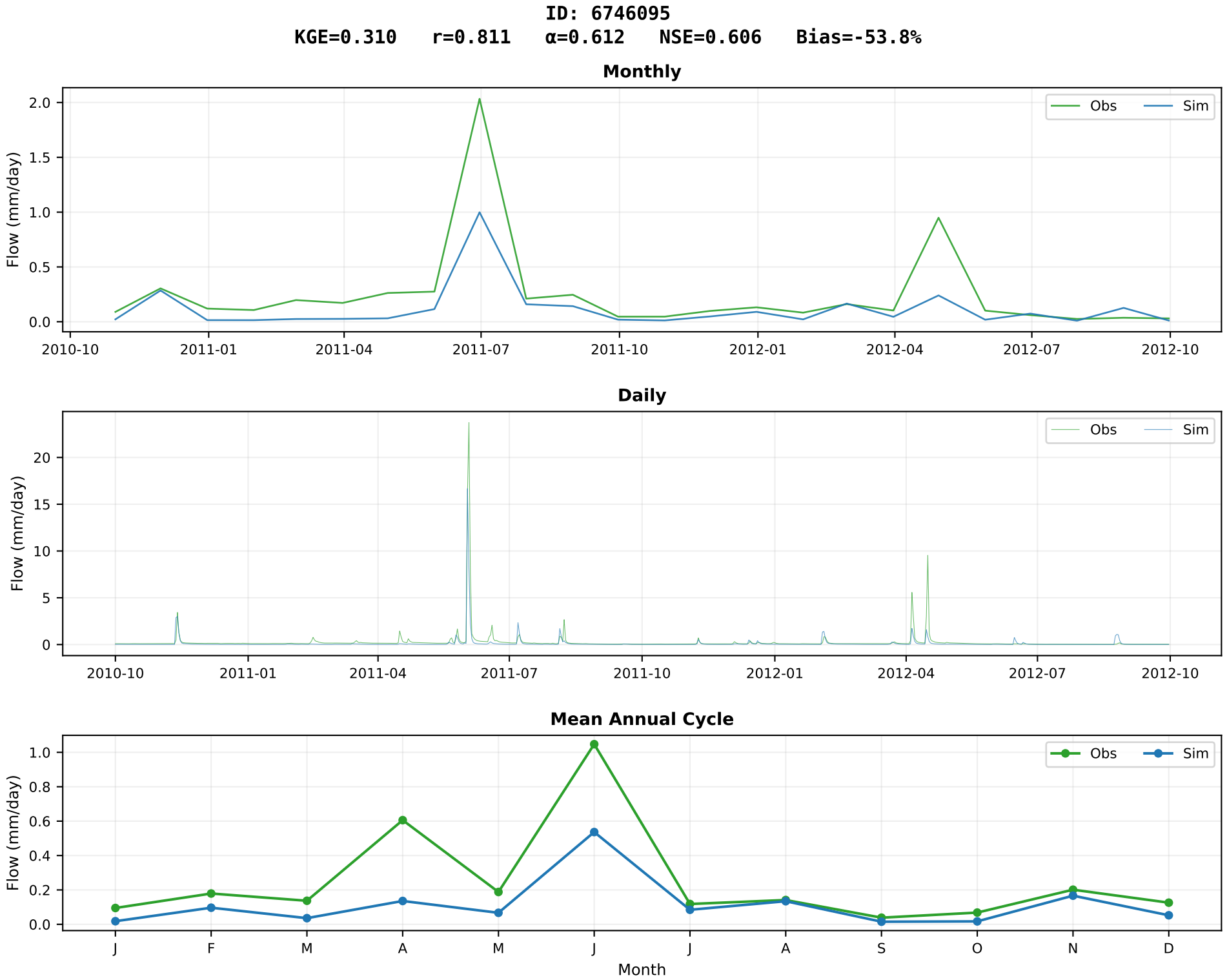

ID: 2070000
KGE=0.337 r=0.821 α=1.365 NSE=0.187 Bias=+52.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 6614800
KGE=0.335 r=0.541 α=1.462 NSE=-0.559 Bias=-13.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D


ID: 2046000
KGE=0.334 r=0.751 α=1.349 NSE=0.011 Bias=+51.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 8271000
KGE=0.331 r=0.904 α=1.620 NSE=0.233 Bias=+23.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

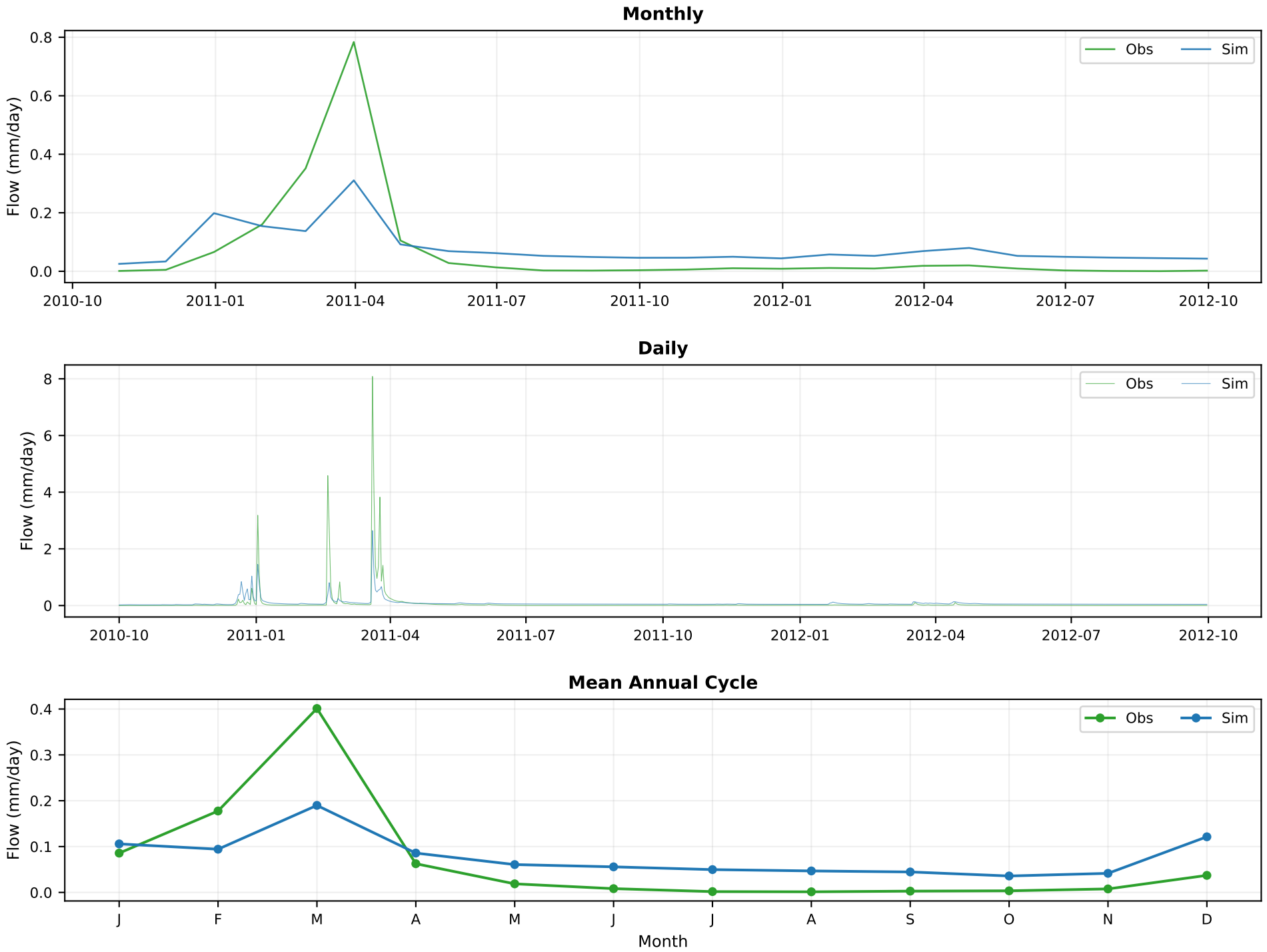

ID: 8158700
KGE=0.311 r=0.755 α=0.799 NSE=0.530 Bias=+61.1%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
7
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
J
F
M
A
M
J
J
A
S
O
N
D
Month

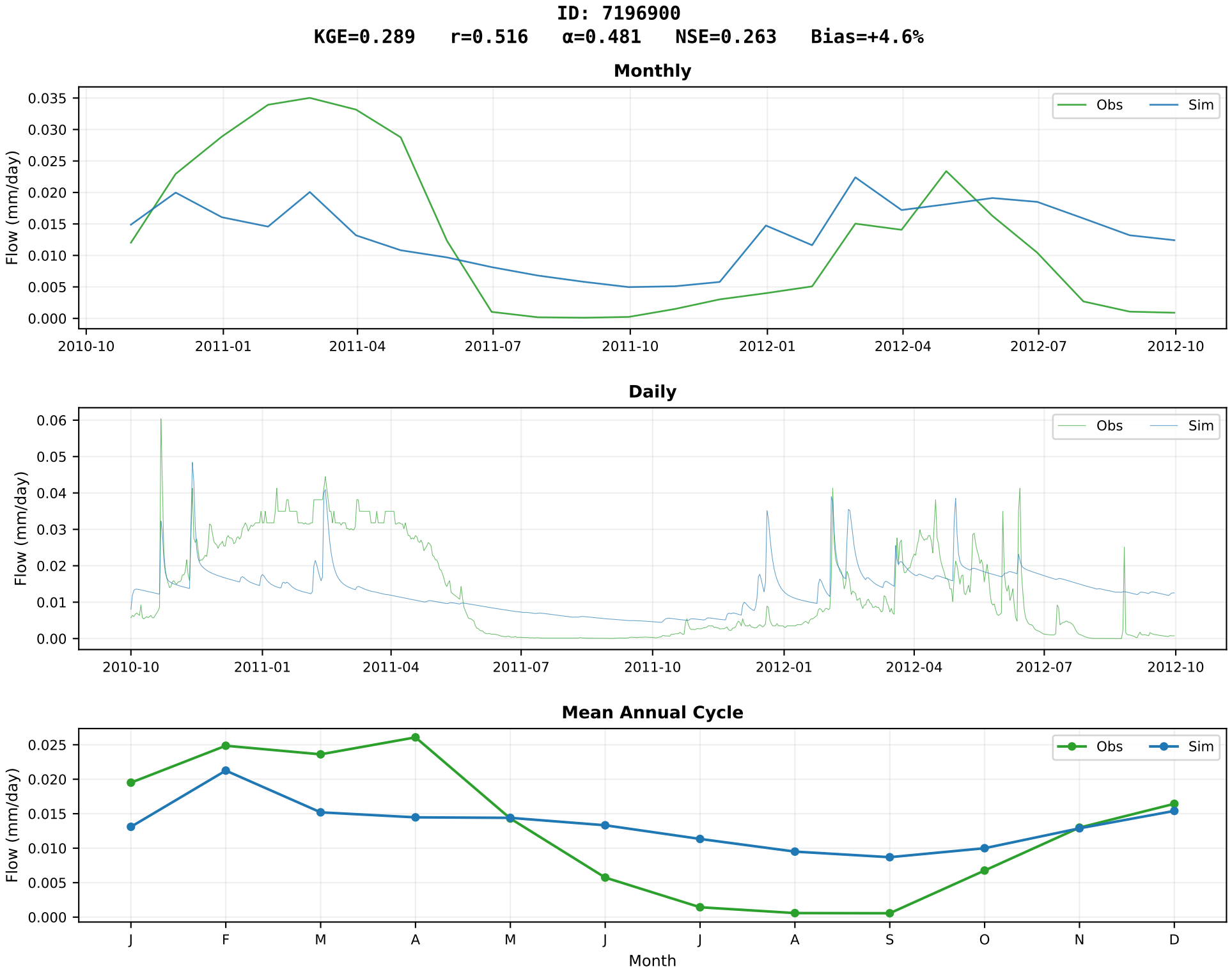

ID: 6746095
KGE=0.310 r=0.811 α=0.612 NSE=0.606 Bias=-53.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

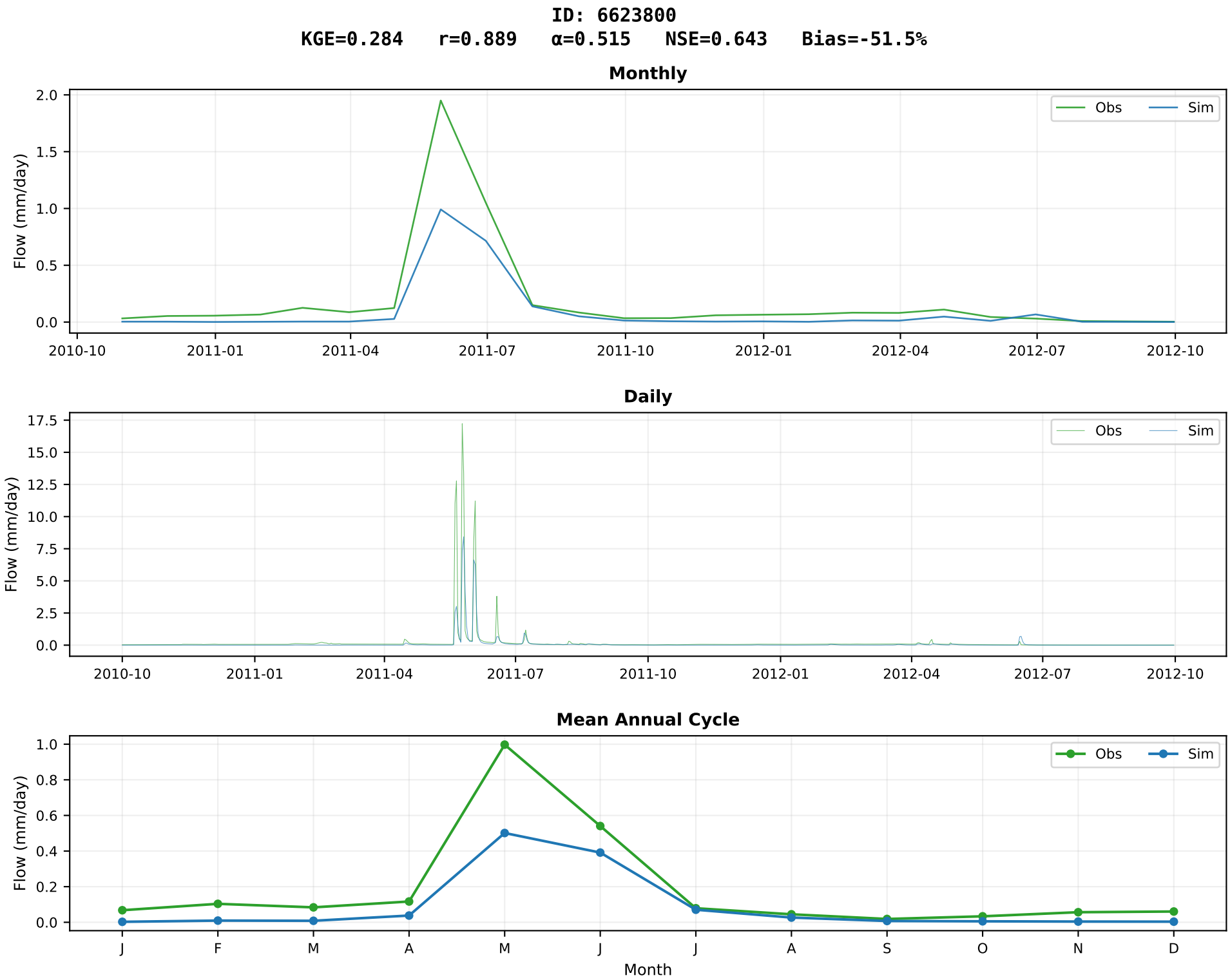
ID: 8324000

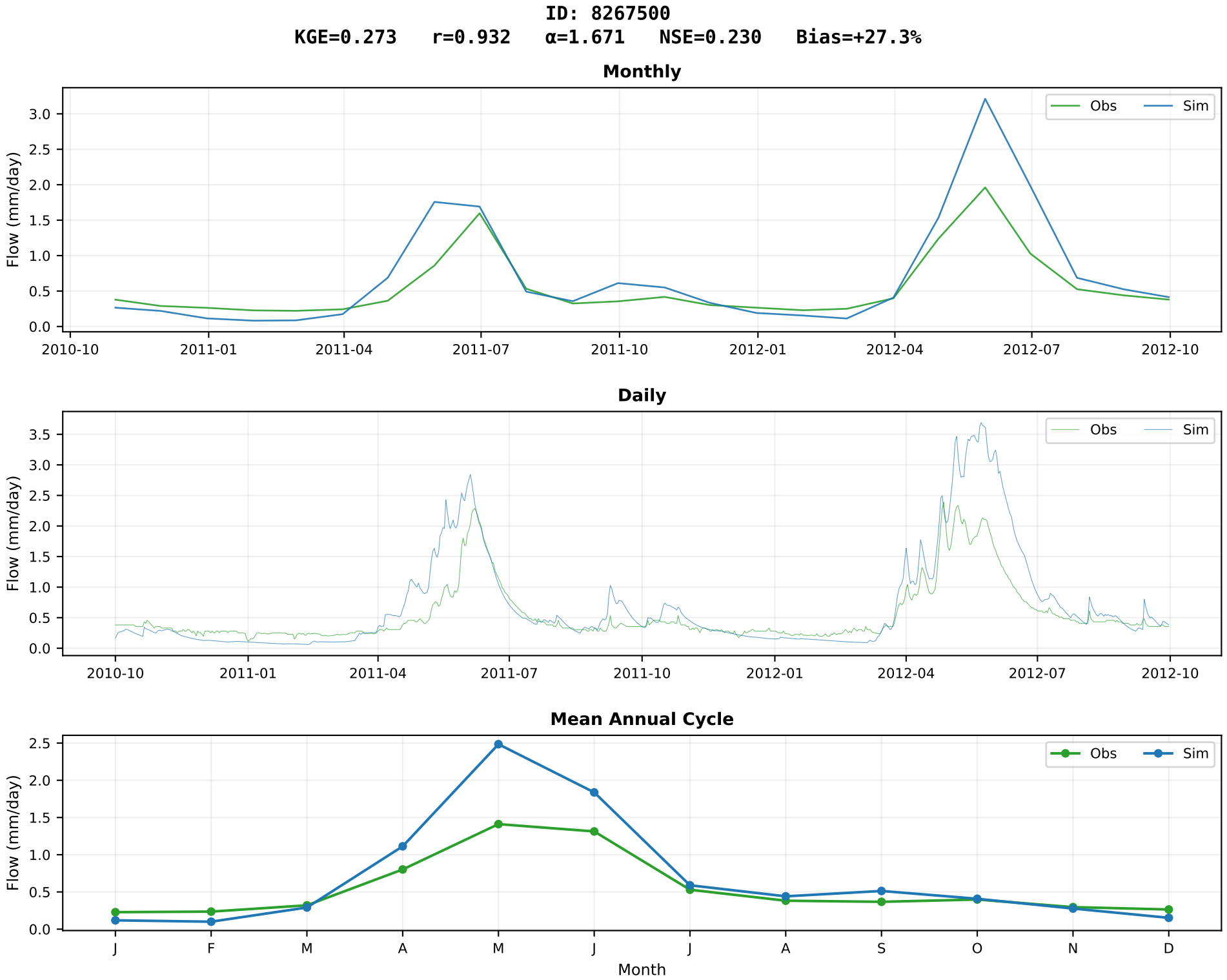
KGE=0.299 r=0.913 α=1.296 NSE=0.415 Bias=+62.9%

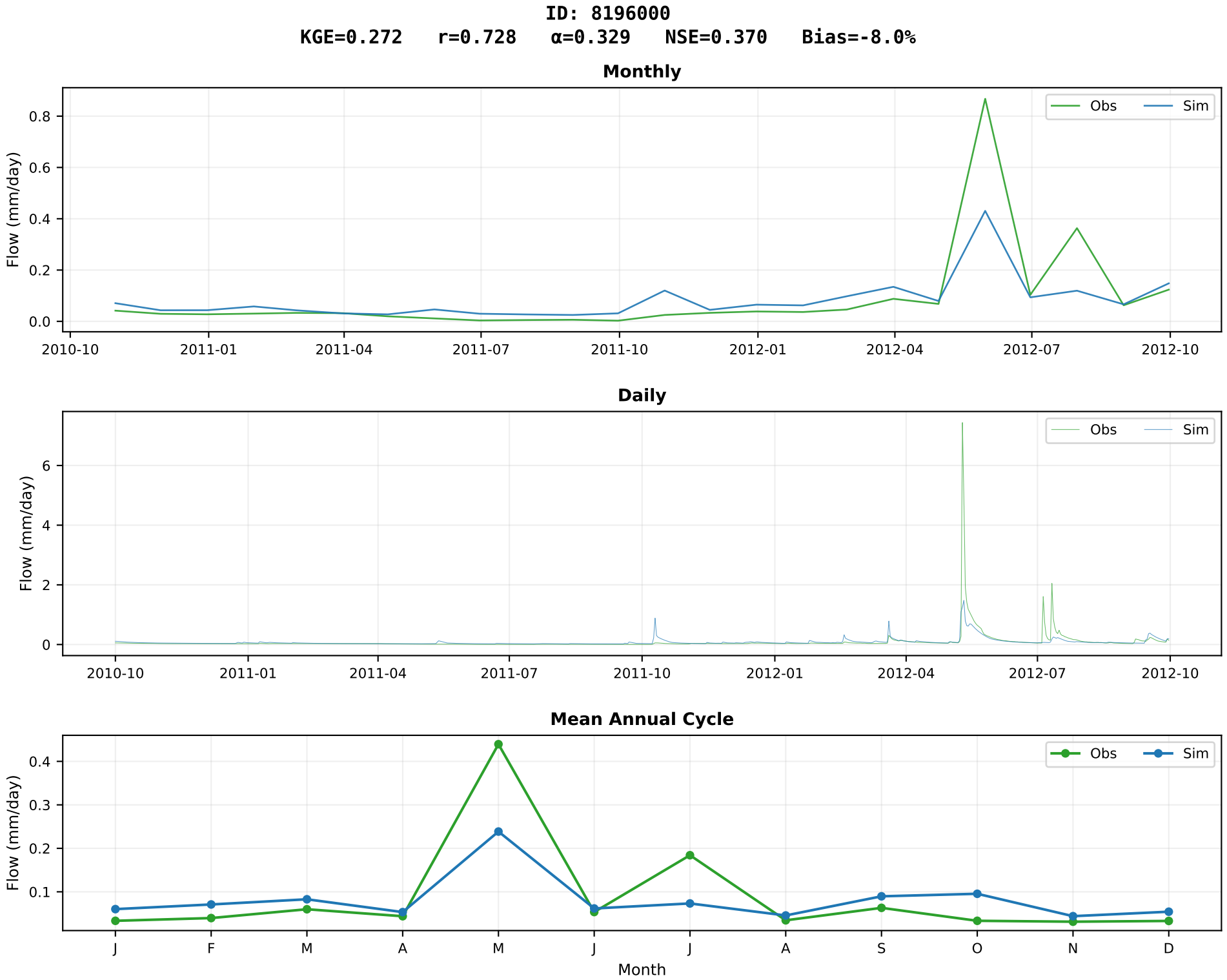


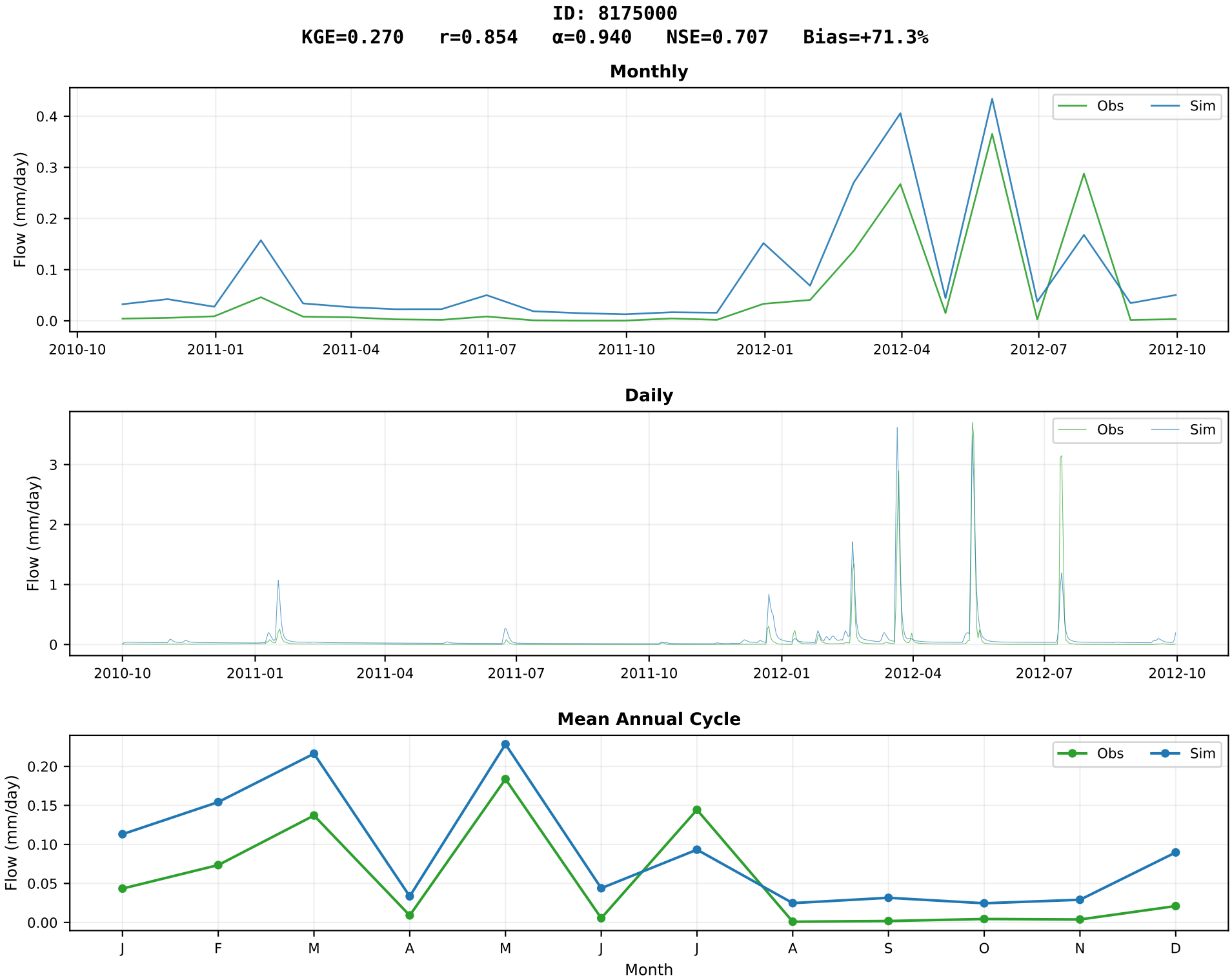

QUEETS RIVER NEAR CLEARWATER, WA (ID=12040500)
KGE=0.294 r=0.850 α=0.328 NSE=0.450 Bias=+15.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

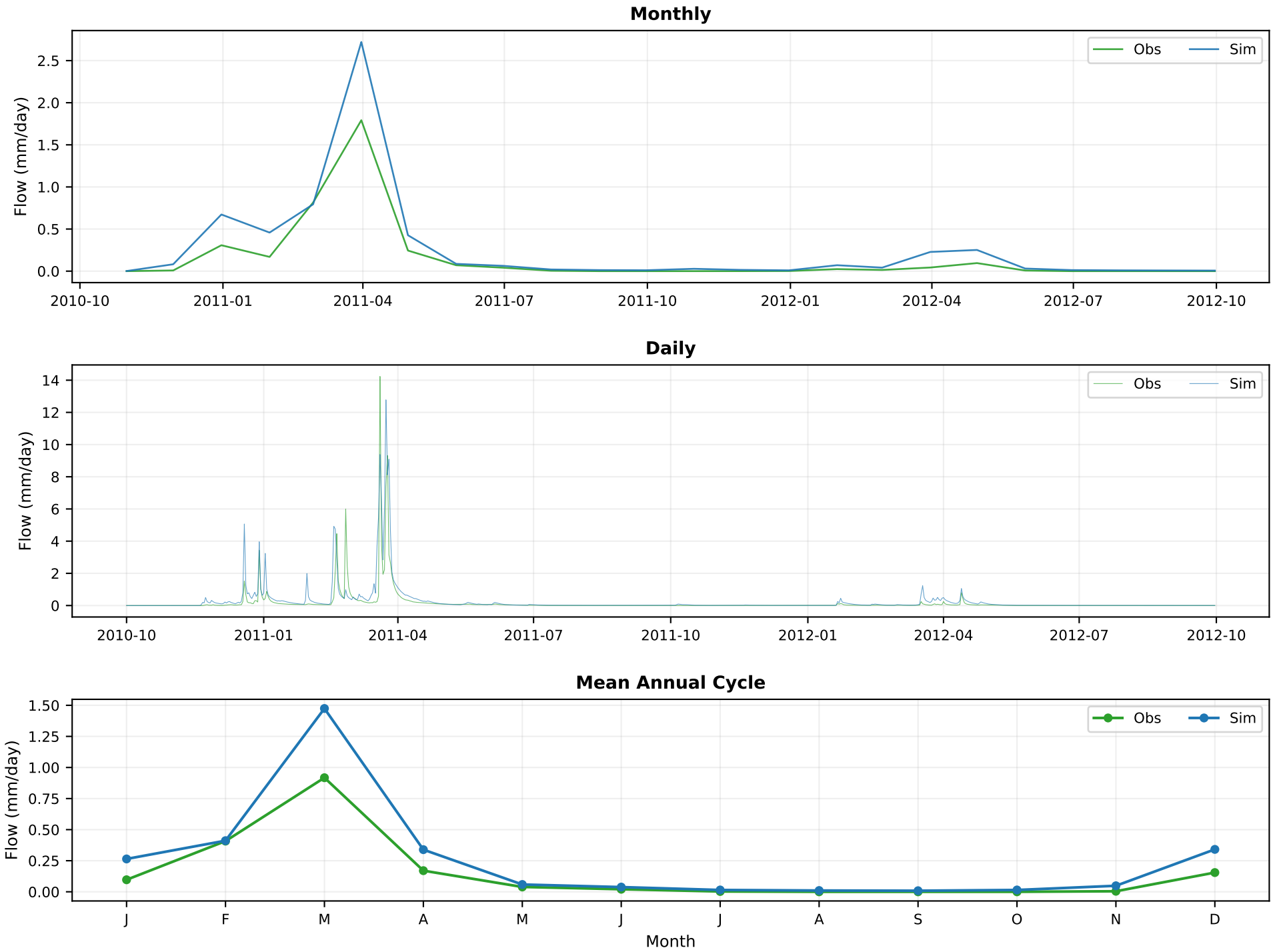

ID: 7196900
KGE=0.289 r=0.516 α=0.481 NSE=0.263 Bias=+4.6%
Monthly
Flow (mm/day)
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
0.025
0.020
0.015
0.010
0.005
0.000
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

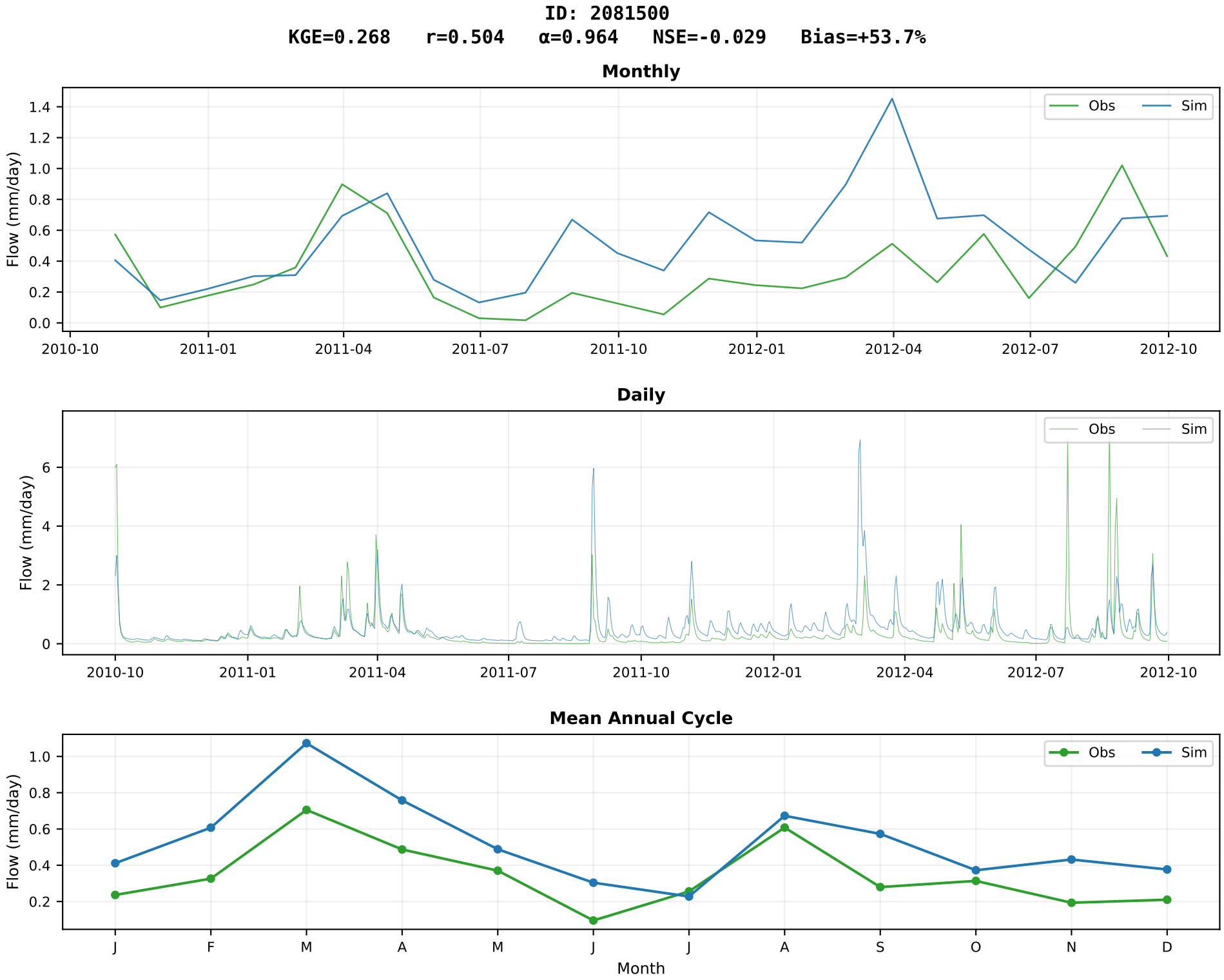

ID: 6623800
KGE=0.284 r=0.889 α=0.515 NSE=0.643 Bias=-51.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

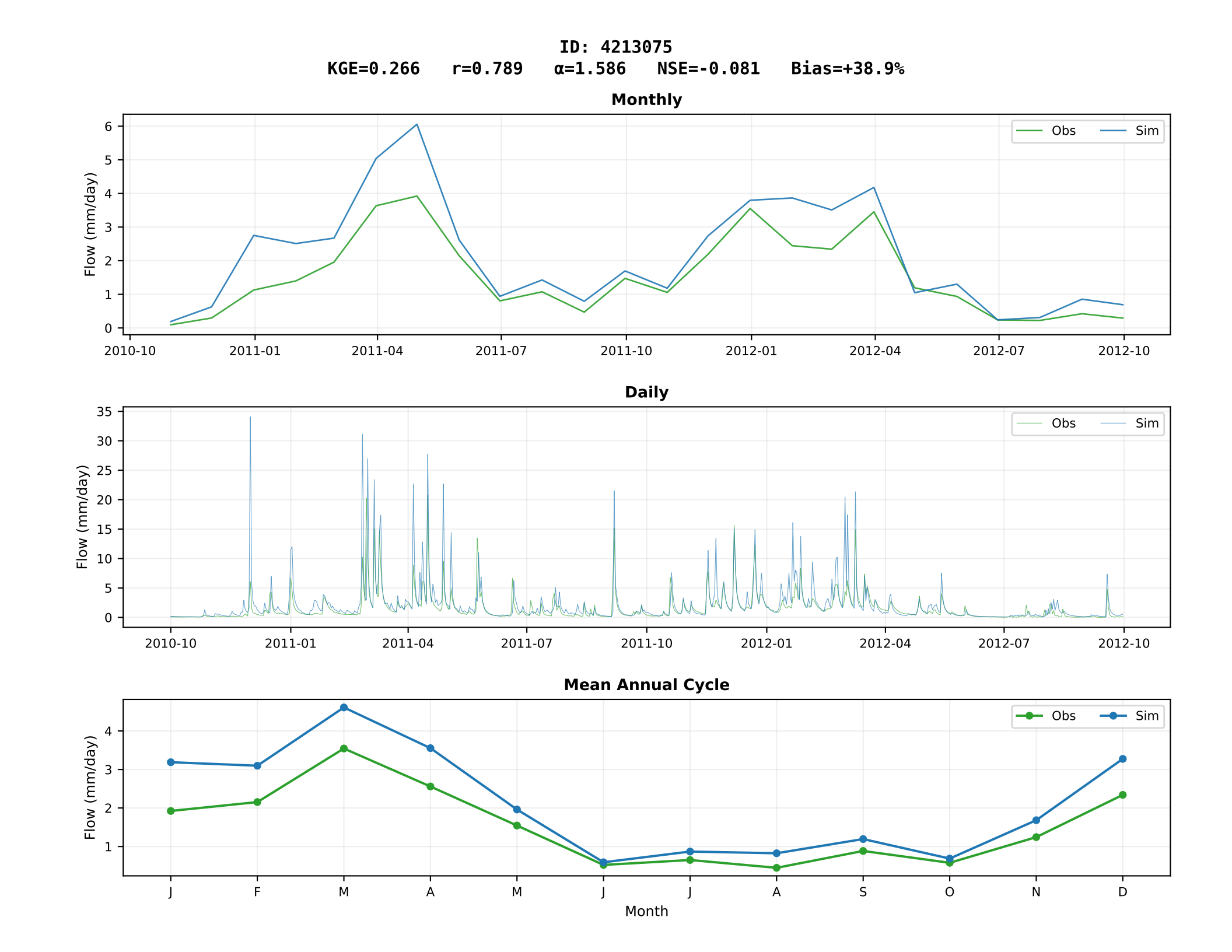

ID: 8267500
KGE=0.273 r=0.932 α=1.671 NSE=0.230 Bias=+27.3%
Monthly
Obs
Sim
Flow (mm/day)
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

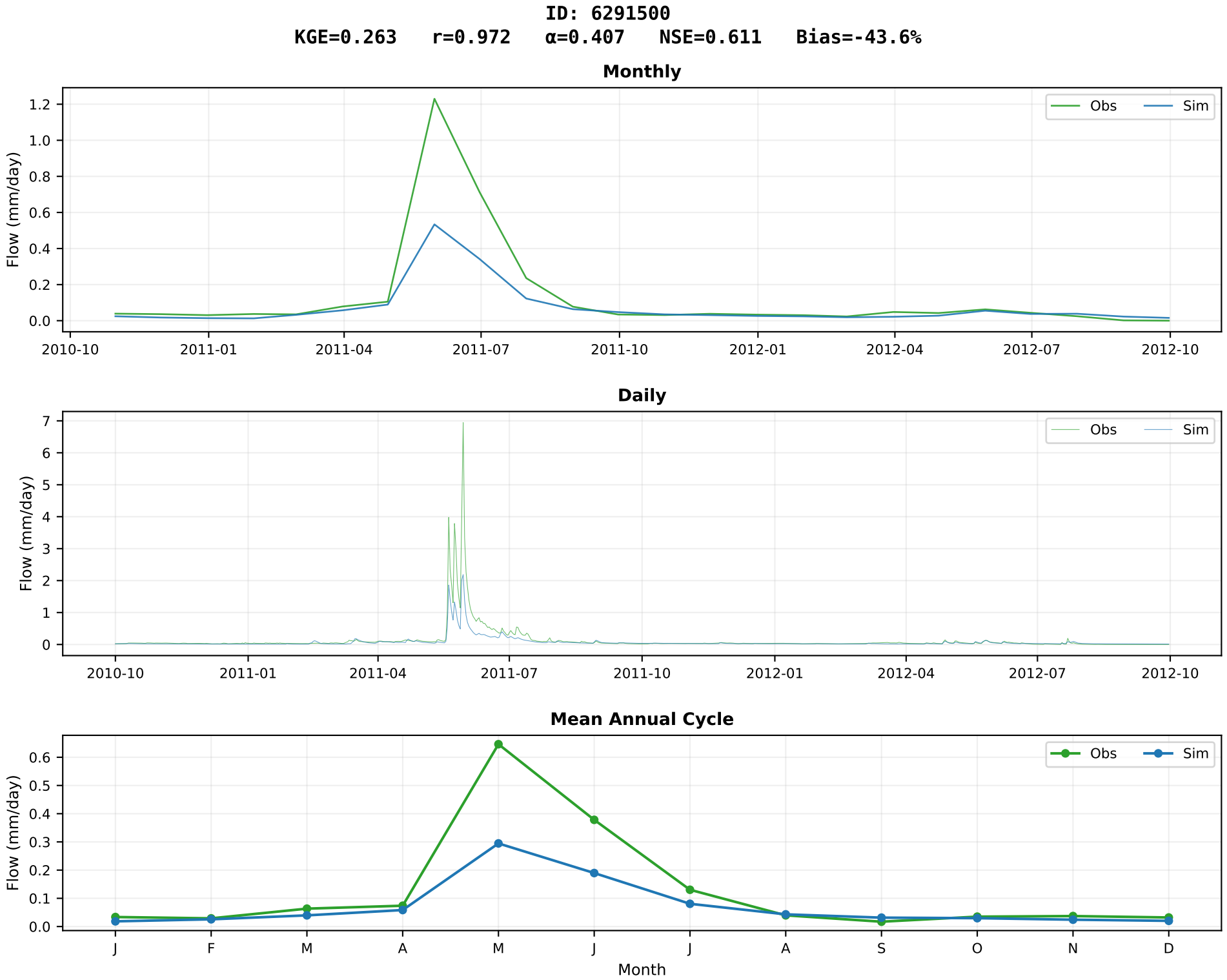

ID: 8196000
KGE=0.272 r=0.728 α=0.329 NSE=0.370 Bias=-8.0%
Monthly
Flow (mm/day)
Obs
Sim
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0
2
4
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.1
0.2
0.3
0.4
J
F
M
A
M
J
J
A
S
O
N
D
Month

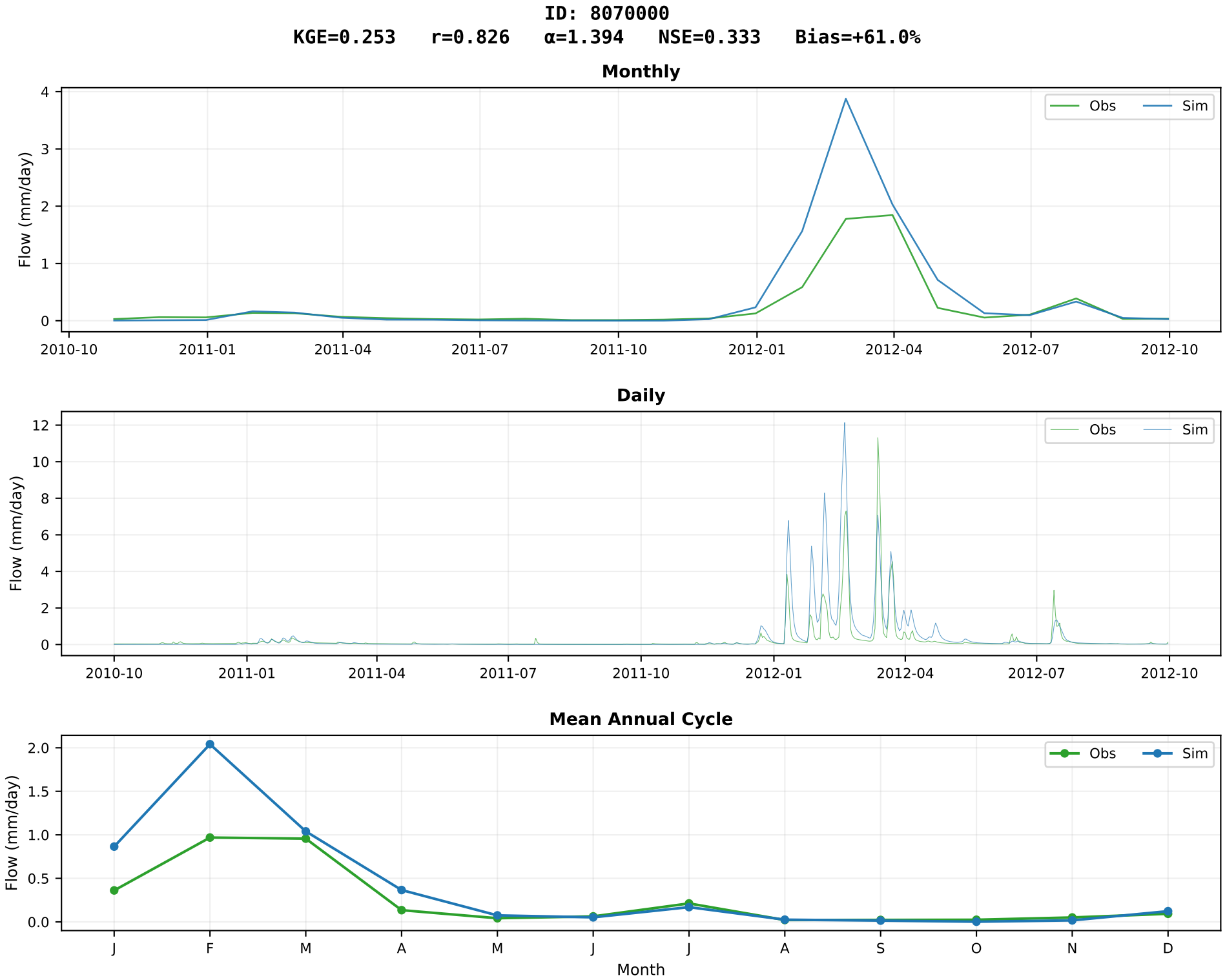

ID: 8175000
KGE=0.270 r=0.854 α=0.940 NSE=0.707 Bias=+71.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

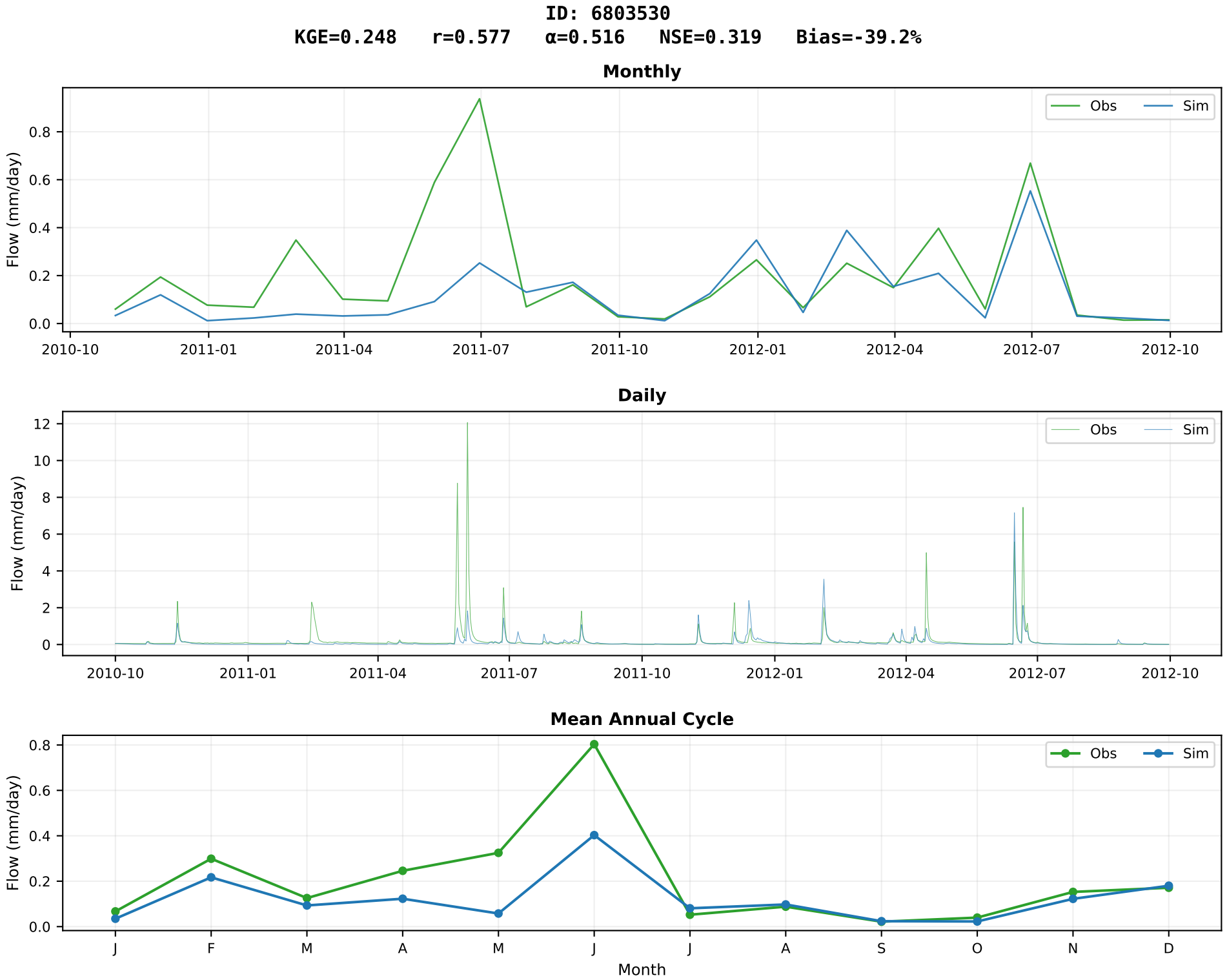

HOH RIVER AT US HIGHWAY 101 NEAR FORKS, WA (ID=12041200)
KGE=0.268 r=0.800 α=1.192 NSE=0.470 Bias=+67.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

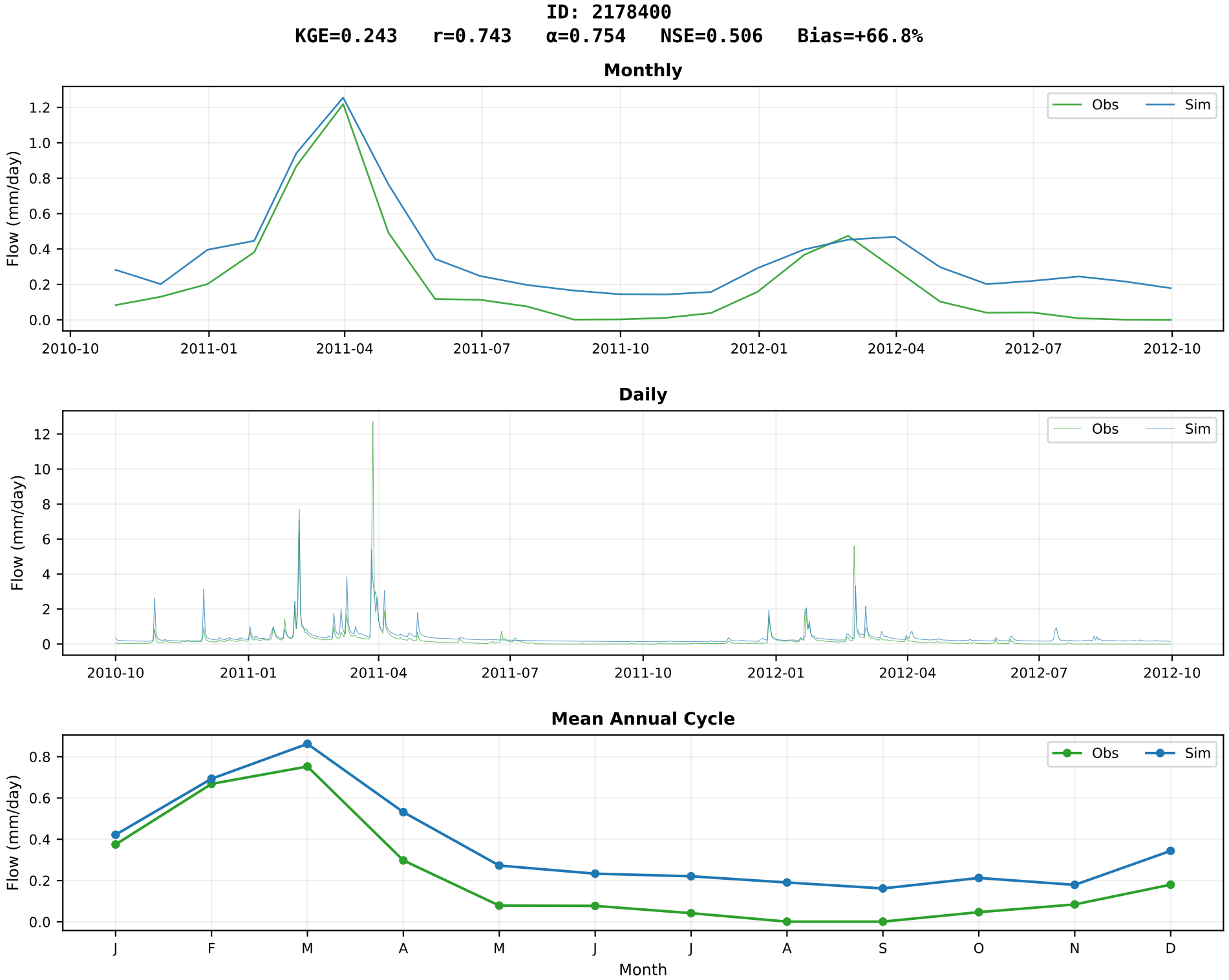

ID: 2081500
KGE=0.268 r=0.504 α=0.964 NSE=-0.029 Bias=+53.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

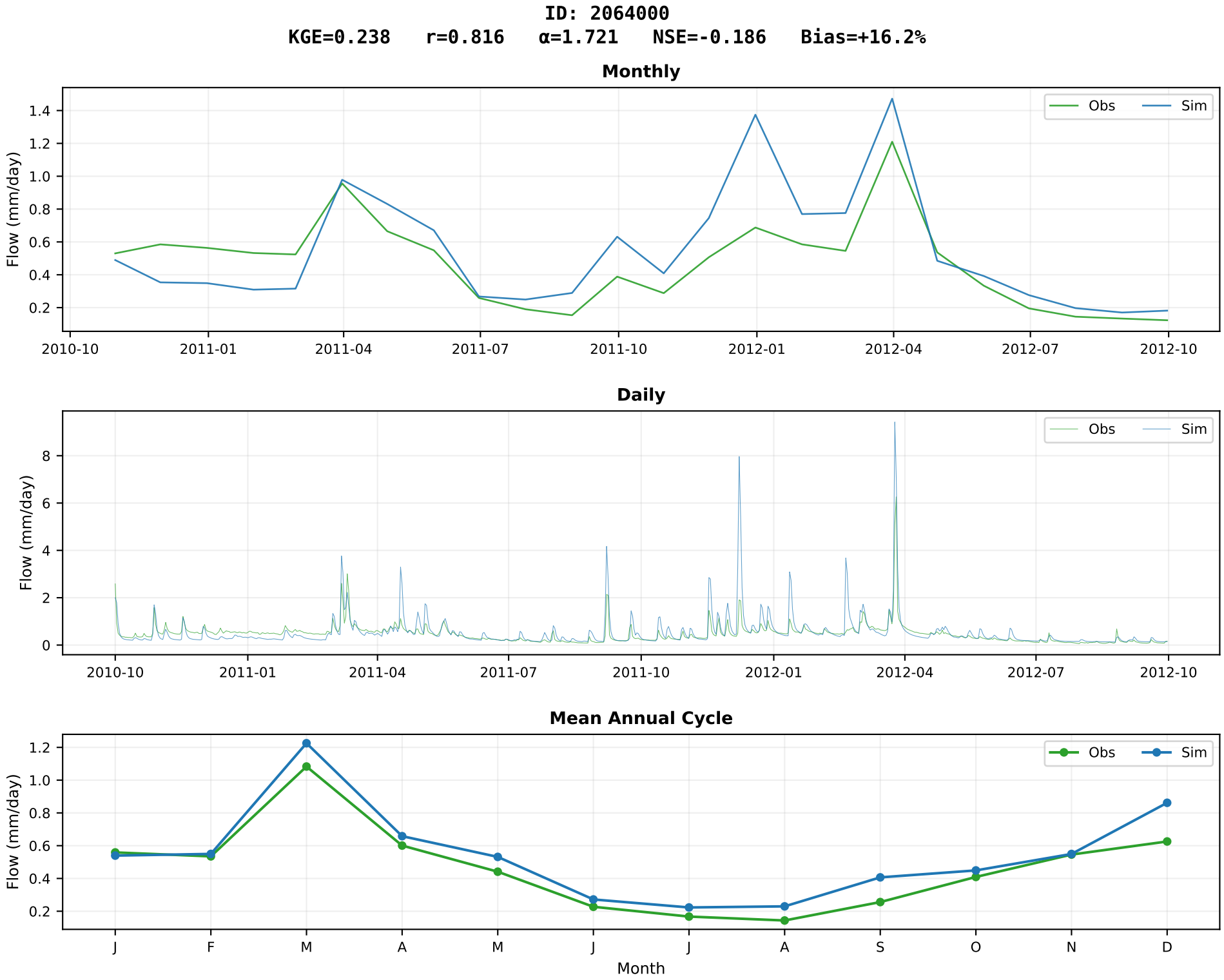

ID: 4213075
KGE=0.266 r=0.789 α=1.586 NSE=-0.081 Bias=+38.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

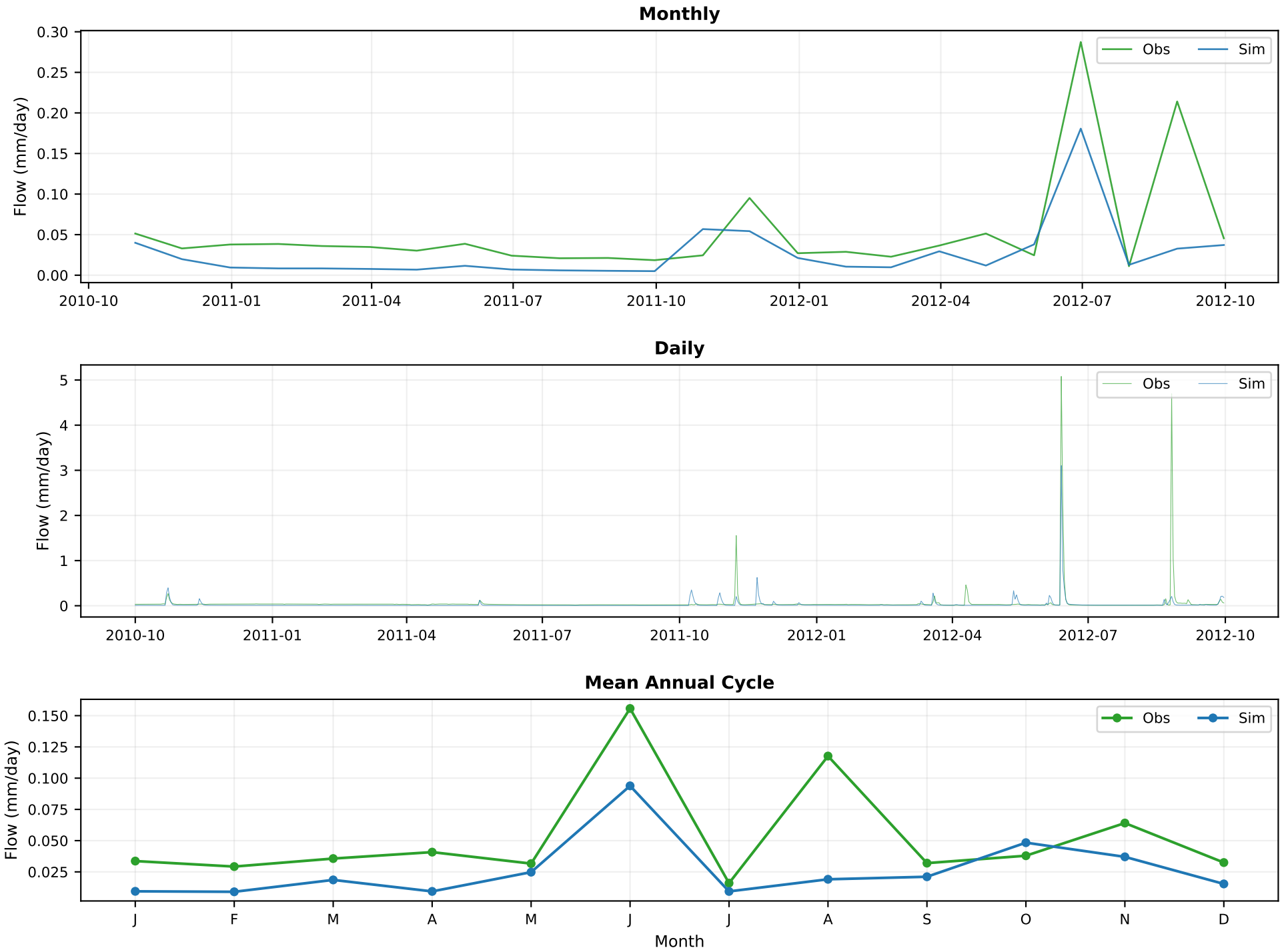

ID: 6291500
KGE=0.263 r=0.972 α=0.407 NSE=0.611 Bias=-43.6%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

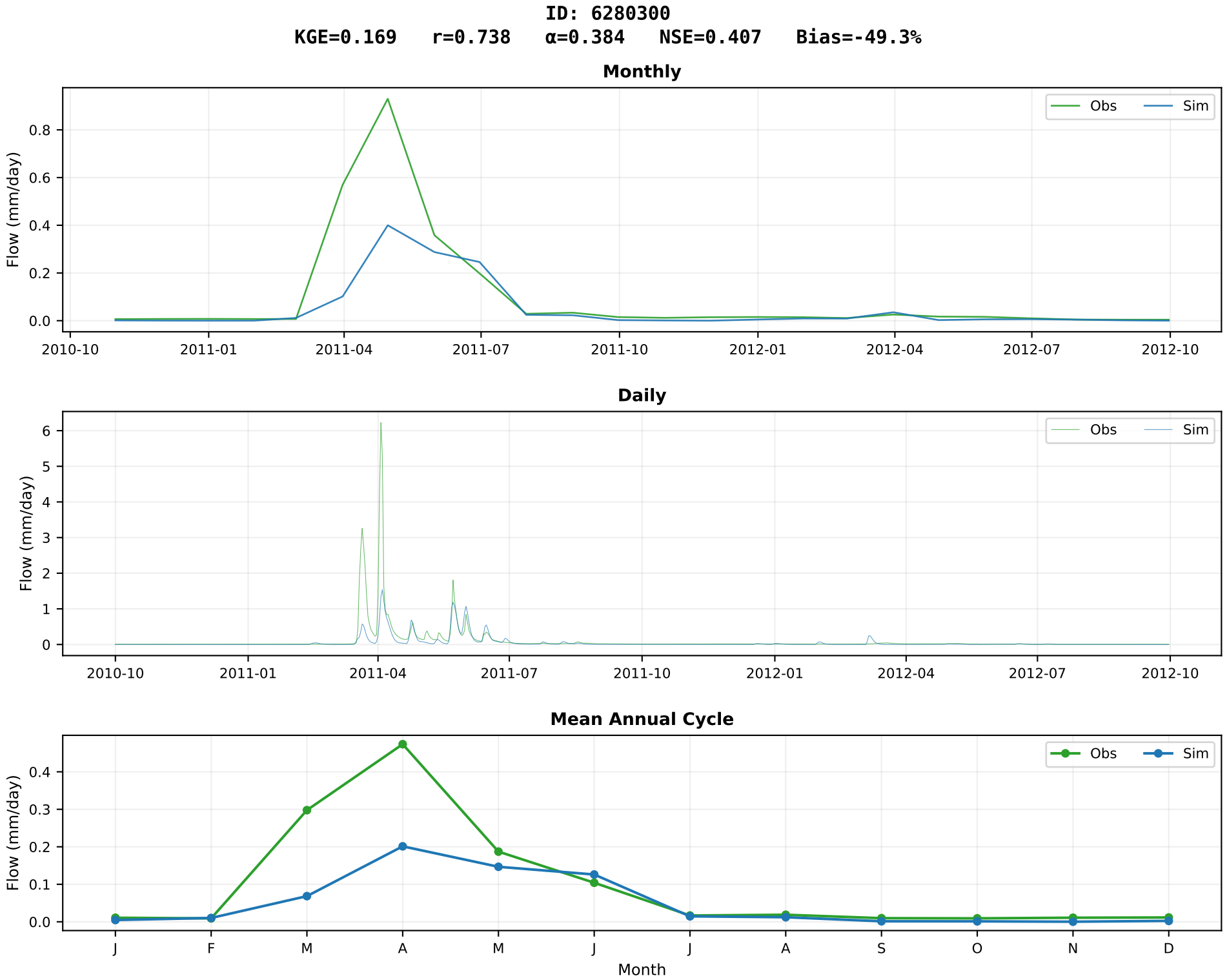

ID: 8070000
KGE=0.253 r=0.826 α=1.394 NSE=0.333 Bias=+61.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

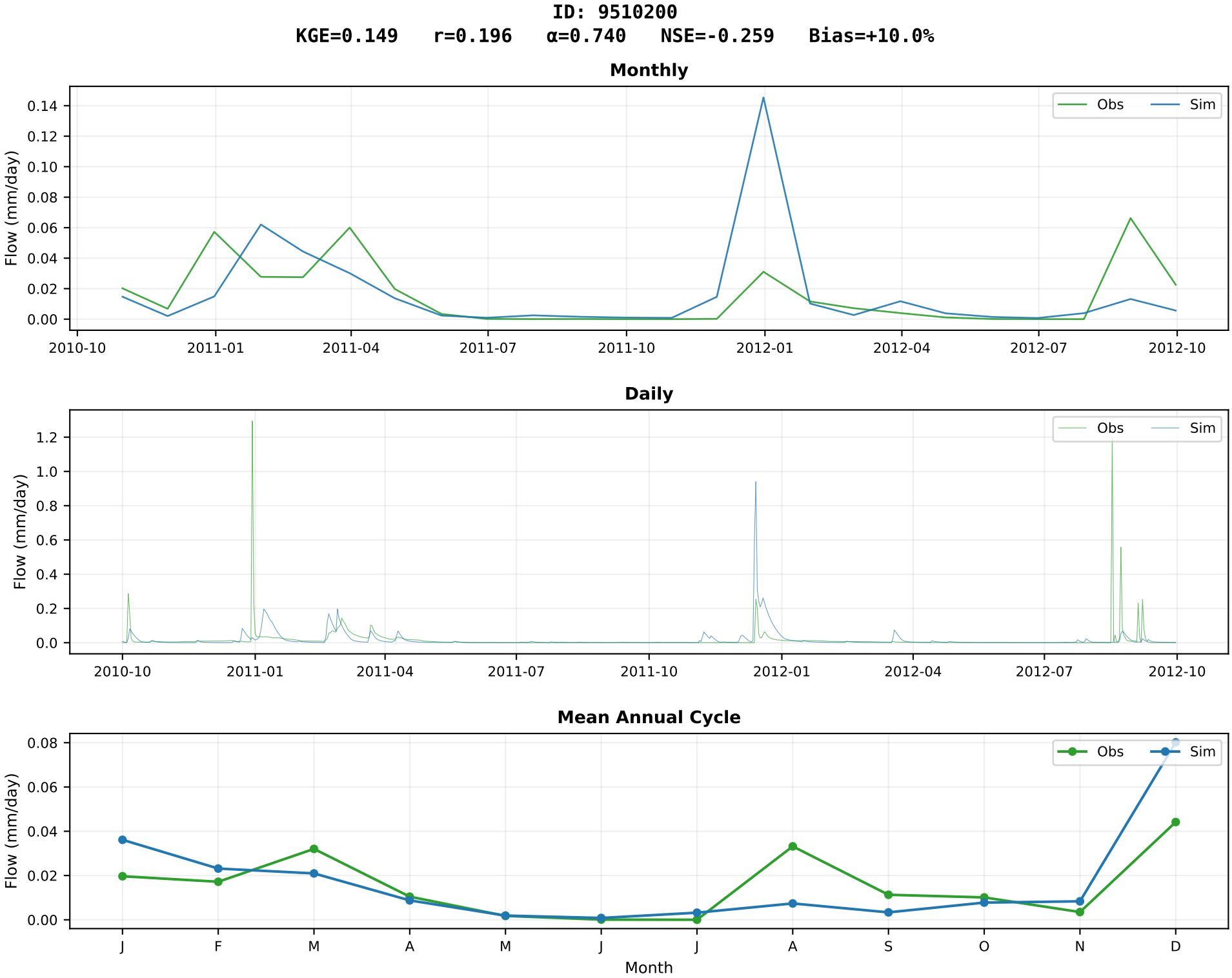

ID: 6803530
KGE=0.248 r=0.577 α=0.516 NSE=0.319 Bias=-39.2%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
12
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2178400
KGE=0.243 r=0.743 α=0.754 NSE=0.506 Bias=+66.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D


ID: 2064000
KGE=0.238 r=0.816 α=1.721 NSE=-0.186 Bias=+16.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

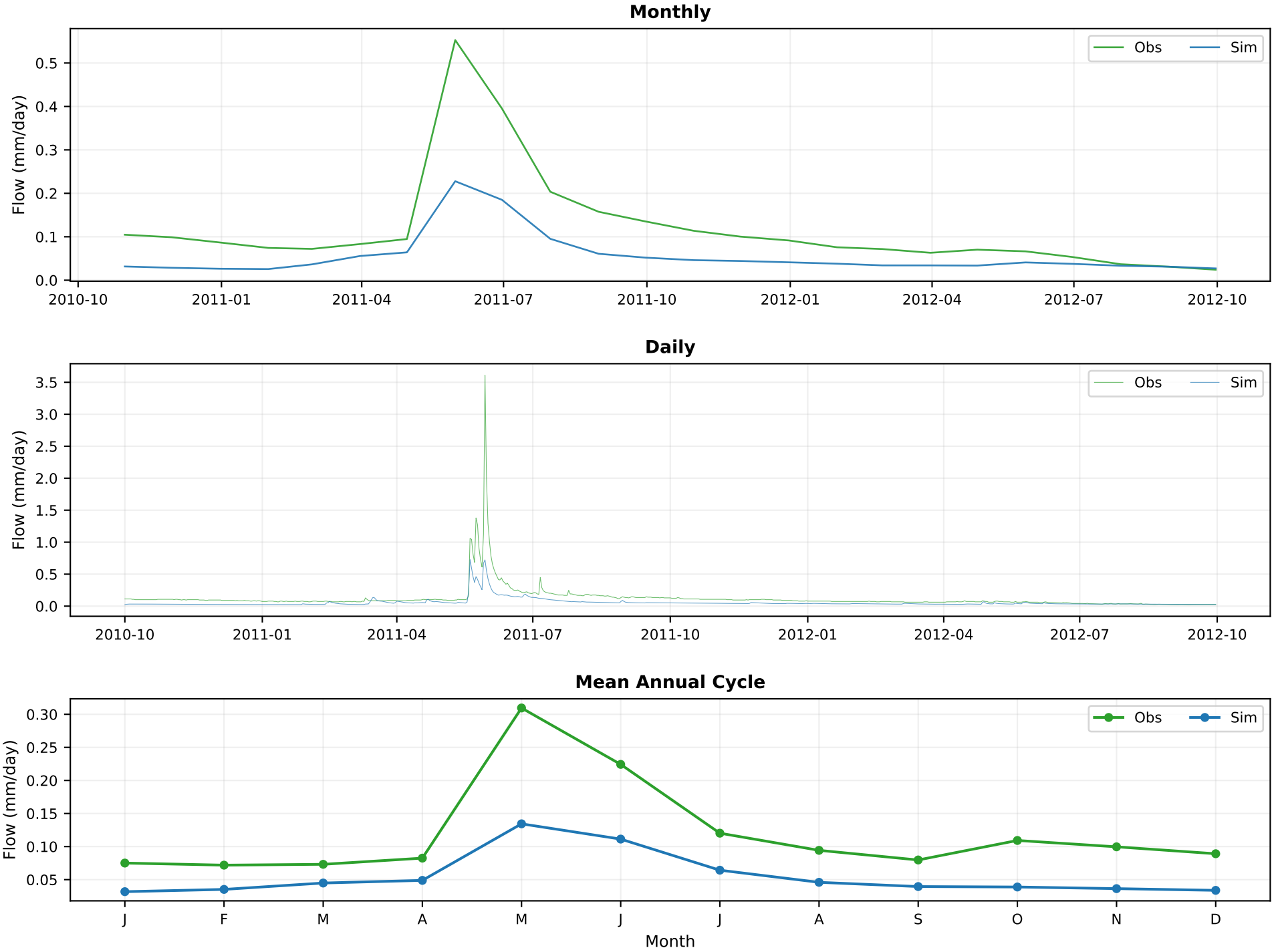

ID: 7195800
KGE=0.219 r=0.733 α=0.460 NSE=0.454 Bias=-49.7%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
Month
J
F
M
A
M
J
J
A
S
O
N
D

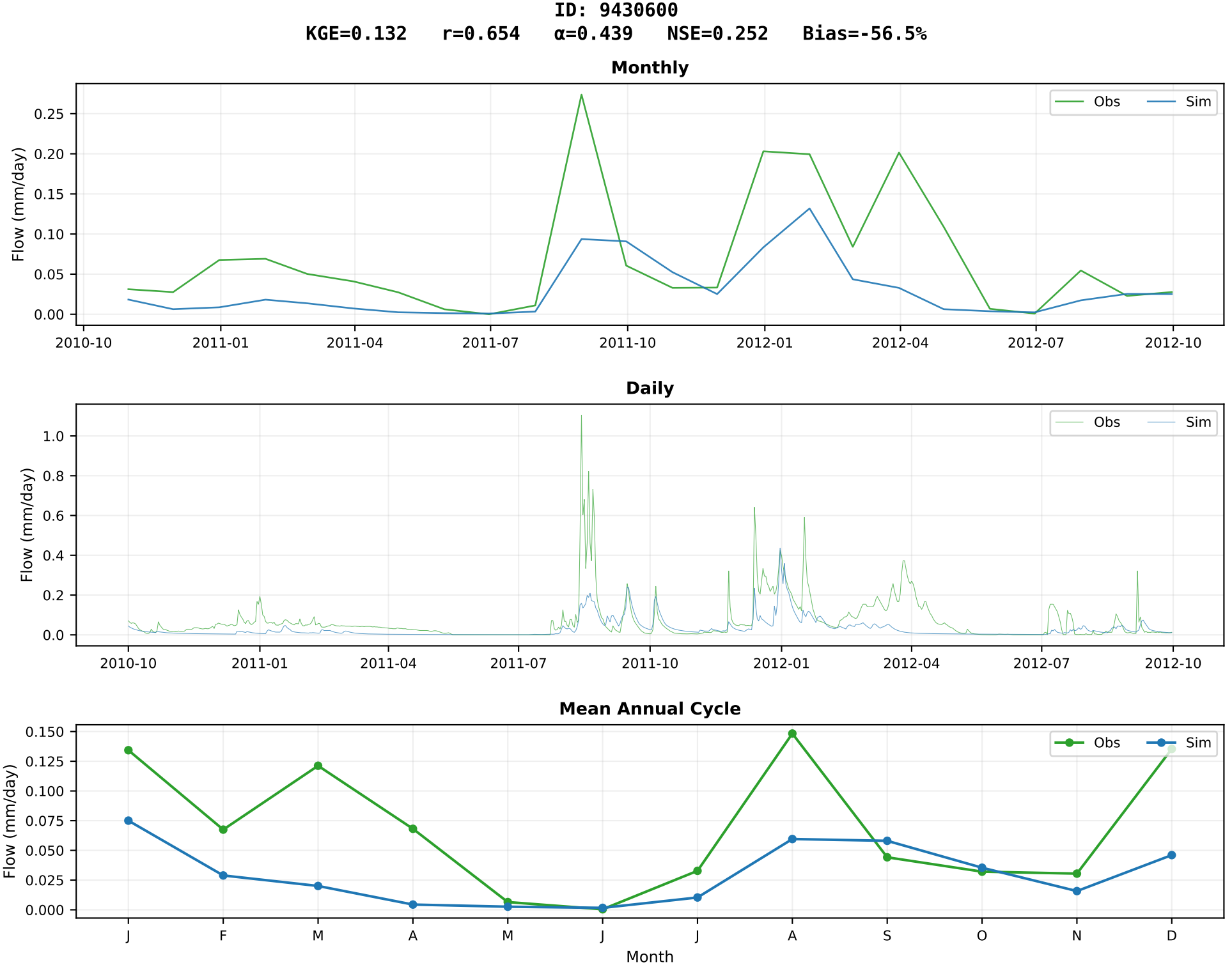

ID: 6280300
KGE=0.169 r=0.738 α=0.384 NSE=0.407 Bias=-49.3%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
J
F
M
A
M
J
J
A
S
O
N
D
Month

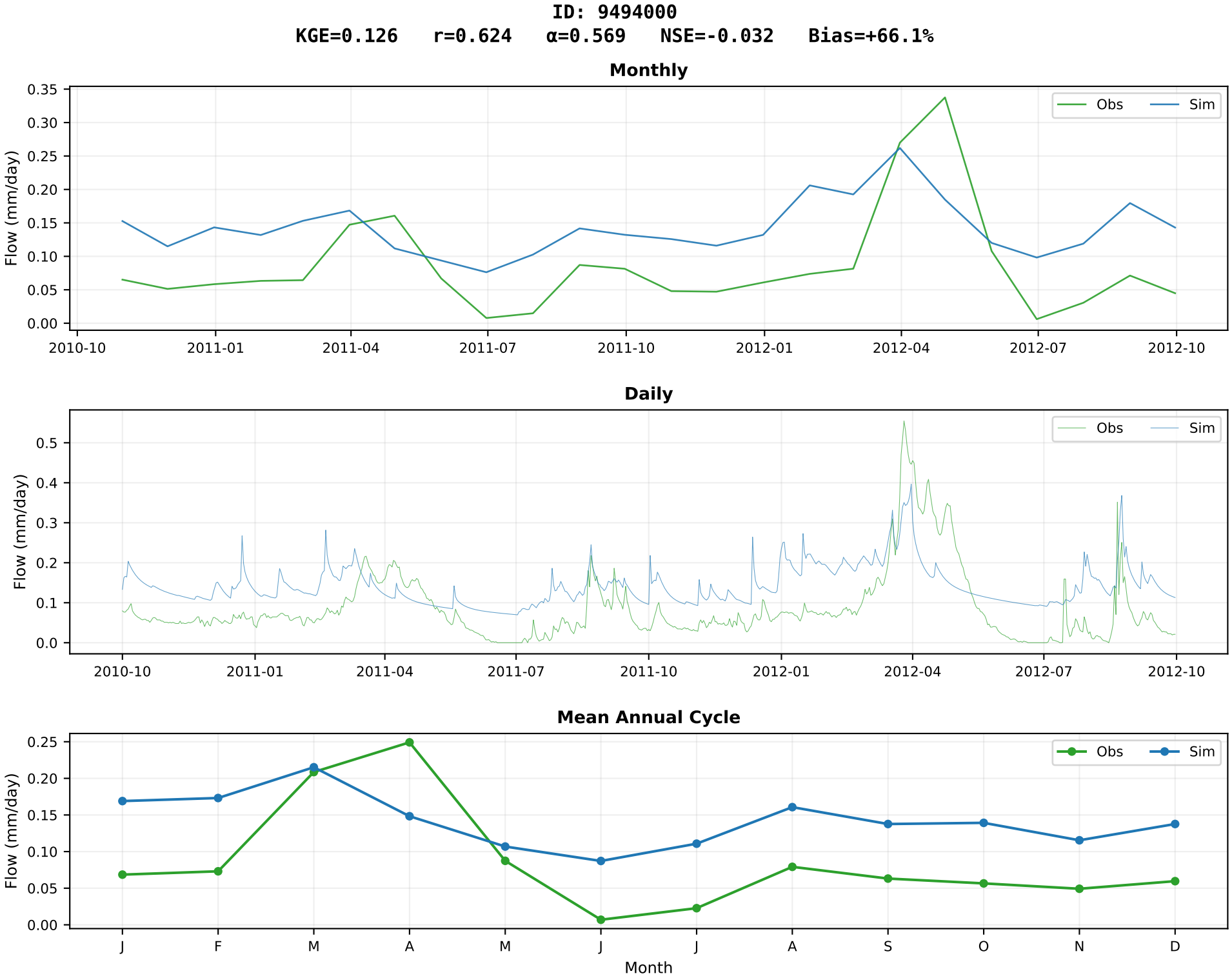

ID: 9510200
KGE=0.149 r=0.196 α=0.740 NSE=-0.259 Bias=+10.0%
Monthly
Flow (mm/day)
Obs
Sim
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.00
0.02
0.04
0.06
0.08
J
F
M
A
M
J
J
A
S
O
N
D
Month

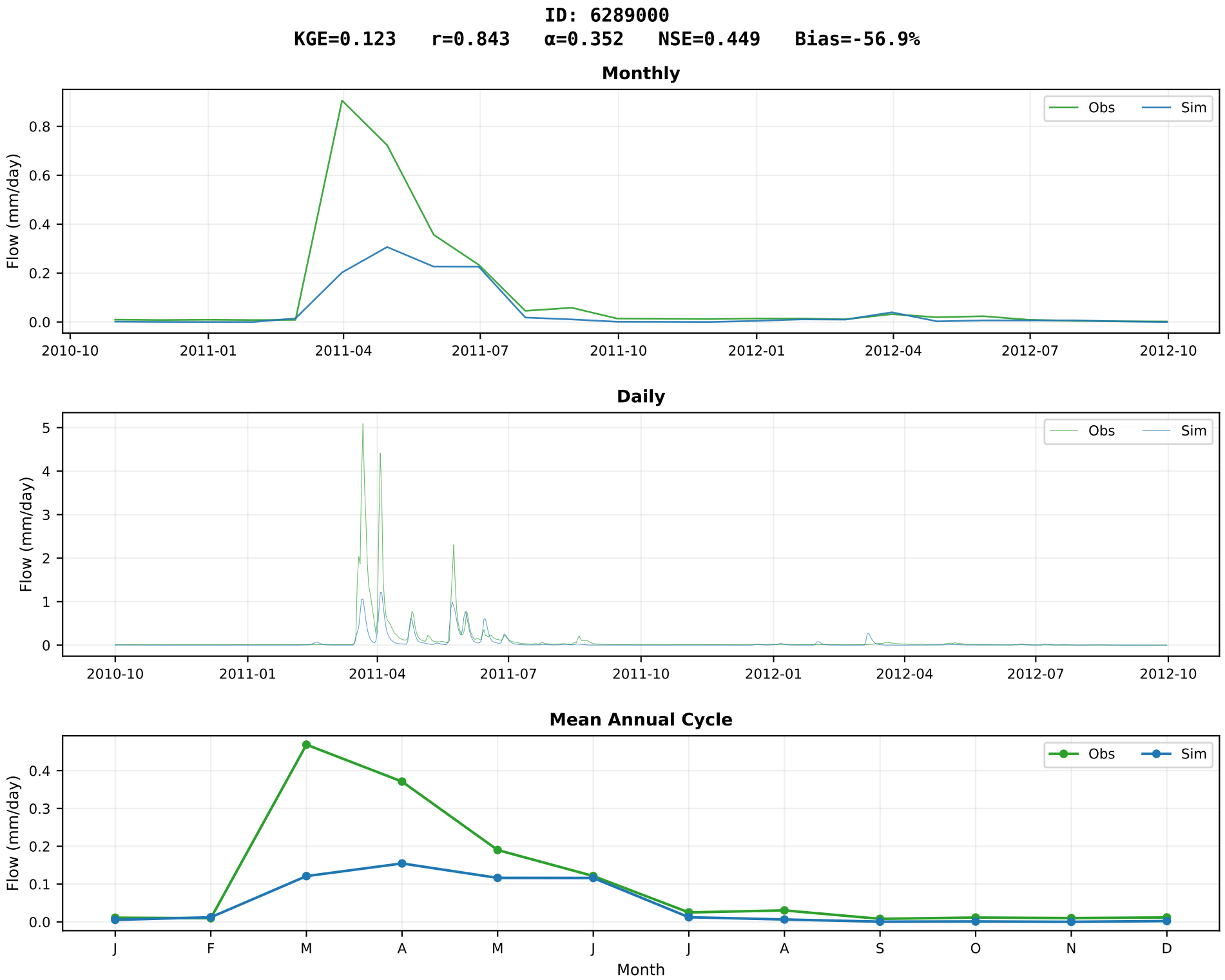
ID: 6311000

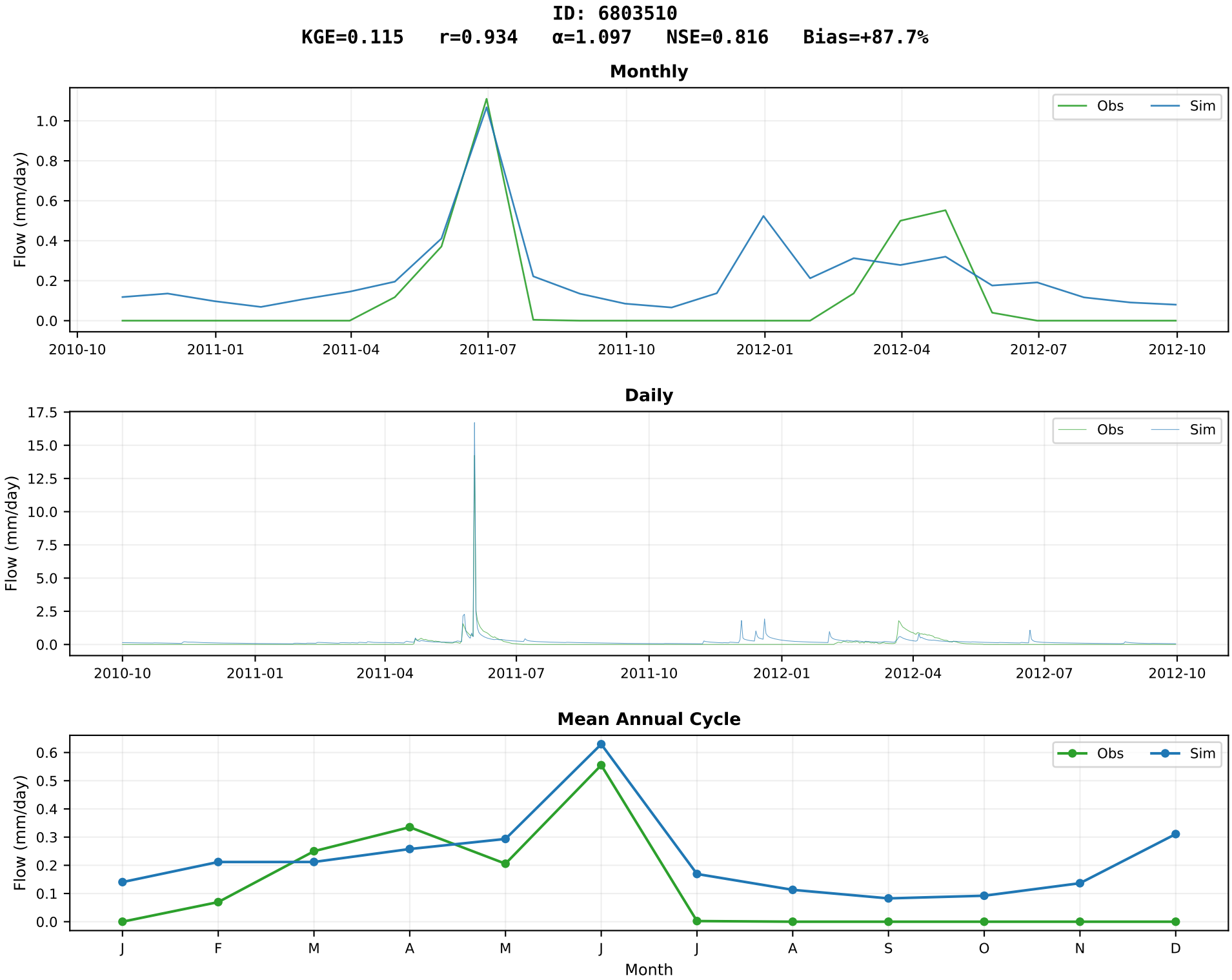
KGE=0.149 r=0.885 α=0.348 NSE=0.394 Bias=-53.5%

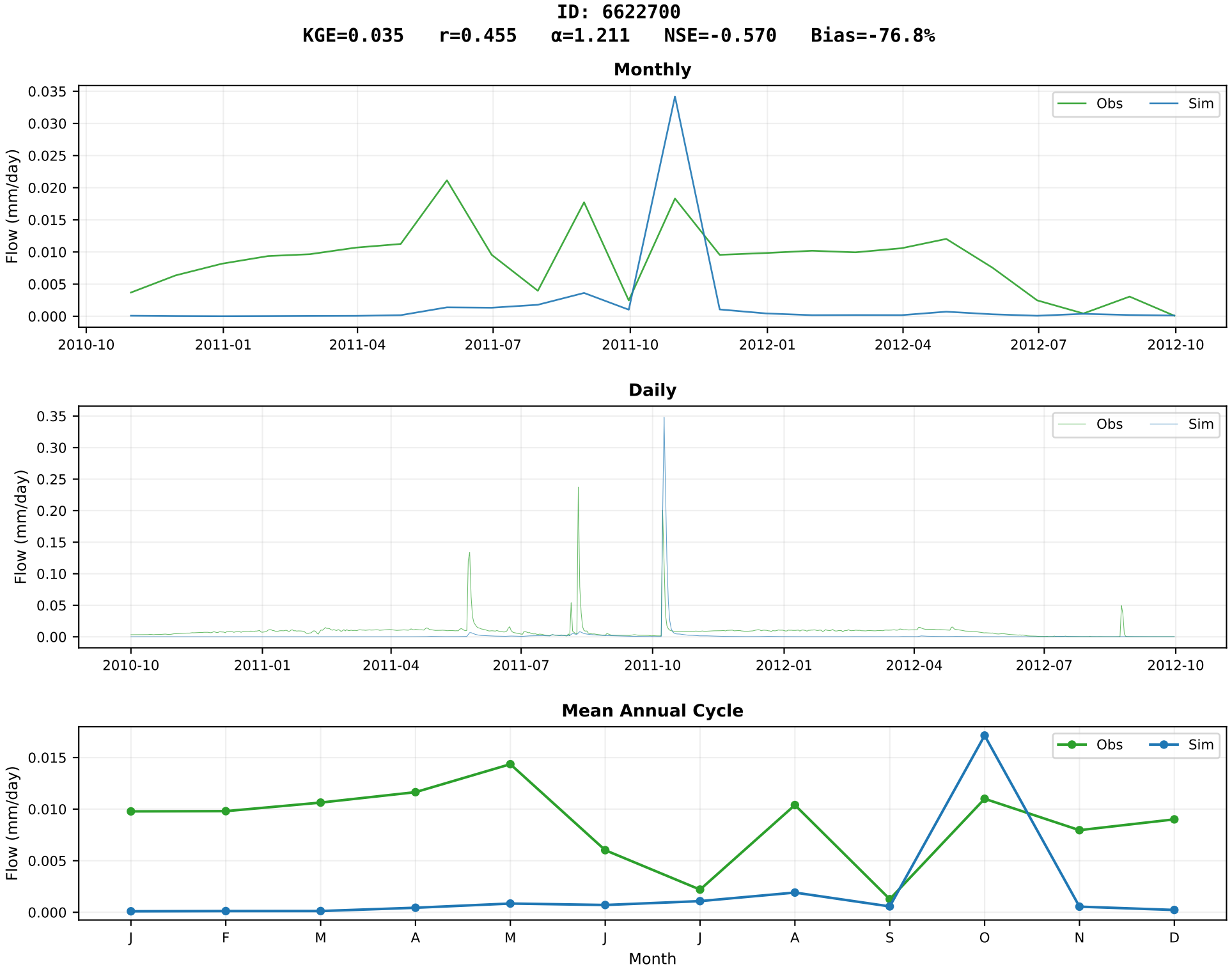


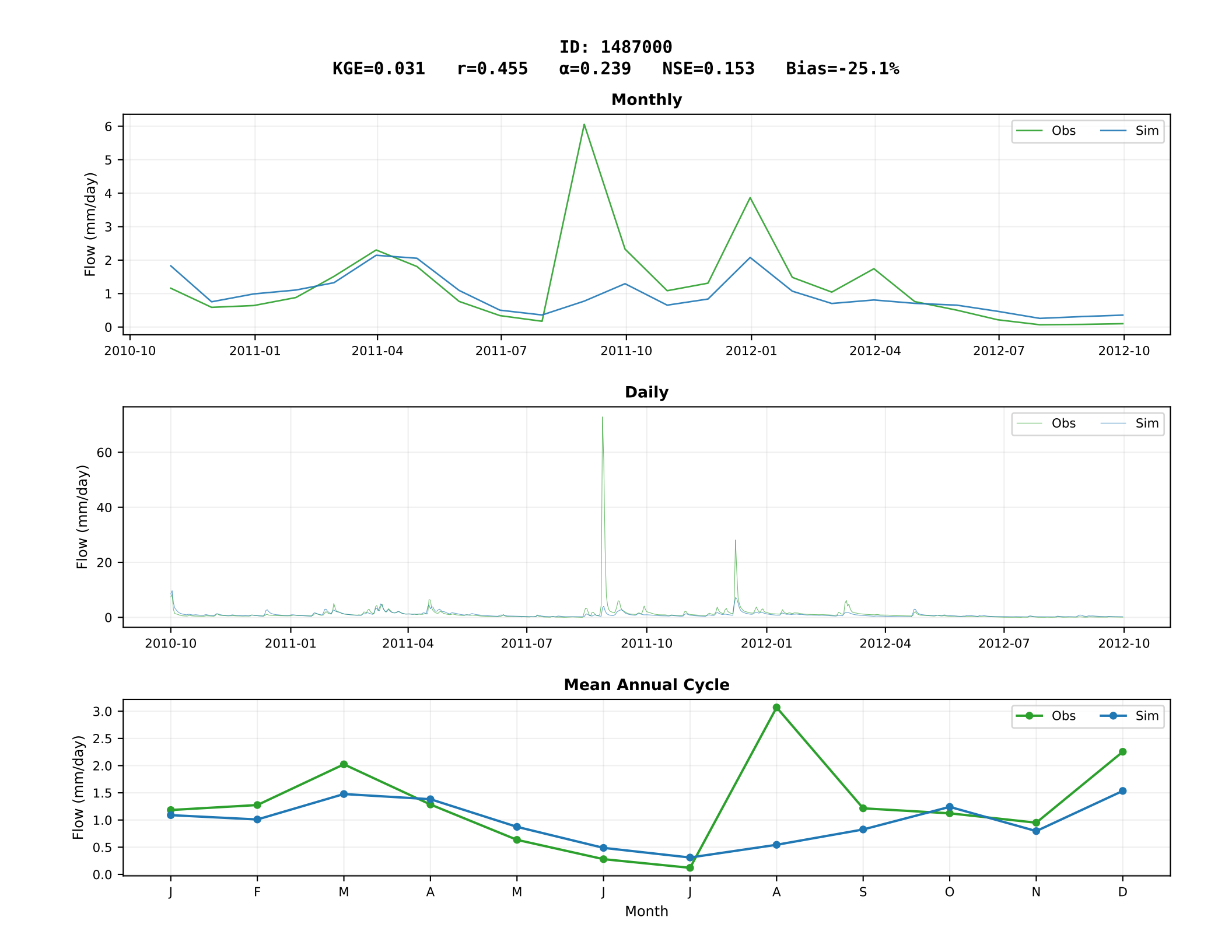

ID: 9430600
KGE=0.132 r=0.654 α=0.439 NSE=0.252 Bias=-56.5%
Monthly
Obs
Sim
Flow (mm/day)
0.25
0.20
0.15
0.10
0.05
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
1.0
0.8
0.6
0.4
0.2
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
J
F
M
A
M
J
J
A
S
O
N
D
Month

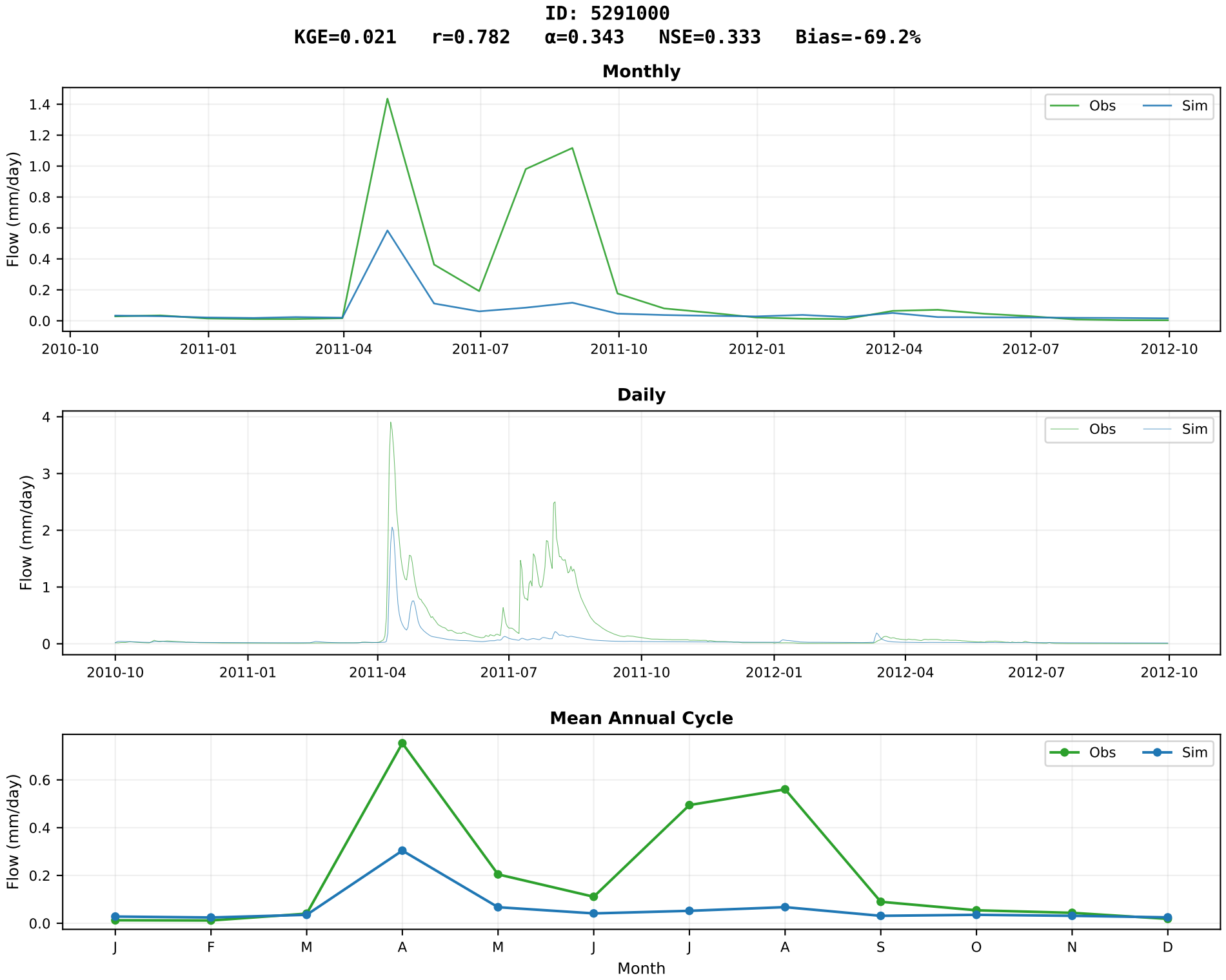

ID: 9494000
KGE=0.126 r=0.624 α=0.569 NSE=-0.032 Bias=+66.1%
Monthly
Flow (mm/day)
Obs
Sim
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0.5
0.4
0.3
0.2
0.1
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.25
0.20
0.15
0.10
0.05
0.00
J
F
M
A
M
J
J
A
S
O
N
D
Month

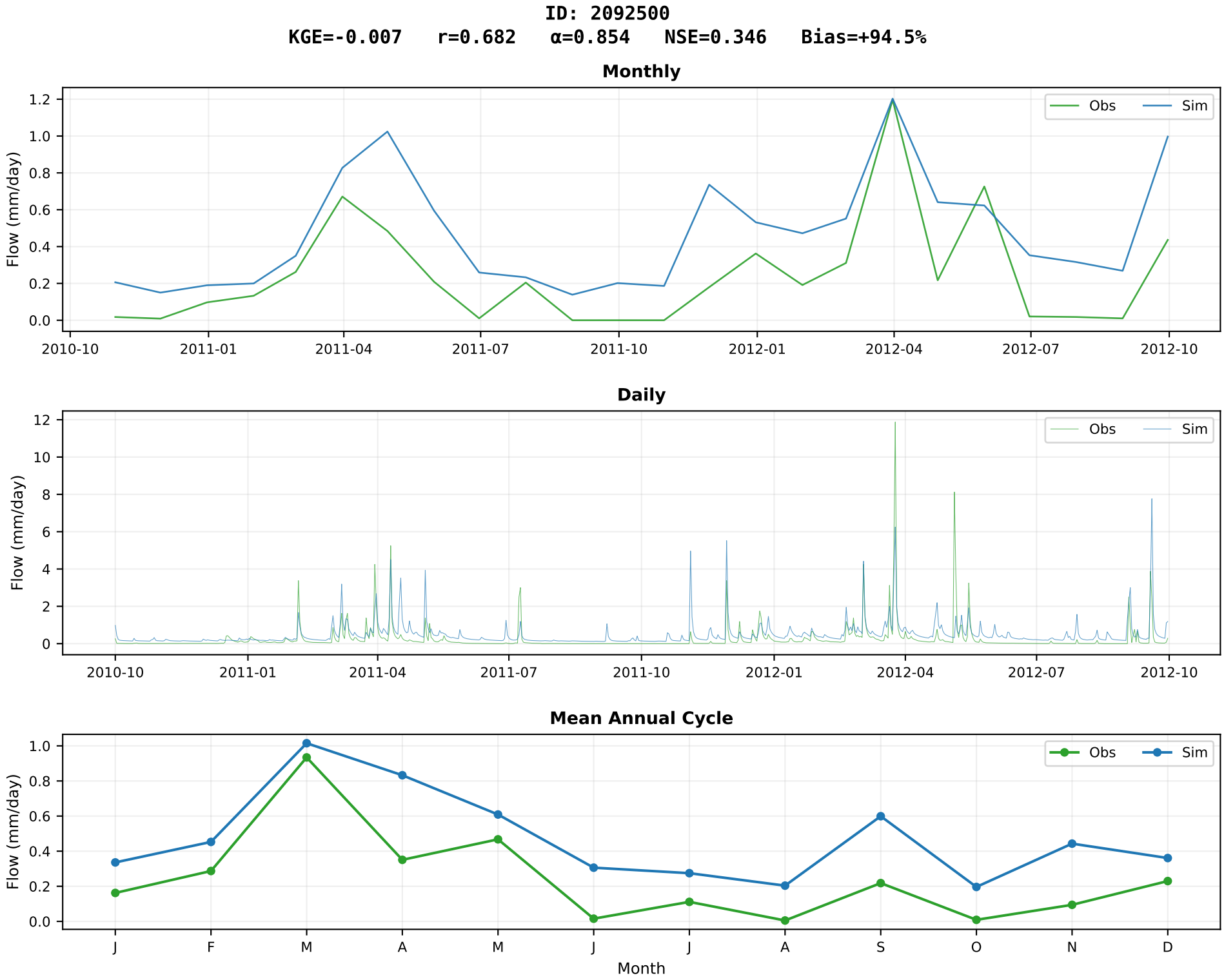

ID: 6289000
KGE=0.123 r=0.843 α=0.352 NSE=0.449 Bias=-56.9%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
J
F
M
A
M
J
J
A
S
O
N
D
Month

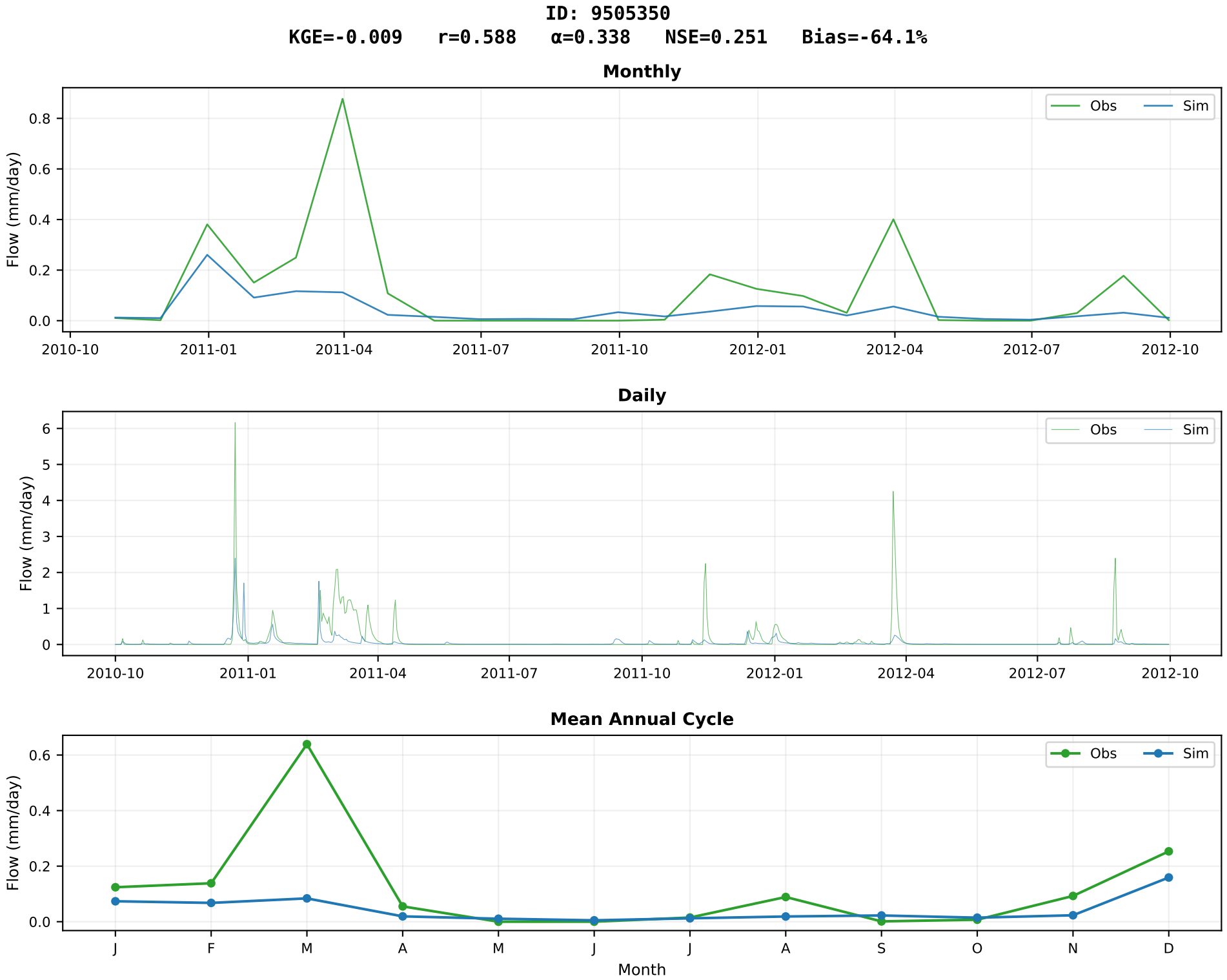

ID: 6803510
KGE=0.115 r=0.934 α=1.097 NSE=0.816 Bias=+87.7%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 6622700
KGE=0.035 r=0.455 α=1.211 NSE=-0.570 Bias=-76.8%
Monthly
Flow (mm/day)
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
0.015
0.010
0.005
0.000
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 1487000
KGE=0.031 r=0.455 α=0.239 NSE=0.153 Bias=-25.1%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

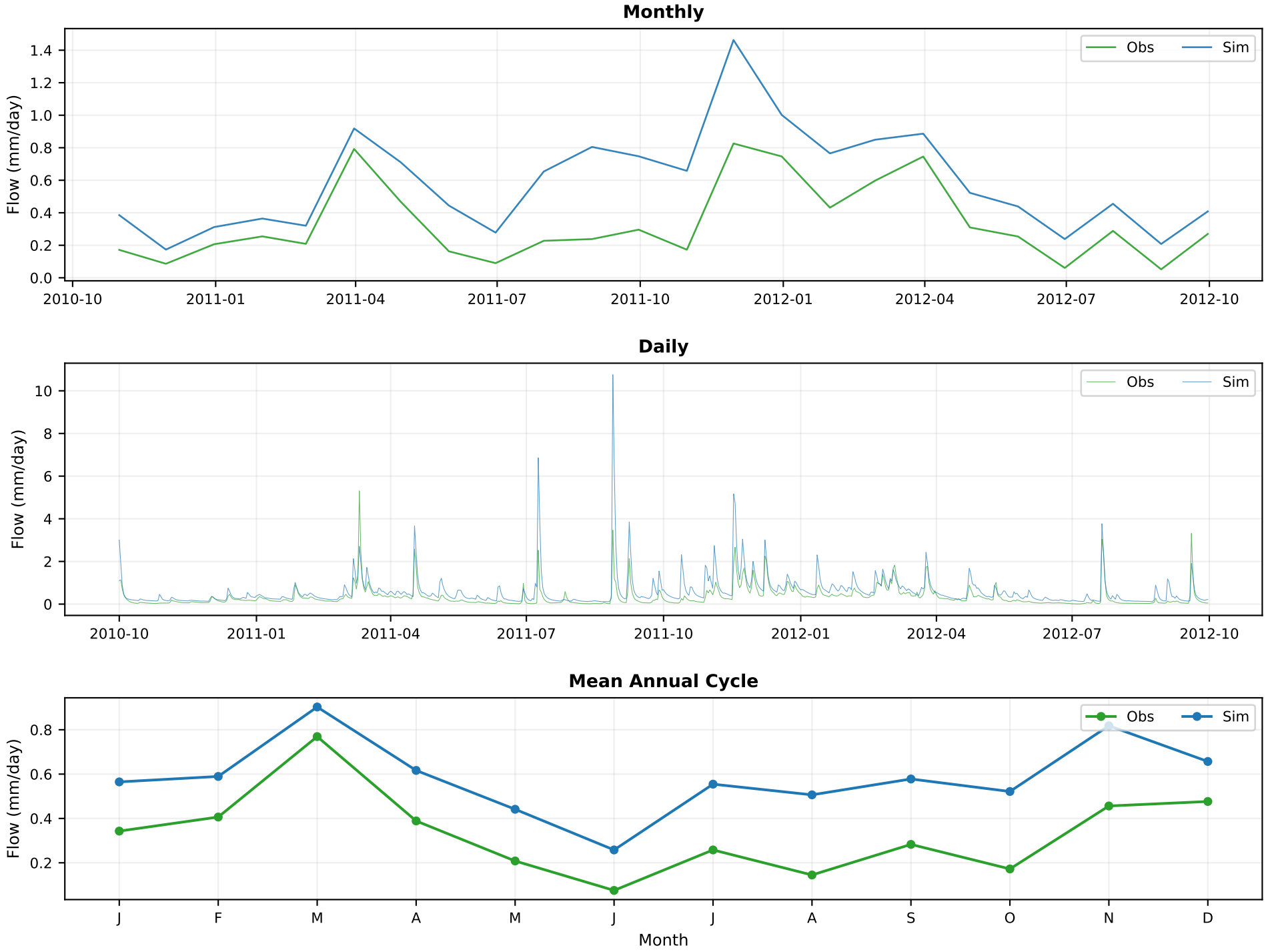

ID: 5291000
KGE=0.021 r=0.782 α=0.343 NSE=0.333 Bias=-69.2%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

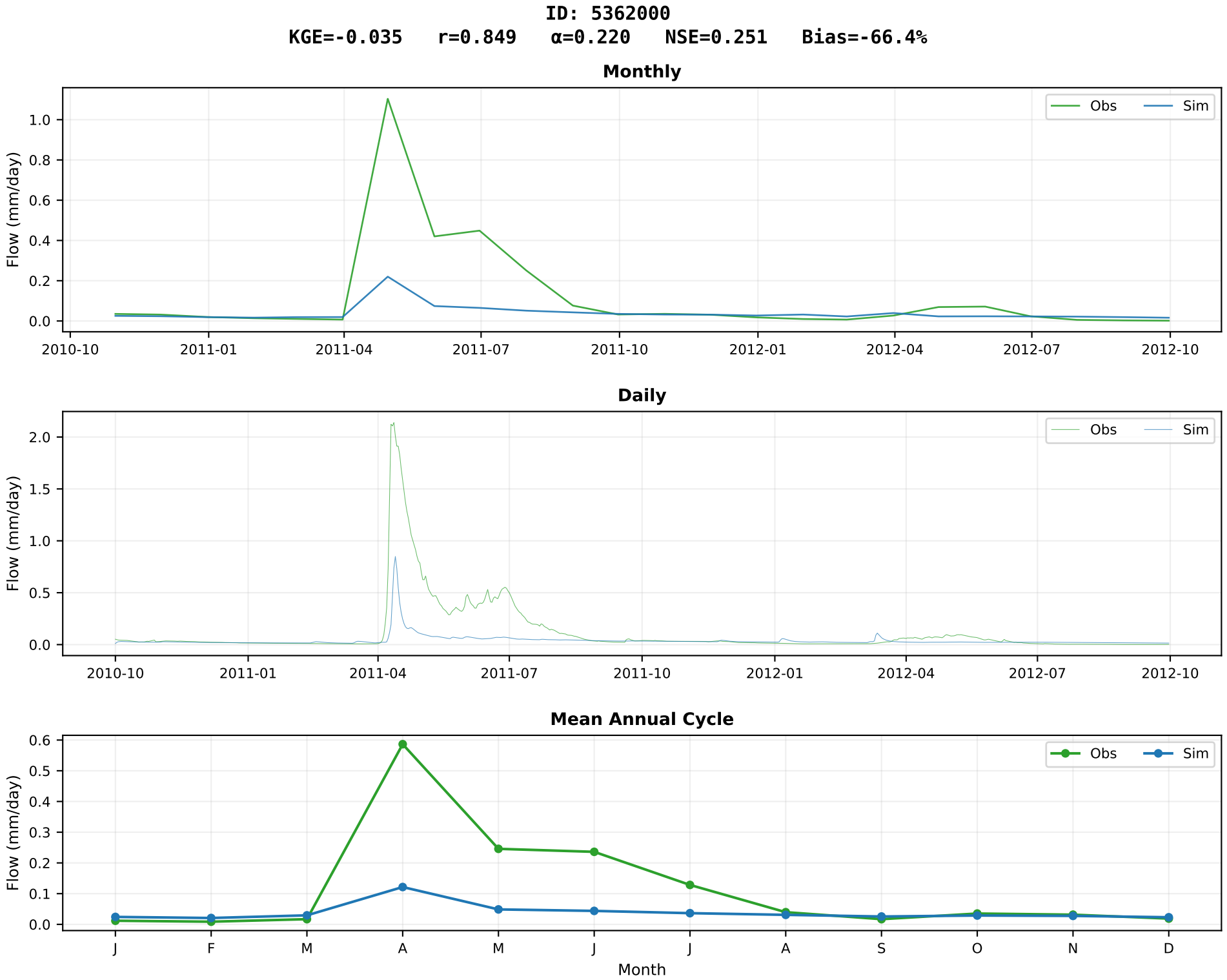

ID: 2092500
KGE=-0.007 r=0.682 α=0.854 NSE=0.346 Bias=+94.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

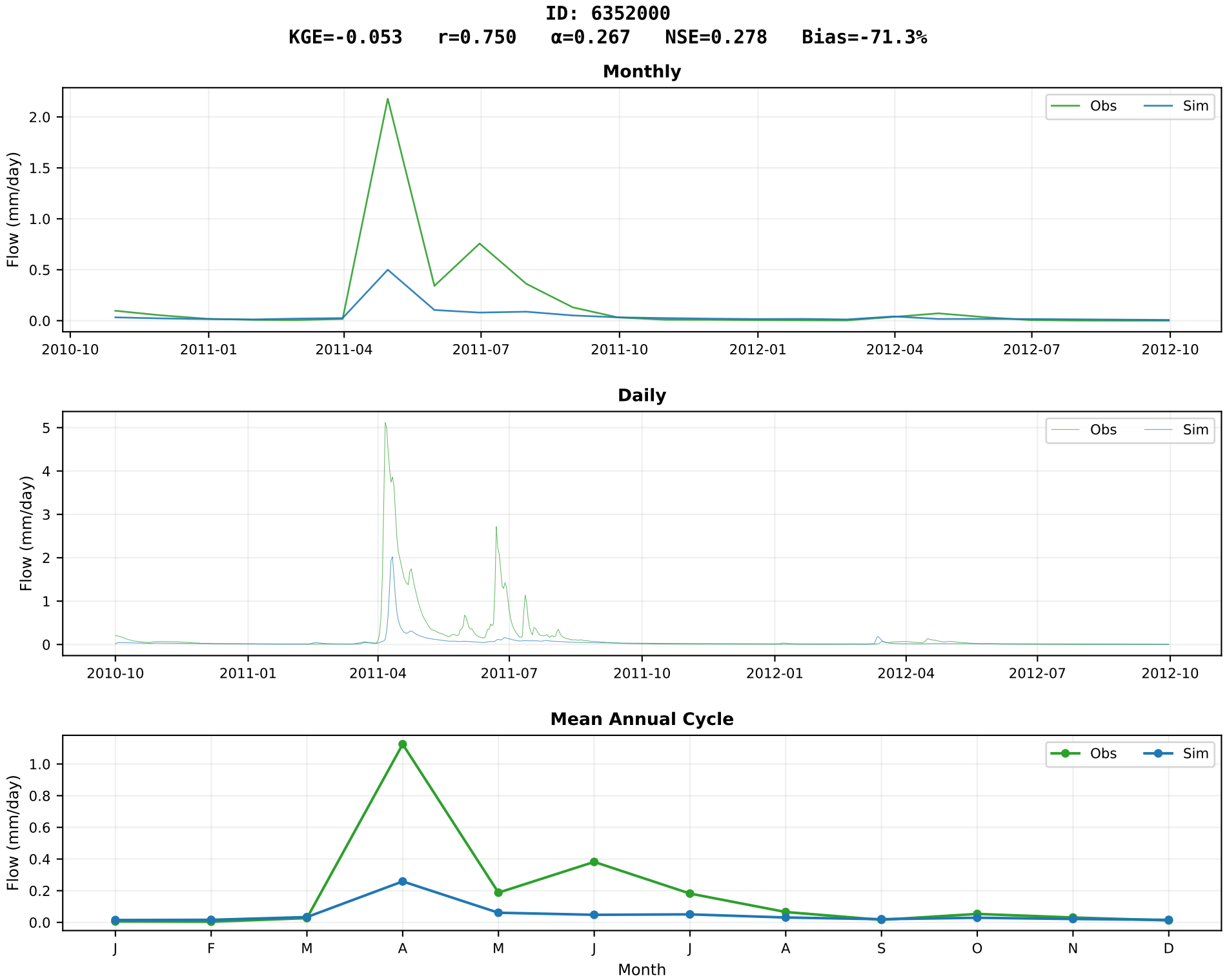

ID: 9505350
KGE=-0.009 r=0.588 α=0.338 NSE=0.251 Bias=-64.1%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
5
6
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month

ID: 2038850

KGE=-0.013 r=0.794 α=1.636 NSE=-0.374 Bias=+76.1%

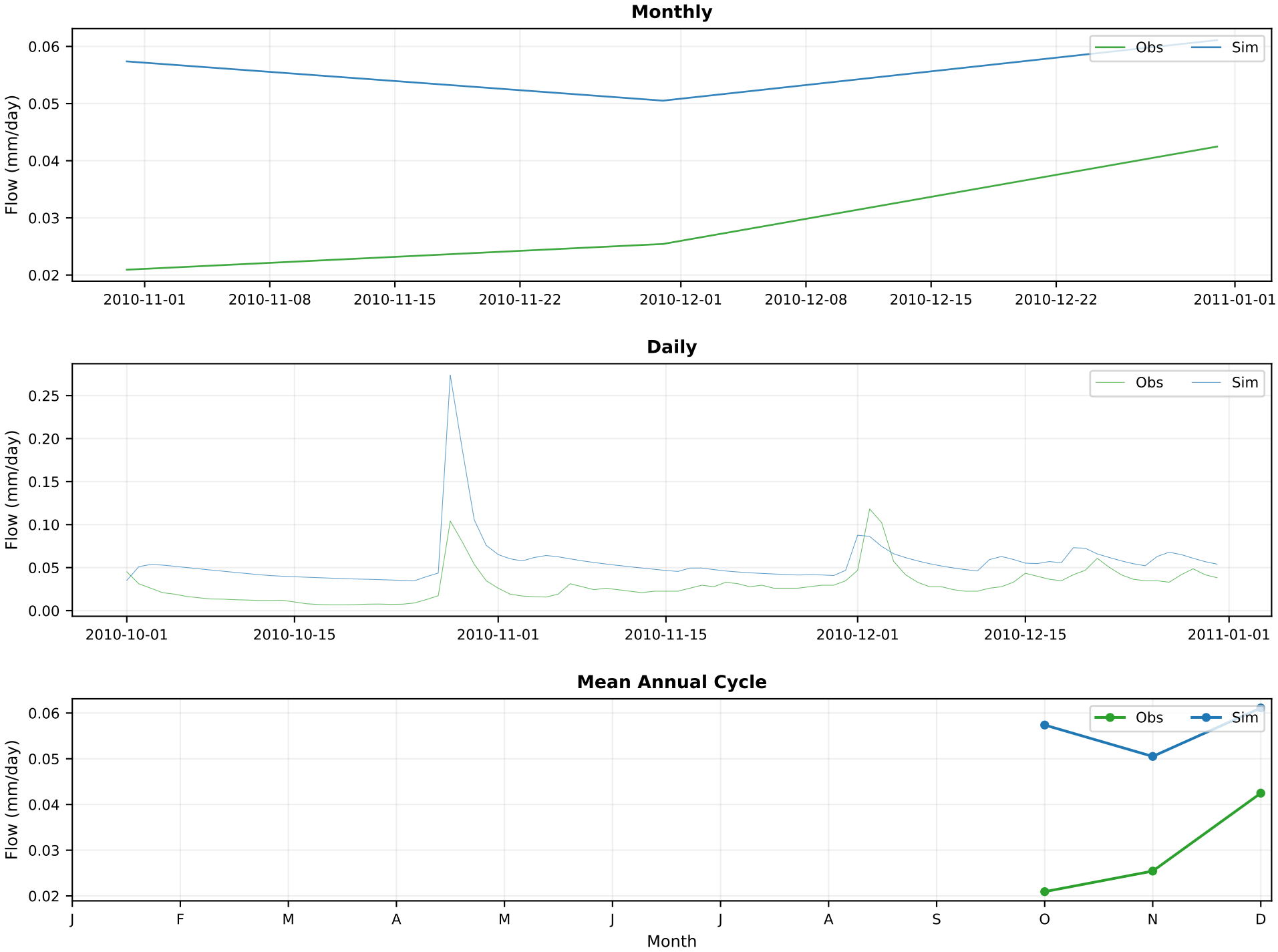


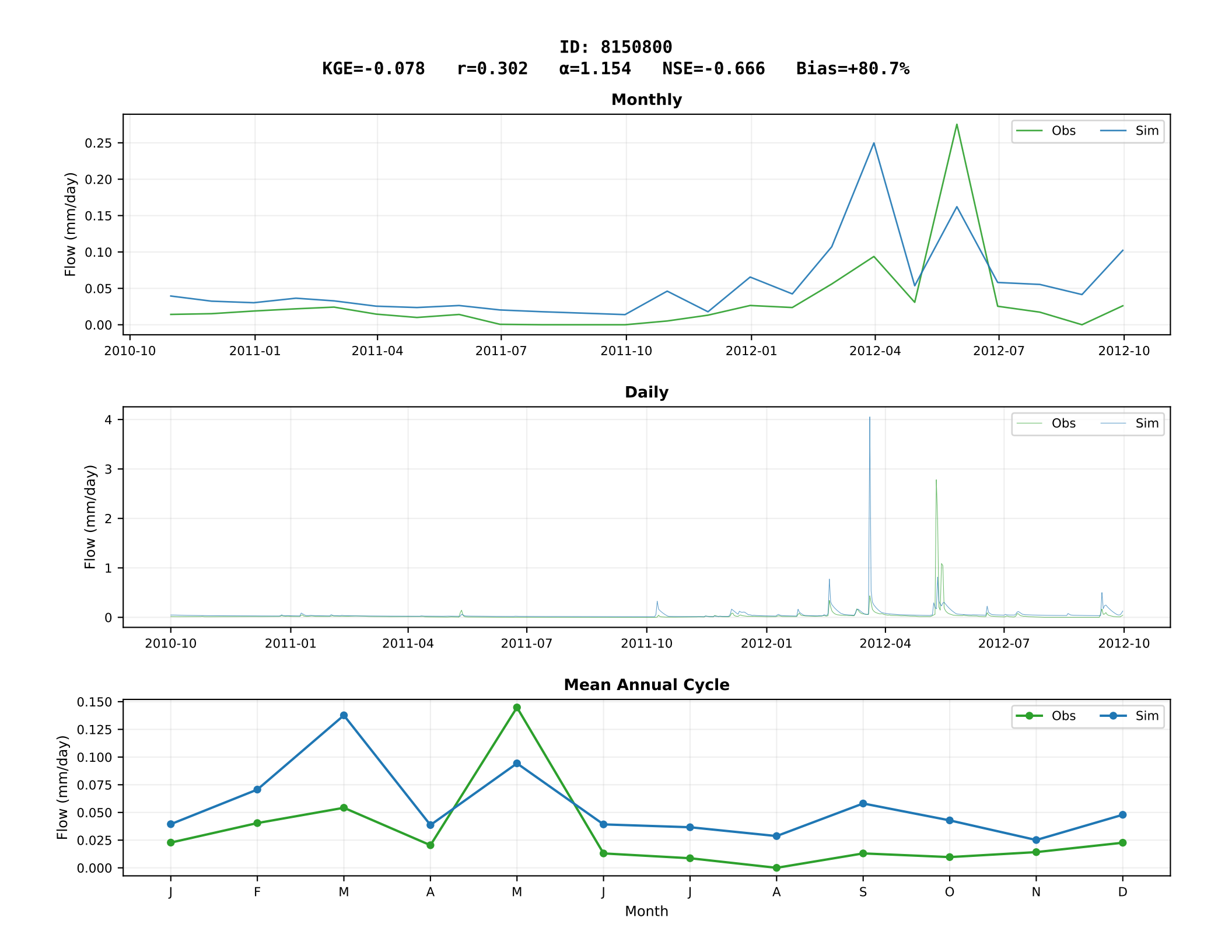

ID: 5362000
KGE=-0.035 r=0.849 α=0.220 NSE=0.251 Bias=-66.4%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month

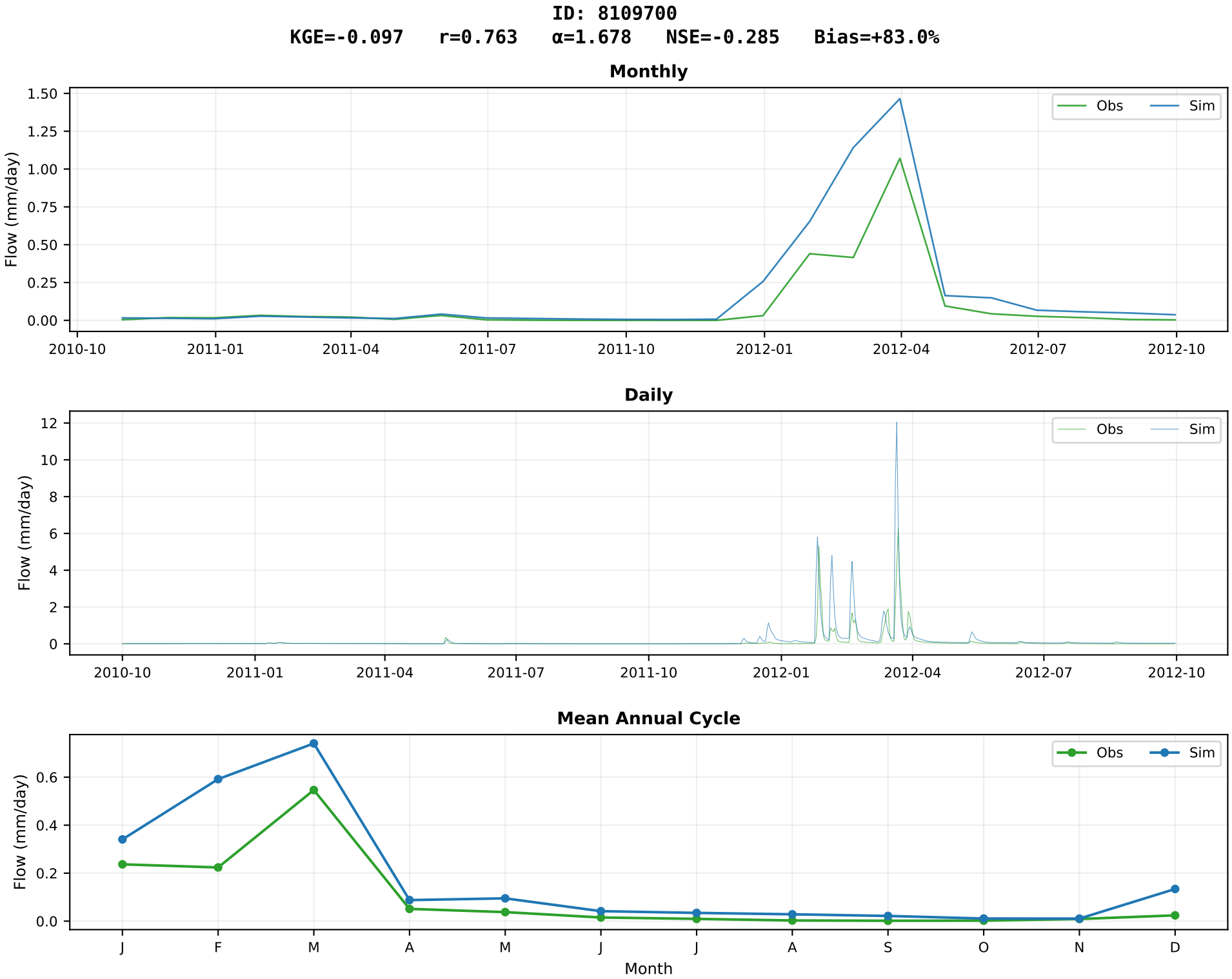

ID: 6352000
KGE=-0.053 r=0.750 α=0.267 NSE=0.278 Bias=-71.3%
Monthly
Obs
Sim
Flow (mm/day)
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
5
4
3
2
1
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
1.0
0.8
0.6
0.4
0.2
0.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

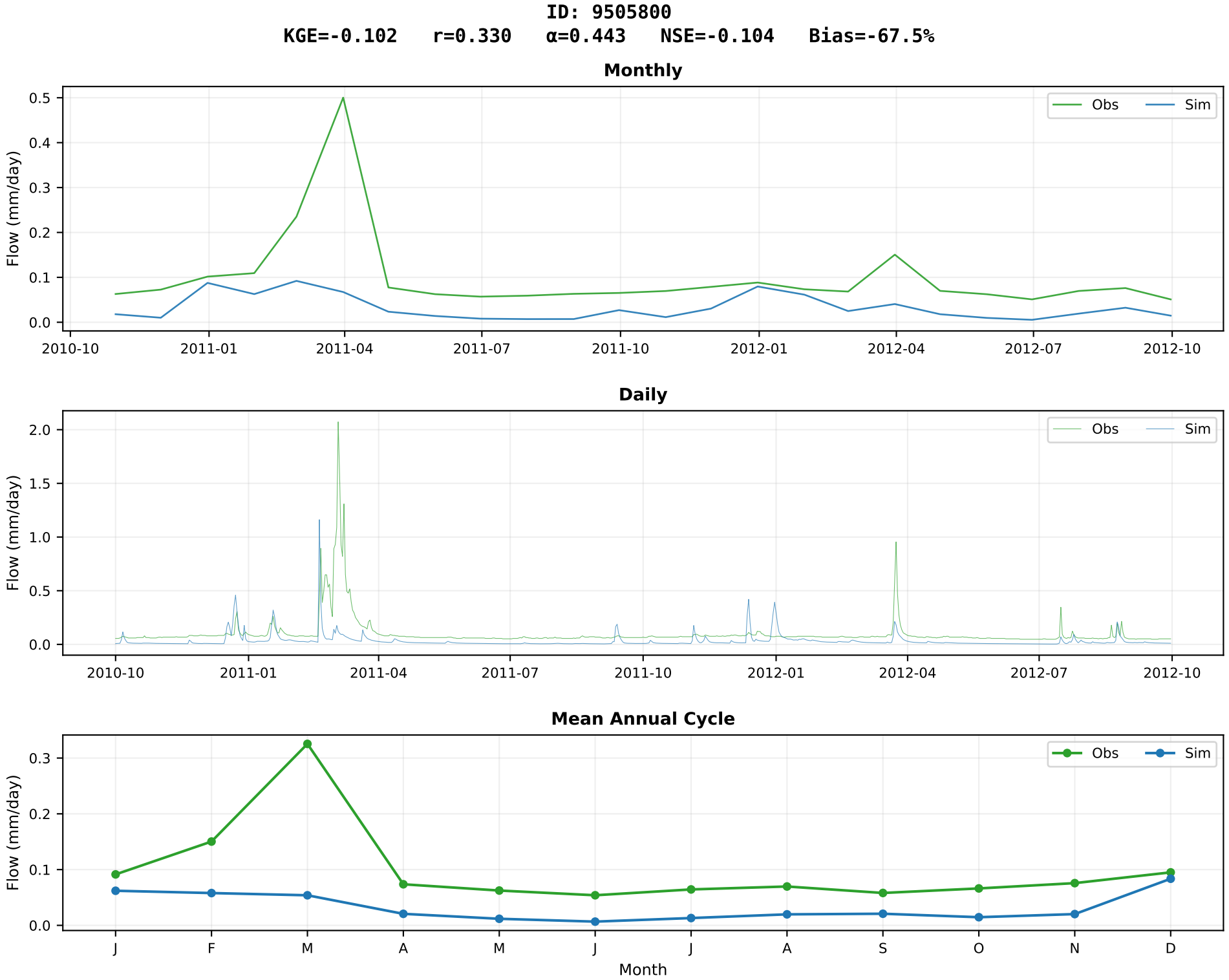
ID: 2193340

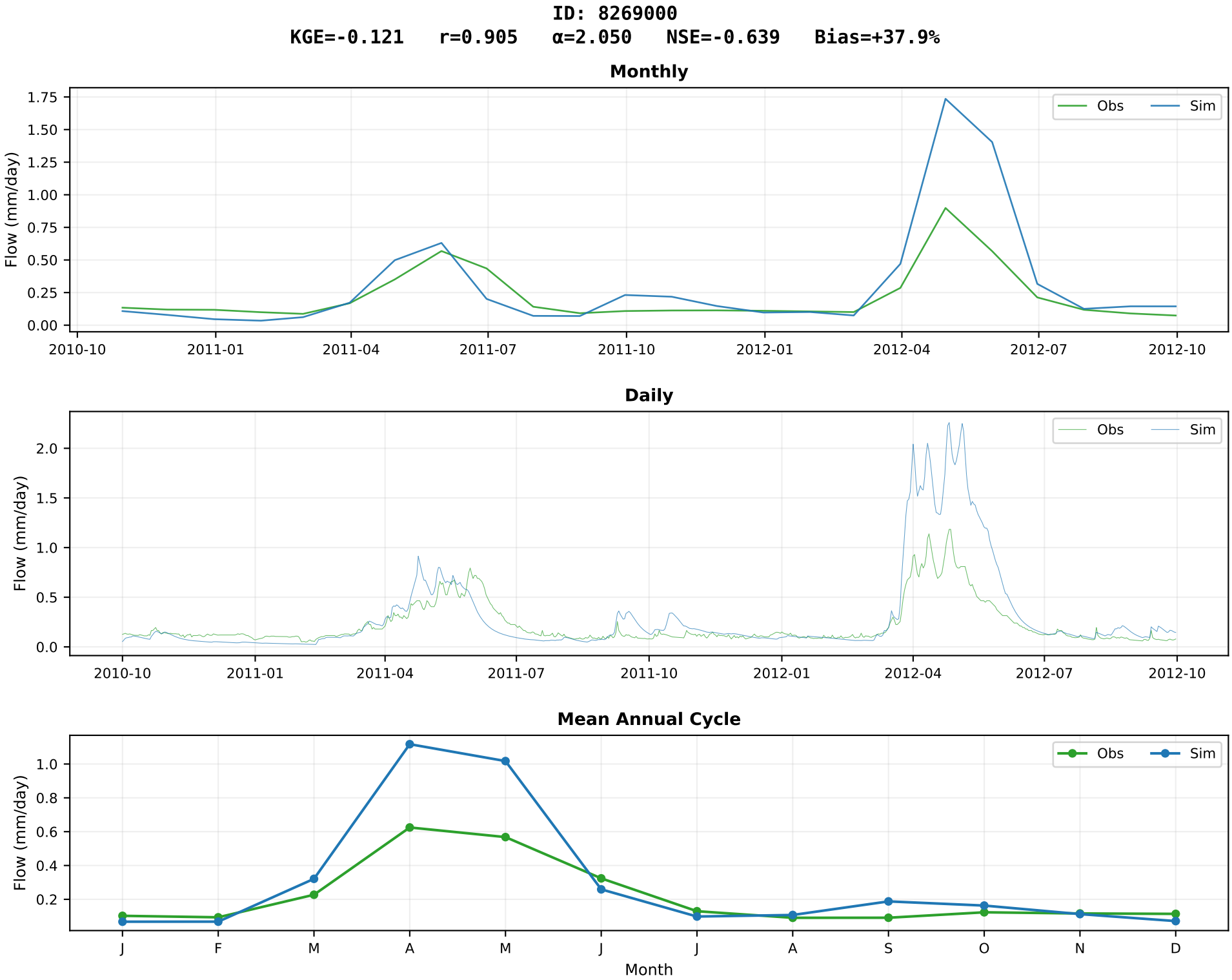
KGE=-0.071 r=0.687 α=1.486 NSE=-1.979 Bias=+90.2%

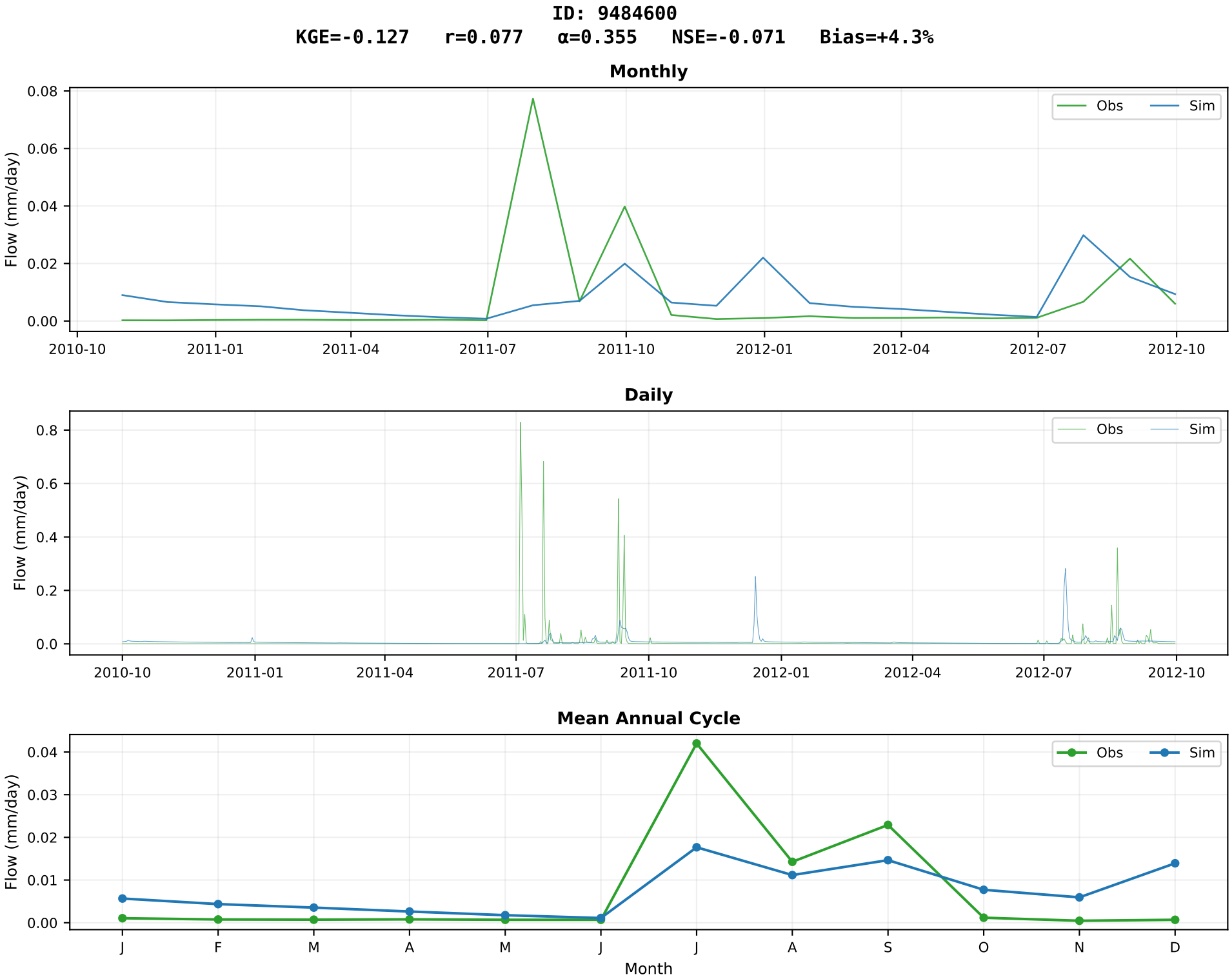


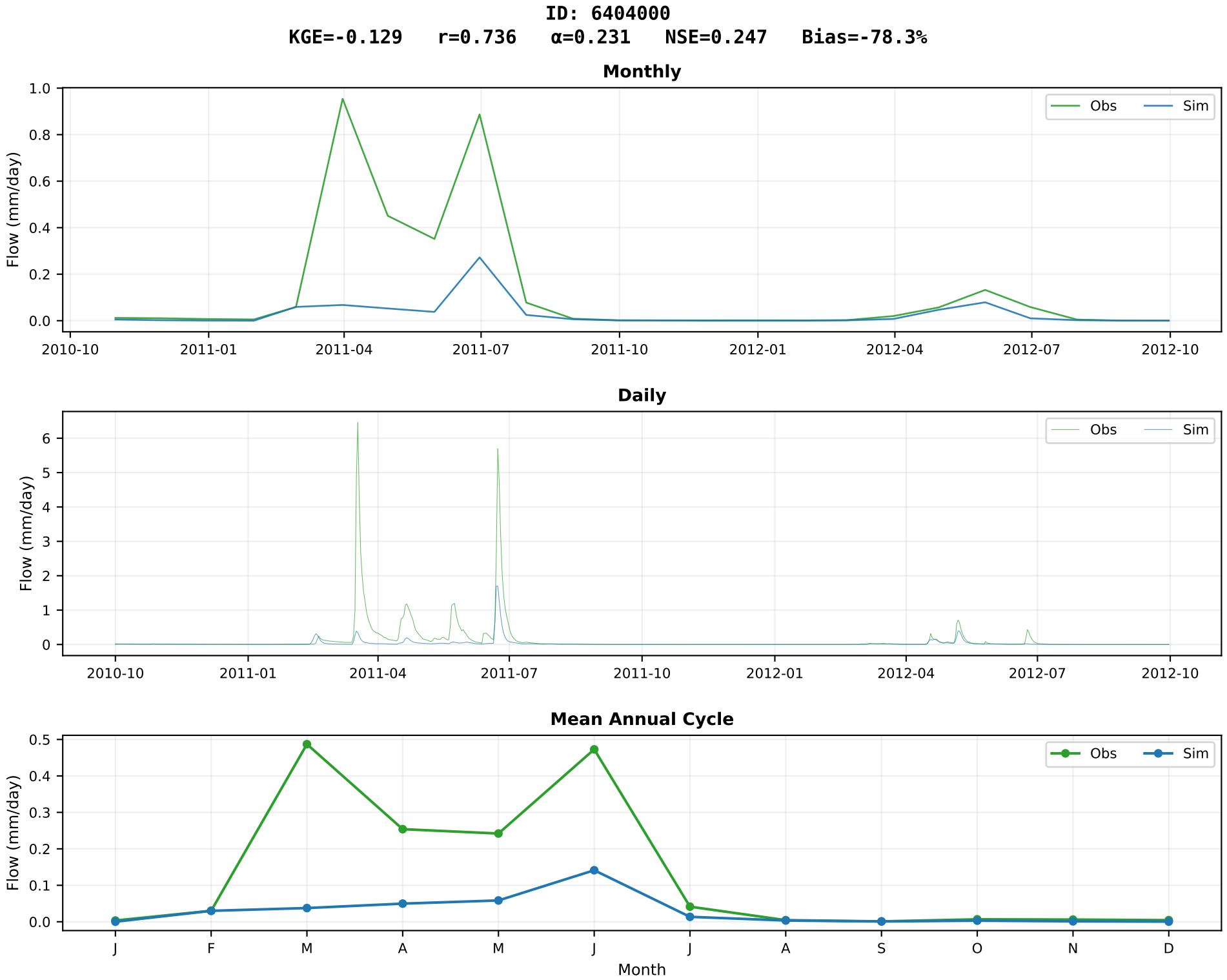

ID: 8150800
KGE=-0.078 r=0.302 α=1.154 NSE=-0.666 Bias=+80.7%
Monthly
Flow (mm/day)
Obs
Sim
0.00
0.05
0.10
0.15
0.20
0.25
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.000
0.025
0.050
0.075
0.100
0.125
0.150
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 8109700
KGE=-0.097 r=0.763 α=1.678 NSE=-0.285 Bias=+83.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 9505800
KGE=-0.102 r=0.330 α=0.443 NSE=-0.104 Bias=-67.5%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
J
F
M
A
M
J
J
A
S
O
N
D
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10

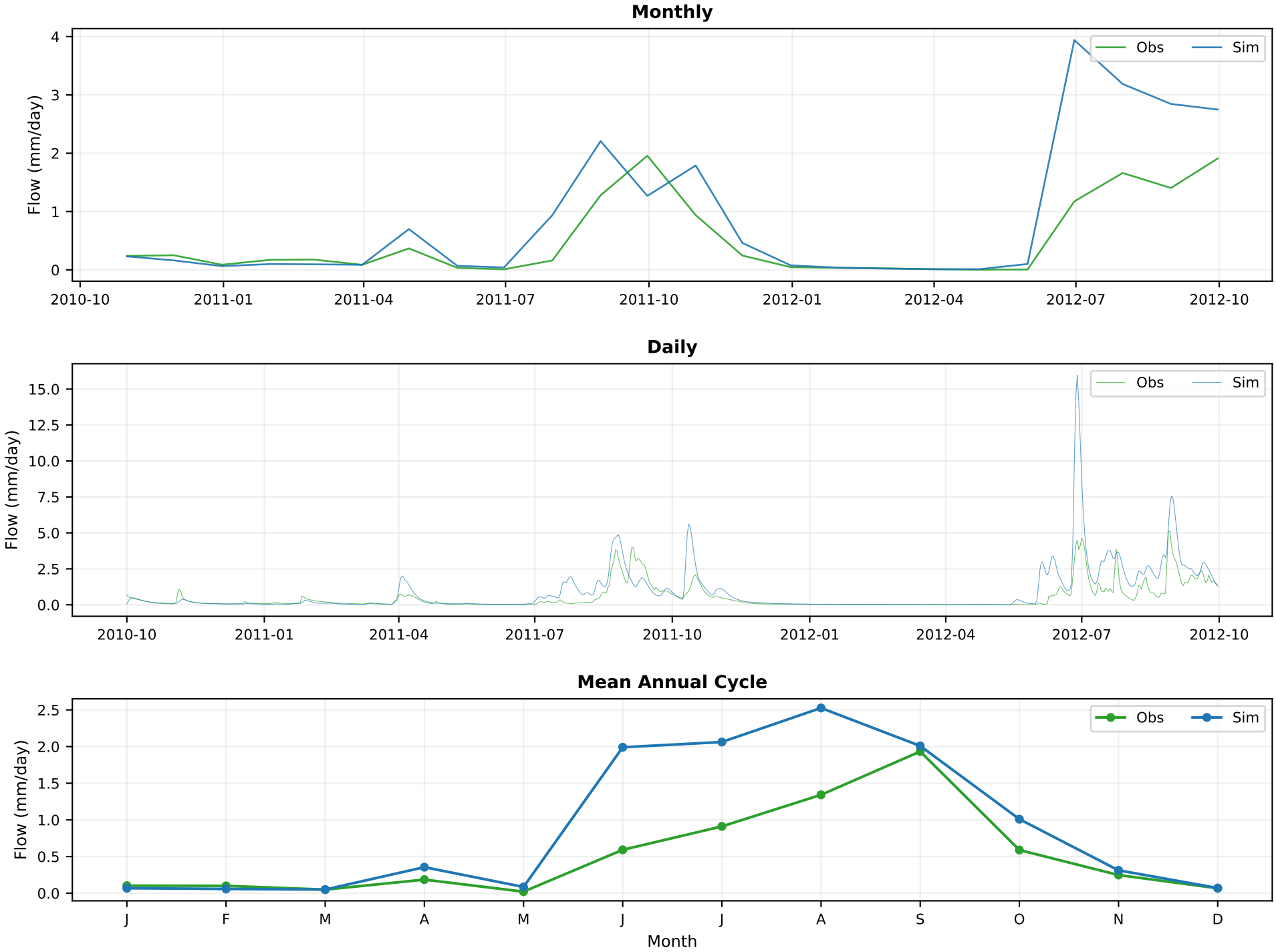

ID: 8269000
KGE=-0.121 r=0.905 α=2.050 NSE=-0.639 Bias=+37.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

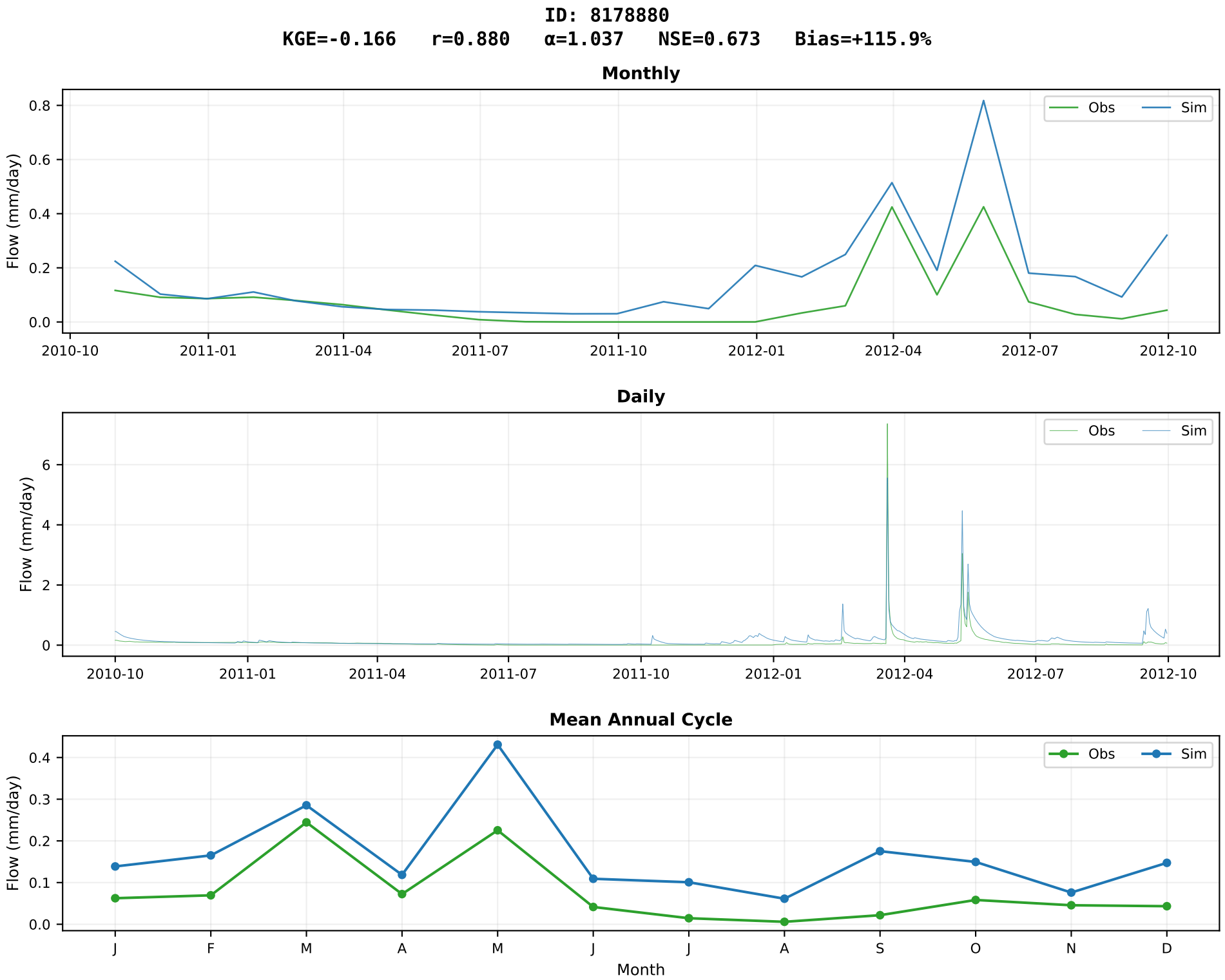

ID: 9484600
KGE=-0.127 r=0.077 α=0.355 NSE=-0.071 Bias=+4.3%
Monthly
Flow (mm/day)
Obs
Sim
0.00
0.02
0.04
0.06
0.08
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.00
0.01
0.02
0.03
0.04
J
F
M
A
M
J
J
A
S
O
N
D
Month

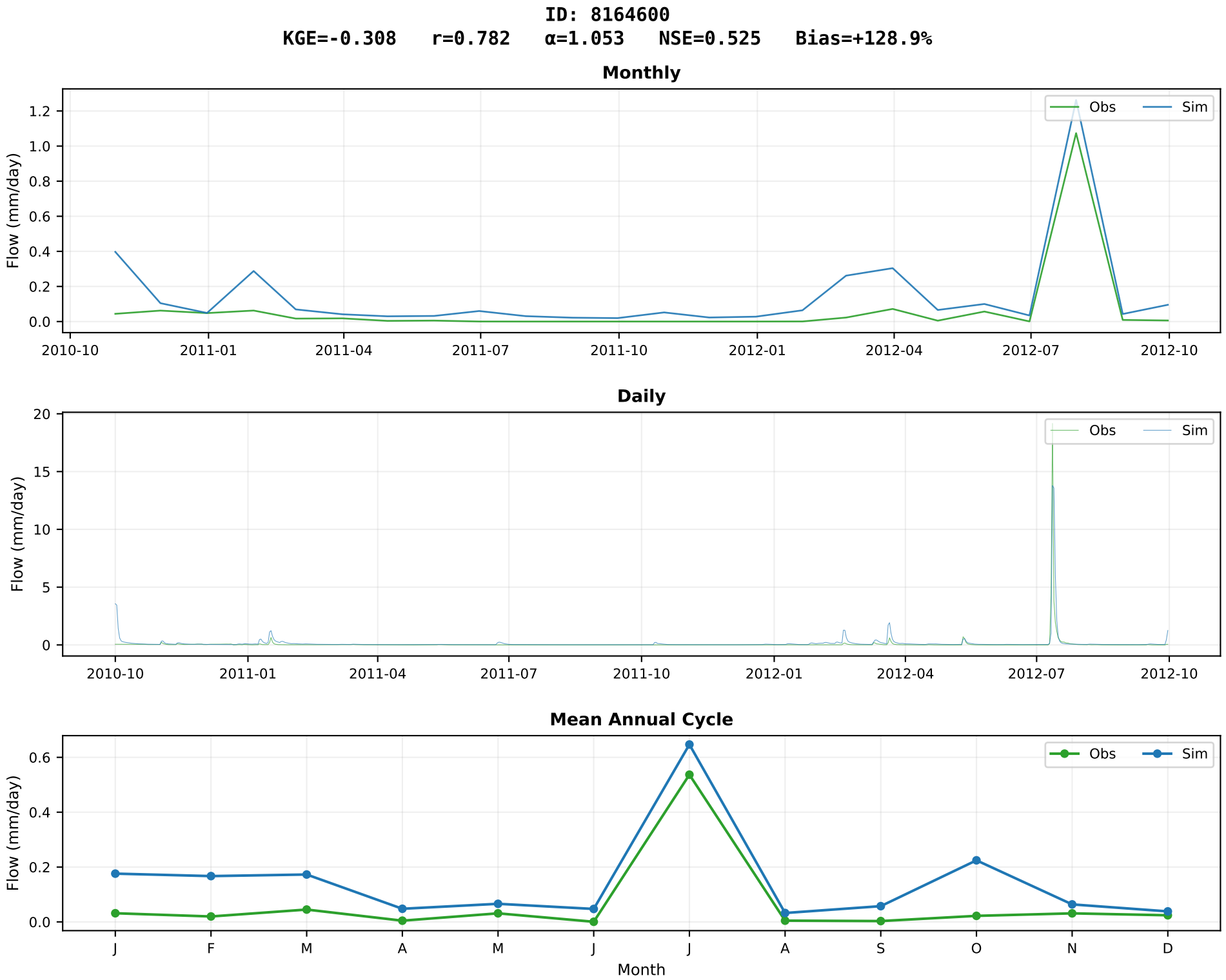

ID: 6404000
KGE=-0.129 r=0.736 α=0.231 NSE=0.247 Bias=-78.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

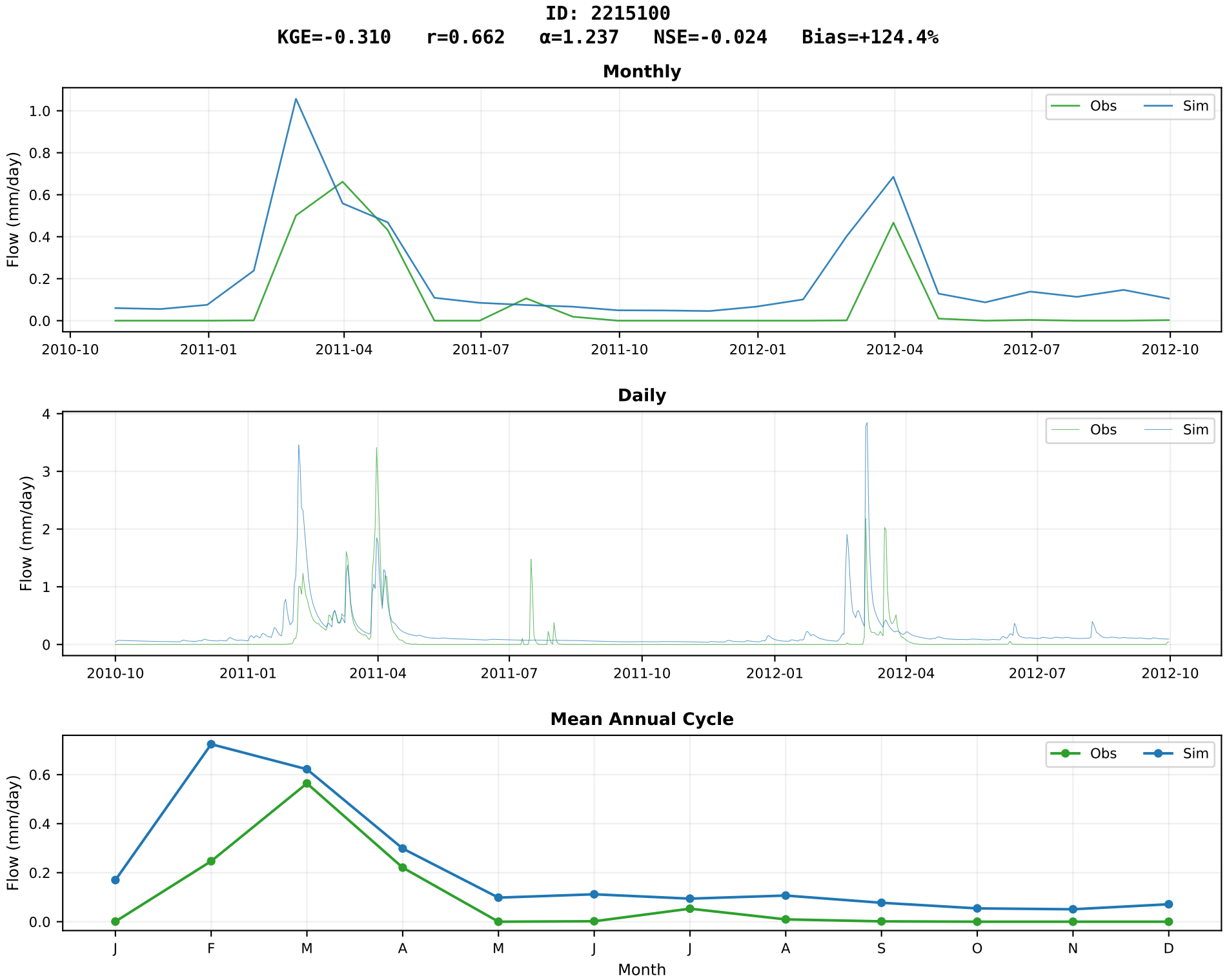
ID: 2297155

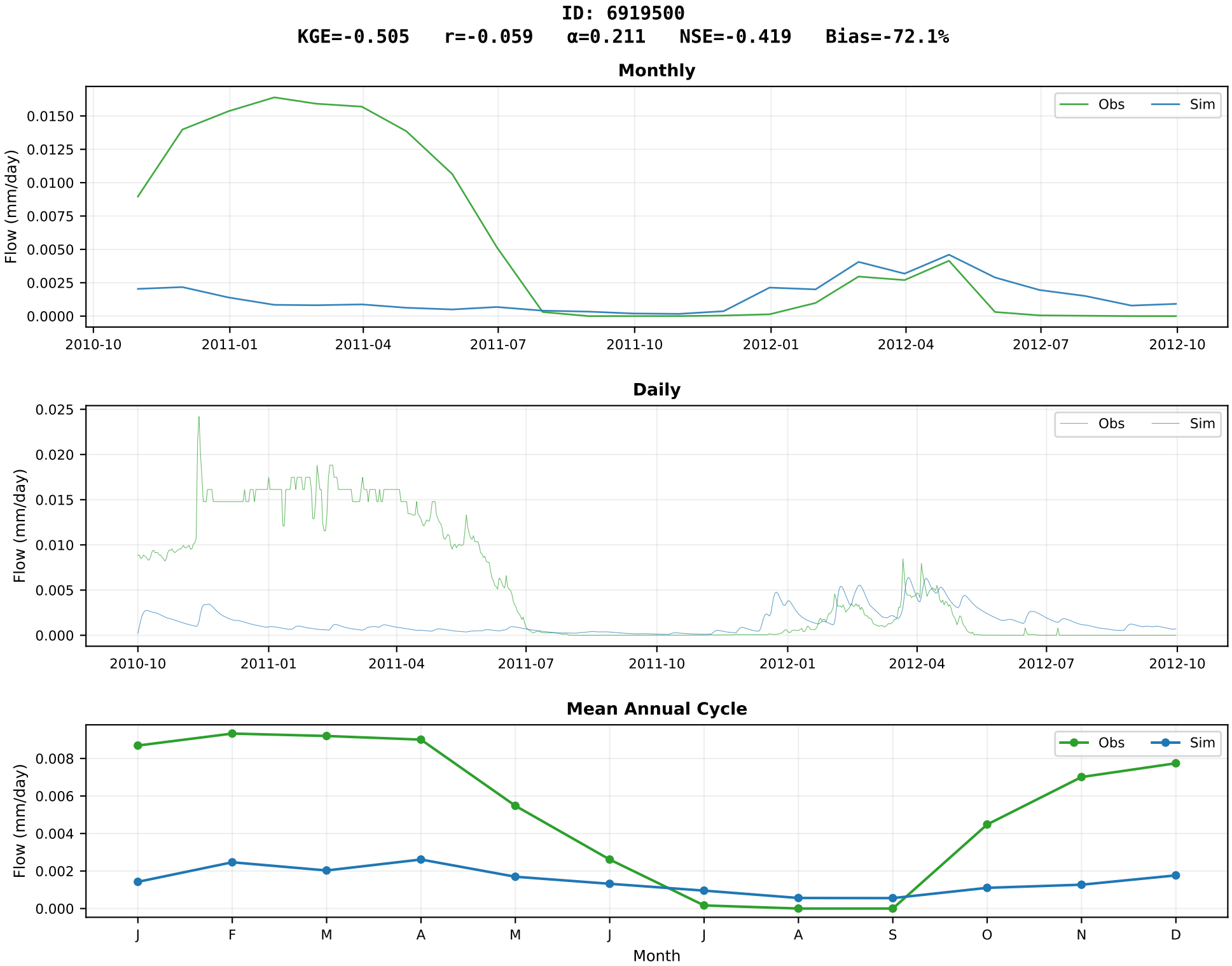
KGE=-0.155 r=0.807 α=1.874 NSE=-0.670 Bias=+73.0%

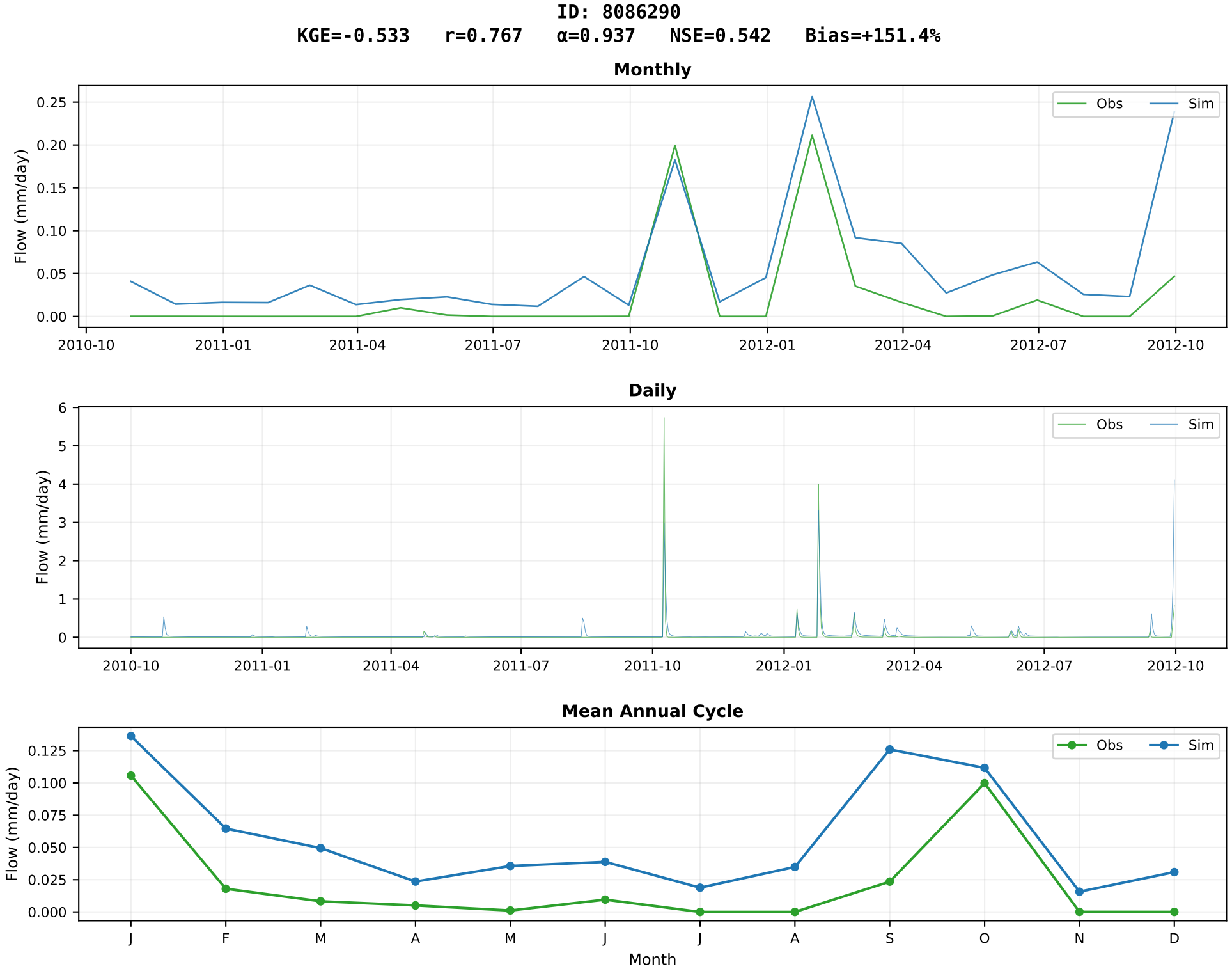


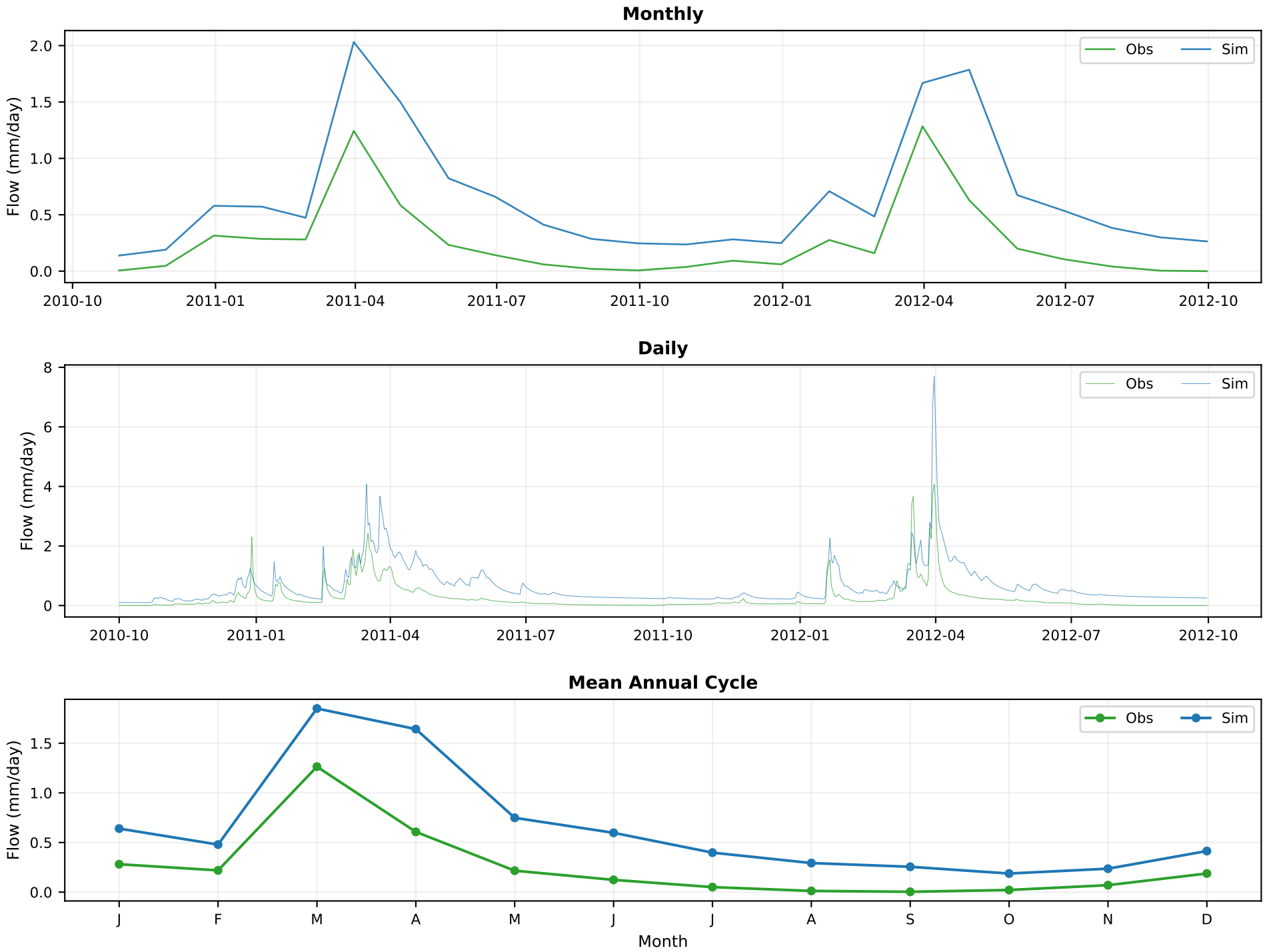

ID: 8178880
KGE=-0.166 r=0.880 α=1.037 NSE=0.673 Bias=+115.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

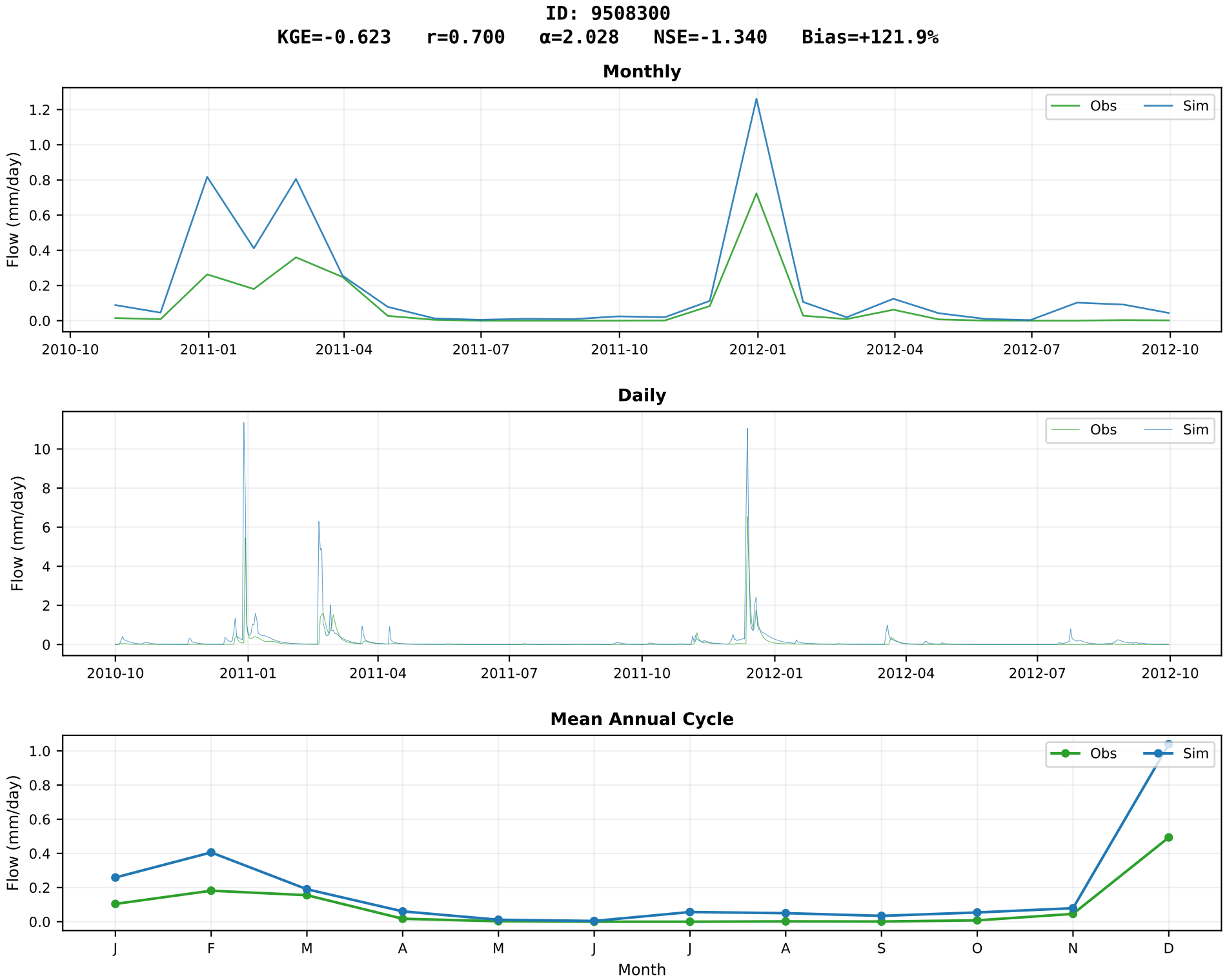

ID: 8164600
KGE=-0.308 r=0.782 α=1.053 NSE=0.525 Bias=+128.9%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

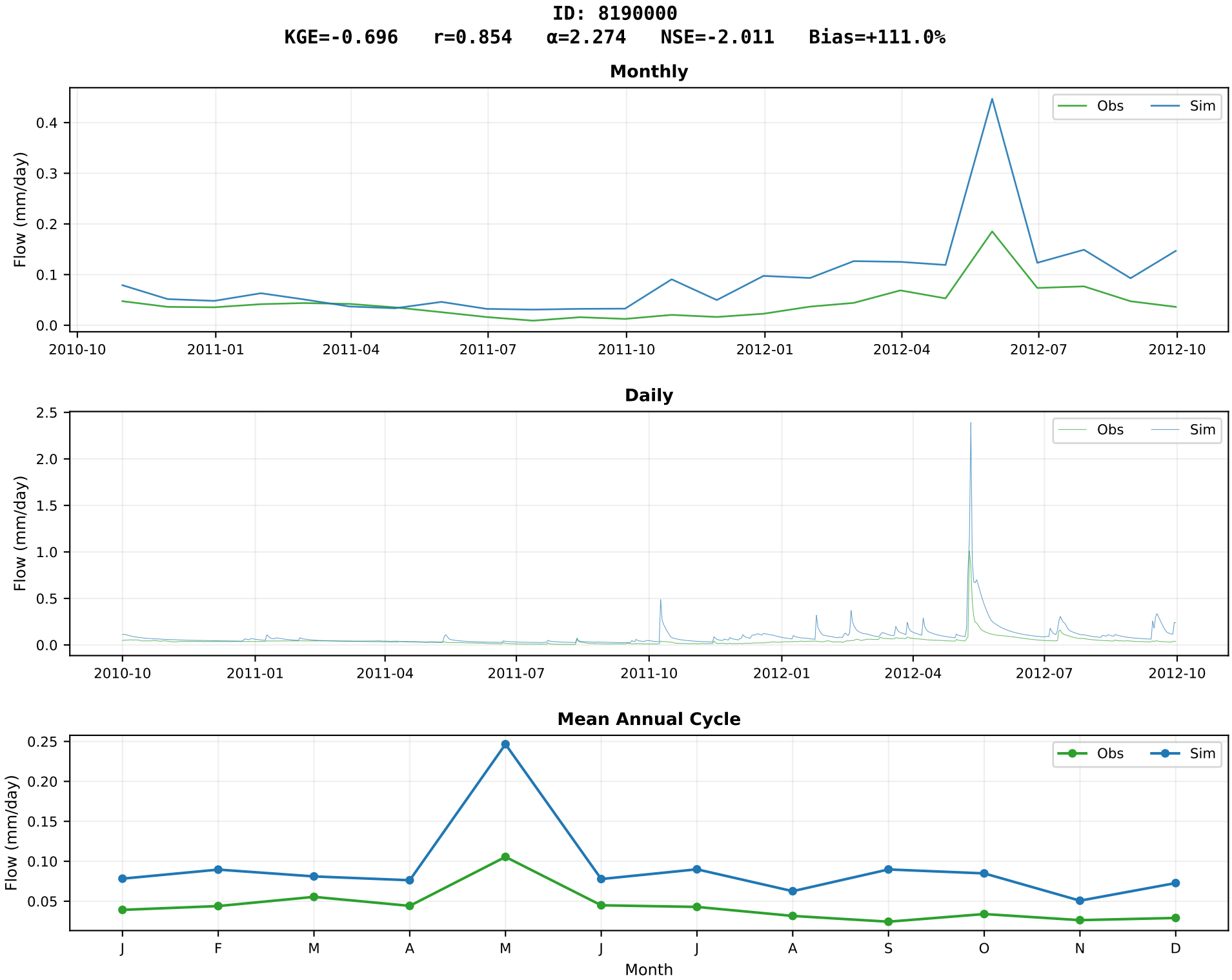

ID: 2215100
KGE=-0.310 r=0.662 α=1.237 NSE=-0.024 Bias=+124.4%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
1
2
3
4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
J
F
M
A
M
J
J
A
S
O
N
D
Month

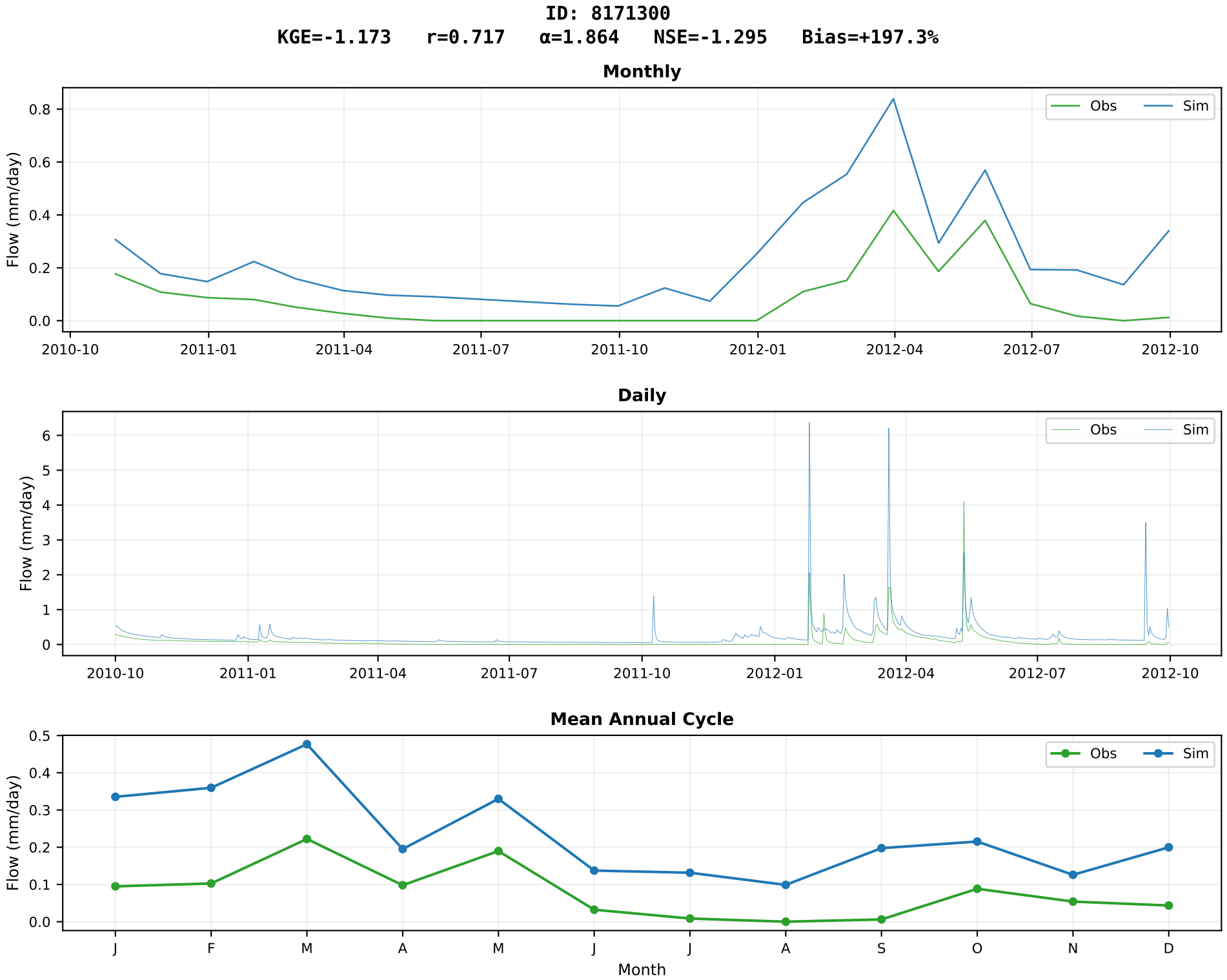

ID: 6919500
KGE=-0.505 r=-0.059 α=0.211 NSE=-0.419 Bias=-72.1%
Monthly
Obs
Sim
Flow (mm/day)
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.025
0.020
0.015
0.010
0.005
0.000
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.008
0.006
0.004
0.002
0.000
J
F
M
A
M
J
J
A
S
O
N
D
Month

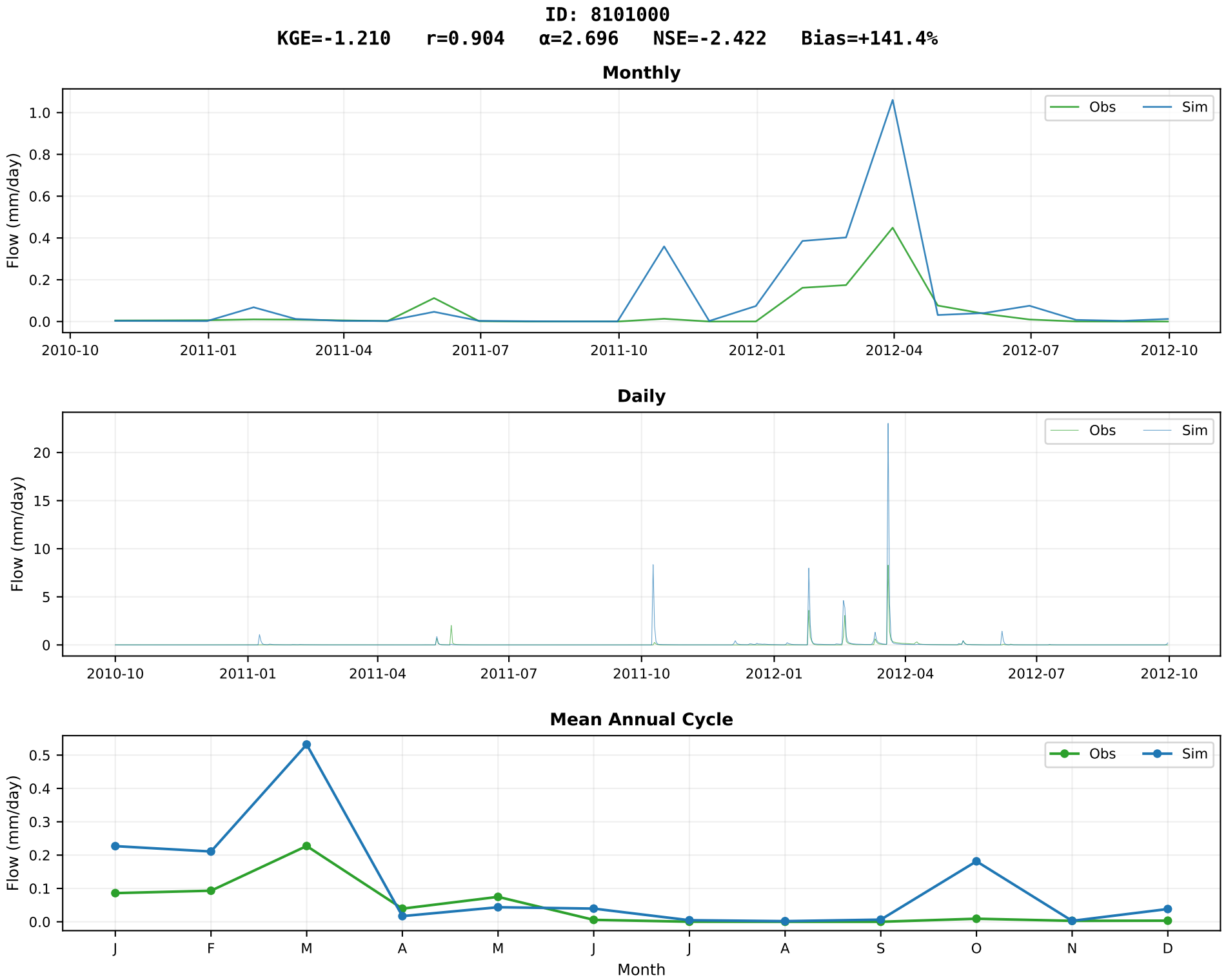

ID: 8086290
KGE=-0.533 r=0.767 α=0.937 NSE=0.542 Bias=+151.4%
Monthly
Flow (mm/day)
Obs
Sim
0.25
0.20
0.15
0.10
0.05
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
6
5
4
3
2
1
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.125
0.100
0.075
0.050
0.025
0.000
J
F
M
A
M
J
J
A
S
O
N
D
Month

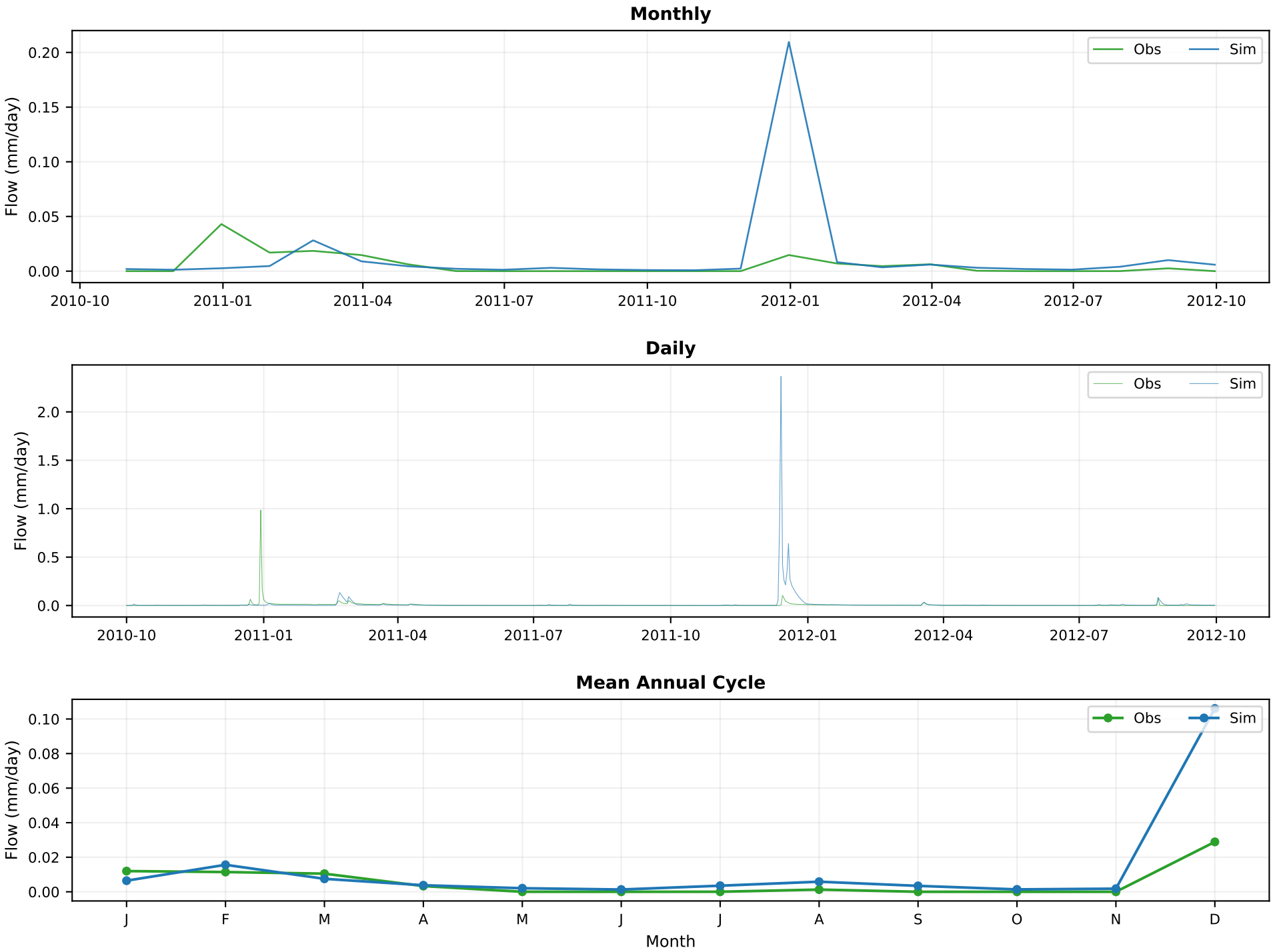

SF TRINITY R BL HYAMPOM CA (ID=11528700)
KGE=-0.613 r=0.866 α=1.499 NSE=-0.348 Bias=+152.8%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

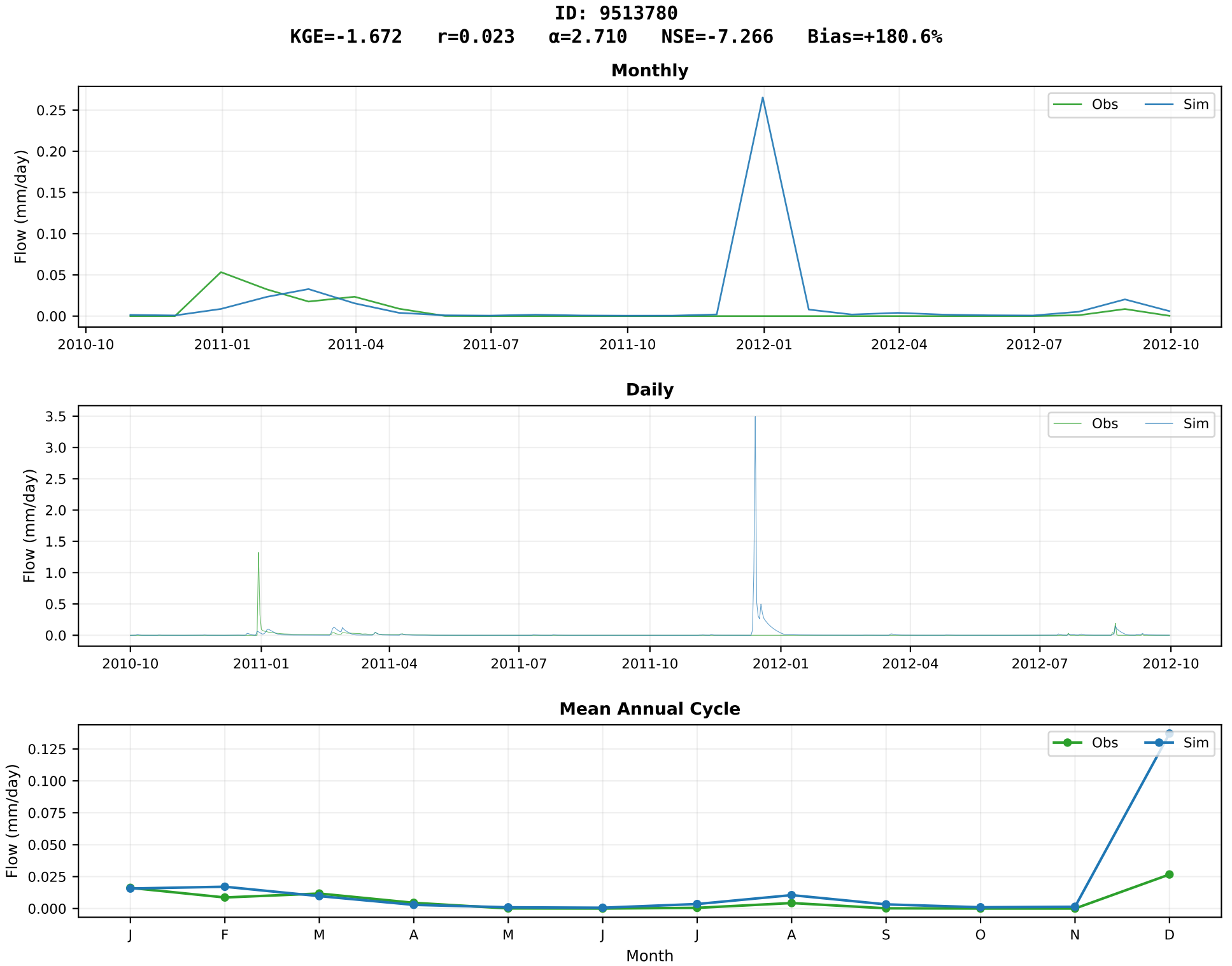

ID: 9508300
KGE=-0.623 r=0.700 α=2.028 NSE=-1.340 Bias=+121.9%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0
2
4
6
8
10
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0
J
F
M
A
M
J
J
A
S
O
N
D
Month

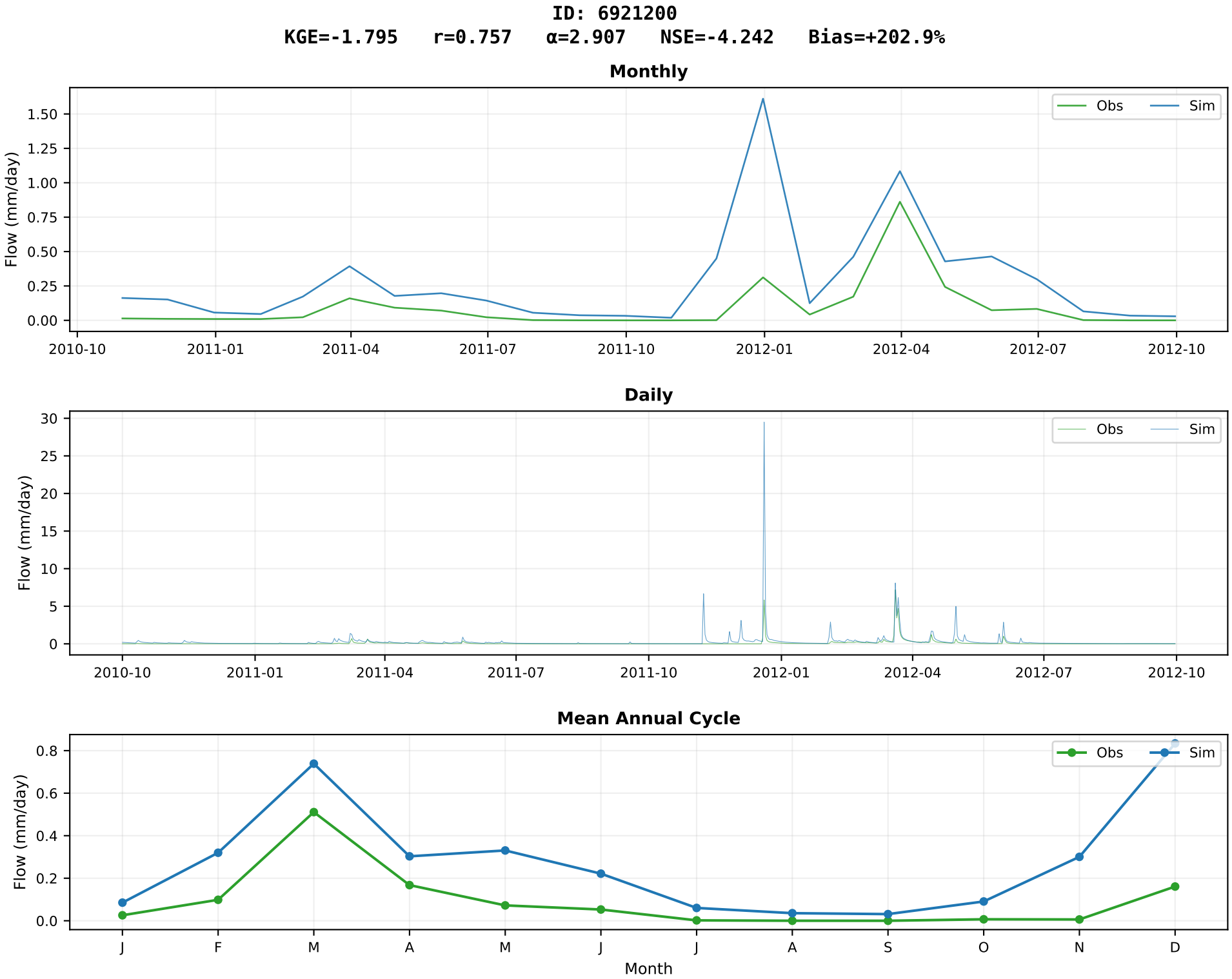

ID: 8190000
KGE=-0.696 r=0.854 α=2.274 NSE=-2.011 Bias=+111.0%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

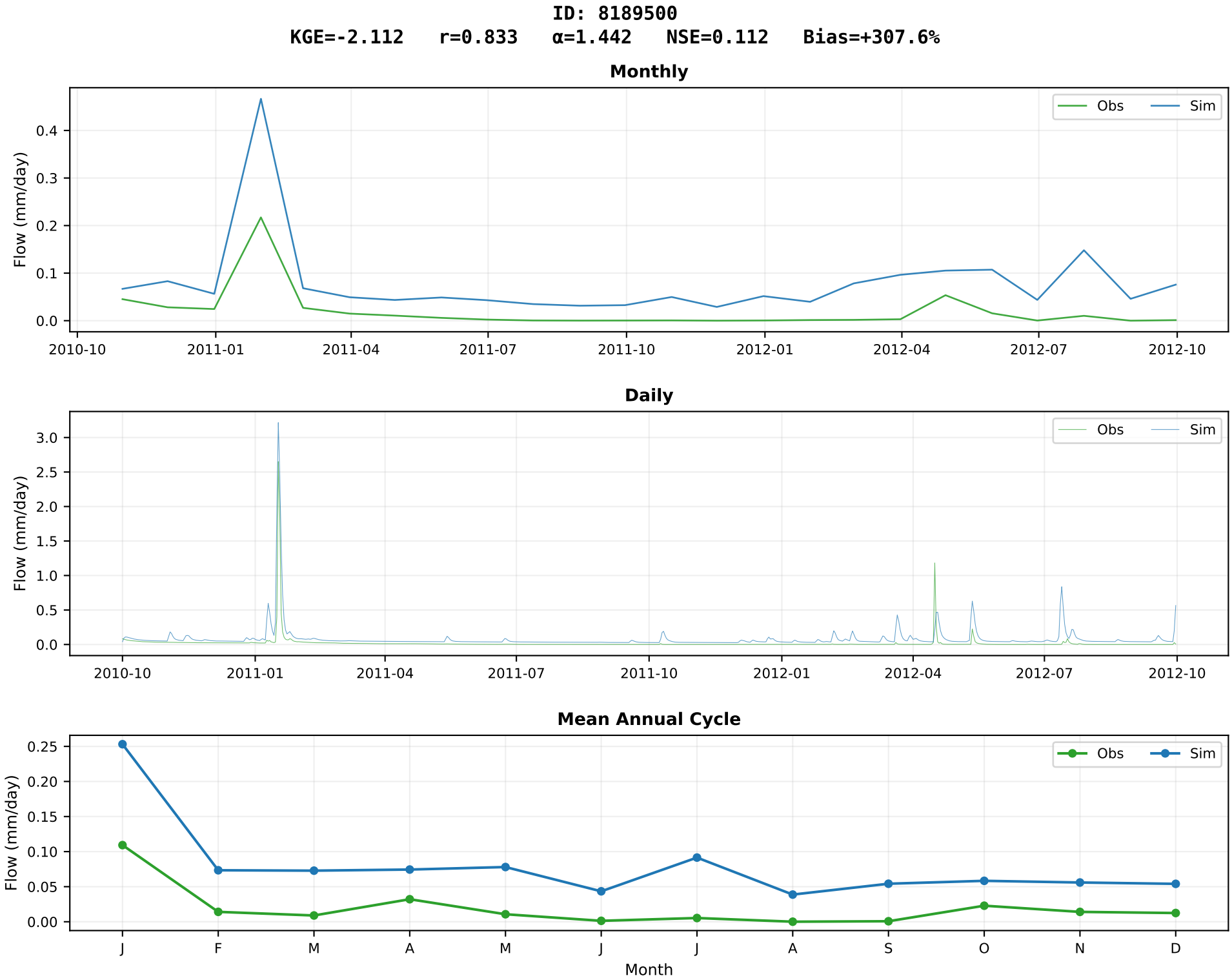

ID: 8171300
KGE=-1.173 r=0.717 α=1.864 NSE=-1.295 Bias=+197.3%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

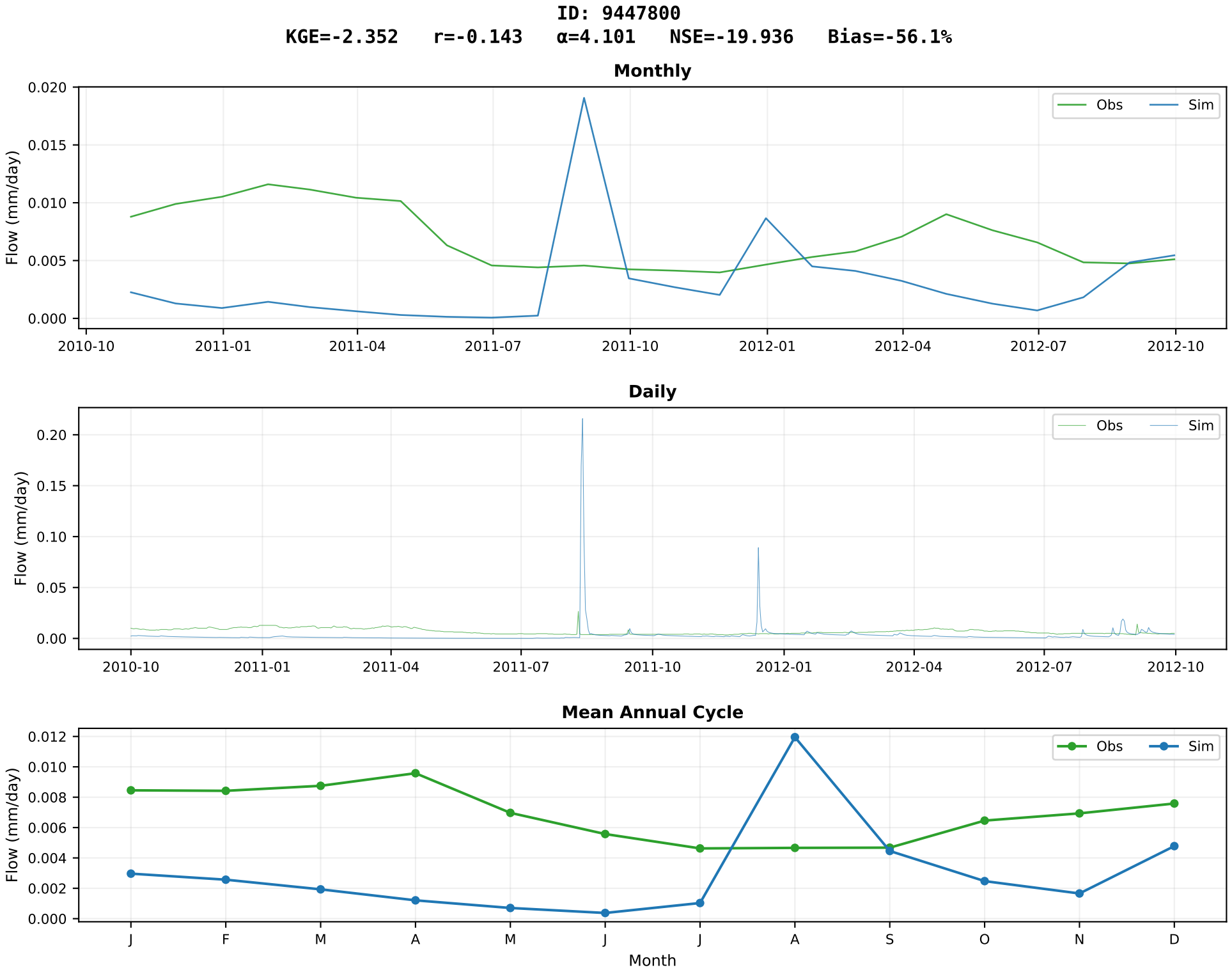

ID: 8101000
KGE=-1.210 r=0.904 α=2.696 NSE=-2.422 Bias=+141.4%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month

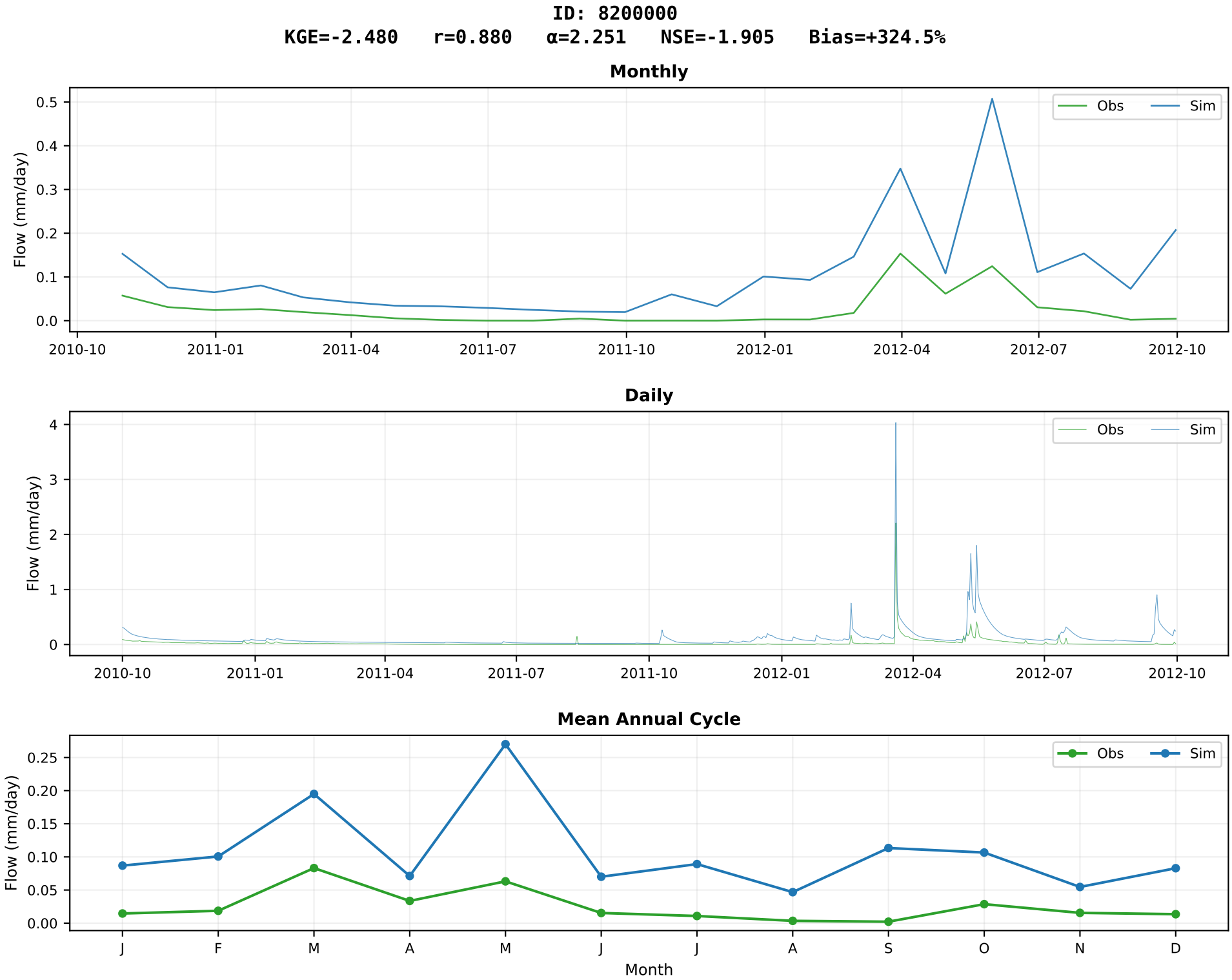

ID: 9512280
KGE=-1.334 r=0.050 α=2.635 NSE=-6.721 Bias=+136.8%
Monthly
Daily
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
Month
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
J
F
M
A
M
J
J
A
S
O
N
D

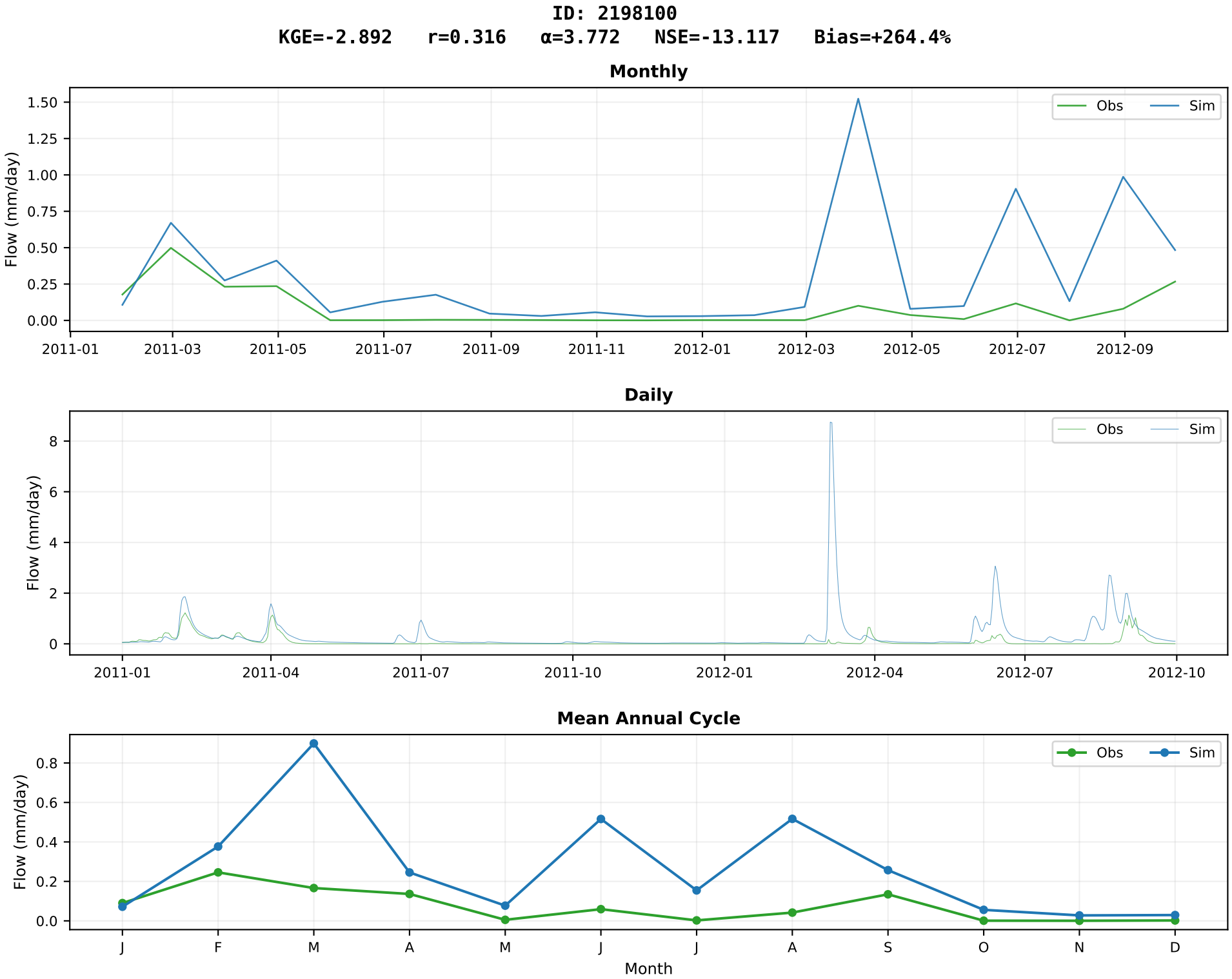

ID: 9513780
KGE=-1.672 r=0.023 α=2.710 NSE=-7.266 Bias=+180.6%
Monthly
Flow (mm/day)
Obs
Sim
0.25
0.20
0.15
0.10
0.05
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.125
0.100
0.075
0.050
0.025
0.000
J
F
M
A
M
J
J
A
S
O
N
D
Month

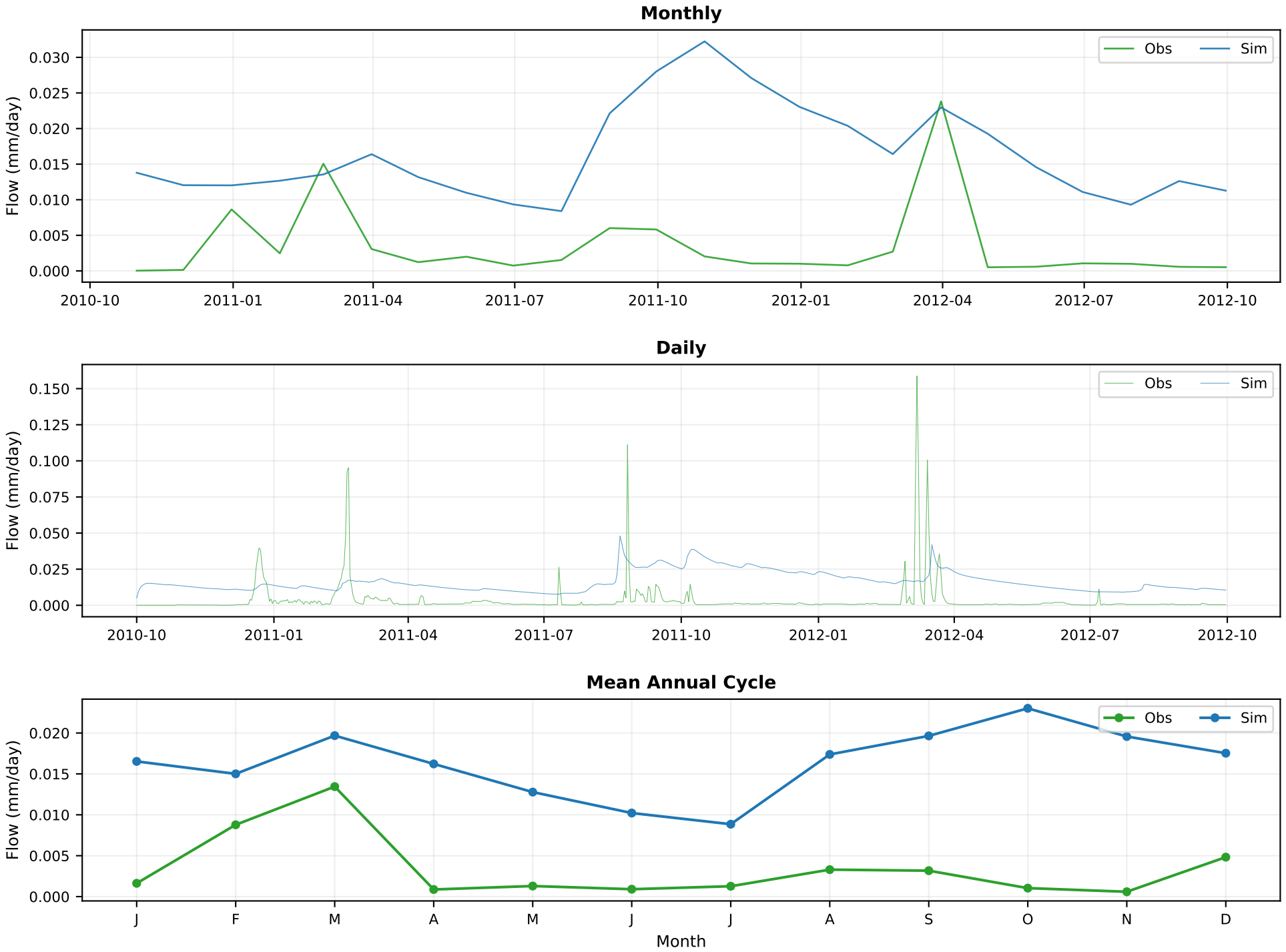

ID: 6921200
KGE=-1.795 r=0.757 α=2.907 NSE=-4.242 Bias=+202.9%
Monthly
Obs
Sim
Flow (mm/day)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
30
25
20
15
10
5
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.8
0.6
0.4
0.2
0.0
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 8189500
KGE=-2.112 r=0.833 α=1.442 NSE=0.112 Bias=+307.6%
Monthly
Flow (mm/day)
Obs
Sim
0.0
0.1
0.2
0.3
0.4
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0.0
0.5
1.0
1.5
2.0
2.5
3.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.00
0.05
0.10
0.15
0.20
0.25
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 9447800
KGE=-2.352 r=-0.143 α=4.101 NSE=-19.936 Bias=-56.1%
Monthly
Flow (mm/day)
0.020
0.015
0.010
0.005
0.000
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
0.20
0.15
0.10
0.05
0.00
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
0.012
0.010
0.008
0.006
0.004
0.002
0.000
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month

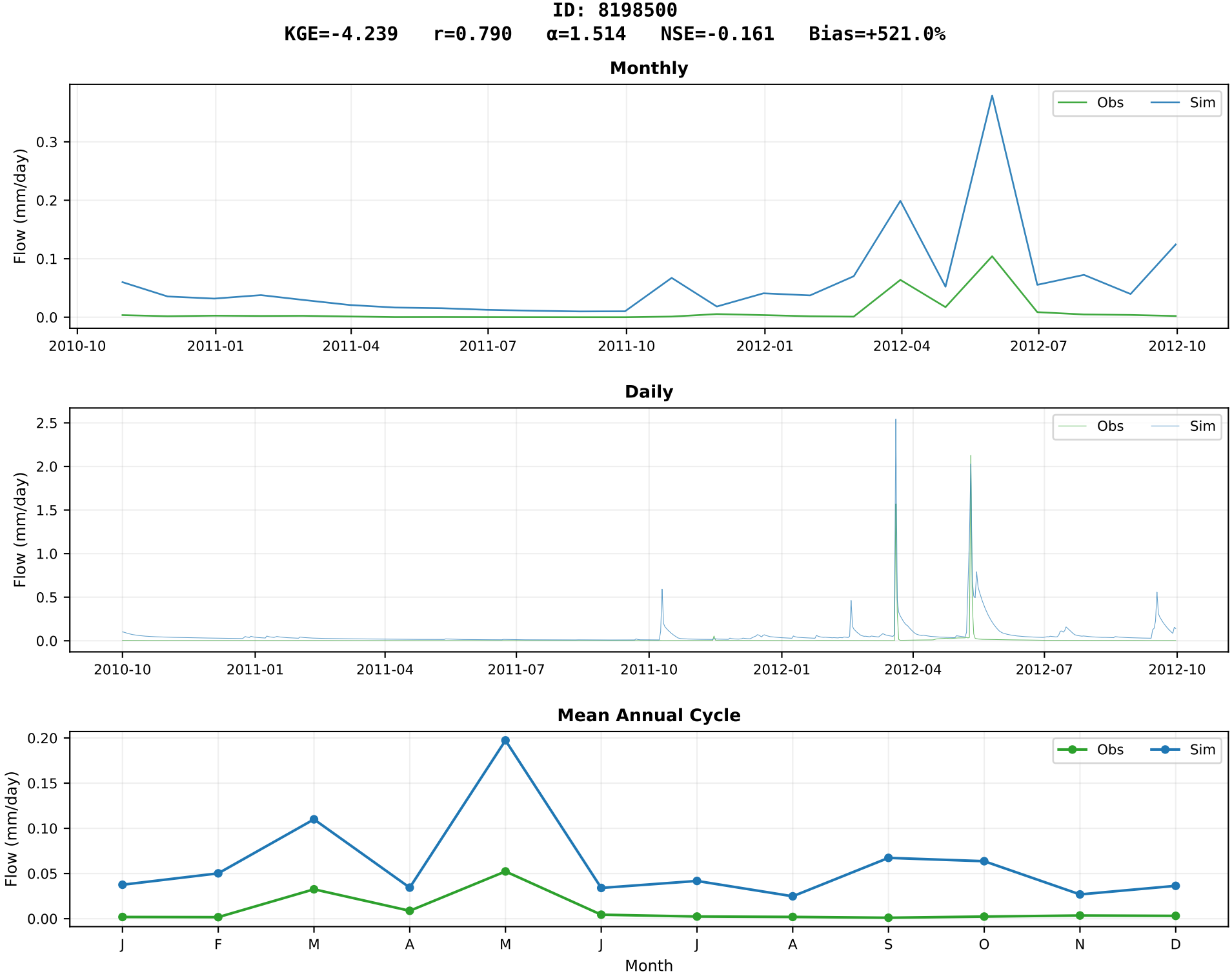

ID: 8200000
KGE=-2.480 r=0.880 α=2.251 NSE=-1.905 Bias=+324.5%
Monthly
Flow (mm/day)
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Mean Annual Cycle
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2198100
KGE=-2.892 r=0.316 α=3.772 NSE=-13.117 Bias=+264.4%
Monthly
Flow (mm/day)
Obs
Sim
Daily
Mean Annual Cycle
Month
J F M A M J J A S O N D

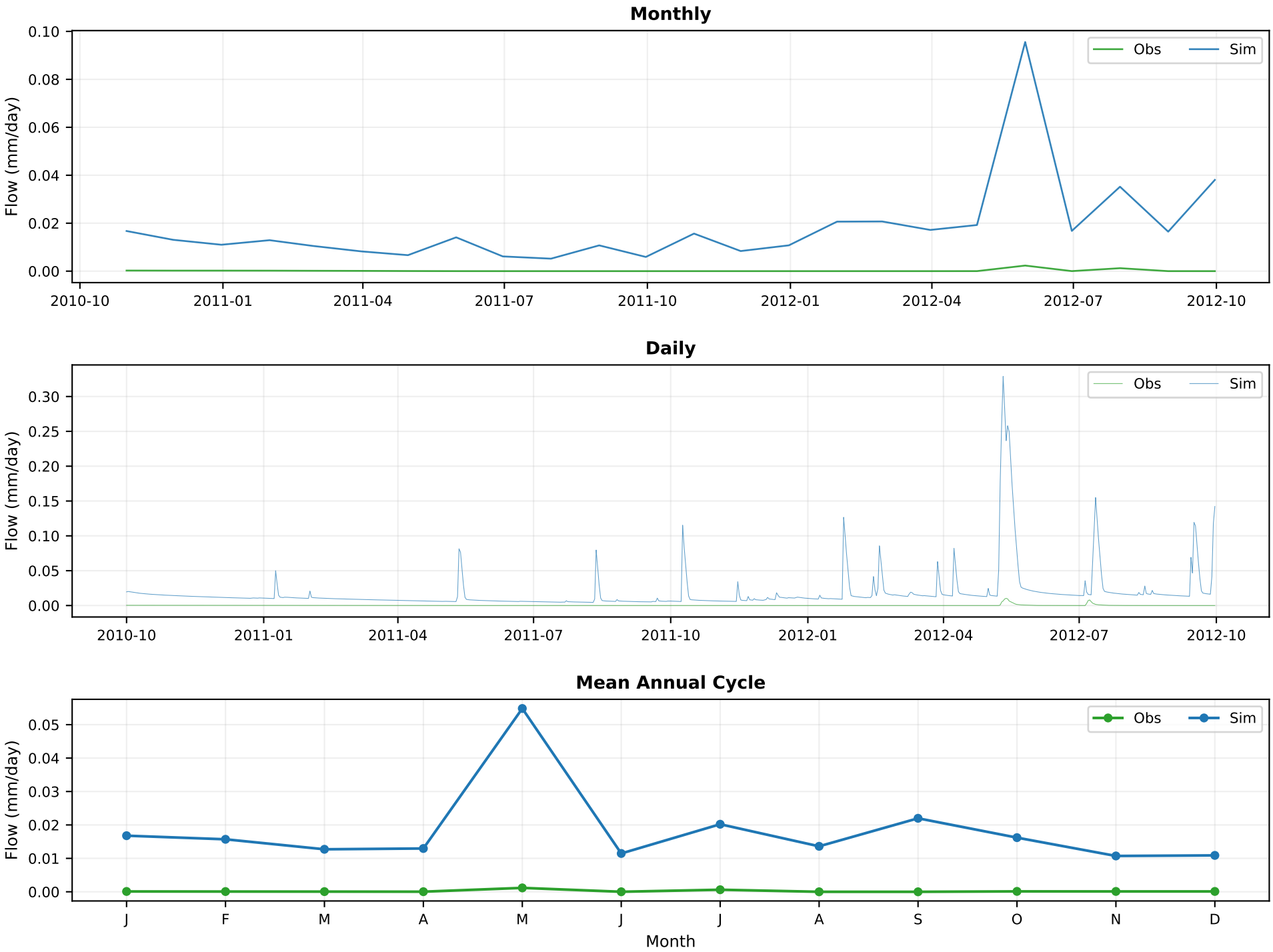

ID: 9386900
KGE=-2.926 r=0.103 α=0.600 NSE=-1.457 Bias=+380.1%
Monthly
Obs
Sim
Flow (mm/day)
0.030
0.025
0.020
0.015
0.010
0.005
0.000
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.150
0.125
0.100
0.075
0.050
0.025
0.000
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.020
0.015
0.010
0.005
0.000
J
F
M
A
M
J
J
A
S
O
N
D
Month

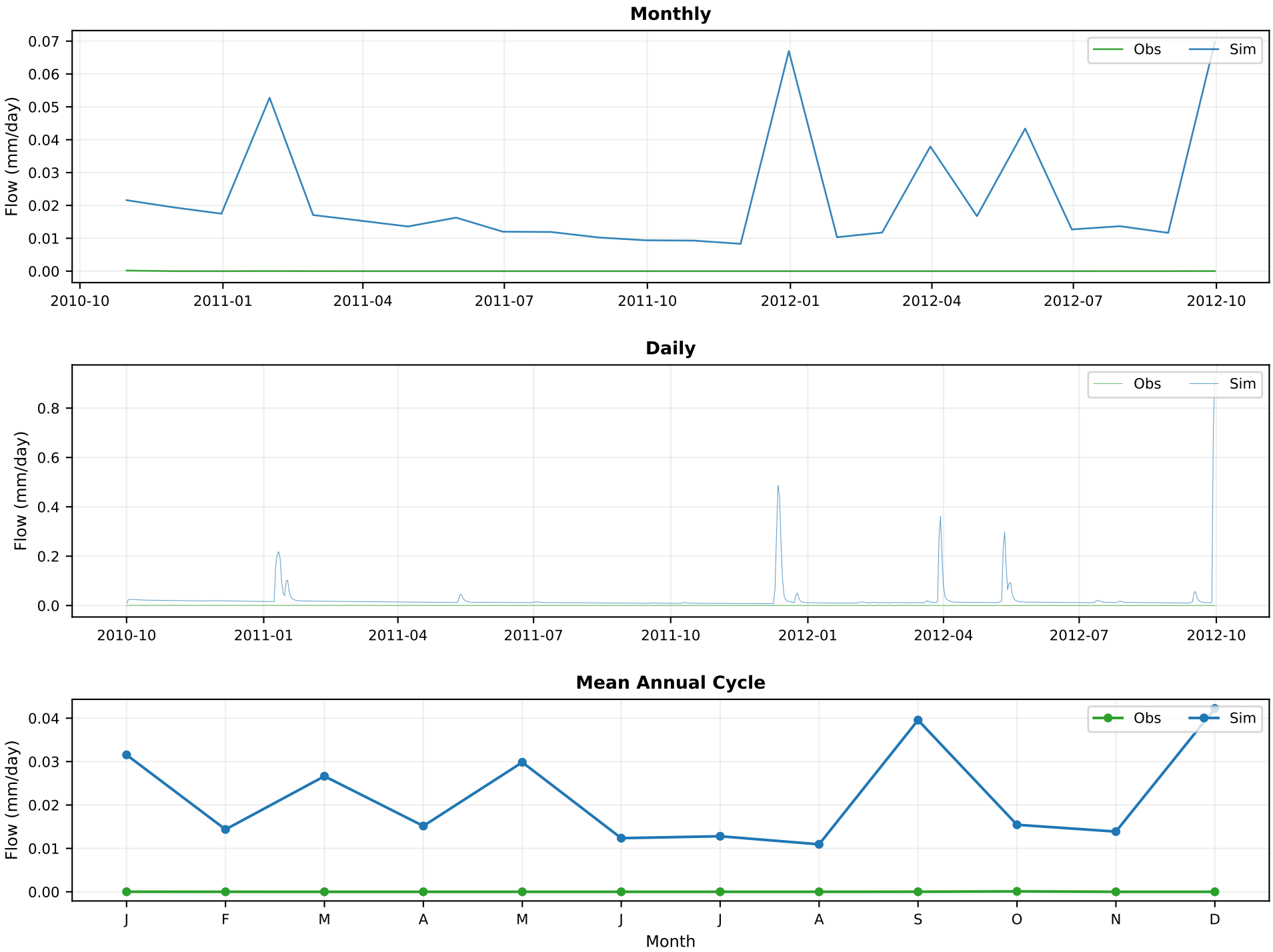

ID: 8176900
KGE=-3.097 r=0.570 α=4.525 NSE=-15.726 Bias=+204.3%
Monthly
Obs
Sim
Flow (mm/day)
0.00
0.05
0.10
0.15
0.20
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.00
0.05
0.10
0.15
0.20
J
F
M
A
M
J
J
A
S
O
N
D
Month

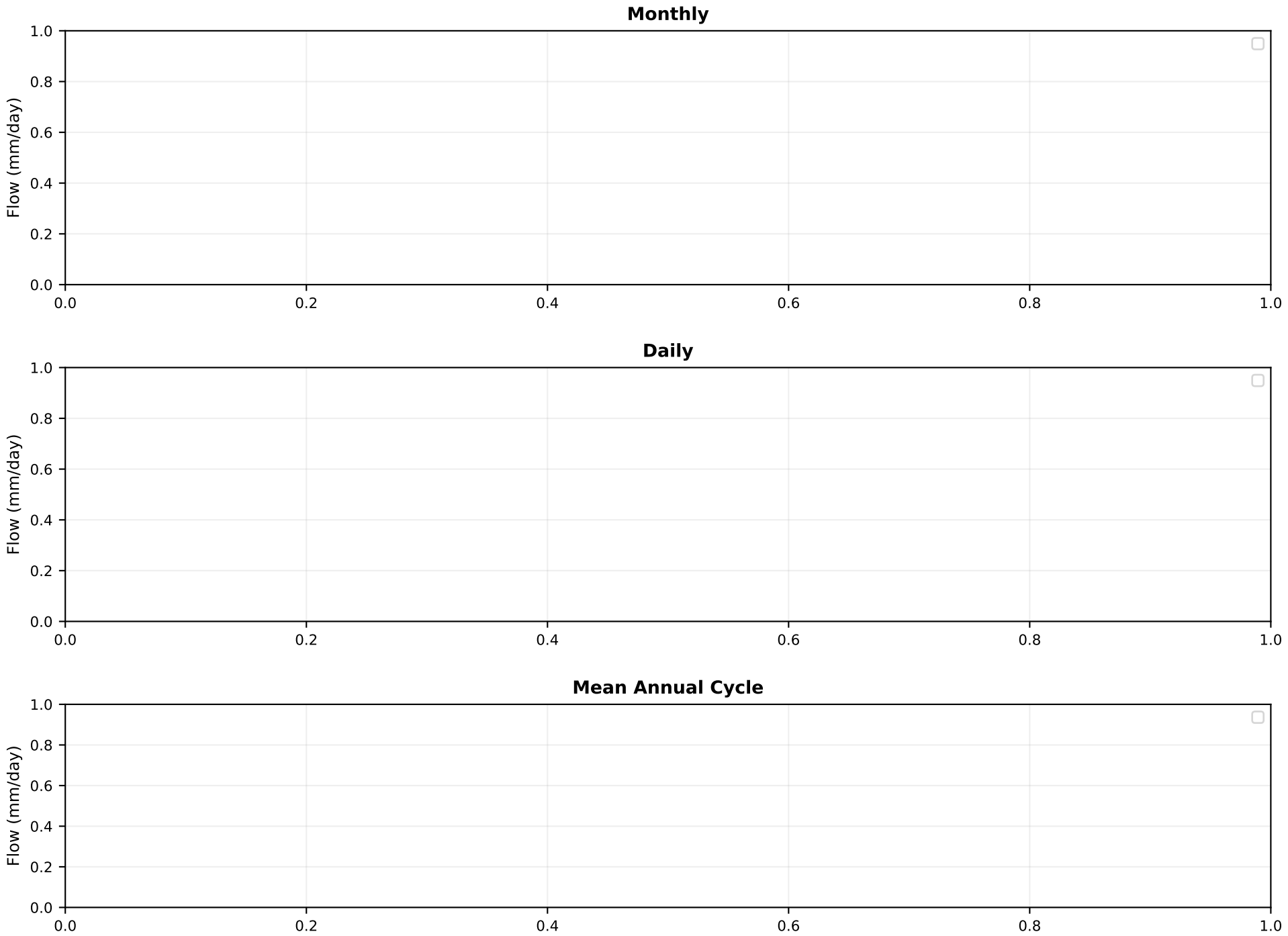

ID: 8086212
KGE=-4.198 r=0.753 α=2.496 NSE=-2.619 Bias=+497.2%
Monthly
Flow (mm/day)
Obs
Sim
0.25
0.20
0.15
0.10
0.05
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
5
4
3
2
1
0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.125
0.100
0.075
0.050
0.025
0.000
J
F
M
A
M
J
J
A
S
O
N
D
Month

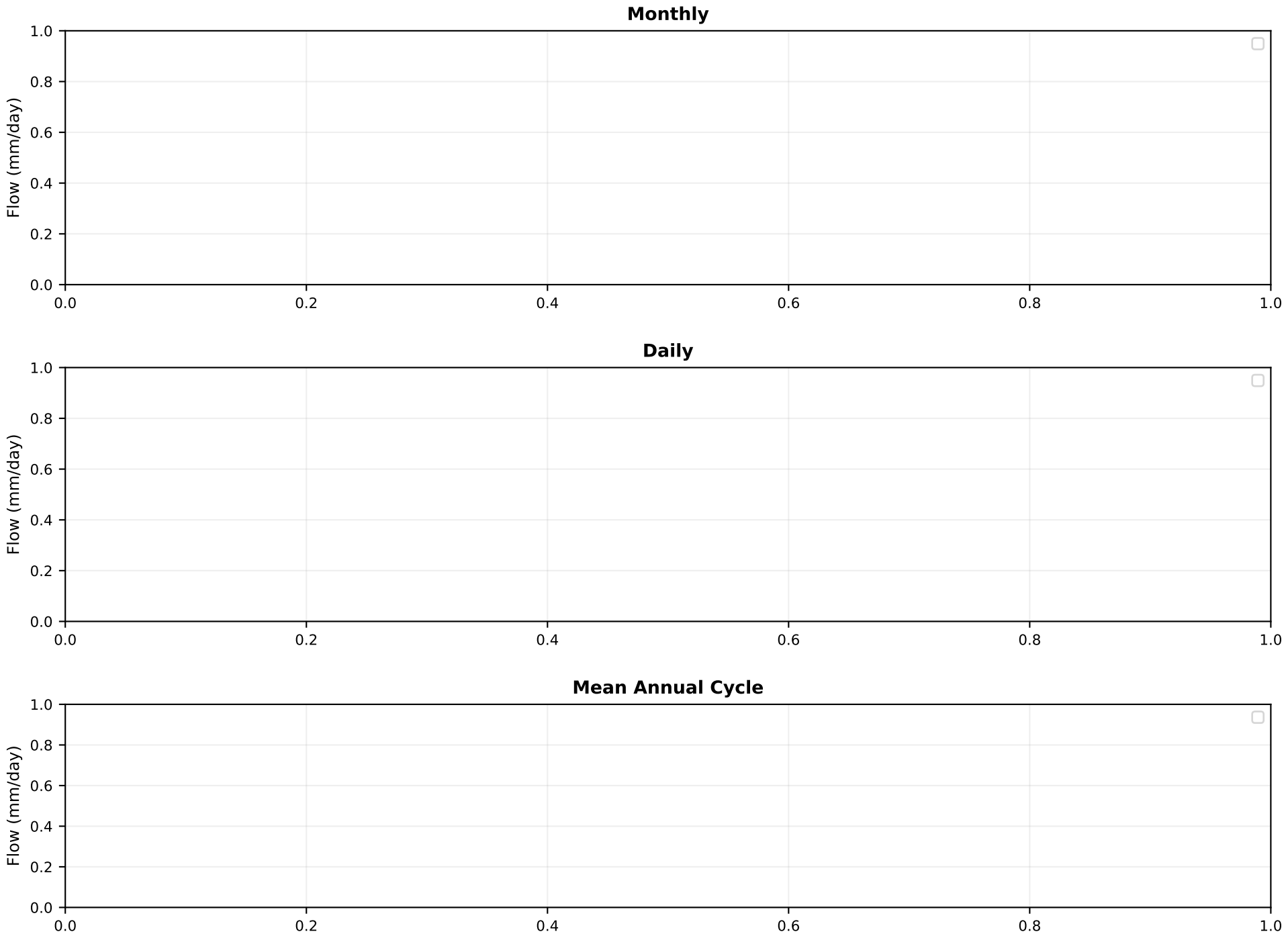

ID: 8198500
KGE=-4.239 r=0.790 α=1.514 NSE=-0.161 Bias=+521.0%
Monthly
Obs
Sim
Flow (mm/day)
0.0
0.1
0.2
0.3
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
0.0
0.5
1.0
1.5
2.0
2.5
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.00
0.05
0.10
0.15
0.20
J
F
M
A
M
J
J
A
S
O
N
D
Month

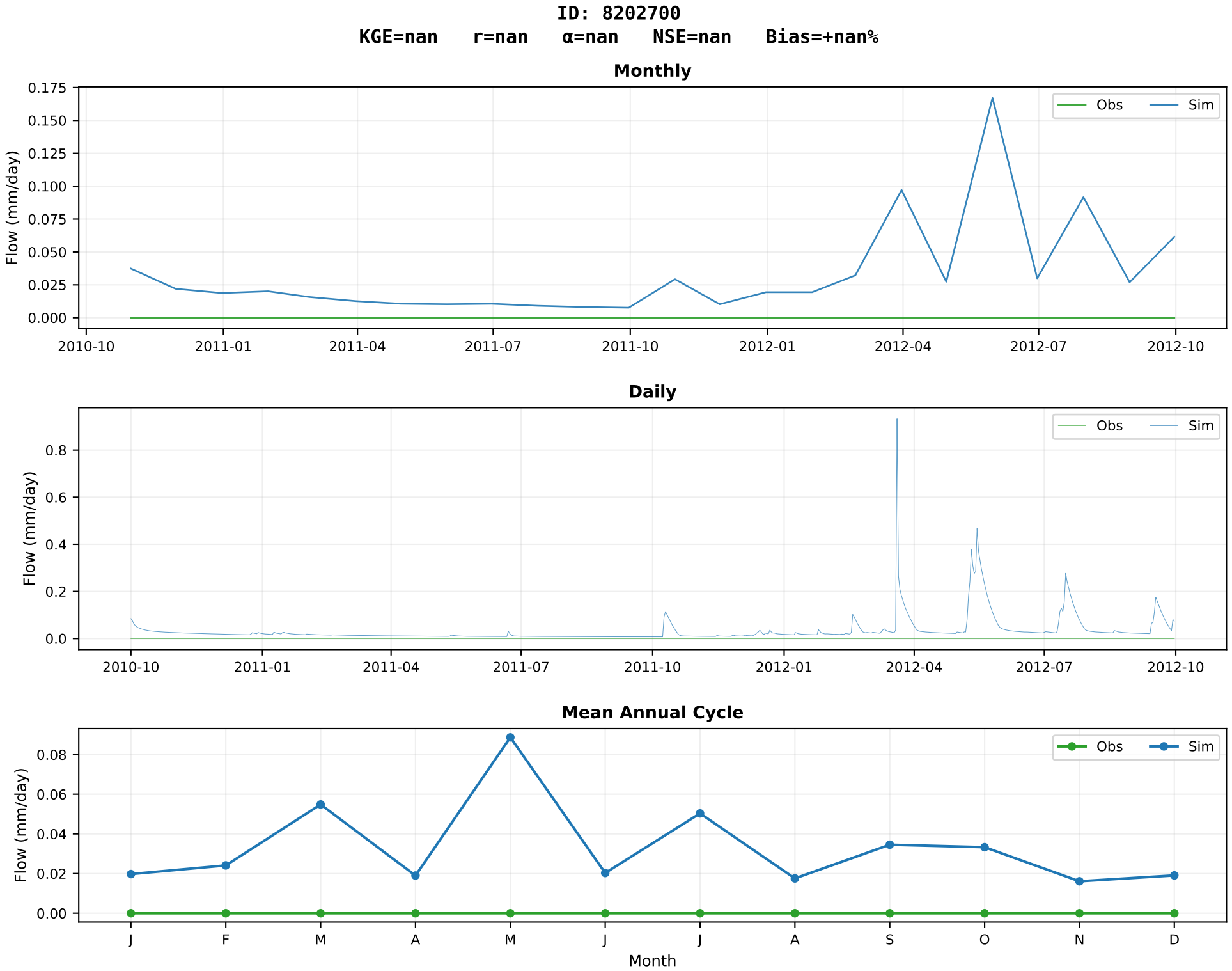

ID: 8082700
KGE=-6.729 r=0.085 α=3.240 NSE=-10.172 Bias=+734.1%
Monthly
Obs
Sim
Flow (mm/day)
0.25
0.20
0.15
0.10
0.05
0.00
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Obs
Sim
Flow (mm/day)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Obs
Sim
Flow (mm/day)
0.15
0.10
0.05
0.00
J
F
M
A
M
J
J
A
S
O
N
D
Month

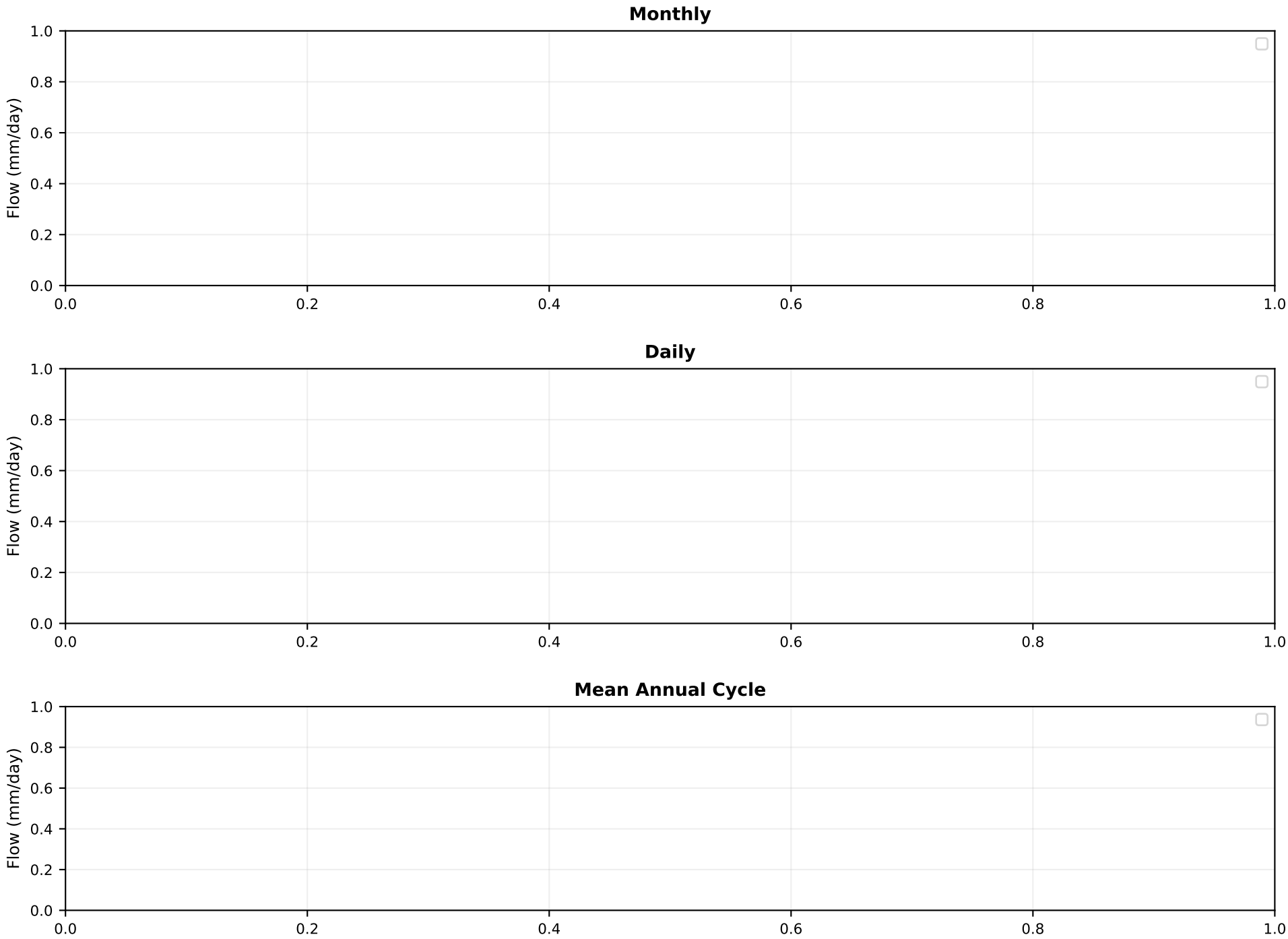

ID: 8190500
KGE=-94.152 r=0.707 α=33.793 NSE=-1470.916 Bias=+8932.2%
Monthly
Flow (mm/day)
0.10
0.08
0.06
0.04
0.02
0.00
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
0.05
0.04
0.03
0.02
0.01
0.00
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 8194200
KGE=-2304.626 r=0.056 α=925.339 NSE=-983518.250 Bias=+211222.8%
Monthly
Flow (mm/day)
Obs
Sim
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
Obs
Sim
0.0
0.2
0.4
0.6
0.8
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
Obs
Sim
0.00
0.01
0.02
0.03
0.04
J
F
M
A
M
J
J
A
S
O
N
D
Month


ID: 2118500
KGE=nan r=nan α=nan NSE=nan Bias=+nan%
Monthly
Daily
Mean Annual Cycle
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0


ID: 5057200
KGE=nan r=nan α=nan NSE=nan Bias=+nan%
Monthly
Daily
Mean Annual Cycle
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0


ID: 8202700
KGE=nan r=nan α=nan NSE=nan Bias=+nan%
Monthly
Flow (mm/day)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Daily
Flow (mm/day)
0.8
0.6
0.4
0.2
0.0
Obs
Sim
2010-10
2011-01
2011-04
2011-07
2011-10
2012-01
2012-04
2012-07
2012-10
Mean Annual Cycle
Flow (mm/day)
0.08
0.06
0.04
0.02
0.00
Obs
Sim
J
F
M
A
M
J
J
A
S
O
N
D
Month


LITTLE NORTH SANTIAM RIVER NEAR MEHAMA, OR (ID=14182500)
KGE=nan r=nan α=nan NSE=nan Bias=+nan%
Monthly
Daily
Mean Annual Cycle
Flow (mm/day)
0.0
0.2
0.4
0.6
0.8
1.0